Philips Medical Systems

AcQSim CT Service Documentation

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PHILIPS

Service Publication, Philips Medical Systems (Cleveland) Inc. Highland Hts., OH 44143

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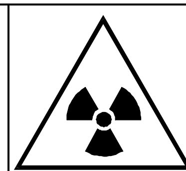
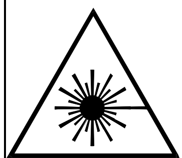
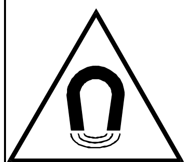
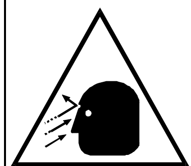
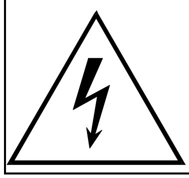
SYMBOL DESCRIPTIONS

	Attention symbol.		Radiation warning symbol.
	Laser warning symbol.		Biohazard warning symbol.
	Magnetism warning symbol.		Projectile warning symbol.
	Electrical warning symbol.		

REVISION HISTORY

REVISION	DATE	COMMENTS
–1	10/10/00	Preliminary version.
–2	12/20/01	Updated for Philips. Updated AcQImage power supply replacement, added gantry display assembly replacement. Added lasers part numbers. Updated laser calibration procedure.
A	08/12/04	Changed manual part number from T55ED–1034 (DRS No. C877:C) to 4535 671 03491. Added backup timer test to Performance section.
B	02/21/05	Added Z logs to Diagnostics section. Updated Horizontal Resolver replacement
C	07/15/05	Added X–ray tube change process links, updated tape switch part number

NOTICE

THE INFORMATION CONTAINED IN THIS MANUAL CONFORMS WITH THE CONFIGURATION OF THE EQUIPMENT AS OF THE DATE OF MANUFACTURE. REVISIONS TO THE EQUIPMENT SUBSEQUENT TO THE DATE OF MANUFACTURE WILL BE ADDRESSED IN SERVICE UPDATES DISTRIBUTED TO PHILIPS MEDICAL SYSTEMS TECHNICAL SERVICE ORGANIZATION.

TO THE USER OF THIS MANUAL

THE USER OF THIS MANUAL IS DIRECTED TO READ AND CAREFULLY REVIEW THE INSTRUCTIONS, WARNINGS AND CAUTIONS CONTAINED HEREIN PRIOR TO BEGINNING INSTALLATION OR SERVICE ACTIVITIES. WHILE YOU MAY HAVE PREVIOUSLY INSTALLED OR SERVICED EQUIPMENT SIMILAR TO THAT DESCRIBED IN THIS MANUAL, CHANGES IN DESIGN, MANUFACTURE OR PROCEDURE MAY HAVE OCCURRED WHICH SIGNIFICANTLY AFFECT THE PRESENT INSTALLATION OR SERVICE.

THE INSTALLATION AND SERVICE OF EQUIPMENT DESCRIBED HEREIN IS TO BE PREFORMED BY AUTHORIZED, QUALIFIED PHILIPS MEDICAL SYSTEM PERSONNEL. ASSEMBLERS AND OTHER PERSONNEL NOT EMPLOYED BY NOR DIRECTLY AFFILIATED WITH PHILIPS MEDICAL SYSTEMS TECHNICAL SERVICES ARE DIRECTED TO CONTACT THE LOCAL PHILIPS MEDICAL SYSTEMS OFFICE BEFORE ATTEMPTING INSTALLATION OR SERVICE PROCEDURES.

INSTALLATION AND ENVIRONMENT

EXCEPT FOR INSTALLATIONS REQUIRING CERTIFICATION BY THE MANUFACTURER PER FEDERAL STANDARDS, SEE THAT A RADIATION PROTECTION SURVEY IS MADE BY A QUALIFIED EXPERT IN ACCORDANCE WITH NCRP 102, SECTION 7, AS REVISED OR REPLACED IN THE FUTURE. PERFORM A SURVEY AFTER EVERY CHANGE IN EQUIPMENT, WORKLOAD, OR OPERATING CONDITIONS WHICH MIGHT SIGNIFICANTLY INCREASE THE PROBABILITY OF PERSONS RECEIVING MORE THAN THE MAXIMUM PERMISSIBLE DOSE EQUIVALENT.

Diagnostic Imaging Systems – MECHANICAL-ELECTRICAL WARNING

ALL OF THE MOVEABLE ASSEMBLIES AND PARTS OF THIS EQUIPMENT SHOULD BE OPERATED WITH CARE AND ROUTINELY INSPECTED IN ACCORDANCE WITH THE MANUFACTURER'S RECOMMENDATIONS CONTAINED IN THE EQUIPMENT MANUALS.

ONLY PROPERLY TRAINED AND QUALIFIED PERSONNEL SHOULD BE PERMITTED ACCESS TO ANY INTERNAL PARTS. LIVE ELECTRICAL TERMINALS ARE DEADLY; BE SURE LINE DISCONNECTS ARE OPENED AND OTHER APPROPRIATE PRECAUTIONS ARE TAKEN BEFORE OPENING ACCESS DOORS, REMOVING ENCLOSURE PANELS, OR ATTACHING ACCESSORIES.

DO NOT UNDER ANY CIRCUMSTANCES, REMOVE THE FLEXIBLE HIGH TENSION CABLES FROM THE X-RAY TUBE HOUSING OR HIGH TENSION GENERATOR AND/OR THE ACCESS COVERS FROM THE GENERATOR UNTIL THE MAIN AND AUXILIARY POWER SUPPLIES HAVE BEEN DISCONNECTED. FAILURE TO COMPLY WITH THE ABOVE MAY RESULT IN SERIOUS OR FATAL BODILY INJURIES TO THE OPERATOR OR THOSE IN THE AREA.

ELECTRICAL-GROUNDING INSTRUCTIONS

THE EQUIPMENT MUST BE GROUNDED TO AN EARTH GROUND BY A SEPARATE CONDUCTOR. THE NEUTRAL SIDE OF THE LINE IS NOT TO BE CONSIDERED THE EARTH GROUND. ON EQUIPMENT PROVIDED WITH A LINE CORD, THE EQUIPMENT MUST BE CONNECTED TO PROPERLY GROUNDED, THREE-PIN RECEPTACLE. DO NOT USE A THREE-TO-TWO PIN ADAPTER.

Diagnostic Imaging Systems – RADIATION WARNING

X-RAY AND GAMMA-RAYS ARE DANGEROUS TO BOTH OPERATOR AND OTHERS IN THE VICINITY UNLESS ESTABLISHED SAFE EXPOSURE PROCEDURES ARE STRICTLY OBSERVED.

THE USEFUL AND SCATTERED BEAMS CAN PRODUCE SERIOUS OR FATAL BODILY INJURIES TO ANY PERSONS IN THE SURROUNDING AREA IF USED BY AN UNSKILLED OPERATOR. ADEQUATE PRECAUTIONS MUST ALWAYS BE TAKEN TO AVOID EXPOSURE TO THE USEFUL BEAM, AS WELL AS TO LEAKAGE RADIATION FROM WITHIN THE SOURCE HOUSING OR TO SCATTERED RADIATION RESULTING FROM THE PASSAGE OF RADIATION THROUGH MATTER.

THOSE AUTHORIZED TO OPERATE, PARTICIPATE IN OR SUPERVISE THE OPERATION OF THE EQUIPMENT MUST BE THOROUGHLY FAMILIAR AND COMPLY COMPLETELY WITH THE CURRENT ESTABLISHED SAFE EXPOSURE FACTORS AND PROCEDURES DESCRIBED IN PUBLICATIONS, SUCH AS: SUBCHAPTER J OF TITLE 21 OF THE CODE OF FEDERAL REGULATIONS, "DIAGNOSTIC X-RAY SYSTEMS AND THEIR MAJOR COMPONENTS", AND THE NATIONAL COUNCIL ON RADIATION PROTECTION (NCRP) NO. 102, "MEDICAL X-RAY AND GAMMA-RAY PROTECTION FOR ENERGIES UP TO 10 MEV-EQUIPMENT DESIGN AND USE", AS REVISED OR REPLACED IN THE FUTURE.

THOSE RESPONSIBLE FOR PLANNING OF X-RAY AND GAMMA-RAY EQUIPMENT INSTALLATIONS MUST BE THOROUGHLY FAMILIAR AND COMPLY COMPLETELY WITH NCRP NO. 49, "STRUCTURAL SHIELDING DESIGN AND EVALUATION FOR MEDICAL OF X-RAYS AND GAMMA-RAYS OF ENERGIES UP TO 10 MEV", AS REVISED AND REPLACED IN THE FUTURE. FAILURE TO OBSERVE THESE WARNINGS MAY CAUSE SERIOUS OR FATAL BODILY INJURIES TO THE OPERATOR OR THOSE IN THE AREA.

TABLE OF CONTENTS

ACQSIM CT INSTALLATION – INTRODUCTION

This AcQSim CT Installation Manual contains all the information necessary to physically install, power up, and do the initial checks for a Philips Medical Systems AcQSim CT Computed Tomographic (CT) system.

Additional items, such as Laser Cameras, Remote Diagnostic Stations, Patient Support pads and rails, injectors, or options from Health Care Products, are <u>not</u> covered in this document.

WHAT THIS MANUAL CONTAINS

This Section – introduces you to the material in this manual and contains the 'model' for the installation (i.e., the basic installation steps). This section contains enough information for the experienced FSE to install a scanner.

Section One – contains general descriptions of the AcQSim CT scanner and descriptions of the individual components

Section Two – describes the site preparations required prior to the actual installation. Uncrating, handling, and placement of components is described.

Section Three – contains the step–by–step installation procedure and connection to the in–house utilities.

Section Four – explains power–up procedures.

Section Five – Section Five contains quick checks and the X–ray tube alignment procedure to be performed prior to system image calibration.

ADDITIONAL DOCUMENTATION REQUIRED

An additional manual is necessary for the installation of this scanner:

- Service Applications (Calibration, Configuration and Performance Sections)

The Service Publications department supplies complete documentation, including these manuals, with each scanner. The actual documentation shipped depends on the type of scanner and the ordered options. See the appropriate documentation shipping list for the documents shipped with your scanner.

WHO THIS MANUAL IS FOR

This manual has been written for currently employed Philips Medical Systems International personnel, or those individuals who have been trained by Philips Medical Systems, for the express purpose of maintaining, servicing, or repairing Philips Medical Systems AcQSim CT scanners.

HOW TO USE THIS MANUAL

- **Step–by–Step Information** – Begin with this section and continue through the manual, doing each section in sequence.

INSTALLATION SEQUENCE

The flow chart on the following page shows that not all procedures in this manual or the referenced manuals are required for a correct and proper installation.:

THINGS TO DO TO MAKE THE INSTALL GO EASIER

1. Visit the site ahead of time to ensure the suite is being prepared properly and on time. Check:
   a. Is floor level?
   b. Is power/power conditioner installed?
   c. Have proper length Gantry–to–Recon Tower cables been ordered (50 ft. or 75 ft.)?
2. Are the installation tools (drills, anchors, etc.) available and ready to go?
3. Is the documentation available or being shipped to the proper location?
4. Find a source of distilled water or purchase some.
5. Find the phantom and fill it with the distilled water. Set it aside during the installation so that the bubbles have a chance to clear out (during the installation time) so that it is ready to go when you need it.
6. Bring your calibration and diagnostics manuals.
7. Follow the installation model.

ORDERING PARTS

Replaceable parts for AcQSim CT scanner are listed in the AcQSim CT Parts manual.

EQUIPMENT DESCRIPTION

The AcQSim CT scanner consists of three major groups: Patient Support, Gantry and Operator’s Console. Figure 1 illustrates a typical AcQSim CT system.

GANTRY

The Gantry contains the ”front end” electronics to collect Patient data as well as the X–ray system, which is mounted around the rotating frame. As a general statement, the Gantry ’electronics’ consists of the following sub–assemblies:

- X–ray Tube
- X–ray Generator
- Detectors
- Main and Secondary motor drives
- Gantry card file
- Power supplies for Cardfile and Detector Ring

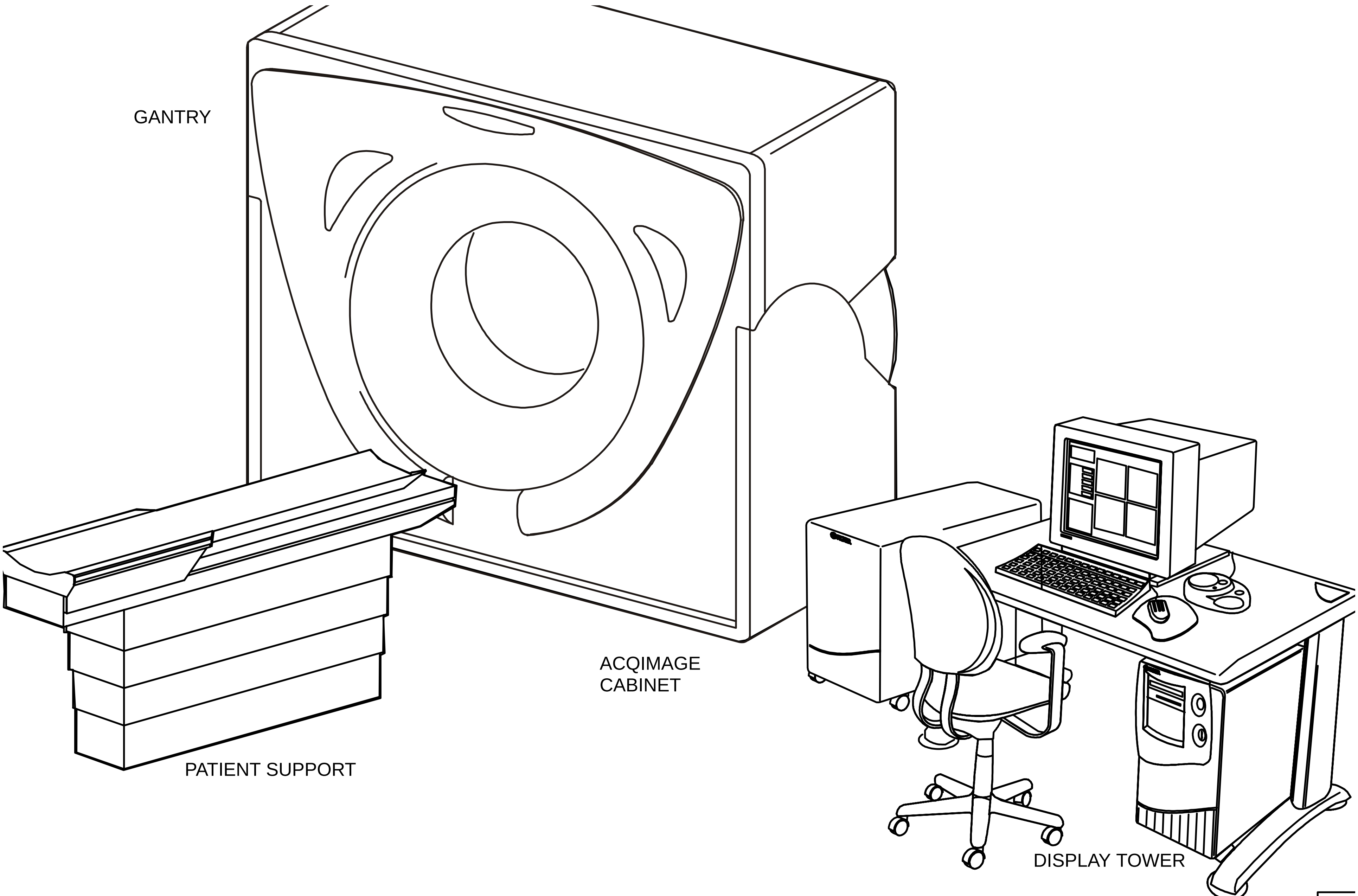


FIGURE 1 EXAMPLE OF ACQSIM CT SYSTEM

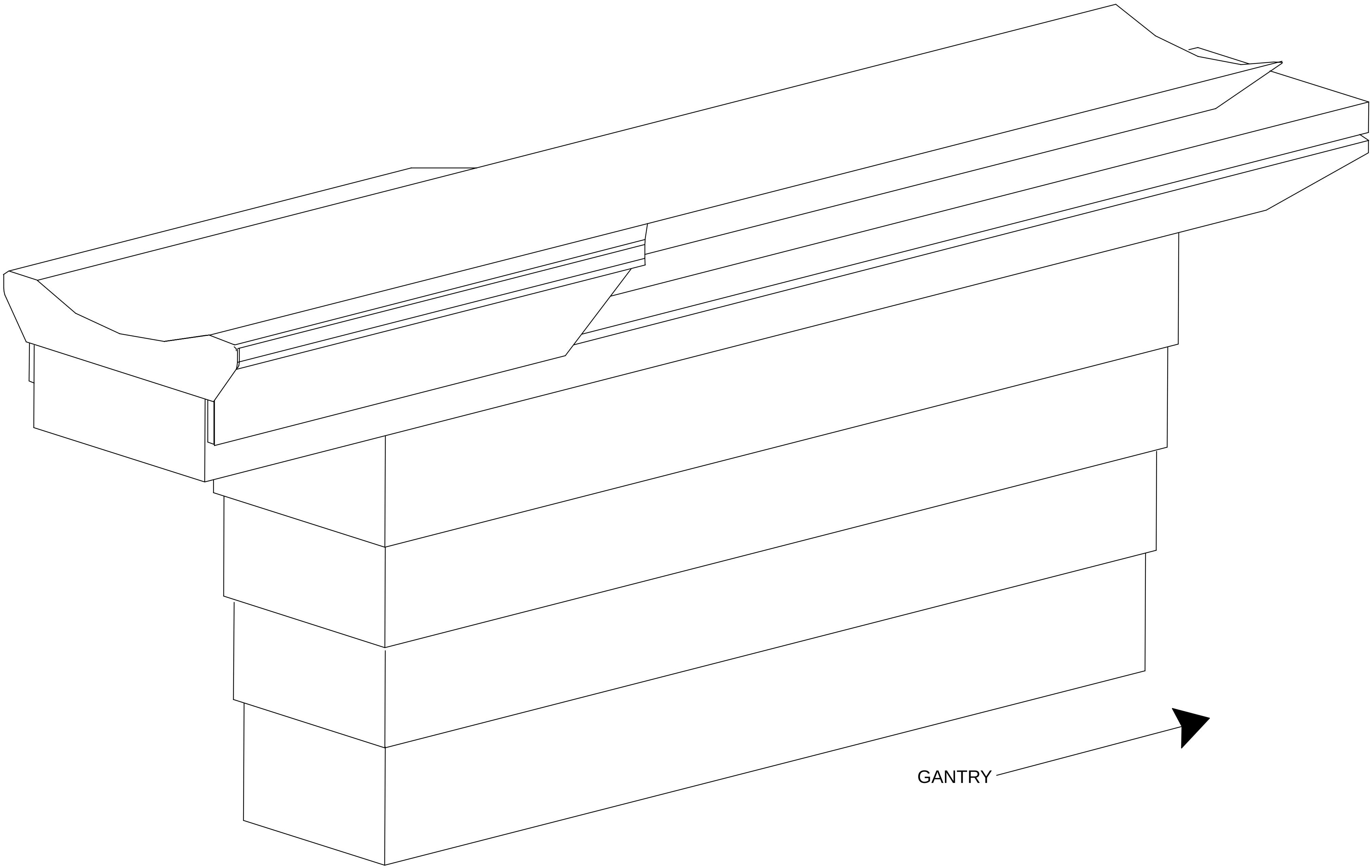


FIGURE 2 PATIENT SUPPORT, FULLY ASSEMBLED

OPERATOR’S CONSOLE

(SHOWN WITHOUT ACQIMAGE CABINET)

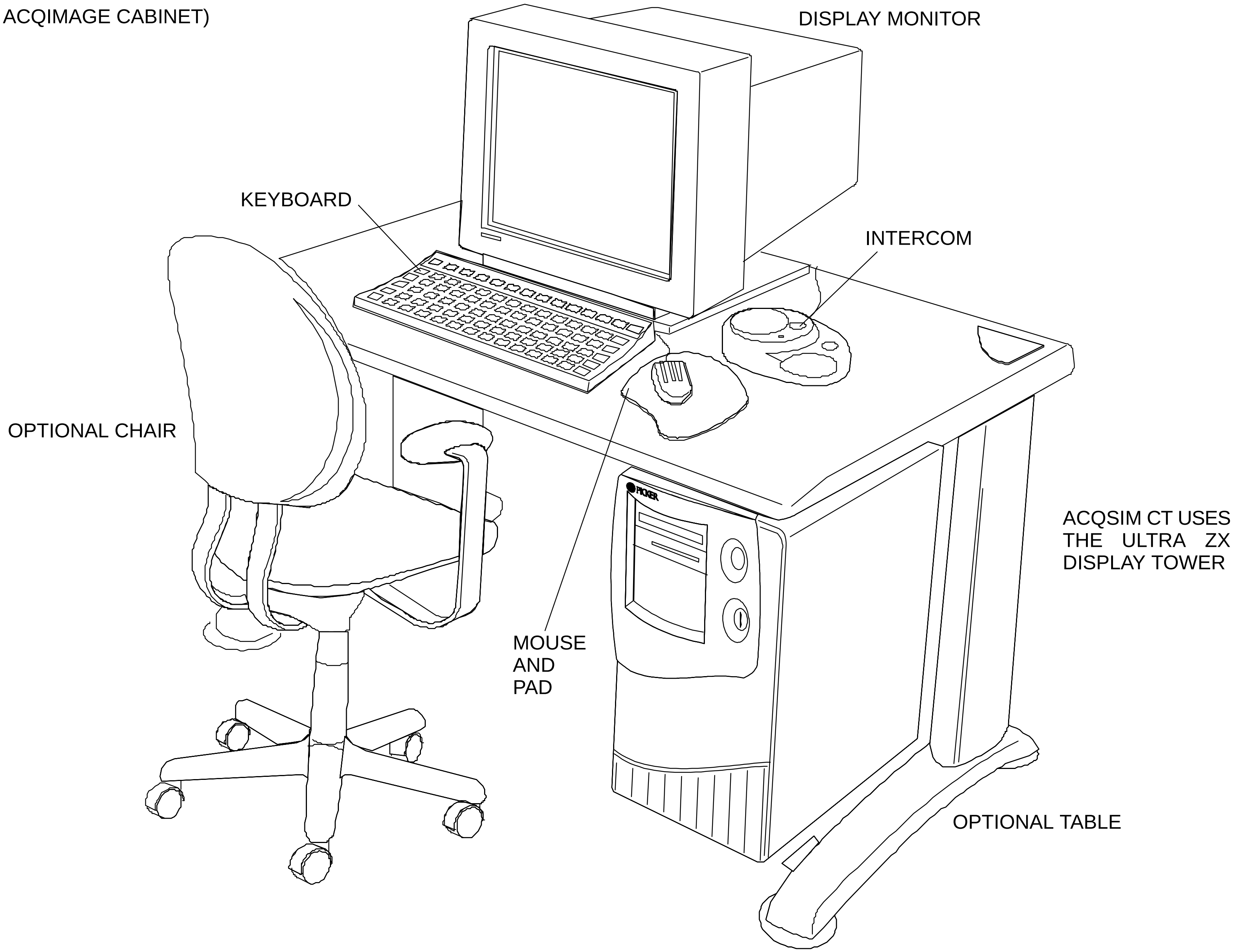


Figure 3 shows the standard boards and their locations for the Recon Tower:

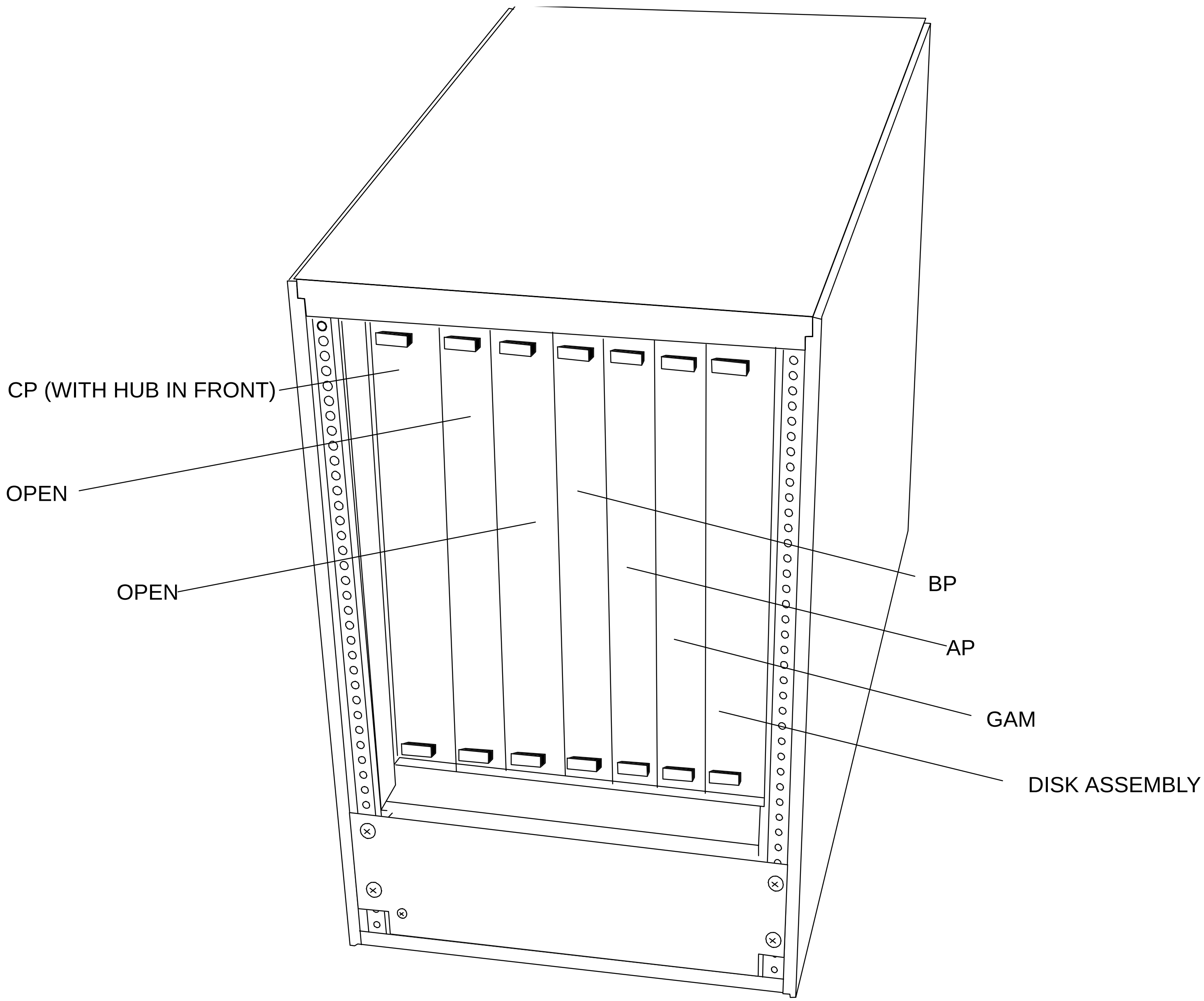


FIGURE 3 ACQIMAGE CABINET BOARD LOCATIONS – TYPICAL CONFIGURATION

DIMENSIONS AND WEIGHTS

All dimensions (width, depth, height) are in inches and millimeters and weights are in pounds and kilograms.

ACQSIM CT DIMENSIONS AND THERMAL OUTPUTS					
ASSEMBLY	**WIDTH**	**DEPTH**	**HEIGHT**	**WEIGHT**	**BTU/HR**
Display Tower	13" / 33 CM	21.5" / 54.6 CM	25.5" / 64.8 CM	115 LBS / 52.2 KG	300
AcQImage Cabinet	14.5" / 36.8 CM	31" / 78.7 CM	27" / 68.6 CM	150 LBS. / 68 KG	1200
Gantry	__ IN. / 223.5 CM	__ IN. / 86.4 CM	__ IN. / 195.6 CM	___ LB / 1545 KG	____
Patient Support	22 IN. / 55.9 CM	101 IN. / 256.5 CM	40 IN. / 101.6 CM	600 LB / 273 KG	

PRIMARY POWER REQUIREMENTS – Incoming and Utilization Power

Volts AC	Phase	Voltage Regulation	Momentary kVA/Amps	Utilization Voltage
480	3	+/– 5 %	75/90	432 to 528

> **NOTE**
>
> Line–to–line voltages given are limits of any one phase set (A–B, B–C, C–A). Phase–to–phase voltages must be within 5% of each other.

Isolation Transformer And Wire Sizing. In all cases, the site planning guide and requirements of the Site Planning department shall prevail.

Voltage Variation	Distribution Transformer	Wire Size Require vs. Distance (ft./m)				
		50/15	100/30	200/61	Neutral	Ground
–5% / +10%	75 kVA	#2	#2	1/0	#6	(1) Same as power

ENVIRONMENTAL REQUIREMENTS

68 to 75 Degrees F. (20 TO 23.9 Degrees C.), 30 to 60% humidity (non–condensing).

Important: Via ServeCom, refer to http://servecom.picker.com/forms/ and complete the appropriate form.

UNCRATING, HANDLING AND PLACEMENT PROCEDURE

This section provides information on:

- physically handling the scanner
- requirements for any lifting equipment used
- uncrating the Gantry, Patient Support, AcQImage Cabinet and Display Tower
- removing the Gantry shipping bracket
- visually checking the placement of the Gantry and the Patient Support

⚠ **WARNING**

LIFTING HAZARD. INSTALLATION PERSONNEL MUST USE DISCRETION WHEN LIFTING OR MOVING CRATES OR COMPONENTS. MANUAL LIFTING OF ITEMS THAT ARE TOO HEAVY CAN RESULT IN SERIOUS INJURY TO PERSONNEL AND DAMAGE TO THE EQUIPMENT OR BOTH. INSTALLATION PERSONNEL

⚠ **CAUTION**

POSSIBLE EQUIPMENT DAMAGE. DO NOT USE FORKLIFT TO MOVE OR LIFT GANTRY. MOVE THE GANTRY ON THE TRANSPORT CASTERS OR ATTACH A PROPER LIFTING DEVICE TO THE CHAIN SHACKLES. USING A FORKLIFT WILL DAMAGE THE GANTRY.

THE GANTRY MUST BE LIFTED OFF THE SKID BY A LIFTING MACHINE CAPABLE OF HANDLING AT LEAST 6.000 POUNDS (2730 KG.). FAILURE TO HAVE THE CORRECT LIFTING MACHINERY CAN RESULT IN DAMAGE TO THE GANTRY.

NOTE

The AcQSim CT is shipped with the side covers NOT installed.
Refer to the **GANTRY COVERS** procedure on the next page to install them correctly.

GANTRY COVERS

The following procedure shows how to open/close, install, remove/replace and adjust the various AcQSim CT Gantry covers.

Front Upper Cover

1. Loosen the captive screws on the front cover, lift up and secure. Refer to Figure 4.

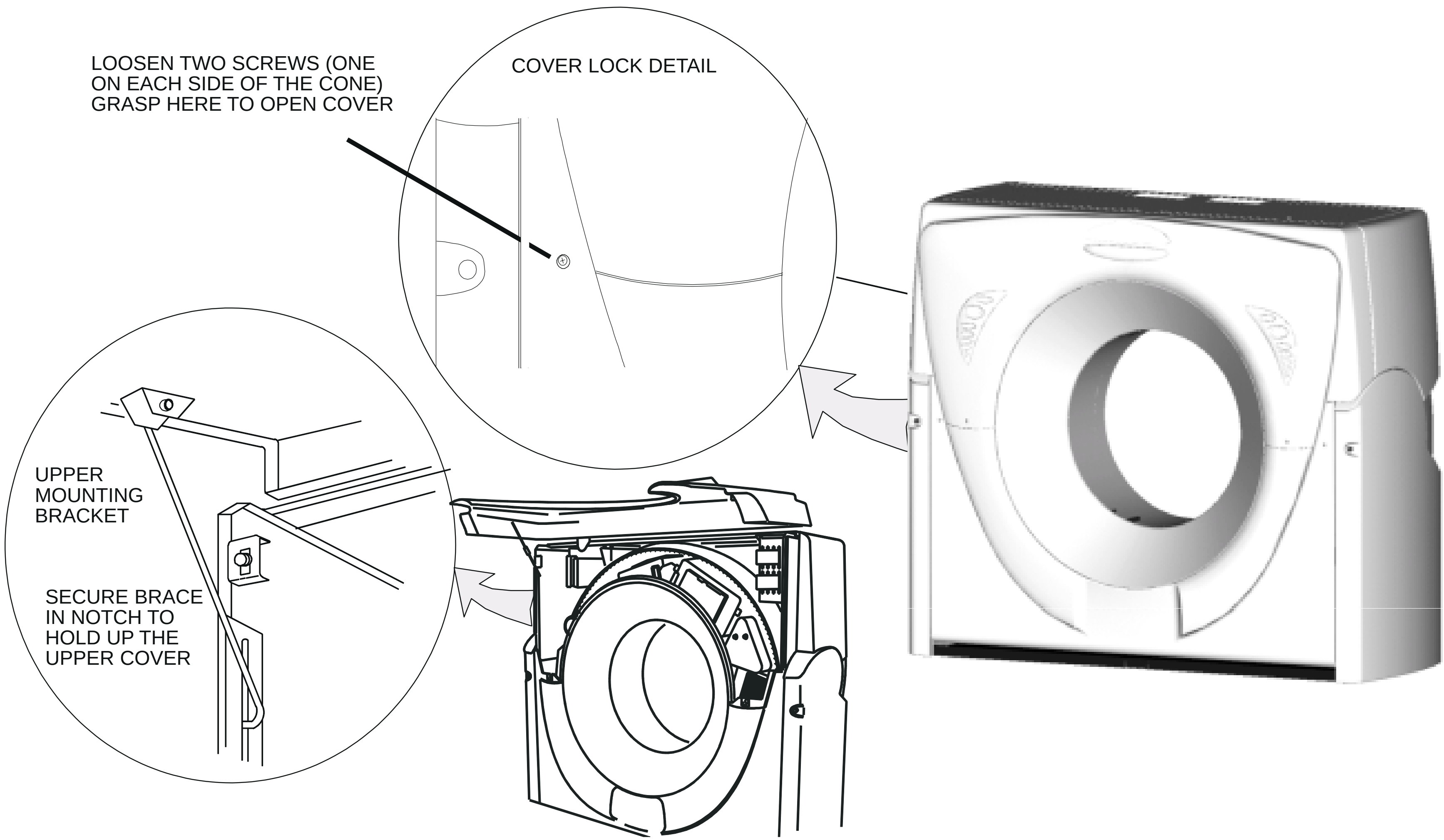


FIGURE 4 GANTRY UPPER PANEL OPENING

Lower Front Cover

1. Remove the screws that secure the lower Gantry cover. Lift the cover up and off the bottom latches and slide the cover away from Gantry. Store the cover in a secure place.. Refer to Figure 5.

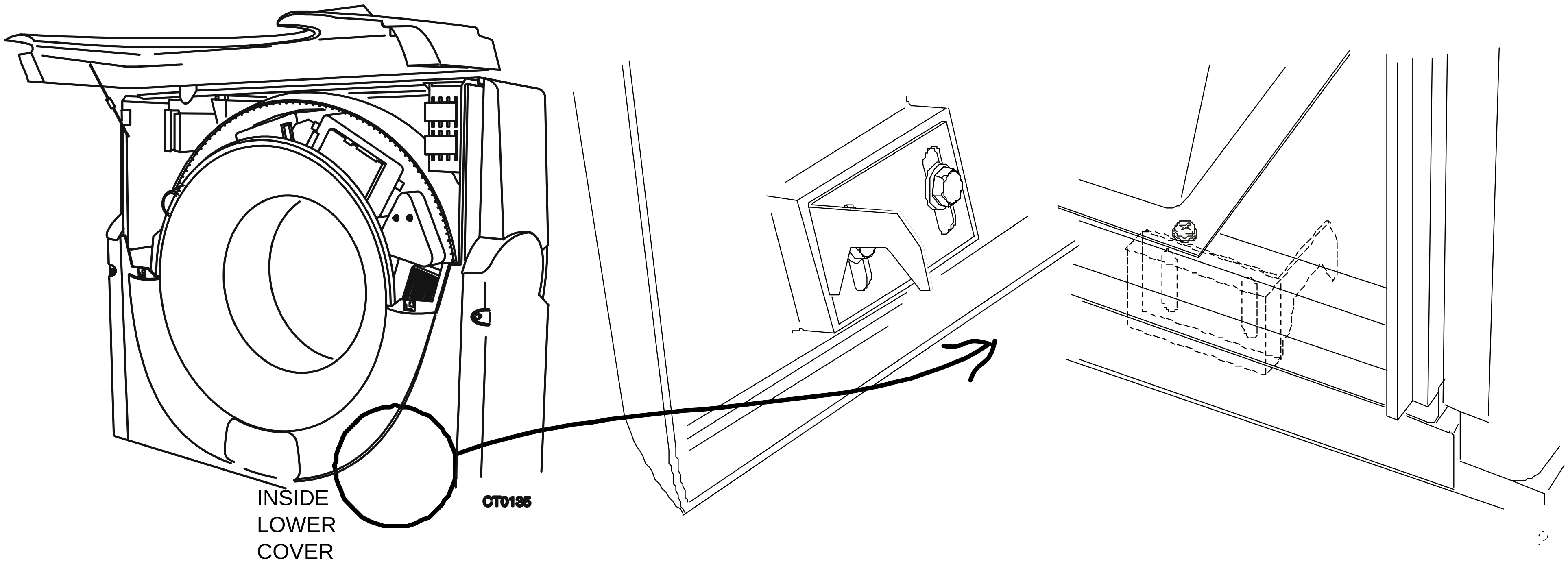


FIGURE 5 GANTRY FRONT LOWER COVER – DETAIL

Rear Covers

> ⚠ **CAUTION**
>
> Before removing the rear covers, unplug the two connectors from the cables that lead to the speaker and x–ray light. Failure to do this will result in damage to the cables and/or connectors.

1. Remove the screws that secure the rear Gantry covers. Lift up and away on each of the covers and set aside in a safe place. Refer to Figure 6.

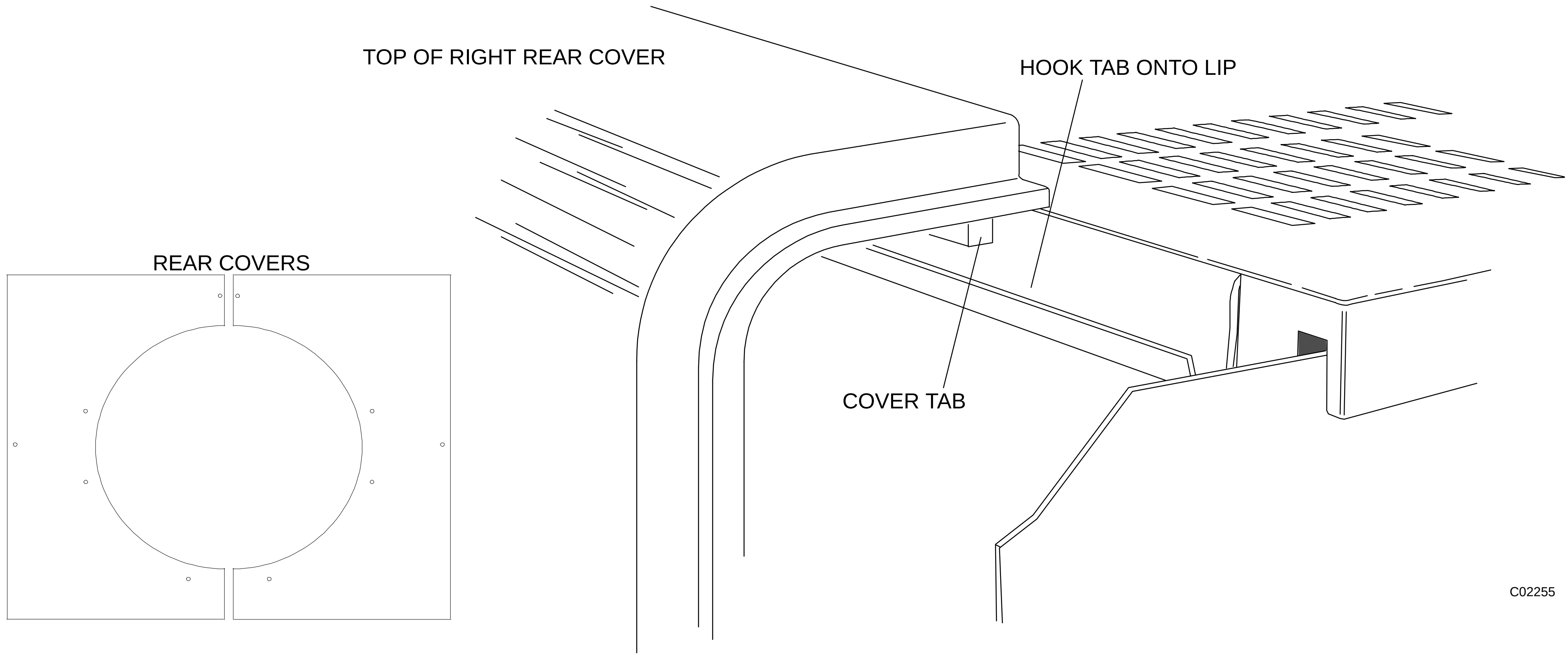


FIGURE 6 REAR COVER REMOVAL

Lower Side Cover Installation

1. Remove Lower Right Column Cover (p/n 312863) from shipping box.
2. Remove the small bag attached to the inside surface of the cover, which contains the mounting screws and washers. Set it aside.
3. Open the Front Upper Cover and lock in place. Uninstall the E–Stop Switch located on the right side column.
4. Install the E–Stop Switch in the Lower Right Side Column. Refer to Figure 7.

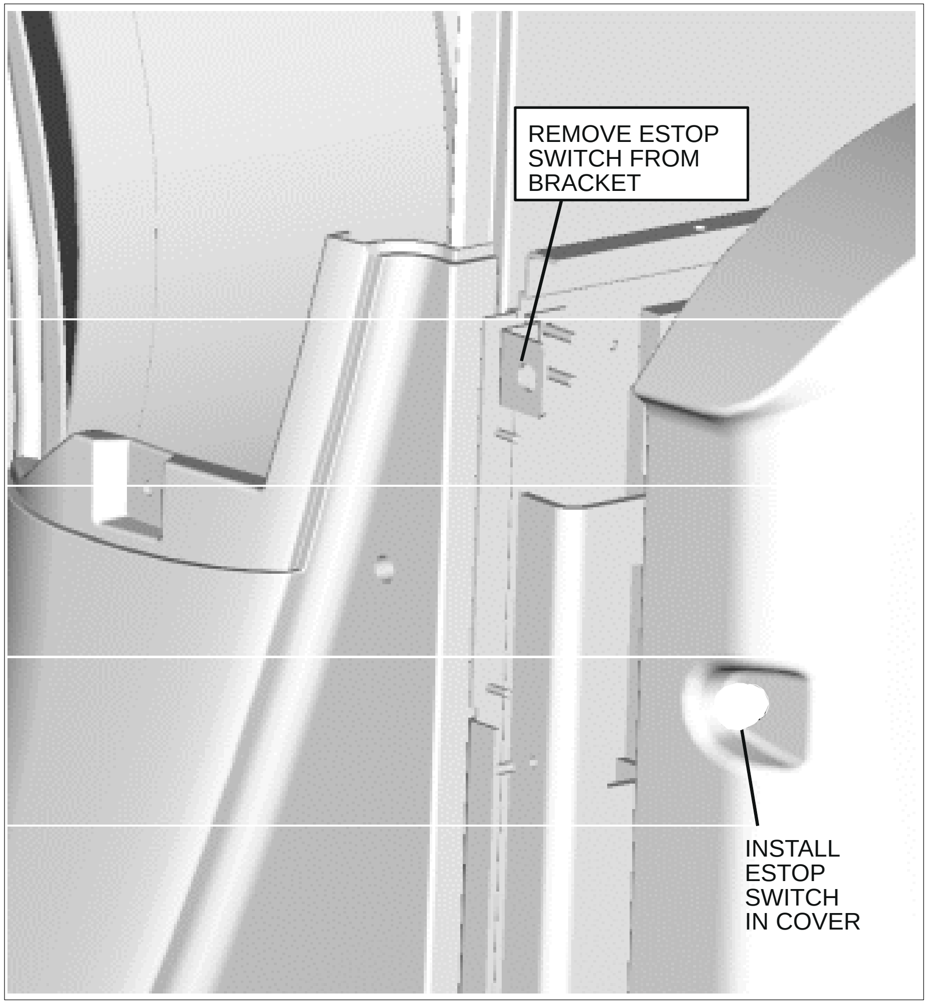


FIGURE 7   ESTOP SWITCH LOCATION

5. Install the Lower Right Column Cover onto the right side column. Align and insert Locating pins on Lower Right Column Cover with the V groove features on the Side Column. Refer to Figure 8.

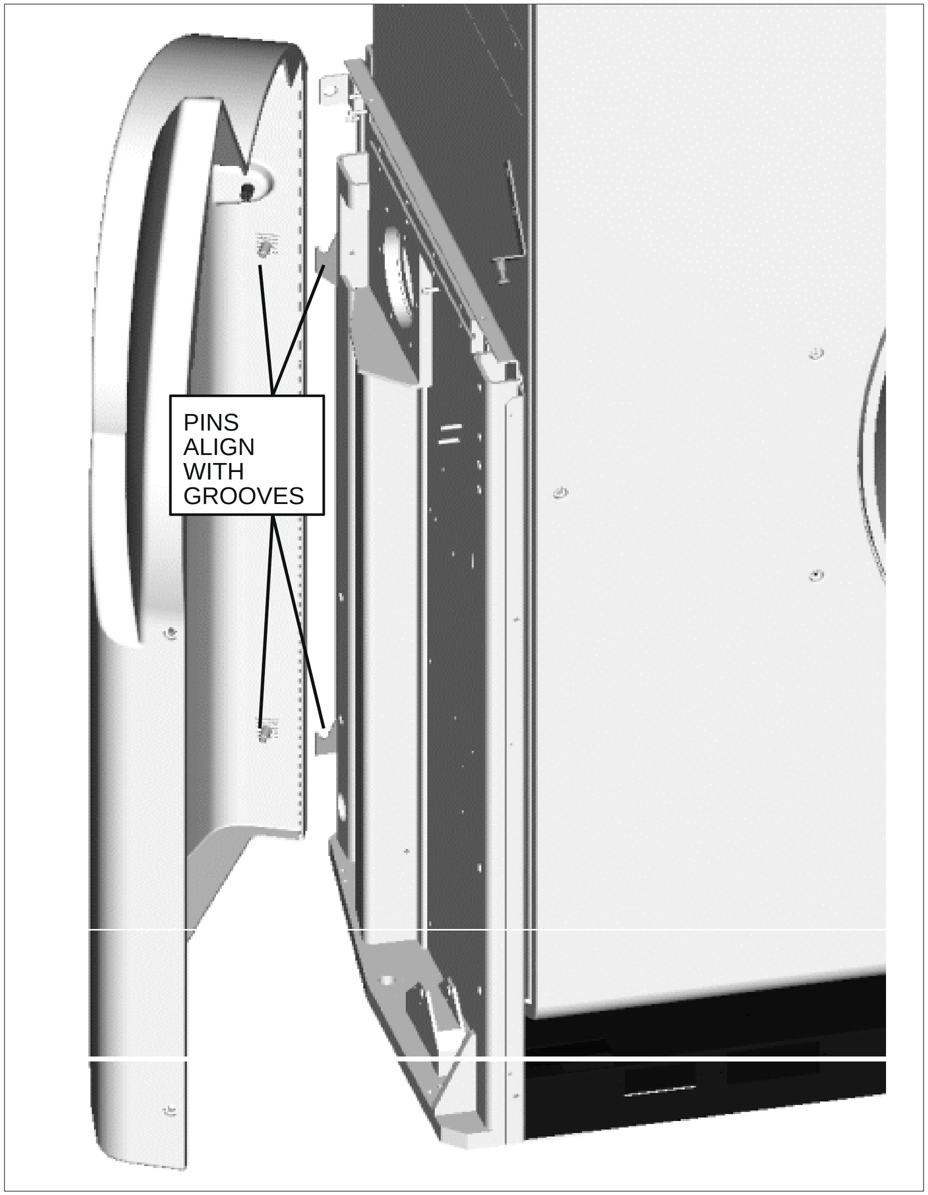


FIGURE 8 LOWER RIGHT COLUMN COVER

6. Fasten the Lower Right Column Cover to the Gantry Column using the two screws (1/4”–20 x 3/4”, Pan Head) and two nylon washers provided. Maintain a gap between the top of the Column Cover and Gantry Side Panel in the range of 0.050 to 0.250 inch. Refer to Figure 9.

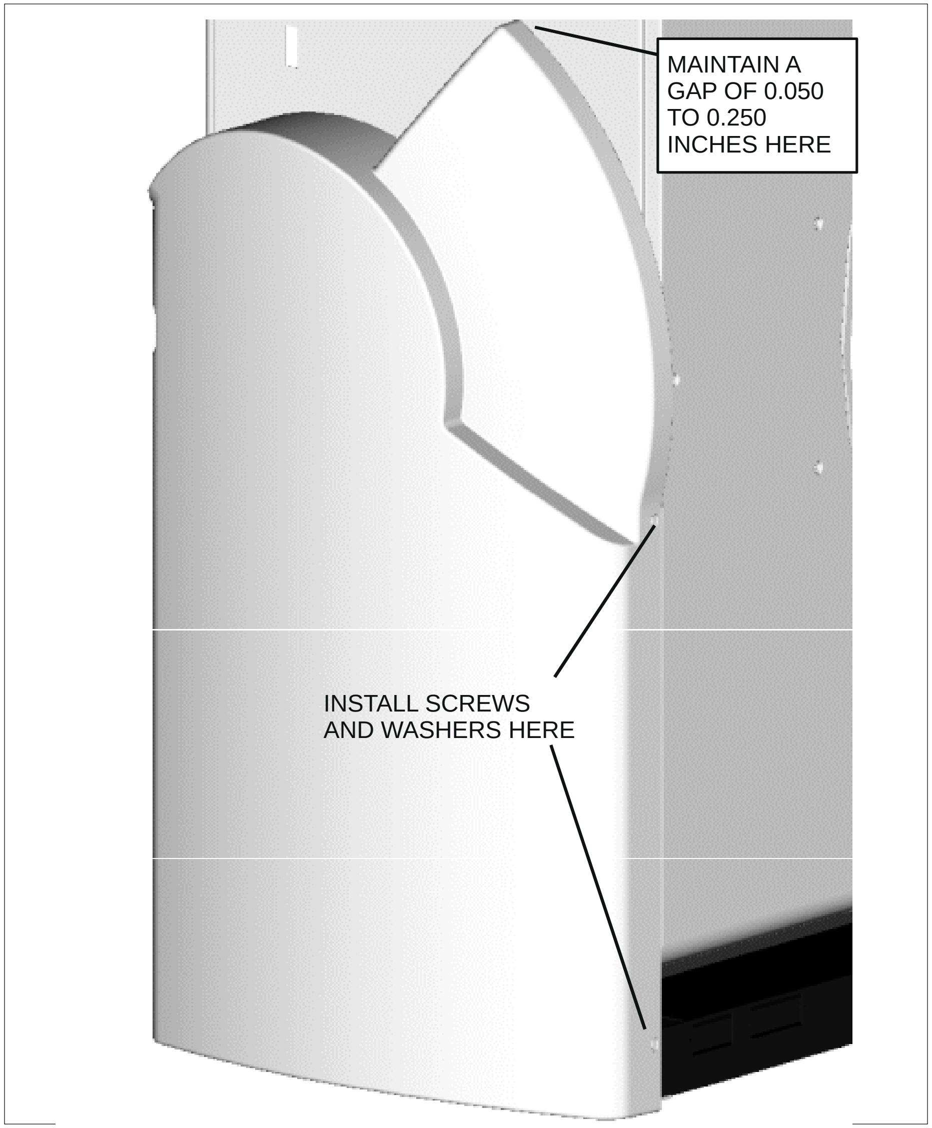


FIGURE 9 FASTENING LOWER SIDE COVER

7. Lower the Front Upper Cover. Make sure the Front Upper Cover is seated firmly against the Lower Front Cover.

8. Check the clearance between the Lower Right Column Cover and the Front Upper Cover as shown in Figure 10. It should measure between 0.100 to 0.250 inch. If the gap measures within the allowable range, assemble the Lower Left Side Cover (p/n 312865) in the same fashion as the right cover. If not, continue to step 9.

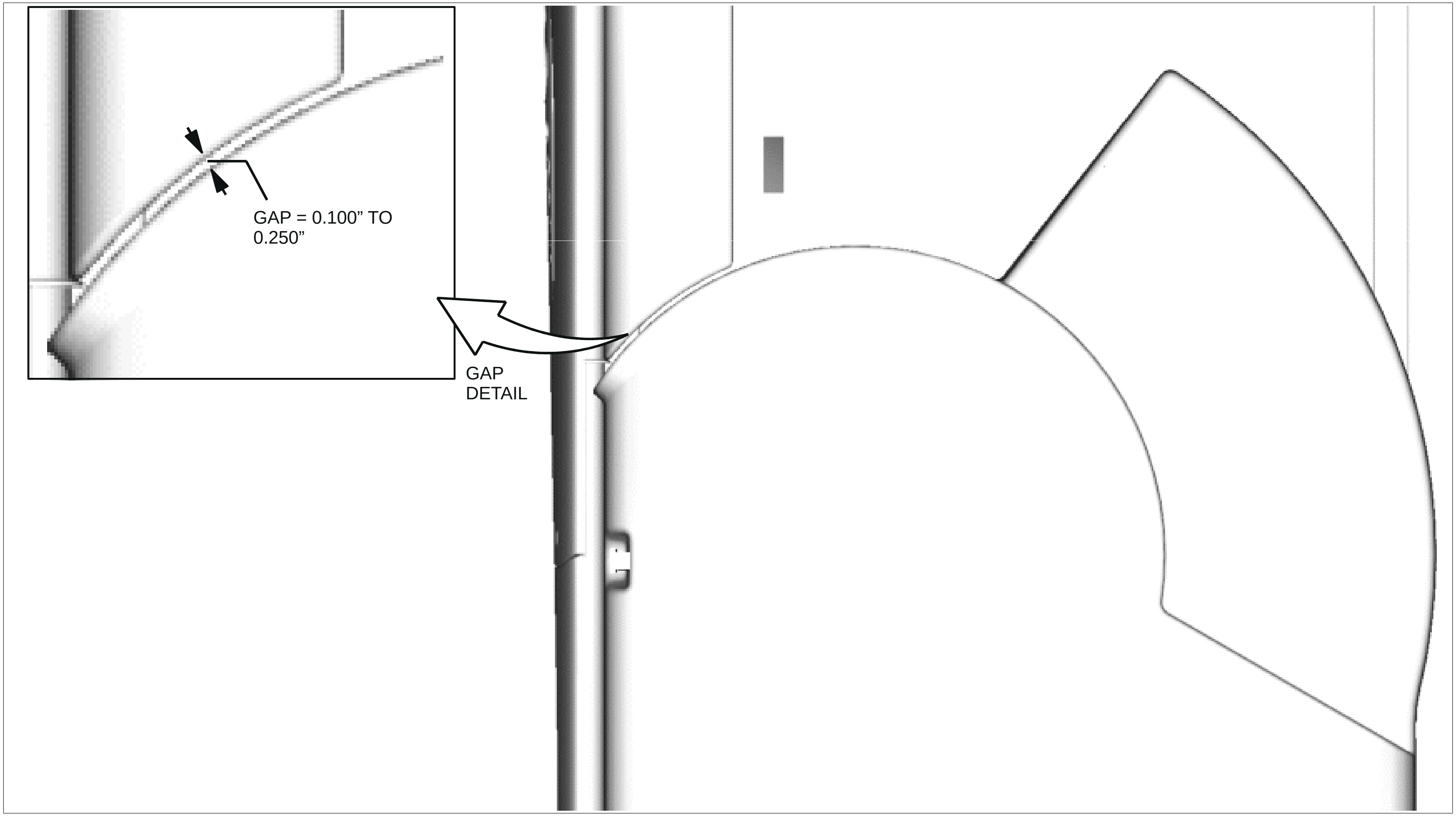


FIGURE 10   GAP BETWEEN SIDE COVER AND FRONT UPPER COVER

9. If the gap is not within the allowable range, raise the Front Upper Cover, remove the Lower Right Column Cover, and adjust the height of the V–groove Mounting Bracket. Refer to Figure 11. Continue to check the clearance between the Lower Right Column Cover and the Front Upper Cover and make sure it is within the allowable range of 0.100 to 0.250 inch.

10. Prior to final installation of Lower Right Side Cover, make sure wires are reconnected to E–Stop Switch.

11. Assemble the Lower Left Side Cover (p/n 312865) in the same fashion as the right cover.

FIGURE 11 V–GROOVE BRACKET ADJUST

UPPER SIDE COVER INSTALLATION

1. Remove Upper Right Column Cover (p/n 362096) from shipping box.
2. Align and insert the pins on the Upper Right Column Cover into the holes on the Cover Mounting Bracket which is a attached to the Gantry Side Panel. Refer to Figure 12.

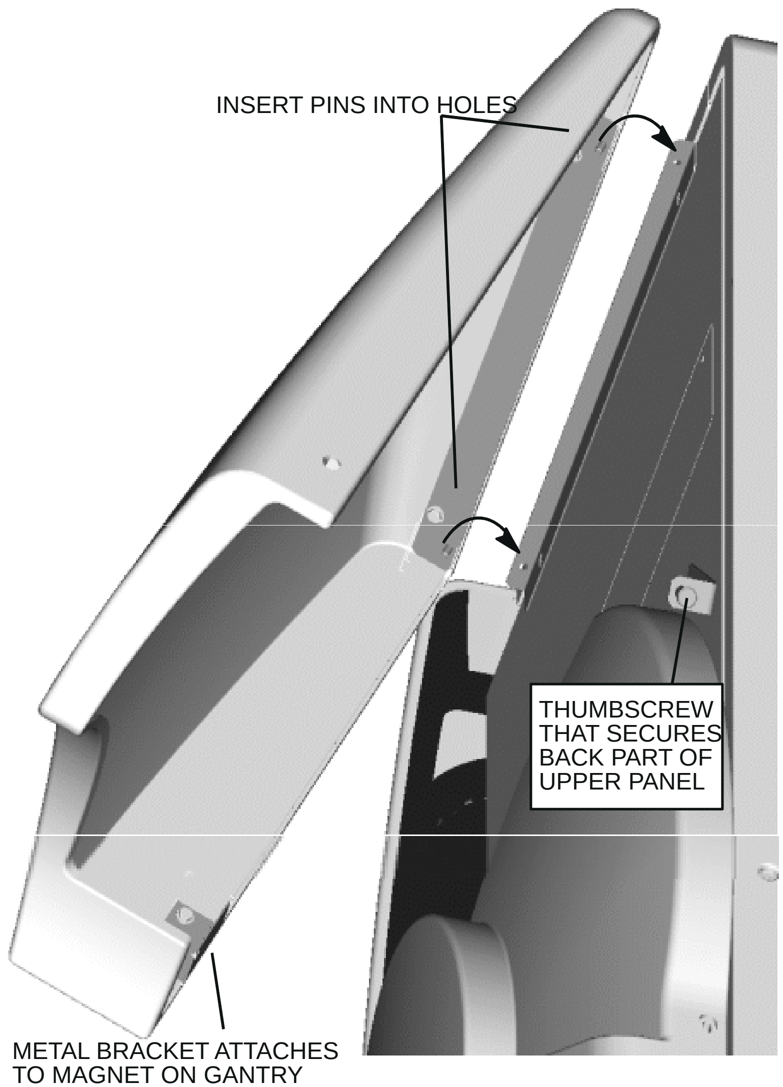


FIGURE 12 UPPER SIDE COVER

3. Swing the lower side of the Upper Right Column Cover down until it comes into contact with the Gantry Side Panel. A magnetic latch should attract and hold the cover in place.

4. Visually locate the thumbscrew through the hole on the back side of the Upper Right Column Cover. Insert a long, narrow slotted screwdriver into the hole and lightly tighten the thumbscrew. This is a blind connection, so ensure you get the right “feel” of the captive screw starting correctly. Refer to Figure 13.

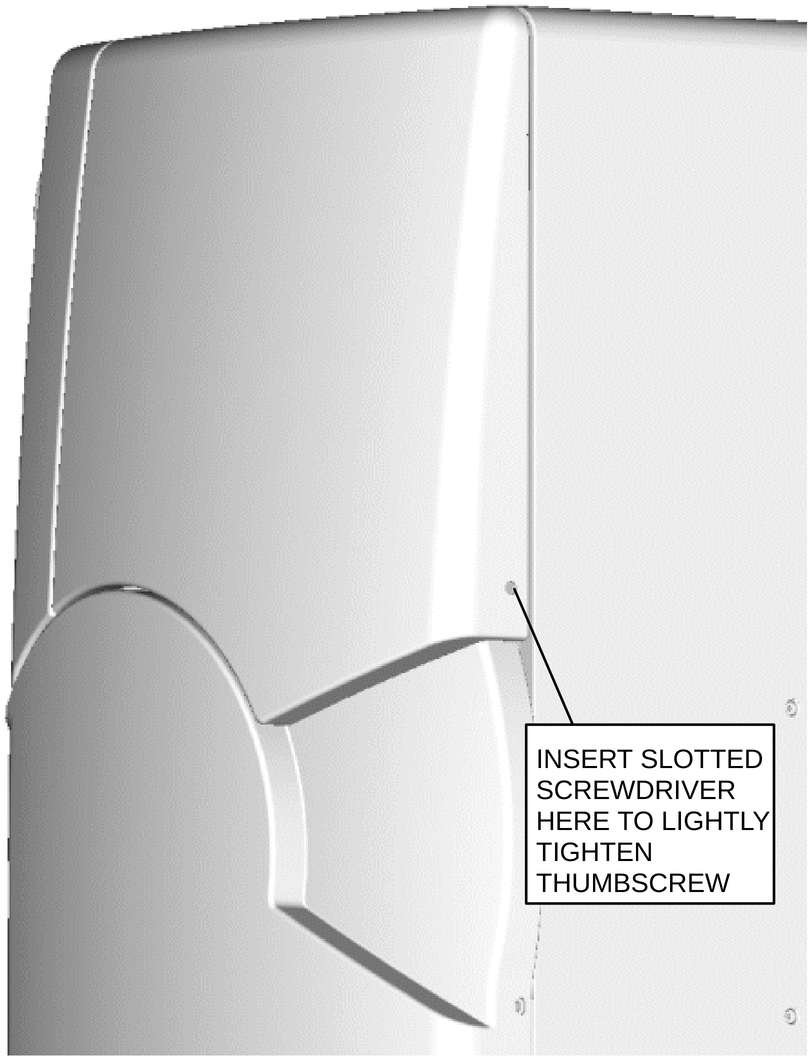


FIGURE 13 THUMBSCREW LOCATION

5. Check clearance "A" between the Upper Right Column Cover and Lower Right Column Cover. Also check the uniformity of the gap between the Upper Right Column Cover and the Upper Front Cover. Refer to Figure 14.

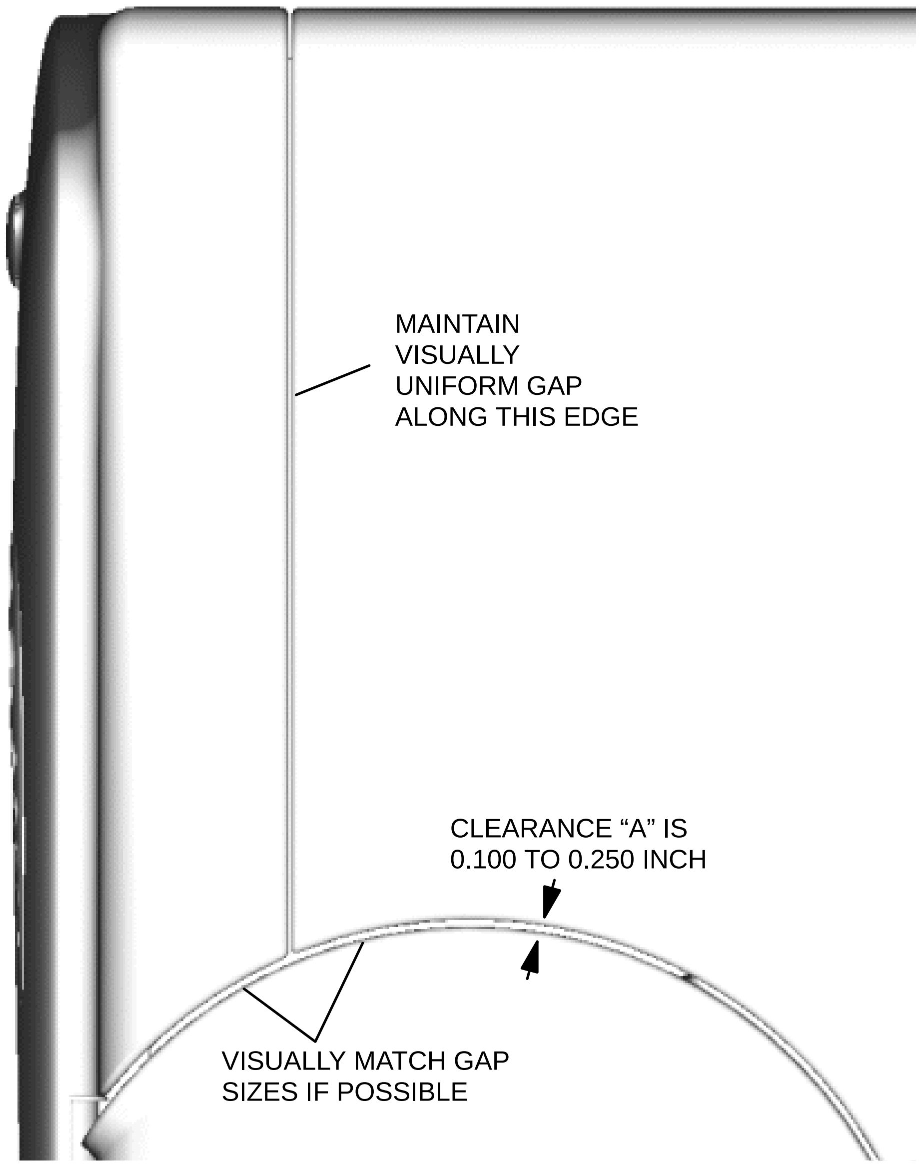


FIGURE 14 CLEARANCE "A" CHECK

6. Check clearance ”B”, between the Upper Right Column Cover and the Lower Right Column Cover. Refer to Figure 15.

7. If clearances ”A” and ”B” are within the allowable ranges and the gap uniformity looks good, than skip to the Hole Plug Installation procedure. If the clearances are not within tolerance, continue to the next page.

FIGURE 15 CLEARANCE “B” CHECK

8. If clearance ”A” is outside the allowable range or the gap between the Upper Right Column Cover and the Upper Front Cover does not visually appear uniform, remove the Upper Right Column Cover, and adjust the Cover Mounting Bracket by loosening the two bolts as shown below.

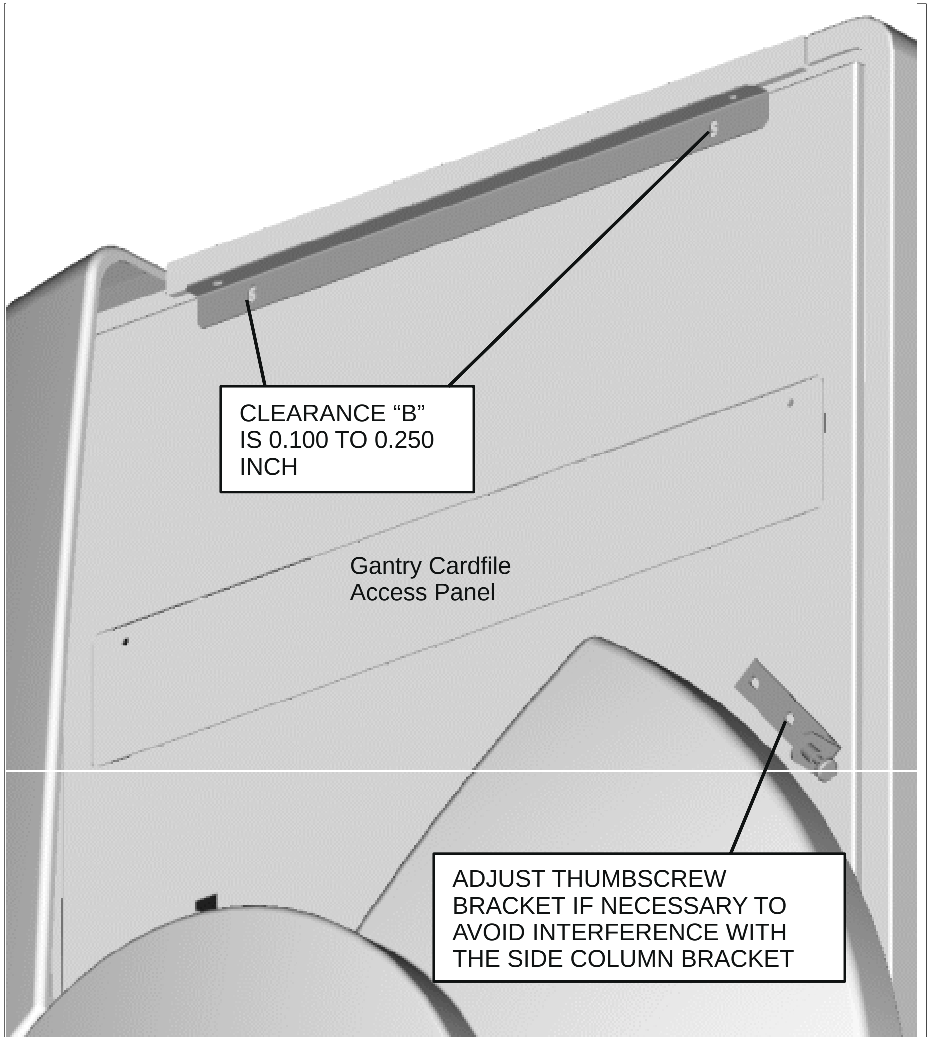


FIGURE 16 ADJUSTING CLEARANCE “A”

9. If clearance ”B” is outside the allowable range, than remove the Upper Right Column Cover, and adjust the Mounting Brackets located inside the Upper Right Column Cover by loosening the bolts as shown below.
10. Assembly of the Upper Left Column Cover (p/n 362097) follows this same procedure.

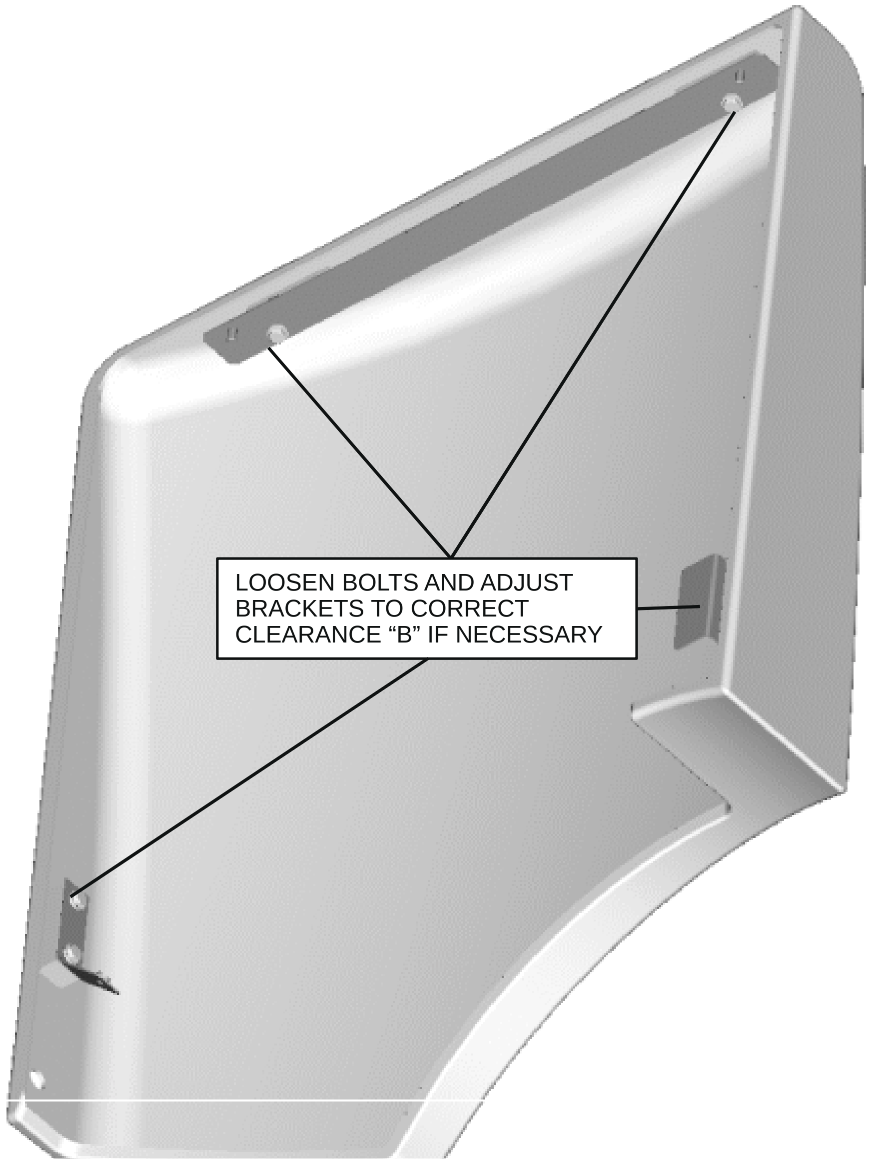


FIGURE 17 ADJUSTING CLEARANCE “B”

HOLE PLUG INSTALLATION

1. Install the Hole Plugs (p/n T39A–1013) at the four locations shown in Figure 18.

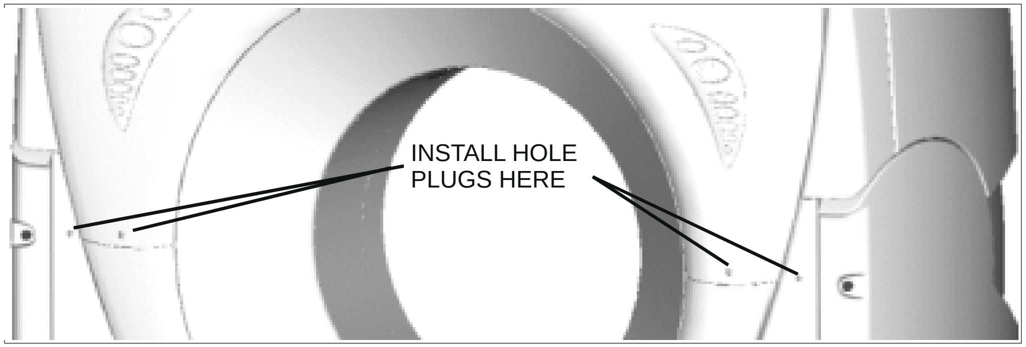


FIGURE 18 FRONT HOLE PLUG INSTALLATION

2. Install the Hole Plugs (p/n T39A–1013) at the two locations shown in Figure 19.

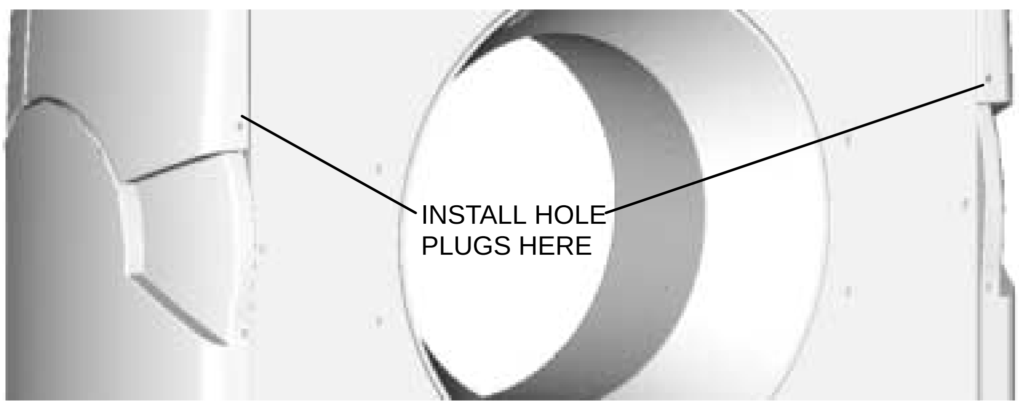


FIGURE 19 REAR HOLE PLUG INSTALLATION

CARDFILE ACCESS REMOVAL:

1. Remove the upper side cover. Loosen the two screws and remove the access cover. Refer to Figure 16.

CARDFILE ACCESS REPLACEMENT:

2. Set the access cover over the opening and secure with the two screws. Install the upper side cover.

DOMESTIC SHIPMENTS

There are FOUR major items to a domestic AcQSim CT CT scanner system:

1. Gantry, on casters (X–ray tube installed).
2. Patient Support, on dollies with returnable container, and accessories.
3. AcQImage Cabinet.
4. Display Tower

The chair and table (optional), any cameras, Patient Support rails, etc. are in additional cartons. All laser imagers are direct–shipped to the site. RDS and Voxel options are additional items not covered in this manual.

PERSONNEL NEEDED

At least four persons are needed to move the Gantry. Personnel will guide and control the speed of the units being moved, but must not attempt to directly lift any of the loaded equipment. Personnel should work as a team to keep the Gantry (or Patient Support, etc) from hitting and damaging walls, doorways, etc. All ramps should be verified as to load carrying ability. The Patient Support can be handled by two persons. The AcQImage Cabinet and Display Tower can be handled by one person. Overall installation requires two people.

UNCRATE THE SYSTEM

> ⚠ **WARNING**
>
> **WHEN STRAPS ARE CUT TO REMOVE PACKING, CASTERS CAN DETACH FROM COUCH. DO NOT REMOVE BANDING FROM COUCH UNTIL IT IS READY FOR FINAL POSITIONING IN THE SCAN SUITE. FAILURE TO COMPLY CAN RESULT IN SERIOUS INJURY.**

Undo the items wrapped on the skid. Obtain a cart or other device, as necessary, and move the X–ray tube (if packed separately) to the scan room; move the AcQImage and Display Tower crate to the control room.

International Shipments

The following paragraphs provide information on the numbering of international shipments and how they must be uncrated. The following information on international shipments excludes the overall international shipping container. The actual appearance of the shipment you receive can vary from this description. The following information refers to the ”internal” containers of any international shipments.

Numbering of Crates

The crates are clearly numbered. Contents of each is identified in a packing list attached to the bill of lading.

The AcQSim CT CT scanner shipments typically consist of three crates:

1. Gantry
2. AcQImage Cabinet and Display Tower, including keyboard, mouse and pad, monitor and cabling, and X–ray tube. For international shipments, the tube is typically shipped uninstalled.
3. Patient Support with returnable container and accessories (except Germany and Austria) The returnable container contains Gantry casters, dollies, etc. All three crates are typically supported by shock absorbing plastic springs.

The chair, table, any cameras, Patient Support rails, etc. are in additional cartons.

– All laser imagers and the digital interface are direct–shipped to the site.

UNCRATING

> ⚠ **WARNING**
>
> **WHEN STRAPS ARE CUT TO REMOVE PACKING, CASTERS CAN DETACH FROM COUCH. DO NOT REMOVE BANDING FROM COUCH UNTIL IT IS READY FOR FINAL POSITIONING IN THE SCAN SUITE. FAILURE TO COMPLY CAN RESULT IN SERIOUS INJURY.**

Unbolt the boxes to gain access to the contents in the crate.

> ⚠ **CAUTION**
>
> THE GANTRY MUST BE LIFTED OFF THE SKID BY A LIFTING MACHINE CAPABLE OF HANDLING AT LEAST 6.000 POUNDS (2730 KG.). FAILURE TO HAVE THE CORRECT LIFTING MACHINERY CAN RESULT IN DAMAGE TO THE GANTRY

> ⚠ **WARNING**
>
> **LIFTING HAZARD.. USE ONLY THE PROVIDED CHAIN SLING WHEN LIFTING THE GANTRY. FAILURE TO COMPLY CAN RESULT IN EQUIPMENT OR PERSONNEL INJURY.**

> ⚠ **WARNING**
>
> **GANTRY DAMAGE HAZARD. DO NOT USE A FORK LIFT TO LIFT THE GANTRY. MOVE THE GANTRY ON THE CASTERS (WHEELS ATTACHED TO EACH END–STAND) OR BY LIFTING AND MOVING WITH THE CHAIN SHACKLES (TOP OF GANTRY). USE OF A FORK LIFT WILL RESULT IN DAMAGE TO THE GANTRY.**

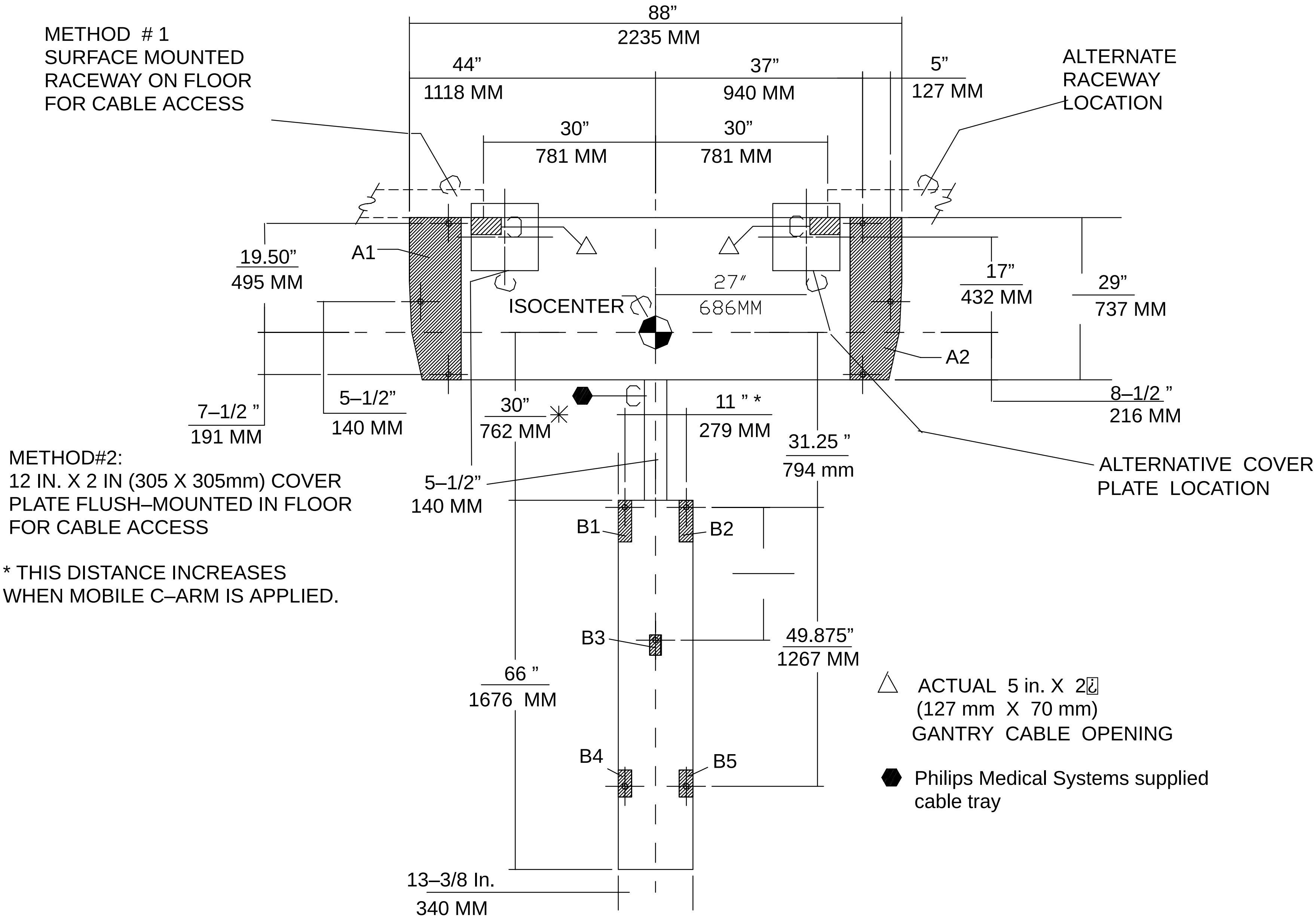
EQUIPMENT POSITIONING

1. ***Prior to the delivery of the system*** the General Contractor shall drill the mounting holes and install 1/2”–13 anchoring expansion nuts for anchoring Gantry (6 places) and Patient Support (5 places).
2. The nuts are to be flush with the finish of the floor surface. A template (P/N 174214) to lay out the hole centers may be obtained from the local Philips Medical SystemsService office.
3. Philips Medical Systems International requires the use of 1/2” – 13 anchors for the Gantry and couch with a pull out force of 8,000 LBS. (3640 kg.) <u>Lead anchors are not acceptable.</u> Acceptable anchors are:

   Red Head, Hilti, Parabolt by Molly

> **NOTE**
>
> Carbide drill bits or ”core drills” are not recommended to drill holes for Red Head anchors. These drill bits cause holes to be drilled outside of positional tolerances; therefore, holes in the floor may not match the holes in the Gantry.

Figure 20 is a representation of the template used to place holes for equipment anchoring. Example weight specifications given in Figure 20. Actual weights for your system are defined in the Architectural Planning Guide for your site.



CONTINUED ON NEXT PAGE:

CONTINUED FROM FIGURE 20

SCANNER GANTRY WEIGHT:	PATIENT SUPPORT:
3,400 LBS	550 LBS
1542 KG	249 KG
FLOOR AREA: 2,552 SQ. IN.	FLOOR AREA: 883 SQ. IN.
CONCRETE LOAD AREA:	CONCRETE LOAD AREA:
A1 + A2 = 393.5 SQ. IN. (2539 SQ CM.)	B1 + B2 + B3 + B4 + B5 = 49.75 SQ. IN. (321 SQ CM.)
DYNAMIC FORCES: NEGLIGIBLE	DYNAMIC FORCES: NEGLIGIBLE

CONTRACTOR SHALL DRILL MOUNTING HOLES AND INSTALL 1/2 INCH–13 ANCHORING EXPANSION NUTS FOR SCANNER GANTRY AND PATIENT SUPPORT. NUTS TO BE FLUSH WITH FINISHED FLOOR SURFACE. A TEMPLATE TO LAY OUT THE HOLE CENTERS MAY BE OBTAINED FROM PHILIPS MEDICAL SYSTEMS SERVICE DIVISION.

PHILIPS MEDICAL SYSTEMS REQUIRE THE USE OF READ HEAD ANCHORS #S12 (RATED PULL–OUT FORCE OF 8500 LBS (37,808 N), OR THROUGH–BOLTING WILL BE REQUIRED (LEAD ANCHORS ARE NOT ACCEPTABLE)

IN ORDER TO MEET ELECTRICAL GROUND ISOLATION REQUIREMENTS, ALL METHODS OF SCANNER GANTRY AND PATIENT SUPPORT BOLTING MUST NOT COME IN CONTACT WITH ANY BUILDING STEEL STRUCTURE OR RE-INFORCEMENT. MODULAR BUILDING FLOOR STRUCTURE MAY REQUIRE ADDITIONAL BRACING BENEATH SCANNER GANTRY AND PATIENT SUPPORT IN ORDER TO ELIMINATE VIBRATION THAT CAN AFFECT CLINICAL IMAGES.

FIGURE 20 MOUNTING TEMPLATE (P/N 174214)

GANTRY POSITIONING

> **⚠ WARNING**
>
> **PINCH HAZARD. FINGERS CAN BE CAUGHT IN THE CLOSE CLEARANCES BETWEEN THE CASTERS AND THE 'DISKS."**

1. After all floor anchors are in place, roll the Gantry into position above them. (If for some reason the Gantry is not supported by the casters, raise the Gantry onto the casters by rotating the circular portion (caster disk) of the caster assembly in a clockwise direction.)

> **⚠ WARNING**
>
> **OPEN GANTRY COVERS MUST BE PROPERLY SECURED OR REMOVED FROM THE IMMEDIATE AREA. FAILURE TO DO SO CAN RESULT IN SERIOUS INJURY TO PERSONNEL.**

> **⚠ WARNING**
>
> **INJURY HAZARD. THE FRONT AND REAR UPPER GULL WINGS HAVE SHARP EDGES. USE CAUTION WHEN SERVICING EQUIPMENT WHILE UPPER GULL WINGS ARE OPEN. FAILURE TO COMPLY CAN RESULT IN SERIOUS INJURY TO PERSONNEL.**

2. Open the panel to full–position and lock in place using the support rod. Refer to NO TAG.

> **NOTE**
>
> Gantry Cover procedure is found in the first chapter of the Gantry section of the Repair Manual.

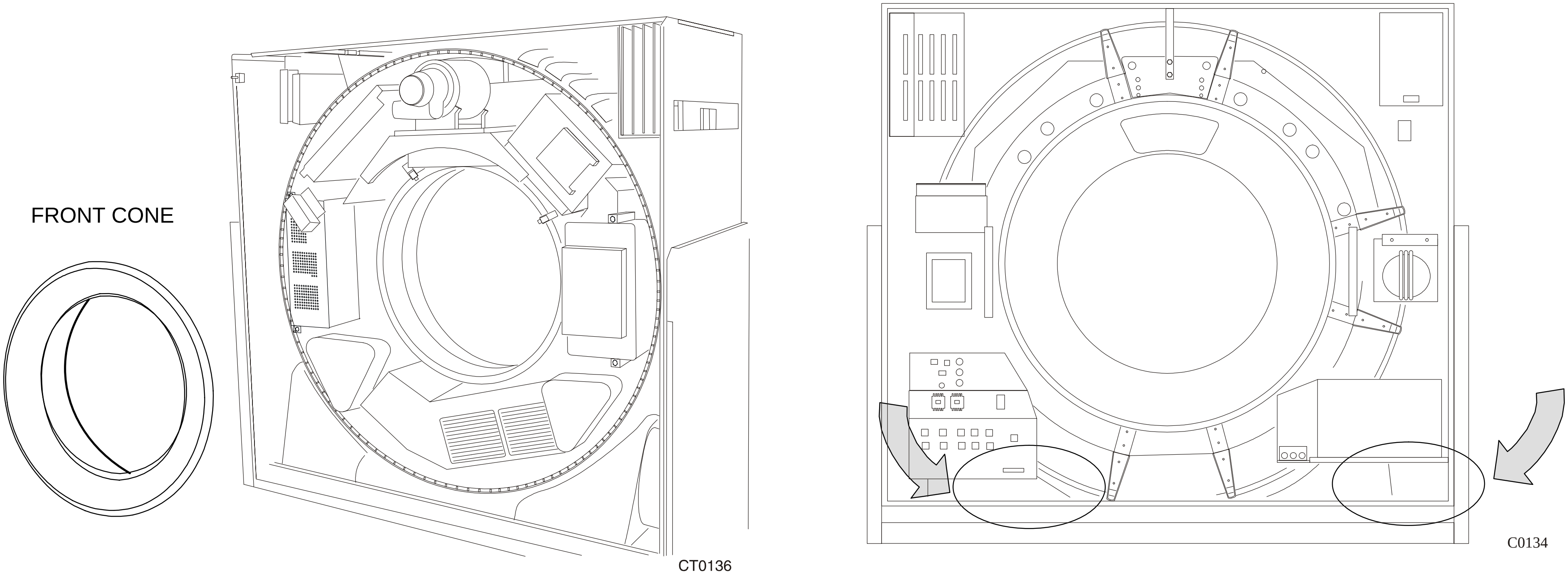


FIGURE 21 AcQSim CT GANTRY FRONT CONE AND CABLE ACCESS COVERS (REAR)

> ⚠ **CAUTION**
>
> THE PLASTIC EDGES OF THE CONE THAT FIT INTO THE SLIP RING ARE VERY FRAGILE. DO NOT LEAN THE CONE AGAINST THE WALL ON THIS EDGE. FAILURE TO COMPLY MAY RESULT IN DAMAGE TO THE CONE RESULTING IN THE NEED FOR REPLACEMENT.

3. Remove the two cable access covers at the lower rear corners of the Gantry base to gain access to the bolt holes. Remove the shipping covers in the bottom of the cable trough. See Figure 21.

> ⚠ **CAUTION**
>
> IN THE FOLLOWING STEP USE THE SUPPLIED 2–INCH BOLTS UNLESS DIRECTED OTHERWISE. IMPROPER LENGTH BOLTS CAN BE TORQUED TO SPECIFICATION (30–40 FT–LBS) BUT WILL NOT SECURE THE GANTRY.

4. Carefully lower the Gantry one end at a time by rotating the caster disks in unison. Lower the Gantry over the holes in the floor (already drilled – see Equipment Positioning) until the Gantry base is about 1/2–inch above the floor. Position the Gantry over the holes and install – but **DO NOT TIGHTEN** – the six 1/2–inch bolts (S617–129) and 1/8–inch thick washers (T11–447) provided in the anchor hardware kit. Figure 22 shows the initial Gantry bolt–down positions. *Lower the Gantry to the floor.*

> **NOTE**
>
> In most installations, all six bolt holes are now accessible. In some situations it will be necessary to power the Gantry (later, in this procedure) and tip it to +5 degrees for access to the front two holes. ***Ensure that the Gantry anti–tilt shipping bracket is removed before tilting. See Figure 23.***

> **NOTE**
>
> Final tightening of Gantry and Patient Support floor anchor bolts and removal of shipping caster assemblies will be done after power is supplied. The Patient Support must be raised to tighten the floor bolts and to do final alignment of Patient Support to Gantry. This requires electrical power: these steps are detailed in Section 4 of this manual, System Installation

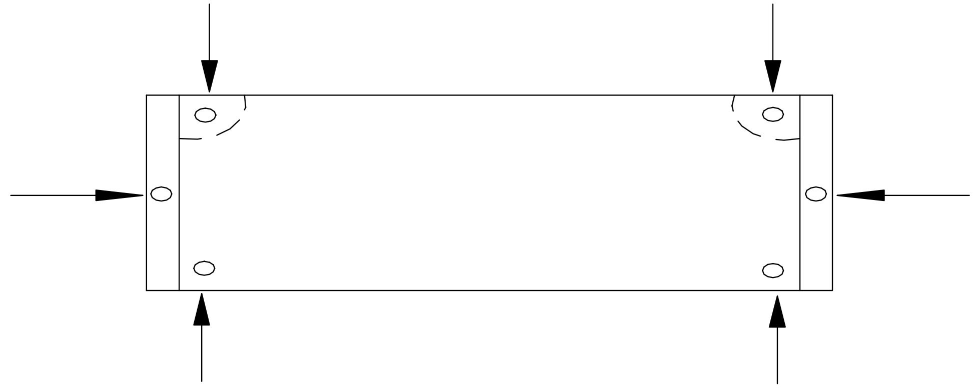


FIGURE 22 INITIAL GANTRY BOLT–DOWN POSITION

> **NOTE**
>
> DO NOT remove the Gantry caster assemblies at this time. They will be removed later in the installation procedure.

5. Use the appropriate wrench (i.e., for hex–head or Allen–head) and remove the Gantry tilt bracket located in the lower center portion in the back of the Gantry. This bracket prevents inadvertent Gantry tilt during shipment. Refer to Figure 23. **HINT**: Sometimes the fit is very tight and the bracket cannot be removed after unbolting. Later, after power is applied, tilt the Gantry slightly to release the pressure on the anti–tilt bracket.

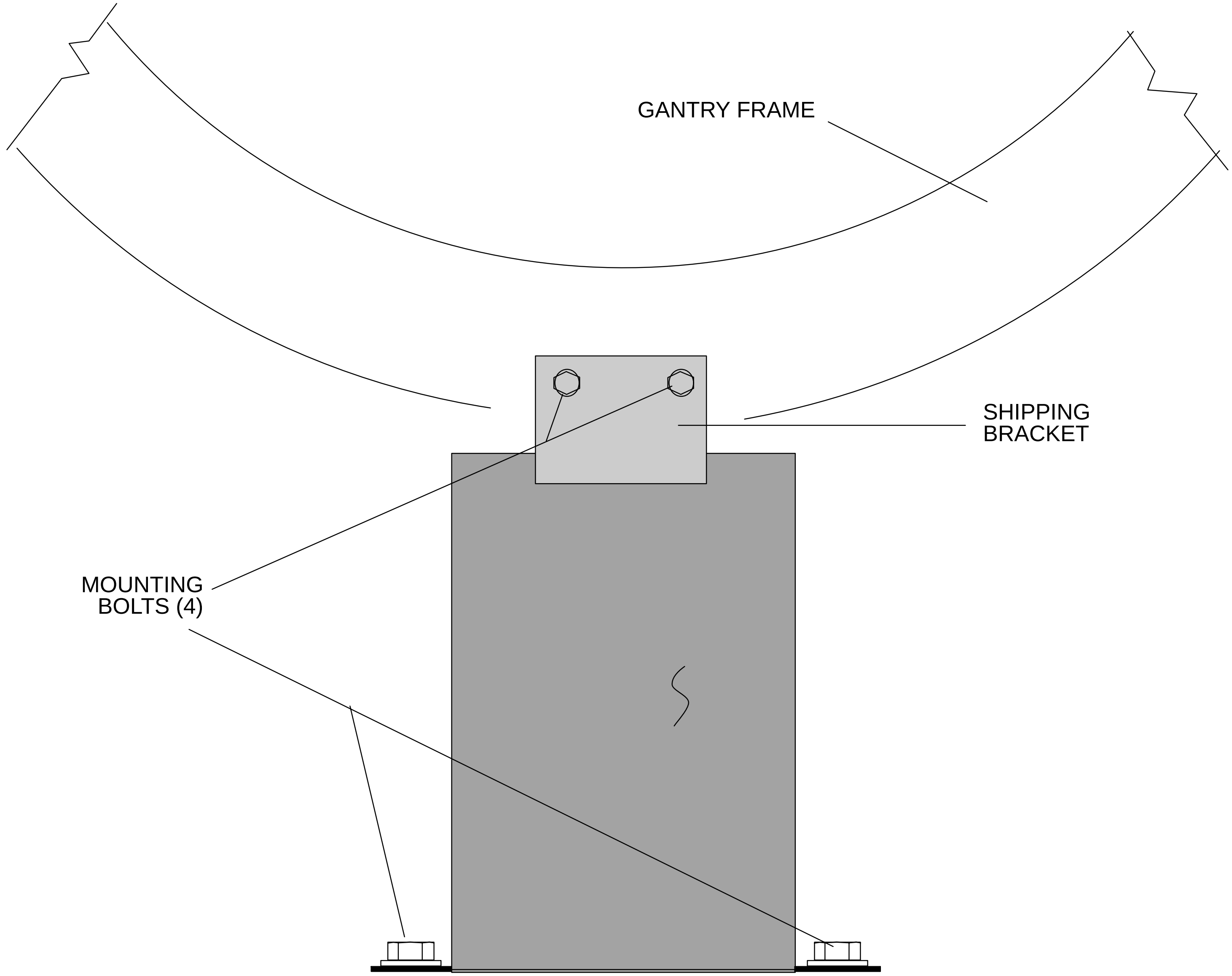


FIGURE 23 GANTRY SHIPPING BRACKET

6. Remove the Connector Plate by removing the two bolts that secure it. The Column Support bar will be removed AFTER the gantry is aligned and the casters are removed. DO NOT attempt to remove them now. Refer to Figure 24.

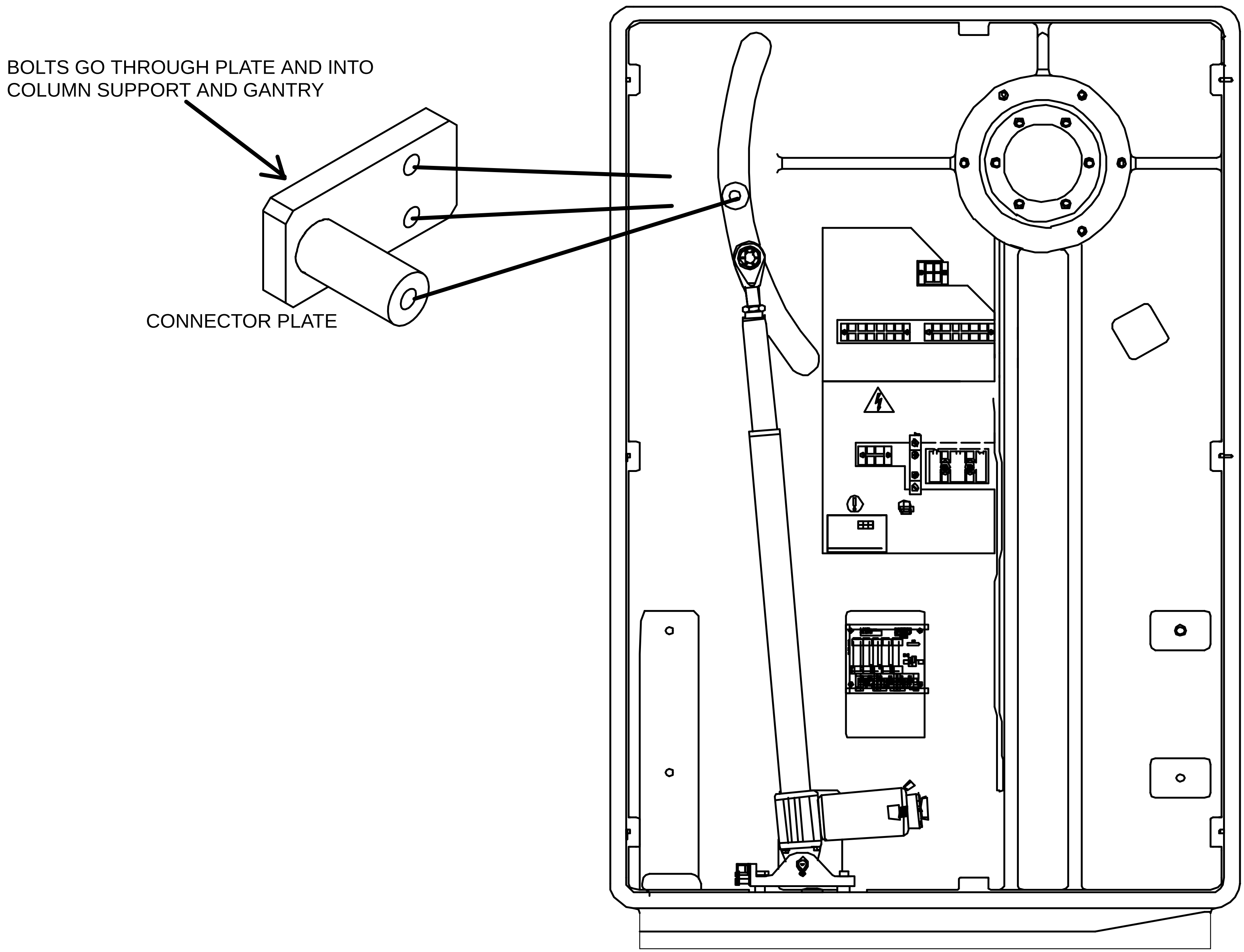


FIGURE 24 COLUMN CONNECTOR PLATE AND COLUMN SUPPORT LOCATIONS – (Approx. first 14 systems)

POSITION PATIENT SUPPORT

NOTE

The Gantry **must** be in its final position before you begin these steps. The rear and center anchor bolts in each support column must be partially threaded into the floor anchor nuts. The front two bolts may already be in or may be inserted later when the Gantry can be tilted 5 degrees forward.

1. Remove the top bolts of the column support on each gantry column. Remove the connector plate that is attached to the column support and has a pin extending into the tilt arm channel.
2. At the front of the Gantry, pull out cables L7272 and L7161. HINT: remove the rear, Gantry base access plate; you can then push the cables through to the front.

⚠ WARNING

LIFTING HAZARD. THE PATIENT SUPPORT IS VERY HEAVY AND SHOULD BE MANEUVERED BY TWO PEOPLE. USE CAUTION WHEN MOVING THIS ITEM. FAILURE TO COMPLY CAN RESULT IN INJURY TO PERSONNEL.

3. Maneuver the Patient Support into the approximate position over the anchor holes and cut the banding straps. (This requires two people). While one person lifts one end of the Patient Support, the other person can remove the dolly. Do the same to the other end.
4. Remove the top two leaves of the telescoping covers to gain access to the anchoring holes (see Figure 25) by doing the following:
   a. Remove the four screws that bolt the two panels together.
   b. Remove the four screws and washers that attach the telescoping base cover panels to the subframe.
   c. Remove the ground wire from each panel.
5. Place the cable trough between the Gantry and Patient Support. The section with a tape switch is the bottom of the trough. Orient the cable trough with the connector end toward the Patient Support. Refer to Figure 26. Do not remove the double–sided tape at this time. Remove when the alignment section is completed.
6. Connect cables L7161 and L7272 to the Patient Support. Refer to Figure 27 for cable connections.

NOTE

For hospital locations that have ordered the additional plastic elevator plate for high fluid levels, refer to Figure 27 for instructions.

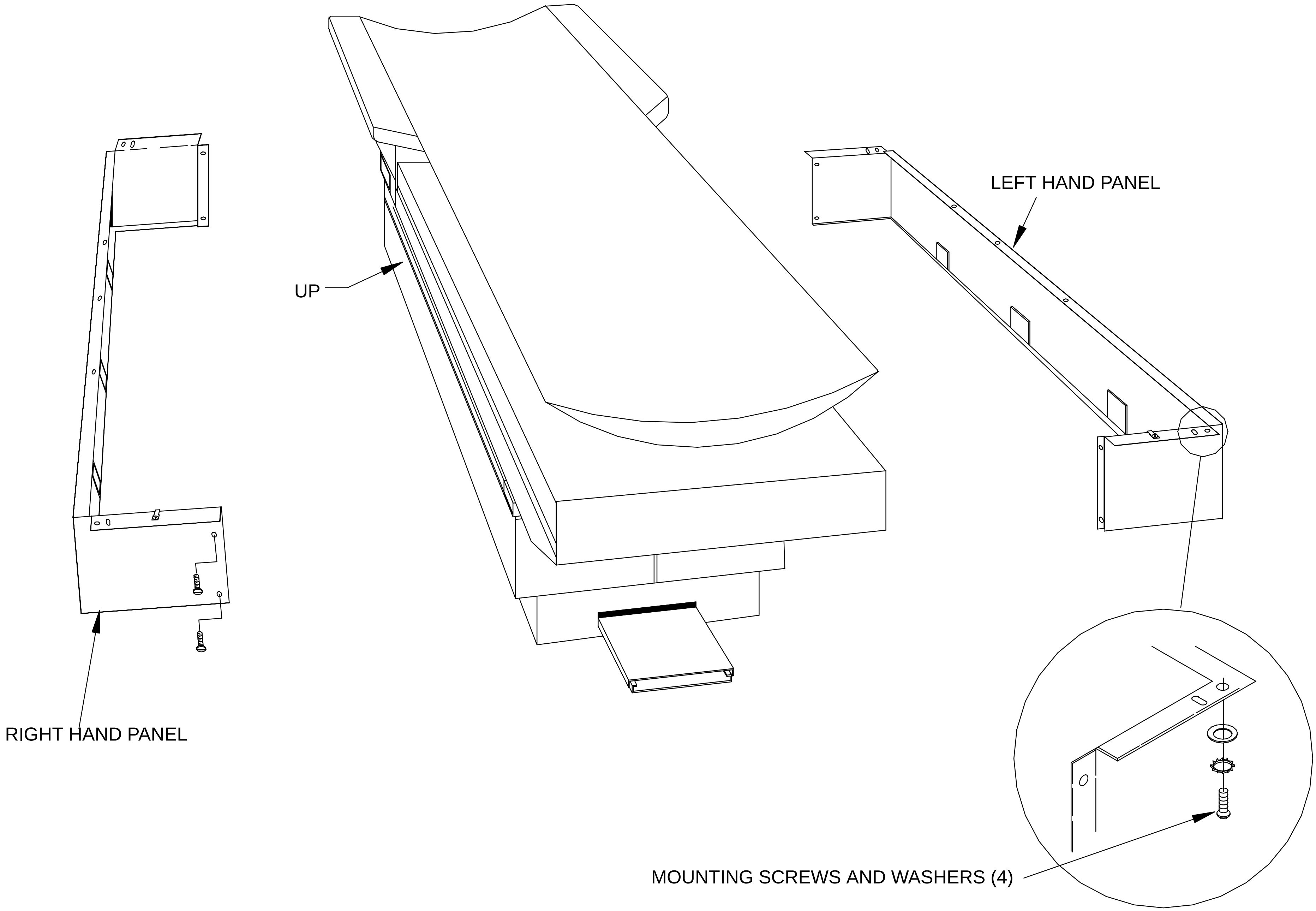


FIGURE 25 PATIENT SUPPORT TELESCOPING BASE COVERS

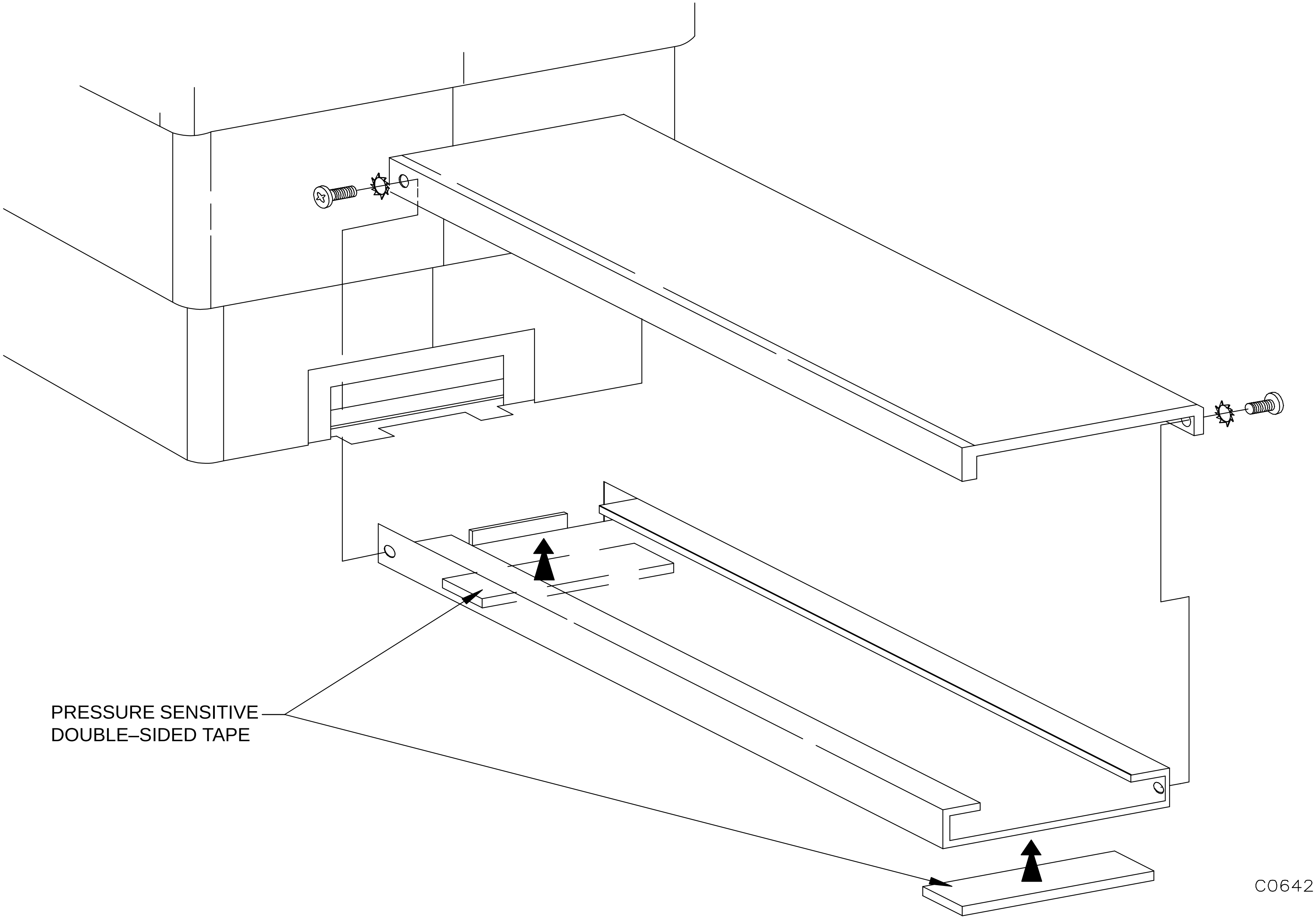


FIGURE 26 CABLE TROUGH WITH FOOT SWITCH

INSTALLATION INSTRUCTIONS:

1. Slide the 18 inch plastic elevator plate under the cables.
2. Join two tyraps together in order to lengthen. Make three assemblies.
3. Bind cables to plastic elevator plate using the joined tyraps as shown.
4. Trim tyraps.
5. Make certain the couch foot switch top does not interfere with connectors.
6. Test foot switch for proper operation.

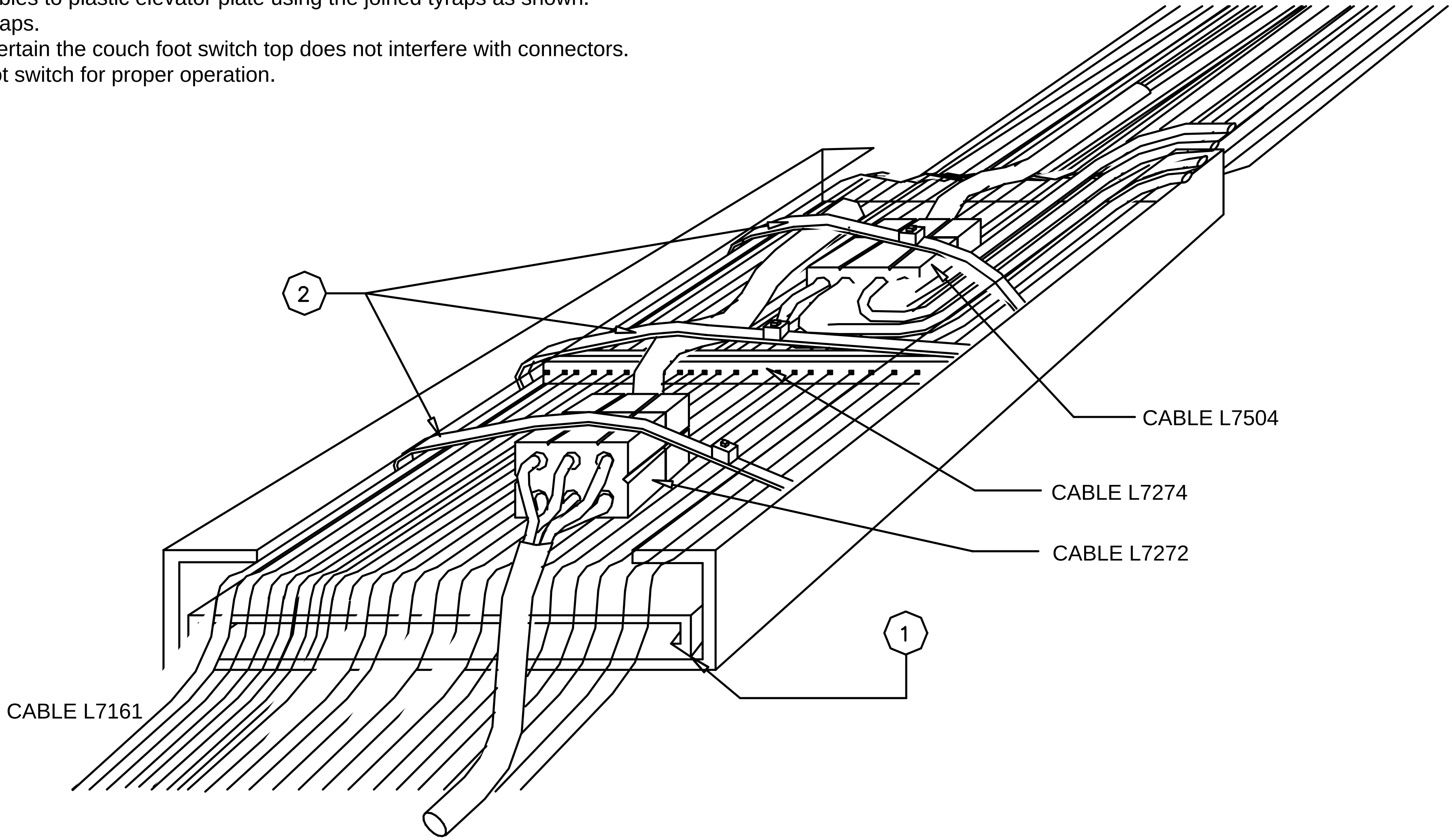


FIGURE 27 OPTIONAL PLASTIC ELEVATOR INSTALLATION

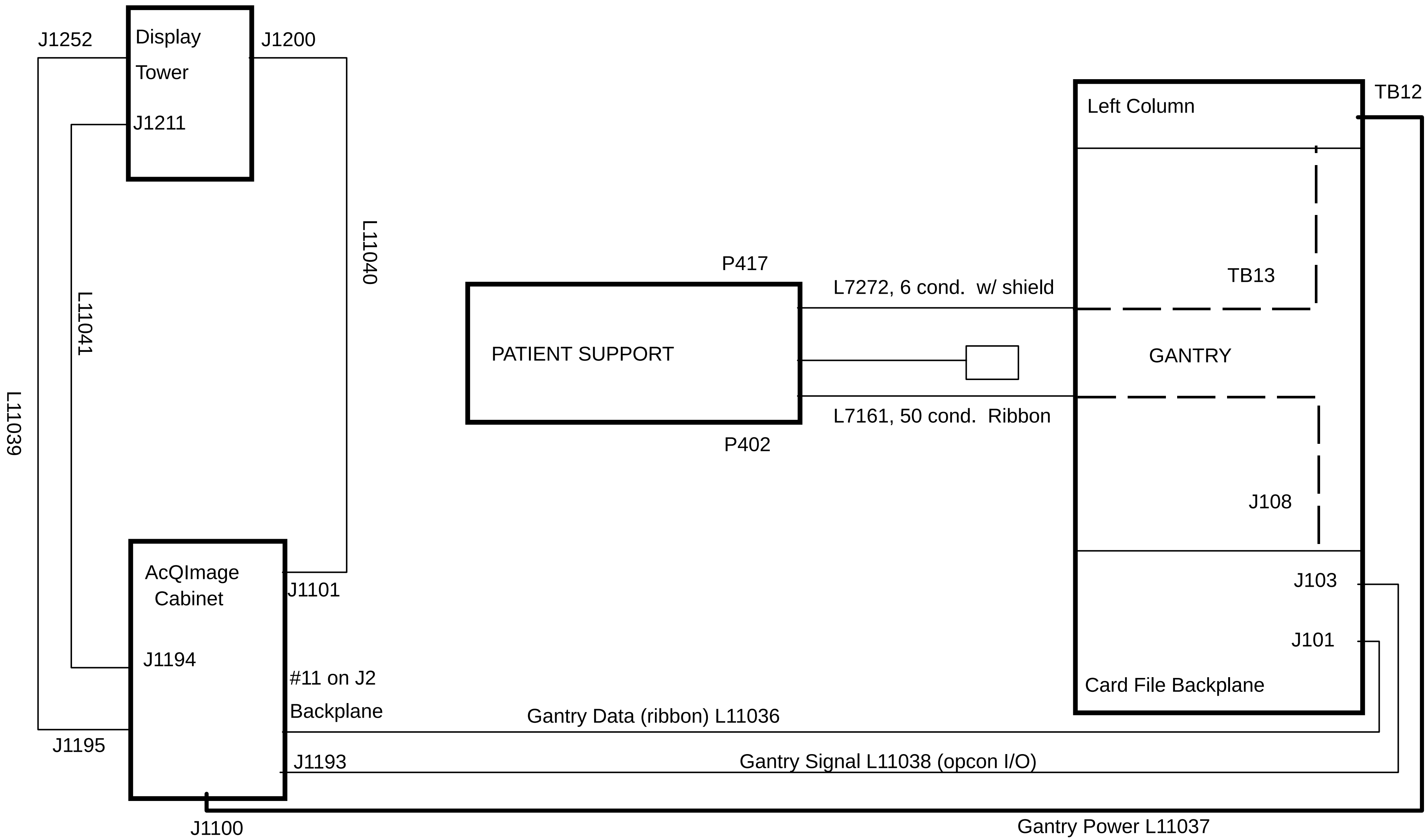


FIGURE 28 SYSTEM INTERCONNECT DIAGRAM

NOTE: Do not wire at this time – the next section covers this in detail.

7. Plug in the Tape Switch connector. If possible, install the safety ground (green and yellow stripped wire,) from the Gantry to the Patient Support; otherwise, do this in the Installation section. ***This connection must be made in the Patient Support!***

8. After the Patient Support is bolted down, place the cable trough cover over the bottom trough.

> **NOTE**
>
> The installation and final tightening of the Gantry and Patient Support floor anchor bolts and the removal of Gantry shipping caster assemblies will be done after power is applied. The Patient Support must be electrically raised to install and tighten bolts, and to do final alignment of the Patient Support to the Gantry. These steps are detailed in **System Installation** section**.**

POSITION ACQIMAGE CABINET & DISPLAY TOWER

1. Remove the AcQImage Cabinet and Display Tower from the shipping palette. Clear away any packing materials and unlock the wheels on both units. They can now be rolled to their respective installation locations.
2. Locate the AcQImage Cabinet in either the Scan Control room, Gantry room or elsewhere per site planning instructions. Ensure you lock the wheels when in place. See SEISMIC CONSIDERATIONS if applicable on the following page.
3. Locate the Display Monitor, Keyboard, Mouse, Mousepad and Intercom Assembly to their respective locations. Refer to Figure 29 for a typical setup. Roll the Display Tower to its general location. (See Canadian Note.). To install the wrist pad on the new curved–top style keyboard (P/N 178918), turn the keyboard and wrist rest over. Place the wrist rest's four capture posts over the keyboard's four capture points and snap into place.
4. Install the anti–tip device on each of the wheels of both cabinets if not factory installed.

THIS FIGURE SHOWS A TYPICAL SETUP. THE DISPLAY TOWER CAN BE ROLLED UNDER THE TABLE.

MAKE SURE THE EQUIPMENT IS LOCATED IN A SPOT WHERE THE CABLES CAN BE INSTALLED BEFORE ROLLING UNDER THE TABLE.

LOCK WHEELS AFTER POSITIONING.

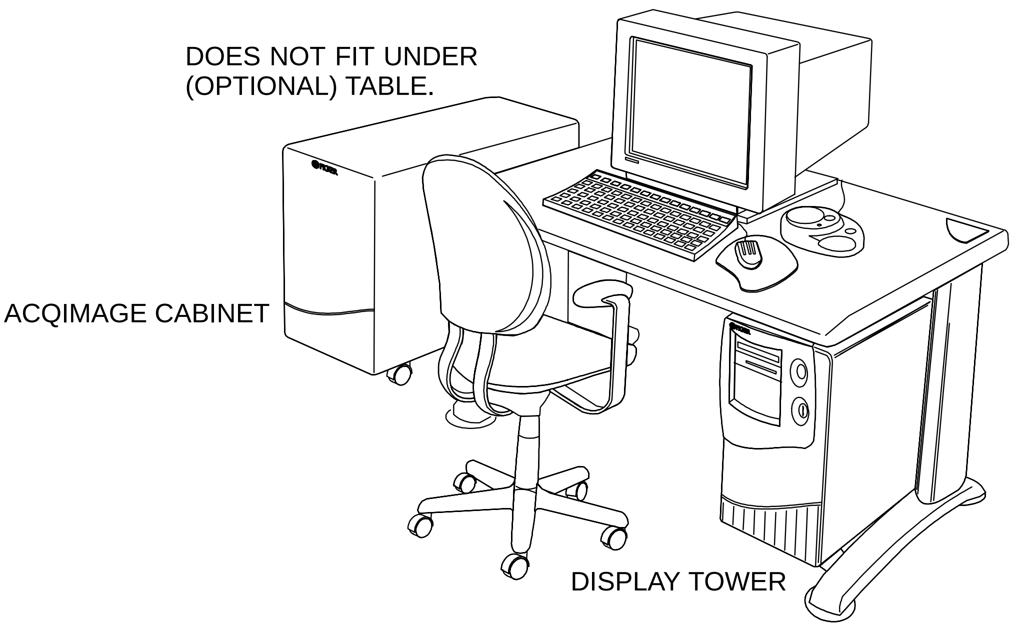


FIGURE 29 ACQIMAGE CABINET AND DISPLAY TOWER SETUP (optional table and chair)

SEISMIC CONSIDERATIONS

The following procedure is for systems that require special seismic anchoring installation. A P/N 178172 Seismic Mounting Package Assembly is required to complete this type of installation for Display Console/AcQImage Cabinets in California or other locations prone to seismic disturbances.

Display Tower / AcQImage Cabinet

1. Set the Display Tower on its left side (when facing the front).
2. Remove the four castors by loosening the 1/2” nut that is just above each castor.
3. Remove each castor and its washer.
4. Install the spacers and washers as shown in Figure 31 for the Display Tower.
5. Put the castors back in and tighten each of the nuts. Tip the cabinet back onto its castors. Roll into the area where they will be located.
6. Repeat this procedure for the AcQImage cabinet. Refer to Figure 30.

To slide either cabinet out for service:

1. Remove both anchor bolts from each of the two FRONT seismic brackets.
2. Unlock the castors and slide the cabinet out (the rear seismic brackets do not move).
3. Slide back in and re–assemble the two front brackets.

NOTES

1. Supporting structure (floor/wall/ceiling) is responsibility of architect or engineer of record at site specific project.
2. For the state of California, expansion anchors shall have special inspection during installation and be tested in accordance with OSHPD requirments.

3. Anchor base with 8 each HSL MB/20 expansion anchors embedded 2 5/8” into 4 3/4” min. thickness = 2000PSI normal wt. concrete. Minimum concrete edge distance = 7”. Install per ICBO ER–3987. 50% of the anchors shall be tested after a minimum of 24 hours of their installation by a torque test of 20 ft–lbs. The specified torque shall be reached with one half turn of the head. Should any anchor fail, test the remainder.
4. Equivalent expansion anchors in lieu of Hilti, may be used provided allowable loads are equal or greater and are approved by ICBO.
5. Anchor Forces
Max. shear (AcQImage Cabinet) = 37 lbs. (Display Tower) = 31 lbs.
Max. tension (AcQImage Cabinet) = 311 lbs. (Display Tower) = 245 lbs.
6. Seismic drilling templates for location of mounting holes are included in P/N 178172 seismic mounting package assembly.

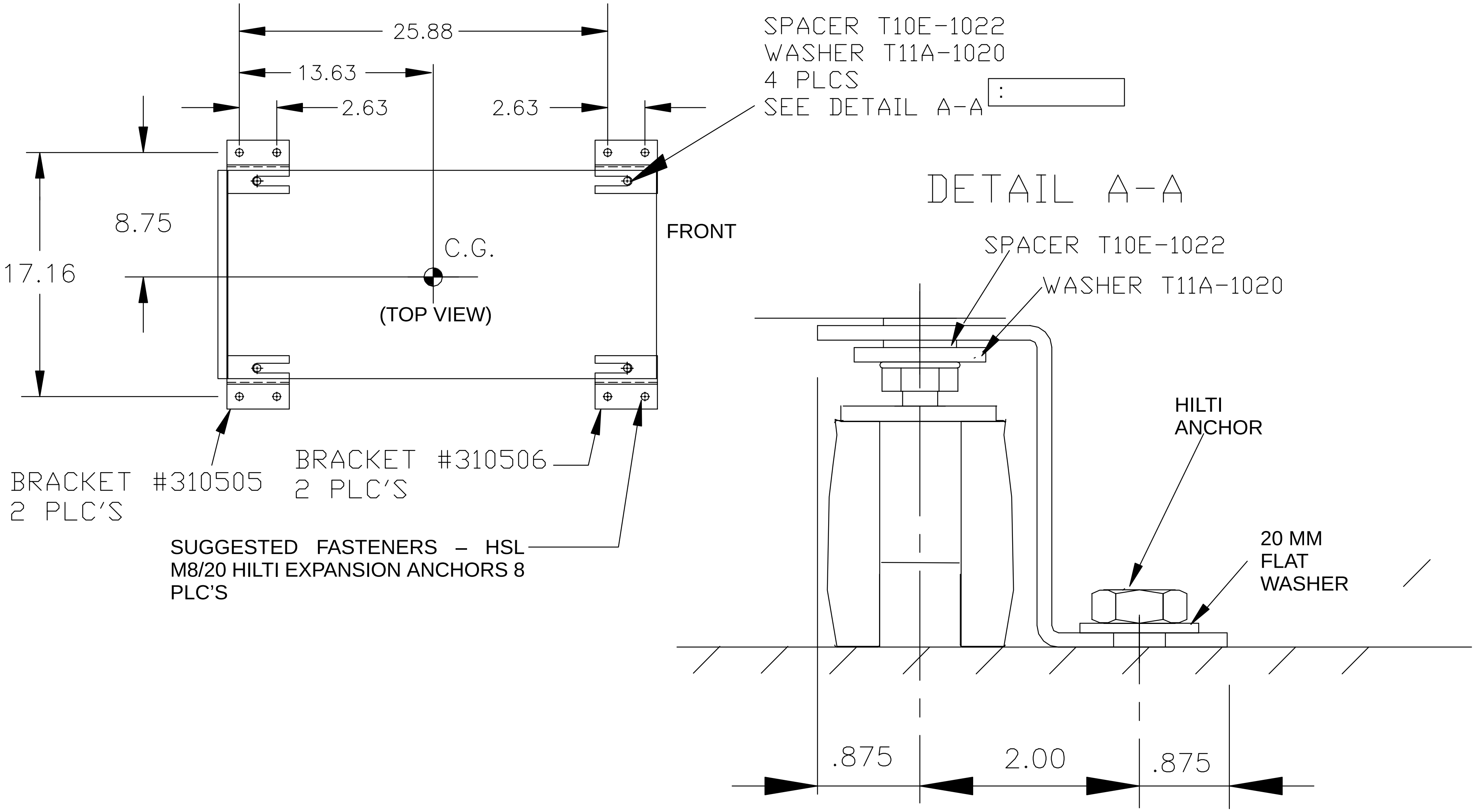


FIGURE 30 SEISMIC BRACKET – ACQIMAGE CABINET

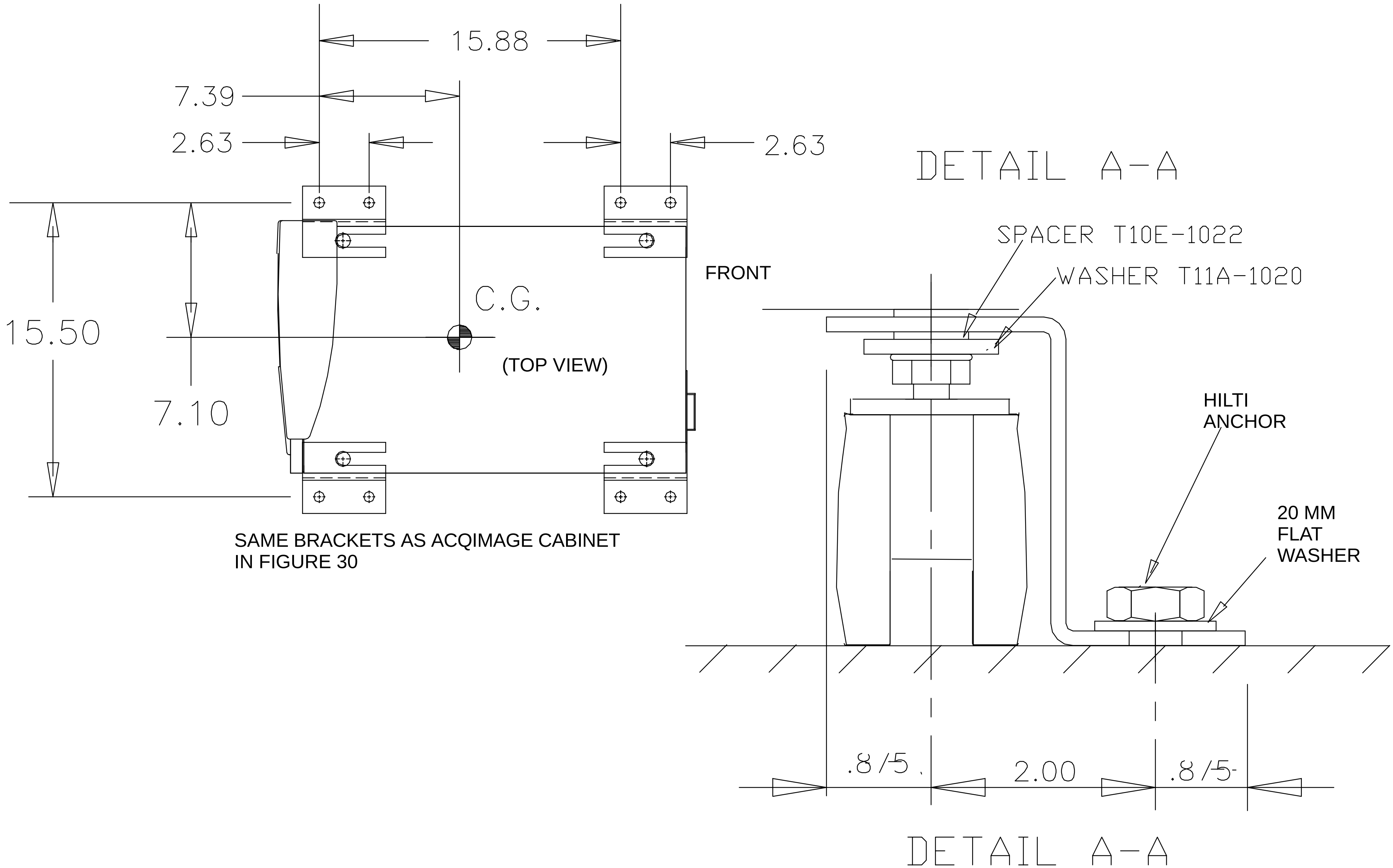


FIGURE 31 SEISMIC BRACKET – DISPLAY TOWER

> **NOTE**
>
> To meet Canadian regulatory requirements, a French–Canadian warning message (83980) is available for any systems to be shipped to Canada. It is part of the ship–with items (311264). The label is applied over the keyboard label (83926) on the older style keyboard. It is placed to the right of the English warning message in the blank area of the label.
>
> For the new curved–top style keyboard, P/N 178918, remove the existing label and install the new label (P/N 312729) at the top of the keyboard.

NAMEPLATE LOCATIONS

It is necessary to record all equipment serial numbers and part numbers from the equipment nameplates on the FDA Reporting forms and on the system performance evaluation sheets. Use Figure 32, Figure 33, and Figure 34 to locate all nameplates and record the proper information. The table below will assist you in loacting and identifying the labels.

1	88856	NAMEPLATE, CATALOG – COLLIMATOR
2	178620	SERIAL TAG PAGE FOR SYSTEM 2007
3	T32–616	LASER CAUTION LABEL
4	T32–703	LASER CAUTION LABEL – CLASS II
5	T32–707	LASER CAUTION LABEL – CLASS II
6	T32–712	LASER WARNING – GERMAN
7	T60C–551	INSTALLATION INSTRUCTIONS – GERMAN LASER WARNING
8	T63G–1150	INSTALLATION INSTRUCTION – LASER CAUTION – FRENCH
9	T63G–1154	INSTALLATION INSTRUCTION – LASER CAUTION – FRENCH
10	T92C–288	FDA CERTIFICATION LABEL
11	T92C–288	LASER RADIATION CAUTION LABEL – FRENCH
12	T92C–289	CAUTION LABEL – FRENCH

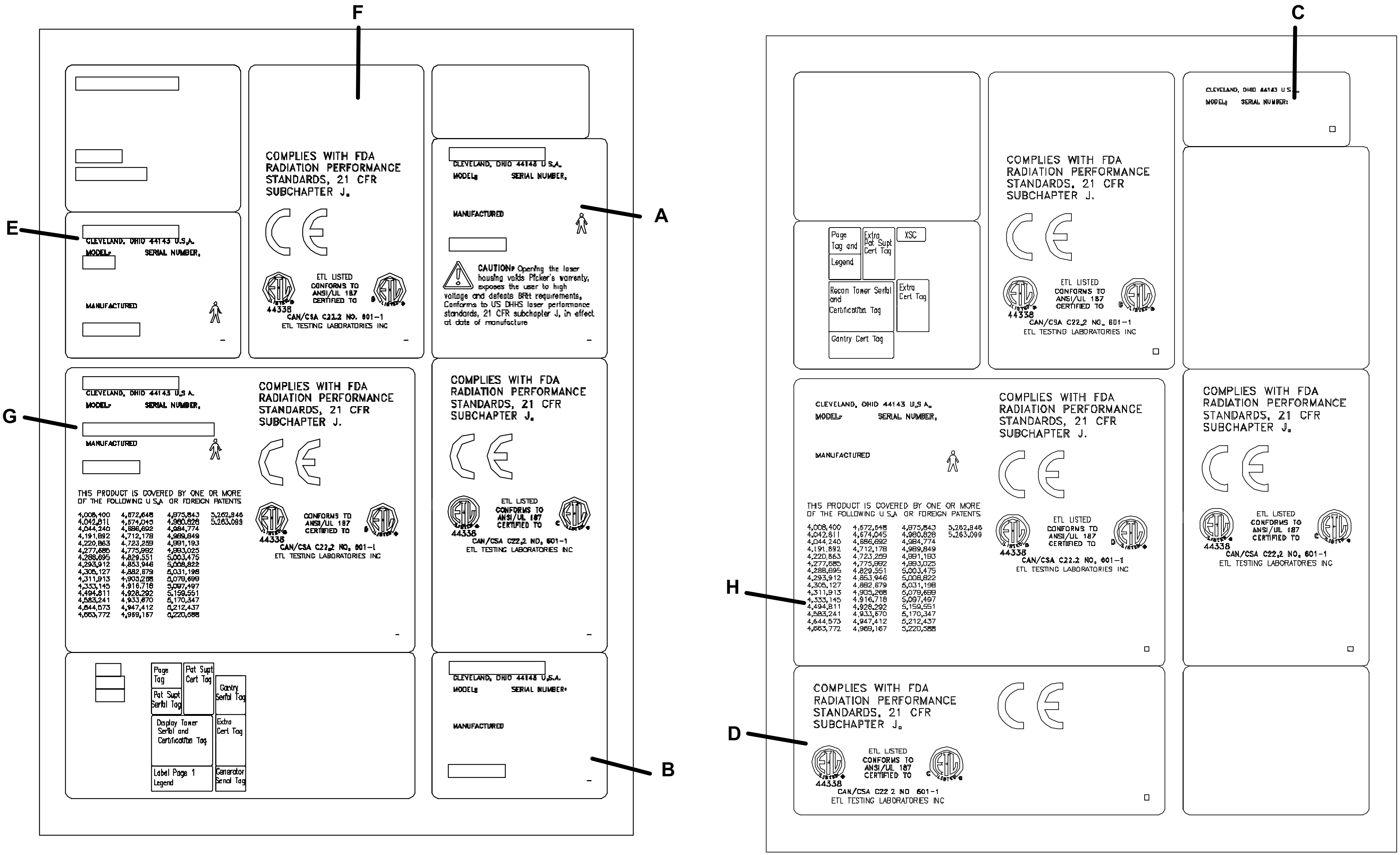


FIGURE 32 LABELS (LETTER CALLOUTS WILL BE IN NEXT 3 FIGURES)

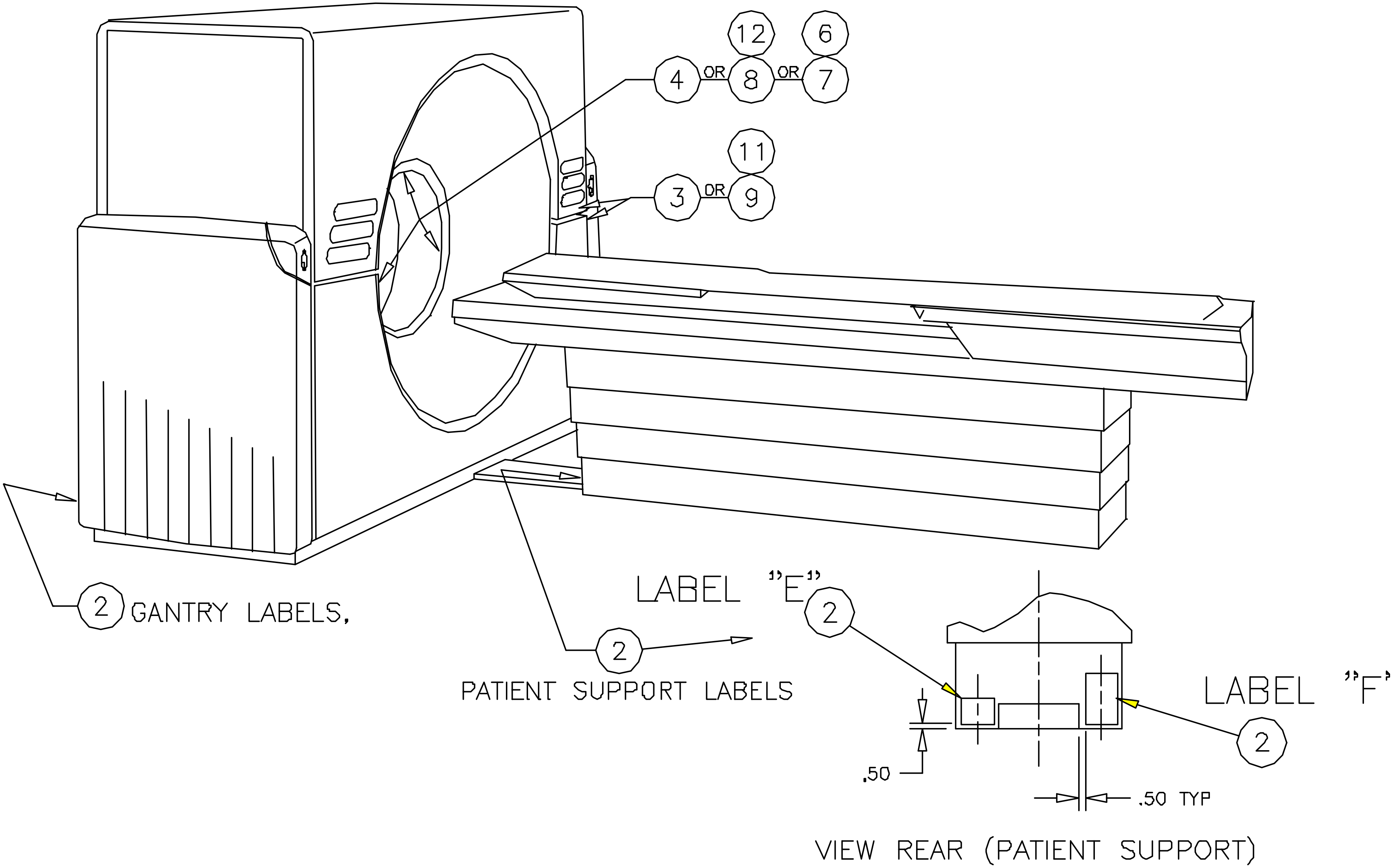


FIGURE 33 GANTRY / PATIENT SUPPORT LABELS

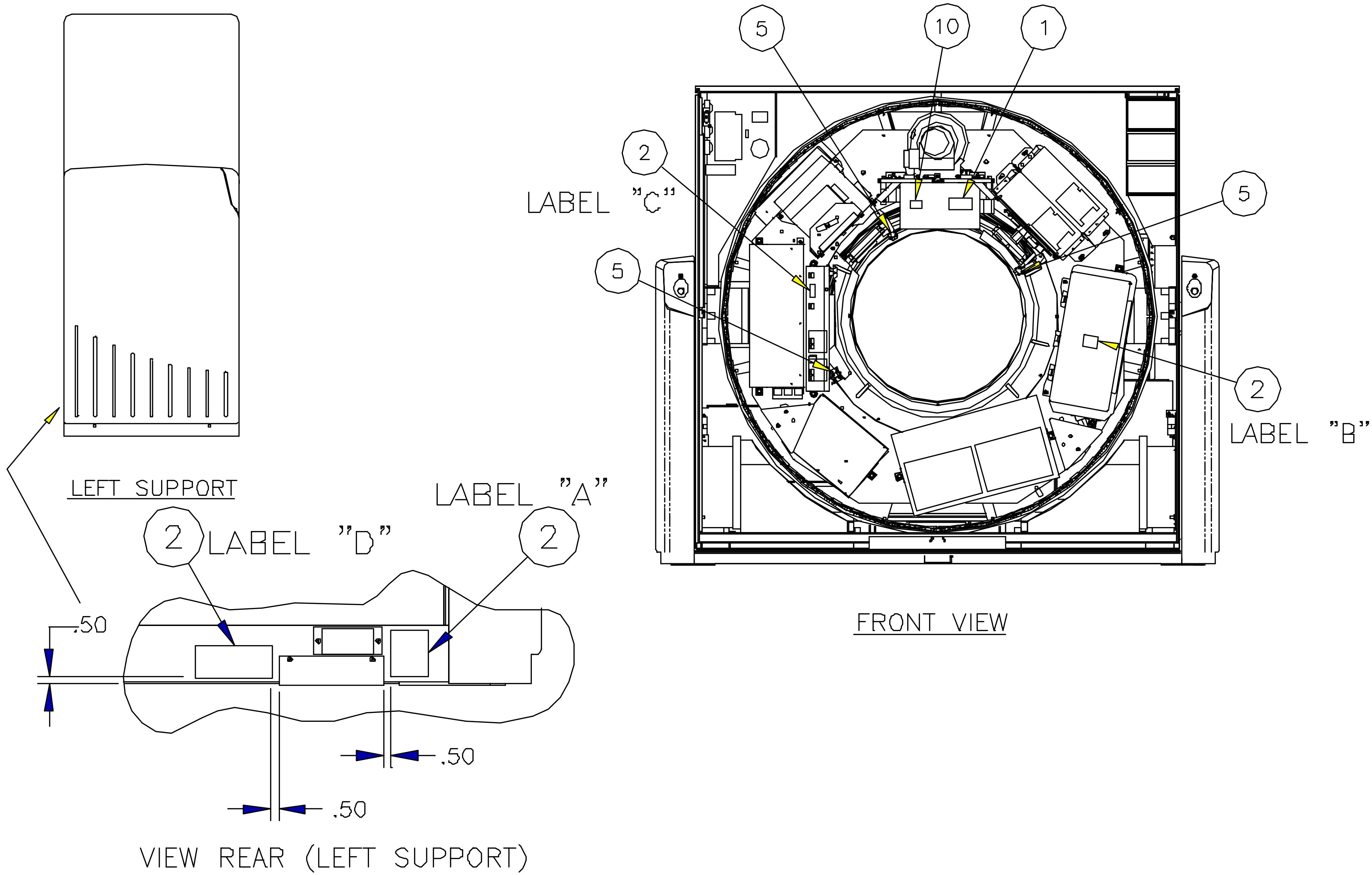


FIGURE 34 GANTRY LABELS

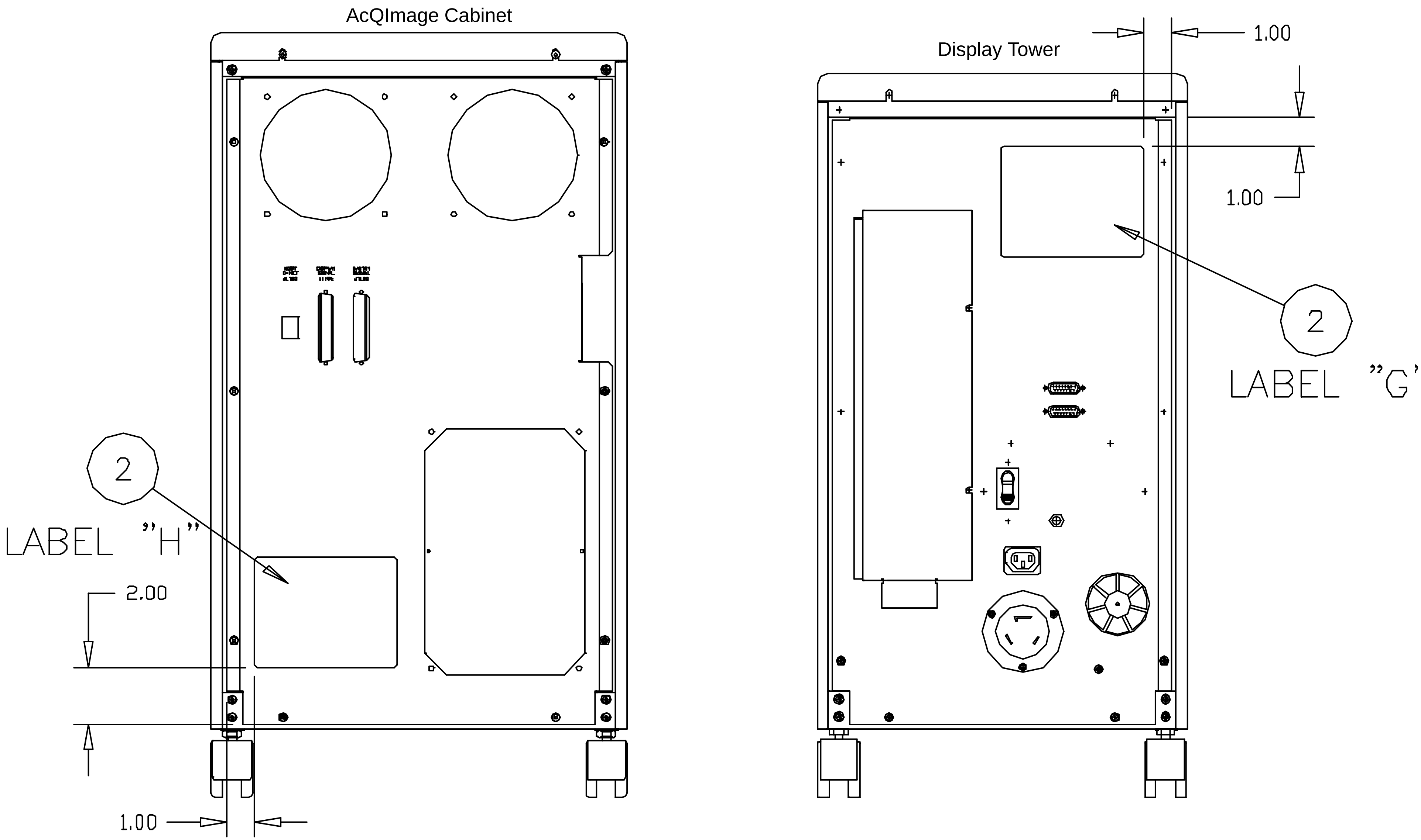


FIGURE 35 ACQIMAGE CABINET / DISPLAY TOWER LABELS

Place the castors, etc back into the crate they came in.   Pack and return the column connector plate and column supports. All must be returned to WHQ.

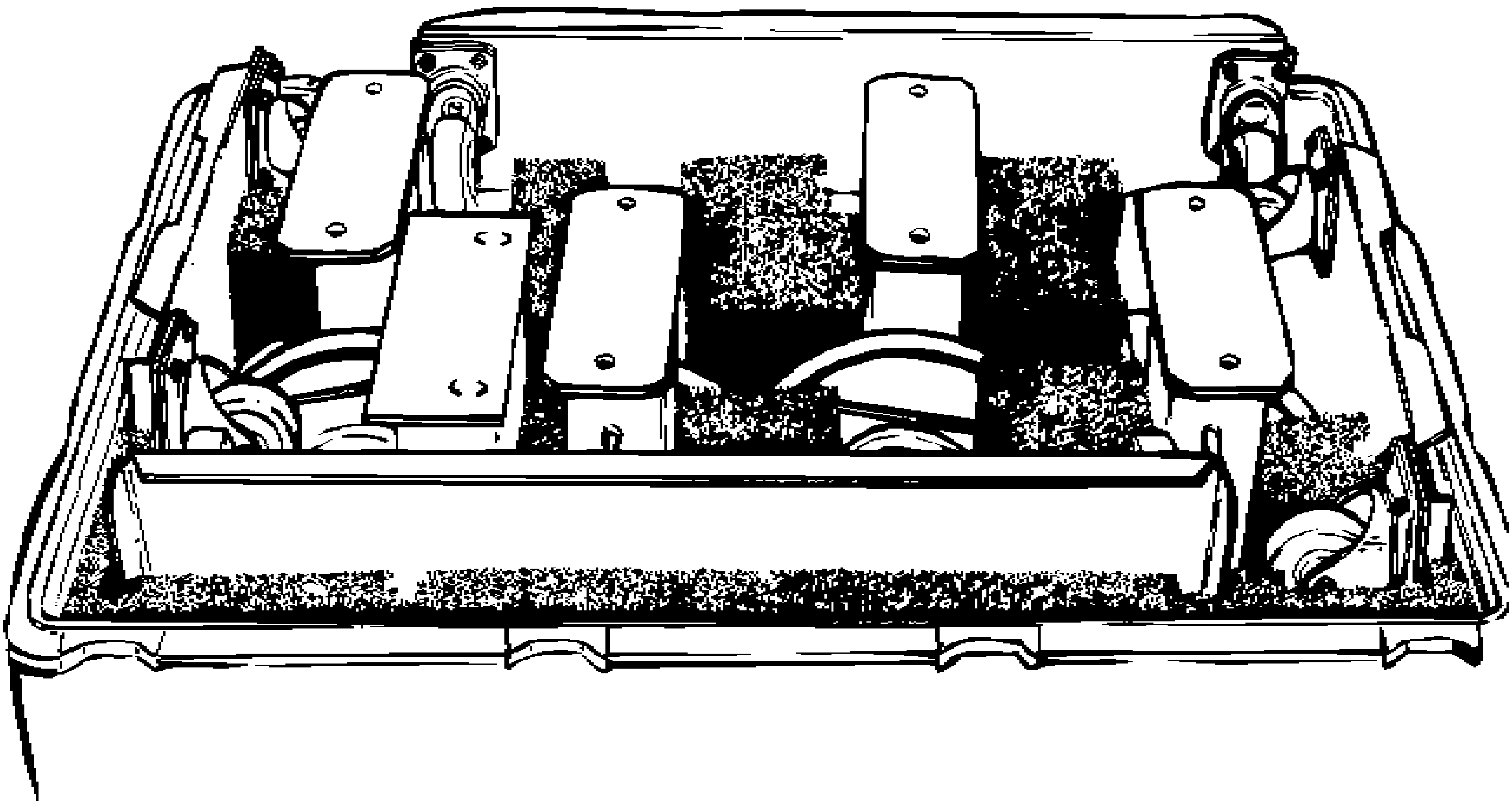


FIGURE 36   SHIP–TO SUPPLIES PACKED FOR RETURN SHIPMENT

SYSTEM INSTALLATION PROCEDURE

INTRODUCTION

In this section you:

- connect power to the Gantry, Patient Support, AcQImage Cabinet and Display Tower
- verify power phasing
- interconnect the Gantry to the AcQImage Cabinet
- interconnect the AcQImage Cabinet with the Display Cabinet
- align the Patient Support to the Gantry
- install the X–ray tube (as required)
- install the monitor

INSTALL POWER AND SIGNAL CABLES

In this section you:

- install and connect the main power cable into the Gantry
- install and connect the Gantry–to–AcQImage Cabinet cables, the AcQImage Cabinet to Display Tower cables and the Display Tower to Operator Console cables.

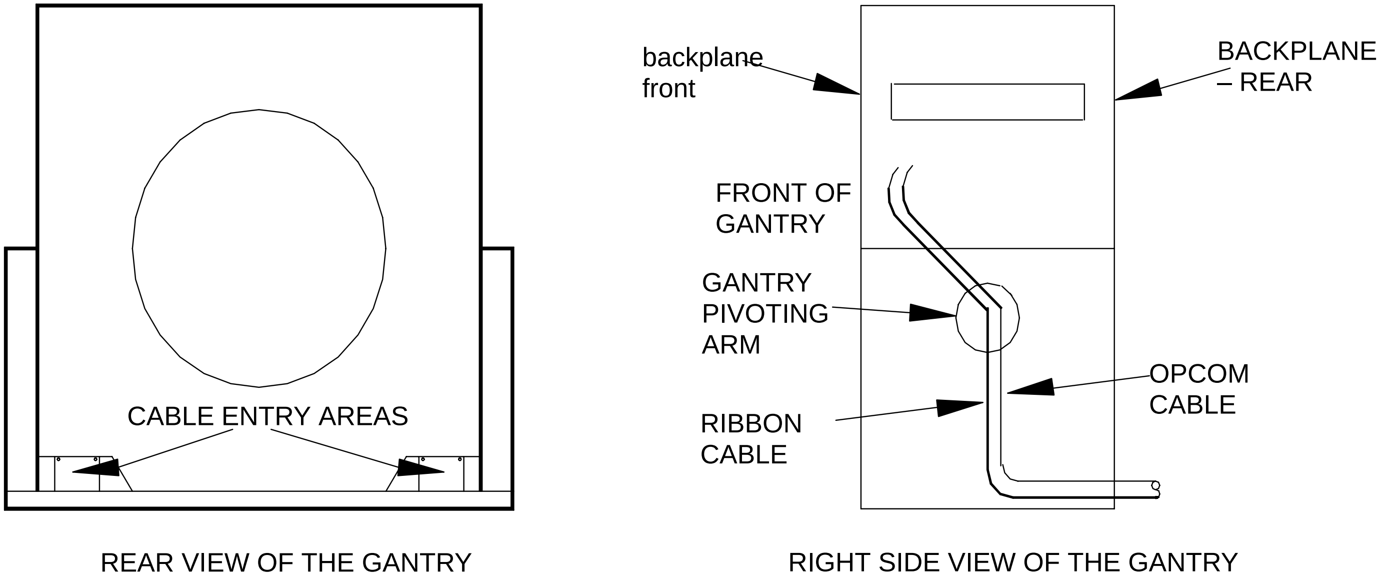


FIGURE 37 CABLE ENTRY AREAS

TABLE 1   INTERCONNECT CABLES

CABLE NUMBER	CABLE LENGTH	CABLE DESCRIPTION	FROM LOCATION AND CONN.	TO LOCATION AND CONN
L11038	50 FT.	Gantry Signal Cable	Gantry Backplane, J103	AcQImage CabinetJ1193 (Use P1193 end)
L11038–A	25 FT.			
L11036	50 FT.	Gantry Data (Ribbon)	Gantry Backplane, J101	AcQImage Cabinet Backplane, #11 on J2
L11036–A	25 FT.			
L11037	50 FT.	Gantry Power Cable	Gantry Left Column, TB12	AcQImage Cabinet, J1100
L11037–A	25 FT.			
L11039	50 FT.	Ethernet Cable –100 Base T4	AcQImage Cabinet, J1195	Display Cabinet, SUN Motherboard, J1252 (Run inside cabinet)
L11039–A	10 FT.			
L11040	50 FT.	Power Cable	AcQImage Cabinet, J1101	Display Cabinet, J1200
L11040–A	10 FT.			
L11041	50 FT.	Display Signal Cable	AcQImage Cabinet, J1194 (Use P1194 end)	Display Cabinet, J1211
L11041–A	10 FT.			
*L7161	12 FT.	Couch Signal Cable	Couch Control Assembly PCB, P402	Gantry Backplane, J108
*L7272	16 FT.	Couch Power Cable	Couch Control Assembly PCB, P417	Gantry Left Column, TB13

* Completed previously in the Uncrating section of this manual.

NOTE

DO NOT bend the ethernet cables sharply. DO NOT place the ethernet cables near the electronics of the system components.

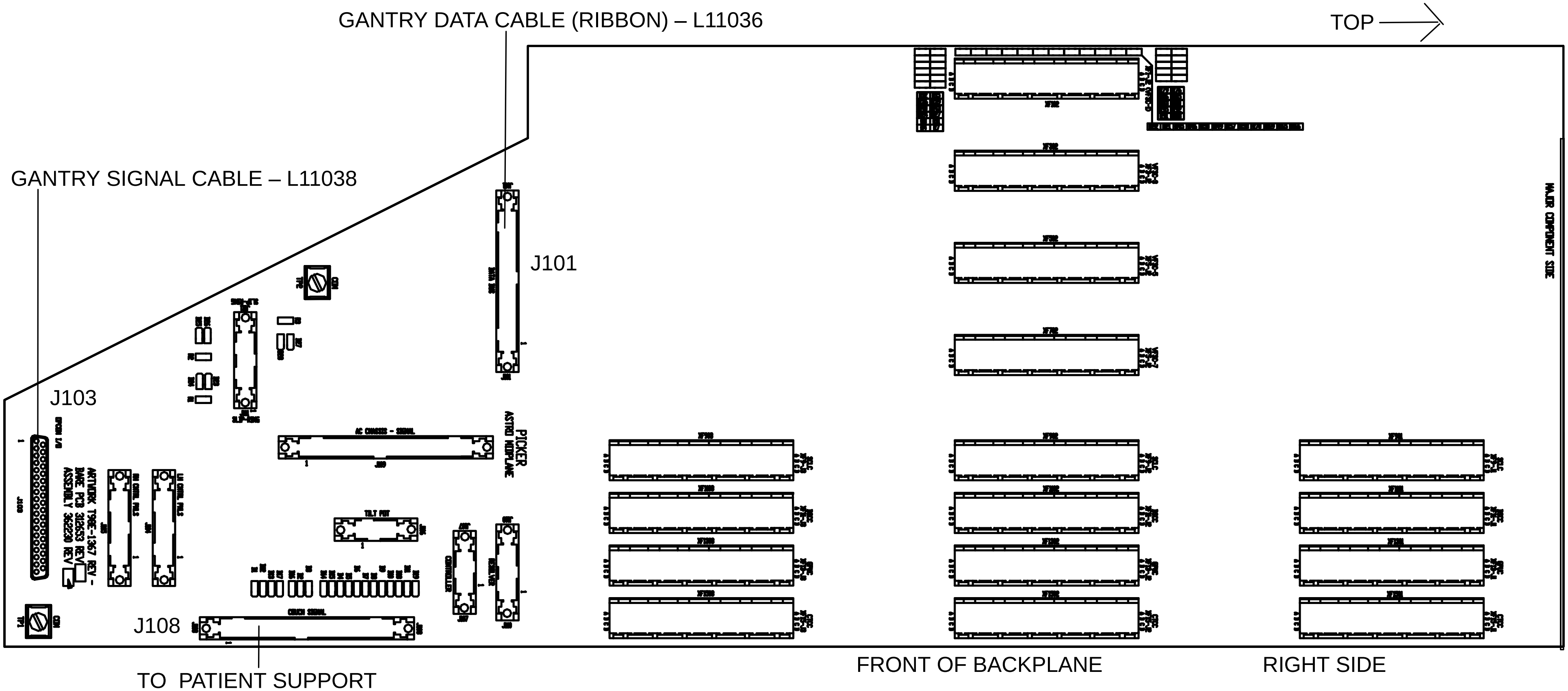


FIGURE 38 GANTRY CARDFILE BACKPLANE – FRONT

1. Connect the Suite Annunciator (if being used) to TB20 in the left column of the Gantry. See Figure 39 for the location of TB20.
2. Connect the Door Interlock (if being used) to TB30 in the <u>right</u> Gantry column.

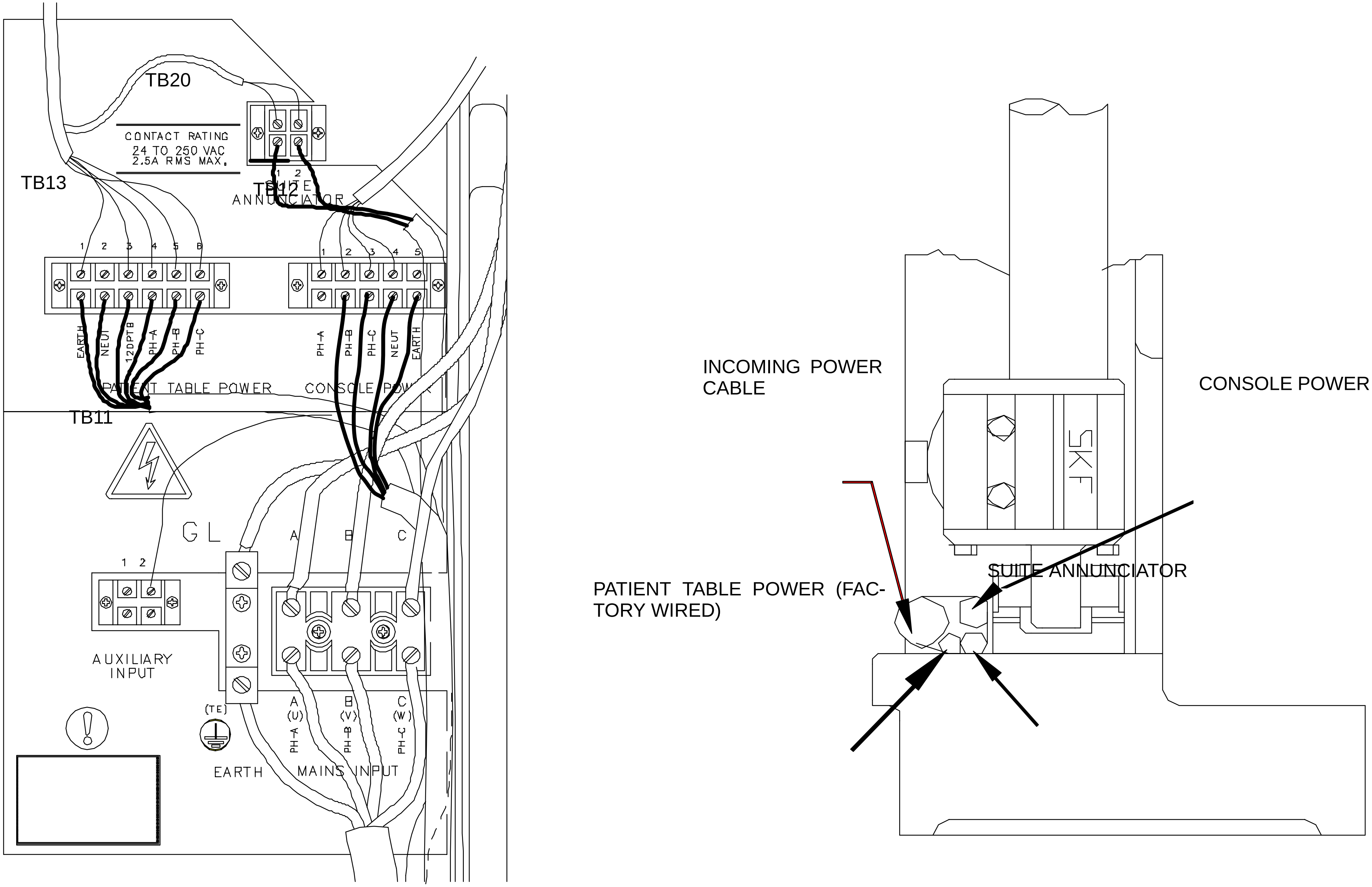


FIGURE 39 LEFT COLUMN WIRING

3. Install/Connect System Main Power At Gantry: Route the incoming main power lines through the access holes located at the rear of either Gantry column. Refer to Figure 37.

4. Secure the power lines to TB10 and the ground lug (GL1) to ground stud (GS) as shown in Figure 39. The thick power line can be routed under the tilt arm. More flexibility can be obtained by carefully removing the black outer cover to expose the four individual wires. Ensure that the line is not located where the column cover will slide down.

5. Wire the Recon cable to TB12. From left to right: Open, Black, Red, White, Green. Wire the couch cable to TB13. See Figure 39. **For detailed information**, refer to the AcQImage Cabinet to Gantry instructions.

6. Put on anti–static wrist strap, grounding it to Gantry card file chassis.

> ⚠ **CAUTION**
>
> STATIC ELECTRICITY HAZARD. PRINTED CIRCUIT BOARDS MUST BE HANDLED WITH ANTI–STATIC TECHNIQUES. DISCHARGE YOUR BODY BEFORE HANDLING ANY PCB. KEEP ALL PCBS IN ANTI–STATIC BAGS WHEN NOT INSTALLED IN THE EQUIPMENT. FAILURE TO COMPLY CAN CAUSE PERMANENT DAMAGE TO COMPONENTS ON PRINTED CIRCUIT BOARDS.

> **NOTE**
>
> The threaded ends of the gantry cardfile cover screw holes are plastic. This helps to hold the screws in place, and are therefore labeled “captive.” However, the screws can come completely out if loosened too far.

7. Remove the gantry cardfile covers by loosening the four captive screws on each cover. Reseat all boards in the Gantry card file rack — merely press each board into the card file, do not remove any! Replace the cardfile covers.

> ⚠ **WARNING** ⚠
>
> **ELECTRIC SHOCK HAZARD. USE EXTREME CAUTION WHEN PERFORMING GANTRY INSTALLATION PROCEDURES WITH THE COVERS OPEN. 480 VAC, 208 VAC AND 120 VAC, AS WELL AS 5–18 VDC ARE EXPOSED IN MANY LOCATION. FAILURE TO COMPLY CAN RESULT IN SERIOUS INJURY OR DEATH.**

8. Visually inspect the power <u>and</u> signal slip ring brushes. All brush wires must be down in their respective grooves. A "brush" with one or two damaged wires can have a wire cut. ***Brushes must have at least 19 wires!***

TECHNICAL INFORMATION: Reverse rotation of the scan frame (clockwise) can result in one or more brush wires being bent/distorted. Severely bent wires in one groove can come in contact with other brush components in adjacent grooves. Because of the potential on these brushes, 480 (or 208) vac for the X–ray power supply and 120 vac for the lasers and the XSC, damage will almost certainly occur to the brush holder; or, on the signal slip rings, damage will most certainly occur to any card associated with the affected 5 volt line. BE AWARE that *reverse rotation can occur during shipment because the scan frame is not locked.*

9. Perform **DC power check**. Refer to Figure 40.
   a. Remove main drive fuses F310, F311, F312.
   b. On the Gantry AC Control Panel, turn DRIVES switch to ○, TILT/VERTICAL DRIVES to **|** (on), CONTROL ELECTRONICS to **|**, and POWER SUPPLY to **|**. (These are not "1's" but are vertical bars.)
   c. Select SERVICE mode on the AC Control Panel.
   d. Turn Wall Power ON and press START button.
   e. Check the Power Monitor board indicators at the right rear of the gantry (All <u>6</u> green indicators should be ON and the green DC OK LED on the AC Control Panel should be ON. If a condition other than this exists, check Wall Power Supply connections and the card file 5–output DC power supply. Refer to Figure 41 for locations.
   f. Press EMERGENCY STOP button. Turn Wall Power OFF.

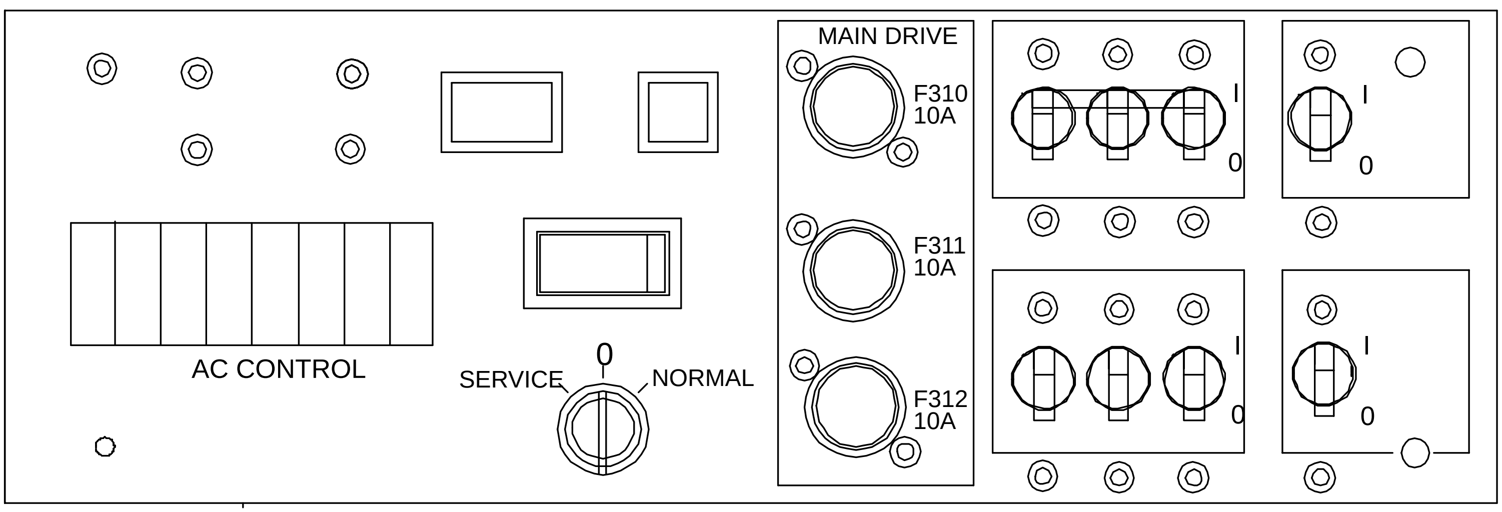


FIGURE 40 AC CONTROL PANEL

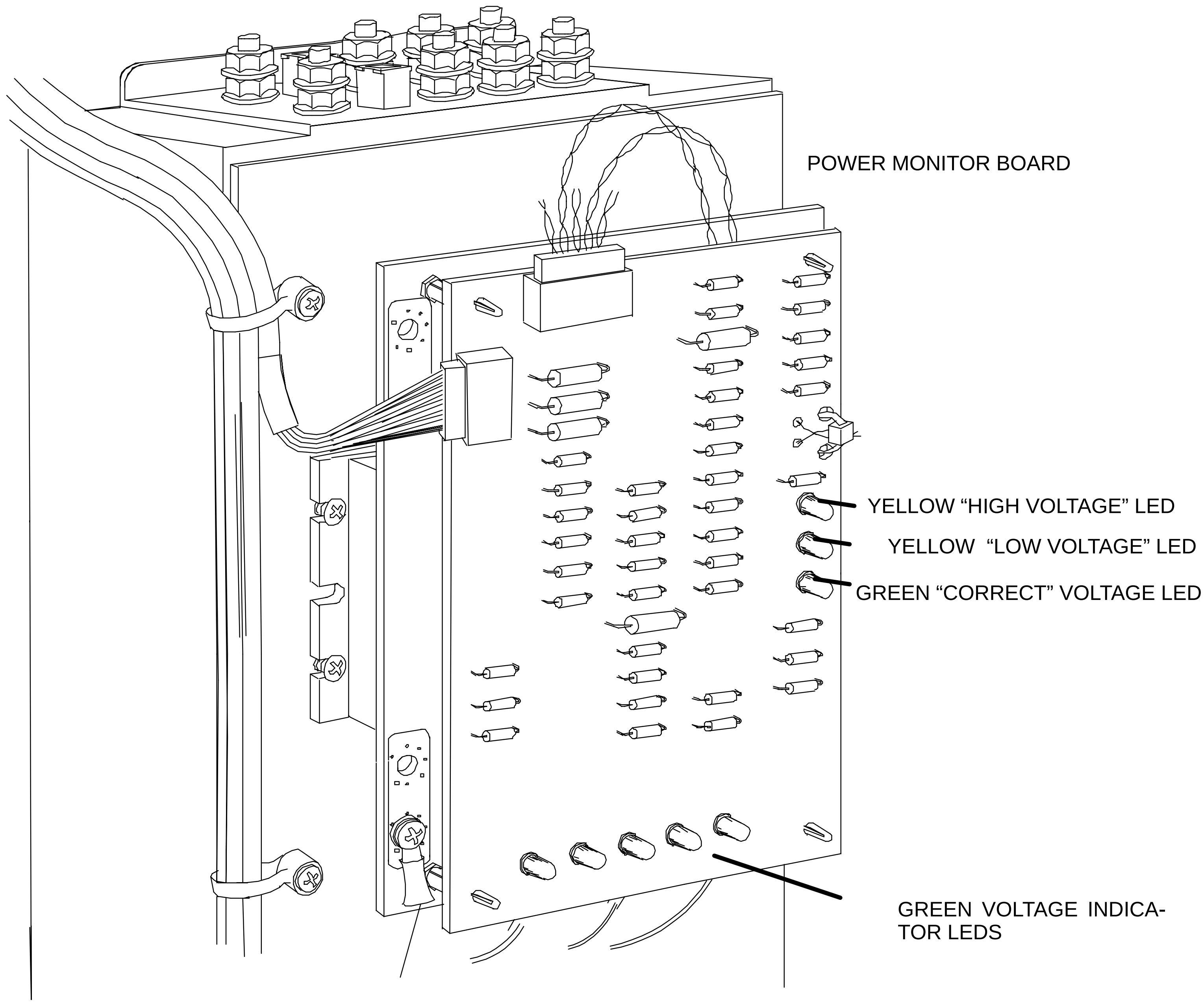


FIGURE 41 POWER MONITOR BOARD

WARNING

ELECTRIC SHOCK HAZARD. USE EXTREME CAUTION WHEN PERFORMING GANTRY INSTALLATION PROCEDURES WITH THE COVERS OPEN. 480 VAC, 208 VAC AND 120 VAC, AS WELL AS 5–18 VDC ARE EXPOSED IN MANY LOCATION. FAILURE TO COMPLY CAN RESULT IN SERIOUS INJURY OR DEATH.

10. Turn ON Wall Power. Press the START button located on the Gantry AC Control Panel.
11. Lower the front gull–wing panel halfway for access to the Gantry Control panels.

VERIFY MAIN POWER PHASING

In this section you will verify main power phasing by exercising the Patient Support.

⚠ WARNING ⚠

ELECTRIC SHOCK HAZARD. USE EXTREME CAUTION WHEN WORKING INSIDE THE PATIENT SUPPORT WITH THE COVERS OPEN. 480 VAC, 208 VAC AND 120 VAC, +17 VDC, –17 VDC AND 15 VDC ARE EXPOSED. FAILURE TO COMPLY CAN RESULT IN SERIOUS INJURY OR DEATH.

⚠ WARNING

CRUSH HAZARD. SCISSOR–ACTION MACHINERY IS EXPOSED WHEN THE PATIENT SUPPORT COVERS ARE REMOVED. KEEP EQUIPMENT AND PERSONNEL AWAY FROM THE MECHANICAL ACTION OF THESE PARTS. FAILURE TO DO SO CAN CAUSE SERIOUS DAMAGE TO EQUIPMENT AND SERIOUS INJURY OR DEATH TO PERSONNEL.

⚠ CAUTION

THE PATIENT SUPPORT DOWN LIMIT SWITCHES ARE NOT ENABLED WHEN THE PATIENT SUPPORT IS TRAVELING IN THE UP DIRECTION. DO NOT EXERCISE THE UPWARD TOUCH COMMAND AT THE GANTRY CONTROL PANEL AT THIS TIME. FAILURE TO COMPLY CAN CAUSE DAMAGE TO THE BALL SCREW AND BEARING.

1. Check the phase rotation of the incoming main power. Exercise the vertical motion of the Patient Support by a downward touch command at the Gantry touch panel. **If the downward command causes the Patient Support to move upward, correct the phasing immediately.** Correct this condition by turning OFF main power at the wall box and reversing any two of the three line conductors supplying power to the system (in the Gantry column at TB10). Then turn main power ON.
2. Raise the Patient Support about 1/2–way up (approximately 680 mm on the Gantry display panel. ***Turn power off at the wall box.***

3. Install the safety ground (green & yellow striped wire) from the Gantry to the Patient Support. ***This connection must be made in the Patient Support, not in the cable trough.*** Remove all wires from the grounding stud in the Patient Support. Put the incoming wire from the Gantry on the bottom. Reinstall the ground wires. Connect ground wire from the Patient Support ground stud to the ground stud on the cable trough cover. Refer to Figure 42.

4. Install cable trough cover on the cable trough.

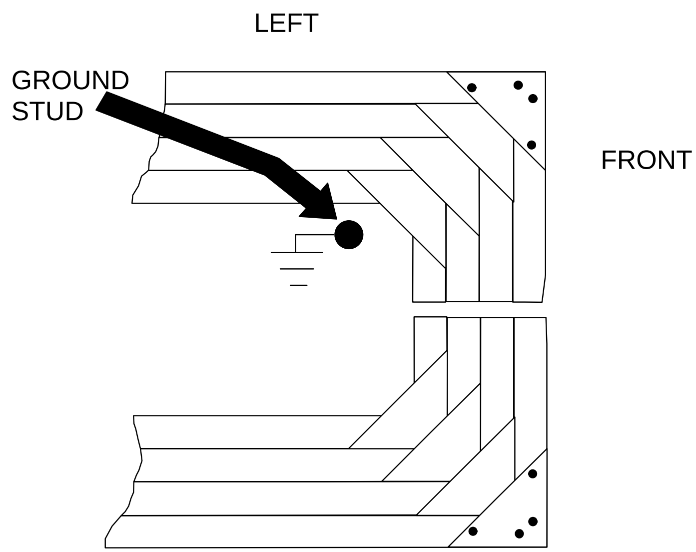


FIGURE 42 PATIENT SUPPORT GROUND STUD

ALIGN GANTRY COLUMNS AND BOLT DOWN

This section details aligning the Gantry columns and then bolting the Gantry to the floor.

Proper Gantry operation requires that the Gantry support columns be aligned with each other, be rigidly secured to the floor, that there are no gaps between a column and the floor at the six hold down bolt holes, and that correct torque is applied to the mounting bolts. Failure to properly mount the columns can cause: a Gantry side panel to contact its nearby column during tilt movement, inadvertent opening of a tilt actuator switch, or Gantry sway in excess of the specified +/– .002–inch limit at speed 1.

Use the Gantry shims included in the Anchor Hardware kit (176360) in the following steps, as required.

ALIGNMENT AND BOLT DOWN

> ⚠ **CAUTION**
>
> USE THE SUPPLIED 2–INCH BOLTS (S617–129) UNLESS DIRECTED OTHERWISE. BOLTS THAT ARE TOO LONG FOR THE ANCHORS CAN BE TORQUED TO SPECIFICATION (30–40 FT–LBS) BUT WILL NOT SECURE THE GANTRY.

1. At this point, all six bolts with 1/8–inch thick hardened washers should be installed in the columns. The bolts should be backed off about 1/4–inch for clearance so you can raise the Gantry during the aligning procedure. If the front two bolts are not installed, tilt the Gantry to +5° and install them now.

2. Turn main power OFF at the wall box.
3. Place a level — oriented front to rear — on top of one column. Note the level reading.
4. Move the level to the other column. Again note the level reading. Compare this with the first observation.

   <u>If level readings of both columns are the same:</u>

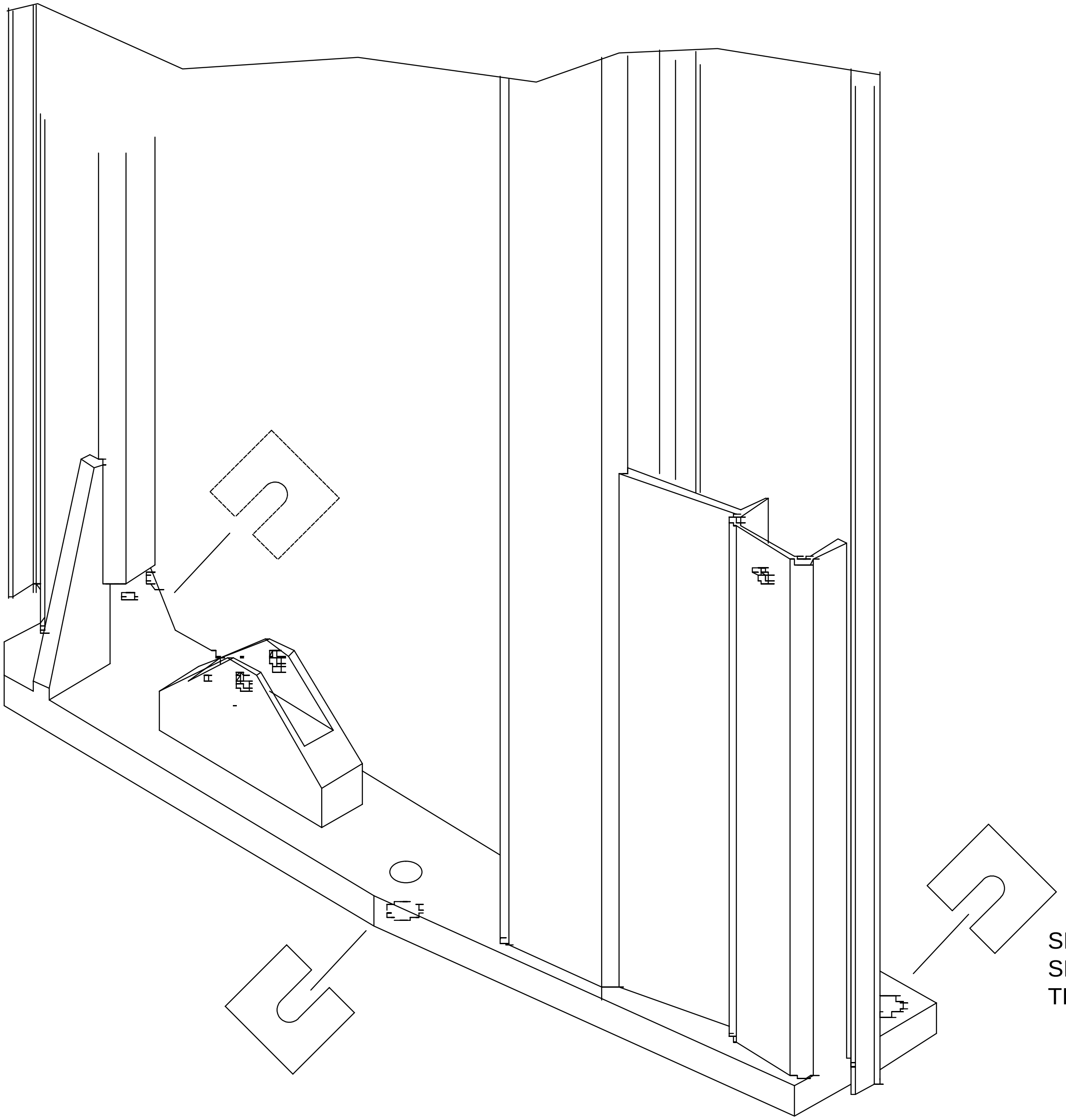
   a. column alignment is acceptable
   b. proceed to Step 6.

   <u>If the level readings are **not** the same:</u> continue at the next step.

5. Do the following steps to bring the most out–of–level column into parallel with the other column.

> **NOTE**
>
> Although, it is generally not necessary that the column tops are level, it is better that they are. All AcQsim sites **<u>must</u>** be level for proper system performance.

   a. Use the caster discs to raise the column for shim placement.
   b. Place shims under the front or rear of the column being adjusted so that the level reads the same on both columns. Refer to Figure 43. The shims have a slot that straddles the floor bolt. Place the shim so that it extends outward from the bolt such that it supports the entire width of the column. Decide if sliding the shim from the inside or outside of the column is easier.
   c. The gap between the gantry and each column must be even from top to bottom on both sides OR rubbing may occur. Refer to the <u>Gantry / Column Gap Alignment Procedure</u> and Figure 44 for details.



SHIM STRADDLES BOLT. PLACE SHIM SO IT SUPPORTS WIDTH OF THE COLUMN.

FIGURE 43 SHIM PLACEMENT AT GANTRY MOUNTING BOLTS

6. Lower the Gantry to the floor and check for gaps between each column and the floor at all bolt locations. Use shims to eliminate gaps. Do this for all Gantries; even if no column alignment was done. (Use the 3–inch shims at the center bolt locations.)
7. Ensure that the gap between the gantry and each column is the same and parallel from top to bottom. If this is not the case, rubbing of the gantry can occur. To remedy this problem:

Gantry / Column Gap Alignment Procedure (if necessary)

8. Attach a level on the top inside of the gantry. Using the casters, raise or lower them to get the system level. (Do not watch the gap at this time since the casters cause the column to distort inward slightly).
9. Place shims under the column that needs them. Refer to Figure 43. Loosen the casters so the system is resting only on the floor or shims. The bolts should be installed, but not tight at this time. Turn on power at the Wall Box and tilt the gantry back and forth once (about 10–15 deg) to allow it to center itself. Now check the gap between the gantry and columns. They must be parallel and even. Refer to Figure 44. Exercise the Gantry tilt function several times to ensure there is no binding or rubbing of the Gantry covers against the columns.

> ⚠ **CAUTION**
>
> IN THE NEXT STEP, DO NOT EXCEED THE MAXIMUM TORQUE OF 40 FT–LBS. EXCESSIVE TORQUE CAN PULL THE ANCHORS OUT OF THE FLOOR.

10. Ensure that the hardened, 1/8–inch thick washers and lock washers are installed between the bolt heads and Gantry base. ***Torque all Gantry floor bolts to 30–40 ft–lbs.***
11. Turn off wall power.

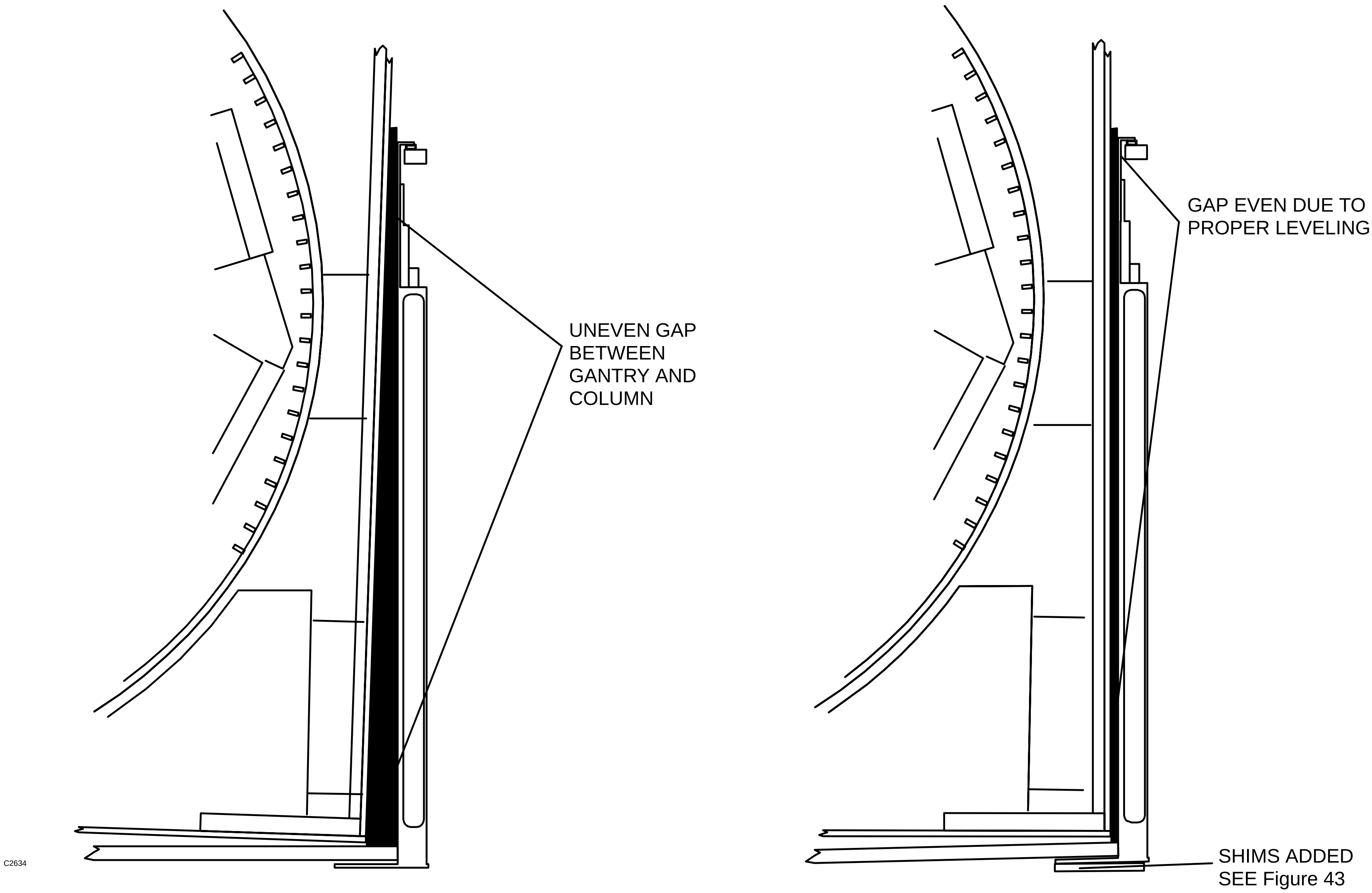


FIGURE 44 GANTRY / COLUMN GAP VERTICAL ALIGNMENT

CAUTION

WHEN REMOVING THE REAR CASTERS, HOLD THE LONG STIFFENING BAR BEFORE REMOVING THE BOLTS OR IT MAY FALL ON YOU. FAILURE TO COMPLY CAN RESULT IN INJURY TO SERVICE PERSONNEL.

12. Use a 3/4–inch wrench to remove all four caster assemblies from the Gantry.
13. Place caster assemblies (and chain from Gantry) in the installation container for return shipment to the factory. Figure 36 shows the placement of these items.
14. Install cable access covers (2) on the bottom rear of the Gantry. Do NOT re–install the L–shaped covers that go under the trough.

INSTALL PATIENT SUPPORT

> **⚠ WARNING ⚠**
>
> **USE EXTREME CAUTION WHEN PERFORMING INSTALLATION PROCEDURES WITH THE PATIENT SUPPORT COVERS OPEN. 480 VAC AND 120 VAC, AS WELL AS DC VOLTAGES ARE EXPOSED IN MANY LOCATIONS. FAILURE TO COMPLY CAN RESULT IN SERIOUS INJURY OR DEATH TO PERSONNEL.**

1. Turn the wall power ON.
2. Power up the Gantry in SERVICE mode, by pressing the START button on AC Control Panel.
3. Verify that the floor tape switch (foot switch) between the Patient Support and the Gantry is not binding, by activating the foot switch several times.
4. Lift the cover of the foot switch and push the cable trough squarely against the base panel of the Gantry. Make sure the foot switch is centered about the cut out in the base panel.
5. Scribe a line along the floor at the outer edge of the trough. Make sure that the floor under the cables is clear of dirt and wax. Clean this area if necessary. (This is to assure that the pressure sensitive tape used to secure the trough makes good contact with the floor).
6. Remove the protective strip from the pressure sensitive tape. Refer to Figure 45. Place the bottom trough along the scribe line and press down to secure it in place. (It may be necessary to slide the Patient Support back slightly to have adequate room to maneuver the trough. Slide the Patient Support back into place after you secure the trough in place.)

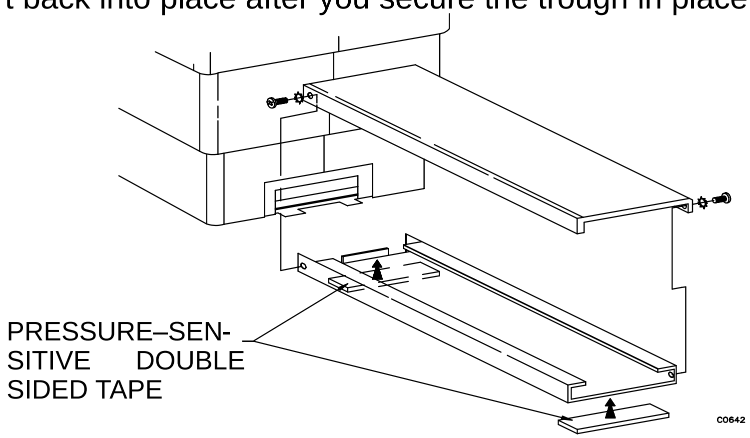


FIGURE 45 FOOT SWITCH ARRANGEMENT

7. Place the top cover onto the trough. Insert the screws (2, one at left front and one at right back) in the top trough cover. These will loosely engage the top cover to the base. Activate the foot switch several times to ensure its proper operation.

> **NOTE**
>
> The cover MUST be free to move up after it has been pressed and released. If the trough cover binds in the ”pressed” condition the top of the Patient Support will ”free float”.

8. Press an E–Stop button, then turn the wall power OFF.

> ⚠ **WARNING** ⚠
>
> **USE EXTREME CAUTION WHEN PERFORMING GANTRY INSTALLATION PROCEDURES WITH THE COVERS OPEN. 480 VAC, 208 VAC AND 120 VAC, AS WELL AS 5–18 VDC ARE EXPOSED IN MANY LOCATIONS. FAILURE TO COMPLY CAN RESULT IN SERIOUS INJURY OR DEATH TO PERSONNEL.**

> **NOTE**
>
> The Patient Support was designed to provide maximum stability to the patient while scanning. It is, therefore, rigidly and tightly assembled and still allows proper motion. Forcing a twist in the Patient Support by incorrectly bolting it to the floor can cause it to function improperly. Use the leveling screws in the base plate to allow for variations and unevenness in the suite floor.

9. Rotate the leveling screws down until they touch the floor, plus one full rotation. DO NOT ROTATE SCREWS EXCESSIVELY. – this may cause the baseplate to lift and introduce an undesired ”twist”.
10. Place a level on the flat surface of the subframe and check the Patient Support for being ”level” in both the transverse (left–right) and longitudinal (front–rear) directions. **If the pre–installation inspection determined that the floor is out of specification (ref: Architectural Planning Guide) then use the leveling screws and level the Patient Support. Place the level on the flat surface of the subframe.
11. Install and **hand–tighten** the five 1/2–13 x 1–1/4–inch Patient Support mounting bolts (S617–126) and flat washers (S612–16). Later, use a 3/8–inch Allen to torque the mounting bolts to 30–40 ft–lb. Ensure all washers and lock washers are installed between bolt heads and the Patient Support base.

This completes Patient Table Installation with the exception of final torque of the bolts, performed in the System Checks section.

VERIFY GANTRY TILT

The Gantry tilt drive consists of two linear actuators, one in each column. Ideally, each actuator shares an equal amount of the Gantry load during tilt operation. In the *extending* mode (forward tilt) the actuators automatically share the load. If one actuator speeds up, it must also carry a higher share of the load. Because DC motors slow down when loaded, this actuator slows down allowing the other actuator to regain its share of the load. Both actuators then run at the same speed.

In the *retracting* mode (backward tilt) the motors do not carry the load, rather they only need to provide enough torque to overcome their internal back–drive friction. This friction prevents the actuators from rotating when power is off.

To control the speed of the motors in the retracting mode (that is, torque load during negative tilt), a Hall–effect device senses motor rotation and an up/down counter toggles power to the faster motor. This causes both motors to supply torque within specified limits.

However, if, for any reason, one of the actuators slows down, or shares less than 40% of the retracting force, the opposite actuator must then provide 60% or more of the retracting force. As a result, the faster of the two actuators lifts on its spring–loaded pivot. A safety limit switch, located on the actuator pivot at the bottom of the column, detects this action and interrupts power to the control circuit stopping all tilt action. If an actuator fails during retracting, the circuitry removes power from the 'good' actuator and any further Gantry tilt motion ceases.

It is imperative that this safety switch perform as designed. Because an uneven floor could also cause the safety switch to break its electrical continuity, the Gantry Shimming Procedure in the Gantry Alignment performed earlier ensures the columns make solid contact with the floor.

> ⚠ **CAUTION**
>
> DO NOT ATTEMPT TO PUT A PLASTIC CABLE TIE UNDER THE SWITCH TO ACTIVATE IT. THIS CAN CAUSE DAMAGE TO THE SWITCH.

1. Turn the power ON at the wall box and exercise the Gantry tilt function. If the Gantry tilt is not functioning, jumper the contact points in the AC Control Chassis to restore continuity. Tilt the Gantry to +10° to seat the Gantry tilt actuator socket and then remove the jumper. Position the Gantry to 0° Gantry tilt.
2. Lower the Patient Support to the lowest position.
3. Verify Patient Support vertical up and down movement using Gantry front touch panel control. Move the carbon top out (away from Gantry) to furthest horizontal position. Raise the Patient Support to full UP vertical position.
4. Turn power OFF at the wall box.

END OF “VERIFY GANTRY TILT”

**** WHAT TO DO NEXT ! ****	
If the Gantry was received . . .	**Go To. . .**
With the X–ray tube installed	GO TO: OPERATOR DISPLAY CONSOLE & ACQIMAGE CABINET HARDWARE INSTALLATION
With the x–ray tube shipped separately	INSTALL X–RAY TUBE AND HEAT EXCHANGER (See next section).

INSTALL X–RAY TUBE AND HEAT EXCHANGER

This section explains how to install the X–ray tube and heat exchanger on systems where these items were shipped separately.

Required Tools:

This process requires the following tools, which may be in addition to your standard tool kit:

9/16” Socket, Ratchet or a closed–end wrench

5/32” Hex Key

Hoist Assembly (7” tube beam installation)

Required Materials:

Transformer oil

Protective cover for Patient Support couch top.

INSTALLATION NOTES, WARNINGS AND CAUTIONS

> ⚠ **WARNING**
>
> **USE EXTREME CAUTION WHEN SERVICING EQUIPMENT WHILE UPPER GULL WING PANEL IS OPEN. THESE PANELS HAVE SHARP EDGES. FAILURE TO COMPLY COULD RESULT IN SERIOUS INJURY.**

> ⚠ **WARNING** ⚠
>
> **ELECTROCUTION HAZARD. USE EXTREME CAUTION WHEN OPERATING OR WORKING ON EQUIPMENT WITH COVERS REMOVED. DANGEROUS VOLTAGES ARE EXPOSED WHEN THE GATNRY COVERS ARE REMOVED. CONTACT WITH THESE VOLTAGES MAY CAUSE SERIOUS INJURY OR DEATH TO PERSONNEL.**

WARNING

CRUSH HAZARD. SCISSOR–ACTION MACHINERY IS EXPOSED WHEN THE PATIENT SUPPORT COVERS ARE REMOVED. KEEP EQUIPMENT AND PERSONNEL AWAY FROM THE MECHANICAL ACTION OF THESE PARTS. FAILURE TO DO SO CAN CAUSE SERIOUS DAMAGE TO EQUIPMENT AND SERIOUS INJURY OR DEATH TO PERSONNEL.

XRAY TUBE COMPONENT WEIGHTS		
Tube Type	**Tube With Cradle**	**Heat Exchanger**
7 INCH	132 pounds (60 kg)	45 pounds (20.5 kg)

CAUTION

ROTATE THE SCAN FRAME COUNTER–CLOCKWISE ONLY. CLOCKWISE ROTATION OF THE SCAN FRAME MAY CAUSE DAMAGE TO THE SLIP RING BRUSHES.

INSTALL X–RAY TUBE AND HEAT EXCHANGER

At this time:

The front and rear upper Gantry covers are open and the rear Gantry covers are off.

The scan frame should be pinned at the 12 o'clock position. Refer to Figure 46.

System power is off at the wall box.

Prepare Gantry and Patient Support for Installation of Tube and Heat Exchanger

> ⚠ **WARNING**
>
> **MAKE NO ATTEMPT TO REMOVE THE HIGH VOLTAGE FEDERAL CONNECTORS FROM THE X–RAY TUBE. THEY ARE PERMANENTLY ATTACHED AND STAY WITH THE TUBE. REMOVING THESE CONNECTORS WILL CAUSE SERIOUS DAMAGE AND/OR DESTRUCTION OF THE TUBE.**

1. Assemble the hoist and secure it to the Patient Support couch top. Screw the *knob bolts* in securely and make sure the Clamp Base seats firmly in position on the Patient Support. See Figure 47.

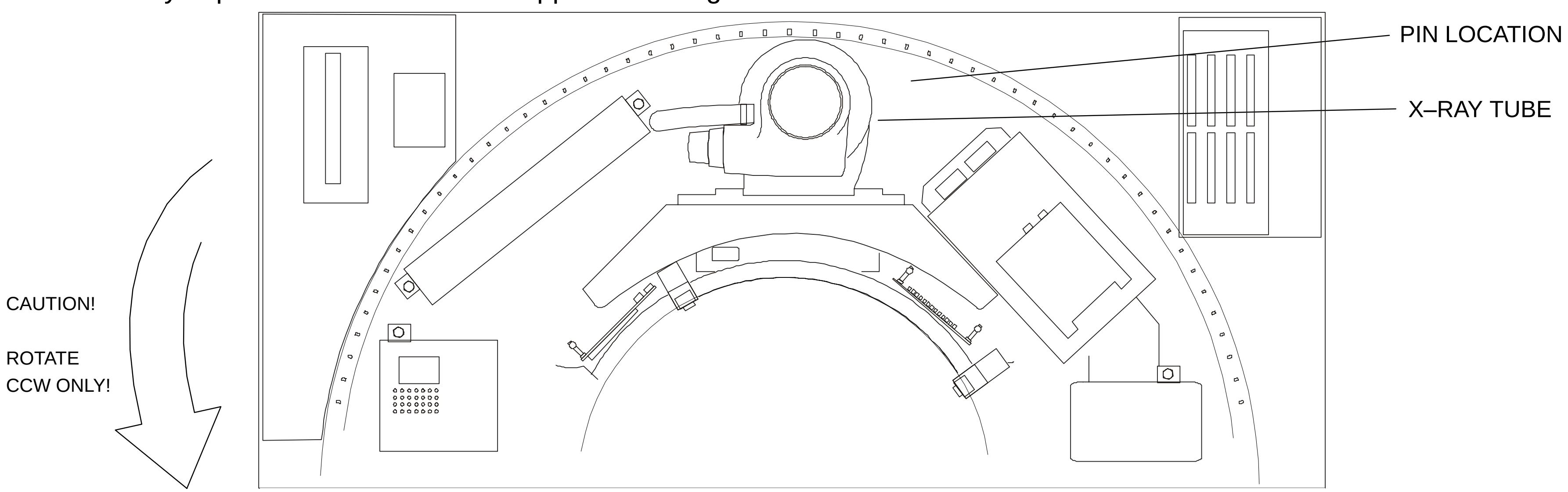


FIGURE 46   SCAN FRAME PINNED WITH TUBE AT 12 O'CLOCK

2. Place a protective cover on the Patent Support (such as a towel or several layers of a sheet).
3. Remove the four Cradle Locking Hex Head Bolts enough to slide Mounting Rails against Cradle and Tube Shelf. Install the Tray Table Slide to the Tube Shelf. Refer to Figure 48.
4. Remove the Cradle–locking Hex Head Bolts and Washers. Refer to Figure 49. (This figure shows the tube in its installed position.)

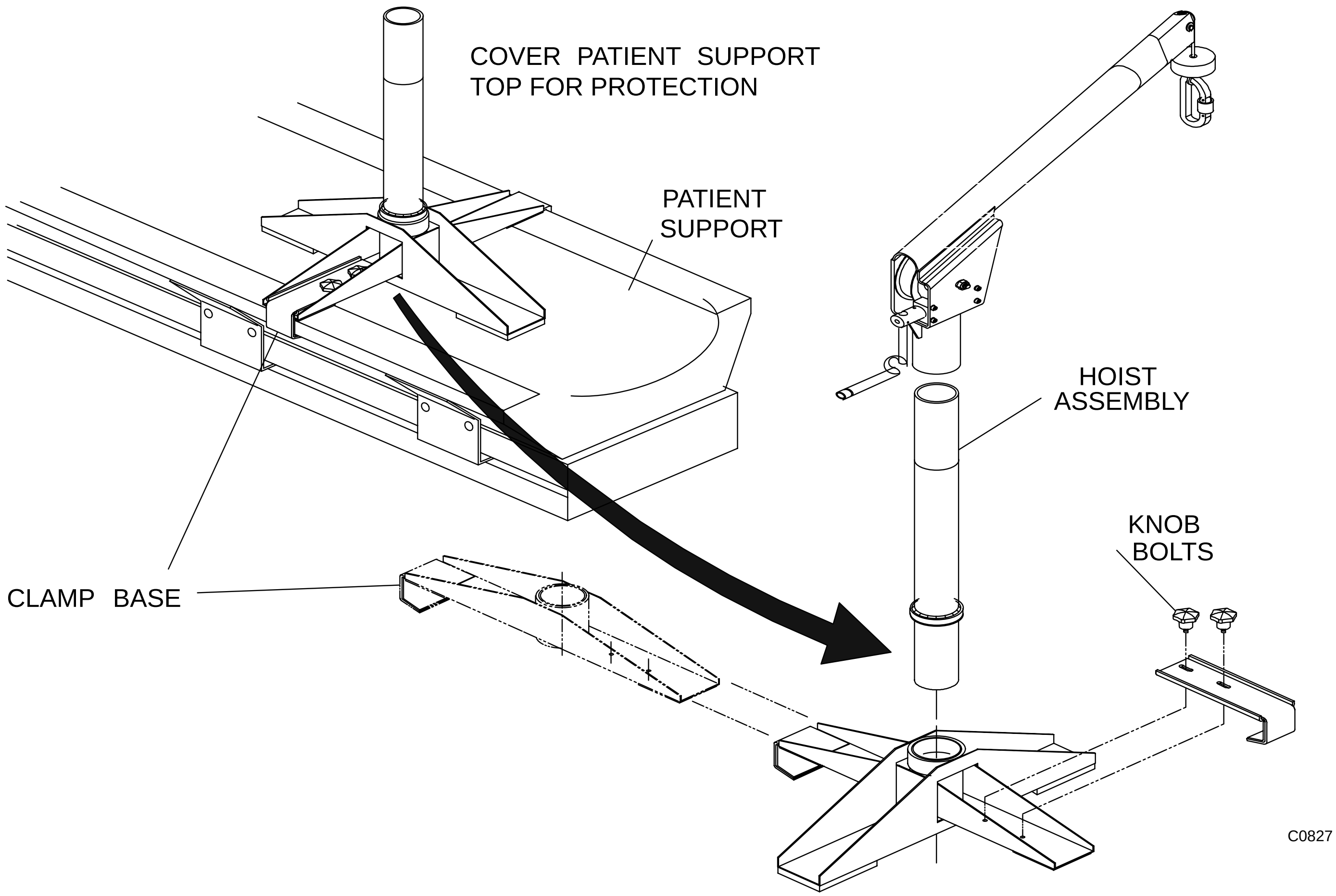


FIGURE 47 HOIST ASSEMBLY

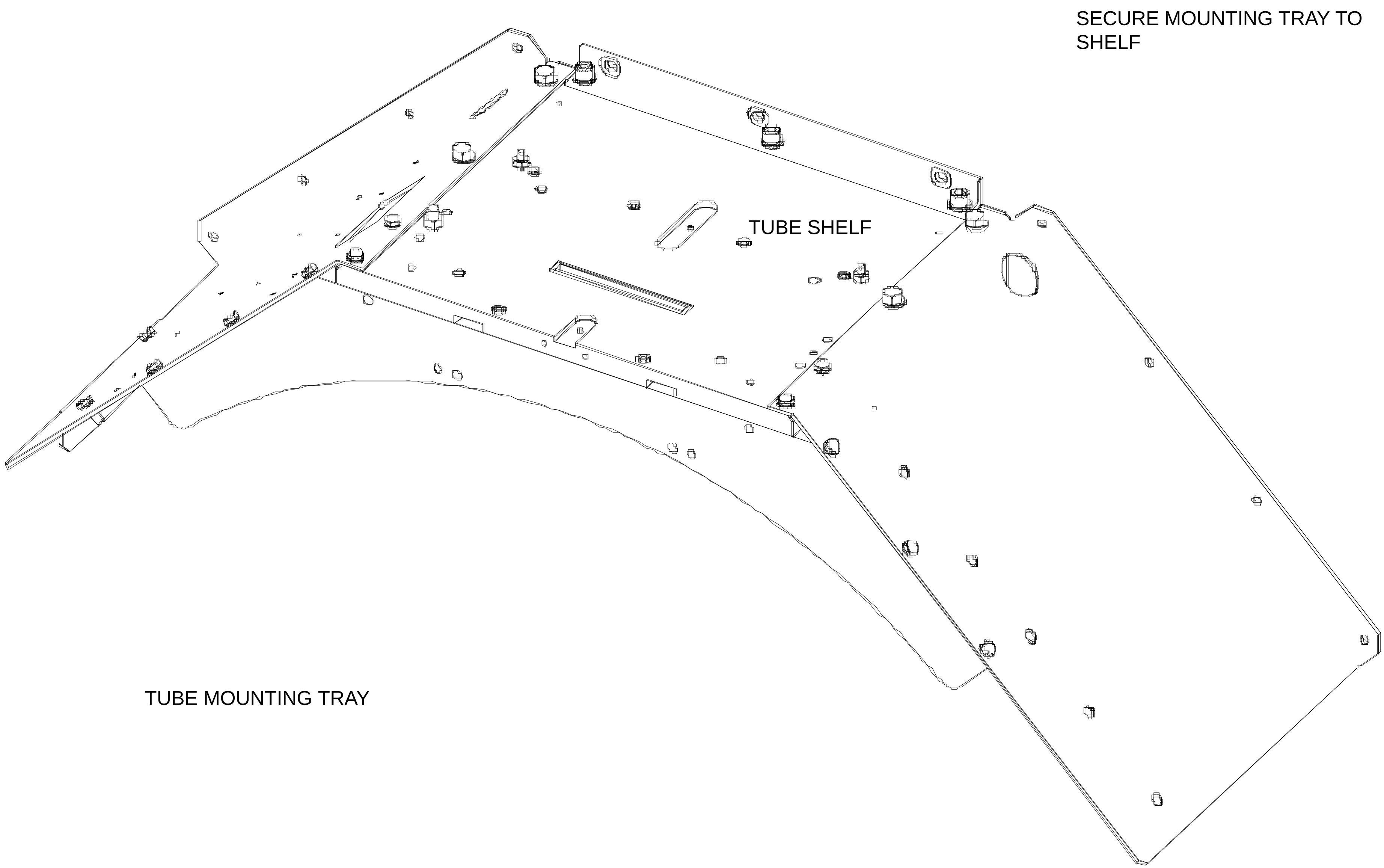


FIGURE 48 XRAY TUBE MOUNTING TRAY ON TUBE SHELF

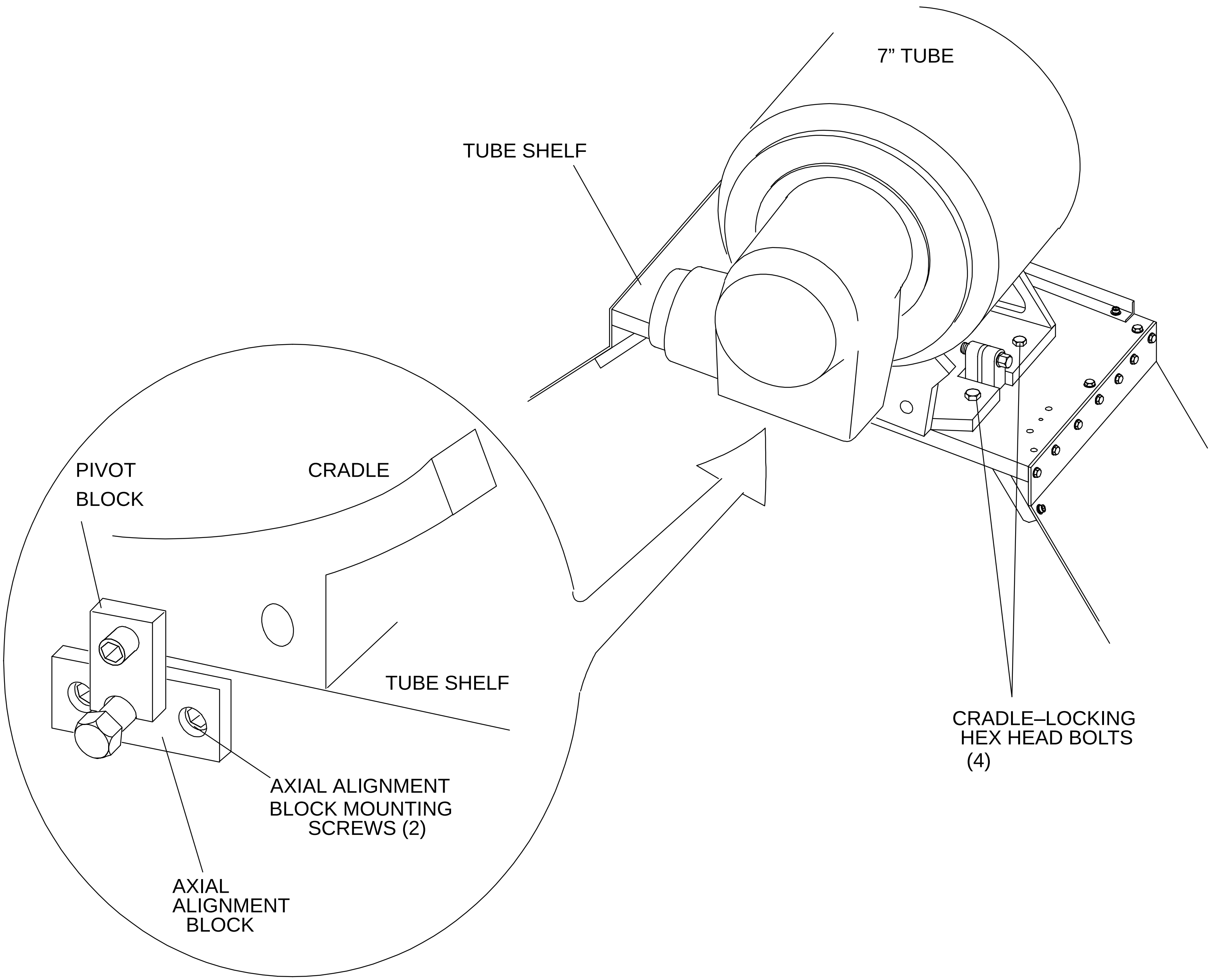


FIGURE 49 TUBE MOUNTED ON SHELF

NOTE

Recheck the mounting tray to ensure it is properly and securely installed.

5. Loosen one of the two Axial Alignment Block Mounting Socket Head Capscrews and remove the other. Let the Block swing down to allow removal of the X–Ray Tube Cradle. Refer to Figure 50.

NOTE

This method minimizes the need for axial adjustment of the X–Ray Tube if the *Axial Pivot block* is not loosened. However, later, care must be exercised not to damage the Patient Support with the protruding *Axial Pivot block*. Place a protective cover on the Carbon Top before setting the Tube and Cradle on it.

6. Install the XSC support rods into the scan frame so they can support the Heat Exchanger chassis.

Move Xray Tube Onto Patient Support

NOTE

The Couch Top must have a protective cover to prevent damage by the Axial Alignment Pivot Block attached to the tube cradle.

7. Place box with X–Ray Tube on left side of Patient Support. Open shipping container and remove packaging material.

⚠ WARNING

LIFTING HAZARD. THE HEAT EXCHANGER WEIGHS 45 LBS (20.5 KG) USE CAUTION WHEN HANDLING THIS ITEM. FAILURE TO COMPLY CAN RESULT IN EQUIPMENT DAMAGE AND/OR SERIOUS INJURY TO PERSONNEL.

8. Lift heat exchanger from container and place it on floor near front of Gantry.

> ⚠ **WARNING**
>
> **LIFTING HAZARD. THE X–RAY TUBE WEIGHS 83 LBS (37.7KG) IN THE CRADLE. USE EXTREME CAUTION WHEN HANDLING THIS ITEM. FAILURE TO COMPLY MAY RESULT IN SEVER DAMAGE TO THE EQUIPMENT AND/OR INJURY TO PERSONNEL.**

9. Place Lifting Straps through eyelets of Tube. Lift the X–Ray Tube from the shipping container and set it on the Patient Support with its emission port up.
10. With the X–Ray Tube Shutter facing upwards, remove protective covering. Check for the plastic covering over Shutter opening and check for dirt/dust. Remove any debris found on the plastic. **DO NOT REMOVE THE PLASTIC !**

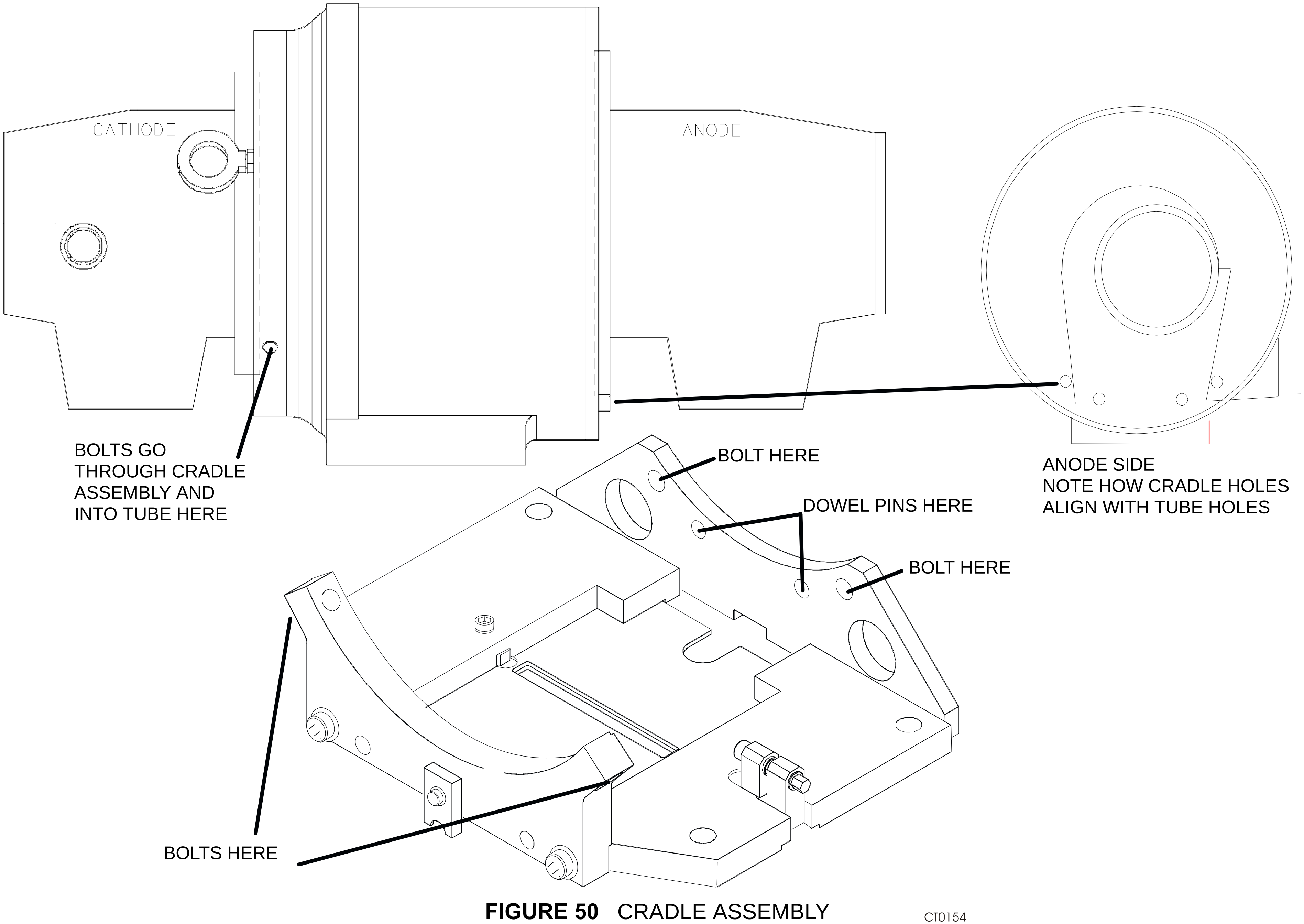


FIGURE 50 CRADLE ASSEMBLY

11. Attach Cradle Assembly to Xray Tube. Follow illustration in Figure 50 for proper installation. Do not tighten Hex Head Bolts until Cradle End Plate is firmly seated on Tube and Base Plate Alignment Pins.

   - Torque Mounting Hex Head and Socket Head Screws to 15 ft–lbs.

12. Wipe off the X–Ray Tube with cloth and check for container–debris.
13. (If possible) Rotate the X–Ray tube so that the tube cradle side faces down and the oil hoses face left on the Patient support.
14. Remove the lifting straps from the eyelets and reposition as follows:

   a. Reposition one strap through one eyelet, under the X–ray tube and up the other side through the other eyelet.
   b. Route second strap through one cradle hole, under the x–ray tube and through hole on opposite side. Refer to Figure 51.

   FIGURE NOTE: Figure 51 shows the tube on the cradle, but you route the straps in the same manner while the tube is on the Patient Support couch top.

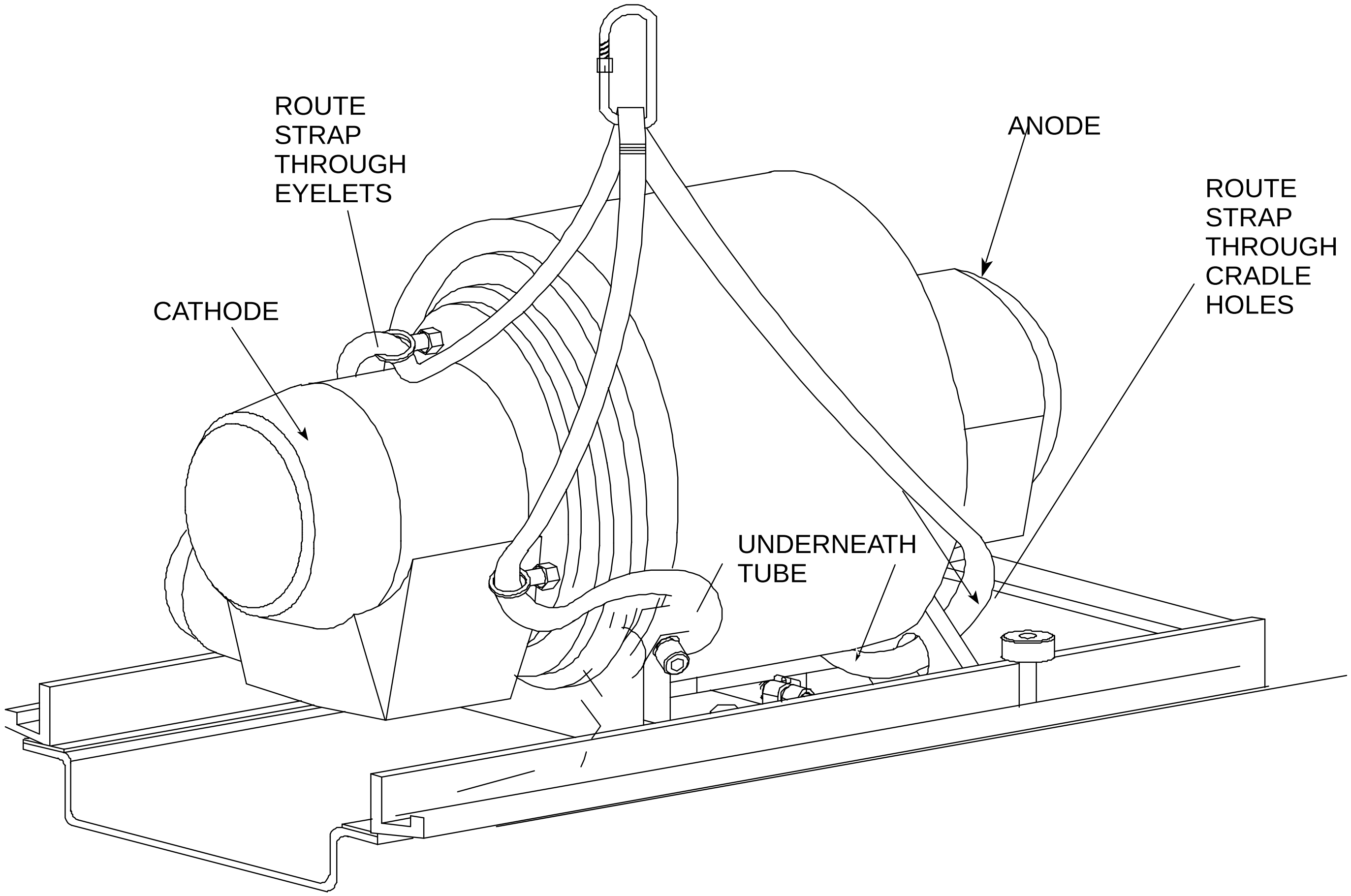


FIGURE 51 LIFTING STRAP ROUTING
MOUNTING TRAY

INSTALL TUBE INTO GANTRY

> ⚠ **WARNING**
>
> **CRUSH HAZARD. USE EXTREME CAUTION WHEN MOVING OR HANDLING OR IN GENERAL VICINITY OF X–RAY TUBE. SEE TUBE WEIGHTS IN THE TABLE AT FRONT OF THIS SECTION. FAILURE TO COMPLY CAN RESULT IN DAMAGE TO THE SCANNER OR X–RAY TUBE AND CAN RESULT IN SERIOUS INJURY OR DEATH TO PERSONNEL.**

15. Hoist the Xray Tube to the Guide Rails. Push Tube onto Rails until back of Cradle Assembly is flush with front of Tube Shelf.

> ⚠ **WARNING**
>
> **USE EXTREME CAUTION AND VERIFY THAT THE X–RAY TUBE IS FULLY SUPPORTED BY THE MOUNTING RAILS BEFORE REMOVING THE LIFTING STRAPS. FAILURE TO COMPLY MAY RESULT IN EQUIPMENT DAMAGE AND/OR SERIOUS INJURY TO PERSONNEL.**

16. Remove Lifting Straps from around Xray Tube and move the Hoist Assembly out of the way.
17. Ensure that all cables are in the clear and that none will be pinched as you slide the tube into the cradle. Push the Xray Tube all the way back onto the Tube Shelf.
18. Replace the Cradle Locking Bolts and fasten the Cradle down enough to relieve pressure from the Mounting Rails.

19. Remove Mounting Rails and secure Axial Alignment Block in its original position with the Socket Head Capscrew removed earlier.
    a. Apply Loc–Tite to Alignment Block Screws and tighten to hand–tight.
    b. Tighten Cradle hardware to 15 ft–lbs.

> ⚠ **WARNING**
>
> **LIFTING AND CRUSH HAZARD. THE HEAT EXCHANGER WEIGHS 45 LBS. USE CAUTION WHEN HANDLING THIS ITEM. FAILURE TO COMPLY MAY RESULT IN EQUIPMENT DAMAGE AND/OR SERIOUS INJURY TO PERSONNEL.**

INSTALL HEAT EXCHANGER INTO GANTRY

20. Lift the Heat Exchanger on XSC+ standoffs and push it back into position. Place the Cables behind the Heat Exchanger.

> **NOTE**
>
> Ensure the Cables rest on top the dogbone counterweights and that the cable–protecting rubber grommet is in place on top of the right heat exchanger mounting bracket.

21. Secure the Heat Exchanger to the scan frame. Refer to Figure 52.
22. Secure Oil Hoses with Hose Clamps. DO NOT OVERTIGHTEN HEX NUTS.
23. Connect Quick Disconnects J4 (Rotor Cable), J5 (Heat Exchanger Power Cable), and J6 (Thermal Switch Cable).

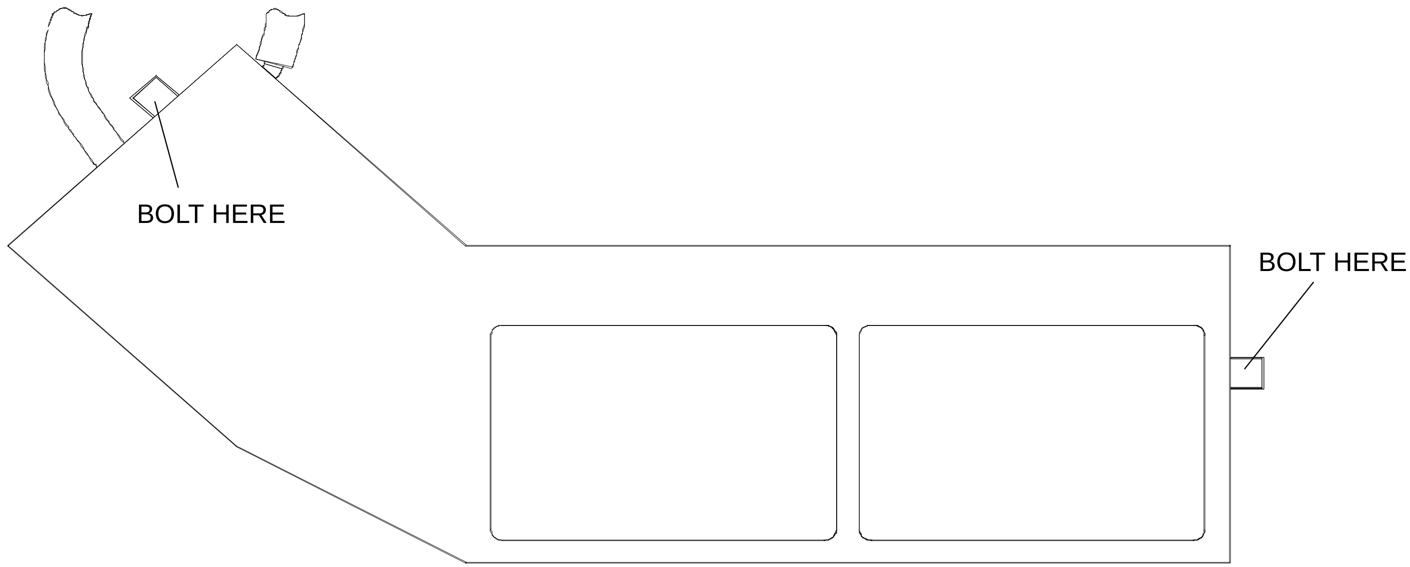


FIGURE 52 HEAT EXCHANGER MOUNTING

24. Install the Anode and Cathode High Voltage Federal Connectors. Refer to the Cable Cleaning and Installation procedure in the Repair manual.

25. VERIFY ALL CABLES AND HOSES ARE PROPERLY SECURED!

26. Remove the protective rubber strip over the detectors and secure it in the bottom of the Gantry. Remove the scan fame locking pin.

27. Rotate the Scan Frame counterclockwise by hand while checking for loose Cables. Also check for clearance over the entire range of travel between the Rotating Frame Cables and the Stationary Frame.

28. Align the X–Ray Tube. Refer to the Xray Tube Mechanical Alignment procedure in the MANEXP section of the Diagnostic Gui manual.

⚠ **CAUTION**

ROTATE THE SCAN FRAME COUNTERCLOCKWISE ONLY. CLOCKWISE ROTATION OF THE SCAN FRAME MAY CAUSE DAMAGE TO THE SLIP RING BRUSHES.

End of X–ray tube installation section. Continue to next page.

OPERATOR DISPLAY CONSOLE & ACQIMAGE CABINET HARDWARE INSTALLATION

The AcQImage Recon Cabinet (referred to as the AcQImage Cabinet) and Display Tower will generally be placed at the user's desired location by Service personnel. Considerations have already been taken into account regarding cable lengths from Gantry to AcQImage Cabinet and AcQImage Cabinet to Display Tower. Refer to the installation arrangement provided by Site Planning for cable routing, ducting and unit locations. Use the following procedure to place the two units at their desired locations.

> **NOTE**
>
> This procedure documents installation using the optional table (P/N 83847). The customer may elect to locate the table top display console on their own shelf or table, therefore a customized installation may vary slightly from the following procedure.

At this time:

The AcQImage Cabinet should already be located. See: POSITION ACQIMAGE & DISPLAY TOWERS.

1. The AcQImage Cabinet may be co–located close to the display components (similar to Figure 53) or in the Gantry room (or elsewhere) as far as 50 feet away from the Display Tower. This consideration should have been decided when choosing the cable lengths: Display Tower to AcQImage Cabinet – 10 or 25 ft., AcQImage Cabinet to Gantry – 10 or 50 ft. cable sets. Refer to Table 1 for details.
2. Each portion of the entire system cabling, Operator Console, Display Tower to AcQImage Cabinet and AcQImage Cabinet to Gantry is detailed in Figure 57, Figure 58 and Figure 59 respectively.
3. Remote access to the system is accomplished via the Site Manager.
4. **After** the two cabinets are in place, press down on the wheel locks to keep them from rolling. (Do not do this at this time.)
5. Locate the table and unpack the optional chair and roll to its location. Assemble the operator table, part number 83847, which is 54" wide by 30" deep in the Scan Control room, using the instructions that came with the table. The tools required should be included in a standard FSE toolkit.
6. Whether or not the optional table and chair are being used, the callouts in Figure 53 should be followed during installation.

> ⚠ **WARNING** ⚠
>
> **SHOCK HAZARD. ENSURE THAT THE DISPLAY CABINET KEYSWITCH IS TURNED OFF AS WELL AS THE CIRCUIT BREAKERS ON THE ACQIMAGE AND DISPLAY CABINETS. FAILURE TO COMPLY MAY RESULT IN SERIOUS INJURY OR DEATH.**

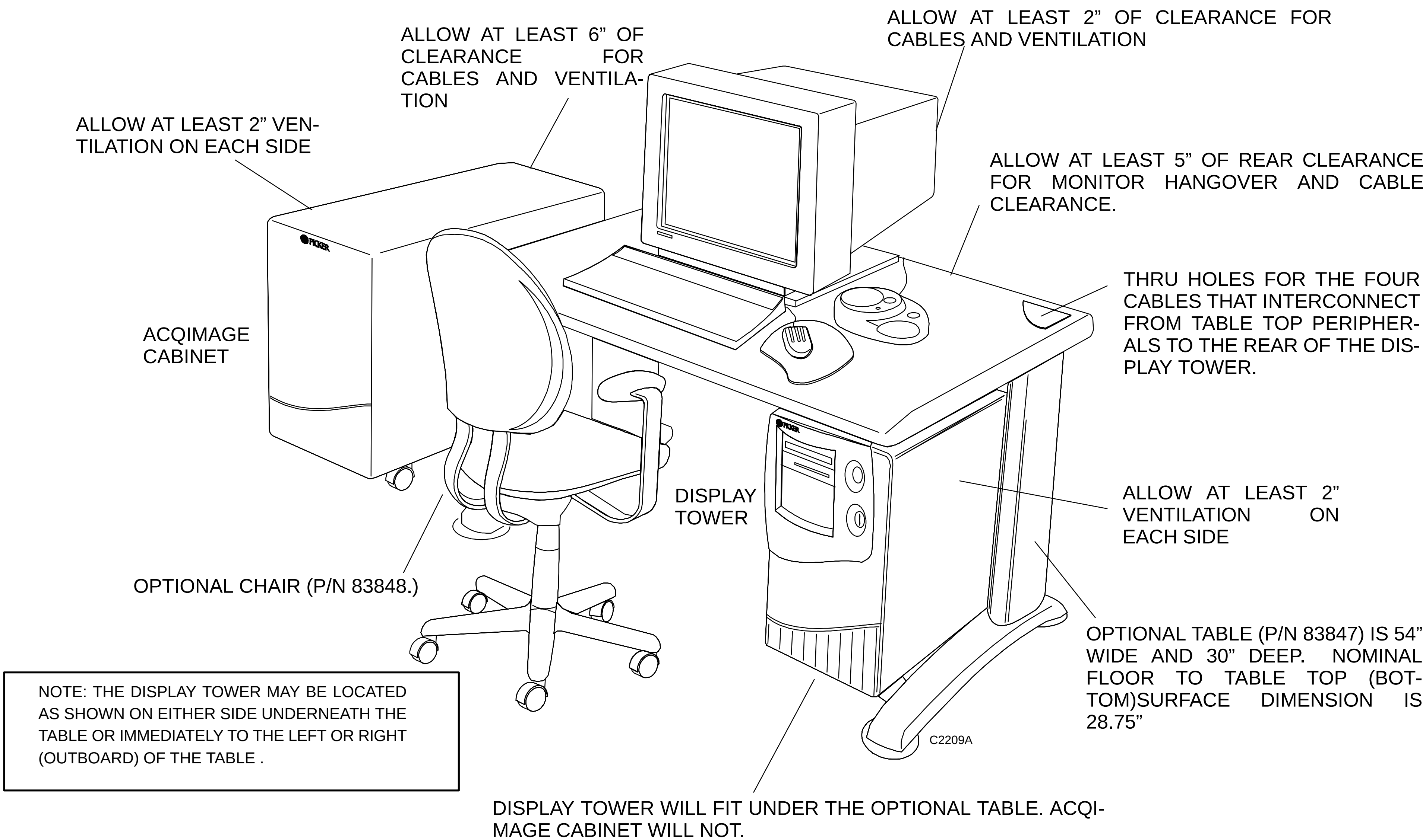


FIGURE 53 ACQIMAGE CABINET and DISPLAY CONSOLE (optional table and chair)

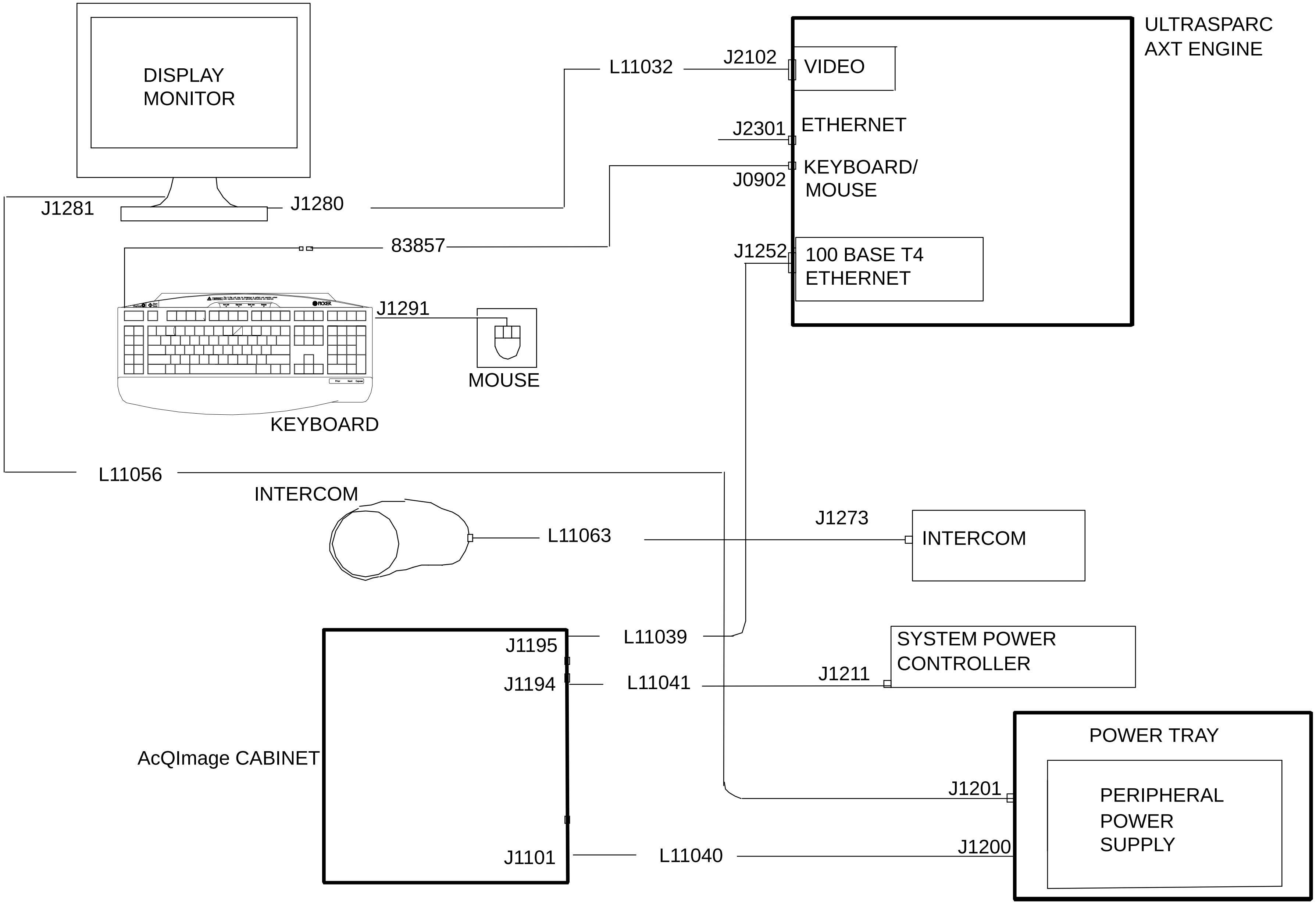


FIGURE 54 CABLE CONFIGURATION – OPERATOR CONSOLE

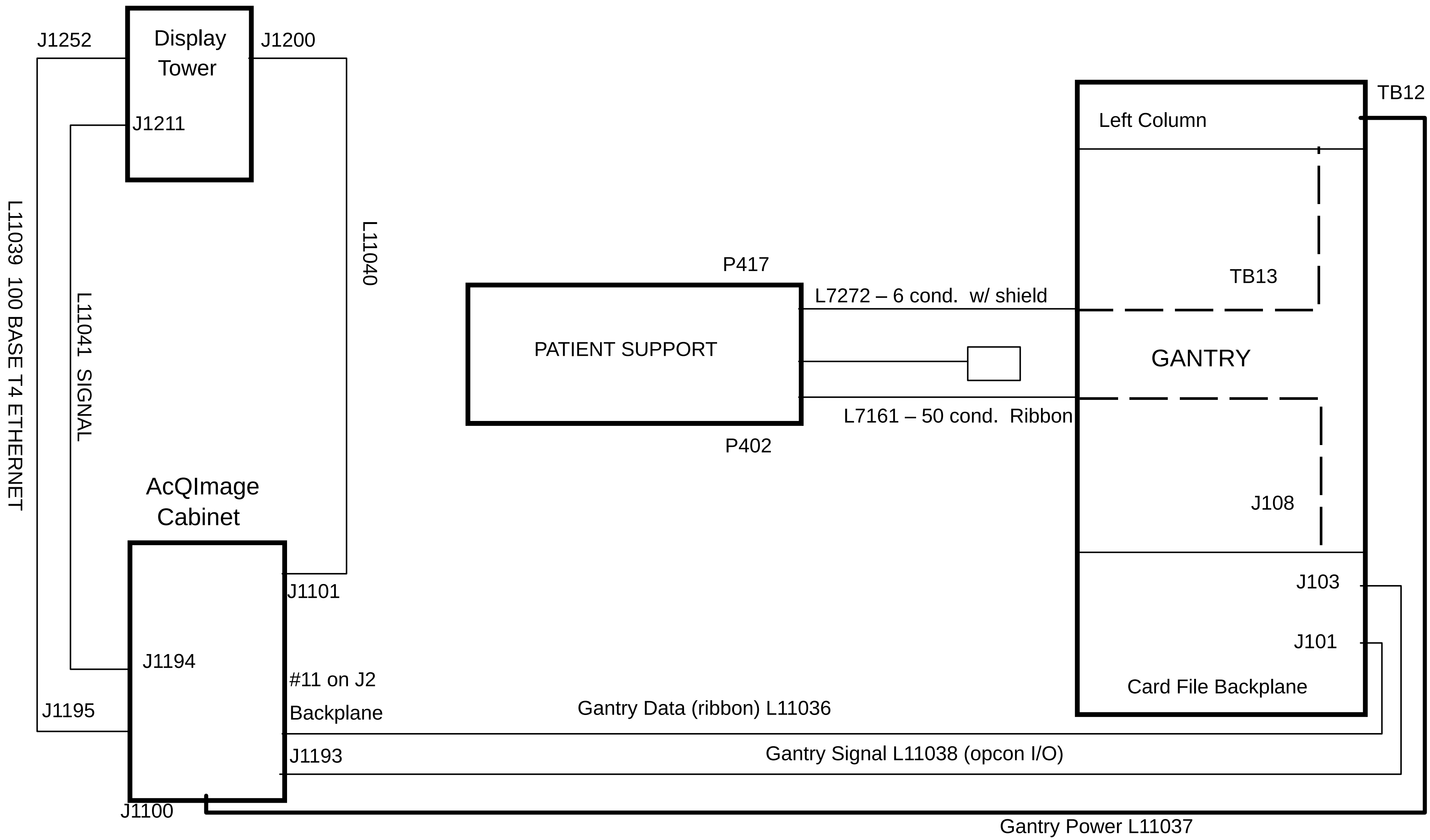
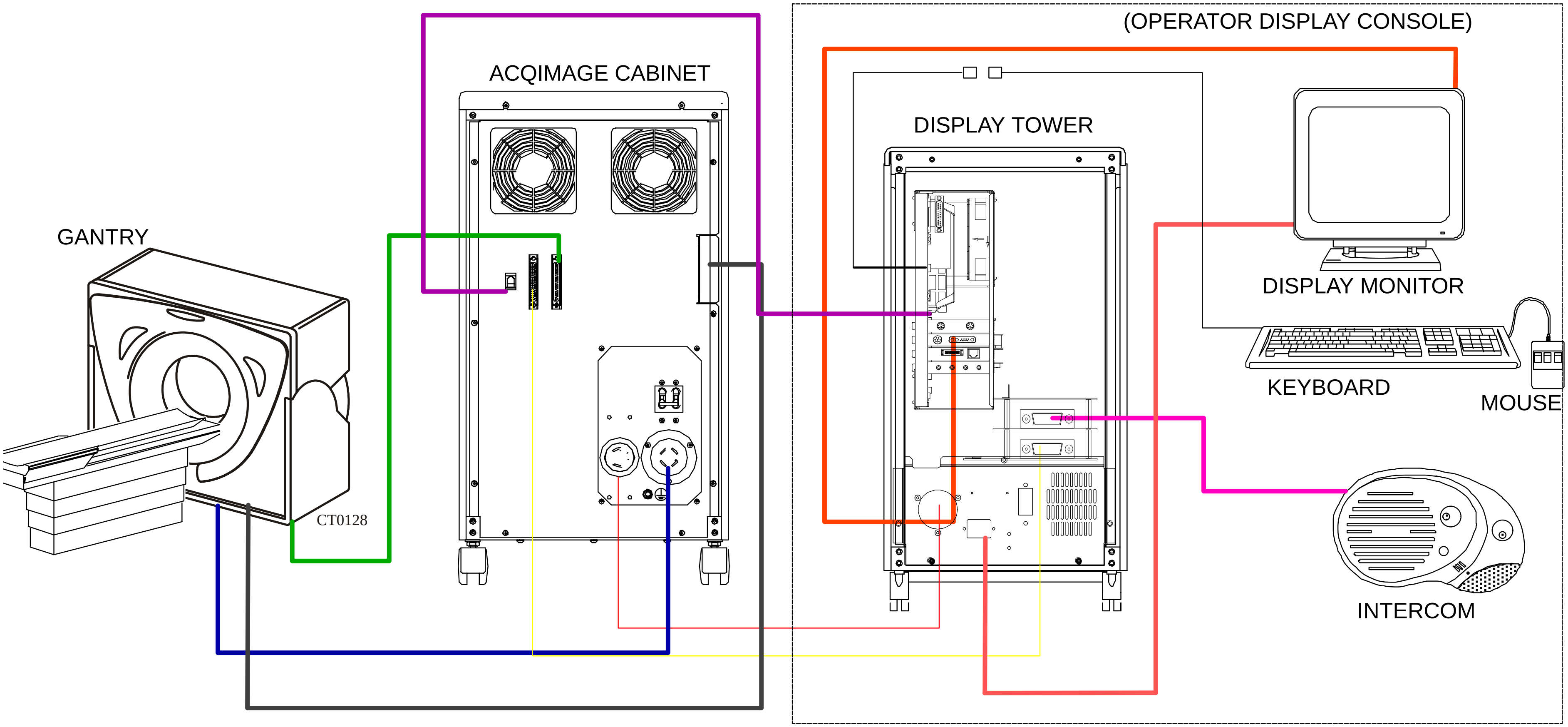


FIGURE 55 SYSTEM INTERCONNECT DIAGRAM

NEED NEW DISPLAY TOWER REAR VIEW. UPPER REAR COVER NEEDS TO BE RE-MOVED.



FOR ADDITIONAL CONFIGURATION DRAWINGS, SEE Figure 38, Figure 39 and Figure 54.

ACQIMAGE CABINET TO GANTRY. REFER TO FIGURE 59 FOR DETAIL

DISPLAY TOWER TO ACQIMAGE CABINET. REFER TO FIGURE 58 FOR DETAIL.

OPERATOR CONSOLE. REFER TO FIGURE 57 FOR DETAIL.

FIGURE 56 CABLING FOR ACQIMAGE, DISPLAY CABINETS, OPERATOR CONSOLE & GANTRY

OPERATOR CONSOLE

1. Ensure all power has been removed from the system. Move the Display Tower to a clear position to permit cable installation. Use Figure 57 and **Table 1** for reference to this text.

> ⚠ **WARNING** ⚠
>
> **ELECTRIC SHOCK HAZARD. USE EXTREME CAUTION WHEN WORKING INSIDE THE OPERATOR CONSOLE WITH THE COVERS OPEN. 120 VAC, +/–12 VDC, 5 VDC AND 3.3 VDC ARE EXPOSED. FAILURE TO COMPLY CAN RESULT IN SERIOUS INJURY OR DEATH.**

2. Remove the covers and make a visual inspection of the boards and connectors, verifying they are secure and properly seated. Cover removal instructions can be found in the AcQImage Cabinet Covers Removal/Replace procedure in the Repair Manual.

3. Attach all of the cabling according to the diagram in Figure 57 and Table 1. *Refer to the following note before running the cables.*

> **NOTE**
>
> **Display Tower**: The cables (312690, L11039, 83857, recon cable, intercom cable) must be routed inside the Display Tower. Remove the upper rear panel by removing the six screws. Make sure the cables run through and are secured by the hook and loop strain relief on the AC Chassis. Refer to Figure 57 and Figure 58.

> **NOTE**
>
> DO NOT bend the ethernet cables sharply. DO NOT place the ethernet cables near the electronics of the system components.

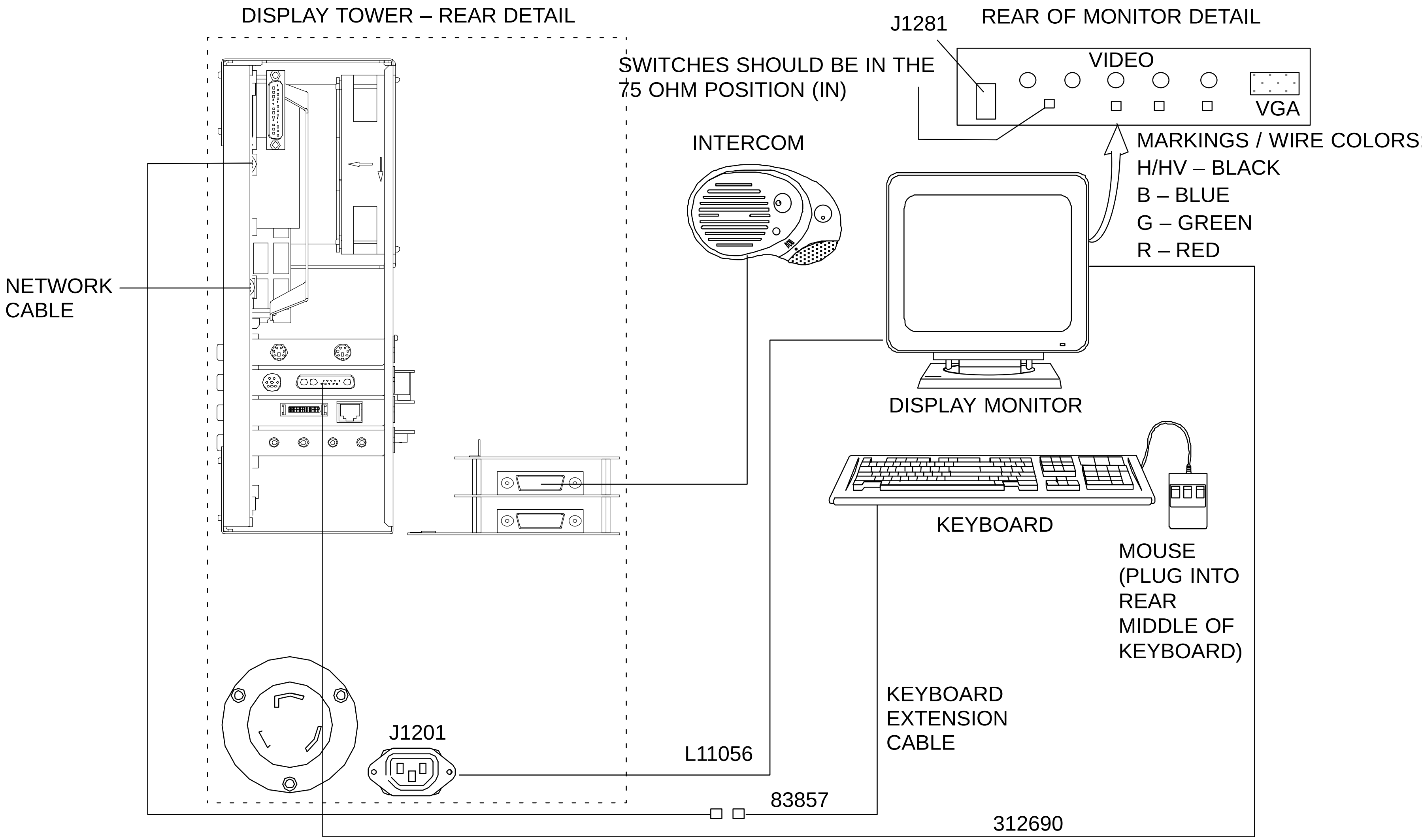


FIGURE 57 OPERATOR CONSOLE CABLING DETAIL

DISPLAY TOWER TO ACQIMAGE CABINET

1. Ensure all power has been removed from the system! Move the AcQImage Cabinet as required to a clear location to permit cable installation. Use Figure 58 and **Table 1** for reference to this text.

> **WARNING**
>
> **ELECTRIC SHOCK HAZARD. USE EXTREME CAUTION WHEN WORKING INSIDE THE DISPLAY TOWER AND ACQIMAGE CABINET WITH THE COVERS OPEN. 120 VAC, +/–12 VDC, 5 VDC AND 3.3 VDC ARE EXPOSED. FAILURE TO COMPLY CAN RESULT IN SERIOUS INJURY OR DEATH.**

2. Cable lengths and routing have been established per site planning instructions. If not, then route the cables as necessary to accomplish an orderly, safe installation. Refer to Figure 58 and table 1.

> **NOTE**
>
> **Display Tower**: The cables (312690, L11039, 83857, L11041, intercom cable) must be routed inside the Display Tower. Remove the upper rear panel by removing the six screws. Make sure the cables run through and are secured by the hook and loop strain relief on the AC Chassis. Refer to Figure 57 and Figure 58.

> **NOTE**
>
> DO NOT bend the ethernet cables sharply. DO NOT place the ethernet cables near the electronics of the system components.

3. Attach the Display Signal cable (P/N L11041) to J1194 on the rear of the AcQImage Cabinet. Attach the other end to the Display Cabinet J1211.
4. Attach the AcQImage Power cable (P/N L11040) to J1101 on the Rear of the AcQImage Cabinet. Attach the other end to the Display Tower J1200.
5. Remove the small rear panel on the Display Tower by removing the two screws.

6. Run the 100 Base T4 Ethernet cable (P/N L11039) through the hole in the upper left of the rear cover to J1195 which is INSIDE the AcQImage cabinet (on a short adapter on the 100mb board.) Attach the other end to the Display Tower Ethernet port (J1252) on the Sparc motherboard as seen in Figure 58.

> **NOTE**
>
> The network between the AcQImage and Display Cabinets is a private network for their intercommunication. The high–speed data link cable (P/N L11039) is a specially constructed 100MB/sec twisted pair cable to be used ONLY as a connection between the two units. **Do not** use this cable for standard ethernet hook–ups or substitute it for a standard ethernet cable.

7. Replace the covers on the Display tower and roll into its final position. Lock the wheels. Refer to Figure 53 for considerations.
8. Proceed to the AcQImage Cabinet to Gantry cabling instructions.

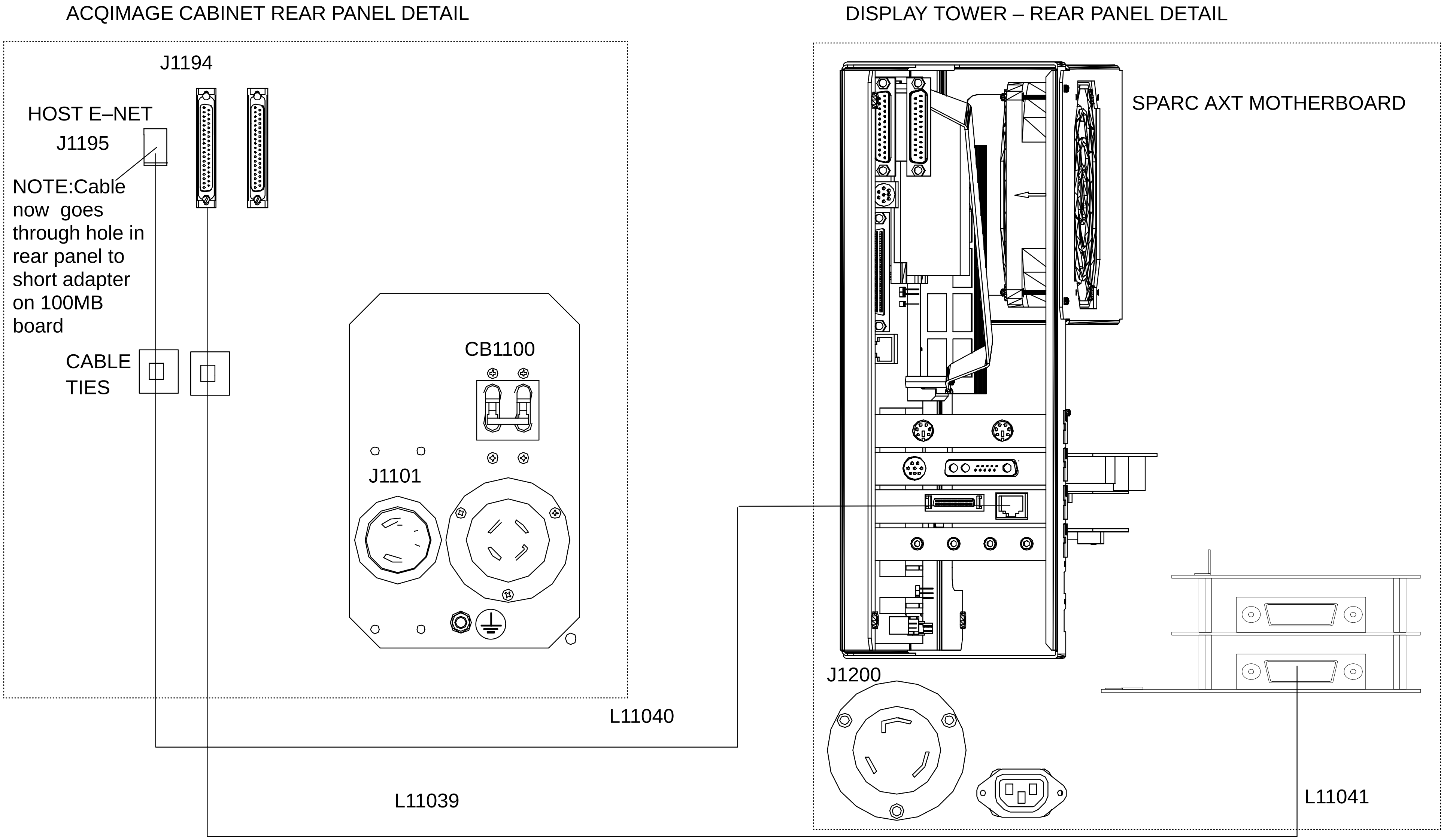


FIGURE 58 DISPLAY TOWER TO ACQIMAGE TOWER CABLING DETAIL

ACQIMAGE CABINET TO GANTRY

1. Ensure all power has been removed from the system! Move the AcQImage cabinet to a clear location to permit cable installation. Use Figure 59 and **Table 1** for reference to this text.

> ⚠ **WARNING** ⚠
>
> **ELECTRIC SHOCK HAZARD. USE EXTREME CAUTION WHEN WORKING INSIDE THE ACQIMAGE CABINET WITH THE COVERS OPEN. 208 VAC, 120 VAC, +/–12 VDC, 5 VDC AND 3.3 VDC ARE EXPOSED. FAILURE TO COMPLY CAN RESULT IN SERIOUS INJURY OR DEATH.**

2. Remove the covers and make a visual inspection of the boards and connectors, verifying they are secure and properly seated. Cover removal instructions can be found in the AcQImage Cabinet Covers Removal/Replace procedure in the Repair Manual.

> **NOTE**
>
> It is a good idea to record the numbers associated with each of the AcQImage boards. Store this information in a safe place as you may have to refer to it at a later date.

3. Open and secure the front upper gantry panel The gantry cardfile backplane is located at the upper right, just below the cardfile. Remove the left column cover to expose TB12.
4. Loosen the screws on the AcQImage Cabinet's right side strain relief. Feed the ribbon cable (P/N L1036) in and attach to the #11 connector of the J2 backplane. Ensure latches are in the locked position. Tighten the strain relief to secure the cable. Details are shown in Figure 60. Replace the rear cover. Attach the other end to J101 of the Gantry cardfile backplane.
5. Attach the Gantry Signal cable (P/N L11038) to J1193 on the rear of the AcQImage Cabinet. Attach the other end to J103 of the Gantry cardfile backplane.
6. Plug in the Gantry Power cable (P/N L11037) into J1100 on the rear of the AcQImage Cabinet. Attach the other end to TB12 of the left Gantry column. Note the proper terminal strip connections as shown in Figure 59.
7. Replace covers on the AcQImage Cabinet and roll into its final position. Lock the wheels. Refer to Figure 53 for placement considerations.
8. With no power applied to the system from the wall breaker, ensure that the Display Tower keyswitch is turned OFF and the circuit breakers (CB1100 – AcQImage, CB1200 – Display) on both units are ON. Hardware installation of the Operator Console and AcQImage Cabinet is now complete!

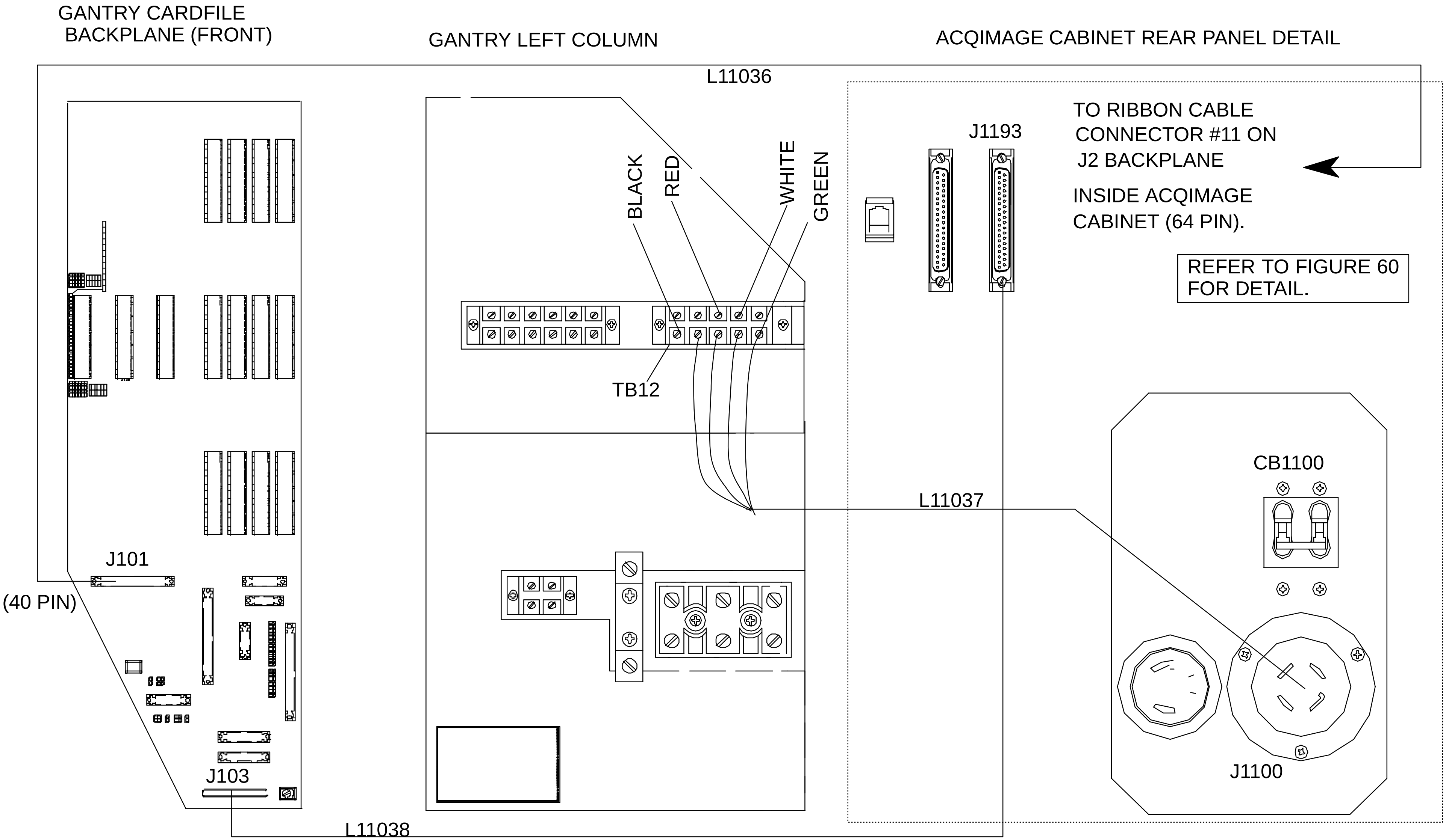


FIGURE 59 ACQIMAGE CABINET TO GANTRY CABLING DETAIL

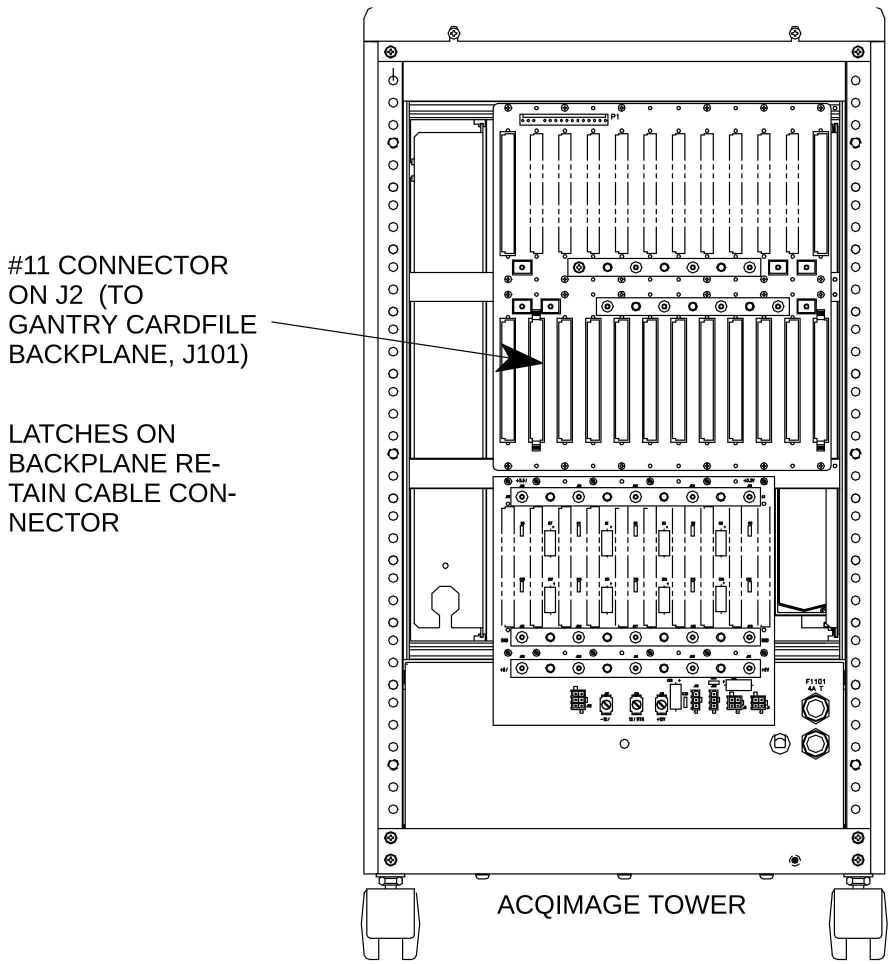


FIGURE 60 L11036 GANTRY DATA CABLING AND STRAIN RELIEF DETAIL

TABLE 1 ACQIMAGE TOWER AND DISPLAY CABINET CABLING CONFIGURATION

Operator Console Cabling: (Figure 57)

CABLE NUMBER	CABLE LENGTH	CABLE DESCRIPTION	FROM LOCATION AND CONN.	TO LOCATION AND CONN
83857	8 FT.	Keyboard and Extension Cable	Keyboard	Display Tower, SUN Motherboard, J____
312690	10 FT.	Monitor Video Cable	Display Monitor, J1280	Display Tower, SUN Motherboard, J____
L11056	10 FT.	Monitor Power	Back of display Monitor, J1281	Display Tower, J1201
L11063	12 FT.	Op intercom connect	Rear I/O Display Tower, J1273	Operator Intercom

Display Tower to AcQImage Cabinet: (Figure 58) **

L11039	50 FT.	Ethernet Cable –100 Base T4	AcQImage Cabinet, J1195	Display Tower, SUN Motherboard, J____
L11040	50 FT.	Power Cable	AcQImage Cabinet, J1101	Display Cabinet, J1200
L11041	50 FT.	Display Signal Cable	AcQImage Cabinet, J1194, (P1194 end)	Display Cabinet, J1211

AcQImage Cabinet to Gantry: (Figure 59) *

L11036	50 FT.	Gantry Data Ribbon	Gantry Backplane, J101	AcQImage Cabinet Backplane, #11 on J2
L11037	50 FT.	Gantry Power Cable	Gantry Left Column, TB12	AcQImage Cabinet, J1100
L11038	50 FT.	Gantry Signal Cable	Gantry Backplane, J103	AcQImage Cabinet, J1193, (Use P1193 end)

** – The "–A" suffix on L11039, L11040 and L11041 indicates a 10 ft. length.

* – The "–A" suffix on L11036, L11037 and L11038 indicates a 25 ft. length.

System component part numbers:

Description	Part number	Description	Part Number
AcQImage Cabinet	178220	Op. Intercom	XXXX
Display TOWER	178927	Keyboard	xxxxx
Mouse	xxxxxx	Mouse Pad	xxxx
Display Monitor	xxxxx		

SOFTWARE INSTALLATION AND CONFIGURATION

The system is shipped with operating software installed on the Display Tower disk drive 0. New software will be loadable on the system via factory releases on CD–ROM disk(s). A CD–ROM drive is standard on all Display Towers for this procedure. New or replacement disk drives will be initialized via utility programs located on a CD–ROM disk.

The system is shipped with standard configuration parameters such as hospital name, system serial numbers, default network configuration, etc. for stand–alone scanner operation already loaded.

Each installation will be unique. However, in light of the fact that many options are attached via ethernet network connection. Additional site specific configuration information will likely need to be entered at installation.

IF 1.0 SOFTWARE MUST BE INSTALLED, SEE THE ACQSIM CT 1.0 SOFTWARE INSTALLATION INSTRUCTIONS.

> **NOTE**
>
> Refer to the Peripherals Configuration manual for disk drive initialization, installation and software installation.

PERIPHERALS INSTALLATION AND CONFIGURATION

The Display Tower and AcQImage Cabinets are shipped from the factory with standard and optional peripherals such as disk, tape, optical and CD–ROM installed and configured.

The Display Tower peripherals are connected to the Sparc host computer via a SCSI–2 bus cable located within the tower. The volume data disks in the AcQImage Cabinet are connected to the GAM board via a separate internal Fast–Wide SCSI bus cable.

If new options need to be added or repair/replacement is required, it will be necessary to install and configure in the field.

> **NOTE**
>
> Refer to the Peripherals Configuration manual for mechanical details on disk drive replacement/installation and also for information on SCSI assignments and jumper settings.

DISPLAY MONITOR INSTALLATION SETUP

NOTE

The Display Monitor may need to be adjusted to local ambient lighting conditions and customer preference. Refer to the Display Monitor Calibration section of the Mechanical Calibration Manual and the Operator's Manual for instructions on monitor setup and adjustment.

ACQIMAGE BOARD INFORMATION

IMPORTANT NOTE

The information on the front of the AcQImage boards is very important. ****Write down the IP address, netmask, router Ethernet address and node name of each board and affix somwhere safe** (inside of AcQImage Cabinet) This information is contained within the Configure | Network/Camera option in the Service Application. If this information is lost, the only place to restore it will be your hardcopy record.

NETWORK CONFIGURATION

NOTE

The AcQSim CT must be configured (IP addresses, network camera, etc.) Refer to the

CONFIGURE chapter of the Service Application manual.

LI PLUS FILM SERVER INSTALLATION

NOTE

Your specific configuration may require the installation of the LI Plus Film Server. If so, refer to the LI PLUS manual (in Ultra Z peripherals) for installation/configuration details.

(*continue to the next page for more installation instructions*).

CREATE A SITE TAPE

You should create a site tape at this time. This is done via the Main Startup Menu. Click in the lower left corner and type the word “site” (lower case) and press <CR>. After the Save Site / Restore Site menu appears, push the tape in, wait for the green light to stop flashing and select “1” from the menu to Save Site. The system will eject the tape and reboot the system when finished. (Restore Site is done in a similar fashion.)

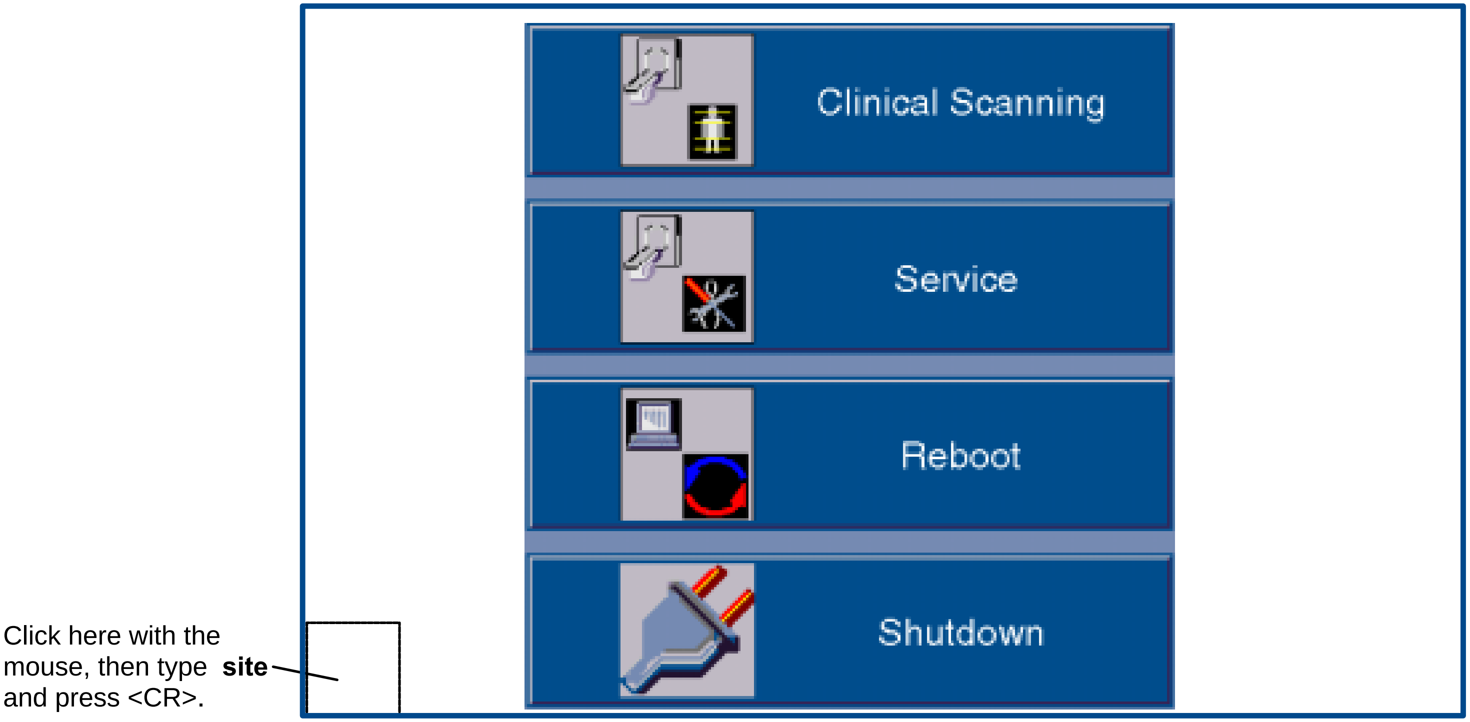


FIGURE 61   Starting the Save Site / Restore Site Menu

ALIGN LASERS

> **NOTE**
> Laser Alignment is performed in the Mechanical Calibration manual.

AUTOVOICE CONFIGURATION

NOTE

Refer to the *Operator’s Manual* for the Autovoice Configuration, recording and playback instructions. Autovoice Diagnostics are part of the Diagnostic GUI.

INTERCOM INSTALLATION SETUP

NOTE

Refer to the *Operator’s Manua*l for instructions on the Intercom setup and operation.

ACQSIM CT 1.0 SOFTWARE INSTALLATION INSTRUCTIONS

The following kit instructions should be used when installing 1.0 software onto an AcQSim CT system.

> **NOTE**
>
> This procedure in intended for the first series of software installations on the AcQSim CT and is not intended to be part of a software upgrade kit.

1. Make a Site tape immediately before installation. If in the Console or Service Application, select Shutdown Console from the General Utilities button. When the main menu appears, click in the lower left corner of the Main Startup menu and type the word “**site**” (lower case) and press <CR>.
2. After the Save Site / Restore Site menu appears, push the tape in, wait for the green light to stop blinking and select “1” to Save Site. The system will eject the tape when finished.
3. Exit the Save Site Utility. When the Shutdown button becomes available, press it and wait for the system to shut down.
4. Open the CD–ROM drive and insert CD #1. Close the drive.
5. When the ok prompt appears, type **boot cdrom** and press <CR>. After a few minutes, the password prompt will appear. Type **picker** and press <CR>.
6. The system then checks the disk configuration and prompts you with:

```
Enter the type of system installation:

1. RDS                  (Note: “RDS” refers to an Ultra Z Diagnostic Workstation, UDW)
2. Scanner Console

q. quit
```

7. Select option “2,” *Scanner Console* from the menu. You will see the following menu.

```
Enter a Console Installation option:

1. Install the OS and Application Software
2. Configure a Replaced Disk(s)
3. Upgrade Application Software Only

q. quit
```

8. Select option “1,” *Install the OS and Application Software* from the menu. Answer “**y**” to the next question. The following will appear:

Main Menu Option #1 – *Reformat and Configure entire system:*

```
Enter IP Address [default xxx.xxx.xx.xx]:
Enter Netmask [default 255.255.255.0]:
Enter Router [default xxx.xx.xx.xx]:
Enter Nodename [default consolexxxx]:
```

9. Press <CR> to accept the default values. Make sure the Netmask number is 255.255.255.0. After the information is entered, the system will display your choices as you entered them:

```
IP Address:
Netmask:
Router:
Nodename:
```

10. After the "Ready to run the installation. Are you satisfied? (y/n) [n]?" prompt appears:
    If you answer NO to "satisfied," the installation will cancel and take you back to the first sub–menu where you can quit the installation program, or continue and adjust the network information.

    If installation information is correct, choose "y" to "satisfied."

11. The software will now begin installing from the first CD. This will take about 20 minutes. When the "serial/parallel" change menu appears, choose "q" and press <CR>. The following menu will appear:

```
1. Install SCANNER Console Application Software
2. Eject the CD.
```

Select option "2," *Eject the CD*. Remove the first CD and put the second CD in, then close the drive door.

12. Select option "1," to continue with the installation. You will be prompted with the "satisfied" question again. Select "y" and the system will continue with the software installation. After the installation is complete, a menu will appear:

```
1. View the installation error log  [/cd_errlog]
2. Eject the CD  (it's OK to eject it now)
3. Continue booting the machine

Choice:
```

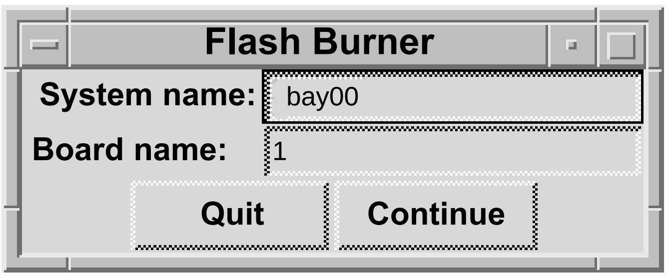
NOTE

After the installation has completed, additional messages may be displayed on the screen, pushing the above menu out of view. Simply press <CR> to see it again.

13. View the error log by choosing option **1**. There should not be any errors in the error log at this time. After viewing, the menu will be displayed again. Select option **2** to eject the CD. Remove the CD and close the drive door. Then select option **3** to continue booting to the application software.

14. Perform a Restore Site.

15. After the 1.0 software installation is done and the system has booted, start the Diagnostic GUI. (To start Diag GUI: Go to General Utilities, select Shutdown Console to get to the Startup men. Click on the lower left hand corner, type PICKER and press <CR>.)



16. Run the Flash Burner from the Diag GUI. The flash burner is a utility for transferring executable code from a directory on the SPARC disk to the Flash of the CP, AP, BP or GAM.

17. Enter the board number (1 for CP).

18. An Xterm will be displayed. Select option 1.) Transfer OS. When completed, the board will reboot.

19. Restart the the Flash burner from the GUI again on the same board. Then select option 2.) Transfer Diags.

20. When completed, the board will reboot. Now one board is done. Repeat this process on the remaining three boards. Board numbers are: 2 for GAM, 3 for DSP/AP, 4 for BP. You can view the date of the Flash OS via a Telnet session. At the pSH+> prompt, type **romver** and press <CR>.

21. This completes the 1.0 scanner software portion of the upgrade. After the initialization is complete, run a pilot, axial and spiral scan to further ensure system functionality. This completes the 1.0 installation. Reboot the system.

POWER UP / DOWN PROCEDURE

This section describes the system start–up/shutdown procedures. **Power–UP** is accomplished via the keyswitch on the Display Tower. **Power DOWN** of the entire system is accomplished via the General Utilities GUI. This is accessed via the “Service” option from the opening bootup GUI. This procedure covers the Service Application mode. *Scanning Application mode procedure is in the OP manual*.

SYSTEM POWER UP

1. Make sure the Gantry keyswitch is in the normal position. Turn the Display Tower keyswitch to the **ON** position. After the system boots, you will be prompted with the startup application display (or “4 Button Screen.”) You have **4 boot options**. Refer to Figure 62.:
   A.) Select Clinical Scanning (or do nothing) and the system will start and connect to the acquisition server. The acquisition server will then power up the AcQImage Cabinet, and if that is done successfully, the acquisition server will power up the gantry and X–ray system.
   B.) Select the Service Application option (within 15 seconds) by clicking on the Service button with the mouse pointer.
   C.) Reboot the system. **D**.) Start the Diagnostic GUI (within 15 seconds) by clicking once in the bottom left corner, then entering PICKER <CR>. (All caps). The AcQImage board LEDs will display a power up test sequence. Refer to the LED Interpretations.

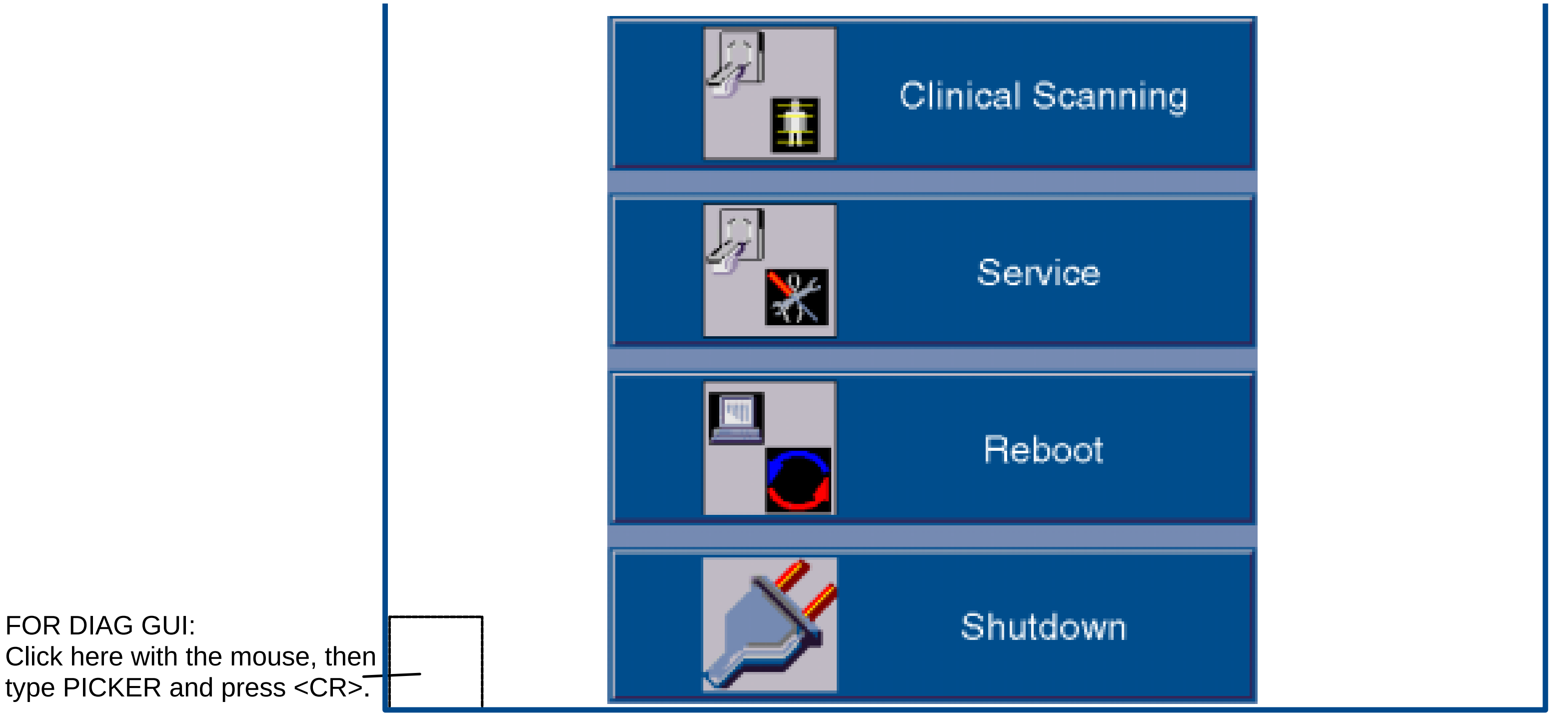


FIGURE 62 4 BUTTON SCREEN

PREPARE THE SYSTEM FOR USE (X–Ray Tube Warmup)

2. The X–Ray system needs to be properly warmed up prior to executing a scan procedure. X–Ray System Warmup must be completed at the start of every scan day or when the tube heat is below 6%. Tube heat is displayed in the Tube Warmup window.

3. To perform a tube warm–up, click on the Tube Warm Up button with the mouse pointer. Refer to Figure 62. This will initiate the warm–up sequence and **X–RAYS WILL BE EMITTED**. The tube warm–up status is displayed is displayed in the Tube Warmup Window. Refer to Figure 63.

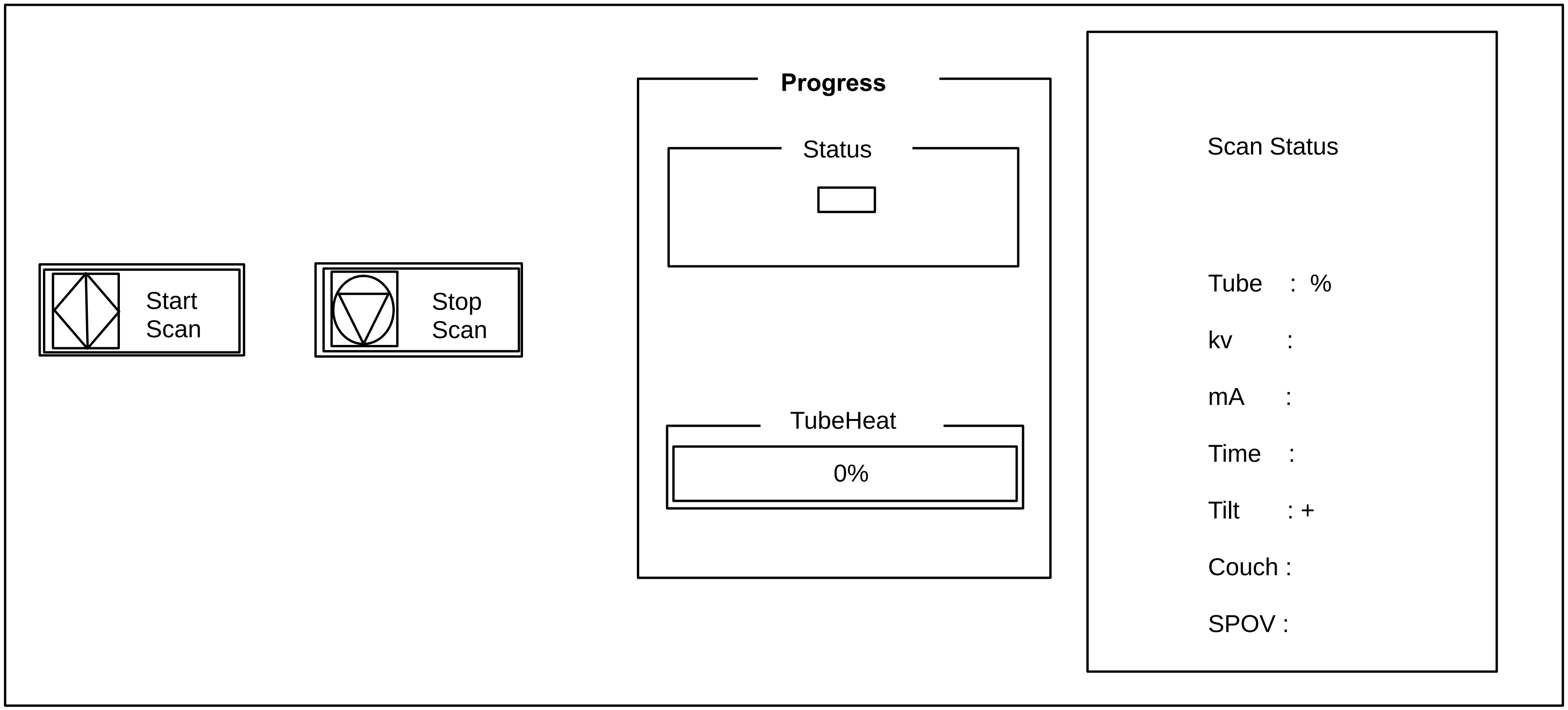


FIGURE 63 TUBE WARM–UP STATUS WINDOW

SYSTEM POWER DOWN

1. From the 4 Button Screen (Figure 62) click on SHUTDOWN with the mouse pointer.

> ⚠ **CAUTION**
>
> DO NOT POWER DOWN THE SYSTEM WITH THE TUBE HEAT AT OR ABOVE 35%. FAILURE TO COMPLY MAY RESULT IN SHORTENED TUBE LIFE AND/OR ERRATIC TUBE BEHAVIOR.

System shutdown ensures that all images on disk are properly closed, all image maintenance functions properly terminated, the gantry parked, the X–Ray system sufficiently cooled and all films printed.

There are three shutdown selections: Shutdown Gantry, Shutdown Console, Shutdown System and Switch to Console as seen in Figure 64.

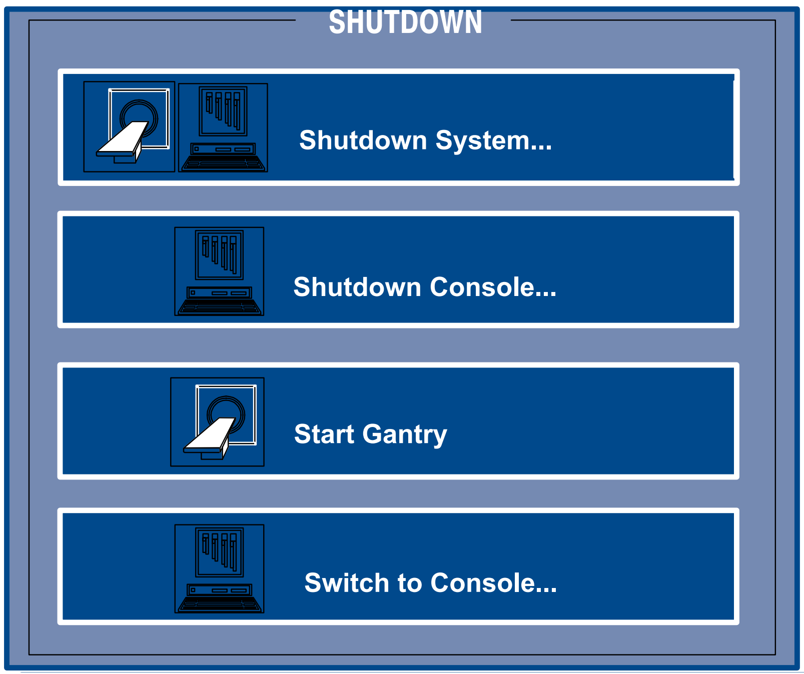


FIGURE 64 SHUTDOWN SCREEN

Shutdown Gantry: This option should be used, if possible, in place of the e–stop switches. An orderly shutdown of the gantry will be performed. Display Tower and/or AcQImage Cabinet operation may continue after selected. To cancel shutdown, click the Abort button.

Shutdown System: Performs an orderly shutdown of the entire scanner system. To cancel shutdown, click the Abort button.

Shutdown Console: This option shuts down only the console units. An orderly shutdown of the image archiver, filming and file system will be done.

Start Gantry: Toggles Start/Shutdown Gantry depending on status of Gantry power.

Switch to Console: Changes the GUI to the Console Application (Operator) screen.

WARNING

THE GUI SHUTDOWN REMOVES POWER FROM THE SYSTEM, BUT NOT AT THE INCOMING POWER LINES. DANGEROUS VOLTAGES ARE EXPOSED WHEN THE GANTRY COVERS ARE REMOVED. BEFORE SERVICING EQUIPMENT, REMOVE POWER FROM THE WALL BOX. FAILURE TO COMPLY CAN CAUSE SERIOUS INJURY OR DEATH TO PERSONNEL.

2. When prompted via a dialog box, turn the Display Tower keyswitch to the **OFF** position.
3. On the AcQImage Cabinet and Display Tower, ensure the rear breaker switch is turned off.

AcQImage Boards – Power Up Test LED Interpretations

```
AP, PB, GAM Board Diagnostic Flash Version 1.1.2

    0 Hardware Initialization
    1 CPU Version Check
    2 Diagnostic Flash CRC Check
    3 Operating System Flash CRC Check
    4 CPU Memory Test
    5 Data Cache Test
    6 Unexpected Exception Test
    7 IBC Test
    8 NVRAM Single Location Test
    9 PCI Bus Select Test
    A Ethernet Loopback Test
    B UART Loopback Test
    C apmem for AP, pbmem for PB, gamem for GAM

    C CIO Test (CP Board Diagnostic Flash Version 1.1.2)
    D Synchronous Serial Chip Loopback Test (CP Board Diagnostic Flash Version 1.1.2)
```

GANTRY TILT CHECKS

> ⚠ **WARNING**
>
> **THE FRONT UPPER GULL WING HAS SHARP EDGES. USE EXTREME CAUTION WHEN SERVICING EQUIPMENT WHILE UPPER GULLWING IS OPEN. FAILURE TO COMPLY CAN RESULT IN INJURY TO PERSONNEL.**

1. Remove the left rear Gantry cover.
2. Set a protractor against the Gantry scan frame. Refer to Figure Figure 65.

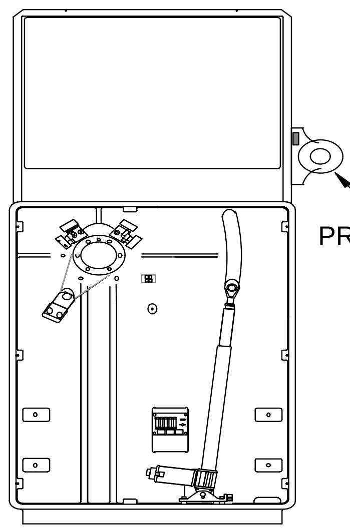


FIGURE 65 GANTRY AT ZERO DEGREES

3. Tilt the Gantry until the protractor bubble level indicates that the Gantry is at **zero** degrees.

> **NOTE**
>
> The protactor supplied in your kit may be different from the one shown here. If your protractor indicates 90^{o} when the Gantry is at 0, use the following table to find the correct Gantry angle.

Gantry Tilt (deg.)	0	10	20	30
Protractor Reading (deg.)	90	80	70	60

4. Observe the Tilt position readout and verify that it displays a reading of zero (**0.0**). If the readout <u>does not</u> display zero (**0.0** +/– .5mm), reference the Calibration manual and calibrate the Gantry tilt. (The Gantry tilt readout displays only to the nearest 1/2–degree: i.e., 0.5, 1.0, 1.5, etc.)
5. Tilt the Gantry **backwards** until the protractor indicates that the Gantry is at **–30** degrees.
6. Observe the Tilt position readout and verify that it displays a reading of **–30.0**. If the readout <u>does not</u> display **–30.0** +/–.5 mm, reference the Calibration manual and calibrate the Gantry tilt.
7. Tilt the Gantry **forward** until the protractor indicates that the Gantry is at **+30** degrees.
8. Observe the Tilt position readout and verify that it displays a reading of **+30.0**. If the readout <u>does not</u> display **+30.0** +/–.5 mm, reference the Calibration manual and calibrate Gantry tilt.

X–RAY TUBE MECHANICAL ALIGNMENT

> **NOTE**
>
> For domestic shipments the x–ray tube is shipped installed; therefore, this procedure should not be necessary at installation. If you installed the tube, you must perform this procedure.

Introduction

This procedure explains how to mechanically align the X–ray tube so that the X–ray beam is centered on the detectors. This procedure must be done whenever an X–ray tube is replaced/installed. You will be working with the Gantry covers open while performing this procedure and will be initiating X–rays.

Tools Required

1. FSE tool kit
2. 3/8–inch drive socket wrench set

Materials Required

1. X–ray film – part number 97806

XRAY TUBE ALIGNMENT PROCEDURE

⚠ WARNING ⚠

SHOCK HAZARD. USE EXTREME CAUTION WHEN SERVICING THE GANTRY WITH THE COVERS OPEN. 480 VAC, 208 VAC, 120 VAC AS WELL AS 5VDC AND 15 VDC ARE EXPOSED IN MANY LOCATIONS. FAILURE TO COMPLY CAN RESULT IN SERIOUS INJURY OR DEATH.

1. Remove Gantry power. Refer to the Power Up/Down procedures in this manual.
2. Open and secure the upper Gantry cover. Remove the lower cover cone. Store the front cone out of the way. Do not lean the cone plastic inner edge of the cone against a wall as this might damage it.
3. Rotate the scan frame until the X–ray tube is at the 12 o'clock position. See Figure 66.

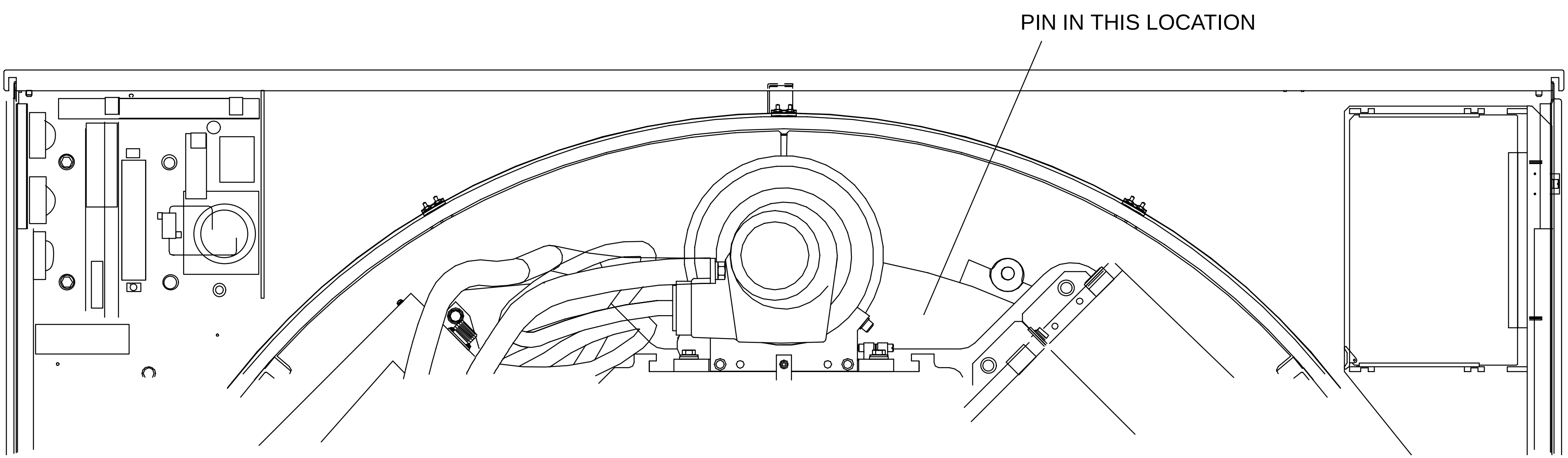


FIGURE 66 ROTATING FRAME WITH TUBE PINNED AT 12 O'CLOCK

4. Loosen the fourtube cradle mounting bolts but do not remove them. Carefully turn the lateral adjustment screw until the hash mark on the cradle lines up with the hash mark on the base plate. See Figure 67.

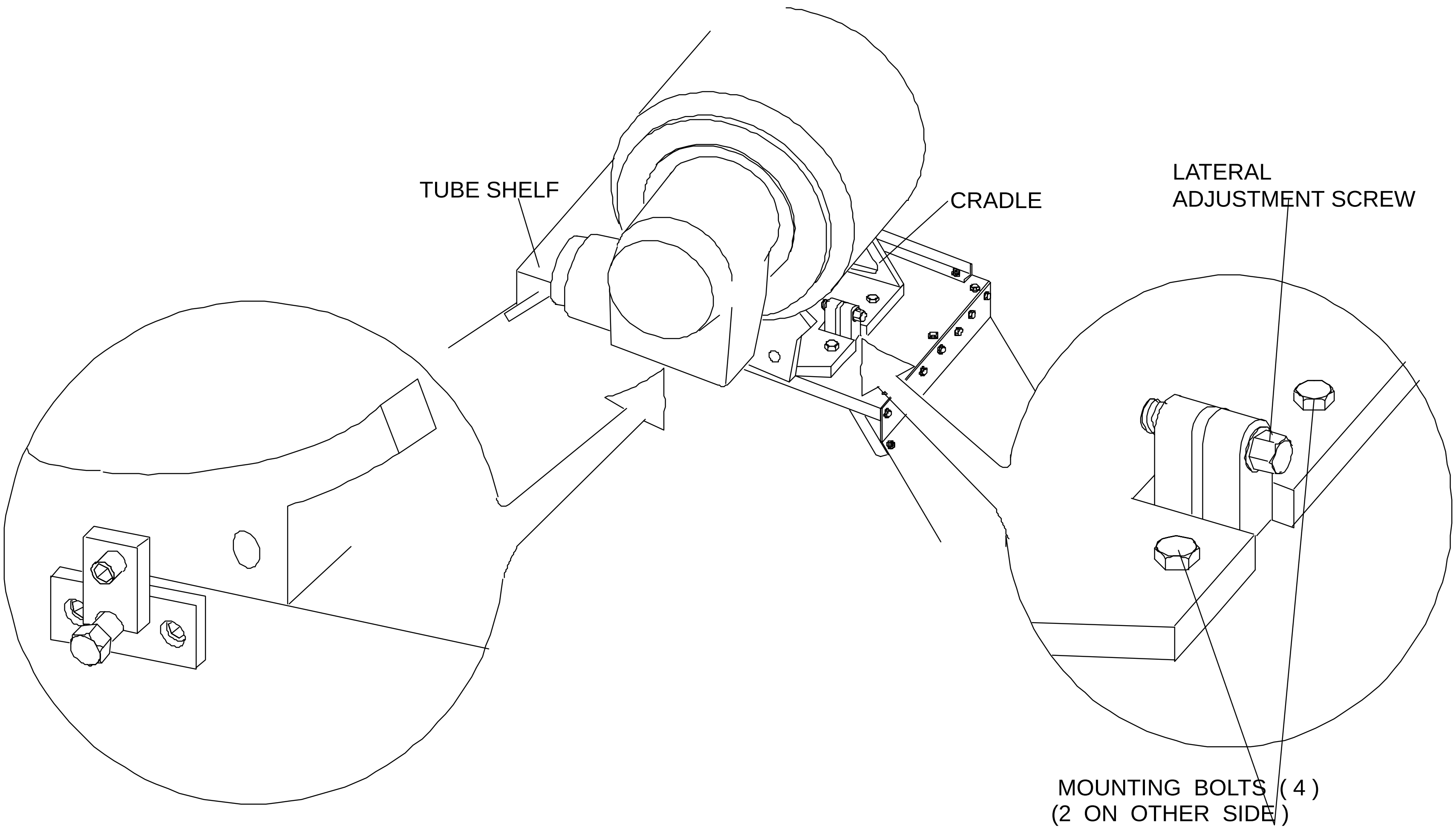


FIGURE 67 X–RAY TUBE ALIGNMENT

5. Tighten the four (4) cradle mounting bolts, in a cross pattern, to 15 ft–lbs.
6. Apply Gantry power. Turn the Display Tower and AcQImage Cabinet ON. Check that the main scan drives are off.
7. Do a Warm–up from the Scanner Utilities selection of the User Interface.
8. Disable the scan drives. When the upper front cover is open, the cover interlock switch automatically disables the scan drives. **DO NOT DEFEAT THE INTERLOCK SWITCH.**

9. Place a 4x5–inch film on the bottom of the inside of the Gantry aperture. Configure a reference marker in the orientation shown in Figure 68, centered in the x–ray field.

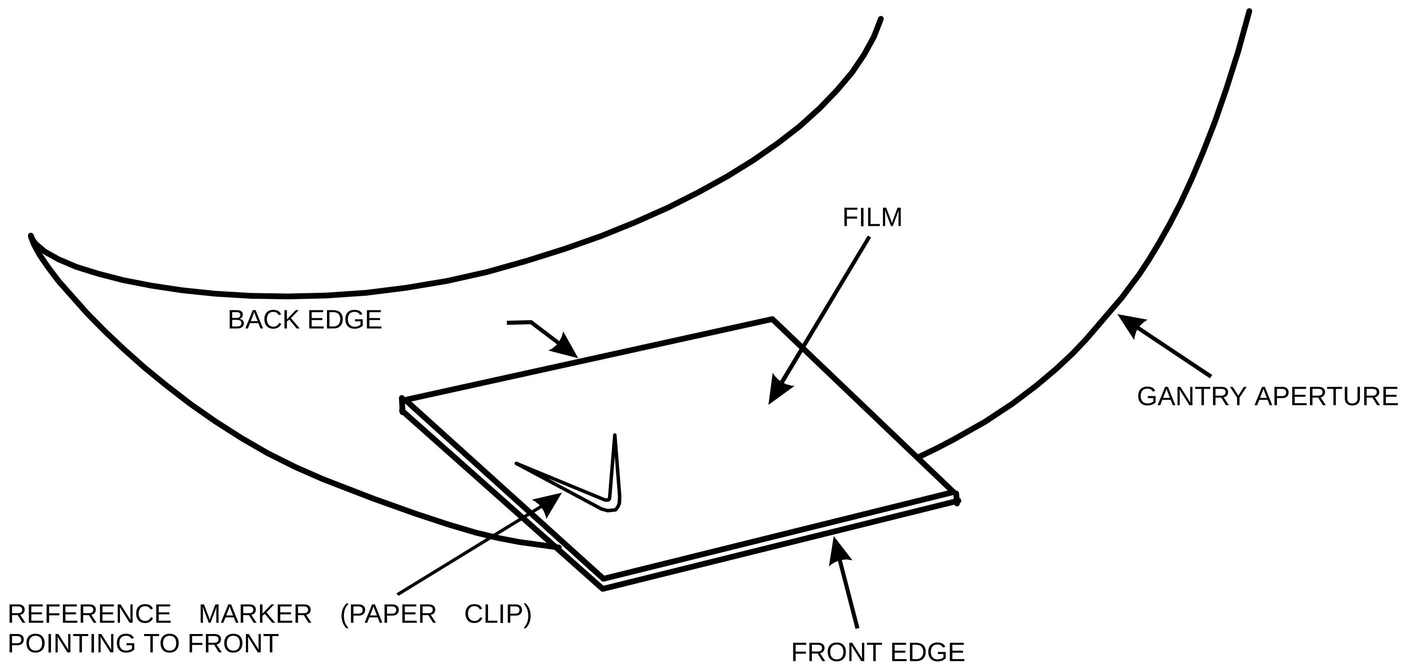


FIGURE 68 FILM AND REFERENCE MARKER

10. Turn scan drives on.
11. Position the X–ray tube at the 12 o'clock position (facing detector 1800).
12. Perform a stationary x–ray exposure (no scan frame motion). Refer to the new MANEXP procedure.

NOTE 1: The value for length of exposure is film speed dependent. Use an initial value of 200.

NOTE 2: The "tube position" option lets you use the rotate motor to position the x–ray tube if the scan frame drive is enabled. This number is the detector position where the tube will be located. Use the following table for examples:

MOVE TO DETECTOR	1800	600
..FACE DETECTOR ..	600	1800
Do Not Move Rotating Frame	–1	

13. Re–position the X–ray tube at the 6 o'clock position (facing detector = 600). Be careful to keep the film stationary.
14. Perform a stationary x–ray exposure (no scan frame motion) using MANEXP .
15. Note the film orientation with the reference object, so that when the film is developed, the image can be positioned correctly to make the centering measurements required.
16. Develop the film and inspect for a narrow, darker band superimposed on a wider, lighter band.

> **NOTE**
>
> The shadow of the X–ray beam on the film should show an even penumbra (fade off) on both sides to ensure mechanical centering.

17. Measure the distance from the front edge (see Figure 69) of the narrow, darker band to the front edge of the wide, lighter band (referred to as distance D1).
18. Measure the distance from the back edge of the narrow, darker band to the back edge of the wide, lighter band (referred to as distance D2).
19. Determine the distance between the band centers by the following equation:

$$\text{Distance between band centers (D)} = \frac{D1 - D2}{2}$$

NOTE

One full turn of the adjustment screw corresponds to a 0.025” (0.63 mm) movement of the X–ray tube.

20. Multiply the distance between band centers (D) by 0.417 to determine the distance the X–ray tube must move. Refer to the table below Figure 69) to determine which way to rotate the alignment screw.

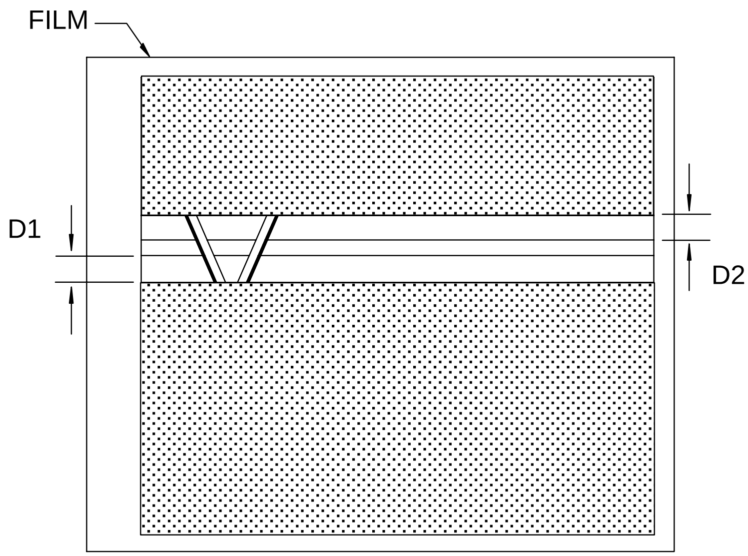


FIGURE 69 EXPOSED FILM

If the band is shifted to the FRONT,	rotate the screw CCW.
If the band is shifted to the REAR,	rotate the screw CW.

21. Loosen the four (4) tube cradle mounting bolts but do not remove them. See Figure 70.
22. Carefully turn the axial adjustment screw by hand, in the direction determined above, until you remove thread backlash; that is, any screw rotation causes movement of the tube base.
23. Look at and note the index position on the axial adjustment screw. Turn it the number of turns and in the direction indicated:

If the band is shifted to the FRONT,	rotate the screw CCW.
If the band is shifted to the REAR,	rotate the screw CW.

24. Tighten the four (4) cradle mounting bolts to 15 ft–lbs, in a cross pattern.
25. Repeat steps 10 through 19. Verify that the centers of the bands on the film are less than 0.25 mm apart.
26. Turn the Console OFF. Remove Gantry power. Refer to the Power Up/Down procedures in this manual.

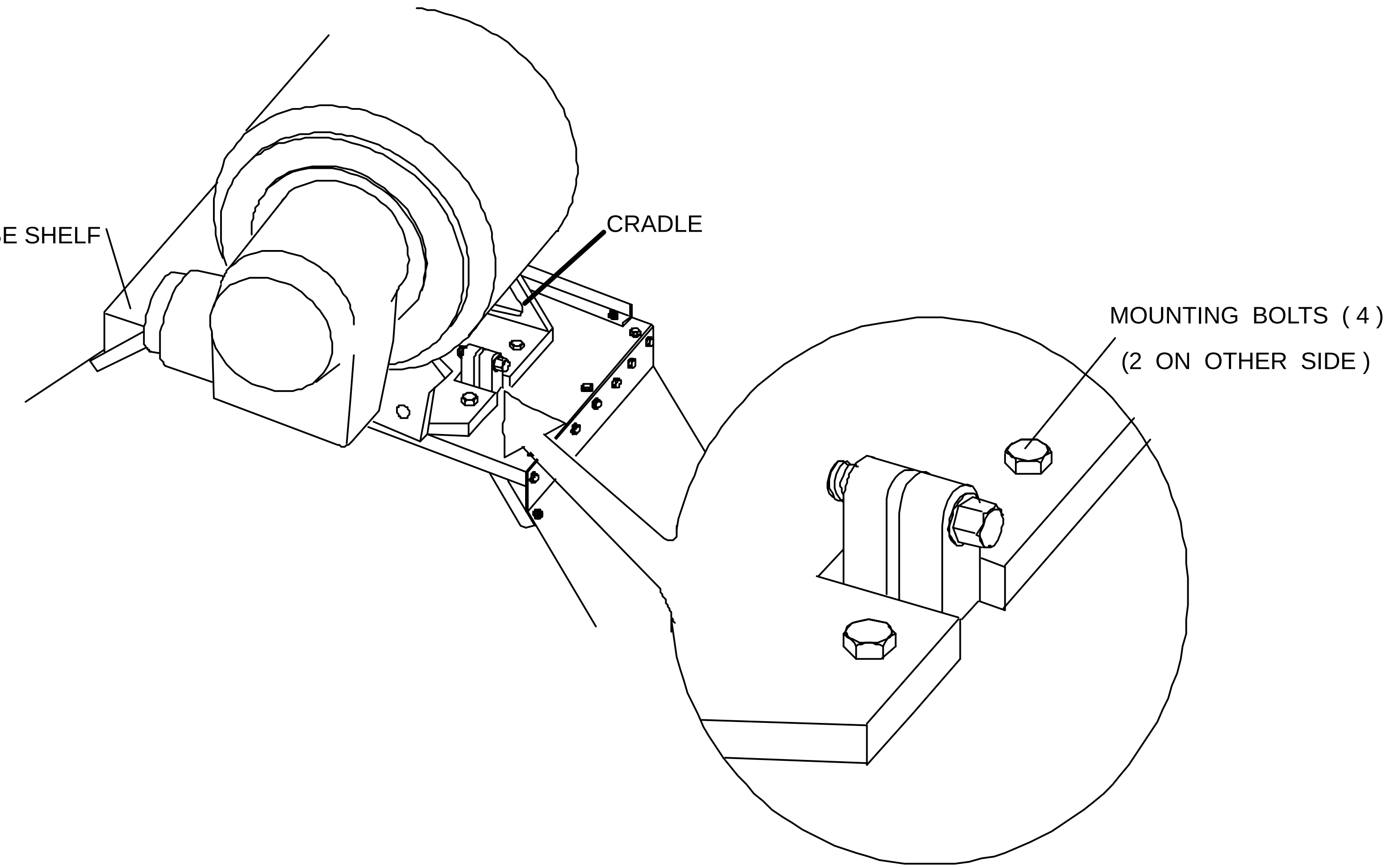


FIGURE 70 TYPICAL TUBE CRADLE ADJUSTMENT

COVERS

27. Gantry cover documentation can be found in the Repair Manual.
28. Apply Gantry power. Refer to the Power Up/Down procedures in this manual. Allow 5–10 minutes for detector stabilization before continuing.
29. Run the Tube Centering procedure: first lateral centering (mechanical); then center calibration Refer to the Auto–Calibration manual.

30. If centering deviates by more than +/– 2 ray sums, adjust the offset of the scan drive fine resolver using the Scan Drive Calibration procedure in the Calibration manual.
31. Run the CNA noise test (X ray noise test) and record the value obtained (for 50–60 Hz range). Refer to the Auto–Calibration manual.
32. Perform the full Auto–Calibration procedure.
33. Remove Gantry power. Refer to the Power Up/Down procedures in this manual.

COUCH–TO–GANTRY ALIGNMENT CHECKS

Prior to checking the relative positioning of the table or the alignment of the phantom, the table must be in the center of the scan circle. The following procedure quickly checks for Patient Support centering.

Verify Patient Support is Perpendicular to the Gantry

1. Measure from the right side, upper corner of the foot end of the Patient support to the top retaining pin of the right side gantry column. See Figure 71. Use a metric tape measure. **Do not use a string.**

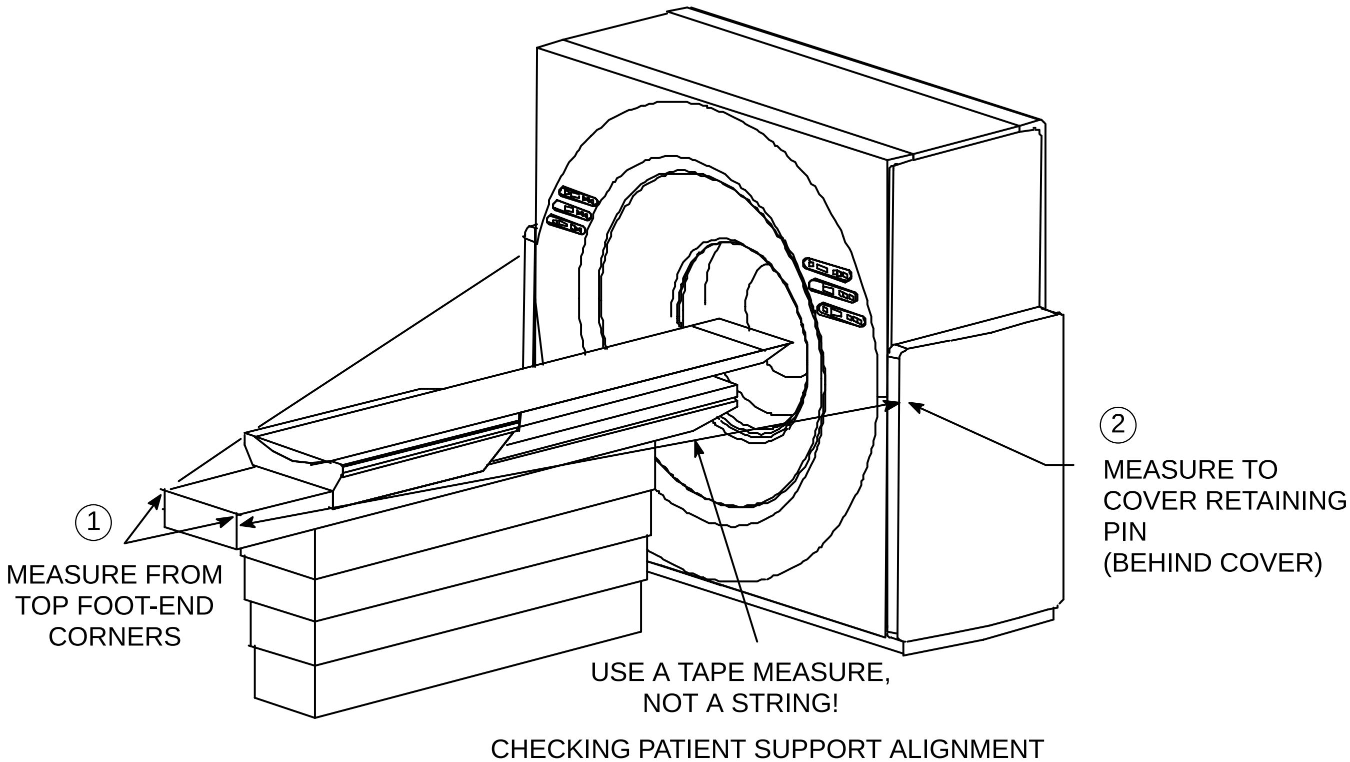


FIGURE 71 HOW TO SET PATIENT SUPPORT PERPENDICULAR TO THE GANTRY

2. Repeat this for the left side of the Patient Support top.

Fill In this Table:

Right Side Measurement	
Left Side Measurement	
Difference (within ± 3 mm)	

This distance should equal the corresponding distance measured on the right side: ± 3 millimeters. If the two measurements are not within this limit, reposition the rear of the Patient Support (maintain the front edge measurement !!), and verify.

3. Recheck levelness of Patient Support.

Scan and Verify Patient Support is Centered in Scan field

1. Power up the system. Push the table top into the scan circle.
2. Take an axial scan of the table top. Measure the distance from each edge of the table top to the edge of the scan field. Select the Help pull down menu for the Operator's On–Line manual if necessary.
3. Take a vertical pilot scan as well; this will ensure perpendicularity.
4. You can make minor adjustments to the Patient Support because it has not been torqued to the floor. Re–scan as necessary.
5. After the Patient Support is in the center of the scan circle:
   a. ensure all washers and lock washers are installed between the bolt heads and the Patient Support base
   b. then use the appropriate socket or drive and <u>torque the bolts to 30 to 40 ft–lbs.</u>
6. Press the E–STOP button. Turn power OFF at the wall box.

> ⚠ **WARNING** ⚠
>
> **USE EXTREME CAUTION WHEN PERFORMING INSTALLATION PROCEDURES WITH THE GANTRY COVERS OPEN. 480 VAC, 208 VAC AND 120 VAC AS WELL AS 5 VDC AND 15 VDC ARE EXPOSED IN MANY LOCATIONS. FAILURE TO COMPLY CAN RESULT IN SERIOUS INJURY OR DEATH.**

7. Refer to Figure 72. Install Patient Support covers. Place cover over sub–frame lips, and reinstall fastening screws at each end, installing top screws first. Fasten top cover panels to couch top, using four screws and washers, as shown.

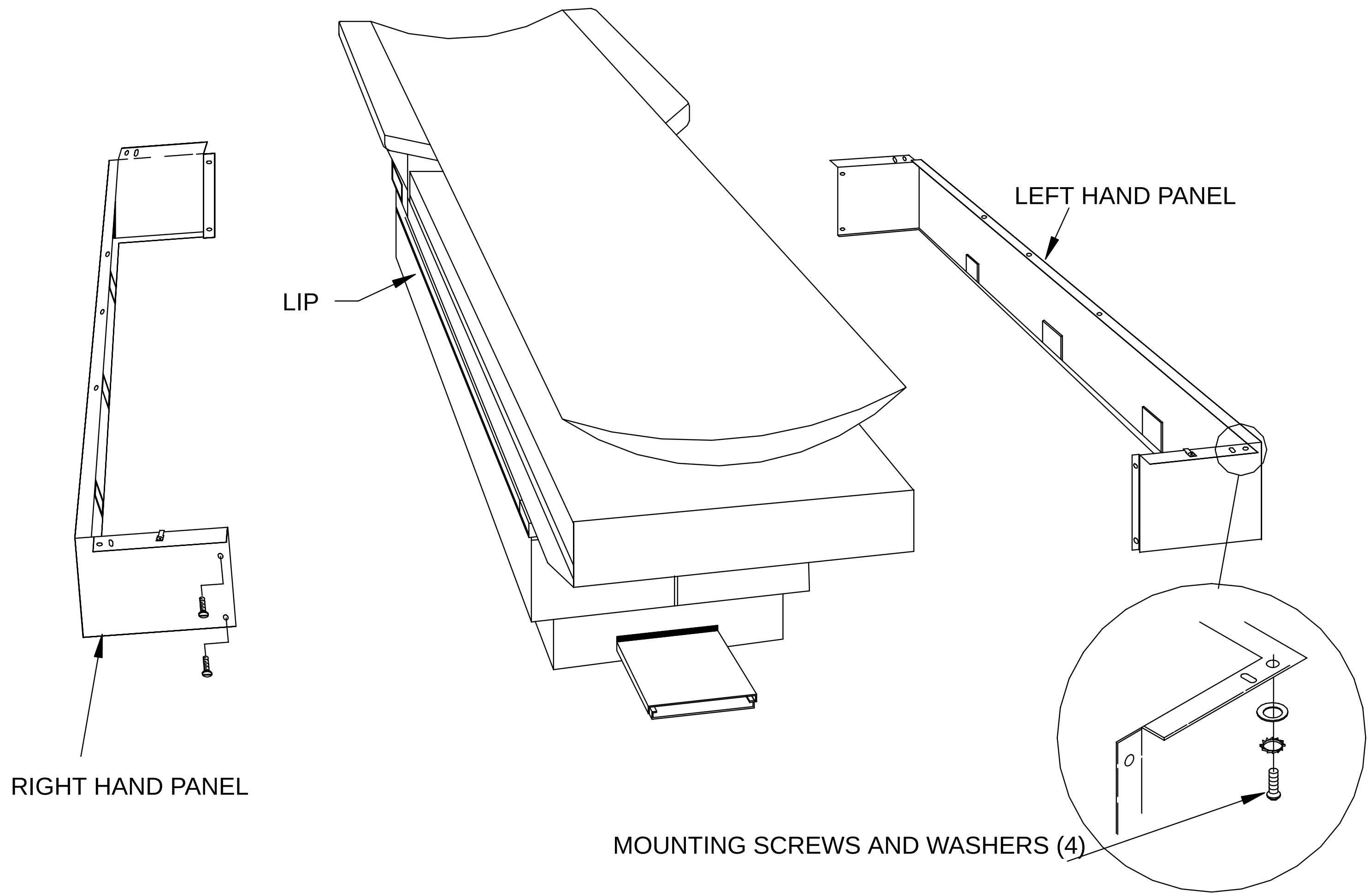


FIGURE 72 PATIENT SUPPORT COVER INSTALLATION

PATIENT SUPPORT VERTICAL/HORIZONTAL CHECKS

Do this procedure to verify Patient Support position display on the Gantry.

Vertical Checks

Perform these checks by measuring the height of the Patient Support sub–frame above the floor, not the couch height above the floor.

1. Position the table top by hand against its rear mechanical stop.
2. Apply Gantry power. Refer to the Power Up/Down procedures in this manual.
3. Position the Gantry to **zero degrees** tilt with the Gantry control panel switches.
4. Position the Patient Support at a vertical elevation of **10.0 mm as displayed on the Gantry**. Measure this distance with a metric tape measure. See Figure 73. ***Record the value.***

   This measured distance should be 458 mm (+/– 3 mm). If the measurement is out of this range, refer to the Calibration manual and calibrate the position of the Patient Support.

> **NOTE**
>
> For consistency, table height measurements are referenced as follows: place a straight–edge across the top of the sub–frame at a point 55.5 inches from the front edge; measure the table top height from the rear side of the straight–edge to the floor.

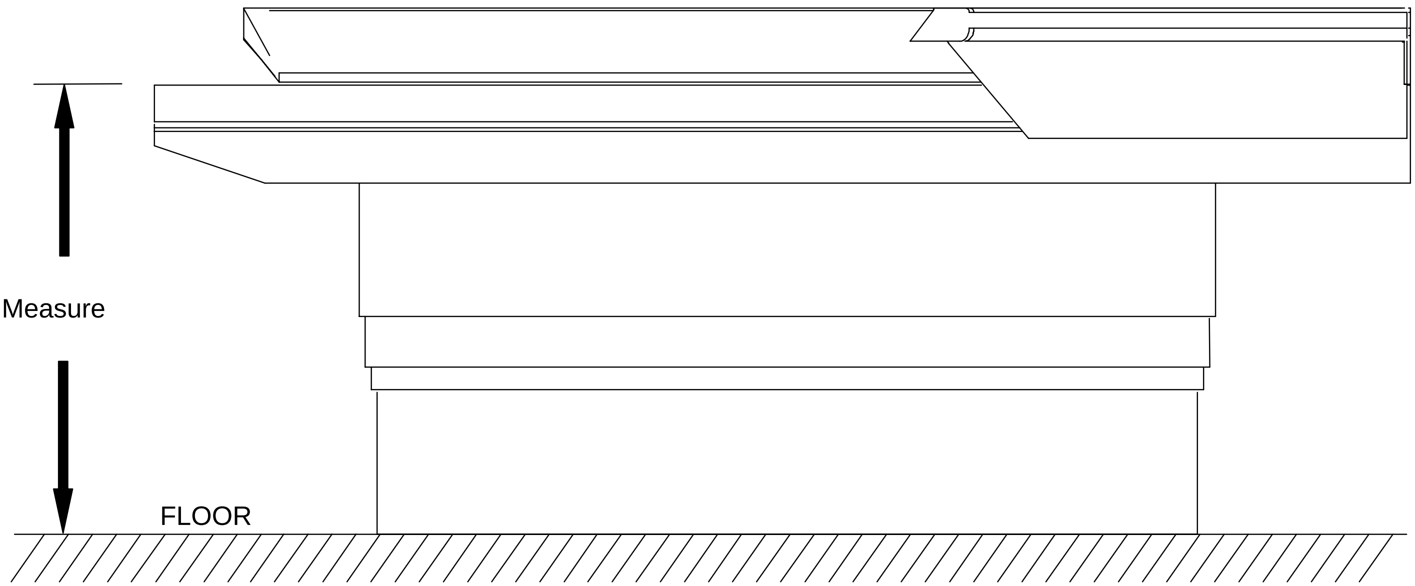


FIGURE 73 PATIENT SUPPORT WITH GANTRY DISPLAY AT 10.0 MM

5. Raise the Patient Support to **475 mm as displayed on the Gantry display panel.** Measure the distance from the floor to the sub–frame with the same tape measure See Figure 74. ***Record the value.***

6. Subtract the value (recorded vertical height) of Step 4 from the measurement in Step 5. The difference should be **465 [2 mm**. If the difference is out of this range, refer to the Vertical Position Calibration and calibrate the vertical position of the Patient Support.

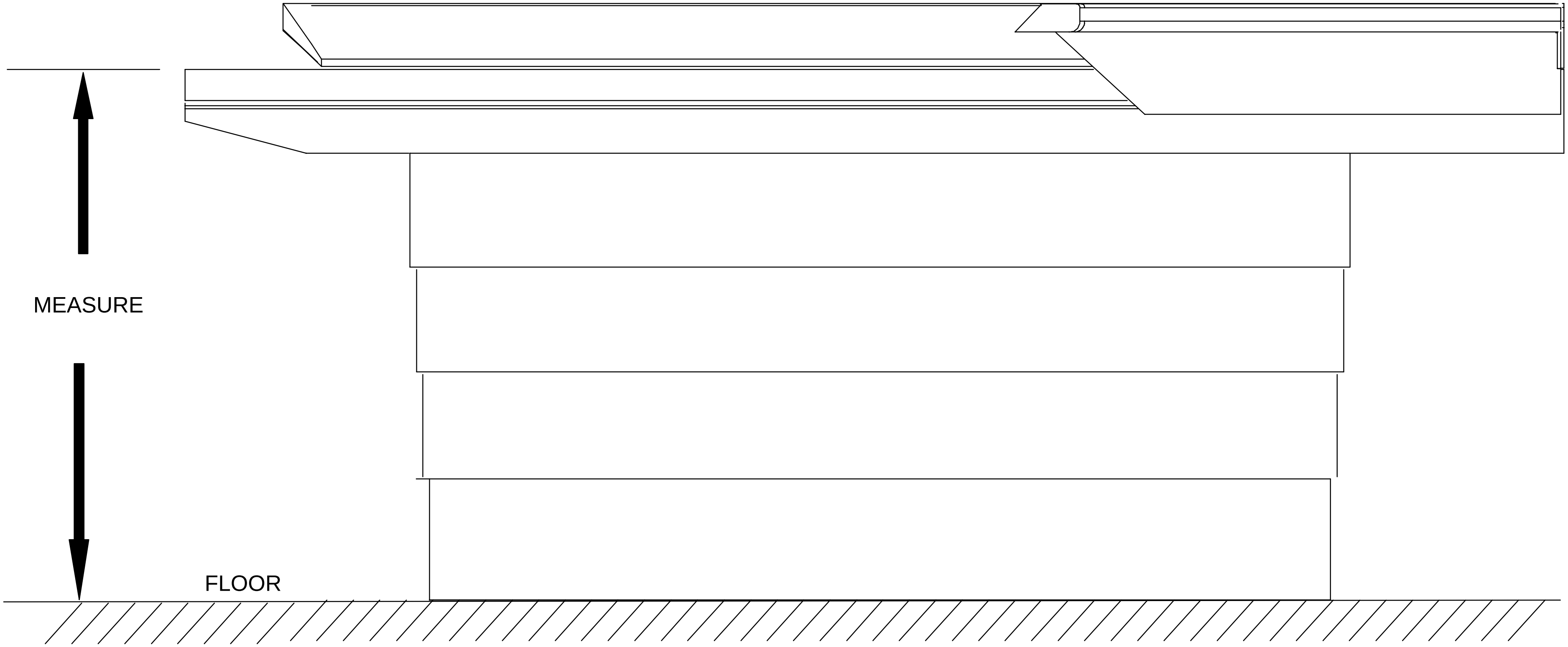


FIGURE 74 PATIENT SUPPORT WITH GANTRY DISPLAY AT 475.0 mm

HORIZONTAL CHECKS

1. Move the table top forward **180 mm** from its rear mechanical stop. See Figure 75. Use a metric tape measure for this measurement. This position will be referred to as the Table Top ”OUT” Reference Mark.

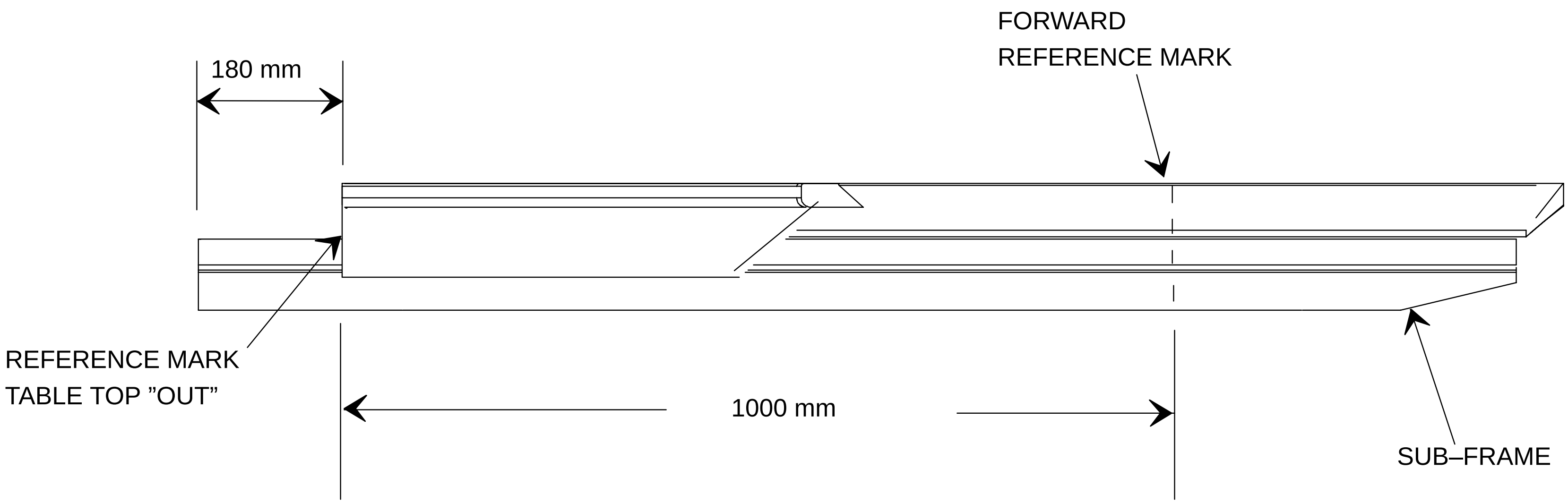


FIGURE 75 TABLE TOP “OUT” REFERENCE MARK

2. Press the Relative Zero switch on the Gantry control panel.

3. Observe the horizontal position readout and verify that it displays a reading of zero (**0.0**). If the readout <u>does not</u> display **0.0**, a failure is present in the Gantry. Utilize Questor in this case.

4. Move the table top forward **1000 mm** from the Table Top ”OUT” Reference Mark established in Step 1. Measure this distance with the same tape measure used in Step 1.

5. Observe the horizontal position readout and verify that it displays a reading of **1000.0**. If the readout <u>does not</u> display **1000.0** +/–0.5 mm, refer to the GPUSPY procedure called CC.

COUCH/GANTRY RELATIVE POSITIONING CHECK

The interference monitor assumes a certain relative position between the table top and the Gantry. If this is not maintained, the performance of the interference monitor may be compromised. A simple check, outlined below, assures the correct positioning.

Two positions must be verified to assure the proper table top–to–Gantry relative position (one for vertical and one for horizontal). It is assumed that the table top and Gantry are calibrated prior to performing these checks.

Check 1 — Relative Vertical Positioning

1. Place the table top at 20 mm horizontal (fully moved away from the Gantry) and 215.0 mm vertical.
2. Manually move the table top forward into the Gantry aperture.
3. The bottom of the table top should touch the bottom of the Gantry aperture or be within 3 mm.

Check 2 — Relative Horizontal Positioning

1. Raise the table top to the approximate center of the aperture.
2. Horizontally position the table top to a display readout of 507.0 mm (This should place the tip of the table top at the scan plane of the Gantry.)
3. Turn on the Gantry lasers and verify that the laser beams are within 3 mm of the tip of the table top.

PHANTOM ALIGNMENT PROCEDURE

Use the following procedure to physically center the phantom within the phantom box. Once this is accomplished, centering the phantom in the field should only require the Service Engineer to place the table top at the positions recorded in this procedure.

1. Place the phantom box on the table top so that the front of the phantom box is flush with the end of the table top. Place the phantom in the phantom holder.
2. Align the phantom in the field using the laser so that the 8” water phantom is in the beam.
3. Scan the phantom using the center cal or H–F (half–field) water protocol from the Test Protocols screen of the User Interface. Use the default mA, no comp.
4. *Need phantom check procedure here.*

LEFT/RIGHT ADJUSTMENT

5. Loosen the screws on **one** of the phantom holder wedges so that the phantom can be moved in the required direction. Refer to Figure Figure 76.
6. Move the wedge the desired distance, tighten the screw and then slide the phantom into its new location.

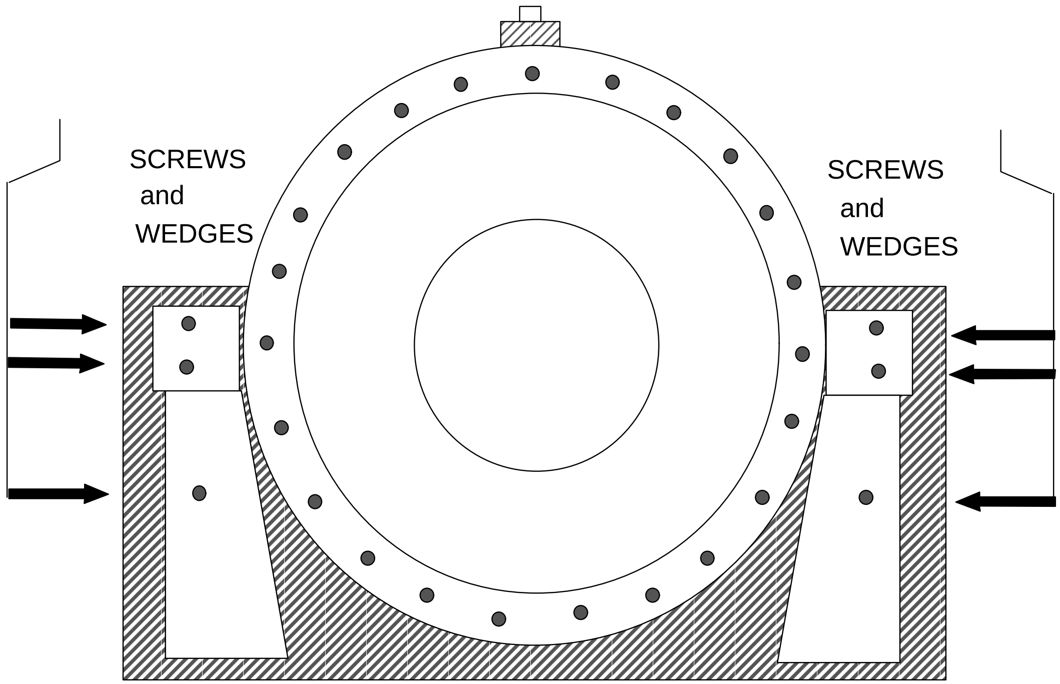


FIGURE 76   MOUNTING SCREWS FOR PHANTOM WEDGES

UP/DOWN ADJUSTMENT

7. Use the Gantry control panels to move the table top into the correct vertical location.
8. Re–scan the phantom as many times as needed to adjust the phantom to proper centering position. When the phantom is within the +/–1 mm specification of the **check phantom(??)** program, record the table top horizontal and vertical locations and use these locations when it is necessary to scan a centered phantom.

NEW PHANTOM TEST KIT *(STARTING SEPTEMBER 1997)*

A Beam Analysis kit has been provided with the scanner systems. This kit is used to verify the calibration and performance of the system. The kit is also used as part of regularly scheduled quality control checks.

The test kit contains: phantom, phantom support, centering pin and two bottles of phantom refill solution. The illustration shows the test kit as it appears packed in its shipping and storage box. The phantom support is always the topmost piece in the kit.

The **phantom support** attaches to the patient support and positions the phantom and centering pin.

The **phantom** is used for evaluating image quality and has five separate scannable sections:

12” water... used for full field.

8” water... used for half field.

6” water... used for low contrast measurements.

6” low contrast... used for low contrast measurements.

6” slice thickness/MTF.

Proper scanner calibration requires the phantom have no air bubbles in any scan section. Additional distilled water can be added to the phantom via the expansion chamber which should be 3/4 filled.

The **centering pin** is used for calibrating global and individual centering constants.

NEW PHANTOM MOUNTING PROCEDURE

1. Remove the phantom support from the storage box.
2. Loosen the adjustable foot knob to its maximum open position by turning the knob counterclockwise. Refer to Figure 77.

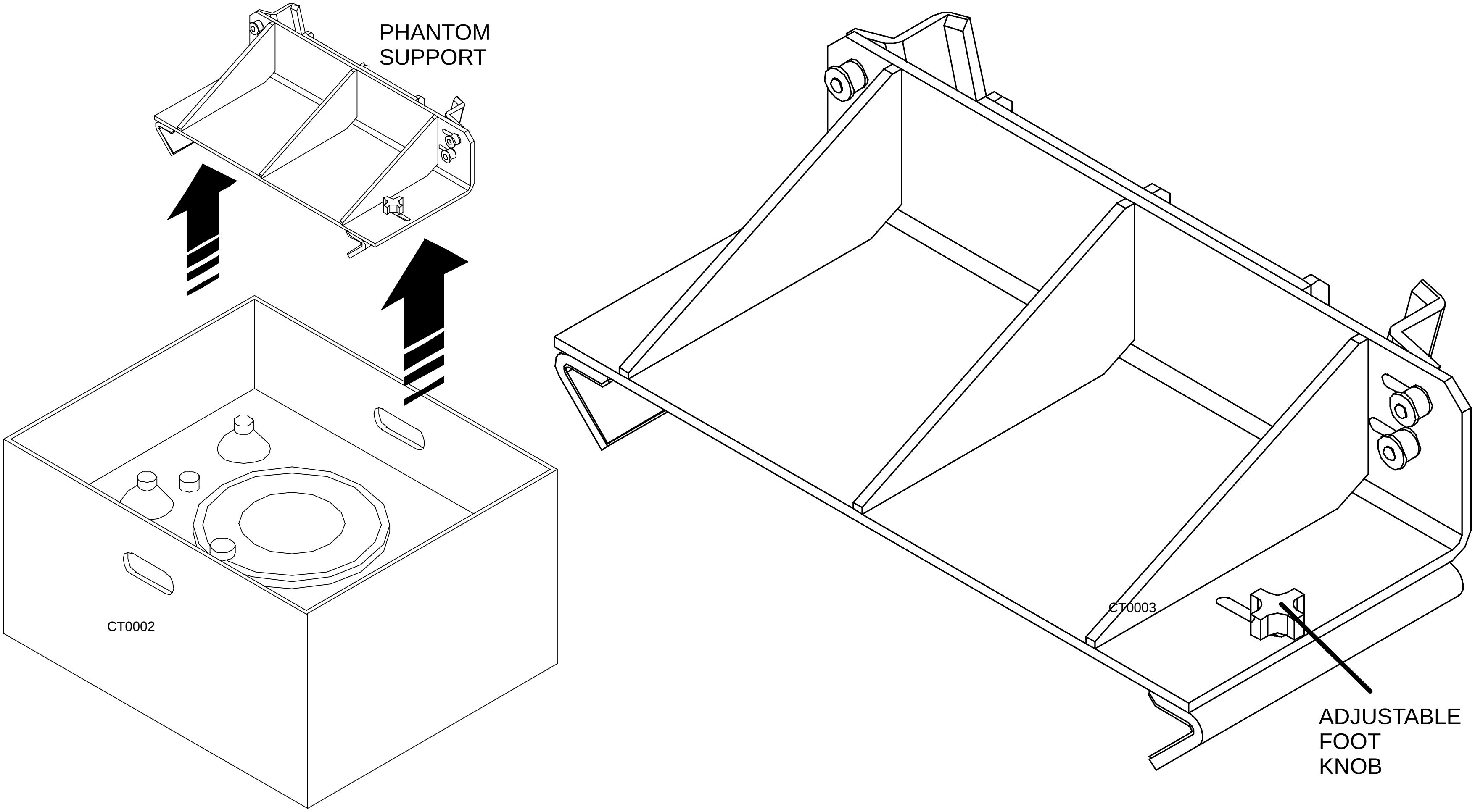


FIGURE 77 NEW PHANTOM SUPPORT

3. Stand on the right side of the patient couch. Place the phantom support on the couch by sliding the supports' feet onto the couch until the phantom support overhang on the front of the support butts up flush against the front of the patient couch. Refer to Figure 78.

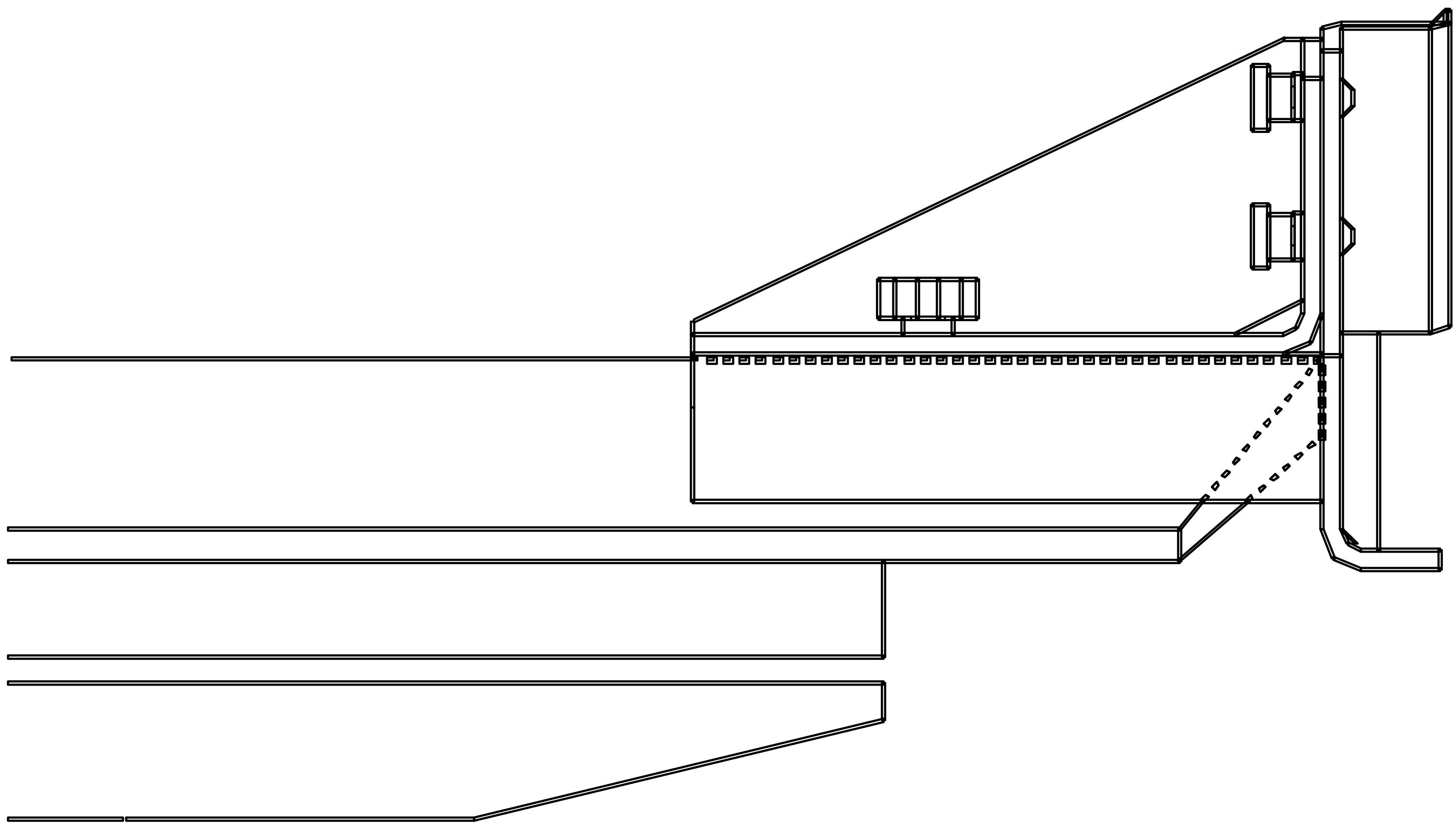


FIGURE 78 NEW FLUSH MOUNTED PHANTOM SUPPORT

4. With both feet underneath the edge of the couch top and the front of the support flush to the patient couch, place your left hand on the left corner of the support and slide unit toward you. Make sure the front overlay piece of the support remains flush to the front of the couch. Refer to Figure 79.

5. The front and left side of the phantom support should now fit snug to the patient couch. Hold the phantom support in place with your left hand. With the right hand, push the right foot bracket in toward the patient couch and tighten the right foot by turning the adjustable knob clockwise until the foot fits snug to the right side of the patient support. Refer to Figure 79.

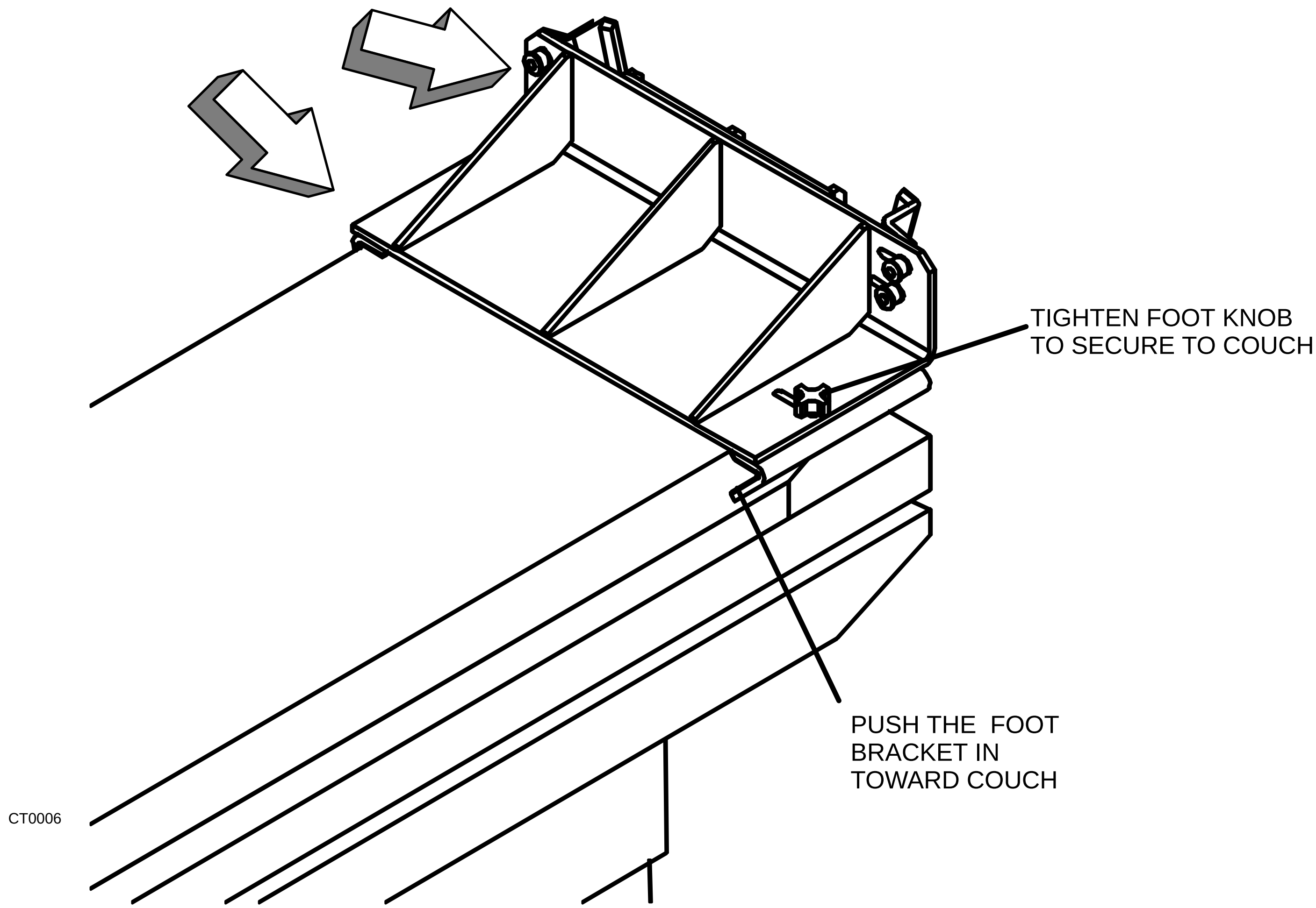


FIGURE 79 SECURING NEW PHANTOM SUPPORT TO PATIENT SUPPORT

Place New Phantom in Retainer Brackets

6. Remove the phantom from the storage box and mount it on the phantom support by inserting the phantom's flange (the part with the aluminum screw posts) behind the retainer holding brackets and position the overflow chamber at the top. Make sure the phantom rests securely between the left/right adjustable holding brackets by turning the screws to tighten brackets.

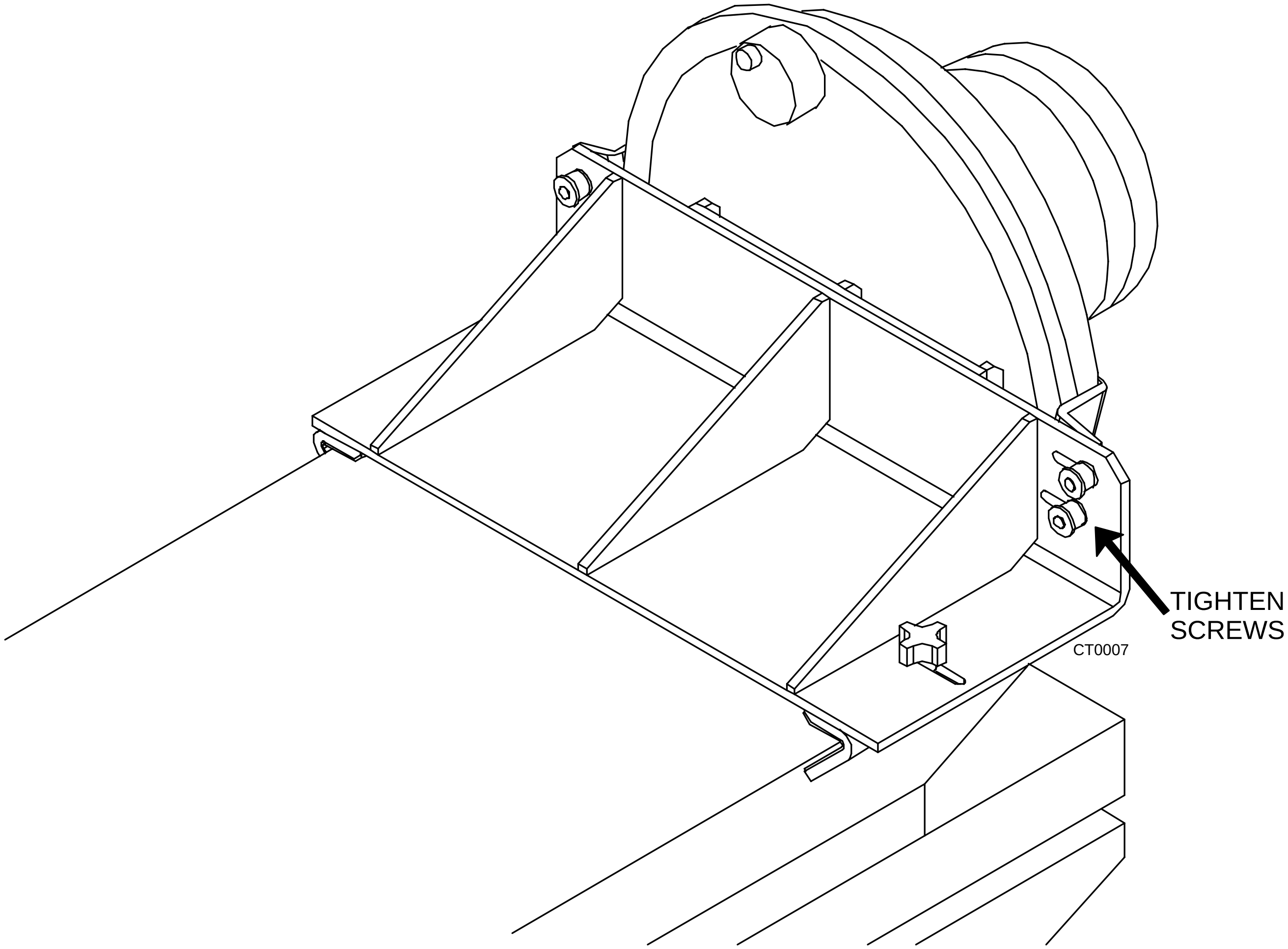


FIGURE 80 PHANTOM MOUNTED ON PATIENT SUPPORT

7. With the phantom sitting in the support, adjust the left/right centering by loosening the screws and sliding the brackets as needed in their slots. Turn the screws clockwise to re–tighten them after making the centering adjustment and rescan to see if centered in the field of view. Refer to Figure 81.

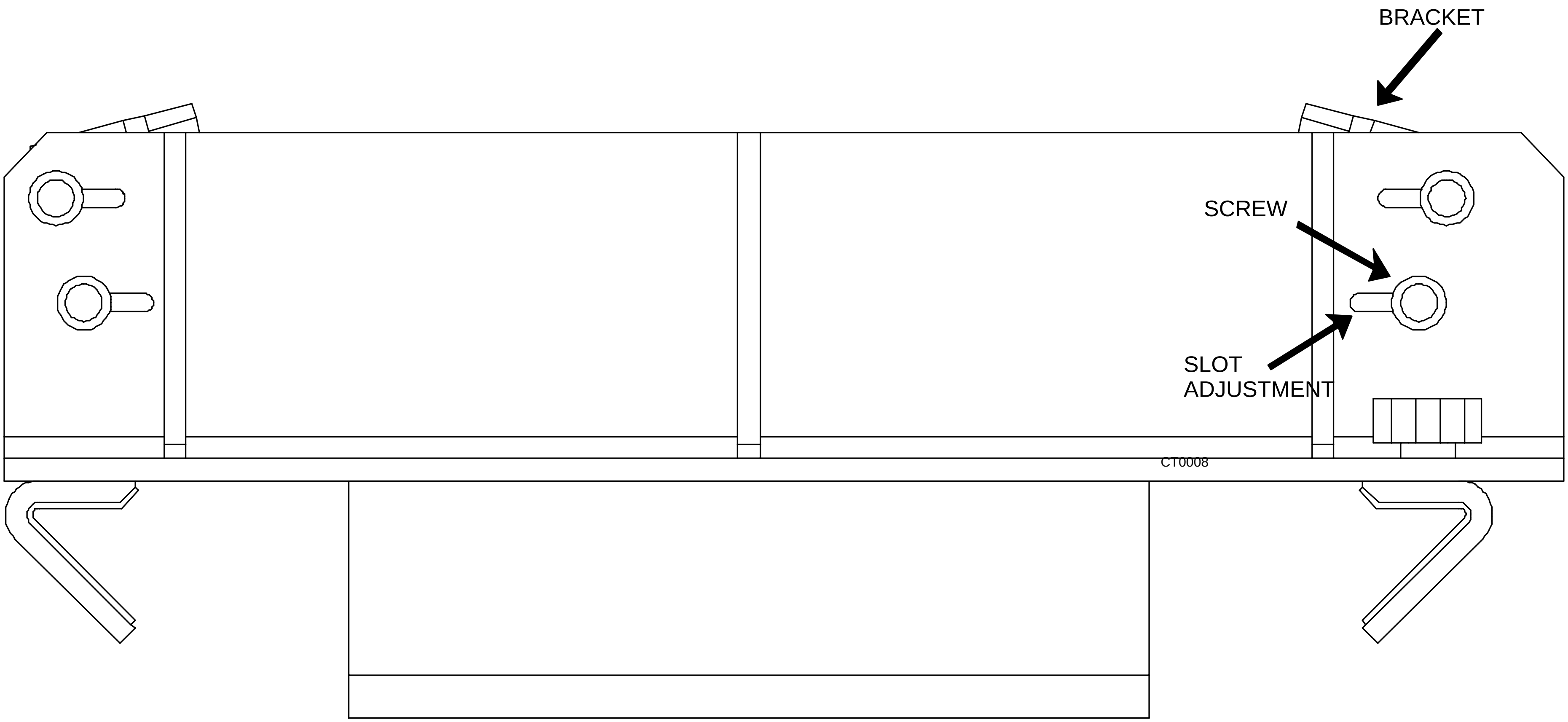


FIGURE 81 RETAINER BRACKET ADJUSTMENT SLOTS

The phantom should now be securely held between the retainer brackets on the phantom support.

Phantom Support and Phantom Positioning

The phantom is properly positioned for the scan when the phantom support is flush with the end of the patient couch, fits snugly to the sides of the patient couch, the phantom overflow chamber is at the top and the centering pin is inserted. Refer to Figure 82.

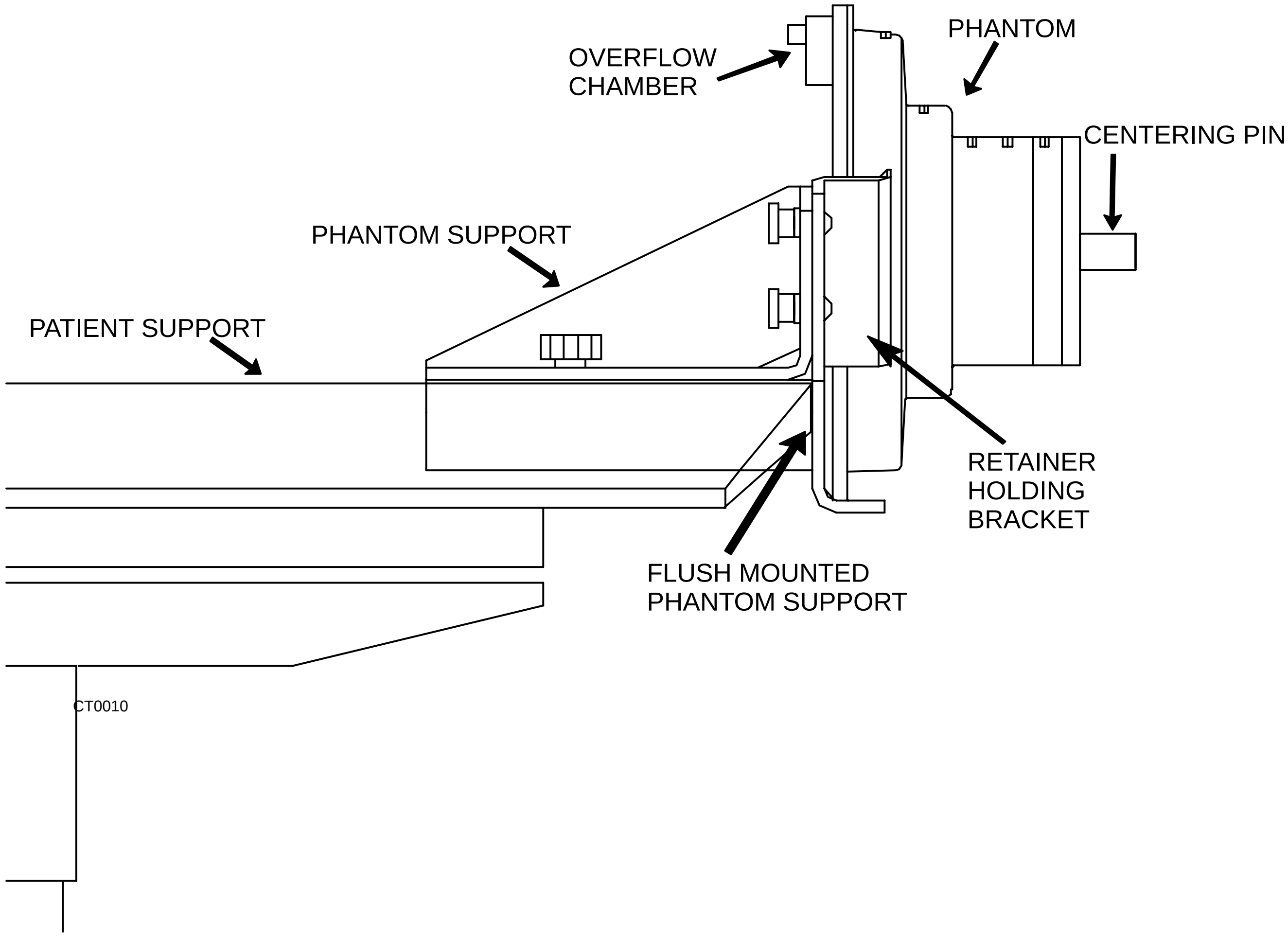


FIGURE 82 PROPERLY MOUNTED SUPPORT AND PHANTOM

DISPLAY CONSOLE AND ACQIMAGE CABINET CHECKS

1. With power off remove air filter in Display Tower and check to see that it is clean. Replace air filter (Refer to the Planned Maintenance Manual for details)
2. With power off remove air filter in AcQImage Cabinet and check to see it is clean. Replace air filter (Refer to the Planned Maintenance Manual for details).

After keyswitch power on of system, check:

3. Tape drive(s) LEDs blink through an initialization cycle and settle to a green go condition. This may take over 30 seconds to complete.
4. Check for audible sound of fans running in the Display Tower tower and light airflow exiting from upper side vents on each side.
5. Display Monitor video screen should present a video output within one minute. Most monitors will show status of video on LED indicator panel or video screen after power up as well.
6. Sparc computer within the Display Tower performs a power on self–test. Indication of status is given on various keyboard LEDs through the cycle and Display Monoitor should remain blank. Test should complete in less than 20 seconds and all LED's should be off. If a malfunction is detected one or more LED's will remain on.
7. Intercom will become operational after approximately 5 seconds. Verify operation by by pressing PTT (Push to talk) button and checking for indication on PTT LED.
8. Check for audible sound of fans running in the AcQImage Cabinet and airflow exiting from upper side vents on each side. There are no visual power on LEDs so this will be your best indicator of power to the unit.

X–RAY TUBE SEASONING PROCEDURE

Each x–ray tube must be seasoned to ensure exposure consistency by helping to eliminate anode–to–cathode tube arcs caused by air molecules. Seasoning activates the device internal to the tube known as the 'getter' which collects these air molecules. Tubes are seasoned by initiating a series of exposures starting with a low energy technique and moving up to the maximum technique. It is extremely important to ensure that the wait times, delay times, and exposure times are followed accurately.

Tube Seasoning instructions are located in the MANEXP manual.

COMPLETE FIELD INSTALLATION

Performance Results

1. After you calibrate the scanner (if needed), test it, and find it to be ready for turn–over to the customer, the latest set of performance archiver results should be stored as the record of and at field installation.

Pick Up / Clean Up

2. Do a 'sweep' of the area. Pick up all tools. clean up all shipping and wrapping papers. Use the supplied cleansers to clean the Console top, the Console, the Gantry and the Patient support. Use Philips spray paint as necessary to cover nicks and scratches.

THIS COMPLETES THE SCANNER INSTALLATION

3. Finish up by clearing all debris, making sure the scanner is clean (there are no smears or chipped paint) and the system is operable.
4. Fill in appropriate local paper work.

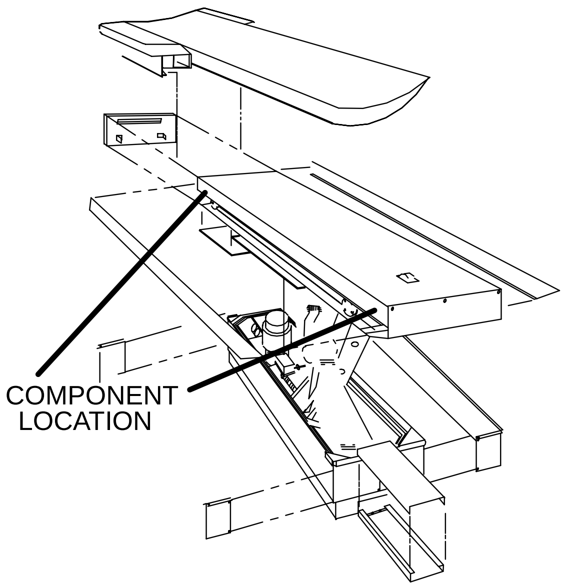
AcQSim CT Mechanical Component Calibration

HORIZONTAL TRAVEL – LIMIT SWITCHES

Introduction: This procedure is used to adjust the Patient Support Top horizontal travel limit switches. The procedure is the same for both the front and the rear switches.

Tools required: FSE tool kit

Estimated time: 1 Hour



ADJUSTMENT PROCEDURE:

1. Apply Gantry power. Refer to the Power Up/Down procedure.
2. Position the table top by hand against its rear stop.
3. Using the gantry control panel switches, raise the Patient Support to its highest position.

WARNING

480 VOLTS AC, 120 VOLTS AC, +17 VOLTS DC, AND –17 VOLTS DC ARE EXPOSED WHEN THE GANTRY COVERS ARE OPEN. ACCIDENTAL CONTACT WITH THESE VOLTAGES MAY CAUSE SERIOUS INJURY OR DEATH TO THE SERVICE PERSON.

4. Remove the right carbon top side cover and the right side rail cover. Refer to the J–Bracket Cover and Rail Cover removal in the Patient Support Cover Removal/Replacement procedures at the beginning of this section.
5. Move the table top by hand as far as it will go toward the front and then move it back 3/8 inch [9.525 mm].
6. Refer to Figure 83. Loosen the screw that attaches the limit switch bracket to the table frame and position the switch so that it just actuates. Tighten the limit switch in place.
7. Move the table top away from the switch and then back to engage the switch to verify proper mechanical operation.
8. Move the table top by hand as far as it will go toward the rear and then move it back 1/8 inch [3.175 mm].
9. Refer to Figure 83. Loosen the screw that attaches the limit switch bracket to the table frame and position the switch so that it just actuates. Tighten the limit switch in place.
10. Move the table top away from the switch and then back to engage the switch to verify proper mechanical operation.
11. Move the Patient Support table top by hand to the center of its travel.
12. Apply Gantry power. Refer to the Power Up/Down procedure.
13. Move the table top forward and backward with the Gantry control panel switches to verify limit switch operation.
14. Position the Patient Support table top by hand against its rear stop.
15. Remove Gantry power. Refer to the Power Up/Down procedure.
16. Install the right side rail cover and right J–bracket cover. Refer to the J–bracket cover and Rail Cover replacement in the Patient Support Cover Removal/Replacement procedures at the beginning of this section.

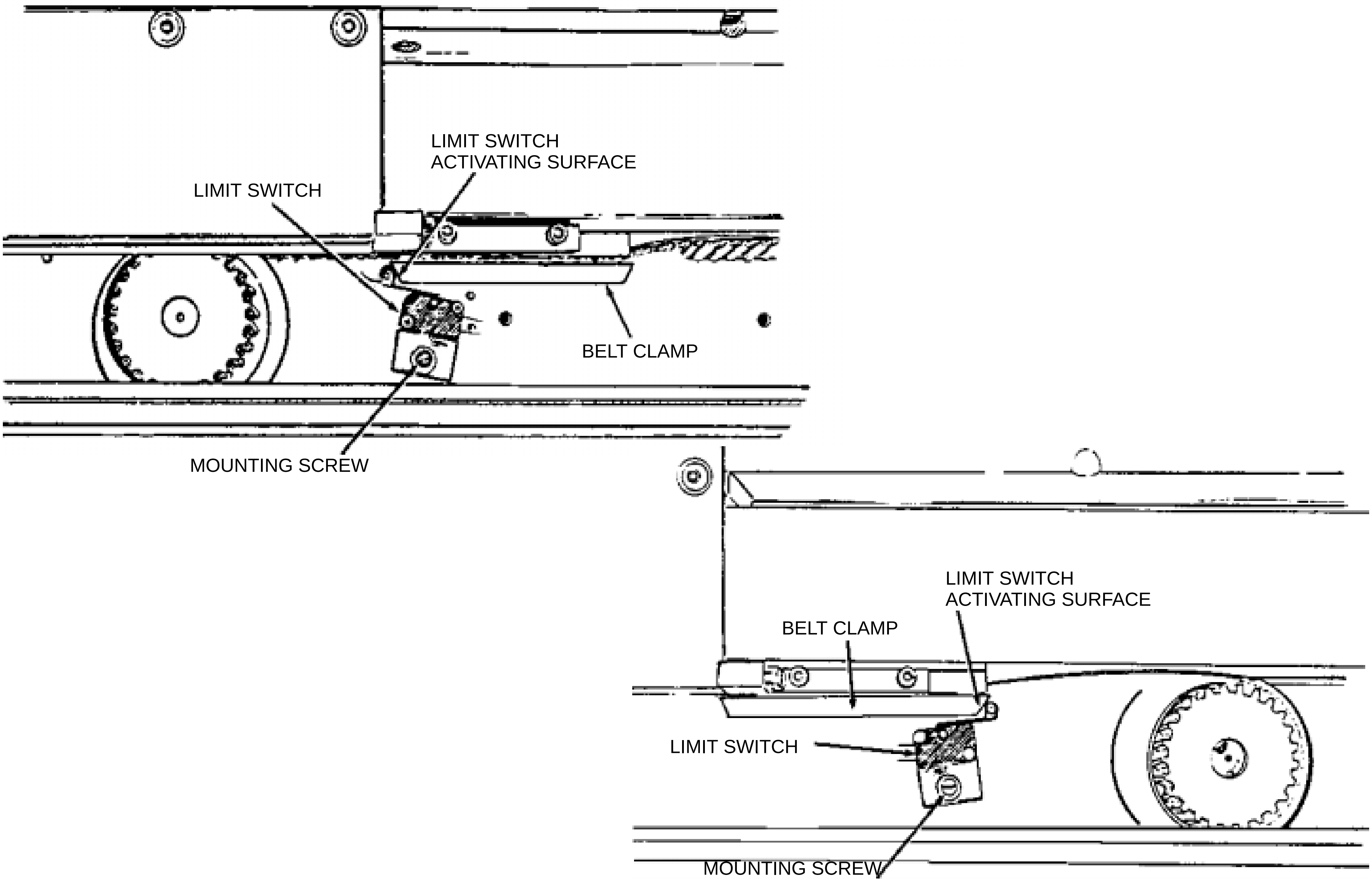


FIGURE 83 FRONT AND REAR HORIZONTAL LIMIT SWITCHES

VERTICAL TRAVEL LIMIT SWITCH

__________**Do not perform this procedure unless ABSOLUTELY NECESSARY.**

Introduction:	This procedure is used to adjust the Patient Support vertical travel limit switches.
Tools required:	FSE tool kit
Estimated time:	1.5 hours

VERTICAL TRAVEL LIMIT SWITCH ADJUSTMENT:

1. Apply Gantry power. Refer to the Power Up/Down procedure.
2. Lower the Patient Support to its lowest position with the Gantry control panel switches.

> ⚠ **WARNING** ⚠
>
> **480 VOLTS AC, 120 VOLTS AC, ARE EXPOSED WHEN THE PATIENT SUPPORT TELESCOPING BASE COVERS ARE OPEN (INCLUDING THE LIMIT SWITCHES). ACCIDENTAL CONTACT WITH THESE VOLTAGES MAY CAUSE SERIOUS INJURY OR DEATH OF THE SERVICE PERSON.**

3. Remove the Patient Support telescoping base covers from the subframe. Completely remove all telescoping panels. Refer to Telescoping Base Cover procedure.

> ⚠ **WARNING**
>
> **SCISSOR–ACTION MECHANISM IS EXPOSED WHEN THE PATIENT SUPPORT COVERS ARE REMOVED. KEEP EQUIPMENT AND PERSONNEL AWAY FROM THE MECHANICAL ACTION OF THESE PARTS. FAILURE TO DO SO MAY CAUSE SERIOUS DAMAGE TO EQUIPMENT AND SERIOUS INJURY OR DEATH TO PERSONNEL.**

4. Raise the Patient Support to the center of its vertical travel with the Gantry control panel switches.

5. Remove Gantry power. Refer to the Power Up/Down procedures.

6. Loosen the lock nuts on the four vertical limit switch adjustment screws. Turn the soft limit switch adjusting screws fully into the frame. Turn the hard limit adjusting screws several turns into the frame. Refer to Figure 84.

> ⚠ **CAUTION**
>
> THE ALIGNMENT OF THE VERTICAL SUPPORT RODS AND THE CARRIAGE IS CRITICAL. DO NOT PERMIT THE CARRIAGE MECHANISM TO COLLIDE WITH THE STOP BLOCKS. FAILURE TO COMPLY MAY RESULT IN MISALIGNMENT AND SUBSEQUENT SERIOUS DAMAGE TO THE EQUIPMENT.

7. Apply Gantry power. Refer to the Power Up/Down procedures.

8. Lower the Patient Support with the Gantry control panel switches until the distance between the carriage and either stop block (whichever is closest) is 1/16 in. Refer to Figure 84.

9. Adjust the ”down” hard limit switch actuating screw until the hard limit switch just actuates.

10. Raise the Patient Support with the Gantry control panel switches to allow tool access to the limit switch actuating screw.

11. Hold the ”down” hard limit switch actuating screw with an allen wrench and tighten the lock nut.

12. Using the gantry control panel switches, raise the Patient Support a little and then lower it until the distance between the carriage and either linear bearing stop block (whichever is closest) is 1/8 in.

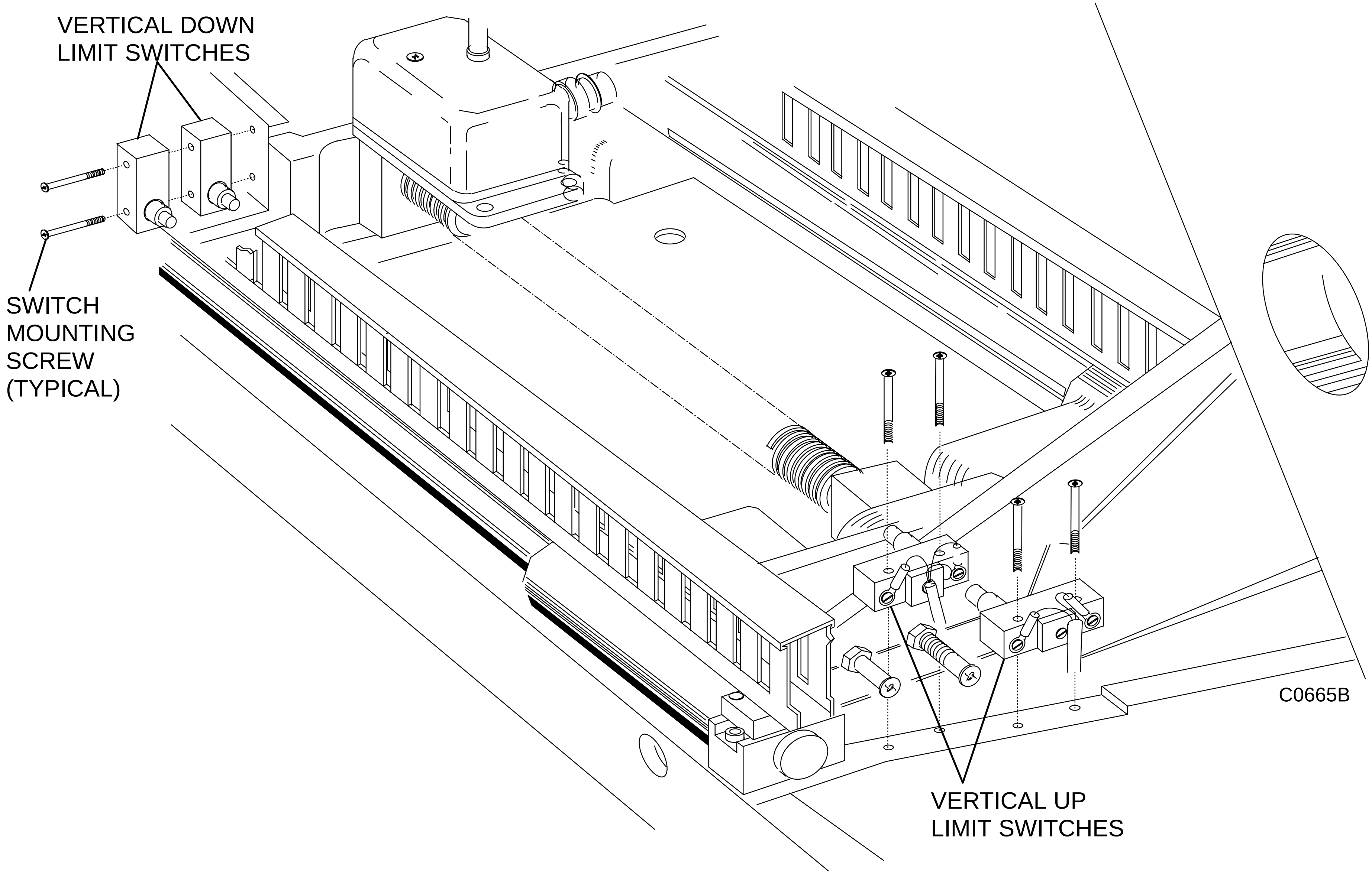


FIGURE 84 VERTICAL LIMIT SWITCHES

13. Adjust the ”down” soft limit switch actuating screw until the soft limit switch just actuates.

14. Using gantry control panel switches, raise the Patient Support to allow tool access to the limit switch actuating screw.

15. Hold the ”down” soft limit switch actuating screw with an allen wrench and tighten the lock nut.
16. Check the ”down” soft limit by using the Gantry control panel switches to lower the Patient Support until the soft limit switch is actuated. Verify the switch activateswhen the carriage is 1/8” from the stop block. The carriage may ”coast” past the point of actuation.
17. Using the Gantry control panel switches, raise the Patient Support until the distance between the carriage and either stop block (whichever is closest) is 1/16 in. Unhook the vertical transducer line if necessary.
18. Adjust the ”up” hard limit switch actuating screw until the hard limit switch just actuates.
19. Using the Gantry control panel switches, lower the Patient Support to allow tool access to the limit switch actuating screw. Pull on the vertical transducer line if necessary.
20. Hold the ”up” hard limit switch actuating screw with an allen wrench and tighten the lock nut.
21. Check the ”up” hard limit by using the Gantry control panel switches to raise the Patient Support until the ”up” hard limit switch is actuated. Verify that the vertical motion stops and the distance between the carriage and the stop block is 1/16 in.
22. Using the Gantry control panel switches, lower the Patient Support until the distance between the carriage and either stop block (which ever is closest) is 1/8 in.
23. Adjust the ”up” soft limit switch adjusting screw until the soft limit switch just actuates.
24. Using the Gantry control panel switches, lower the Patient Support with the Gantry control panel switches to allow tool access to the limit switch actuating screw.
25. Hold the ”up” soft limit switch actuating screw with an allen wrench and tighten the lock nut.
26. Check the ”up” soft limit by using the Gantry control panel switches to raise the Patient Support until the soft limit switch is actuated. Verify that the vertical motion stops and that there is 1/8 in. between the carriage and the stop blocks.
27. Using the Gantry control panel switches, lower the Patient Support to its lowest position.
28. Remove Gantry power. Refer to the Power Up/Down procedures in the Introduction section of this manual.
29. Install the Patient Support telescoping base covers. Refer to Telescoping Base Cover procedure.

VERTICAL POSITION

Introduction: This procedure is used to calibrate the Patient Support vertical movement measurement circuitry.

Tools Required: FSE tool kit
Metric tape measure (200 cm)
Metal 6” ruler

Estimated Time: 30 min.

> **NOTE**
>
> Before beginning this procedure, it is important to confirm the Patient Support table top is level.

1. With system power OFF, position the carbon table top by hand against its rear mechanical stop.
2. Apply system power and allow the system to autoboot into the Clincal Scanning mode. Refer to the Power Up/Down procedure.
3. Position the Gantry to zero degrees tilt with the Gantry control panel switches.
4. Loosen the two captive screws and remove the small Gantry cardfile access cover on the right side of the upper gantry column.

> ⚠ **WARNING** ⚠
>
> **USE EXTREME CAUTION WHEN OPERATING OR WORKING ON EQUIPMENT WITH COVERS REMOVED. HAZARDOUS VOLTAGES ARE EXPOSED WHEN THE GANTRY COVERS ARE REMOVED. ACCIDENTAL CONTACT WITH THESE VOLTAGES MAY CAUSE SERIOUS INJURY OR DEATH TO SERVICE PERSONNEL.**

5. Turn the Vertical Zero clockwise (R13) and Vertical Scale (R4) counterclockwise on the CTCC fully (approx. 10 turns).
6. For consistency, table height measurements will be referenced as follows: Tape a 6 inch ruler across the subframe at a point 20 inches from the front edge; measure the table height from the underside of straight-edge to the floor.
7. Position the Patient Support 448 mm above the floor. Measure this distance with a tape measure. See Figure 85.
8. Adjust the Vertical Zero pot (R13) until the vertical gantry display indicates zero.

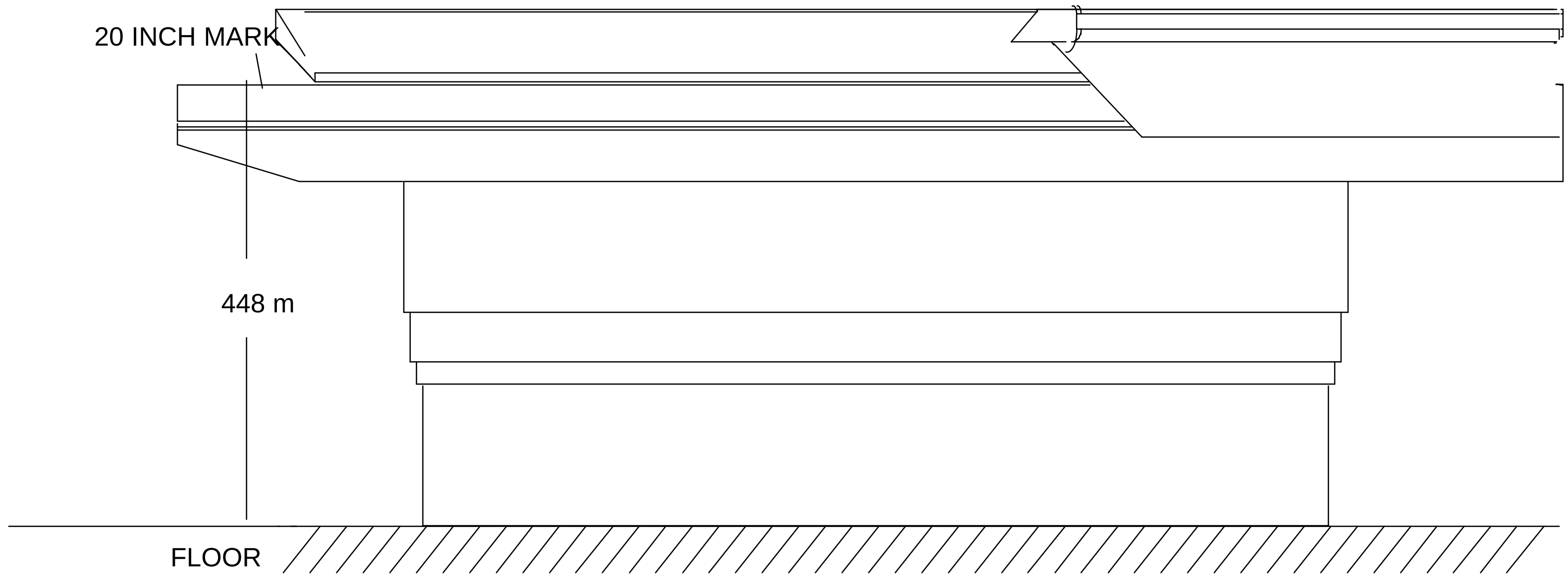


FIGURE 85 PATIENT SUPPORT AT 448 mm

9. Position the Patient Support 923mm above the floor. Measure the height with a tape measure. Adjust the Vertical Scale pot (R14) on the CTCC card until the gantry displays 475.
10. Use the Gantry control panel switches to verify that the vertical position display indicates 475 +/– 2 mm when the Patient Support is 923mm above the floor and zero when the Patient Support is 448 above the floor.
11. If necessary, readjust R13 to make the gantry vertical display read zero. Raise the patient support again. The display should read 455 +/– 2mm.
12. Close and secure the Gantry cardfile access cover.

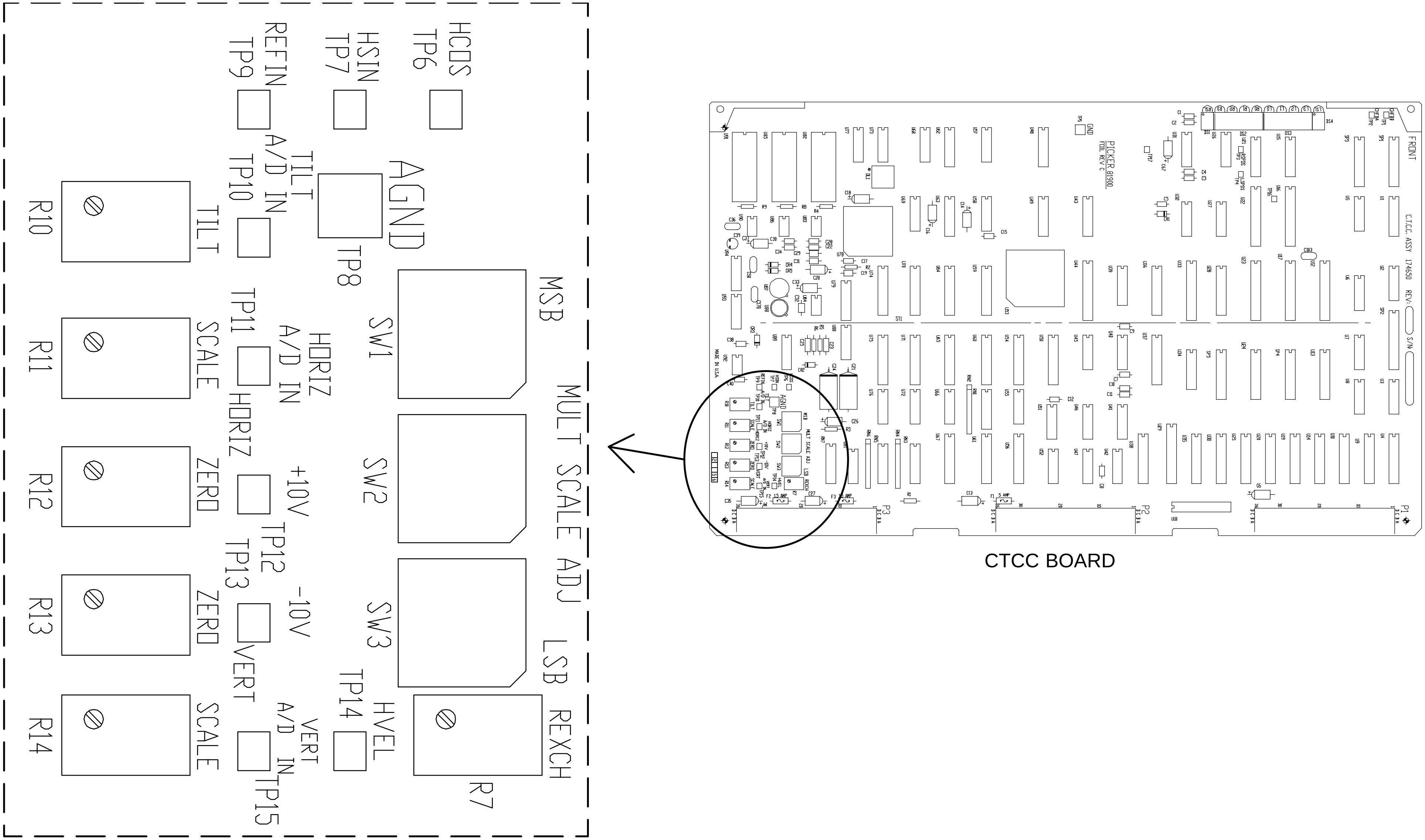


FIGURE 86 CTCC – POTENTIOMETER LOCATIONS

HORIZONTAL COUCH CALIBRATION

NOTE

Patient Support Horizontal Calibration is done via Gputty+, run via the Diagnostic GUI.

HORIZONTAL DRIVE BELT TENSION AND RUN–OUT ADJUSTMENT

Introduction: This procedure is used to adjust (set) the tension on the patient support horizontal drive belt.

Tools Required: FSE tool kit

Estimated Time: 60 min.

1. Raise the Patient Support to its highest position with the Gantry control panel switches.
2. Remove Gantry power. Refer to the Power Up/Down procedure.
3. Remove the Patient Support rear cover, right carbon top side cover and right rail cover. Refer to the Cover Removal procedure.
4. Manually move the table top until the horizontal drive belt block is centered between the front and rear pulleys.
5. Loosen (but do not remove) the four mounting screws that secure front pulley mounting plate to the right side rail. Loosen these screws just enough to permit movement of the mounting plate. Refer to Figure 87.
6. Disconnect the horizontal drive belt block from the table top. See Figure 88.

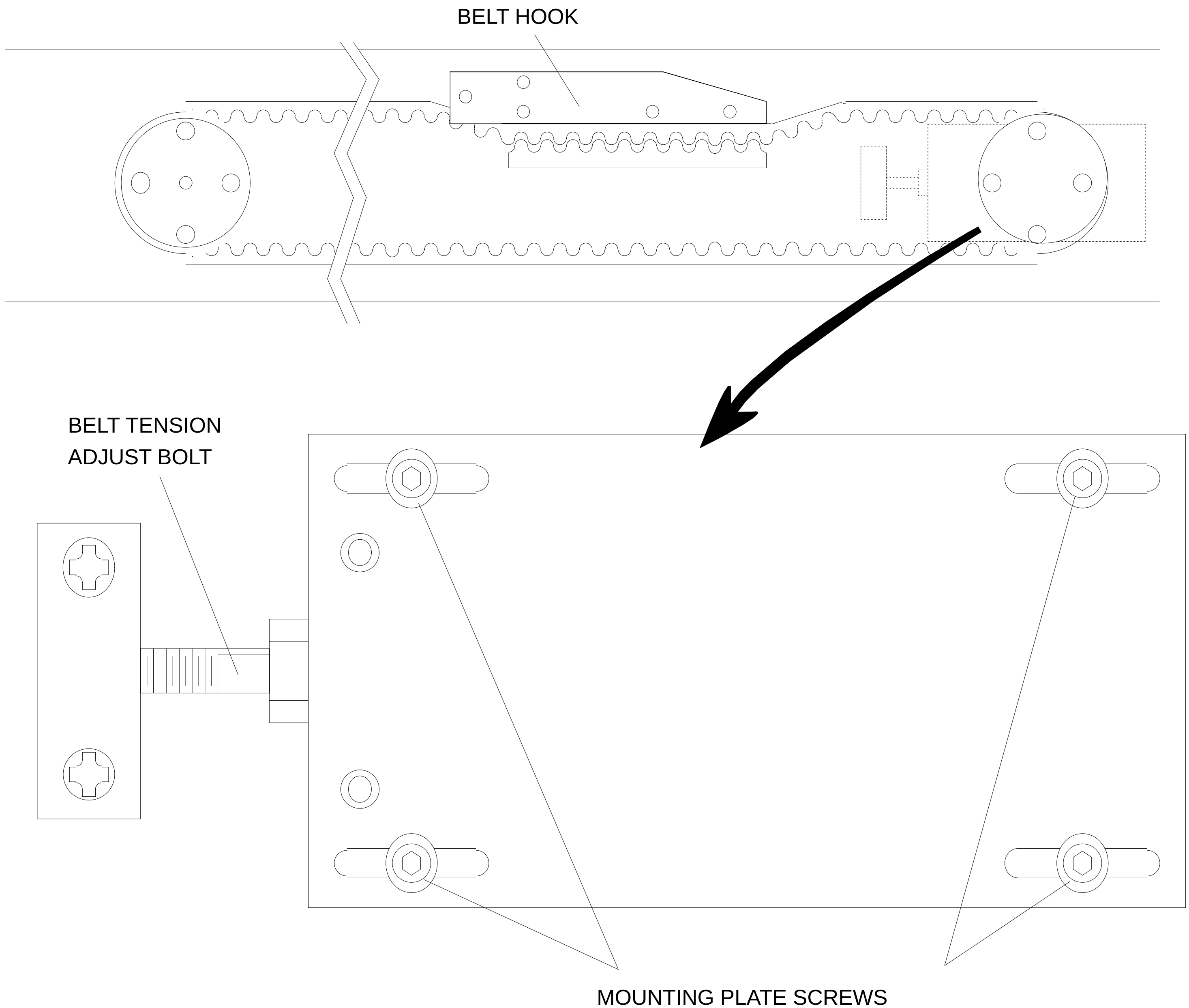


FIGURE 87 HORIZONTAL DRIVE BELT TENSION ADJUSTMENT

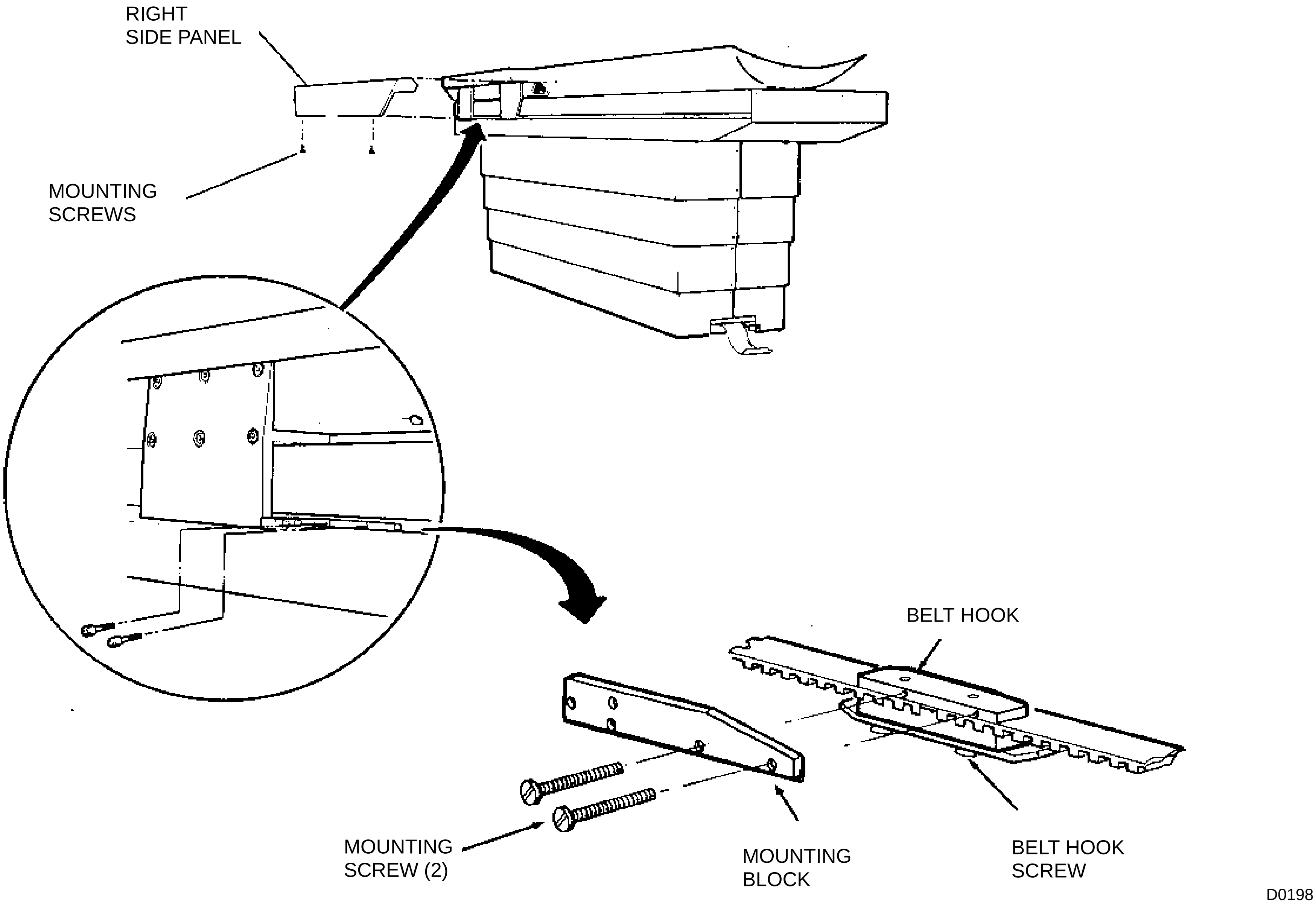


FIGURE 88 HORIZONTAL DRIVE BELT BLOCK

7. Release the tension on the belt by turning the belt tension adjusting bolt **clockwise** (facing couch horizontal drive motor). Turn the adjusting screw until the belt clamp just touches the bottom belt surface. Refer to Figure 89.

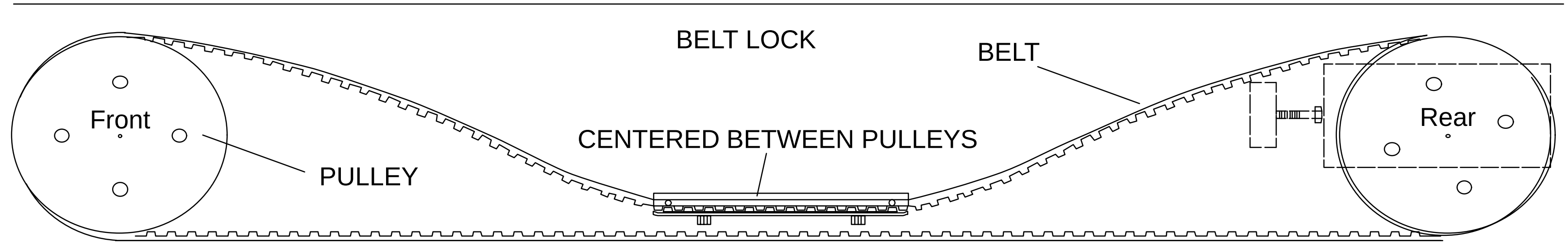


FIGURE 89 HORIZONTAL BELT BEFORE ADJUSTMENT

8. Mark the belt tension adjusting screw so that the number of turns can be counted and turn it 2-1/2 to 2-3/4 full turns **counter -clockwise** (facing couch horizontal drive motor) to tighten the belt.

> ⚠ **CAUTION**
>
> DO NOT TIGHTEN THE BELT MORE THAN THE SPECIFIED AMOUNT. IF THE BELT IT TIGHTENED TOO MUCH, THE BELT AND THE SPROCKETS MAY BE DAMAGED.

9. Reconnect the horizontal drive belt block to the table top. See Figure 88.

10. If the pulley does not have a rim to retain the drive belt, manually, verify that the belt rides true on the front pulley. If not, adjust the set screws on the pulley block (See Figure 90). (Turn the set screws clockwise to move the rear of the block away from the Patient Support). Moving the rear of the block away from the subframe will cause the drive belt to move away from the subframe.

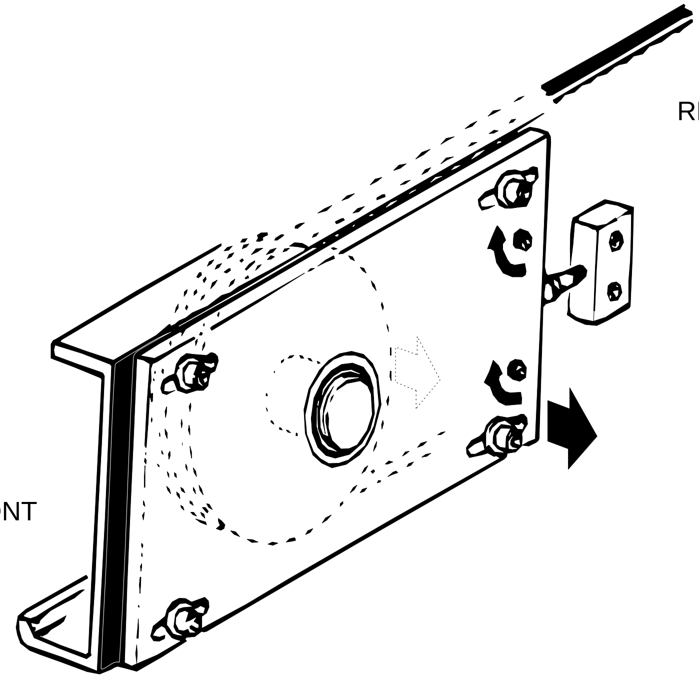


FIGURE 90 BELT RUN OUT ADJUSTMENT

11. Tighten the four mounting screws that secure the pulley assembly to the right side rail.
12. Replace the Patient Support rear cover, carbon top side cover and right rail cover. Refer to the Cover Replacement procedure.

X–RAY TUBE AXIAL MECHANICAL ALIGNMENT

Introduction: The purpose of this procedure is to mechanically align the X–ray tube so it is centered on the detectors. This procedure must be done whenever an X–ray tube is replaced/installed. The FSE will be working with the Gantry covers open while performing this procedure and will be initiating X–rays.

> **NOTE**
>
> A new Xray Tube Mechanical Alignment procedure is available via MANEXP.

LASER ALIGNMENT CALIBRATION

Introduction: The purpose of this procedure is to align the positioning lasers so that they properly indicate the center of the X–ray beam.

Tools Required: FSE Tool Kit, 3 index cards (approx. 4 x 6 in.), transparent or masking tape, mechanical pencil, black electrical tape

Estimated Time: 60 min.

> **NOTE**
>
> The 3 Gantry Lasers are essentially the same for Ultra Z and AcQSim–CT with one small difference. The 9:00 O'clock laser has been split into (2) separate lasers on AcQ-Sim–CT. The 'X' direction is at 10:00 O'clock and the 'Z' direction is placed at 9:00 O'clock. See Figure 92.

LASER ALIGNMENT PROCEDURE:

1. Remove Gantry power. Refer to the Power Up/Down procedure.
2. Open the top front Gantry gull wing. Remove the lower front Gantry cover and the front Gantry cone. Refer to the Gantry Cover procedures in the Repair Manual.
3. Enable the scan drives by pulling the gullwing interlock switch out and turn on Gantry power. Refer to the Power Up/Down procedure.

> ⚠ **WARNING**
>
> **THE SCAN FRAME WILL ROTATE WHEN GANTRY IS POWERED UP. USE CAUTION WHEN POWERING UP THE GANTRY. FAILURE TO COMPLY MAY RESULT IN INJURY TO PERSONNEL.**

4. In the Gscope program, position the collimator to the 1mm position by typing **cl 10** and press <CR>. Refer to the Gscope Service Manual for more details.
5. Depress the laser button to turn the lasers ON. (You may have to lower the gullwing momentarily to do this.)
6. Cover the 3 and 9 o'clock lasers (two lasers at 9 and 10 o'clock for AcQSim CT) openings over the laser opening with black electrical tape. Refer to Figure 91.

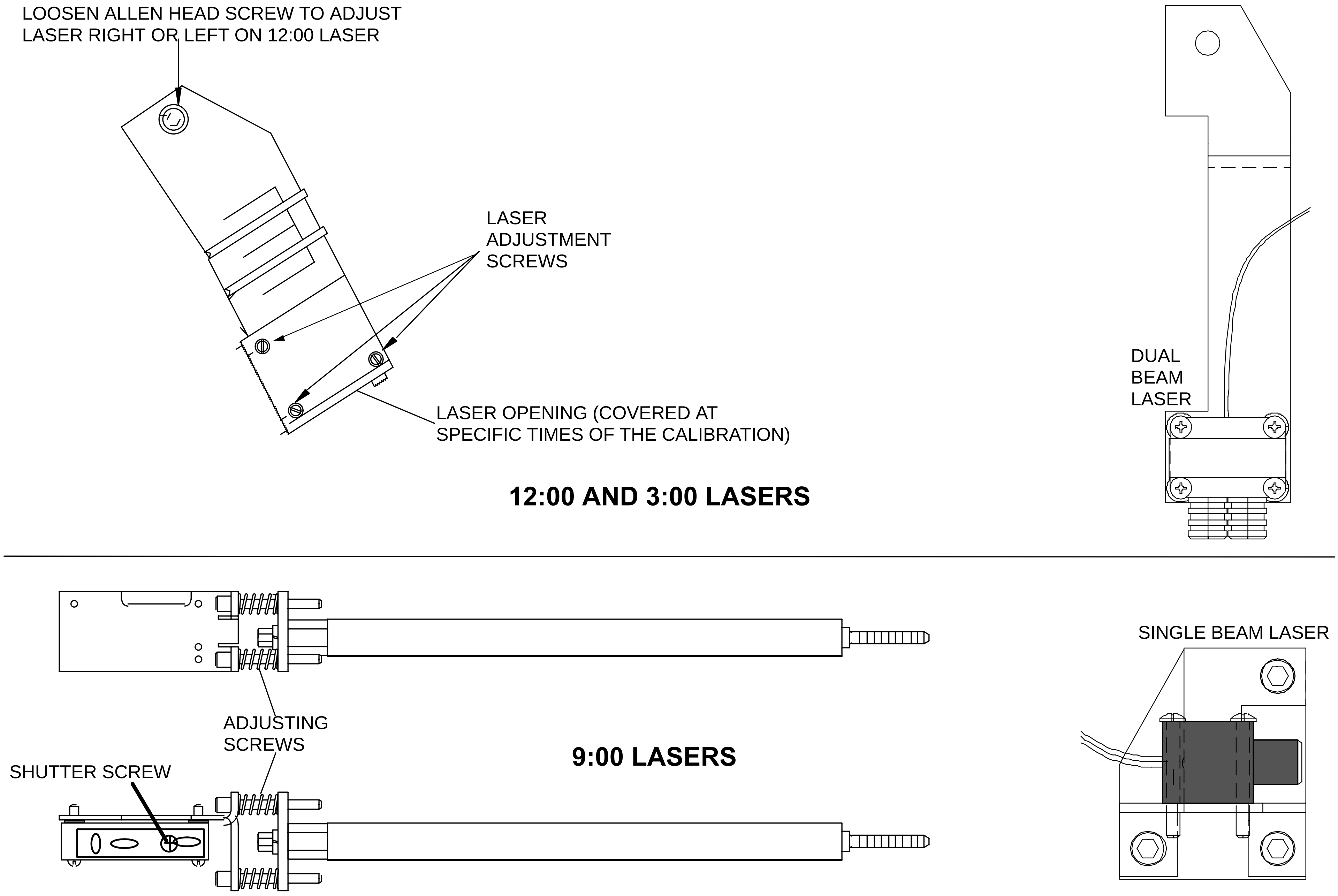


FIGURE 91 LASER ALIGNMENT SCREWS

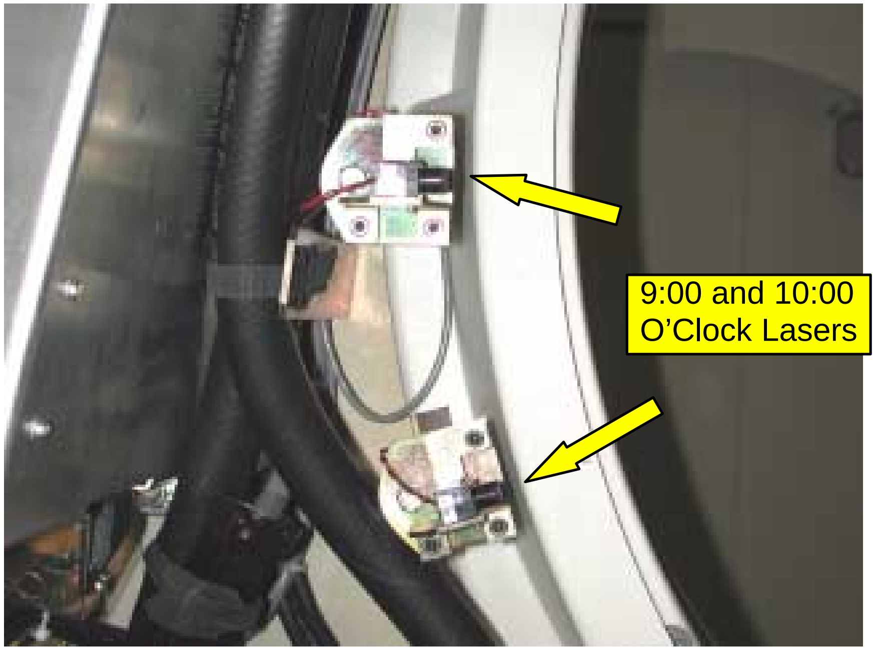


FIGURE 92 9:00 and 10:00 O'Clock Split Laser on AcQSim–CT Scanframe

7. Adjust the 12 o'clock laser so the beam is aligned between detector channel **1799** and **1800** at the bottom of the gantry. To adjust the laser, you must loosen the allen head screw at the mounting bracket stud, then move the laser right or left as needed. Tighten after adjusting. Refer to Figure 91.

> ⚠ **WARNING** ⚠
>
> **USE EXTREME CAUTION WHEN OPERATING OR WORKING ON EQUIPMENT WITH COVERS REMOVED. DANGEROUS VOLTAGES ARE EXPOSED WHEN THE GANTRY COVERS ARE REMOVED. FAILURE TO COMPLY CAN CAUSE SERIOUS INJURY OR DEATH TO PERSONNEL.**

8. Turn the lasers OFF. Disable the Gantry scan drives by placing the upper cover interlock switch in the middle position.
9. Manually rotate the scan frame ccw until the x–ray tube and collimator are at the 8:00 position.
10. Lay an index card across collimator opening and tape it to the slip ring.

11. Put two marks on each edge of the card that line up with the collimator opening. With a ruler, draw a line connecting the two marks. Using a mechanical lead pencil will produce a line approx. .5mm thick.

12. Move the collimator to approximately the 6:00 position. Repeat steps 10. and 11. Then rotate the scan frame around ccw so the collimator is at the 4:00 position and repeat the card marking steps. You should now have a marked card taped at the 8:00, 6:00 and 4:00 position. Cards in Figure 93 are colored gray.

> ⚠ **WARNING**
>
> **THE SCAN FRAME WILL ROTATE TO THE "HOME" POSITION WHEN THE LASER BUTTON IS PRESSED. USE CAUTION WHEN POWERING UP THE GANTRY. FAILURE TO COMPLY MAY RESULT IN INJURY TO PERSONNEL.**

13. Enable the scan drives by pulling the interlock switch out. Press the laser button on the front of the gantry.

14. Disable the main drive. Adjust the 12:00 laser so the line falls on the line of all three cards. Figure 91 shows the the spring–mounted laser adjustment screws. Note that adjustment screws interact with each other.

15. Rotate the scan frame ccw until the 12:00 laser is directly below the 6:00 card. The laser line should fall directly on the card line. Adjust the 12:00 laser as necessary. While rotating, you can quickly see if the laser line is falling on the 8:00 and 4:00 cards correctly. Repeat this step until the laser line falls directly on the card line at both top and bottom positions.

16. Now cover the laser opening of the 12:00 laser. Uncover the 3:00, 9:00, and 10:00 laser openings. Rotate the scan frame ccw until the 12:00 laser is near the top (this only makes the next adjustment easier and does not have to be exact). Lasers should still be on.

17. Check the line of the 3:00 laser and ensure that the horizontal beam falls directly on the horizontal laser port of the 9:00 laser. Check the 9:00 laser in the same manner on the 3:00 laser and adjust either laser as necessary.

18. Cover both 9:00 and 10:00 laser openings. Rotate the scan frame so the 3:00 laser is pointing down at the 6:00 card. Adjust the 3:00 laser as necessary. While rotating, you can quickly see if the laser line is falling on the 8:00 and 4:00 cards correctly. Repeat this step until the laser line falls directly on the card line at both top and bottom positions.

19. After the 3:00 laser is aligned, cover its laser opening and uncover the opening of the 10:00 laser. Rotate the scan frame so the 10:00 laser is pointing down at the 6:00 card. Adjust the 10:00 laser as necessary. While rotating, you can quickly see if the laser line is falling on the 8:00 and 4:00 cards correctly.

⚠ WARNING

THE SCAN FRAME WILL ROTATE TO THE “HOME” POSITION WHEN THE LASER BUTTON IS PRESSED. USE CAUTION WHEN POWERING UP THE GANTRY. FAILURE TO COMPLY MAY RESULT IN INJURY TO PERSONNEL.

20. Enable the main drive. Press the laser button so the gantry rotates to “home” position. Disable the main drive and uncover lasers.
21. Align a phantom on a centered couch. Wrap a piece of paper around the 6” section of the phantom and secure in place. Push the couch in until the laser line falls on the paper. If properly aligned, you should see a continuous, solid line all the way around the paper. If a laser is out of alignment, you will see a blurring or two solid lines. If this is the case, locate the misaligned laser and re–adjust.

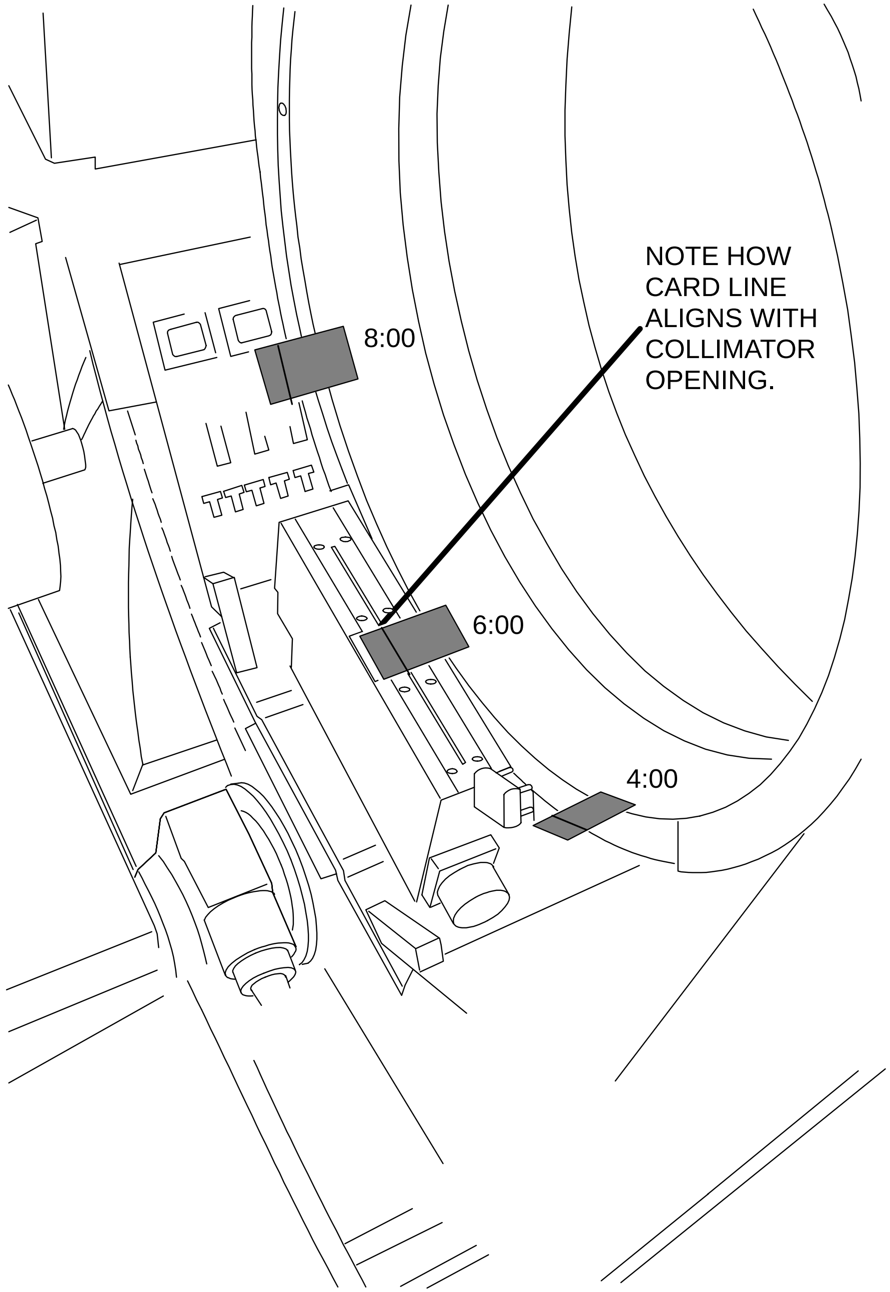


FIGURE 93 CARD ALIGNMENTS – 8:00, 6:00 AND 4:00 POSITIONS

LASER ALIGNMENT VERIFICATION

22. Keep the paper on the phantom. Once you are confident that lasers are aligned, remove the alignment cards and CAREFULLY replace the cone (it is easy to bump the 12:00 and 3:00 lasers when doing this).
23. Verify the laser alignment on the paper once more. Run the following procedure for a final verification:
24. Unfold and cut an ordinary paper clip to a one inch length . Tape the clip in the groove between the MTF and Low Contrast sections (approx. 70 mm from the tip of the peg.)
25. Turn on the lasers and make sure the beam is directly on the paper clip.
26. From the Clinical Scanning GUI, create a new test patient entry.
27. Select the Protocol Selection option, click on the phantom icon and choose the Half Water protocol. Press Select.
28. Select the Planning option at the bottom of the screen.
29. The tube must be warmed up. In Pilot Planning, select the OM Landmark and Confirm with the mouse. Another Pilot screen will appear. Select 60mA. Click on Confirm.
30. Click on the Advance to Scan Button. Click on Start scan. The pilot will appear. Click on the Extent Tab under Scan Parameters Selection, and select a slice thickness of 1mm (2mm on AcQSim CT). Make sure the Scan Extents (mm) are set to +1.0 Start and +1.0 End so the couch will not move when it scans.
31. Click on the Scanning/Recon button. Click on Advance to Scan, then Start Scan.
32. Observe the resulting axial image. The lasers are positioned properly if the clip is visible in the image. If this is not the case, the calibration procedure should be done again.

BRUSH BLOCK ALIGNMENT

Introduction: The purpose of this procedure is to mechanically align the slip ring brush blocks. The brush blocks must be aligned axially, ensuring that the brushes are centered in each ring groove; and radially, ensuring that the proper pressure is applied to each brush.

Tools Required: FSE tool kit
Alignment Pins P/N 95341

Estimated Time: 20 min.

ALIGNMENT PROCEDURE:

> ⚠ **WARNING** ⚠
>
> **USE EXTREME CAUTION WHEN OPERATING OR WORKING ON EQUIPMENT WITH COVERS REMOVED. HAZARDOUS VOLTAGES ARE EXPOSED WHEN THE GANTRY COVERS ARE REMOVED. ACCIDENTAL CONTACT WITH THESE VOLTAGES MAY CAUSE SERIOUS INJURY OR DEATH TO SERVICE PERSONNEL.**

1. Remove Gantry power. Power **MUST BE REMOVED AT THE WALL BOX**. Refer to the Power Up/Down procedure.

> ⚠ **WARNING**
>
> **THE FRONT UPPER GANTRY COVER HAS SHARP EDGES. USE EXTREME CAUTION WHEN SERVICING EQUIPMENT WHILE UPPER GULL WINGS ARE OPEN. FAILURE TO COMPLY MAY RESULT IN SERIOUS INJURY TO SERVICE PERSONNEL.**

2. Open and secure the upper front Gantry cover. Remove the lower front cover and remove the front cone. .Refer to the Gantry Cover procedures of the Repair Manual.

3. Move the rotating scan frame ccw by hand to position the X–ray tube at the 12:00 position. Lock the scan frame with the locking pin. Refer to Figure 94.

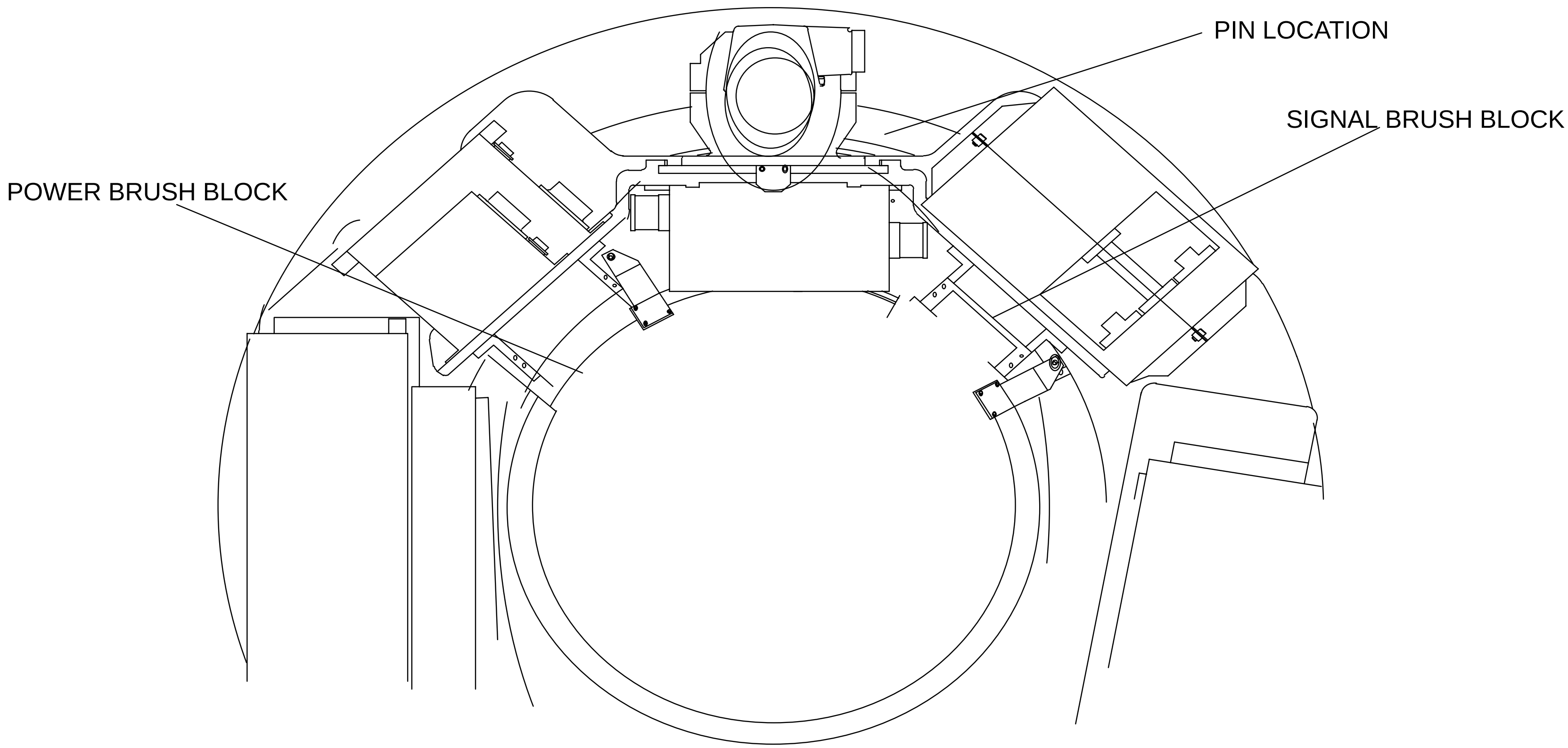


FIGURE 94 SCAN FRAME PINNED WITH TUBE AT 12:00

SIGNAL BRUSH BLOCK

Refer to Figure 95 and Figure 96.

1. Loosen the two **radial** mounting screws. Loosen the four screws that secure the mounting bracket to the underside of the shelf. Refer to Figure 95.
2. Move the brush block assembly so that the brushes are centered in the groove, as shown in Figure 96. Slowly tighten the four screws that secure the mounting bracket to the underside of the shelf. Monitor the position of the brushes as the screws are tightened to ensure they remain centered in the grooves.

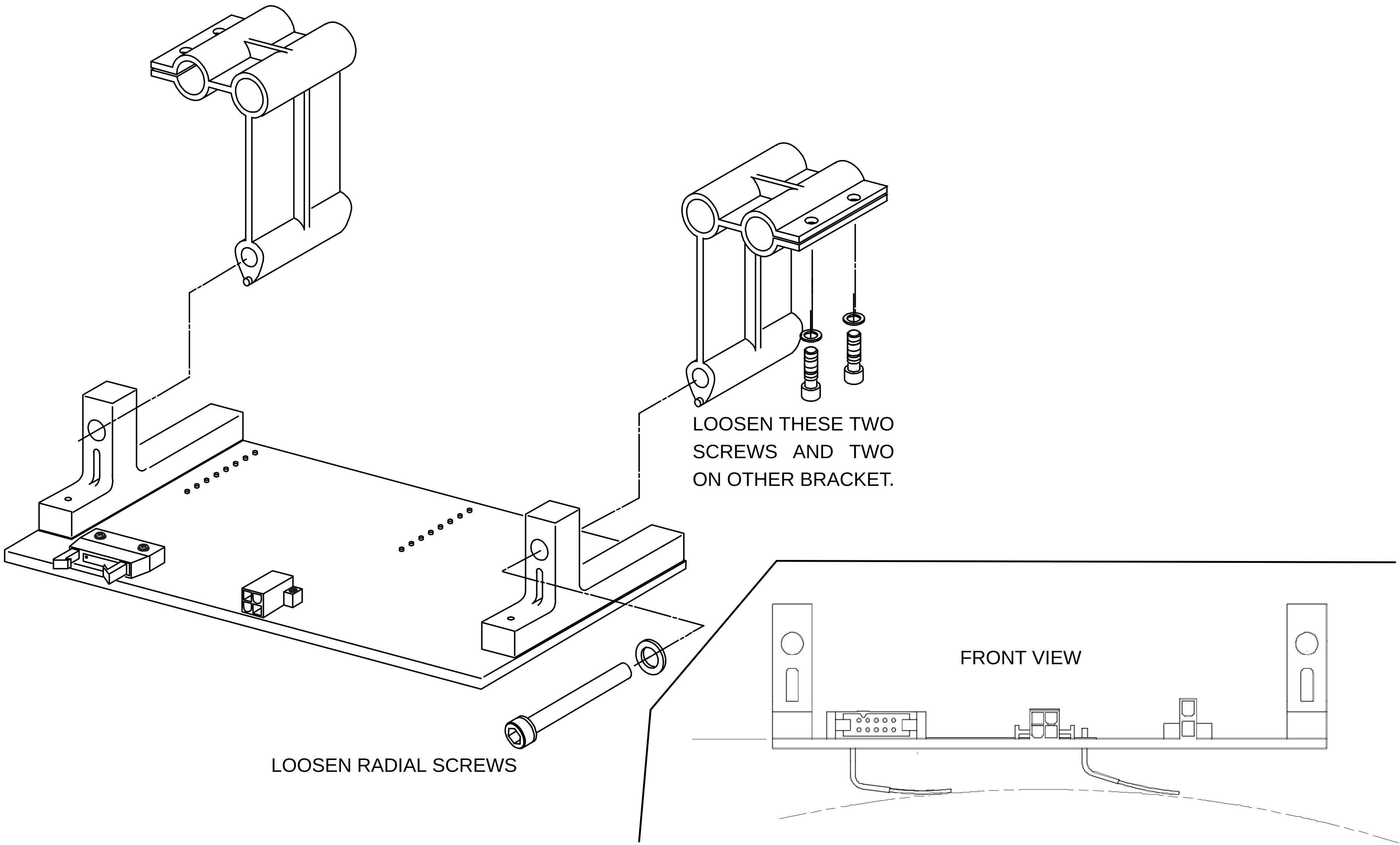


FIGURE 95 SIGNAL BRUSH BLOCK

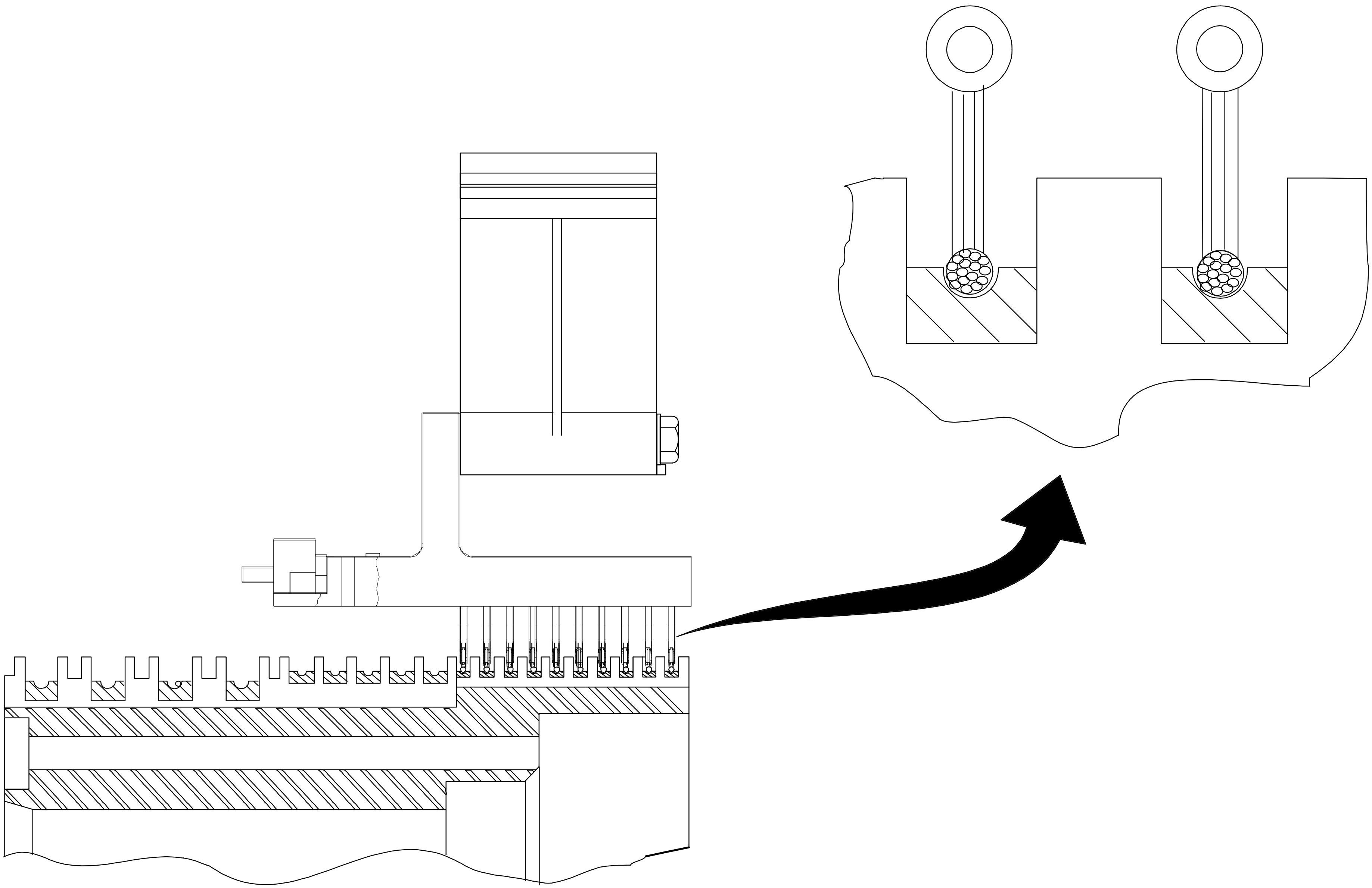


FIGURE 96 ALIGNMENT OF ASSEMBLY AND SIGNAL BRUSHES

3. Slowly rotate the Gantry scan frame counterclockwise by hand. Listen to and observe the brush in each slip ring groove to determine that alignment is maintained throughout 360 degrees of rotation. Make sure the assembly tracks smoothly and encounters no obstructions or collisions during the rotation.
4. Replace the laser or the relay card cover, as appropriate. If the laser was removed, refer to the Laser Calibration section of this manual after its installation.
5. Rotate the scan frame by hand 360 degrees counterclockwise and make sure all cables are safely secured.
6. Apply system power. Refer to the Power Up/Down procedure.
7. Perform a warm–up of the x–ray system (if necessary) and run a typical water phantom scan to verify proper system operation.
8. Refer to the RING procedure in the Diagnostic GUI manual and execute RING for 2 and 4 second scan speeds. The Ring program tests the integrity of the slip rings themselves. However, if the signal brush block is not properly aligned, communication errors will occur during normal operation. Common errors that occur if the brush block is misaligned are: ACK/NACK BREAK (ANB) error, Drives Disable errors and X–ray control resets (contactor drops out or shutdown faults).
9. If any of the listed errors occur, either the alignment of the brush block assembly is bad or a brush tip or slip ring is bad. If no errors, exit the diagnostic.
10. Remove system power. Refer to the Power Up/Down procedure.
11. Replace the front cone and close the Gantry covers. Refer to the Gantry Cover procedures of the Repair Manual.

POWER BRUSH BLOCK

1. Power should be removed from the system **AT THE WALL BOX**. In order to remove the **power brush block,** you must first remove the laser assembly. Refer to the Laser Removal/Replacement procedure in the Repair manual for complete details.
2. Insert the two alignment pins into the holes provided on the brush block assembly. Refer to Figure 97. These two pins should fall into the inner air gap on the slip ring assembly.
3. Loosen the two **radial** mounting screws (Figure 97).
4. Move the brush block assembly so that the brushes are centered in the groove, as shown in Figure 98. Slowly tighten the four screws that secure the mounting bracket to the underside of the shelf. Monitor the position of the brushes as the screws are tightened to ensure they remain centered in the grooves.
5. Remove the two **radial** mounting screws (Figure 95) one at a time, apply a drop of Loc–Tite to the threads of each screw, reinstall, but do not tighten them.

6. Move the brush block assembly so that the alignment pins are centered in the groove, as shown in Figure 96. Slowly tighten the four screws that secure the mounting bracket to the underside of the shelf. Monitor the position of the alignment pins as the screws are tightened to ensure the brush tips remain centered in the slip ring grooves.

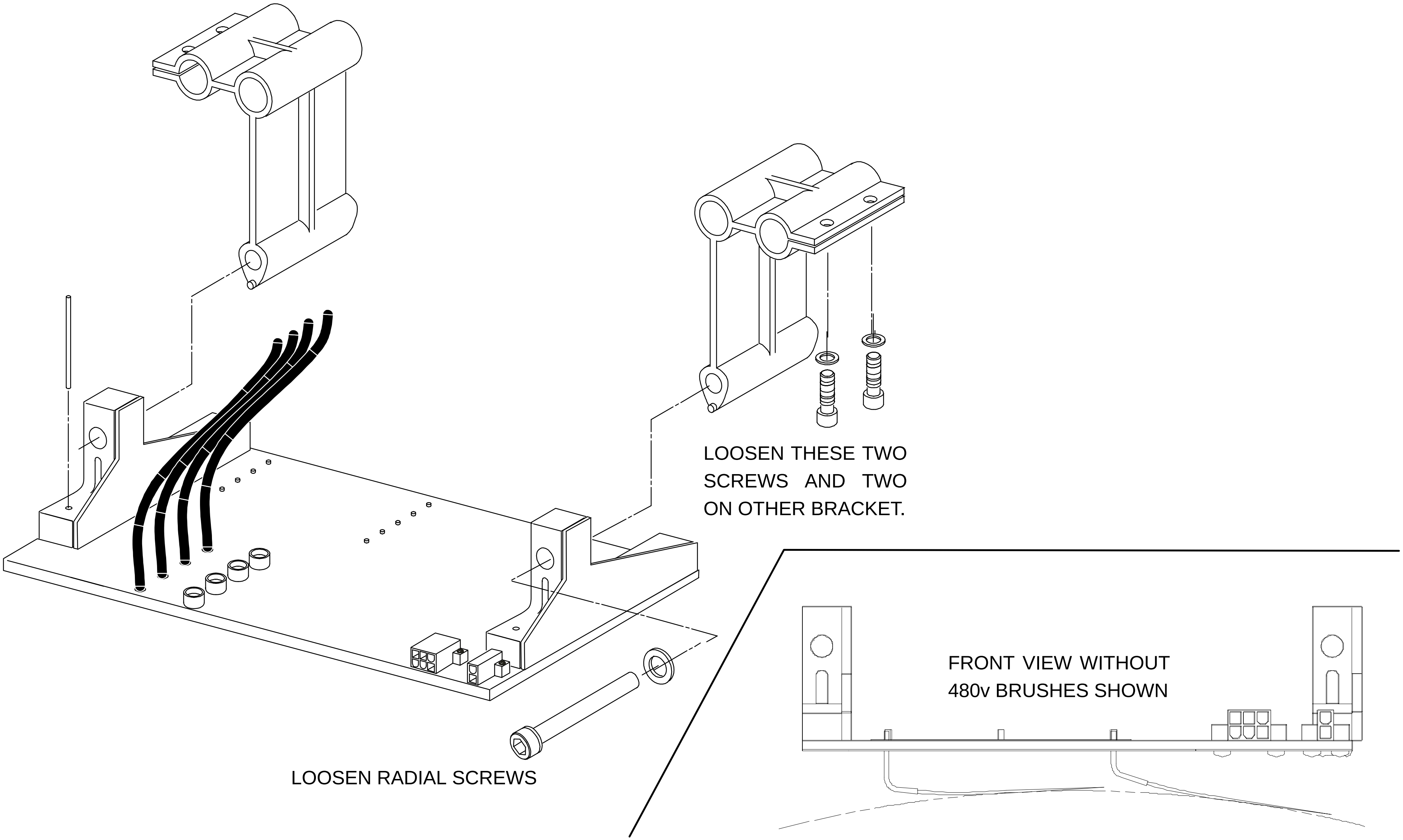


FIGURE 97 POWER BRUSH BLOCK

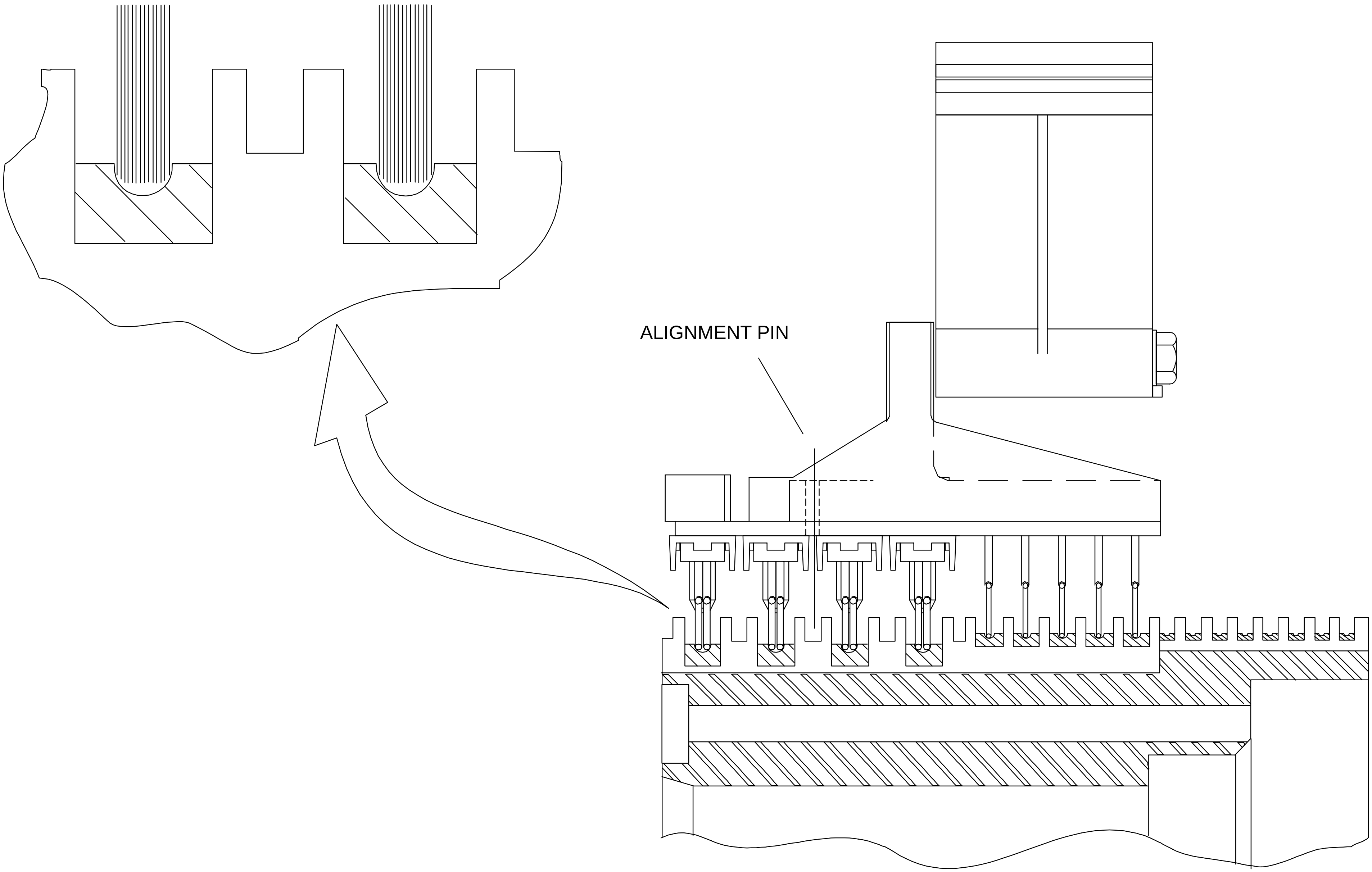


FIGURE 98 ALIGNMENT OF ASSEMBLY AND BRUSH FIBER TIPS (POWER)

7. Slowly rotate the Gantry scan frame counterclockwise by hand. Listen to and observe the brush in each slip ring groove to determine that alignment is maintained throughout 360 degrees of rotation. Make sure the assembly tracks smoothly and encounters no obstructions or collisions during the rotation.

8. Replace the laser or the relay card cover, as appropriate. If the laser was removed, refer to the Laser Calibration section of this manual after its installation.

9. Rotate the scan frame by hand 360 degrees counterclockwise and make sure all cables are safely secured.

10. Apply system power. Refer to the Power Up/Down procedure.

11. Perform a warm–up of the x–ray system (if necessary) and run a typical water phantom scan to verify proper system operation.

12. Refer to the Gscope manual and execute the **@ring** macro for 2 and 4 second scan speeds. The **@ring** macro was created to test the integrity of the slip rings themselves. However, if the signal brush block is not properly aligned, communication errors will occur during normal operation. Common errors that occur if the brush block is misaligned are: ACK/NACK BREAK (ANB) error, Drives Disable errors and X–ray control resets (contactor drops out or shutdown faults).

13. If any of the listed errors occur, either the alignment of the brush block assembly is bad or a brush tip or slip ring is bad. If no errors, exit the diagnostic.

14. Remove system power. Refer to the Power Up/Down procedure.

15. Replace the front cone and close the Gantry covers. Refer to the Gantry Cover procedures of the Repair Manual.

TILT CALIBRATION

Introduction:	The following procedure is used to calibrate the Gantry tilt position and read–out.
Tools Required:	Combination protractor head
	FSE tool kit
Estimated Time:	30 min.

1. Remove Gantry Power. Refer to the Power Up/Down procedure.

> ⚠ **WARNING**
>
> **THE FRONT UPPER GANTRY COVER HAS SHARP EDGES. USE EXTREME CAUTION WHEN SERVICING EQUIPMENT WHILE THE UPPER GULL WING IS OPEN. FAILURE TO COMPLY MAY RESULT IN SERIOUS INJURY TO SERVICE PERSONNEL.**

2. Refer to the Gantry Cover procedures of the Repair Manual to:
   - c. Open the upper front Gantry cover and secure the upper cover in the full–open position.
   - d. Remove the right Gantry column cover.
   - e. Remove the Gantry cardfile access cover.

> ⚠ **WARNING** ⚠
>
> **480 VOLTS AC, 120 VOLTS AC, 5 VOLTS DC, AND 15 VOLTS DC ARE EXPOSED WHEN THE GANTRY COVERS ARE OPEN. ACCIDENTAL CONTACT WITH THESE VOLTAGES MAY CAUSE SERIOUS INJURY OR DEATH TO THE SERVICE PERSON.**

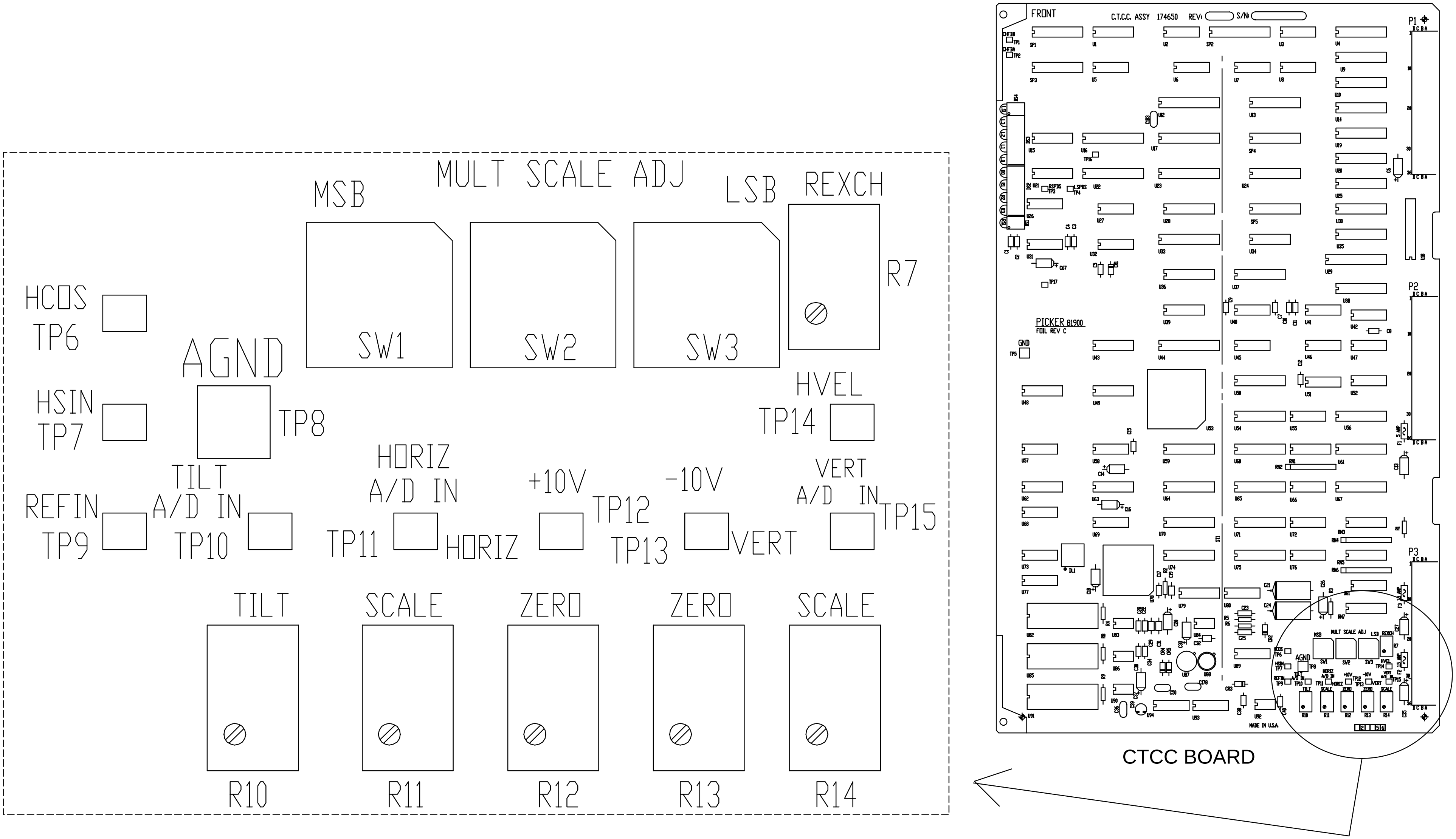


FIGURE 99 CTCC ADJUSTMENT LOCATIONS

3. Apply Gantry power. Refer to the Power Up/Down procedure.

4. Set the protractor to indicate zero (0) degrees and tape it against the side of the collimator (non–magnetic metal). Tilt the Gantry until the protractor bubble level indicates that the Gantry is at zero degrees. Refer to Figure 100.

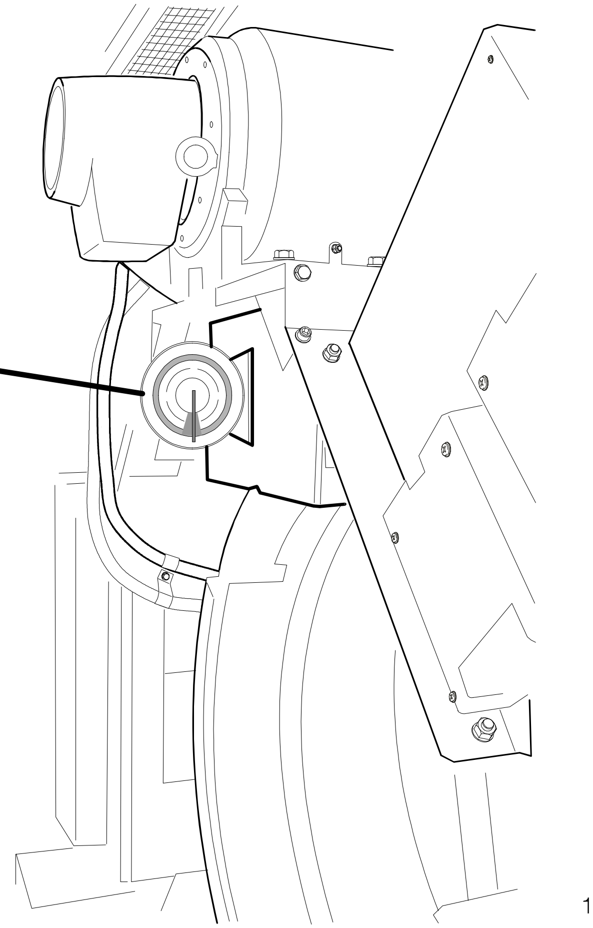


FIGURE 100 PROTRACTOR ON COLLIMATOR AT ZERO DEGREES

5. Remove Gantry power. Refer to the Power Up/Down procedure.
6. On the CTCC card, connect pos (+) lead of a digital Multi–meter (DMM) to TP10 and the black (–) lead to TP8. Refer to Figure 99.

⚠ **WARNING** ⚠
120 VAC IS PRESENT ON THE TILT LIMIT SWITCHES. AVOID CONTACT WITH THE TILT LIMIT SWITCH WIRING. FAILURE TO COMPLY MAY CAUSE SERIOUS INJURY OR DEATH TO THE SERVICE PERSON.

⚠ WARNING

MOVING MACHINERY. KEEP BODY PARTS AND ALL OBJECTS AWAY FROM MOVING PARTS OF THE GANTRY. FAILURE TO COMPLY MAY CAUSE SERIOUS DAMAGE TO THE EQUIPMENT AND SERIOUS INJURY TO PERSONNEL.

7. Apply Gantry power. Refer to the Power Up/Down procedure. Rotate the pot output shaft until the voltmeter indicates 0.0 VDC. The pot output can be adjusted by carefully turning the pot, allowing the belt to slip around the smooth Gantry pulley. Refer to Figure 101.

8. Disconnect the DMM from the CTCC. Use the gantry control panel switches to tilt the gantry in the positive direction. While the Gantry is tilting, activate each positive limit switch to verify it interrupts the tilting motion. Refer to Figure 101 for limit switch location.

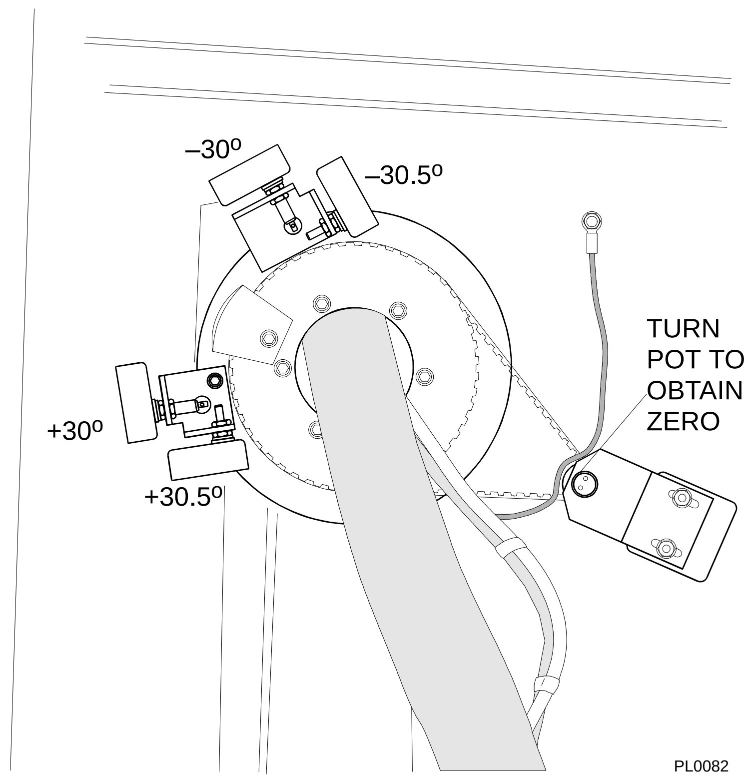


FIGURE 101 TILT POT AND LIMIT SWITCHES – RIGHT GANTRY COLUMN

9. Use the gantry control panel switches to tilt the gantry in the negative direction. While the Gantry is tilting, carefully activate each negative limit switch to verify that it interrupts the tilting motion.
10. Tilt the gantry to a –30 degree position, using the protractor.
11. Adjust the tilt scale pot R11 on the CTCC card until the front display indicates –29.5 degrees. Refer to Figure 99. With the protractor, verify that the Gantry will tilt to –30.5 degrees and that the software limit stops the the tilt at –30.5 degrees.
12. Repeat steps 14. through 11., adjusting the positive limit switch (+30).
13. Tilt the Gantry to zero degrees.
14. Using a protractor against the scan frame as the means of measurement, tilt the gantry to –10 degrees. Adjust the CTCC potentiometer R10 until the gantry touch panel displays –10 degrees. Refer to Figure 99.
15. Tilt the gantry to a –20 degree position, using the display as the means of measurement. Verify with the protractor that the gantry position is accurate.
16. Remove Gantry power. Refer to the Power Up/Down procedure.
17. Refer to the Gantry Cover procedures of the Repair Manual to:
    a. Close the upper front Gantry cover and secure the upper cover in the full–open position.
    b. Close the right Gantry column cover.
    c. Replace the small carfdfile access cover back into position.

+5, 15 +12, –20 VOLT DC POWER SUPPLY VERIFICATION

Introduction: The purpose of this procedure is to verify the accuracy of the +5, +/–15, +12, and –20 VDC power supplies. There is no calibration adjustment on these supplies.

Tools required: FSE tool kit

Estimated time: 10 Minutes

+5, 15 +12, –20 POWER SUPPLY VERIFICATION:

1. Remove system power. Refer to the Power Up/Down procedure. Remove the left smaller rear gantry cover. Refer to the Gantry Cover procedures of the Repair Manual.

> **WARNING**
>
> **480 VOLTS AC, 120 VOLTS AC, 5 VOLTS DC, ”15 VOLTS AND ”20 VOLTS DC ARE EXPOSED WHEN THE GANTRY COVERS ARE OPEN. ACCIDENTAL CONTACT WITH THESE VOLTAGES MAY CAUSE SERIOUS INJURY OR DEATH TO THE SERVICE PERSON.**

2. Disable the Console by positioning the key switch on the AC control chassis to the SERVICE position. Refer to Figure 102.
3. Set the DMM to a 20 volt or greater DC range. Connect the leads while observing proper polarity to the proper test points (+5, +15, –15, +20 and –20vdc with respect to ground). Refer to Figure 103.

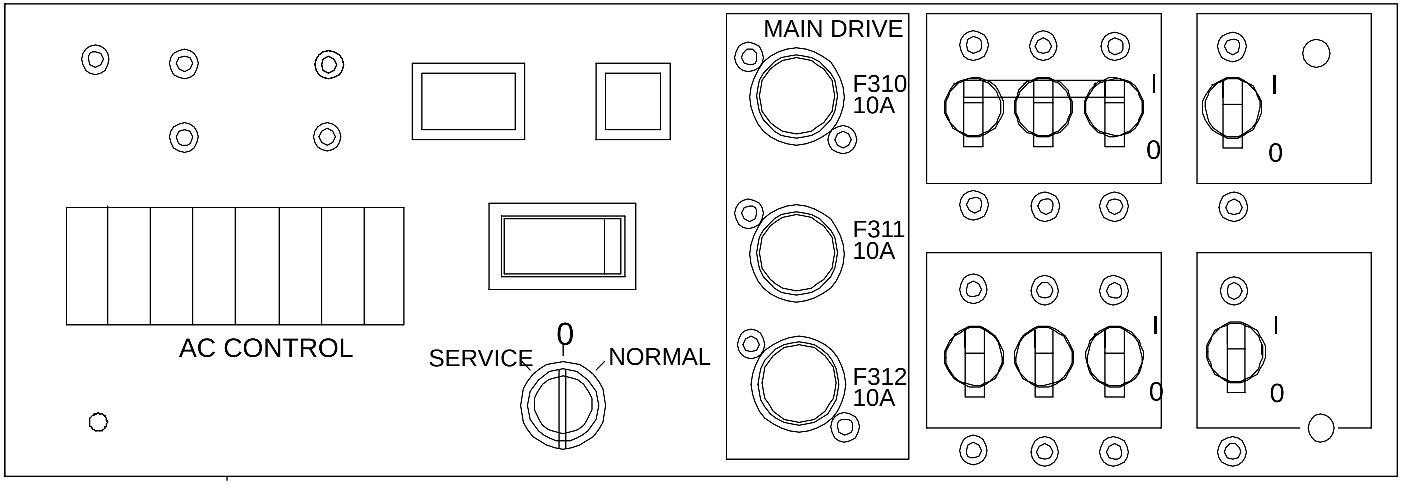


FIGURE 102 AC CONTROL CHASSIS KEY SWITCH

Under Normal Conditions:

Power Supply	**Voltage Output***	**Load Current**
+5vdc	4.85 – 5.15v	50 – 60A
+15vdc	14.55 – 15.45v	8 – 10A
–15vdc	–14.55 to –15.45v	6 – 6.7A
+12vdc	12.36 – 11.64v	7.5 – 8.3A
–20vdc	–19.40 to –20.60v	9 – 10A
*absolute values		

If the power supply voltages are not within these ranges, the supply must be <u>replaced</u>.

4. Return the service key switch on the AC control chassis to the NORMAL position. Replace the Gantry covers. Refer to the Gantry Cover procedures of the Repair Manual.

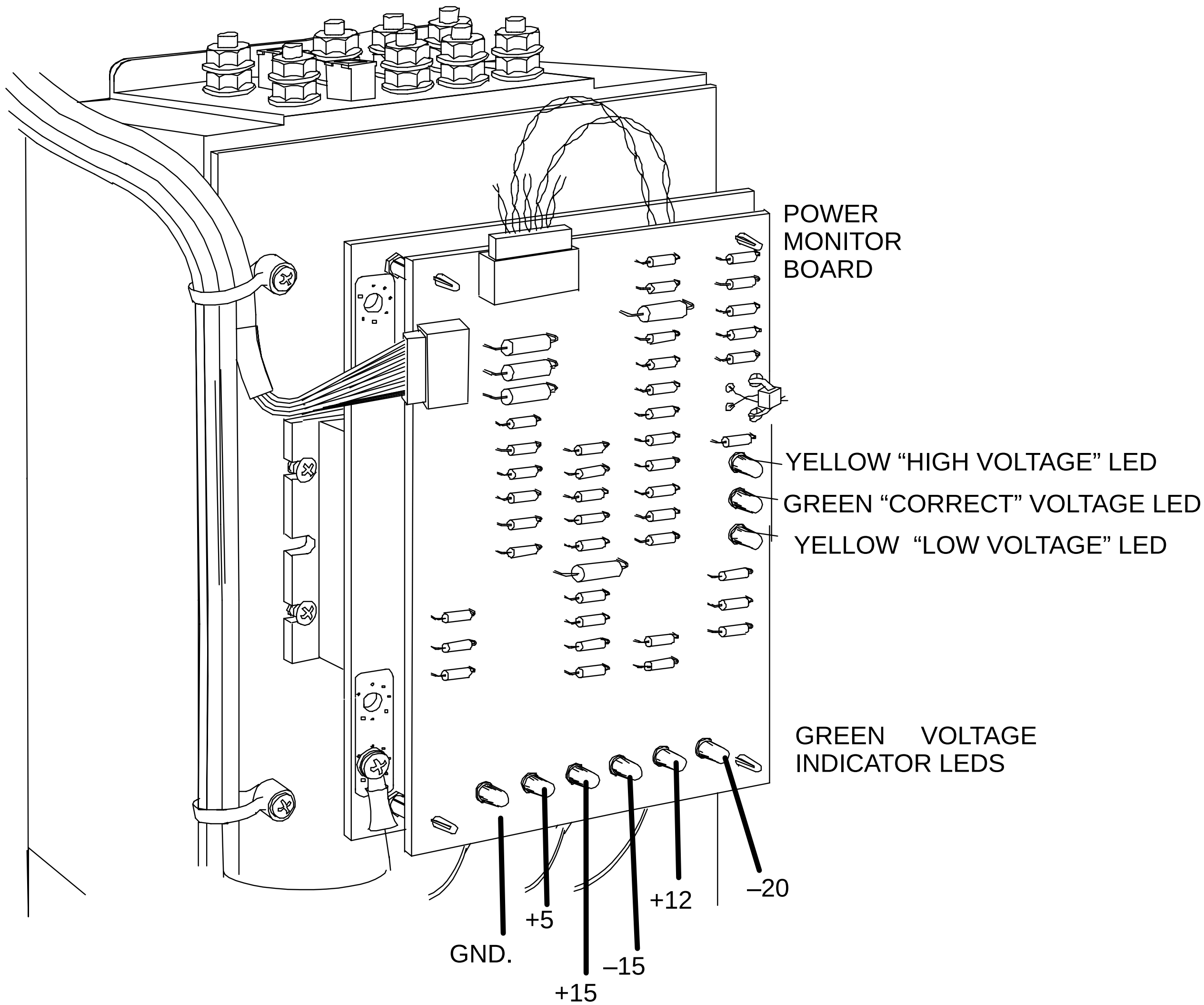


FIGURE 103 POWER SUPPLY TEST POINTS

MAIN DRIVES CALIBRATION

> **NOTE**
>
> The Main Drives Calibration can be found in the Gputty+ section of the Diagnostic GUI manual. The procedure is called MC. Click on the MC link to access the procedure now.

+3.3, +5, +/–12 VOLT DC ACQIMAGE CABINET POWER SUPPLY VERIFICATION/CAL

Introduction: The purpose of this procedure is to verify the accuracy of the +3.3 VDC, +5 VDC and +/–12VDC AcQImage Cabinet power supplies.

Tools required: FSE tool kit

Estimated time: 15 Minutes

1. Remove system power. Refer to the Power Up/Down procedure.
2. Remove the front cosmetic cover and inside front cover. Remove the top cover. Refer to the Cover Removal/Replacement procedures in the Repair manual. Apply power. Refer to the Power Up/Down procedure.

> ⚠ **WARNING** ⚠
>
> **120 VOLTS AC, 5 VOLTS DC and +/–12 VOLTS DC ARE EXPOSED WHEN THE DISPLAY TOWER COVERS ARE OPEN. ACCIDENTAL CONTACT WITH THESE VOLTAGES MAY CAUSE SERIOUS INJURY OR DEATH.**

3. Set the DMM to a 20 volt or greater DC range. Connect the leads while observing proper polarity to the proper test points (+3.3, +5, and +/–12vdc with respect to ground). Refer to Figure 104.

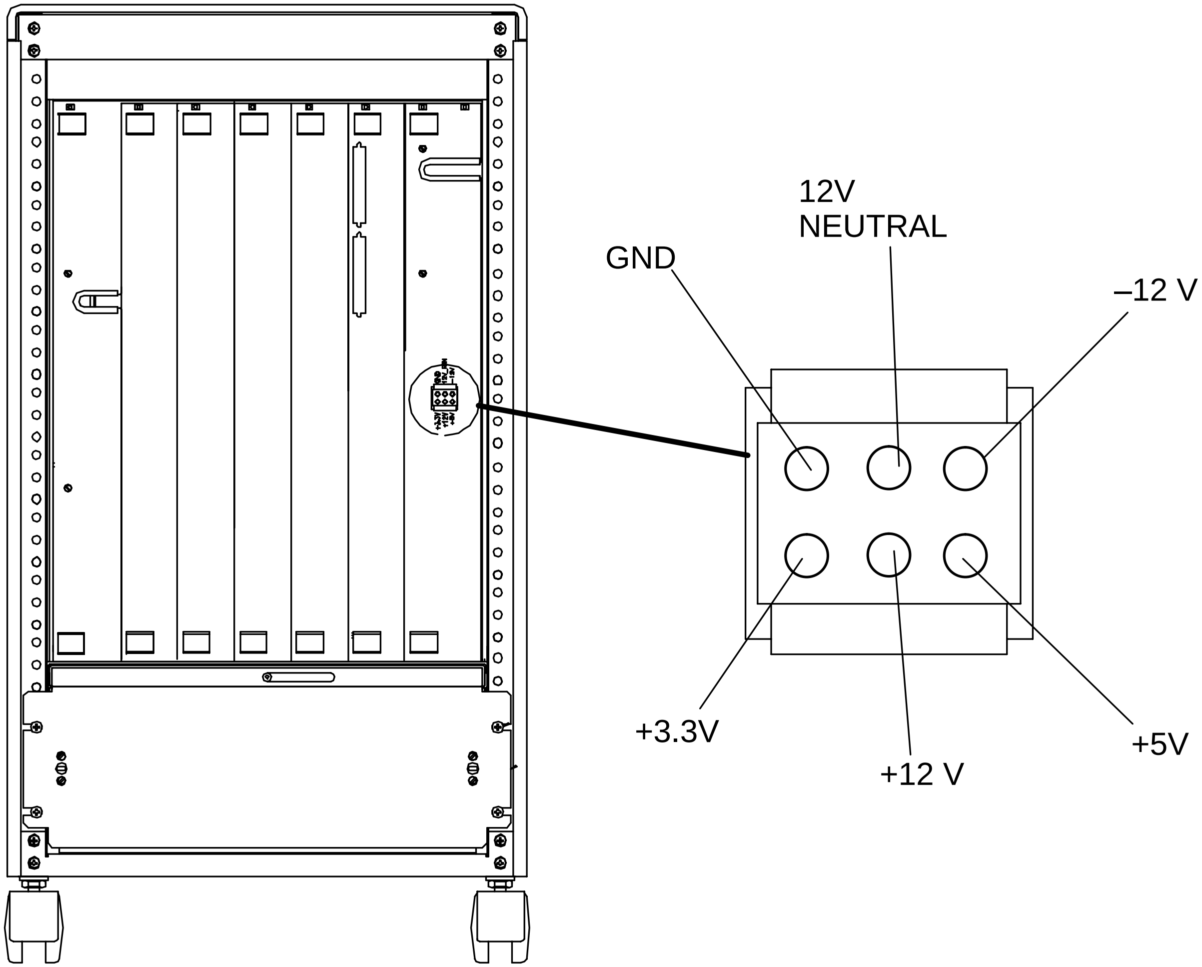


FIGURE 104 POWER SUPPLY VOLTAGE TEST POINTS

Under Normal Conditions:

POWER SUPPLY	MIN. VOLTAGE	MAX. VOLTAGE
+3.3 VDC	+3.25	+3.45
+5 VDC	+4.875	+5.25
+12 VDC	+11.64	+12.6
–12 VDC	–11.64	–12.6

If the voltages are out of range, adjust using the following procedure:

+3.3 VDC Supply:

4. Using a small screwdriver, adjust the pot until the voltage falls within the acceptable range. Refer to Figure 105.

+5 VDC and +/–12 VDC Supply:

5. Using a small screwdriver, adjust the pot until the voltage falls within the acceptable range. Refer to Figure 106. Both the +12vdc and –12vdc are adjusted with the same pot.

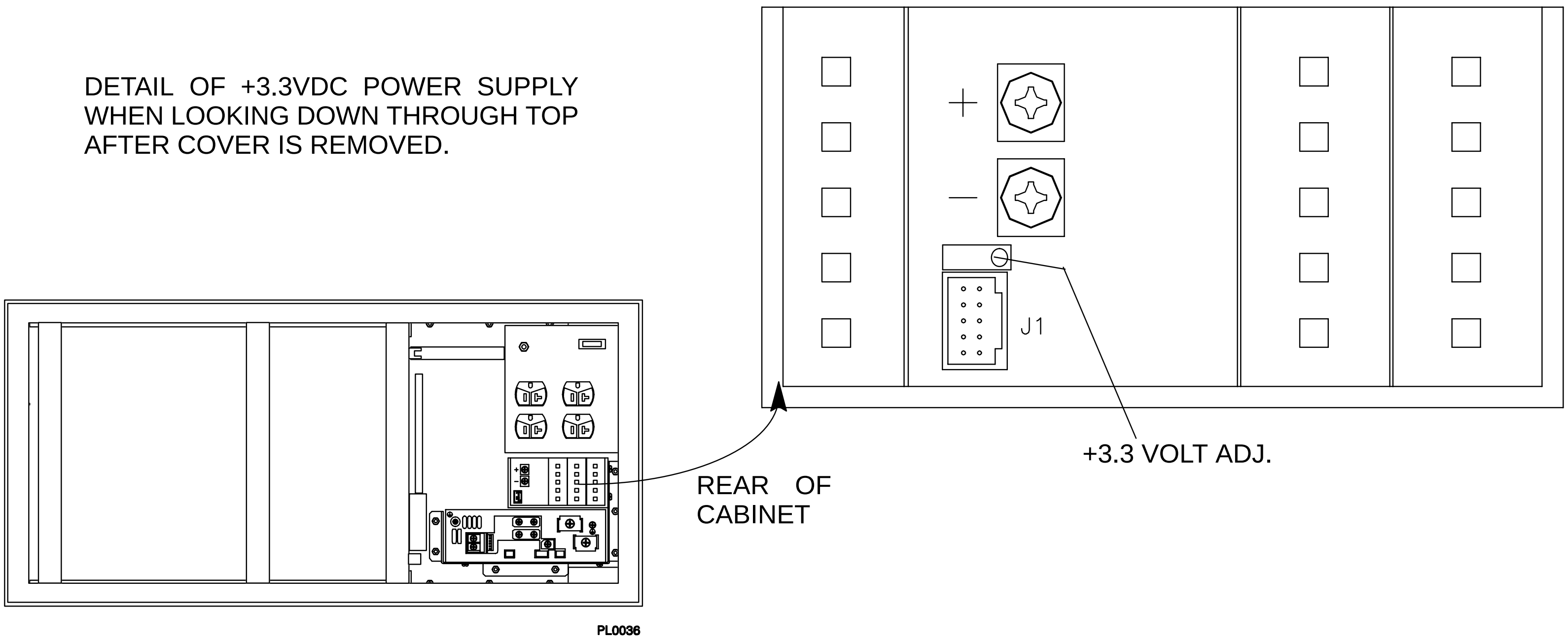


FIGURE 105 +3.3 VDC POWER SUPPLY ADJUSTMENT LOCATION

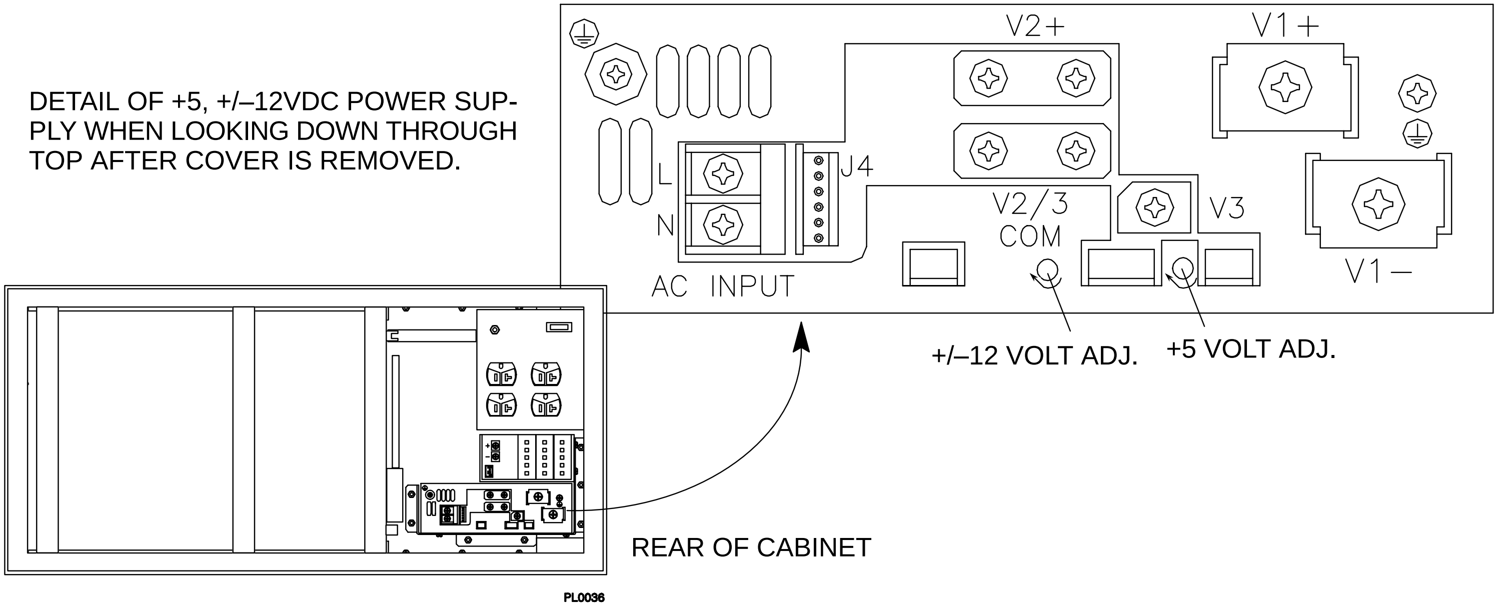


FIGURE 106 +5, +/– 12VDC POWER SUPPLY ADJUSTMENT LOCATIONS

6. When finished, attach the inside front and outside cosmetic covers and the top cover.. Refer to the Cover Removal/Replacement procedures in the Repair manual.

ULTRA ZX +5, +12, –12, +3.3 VOLT DC DISPLAY TOWER POWER SUPPLY VERIFICATION/CALIBRATION

Introduction: The purpose of this procedure is to verify the accuracy of the +5, +12, –12 and 3.3VDC Display Tower power supply.

Tools required: FSE tool kit

Estimated time: 15 Minutes

1. Remove system power. Refer to the Power Up/Down procedure.

> ⚠ **WARNING** ⚠
>
> **120 VOLTS AC, +5, +12, –12 and 3.3 VOLTS DC DC ARE EXPOSED WHEN THE DISPLAY TOWER COVERS ARE OPEN. ACCIDENTAL CONTACT WITH THESE VOLTAGES MAY CAUSE SERIOUS INJURY OR DEATH TO THE SERVICE PERSON.**

2. Remove the left side and top cover. Refer to the Display Tower Cover Removal procedure in the Repair Manual.
3. Refer to Figure 107 and locate the power supply voltage check points and adjustment locations.
4. Set the DMM to a 20 volt or greater DC range. Connect the leads while observing proper polarity to the proper test points.

> **NOTE**
>
> The +5V adjustment access hole is located at the TOP of the power supply. You must temporarily move the SPC and boards above it to gain access.

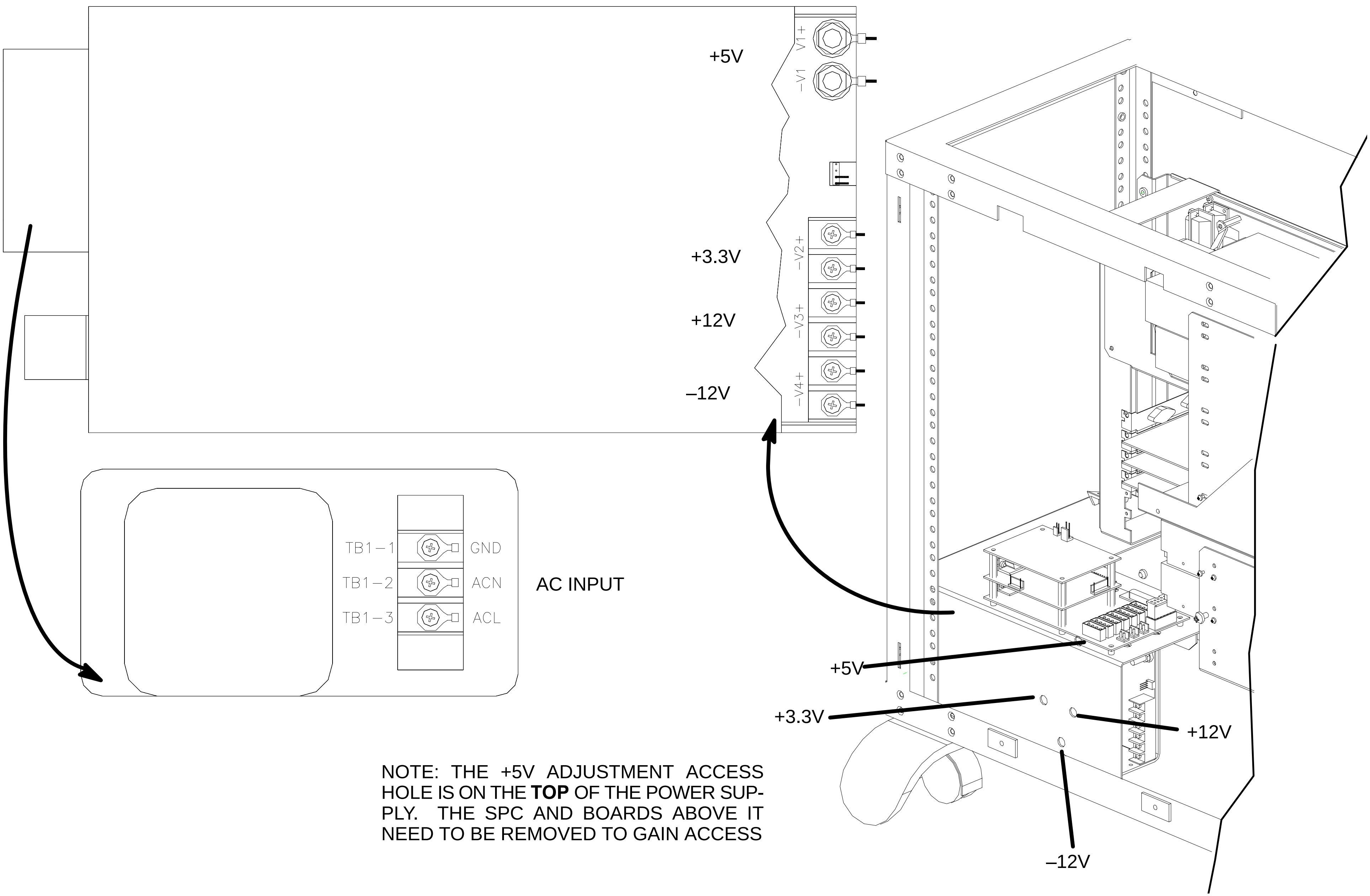


FIGURE 107 SYSTEM POWER SUPPLY – VOLTAGE TEST POINTS AND ADJUSTMENT LOCATIONS

Under Normal Conditions:

POWER SUPPLY	MIN. VOLTAGE	MAX. VOLTAGE
+5 VDC	+4.75	+5.25
+12 VDC	+11.4	+12.6
–12 VDC	–11.4	–12.6
+3.3 VDC	+3.465	+3.135

If the voltages are out of range, adjust the pot with a small screwdriver until the voltage falls within an acceptable range. Refer to the table above.

DISPLAY MONITOR CALIBRATION

Introduction: The purpose of this procedure is to calibrate the display monitor to factory specifications using the SMPTE pattern.

Tools required: FSE tool kit

Estimated time: 5 Minutes

> **NOTE**
>
> The display monitor cal via the SMPTE pattern is initiated via the Diagnostic GUI.

MANUAL IMAGE CALIBRATION

Image Calibration not run via AutoCalibration is run via the REASON pulldown menu of the Calibration selection of the Service Application.

Refer to the Service Application manual for more details.

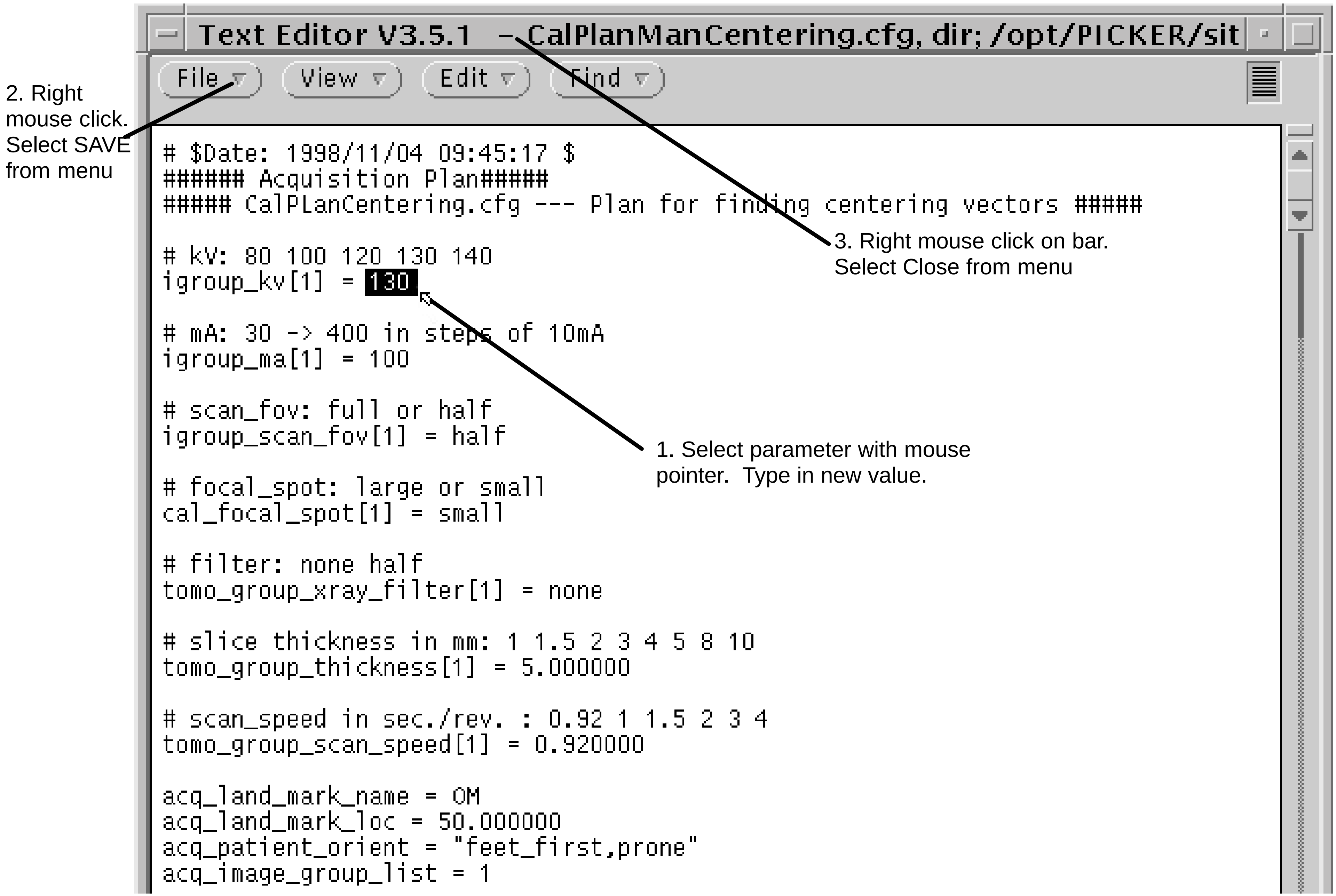
Refer to the table below for the Manual Calibration/Performance pulldown menu. (Manual Calibration routines are in **bold**).

Reason Pull Down Menu – MANUAL CALIBRATION

REASON	DESCRIPTION OF ACTION PERFORMED
ManPerform_Lateral_Centering	
ManFind_Centering	
ManFind_DAC	
MandFind_Air	
MandFind MA	
ManFind_Flatts	
ManFind_FlattsFF	
ManFind_Level	
ManFind_LevelFF	
ManFind_Level2	
ManFind_Level2FF	
ManFind_PilotAir	
ManPerform_MTF	
Clear_BadDetsList	

How to Run Manual Performance/Calibration

1. After the reason is selected, a text editor appears, containing the calibration/performance routine text of the Reason that was selected. Find and edit the appropriate text parameters for the calibration or performance routine you selected. To edit a routine, highlight the text value (mA, voltage, etc.) with the mouse pointer. (Click on the text, hold and drag across the text of choice until the text displays in reverse video.) Refer to Figure 110.
2. After the text is highlighted, type in the new value. For example, select 130kV and type in 120kV. After the text is edited, save the file and close the text editor window. When closing the editor, you will be prompted with a information about the steps you should perform. Refer to Figure 111. (In this example, the ManFind_Centering Reason was chosen.) Click on OK to continue.
3. From the scanning screen, click on the Proceed to Scan button, then the Start Scan button to continue with the Calibration or Performance routine.




```
# $Date: 1998/11/04 09:45:17 $
###### Acquisition Plan#####
##### CalPLanCentering.cfg --- Plan for finding centering vectors #####

# kV: 80 100 120 130 140
igroup_kv[1] = 130

# mA: 30 -> 400 in steps of 10mA
igroup_ma[1] = 100

# scan_fov: full or half
igroup_scan_fov[1] = half

# focal_spot: large or small
cal_focal_spot[1] = small

# filter: none half
tomo_group_xray_filter[1] = none

# slice thickness in mm: 1 1.5 2 3 4 5 8 10
tomo_group_thickness[1] = 5.000000

# scan_speed in sec./rev. : 0.92 1 1.5 2 3 4
tomo_group_scan_speed[1] = 0.920000

acq_land_mark_name = OM
acq_land_mark_loc = 50.000000
acq_patient_orient = "feet_first,prone"
acq_image_group_list = 1
```

FIGURE 108 MANUAL PERFORMANCE / CALIBRATION TEXT EDITOR

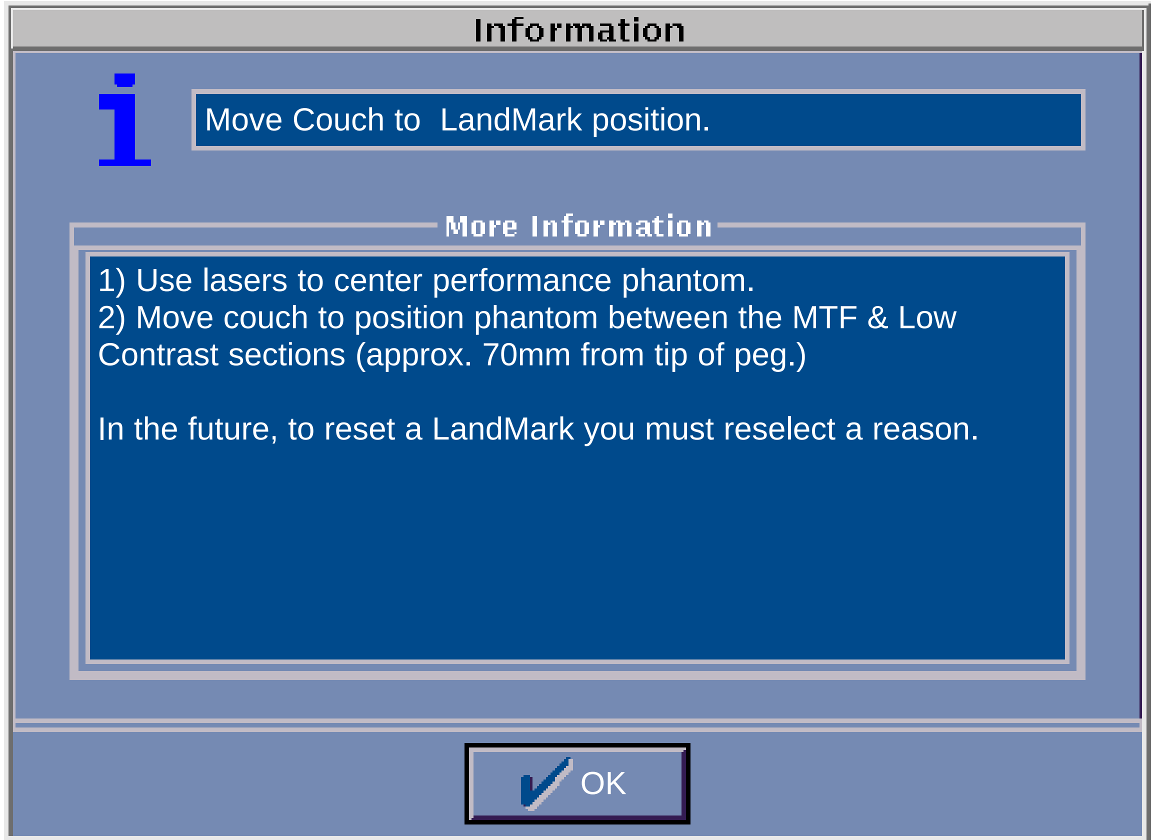


FIGURE 109 MANUAL PERFORMANCE / CALIBRATION INFORMATION WINDOW

Manual Calibration Protocols

Scan parameters for 3rd party service manual calibration of an Astro scanner.

	Vector Name	kv	filter	spot	scan_fov	thick	ma	speed
CAL:	Lateral	130	none	large	full	5	100	1.5
CAL:	Centering	130	none	small	full	5	100	1
CAL:	Centering	130	none	small	full	5	100	1.5
CAL:	Centering	130	none	small	full	5	100	2
CAL:	Centering	130	none	small	full	5	100	3
CAL:	Centering	130	none	small	full	5	100	4
CAL:	Centering	130	none	large	full	5	100	1
CAL:	Centering	130	none	large	full	5	100	1.5
CAL:	Centering	130	none	large	full	5	100	2
CAL:	Centering	130	none	large	full	5	100	3
CAL:	Centering	130	none	large	full	5	100	4
CAL:	DAC	130	none	small	full	8	100	4
CAL:	DAC	130	none	large	full	8	100	4
CAL:	Air	80	none	small	full	5	100	4
CAL:	Air	80	half	small	full	5	100	4
CAL:	Air	80	none	large	full	5	100	4
CAL:	Air	80	half	large	full	5	100	4
CAL:	Air	100	none	small	full	5	100	4
CAL:	Air	100	half	small	full	5	100	4
CAL:	Air	100	none	large	full	5	100	4
CAL:	Air	100	half	large	full	5	100	4
CAL:	Air	120	none	small	full	5	100	4
CAL:	Air	120	half	small	full	5	100	4
CAL:	Air	120	none	large	full	5	100	4

CAL:	Air	120	half	large	full	5	100	4
CAL:	Air	130	none	small	full	2	100	4
CAL:	Air	130	none	small	full	3	100	4
CAL:	Air	130	none	small	full	4	100	4
CAL:	Air	130	none	small	full	5	100	4
CAL:	Air	130	none	small	full	8	100	4
CAL:	Air	130	half	small	full	2	100	4
CAL:	Air	130	half	small	full	3	100	4
CAL:	Air	130	half	small	full	4	100	4
CAL:	Air	130	half	small	full	5	100	4
CAL:	Air	130	half	small	full	8	100	4
CAL:	Air	130	none	large	full	2	100	4
CAL:	Air	130	none	large	full	3	100	4
CAL:	Air	130	none	large	full	4	100	4
CAL:	Air	130	none	large	full	5	100	4
CAL:	Air	130	none	large	full	8	100	4
CAL:	Air	130	half	large	full	2	100	4
CAL:	Air	130	half	large	full	3	100	4
CAL:	Air	130	half	large	full	4	100	4
CAL:	Air	130	half	large	full	5	100	4
CAL:	Air	130	half	large	full	8	100	4
CAL:	Air	140	none	small	full	5	100	4
CAL:	Air	140	half	small	full	5	100	4
CAL:	Air	140	none	large	full	5	100	4
CAL:	Air	140	half	large	full	5	100	4

CAL:	Ma	100	none	large	full	5	30	2
CAL:	Ma	100	none	large	full	5	40	2
CAL:	Ma	100	none	large	full	5	50	2
CAL:	Ma	100	none	large	full	5	60	2
CAL:	Ma	100	none	large	full	5	70	2
CAL:	Ma	100	none	large	full	5	80	2
CAL:	Ma	100	none	large	full	5	90	2

CAL:	Ma	100	none	large	full	5	100	2
CAL:	Ma	100	none	large	full	5	110	2
CAL:	Ma	100	none	large	full	5	120	2
CAL:	Ma	100	none	large	full	5	130	2
CAL:	Ma	100	none	large	full	5	140	2
CAL:	Ma	100	none	large	full	5	150	2
CAL:	Ma	100	none	large	full	5	160	2
CAL:	Ma	100	none	large	full	5	170	2
CAL:	Ma	100	none	large	full	5	180	2
CAL:	Ma	100	none	large	full	5	190	2
CAL:	Ma	100	none	large	full	5	200	2
CAL:	Ma	100	none	large	full	5	210	2
CAL:	Ma	100	none	large	full	5	220	2
CAL:	Ma	100	none	large	full	5	230	2
CAL:	Ma	100	none	large	full	5	240	2
CAL:	Ma	100	none	large	full	5	250	2
CAL:	Ma	100	none	large	full	5	260	2
CAL:	Ma	100	none	large	full	5	270	2
CAL:	Ma	100	none	large	full	5	280	2
CAL:	Ma	100	none	large	full	5	290	2
CAL:	Ma	100	none	large	full	5	300	2
CAL:	Ma	100	none	large	full	5	310	2
CAL:	Ma	100	none	large	full	5	320	2
CAL:	Ma	100	none	large	full	5	330	2
CAL:	Ma	100	none	large	full	5	340	2
CAL:	Ma	100	none	large	full	5	350	2
CAL:	Ma	100	none	large	full	5	360	2
CAL:	Ma	100	none	large	full	5	370	2
CAL:	Ma	100	none	large	full	5	380	2
CAL:	Ma	100	none	large	full	5	390	2
CAL:	Ma	100	none	large	full	5	400	2

CAL:	PilotAir	130	none	large	full	2	30	4

CAL:	FlattsFF	80	none	small	full	8	100	4
CAL:	FlattsFF	80	half	small	full	8	100	4
CAL:	FlattsFF	80	none	large	full	8	100	4
CAL:	FlattsFF	80	half	large	full	8	100	4
CAL:	FlattsFF	100	none	small	full	8	100	4
CAL:	FlattsFF	100	half	small	full	8	100	4
CAL:	FlattsFF	100	none	large	full	8	100	4
CAL:	FlattsFF	100	half	large	full	8	100	4
CAL:	FlattsFF	120	none	small	full	8	100	4
CAL:	FlattsFF	120	half	small	full	8	100	4
CAL:	FlattsFF	120	none	large	full	8	100	4
CAL:	FlattsFF	120	half	large	full	8	100	4
CAL:	FlattsFF	130	none	small	full	8	100	4
CAL:	FlattsFF	130	half	small	full	8	100	4
CAL:	FlattsFF	130	none	large	full	8	100	4
CAL:	FlattsFF	130	half	large	full	8	100	4
CAL:	FlattsFF	140	none	small	full	8	100	4
CAL:	FlattsFF	140	half	small	full	8	100	4
CAL:	FlattsFF	140	none	large	full	8	100	4
CAL:	FlattsFF	140	half	large	full	8	100	4

CAL:	LevelFF	80	none	small	full	8	100	4
CAL:	LevelFF	80	half	small	full	8	100	4
CAL:	LevelFF	80	none	large	full	8	100	4
CAL:	LevelFF	80	half	large	full	8	100	4
CAL:	LevelFF	100	none	small	full	8	100	4
CAL:	LevelFF	100	half	small	full	8	100	4
CAL:	LevelFF	100	none	large	full	8	100	4
CAL:	LevelFF	100	half	large	full	8	100	4
CAL:	LevelFF	120	none	small	full	8	100	4
CAL:	LevelFF	120	half	small	full	8	100	4
CAL:	LevelFF	120	none	large	full	8	100	4

CAL:	LevelFF	120	half	large	full	8	100	4
CAL:	LevelFF	130	none	small	full	8	100	4
CAL:	LevelFF	130	half	small	full	8	100	4
CAL:	LevelFF	130	none	large	full	8	100	4
CAL:	LevelFF	130	half	large	full	8	100	4
CAL:	LevelFF	140	none	small	full	8	100	4
CAL:	LevelFF	140	half	small	full	8	100	4
CAL:	LevelFF	140	none	large	full	8	100	4
CAL:	LevelFF	140	half	large	full	8	100	4

CAL:	Level2FF	80	none	small	full	8	100	4
CAL:	Level2FF	80	half	small	full	8	100	4
CAL:	Level2FF	80	none	large	full	8	100	4
CAL:	Level2FF	80	half	large	full	8	100	4
CAL:	Level2FF	100	none	small	full	8	100	4
CAL:	Level2FF	100	half	small	full	8	100	4
CAL:	Level2FF	100	none	large	full	8	100	4
CAL:	Level2FF	100	half	large	full	8	100	4
CAL:	Level2FF	120	none	small	full	8	100	4
CAL:	Level2FF	120	half	small	full	8	100	4
CAL:	Level2FF	120	none	large	full	8	100	4
CAL:	Level2FF	120	half	large	full	8	100	4
CAL:	Level2FF	130	none	small	full	8	100	4
CAL:	Level2FF	130	half	small	full	8	100	4
CAL:	Level2FF	130	none	large	full	8	100	4
CAL:	Level2FF	130	half	large	full	8	100	4
CAL:	Level2FF	140	none	small	full	8	100	4
CAL:	Level2FF	140	half	small	full	8	100	4
CAL:	Level2FF	140	none	large	full	8	100	4
CAL:	Level2FF	140	half	large	full	8	100	4

PERFORMANCE / VERIFICATION

Performance/Verification is run via the REASON pulldown menu of the Calibration selection of the Service Application.
For *system performance tests, test methods,* and *performance specifications*, refer to the Performance Specifications section of this manual.

Manual Performance is run the same way as auto, except the REASON menu is different to allow only certain calibration/performance routines. Refer to the table below for the Manual Calibration/Performance pulldown menu. (Performance routines are in bold).

Reason Pull Down Menu – MANUAL CALIBRATION

REASON	DESCRIPTION OF ACTION PERFORMED
ManPerform_Lateral_Centering	
ManFind_Centering	
ManFind_DAC	
MandFind_Air	
MandFind MA	
ManFind_Flatts	
ManFind_FlattsFF	
ManFind_Level	
ManFind_LevelFF	
ManFind_Level2	
ManFind_Level2FF	
ManFind_PilotAir	
ManPerform_MTF	
Clear_BadDetsList	

How to Run Manual Performance / Calibration

1. After the reason is selected, a text editor appears, containing the calibration/performance routine text of the Reason that was selected. Find and edit the appropriate text parameters for the calibration or performance routine you selected. To edit a routine, highlight the text value (mA, voltage, etc.) with the mouse pointer. (Click on the text, hold and drag across the text of choice until the text displays in reverse video.) Refer to Figure 110.
2. After the text is highlighted, type in the new value. For example, select 130kV and type in 120kV. After the text is edited, save the file and close the text editor window. When closing the editor, you will be prompted with a information about the steps you should perform. Refer to Figure 111. (In this example, the ManFind_Centering Reason was chosen.) Click on OK to continue.
3. From the scanning screen, click on the Proceed to Scan button, then the Start Scan button to continue with the Performance routine.

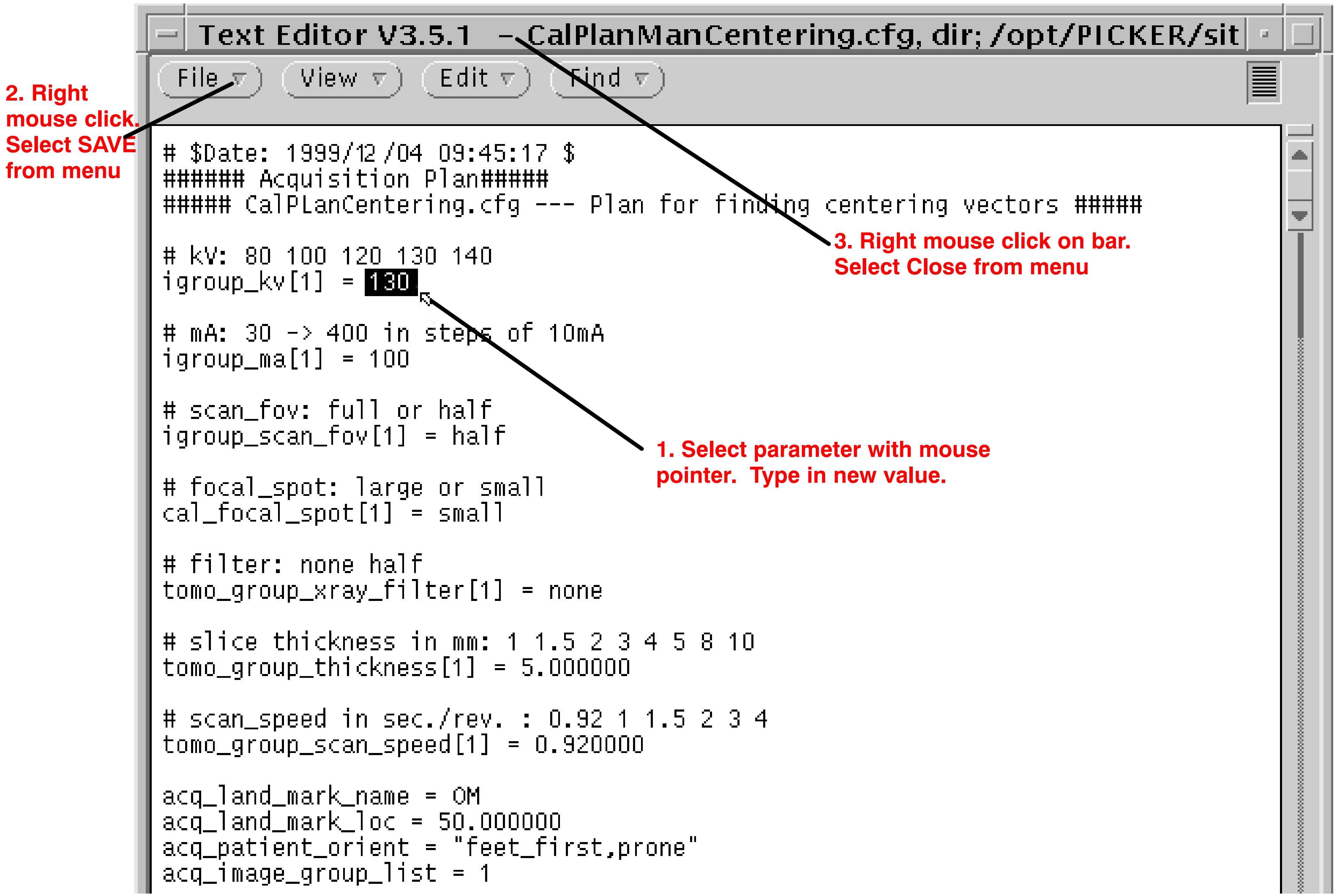


FIGURE 110 MANUAL PERFORMANCE / CALIBRATION TEXT EDITOR

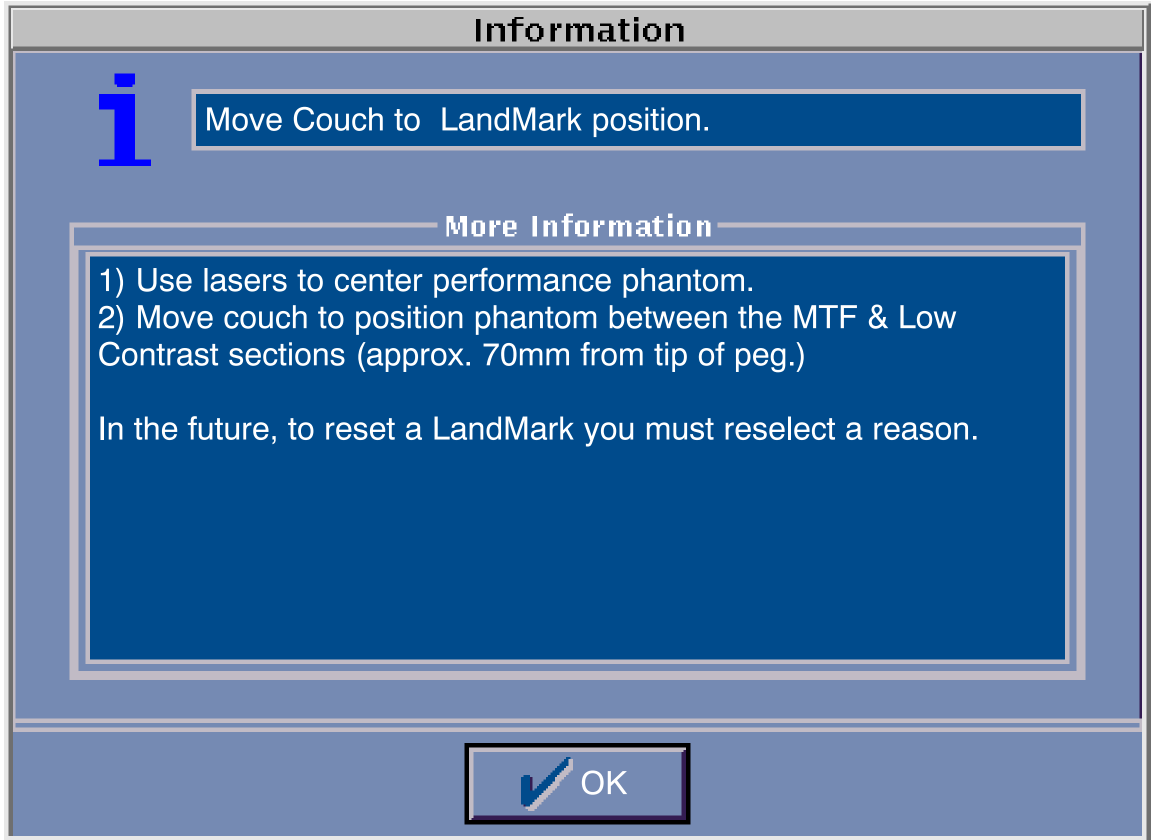


FIGURE 111 MANUAL PERFROMANCE / CALIBRATION INFORMATION WINDOW

PERFORMANCE SPECIFICATIONS

INTRODUCTION

This document provides system performance tests, test methods, and performance specifications for the AcQSim CT CT System. The test methods assume familiarity with the operation of the AcQSim CT System. A AcQSim CT System satisfying these performance specifications is a necessary but not sufficient condition for acceptable system performance.

TEST PHANTOM

The test phantom used for the majority of these tests is the Philips Medical Systems Performance Phantom. This phantom is supplied with the AcQSim CT System. This phantom is specially designed to evaluate the image performance of Philips Medical Systems CT systems and consists of the six separate scanable sections as described below. The standard phantom landmark for many of the tests is the crease between the fourth and fifth sections.

1. A 12 inch diameter uniform water section is used for full field uniformity and noise.

2. A 8 inch diameter uniform water section is used for half field uniformity and noise.

3. A 6 inch diameter uniform water section is used for low contrast measurements.

4. A 6 inch diameter low contrast section contains a low contast plastic insert (.35% contrast compared to water) drilled with 7 rows of four holes (6mm, 5mm, 4mm, 3mm, 2.5mm, 2mm, and 1.5mm). This plastic section is surrounded by water and also contains a large acrylic peg and large polyethylene peg for measuring the linearity of the CT numbers.

5. A 6 inch diameter slice thickness / MTF section is a plastic section which contains two foil strip ramps or wedges whose angle is 14 degrees from the phantom's central axis. These ramps are used for measuring slice thickness. Also imbedded in this plastic section is a .05mm diameter tungsten wire used for line spread function measuremnts leading to MTF results.

6. The last section is a removeable 1 inch diameter plastic cylinder which contains a central pin which is useful for phantom centering and alignment with the systems central axis of rotation.

DESIGN

The following list contains descriptions of the AcQSim CT System's design and notes differences from the Ultra Z product (version 1.0) that are relevant for this T95 document:

<u>Detectors and Detector Modules</u>:

The gantry ring has 2400 individual CdWO4 detector elements. These detector elements are divided into 120 detector modules.
Each Module contains two (10 element) crystal arrays (29mm long and 2.26mm thick);
Each Crystal array (21.92mm wide) has 10 defined channels ;
Each Channel has a 1.58mm detector space;
Detector (center–to–center) distance is 2.17mm.

SDD:
The radial distance (863mm) of the detector hoop is maintained from the distance established by the Ultra Z. The SCD (focal spot to imaging isocenter) is extended with respect to the Ultra Z from 635mm to 651mm for Rhino 4/5/6.5 X–ray tubes.

Dose Efficiency:
Assuming negligible spacing between modules and crystals:
Detector Dose EfficiencyAcQSim CT = Q.D.E.*Geometric Efficiency
QDE~100%
Geometric Efficiency= Detector space/(Module space + Intermodule dead space)
Geometric Efficiency~ 15.8/(21.9+ (2*Pi*13/240)) ~ 15.8/22.3= 70%
Geormetric Does Efficiency: Since the tube has been moved out to accomodate the larger bore, the dose to the patient will be reduced (with respect to the Ultra Z).

Sampling:
Half Field scans are not supported by the AcQSim CT system.
In Full–field, 3 samples per detector angle of rotation (.15 degrees) for 512 detectors.

FOV
The Full Field field of view is extended from 48 cm (Ultra Z) to 60 cm.

PREREQUISITES

1. The factory laser alignment procedure has been successfully completed.

2. Couch calibration has been completed and tilt position is correct.

3. A complete calibration, i.e. changed tube calibration, has run successfully. This complete calibration means that electronic centering, DAC, air calibration, MA curves, pilot cals, spectrum and scaling constants have been generated. It also means that mechanical alignment of the x–ray tube, both lateral and axial have been generated.

4. This procedure is intended to be used only on systems running with AcQSim CT 1.0 Software (or higher) system software and AcQSim CT 1.0 (or higher) diagnostic software.

5. All diagnostic programs and software tools referenced within this document, or their functional equivalent, must be resident in the software (i.e. cna baco, cna resolver, cna water, mtf, kvhvl, lowcon, slice thickness, and uniformity).

6. AutoVoice and auto–archive must be ”OFF” for all tests. Data Save must be off unless otherwise specified.

7. Prior to beginning this performance test, it must be verified that an open circuit (>10 Mohm) exists between the ground block (DC Common) and the AC Common wire. This can be performed simply by removing the green/yellow striped wire from the ground block located on the cardfile and measuring with an ohmmeter (20M range).

8. Ensure that the face of the phantoms used during these tests are positioned parallel to the central xray of the fan beam. Run Center_Phantom to insure correct positioning of the Philips Medical Systems performance test phantom.

9. The number and location of bad detecters is within the system specification for bad detecters. See below.

System Spec for Bad Detectors

a. There can be no more than 16 bad detectors per system / else replace the module with the most bad detectors.
b. There can be no more than 3 bad channels per detector module / else replace the module.
c. There can be no more than 2 adjacent bad detectors, either in a module or in adjacent modules / else replace the module.
d. No defective channel can adversely effect other channels in a quadrature manner / else replace the defective component causing the adverse effect.
e. Any module with a defective ARDAS channel must be replaced.

****** VERIFICATION ******

1. A permanent record (as a logfile or data sheet) of all system performance data of the scanner must be stored via digital archive or hard copy.

TEST 1 – LASER ALIGNMENT TEST – FIRST PILOT –AUTO POSITIONING

Purpose: To verify that the alignment laser beams align with the transaxial x–ray beam. Also to verify the pilot scan operation and the couch positioning are correct.   Test Parameters and Method: Manual

1. Enter the Clinical Scanning or Console mode of operation.
2. Select the Adult Brain protocol.
3. Center and align the 6 inch water section of the Philips Medical Systems Performance Phantom on the CT System couch and gantry using the patient alignment lasers. The laser lights should bisect the phantom.
4. Perform a vertical pilot scan of the phantom with a pilot range of +150 to –150 mm. After the pilot image is displayed, note that the image of the phantom is centered in he the pilot scanned area. Adjust the window setting and zoom the image to optimally view the phantom.
5. In the pilot scan find the dark rectangular portion in the last section of the phantom that is a slot for hanging the centering rod or additional phantoms. Position the plan line at the exact bottom of this rectangular region which is between the endplate and the slice thickness/mtf section.
6. Select a Cine scan type and record the the co–ordinate location of the plan line under Extent: __________________.
7. Start the Cine scan and observe the resulting axial image. Set the level to 100 and the window to 200. The table is positioned properly if the two parallel bars (slice thickness wedges) and the oval portion (bisecting the upper bar) are visible in the image. Note the final couch position recorded under scan status: _________________.
8. Turn on the lasers and verify that the beams fall directly on the seam between the mtf/slice thickness section and the endplate. If necessary, adjust laser positions on the x–ray frame and repeat the laser alignment test if one or more of the lasers do not fall on the seam.
9. Record all test results below or on a system performance data sheet.

Results and Specification:

Pilot Scan Operation [pilot image centered]	(ok)
Pre Cine Scan Plan Location [extent position]	
Post Scan Couch Location (within .5mm of previous line)	
Couch Positioning (Parallel bars and oval visible in image)	(ok)
Laser / X–Ray Beam Alignment (Laser line falls on phantom seam correctly)	(+/– 0.5mm)

TEST 2 – CHECK FOR BAD DETECTORS

Tests and Purposes: **BadDetsBaco**: Determine each detector's background noise and offset level and compare against specification. **BadDetsWater**: Determine each detector's noise level on a subtracted 12 inch water phantom scan and compare against specification. Insure that there is no air bubble in the phantom as this could cause the test to fail.
BadDetsAir: Determine each detectors dc response level for an air scan and compare against specification.
BadDetsPEG: Determine each detector's pin area on a subtracted peg from pin scan and compare against specification.

Test Parameters: (All set automatically by the service application software test method)

KV	100	130	130	130
Filter	0	0	0	0
Focal Spot	Small	Small	Small	Small
Scan Field	Full	Full	Full	Full
Slice Thickness	8mm	8mm	8mm	5mm
MA	30	130	90	70
Frame Speed	2	4	4	4
Matrix_Y	512	512	512	512
Recon_type	Normal	Over	Over	Over
Recon_Field	na	600	600	600
Couch	na	+106mm #1 +108mm #2	–160mm	–39mm for scan #1 –60mm for scan #2
Conditions				
Acq_chspd	0	0	0	0
Samples/unit	3	3	3	3
Src_active_dets	648	648	648	648
Calcium_correction	Off	Off	Off	Off
Scan_Start_Degrees	Variable	Variable	Variable	Variable
Off set Frequency	12 KHz	na	na	na
Scan Type	na	Scan	Scan	Scan
Command_type	Scan	Scan	Scan	Scan

Test Method:
· Service Application
· Calibrate WorkStep
· Center_Phantom
· Find_BadDets Reason

–BadDetsBaco
–BadDetsWater
–BadDetsAir
–BadDetsPeg

Results and Specification:

BadDetsBaco	**BadDetsWater**
Ave offset: ______________________(ref. e–02v)	Water Ave. Noise:____________________(ref. e–04v)
Max offset: ______________________ (–3.50e–02 v)	Water Max Noise: ____________________(6.00e–04 v)
Min offset: ______________________ (–2.50e–02 v)	Water Min Noise: ____________________(≥ 1.70e–04 v)
Ave noise: ______________________ (ref. e–05v)	
Max noise: ______________________ (7.00e–05 v)	
Min noise: ______________________ (≥ 1.5.e–05 v)	

BadDetsAir	**BadDetsPeg**
Air Ave. Response:____________________(ref. +00v)	Peg Area Ave. : ______________ (ref. e +04)
Air Max Response : ____________________(–6.00e+00 v)	Peg Area Max. :______________ (ref. e +04)
Air Min Response: ____________________(–2.00e+00 v)	Peg Area Min. :______________ (ref. e +04)

All detector Peg Area Values must be within +/–12% of the system average, else place the offending detector on the bad list.
Specification: All BadDets tests must run to completion without errors.

TEST 3 – PERFORM CNA (COHERENT NOISE ANALYSIS)

Tests and Purposes: **Perform CNABaco**: Perform spectrum analysis on detector fans for a background scan and verify that background noise is within specifications.
Perform CNAWater: Perform Spectrum Analysis on detector fans of two subtracted 8 inch water scans and verify that the noise amplitudes are within specification.
Perform CNAResolver: Perform spectrum analysis of detector fan data of a digital check scan and verify that the peak to peak arc–sec amplitudes are within specification.
Key Test Parameters: (All set automatically by software method)

KV	130	130	130
Filter	0	0	0
Focal Spot	Large	Large	Large
Slice Thickness	8mm	8mm	8mm
MA	120	150	120
Scan Field	Full	Full	Full
Frame Speed	2	2	2
Matrix_Y	512	512	512
Recon_type	Normal	Over	Norm
Conditions			
Samples/unit	3	3	3
Src_active_dets	648	648	648
Acq_chspd	3	3	3
Command_type	Scan	Scan	Scan
Scan_Start_Degrees	Variable	0 degrees	Variable
Scan Digital Check Freq.	na	na	2MHz
Background Freq	12kHz	na	na

Test Method:
· Service Application
· Calibrate WorkStep
· Perform _CNA Reason
–CnaBaco
–CnaWater
–CnaResolver

Test Results and Specifications:

CnaBaco		
Frequency	Measured Result	Specification
Any one ≥ 30 hz	Highest Value =	≤ 4.00e–06 vp
White Noise (Total)		≤ 5.0e–05 vp

CnaWater		
Frequency Range (hz)	Highest Measured Value	Specification (vp)
10 – 100hz		≤ 3.5e–04 vp
100 – 399hz		≤ 9.0e–04vp
400 – 650hz		≤ 3.7e–04vp
> 650hz		≤ 1.4e–04vp
White Noise (Total)		≤ 5.0e–03 vp

CnaResolver		
Frequency	Measured Result	Specification (arc sec)
450 cycles/rev		< 3.35 arc–sec p–p
Any frequency other than 450cycles/rev	Highest Value =	≤ 1.67 arc–sec p–p
White Noise (Total)		< 2.0e+01 arc–sec p–p

TEST 4 – BEAM QUALITY – KVHVL

Purpose: Measure the half value layer of the system and verify that it meets specification.

Test Parameters: : (All set automatically by the service application software test method:

Compensator	None
KV	130
Filter	0
Focal Spot	Large
Scan Field	Full
Slice Thickness	2mm
MA	130
Frame Speed	4
Matrix_Y	512
Recon_type	Over
Recon_Field	240
Algorithm	Smooth
Couch	Move couch –159mm relative to phantom 0 for scan #1 with kvhvl fixture Move couch –161mm relative to phantom 0 for scan #2 w/o kvhvl fixture
Conditions	
Acq_chspd	0
Scan Type	Scan
Samples/unit	3
Src_active_dets	648
Calcium_correction	Off
Command_type	Scan
Scan_Start_Degrees	0

Test Method:
· Service Application
· Calibrate WorkStep
· Perform_KVHVL Reason

Test Results and Specifications:

HVL (@130 KV)	________________________(3.7 to 4.5 mm of aluminum)
HVL (@100 KV)	________________________extrapolated from 130 KV measurement (2.8 to 3.7 mm)
Calculated KVp	________________________(For reference only; approx. 130KVp)

TEST 5 – MTF

Purpose: Determine the Modulation Transfer Function (MTF) by scanning a thin wire and verifiying that the system meets specification.

Test Parameters: : All set automatically by the service application software test method:

3 MTF Measurements are made:		#1	#2	#3
KV	130			
Filter	0			
Focal Spot		Small	Large	Small
Scan Field	Full			
Slice Thickness	8mm			
MA	130			
Frame Speed		4	4	1
Matrix_Y	512			
Scan Type	Scan			
Recon_type	NORM			
Recon_Field	60mm			
Algorithm	MTF			
Couch	Move couch to –9mm relative to phantom 0 for scan #1, #2, #3.			
Conditions				
3 MTF Measurements are made:				
Command_type	Scan			
Scan_Start_Degrees	Variable			
Samples/unit	3			
Src_active_dets	648			
Calcium_correction	Off			
Acq_chspd	0			

Test Method:
· Service Application
· Calibrate WorkStep
· Perform_MTF Reason

Test Results and Specifications:

	Scan #1 (Small Spot)	Scan #2 (Large Spot)	Scan #3 (1 sec; Small Spot)
50% modulation	__________ (7.0 lp/cm)	___________ (6.8 lp/cm)	___________ (7.0 lp/cm)
10% modulation	__________ (12.0 lp/cm)	___________ (11.5 lp/cm)	___________ (11.5 lp/cm)
5% modulation	__________ (13.0 lp/cm)	___________ (12.6 lp/cm)	___________ (12.6 lp/cm)

TEST 6 – SLICE THICKNESS

Purpose: Measure various slice thicknesses and verify that the system meets the specification.

Test Parameters: All set automatically by the service application software test method.

KV	130	130	130	130
Filter	0	0	0	0
Focal Spot	Large	Large	Large	Large
Scan Field	Full	Full	Full	Full
Slice Thickness	8mm	5mm	3mm	2mm
MA	100	100	100	100
Frame Speed	4	4	4	4
Matrix_Y	512	512	512	512
Recon_type	Norm	Norm	Norm	Norm
Recon_Field	60mm	60mm	60mm	60mm
Conditions:				
Acq_chspd	0	0	0	0
Scan Type	Scan	Scan	Scan	Scan
Samples/unit	3	3	3	3
Src_active_det	648	648	648	648
Calcium_correction	Off	Off	Off	Off
Command_type	Scan	Scan	Scan	Scan
Scan_Start_Degrees	Variable	Variable	Variable	Variable

Test Method:

Service Application
Calibration Workstep
Perform_

Test Results and Specifications:

Nominal Slice Thickness	Results	Specification
8mm		7.2 to 8.8mm
5mm		4.5 to 5.5mm
3mm		2.5 to 3.5mm
2mm		1.5 to 2.5mm

TEST 7 – WATER UNIFORMITY AND NOISE

KV	80	100	120	130	140
Scan Field	Full	Full	Full	Full	Full
Filter	0	0	0	0	0
Focal Spot	Large	Large	Large	Large	Large
Algorithm	Soft Tissue	Soft Tissue	Soft Tissue	Soft Tissue	Soft Tissue
MA	130	130	130	130	130
Frame Speed	4	4	4	4	4
Slice Thickness	8mm	8mm	8mm	8mm	8mm
Recon Field	480 mm	480 mm	480 mm	480 mm	480 mm

Test Method:

· Service Applications
· Calibrate WorkStep
· Perform_Uniformity: Measurements made on center of fov, at 1/2 radius for peripherals. Measurements using a 2000mm^2 square ROL.

Results and Specifications:

FF Uniformity and Noise (Filter 0)	80KV	100KV	120KV	130KV	140KV
Center Value (0) = (0 +/– 2.0)					
Highest Perif Value (0) =(C.V. +/–3.0)					
Full Field Noise Spec (x10)	7.5	7	7	7	7
(Max. Std. Dev. in HU)					

TEST 8 – LOW CONTRAST RESOLUTION

Purpose: To measure the low contrast resolution of the system under certain clinical parameters and determine if the low contrast resolution is within specification.

Test Parameters: (All set automatically by the service application software test method)

KV	130
Filter	0
Focal Spot	Small
Scan Field	Full
Slice Thickness:	8mm
MA	160
Frame Speed	4
Matrix_Y	512
Recon Type	Over
Recon Field	250
Algorithm	Soft Tissue
Couch	Move couch +16mm relative to phantom 0 for scan #1 Move couch +40mm relative to phantom 0 for scan #2
Conditions:	
Acq_chsp	0
Scan Type	Scan
Samples/Unit	3
Src_active_dets	648
Calcium_correction	Off
Command_type	Scan
Scan_start_degrees	Variable

Test Method:
· Service Applications
· Calibrate WorkStep
· Perform_LowCon_Reason

Test Results and Specifications:

Low Contrast Measurement		0.25 to 0.55%
Noise in Low Con Section		</= 1.55 HU std. dev. of water
Hole Size at 0.35% contrast		</= 3.5mm
CT Number of Water		0 +/– 5 HU
CT NUmber of Acrylic (peg1)		120 +/– 10 HU
CT Number of Polyethylene (peg2)		–96 +/– 5 HU

TEST 9 – MULTI–SLICE STUDY CYCLE TIME

Purpose: Measure the study cycle time for a common time consuming multi–slice study and verify that the system meets the timing specification.

Test Parameters: This is a manual test.

Test Object	8" Water Phantom
Protocol	Abdomen
Algorithm	Soft Tissue
Scan Time	1.5 sec
mA	120
KV	130
Slice Thick	8mm
Scan Angle	normal
Couch Increment	8mm
Data save	Off

Test Method:

1. Verify that the starting tube heat is less than 30%.
2. Plan an axial scan sequence of 30 axial images and Plan / Edit any of the specified scan parameters of the protocol to match the above specified parameters.
3. Total study time is measured from x–ray on of the first scan to image preview complete for the 30 th scan.

Results and Specification:

Time from first to last exposure =	____________ ($\leq$ 85sec.)
Total time from first exposure to last image review =	____________ ($\leq$ 90sec.)

TEST 10 – SPIRAL SCAN STUDY CYCLE TIME

Purpose: Measure the spiral study cycle time for a common time consuming spiral study and verify that the system meets the timing specification.

Test Parameters: This is a manual test.

Test Object	8” Water Phantom
Protocol	Abdomen
Algorithm	Soft Tissue
Scan Time	1.5 sec per revolution
mA	120
KV	130
Slice Thick	8 mm
Index	8 mm
Pitch	1
Interpolation	Standard
PVC	Off
Data save	Off

Test Method:

1. Verify that the starting tube heat is less than 30%.
2. Plan a spiral scan sequence of 29 axial images and Plan/Edit any of the specified scan parameters of the protocol to match the above specified parameters.
3. Total study time is measured from x–ray on of the first scan to image preview complete for the 29th scan.

   Use either the system’s line profile and distance measurement utilities or the background to disappearing ramp method. See the CTDI document for a detailed description of this latter method].

Time from first to last exposure =	_____________ ($\leq$ 48sec.)
Total time from first exposure to last image review =	_____________ ($\leq$ 80sec.)

TEST 11 – BACKUP TIMER (OPTIONAL)

PURPOSE

This procedure defines the methods to be followed to verify the scan time, pilot time and x–ray back–up timer are within design parameters.

EQUIPMENT REQUIRED

Standard Test Technician Tool Box.

Gould RS 3200 strip chart recorder.

> ⚠ **CAUTION**
>
> WHEN HANDLING PRINTED CIRCUIT BOARDS OR ANY ESD SENSITIVE DEVICES USE APPROPRIATE ESD GROUNDING PRECAUTIONS.

> **NOTE**
>
> Unless otherwise specified the system should be powered up and the computer booted up.

> **NOTE**
>
> Ensure the tube has been properly warmed up prior to performing sections that produce x–rays.

PROCEDURE

1. Raise the patient support to a height that will allow the carbon top to pass through the center of the scanner before starting the test.
2. Put the GPUC on an extender card.
3. Attach a chip clip to U13.

⚠ CAUTION

THE LENGTH OF THE CHIP CLIP CAUSES THE PINS TO BE VERY CLOSE TO THE GANTRY SIDE COVER. CARE MUST BE TAKEN THAT THE CHIP CLIP LEADS DO NOT SHORT AGAINST THE GANTRY SIDE COVER.

4. On the Gould chart recorder set the D.C. amplifier (channel 1) as follows:

Full Scale Voltage	.05
Sensitivity	x100
Filter	OFF

5. Connect both leads of channel 1 of the chart recorder to TP1 (D.C. common on the GPUC card).
6. Start the chart recorder at a speed of 5 mm per sec, center the ink pen on the graph, and stop the chart recorder.
7. Connect the signal lead of channel 1 to the x–ray start/ stop pulse (XRSTSP) located at U13 pin 3 on the GPUC board. Connect the negative lead to TP1.
8. Modify **CENTER CAL** protocol in the CLINICAL mode as follows:

   SCAN TIME: **4** FOV: **LARGE** SCAN ANGLE: **OVER**

9. Perform a scan. Prior to clicking the 'start' button on the screen, start the chart recorder at a speed of 100mm/sec.
10. At the completion of the scan, stop the recorder and compute the time between the two "XRSTSP" pulse transitions. Record this time on data sheet.
11. Modify **CENTER CAL** protocol in the CLINICAL mode as follows:

    SCAN TIME: **2** FOV: **LARGE** SCAN ANGLE: **NORM**

12. Perform a scan. Prior to clicking the 'start' button on the screen, start the chart recorder at a speed of 100mm/sec.
13. At the completion of the scan, stop the recorder and compute the time between the two "XRSTSP" pulse transitions. Record this time on data sheet.

14. Modify **CENTER CAL** protocol in the CLINICAL mode to perform a pilot scan. Position the patient support top to allow for a travel of 256mm on each side of the landmark. Change the PILOT RANGE to:

    START: –256 END: +256

15. Perform a pilot scan. Prior to clicking the 'start' button on the screen, start the chart recorder at a speed of 100mm/sec.
16. At the completion of the pilot scan, stop the recorder and compute the time between the two "XRSTSP" pulse transitions. Record this time on data sheet.
17. Move the carbon top to its outermost position. Ensure there are no obstructions in the scan field.
18. Switch from CLINICAL to DIAGNOSTIC MODE.
19. On the Gould chart recorder set the D.C. amplifier (channel 1) as follows:

Full Scale Voltage	.25
Sensitivity	x100
Filter	OFF

20. Connect both leads of channel 1 of the chart recorder to DC common.
21. Start the chart recorder at a speed of 5 mm per sec, center the ink pen on the graph, and stop the chart recorder.
22. Remove the J1 connector from VFSC board 5 (For AcQSim CT use VFSC board 7). Insert an extender pin in J1 pin 4 and J1 pin 1. Connect the chart recorder positive lead to pin 1 and the common lead to pin 4.
23. From an Xterm window change the directory by typing: cd/opt/PICKER/picr/service_util/diagnostics/pq2000p/olevel/bin
24. At the prompt type: **backup_timer**

25. Execute macro **@povt4**. Start the chart recorder at a speed of 100mm/sec when instructed to do so. This macro will generate an exposure to verify the back–up timer.
26. Stop the chart recorder once the x–ray over time error occurs.
27. Compute the actual exposure time from the strip chart read–out. Record this time on data sheet.
28. Execute macro **@povt2.** Start the chart recorder at a speed of 100mm/sec when instructed to do so. This macro will generate an exposure to verify the back–up timer.
29. Stop the chart recorder once the x–ray over time error occurs.
30. Compute the actual exposure time from the strip chart read–out. Record this time on data sheet.
31. Return the system to the original configuration.

SYSTEM PERFORMANCE DATA SHEET **Date:______________** **System I.D.:____________________________**

TEST 1: COUCH/LASER POSITION

Pilot Scan Operation [pilot image centered]	__________ (ok)
Pre Cine Scan Plan Location [extent position]	__________
Post Scan Couch Location [within .5mm of previous line]	__________
Couch Positioning [Parallel bars and oval visible in image]	__________ (ok)
Laser / X–Ray Beam Alignment [Laser line falls on phantom seam correctly]	__________ (+/– .5mm)

TEST 2: CHECK FOR BAD DETECTORS

BadDetsBaco	Ave offset	____________________________ (ref. e–02v)
	Max offset	____________________________ (–3.50e–02 v)
	Min offset	____________________________ (–2.50e–02 v)
	Ave noise	____________________________ (ref. e–05v)
	Max noise	____________________________ (7.00e–05 v)
	Min noise	____________________________ (≥ 1.5.e–05 v)

BadDetsWater	Water Ave. Noise	____________________________(ref. e–04 v)
	Water Max Noise	___________________________ (6.00e–04 v)
	Water Min Noise	____________________________ (≥ 1.70e–04 v)

BadDetsAir	Air Ave. Offset	____________________________ (ref. +00v)
	Air Max Offset	____________________________ (–6.00e+00 v)
	Air Min Offset	____________________________ (–2.00e+00 v)

BadDetsPeg	Peg Area Ave. Offset	____________________________ (ref. e +04)
	Peg Area Max. Offset	____________________________ (ref. e +04)
	Peg Area Min. Offset	____________________________ (ref. e +04)

All detector Peg Area Values must be within +/– 12% of the system average, else place the offending detector on the bad list.
Specification: All BadDets tests must run to completion without errors.

TEST 3: COHERENT NOISE ANALYSIS (CNA)

CnaBaco	Frequency	Measured Result	Specification
	Any one $\geq$ 30 hz	Highest Value =	$\leq$ 4.00e–06 vp
	White Noise (Total)		$\leq$ 5.0e–05vp

CnaWater	Frequency Range (hz)	Highest Measured Value	Specification (vp)
	10 – 100hz		$\leq$ 3.5e–04 vp
	100 – 399hz		$\leq$ 5.0e–04vp
	400 – 650hz		$\leq$ 3.7e–04vp
	> 650hz		$\leq$ 1.4e–04vp
	White Noise (Total)		$\leq$ 5.0e–03 vp

CnaResolver	Frequency	Measured Result	Specification (arc sec)
	450 cycles/rev		< 3.35 arc–sec p–p
	Any freq other than 450cycles/rev	Highest Value =	$\leq$ 1.67 arc–sec p–p
	White Noise (Total)		< 2.0e+01 arc–sec p–p

TEST 4: BEAM QUALITY – KVHVL

HVL (@130 KV) ________________________	(3.7 to 4.5 mm of aluminum)
HVL (@100 KV) ________________________	extrapolated from 130 KV measurement (2.8 to 3.7 mm)
Calculated KVp __________________________	For reference Only; approx. 130KV)

TEST 5: MTF

	Scan #1 (Small Spot)	Scan #2 (Large Spot)	Scan #3 (1 sec; Small)
50% modulation	__________ (7.0 lp/cm)	___________ (6.8 lp/cm)	__________ (6.8 lp/cm)
10% modulation	__________ (12.0 lp/cm)	___________ (11.5 lp/cm)	__________ (11.5 lp/cm)
5% modulation	__________ (13.0 lp/cm)	___________ (12.6 lp/cm)	__________ (12.6 lp/cm)

TEST 6: SLICE THICKNESS

Nominal Slice Thickness	Results	Specification
8mm		7.2 to 8.8mm
5mm		4.5 to 5.5mm
3mm		2.5 to 3.5mm
2mm		1.5 to 2.5mm

TEST 7: WATER UNIFORMITY AND NOISE

FF Uniformity and Noise (Filter 0)	80KV	100KV	120KV	130KV	140KV
Center Value (0) = (0 +/– 2.0)					
Highest Perif Value (0) =(C.V. +/– 3.0)					
FF (0) Noise Spec (x10)	7.5	7.00	7.00	7.0	7.00
(Max. Std. Dev. in HU)					

TEST 8: LOW CONTRAST RESOLUTION

Low Contrast Measurement	________________ (0.25 to 0.55%)
Noise in Low Con Section	________________ (≤ 1.5 HU std. dev. of water)
Hole Size at 0.35% contrast	________________ (≤ 3.5mm)
CT Number of Water	________________ (0 +/– 5 HU)
CT Number of Acrylic (peg 1)	________________ (120 +/– 10 HU)
CT Number of Polyethylene (peg 2)	________________ (–96 +/– 5 HU)

TEST 9: MULTI–SLICE STUDY CYCLE TIME (30 Axial Slice Sequence)

Time from first to last exposure =	___________ (≤ 85sec.)
Total time from first exposure to last image review =	___________ (≤ 90sec.)

TEST 10: SPIRAL SCAN STUDY CYCLE TIME (30 Spiral Slice Sequence)

Time from first to last exposure =	____________ (≤ 48sec.)
Total time from first exposure to last image review =	____________ (≤ 80sec.)

TEST 11: BACKUP TIMER (OPTIONAL)

TIME SPECIFICATIONS

Description	**Results Obtained**	**Specification**	**Initials**	**Date**
4.2 sec. Scan time	sec	3.79 to 4.61 secs.	NA	NA
2 sec. Scan time	sec	1.80 to 2.19 secs.	NA	NA
Full field pilot	sec	4.20 to 6.00 secs.	NA	NA
Back–up timer	sec	≤ 4.45 secs.	NA	NA
Back–up timer	sec	≤ 2.225 secs	NA	NA

ACQSIM CT PLANNED MAINTENANCE

This manual provides the operational guidelines and planned maintenance checks to be done on Philips Medical System's AcQSim CT system. All system values measured must be within the allowable tolerances noted in the procedure sections. Repair action description and the completion date of the action should also be entered in the Data Sheet. Data sheets are provided separately on paper.

GENERAL

Some procedure intervals (e.g. air filters) may vary slightly per system environmental conditions. Maintenance time should be prearranged with the customer to assure ample time for the system checks without any loss of patient scan time.

The following sections consist of the maintenance procedures to be done during the PM checks. Any parameter measurements made should be within the denoted tolerance, and checked off in the system log book once within specifications. PM frequency is quarterly, but may vary slightly according to specific conditions and customer use.

POWER UP/DOWN PROCEDURE

Power up/power procedures are accessible via a hyperlink to that procedure: Power Up/Down

MA CURVE CALIBRATION

Executed from the AutoCalibration Service Application. Refer to the AUTOCAL Manual.

SPECTRUM (LEVEL) PROGRAM

Executed from the AutoCalibration Service Application. Refer to the AUTOCAL Manual.

ERROR LOGGER

The error log should be reviewed and cleared at each PM. This is done via the Service Applications GUI. The report generated should then be viewed for the validity, frequency, and severity of the errors. Any errors should be addressed during the service call.

PERFORMANCE TESTS

The Quarterly Planned Maintenance for the AcQSim CT requires the Service Engineer to run Standard Performance through Autocal. Refer to the AUTOCAL Manual for procedures and specifications.

Archiver Transfer and Setup

The phone number of WHQ Performance Archiver database is "dictated" by the .MAIN_PHONEHOME_CONF file. This allows changes to the telephone number directly from WHQ Engineering. Use the PhoneHome program to set the performance archiver transfer for once every three months.

GANTRY CLEANING–INTERNAL

Introduction: This procedure is used to perform a general cleaning of the Gantry interior.

Tools Required: FSE Tool Kit
Vacuum Cleaner

Materials Required: P/N 311275

Estimated Time: 15 Minutes

Interior

> ⚠ **WARNING** ⚠
>
> **USE EXTREME CARE WHEN WORKING INSIDE THE GANTRY. 480 & 208 AC, 5 AND 15 VOLTS DC ARE EXPOSED WHEN THE GANTRY COVERS ARE OPEN. ACCIDENTAL CONTACT WITH THESE VOLTAGES MAY CAUSE SERIOUS INJURY OR DEATH TO THE SERVICE PERSONNEL.**

1. Remove Gantry power. Refer to the Power Up/Down procedure. Remove Gantry power at the wall box.
2. Open the top front gantry cover. Remove the bottom front gantry cover. Refer to the Gantry Cover Removal procedure.

> ⚠ **WARNING**
>
> **GANTRY COVERS ARE VERY HEAVY. OPEN GANTRY COVERS MUST BE PROPERLY SECURED OR REMOVED FROM THE IMMEDIATE SERVICE AREA. FAILURE TO COMPLY COULD RESULT IN SERIOUS PERSONAL INJURY.**

3. Use mild, non–abrasive cleaner applied to a cloth to remove soil marks from the painted surfaces.
4. Continue to the Slip Ring Cleaning procedure.

SLIP RING CLEANING (QUARTERS 2 AND 4)

Introduction : This procedure covers cleaning the slip–rings using a vacuum cleaner.

Tools Required: FSE Tool Kit, Vacuum Cleaner

WARNING

USE EXTREME CARE WHEN WORKING INSIDE THE GANTRY. 480 & 208 AC, 5 AND 15 VOLTS DC ARE EXPOSED WHEN THE GANTRY COVERS ARE OPEN. ACCIDENTAL CONTACT WITH THESE VOLTAGES MAY CAUSE SERIOUS INJURY OR DEATH TO THE SERVICE PERSONNEL.

1. Remove Gantry power. Refer to the Power Up/Down procedure. Remove Gantry power at the wall box.
2. Open and secure the top front gantry cover and remove the bottom cover. Refer to the Gantry Cover Removal procedure. Remove the gantry cone. Refer to the Front Cone Removal procedure in the same chapter.

NOTE

The Slip Ring Guard Access Window Cover is held in place by velcro strips. Do not cut the single cable tie that holds the cover.

3. Lift the Slip Ring Guard Access Window Cover and move it aside. Manually rotate the Scan Frame in the CCW direction and vacuum any dust or debris that may have collected in the area.

CAUTION

ONLY ROTATE THE SCAN FRAME IN THE CCW DIRECTION. FAILURE TO COMPLY MAY DO DAMAGE TO THE SLIP RING BRUSHES.

4. Replace the cone and top and bottom covers and return system to normal.

NOTE

The only approved cleaning solvent for the Poly Sci Slip Ring is **90%** isopropyl alcohol. Cramolin Deoxit Wipes and Pink Pearl Erasers are no longer approved.The alcohol is to be purchased locally and must not contain any impurities, such as camphor, menthol.

GANTRY CLEANING EXTERNAL

Materials Required:
1. Spray touch up paint – Philips White P/N T53ZG–77, Philips Blue P/N T53ZG–76
2. General purpose, mild, non–abrasive solvent (409, Fantastic, or Glass Plus.)
3. Cleaning cloth or paper towels
4. 20 x 25 x 1 Air Filter

Estimated Time: 15 minutes

General Cleaning

1. Remove Gantry power. Refer to the Power Up/Down procedure. Remove Gantry power at the wall box.
2. Use solvent cleaner to remove soil marks from painted Gantry surfaces.

Paint Touch Up

1. Clean any remaining surfaces and inspect for scratches in the paint. Touch up any blemishes found using the paints listed above. Be careful not to overspray. To repair small scratches, apply paint using a small brush or cotton swab.

Air Filter

1. Locate the air filter access slots in the rear of the gantry base.
2. Apply Gantry power. Refer to the Power Up/Down procedure.
3. Remove the gantry front bottom cover. (Filters are removed from the front).
4. Position the new air filter into the slot in the gantry base and slide the filter in until it stops.
5. Remove gantry power.

Cables

1. Check cables and harness drapes/routings for possible damage from heat and/or moving components. Reposition and/or secure as required.
2. Check cables for loose or damaged connectors on all power and signal cable/wires. Tighten as required.
3. Check for oil leaks around Federal Connectors. Remove, clean, re–oil both Federal Connectors.
4. Ensure no cables can be moved to touch the detector modules or mother boards.

Tilt Position Drive Belt

1. Check the tilt position drive belt for tension and signs of wear. Belt tension should be snug but not tight. Approximately 1/8” slack is appropriate for belt tension.

Gantry Control Panel

1. Apply gantry power. Refer to the Power Up/Down procedure. Open and secure the top front gantry cover.

> ⚠ **WARNING**
>
> **ROTATING MACHINERY. STAND CLEAR OF ROTATING FRAME WHEN EQUIPMENT IS OPERATED. FAILURE TO COMPLY MAY CAUSE SERIOUS INJURY OR DEATH.**

> ⚠ **WARNING**
>
> **LASER SOURCE PRESENT. DO NOT LOOK DIRECTLY INTO LASER SOURCE. FAILURE TO COMPLY MAY RESULT IN SERIOUS EYE DAMAGE OR BLINDNESS.**

2. Verify the following:
    d. Both Gantry blowers operate.
    e. Vertical, Horizontal, Tilt and Laser displays and controls operate correctly.
    f. Verify that the X–ray Tube Heat Exchanger fans are operating.

GANTRY – VISUAL INSPECTION

1. Visually inspect for loose components.
2. Visually inspect for loose bolts, i.e., lock washers not in compression.

> **NOTE**
>
> If any bolts are in need of tightening, refer to the Repair Manual for torque specs.

Laser Radiation Caution Labels

1. Inspect the four laser radiation caution labels. If any are missing or illegible, they must be replaced.

GANTRY – LUBRICATION (QUARTERS 1 AND 3)

Introduction:	This procedure outlines the steps to be followed for lubrication of the gantry subsystem which is to be done on six month intervals.
Introduction:	
Tools Required:	1. FSE tool kit. 2. Grease Gun with swivel tip.
Material Required:	Mobilgrease 28 – P/N – 95749
Estimated Time:	25 minutes

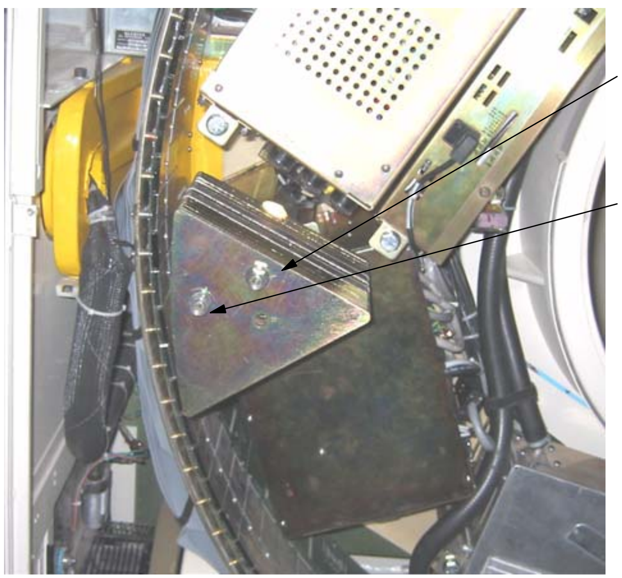
Main Bearing – Grease

1. Remove Gantry power. Refer to the Power Up/Down procedure. Remove Gantry power at the wall box. Manually position the gantry frame such that the x–ray tube is at 12 o'clock position The access hole to the grease fitting is found at the 10 o'clock position (rear of gantry).
2. Using a grease gun, inject a small amount of grease into the fitting. Manually rotate the frame 60 degrees and add another small amount of grease, continue 60 degree rotation and greasing until 360 degrees are covered.

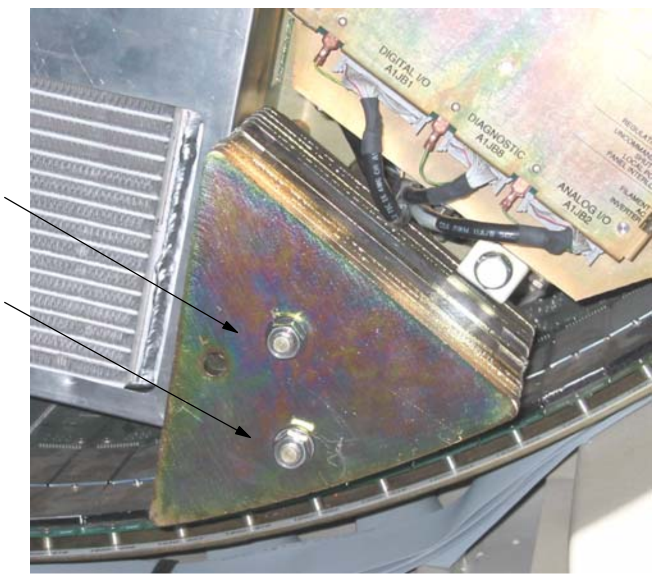
Z–BALANCE COUNTERWEIGHT TORQUE CHECK

Physically check the torque of the 3/4 in. (19.05mm) nuts securing the Z–Balance (triangular–shaped) counterweights to the scanframe standoffs of the AcQSim CT Scanner. The left and right–hand Z–balance counterweight assemblies are shown below. This torque check should be performed at **<u>EVERY</u>** preventive maintenance visit.

1. Turn gantry power off completely.
2. Open the front gantry gullwing cover.
3. Remove the front cone.
4. Locate the Z–Balance (triangular) weights. There are two sets of these triangular weights held in place with two nuts each, 4 total. **Check the torque of each nut and ensure 80 ft–lb. (108 Nm)** These do not require loctite. Ensure the lockwasher is in place and compressed.
5. Install the front cone.
6. Close the front gullwing cover.



ENSURE THESE FOUR NUTS SHOWN ON THE Z–BALANCE COUNTER-WEIGHTS ARE TIGHTENED TO 80 FT–LB (108 N/m AT EVERY PM INTERVAL



ACQIMAGE CABINET AIR FILTER CHECK

Introduction:	This procedure is used to clean or replace the AcQImage Cabinet air filter.
Tools Required:	FSE Tool Kit
Material Required:	AcQImage Cabinet Air filter, P/N 83885 (If replacement is needed)
Estimated TIme:	15 Minutes

1. Remove AcQImage power. Refer to the Power Up/Down procedure.
2. Remove the outside cosmetic cover and inside steel cover. Refer to the AcQImage Cover Removal procedure.
3. Remove the air filter by grasping it's front tab and sliding it out forward. Apply power and verify that the cooling fans are running.
4. The filter is designed to be cleaned and washed. Inspect the filter. If it has an excessive amount of dirt, replace it, otherwise simply clean it. Ensure that the filter is dry before replacing.
5. Replace the covers on the cabinet. Refer to the AcQImage Covers Removal procedure in the Repair manual.

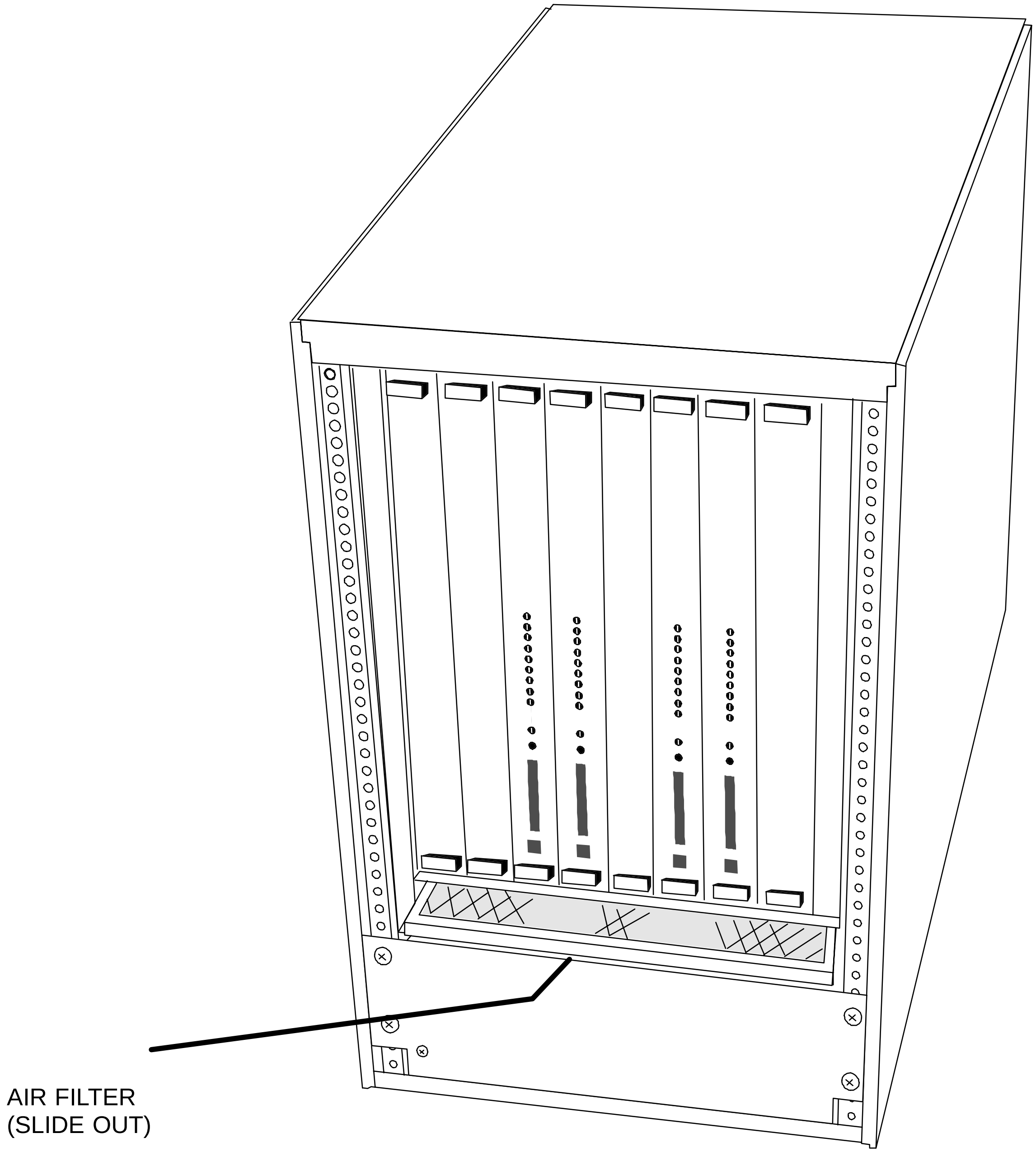


FIGURE 112 ACQIMAGE CABINET AIR FILTER LOCATION

ACQIMAGE CABINET CLEANING – EXTERNAL

Introduction: This procedure is for the cleaning operation to be done on the AcQImage Cabinet.

Materials Required:

1. Spray touch up paint – Light Pearl P/N T53ZG–33 (variation -), Philips White P/N T53ZG–77 (variation A)
2. General purpose, mild, non–abrasive solvent (409, Fantastic, or Glass Plus.)
3. Cleaning cloth or paper towels

Estimated Time: 15 minutes

1. Remove Display Tower power. Refer to the Power Up/Down procedure.
2. Use solvent cleaner to remove soil marks from painted surfaces.

Paint Touch Up

1. Clean any remaining surfaces and inspect for scratches in the paint. Touch up any blemishes found using the paints listed above. Be careful not to overspray. To repair small scratches, apply paint using a small brush or cotton swab.

> ⚠ **CAUTION**
>
> DO NOT APPLY CLEANING SOLUTION DIRECTLY TO THE FACE OF THE SCREEN, THE FACE OF THE CRT MONITOR, KEYBOARD OR TABLE TOP. APPLY THE CLEANER TO A TOWEL FIRST THEN WIPE.

DISPLAY TOWER AIR FILTER CHECK

Introduction: This procedure is used to clean or replace the Display Tower air filter.

Tools Required: None

Material Required: Display Console Air Filter, P/N 83942 (If replacement is needed)

Estimated TIme: 5 Minutes

1. If the Display Tower is close to a wall on the right side, unlock the wheels and roll it out a bit to gain access to the air filter, located on the rear panel.

> **NOTE**
>
> For units that have seismic brackets installed, the supplied template allows for right side air filter clearance of 14”.

2. Grab the air filter and slide it up and out. Verify that both fans are running.
3. The filter is designed to be cleaned and washed. Inspect the filter. If it has an excessive amount of dirt, replace it, otherwise simply clean it. Ensure that the filter is dry before replacing.
4. Roll the Display Tower back into place and lock the wheels.

DISPLAY TOWER CLEANING – EXTERNAL

Introduction:	This procedure is used to perform a general cleaning of the Display Tower exterior and related components.
Tools Required:	FSE Tool Kit
Materials Required:	Spray touch up paint – Philips White P/N T53ZG–77. Soft, lint free, pre–wetted cleaning towelettes. 8mm Tape Cleaning Kit – P/N 95877
Estimated TIme:	15 Minutes

1. Clean any surfaces and inspect for scratches in the paint. Touch up any blemishes found using the paints listed above. Be careful not to overspray. To repair small scratches, apply paint using a small brush or cotton swab.

8mm TAPE DRIVE

2. Inspect the tape drive, ensure that the drive has been cleaned by the operators.

> **NOTE**
>
> The tape drive's top and bottom lights will flash when tape cleaning is required.

3. If the lights indicate that cleaning is needed, insert a cleaning tape and head cleaning will automatically start. The lights will stop flashing and the cleaning tape will be ejected when the cleaning cycle is completed.

CRT MONITOR

4. Clean the viewing monitor with pre–wetted paper towelettes.

KEYBOARD / MOUSE

5. Vacuum the keyboard using a soft bristle brush. Clean the keyboard using pre–wetted paper towelettes.
6. Open the bottom of the mouse (see figure on right) and clean the ball using tap water. If needed, use a mild detergent. Dry with a lint–free cloth. Clean the ball cage inside the mouse. Close your eyes and blow into the ball cage to remove dust.

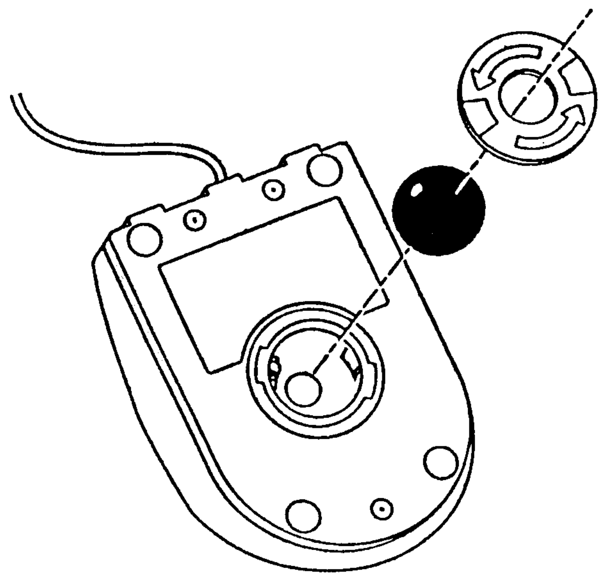


TABLE TOP

7. Clean the table top where required, using pre–wetted paper towelettes.

PAINT TOUCH UP

8. Clean any remaining surfaces and inspect for scratches in the paint. Touch up any blemishes found using the above listed paint. Avoid runs by applying thinner coats more times, rather than trying to cover all in one coat. Be careful of any overspray that may occur. For small scratches, a small brush or cotton swab may be used to apply the paint.

PATIENT SUPPORT CLEANING–EXTERNAL

Introduction: This procedure is used to perform general Patient Support cleaning.

Tools Required: 1. FSE Tool Kit

Materials Required:
1. Spray Paint - Gray P/N T55ZG-33 and Philips White P/N T55ZG–77.
2. General purpose, mild , non–abrasive solvent, such as Formula 409, etc.
3. Cleaning cloth or paper towels.

ESTIMATED TIME: 15 Minutes

General Cleaning

1. Apply cleaning spray to a paper towel to remove soil marks from the exterior of the Patient Support system. Pay specific attention to the side panels that collapse at the couch base.

Paint Touch up

2. Use spray paint to repair any scratches to exterior paint.

PATIENT SUPPORT CLEANING – INTERIOR

Introduction: This section describes the process of cleaning the patient support subsystem. General surface cleaning should be performed.

Tools Required: FSE Tool Kit

Materials Required:
1. Spray Touch Up Paint – Gray P/N T53ZG–33, Philips White P/N T53ZG–77.
2. General purpose, mild , non–abrasive solvent such as Formula 409, etc.
3. General purpose cleaning cloth or paper towels.

Estimated Time: 15 Minutes.

> ⚠ **WARNING**
>
> **THE TELESCOPING BASE COVERS ARE HEAVY AND ARE HELD ONTO THE SUB–FRAME BY FOUR SCREWS. REMOVE THE MOUNTING SCREWS ONLY WHEN THE PATIENT SUPPORT IS AT ITS LOWEST POSITION OR ONLY AFTER THE TELESCOPING BASE COVERS ARE PROPERLY SECURED AND BLOCKED. FAILURE TO COMPLY MAY RESULT IN SERIOUS INJURY TO SERVICE PERSONNEL AND MAY CAUSE EQUIPMENT DAMAGE.**

1. Remove system power. Refer to the Power Up/Down procedure.
2. Remove Patient Support covers. Refer to the Patient Support Covers instructions.
3. Apply system power. Refer to the Power Up/Down procedure.
4. Raise the Patient Support to its maximum height with the Gantry Control panel switches.
5. Remove system power. Refer to the Power Up/Down procedure.
6. Use mild, non–abrasive cleaner applied to a cleaning cloth to remove soil marks from painted interior surfaces.

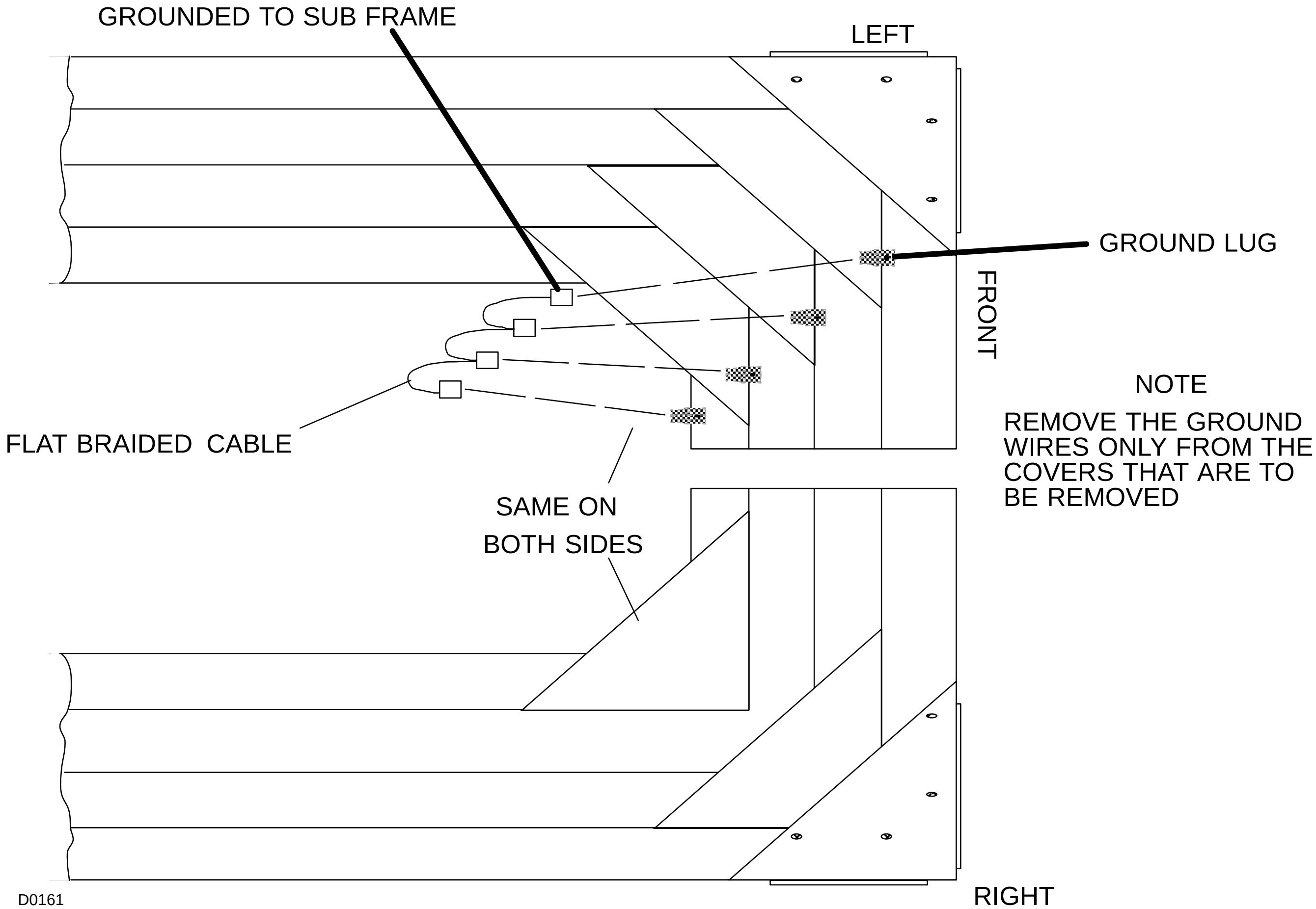


FIGURE 113 TELESCOPING BASE COVERS – GROUND CONNECTIONS

Table Rails and Rollers

> **NOTE**
>
> The tabletop must move freely and smoothly in and out of the Gantry when power is off to the Patient Support, or when power is off to the drive motor. After time the bearings for the carbon top may require additional grease.

PATIENT SUPPORT – INSPECTION

Introduction: This procedure describes the process for inspecting the patient support subsystem for any problems that may have developed over time. This check should include rollers, belt tensions, limit switch assemblies and the scissor/slide shaft parts.

Tools Required: FSE Tool Kit

Materials Required: None

Estimated Time: 15 Minutes

> ⚠ **WARNING** ⚠
>
> **DANGEROUS VOLTAGES MAY BE PRESENT. ENSURE ALL POWER IS REMOVED FROM THE PATIENT SUPPORT SUBSYSTEM BEFORE CONTINUING. FAILURE TO COMPLY MAY RESULT IN SERIOUS INJURY.**

Hardware

1. Remove system power. Refer to the Power Up/Down procedure.
2. Look into the support pedestal area for loose or missing hardware and tighten or replace as needed.

Cables

3. Inspect all cable routings for proper placement and worn or damaged cables. Ensure all connectors are properly seated.

Belts

4. Look up under the support table top and inspect the drive belt in the table resolver assembly for any excessive wear or damage such as grooves or slipping. Refer to Figure 114. Repair/replace as needed.
5. Inspect for dirt or dust on the drive belt and pulleys. Clean drive belt and pulleys with a clean dry cloth if needed.

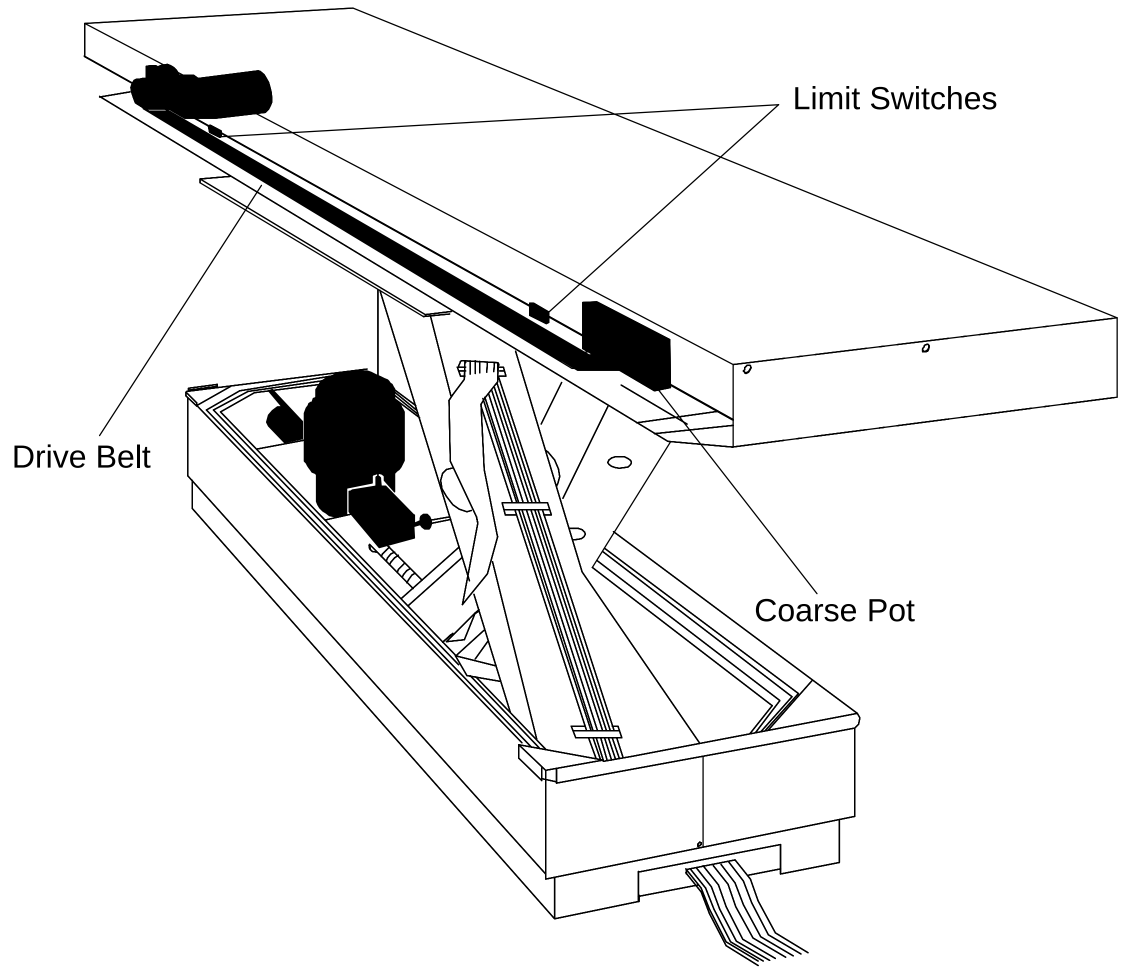


FIGURE 114 INSPECTION POINTS

Table Top Rails and Rollers

With the gantry power OFF and the table raised high enough so that the top can be extended through the patient aperture of the gantry, move the top manually through its entire travel. The action of the table should be smooth, quiet and easy throughout its entire travel.

1. Remove the rail covers. Refer to the Patient Support Covers procedure.
2. Visually inspect the rails for defects or debris.
3. Move the table top fully into then out of the patient aperture. If the carbon top is difficult to move or it sounds like the linear bearings for the top need grease, then remove the front and rear covers and the mechanical stops at the end of the rails. Refer to the Patient Support Covers procedure.
4. Move the carbon top to the rear of the Patient Support to make the grease fittings accessible on the rear linear bearing blocks. Use a grease gun with Mobilgrease 28 and proper fittings and grease the bearing blocks.
5. Move the carbon top to the front of the patient support and repeat the greasing procedure for the front linear bearing blocks..
6. Move the Carbon top fully out of the patient aperture and replace the mechanical stops and the front and rear covers.

PATIENT SUPPORT – LUBRICATION – (ANNUALLY)

Introduction	This procedure explains the procedure for lubrication of the Patient Support Subsystem.
Tools Required:	FSE Tool Kit
Materials Required:	1. Paper towels 2. Dow Corning G–n paste lubricant – P/N 81709 3. Mobilgrease 28 – P/N 95749
Estimated Time:	15, Minutes

> **NOTE**
>
> Use a paper towel or some other suitable material or tool to apply grease to the ball–screw.

BALL SCREW/BEARING

1. Remove system power. Refer to the Power Up/Down procedure.
2. Use a paper towel to remove any old, dirty lubricant from the ballscrew and bearing assembly shown in Figure 115. Use a paper towel to apply a conservative amount of Mobilgrease 28 to the ballscrew and wipe away the excess.

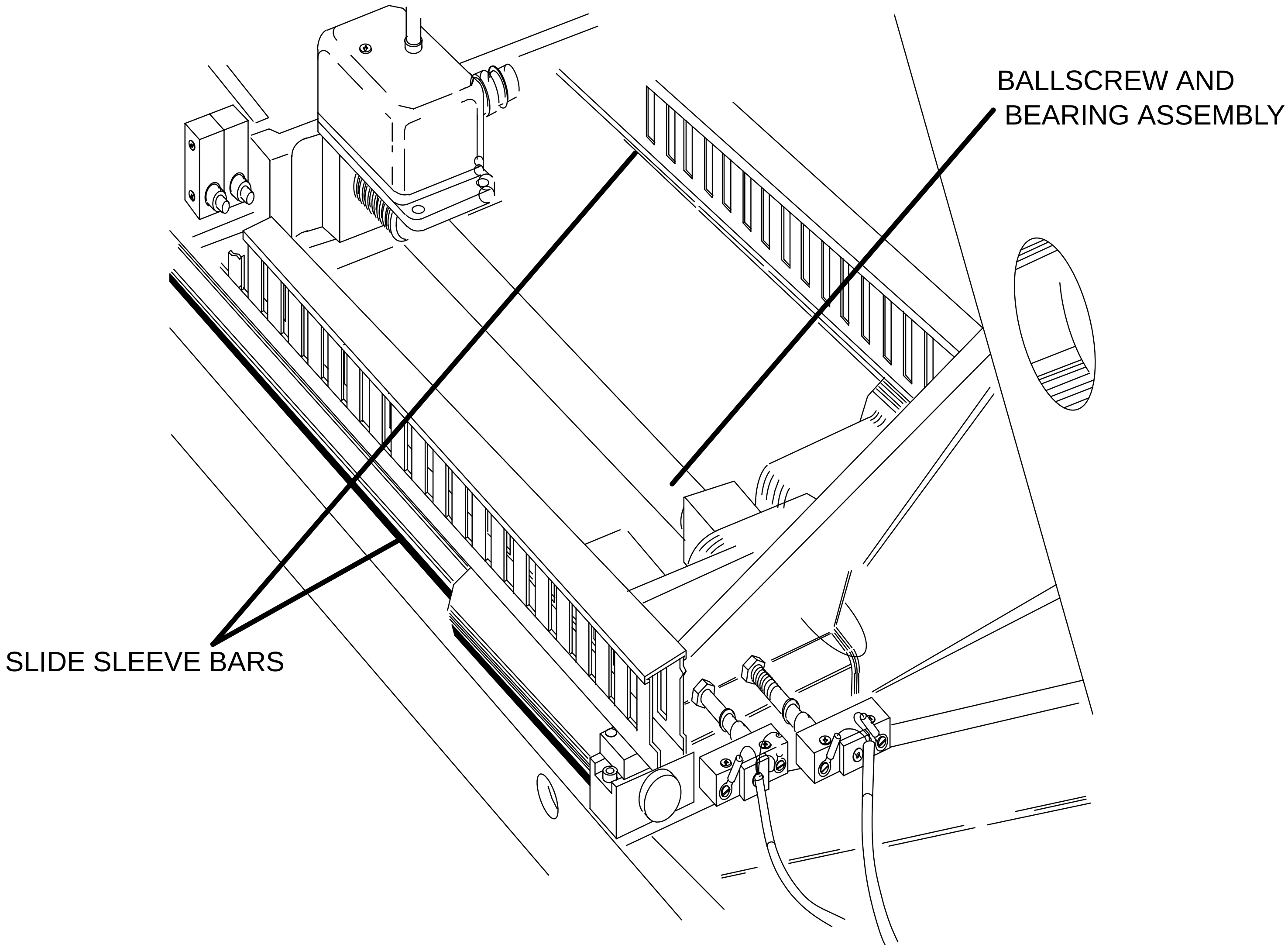


FIGURE 115 DRIVE SYSTEM LUBRICATION

Slide Sleeve Bars

3. Use a paper towel to wipe the slide sleeve bars completely.

4. Either brush or wear rubber gloves and apply by hand a thin layer of Dow Corning G–n paste to all surfaces of the slide sleeve bars. Refer to Figure 115.

VERTICAL TRAVEL LIMIT SWITCHES – (quarters 2 and 4)

1. Check the Up and Down Limit Switches by depressing the Gantry Control Panel switches and as the table is moving depress the limit switch (use a rod to reach the limit switch) and verify that motion stops.

> **NOTE**
>
> An alterative method of testing the limit switches is to apply a cable tie around the switch so that the switch is closed and verifying no motion occurs when the respective Gantry Control Switches are depressed.

COVER RE–ASSEMBLY

1. Lower the Patient Support to its lowest position.
2. Remove system power. Refer to the Power Up/Down procedure.
3. Carefully replace the side panels. Connect the ground wires to the side panel surfaces. Refer to Figure 113.
4. Replace the 2 screws in both the outer cover ends. Front and rear (4 total). Refer to Figure 116.
5. Replace the four phillips head screws and washers that attach the telescoping base cover panels to the subframe. Refer to Figure 116.

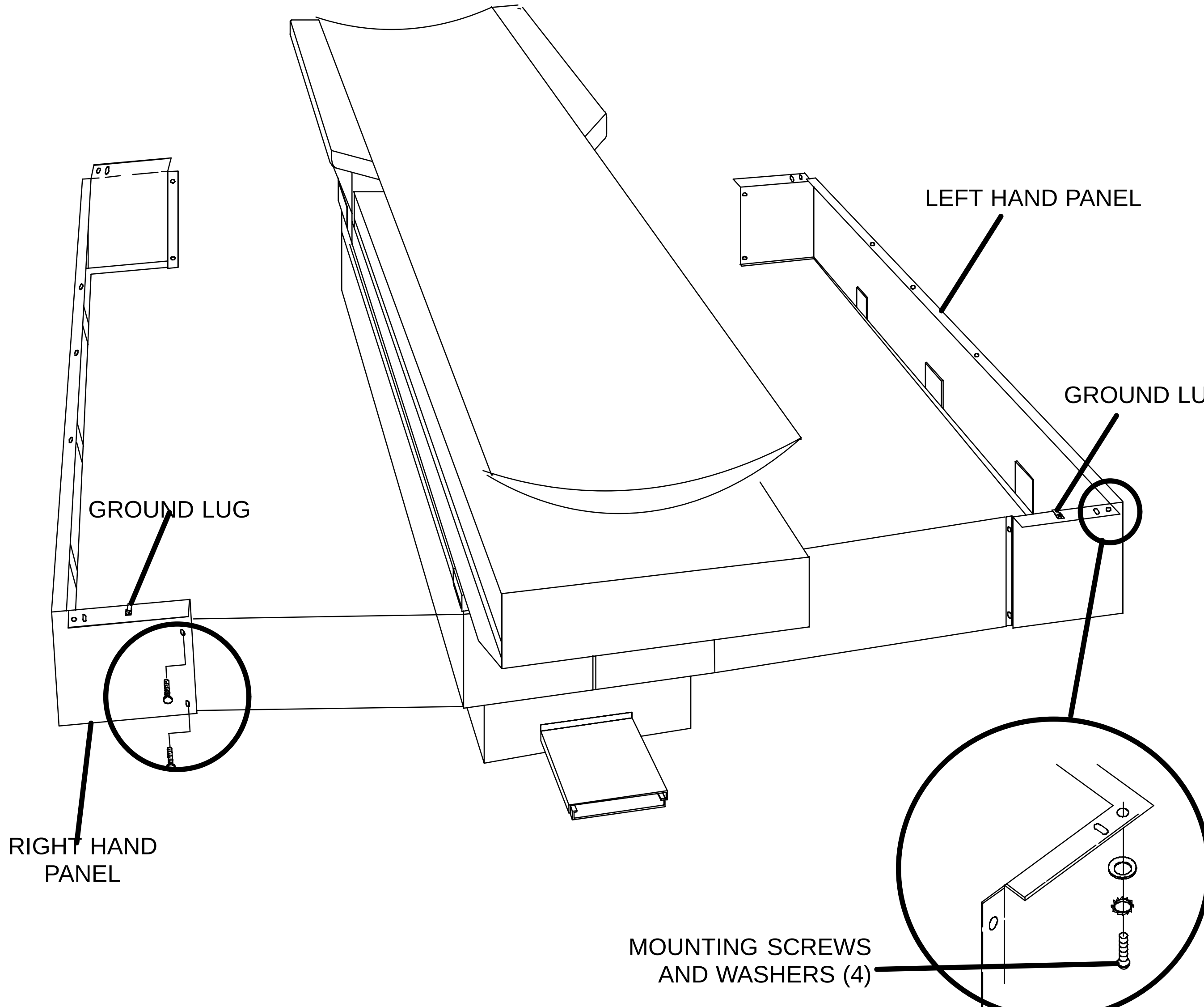


FIGURE 116 PATIENT SUPPORT TELESCOPING BASE COVERS

PM LOG

Introduction

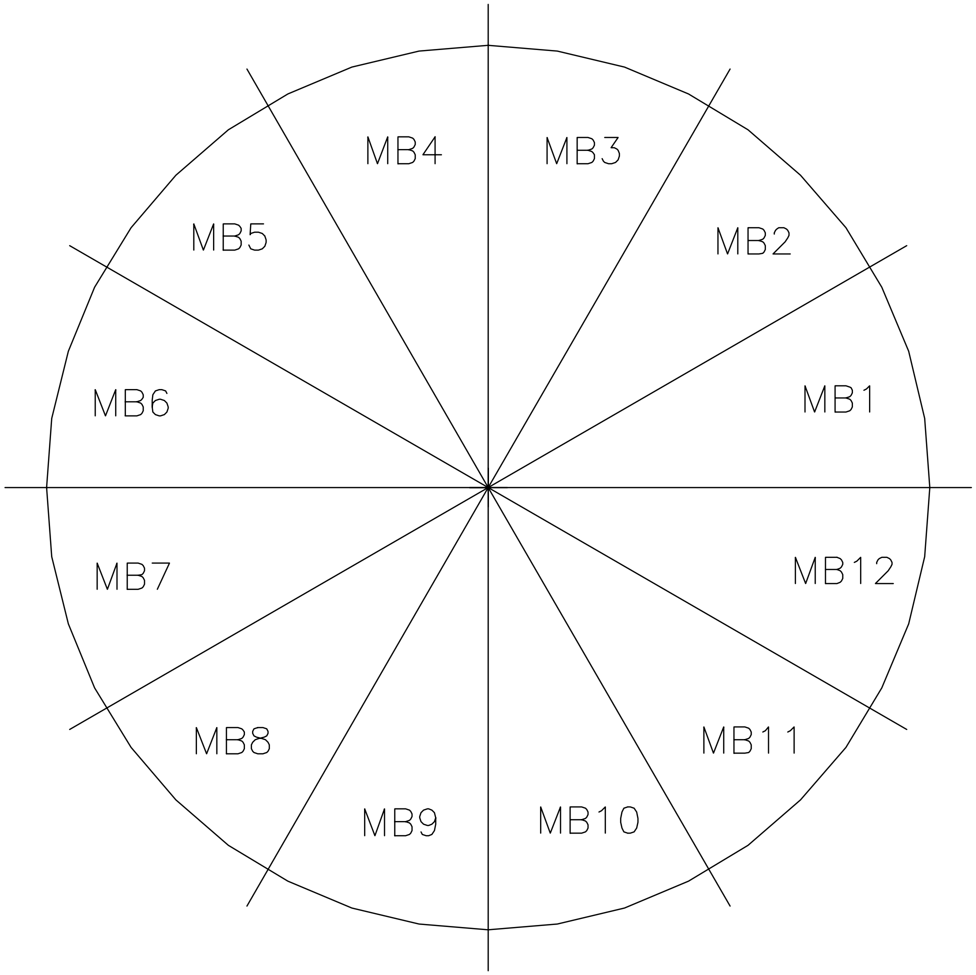
Results of the PM procedures should be recorded on the data sheets provided in the Data Section. If a parameter measurement is required, the value measured should be noted in the comments and calibrated to specification as needed. When action is required, a check mark should be entered in the ”Action Item” column on the Data Sheet. When the action item is completed, the date should be entered in the ”Data Corrected” column. Any additional comments should be recorded on the data sheets.

A copy of the completed log sheets is to be kept in the log book at the customer site and a copy is to be sent to the District Office.

Each quarterly PM will take approximately 4–5 hours to complete. This time will vary from unit to unit dependent upon problems found.

PM Log Sheet – AcQSim CT

AcQSim CT Log Sheet		Q1 Date Initials: ________	Q2 Date Initials: ________	Q3 Date Initials: ________	Q4 Date: Initials: ________	Date Corrected	Action Item
No.	**PROCEDURE**						
1	Boot System						
2	MA Curve Calibration						
3	Spectrum (Level) Prog.						
4	Error Logger						
5	Performance Tests						
6	Internal Gantry Cleaning						
7	Slip Ring Cleaning						
8	External Gantry Cleaning						
9	Gantry Visual Inspection						
10	Gantry Lube (annual)						
11	Z–Balance CW Check						
12	AcQImage Cabinet Filter						
13	AcQ. Cabinet Cleaning						
14	Display Tower Filter						
15	Display Tower Cleaning						
16	Couch External Cleaning						
17	Couch Internal Cleaning						
18	Couch – Inspection						
19	Couch – Lube (Annual)						

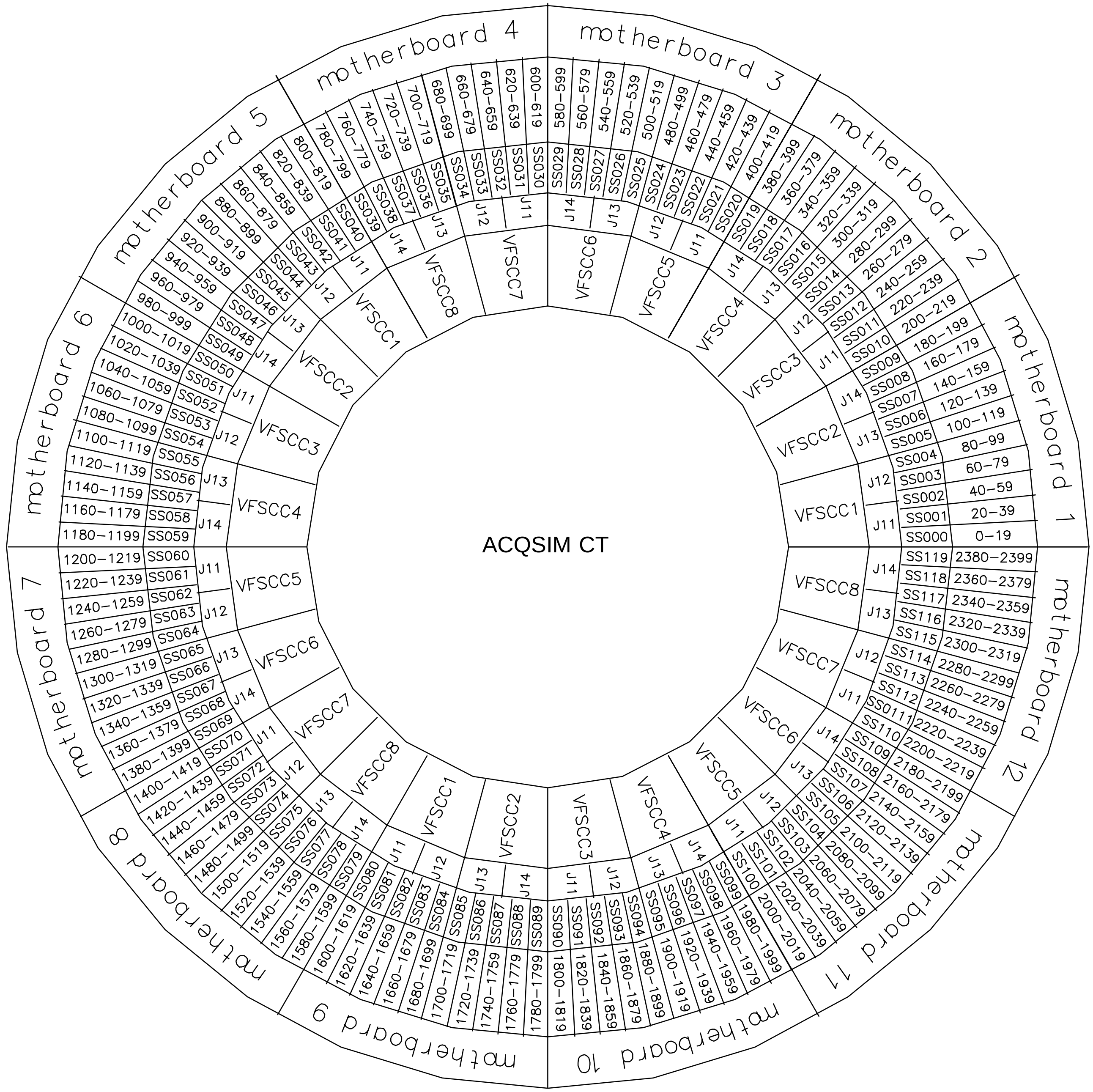


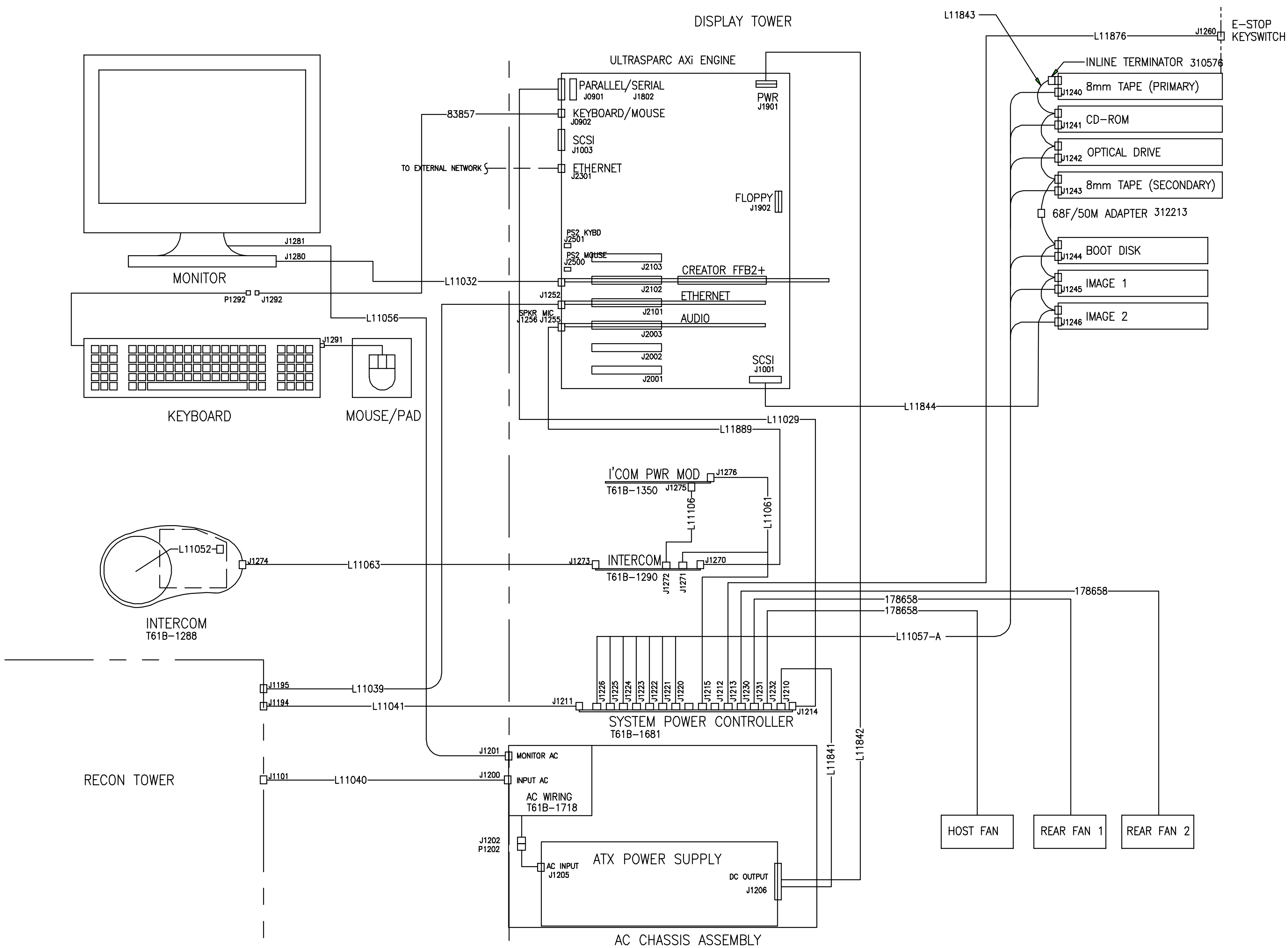
BANK A = MB1, MB5, MB9

BANK B = MB2, MB6, MB10

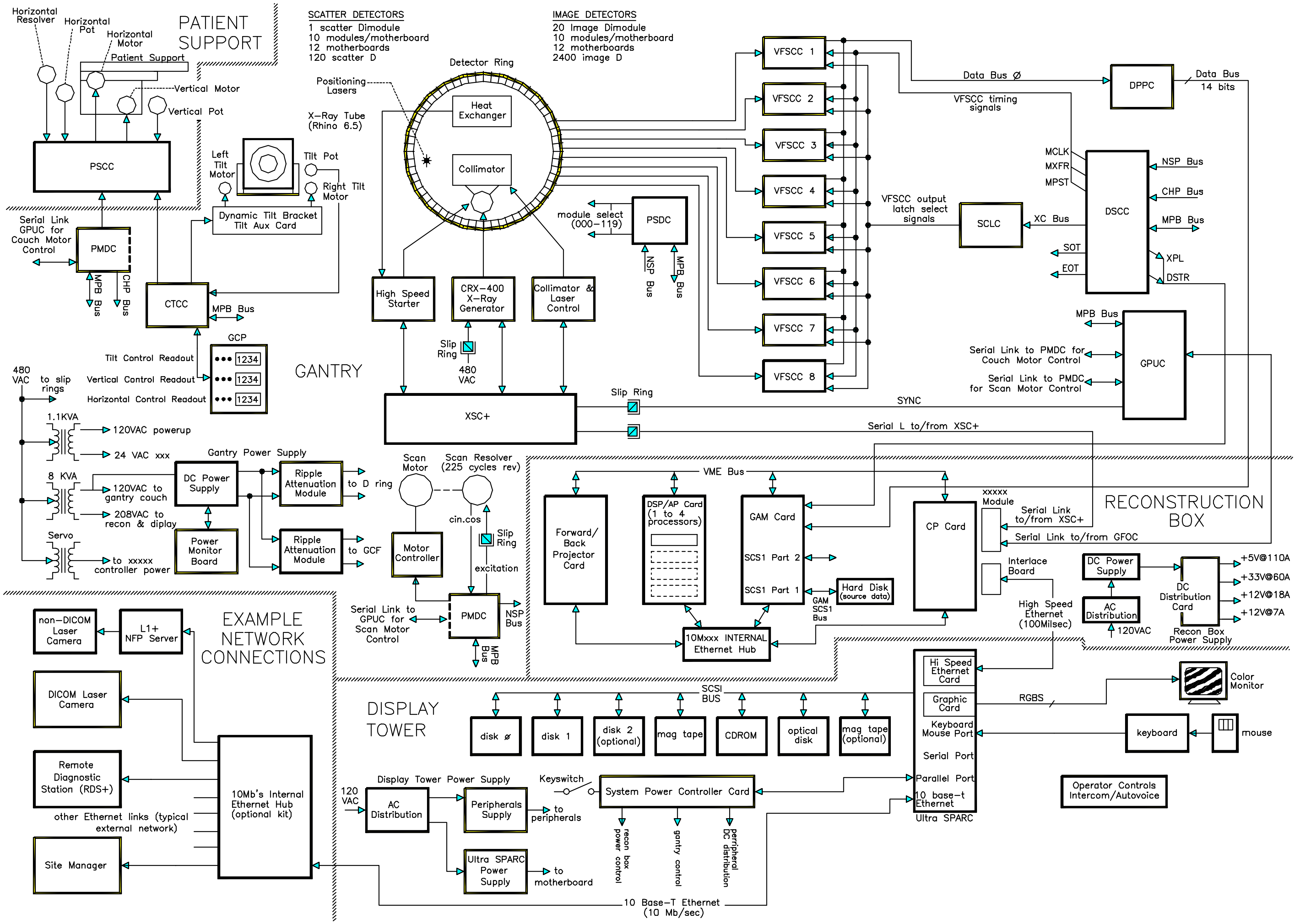
BANK C = MB3, MB7, MB11

MOTHERBOARD BANK DEFINITION
FOR POWER MONITORING


ACQSIM CT
motherboard 1
motherboard 2
motherboard 3
motherboard 4
motherboard 5
motherboard 6
motherboard 7
motherboard 8
motherboard 9
motherboard 10
motherboard 11
motherboard 12
VFSCC1
VFSCC2
VFSCC3
VFSCC4
VFSCC5
VFSCC6
VFSCC7
VFSCC8
J11
J12
J13
J14



DISPLAY TOWER
ULTRASPARC AXi ENGINE
PARALLEL/SERIAL
J0901
J1802
KEYBOARD/MOUSE
J0902
SCSI
J1003
ETHERNET
J2301
TO EXTERNAL NETWORK
PWR
J1901
FLOPPY
J1902
PS2 KYBD
J2501
PS2 MOUSE
J2500
J2103
CREATOR FFB2+
J2102
ETHERNET
J2101
AUDIO
J2003
J2002
J2001
SCSI
J1001
SPKR MIC
J1256 J1255
J1252
83857
L11032
L11056
MONITOR
J1281
J1280
P1292
J1292
KEYBOARD
J1291
MOUSE/PAD
L11843
L11876
J1260
E-STOP
KEYSWITCH
INLINE TERMINATOR 310576
J1240
8mm TAPE (PRIMARY)
J1241
CD-ROM
J1242
OPTICAL DRIVE
J1243
8mm TAPE (SECONDARY)
68F/50M ADAPTER 312213
J1244
BOOT DISK
J1245
IMAGE 1
J1246
IMAGE 2
L11844
L11029
L11889
I'COM PWR MOD
T61B-1350
J1276
J1275
L11106
L11061
L11052
J1274
L11063
J1273
INTERCOM
T61B-1290
J1272
J1271
J1270
INTERCOM
T61B-1288
178658
178658
178658
L11057-A
J1195
L11039
J1194
L11041
J1211
J1226
J1225
J1224
J1223
J1222
J1221
J1220
J1215
J1212
J1213
J1230
J1231
J1232
J1210
J1214
SYSTEM POWER CONTROLLER
T61B-1681
L11841
L11842
RECON TOWER
J1201
MONITOR AC
J1101
L11040
J1200
INPUT AC
AC WIRING
T61B-1718
J1202
P1202
ATX POWER SUPPLY
AC INPUT
J1205
DC OUTPUT
J1206
AC CHASSIS ASSEMBLY
HOST FAN
REAR FAN 1
REAR FAN 2


PATIENT SUPPORT
Horizontal Resolver
Horizontal Pot
Horizontal Motor
Patient Support
Vertical Motor
Vertical Pot
PSCC
Serial Link GPUC for Couch Motor Control
PMDC
MPB Bus
CHP Bus
CTCC
MPB Bus
Left Tilt Motor
Tilt Pot
Right Tilt Motor
Dynamic Tilt Bracket Tilt Aux Card
SCATTER DETECTORS
1 scatter Dimodule
10 modules/motherboard
12 motherboards
120 scatter D
IMAGE DETECTORS
20 Image Dimodule
10 modules/motherboard
12 motherboards
2400 image D
Detector Ring
Positioning Lasers
X-Ray Tube (Rhino 6.5)
Heat Exchanger
Collimator
module select (000-119)
PSDC
NSP Bus
MPB Bus
High Speed Starter
CRX-400 X-Ray Generator
Collimator & Laser Control
Slip Ring
480 VAC
GCP
Tilt Control Readout
Vertical Control Readout
Horizontal Control Readout
1234
GANTRY
480 VAC
to slip rings
1.1KVA
120VAC powerup
24 VAC xxx
8 KVA
120VAC to gantry couch
208VAC to recon & diplay
Servo
to xxxxx controller power
Gantry Power Supply
DC Power Supply
Power Monitor Board
Ripple Attenuation Module
to D ring
to GCF
XSC+
Slip Ring
SYNC
Serial L to/from XSC+
VFSCC 1
VFSCC 2
VFSCC 3
VFSCC 4
VFSCC 5
VFSCC 6
VFSCC 7
VFSCC 8
Data Bus Ø
DPPC
Data Bus 14 bits
VFSCC timing signals
MCLK
MXFR
MPST
DSCC
NSP Bus
CHP Bus
MPB Bus
XPL
DSTR
SOT
EOT
VFSCC output latch select signals
SCLC
XC Bus
MPB Bus
GPUC
Serial Link to PMDC for Couch Motor Control
Serial Link to PMDC for Scan Motor Control
Scan Motor
Scan Resolver (225 cycles rev)
cin.cos
Slip Ring
excitation
Motor Controller
PMDC
Serial Link to GPUC for Scan Motor Control
NSP Bus
MPB Bus
VME Bus
Forward/ Back Projector Card
DSP/AP Card (1 to 4 processors)
GAM Card
SCS1 Part 2
SCS1 Part 1
GAM SCS1 Bus
Hard Disk (source data)
CP Card
xxxxx Module
Serial Link to/from XSC+
Serial Link to/from GFOC
Interlace Board
RECONSTRUCTION BOX
10Mxxx INTERNAL Ethernet Hub
High Speed Ethernet (100Milsec)
DC Power Supply
AC Distribution
120VAC
DC Distribution Card
+5V@110A
+33V@60A
+12V@18A
+12V@7A
Recon Box Power Supply
EXAMPLE NETWORK CONNECTIONS
non-DICOM Laser Camera
L1+ NFP Server
DICOM Laser Camera
Remote Diagnostic Station (RDS+)
other Ethernet links (typical external network)
Site Manager
10Mb's Internal Ethernet Hub (optional kit)
DISPLAY TOWER
SCSI BUS
disk ø
disk 1
disk 2 (optional)
mag tape
CDROM
optical disk
mag tape (optional)
Hi Speed Ethernet Card
Graphic Card
Keyboard Mouse Port
Serial Port
Parallel Port
10 base-t Ethernet
Ultra SPARC
RGBS
Color Monitor
keyboard
mouse
Operator Controls Intercom/Autovoice
Display Tower Power Supply
120 VAC
AC Distribution
Peripherals Supply
to peripherals
Ultra SPARC Power Supply
to motherboard
Keyswitch
System Power Controller Card
recon box power control
gantry control
peripheral DC distribution
10 Base-T Ethernet (10 Mb/sec)

ACQSIM CT DIAGNOSTICS

DISPLAY CONSOLE DIAGNOSTICS

The AcQSim CT Display Console is equipped with several different peripherals (e.g., CD–ROM, 8mm tape, etc). Diagnostics are detailed here to help you verify / troubleshoot a problem associated with some of the peripherals and other hardware areas. The 8mm tape, CD–ROM drive verifications and 10MB ethernet port and network tests may be used to troubleshoot/diagnose RDS+ problems.

The following verification procedures are:

8MM TAPE DRIVE TEST

> **NOTE**
>
> This test will verify that the 8mm tape drive hardware is properly installed and functioning. It will verify that a file can be transferred to and read back from the tape drive.

> **NOTE**
>
> This test requires at least a functional Display Tower or RDS+ loaded with applications software on disk0. Only use 8mm tapes that have not been written to by older 8mm tape drives on other systems.

Required: 96033 – 8mm data cartridge, 54 meter

1. From the Service Diagnostic GUI select **XTERM** from the UTILITIES menu. To start the Diagnostic GUI, refer to the Service Application manual.
2. Install a non–write protected 8mm tape into the tape drive and wait for the green "ready" LED.

COPY FILE TO TAPE:

> **NOTE**
>
> This creates a new tape. It will write over any data currently on the tape.

If testing the primary 8mm tape drive:	**If testing the secondary 8mm tape drive**:
At the % prompt, type **tar –cvf /dev/rst29 /kernel/genunix** and press <CR>.	At the % prompt type **tar –cvf /dev/rst28 /kernel/genunix** and press <CR>.

This will copy the file named *genunix* from the kernel directory onto the 8mm tape.

COPY FILE FROM TAPE TO DIR:

3. At the % prompt, type **cd /tmp** and press <CR>. This changes to the **tmp** directory. The tmp directory will be used to copy the file from the tape to this directory.

> **NOTE**
>
> The /tmp directory is always emptied at boot up.

To test the primary 8mm tape drive:	**To test the secondary 8mm tape drive**:
At the % prompt, type **tar –xvf /dev/rst29 genunix** and press <CR>.	At the % prompt type **tar –xvf /dev/rst28 genunix** and press <CR>.

This will extract the file *genunix* to the tmp directory.

6) At the % prompt type **ls** and press <CR>. This will print the tmp directory to the screen. You will see the genunix file listed.

VERIFY FILE TRANSFERRED SUCCESSFULLY

4. At the % prompt type **cmp /kernel/genunix genunix** and press <CR>.

The **cmp** command will compare the files from the original to the one transferred from the 8mm tape. Any difference in the two files will show up on the screen. (Files differ: char x line x; x is the position in the file). There should be no differences.

5. The test is now complete. You may eject the tape. At the **%** prompt type **exit** and press <CR> key to leave the xterm window. You may now exit the Diagnostic GUI.

OPTICAL DRIVE TEST

> **NOTE**
>
> This test will verify whether the magneto–optical disk hardware is properly installed and functional. It will verify that files can be transferred to the optical drive and read back successfully. This test requires at least a functional display tower with application software running on disk0.

Requires: Magneto–optical media 83977

1. Insert a magneto–optical disk into the Optical drive.
2. Launch the Diagnostic GUI. To start the Diagnostic GUI, refer to the Service Application manual. Select **xterm** from the UTILITIES menu.
3. Enter superuser mode by typing **su** and press the RETURN key. At the password prompt, type **picker** and press <CR>.
4. If the magneto–optical media has a file system already, then go to the **Verify Optical Drive** section. If you are unsure, perform the following procedure to determine:

   At the # prompt type **mkdir /optical** and press <CR>.
   At the # prompt type **mount /dev/dsk/c0t3d0s2 /optical** and press <CR>.

   If the # prompt appears without errors go to **Verify Optical Drive**.

   If you get an error then you must perform the **Create a File System** procedure.

Create a File System

> **NOTE**
>
> This procedure is run if errors were encountered in the file system procedure in step 4. If no errors were encountered and you know the magneto–optical has a file system already, then go to the **Verify Optical Drive** section.

1. At the # prompt type **format** and press <CR>.

2. At the *Available disk* selection prompt, specify the disk. Enter the selection for **c0t3d0** from the menu (Usually selection **2**).
3. At the *Available drive types* prompt, select the number for “**other.**” (Usually **16**)
4. At the *Enter number of data cylinders:* prompt, type **1773** and press <CR>.
5. At the *Enter number of alternate cylinders: [2]* prompt, press <CR>.
6. At the *Enter number of physical cylinders: [1775]* prompt, press <CR>.
7. At the *Enter number of heads*: prompt, type **16** and press <CR>.
8. At the *Enter physical number of heads [default]:* prompt, press <CR>.
9. At the *Enter number of data sectors/track:* prompt, type **78** and press <CR>.
10. At the *Enter number of physical sectors/track [default]:* prompt, press <CR>.
11. At the *Enter RPM of drive [3600]:* prompt, press <CR>.
12. At the *Enter format time [default]:* prompt, press <CR>.
13. At the *Enter cylinder skew [default]:* prompt, press <CR>.
14. At the *Enter track skew [default]:* prompt, press <CR>.
15. At the *Enter tracks per zone [default]:* prompt press <CR>.
16. At the *Enter alternate tracks [default]:* prompt press <CR>.
17. At the *Enter alternate sectors [default]:* prompt, press <CR>.
18. At the *Enter cache control [default]:* prompt, press <CR>.
19. At the *Enter prefetch threshold [default]:* prompt, press <CR>.
20. At the *Enter minimum prefetch [default]:* prompt, press the <CR>.
21. At the *Enter maximum prefetch [default]:* prompt, press <CR>.
22. At the *Enter disk type name (remember quotes):* prompt, type **“C1113F_1GB”** and press <CR>.

The prompt should be at the FORMAT menu

23. Type **partition** and press <CR>.

24. At the *partition* prompt, enter a **2** and press <CR>.
25. At the *Enter partition id tag [backup]:* prompt, press <CR>.
26. At the *Enter partition permission flags [wu]:* prompt, press <CR>.
27. At the *Enter new starting cy [0]:* prompt, press <CR>.
28. At the *Enter partition size [0b, 0c, 0.00mb]:* prompt, type **2212704b** and press <CR>.
29. At the *partition* prompt, type **quit** and press <CR>.

The prompt should be at the FORMAT menu.

30. At the format prompt, type **label** and press <CR>.
31. When the prompt *Ready to label disk, continue?* appears, type **y** and press <CR>.

The label will be written to the magneto–optical disk.

32. At the *format* prompt, type **quit** and press <CR>. The # prompt should appear.
33. At the # prompt, type **newfs /dev/rdsk/c0t3d0s2** and press <CR>.
34. At the prompt *newfs: Construct a new file system /dev/dsk/c0t3d0s2 (y/n)?,* type **y** and press <CR>.

The file system creation takes about 3 minutes. You will see the block numbers as the file system is created. At completion the # prompt will appear again.

Verify Optical Drive

> **NOTE**
>
> If you have performed the step to determine if you have a file system on your magneto–optical media then skip steps 1. and 2.

1. Type **mkdir /optical** and press <CR>. This will create a directory named optical. It will be used as the mount point for the optical disk.
2. Type **mount /dev/dsk/c0t3d0s2 /optical** and press <CR>. This mounts the optical disk to the /optical directory.
3. At the # prompt, type **cp /etc/hosts /optical** and press <CR>. This copies the file named hosts from the /etc directory to the magneto–optical disk in /optical.
4. At the # prompt, type **ls /optical** and press<CR>. This will let you see the contents of directory /optical. You should see hosts as one of the files.
5. At the # prompt, type **cp /optical/hosts /tmp/tmphosts** and press <CR>. This will copy the file on the optical disk to the /tmp directory as a file named tmphosts.
6. At the # prompt, type **cmp /etc/hosts /tmp/tmphosts** and press <CR>. This will compare the original file hosts with the file copied to and from the magneto–optical disk named tmphosts in the /tmp directory.
7. If no errors are seen, the compare files command passes and the optical drive is functional. Any errors will show a difference in the two files. (files differ: char x, line x, where x is the char and line number)
8. At the # prompt, type **rm /optical/hosts** and press <CR>. This will delete the file hosts from the magneto–optical disk.
9. At the # prompt, type **rm /tmp/tmphosts** and press <CR>. This will delete the file copied from the magneto–optical disk to the /tmp directory.
10. At the # prompt, type **umount /optical** and press <CR>to unmount the optical drive from the directory /optical.
11. At the # prompt, type **rm –r optical** and press <CR>. This removes the directory named *optical* from the hard disk.
12. Press the eject button on the optical drive to eject the media. If that doesn’t work, type **eject /dev/dsk/c0t3d0s2** and press <CR>.
13. At the # prompt, type **exit** and press <CR> to leave the xterm window. You may now exit the Diagnostics GUI.

CD–ROM DRIVE VERIFICATION

> **NOTE**
>
> This test will verify whether the CD–ROM drive hardware is properly installed, functioning or not. It will verify that the drive can read data from a CD–ROM disk.

1. On boot–up of the system. When the video begins to appear, you will see:

   ```
   initializing memory.....
   ```

2. At that point, press the **STOP** and **A** keys on the keyboard. The “ok” prompt will appear.
3. Press the eject button on the CD–ROM drive to open its drawer. Insert a Software install Release CD and press the button again to close the drawer.
4. At the “ok” prompt, type **boot cdrom** and press <CR>. The disk should start to be read.
5. Verification is accomplished if the CD–ROM drive reads the CD and the Software Installation main menu appears on the screen.
6. To exit, eject the CD, remove the CD and close the CD drawer.
7. Press **STOP** and **A**. At the “ok” prompt, type boot and press <CR>. This will reboot the system.

AUTOVOICE TEST

> **NOTE**
>
> The AutoVoice is tested via the Diagnostic GUI.

10MB ETHERNET PORT AND NETWORK TESTS

> **NOTE**
>
> This test will verify whether the 10mbit external Ethernet port is functional and can communicate with Ethernet devices at external network IP addresses.
>
> This test assumes that the Display Console Diagnostic GUI is running (or the RDS+ Diagnostic GUI is running if the test is being performed on an RDS+ workstation). The Ethernet port must be locally configured and connected to a working network.

1. Enter the Diagnostics GUI Utility menu. To start the Diagnostic GUI, refer to the Service Application manual. Select **XTERM** from the UTILITIES menu. This will open up a terminal screen.
2. At the % prompt, enter the command **ping** and the IP address of another working node on the network and press <CR>. (Example– **ping** *199.99.1.1* <CR>). The node will respond with a message conveying that it is “alive” if the ping command was received. This will confirm that your Ethernet port is working and the path to the other node is working as well.

> **NOTE**
>
> Do not use your own local IP address as a destination for a ping as this will not adequately test your local SPARC port although it may show an “is alive” message.

3. To perform a more rigorous test of your network port and the network connections, you may execute the following repetitive test. This test will do a multiple pass ping test with a larger size packet. This test will take a few minutes to complete. You can watch the progress on your screen as it goes through the multiple passes. At the end, there will be a summary report presented outlining the results of the test. This may help in diagnosing occasional or intermittent failures.

To launch the test at the % prompt, enter:

ping –s *IP address* **1000 100** and press <CR>.

This will cause a message packet size of 1000 to be sent 100 times.

4. At the # prompt, type **exit** and press <CR> to leave the xterm window. You may now exit the Diagnostics GUI.

100MB ETHERNET LINK TESTS

> **NOTE**
>
> This test will verify whether the 100mbit link hardware is functional between the Display Console and the AcQImage cabinet.
>
> This test assumes that the Display GUI is running and the AcQImage cabinet has been powered.

1. Enter the Diagnostics GUI Utility menu. To start the Diagnostic GUI, refer to the Service Application manual. Select xterm from the UTILITIES menu. This will open up a terminal screen.
2. At the % prompt, type **ping 144.54.50.2** and press the RETURN key.
3. The 100mb hardware link is functional if an "*is alive*" message is returned.
4. To perform a more rigorous test of your link you may execute the following repetitive test. This test will do a multiple pass ping test with a larger size packet. This test will take a few minutes to complete. You can watch the progress on your screen as it goes through the multiple passes. At the end, there will be a summary report presented outlining the results of the test. The test should report 100% functionality. This may help in diagnosing occasional or intermittent failures.

To launch the test at the % prompt, enter:

ping –s 144.54.50.2 1000 100 and press <CR>.

This will cause a message packet size of 1000 to be sent 100 times.

5. To exit, close the xterm window by typing **exit** and press <CR>. You may now exit the Diagnostic GUI.

FORTH–BASED PROM DIAGNOSTICS (SPARC)

Introduction

The AcQSim CT Display Tower is powered by an Ultra Sparc series computer. In the event that system diagnostics cannot be run due to the system not booting, Forth–Based PROM Diagnostics can be run.

1. Forth Based diagnostics are started just after the system is powered. Ensure that the monitor power is turned ON. Just as the system starts to boot, you will see a revolving cursor – press the **STOP and A keys**. The system will stop booting and the "ok" prompt will be displayed.

> ⚠ **WARNING**
>
> **THE DIAG–SWITCH? MUST BE SET TO FALSE AFTER THE DIAGNOSTICS ARE COMPLETED. FAILURE TO COMPLY CAN RESULT IN UNCOMMANDED GANTRY MOTION AND POSSIBLE INJURY OR DEATH TO SERVICE PERSONNEL. (SEE DIAG–SWITCH? INSTRUCTIONS BELOW)**

The Diag–switch? parameter must be set to **false** after the diagnostics are completed. This can be be confirmed as follows:

1. At the OK prompt, type **printenv** and press <CR>.
2. Press the space bar to scroll through all of the environment settings. The last environment parameter in the list is the diag–switch? status.
3. The first column will display either true or false. Set the diag–switch? parameter to false by typing **setenv diag–switch? false** and press <CR>. The response will be "diag–switch? = false."

On–Board Diagnostics Available

The tables below list useful Forth–Based PROM Diagnostics on the AcQSim CT:

Test	Description	Preparation	When to Use
probe–scsi	Returns the SCSI devices (internal/ external) and their SCSI targets connected to the built–in SCSI port.	Connect external SCSI devices to the system and turn on their power.	Determine if SCSI periph. is “talking” to the system.

1. At the OK prompt, type **probe–scsi** and press <CR>.
2. The following note will be displayed:

```
This command may hang the system if a Stop–A or halt command has been executed.
Please type reset–all to reset the system before executing this command.
Do you wish to continue?  (y/n)
```

3. Press **y** and <CR>. After a pause (up to 30 sec), similar results will be displayed:

```
Target 0
  Unit 0    Disk                    HP          C3725S      5153
Target 1
  Unit 0    Disk                    HP          C3725S      5153
Target 3
  Unit 0    Removeable Disk         HP          C1113F      1.48
Target 5
  Unit 0    Removeable Tape         Exabyte     EXB–8505HE–00000085
Target 6
  Unit 0    Removeable Read Only Device         Plextor CD–ROM PX–12TS  1.02
```

Test	Description	Preparation	When to Use
test screen	Tests the system video graphics hardware and monitor	The diag–switch? NVRAM parameter must be set to true.	When video is not fully operational.

1. Type test screen and press <CR>.
2. A similar screen will be displayed:

```
Testing cgsix
FBC register test
Font test
            Data lines test – succeeded.
            Address quick test – succeeded.
            Data size test – succeeded.
            Data bits test
```

The screen will change to all white and return the OK prompt when complete.

Test	Description	Preparation	When to Use
test keyboard	Executes a keyboard self test. The four LEDs should flash at once, and a “Keyboard present” message should appear.	Keyboard must be connected.	When keyboard seems to be malfunctioning.

1. Type **test keyboard** and press <CR>.
2. The four keyboard LEDs will flash once and the screen will return a “keyboard present” message.

Test	Description	Preparation	When to Use
test net–tpe	Performs an internal and external loopback test on the twisted pair Ethernet (TPE) interface.	A cable must be connected to the system TPE port and to a TPE hub or the test will fail the external loopback phase. If the tpe–link–test? parameter is false (disabled), the external loopback test will appear to pass even if a cable is not connected.	Helps diagnose network problems.

1. Type **test net–tpe** and press <CR>.
2. The following will be displayed on the screen:

```
Using TP Ethernet Interface
Lance Register test – succeeded.
Internal loopback test – succeeded.
External loopback test – succeeded.
```

<table>
<tr><th>Test</th><th>Description</th><th>Preparation</th><th>When to Use</th></tr>
<tr><td>test net</td><td>Performs an internal and external loopback test on the auto–selected system Ethernet interface.</td><td>A cable must be connected to the system & to Ethernet tap hub or external loopback test will fail.</td><td>Helps diagnose network problems.</td></tr>
<tr><td colspan="4">1. The AUI ethernet connection and the TP ethernet connection are both tested.

2. At the OK prompt, type test net and press <CR>.

3. The following will be displayed on the screen:<pre>
Using AUI Ethernet Interface
Lance Register test – succeeded.
Internal loopback test – succeeded.
External loopback test – Lost carrier. (Transceiver cable problem?)

Using TP Ethernet Interface
Lance Register test – succeeded.
Internal loopback test – succeeded.
External loopback test – succeeded.
</pre>Note: Only a TP connection is used. This will cause unconnected AUI port to fail external loopback test.</td></tr>
</table>

<table>
<tr><th>Type of Test</th><th>Description</th><th>Preparation</th><th>When to Use</th></tr>
<tr><td>watch–tpe</td><td>Monitors broadcast Ethernet packets (10Base5T—Twisted Pair Ethernet) on the Ethernet cable(s) connected to the system.</td><td>Connect the system to the network via the desired Ethernet port.</td><td>Help verify Ethernet operation.</td></tr>
<tr><td colspan="4">See watch–aui for typical test results.</td></tr>
</table>

Test	Description	Preparation	When to Use
watch–aui	Monitors broadcast Ethernet packets (10Base5–Thicknet) on the Ethernet cable(s) connected.	Connect the system to the network via the desired Ethernet port.	Help verify Ethernet operation.

1. At the OK prompt, type **watch–aui** and press <CR>.
2. The following messages will be displayed:

```
Using AUI Ethernet Interface
Lance Register test – succeeded.
Internal loopback test – succeeded.
External loopback test – Lost carrier. (Transceiver cable problem?)

Looking for Ethernet packets.
’.’ is a good packet.  ’X’ is a bad packet.
Type any key to stop.
```

3. Until you press a key, dots or Xs will appear as long as the packets appear on the network.

Test	Description	Preparation	When to Use
watch–clock	Displays seconds from the system’s Time–of–Day chip.	None.	Verifies Time–of_day chip operation.

1. At the OK prompt, type **watch–clock** and press <CR>.
2. The following messages will be displayed:

```
Watching the ’seconds’ register of the real time clock–chip.
It should be ’ticking’ once a second.
Type any key to stop.
[ ] display seconds
```

DIAGNOSTIC GRAPHICAL USER INTERFACE (GUI)

Access to the various diagnostic programs created for the AcQSim CT is accomplished via the Diagnostic GUI. Each selection is shown briefly, with more detail on following pages. Blue underlined text indicates a hyperlink to an area with more detail on that respective Diag GUI option.

Starting the Diag GUI

At the main bootup screen, click the mouse once in the lower left corner. Then type PICKER (all caps) and press <CR>. Refer to Figure 117. The Board Status window will also be displayed.

At system bootup, the main system bootup screen will appear. See Figure 117. If you are already at the Operator Application screen or Service Application screen, first select SHUTDOWN from the GENERAL UTILITIES menu. From the Shutdown menu, select SHUTDOWN CONSOLE. After a moment, the main bootup screen will appear. The Hardware Status Screen will also be displayed.

WARNING: If the Gantry is powered, it will spin up at the start of the Diag GUI !

FOR DIAG GUI:
Click here with the mouse, then type PICKER and press <CR>.

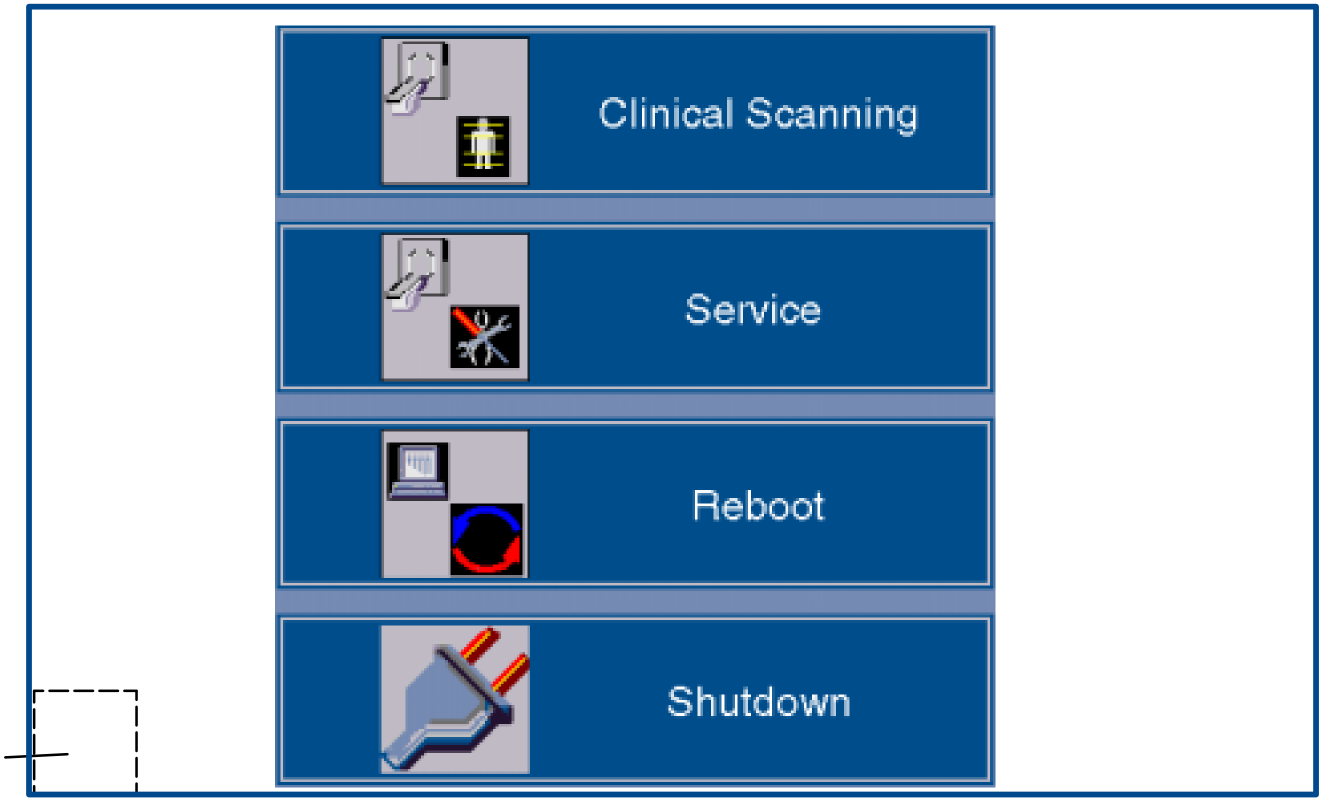
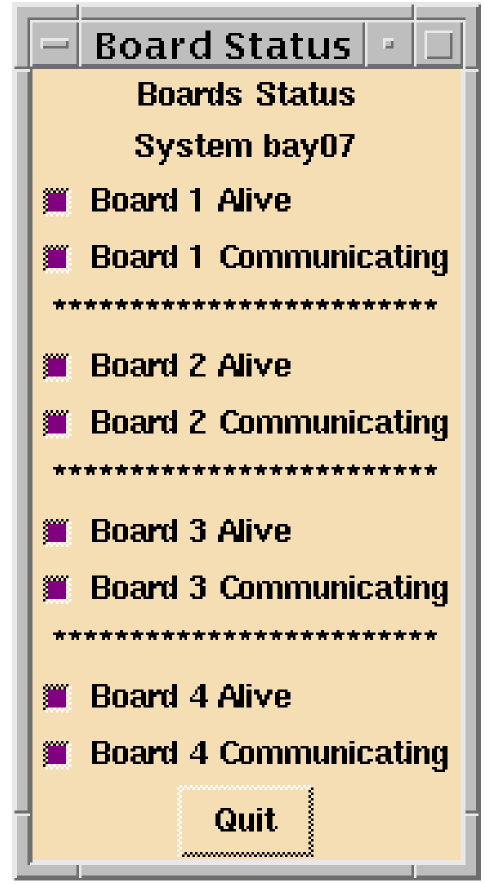


FIGURE 117 MAIN BOOTUP SCREEN –DIAG GUI STARTUP



Before running diagnostics. wait until all boards are alive and communicating as shown above.

FIGURE 118 MAIN DIAGNOSTIC GUI – with BOARD STATUS WINDOW

Click the respective Diagnostic selection in Figure 118 for description, procedure and expected results.

Acqim

- CP ▷
- GAM ▷
- AP ▷
- Projector ▷

The ACQIM (AcQImage) Top Level button yields the following diagnostic selections: **CP**, **GAM**, **AP** and **Projector**. Each selection has sub–selections. Click on the arrow next to the board of choice. The following illustrations detail the four options under the **Acqim** selection:

CP ▷
- Flash Diagnostics

GAM ▷
- SCSI
- GAM Memory
- DSP Diagnostics
- VME Test
- Flash Diagnostics

AP ▷
- AP Memory
- DSP Diagnostics
- VME Test
- Flash Diagnostics

Projector ▷
- Projector Memory
- Projector Pipeline
- Projector Broadcast
- VME Test
- Asic Test
- Flash Diagnostics

The following illustrations detail the three remaining Top Level options: **Gantry**, **Sun** and **Utilities**:

Gantry
- Manexp
- Gscope
- GPU / XSC Downloader
- Gputty+
- Gena
- Ring

Sun
- SUN Diagnostics
- Smpte

Utilities
- Hardware Status / Control
- Flash Burner
- Enet
- Dicom–Echo
- Telnet Session
- Kermit2
- Xterm
- Error Logs
- Autovoice
- Set Color
- Board Status

The following pages detail the menu options. In EDOC, simply click on the blue underlined option for detail.

AcQIM – CP

> **NOTE**
>
> You must have a null modem serial cable plugged into the CP board and port A on the SUN to run this test. (Type exit <CR> at the pSH+> prompt to exit the Xterm window.)
>
> Ensure the Num Lock key is de–activated (LED off) before opening multiple windows on the display. Having the Num Lock key active with multiple windows open could cause system lock–up.

Description: A collection of tests that allow various hardware components of the AP, the PB, the CP, and the GAM to be tested on power–up and after power–up. Power–up test results are communicated via the diagnostic display on the AP, the PB, and the GAM. The CP uses the board fail LED. Tests embedded in flash can also be accessed after power–up and run by the operator.

```
BOARD LEVEL TEST/UTILITY MENU
1.) Diagnostics
2.) Environmental Stress Screening
3.) NVRAM maintenance
4.) Exit
Select option:
```

BOARD LEVEL TEST/UTILITY MENU – Option 1: Diagnostics Menu

```
DIAGNOSTICS MENU
 1.) CPU Version
 2.) Diagnostic FLASH CRC
 3.) Operating System FLASH CRC
 4.) Select
 5.) Ethernet
 6.) Serial 16C552
 7.) Serial 16C552 with external loop back plug
 8.) VME chip2
 9.) CIO Z8536
10.) NVRAM
11.) Instruction cache enable
12.) Instruction/data cache disable
13.) Access utility
14.) Return
```

More details about flash can be seen in the GAM Flash. (Type exit <CR> at the pSH+> prompt to exit the Xterm window.)

AcQIM – GAM / SCSI

Description: The SCSI diagnostics run on the CP and GAM boards. Diagnostics designed for the CP will not run on the GAM and vice versa due to differences in the board support packages. The SCSI diagnostics consist of two separate parts: Bus level and Board level diagnostics. The bus level diagnostic contains all SCSI bus tests and its agents (disk drives) on the bus. The bus level diagnostic makes use of the SCSI driver supplied by the operating system. The board level diagnostic contains the tools for testing the SCSI controller chip.

1. After the tests are initiated, the following menu is displayed:

```
SCSI DIAGNOSTIC MAIN MENU

1. Board Level Diagnostic
2. Bus Level Diagnostic
3. Quit

Please enter your selection (1–3):
```

2. Select option **1** for **Board Level Diagnostics.** The sub menu will appear:

```
BOARD LEVEL DIAGNOSTIC MENU

1.          All Tests
2.          PCI Test
3.          Device Access Test
4.          Register Access Test
5.          SCSI FIFO Test
6.          DMA FIFO Test
7.          SCRIPTS Processor Test
8.          Interrupt Test
9.          Return to Main Menu

Please enter your selection (1–9):
```

The following details each of the possible selections (2–8) in the Board Level Diagnostics menu:

PCI Test	tests the ability to read the PCI configuration registers and and verifies their content.
Device Access Test	tests the accessibility of the controller
Register Access Test	tests the register functionality by witting and reading to/from different registers of the NCR.
SCSI FIFO Test	fills the SCSI FIFO with data, reads the status, reads the data out of the FIFO, verifies their accuracy and checks the status again.

DMA FIFO Test	writes lane by lane, verifies if lanes are full, reads lane by lane, checks accuracy of the data and checks if the lanes are empty.
SCRIPTS Processor Test	not implemented yet.
Interrupt Test	not implemented yet.

3. If you select option 2, PCI Test a similar message will be displayed:

```
running pci access test
PCI ACCESS TEST COMPLETED SUCCESSFUL – NO ERRORS
```

4. The Board Level Diagnostic menu is displayed again. Similar messages are displayed for each of the individual tests. If you select option 1, All Tests, similar messages will be displayed

```
DEVICE ACCESS TEST COMPLETED – NO ERRORS
REGISTER ACCESS TEST COMPLETED – NO ERRORS
WRITE SCSI FIFO COMPLETED – NO ERRORS
READ SCSI FIFO COMPLETED – NO ERRORS
SCSI FIFO TEST COMPLETED – NO ERRORS
DMA FIFO TEST COMPLETED – NO ERRORS
running scripts processor test
running interrupt test
```

5. Press **9** to return to the main menu.

6. From the main menu, select option **2** for **Bus Level Diagnostics**. The program searches for devices on the SCSI bus and presents a similar list of devices (disks) found:

```
ID: 0       NO DEVICE FOUND
ID: 1       NO DEVICE FOUND
ID: 2       Model: 11   Vendor: DSHW
BID: 3      Model: 11   Vendor: DSHW
Blocksize:  512 # of Blocks: 8080524
Blocksize:  512 # of Blocks: 8080524
ID: 4       Model: SA2654 Vendor: Fujitsu
Blocksize:  512 # of Blocks: 444020
ID: 5       Model: SA2654 Vendor: Fujitsu
Blocksize:  512 # of Blocks: 444020
ID: 6       NO DEVICE FOUND
```

7. After the system looks for the SCSI devices, you are prompted with the following menu:

```
SCSI DEVICE SELECTION MENU

1.          SCSI ID: 2  VENDOR: DHSW  MODEL 11
2.          SCSI ID: 3  VENDOR DHSW MODEL 11
3.          Return to Main Menu

Please enter your selection (1–3):
```

8. Select any of the listed devices to test. The following will be displayed:

```
BUS LEVEL DIAGNOSTIC MENU

1.          All non destructive tests
2.          Test Unit Ready
3.          Write/Read RAM Buffer Test
4.          Oscillating Read Test
5.          Write/Read Test – non destructive test
6.          Write/Read Test – destructive test
7.          Format Disk Drive
8.          Return to SCSI DEVICE SELECTION MENU

Please enter your selection (1–8):
```

The following details each of the possible selections (2–7) in the Bus Level Diagnostic menu:

Test Unit ready test	Tests if the drive is spun up and ready to accept commands.
Write/Read RAM Buffer	writes to the on disk RAM buffer, reads and verifies the correctness of the read data. Does not alter the contents of the disk.
Oscillating Read Test	is a stress test for the read head and verifies the disks ability to position the head correctly by oscillating between track 0 and all the other tracks on the disk, up to the last track.
Read/Write non destr. test	Reads the information on the disk, backs it up,writes a pattern, reads and verifies that pattern and restores original information.
Write/Read destr. test	writes a data pattern disk without backing up original data, reads and verifies the pattem. Test alters the contents of the disk
Format disk.	allows the user to format the current disk drive. Please note that the formatting process is not to be interrupted.

9. You may select individual tests here or run all of the non–destructive tests. Progress and results will be updated on the screen. The non–destructive READ/WRITE test will prompt you to use the default configuration or to customize the test:

```
Starting test at block address: 0
Default configuration: Ending test at highest block number: 8812868

1. Use default test configuration
2. Customize the test
```

10. Custom start and end block numbers should be entered in hex. The default test configuration will take a long time to run. The display will inform you of the ”Percent of Reading” that has completed. The menu will return after the test is completed.

11. Press **8** to return to the SCSI DEVICE SELECTION MENU.

12. You may test another device now or return to the main menu. Press **3** to exit the SCSI tests.

13. The Main Menu also contains two hidden options – 4 and 5:

Option 4 is a runtime DEBUG flag which will cause the printout of main status information throughout the program. To turn it off, go back to the main menu and select option 4 again. When the program is restarted, this flag will automatically be reset.

Option 5 allows the user to switch to different languages, the menus will be presented in English (default), German or French. The error messages remain in English. Please note, that in the current configuration up to 20 languages could be supported.

(Type exit <CR> at the pSH+> prompt to exit the Xterm window.)

ERROR MESSAGES

In the event of an error, an error message is displayed on the screen, but not logged to the error logger. The following error messages will be printed if the driver call fails:

```
STAT_CHECKCOND – target wants to give some info
STAT_ERR – SCSI error that may be retried
STAT_TIMEOUT – Target selection timed out – device may not be present on SCSI bus
STAT_BUSY – target busy try again
STAT_SEMFAIL – semaphore call failed
STAT_NOMEM – no memory available for request
STAT_RETRYEXC – Failed after allotted retries
STAT_RESET – SCSI bus reset (should retry)
STAT_BADSIZE – Drive shows no blocks
STAT_BUSY – target busy try again
STAT_SEMFAIL – semaphore call failed
STAT_NOMEM – no memory available for request
STAT_RETRYEXC – Failed after allotted retries
STAT_RESET – SCSI bus reset (should retry)
STAT_BADSIZE – Drive shows no blocks
STAT_NOMEDIA – Removeable disk not in drive
STAT_BLANK – end of recorded data
STAT_BAD_CMD – Target reports Illegal Request
STAT_NO_SENSE – request sense returned no sense
EERR_NODR – No driver provided – requested driver not present on system
RR _IODN – Illegal device (major) number – wrong device or driver selected
```

Test Unit Ready Test – Errors

If the Test Unit Ready Test fails, it will report:

```
DEVICE at SCSI ID: X is NOT READY – Bad Disk Drive
```

RAM Buffer Test – Errors

If errors are encountered during the RAM Buffer Test, the RAM address as well as expected data and actual data are displayed:

```
ERROR – DATA at RAM address: XXXXXXX  DO NOT MATCH,

            EXPECTED:   XXXXXXX RECEIVED: XXXXXXXX
```

Oscillating Read Test – Errors

The oscillating read test will display the block address where the error occurred:

```
ERROR – at BLOCKADDRESS: XXXXXXXX
```

Read/Write destructive and non destructive Test – Errors

Both tests report the blockaddress, the expected and actual data.

```
ERROR – BLOCKADDRESS: XXXXXXXX DATA DO NOT MATCH

            EXPECTED: XXXXXXXX  RECEIVED: XXXXXXXX
```

Format Disk Drive

If the formatting fails, a driver error will be displayed along with:

```
ERROR – COULDN'T FORMAT DISK DRIVE AT ID: XX
```

Board Level Test Error Messages

All Board Level messages report the register which failed the test, the expected and actual contents. The FIFO tests report in addition the lane number or FIFO status if an error occurred.

Safety

A safety analysis does not apply.

AcQIM – GAM / GAM Memory

Description: Tests the internal SRAM of each DSP and external DRAM of the DSP cluster for the Gantry Acquisition Module board. This test does not test the CPU memory. All of the CPU memory is tested on power–up via the flash based tests.

1. After the test is initiated, the following Main Menu will appear:

```
GAM BOARD MEMORY TEST

0.) All listed memories
1.) DSP internal SRAM
2.) DSP external DRAM
3.) Exit

Select option:
```

2. Option 0 will test all of the available internal SRAM and external DRAM and display the results.

3. Option 1 responds with:

```
0.) All listed memories
1.) DSP one internal SRAM
2.) DSP two internal SRAM
3.) DSP three internal SRAM
4.) Return

Select option:
```

Results of the three DSP tests above:

```
Testing Memory Starting @ e0280000
Fill with pattern
Read pattern, write invert – PASSED
Read invert, writer pattern – PASSED
Test the set to clear transition – PASSED

The test passed!
```

4. Press **4** to return to the Main Menu.

5. Option 2 (DSP external DRAM) of the Main Menu responds with:

```
0.) All listed tests
1.) DSP external DRAM word test
2.) DSP external dRAM byte test
3.) Return

Select option:
```

6. Options 1 and 2 will perform write and read sequences in the memory. When finished, the test will return a pass or fail notice.
7. Press **3** to return to the previous menu and **3** again to return to the main menu.
8. (Type exit <CR> at the pSH+> prompt to exit the Xterm window.)

AcQIM – GAM / DSP Diagnostics

1. After the test is selected, the following Main Menu appears:

```
Process all options
PCI Setup Test
Cluster Interface Test
IOP Register Read/Write
Slave DMA Test
DSP download
DSP Cluster Interrupts
DSP Cluster Error Interrupts
DSP Cluster External RAM DMA Test
DSP Prefetch FIFO Test
GAM Internal Link Port
GAM PCI Master Accesses
DSP Core R/W of External Memory
Exit
```

2. Enter the selection of the test(s) that you want or enter ”0” to perform all tests. When a selection is made, you are prompted to select the following options:

```
Test Control Mode is [STOP ON FAILURE]
Loop Count is [20]
Test Reporting Mode is [TERSE]
Enter <CR> to accept, any other key to change>
```

Note: Selection ”A” must be capped.

3. If the options listed are correct, press <CR>. If you want to change an option, press any key to get the following option screen:

```
1 – STOP ON FAILURE
2 – LOOP ON FAILURE
3 – LOOP UNTIL FAILURE
4 – CONTINUE ON FAILURE
5 – LOOP ON TEST
6 – STEP THROUGH TEST
7 – LOOP ON TEST SUITE
<CR> for [STOP ON FAILURE]
```

4. Enter your selection and press <CR>. You will be prompted to enter the Loop Count number:

```
Specify the Loop Count [20]:
```

5. Enter a new number or <CR> to keep the present value. You will be prompted to enter the Test Reporting Mode:

```
Test Reporting Mode is [TERSE]
1 – VERBOSE
2 – TERSE
3 – PASS/FAIL ONLY
<CR> for [TERSE]
```

6. The Verbose mode prints to the screen a large amount of information, the Terse mode prints what test has passed or failed and the pass/fail mode merely prints if the selected tests passed or failed. Select the Test Reporting Mode and press <CR>.
7. The test(s) will proceed and display the results in the mode chosen.
8. (Type exit <CR> at the pSH+> prompt to exit the Xterm window.)

AcQIM – GAM / VME Test

Description: Allows two boards populated with the Universe VME interface chip to exchange data sets over the VME bus backplane. This includes interaction between the Array Processor, Gantry Acquisition Module, and the Projector Boards. This excludes the CP board. The VME test must be run on ALL boards (except CP), then two boards are selected – one as a Master and one as a Slave. For the board that is not participating in the current test, the VME test is started, but the Master/Slave selection is not made. After the test on the first two boards completes, the operator must then reverse the Master/Slave relationship and run the test on those two boards again. This process is repeated on the remaining two combinations of boards. A table of these combinations is provided below.

1. After the VME Test is initiated, the following menu will be displayed:

```
VME Bus Test – Main Menu
========================================================
M – Set this board to be Test Master (Sets up PCI Slave)
S – Set this board to be Test Slave (Sets up VME Slave)
Q – Quit
========================================================
Advanced Universe Configuration
U – Toggle Local Universe Slave Posted Writes/PreReads Current = 0  (1=ON, 0=OFF)
```

2. DO NOT make a menu selection at this time. Run the VME test again on the other two boards from their respective pulldown menus (in this case, the AP and Projector).

3. Press **S** to set up the first board as the slave. The slave window will appear at the bottom half of the monitor and the following will be displayed:

```
Vme Bus Test Slave Enabled (VME Slave is Active)
Press ANY KEY to Disable and Return to Main Menu
```

4. Set up the other board as the Master . Move the forefront window out of the way. Press **M** <CR>:

```
TEST MASTER MENU
===========================================================
R – Run memory test
A – Address range to test (VME Base Address and Block Size)
Q – Return to Main menu
Advanced Settings:
C – retry Count.             Current = 10
P – Pass count.              Current = 1
T – Cycle Type               Current = 0 (0=BOTH, 1=SINGLE, 2=DMA)
L – Loop to beginning on FAILURE Current = 0 (1=ON, 0=OFF)
V – Verbose mode Toggle      Current = 0 (1=ON, 0=OFF)
E – Enhanced Test Toggle     Current = 0 (1=ON, 0=OFF)
U – Toggle Posted Writes     Current = 0 (1=ON, 0=OFF)
M – Menu Display Toggle      Current = 1 (1=ON, 0=OFF)
S – Single cycle Stop on error   Current = 1 (1=ON, 0=OFF)
W – What to do if the test hangs
NOTES:      1. You must Set A Test Master on another board before running!
            2. Before using Enhanced Test (choice E), read the instructions in choice W.
===========================================================
Enter R, A, Q, C, P, T, L, V, E, U, M, S or W  and Press ENTER:
```

5. After the Master is set up, press **r** <CR> to run the test. If this portion of the test passes, the Slave and Master roles must be reversed. Quit the test on the two participating boards from their respective menus (”Q” on Master menu, any key then “Q” on the Slave menu.)
6. Run the test again on the same two boards, but REVERSE their roles – make the Slave the Master and the Master the Slave. After the test is complete, quit the test on the participating boards. Restart the VME test on these two boards again.
7. Set up one of the previously tested boards as the Slave. Select the remaining untested board as the Master. Once this test completes, reverse their roles as you did in step 5. Run the Slave/Master test one more time on the remaining combination of boards, reversing their relationships as before. The table below will give you a better idea of the (three) two–pair board combinations and how each board should be set up with respect to the other:

AP	GAM	Projector	AP	GAM	Projector
Slave	Master	not tested	not tested	Slave	Master
Master	Slave	not tested	Master	not tested	Slave
not tested	Master	Slave	Slave	not tested	Master

Detail of options from the TEST MASTER MENU:

A:

```
VME Bus Test – VME Bus Range Menu
=============================================================
B – Set VME Base Address – Current Base = 0x02000000
L – Set VME block Size   – Current Size = 0x00100000
Q – Return to previous menu
=============================================================
Enter B, L, or Q and Press ENTER:
```

8. Set new VME base address and/or VME block size here, otherwise, press Q to use the current values and return to the previous menu.

Q:

```
Returns to the Main Menu

Advanced settings:
```

C:

```
Master Retry count = 10
Enter the number of times for the Test Master to attempt to make a connection with the test slave.
Type a number of passes then press ENTER:
```

P:

```
Pass count = 1
Type number of passes then press ENTER:
```

T:

```
Cycle Type = 0
Type 0 = Both, 1 = Single or 2 = DMA
Then press Enter
```

If option 2 = DMA is selected, the following submenu will be displayed:

```
0.) All
1.) Random
2.) Alternating 64 bit 0x00’s and 0xf’s
3.) Alternating 64 bit 0x5a’s and 0x5’s
4.) Walking 1 in a field of 0’s
5.) Walking 0 in a field of 1’s
6.) Alternating 32 bit 0x00’s and 0xff’s
7.) Alternating 32 bit 0x00’s and 0xff’s
```

L:

```
Test will loop to beginning on failure!
```

V:

```
Verbose mode is ON

Press any key to continue
```

E:

```
Enhanced Mode is ON
WARNING: Test will run considerably slower

Press any key to continue
```

M:

```
Menu Display is OFF
Press any key to continue

(Note that if this is enabled only the lettered options will be displayed.)
```

S:

```
Single cycle stop on error handling is OFF
Press any key to continue

When stop on error is OFF and the operator selects S, they get:

Single cycle stop on error handling is ON

Press any key to continue
```

W:

```
What to do when the test hangs

If Verbose mode is ON and no new status messages appear for two or more minutes then follow the instructions below:

A. Record the base address from the last status message displayed
B. Manually reset the Test Master and Test Slave boards
C. From the Test Master Menu, Change the Address range to the base address you recorded in step A and a size of
0x0010000
D. From the Test Master Menu, set the Posted Writes option to OFF
E. Run the test again

NOTE: Enhanced mode will considerably slow test performance
Press any key to continue
```

R:

```
Runs the test and responds with a similar screen:
VME Test running – Press any key to exit...
Pass Number: 1
Testing block 1 of 1, Base Address: 0x02000000 Block Size: 0x00100000
VME BUS Test PASSED!

Test completed!
```

U – Toggle Posted Writes

```
Used to make a change to the configuration of the VME bus interface chip Universe II.
```

D – Disable all VME/PCI Slaves

```
Disable a Picker board (BP, AP, or GAM) from using the VME Bus.
```

If you run the VME Bus test between a BP and an AP, you want to make sure that the GAM doesn't try to use the VME bus at the same time. This option tells the GAM to not use the bus. Additionally, if you get the following message, then choose option D for all Picker Boards on the system before running the test.

```
"ERROR – Could not establish communication with Test Slave!"
```

(Type **exit** and press <CR> at the pSH+> prompt to exit the Xterm window.)

AcQIM – GAM / Flash Diagnostics

Description: A collection of tests that allow various hardware components of the AP, the PB, the CP, and the GAM to be tested on power–up and after power–up. Power–up test results are communicated via the diagnostic display on the AP, the PB, and the GAM. The CP uses the board fail LED. Tests embedded in flash can also be accessed after power–up and run by the operator.

> **NOTE**
>
> You must have a null modem serial cable plugged into the GAM board and port A on the SUN to run this test.

1. Press **5** to get the board level diagnostics main menu:

```
BOARD LEVEL TEST/UTILITY MENU

1.) Diagnostics
2.) Environmental Stress Screening
3.) NVRAM maintenance
4.) Exit

Select option:
```

*(The following section refers to tests done in the **TEST** mode, which is intended for field service use. This is set via option 3 of the main menu – NVRAM Maintenance. Some of the tests will vary depending on the present mode, DIAG or TEST. Default setting is TEST.)*

BOARD LEVEL TEST/UTILITY MENU – Option 1: Diagnostics Menu

```
DIAGNOSTICS MENU

 1.) Diagnostic FLASH CRC
 2.) Operating System FLASH CRC
 3.) Select
 4.) Ethernet
 5.) NVRAM
 6.) Cluster memory
 7.) SCSI chip one
 8.) SCSI chip two
 9.) Instruction cache enable
10.) Instruction/data cache disable
11.) Access utility
12.) Return
```

DIAGNOSTICS MENU – Options 1 and 2: will pause and let you know if the test passed or failed:

```
The test passed!

Hit return to continue
```

If an error occurs:

```
There is an error on the board

The test failed!

Hit return to continue
```

DIAGNOSTICS MENU – Option 3:

```
The test passed!

Hit return to continue
```

DIAGNOSTICS MENU – Option 4: Ethernet

This test checks the ethernet port of the respective board. Successful transmission will return:

```
The test passed!

Hit return to continue
```

DIAGNOSTICS MENU – Option 5: NVRAM

This test .....

DIAGNOSTICS MENU – Option 6: Access Utility

```
1.) Write
2.) Read
3.) Write/Read
4.) Quit
```

All three options above will prompt you for the same set of input:

```
1.) I/O enabled
2.) I/O disabled

Select option:

1: I/O enabled will return information to the screen about what the test is currently doing.  However, this
will slow down the test.
2. I/O disabled will run faster but will not return information to the screen.
```

Select either I/O enabled or disabled. This will return:

```
Select loop quantity:
```

Loop quantity can be any hex number from 1 to FFFFFFFF. Use ”0” if you prefer infinite loop testing. Power down or reset is needed to end infinite loop testing. The next menu is displayed:

```
1.) Eight bits
2.) Sixteen bits
3.) Thirty–two bits
```

Select the size of the data transfer. The next prompt is displayed:

```
Select Data:
```

Enter the actual data to be used in the transaction. Refer to the previous choice when deciding. For example, if eight bits was chosen, select 1C (eight bits), etc.   The next prompt is displayed:

```
Select address:
```

Enter the address where the transaction will begin (e.g. 600000). The next prompt is displayed:

```
Select expanse:
```

Indicates the range of addresses where the transaction will occur, starting from the above selected address.

1. Press 4 to return to the BOARD LEVEL/UTILITY MENU.

BOARD LEVEL TEST/UTILITY MENU – Option 2: Environmental Stress Screening

If you simply press 2, the following will be displayed:

```
ENVIRONMENTAL STRESS SCREENING

Start Time: Start Date: Total Iterations: 0

No tests were selected for execution through the ESS test selection menu!

Hit return to proceed
```

To prevent the above message, set up the ESS options before execution:

1. Press <CR> if you had initiated the ESS by pressing 2.
2. Select Option **3**, NVRAM Maintenance.
3. Select Option **2**, ESS control/information:

```
DISABLED    1.) Diagnostic FLASH CRC
DISABLED    2.) Operating System FLASH CRC
DISABLED    3.) Select
DISABLED    4.) Serial 16C552 internal
DISABLED    5.) Serial 16C552 external
DISABLED    6.) NVRAM
DISABLED    7.) Continue

Select option:
```

4. Selecting the respective number will toggle DISABLED to ENABLED. Hitting it again will cause ENABLED to be toggled to DISABLED. The state selected will survive powerdowns of the board. After the tests are set, press **7** to return. Press **3** to return to the NVRAM MAINTENANCE MENU.
5. Press **6** to return to the BOARD LEVEL TEST/UTILITY MENU. If an ESS test was enabled, you can press **2**, Environmental Stress Screening, to execute the enabled test(s). This basically can run some or all of the options from the main Diagnostics menu. The screen will report on any errors.

BOARD LEVEL TEST/UTILITY MENU – Option 3: NVRAM Maintenance Menu

```
1.) Error Log Maintenance
2.) ESS control/information
3.) Diagnostic remote download
4.) Remote test location
5.) Test/diagnostic mode   [TEST]
6.) Return
```

1. Press **1** from NVRAM Maintenance Menu:

```
ERROR LOG MAINTENANCE MENU

1.) Display error log
2.) Clear error log
3.) Error handling      [CONT]
4.) Error log handling  [STOP WHEN BUFFER FULL]
5.) Error log length    [16]
6.) Return
```

1: Display error log: Shows the contents of the error log

2: Clear error log: Will delete the current contents of the error log

3: Error handling: Returns three options to handle the testing after an error: **Continue**: running other tests, **Stop**: report error and stop all testing, or **Skip**: report error, do not execute failing test again.

4: Error log handling returns three options for error logging: Circular buffer: errors are written to the log so that the oldest entry is overwritten. Stop logging when buffer is full: self–explanatory. Logging off: used to prevent loss of information in the log.

5: Error log length – choose number from 1 to 16.

2. Press **6** to return to **NVRAM Maintenance Menu**.
3. Press **2** from the NVRAM Maintenance Menu:

```
ESS CONTROL/INFORMATION MENU

1.) ESS test selection
2.) Display ESS statistics
3.) Return
```

The available tests are dependent upon what is present in FLASH and the particular board.

1: ESS test selection returns three options that let you disable or enable Diagnostic FLASH CRC or Operating System Flash CRC or Select (from the DIAGNOSTICS MENU.)

2: Show the start time, date and total iterations. Displays statistics on a previous ESS run.

4. Press **3** to return.

NVRAM Maintenance Menu – Option 3: Diagnostic remote download – (Currently not implemented)

NVRAM Maintenance Menu – Option 4: Remote test location

```
1.) Host IP address  [Currently not implemented]
2.) Download pathname  [Currently not implemented]
3.) Return
```

1. Press **3** to return.

NVRAM Maintenance Menu – Option 5: Test/Diagnostic mode – (Toggles DIAG and TEST modes)

1. Press **6** to Return to the BOARD LEVEL TEST/UTILITY MENU. (end of TEST mode FLASH diagnostics)
   (Type exit <CR> at the pSH+> prompt to exit the Xterm window.)

==

(The following section refers to tests done in the DIAG mode, which was intended for factory and engineering use. This is set via option 3 of the BOARD LEVEL TEST/UTILITY MENU – NVRAM Maintenance. Some of the tests will vary depending on the present mode, DIAG or TEST.)

BOARD LEVEL TEST/UTILITY MENU – Option 1: Diagnostics Menu

```
DIAGNOSTICS MENU

1.) Diagnostic FLASH CRC
2.) Operating system FLASH CRC
3.) Select
4.) Ethernet
5.) NVRAM
6.) Access Utility
7.) Return
```

DIAGNOSTICS MENU – Option 1 and 2: will pause and let you know if the test passed or failed:

```
The test passed!

Hit return to continue
```

If an error occurs:

```
There is an error on the board

The test failed!

Hit return to continue
```

DIAGNOSTICS MENU – Option 3: SELECT

```
Devices Found

Device #    Vendor      Device Type                         Select Test

===================================================

11 Intel SIO ISA Bridge              Passed
12 NCR 53C825 SCSI Bridge            Passed
13 Newbridge Univers VME Interface  Passed
14 DEC Ethernet Controller           Passed
15 Picker GAM DSP Cluster            Passed
16 NCR 53C825 SCSI Bridge            Passed
==================================================

To see the CONFIGURATION SPACE REGISTERS for a DEVICE, type the number of the DEVICE and press the ENTER key

To return to the previous menu, type Q and press the ENTER key.  To run the select test again, type R and presS
the ENTER key
```

1. Press Q and <CR> to see if the test passed or failed.

DIAGNOSTICS MENU – Option 4:Ethernet

DIAGNOSTICS MENU – Option 5: NVRAM

```
Machine Check Excep                Running:  *****  ISR *****
------------------------------------------------------------------
```

DIAGNOSTICS MENU – Option 6: Access Utility

```
1.) Write
2.) Read
3.) Write/Read
4.) Quit
```

All three options above will prompt you for the same set of input:

```
1.) I/O enabled
2.) I/O disabled

Select option:
```

1: I/O enabled will return information to the screen about what the test is currently doing. However, this will slow down the test.
2. I/O disabled will run faster but will not return information to the screen.

Select either I/O enabled or disabled. This will return:

```
Select loop quantity:
```

Loop quantity can be any hex number from 1 to FFFFFFFF. Use ”0” if you prefer infinite loop testing. Power down or reset is needed to end infinite loop testing. The next menu is displayed:

```
1.) Eight bits
2.) Sixteen bits
3.) Thirty–two bits
```

Select the size of the data transfer. The next prompt is displayed:

```
Select Data:
```

Enter the actual data to be used in the transaction. Refer to the previous choice when deciding. For example, if eight bits was chosen, select 1C (eight bits), etc.   The next prompt is displayed:

```
Select address:
```

Enter the address where the transaction will begin (e.g. 600000). The next prompt is displayed:

```
Select expanse:
```

Indicates the range of addresses where the transaction will occur, starting from the above selected address.

1. Press **4** to return to the BOARD LEVEL/UTILITY MENU.

BOARD LEVEL TEST/UTILITY MENU – Option 2: Environmental Stress Screening

If you simply press 2, the following will be displayed:

```
ENVIRONMENTAL STRESS SCREENING

Start Time: Start Date: Total Iterations: 0
No tests were selected for execution through the ESS test selection menu!
Hit return to proceed
```

To prevent the above message, set up the ESS options before execution:

1. Press <CR> if you had initiated the ESS by pressing 2.
2. Select Option **3**, NVRAM Maintenance.

3. Select Option **2**, ESS control/information:

```
DISABLED    1.) Diagnostic FLASH CRC
DISABLED    2.) Operating System FLASH CRC
DISABLED    3.) Select
DISABLED    4.) Serial 16C552 internal
DISABLED    5.) Serial 16C552 external
DISABLED    6.) NVRAM
            7.) Continue

            Select option:
```

4. Selecting the respective number will toggle DISABLED to ENABLED. After each test is set, press **7** to return. Press **3** to return to the NVRAM MAINTENANCE MENU.

5. Press **6** to return to the BOARD LEVEL TEST/UTILITY MENU. If an ESS test was enabled, you can press **2**, Environmental Stress Screening, to execute the enabled test(s). The screen will report on any errors. Reset button will stop the test (reboot the board).

BOARD LEVEL TEST/UTILITY MENU – Option 3: NVRAM Maintenance Menu

```
1.) Error Log Maintenance
2.) ESS control/information
3.) Diagnostic remote download
4.) Remote test location
5.) Test/diagnostic mode   [TEST]
6.) Return
```

1. Press **1** from NVRAM Maintenance Menu:

```
ERROR LOG MAINTENANCE MENU

1.) Display error log
2.) Clear error log
3.) Error handling      [CONT]
4.) Error log handling  [STOP WHEN BUFFER FULL]
5.) Error log length    [16]
6.) Return
```

1: Display error log: Shows the contents of the error log

2: Clear error log: Will delete the current contents of the error log

3: Error handling: Returns three options to handle the testing after an error: Continue, Stop or Skip.

```
4.): Error log handling returns three options for error logging.

5.): Error log length – choose number from 1 to 16.
```

2. Press **6** to return to **NVRAM Maintenance Menu**.

3. Press **2** from the NVRAM Maintenance Menu:

```
ESS CONTROL/INFORMATION MENU

1.) ESS test selection
2.) Display ESS statistics
3.) Return
```

1: ESS test selection returns three options that let you disable or enable Diagnostic FLASH CRC or Operating System Flash CRC or Select (from the DIAGNOSTICS MENU.)

2 :Show the start time, date and total iterations. Displays statistics on a previous ESS run.

4. Press **3** to return.

5. **NVRAM Maintenance Menu – Option 3**: Diagnostic remote download

   (Currently not implemented)

6. **NVRAM Maintenance Menu – Option 4**: Remote test location

```
1.) Host IP address     [Currently not implemented]
2.) Download pathname   [Currently not implimented]
3.) Return
```

7. Press **3** to return.

8. **NVRAM Maintenance Menu – Option 5**: Test/Diagnostic mode

   (Toggles DIAG and TEST modes)

9. Press **6** to Return

(end of NVRAM Maintenance menu)
(Type exit <CR> at the pSH+> prompt to exit the Xterm window.)

AcQIM – AP / AP Memory

Description: Tests the internal SRAM of each DSP and external DRAM of the DSP cluster(s) for the Array Processor board. This test does not test the CPU memory. All of the CPU memory is tested on power–up via the flash based tests.

After the test is initiated, the following Main Menu will appear:

```
ARRAY PROCESSOR BOARD MEMORY TEST

0.) All listed memories
1.) Cluster one memories
2.) Exit

Select option:
```

Option 1 responds with:

```
ARRAY PROCESSOR BOARD MEMORY TEST

0.) All listed memories
1.) DSP internal SRAM
2.) DSP external DRAM
3.) Return

Select option:
```

Option 0 will test ALL DSP internal SRAM and external DRAM memory.

Option 1 tests the DSP internal SRAM memory. Responds with:

```
ARRAY PROCESSOR BOARD MEMORY TEST

0.) All listed memories
1.) DSP one internal SRAM
2.) DSP two internal SRAM
3.) DSP three internal SRAM
4.) DSP four internal SRAM
5.) DSP five internal SRAM
6.) DSP six internal SRAM
7.) Return

Select option:
```

Select any individual tests or ”0” to test all of the listed memories. The menu will return after the test has completed. Press 7 to return:

```
ARRAY PROCESSOR BOARD MEMORY TEST

0.) All listed memories
1.) DSP internal SRAM
2.) DSP external DRAM
3.) Return

Select option:
```

Option 2 tests the DSP external DRAM memory on the AP. When selected, the test runs immediately and will not prompt for an additional menu. Will return the menu when complete. This test takes approximately about 35 seconds.

Press **3** to return to the main menu.

(Type exit <CR> at the pSH+> prompt to exit the Xterm window.)

AcQIM – AP / DSP Diagnostics

1. After the option is selected, the following Main Menu appears:

```
Process all options
PCI Setup Test
Cluster Interface Test
IOP Register Read/Write
Slave DMA Test
DSP download
DSP Cluster Interrupts
DSP Cluster Error Interrupts
DSP Cluster External RAM DMA Test
DSP Prefetch FIFO Test
GAM Internal Link Port
GAM PCI Master Accesses
DSP Core R/W of External Memory
GAM/Gantry Link Port
Exit
```

2. Enter the selection of the test(s) that you want or enter ”0” to perform all tests. When a selection is made, you are prompted to select the following options:

```
Test Control Mode is [STOP ON FAILURE]
Loop Count is [20]
Test Reporting Mode is [TERSE]
Click on Change or Continue

Note: Selection ”A” must be capped.
```

3. If the options listed are correct, press <CR>. If you want to change an option, press any key to get the following option screen:

```
1 – STOP ON FAILURE
2 – LOOP ON FAILURE
3 – LOOP UNTIL FAILURE
4 – CONTINUE ON FAILURE
5 – LOOP ON TEST
6 – STEP THROUGH TEST
7 – LOOP ON TEST SUITE
<CR> for [STOP ON FAILURE]
```

4. Enter the selection of your choice and press <CR>. You will be prompted to enter the Loop Count number:

```
Specify the Loop Count [20]:
```

5. Enter a new number or <CR> to keep the present value. You will be prompted to enter the Test Reporting Mode:

```
Test Reporting Mode is [TERSE]
1 – VERBOSE
2 – TERSE
3 – PASS/FAIL ONLY
<CR> for [TERSE]
```

6. The Verbose mode prints to the screen a large amount of information, the Terse mode prints what test has passed or failed and the pass/fail mode merely prints if all selected tests passed or failed. Select the Test Reporting Mode and press <CR>.
7. The test(s) will proceed and display the results in the mode chosen.
8. (Type exit <CR> at the pSH+> prompt to exit the Xterm window.)

AcQIM – AP / VME Test

Description: Allows two boards populated with the Universe VME interface chip to exchange data sets over the VME bus backplane. This includes interaction between the Array Processor, Gantry Acquisition Module, and the Projector Boards. This excludes the CP board. The VME test must be run on ALL boards (except CP), then two boards are selected – one as a Master and one as a Slave. For the board that is not participating in the current test, the VME test is started, but the Master/Slave selection is not made. After the test on the first two boards completes, the operator must then reverse the Master/Slave relationship and run the test on those two boards again. This process is repeated on the remaining two combinations of boards. A table of these combinations is provided below.

1. After the VME Test is initiated, the following menu will be displayed:

```
VME Bus Test – Main Menu
========================================================
M – Set this board to be Test Master (Sets up PCI Slave)
S – Set this board to be Test Slave (Sets up VME Slave)
Q – Quit
========================================================
Advanced Universe Configuration
U – Toggle Local Universe Slave Posted Writes/PreReads Current = 0  (1=ON, 0=OFF)
```

2. DO NOT make a menu selection at this time. Run the VME test again on the other two boards from their respective pulldown menus (in this case, the GAM and Projector).

3. Press **S** to set up the first board as the slave. The slave window will appear at the bottom half of the monitor and the following will be displayed:

```
Vme Bus Test Slave Enabled (VME Slave is Active)
Press ANY KEY to Disable and Return to Main Menu
```

4. Set up the other board as the Master . Move the forefront window out of the way. Press **M** <CR>:

```
TEST MASTER MENU
===========================================================
R – Run memory test
A – Address range to test (VME Base Address and Block Size)
Q – Return to Main menu
Advanced Settings:
C – retry Count.             Current = 10
P – Pass count.              Current = 1
T – Cycle Type               Current = 0 (0=BOTH, 1=SINGLE, 2=DMA)
L – Loop to beginning on FAILURE Current = 0 (1=ON, 0=OFF)
V – Verbose mode Toggle      Current = 0 (1=ON, 0=OFF)
E – Enhanced Test Toggle     Current = 0 (1=ON, 0=OFF)
U – Toggle Posted Writes     Current = 0 (1=ON, 0=OFF)
M – Menu Display Toggle      Current = 1 (1=ON, 0=OFF)
S – Single cycle Stop on error   Current = 1 (1=ON, 0=OFF)
W – What to do if the test hangs
NOTES:      1. You must Set A Test Master on another board before running!
            2. Before using Enhanced Test (choice E), read the instructions in choice W.
===========================================================
Enter R, A, Q, C, P, T, L, V, E, U, M, S or W  and Press ENTER:
```

5. After the Master is set up, press **r** <CR> to run the test. If this portion of the test passes, the Slave and Master roles must be reversed. Quit the test on the two participating boards from their respective menus ("Q" on Master menu, any key then "Q" on the Slave menu.)
6. Run the test again on the same two boards, but REVERSE their roles – make the Slave the Master and the Master the Slave. After the test is complete, quit the test on the participating boards. Restart the VME test on these two boards again.
7. Set up one of the previously tested boards as the Slave. Select the remaining untested board as the Master. Once this test completes, reverse their roles as you did in step 5. Run the Slave/Master test one more time on the remaining combination of boards, reversing their relationships as before. The table below will give you a better idea of the (three) two–pair board combinations and how each board should be set up with respect to the other:

AP	**GAM**	**Projector**	**AP**	**GAM**	**Projector**
Slave	Master	not tested	not tested	Slave	Master
Master	Slave	not tested	Master	not tested	Slave
not tested	Master	Slave	Slave	not tested	Master

Detail of options from the TEST MASTER MENU:

A:

```
VME Bus Test – VME Bus Range Menu
=============================================================
B – Set VME Base Address – Current Base = 0x02000000
L – Set VME block Size   – Current Size = 0x00100000
Q – Return to previous menu
=============================================================
Enter B, L, or Q and Press ENTER:
```

8. Set new VME base address and/or VME block size here, otherwise, press Q to use the current values and return to the previous menu.

Q:

```
Returns to the Main Menu

Advanced settings:
```

C:

```
Master Retry count = 10
Enter the number of times for the Test Master to attempt to make a connection with the test slave.
Type a number of passes then press ENTER:
```

P:

```
Pass count = 1
Type number of passes then press ENTER:
```

T:

```
Cycle Type = 0
Type 0 = Both, 1 = Single or 2 = DMA
Then press Enter
```

If option 2 = DMA is selected, the following submenu will be displayed:

```
0.) All
1.) Random
2.) Alternating 64 bit 0x00’s and 0xf’s
3.) Alternating 64 bit 0x5a’s and 0x5’s
4.) Walking 1 in a field of 0’s
5.) Walking 0 in a field of 1’s
6.) Alternating 32 bit 0x00’s and 0xff’s
7.) Alternating 32 bit 0x00’s and 0xff’s
```

L:

```
Test will loop to beginning on failure!
```

V:

```
Verbose mode is ON

Press any key to continue
```

E:

```
Enhanced Mode is ON
WARNING: Test will run considerably slower

Press any key to continue
```

M:

```
Menu Display is OFF
Press any key to continue

(Note that if this is enabled only the lettered options will be displayed.)
```

S:

```
Single cycle stop on error handling is OFF
Press any key to continue

When stop on error is OFF and the operator selects S, they get:

Single cycle stop on error handling is ON

Press any key to continue
```

W:

```
What to do when the test hangs

If Verbose mode is ON and no new status messages appear for two or more minutes then follow the instructions below:

A. Record the base address from the last status message displayed
B. Manually reset the Test Master and Test Slave boards
C. From the Test Master Menu, Change the Address range to the base address you recorded in step A and a size of
0x0010000
D. From the Test Master Menu, set the Posted Writes option to OFF
E. Run the test again

NOTE: Enhanced mode will considerably slow test performance
Press any key to continue
```

R:

```
Runs the test and responds with a similar screen:
VME Test running – Press any key to exit...
Pass Number: 1
Testing block 1 of 1, Base Address: 0x02000000 Block Size: 0x00100000
VME BUS Test PASSED!

Test completed!
```

U – Toggle Posted Writes

```
Used to make a change to the configuration of the VME bus interface chip Universe II.
```

D – Disable all VME/PCI Slaves

```
Disable a Picker board (BP, AP, or GAM) from using the VME Bus.
```

If you run the VME Bus test between a BP and an AP, you want to make sure that the GAM doesn't try to use the VME bus at the same time. This option tells the GAM to not use the bus. Additionally, if you get the following message, then choose option D for all Picker Boards on the system before running the test.

```
”ERROR – Could not establish communication with Test Slave!”
```

(Type **exit** and press <CR> at the pSH+> prompt to exit the Xterm window.)

AcQIM – AP / Flash Diagnostics

Description: A collection of tests that allow various hardware components of the AP, the PB, the CP, and the GAM to be tested on power–up and after power–up. Power–up test results are communicated via the diagnostic display on the AP, the PB, and the GAM. The CP uses the board fail LED. Tests embedded in flash can also be accessed after power–up and run by the operator.

> **NOTE**
>
> You must have a null modem serial cable plugged into the AP board and port A on the SUN to run this test.

BOARD LEVEL TEST/UTILITY MENU – Option 1: Diagnostics Menu

```
DIAGNOSTICS MENU

 1.) CPU Version
 2.) Diagnostic FLASH CRC
 3.) Operating System FLASH CRC
 4.) Select
 5.) Ethernet
 6.) NVRAM
 7.) Cluster Memory
 8.) Instruction cache enable
 9.) Instruction/data cache disable
10.) Access utility
11.) Return
```

(Type exit <CR> at the pSH+> prompt to exit the Xterm window.)

AcQIM – Projector / Projector Memory

Description: Tests the shape table memory, the output memory, and all of the memory internal to each of the Panther ASICs of the mother board/daughter card(s) for the Projector board. This test does not test the CPU memory. All of the CPU memory is tested on power– up via the flash based tests.

After the test is initiated, you are prompted with the Main Menu:

```
PROJECTOR BOARD MEMORY TESTS

0.) All listed memories (DO NOT DO AT THIS TIME!)
1.) Shape Table
2.) Output memories
3.) Daughterboard memories
4.) Exit

Select option:
```

Main Menu options are detailed below:

1.) Shape Table: Will return “The Test Passed” or print the associated error.
2.) Output Memories: (Brian?) This option will display a sub–menu:

```
PROJECTOR BOARD MEMORY TEST

0.) All listed memories
1.) Output memory accumulator bank zero
2.) Output memory accumulator bank one
3.) Output memory buffer
4.) Return
```

Each of these tests will run and return with a “Test Passed” message or an associated error.

3.) Daughter board memories: This option will display a sub–menu: The following only shows the subsequent menus. The methodfor choosing the options follows immediately after the menus.

```
0.) All listed memories
1.) Daughter board zero memories   (FUP)
2.) Daughter board one memories
3.) Daughter board two memories
4.) Daughter board three memories
5.) Daughter board four memories
6.) Daughter board five memories
7. Return

Select option:
```

Each of these seven options (0–6) will return the following sub–menu. You will need to know what Asics are physically populating the respective board to correctly run this test:

```
0.) All listed memories
1.) ASIC zero
2.) ASIC one
3.) ASIC two
4.) ASIC three
5.) Return

Select option:
```

Each one of the five options (0–4) will return the following sub–menu:

```
0.) All listed memories
1.) Pipeline Zero
2.) Pipeline One
3.) Return

Select option:
```

(Type exit <CR> at the pSH+> prompt to exit the Xterm window.)

AcQIM – Projector / Projector Pipeline

Description: Tests the Projector board's ability to perform pipeline operations on different data sets designed to test various capabilities of the board. Input data sets are manipulated by hardware and the results are compared with known good values.

1. After the test is initiated, the following Main Menu will be displayed:

```
PROJECTOR BOARD PIPELINE UTILITY

0.) Pipeline operations
1.) Configuration
2.) Exit

Select option:
```

2. Select option **1**, Configuration:

```
PROJECTOR BOARD PIPELINE UTILITY

1.) File I/O
2.) Registers
3.) Miscellaneous
4.) Return

Select option: 1
```

3. Select option **1**, File I/O:

```
PROJECTOR BOARD PIPELINE UTILITY

 1.) Coefficient files [/../common/coefff.bin]
 2.) Shape table [/../common/shape.dat]
 3.) Table one [/]
 4.) Table two [/../common/ffm2t2.bin]
 5.) Table three [/]
 6.) Output memory accumulator [/omacc.hrd]
 7.) Output memory buffer [/ombuf.0]
 8.) Output memory init [/]
 9.) View memory [/views.bin]
10.) FP scale [/]
11.) Host IP address [0x90362e0a]
12.) Host directory [/client/picker/bp_aqi/BAY3/waterFF]
13.) Return

Select option:
```

4. (Explain choices 1–12 here.) Press **13** to return:

```
PROJECTOR BOARD PIPELINE UTILITY

1.) File I/O
2.) Registers
3.) Miscellaneous
4.) Return

Select option:
```

5. Select option **2**, Registers:

```
PROJECTOR BOARD PIPELINE UTILITY

1.) Narrowcast
2.) Broadcast
3.) Return

Select option:
```

6. Select from the two choices:

```
Narrowcast: (explain here)

Broadcast: (explain here)
```

Either of the choices will prompt you to select a daughterboard and asic number:

```
Select daughter board [4–5]:

Select ASIC [0–3]:
```

7. After the daughterboard and asic are selected, you will be prompted to enter values for __?_:

```
PROJECTOR BOARD PIPELINE UTILITY

1.) FP_SCALE [0xffffffff]
2.) SIG_AN [0xffffffff]
3.) OMINIT [0xffffffff]
4.) OFFSET [0xffffffff]
5.) VIEWLEN [0xffffffff]
6.) CONFIG1 [0xffffffff]
7.) CONFIG2 [0xffffffff]
8.) CONSTAT [0x0]
9.) Return

Select option:
```

8. Each of the options (1–8) will prompt you to enter a value.:

```
Enter value:
```

9. After the values are entered, press **9** to return:

```
PROJECTOR BOARD PIPELINE UTILITY

1.) Narrowcast
2.) Broadcast
3.) Return

Select option:
```

10. Option 2, Broadcast will prompt for the same input as Narrowcast. Press **3** to return:

```
PROJECTOR BOARD PIPELINE UTILITY

1.) File I/O
2.) Registers
3.) Miscellaneous
4.) Return

Select option:
```

11. Select option **3**, Miscellaneous:

```
PROJECTOR BOARD PIPELINE UTILITY

1.) Number of views [2400]
2.) Database header size [0x0]
3.) View header size [0x0]
4.) Operation mode [HARDWARE]
5.) View size [1024]
6.) Return

Select option:
```

12. Each of the options (1, 2, 3 and 5) will prompt you to enter a value. Option 4 will toggle between HARDWARE and SIMULATION. After the values are entered, press **6** to return:

```
PROJECTOR BOARD PIPELINE UTILITY

1.) File I/O
2.) Registers
3.) Miscellaneous
4.) Return

Select option:
```

13. Press **4** to return to the Main Menu.
14. Press **2** from the Main Menu if you wish to exit the PROJECTOR BOARD PIPELINE UTILITY.
15. (Type exit <CR> at the pSH+> prompt to exit the Xterm window.)

AcQIM – Projector / Projector Broadcast

Description: Tests the broadcast capability of the Projector board's Panther ASIC's.

1. After the option is selected, you will see the following display:

```
A (motherboard with one daughter card)
B (motherboard with two daughter cards)
C (motherboard with three daughter cards)
D (motherboard with four daughter cards)
E (motherboard with five daughter cards)
  None (motherboard with no daughter cards)
```

Make a selection based on the system's AcQImage board.

```
PROJECTOR BOARD BROADCAST TEST

    The test has begun..
    The test passed!
    Hit return to proceed
or
    PROJECTOR BOARD BROADCAST TEST

    The test has begun..
    The test failed!
    Hit return to proceed

Broadcast failure!  Write: YYYYYYYY Read: XXXXXXXX Address: XXXXXXXX
```

Where YYYYYYYY could be 00000000, ffffffff, 55555555, or aaaaaaaa and XXXXXXXX could be anything.

(Type exit <CR> at the pSH+> prompt to exit the Xterm window.)

AcQIM – Projector / VME Test

Description: Allows two boards populated with the Universe VME interface chip to exchange data sets over the VME bus backplane. This includes interaction between the Array Processor, Gantry Acquisition Module, and the Projector Boards. This excludes the CP board. The VME test must be run on ALL boards (except CP), then two boards are selected – one as a Master and one as a Slave. For the board that is not participating in the current test, the VME test is started, but the Master/Slave selection is not made. After the test on the first two boards completes, the operator must then reverse the Master/Slave relationship and run the test on those two boards again. This process is repeated on the remaining two combinations of boards. A table of these combinations is provided below.

1. After the VME Test is initiated, the following menu will be displayed:

```
VME Bus Test – Main Menu
========================================================
M – Set this board to be Test Master (Sets up PCI Slave)
S – Set this board to be Test Slave (Sets up VME Slave)
Q – Quit
========================================================
Advanced Universe Configuration
U – Toggle Local Universe Slave Posted Writes/PreReads Current = 0  (1=ON, 0=OFF)
```

2. DO NOT make a menu selection at this time. Run the VME test again on the other two boards from their respective pulldown menus (in this case, the AP and GAM).

3. Press **S** to set up the first board as the slave. The slave window will appear at the bottom half of the monitor and the following will be displayed:

```
Vme Bus Test Slave Enabled (VME Slave is Active)
Press ANY KEY to Disable and Return to Main Menu
```

4. Set up the other board as the Master . Move the forefront window out of the way. Press **M** <CR>:

```
TEST MASTER MENU
==========================================================
R – Run memory test
A – Address range to test (VME Base Address and Block Size)
Q – Return to Main menu
Advanced Settings:
C – retry Count.             Current = 10
P – Pass count.              Current = 1
T – Cycle Type               Current = 0 (0=BOTH, 1=SINGLE, 2=DMA)
L – Loop to beginning on FAILURE Current = 0 (1=ON, 0=OFF)
V – Verbose mode Toggle      Current = 0 (1=ON, 0=OFF)
E – Enhanced Test Toggle     Current = 0 (1=ON, 0=OFF)
U – Toggle Posted Writes     Current = 0 (1=ON, 0=OFF)
M – Menu Display Toggle      Current = 1 (1=ON, 0=OFF)
S – Single cycle Stop on error   Current = 1 (1=ON, 0=OFF)
W – What to do if the test hangs
NOTES:      1. You must Set A Test Master on another board before running!
            2. Before using Enhanced Test (choice E), read the instructions in choice W.
==========================================================
Enter R, A, Q, C, P, T, L, V, E, U, M, S or W  and Press ENTER:
```

5. After the Master is set up, press **r** <CR> to run the test. If this portion of the test passes, the Slave and Master roles must be reversed. Quit the test on the two participating boards from their respective menus ("Q" on Master menu, any key then "Q" on the Slave menu.)
6. Run the test again on the same two boards, but REVERSE their roles – make the Slave the Master and the Master the Slave. After the test is complete, quit the test on the participating boards. Restart the VME test on these two boards again.
7. Set up one of the previously tested boards as the Slave. Select the remaining untested board as the Master. Once this test completes, reverse their roles as you did in step 5. Run the Slave/Master test one more time on the remaining combination of boards, reversing their relationships as before. The table below will give you a better idea of the (three) two–pair board combinations and how each board should be set up with respect to the other:

AP	GAM	Projector	AP	GAM	Projector
Slave	Master	not tested	not tested	Slave	Master
Master	Slave	not tested	Master	not tested	Slave
not tested	Master	Slave	Slave	not tested	Master

Detail of options from the TEST MASTER MENU:

A:

```
VME Bus Test – VME Bus Range Menu
=============================================================
B – Set VME Base Address – Current Base = 0x02000000
L – Set VME block Size   – Current Size = 0x00100000
Q – Return to previous menu
=============================================================
Enter B, L, or Q and Press ENTER:
```

8. Set new VME base address and/or VME block size here, otherwise, press Q to use the current values and return to the previous menu.

Q:

```
Returns to the Main Menu

Advanced settings:
```

C:

```
Master Retry count = 10
Enter the number of times for the Test Master to attempt to make a connection with the test slave.
Type a number of passes then press ENTER:
```

P:

```
Pass count = 1
Type number of passes then press ENTER:
```

T:

```
Cycle Type = 0
Type 0 = Both, 1 = Single or 2 = DMA
Then press Enter
```

If option 2 = DMA is selected, the following submenu will be displayed:

```
0.) All
1.) Random
2.) Alternating 64 bit 0x00’s and 0xf’s
3.) Alternating 64 bit 0x5a’s and 0x5’s
4.) Walking 1 in a field of 0’s
5.) Walking 0 in a field of 1’s
6.) Alternating 32 bit 0x00’s and 0xff’s
7.) Alternating 32 bit 0x00’s and 0xff’s
```

L:

```
Test will loop to beginning on failure!
```

V:

```
Verbose mode is ON

Press any key to continue
```

E:

```
Enhanced Mode is ON
WARNING: Test will run considerably slower

Press any key to continue
```

M:

```
Menu Display is OFF
Press any key to continue

(Note that if this is enabled only the lettered options will be displayed.)
```

S:

```
Single cycle stop on error handling is OFF
Press any key to continue

When stop on error is OFF and the operator selects S, they get:

Single cycle stop on error handling is ON

Press any key to continue
```

W:

```
What to do when the test hangs

If Verbose mode is ON and no new status messages appear for two or more minutes then follow the instructions below:

A. Record the base address from the last status message displayed
B. Manually reset the Test Master and Test Slave boards
C. From the Test Master Menu, Change the Address range to the base address you recorded in step A and a size of
0x0010000
D. From the Test Master Menu, set the Posted Writes option to OFF
E. Run the test again

NOTE: Enhanced mode will considerably slow test performance
Press any key to continue
```

R:

```
Runs the test and responds with a similar screen:
VME Test running – Press any key to exit...
Pass Number: 1
Testing block 1 of 1, Base Address: 0x02000000 Block Size: 0x00100000
VME BUS Test PASSED!

Test completed!
```

U – Toggle Posted Writes

```
Used to make a change to the configuration of the VME bus interface chip Universe II.
```

D – Disable all VME/PCI Slaves

```
Disable a Picker board (BP, AP, or GAM) from using the VME Bus.
```

If you run the VME Bus test between a BP and an AP, you want to make sure that the GAM doesn't try to use the VME bus at the same time. This option tells the GAM to not use the bus. Additionally, if you get the following message, then choose option D for all Picker Boards on the system before running the test.

```
"ERROR – Could not establish communication with Test Slave!"
```

(Type **exit** and press <CR> at the pSH+> prompt to exit the Xterm window.)

AcQIM – Projector / Asic Test

Description: Tests various aspects of the Projector board's Panther ASIC's.

1. After the ASIC tests are initiated from the Diag GUI, you are prompted to enter the Board Assembly number from the board face plate: 178064 – AP, / **178142 – Proj**, / 178141 – GAM.
2. You then pick the Version Specifier, which indicates to the diagnostic how many daughterboards are on the respective board.
3. Then, the following Main Menu appears:

```
ASIC TESTS  (BP)
================================================================
1 – Auto Mode          – Coefficient Memory
2 – Manual Mode        – Coefficient Memory
3 – Auto Mode          – View Memory
4 – Manual Mode        – View Memory
5 – Walking Bit        – Register Tests: OMINIT, SIG_AN, FP_SCALE.......
6 – Cubic Spine        – Pipe One, View Memory
7 – Cubic Spine        – Pipe Two, View Memory
8 – Forward Proj.      – Table 3
9 – Forward Proj.      – Table 2 Part 1
10– Forward Proj.      – Table 2 Part 2
11– Forward Proj.      – Table 2 Part 3
12– Forward Proj.      – Table 2 Part 4
13– Forward Proj.      – Table 2 Part 5
14– Forward Proj.      – Table 2 All Parts
M – More Tests
T – Test Options Menu
Q – Quit Tests
==============================================================
Enter Choice then press ENTER:
```

4. Press M and <CR> to show the rest of the available tests:

```
ASIC TESTS
==============================================================
15– Table Two Mem Rnd   – Use Output Bank 0 Mem A
16– Table Two Mem Rnd   – Use Output Bank 0 Mem B
17– Table Two Mem Rnd   – Use Output Bank 1
18– Table Two Mem Rnd   – Use Output Buffer
19– Signature Analyzer  – Only Valid for ASIC: 3 DB: 5
20– Signature Analyzer  – ALL ASICs

M – More Tests
T – Test Options Menu
Q – Quit Tests
=============================================================
Enter Choice then press ENTER:
```

5. Press **T** to enter the Test Options Menu:

```
TEST OPTIONS
======================================================================
S – SET PASS COUNT                  Current = 1
R – RUN ALL TESTS
A – Asic to test                    Current DB:5 ASIC: 5
D – DISPLAY MENU ON/OFF             Current = 1 (1=ON 0=OFF)
C – CONTINUE/HALT TESTING AFTER FAILURE Current =1 (1=HALT 0=CONT.)
L – Loop
======================================================================
```

6. Type a TEST OPTION then press ENTER: (Test Option descriptions are on following page).

7. After you select the Daughterboard and ASIC, the screen will inform you of your selection and prompt you to press ENTER.

8. After pressing <CR>, the screen will return to the Main Menu. If you wish to set another TEST OPTION, press T.

9. If you want to test all of the ASICs, see option 6.

10. After all of the Test Options have been selected, press ”R” to run all tests. You will begin to see each of the tests and an indication whether it PASSED or FAILED. The tests (all) take about 1 minute. When finished, the screen will display:

```
Tests Completed – Press ENTER to Continue
```

11. Press Q to quit the ASIC tests. (Type exit <CR> at the pSH+> prompt to exit the Xterm window.)

Description of TEST OPTIONS:

1 – SET PASS COUNT: sets how many times the test(s) will be run.

2 – RUN ALL TESTS: Press ”R” to run every test from the menu. (As opposed to running specific tests from the Main Menu)

3 – Asic to Test: Press ”A” to Choose a Daughterboard:
Choose a DAUGHTERBOARD NUMBER 0, 1, 2, 3, 4 or 5 : Choose an ASIC NUMBER 0, 1, 2 or 3 :

4 – DISPLAY MENU ON/OFF: Will inform you that the menus will not be displayed. You will be given the first letter of each test:
Menus will NOT be displayed
Press ENTER to Continue
Enter Test Number, S,R,D,C,T,M,A,L OR Q then press ENTER :

5 – CONTINUE/HALT TESTING AFTER FAILURE: Useful if you do not want the tests to stop after an error has occurred.

6 – LOOP THROUGH ALL EXISTING ASICS: Tests all ASICS, so option 3 above will not have to be selected. When this selection is toggled by pressing ”L” you will be informed that the test will or will not loop through the ASICs.

This option is toggled but not indicated on the Test Options menu, so it is important to watch the response to your ”L” command.

AcQIM – Projector / Flash Diagnostics

Description: A collection of tests that allow various hardware components of the AP, the PB, the CP, and the GAM to be tested on power–up and after power–up. Power–up test results are communicated via the diagnostic display on the AP, the PB, and the GAM. The CP uses the board fail LED. Tests embedded in flash can also be accessed after power–up and run by the operator.

> **NOTE**
>
> You must have a null modem serial cable plugged into the BP board and port A on the SUN to run this test.

BOARD LEVEL TEST/UTILITY MENU – Option 1: Diagnostics Menu

```
DIAGNOSTICS MENU

 1.) CPU Version
 2.) Diagnostic FLASH CRC
 3.) Operating System FLASH CRC
 4.) Select
 5.) Ethernet
 6.) NVRAM
 7.) Memory
 8.) Broadcast
 9.) Instruction cache enable
10.) Instruction/data cache disable
11.) Access utility
12.) Return
```

(Type exit <CR> at the pSH+> prompt to exit the Xterm window.)

GANTRY – MANEXP

Description: The MANEXP program allows you to season the tube, perform stationary exposures and warm up the tube.

One default optable will be stored within the program for modification by the user or automatic modification by the seasoning or warmup program.

In the interactive mode the user will be presented with a menu which contains 4 buttons:

MANUAL EXPOSURE BUTTON: Starts and allows only modification of kV, mA, Time and Scanspeed.

TUBE SEASONING BUTTON: The Tube Seasoning file is read from the disk and the exposures are performed accordingly without any user interaction. The default optable is modified with the parameters in the seasoning file.

WARMUP BUTTON: The Warmup file is read from the disk and the exposures are performed accordingly without any user interaction. The default optable is being modified with the parameters in the seasoning file.

ENGINEERING OPTABLE EDITOR BUTTON: A START button will be painted on the screen and if pressed, the optable will be converted to a scantable and submitted to the gantryserver, where the exposure is commanded. The optable editor contains a status window, where the COMPLETE or ERROR Message will be displayed if the exposure completes or fails. No other requests can be sent until the status of the operation is received. Please note that a –1 in the scanspeed field does not move the scanframe and a stationary exposure is taken.

> **NOTE**
>
> The engineering optable editor button is password protected since the optable editor allows to edit all fields in the optable.

When the program is initiated, the following Main Screen will appear (Figure 119).

For MANUAL EXPOSURE, refer to Figure 120.

For TUBE SEASONING, refer to Figure 121.

For TUBE WARMUP, refer to Figure 122.

For MECHANICAL TUBE ALIGNMENT, refer to Figure 123.

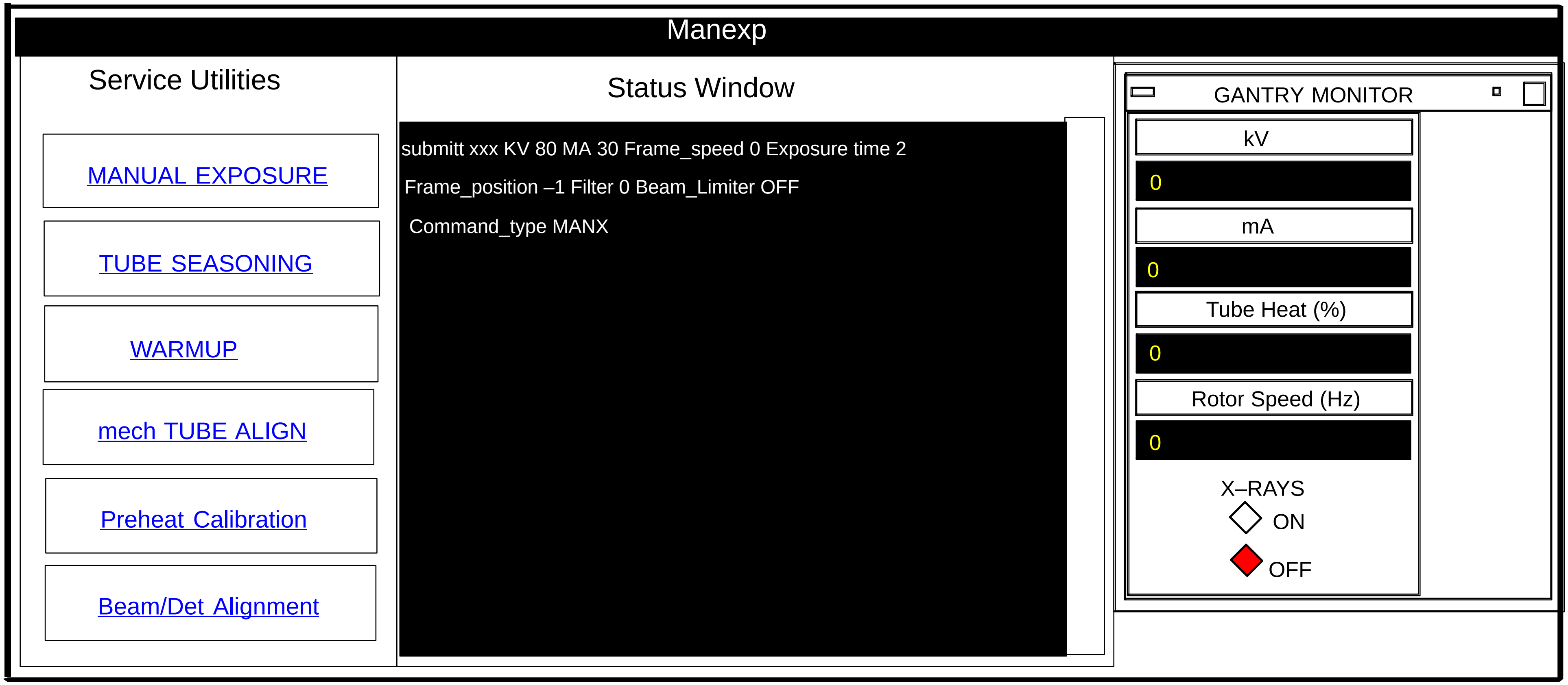


FIGURE 119 MANEXP MAIN SCREEN

Click on the left hand *Service Utilities* to hyperlink to their respective screens.

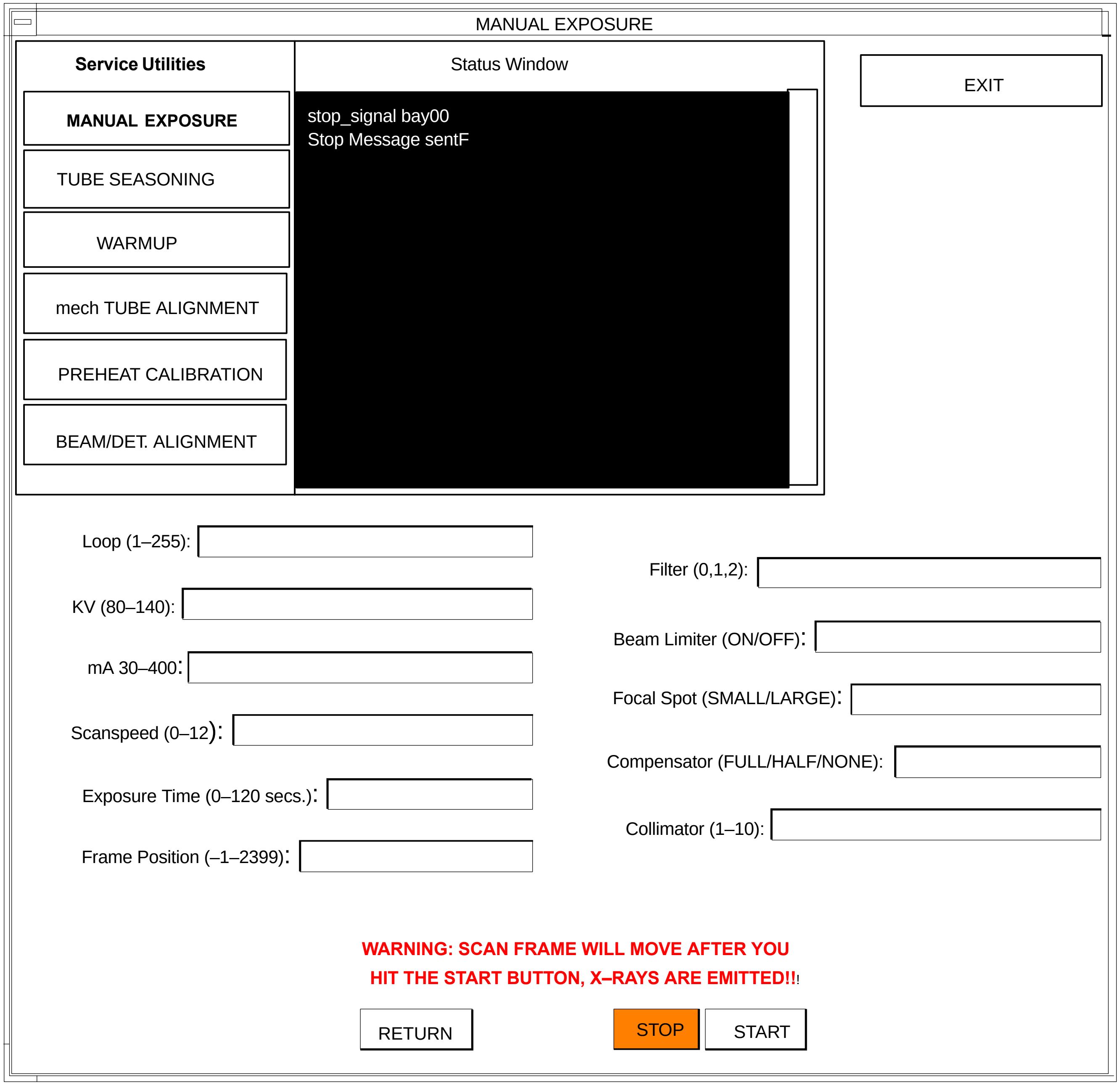


FIGURE 120 MANUAL EXPOSURE SCREEN

FIGURE 121 TUBE SEASONING SCREEN

FIGURE 122 TUBE WARMUP SCREEN

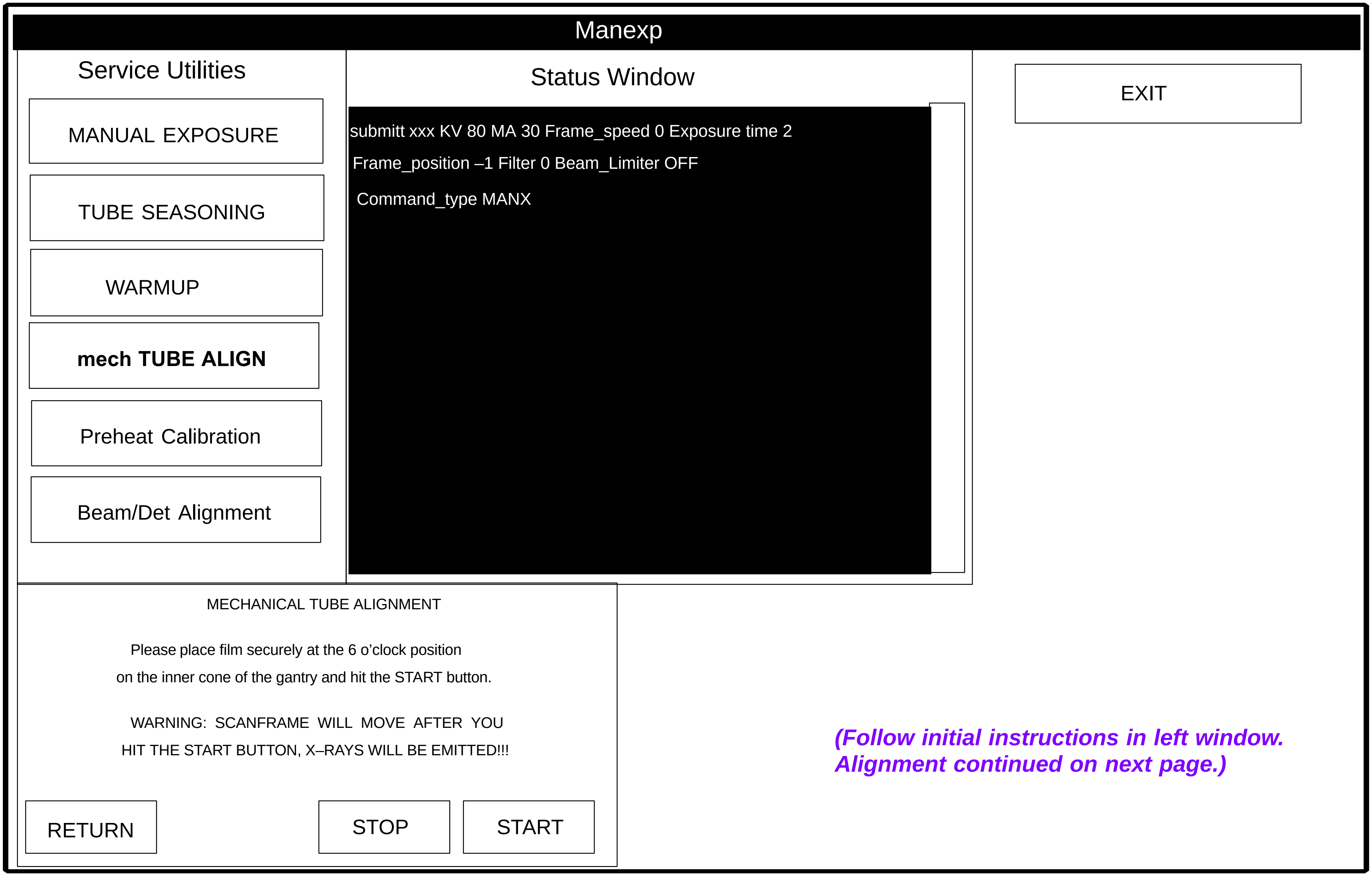


FIGURE 123 MECHANICAL TUBE ALIGNMENT SCREEN

Tube Alignment Procedure

1. After following the initial instructions on the Manexp window and both scans are complete, develop the film and inspect for a narrow, darker band superimposed on a wider, lighter band. (Type 52 Polaroid film needs the Polaroid Model 545 Film holder for developing). Measure the distance from the front edge (see Figure 124) of the narrow, darker band to the front edge of the wide, lighter band (referred to as distance D1).
2. Measure the distance from the back edge of the narrow, darker band to the back edge of the wide, lighter band (referred to as D2).

FIGURE 124 EXPOSED FILM

3. Determine the distance between the band centers by the following equation:

$$\text{Distance between band centers (D)} = \frac{\mathbf{D1 - D2}}{\mathbf{2}}$$

4. Multiply the distance between band centers (D) by 0.417 to determine the distance the X–ray tube must move:

If “D” is **positive**, rotate axial adj screw **CCW**
If “D” is **negative**, rotate axial adj screw **CW**

> **NOTE**
>
> Approx. 4 turns spans the width of the D1, D2 gray areas on the film.

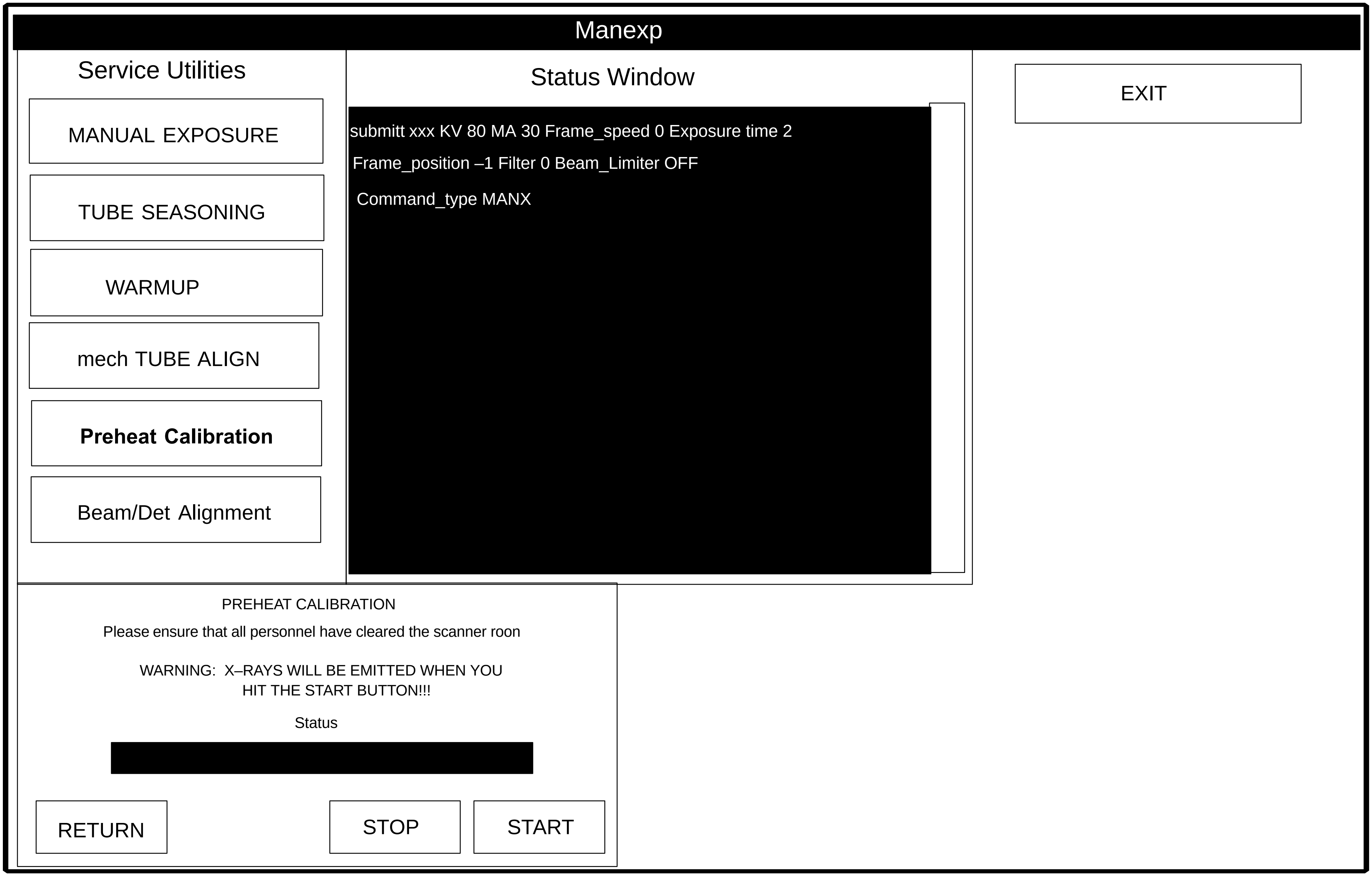


FIGURE 125 PREHEAT CALIBRATION SCREEN

FIGURE 126 BEAM / DETECTOR ALIGNMENT SCREEN – (MANUF. PROCEDURE ONLY)

GANTRY – GSCOPE

Description: GSCOPE was created to give the AcQSim CT the ability to collect data using the GAM and to provide *IQOUT* functionality similar to what Field Service was accustomed.

Gantry Scope Lite, now referred to as GScope, bypasses the server architecture of the AcQSim CT. One benefit is that GScope can run and save data to disk, WITHOUT requiring Recon software to be functional.

Gscope is a Diagnostic tool, thus software defects in other parts of the system should not affect its functionality

GScope works independent of the normal data flows of the system: Commands are sent and intercepted by using the Gserv Commando for communication with the Gantry. Data is collected on the GAM using the GAM diagnostics routines (which are used to test the GAM at Manufacturing.)

Gscope commands that are executed via remote access are limited. Operations such as making xrays and gantry or couch movement are disabled. The following commands may be executed remotely:

BG, BS, CR, DD, DF, DV, ES, FD, GA, GB, GE, GL, GS, GT, GV, GX, LS, NR, SA, SC,SD, SS, TW, WR, OL, XE, XS and XV. Details on each of these may be obtained from typing “help xx” at the CMD> prompt (where “xx” is the command.)

Special Notes

⚠ **WARNING**

GSCOPE HAS THE ABILITY TO COMMAND THE SYSTEM TO GENERATE X–RAYS. USE APPROPRIATE RADIATION SAFEGUARDS. FAILURE TO COMPLY CAN RESULT IN SERIOUS INJURY OR DEATH.

⚠ **WARNING**

GSCOPE HAS THE ABILITY TO COMMAND GANTRY AND COUCH MOTION. STAND CLEAR OF EQUIPMENT WHEN MOTION IS COMMANDED. FAILURE TO COMPLY CAN RESULT IN SERIOUS INJURY OR DEATH.

To exit Gscope, type **qu** in the COMMAND window.

START GSCOPE

1. Start Gscope from the Gantry pulldown menu of the Diagnostic GUI. Type **qu** and press <CR> at the CMD> prompt to exit.

```
STATUS Window
9:3:41
GSERV STATUS:
GSRV:     0   SCNS:     0   FUNS:     0   CONT:     1   ROTR:     0   FLMT:     0
XRAY:     0   FLTR:     0   COMP:     0   COLL:     0   LASR:     0   SHUT:  ffff
SPOT:     0   BLIM:     0     KV:    82     MA:    14   HEAT:     0   FSPD:     0
FPOS:  ab32   CSPD:     0   CHOR:   320   CVER:   21b   TILT:    80   BACK:     0
```

STATUS Window: The Status of the Gantry Server is displayed in this window and updated only when a change occurs, and change monitored twice a second.

The following indicators are used for marking the currently active system (from left to right, top to bottom in the Status Window above):

GSRV:

IN_ERR	0x8000	Set Bit to mark GPU in error
RES_IN_ERR	0x7FFF	Reset Bit to mark GPU in error
ERR_LOCK	0x4000	Set Bit to lock errarry[]
RES_ERR_LOCK	0xBFFF	Reset lock to errarry[]
IN_ERROR	0xC000	Bits to mark GPU in error
INIT	–1	GPU in soft/hardware startup
GPUIDLE	0	Nothing happening
SCANACTV	1	Scan system active
PILTACTV	2	Pilot system active
VOLMACTV	3	Volume scan system active
BACKACTV	4	Background in progress
STADACTV	5	Stationary data in progress
MAX_GSTATE	5	For ascii error string use

scns: Active scan/pilot system state:

0: Scan Idle
1: In Pre–scan
2: At start button
3: In image acquisition sequence before data collection
4: Data collection active
5: In image acquisition sequence after data collection
6: In Post–scan

funs:

Function status – unused at this time.

cont:

Contactor status: 0 = off, 1 = on

rotr:

Rotor status: 0 = off, rotor speed, (in Herz) x78 = 120

flmt:

Filament status: 0 = off, 1 = on

xray:

Xray status: 0 = Xrays off, 1 = Xrays on

fltr:

Filter Position: 0 = no filtration, 1 = x.x mm Al, 2 = x.x mm Al

comp:

Compensator Position: 0 = none, 1 = Full Field, 2 = Half Field

coll:

Collimator Position: 10 to 100 (tenths of a mm)

lasr:

Laser status: 0 = off, 1 = on

shut:

Beam Shutter state: 0 = closed, 1 = open

spot:

Focal Spot status: 0 = small spot, 1 = large spot

blim:

Beam Limiter status: 0 = off, 1 = on

kv:

kV status : actual kV (in kilovolts)

ma:

mA status : actual mA (in milliamps)

heat:

Tube Heat status: percent tube heat

fspd:

Scan Speed: 0 = off, 10–120 (tenths of a second/revolution/bit)

fpos:

Scan Position: Range: 0–xxxx ticks (1/16 detecotr/bit)

cspd:

Couch Speed: Range: 0–3200 ticks (1/32 mm/sec/bit)

chor:

Couch Horizontal Position Range: 0–57696 bits (0–1803 mm)

cver:

Couch Vertical Position: Range: 0–1016 bits (0–508 mm) (Soft Limit = 484 mm)

tilt:

Gantry Tilt Position: Range: –30.0 to +30.0 degrees (1/4 degree/bit MSB active = forward)

back:

Background Taken: 0 = incomplete, 1 = background complete

COMMAND Window: The commands a user types to control the GantryScope are entered in this window. Every command is time stamped, and a history of 1000 lines is kept. Command definitions are covered here:

All commands and responses are in the following format:

AA (xx (yy (zz))) ()denotes command parameters.

Where: AA – 2 byte ASCII mnemonic
A – xx, yy, zz — integer parameters

The parameters are interpreted as either hex or decimal (hex being the default) depending on the current I/O selected. The third letter of the screen prompt will be either an ”H” or ”D”, indicating either hexadecimal or decimal, respectively. This applies to the user entry screen only. The real time data screen always displays in the hex mode. A command is entered by simply entering the proper mnemonic. The mnemonics are all described in Command Definitions.

Commands may be entered in small or capital letters.

If a necessary parameter is not entered with a command, a default parameter of **0** is then used. Many commands don't have direct replies, but effect the state of the system, such as collimator movement. A response for these will not be displayed on the reply screen, but the change will be reflected in the Response Window.

There is no longer a STOP button. The **nowait** command causes a command (like TS – take scan) to stop an infinite wait for a response. You can use this command when it seem no response is coming back.

NOTE

The “nowait” command is NOT a software substitute for a mechanical STOP button.

Commands may be entered into a standard ASCII file and executed in a macro sequence. Refer to the macro description for more detail.

A command (or macro) may be repeated by typing ’!’ followed by a carriage return.

GScope commands are listed under their respective HELP menus. Simply type:

help software
help gpu
help xsc
help galil

to get the commands displayed (detailed screens follow).

Gscope has four different categories of commands:

SOFTWARE
GPU
XSC
GALIL

```
Help on Help for Gscope
------------------------------
help soft   Get help on everything
help gpu    Get help on the GantryScope Software commands
help xsc    Get help on the XSC+ commands
help galil  Get help on the Galil mode commands

All commands are case sensitive
```

Gscope provides online help for all commands. In the COMMAND window, type **help** and press **<cr>**. The following will appear in the RESPONSE window:

> **NOTE**
>
> Help screens reflect all commands no matter what mode you are in (local or remote). If you try to use a local command when in the remote mode, the message ”Unknown or Disabled function” will be displayed.

Gscope commands are listed under their respective HELP menus.

Software Commands (type **help soft** in the COMMAND window):

```
DL  secs                            Delay <secs>
QU                                  Quit  OR exit Galil mode
EXIT                                Exit looping or macro
SEXIT                               Exit nested lopping & macros.
bail                                Quit
EE <0–off 1–on>                     Error exit lopping & macros
GC                                  Change to Galil Couch mode
GF                                  Change to Galil Frame Mode
DE                                  Set Display to DEcimal mode
LP <count> <delay>                  Loop the next command CP board <count> times, delay <delay .01 secs>
                                    Any subseqent xsc/gpu command will stop the looping. Only one respnse.
ML  <count> <delay>                 Macro Loop the next command <count> times,  with <delay secs.>
                                    All responses returned. exit will stop looping.
HE                                  Set Display to HEx mode
GAMLOAD                             Load GAM Diagnostics on DSP
GAMCOLLECT <#fans><fansize><timeout> Collect Data to GAM Memory
GAMSAVE    <file><fansize>          Save data collected to file, specify fansize in 32 bit words
LDIR                                DIRectory of Local (SUN) drive
NOWAIT                              Cause a command(like TS–take scan, IJ–on/off) to stop an infinite wait for a
                                    response. Try this when no responses are being received.
system                              execute unix command
systemr                             as above but output to reponse window
@macro                              execute built–in macro read (file path/macro_file_name) execute user macro
GSTAT                               Toggle status display on/off in single screen remote access mode
ME                                  Message to response window';
```

For details on each command, type **help xx** and press **<cr>** where “xx” is the command.

GPU Commands (type **help gpu** in the COMMAND window):

```
GPU Commands
---------------------------------------
AL   boolean          autolight          BG  boolean             take background
BS   boolean          bs to gpu          CA  speed position      abs couch move
CR   speed distance   rel couch   move   CW  register.val        load count word
DD   device boolean   gcp disable        DF  num.detectors       det fan size
DV   register.val     load DPPC VFSC     ES  ticks               set eof
FD   detector         first detector     GA  detector.num        set gapa
GB   detector         set gapb           GE                      info request
GL   num.detectors    set gap length     GR  boolean             reset
GS                    status update      GT  horz+ vert+         Max tilt request
GV                    soft revision      GX  boolean             xray status to GP
IJ   boolean          Injector ON/OFF    IP  1/lOmsecs           int period
LS   boolean          laser status       NR  revs                num revs
SD                    stationary data    SA  sampling/det        sampling
SS   ticks            sof                SN  speed(1/10sec/rev)  Scan frame on
SF   stop.position    scan frame off     TP  speed length        take pilot
TS                    take scan          TW  register–val        set time word
SC   num.detectors    set scatter len    TV                      take volume
WR   group word       word read          WW  group word value    word write
```

For details on each command, type **help xx** and press **<cr>** where “xx” is the command.

XSC Commands (type **help xsc** in the COMMAND window):

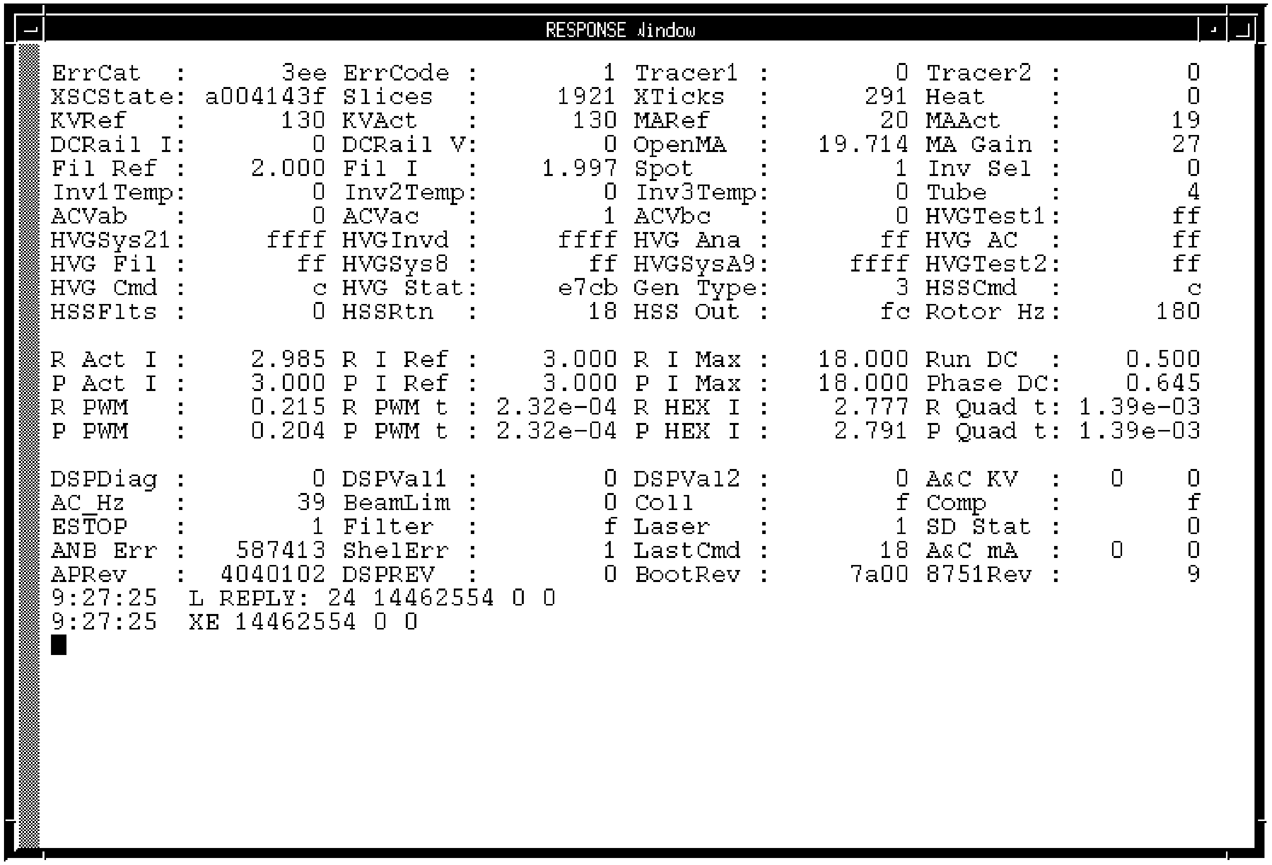
```
XSC Commands
---------------------------------------
BL boolean               Beam limiter          CL  position          Move Collimator
CM position              Move Compensator      CP  kv ma exptime     Change param
DP                       Default preheat       ER  ticks.per.slice   request
FN                       Turn Filament on      FF                    filament off
FL pos openflag          Set filter/shutter    FP  delta             fudge param
FS size                  Focal spot            LF                    laser off
LN                       Turn Laser on         OL  expr expr         open loop
PC                       Paramter check        RN freq timeout warm  Turn Rotor on
RF                       rotor off             XA                    Xray arm
XE                       info request          XF                    Xray off
xi                       init                  XN                    Xray on
XS device                xray status           XR                    Xray reset
xv                       soft revision         DL                    Delay
```

For details on each command, type **help xx** and press **<cr>** where "xx" is the command.

GALIL Commands (type **help galil** in the COMMAND window):

```
Galil Commands
---------------------------------------
AB                    abort               AC  expr            accel
BG                    begin               DB  expr            deadband
DF                    dir forward         DH                  define home
DR                    dir reverse         DS  expr            dir on switch
ER  expr              err limit           ES  expr            stop switch
FE                    find edge           GN  expr            gain
im                    inc mode            IP  expr            inc position
KI  expr              kill                MO                  motor off
OE  expr              stop on error       OF  expr            offset
PA  expr              position abs        PL  expr            pole
PR  expr              position rel        RD  expr            report done
RP  expr              repeat              RR  expr            repeat revs
RS  expr              reset               SH                  servo here
SM  expr              pulse width         SN  expr            stop from run
SP  expr              speed               SS  expr            start on sw
ST                    stop                sv                  servo mode
TC                    tell stop code      TE                  tell error
TI                    tell status         TL  expr            torque limit
TM  expr              sample time         TO  expr            stop on timer
TP                    tell position       TQ  expr            torque
TT                    tell torque         TV                  tell velocity
vm                    velocity mode       WT  expr            wait
ZR  expr              zero
```

For details on each command, type **help xx** and press **<cr>** where “xx” is the command.



```
RESPONSE Window
ErrCat    :       3ee ErrCode :        1 Tracer1 :        0 Tracer2 :        0
XSCState: a004143f Slices  :     1921 XTicks  :      291 Heat    :        0
KVRef    :       130 KVAct   :      130 MARef   :       20 MAAct   :       19
DCRail I:         0 DCRail V:        0 OpenMA  :   19.714 MA Gain :       27
Fil Ref :     2.000 Fil I   :    1.997 Spot    :        1 Inv Sel :        0
Inv1Temp:         0 Inv2Temp:        0 Inv3Temp:        0 Tube    :        4
ACVab    :         0 ACVac   :        1 ACVbc   :        0 HVGTest1:       ff
HVGSys21:      ffff HVGInvd :     ffff HVG Ana :       ff HVG AC  :       ff
HVG Fil :        ff HVGSys8 :       ff HVGSysA9:     ffff HVGTest2:       ff
HVG Cmd :         c HVG Stat:     e7cb Gen Type:        3 HSSCmd  :        c
HSSFlts :         0 HSSRtn  :       18 HSS Out :       fc Rotor Hz:      180

R Act I :     2.985 R I Ref :    3.000 R I Max :   18.000 Run DC  :    0.500
P Act I :     3.000 P I Ref :    3.000 P I Max :   18.000 Phase DC:    0.645
R PWM   :     0.215 R PWM t : 2.32e-04 R HEX I :    2.777 R Quad t: 1.39e-03
P PWM   :     0.204 P PWM t : 2.32e-04 P HEX I :    2.791 P Quad t: 1.39e-03

DSPDiag :         0 DSPVal1 :        0 DSPVal2 :        0 A&C KV  :   0     0
AC_Hz   :        39 BeamLim :        0 Coll    :        f Comp    :        f
ESTOP   :         1 Filter  :        f Laser   :        1 SD Stat :        0
ANB Err :    587413 ShelErr :        1 LastCmd :       18 A&C mA  :   0     0
APRev   :   4040102 DSPREV  :        0 BootRev :     7a00 8751Rev :        9
9:27:25  L REPLY: 24 14462554 0 0
9:27:25  XE 14462554 0 0
```

RESPONSE Window: Every GantryScope command has a response. and is displayed here. (Response of the “xe,” xsc info request command is shown here.)

```
ERROR Window
9:3:26
WARNING-ERROR STATUS: EROR code: 2012
```

ERROR Window: Errors and Help to the user are displayed here.

MACRO–COMMAND EXECUTION

The commands documented may be entered into a preset sequence in a standard ASCII file and executed by typing an **@** followed by the name of the file. One command is decoded per line. Any line that starts with the character **;** is ignored. The **;** line may be useful for comments.

Macro names MUST contain only upper case letters or numbers and must reside in the /opt/picr/service_util/diagnostics/pq2000p/gscope/macro dir. Names cannot be longer than 6 characters.

Any UNIX standard editor may be used to create macros. The **microemacs** program is installed on the AcQSim CT scanner and may be used for this purpose. To start it, type **me** at the % prompt and press <cr>. You may refer to the MicroEmacs manual.

Macros may be looped on with the **ml** command with the delay executed (if non–zero) between the execution of the complete file. When complete, the prompt will be redisplayed. The STOP button may be used to terminate macro execution. The following is an example of a macros written to command X–rays.

1. From the MicroEmacs screen, enter the command sequence. For X–rays it is as follows:

```
CP 130 20 400
XR
ER
FN
DL 3
XA
XN
DL 4
XF
FF
```

This sequence will command an exposure at 130 kV, 20 mA for 4 seconds.

2. Save the file by typing: **CONTROL X**, **CONTROL S**.
3. Exit the emacs editor by typing: **CONTROL X**, then **CONTROL C**.
4. Enter the Gscope program. To execute this macor: type **@XRAY** and press <**CR**>.

CURRENT STANDARD MACROS

@ring: A new macro has been written (3–16–98): @ring will perform a scanner operation similar to the RING program on the Q–Series scanners. This procedure is required for operations such as Brush Block Alignment. However, the RING program can now be executed viat the Diagnostic GUI. See RING hyperlink.

@XC_INIT and **@AUTOXCTEST**: These setup to take a transfer counter scan and get transfer counter data (incrementing channel #'s) from the gantry. The command: system $PICR_ROOT_DIR/service_util/diagnostics/pq2000p/gscope/bin/data_check /client/pic/data/xctest.dat address. In autoxctest, the data in the file /client/pic/data/xctest.dat. is checked.

The Service Person must look in the window or xterm where gscope was started to determine whether or not it passed. If it passed it prints:

```
data_check complete with no errors found
```

on error prints 1 error:

```
Error detected
Source fan
error #, error text
Expected #
Condition #
Error location #
```

xcinit macro listing:

```
he
gr 1
gamload
ss 18
ww 4 1 18
nr 1
fd 0
df 200
sa 4
dv 400
sc 0
ww e 1 0
@autoxctest
```

autoxctest macro listing:

```
he
ww e 1 0
de
gamcollect 9600 259 0
ww 1 3 0
dl 5
gamsave 99.0/pic/data/xctest.dat 259
system $PICR_ROOT_DIR/service_util/diagnostics/pq2000p/gscope/bin/data_check/client/pic/data/xctest.dat address error

restart
```

STANDARD COMMAND SEQUENCES

When using Gscope to make X–rays, perform a stationary data transfer, take a scan, expose film or take a pilot scan, a sequence of commands is followed. Use the following command sequences when trying to do one of these procedures:

Make X–Rays

RN	Rotor On (ensure this command is entered with either a xxx dec or xx hex
DL 6	Allows a six second delay for the rotor to get up to speed.
CP	Change parameters (set desired kV, mA, time)
XR	X–ray reset (to ensure generator is reset)
ER	Exposure Request (look for a reply of ER 0 0 0 before proceeding)
FN	Filament On (if being entered into a macro sequence, give 6 second delay.
DL 6	Allows a six second delay for the filament to heat up.
XA	X–ray arm (6 second timeout)
XN	X–rays On
DL xx	Enter the same amount of time as entered in the CP command.
XF	X–rays Off
FF	Filament Off
FR	FIFO read

Stationary Data Transfer

IP	Integration Period (entered as .1ms/bit)
FD	First Detector (Set desired starting detector)
DF	Detector Fan (Set desired detector fan size)
SD	Stationary Data (Executes the transfer)

Take Scan

SS	Start of Field (entered as [1/32 detector/bit for IQ] [1/8 detector/bit for PQ])
ES	End of Field (entered as [1/32 detector/bit for IQ] [1/8 detector/bit for PQ])
NR	Number of Revolutions (number of revs before EOF interrupt)
SN	Scan Frame Speed
DF	Detector Fan Size (use approximately 5)
WC	Write to control reg. (Enter 200 to set scan enable bit)
TS	Take Scan
FR	FIFO Read

Pilot Scan

SF	Set first detectors
DF	Detector fan size
WC	Write to control reg. (set to 200 to set scan enable bit)
TP	Take Pilot

Expose Film

DE	(Places IQOUT in decimal mode)
RN 60[PQ=180]	(Turns rotor on)
DL 6	(Allows a six second delay)
CP 80 20 300	(changes parameters to 80kV, 20mA and 3 sec. exposure)
XR	(XSC Reset)
ER	(Exposure request)
FN	(Filament on)
DL 6	6 sec. delay for filament)
FL 0 1	(no filter, shutter open)
DL 2	(2 sec. delay)
CL 100	(collimator to 10 mm)
DL 2	(2 sec. delay)
CM 0	(no compensator)
DL 2	(2 sec. delay)
XA	(X–ray arm)
XN	(X–ray on)
DL 3	(3 sec. delay)
XF	(X–ray off)
FF	(Filament off)

A number of preprogrammed macros are included with Gscope. These macros force hexadecimal mode. These are:

DSx	This macro sets and displays the chosen display switch continuously. ’x’ is the switch number. Valid entries are 1 to 9.
BZ	Loads zeroes into the background RAM’s.
BG	Loads normal values into the background RAM’s
DC	Takes a digital check scan at the frequency ’x’. Valid entries are 10K,20K,1M, & 2M.

NMSCAN Takes a normal scan (a very short one) without X–rays. The background RAM's are zeroed so the data is the background value.

NMSX This is a normal mode scan, with X–rays (130kV, 20 mA), for approximately 1 second. The rotor must already have been spun up.

XCSCAN Performs a maintenance mode scan (no gaps).

XCTST Performs a transfer counter test data transfer with 5 detector gaps.

GANTRY – GPU / XSC DOWNLOADER

Downloads the latest software to the GPU and XSC from disk. Used after a recent software installation.

After entering the CP board name, the downloader option screen will be displayed. Refer to Figure 128.

Select the download you wish to perform. During the download, a progress screen in displayed. The the system reboots the Gantry server for you (you will not be prompted and will not have to do a reboot).

FIGURE 127   GPUXSC Downloader Window

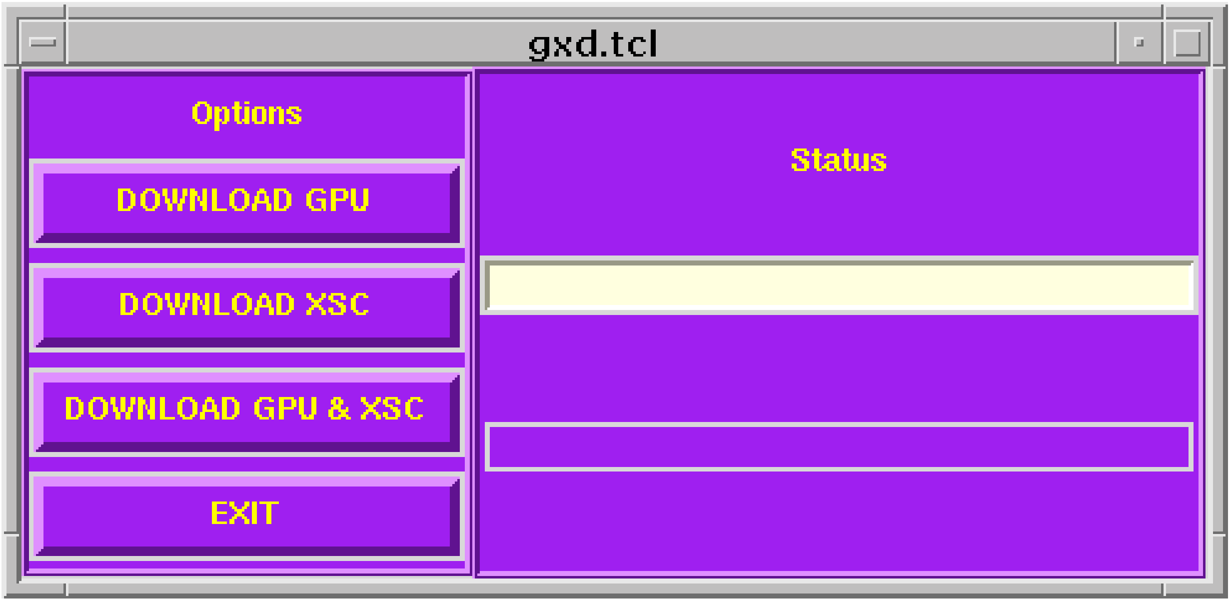


FIGURE 128 OPTION / STATUS WINDOW

GANTRY – GPUTTY+

Description: The GPUSPY diagnostics are resident on the GPU. A program called ***gena*** checks to see if the diagnostics are locked and if not locked, enables GPUSPY. Gputty+ actually consists of calibration utilities as well as diagnostics.

Upper and lower case input with commands and spaces are delimiters are allowed. All commands are two letters followed by appropriate parameters. Data checks are made to help prevent illegal inputs from the keyboard.

Running Gputty+

After the communication with the GPU is established, you will see the AcQSim CT> prompt.

Press the letter "**r**" to run the program. Enter a ? to display the available options:

```
Shift Control E return to Quit
MM(maintenance reg) SS(scan sequence diags) ?(help)
PA(use) CT(cntrl) DD(main drive displays)
TX(temp xducers CC(couch horz. cal)
AD (starts a –> d convert)
MC(main drive cal)
BA(main drive balancing software)
BT(main drive bearing test)
EF(enables Fifo) RF(reads fifo)
GL(galil direct comm mode) GE(GPU status)
SO(set overspeed control bit)PS(PSDC status registers)
```

REMINDER: Indicator beeps will be heard when your adjustment is in range. Many of the steps DO NOT require pressing <SPACEBAR> <CR> and will time out automatically.

Below are descriptions and procedures for running the various Gputty+ options:

Gpuspy command set

All commands are two letters followed by the appropriate parameter. Data checks are made to prevent illegal inputs from the keyboard. Calibration/Utilities are <u>underlined.</u>

Command	**Description**
AD	Starts Analog to Digital conversion.
<u>BA</u>	Main drive balancing utility.
<u>BT</u>	Main drive bearing test.
<u>CC</u>	Couch horizontal calibration utility.
CD	Displays ds4, ds5, ds6, st 8a and couch horizontal position. The display switches and status register are always displayed in hex and couch position is in mm's to the left of the decimal point and in .0625mm to the right.
CM	Performs the command requested. In loop mode the repetition rate is every few microseconds unless an additional parameter is added to the command line. That will cause a delay of 10 milliseconds per bit between commands. A delay of up to two seconds may be entered. Example; cm 18 10 loops at .1 second rate.
CO	Puts control loops into continuous mode. Once called, any process will loop until a space bar and a return is entered
CT	Performs a control register operation. The valid registers are 0x40 through 0x50. Typing 'CT xx' where xx is any valid register, causes that registers' value to be displayed in the selected format. Typing 'CT xx xxxx' causes the second the second parameter to be loaded into the register defined by the first parameter. Typing 'CT' displays the last selected registers' contents.
DD	Displays ds1, ds2, ds3, and normal scan position simultaneously. The display switches appear in hex format only. Nsp is in detector number to the left of the decimal point and 1/8's of a detector number to the right of the decimal point for a 1200 detector machine. For a 4800 detector machine the nsp display is the detector number divided by four. Normal scan position is in detector number to the left of the decimal point and .125 of a detector to the right.
DE	Causes integer display format
<u>DI</u>	Calls up the acquisition diagnostic menu

DS	Reads the current value of the selected display switch. The display switch is selected by entering it after the command. (e.g. DS 4 <CR> will output the contents of display switch 4.)
DT	Initiates a data transfer at the selected channel. The user must control the integration period, transfer length, and data acquisition mode. The user must have Scan Data Enable (WC 200 in HEX mode, IQOUT) active on the SIF or the data cable disconnected, otherwise a data transfer timeout will occur. The maximum repetition rate is 70 transfers a second decreasing as the integration period is made longer. There is a maximum transfer length that must be set due to the way the starting address is calculated. The center channel has half of the transfer length subtracted from it and the address is loaded to CT45. The grabber is loaded with the channel plus four for log mode, plus two for linear mode transfers. The minimum transfer length in log mode is three and in linear mode it is actually zero. Error messages are: ”Too many parameters” if command line is wrong. ”Invalid detector” if the requested detector number is invalid. ”Invalid ip” if the integration period is set to 0.
EF	Enables FIFO.
GE	GPU status
GL	Galil direct comm mode
HE	Causes HEX display format.
<u>MC</u>	Main Drive Calibration.
MM	Loads the maintenance mode position register and connects the register to the position bus. The format is MM xxxx, where xxxx is the desired position. Continuous writes are permitted, but there is no provision for readback in hardware or software.
PA	Accepts a parameter for programmed pauses to support slow terminals. The valid range is from 0 to 1 second with a resolution of 10 msec. Most slow terminals will work with a pa 10 or less and a VT320 or equivalent will work with a pa 0. The pause period is appended to any other operation being done at the time. For example, this will slow down data transfers.
PS	PSDC status registers
RD	Sends a message to the read exchange. You are prompted for input by the display. Enter, in hex, command p1 p2 p3.
RF	Reads FIFO.

RS	Issues a general reset to the Gantry electronics, re–initializes the galils and if followed by a 1, reloads the control words to their default values.
SE	Sends a message to the ‘GPMN’ exchange. You are prompted for an input by the display. Enter, in hex, command p1 p2 p3.
SG	Puts control loops into single mode.
SO	Set overspeed control bit
SS	Calls up a general diagnostic menu.
ST	Performs a status register operation. Usage is similar to the ’CT’ command, except ’ST’ is a read only operation. Valid registers are 0x80 through 0x90
TX	Causes the temp monitor reading to be displayed. Also works in continuous mode.

UTILITIES

The following GPUSPY utility commands are available:

CD (Couch Displays)

```
ds4 =  22   ds5=  633   ds6=  218C  st8a=  C7CE chp = 418.14
```

DD (Main Drive Displays)

```
ds3= CE3D   nsp= 2199.21
```

TX (Temperature Xducers)

```
IN DEGREES CELCIUS

TX1 =       TX2 =       TX3 =       TX4 =0    0    31   27
```

The following procedures cover calibrations or utilities and are listed in alphabetical order.

BA (MAIN DRIVE BALANCING UTILITY)

> ⚠ **WARNING**
>
> **THIS UTILITY CAUSES THE SCAN FRAME TO MOVE. STAY CLEAR OF THE GANTRY. FAILURE TO COMPLY CAN RESULT IN INJURY OR DEATH TO PERSONNEL.**

1. Type **ba** at the prompt to display:

```
!! WARNING, ANSWERING YES WILL CAUSE MAIN DRIVES TO
!! MOVE AT SPEED TWO, ANSWER NO TO ABORT
!! ENTER ONE FOR YES, ZERO TO ABORT
```

2. Press **1** to continue:

```
WAITING FOR DRIVES TO SETTLE
ACQUIRING DATA FROM GALIL
sorting....
DOING DFT ON FILTERED DATA
AVERAGE VOLTAGE =’s 1.699
ANGLE OF PEAK POSITIVE TORQUE =’s 242.899
PEAK –> PEAK AMPLITUDE OF FUNDAMENTAL =’s 0.031 VOLTS
IMBALANCE IS = 0.8 FOOT–LBS.
SCAN FRAME IN SPEC   (A)
!! WARNING, ANSWERING YES WILL CAUSE MAIN DRIVES TO
!! MOVE , ANSWER NO TO SKIP
ENTER ONE FOR YES, ZERO TO SKIP
```

3. Press **1** to continue with balancing, or **0** if frame is within spec (see "A" above)

```
POSITIONING FRAME SO THAT HEAVIEST PART IS NEXT TO
DETECTOR ZERO
IT WOULD TAKE x.x POUNDS ADDED BY DET 1200 TO BALANCE
OR THE SAME AMOUNT REMOVED BY DET ZERO
```

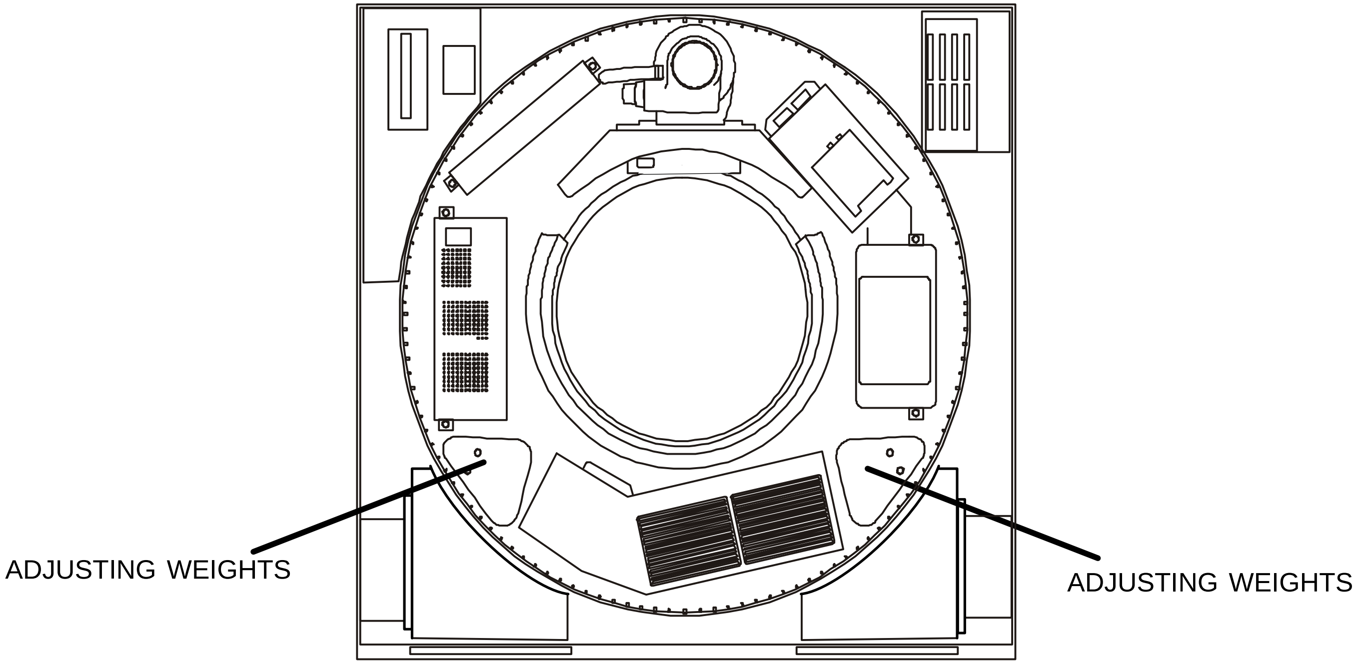


FIGURE 129 SCAN FRAME ADJUSTMENT WEIGHT LOCATIONS

BT (MAIN DRIVE BEARING TEST)

1. Type **bt** at the prompt to display:

```
DO YOU WISH TO EXIT NOW?
ENTER ONE FOR YES, ZERO TO CONTINUE
>0

!! WARNING, ANSWERING YES WILL CAUSE MAIN DRIVES TO
!! MOVE AT SPEED ONE, ANSWER NO TO ABORT
ENTER ONE FOR YES, ZERO TO ABORT

>1
ABORTING MOTOR, ACQUIRING DATA
```

The system will pause and then plot the coastdown curve. It should look similar to the following:

```
| XX
|   XX
|     X
|      XX
|        XX
|          XX
|            XX
|              XX
|                XXX
|                   XX
|                     XX
|                       XXX
|                          XXX
|                             XXXX
|                                 XXX
|                                    XXXX
|                                        XXXX
|                                            XXXX
|                                                XXXXX
|                                                     XXXXXX
|                                                           XXXXXXX
|                                                                  XXXXXXX
|_________________________________________________________________________XXXXXX
PLOT OF COAST DOWNMAX SPEED =’s 60 RPMTOTAL TIME =’s 16.7AVERAGE FRICTION =’s 62.1 FT.–LBS.SPACEBAR RETURN TO EXIT

      !!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!
               !!! REMEMBER TO REPLACE FUSE !!!!
      !!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!
```

CC (HORIZONTAL COUCH CALIBRATION)

1. Before this procedure can be run, you must first power down the system. Refer to the PowerUp/Down procedure.
2. Lower the telescoping covers on the couch and remove the panel at the underneath rear of the couch. Refer to the telescoping base cover removal procedure.
3. Remove the bottom rear panel from the couch. Disconnect the red and black connector that is attached to the couch servo motor leads.
4. Remove the Gantry cardfile access cover.
5. Apply power to the system. Refer to the PowerUp/Down procedure. At the 3 Button Screen, start the Diagnostic GUI.
6. Select **Gputty+** from the GANTRY pulldown menu. Press the letter "**r**" to execute (Shift Control E to exit at any time).

> **NOTE**
>
> <sb cr> refers to pressing the Spacebar and Return key. Beeps indicate adjustment within range.

> **NOTE**
>
> A step that does not tell you to "<sb cr>" will automatically time out. Adjustments should be made quickly since you cannot go back a step.

7. Type **cc** at the prompt to display:

```
SPACE BAR AND RETURN STEPS THROUGH TESTS, YOU WILL SEE A <sb, cr> PROMPT
A –> SIGN PROMPTS YOU TO SETUP OR ADJUSTMENT
HINT:  0x BEFORE A NUMBER IS A HEXIDECIMAL NUMBER !!!!!
DISCONNECT OR DISABLE COUCH HORZ. MOTOR
!!!!!!!!!!!!!!!!!
DO YOU WISH TO CONTINUE ? 1=YES, 0= EXIT>
```

8. Press **1** to continue. Push the couch toward the scan frame and mark two positions from the back seam of the patient support with a pencil: 180mm and 1180mm. Refer to Figure 130.
9. The pots are 20 turn pots. Turn them at least 20 times ccw or until you hear or feel them click against their stops.
10. Refer to Figure 132 for location of adjustment pots on the CTCC. **Follow the on–line instructions**:

```
TURN BOTH HORIZ ZERO (R12) AND
SCALE POTS (R11) FULLY CCW, THEN <sb cr>

SET THE MULT SCALE ADJ. SWITCHES (SW1, 2 AND 3) TO 0x800, THEN <sb cr>
```

11. Refer to Figure 132:

```
*** PUSH VERTICAL ENABLE ON GCP WHEN ADJUSTMENT IS COMPLETE
MOVE TOP SO DS5 READS BETWEEN 0xE80 AND 0x1180
AND ADJUST RESOLVER FEEDBACK (R7) FOR 2.0 V RMS
BEEPER WILL SOUND AT CORRECT ADJUSTMENT
            DS5         Vrms
            1038        1.97
```

12. Refer to Figure 131:

```
PUT END OF COUCH TOP 180MM (7 1/16 IN) FROM SEAM
THEN <sb cr>
PUSH VERTICAL ENABLE ON GCP TO LEAVE LOOP

SET HORZ POT TO –1.0 ––> –1.5 VOLTS  THEN TIGHTEN
           –1.475

PUSH VERTICAL ENABLE ON GCP TO LEAVE LOOP
```

> ⚠ **CAUTION**
>
> EXERCISE EXTREME CARE WHILE MOVING AND ALIGNING RESOLVER TO PREVENT ANY BENDING OR MISALIGNMENT ON RESOLVER SHAFT.

13. Without moving the carbon top, carefully loosen 3 cleat screws just enough to rotate the housing.
14. Turn resolver body housing until DS5 output reads zero +/– –$0x10_{16}$.

```
TURN RESOLVER BODY HOUSING TO READ ZERO +/– 0x10 AND RETURN
            DS5         10
```

15. Tighten the resolver mounting cleat screws ensuring the rounded portion of all three retaining elements are engaged in the resolver body. Torque cleat screws to 3–4 lb–in.
16. Verify that the display word still reads 000 + / – decimal after torque.

17. Repeat adjustment procedure if necessary.

```
PUSH VERTICAL ENABLE ON GCP WHEN ADJUSTMENT IS COMPLETE
ADJUST COARSE ZERO POT. DS4 SHOULD BE 0x10
            DS4         FF
```

This step will automatically time out (no <sb cr>).

```
MOVE COUCH IN UNTIL READOUT IS ABOUT –9.49
BEEP  –8.67  BEEPER WILL SOUND WHEN IN RANGE
```

This step will automatically time out (no <sb cr>).

```
NOW MOVE UNTIL READOUT IS ABOUT 0x0  +/– 0x20
            DS5         0
```

This step will automatically time out (no <sb cr>).

```
ADJUST HORIZ SCALE POT (R11) TO READ 0x90
            DS4         90
```

This step will automatically time out (no <sb cr>).

```
PUT END OF COUCH TOP 180MM (7 1/16 IN) FROM SEAM
THEN <sb cr>

ADJUST HORZ ZERO POT (R12)  DS4 SHOULD BE 0x10
            DS4         10
```

This step will automatically time out (no <sb cr>).

```
MOVE COUCH IN UNTIL READOUT IS ABOUT –9.47
DS5         –9.47

BEEP  –9.47  BEEPER WILL SOUND WHEN IN RANGE
```

This step will automatically time out (no <sb cr>).

```
NOW MOVE UNTIL READOUT IS ABOUT 0x0  +/– 0x20
            DS5         0
```

This step will automatically time out (no <sb cr>).

```
ADJUST HORIZ SCALE POT (R11) TO READ 0x90.
            DS4         90
```

This step will automatically time out (no <sb cr>).

```
MOVE TO EXACTLY 1180MM FROM SEAM (47 7/16 IN) THEN <sb cr>

SET MULT SCALE ADJ SWITCHES (SW1, 2 AND 3) TO xxxx THEN <sb cr>
```

Refer to Figure 132 for switch locations.

```
HORIZONTAL CAL COMPLETE
REMEMBER TO RECONNECT THE MOTOR
```

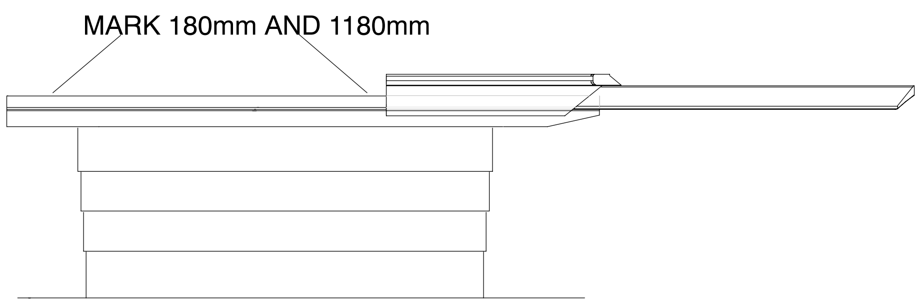


FIGURE 130 180mm and 1180mm COUCH MARKINGS

CAUTION: EXERCISE EXTREME CARE WHILE MOVING AND ALIGNING RESOLVER TO PREVENT ANY BENDING OR MISALIGNMENT ON RESOLVER SHAFT.

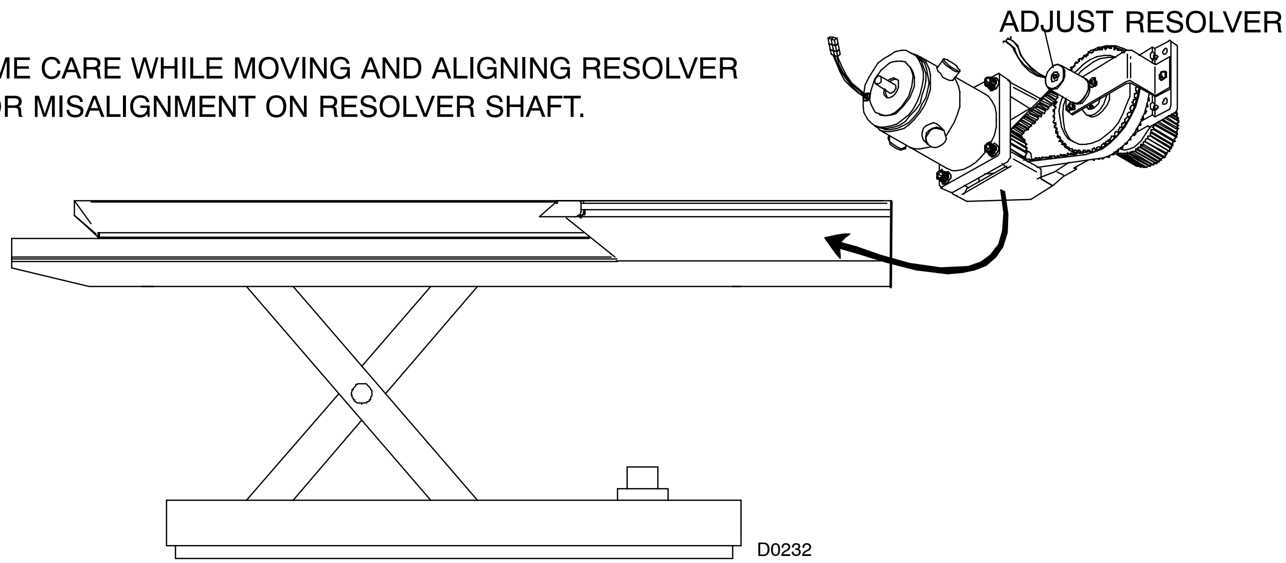


FIGURE 131 COUCH RESOLVER AND POTENTIOMETER LOCATION

Refer to Figure 132.

- The ”old” MSB hex switch has been hardwired to a value of ”C.” SW1 is the second MSB, SW2 and SW3 follow.

18. After the calibration procedure is complete, exit from the CC program. At the AcQSim CT> prompt, press Shift Control E to exit Gputty+.
19. Exit the Diagnostic GUI and remove power from the system. Refer to the PowerUp/Down procedure.
20. Attach the red and black connector that is attached to the couch servo motor leads.
21. Replace the telescoping couch base covers and rear panel. Refer to the telescoping base cover removal procedure.

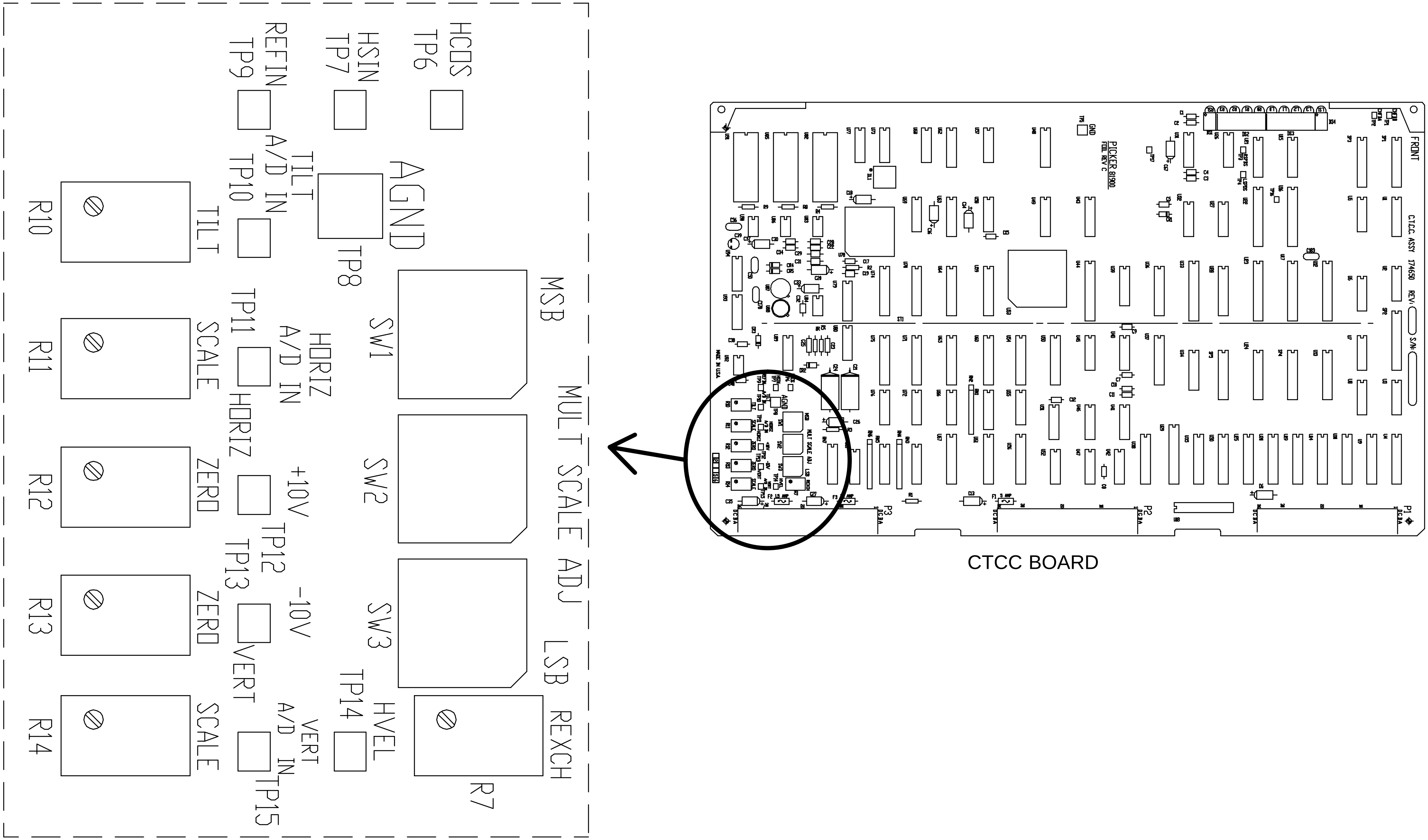


FIGURE 132 CTCC BOARD COMPONENT LOCATIONS DETAIL

DI (DATA ACQUISITION MENU)

1. Type **di** at the prompt to display:

```
DATA ACQUISITION DIAGNOSTICS

0 – QUIT                1 – LIN, XC & LOG
2 – BACKGROUND          3 – DELTA DATA
4 – AVERAGING           5 – DIGITAL CHECK
6 – STATIONARY DATA     7 – ALL TESTS IN ORDER

Enter your choice >
```

2. Enter a **7** to run all tests in order:

```
LINEAR
LINEAR PASSED !
XC TEST
TEST PASSED !
LOGTEST
HARDCODED TESTS PASSED
TEST ONE, COUNT = 1, TIME INCREMENTED
TEST TWO, TIME = 3FFF, COUNT INCREMENTED
ALL TESTS PASSED !
BKGD
HARDCODED TESTS
TEST PASSED !
ADDRESS TEST
TESTING RAM !
TEST PASSED !
TEST ONE, TIMEWORD = 0x3FFF
TEST TWO, TIMEWORD = 0x3FFF
ALL TESTS PASSED
DELTAD
COUNT TEST
COUNT TEST PASSED !
LOG TEST
LOG TEST PASSED !
RAM ADDRESS TEST
DELTA DATA TEST PASSED !
AVG
ADDER AND MUX TEST
```

```
LOADING RAM
TESTING RAM
LOADING RAM
TESTING RAM
LOADING RAM
TESTING RAM
AVGING PASSED!
DIGCHECK
DIGITAL CHECK PASSED !
CHANNELS WITHIN 5000 AND 11000 Hz. CONSIDERED GOOD
USUAL RESULTS 8 KHZ +/– 15%
1858 out of usual range, freq = 6022


SPACE BAR AND RETURN TO QUIT
```

3. Press <SPACE BAR><CR> to return to the main menu.

MC (MAIN DRIVE CALIBRATION)

> **NOTE**
>
> The following procedure utilizes the SCLC cable (P/N L179528) and adapter (P/N L11413) in order to perform Main Drive Calibration.

1. Remove power from the system. Refer to the PowerUp/Down procedure.
2. Disable the main drive by placing the drives switch to the ”0” position. Also, remove fuses F310, F311, F312 on the AC Control Panel. The software will not continue if drives are not enabled.
3. Apply power to the system. Refer to the PowerUp/Down procedure. At the 3 button screen, start the Diagnostic GUI. Select **Gputty+** from the GANTRY pulldown menu.
4. Type **mc** at the AcQSim CT> prompt to display:

```
PMDC / MAIN DRIVE CAL

SPACE BAR & RETURN STEPS THROUGH TESTS, YOU WILL SEE A <sb,cr> PROMPT
HINT: 0x BEFORE A NUMBER IS A HEXADECIMAL NUMBER
!!!!!!!!!!!!!!!!!!!!
WARNING: DISABLE DRIVE SWITCH ON AC CONTROL PANEL
ALSO REMOVE FUSES F310, F311, F312 ON AC CONTROL PANEL
IT IS BEST TO TURN OFF GANTRY, REMOVE FUSES, AND RESTART PROCEDURE
!!!!!!!!!!!!!!!!!!!!
DO YOU WISH TO SKIP TO MOTOR POLES ?
1 = Yes, 0 = No
>
```

5. Refer to Figure 134 for switch and pot locations. Answer **NO** (0) to this question. The following will appear:

```
SET POS OFFSET (SW3 and SW4) SWITCHES TO 0x00 THEN <sb cr>

MOVE FRAME COUNTERCLOCKWISE SO THAT DSW3 READS 0x40 +/- 0x20
AND ADJUST RESOLVER FEEDBACK POT (EXC, R28) FOR 2.0 Vrms

THE SYSTEM WILL BEEP AND AUTOMATICALLY ADVANCE WHEN THE ADJUSTMENT IS IN SPEC

       DSW3:      SIN FEEDBACK VOLTS RMS
        55              1.99

RESOLVER FEEDBACK NOW IN SPEC
```

This step will automatically time out (no <sb cr>).

> **NOTE**
>
> Reading must be initially below 2.0 for beep to occur.

PHASE REFERENCE ADJUSTMENT

```
TURN FINE REF ADJ (R27) COUNTERCLOCKWISE
UNTIL DS1 LED (SYNC POINT) TURNS OFF,
THEN TURN CLOCKWISE UNTIL DS1 LED JUST TURNS FULLY ON THEN STOP ADJUSTMENT  <sb,cr>

TESTING ZERO GATE OPTO INTERRUPTER
–> ROTATE DRIVE CCW TO CENTER ZERO GATE FLAG IN INTERRUPTER <sb cr>
```

Refer to Figure 133 for flag interrupter reference:

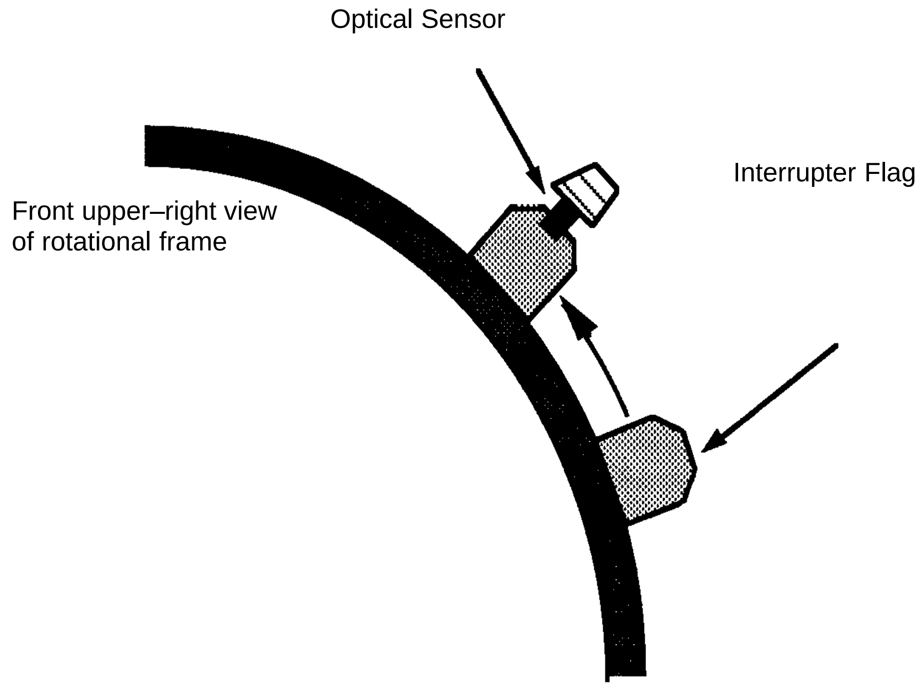


FIGURE 133 FLAG INTERRUPTER THROUGH ZERO GATE

```
-> NOW MOVE FLAG CLEAR OF OPTO INTERRUPTER <sb cr>


-> NOW ROTATE DRIVE CCW SO FLAG PASSES THROUGH ZERO OPTO INTERRUPTER

SET POS OFFSET SWITCHES (SW3 and SW4) TO AF, THEN <sb cr>


-> NOW ROTATE MAIN DRIVE CCW SO FLAG PASSES THROUGH ZERO OPTO INTERRUPTER TWICE
-> THE FIRST REV INITIALIZES NSP WITH SWITCH SETTINGS
-> THE SECOND REV CHECKS NSP AT CENTER OF FLAG TO BE CLOSE TO ZERO

SET POS SWITCHES (SW3 AND SW4) TO XX THEN <sb cr>

NOW ROTATE TWO REVS CCW SO THE FLAG GOES THROUGH GATE
```

This step will automatically time out (no <sb cr>). The display will read:

```
FOUND EDGES XXA7 FOR CCW AND XX FOR CW
WIDTH OF FLAG ='S AB
NSP AT CENTER OF FLAG ='S 95FC
SCAN POSITION IN SPEC
PLEASE SET PHASE ADJ SWITCHES (SW1 AND SW2) TO 0x00  <sb, cr>

DOING MOTOR POLE OFFSET, SETTING SATURATED GALIL COMMAND (DERSIG)
OF PMDC TO +10 VOLTS  PROGRAM ISSUES AB, MO, SM GALIL COMMANDS

DO YOU WISH TO USE SCLC CABLE FOR MOTOR POLES?
1 = YES, 0 = NO
```

If you have the SCLC cable, answer YES (1).

```
MAKE SURE CABLE IS CONNECTED PROPERLY

PLUG SIGNAL END CONNECTOR INTO SCLC P4
PUT BLACK CLIP LEAD ON MOTOR NEUTRAL, LOCATED AT TB616-4
PUT RED CLIP LEAD ON MOTOR PHASE A, TB616-3

-> SPIN DRIVE CCW AT ABOUT 6 SECONDS / REV, BEEPER WILL SOUND
```

6. Make connections of special cable according to above display. Refer to Figure 135 for detail. Make sure cable is not within scan frame. After rotating the scan frame ccw at approx. 6 sec/rev, the following will be displayed:

```
ACQUISITION DONE!
SET PHASE ADJ SWITCHES (SW1 AND SW2) TO xx THEN <sb cr>

NOW WE CHECK SETTINGS—
SPIN DRIVE CCW AT ABOUT 6 SEC/REV

PHASE ADJUSTMENT CORRECT!
DISCONNECT CABLE NOW, THEN <SB CR>


REMEMBER TO CHECK COUCH RESOLVER FEEDBACK CAL
```

7. If an error occurs, you will see a similar display:

```
PHASING DID NOT OCCUR PROPERLY,FOUND 60 MISMATCHES
I SUGGEST RETRYING OR USING SCOPE METHOD
DISCONNECT CABLE NOW, THEN <sb cr>
```

> **NOTE**
>
> After completing the Main Drives Calibration procedure, complete the Resolver Error Correction Calibration procedure in the next paragraph.

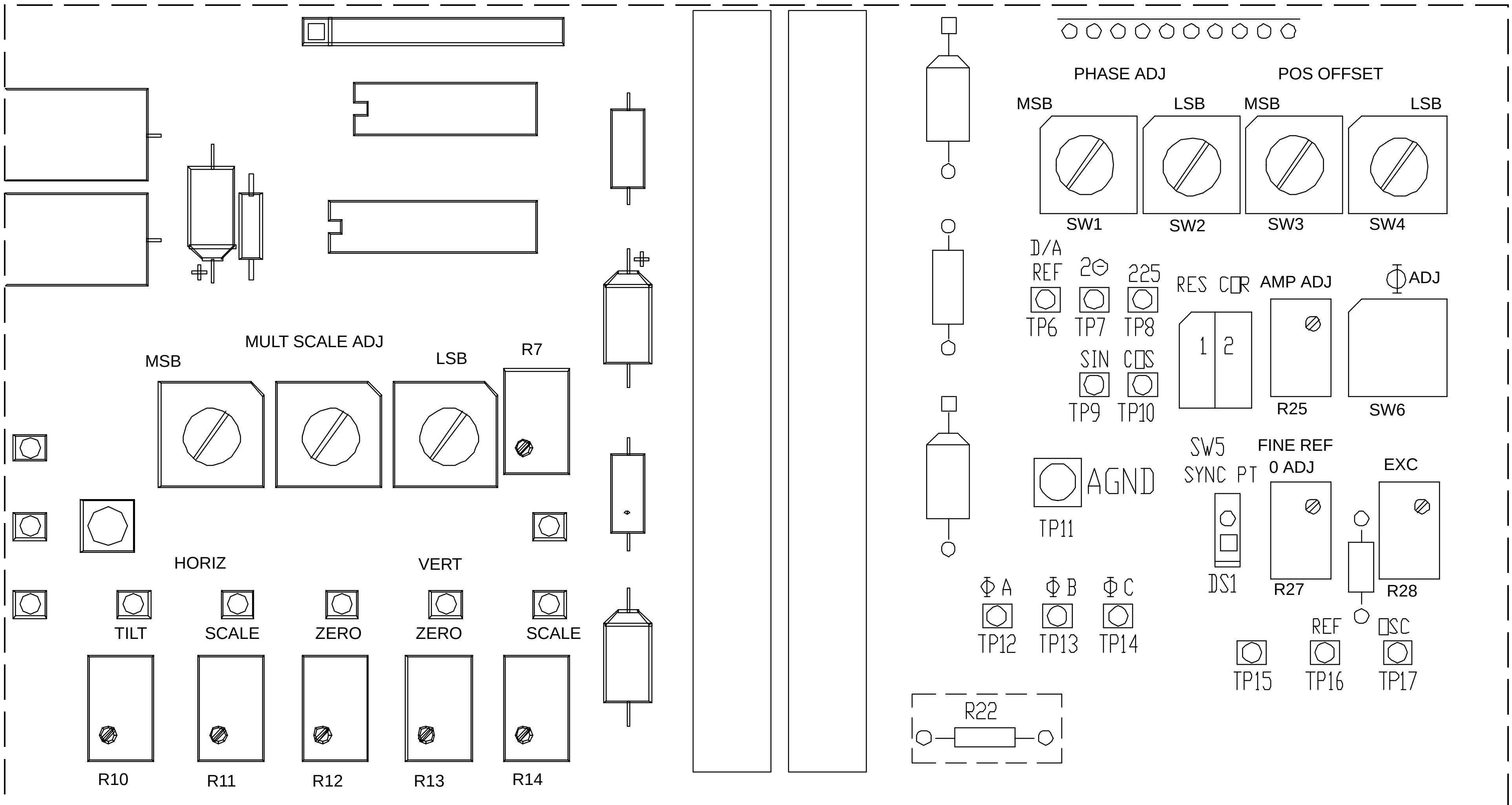


FIGURE 134 PMDC SWITCH SETTING LOCATIONS – (via access panel)

TB616 or TB16:
4= GROUND (BLACK)
3= HOT (WHITE)

(PLUG END GOES TO P4 ON THE SCLC)

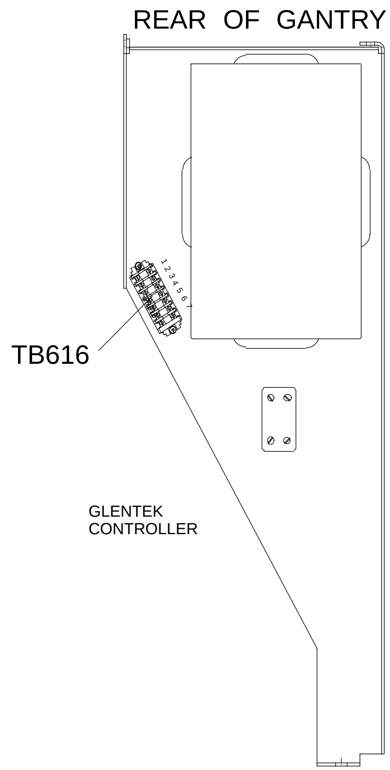


FIGURE 135 SCLC CABLE HOOKUP TO TB616

RESOLVER ERROR CORRECTION CALIBRATION PROCEDURE

> **NOTE**
>
> This procedure should be performed after completing the Main Drives Calibration procedure.

1. At the PMDC board, attach an oscilloscope probe to TP8 (the 225 Hz signal). Set the scope for DC coupled, 0.5 V/div, and a time base of 2 ms/div.
2. On the PMDC board, make sure that switch 1 of SW5 is in the ”ON”, or closed, position. This is the normal position of the switch.
3. With the Diagnostic GUI on your screen. in Board Status box, select X–Ray Momentary enable in the Hardware Control Status box, then CLICK HERE TO CONTINUE. Select Gantry then Ring, continue,continue, and OK to start the ring diagnostic program. Type r (Enter) to run the program then when asked to enter Scan Frame Velocity, type 2 to rotate the scan frame at 2 sec/rev.
4. While observing the waveform on the scope, adjust potentiometer R25 in a clockwise direction. Adjust the potentiometer to obtain the smallest amplitude possible.
5. While observing the waveform on the oscilloscope, adjust hex switch SW6 in single digit increments. If the amplitude of the waveform increases, decrease the setting of the hex switch. The hex switch should be adjusted to produce the smallest amplitude possible on the scope. Adjust the scope vertical deflection setting if needed.
6. After adjusting the hex switch, again adjust potentiometer R25 to fine–tune the waveform to its smallest amplitude. Alternate adjustments of the hex switch and potentiometer to achieve the smallest waveform possible.
7. Press (Enter) at the console to stop the scan frame and Q to exit the Ring program. Next type exit (Enter) to close the unix window.
8. Shutdown the Gantry.
9. Disconnect the test equipment from the scanner and the circuit boards. Select the Exit tab to close Diags and click yes.
10. After the resolver error correction circuit has been calibrated, run the Perform_Lateral_ Centering calibration procedures. Exit out of Diagnostics. At the four button screen, select Service.
11. Start the Gantry. If the X–ray tube has not been previously warmed up, do it now.
12. Select the Calibrate button, click on the Please Select Reason button and select the Perform_Lateral_Centering reason, next click Advance to Scan then Start Scan and the calibration will run automatically displaying complete when finished.
13. Select General Utilities from the bottom tool bar, then Shutdown Console from the shutdown menu.

SS (SCAN SEQUENCE)

1. Type **ss** at the prompt to display:

```
GENERAL DIAGNOSTICS

0 – QUIT                      1 – CONTROL WORDS
2 – SCAN SEQUENCE             3 – IP TIMER AND PD2
4 – PD4 & PD5                 5 – AVERAGING INITIAL VIEWS
6 – DSCC SKIP                 7 – XFR LENGTH TEST
8 – GATE ARRAY DELTA D        9 – V–>F LINEARITY
10 – BAC0 DELTA DATA         11 – PSDC SELECTS
12 – ALL TESTS

Enter your choice >
```

Option #**1** will return:

```
CONTROL WORD TEST
PASSED!
```

Option #**2** will return:

```
PPI
SCAN SEQUENCE
START ADDRESS
0
1
2
3
ALL TESTS PASSED
```

Option #**3** will return:

```
DI TIMER TEST, .1 –> 16.4 msec
IP TIMING TEST, .025 –> 3.28 sec.
IP TIMING PASSED !
```

Option #**4** will return:

```
TESTING PD4, PRIMARY START ADDRESS
TESTING OVERRIDING LOAD OF START ADDRESS COUNTERS
TESTING PD5, SCATTER START ADDRESS
TESTS PASSED !
```

Option #**5** will return:

```
AVERAGING INITIAL XFER TESTS
PASSED
```

Option #**6** will return:

```
DATA SKIP DSCC TEST
TESTING SKIP, XFR LENGTH = 282, SKIP LENGTH = 2
START ADDRESS INCREMENTED
SKIP TEST WITH SKIP LENGTH INCREMENTED FROM 2 TO 255
FIXED LOCATIONS AND FAN LENGTH
SKIP PASSED
```

Option #**7** will return:

```
PRIMARY RING XFR LENGTH TEST
SCATTER RING XFR LENGTH TEST
SCATTER TRANSFER LENGTH TEST PASSED !
ALL TESTS PASSED
```

Option #**8** will return:

```
FIRST TRANSFER SUPPRESSION TEST
PRIMARY RING GATE ARRAY DELTA DATA TEST
TEST 0  CT48 = C037 TESTING FOR 0x1000
TEST 1  CT48 = C036 TESTING FOR 0x1800
TEST 2  CT48 = C034 TESTING FOR 0x4D27
TEST 3  CT48 = C035 TESTING FOR 0x4527
PASSED
```

Option #**9** will return:

```
V –> F LINEARITY TEST, PRIMARY AND SCATTER
TESTING QUADRANT 0
TESTING QUADRANT 1
TESTING QUADRANT 2
TESTING QUADRANT 3
PASSED!
```

Option #**10** will return:

```
PRIMARY RING BAC0 DELTA DATA NOISE TEST, 300 usec IP, 12 KHZ OFFSET
SCATTER RING BAC0 DELTA DATA TEST, 1.2 msec IP, 12 KHZ
ACQUISITION DONE
CHECKING PRE DATA TAGS
SHUFFLING
ANALYZING NOISE
PASSED
```

Option #**11** will return:

```
TESTING SELECTS, GPU SELECT MODE
TESTING START ADDRESS HALF OF SELECT CIRCUITRY
TESTS PASSED !
```

Option #**12** will return the results of all 11 tests.

Option #**0** will quit the diagnostics.

GANTRY – GENA

Gena is a utility that enables the built–in GPU diagnostics. This is used when a service person has a vt100 compatible terminal or laptop connected to the GPU board.

Enabling Gputty+ via the laptop (GENA Selection from GANTRY pulldown)

1. Connect a serial cable from the DB25 connector at the rear left bottom of the gantry to the serial port on the laptop computer.
2. When using a laptop, a communication program capable of emulating an ANSI standard terminal is required (e.g. HyperTerminal). Set the communication port on the laptop to 9600 baud, 8 data bits, no parity, 1 stop bit, flow control Xon/Xoff, ANSI or VT100 emulation and select Com1 or Com2 depending on the connector into which you have plugged the serial cable.
3. Enable Gena from the GANTRY pulldown menu. See Figure 136. Click on Continue and OK at the screen prompts. After completion, the screen will indicate “GPU Enable done.”
4. Click in the blue window to continue. The AcQSim CT> prompt should appear on the laptop screen. Enter a ? to display the available options. Details of the program can be seen in the Gputty+ description.
5. (Type exit <CR> at the pSH+> prompt to exit the Xterm window.)
6. To EXIT GPUSPY, press <Ctrl><Shift>E. (At this time, this does not work – instead, exit the terminal program.)

For a laptop connection, the following serial cable wiring configuration is used:

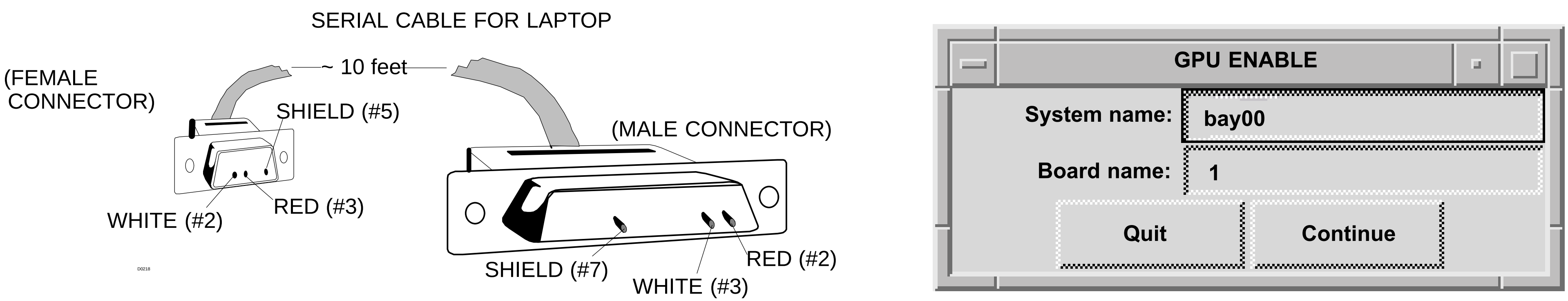


FIGURE 136 SERIAL CABLE CONFIGURATION / GPU ENABLE WINDOW

GANTRY – RING

RING tests the ability of the XSC (Xray System Control board) to communicate over the slip ring to the CP board. It will report ACK NAK BREAK Errors with respect to frame position(detector number). The intent here is to try and determine where on the communications slip ring there is a problem. It is assumed that there is no problem with the communications between the GPU and the CP boards. The test will also print out a coverage listing. Test Coverage shows the number of times a frame position(detector number) was received up to 9, + indicates > 9 times. Each row is 100 detectors wide. A column is 24 deep, therefore 2400 detectors.

1. Ensure the AcQImage Cabinet, Gantry Power are enabled before running the test.
2. Launch the RING program from the GANTRY pulldown menu. When the window appears, click on Continue. See Figure 137.

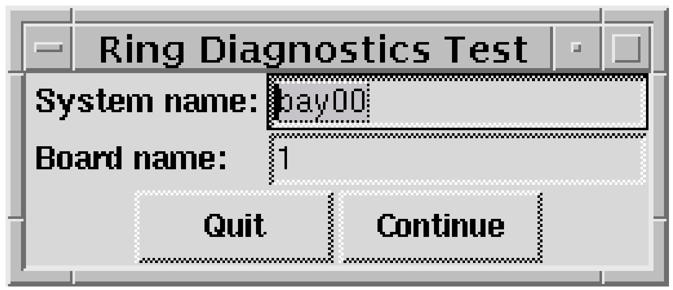


FIGURE 137 Ring Diagnostic Window

3. A warning window will remind you that they gantry and xray power must be on and the main drives are enabled to run this test. Click on OK in the warning window. The Ring program will start. Select “R” to run Ring. Refer to Figure 138.
4. Enter a scan frame velocity of 0 or 0.92 to 7 secs/rev. (8=abort). If “0” is entered, the scan frame will not rotate. You will be warned that the scan frame will move after <CR> is pressed.
5. Press any key to stop scan frame motion. The coverage will be reported with each individual detector status represented as a “1” or a “0.” Refer to Figure 139 for a sample status window.
6. Type exit <CR> at the pSH+> prompt to exit the Xterm window. (Letters will not display when typing exit).

Beginning ring diagnostic test
Turning rotor on, this may take more than 15 seconds
Don not push the Emergency Stop Button during rotor spin up

PICKER XSC RING COMMUNICATIONS TEST
VERSION 3.00
Picker International, Inc. 2/30/98

(Q)UIT
(R)UN

Command: []

FIGURE 138 RING WINDOW

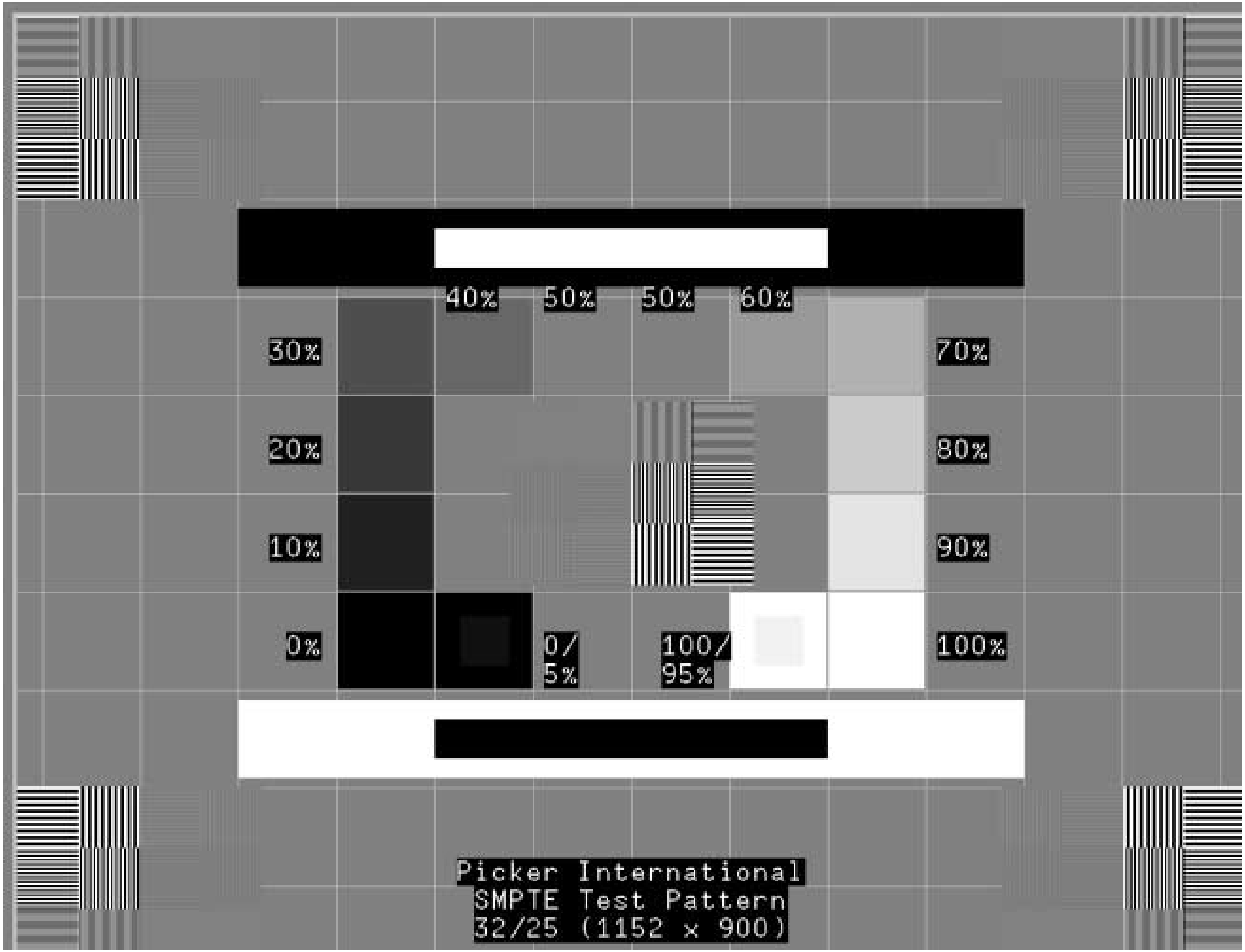
```
Enter scan frame velocity 0 or 0.92-7.00 secs./rev. 8=abort: 3

*** WARNING, SCAN FRAME WILL MOVE AFTER RETURN IS HIT ***

Testing
Hit any key to stop!

Test Coverage shows the number of times a frame position was received up to 9, + indicates > 9 times
Each row is 100 detectors wide
A column is 24 deep, therefore 2400 detectors
             1         2         3         4         5         6         7         8         9
   0123456789012345678901234567890123456789012345678901234567890123456789012345678901234567890123456789

 0 0000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000
 1 0000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000
 2 0000000000000000000000000000000000000000000000000000010000000000000000000000000000000000000000000000
 3 0000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000
 4 0000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000
 5 0000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000
 6 0001000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000
 7 0000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000
 8 0000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000
 9 0000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000
10 0000000000000000000000000000000000000100000000000000000000000000000000000000000000000000000000000000
 1 0000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000
 2 0000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000
 3 0000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000
 4 0000000000000000000000000000000000000000000000000000000000000000000001000000000000000000000000000000
 5 0000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000
 6 0000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000
 7 0000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000
 8 0000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000
 9 0000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000
20 0000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000
 1 0000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000
 2 0000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000
 3 0000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000
Coverage=4/2400=0%
```

FIGURE 139 SAMPLE RING STATUS WINDOW

SUN – SMPTE

The SMPTE pattern is displayed to assist in calibrating the display monitor.

1. After the Smtpe option is selected, the SMPTE test pattern will appear. Adjust the screen height to the full viewable area, then adjust the width so that the squares in the SMPTE are square.
2. To remove it from the screen, click on it with the mouse. The pattern is shown on the next page.


40%
50%
50%
60%
30%
70%
20%
80%
10%
90%
0%
0/
5%
100/
95%
100%
Picker International
SMPTE Test Pattern
32/25 (1152 x 900)

UTILITIES – HARDWARE STATUS CONTROL

> **NOTE**
>
> This test will verify that the System Power Controller Board (board) is able to communicate with the SPARC parallel port. This test also will verify if the **SPC** board is reading and writing status and control signals to the I/O ports. A Display Console with Diagnostics running on disk0 must be available.

1. Verify the Gantry key switch is in normal mode.
2. Gain access to the SPC board L.E.D. status indicators: Remove either the top or left side cover on the Display Console.
3. Enter the Diagnostics GUI.

> **NOTE**
>
> To simplify viewing and operation, any window that appears before the DIAGNOSTIC SOFTWARE TEST MENU should be <u>closed</u>. Do this by clicking on either the OK button, the QUIT button or the upper left hand corner of the window (a dash) and clicking on the Close button.

4. At the DIAGNOSTIC SOFTWARE TEST MENU window select the UTILITIES menu and click on the HARDWARE STATUS/CONTROL selection.
5. The HARDWARE STATUS/CONTROL window will open. Click on the REFRESH button to update the STATUS part of the window.

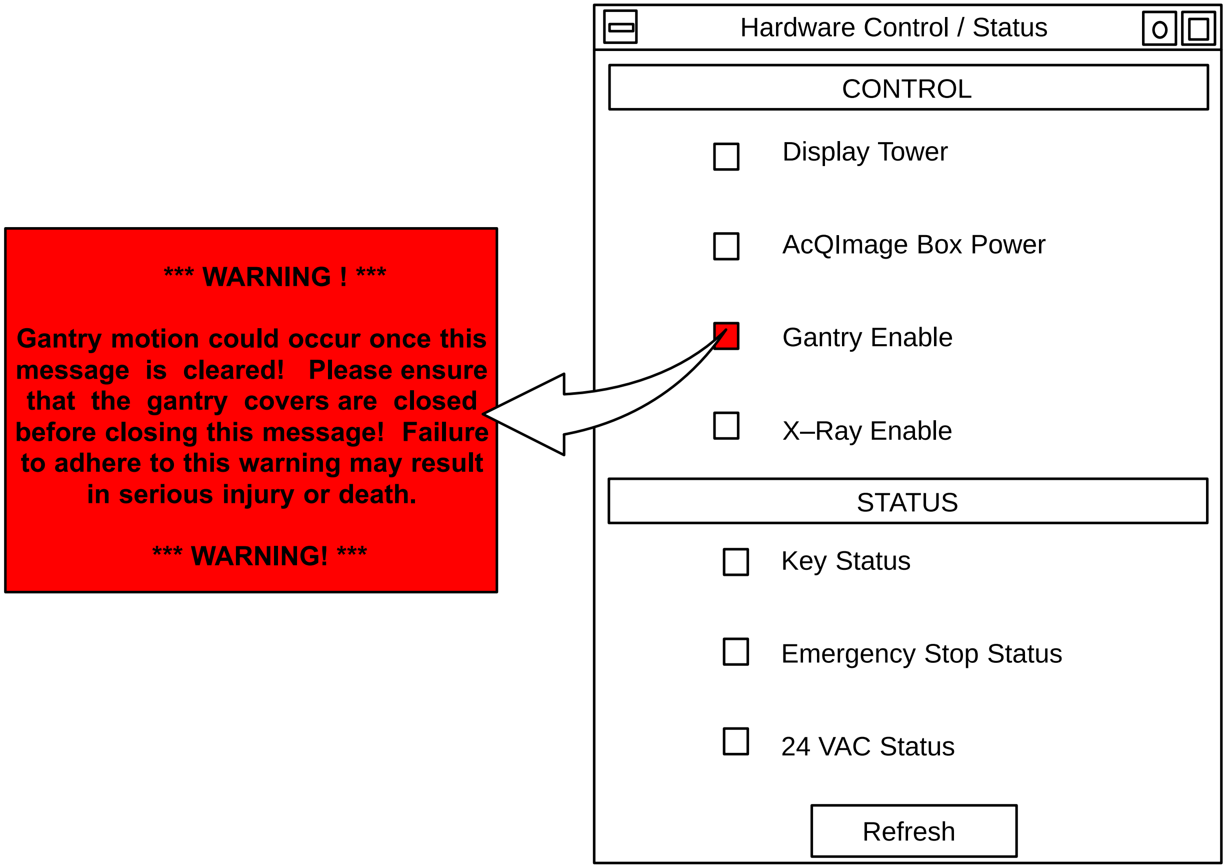


FIGURE 140 Hardware Status Control Window

6. Verify the KEY STATUS and 24 VAC STATUS are enabled (red square). Verify the SPC board's KEY and 24VAC status L.E.D.'s are on.

7. Press and release the Emergency Stop switch on the front panel of the Display Console. As you press it verify the ESTOP status L.E.D. on the SPC board comes on and then turns off when the switch is released.

CAUTION

If the gantry and patient support are turned on at the time of test they will power down when the E–stop switch is pressed.

8. Click the REFRESH button on the HARDWARE STATUS/CONTROL window. Verify the EMERGENCY STOP STATUS indicator is red. Click the REFRESH button. Verify the EMERGENCY STOP STATUS indicator is not red.
9. Click the DISPLAY POWER selection in the CONTROL window. The square should be red. This verifies that the SPARC is writing to the Display Power signal. Click the DISPLAY POWER selection again, the square should not be red.
10. Click the DISPLAY POWER selection again and verify the DISP status L.E.D. on the SPC board comes on and then goes off again after 22 seconds. Verify the DISPLAY POWER status indicator is red. Click the REFRESH button. The DISPLAY POWER status indicator will not be red.
11. Click the ACQIMAGE BOX POWER selection in the CONTROL window. Verify the status indicator is red. Verify the RECON status L.E.D. on the SPC board is lit. Verify the AcQImage cabinet powers up.
12. Click the ACQIMAGE BOX POWER selection again in the CONTROL window. Verify the status indicator is not red. Verify the SPC board RECON status L.E.D. turns off and the ACQImage cabinet powers down.
13. Click the GANTRY ENABLE selection in the CONTROL window.

NOTE

A warning message stating that the gantry may move will appear. Click on it to enable the gantry.

14. Verify the GANTRY ENABLE status indicator in the CONTROL window turns red. Verify the GAN status L.E.D. on the SPC is on.
15. Click the X–RAY ENABLE selection in the CONTROL window.

NOTE

A warning message stating that the gantry may move will appear. Click on it to enable the X–RAY generator power.

16. Verify the X–RAY ENABLE status indicator turns red. Verify the START status L.E.D. on the SPC is on for approximately 2 seconds and turns off. Verify that the X–RAY generator power comes on.
17. Click the REFRESH button. Verify the X–RAY ENABLE selection is not red.
18. Press the Emergency Stop switch on the front of the Display Console. Verify the GAN status L.E.D. on the SPC board turns off. Also the gantry should be powered off. Click on the REFRESH button. Verify the GANTRY ENABLE selection is not red.
19. Exit the HARDWARE STATUS/CONTROL window by clicking on the dash in the upper left corner of the window. Click on CLOSE. The window will close.
20. You may now exit the SOFTWARE DIAGNOSTIC MENU.

UTILITIES – FLASH BURNER

21. After the GPU/XSC downloader is complete, run the *Flash Burner* from the Diag GUI. The flash burner is a utility for transferring executable code from a directory on the SPARC disk to the Flash of a board of interest.

22. Enter the board number (**1 for CP**).

23. An Xterm will be displayed. Select option 1.) Transfer OS. When completed, the board will reboot.

24. Restart the the Flash burner from the GUI in Figure 141again on the <u>same</u> board. Then select option 2.) Transfer Diags.

25. When completed, the board will reboot. Now one board is done. Repeat this process on the remaining three boards. Board numbers are: 2 for GAM, 3 for DSP/AP, 4 for BP. You can view the date of the Flash OS via a Telnet session. At the pSH+> prompt, type **romver** and press <CR>.

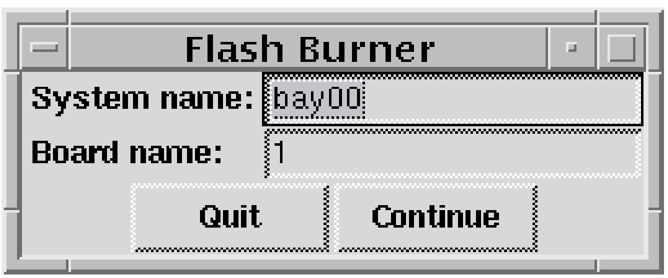


FIGURE 141 Flash Burner

UTILITIES – ENET

ENET OUTPUT

localhost, IP address xxx.x.x.x : localhost is alive
ctscanner1, IP address 1xx.xx.xx.xx : ctscanner1 is alive
ct2a, IP address 1xx.xx.xx.xx : ct2a is alive

FIGURE 142 Enet Window

```
Trying 1xx.xx.xx. .....
Connected to ctscanner1
Escape character is '^]'

pSOSystem  (1xx.xx.xx.xx)

Copyright (c)  Integrated Systems, Inc.,  1992
Welcome to pSOSystem  (NUPPC.2.0.1)...

[cp/sft01–1]  pSH+> []
```

FIGURE 143 Dicom Echo Window

UTILITIES – TELNET SESSION

Allows communication from an Xterm window to a specific board where commands such as reset, etc. can be executed.

When the diag gui first starts, the *Reset Boards* window will appear. Before entering the system number, note the board numbers for the BP, AP and GAM. These are necessary if a specific board reset is needed. The order should be: 1–CP, 2–GAM, 3–DSP/AP, 4–BP.

> **NOTE**
>
> If the AcQImage Cabinet front covers are removed, the LED on the front of the boards will display the board number – 4 (BP), 3 (AP) or 2 (GAM). The CP board is board 1.

To reset a specific board from the Diag gui:

1. Select Telnet session from the Utilities pull down. At the Telnet Session window, select the board you wish to reset and press Continue.
2. The telnet window will appear. At the pSH+> prompt, type **reset** and press <CR>. The specific board will then reset.
3. The telnet session window will not dismiss itself. Move the mouse pointer to the upper bar and press the right mouse button – select **Close** to dismiss the window.

FIGURE 144   Telnet Session Window

UTILITIES – KERMIT2

A terminal connection and file transfer program primarily used on the AcQSim CT to connect to the sun serial ports to run serial port only diags.

```
C-Kermit 5A(189), 30 June 93, Solaris 2.x
Type ? or HELP for help
C-Kermit>? Command, one of the following:
 apc                 ask                 askq                assign
 bug                 bye                 cd                  check
 clear               close               comment             connect
 declare             decrement           define              delete
 dial                directory           disable             do
 echo                enable              end                 exit
 finish              for                 get                 getok
 goto                hangup              help                if
 increment           input               introduction        log
 mail                msend               msleep              news
 open                output              pause               print
 push                pwd                 quit                read
 receive             redial              reinput             remote
 rename              return              run                 script
 send                server              set                 show
 space               statistics          stop                suspend
 take                translate           transmit            type
 version             wait                while               who
 write               xif
or one of the tokens '!#;:@'
C-Kermit>[18~
```

FIGURE 145 Kermit2 Window

UTILITIES – XTERM

Simply provides an Xterm window in which specific commands can be entered via the keyboard.

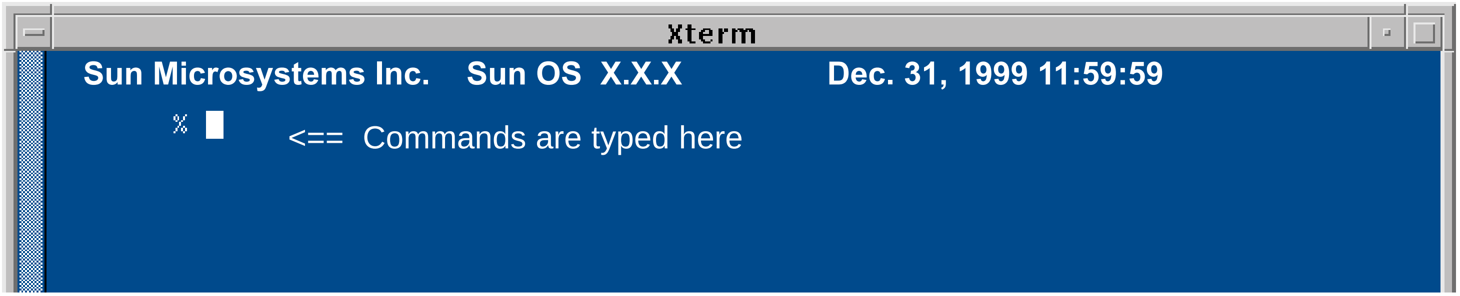


FIGURE 146 Xterm Window

XTERM UTILITIES – WHAT CAN BE RUN VIA AN XTERM WINDOW

There are several utilities you can run directly from the Xterm window. The following are covered:

Ping, Telnet and Reset
Arc Dump
SUN Logs
EEPROM

PING

Tests the status of a node on the network by sending out a quick signal and receiving a quick response if a connection was made.

At the % prompt, type **ping** [and the internet address] of the board or machine you wish to reach and press <CR>. The response will show the status of the ping attempt {address is “alive” or address doesn’t respond].

TELNET

Log onto a board to reset it or check its configuration. At the % prompt, type **telnet bay00–n** , where “n” is the board number:

n = 1 for CP
n = 2 for GAM
n = 3 for AP
n = 4 for BP

At the % prompt, type **board** and press <CR>. A list of board information , including ethernet address will be displayed.

RESET

Reset an individual board. At the % prompt, type **reset bay00–n** , where “n” is the board number:

n = 1 for CP
n = 2 for GAM
n = 3 for AP
n = 4 for BP

ARC DUMP

There is a Sun command that will capture a snapshot of all the log files produced. This snapshot is useful for CT Engineering to help diagnose system problems based on the history of certain events. The progam is contained in picr $path: /opt/PICKER/picr/service_util/diagnostics/pq2000p/diag_gui/bin.

Running arcdump (or logdump) will switch to $LOG_DIR, get info from the pSOS boards & then archive all files to a .tar file, excluding any existing .tar files. It creates the tar file name from the system name, plus date & time and adds the .tar extension, such as:

```
bay00.12–31–1999.11:59.tar
```

To run arcdump at the system:

Open an Xterm window via the Diagnostic GUI. Change directory (*cd*) to /opt/PICKER/picr/service_util/diagnostics/pq2000p/diag_gui/bin. At the % prompt, type **arcdump** and press <CR>. The system will begin generating the arcdump and it will take a few minites to create the file, which is typically in the 5MB range.

To run arcdump remotely:

You can run arcdump remotely via telnet, if the system is set up to receive a telnet command. You will need the IP address of the system, log in as **scanvis** with **picker** as the password. Change directory (*cd*) to opt/PICKER/picr/service_util/diagnostics/pq2000p/diag_gui/bin. At the % prompt, type **arcdump** and press <CR>. The system will begin generating the arcdump and it will take a few minites to create the file, which is typically in the 5MB range. Copy the file to a tape (*tar –cv filename.tar*) and send to CT engineering at WHQ. Or, the .tar file can be uploaded to ServeCom. If you feel you can decifer the individual error logs, you can extract the files and view each one:

To extract the information in the newly created .tar file, change directory (*cd*) to $LOG_DIR (/opt/PICKER/users/scanvis/logs). Create a new directory of your choice using the *mkdir* command (e.g. mkdir arcdump1). Move the newly created .tar file to this directory using the mv command (e.g. mv bay00.12–31–99.11:59.tar /opt/PICKER/users/scanvis/logs/arcdump1 and press <CR>.)

After the file is moved to the new directory, uncompress the file by typing tar –xvf *.tar filename* (The .tar file name will be different each time – for this example, you would type tar –xvf bay00.12–31–99.11:59.tar and press <CR>.) The file will uncompress.

SUN LOGS

The location of SUN log files is /var/adm and the file names begin with *messages*. These files are rolled once a certain size is reached so the most current file is messages. Once the files are rolled, a .<number> suffix is added to each file.

Currently, there is not a method to view these log files other than using the xterm window and paging out the files.

At the % prompt, type **cd /var/adm** and press <CR>. Then type **ls –l /var/adm/messages*.** A similar listing will be displayed:

```
-rw-r--r--   1 root           0 Feb 13 04:05 /var/adm/messages
-rw-r--r--   1 root        6097 Feb 12 16:15 /var/adm/messages.0
-rw-r--r--   1 root       21915 Feb  4 16:28 /var/adm/messages.1
-rw-r--r--   1 root       49461 Jan 29 10:34 /var/adm/messages.2
-rw-r--r--   1 root        7548 Jan 22 14:44 /var/adm/messages.3
```

Below is a printout of one pariticular SUN log:

```
Feb  1 03:29:44 suntan ypbind[2050]: NIS server for domain "ct.picker.com" OK


Feb  1 09:07:21 suntan automountd[153]: server three not responding
Feb  1 09:07:29 suntan last message repeated 1 time
Feb  1 09:47:56 suntan automountd[153]: server openwin not responding
Feb  1 15:02:46 suntan automountd[153]: server openwin not responding
Feb  1 15:03:58 suntan automountd[153]: server openwin not responding
Feb  2 09:09:19 suntan unix: SunOS Release 5.6 Version Generic [UNIX(R) System V Release 4.0]
Feb  2 09:09:19 suntan unix: Copyright (c) 1983-1997, Sun Microsystems, Inc.
Feb  2 09:09:19 suntan unix: vac: enabled
Feb  2 09:09:19 suntan unix: cpu0: FMI,MB86904 (mid 0 impl 0x0 ver 0x4 clock 70 MHz)
Feb  2 09:09:19 suntan unix: mem = 57344K (0x3800000)
Feb  2 09:09:19 suntan unix: avail mem = 53276672
Feb  2 09:09:19 suntan unix: Ethernet address = 8:0:20:73:97:8d
Feb  2 09:09:19 suntan unix: root nexus = SUNW,SPARCstation-5
Feb  2 09:09:19 suntan unix: iommu0 at root: obio 0x10000000
Feb  2 09:09:19 suntan unix: sbus0 at iommu0: obio 0x10001000
Feb  2 09:09:19 suntan unix: espdma0 at sbus0: SBus slot 5 0x8400000
Feb  2 09:09:19 suntan unix: esp0:  esp-options=0x46
Feb  2 09:09:19 suntan unix: esp0 at espdma0: SBus slot 5 0x8800000 sparc ipl 4
Feb  2 09:09:19 suntan unix: sd3 at esp0: target 3 lun 0
Feb  2 09:09:19 suntan unix: sd3 is /iommu@0,10000000/sbus@0,10001000/espdma@5,8400000/esp@5,8800000/sd@3,0...
Feb  2 09:09:19 suntan unix: SUN2.1G cyl 2733 alt 2 hd 19 sec 80>
Feb  2 09:09:19 suntan unix: root on /iommu@0,10000000/sbus@0,10001000/espdma@5,8400000/esp@5,8800000/sd@3
Feb  2 09:09:19 suntan unix: obio0 at root
```

```
Feb  2 09:09:19 suntan unix: zs0 at obio0: obio 0x100000 sparc ipl 12
Feb  2 09:09:19 suntan unix: zs0 is /obio/zs@0,100000
Feb  2 09:09:19 suntan unix: zs1 at obio0: obio 0x0 sparc ipl 12
Feb  2 09:09:19 suntan unix: zs1 is /obio/zs@0,0
Feb  2 09:09:19 suntan unix: cgsix0 at sbus0: SBus slot 3 0x0 SBus level 5 sparc ipl 9
Feb  2 09:09:19 suntan unix: cgsix0 is /iommu@0,10000000/sbus@0,10001000/cgsix@3,0
Feb  2 09:09:19 suntan unix: cgsix0: screen 1152x900, single buffered, 1M mappable, rev 7
Feb  2 09:09:19 suntan unix: cpu 0 initialization complete - online
Feb  2 09:09:19 suntan unix: ledma0 at sbus0: SBus slot 5 0x8400010
Feb  2 09:09:19 suntan unix: le0 at ledma0: SBus slot 5 0x8c00000 sparc ipl 6
Feb  2 09:09:19 suntan unix: le0 is /iommu@0,10000000/sbus@0,10001000/ledma@5,8400010/le@5,8c00000
Feb  2 09:09:19 suntan unix: dump on /dev/dsk/c0t3d0s1 size 262184K
Feb  2 09:09:23 suntan unix: pseudo-device: pm0
Feb  2 09:09:23 suntan unix: pm0 is /pseudo/pm@0
Feb  2 09:09:26 suntan unix: pseudo-device: vol0
Feb  2 09:09:26 suntan unix: vol0 is /pseudo/vol@0
Feb  2 09:17:10 suntan unix: SunOS Release 5.6 Version Generic [UNIX(R) System V Release 4.0]
Feb  2 09:17:10 suntan unix: Copyright (c) 1983-1997, Sun Microsystems, Inc.
Feb  2 09:17:10 suntan unix: vac: enabled
Feb  2 09:17:10 suntan unix: cpu0: FMI,MB86904 (mid 0 impl 0x0 ver 0x4 clock 70 MHz)
Feb  2 09:17:10 suntan unix: mem = 57344K (0x3800000)
Feb  2 09:17:10 suntan unix: avail mem = 53276672
Feb  2 09:17:10 suntan unix: Ethernet address = 8:0:20:73:97:8d
Feb  2 09:17:10 suntan unix: root nexus = SUNW,SPARCstation-5
Feb  2 09:17:10 suntan unix: iommu0 at root: obio 0x10000000
Feb  2 09:17:10 suntan unix: sbus0 at iommu0: obio 0x10001000
Feb  2 09:17:10 suntan unix: espdma0 at sbus0: SBus slot 5 0x8400000
Feb  2 09:17:10 suntan unix:esp0:   esp-options=0x46
Feb  2 09:17:10 suntan unix: esp0 at espdma0: SBus slot 5 0x8800000 sparc ipl 4
Feb  2 09:17:10 suntan unix: sd3 at esp0: target 3 lun 0
Feb  2 09:17:10 suntan unix: sd3 is /iommu@0,10000000/sbus@0,10001000/espdma@5,8400000/esp@5,8800000/sd@3,0
Feb  2 09:17:10 suntan unix: <SUN2.1G cyl 2733 alt 2 hd 19 sec 80>
Feb  2 09:17:10 suntan unix: root on /iommu@0,10000000/sbus@0,10001000/espdma@5,8400000/esp@5,8800000/sd@3...
Feb  2 09:17:10 suntan unix: obio0 at root
Feb  2 09:17:10 suntan unix: zs0 at obio0: obio 0x100000 sparc ipl 12
Feb  2 09:17:10 suntan unix: zs0 is /obio/zs@0,100000
Feb  2 09:17:10 suntan unix: zs1 at obio0: obio 0x0 sparc ipl 12
Feb  2 09:17:10 suntan unix: zs1 is /obio/zs@0,0
Feb  2 09:17:10 suntan unix: cgsix0 at sbus0: SBus slot 3 0x0 SBus level 5 sparc ipl 9
Feb  2 09:17:10 suntan unix: cgsix0 is /iommu@0,10000000/sbus@0,10001000/cgsix@3,0
Feb  2 09:17:10 suntan unix: cgsix0: screen 1152x900, single buffered, 1M mappable, rev 7
Feb  2 09:17:10 suntan unix: cpu 0 initialization complete - online
Feb  2 09:17:10 suntan unix: ledma0 at sbus0: SBus slot 5 0x8400010
```

```
Feb  2 09:17:10 suntan unix: le0 at ledma0: SBus slot 5 0x8c00000 sparc ipl 6
Feb  2 09:17:10 suntan unix: le0 is /iommu@0,10000000/sbus@0,10001000/ledma@5,8400010/le@5,8c00000
Feb  2 09:17:10 suntan unix: dump on /dev/dsk/c0t3d0s1 size 262184K
Feb  2 09:17:14 suntan unix: pseudo-device: pm0
Feb  2 09:17:14 suntan unix: pm0 is /pseudo/pm@0
Feb  2 09:17:18 suntan unix: pseudo-device: vol0
Feb  2 09:17:18 suntan unix: vol0 is /pseudo/vol@0
Feb  2 13:52:51 suntan automountd[155]: server openwin not responding
Feb  3 08:35:19 suntan automountd[155]: server openwin not responding
Feb  3 08:36:59 suntan su: 'su root' failed for aria on /dev/pts/4
Feb  3 08:46:20 suntan automountd[155]: server openwin not responding
Feb  3 10:07:14 suntan su: 'su root' failed for aria on /dev/pts/7
Feb  3 11:36:47 suntan automountd[155]: server openwin not responding
Feb  3 11:57:14 suntan automountd[155]: server openwin not responding
Feb  3 13:28:06 suntan automountd[155]: do_mapent_hosts: ultrawoman: export list: RPC: Unable to receive
Feb  3 13:28:06 suntan last message repeated 5 times
Feb  4 14:09:59 suntan automountd[155]: server openwin not responding
Feb  4 14:32:51 suntan automountd[155]: server openwin not responding
Feb  4 15:42:38 suntan automountd[155]: server openwin not responding
Feb  5 08:06:11 suntan automountd[155]: server openwin not responding
Feb  5 09:58:41 suntan automountd[155]: server openwin not responding
Feb  5 10:36:27 suntan automountd[155]: server openwin not responding
Feb  6 20:32:03 suntan ypbind[5717]: NIS server not responding for domain "ct.picker.com"; still trying
Feb  6 20:32:08 suntan ypbind[5718]: NIS server for domain "ct.picker.com" OK
```

The following are detailed descriptions of common Solaris errors reported via the SUN Logs:

Solaris Common Error Messages

The following are Solaris Common Error Messages (alphabetized) DETAIL DESCRIPTION:

***** FILE SYSTEM WAS MODIFIED *****
====================================
This comment from the fsck(1M) command tells you that it changed the filesystem it was checking.

If fsck was checking the root filesystem, reboot the system immediately to avoid corrupting the / partition. If fsck was checking a mounted filesystem, unmount that filesystem and run fsck again, so that work done by fsck is not undone when in–memory file tables are written out to disk.

** Phase 1— Check Blocks and Sizes
===================================
The fsck(1M) command is checking the filesystem shown in the messages that are displayed before this one. The first phase checks the inode list, finds bad or duplicate blocks, and verifies the inode size and format.

If more than a dozen errors occur during this important phase, you might want to restore the filesystem from backup tapes. Otherwise it is fine to proceed with fsck.

** Phase 1b— Rescan For More DUPS
==================================
The fsck(1M) command detected duplicate blocks while checking a filesystem, so fsck is rescanning the filesystem to find the inode that originally claimedthat block.

If fsck executes this optional phase, you will see additional DUP/BAD messages in phases 2 and 4.

** Phase 2— Check Pathnames
============================
The fsck(1M) command is checking a filesystem, and fsck is now removing directory entries pointing to bad inodes that were discovered in phases 1 and 1b. This phase might ask you to remove files, salvage directories,fix inodes, reallocate blocks, and so on.

If more than a dozen errors occur during this important phase, you might want to restore the filesystem from backup tapes. Otherwise it is fine to proceed with fsck.

** Phase 3— Check Connectivity
================================
The fsck(1M) command is checking a filesystem, and fsck is now verifying the integrity of directories. You mightbeasked to adjust, create, expand, reallocate, or reconnect directories.

You can usually answer yes to all these questions without harming the filesystem.

** Phase 4— Check Reference Counts
====================================
The fsck(1M) command is checking a filesystem, and fsck is now checking link count information obtained in phases 2 and 3. You might be asked to clear or adjust link counts.

You can usually answer yes to all these questions without harming the filesystem.

** Phase 5— Check Cyl groups
==============================
The fsck(1M) command is checking a filesystem, and fsck is now checking the free–block and used–inode maps. You might be asked to salvage free blocks or summary information.

You can usually answer yes to all these questions without harming the filesystem.
451 timeout waiting for input during variable
===
When sendmail(1M) reads from anything that might time out, such as an SMTP connection, it sets a timer to the value of the r processing option before reading begins. If the read doesn't complete before the timer expires, this message appears and reading stops. (Usually this is during RCPT.) The mail message is then queued for later delivery.

If you see this message often, increase the value of the rprocessing option in the /etc/mail/sendmail.cf file. If the timer is already set to a large number, look for hardware problems such as poor network cabling or connections.

550 variable... Host unknown
=============================
This sendmail(1M) message indicates that the destination host machine, specified by the address portion after the @ (at–sign), was not found during DNS (Domain NamingSystem) lookup.

Use the nslookup(1M) command to verify that the destination host exists in that or other domains, perhaps with a slightly different spelling. Failing that, contact the intended recipient and ask for a proper address.

Sometimes thisreturn message indicates that the intended host is merely down, rather than unknown. If a DNS record contains an unknown alternate host, and the primary host is down, sendmail returns a ”Host unknown” message from the alternate host.¤ For uucp mail addresses, the ”Host unknown” message probably means that the destination hostname is not listed in the /etc/uucp/Systems file.

¤ This is a known sendmail version 8.6.7 bug.

550 variable... User unknown
============================
This sendmail(1M) message indicates that the intended recipient, specified by the address portion before the @ (at–sign), could not be located on the destination host machine.

Check the e–mail address and try again, perhaps with a slightly different spelling. If this doesn’t work, contact the intended recipient and ask for a proper address.

554 variable... Local configuration error
===
This sendmail(1M) message usually indicates that the local host is trying to send mail to itself. Check the value of the $j macro in the /etc/mail/sendmail.cf file to ensure that this value is a fully–qualified domain name.
When the sending system provides its hostname to the receiving system (in the SMTP HELO command), the receiving system compares its name to the sender’s name. If these are the same, the receiving system issues this error messageand closes the connection. The name provided in the HELO command is the value of the $j macro.

A command window has exited because its child exited.
===
The argument to a cmdtool(1) or a shelltool(1) window looks like it is supposed to be a command, but the system cannot find the command.

To run this command inside a cmdtool or a shelltool, make sure the command is spelled correctly and is in your search path (if necessary, use a full path name). If you intended this argument as an option setting, use a minus sign (–) at the beginning of the option. Both the cmdtool and the shelltool are OpenWindows terminal emulators.

admintool: Received communication service error 4
==
AdminTool could not start a display method because a remote procedure call timed out, so it can't send the request. This error results when admintool tries to access the NIS or NIS+ tables when networking is not enabled.

Verify the system network status with ifconfig –a to make sure the system is connected to the network. Make sure the ethernet cable is connected and the system is configured to run NIS or NIS+.

answerbook: XView error: NULL pointer passed to xv_set
===
The AnswerBook navigator window comes up, but the document viewer window does not. This message appears on the console, and themessage "Could not start new viewer" appears in the navigator window. This situation indicates that you have an unknown client or a problem with the network naming service.

Run the ypmatch(1) or nismatch(1) command o determine if the client hostname is in the hosts map. If it isn't, add it to to NIS hosts map on the NIS master server. Then make sure the /etc/hosts file on the client contains an IP address and entry for that hostname followed by loghost (reboot if you changed the /etc/hosts file). Check that the ypmatch or nismatch client hosts command returns the same IP host address as in the /etc/hosts file. Finally, quit all existing AnswerBooks and restart.

Arg list too long
=================
The system could not handle the number of arguments given to a command or program when it combined those arguments with the environment's exported shell variables. The argument list limit is the size of the argument list plus the size of the environment's exported shell variables.

The easiest solution is to reduce the size of the parent process environment by unsetting extraneous environment variables. (See the man page for the shell you're using to find out how to list and change your environment variables.) Then run the program again. An argument list longer than ARG_MAX bytes was presented to a member of the exec() family of system calls. The symbolic name for this error is E2BIG, errno=7.

Argument out of domain
======================
This is a programming error or a data input error.

Ask the program's author to fix this condition,or supply data in a different format.

This indicates an attempt to evaluate a mathematical programming function at a point where its value is not defined. The argument of a programming function in the math package (3M) is out of the domain of the function. This could happen when taking the square root, power, or log of a negative number, when computing a power to a non–integer, or when passing an out–of–range argument to a hyperbolic programming function. To help pinpoint a program's math errors, use the matherr(3M) facility. The symbolic name for this error is EDOM, errno=33.

Arguments too long
==================
This C shell error message indicates that there are too many arguments after a command. For example, this can happen by invoking rm * in a huge directory. The C shell cannot handle more than 1706 arguments.

Temporarily start a Bourne shell with sh and run the command again. The Bourne shell dynamically allocates command line arguments. Return to your original shell by typing exit.

assertion failed: variable, file variable, line N
==
A condition in the program that was never expected to happen has happened.

Contact the vendor or author of the program to ask why it failed. If you have the source code for the program, you can look at the file and line number where the assertion failed. This might give you an idea of how to run the program differently.

This message results from a diagnostic macro called assert() that a programmerinserted into the specified line of a source file. The expression that evaluated untrue precedes the file name and line number.

automountd[N]: No network locking on variable: contact admin to install server change
==
See "WARNING: No network locking on variable: contact admin to install server" message for details. If the server is not changed, data loss is possible in applications that depend on locking.

automountd[N]: server variable not responding
==

This automounter message indicates that the system tried to mount a filesystem from an NFS server that is either down or extremely slow to respond. In some cases thismessage indicates that the network link to the NFS server is broken, although that condition produces other error messages as well.

If you are the system administrator responsible for the non– responding NFS server, check it out to see whether the machine needs repair or rebooting. Encourage your user community to report such problems quickly but only once. When the NFS server is back in operation, the automounter will be able to access the requested filesystem.

automount[N]: variable: Not a directory
==
The file specified after the first colon is not a valid mount point because it is not a directory.

Ensure that the mount point is a directory, and not a regular file or a symbolic link.

Bad address
===========
The system encountered a hardware fault in attempting to access a parameter of a programming function.

Check if the bad address resulted from supplying the wrong device or option to a command. If that is not the problem, contact the vendor or author of the program for an update.

This error could occur any time a function that takes a pointer argument is passed an invalid address. Because processors differ in their ability to detect bad addresses, on some architectures passing bad addresses can result in undefined behaviors. The symbolic name for this error is EFAULT,errno=14.

BAD/DUP FILE I=i OWNER=o MODE=m SIZE=s MTIME=t
== CLEAR?=======
While checking inode link counts during phase 4, fsck(1M) found a file (or directory) that either does not exist or exists somewhere else.

To clear the inode of its reference to this file or directory, answer yes. With the –p (preen) option, fsck automatically clears bad or duplicate file references, so answering yes to this question seldom causes a problem.

Bad file number
===============
Generally this is a program error, not a usage error. Contact the vendor or author of the program for an update.

Either a file descriptor refers to no open file, or a read (or write) request is made to a file that is open only for writing (or reading). The symbolic name for this error is EBADF, errno=9.

N BAD I=N
=========
Upon detecting an out–of–range block, fsck(1M) prints the bad block number and its containing inode (after I=).

In fsck phases 2 and 4, you will decide whether ornot to clear these bad blocks. Before committing to repair with fsck, you could determine which file contains this inode by passing the inode number to the ncheck(1M) command: by passing the inode number to the ncheck(1M) command: # ncheck –iinum filesystem

bad module/chip at: variable
============================
This message from the memory management system often appears with parity errors, and indicates a bad memory module or chip at the position listed. Data loss is possible if the problem occurs other than at boot time.

Replace the memory module or chip at the indicated position. Refer to the vendor's hardware manual forhelp finding this location.

BAD SUPER BLOCK: variable
=========================
This message from fsck(1M) indicates that a filesystem's super– block is damaged beyond repair and must be replaced. At boot time (with the –p option) this message is prefaced by the filesystem's device name. After this message comes the actual damage recognized (see Action). Unfortunately fsck does not print the number of the damaged super–block.

The most common cause of this error is overlapping disk partitions. Donot immediately rerun fsck as suggested by the lines that display after the error message. First make sure that you have a recent backup of the filesystem involved; if not, try to back up the filesystem now using ufsdump(1M). Then run the format(1M) command, select the disk involved, and print out the partition information.

format : N > partition > print

Note whether the overlap occurs at the beginning or end of the filesystem involved. Then run newfs(1M) with the –N option to print out the filesystem parameters, including the location of backup super–blocks.

newfs –N /dev/dsk/device

Select a super–block from a non–overlapping area of the disk, but note that in most cases you have only one chance to select the proper replacement super–block, which fsck soon propagates to all the cylinders. If you select the wrong replacement super–block, data corruption will probably occur, and you will have to restore from backup tapes. After you select a new super–block, provide fsck with the new master super–block number:

fsck –o b=NNNN /dev/dsk/device

Specific reasons for a damaged super–block include: a wrong magic number, out of range NCG (number of cylinder groups) or CPG (cylinders per group), the wrong number of cylinders, a preposterously large super–block size, and trashed values in super–block. These reasons are generally not meaningful because a corrupt super–block is usually extremely corrupt.

BAD TRAP
========
A bad trap can indicate faulty hardware or a mismatch between hardware and its configuration information. Data loss is possible if the problem occurs other than at boot time.

If you recently installed new hardware, verify that the software was correctly configured. Check the kernel traceback displayed on the console to see which device generated the trap. If the configuration files are correct, you will probably have to replace the device. In some cases, the bad trap message indicates a bad or down–rev CPU.

A hardware processor trap occurred, and the kernel trap handler was unable to restore system state. This is a fatal error that usually precedes a panic, after which the system performs a sync, dump, and reboot. The following conditions can cause a bad trap: a system text or data access fault, a system data alignment error, or certain kinds of user software traps.

bad trap = N
============
See the message ”BAD TRAP” for details.

/bin/sh: variable: too big
==========================
This Bourne shell message indicates a classic ”no memory” error. While trying to load the program specified after thefirstcolon, the shell noticed that the system ran out of virtual memory (swap space).

See the message ”Not enough space” for information on reconfiguring your system to add more swap space.

Block device required
=====================
A raw (character special) device was specified where a block device was required, such as during a call to the mount(1M) command.

To see which block devices are available, use ls –l to look in /devices. Then specify a block device instead of a character device. Block device modes start with a b, whereas raw character device modes start with a c.

The symbolic name for this error is ENOTBLK, errno=15.

Boot device: /iommu/sbus/variable/variable/sd@3,0
===
This message alwaysappears at the beginning of rebooting. If there is a problem, the system hangs, and no other messages appear. This condition is caused by conflicting SCSI targets for the boot device, which is almost always target 3. The boot device is usually the machine's internal disk drive, target 3. Make sure that external and secondary disk drives are targeted to 1, 2, or 0, and do not conflict with each other. Also, make sure that tape drives are targeted to 4 or 5, and CD drives to 6, avoiding any conflict with each other or with the disk drives. You can set a device's target number using pushbutton switches or a dial on the back near the SCSI cables. If the targeting of the internal disk drive is in question, check it by powering off the machine, removing all external drives, turning the power on, and running the probe–scsi–all or probe–scsi command from the PROM monitor.

Broadcast Message from root (pts/N) on server [date]
==
This message from the wall(1M) command gets transmitted to all users logged into a system. You could see it during a rlogin or telnet session, or on terminals connected to a timesharingsystem.

Carefully read the broadcast message. Often this broadcast is followed by a shutdown warning. See the message ”The system will be shut down in N minutes” for details about system shutdown.

Broken pipe
===========
This condition is often normal, and the message is merely informational (as when piping many lines to the head program). The condition occurs when a write on a pipe does not find a reading process. This usually generates a signal to the executing program, but this message displays when the program ignores the signal.

Check the process at the end of the pipe to see why it exited. The symbolic name forthis error is EPIPE, errno=32.

Bus Error
=========
A process has received a signal indicating that it attempted to perform I/O to a device that is restricted or that does not exist. This message is usually accompanied by a core dump, except on read–only filesystems.

Use a debugger to examine the core file and determine what program fault or system problem led to the bus error. If possible, check the program's output files for data corruption that might have occurred before the bus error.

Bus errors canresult from either programming error or device corruption on your system. Some common causes of bus errors are:invalid file descriptors, unreasonable I/O requests, bad memory allocation, misaligned data structures, compiler bugs, and corrupt boot blocks.

Cannot allocate colormap entry for ”variable”
===
This message from libXt (X Intrinsics library) indicates that the system colormap was full even before the color name specified in quotes was requested. Some applications can continue after this message. Other applications, such as Workspace Properties Color, fail to come up when the colormap is full.

Exit the programs that make heavy use of the colormap, then restart the failed application and try again.

Can't create public message device (Device busy)
==
This message comes from the lp print scheduler, indicating that it is either extremely busy or hung.
If print jobs are coming out of the printer in question, wait until they are finished and then resubmit this print job. If you see this message again, the lp system is probably hung. See the message ”lp hang” for a procedure to clear the queue. If lp is unable to create a device for printer messages, the message FIFO could be already in use, or locked by another print job.

Can’t invoke /etc/init, error N
================================
This message can appear while a system is booting, indicating that the init program is missing or corrupted. Note that /etc/init is a symbolic link to /sbin/init.

Boot the miniroot so you can replace init. Halt the machine by typing Stop–A or by pressing the reset button. Reboot single–user from CDROM, the net, or diskette. For example, type boot cdrom –s at the ok prompt to boot from CDROM. After the system comes up and gives you a # prompt, mount the device corresponding to the original / partition somewhere, with a command similar to the mount command below. Then copy the init program from the miniroot to the original / partition, and reboot the system.

mount /dev/dsk/c0t3d0s0 /mnt # cp /sbin/init /mnt/sbin/init # reboot   If this doesn’t work, other files might be corrupted, and you might need to reinstall the entire system.The error number is 2 if /sbin/init is missing, or 8 if /sbin/init has an incorrect executable format. This is usually followed by a ”panic:icode” message. The system tries to reboot itself, but goes into a loop, because rebooting is impossible without init.

can’t synchronize with hayes
=============================
This message sometimes appears when using a modem that the system regards as a ”Hayes” type modem, which includes most modems manufactured today. The message can be caused by incorrectswitch settings, by poor cable connections, or by not turning the modem on.

Check that the modem is on and that the cables between the modem and your system are securely connected. Check the internal and external modem switch settings. Turn themodem off and then on again, if necessary.

cd: Too many arguments
=======================
The C shell’s cd(1) command takes only one argument. Either more than one directory was specified, or a directory name containing a space was specified. Directory names with spaces are easy to create with File Manager. Use only one directory name. To change to a directory whose name contains spaces, enclose the directory name in double (”) or single (’) quotes, or use File Manager.

Channel number out of range
============================
The system has run out of stream devices. This error results when a stream head attempts to open a minor device that does not exist or that is currently in use.

Check that the stream device in question exists and was created with an appropriate number of minor devices. Make sure that the hardware corresponds to this configuration. If the stream device configuration is correct, try again later when more system resources might be available. The symbolic name for this error is ECHRNG, errno=37.

chmod: ERROR: invalid mode
==========================
This message from the chmod(1) command indicates a problem in the first non–option argument.

If you are specifying a numeric file mode, you can provide any number of digits (although only the final one to four are considered), but all digits must be between 0 and 7. If you are specifying a symbolic file mode, use the syntax provided in the chmod usage message to avoid the ”invalid mode” error message:

Usage: chmod [ugoa][+–=][rwxlstugo] file ...

Note that some combinations of symbolic keyletters produce no error message but fail to have any effect. The first group,ugoa], is truly optional. The second group, [+–=], is mandatory for chmod to have an effect. The third group,[rwxlstugo], is also mandatory for effect, and can be used in combination when that combination does not conflict.

Command not found
=================
The C shell could not find the program you gave as a command. Check the form and spelling of the command line. If that looks correct, echo $path to see if the user’s search path is correct. When communications are garbled, it is possible to unset a search path to such an extent that only built–in shell commands are available. Here is a command to reset a basic search path:

 % set path = (/usr/bin /usr/ccs/bin /usr/openwin/bin .)
If the search path looks correct, check the directory contents along the search path to see if programs are missing or if directories are not mounted.

Connection closed.
==================
This message can appear when using rlogin(1) to another system if the remote host cannot create a process for this user, if the user takes too long to type the correct password, if the user interrupts the network connection, or if the remote host goes down. Data loss is possible if files were modified and not saved before the connection closed. Try again. If the other system has gone down, wait for it to reboot first.

Connection closed by foreign host.
==================================
When a user telnets to another system, this message can appear if the user takes too long to type the correct password, if the remote host cannot create a login for this user,or if theremote host goes down or terminates the connection. Data loss is possible if files were modified and not saved before the connection closed. Just try again. If the other system has gone down, wait for it to reboot first.

[Connection closed.Exiting]
============================
After using the talk(1) command to communicate with another user, the other person enters an interrupt (usually Control–c), and this message appears on your screen.

Sending an interrupt like this is the usual wayof exiting the talk program. The talk session is over and you can return to your work.

Connection refused
==================
No connection could be made because the target machine actively refused it. This happens either when trying to connect to an inactive service or when a service process is not present at the requested address.

Activate the service on the target machine, or start it up again if it has disappeared. If for security reasons you do not intend to provide this service, inform the user community, possibly suggesting an alternative.

The symbolic name for this error is ECONNREFUSED, errno=146.

Connection timed out
====================
This occurs either when the destination host is down or when problems in the network cause lost transmission.

First check the operation of the host system, for example by using ping(1M) and ftp (1), then repair or reboot as necessary. If that doesn't solve the problem, check the network cabling and connections. No connection was established in aspecified time. A connect or send request failed because the destination host did not properly respond after a reasonable interval. (The timeout period is dependent on the communication protocol.)

The symbolic name for this error is ETIMEDOUT, errno=145.

console login:
===========================
This usually occurs because OpenWindows exited abnormally, leaving the system’s keyboard in the wrong mode. The characters that appear when someone attempts to login are garbage transliterations of what someone types.

Find another machine and remote login to this system, then run this command:
$ /usr/openwin/bin/kbd_mode –a
This puts the console back into ASCII mode. Note that kbd_mode is not a windows program, it just fixes the consolemode. The usual reason for this problem occurring is an automated script run from cron that clears out the /tmp directory every so often. Ensure that any such scripts do not remove the /tmp/.X11– pipe or /tmp/.X11–unix directories, or any files therein.

core dumped
===========
A core file contains an image of memory at the point of software failure, and is used by programmers to find the reason for the failure.

To see which program produced a core file, run either the file(1) command or the adb (1) command. The following examples show the output of the file and adb commands on a core file from the dtmail program.

$ file core core: ELF 32–bit MSB core file SPARC Version 1, from‘dtmail’

$ adb core core file = core –– program ‘dtmail’ SIGSEGV 11: segmentation violation ^D (use Control–d to quit the program)

Ask the vendor or author of this program for a debugged version. Some signals, such as SIGQUIT, SIGBUS, and SIGSEGV, produce a core dump. See the signal(5) man page for a complete list. If youhave the source code for the program, you can try compiling it with cc –g, and debugging it yourself using dbx or a similar debugger. The where directive of dbx provides a stack trace.

On mixed networks, it can be difficult to discern which machine architecture produced a particular core dump, since adb on one type of system generally cannot read a core file from another type of system, and will produce an ”unrecognized file” message. Run adb on various machine architectures until youfind the right one. The term ”core” is archaic–– ferrite core memory was supplanted by silicon RAM in the 1970s, although spaceships still employ core memory for its imperviousness to radiation.

For information on saving and viewing crash informationsee the System Administration Guide, Volume II. If you are using the AnswerBook, ”system crash” is a good search string.

Could not initialize tooltalk (tt_open): TT_ERR_NOMP
===
Various desktop tools display or print this message when the ttsession(1) process is not available. The TookTalk service generally tries to restart ttsession if it is not running. So this error indicates that the ToolTalk service is either not installed or is not installed correctly.

Verify that the ttsession command exists in /usr/openwin/bin or /usr/dt/bin. If this command is not present, ToolTalk is not installed correctly. The packages constituting ToolTalk are the runtime SUNWtltk, developer support SUNWtltkd, and the manualpages SUNWtltkm. CDE ToolTalk packages have the same names with ".2" appended. The full TT_ERR_NOMP message string reads as follows: "No ttsession is running, probably because tt_open() has not been called yet. If this is returned from tt_open() it means ttsession could not be started, which generally means ToolTalk is not installed on the system."

Could not start new viewer
==========================
This message appears in the AnswerBook navigator window, along with an XView error messageon the console. See the message "answerbook: XView error: NULL pointer passed to xv_set" for details.

cpio: Bad magic number/header.
==============================
A cpio(1) archive has either become corrupted or was written out with an incompatibleversion of cpio.

Use the –k option to cpio to skip I/O errors and corrupted file headers. This might permit you to extract other files from the cpio archive. To extract files with corrupted headers, try editing the archive with a binary editor such asemacs. Each cpio file header contains a filename as a string. For more information on magic numbers, see magic(4).

Cross–device link
=================
An attempt was made to make a hard link to a file on another device, such as on another filesystem. Establish a symbolic link using ln –s instead. Symbolic links are permitted across filesystem boundaries. The symbolic name for this error is EXDEV, errno=18.

data access exception
=====================
This message can result from running an old version of the operating system that does not support new hardware, or by running an operating system that is not configured for new hardware. It can also result from incorrectly installed DSIMMs or from a disk problem.

Upgrade your operating system toa version that supports the new hardware or machine architecture. For example, upgrading a SPARCstation 2 (with sun4c kernel architecture) to a SPARCstation 0 (with sun4m kernel architecture) requires an operating system upgrade or reconfiguration.

Data fault
==========
This is a kind of bad trap that usually causes a system panic. When this message appears after a bad trap message, a system text or data access fault probably occurred.¤ In the absence of a bad trap message, this message might indicate a user text or data access fault. Data loss is possible if the problem occurs other than at boot time.

Make sure the machine can reboot, then check the log file /var/adm/messages for hints about what went wrong. ¤ See the message ”BAD TRAP” for more information.

Deadlock situation detected/avoided
===================================
A programming deadlock situation was detected and avoided. If the system had not detected and avoided a deadlock, a piece of software would have hung. Run the program again. The deadlock might not reoccur.

This error usually relates to file and record locking, but can also apply to mutexes, semaphores, condition variables, and read/write locks. The symbolic name for this error is EDEADLK, errno=45. See the section on deadlock handling in the System Interface Guide. See the section on avoiding deadlock in the Multithreaded Programming Guide.

Device busy
===========
An attempt was made to mount a device that was already mounted or to unmount a device containing an active file (such as an open file, a current directory, a mount point, or a running program). Thismessage also occurs when trying to enable accounting that is already enabled.

To unmount a device containing active processes, close all the files under that mount point, quit any programs started from there, and change directories out of that hierarchy. Then try to unmountagain.

Mutexes, semaphores, condition variables, and read/write locks set this error condition to indicate that a lock is held. The symbolic name for this error is EBUSY, errno=16.

/dev/rdsk/variable: CAN’T CHECK FILE SYSTEM.
==
The system cannot automatically clean (preen) this file systembecause it appears to be set up incorrectly or is having hard disk problems. This message asks that you run fsck(1M) manually, since data corruption might already haveoccurred.
Run fsck to clean the filesystem in question. See the message ”/dev/rdsk/N: UNEXPECTED INCONSISTENCY; RUN fsck MANUALLY” for proper procedures.

/dev/rdsk/variable: UNEXPECTED INCONSISTENCY; RUN fsck MANUALLY.
===
At boot time the /etc/rcS script runs the fsck(1M) command to check the integrity of filesystems marked ”fsck” in /etc/vfstab. f fsck cannot repair a filesystem automatically, it interrupts theboot procedure and produces this message. When fsck gets into this state, it cannot repair a filesystem without losing one or more files, so it wants to defer this responsibility to you, the administrator. Data corruption has probably already occurred.

First run fsck –n on the filesystem, to see how many and what type of problems exist. Then run fsck again to repair the filesystem. If you have a recent backup of the filesystem, you can generally answer ”y” to all the fsck questions. It’s a good idea to keep a record of all problematic files and inode numbers for later reference. To run fsck yourself, specify options as recommended by the boot script. For example: # fsck /dev/rdsk/c0t4d0s0

Usually the files lost during fsck repair are these that were created just before a crash or power outage, and they cannot be recovered. If you lose important files, you can recover them from backup tapes. If you don’t have a backup, ask an expert to run fsck for you.

Directory not empty
====================
The directory operation that was attempted, such as directory removal with rmdir, can be performed only on an empty directory.
To removethe directory, first remove all the files that it contains. A quick way to remove a non–empty directory hierarchy is with the rm –r command. The symbolic name for this error is ENOTEMPTY, errno=93.

Disc quota exceeded
====================
The user’sdisk limit has been exceeded on a user filesystem, usually because a file was just created or enlarged beyond the limit. This almost always refers to a magnetic disk, and not to an optical disc. Any data created after this condition occurs will be lost.

The user can delete files to bring disk usage under the limit, or the server administrator can use the edquota(1M) command to increase the user's disk limit. The symbolic name for this error is EDQUOT, errno=49.

dumptm: Cannot open '/dev/rmt/variable': Device busy
===
During filesystem backup, the dump program cannot open the tape drive because some other process is holding it open. Find the process that has the tape drive open, and either kill(1) the process or wait for it to finish.

ps –ef | grep /dev/rmt # kill –9 processID
DUP/BAD I=i OWNER=o MODE=m SIZE=s MTIME=t FILE=f REMOVE?
===
During phase 1, fsck(1M) found duplicate blocksor bad blocks associated with the file or directory specified after FILE= whose inode number appears after I= (with other information).

To remove this file or directory, answer yes. If you end up removing more than a few files in this manner, data loss will result, so it might be preferable to restore the filesystem from backup tapes.

N DUP I=N
=========
Upon detecting a block that is already claimed by another inode, fsck(1M) prints the duplicate block number and its containing inode (after I=).

In fsck phases 2 and 4, you will decide whether or not to clear these bad blocks. Before committing to repairwith fsck, you could determine which file contains this inode by passing the inode number to the ncheck(1M) command: # ncheck –iinum filesystem

error: DPS has not initialized or server connection failed
==
This message appears when trying to run AnswerBook with a generic X11 window server or on a generic X terminal.

Running AnswerBook requires Display PostScript (DPS), or a NeWS server, or the Adobe DPS NS remote display software. In addition, a complete LaserWriterII Type–1 font set (including Palatino) should be installed on the X server. To find out if your X server has DPS, run xdpyinfo(1) to verify the presence of an "Adobe–DPS–Extension" line. X servers without this line don't know about DPS.

ERROR: missing file arg (cm3)
==============================
An attempt was madd to run some sccs(1) operation that requires a filename, such as create, edit, delget, or prt. Supply the appropriate filename after the SCCS operation.

ERROR [SCCS/s.variable]: ‘SCCS/p.variable’ nonexistent (ut4)
==
An attempt was made to sccs edit or sccs get a file that is not yet under SCCS control. Run sccs create on that file to place it under SCCS control.

ERROR [SCCS/s.variable]: writable ‘variable’ exists (ge4)
===
An attempt was made to sccs edit a file that is writable, probably because it is already checked out.
Run sccs info to see who has the file checked out. If it is you, go ahead and edit it. If it is somebody else, ask that personto check in the file.

esp0: data transfer overrun
============================
When a user tries to mount a CDROM on a third–party CD drive, mount(1M) fails with the above error, followed by the ”sr0: SCSI transport failed” message. The CD drive probably comes from a vendor unknown to the system.

Third–party CD drives generally have an 8192 block size, as opposed to the 512 block size on supported Sun drives. Check with the vendor to see if any special configuration is possible to allow the drive to operate on a Sun workstation.

Event not found
===============
This C shell message indicates that a user tried to repeat a command from the history list, but that command or number does not exist in the list.

Run the C shell history command to display recent events in the history list. If a user often tries to run commands that have disappeared from the history list, make the list longer by setting history to a higher value. For more information about the C shell, see csh(1).

EXCESSIVE BAD BLKSI=N CONTINUE?
===================================
During phase 1, fsck(1M) found more than 10 bad (out–of–range) blocks associated with the specified inode number. With this many bad blocks, it might be preferable to restore the filesystem from backup tapes.

EXCESSIVE DUP BLKS I=N CONTINUE?
===================================
During phase 1, fsck(1M) found more than 10 duplicate (previously claimed) blocks associated with the specified inode number. With this many duplicate blocks, it might be preferable to restore the filesystem from backup tapes.

Exec format error
=================
This often happens when trying to runsoftware compiled for different systems or architectures, such as when executing Solaris 2.x programs on a SunOS 4.1.x system, or when trying to execute SPARC–specific programs on an x86 machine. On a Solaris 2.x system, it can also occur if the BinaryCompatibility Package was not installed.

Make sure that the software matches the architecture and system you're using. The file(1) command can help you determine the target architecture. If you're using SunOS 4.1.x softwareon a Solaris 2.x system, make sure that the Binary Compatibility Package is installed. You can check for it using this command: $ pkginfo | grep SUNWbcp

A request was made to execute a file that, although it has the appropriate permissions, does not start with a valid format. The symbolic name for this error is ENOEXEC, errno=8. See the a.out(4) man page for a description of executable files.
fd0: unformatted diskette or no diskette in the drive
==
This message appears on the system console to indicate that the floppy driver fd(7) could not read the label on a diskette. Usually this is either because a new diskette has not yet been formatted, or a formatted diskette has become corrupted. This message often appears along with "read failed" and "bad format" messages after volcheck(1) is run.

If you are certain that the diskette contains no data, run fdformat –d to format the diskette in DOS format. (You can also format a diskette in UFS format if you like, although then it is not transportable to most other systems.) When the diskette is formatted, you can write on it, if it was not corrupted beyond repair.

File exists
===========
The name of an existing file was mentioned in an inappropriate context. For example,it is not allowed to establish a link to an existing file, or to overwrite an existing file when the csh(1) noclobber option is set.

Look at the names of files in the directory, then try again with a different name or after renaming or removing the existing file. The symbolic name for this error is EEXIST, errno=17.

File locking deadlock
=====================
This is a programming problem, in some cases unavoidable. All a user can do is restart the program and hope deadlock does not reoccur.

Inthe file locking subsystem, two processes tried to modify some lock at the same time. In the multithreading subsystem, two threads became deadlocked and could not continue. When a program using the threads library encounters this error, it should restart the deadlocked threads. The symbolic name for this error is EDEADLOCK, errno=56.

filemgr: mknod: Permission denied
=================================
File Manager issues this message and fails to come up whenever the /tmp/.removable directory is owned by another user and is not 1777 mode. This can happen, for example, when multiple users share a workstation.

Have the original owner change the mode ((chmod(1)) of this file back to 1777, its default creation mode. Rebooting the workstation also resolves this problem. This is a known problem that was fixed in Solaris 2.4.

File name too long
==================
The specified file name has too many characters. If a file name or path name component is too long, devise a
shorter name. If the totalpath name is longer than PATH_MAX characters, first change to an intermediate directory, then specify a shorter path name. Newly–created data will be lost unless written to another file with a shorter name.
In a UFS or NFS–mounted UFS filesystem, the length of a path name component exceeds MAXNAMLEN (255) characters, or the total length of the path name exceeds PATH_MAX (1024) characters. In a System V filesystem, the length of a path name component exceeds NAME_MAX (14) characters while no–truncation mode is in effect. These values are defined in the /usr/include/limits.h(4) file. The symbolic name for this error is ENAMETOOLONG, errno=78.

FILE SYSTEM STATE IN SUPERBLOCK IS WRONG; FIX?
==
The fsck(1M) command has just checked a filesystem, and has determined that the filesystem is clean. The filesystem's superblock, however, still thinks the filesystem is "dirty" in some way If you believe that the filesystem is adequately repaired, answer yes to mark the filesystem as clean.

Different "dirty" filesystem types are listed in /usr/include/sys/fs/ufs_fs.h, and include FSACTIVE, FSBAD, FSFIX, FSLOG, and FSSUSPEND.

File table overflow
===================
The kernel file table is full because too many files are open on the system. Temporarily, no more files can be opened. New data created under this condition will probably be lost.

Simply waiting often gives the system time to close files. However, if this message occurs often, reconfigure the kernel to allow more open files. To increasethe size of the file table in Solaris 2.x, increase the value of maxusers in the /etc/system file. The default maxusers value is the amount of main memory in MB, minus 2. The symbolic name for this error is ENFILE, errno=23.

File too large
==============
The file size exceeded the limit specified by ulimit(1), or the file size exceeds the maximum supported by the file system. New data created under this condition will probably be lost.

In the C shell, use the limit command to see or set the default file size. In the Bourne or Korn shells, use the ulimit –a command. Even when the shells claim that the file size is unlimited, in fact the system limit is FCHR_MAX (usually 1 gigabyte). The symbolic name for this error is EFBIG, errno=27.

FREE BLK COUNT(S) WRONG IN SUPERBLK SALVAGE?
===
During phase 5, fsck(1M) detected that the actual number of free blocks in the filesystem did not match the superblock's free block count.The df(1M) command accesses this free block count when measuring filesystem capacity. Generally you can answer yes to this question without harming the filesystem.

fsck: Can’t open /dev/dsk/variable
==================================
The fsck(1M) command cannot open the disk device, because although a similar filesystem exists, the partition specified does not. Run the mount(1M) or the format(1M) command to see what filesystems are configured on the machine. Then run fsck again on an existing partition.

fsck: Can’t stat /dev/dsk/variable
==================================
The fsck(1M) command cannot open the disk device, because the specified filesystem does not exist. Run the mount(1M) or the format(1M) command to see what filesystems are configured on the machine. Then run fsck again on an existing filesystem.

giving up
=========
This message appears in the SCSI log to indicate that a read or write operation has been retried until it timed out. With SCSI disk the timeout period is usually 30 seconds; with tape the period is usually 20 attempts. Timeout periods are generally coded into the drivers.

Check that all SCSI devices are connected and powered on. Make sure that SCSI target numbers are correct and not in conflict. Verify all cables are no longer than six meters, total, and that all SCSI connections are properly terminated. The scsi_log(9F) routine usually displays messages on the system console and in the /var/adm/messages file. Run the dmesg(1M) command to see the most recent message buffer.

Graphics Adapterdevice /dev/fb is of unknown type
==
The /dev/fb driver is either missing or corrupted. See ”InitOutput: Error loading module for /dev/fb” for details.

group.org_dir: NIS+ servers unreachable
=======================================
This is the second of three messages that an NIS+ client prints when it cannot locate an NIS+ server on the network. See the message ”hosts.org_dir: NIS+ servers unreachable” for details.

/home/variable: No such file ordirectory
===
An attempt was made to change to a user’s home directory, but either that user does not exist or the user’s fileserver has not shared (exported) that filesystem.

To check on the existence of a particular user, run the ypmatch(1) or nismatch(1) command, specifying the user name and then the passwd map.

To export filesystems from the remote fileserver, become superuser on that system and run the share(1M) command with the appropriate options. If that system is sharing (exporting) filesystems for the first time, also invoke
/etc/init.d/nfs.server start to begin NFS service.

Host is down
============
A transport connection failed because the destination host was down. For example, mail delivery was attempted over several days, but the destination machine was not available during any of these attempts.

Report this error to the system administrator for the host. If you are the person responsible for this system, check to see if the machine needs repair or rebooting. This error results from status information delivered by the underlying communication interface. If there is no known connection to the host, a different message usually results. See "No route to host" for details. The symbolic name for this error is EHOSTDOWN, errno=147.

host name configuration error
=============================
This is an old sendmail message, which replaced "I refuse to talk tomyself" and is now replaced by the "Local configuration error" message. See the message "554 variable... Local configuration error" for details.

hosts.org_dir: NIS+ servers unreachable
=======================================
This is the third of three messages that an NIS+ client prints when it cannot locate an NIS+ server on the network.

If other NIS+ clients are behaving normally, check the Ethernet cabling on the workstation showing this message. On SPARC machines, disconnected network cablingalso produces a series of "no carrier" messages. On x86 machines, the NIS+ messages might be your only indication that network cabling is disconnected.

If many NIS+ clients on the network are giving this message, go to the NIS+ server in question and reboot or repair it, as necessary. When the server machine is back in operation, NIS+ clients will give an "NIS server for domain OK" message.

I can’t read your attachments. What mailer are you using?
===
The SunView mailtool andpre–3.3 OpenWindows mailtool produce this message when they cannot cope with an attachment. The attachment is probably in MIME (Multipurpose Internet Mail Extensions) format, using base64 encoding.

To read a mail message containing MIME attachments, use mailtool(1) from Solaris 2.3 or later. If you are running an earlier version of Solaris, rlogin(1) to a later version of Solaris, set the DISPLAY environment variable back to the first system, and run mailtool remotely. If those options prove impossible, ask the originator to send the message again using mailtool, or using the CDE dtmail compose File–>SendAs–>SunMailTool option.

Standard MIME attachments with base64 encoding, for example, produce this message and fail to display in older mailtools. Look into using metamail, available on the Internet, which allows you to send and receive MIME attachments.

ie0: Ethernet jammed
=====================
This message can appear on SPARCservers or x86 machines with an Intel 82586 Ethernet chip. It indicates that 16 successive transmission attempts failed, causing the driver to give up on the current packet.

If this error occurs sporadically or at busy times, it probably means that the network is saturated. Wait for network traffic to clear. If bottlenecks arise frequently, think about reconfiguring the network or adding subnets.

Another possible cause of this message is a noise source somewhere in the network, such as a loose transceiver connection. Use snoop(1M)or a similar program to isolate the problem area, then check and tighten network connectors as necessary.

ie0: no carrier
================
This message can appear on SPARCservers or x86 machines with an Intel 82586 Ethernet chip. It indicates that thechip has lost input to its carrierdetect pin while trying to transmit a packet, causing the packet to be dropped.

Check that the Ethernet connector is not loose or disconnected. Other possible causes include an open circuit somewhere in the network and noise on the carrier detect linefrom the transceiver. Use snoop(1M) or a similar program to isolate the problem area, then check the network connectors and transceivers, as needed.

Illegal Instruction

A process has received a signal indicating that it attempted to execute an instruction that is not allowed by the kernel. This usually results from running programs compiled for a slightly different machine architecture. This message is usually accompanied by a core dump, excepton read–only filesystems.

If you are booting from CDROM or from the net, check README files to make sure you are using an image appropriate for your machine architecture. Run df to make sure there is enough swap space on the system; too little swap space can cause this error. If you recently upgraded your CPU to a new architecture, replace your operating system with one that supports the new architecture (an operating system upgrade might be required). Sometimes this condition results from programming error, such as when a program attempts to execute data as instructions. This condition can also indicate device file corruption on your system.

Illegal instruction ”0xN” was encountered at PC 0xN

The machine is trying to boot from a non–boot device, or from a boot device for a different hardware architecture.

If you are booting from the net, check README files to make sure you are using a boot image for that architecture. If you are booting from disk, make sure the system is looking at the right disk, which is usually SCSI target 3. Failing these solutions, connect a CD drive to the system and boot from CDROM.

Illegal seek

Using a pipe (”|”) on the command line doesn’t work here. Rather than using a pipe on the command line, redirect the output of the first program into a file and then run the second program on that file.

A call to lseek(2) was issued to a pipe. This error condition can also be fixed by altering the program to avoid using lseek(). The symbolic name for this error is ESPIPE, errno=29.

Image Tool: Unable to open XIL Library.

This message follows multiple multi–line ”XilDefaultErrorFunc” errors, indicating that ImageTool could not locate the X Imaging Library. Many OpenWindows and CDE deskset programs require XIL. Run pkginfo(1) to determine what packages are installed on the system. If the following packages are not present, install them from CDROM or over thenet: SUNWxildg, SUNWxiler, SUNWxilow, and SUNWxilrt.

Inappropriate ioctl for device
==============================
This is a programming error.

Ask the program's author to fix this condition. The program needs to be changed so it employs a device driver that can accept special character device controls.

The ioctl() system call was given as an argument for a file that is not a special character device. This message replaces the traditional but puzzling "Not a typewriter" message. The symbolic name for this error is ENOTTY, errno=25.

INCORRECT BLOCK COUNT I=N (should be N) CORRECT?
===
During phase 1, fsck(1M) determined that the specified inode pointed to a number of bad or duplicate blocks, sothe block count should be corrected to the actual number shown. Generally you can answer yes to this question without harming the filesystem.

inetd[N]: execv /usr/sbin/in.uucpd: No such file or directory
===
This message indicates that the Internet services daemon inetd(1M) tried to start up the UUCP service without the UUCP daemon existing on the system.

The SUNWbnuu package must be installed before the machine can run UUCP. Run pkgadd(1M) to install this package from the distribution CDROM or over the network.

inetd[N]: variable/tcp: unknown service
=======================================
This message indicates that the Internet services daemon inetd(1M) could not locate the TCP service specified after the first colon.

Check the current machine's /etc/services file, and the NIS services map, to see if the service is described. To start this service, add an appropriate entry into the /etc/services file and possibly the services map as well. Note that NIS+ does not consult the local /etc/services file unless you put "files" right after "nisplus" on the services line of the system's /etc/nsswitch.conf file.

If you do not want to start this service, edit the system's /etc/inetd.conf file and delete the entry that tries to start it up.

inetd[N]: variable/udp:unknown service
==

This message indicates that the Internet services daemon inetd(1M) could not locate the UDP service specified after the first colon. See the message ”inetd[N]: variable/tcp: unknown service” for a solution.

inetd: Too many open files
==========================

This message can appear when someone runs a command from the shell or uses a third–party application. The sar(1M) command does not indicate that the system–wide open file limit has been exceeded.

The probable cause for this is that the shell limit has been exceeded. The default open file limit is 64, but can be raised to 256. See the message ”Too many open files” for a solution.

INIT: Cannot create /var/adm/utmp or /var/adm/utmpx
==

This console message indicates that init(1M) cannot write in the /var directory, which is usually part of the / (root) filesystem. Some other messages follow, andthe system usually comes up single–user. The problem is often that / or /var is mounted read–only. Sometimes a brief power outage leaves the system believing that many filesystems are still mounted.

If /var is a separate filesystem on the machine, andis not yet not mounted, mount it now. If the filesystem containing /var is mounted read–only, remount it read–write with a command similar to this:

mount –o rw,remount /

Then type Control–d and try to bring up the system multi–user. If that fails, the root filesystem is probably corrupted. Run fsck(1M) on the root filesystem, halt the machine, power cycle the CPU, and wait for the system to reboot. Should this problem still occur, restore the root filesystem from backup tapes, or re–install the system from net or CDROM to replace the root filesystem.

InitOutput: Error loading module for /dev/fb
==

This fatal X server error message indicates that /dev/fb, the ”dumb frame buffer,” is either missing or corrupted. It is usually followed by a ”giving up” message and a few xinit errors. If other devices on the system are working correctly, the most
likely reason for this error is that the SUNWdfb package was removed or never installed. Insert the installation CD–ROM, change to the

Solaris_2.xdirectory, and run the following command to install the packages SUNWdfbh and SUNWdfb (for your machine architecture):

pkgadd –d .

If other devices on the system are not working correctly, the system might havea corrupt /devices directory. Halt the system and boot using the –r (reconfigure) option. The system will run fsck(1M) if the /devices filesystem is corrupted, most likely fixing the problem.

Interrupted system call
=======================
The user issued an interrupt signal (usually Control–c) while the system was in the middle of executing a system call. When network service is slow, interrupting cd(1) to a remote–mounted directory can produce this message. Proceed with your work, this message is purely informational.

An asynchronoussignal (such as interrupt or quit), which a program was set up to catch, occurred during an internal system call. If execution is resumed after processing the signal, it will appear as if the interrupted programming function returned this error condition, so the program might exit with an incorrect error message. The symbolic name for this error is EINTR, errno=4.

Invalid argument
================
An invalid parameter was specified that the system cannot interpret. For example, trying to mount an uncreated filesystem, printing without sufficient system support, or providing an undefined signal to a signal(3c) library function, can all produce this message.

If you see this message when you are trying to mount a filesystem, make sure that you have run newfs(1M) to create the filesystem. If you see this message when you are trying to read a diskette, make sure that the diskette was properly formatted with fdformat(1), either in DOS format (pcfs) or as a UFS filesystem. If you see this message while you are trying to print, make sure that the print service is configured correctly.

The symbolic name for this error is EINVAL, errno=22.

Invalid null command
====================
This C shell message results from a command line with two pipes (|)in a row or from a pipe without a command afterwards. Change the command line so that each pipe is followed by a command.

I/O error
=========
Some physical Input/Output error has occurred. If the process was writing a file, data corruption is possible. First find out which device is experiencing the I/O error. If the device is a tape drive, make sure a tape is inserted into the drive. When this error occurs with a tape in the drive, it is likely that the tape contains an unrecoverable bad spot.

If the device is a floppy drive, an unformatted or defective diskette could be at fault. Format the diskette, or obtain a replacement. If the device is a hard disk drive, you might need to run fsck(1M) and possibly even reformat the disk. In some cases this error might occur on a call following the one to which it actually applies. The symbolic name for this error is EIO, errno=5.

Is a directory
==============
An attempt was made to read or write a directory as if it were a file.

Look at a listing of all the files in the current directory and try again, specifying a file instead of a directory. The symbolic name for this error is EISDIR, errno=21.

kernel read error
=================
This message appears when savecore(1M), if activated, tries to copy a debugging image of kernel memory to disk but cannot read various kernel data structures correctly. Generally this occurs after a system panic has corrupted main memory. Data corruption on the systemis possible.

Look at the kernel error messages that preceded this one to try to determine the cause of the problem. Error messages such as ”BAD TRAP” usually indicate faulty hardware. Until the problem that caused the kernel panic is resolved, a kernel core image cannot be saved for debugging.

Killed
======
This message is purely informational. If the killed process was writing a file, some data might be lost. Continue with your work.

This message from the signal handler or various shells indicates hat a process has been terminated with a SIGKILL. However, if you don’t see this message and cannot terminate a process with a IGKILL, you might have to reboot the machine to get rid of that process.

kmem_free block already free
============================
This is a programming error,probably from a device driver.

Determine which driver is giving this message and contact the vendor for a software update, as this message indicates a bug in the driver.

This message is from the DDI programming function kmem_free(9F), which releases a block of memory at address addr of size siz that was previously allocated by the DDI function kmem_alloc(9F). Both addr and siz must correspond to the original allocation. If you have source code for the driver, follow kmem_alloc() and kmem_free() in the code to make sure they allocate and free the same chunk of memory.

last message repeated N times
=============================
This message comes from syslog(1M), the facility that prints messages on the console and records them in /var/adm/messages. To reduce the log size and minimize buffer usage, syslog collapses any identical messages it sees during a 20 second period, then prints this message with the number of repetitions.

Look above this message to see which message was repeated so often. Then consider the repeated message and take action accordingly. If repeated log entries such as "su ... failed" appear, consider the possibility of a security breach.

ld.so.1: variable: fatal: relocation error: symbol not found: variable
==
This message from the run–time linker ld.so.1 indicates that in trying to execute the application given after the first colon, the specified symbol could not be found for relocation. The message goes on to say in what file the symbol was referenced. Since this is a fatal error, the application terminates with this message.

Run the ldd –d command on the application to show its shared object dependencies and symbols that aren't found. Probably your system contains an old version of the shared object that should contain this symbol. Contact the library vendor or author for an update.

This error does not necessarily occur when you first bring up an application. It could take months to develop, if ordinary use of the application seldom references the undefined symbol.

ld.so.1: variable: fatal: variable: can’t open file: errno=2
==
This message indicates that the run–time linker, ld.so.1, while running the program specified after the first colon, could not find the shared object specified after the third colon. (A shared object is sometimes called a dynamically linked library.) Error number 2 translates to ”No such file or directory” (ENOENT).

As a workaround, set the environment variable LD_LIBRARY_PATH to include the location of the shared object in question, for example: /usr/dt/lib:/usr/openwin/lib.   Better yet, if you have accessto source code, recompile the program using the –Rpath loader option. Using LD_LIBRARY_PATH is discouraged because it slows down performance.

le0: Memory error!
==================
This message indicates that the network interface encountered an access time–out from the CPU’s main memory. There is probably nothing wrong except system overload.

If the system is busy with other processes, this error can occur frequently. If possible, try to reduce the system load by quitting applications or killing some processes.

The Lance Ethernet chip timed out while trying to acquire the bus for a DVMA transfer. Most network applications wait for a transfer to occur, so generally no data gets lost. However, data transfer might fail after too many time–outs.

le0: No carrier— cable disconnected or hub link test disabled?
===
Standalone machines with no Ethernet port connection get this error when the system triesto access the network. If the Ethernet cable is disconnected, SPARC machines with the sun4m architecture usually display this message, whereas machines with the sun4c architecture usually display the ”le0: No carrier—transceiver cable problem” message instead. If the Ethernet cable is connected, this message could result from a mismatch between the machine’s NVRAM settings and the Ethernet hub settings.

If this message is continuous, try to save any workto local disk. When a machine is configured as a networked system, it must be plugged into the Ethernet with a twisted pair J45 connector. If the Ethernet cable is plugged in, find out whether or not the Ethernet hub does a Link Integrity Test. Then become superuser to check and possibly set the machine’s NVRAM. If the hub’s Link Integrity Test is disabled, set this variable to false.   # eeprom | grep tpe tpe–link–test?=true # eeprom ’tpe–link–test?=false’

The default setting is true. If for some reason tpe–link–test? was set to false,and the hub's Link Integrity Test is enabled, set this variable to true.

le0: No carrier— transceiver cable problem?
==
Standalone machines with no Ethernet port connection get this error when the system tries to access the network. If this message is continuous, try to save any work to local disk. When a machine is configured as a networked system, it must be plugged into the Ethernet with either a twisted pair J45
connector or thicknet 10Base–T connector (depending on the building's Ethernet cable type).

Older workstations have a thicknet connection on the back instead of a twisted pair Ethernet connection, so they require a thicknet to twisted pair transceiver to translate between cabling types.

LINK COUNT FILE I=i OWNER=o MODE=m SIZE=s MTIME=t COUNT... ADJUST?
==
During phase 4, fsck(1M) determined that the inode's link count for the specified file is wrong, and asks if you want to adjust it to the value given. Generally you can answer yes to this question without harming the filesystem.

LL105W: Protocol error detected.
====================================
This error message comes from Lifeline Mail, an unbundled PC compatibility application. The likeliest cause for this problem is that someone set up a user account without a password. Assign the user a password to
solve this problem.

ln: cannot create /dev/fb: Read–only file system
==
During device reconfiguration at boot time, the system cannot link to the frame buffer because /dev is on a read–only filesystem. Check that /dev/fb is a symbolic link to the hardware frame buffer, such as cgsix or tcx. Ensure that the filesystem containing /dev is mounted read–write.

lockd[N]: create_client: no name forinet address 0xN
==
This lock daemon message usually indicates that the NIS hosts.byname and hosts.byaddr maps are not coordinated. Wait a short time

for the maps to synchronize. If they don’t, takesteps to coordinate them.

Login incorrect
===============
This message from the login(1) program indicates an incorrect combination of login name and password. There is no way to tell whether what’s wrong is the login name, the password, or both. Other programs such as ftp(1), rexecd(1M), sulogin(1M), and uucp(1C) alsogive this error under similar conditions.

Check the /etc/passwd file and the NIS or NIS+ passwd map on the local system to see if an entry exists for this user. If a user has simply forgotten the password, su and set a new one with the passwd usernamecommand. This command automatically updates the NIS+ passwd map, but with NIS you’ll need to coordinate the update with the passwd map.

The ”Login incorrect” problem can also occur with older versions of NIS when the user name has more than eight characters. If this is the case, edit the NIS password file, change the user name to have eight or fewer characters, and then remake the NIS passwd map.

If you cannot log in to the system as root, despite knowing the proper password, it is possible that the /etc/passwd file is corrupted. Try to log in as a regular user and su to root If that doesn’t work, see the message ”su: No shell” and follow most of the instructions given there. Instead of changing the default shell however, make the password field blank in /etc/shadow.

lp hang
=======
On a print server, the queue continues to grow but nothing comes out of the printer. The printer daemon is hung. Here is a simple procedure for flushing a hung printing queue:

1. Login or switch user to root.
2. Issue the reject printername command to make sure no one sends any job to the printer.
3. Turn off power to the printer.
4. If the active job appears to be causing the hang, remove it from the print queue with the cancel jobnumber command, and ask the owner to requeue that print job.
5. Shut down the print queue with the /usr/lib/lpshut command.
6. Remove the lock file /var/spool/lp/SCHEDLOCK and the temporary files /var/spool/lp/tmp/*/*.
7. Turn the printer back on.
8. Restart the print queue with the /usr/lib/lpsched command.

mailtool: Can’t create dead letter: Permission denied
===
An attempt was made to send a message with mailtool(1) from a directory where the user does not have write permission, and the user’s home directory is currently unavailable. Change to another directory and start mailtool again, or use chmod(1) to change permissions for the directory (if possible).

mailtool: Could not initialize the Classing Engine
==
When a user runs mailtool(1) on a remote machine, setting the DISPLAY environment back to the local machine, this message might appear inside a dialog box window. The dialog box goes on to say that the Classing Engine must be installed to use Attachments. This problem occurs because rlogin(1) does not propagate the user’s environment.

Exit mailtool and set your OPENWINHOME environment variable to /usr/openwin. Then run mailtool again. The error message will not appear, and you will be able to use Attachments. Classing Engine is a new name for Tool Talk. Earlier versions of mailtool said ”Tool Talk: TT_ERR_NOMP” instead of Classing Engine.

Mail Tool is confused about the state of your Mail File.
==
This message appears in a pop–up dialog box whenever you ask mailtool(1) to access messages after another mail reader has modified your inbox. A request follows: ”Please Quit this Mail Tool.”

Click ”Continue” to close the dialog box, then exit mailtool. If you continue trying to read mail, messages deleted by the other mail reader will never appear, and mailtool will fail to see any new messages.

mail: Your mailfile was found to be corrupted (Content–length mismatch).
===
This message comes from mail(1) or mailx(1) whenever it detects messages with a different content length than advertised. The mail program tells you which message might be truncated or might have another message concatenated to it.

Two common causes of content length mismatches are the simultaneous use of different mail readers (such as mail and mailtool), or using a mail reading program (or an editor) that does not update the Content–Length field after altering a message. The mailx program can usually recover from this error and delineate mail message boundaries correctly. Pay close attention to the message that might be truncated or combined with another message, and to all messages after that one. If a mail file becomes hopelessly corrupted, run it through a text editor to eliminate all Content–Length lines, and ensure that each message has a From (no colon) line for each message, preceded by a blank line. To avoid mailfile corruption, exit from mailtool without saving changes when you are currently running mail or mailx.

Memory address alignment
========================
This message can occur when printing large files on a SPARCprinter attached to a SPARCstation 2 Replace the SPARCstation 2 CPU with one that isat the most recent dash level.

memory leaks
============
An application uses up more and more memory, until all swap space is exhausted.

Many developers have found that third party software (such as Purify) can help identify memory leaks in their applications. If you suspect that you have a memory leak, you can use sar(1) to check on the Kernel Memory Allocation (KMA). Any driver or module that uses KMA resources, but does not specifically return the resources before it exits, can create a memory leak.

mount: /dev/dsk/variable is already mounted, /variable is busy, or...
==
While trying to mount a filesystem, the mount(1M) command received a ”Device busy” (EBUSY) error code.There are several possible reasons: this /dev/dsk filesystem is already mounted on a different directory, the busy path name is the working directory of an active process, or the system has exceeded its maximum number of mount points (unlikely).

Run /etc/mount to see if the filesystem is already mounted. If not, check to see if any shells are active in the busy directory (did the user cd into the directory?), or if any processes in the ps(1) listing are active in that directory. If the reason for the error message isn’t obvious, try using a different directory for the mount point.

mount: giving up on: /variable
==============================
An existing server did not respond to an NFS mount request, so after retrying a number of times (default1000), the mount(1M) command has given up. Nonexistent servers or bad mount points produce different messages.

If the ”RPC: Program not registered” message precedes this one, the requested mount serverprobably did not share (export) any filesystems, so it has no NFS daemons running. Have the superuser on the mount server share(1M) the filesystem, then run

/etc/init.d/nfs.server start to begin NFS service. f the requested mount server is down or slow to respond, check to see whether the machine needs repair or rebooting.

mount: mount–point /variable does not exist.
==
Someone tried to mount a filesystem onto the specified directory, but there is no suchdirectory. If this is the directory name you want,run mkdir(1) to create this directory as a mount point.

mount: the state of /dev/dsk/variable is not okay
===
The system was unable to mount the filesystem that was specified because the super–block indicates that the filesystem might be corrupted. This is not an impediment for read–only mounts.

If you don’t need to write on this filesystem, mount(1M) it using the –o ro option. Otherwise, do as one of the message continuation lines suggests and run fsck(1M) to correct the filesystem state and update the super–block.

/net/variable: No such file or directory
==
A user tried to change directory (for example with cd) to a network partition on the system specified after /net/, but this host either does not exist or has not shared (exported) any filesystem. To gain access to files on this system, try rlogin(1).

To export filesystems from the remote system, become superuser on hat system and run the share(1M) command with the appropriate options. If that system is sharing filesystems for the first time, also run /etc/init.d/nfs.server start to begin NFS service.

Network is down
===============
A transport connection failed because it encountered a dead network.

Report this error to the system administrator for the network. If you are the person responsible for this network, check to see why the network is dead and what repairs are necessary. This error results from status information delivered by the underlying communication interface. The symbolic name for this error is ENETDOWN, errno=127.

Network is unreachable
=====================
An operational error occurred either because there was no route to the network or because negative status information was returned by intermediate gateways or switching nodes.

The returned status is not always sufficient to distinguish between a network that is down and a host that is down. See the ”No route to host” message. Check the network routers and switches to see if they are disallowing these packet transfers. If they areallowing all packet transfers, check network cablingand connections. The symbolic name for this error is ENETUNREACH, errno=128.

NFS getattr failed for server variable: RPC: Timed out
==
This message appears on an NFS client that requested a service from an NFS server whose hardware is failing. Often the message ”NFS read failed” appears along with this message. If the server were merely down or slow to respond, the ”NFS server not responding” message would appear instead. Data corruption on the server system is possible. Because this message usually indicates server hardware failure, initiate repair procedures as soon as possible. Check the memory modules, disk controllers, and CPU board.

nfs mount: Couldn’t bind to reserved port
=======================================
This message appears when a client attempts to NFS mount a filesystem from a server that has more than one Ethernet interface configured on the same physical subnet. Always connect multiple Ethernet interfaces on one router system to different physical subnetworks.

nfs mount: mount: variable: Device busy
=====================================
This message appears when the superuser attempts to NFS mount on top of an active directory. The busy device is actually the working directory of a process. Determine which shell on the workstation is currently located below the mount point, and change out of that directory. Be wary of subshells (such as su shells) that could be in different working directories while the parents remain below the mount point.

NFS mount: /variable mounted OK
==============================
While booting, the system failed to mount the directory specified after the first colon, probably because the NFS server involved was down or slow to respond. The mount ran in the background and successfully contacted the NFS server. This is a purely informative message to let you know that the mount process has completed.

NFS read failed for server variable
=================================
This is generally a permissions problem. Perhaps a directory or file permission was changed while the client held the file open. Perhaps the filesystem's share or netgroup permissions changed. If the server were down or the network saturated, the "NFS server not responding" message would appear instead.

Log in to the NFS server and check the permissions of directories leading to the file. Make certain that the filesystem is shared with (exported to) the client experiencing an NFS read failure.

nfs_server: bad getargs for N/N
=============================
This message comes from the NFS server when it gets a request with unrecognized or incorrect arguments. Typically, it means the request could not be XDR decoded properly. This can result from corruption of the packet over the network, or from an implementation bug causing the NFS client to improperly encode its arguments.

If this message originates from a single client, investigate that machine for NFS client software bugs. If this message appears all over a network, especially accompanied by other networking errors, investigate the network cabling and connectors.

NFS server variable not responding still trying
==
In mostcases this very common message indicates that the system has requested a service from an NFS server that is either down or extremely slow to respond. In some cases this message indicates that the network link to this NFS server is broken, although usually that condition generates other error messages as well. In a few cases this message indicates NFS client set–up problems.

Check the non–responding NFS server to see whether the machine needs repair or rebooting. Encourage your user community to report such problems quickly but only once. Should this message appear when booting a diskless client, make sure that the client's /etc/hosts file and the network naming service (NIS, NIS+, or other /etc/hosts files on the network) have been updated.

NFS server variable ok
=====================
This message is the follow–up to the "NFS server not responding" error. It indicates that the NFS server is back in operation. When an NFS server first comes up, it will be busy fulfilling client requests for a while. Be patient and wait for your client system to respond. Making many extraneous requests only further slows the NFS server response time.

nfs umount:variable: is busy
============================
This message appears when the superuser attempts to unmount an active NFS filesystem. The busy point is the working directory of a process.

Determine which shell (or process) on the workstation is currently located in the remotely mounted filesystem, and change (cd) out of that directory. Be wary of subshells (such as su shells) that could be in different directories while the parent shells remain in the NFS filesystem.

NFS write error on host variable: No space left on device.
==
This console message indicates that an NFS–mounted partition has filled up and cannot accept writing of new data. Unfortunately, software that attempts to overwriteexisting files will usually zero out all data in these files. This is particularly destructive on NFS–mounted /home partitions.

Find the user or process that is filling up the filesystem, and get the out–of–control process stopped as soon as you can. Then delete files as necessary to create more space on the filesystem (large core files are good candidates for deletion). Have users write any modified files to local disk if possible. If this error occurs often, redistribute directories to ease demandon this partition.

NFS write failed for server variable: RPC: Timed out
==
This error can occur when a file system is soft–mounted, and server or network response time lags. Any data written to the server during this period could be corrupted.

If you intend to write on a filesystem, never specify the soft mount option. Use the default hard mount for all the filesystems that are mounted read–write.

NIS+ authentication failure
==========================
This is a Federated Naming Service message. The operation could not be completed because the principal making the request could not be authenticated with the name service involved.

Run the nisdefaults(1) command to verify that you are identified as the correct NIS+ principal. Also check that the system has specified the correct public key source.

No buffer space available
========================
An operation on a transport endpoint or pipe was not performed because the system lacked sufficient buffer space or because a queue was full. The target system probably ran out of memory or swap space. Any data written during this condition will probably be lost.

To add more swap area, use the swap –a command on the target system. Alternatively, reconfigure the target system to have more swap space. As a general rule, wwap space should be two to three times as large as physical memory. The symbolic name for this error is ENOBUFS,errno=132.

No child processes
=================
This message can appear when an application tries to communicate with cooperating process that do not exist.

Restart the parent process so it can create the child processes again. If that doesn't help, this could be the result of programming error; contact the vendor or author of the program for an update. A wait(2) system call was executed by a process that had no existing or unwaited–for child processes. The child process escould have exited prematurely, or might never have been created.

The symbolic name for this error is ECHILD, errno=10.

No default media available
=========================
The volume manager issues this message if a user makes an eject(1) request when the drives containno diskette or CDROM to eject. Insert a diskette or CDROM. If the volume manager is confused and there actually is a diskette or CDROM in a drive, run volcheck to update the volume manager. If the system remains confused, try booting with the –r option to reconfigure devices.

No directory! Logging in with home=/
==================================
The login(1) program could not find the home directory listed in the password file or NIS passwd map, so it deposited the user in the root directory.

Check that the user's home directory is mounted and is owned by and accessible to that user. Perhaps the automounter tried to mount the home directory, but the NFS server did not respond quicklyenough. Try listing the files in /home/username. If the NFS server responds to this request, have the user log out and log in again.

It is possible that the automounter daemon is not running. Run the ps command to see if automountd is present. If not,run the second command; if it appears to be wedged, run both these commands: # /etc/init.d/autofs stop # /etc/init.d/autofs start. When the automounter daemon is running, verify that the /etc/auto_master file has a line like this: /home auto_home

Verify that the /etc/auto_home file has a line like this: +auto_home These entries depend on the NIS auto_home map. It is also possible that the NFS server has not shared (exported) this /home directory, or that the NFS daemons on the server have disappeared.

No message of desired type
==========================
An attempt was made to receive a message of a type that does not exist on the specified message queue. See the msgop(2) man page for details.

This indicates an error in the System V IPC message facility. Generally the message queue is empty or devoid of the desired message type, while IPC_NOWAIT is set. The symbolic name for this error is ENOMSG, errno=35.

No recipients specified
=======================
This message comes from the mailx(1) command whenever a user doesn't provide an address in the To: field. See the message "Recipient names must be specified" for details.

No record locks available
=========================
No more record locks are available. The system lock table is full. The symbolic name for this error is ENOLCK, errno=46.

Perhaps a process called fcntl(2) with the F_SETLK or F_SETLKW option, and the system maximum was exceeded. The system contains several different locking subsystems, including fcntl,the NFS lock daemon, and mail locking, all of which can produce this error. Try again later, when more locks might be available.

No route to host
================
An operational error occurred because there was no route to the destination host, or because of status information returned by intermediate gateways or switching nodes. The returned status is not always sufficient to distinguish between a host that is down and a network that isdown. See the "Network is unreachable" message.

Check the network routers and switches to see if they are disallowing these packet transfers. If they are allowing all packet transfers, check network cabling and connections. The symbolic name for thiserror is EHOSTUNREACH, errno=148.

No shell Connection closed
===========================
A user has attempted to remote login to the system, and has a valid account name and password, but the shell specified for their account is not available on that system. For example, the seventh field could request the GNUBourne–again shell /bin/bash, which does not exist on standard Solaris distributions. If you have a copy of the requested shell, become superuser and install the missing shell on that system. Otherwise, change the user's password file entry (perhaps only in the NIS+ or NIS passwd map) to specify an available shell such as /bin/csh or /bin/ksh.

No space left on device
=======================
While writing an ordinary file or creating a directory entry, there was no free space left on the device. The disk, tape, or diskette is full of data. Any data written to that device during this condition will be lost.

Remove unneeded files from the hard disk or diskette until there is space for all the data you are writing. It might be advisable to move some directories onto another filesystem and create symbolic links accordingly. When a tape is full, continue on another one, use a higher density setting, or obtain a higher–capacity tape. To create multi–volume tapes or diskettes, use the pax(1) or cpio(1) command; tar(1) is still limited to a single volume. The symbolic name for this error is ENOSPC, errno=28.

No such device
==============
An attempt was made to apply an operation to an inappropriate device, such as writing to a nonexistent device. Look in the /devices directory to see why this device does not exist, or why the program expects it to exist. The similar "No such device or address" message tends to indicate I/O problems with an existing device, whereas this message tends to indicate a device that does not exist at all. The symbolic name for this error is ENODEV, errno=19.

No such device or address
=========================
This can occur when a tape drive is off–line or when a device has been powered off or removed from thesystem. For tape drives, make sure the device is connected, powered on, and toggled on–line (if applicable). For disk and CDROM drives, check that the device is connected and powered on.

With all SCSI devices, ensure that the target switch or dial is set to the number where the system originally mounted it. To inform the system of a change to the target device number, reboot using the –r (reconfigure) option.

This message results from I/O to a special file's subdevice that either does not exist or that exists beyond the limit of the device. The symbolic name for this error is ENXIO, errno=6.

No such file or directory
========================
The specified file or directory does not exist. Either the file name or path name was entered incorrectly.

Check the file name and path name for correctness and try again. If the specified file or directory is a symbolic link, it probably points to a nonexistent file or directory. The symbolic name for this error is ENOENT, errno=2.

no such map in server's domain
=============================
A user or an application tried to look up something using Network Information Services (NIS), but NIS has no corresponding database for this request.

Make sure the NIS map name is spelled correctly. To see a list of nicknames for the various NIS maps, run the ypcat –x command. To see a full list of the various NIS maps (databases), run the ypwhich –m command. If the NIS service were not running on the current machine, these commands would result in a "can't communicate with ypbind" message.

No such process
===============
This process cannot be found. The process could have finished execution and disappeared, or it might still be in thesystem under a different numeric ID. Use the ps(1) command tocheck that the process ID you're supplying is correct.

No process corresponds to the specified process ID (PID), light–
weight process ID, or thread_t. The symbolic name for this error is ESRCH, errno=3.

No such user as variable— cron entries not created
===
A file exists in /var/spool/cron/crontabs for the specified user, but this user is not in /etc/passwd or the NIS passwd map. The system cannot

create cron entries for nonexistent users.

To eliminate this message at boot time, remove the cron file for the nonexistent user, or rename it if the user's login name has changed. If this is a valid user, create an appropriate password entry for this name.

Not a directory
===============
A non–directory was specified where a directory is required, such as in a path prefix or as an argument to the chdir(2) system call. Look at a listing of all the files in the current directory and try again, specifying a directory instead of a file. The symbolic name for this error is ENOTDIR, errno=20.

Not enough space
================
This message indicates that the system is running many large applications simultaneously, and has run out ofswap space (virtual memory). It could also indicate that applications failed without freeing pages from the swap area. Swap space is an area of disk set aside to store portions of applications and data not immediately required in memory. Any data written during this condition will probably be lost.

Reinstall or reconfigure the system to have more swap space. A general rule of thumb is that swap space should be two to three times as large as physical memory. Alternatively, use mkfile(1M) and swap(1M) to add more swap area. This example shows how to add 16 MB of virtual memory in the /usr/swap file (any filesystem with enough free space would work): # mkfile 16m /usr/swap # swap –a /usr/swap

To make this automatic at boot time, add the following line to the /etc/vfstab file: /usr/swap – – swap – no –

In calling the fork(2), exec(2), sbrk(2), or malloc(3C) routine, a program asked for more memory than the system could supply. This is not a temporary condition; swap space is a system parameter. The symbolic name for this error is ENOMEM, errno=12.

not found
=========
This message indicates that the Bourne shell could not find the program name given as a command. Check the form and spelling of the command line. If that looks correct, echo $PATH to see if the user's search path is correct. When communications are garbled, it is possible to unset a search path to such an extent that only built–in shell commands are available. Here is a command to reset a basic search path:

$ PATH=/usr/bin:/usr/ccs/bin:/usr/openwin/bin:. If the search path looks correct, check the directory contents along the search path to see if programs are missing or if directories are not mounted.

NOTICE: /variable: out of inodes
=================================
The filesystem specified after the first colon probably contains many small files, exceeding the per–filesystem limit for inodes (file information nodes).

If many small files were created unintentionally, removing them will resolve the problem.

Otherwise, follow these steps to increase filesystem capacity for small files. Make several backup copies of the filesystem on different tapes (for safety), then bring the machine down to single–user mode. Use the newfs(1M) command with the –i option to increase inode density for this filesystem. Here is an example:

newfs –i 1024 /dev/rdsk/partition

Finally, restore the filesystem from a backup tape. Note that increasing the inode density slightly reduces total filesystem capacity.

Not login shell
===============
This message results when a user triesto logout(1) from a shell other than the one started at login time. To quit a non–login shell, use the exit(1) command. Continue doing so until you have logged out.

Not on system console
=====================
A user tried to login(1) to a system as the superuser (uid=0, which is not necessarily root) from a terminal other than the console.

Login to that system as a normal user, then run su(1M) to become superuser. To allow superuser logins from any terminal, comment out the CONSOLE line in /etc/default/login (this is not recommended for security reasons).

Not owner
=========
Either an ordinary user tried to do something reserved for the superuser, or the user tried to modify a file in a way restricted to the file's owner or to the superuser. Switch user to root and try again. The symbolic name for this error is EPERM, errno=1.

Not supported
=============
This version of the system does not support the feature requested, although future versions of the system might provide support.

This is generally not a system message from the kernel, but an error returned by an application. Contact the vendor or author of the application for an update. The symbolic name for this error is ENOTSUP, errno=48. ok ==

This is the OpenBoot PROM monitor prompt. From this prompt, you can boot the system (from disk, CDROM, or net), or you can use the go command to continue where you left off.

If you suddenly see this prompt, look at the messages above it to see if the system crashed. If no other messages appear, and you just typed Stop–A or plugged in a new keyboard, type go to continue. You might need to Refresh the window system from its Workspace Menu.

Never invoke sync from the ok prompt without first running the fsck(1M) command, especially if the filesystem has changed.

operation failed [error 185], unknown group error 0, variable
==
When you use admintool to add a user to a newly–created group, admintool issues this error. Apply patch 101384–05 to fix bug ID 1151837 and to provide a workaround for bug ID 1153087.

Operation not applicable
=======================
This error indicates that no system support exists for some function that the application requested. Ask the system vendor for an upgrade, or contact the vendoror author of the application for an update. This message indicates that no system support exists for an operation. Many modules set this error when a programming function is not yet implemented. If you are writing a program that produces this message while calling a system library, try to find and use an alternative library function. Future versions of the system might support this operation; check system release notes for further information. The symbolic name for this error is ENOSYS, errno=89.

out of memory
=============
Hundreds of different programs can produce this message when the system is running many large applications simultaneously. This message usually means that the system has run out of swap space (virtual memory).

See the message ”Not enough space” for details. Any data written during this condition will probably be lost.

PARTIALLY ALLOCATED INODE I=N CLEAR?
=====================================
During phase 1, fsck(1M) found that the specified inode was neither allocated nor unallocated. The reason is probably that the system crashed in the middle of a sync(2) or write(2) operation.

Should you answer yes to this question, ”UNALLOCATED” messages might result during phase 2, if any directory entries point to this inode. If you are being careful, exit fsck(1M) and run ncheck(1M) (specifying the inode number after the –i option) to determine which file or directory is involved here. You might be able restore this file or directory from another system. It is also possiblethat fsck will copy this file to the lost+found directory in a later phase.

For more information, see the chapter on checking filesystem integrity in the System Administration Guide, Volume I.

passwd.org_dir: NIS+ servers unreachable
==
This is the first of three messages thatan NIS+ client prints when it cannot locate an NIS+ server on the network. See the message ”hosts.org_dir: NIS+ servers unreachable” for details.

Password does not decrypt secret key for unix.uid@variable
==
This message appears at login time when a user’s password is not identical to the user’s keylogin network password. When a system is running NIS+, the login program firstperforms UNIX authentication, and then attempts a keylogin(1) for secure RPC authentication.

To gain credentials for secure RPC, users can run keylogin (after login) and type in their secret key. To stop this message from appearing at login time, users can run the chkey –p command and set their network password to bethe same as their NIS+ password. If a user doesn’t remember the network password, the system administrator should delete and re–create the user’s credentials table entry so the user can establish a new network password with chkey.

Permission denied
=================
An attempt was made to access a file in a way forbidden by the protection system. Check the ownership and protection mode of the file (with a long listing from the ls–l command) to see who is allowed to access the file. Then change the file or directory permissions as needed. The symbolic name for this error is EACCES, errno=13.

Please specify a recipient.
============================
With mailtool, this message comes up in a dialog box whenever a user tries to deliver a message with no address in the To: field. See the message ”Recipient names must be specified” for details.

Protocol not supported
======================
The requested networking protocol hasnot been configured into the system, or no implementation for it exists. (A protocol is a formal description of the messages to be exchanged and the rules to be followed when systems exchange information.) Verify that the protocol is in the /etc/inet/protocols file and in the NIS protocols map, if applicable. If the protocol is not listed, and you want to permit its use, configure the protocol as documented or as required. The symbolic name for this error is EPROTONOSUPPORT, errno=120.

Protocol wrong type for socket
===============================
This message indicates either application programming error, or badly configured protocols. Make sure that the /etc/protocols file corresponds number–for– number with the NIS protocols map. It it does, ask the vendor or author of the application for an update.

A protocol was specified that does not support the semantics of the socket type requested. This amounts to a request for an unsupported type of socket. Look at the source code that made this socket request and check that it requested one of the types specifiedin /usr/include/sys/socket.h. The symbolic name for this error is EPROTOTYPE, errno=98.

Read error from network: Connection reset by peer
===
This message appears when a user is remotely logged into a machine that crashes or gets rebooted during the rlogin(1) or rsh(1) session. Any data changes that were not saved are probably lost. Sometimes this message appears only when the user types something, even though the system went down hours before. Try torlogin again, perhaps after waiting a few minutes for the system to reboot.

Read–only file system
=====================
Files and directories on filesystems that are mounted read–only cannot be changed. If you only modify these files and directoriesoccasionally, rlogin(1) to the servers from which the filesystems are mounted and change the files or directories there. If you change these files and directories frequently, mount(1M) the filesystems read/write. The symbolic name for this error is EROFS, errno=30.

rebooting...
============
This message appears on the console to indicate that the machine is booting, either after the superuser issued a reboot command, or after a system panic if the EEPROM's watchdog–reboot? variable is set to true. Allow the machine to boot itself. In case of a system panic, look above this message for other indications of what went wrong.

Recipient names must be specified
=================================
Somebody sent mail without a valid recipient in the To: field, so sendmail could not deliver the mail message. Using mail(1), the recipient's address might have been specified using spaces or non–alphanumeric characters. The mailtool(1)and mailx(1) commands try to prevent this by issuing "Please specify a recipient" or "No recipients specified" messages instead. If there is at least one valid recipient, each invalid recipient address will generate a "User unknown" message. Look in the sender's dead.letter file for the automatically saved message, andhave the originator send it again, this time specifying a recipient.

Reset tty pgrp from N to N
==========================
The C shell sometimes issues this message when it clears away the window process group after the user exits the window system. This can happen when the window system doesn't clean up after itself. Proceed with your work. This message is purely informational.

Resource temporarily unavailable
================================
This indicates that the fork(2) system call failed because the system's process table is full, or that a system call failed because of insufficient memory or swap space. It is also possible that a user is not allowed to create anymore processes. Simply waiting often gives the system time to free resources. However if this message occurs often on a system, reconfigure the kernel and allow more processes. To increase the size of the process table in Solaris 2.x, increase the value of maxusers in the /etc/system file. The default maxusers value is the amount of main memory in MB, minus 2.

If one user is not allowed to create any more processes, that user has probably exceeded the memorysize limit; see the limit(1) man page for details. The symbolic name for this error is EAGAIN, errno=11.

Result too large
================
This is a programming error or a data input error. Ask the program's author to fix this condition. This indicates an attempt to evaluate a mathematical programming function at a point where its value would overflow or underflow. The value of a programming function in the math package (3M) is not representable within machine precision. This could occur after floating point overflow or underflow (either single or double precision), or after total loss of numeric significance in Bessel functions.

Note that this message can indicate "Result too small" in the case of floating pointunderflow. To help pinpoint a program's math errors, use the matherr(3M) facility. The symbolic name for this error is ERANGE, errno=34.

rmdir: variable: Directory not empty
====================================
The rmdir(1) command can remove empty directories, only. The directory whose name appears after the first colon in the message still contains some files or directories.

Use rm(1) instead of rmdir. To remove this directory and everything underneath it, use the rm –ir command to recursively descend the directory, being asked if you want to delete each element. To remove the directory and all its contents without being asked for approval, use the rm –r command.

ROOT LOGIN /dev/console
=======================
This syslog message indicates that someone has logged in as root on the system console. If you have just logged in as root, don't worry. If this is not you, consider the possibility of a security breach. The best site–wide policy is for all system administrators to su instead oflogging in as root.

ROOT LOGIN /dev/pts/N FROM variable
===================================
This syslog message indicates that someone has remote logged in as root on a pseudo–terminal from the system specified after the FROM keyword. For security reasons, it is a bad idea to allow root logins from anywhere besides the console. To restrict superuser logins to the console, remove the comment from the CONSOLE line in /etc/default/login.

rx framing error
================
Usually this error indicates a hardware problem. Check the Ethernet cabling and connectors to locate a problem.
A framing error occurs when the Ethernet I/O driver receives a non–integral unit of octets, such as 63 bytes and then 3 bits. (Ethernet specifies the use of octets.) Framing errors are caused by corruption of the starting or ending frame delimiters. These can be corrupted by some violation of the encoding scheme.

Framing errors are a subset of CRC errors, which are usually caused by anomalies on the physical media.An ”alignment/framing error” is a type of CRC error where octet boundaries do not line up.

SCSI bus DATA IN phase parity error
====================================
The most common cause of this problem is unapproved hardware. Some SCSI devices for thePC market do not meet the high I/O speed requirements for the UNIX market. Other possible causes of this problem are improper cabling or termination, and power fluctuations. Data corruption is possible but unlikely to occur, because this parity error prevents data transfer.

Check that all SCSI devices on the bus are Sun approved hardware. Then verify that all cables are no longer than six meters, total, and that all SCSI connections are properly terminated. If power fluctuations are occuring, invest in an uninterruptible power supply.

SCSI transport failed: reason ’reset’
======================================
This message indicates that the system sent data over the SCSI bus, but the data never reached its destination because of a SCSI bus reset. The most common cause of this condition is conflicting SCSI targets.¤Data corruption is possible but unlikely to occur, because this failure prevents data transfer. Verify that all cables are no longer than six meters, total, and that all SCSI connections are properly terminated. If power surges are a problem, acquire a surge suppressor or uninterruptible power supply.

The AcQSim CT SCSI IDs are: 0 Boot Disk (standard), 1 Main Image Disk (standard), 2 Secondary Image Disk, (optional) 3 Optical Disk (optional), 4 Secondary Tape (optional), 5 Primary Tape (standard), 6 CD ROM Drive (standard), 7 Host CPU.

If SCSI device targeting is acceptable, memory configuration could be the problem, especially for machines with the sun4c architecture. Ensure that high–capacity memory chips (such as 4MB SIMMs) are in lower banks, while lower–capacity memory chips (such as 1MB SIMMs) are in the upper banks.

Note that SPARC systems do not always support third party CDROM drives, and might generate a similar ”unknown vendor” error message. Check with the CDROM vendor for specific configuration requirements. Some third party disk drives have a read–ahead cache that interferes with Solaris device drivers. Make sure that any existing read–ahead cache facility is turned off.

Segmentation Fault
==================
Segmentation faults usually result from programming error. This message is usually accompanied by a core dump, except on read–only filesystems.

To see which program produced a core file, run either the file(1) command or the adb (1) command. The following examples show the output of the file and adb commands on a core file from the dtmail program. $ file core core: ELF 32–bit MSB core file SPARC Version 1, from ‘dtmail’

$ adb core core file = core –– program ‘dtmail’ SIGSEGV 11: segmentation violation ^D (use Control–d to quit the adb rogram) Ask the vendor or author of this program for a debugged version.

A process has received a signal indicating that it attempted to access an area of memory that is protected or that does not exist. The two most common causes of segmentation faults are attempting to dereference a null pointer or indexing past the bounds of an array.

sendmail[N]: NOQUEUE: SYSERR: net hang reading from variable
==
This is a sendmail message that appears on the console and in the log file /var/adm/messages. If this message occurs once for a particular user, it is possible that a mail message from this user ends with a partial line (having no terminating newline character). If this message appears frequently or at busy times, especially along with other networking errors, it could indicate network problems.
Check the user’s mail spool file to see if a message ends without a newline character. If so, talk with the user and determine how to prevent the problem from occurring again. If these messages are the result of network problems, you could try moving the mail spool directory to another machine with a faster network interface.

During the SMTP receipt of DATA phase, a message–terminating period on a line of its own never arrived, so sendmail timed out and produced this error.

setmnt: Cannot open /etc/mnttab for writing
==
The system is having problems writing to /etc/mnttab. It is possible that the filesystem containing /etc is mounted read–only, or is not mounted at all. Check that this file exists and is writable by root. If so, ensure that the /etc filesystem has been mounted, and is mounted read–write rather than read–only.

share_nfs: /home: Operation not applicable
===
This message usually indicates that the system has a local filesystem mounted on /home, which is where the automounter usually mounts users' home directories. When a systemis running the automounter, do not mount local filesystems on the /home directory. Mount them on another directory, such as /disk2, which on most systems you will have to create.You could also change the automounter auto_home entry, but that is a more difficult

solution. Soft error rate (N%) during writing was too high
===
This message from the SCSI tape drive appears when Exabyteor DAT tapes generate too many soft (recoverable) errors. It is followed bythe advisory "Please, replace tape cartridge" message. Soft errors are an indication that hard errors could soon occur, causing data corruption. First clean the tape head witha cleaning tape as recommended by the manufacturer. If that doesn't work, replace the tape cartridge. You might need to replace the tape drive if the problem still occurs with new tape cartridges.

Soft error rate (retries = N) during writing was too high
==
This message from the SCSI tape drive appears when Archive tapes generate too many soft (recoverable) errors. It is followed by the advisory "Periodic head cleaning required and/or replace tape cartridge" message. Soft errors are an indication that hard errors couldsoon occur, causing data corruption.

First clean the tape head with a cleaning tape as recommended by the manufacturer. If that doesn't work, replace the tape cartridge. Youmight need to replace the tape drive if the problem still occurs with new tape cartridges.

NOTE

Soft Error rate errors are very common with the Exabyte Eliant 820 8mm tape drive, even with a brand new tape.

Stale NFS file handle
====================
A file or directory that was opened by an NFS client was either removed or replaced on the server. If you were editing this file, write it to a local filesystem instead. Try remounting the filesystem on top of itself or shutting down any client processes that refer to stale file handles. If neither of these solutions works, reboot the system. The original vnode isno longer valid. The only way to get rid of this error is to force the NFS server and client to renegotiate file handles. The symbolic name for this error is ESTALE, errno=151.

statd: cannot talk to statd at variable
==
This message comes from the NFS status monitor daemon statd, which provides crash recovery services for the NFS lock daemon lockd. The message indicates that statd has left old references in the /var/statmon/sm and /var/statmon/sm.bak directories. After a user has removed or modified a host in the hosts database, statd might not properly purge files in these directories, which results in its trying to communicate with a nonexistent host.

Remove the file named variable (where variable is the hostname) from both the /var/statmon/sm and /var/statmon/sm.bak directories. Then kill the statd daemon and restart it. If that doesn't get rid of the message, kill and restart lockd as well. If that doesn't work, reboot the machine at your convenience.

stty: TCGETS: Operation not supported on socket
==
This message results when a user tries to remote copy with rcp(1) or remote shell with rsh(1) from one machine to another, but has an stty(1) command in the remote. The solution is to move the stty command to the user's .login (or equivalent) file. Alternatively, execute the stty command in .cshrc only when the shell is interactive. Here is a test to do just that: if ($?prompt) stty ... The rcp andrsh commands make a connection using sockets, which do not support stty's TCGETS ioctl.

su: No shell
============
This message indicates that someone changed the default login shell for root to a program missing from the system. For example, the final colon–separated field in /etc/passwd could have been changed from /sbin/sh to/usr/bin/bash, which does not exist in that location. Possibly an extra space was appended at the end of line. The outcome is that you cannot login as root or switch user to root, and so cannot directly fix this problem.

The only solution is to rebootthe system from another source, then edit the password file to correct this problem. Invoke sync(1M) several times, then halt the machine by typing Stop–A or by pressing the reset button. Reboot single–user from CDROM, the net, or diskette, such as by typing boot cdrom –s at the ok prompt.

After the system comes up and gives you a # prompt, mount the device corresponding to the original / partition somewhere, such as with a mount(1M) command similar to the one below. Then run an editor on the newly–mounted system password file (use ed(1) if terminal support is lacking): # mount /dev/dsk/c0t3d0s0 /mnt # ed /mnt/etc/passwd

Use the editor to change the password file's root entry to call an existing shell, such as /usr/bin/csh or /usr/bin/ksh. To keep the "No shell" problem from happening, habitually use admintool or /usr/ucb/vipw to edit the password file. These tools make it difficult to change password entries in ways that make the system unusable.

su: 'su root' failed for variable on /dev/pts/N
==
The user specified after "for" tried to become superuser, but typed the wrong password.

If the user is supposed to know the root password, wait to see if the correct password is supplied. If the user is not supposed to know the root password, ask why he or she is attempting to become superuser.

su: 'su root' succeeded for variable on /dev/pts/N
===
The user specified after "for" just became superuser by typing the root password.

If the user is supposed to know the root password, this message is purely informational. If the user is not supposed to know the root password, change this password immediately and ask how the user learned it.

syncing file systems...
======================
This indicates that the kernel is updating the super–blocks before taking the system down, to ensure filesystem integrity. This message appears after a halt(1M) or reboot (1M) command. It can also appear after a system panic, in which case the system might contain corrupted data.

If you just halted or rebooted the machine, don't worry–– this message is normal. In case of a system panic, look up the panic messages that appear above this one. Your system vendor might be able to help diagnose the problem. So that you can describe the panic to the vendor, either leave your system in its panicked state or be sure that you can reproduce the problem. Numbers that sometimes display after the three dots in the message show the count of dirty pages that are being written out. Numbers in brackets show an estimate of the number of busy buffers in the system.

syslog service starting.

=======================

During system reboot, this message might appear and theboot seems to hang. After starting syslogd(1M) service, the system runs /etc/rc2.d/S75cron, which in turn calls ps(1). Sometimes after an abrupt system crash /dev/bd.off becomes a link to nowhere, causing the ps command to hang indefinitely.

Reboot single user (for example with boot –s) and run ls –l /dev/bd* to see if this is the problem. If so, remove /dev/bd.off, then run bdconfig off or reboot with the –r (reconfigure) option. This is the most commonly reported situation that causes ps to hang.

tar: /dev/rmt/0: No such file or directory

==

The default tape device /dev/rmt/0, or possibly the device specified by the TAPE environment variable, is not currently connected to the system, is not configured, or its hardware symbolic link is broken.

List the files in the /dev/rmt directory to see which tape devices are currently configured. If none are configured, ensure that a tape device is correctly attached to the system, and reboot with the –r option to reconfigure devices. If tape devices other than /dev/rmt/0 are configured, you could specify one of them after the –f option of tar(1).

tar: directory checksum error

============================

This error message from tar(1) indicates that the checksum of the directory and the files it has read from tape does not match the checksum advertised in the header block. Usually this indicates the wrong blocking factor, although it could indicate corrupt data on tape.

To resolve this problem, make certain that the blocking factor you specify on the command line (after –b) matches the blocking factor originally specified. If in doubt, leave out the block size and let tar determine it automatically. If that doesn't help, tape data could be corrupted.

tar: tape write error
====================
A physical write error has occurred on the tar(1) output file, which is usually a tape, although it could be a diskette or disk file. Look on the system console, where the device driver should provide the actual error condition. This might be a write–protected tape, a physical I/O error, an end–of–tape condition, or a File too large limitation.
In the case of write–protectedtapes, enable the write switch. For physical I/O errors, the best course of action is to replace the tape with a new one. For end–of–tape conditions, try using a higher density if the device supports one, or use cpio(1) or pax (1) for their multi–volume support., When encountering File too large limitations, use the parent shell'slimit(1) or ulimit facility to increase the maximum file size.

Text is lost because the maximum edit log size has been exceeded.
==
This message appears at the beginning of a cmdtool(1) session after 100,000 characters have gone by in the scrolling window. Clicking on the top rectangle of the scrollbar might display this message. No data were lost, but the user cannot scroll back before this wraparound point. To increase the maximum size of the Command Tool log file, use cmdtool with the–M option, specifying more than 100,000 bytes.

THE FOLLOWING FILE SYSTEM(S) HAD AN UNEXPECTED INCONSISTENCY:
===
At boot time the /etc/rcS script runs the fsck(1M) command to check the integrity of filesystems marked "fsck" in /etc/vfstab. If fsck cannot repair a filesystem automatically, it interrupts the boot procedure and produces this message. When fsck gets into this state, it cannot repair filesystems without losing one or more files, so it wants to defer this responsibility to you, the administrator. Data corruption has probably already occurred.
First run fsck –n on the filesystem, to see how many and what type of problems exist. Then run fsck again to repair the filesystem. If you have a backup of the filesystem, you can generally answer "y" to all the fsck questions. It's a good idea to keep a record of all problematic files and inode numbers for later reference. To run fsck yourself, specify options as recommended by the boot script. For example:

fsck /dev/rdsk/c0t4d0s0. Usually, files lost during fsck repair were created just before a crash or power outage, and cannot be recovered. If important files are lost, you can recover them from backup tapes. If you don't havea backup, ask an expert to run fsck for you.

The SCSI bus is hung. Perhaps an external device is turned off.
==
This message appears near the beginning of rebooting, immediately after a "Boot device: ..." message, and then the system hangs. The problem is conflicting SCSI targets for a non–boot device. Having an external device turned off is unlikely to cause this problem.

THE SYSTEM IS BEING SHUT DOWN NOW !!!
======================================
This message means the system is going down immediately and it’s too late to save any changes.

This message is often preceded by messages telling you that the system is going down in 15 minutes, 10 minutes, and so on. When you see these initial broadcast shutdown messages, save all your work, send any e–mail you’re working on, and close your files. Fortunately vi sessions are automatically saved for later recovery, but many otherapplications have no crash protection mechanism. Data loss is likely.

For more information on shutting down the system, see the System Administration Guide, Volume I. If you are using the AnswerBook, ”halting the system” is a good search string.

The system will be shut down in N minutes
==
Thismessage from the system shutdown(1M) script informs you that the superuser is taking down the system.

Save all changes now or your work will be lost. Write out any files you were changing, send any e–mail messages you were composing, and close your files.

This mail file has been changed by another mail reader.
===
This message appears in a pop–up dialog box whenever you start mailtool(1) while another mail reader has the inbox locked. A question follows: ”Do you wish to ask that mail reader to save the changes?” You are given three choices.

If you choose ”Save Changes” mailtool will request the other mail reader to relinquish its lock and write out any changes it has made to your inbox. If you choose ”Ignore” mailtool will read your inbox without locking it. If you choose ”Cancel” mailtool will exit.

Timeout waiting for ARP/RARP packet
====================================
This problem can occur while booting from the net, and indicates a network connection problem. Make sure the Ethernet cable is connected to the network. Check that this system has an entry in the NIS ethers map or locally on the boot server. Then check the IP address of the server and the client to make sure they are on the same subnet. Local /etc/hosts files must agree with each other and withthe NIS hosts map.

If those are not causing the problem, go to the system's PROM monitor ok prompt and run test net to test the network connection. (On older PROM monitors, use test–net instead.) If the network test fails, check the Ethernet port, card, fuse, and cable, replacing them if necessary. Also check the twisted pair port to make sure it is patched to the correct subnet.

For more information on packets, see SPARC: Installing Solaris Software. If you are using the AnswerBook, "ARP/RARP" isa good search string.

Too many links
==============
An attempt was made to create more than the maximum number of hard links (LINK_MAX, by default 32767) to a file. Because each subdirectory is a link to its parent directory, the same error results from trying to create too many subdirectories.

Check to see why there are so many links to the same file. To get more than the maximum number of hard links, use symbolic links instead.

The symbolic name for this error is EMLINK, errno=31.

Too many open files
===================
A process has too many files open at once. The system imposes a per–process soft limit on open files, OPEN_MAX (usually 64), which can be increased, and a per–process hard limit (usually 1024), which cannot be increased.

You can control the soft limit from the shell. In the C shell, use the limit command to increase the number of descriptors. In the Bourne or Korn shells, use the ulimit command with the –n option to increase the number of file descriptors.

If the window system refuses to start new applications because of this error, increase the open file limit in your login shell before starting the window system. The symbolic name for this error is EMFILE, errno=24.

umount: warning: /variable not in mnttab
==
This message results when the superuser attempts to unmount a filesystem that is not mounted. Note that subdirectories of filesystems,such as /var, cannot be unmounted.

Run the mount(1M) or df(1M) command to see what filesystems are mounted. If you really want to unmount one of them, specify the existing mount point.

Unable to install/attach driver 'variable'
==
These messages appear in /var/adm/messages at boot time, when the system tries to load drivers for devices the machine does not have. Despite the alarmist tone, this message is intended as purely informational. You probably don't want all these device drivers, because they make your system kernel larger, requiring more memory.

undefined control
=================
This message, prefaced by the file name and line number involved, is from the C preprocessor /usr/ccs/lib/cpp, and indicates a line starting with a sharp (#) but not followed by a valid keyword such as define or include.

A piece of software might be running the C preprocessor on an initialization file that you thought was interpreted by a shell. In most shells, the sharp (#) indicates a comment. The C preprocessor considers comments to be anythingbetween /* and */ delimiters.

Unmatched ‘
===========
This message from the C shell csh(1) indicates that a user typed a command containing a backquote symbol (‘) without a closeing backquote. Similar messages result from an unmatched single quote (') or an unmatched double quote ("). Other shells generally give a continuation prompt when a command line contains an unmatched quote symbol.

Correct the command line and try again. To continue typing on another line, give the C shell a backslash right before the newline.

UNREF FILE I=i OWNER=o MODE=m SIZE=s MTIME=tCLEAR?
==
During phase 4, fsck(1M) discovered that the specified file was orphaned because the inode had no record of its pathname. In other words, the file was not connected into any directory.

Answer yes to reconnect the file into the lost+found directory. Then contact the file's owner to ask whether they want it back, and where they want you to place it.

Use ”logout” to logout.
=====================
This C shell message might come as a surprise to Bourne or Korn shell users accustomed to logging out with a Control–d.

When ignoreeof is set, the C shell requires users to logout by typing logout or exit. Write any modified files to disk before exiting.

/usr/openwin/bin/xinit: connection to X server lost
===
This means that the xinit(1) program, which sets up X11 resources and starts a window manager, failed to locate the X server process. Perhaps the user interrupted window system startup, or exited abnormally from OpenWindows (for example, by killing processes or by rebooting). It is possible that the X server crashed. Data loss is possible in some cases. Depending on process timing, this message might be normal when OpenWindows exits during a system reboot.

The only solution is to exit and restart OpenWindows. You do not need to reboot the system unless it hangs and fails to give you a console prompt. To exit OpenWindows, select Workspace–>Exit. To restart OpenWindows, type openwin at the system prompt.

Value too large for defined data type
==================================
The user ID or group ID of an IPC object or file system object was too large to be stored in an appropriate member of the caller–provided structure. Run the application on a newer system, or ask the program’s author to fix this condition.

This error occurs only on systems that support a larger range of user or group ID values than a declared member structure can support. This condition usually occurs because the IPC or file system object resides on a remote machine with a larger value of type uid_t, off_t, or gid_t than that of the local system. The symbolic name for this error is EOVERFLOW, errno=79.

WARNING: Clock gained N days–– CHECK AND RESET THE DATE!
===
Each workstation contains an internal clock powered by a rechargeable battery. After the system is halted and turned off, the internal clock continues to keep time. When the system is powered on and reboots, the system notices that the internal clock has gained time since the workstation was halted.

In most cases, especially if the power has been off for less than a month, the internal clock keeps the correct time, and you do not have to reset the date. Use the date(1) command to check the date andtime on your system. If the date or time is wrong, become superuser and use the date(1) command to reset them.

WARNING: No network locking on variable: contact adminto install server change
==
The Solaris 2.x mount(1M) command issues this message whenever it mounts a filesystem that doesn’t have NFS locking, such as a standard SunOS 4.1.x exported filesytem. Data loss is possible in applications that depend on locking.

On the remote SunOS 4.1.x system, install the appropriate rpc.lockd jumbo patch to implement NFS locking. For SunOS 4.1.4, install patch #102264; for SunOS 4.1.3, install patch #100075; for earlier 4.1 releases, install patch #101817.

WARNING: processorlevel 4 interrupt not serviced
==
This message is basically a diagnostic from the SCSI driver. Especially on machineswith the sun4c architecture, it can appear on the console every 10 minutes or so. To reduce the frequency of this message, add this line near the bottom of the /etc/system file and reboot: set esp:esp_use_poll_loop=0 You might also see this message repeatedly after manually removing a CD when it was busy. Don’t do this! To get the system back to normal, reboot the system with the –r (reconfigure) option.

WARNING: Thermal Warning Detected
===
This is an abnormal shutdown of the system caused by overheating of the host CPU. Because this is an abnormal shutdown, data loss/corrup[tion is possible.

The most likely cause for thie shutdown is clogged air filters in the console itself.

WARNING: /tmp: File system full, swap space limit exceeded
===
The system swap area (virtual memory) has filled up. You need to reduce swap space consumption by killing some processes or possibly by rebooting the system. See the message ”Not enough space” for information about increasingswap space.

WARNING: TOD clock not initialized— CHECK AND RESET THE DATE!
==–=====
This message indicates that the Time Of Day (TOD) clock reads zero, so its time is the beginning of the UNIX epoch: midnight 31 December

1969. On a brand–new system, the manufacturer might have neglected to initialize the system clock. On older systems it is more likely that the rechargeable battery has run out and requires replacement.

First replace the batteryaccording to the mnufacturer’s instructions. Then become superuser and use the date(1) command to set the time and date. On SPARC systems the clock is powered by the same battery as the NVRAM, so a dead battery also causes loss of the machine’s Ethernet address and host ID, which are more serious problems for networked systems.

WARNING: Unable to repair the / filesystem. Run fsck
===
This message comes at boot time from the /etc/rcS script whenever it gets a bad return code from fsck(1) after checking a filesystem. The message recommends an fsck command line, and instructs you to exit the shell when done to continue booting. Then the script places the system in single–user mode so fsck can be run effectively.

See ”/dev/rdsk/variable: UNEXPECTED INCONSISTENCY” for information about repairing UFS filesystems. See ”THE FOLLOWING FILE SYSTEM(S) HAD AN UNEXPECTED INCONSISTENCY” for information about repairing non–UFS filesystems.

Watchdog Reset
==============
This fatal error usually indicates some kind of hardware problem. Data corruption on the system is possible.

Look for some other message that might help diagnose the problem. By itself, a watchdog reset doesn’t provide enough information; because traps are disabled, all information has been lost. If all that appears on the console is an ok prompt, issue the PROM command below to view the final messages that occurred just before system failure: ok f8002010 wector p Yes, that word iswector, not vector. The result is a display of messages similar to those produced by the dmesg(1M) command. These messages can be useful in finding the cause of system failure. This message doesn’t come from the kernel, but from the OpenBoot PROM monitor, a piece of Forth software that gives you the ok prompt before you boot UNIX. If the CPU detects a trap when traps are disabled (an unrecoverable error), it signals a watchdog. The OpenBoot PROM monitor detects the watchdog, issues this message, and brings down the system.

Watchdog Reset, Rebooting.
=========================
See the message ”Watchdog Reset” for details. This rebooting message occurs under the same conditions, but when the EEPROM’s watchdog–reboot? variable is set to true, causing the machine to automatically reboot itself. Data corruption on the system is possible.

Who are you?
============
Many networking programs can print this message, including from(1B), lpr(1B), lprm(1B), mailx(1), rdist(1), sendmail(1M), talk(1), and rsh(1). The command prints this message when it cannot locate a password file entry for the current user. This might occur if a user logged in just before the superuser deleted that user's password entry, or if the network naming service fails for a user who has no entry in the local password file.
If a user's password file entry was accidentally deleted, restore it from backups or from another password file. If a user's login name or user ID was changed, ask that user to logout and login again. If the network naming service failed, check the NIS server(s) and repair or reboot as necessary. There is a known problem (bug 1138025) with starting hundreds of rsh processes on another machine. This message appears because rsh hangs while binding to a reserved port, and responds too slowly to interact with the network naming service.

Window Underflow
================
This message often occurs at boot time, sometimes along with a "Watchdog Reset" error. It comes from the OpenBoot PROM monitor, which was passed a processor trap from the hardware. This error indicates that some programtried to access a SPARC register window that wasn't accessible from the processor.

On some system architectures, specifically sun4c, the problem could be that different capacity memory chips are mixed together. Someone might have placed 1MB SIMMs in the same bank with 4MB SIMMs. If this is so, rearrange the memory chips. Make sure to put higher–capacity SIMMs in the first bank(s), and lower–capacity SIMMs inthe remaining bank(s); never mix different capacity SIMMs in the same bank. The problem could also be that cache memory on the motherboard has gone bad and needs replacement. If main memory is installed correctly, try swapping the motherboard. The best way to isolate the problem is to look at the %pc register to see where it got its arguments from, and why the arguments were bad. If you can reproduce the condition causing this message, your system vendor might be able to help diagnose the problem.

X connection to variable:0.0 broken (explicit kill or server shutdown).
===
This means that the client has lost its connection to the X server. The "0.0" represents the display device, which is usually the console. This message can appear when a user is running an X application on a remote system with the DISPLAY set back to the original system and the remote system's X server disappears, perhaps because someone exited X windows orrebooted the machine. It sometimes appears locally when a user exits the window system. Dataloss is possible if applications were killed before saving files. Try to run the application again in a few minutes after the system has rebooted and the window system is running.

xinit: not found
================
OpenWindows was probably not installed properly, and the openwin(1) program could not find xinit(1) to start up the X windows system. If the user is running another version of X windows, such as the MIT X11 distribution, the startx program serves the same function as xinit. Check the PATH environment variable to make sure it contains the appropriate X windows install directory. Verify that xinit is in this directory as an executable program.

XIO: fatal IO error 32 (Broken pipe) on X server ”variable:0.0”
==
This means that I/O with the X server has been broken. The ”0.0” represents the display device, which is usually the console. This message can appear when a user is running Display PostScript applications and the X server disappears or the client is shut down. Data loss is possible if applications disappeared before saving files Try to run the application again in a few minutes after the system has rebooted and the window system is running.

Xlib: Client is not authorized to connect to Server
==
See the message ”Xlib: connection to ... refused by server” for details.

Xlib: connection to ”variable:0.0” refused by server
===
This message is immediately followed by the ”Xlib: Client is not authorized to connect to Server” message. These messages indicate that an X windows application tried to run on the X server specified inside double quotes, which did not allow the request.
The ”0.0” represents the display device, which is usually the console. If no server name appears, the superuser probably tried to run an X application on the current machine in an X session that was owned by somebody else.

To allow this client to connect to the X server, run xhost +clientname on the X server system. Only the owner of the current X session (who is not necessarily the superuser) isallowed to run the xhost command. If somebody else is running X windows on the server, ask them to log out and then start your own X session on that server; remote X connections are usually allowed for the same user ID.

xterm: fatal IO error 32 (Broken Pipe) or KillClient on X server ” variable:0.0”
=============
This means that xterm(1) has lost its connection to the X server. The ”0.0” represents the display device, which is usually the console. This message can appear when a user is running xterm and the X server disappears or the client gets shut down. Data loss is possible if applications were killed before saving files. Try to run the terminal emulator again in a few minutes after the system has rebooted and the

window system is running.

XView warning: Cannot load font set 'variable' (Font Package)
==

This message from the XView library warns that a requested font is not installed on the X server. Often multiple warnings appear about the same font. The set of available fonts can vary from release to release.
To see which fonts are available on the X server, run the xlsfonts(1) program. Then specify another font name that you see in the output of xlsfonts. Sometimes it is possible to locate a similar font from a different vendor.

There are two packages of X windowsfonts: the common but not required fonts (SUNWxwcft), and the optional fonts (SUNWxwoft). Run pkginfo(1) to see if both these packages are installed, and add them to the system as you wish.

ypbind[N]: NIS server for domain "variable" OK
===

This message appears after an "NIS server not responding" message to indicate that ypbind(1M is able to communicate with an NIS server again. Proceed with your work. This message is purely informational.

ypbind[N]: NIS server not responding for domain "variable"; still trying
==

This means that the NIS client daemon ypbind(1M) cannot communicate with an NIS server for the specified domain. This message appears when a workstation running the NIS naming service has become disconnected from the network, or when NIS servers are down or extremely slow to respond.

If other NIS clients are behaving normally, check the Ethernet cabling on the workstation that is getting this message. On SPARC machines, disconnected network cabling also produces a series of "no carrier" messages. On x86 machines, the above message might be your only indication that network cabling is disconnected.

If many NIS clients on the network are giving this message, go to the NIS server in question and reboot or repair as necessary. To locate the NIS server for a domain, run the ypwhich(1) command. When the server machine comes back in operation, NIS clients give an "NIS server for domain OK" message.

ypwhich: can’t communicate with ypbind
======================================
This message from the ypwhich(1) command indicates that the NIS binder process ypbind(1M) is not running on the local machine.

If the system is not configured to use NIS, this message is normal and expected. Configure the system to use NIS if necessary. If the system is configured to use NIS, but the ypbind process is not running, invoke the following command to start it up:
/usr/lib/netsvc/yp/ypbind –broadcast

zsN: silo overflow
==================
This message means that the Zilog 8530 character input silo (or serial portFIFO) overflowed before it could be serviced. The zs(4S) driver, which talks to a Zilog Z8530 chip, is reporting that the FIFO (holding about two characters) has been overrun. The number after zs shows which serial port experienced an overflow:

zs0 – tty serial port 0 (/dev/ttya) zs1 – tty serial port 1 (/dev/ttyb) zs2 – keyboard port (/dev/kbd) zs3 – mouse port (/dev/mouse)

Silo overflows indicate that data in the respective serial port FIFO has been lost. However, consequences of silo overflows might be negligible if the overflows occur infrequently, if data loss is not catastrophic, or if data can be recovered or reproduced. For example, although a silo overflow on the mouse driver (zs3) indicates that the system could not process mouse events quickly enough, the user can perform mouse motions again. Similarly, lost data from a silo overflow on a serial port with a modem connection transferring data using uucp(1C) will be recovered when uucp discovers the loss of data and requests retransmission of the corrupted packet.

Frequent silo overflow messages can indicate a zs hardware FIFO problem, a serial driver software problem, or abnormal data or system activity. For example, the system ignores interrupts during system panics, so mouse and keyboard activity result in silo overflows.

If the serial ports experiencing silo overflows are not being used, a silo overflow could indicate the onset of a hardware problem.

Another type of silo overflow is one that occurs during reboot when an HDLC line is connected to any of the terminal ports. For example, an X.25 network could be sending frames before the kernel has been told to expect them. Such overflow messages can be ignored.

EEPROM

Shows configuration of the Sparc EEPROM. Type **eeprom** <CR> at the % prompt. A sample screen is shown below:

```
tpe-link-test?=true
scsi-initiator-id=7
keyboard-click?=false
keymap: data not available.
ttyb-rts-dtr-off=false
ttyb-ignore-cd=true
ttya-rts-dtr-off=false
ttya-ignore-cd=true
ttyb-mode=9600,8,n,1,-
ttya-mode=9600,8,n,1,-
sbus-probe-list=012
mfg-mode=off
diag-level=min
#power-cycles=135
system-board-serial#=5013084042511
system-board-date=IST_1121
fcode-debug?=false
output-device=screen
input-device=keyboard
load-base=16384
boot-command=boot
auto-boot?=true
watchdog-reboot?=false
diag-file: data not available.
diag-device=disk0
boot-file: data not available.
boot-device=disk0
local-mac-address?=false
ansi-terminal?=true
screen-#columns=80
screen-#rows=34
silent-mode?=false
use-nvramrc?=falsenvramrc: data not available.
security-mode=none
security-password: data not available.
security-#badlogins=0oem-logo=
oem-logo?=true
oem-banner= Rev: AcQSim CT.PICR.1.00X.19980815.1
AcQSim CT.ACQIM.1.00X.19980820.1 oem-banner?=true hardware-revision: data not available. last-hardware-update:
data not available. diag-switch?=false  name=options rds4 %
```

UTILITIES – ERROR LOGS

Errors created during abnormal system operation are recorded in the Error Logger. The contents of the Error Logger can be displayed from the Diagnostic GUI Error Log option. Figure 147 shows the initial Error Log screen. shows the search capabilities screen, and shows the results when searching for a “512” error.

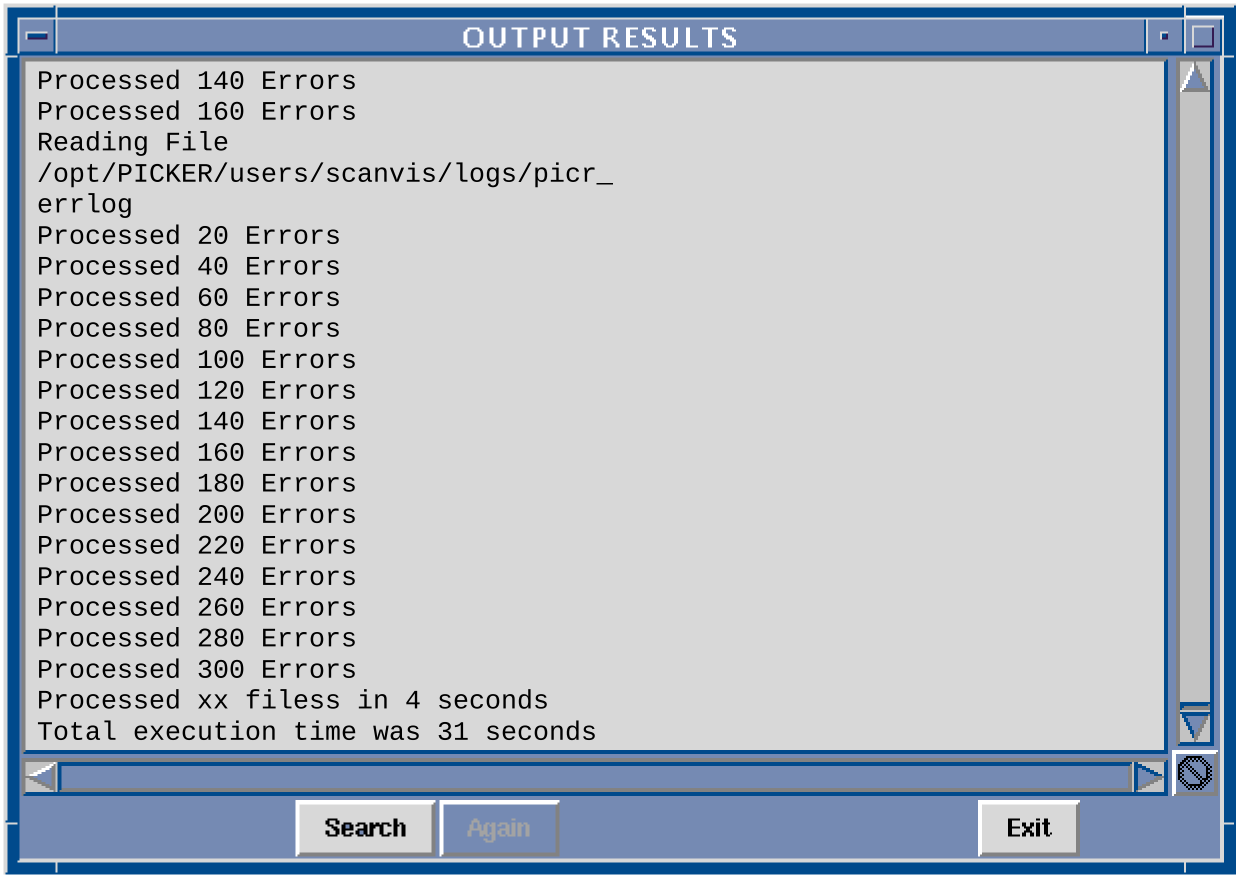


FIGURE 147 Log Viewer Window

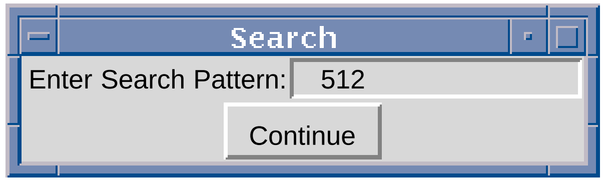


FIGURE 148 Search for a 512 Error

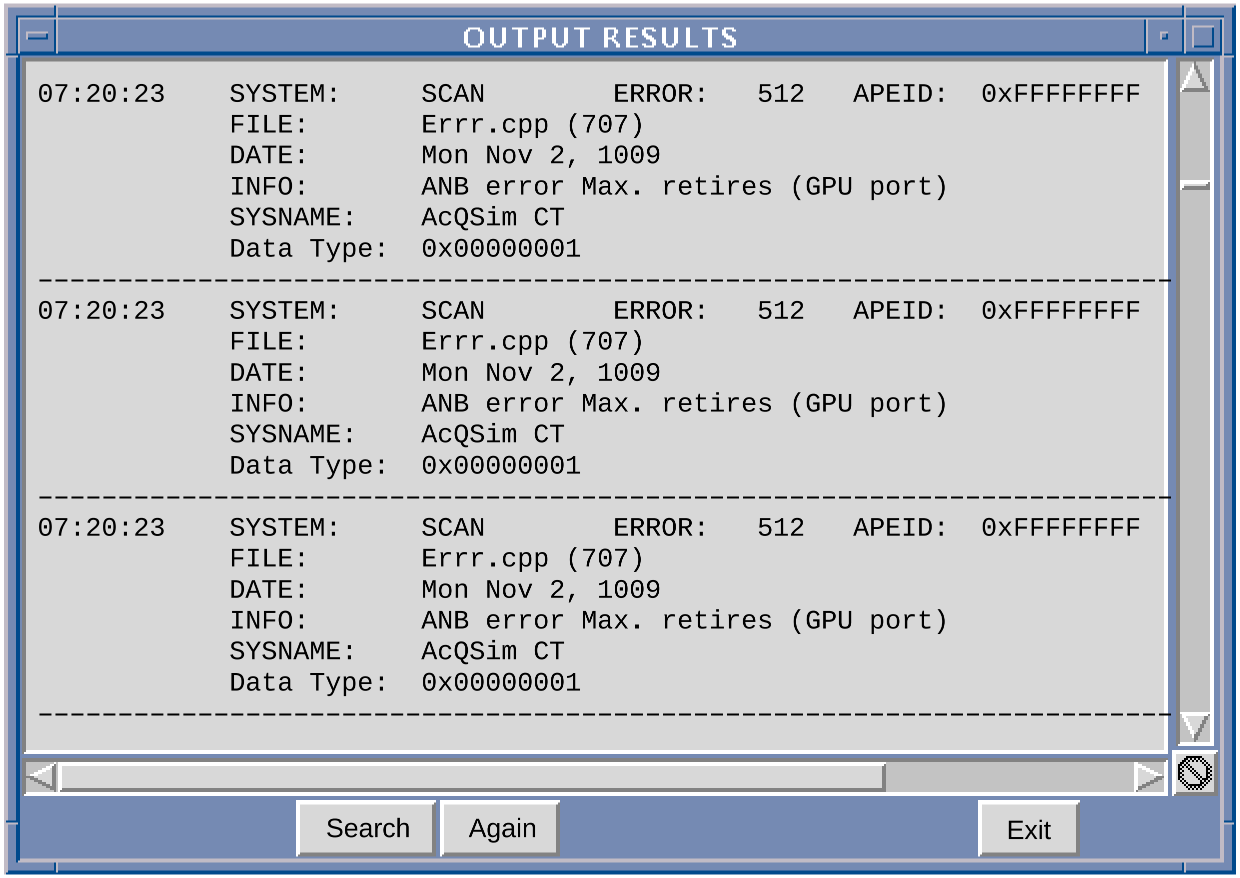


FIGURE 149 Search Results

UTILITIES – AUTOVOICE

> **NOTE**
>
> This test requires at least a functional Display Tower loaded with applications software on disk. The Gantry must be powered up to test the gantry autovoice.

This test will verify that the Autovoice audio channel from the SPARC to the Operator Intercom and to the Gantry are functioning.

> **NOTE**
>
> To simplify viewing and operation any window that appears before the DIAGNOSTIC SOFTWARE TEST MENU should be closed. Do this by clicking on either the OK button, the QUIT button or the upper left hand corner of the window (a dash) and clicking on the close button.

1. Enter the Diagnostic GUI.
2. At the DIAGNOSTIC SOFTWARE TEST MENU window, select the UTILITIES menu and click on ***Autovoice Diagnostics***. The Autovoice Diagnostics window will appear:
3. Choose the amount of record time by sliding the Record Time bar to the left or right. The default is 5 seconds. If required, up to 60 seconds of record time can be selected.
4. Click the Play Location required, either Console or Gantry. A status indicator next to the selection will turn red to indicate the choice.

FIGURE 150   Autovoice Window

5. Click the Record button. Speak into the Operator Intercom microphone. The Record button will stay depressed as long as the Autovoice is in the record mode. When the recording time ends, the Record button will return to the unclicked position.
6. Click on the Play button. The audio will be played back to the Play Location previously selected.
7. Click on the Quit button to exit the Autovoice Diagnostics.

> **NOTE**
>
> Additional Autovoice Testing can be performed using the General Utilities section on Autovoice in the Scanner Application. In addition to record and playback of messages, the playback volume can also be set from minimum to maximum. Refer to the Operator's manual for instructions.

UTILITIES – SET COLOR

If for some reason, the system colors need to be modified, use the Set Color Options menu selection to do so.

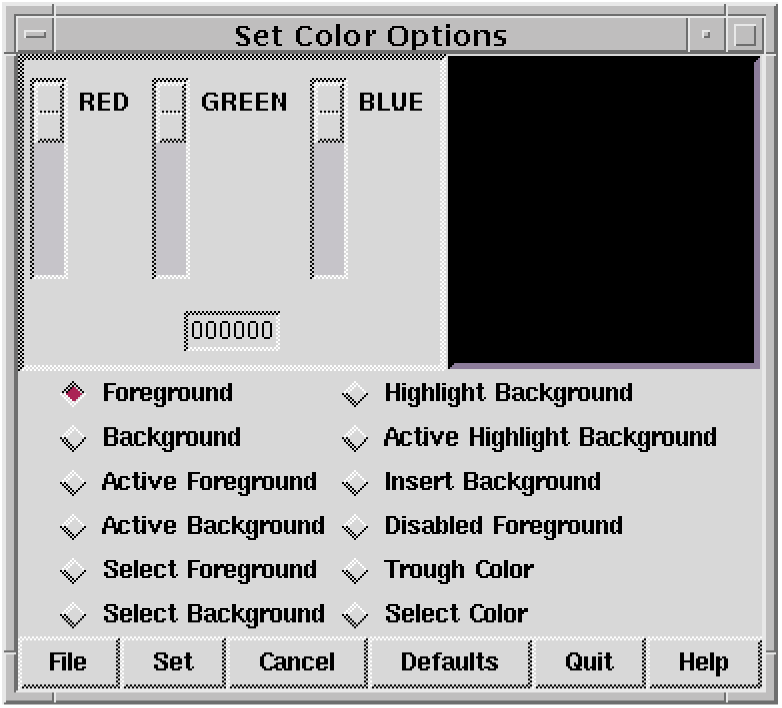


FIGURE 151 SET COLORS SCREEN

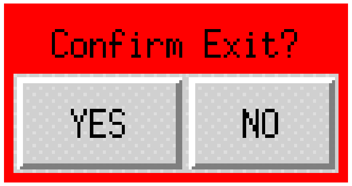
UTILITIES – BOARD STATUS

At the start of the Diagnostic GUI, the status of the AcQImage boards is shown via the Board Status window. This window can be called up at any time from the Utilities pulldown menu.

FIGURE 152   BOARD STATUS WINDOW

EXIT

The EXIT options prompts you before terminating the program:



(This ends the Diagnostic GUI section).

DITA – SEE SERVICE APPLICATIONS MANUAL

Z LOGS

Z Logs is a Log viewing utility available with software release 1.2.3 or higher.

Z Logs provides for the following basic functions:

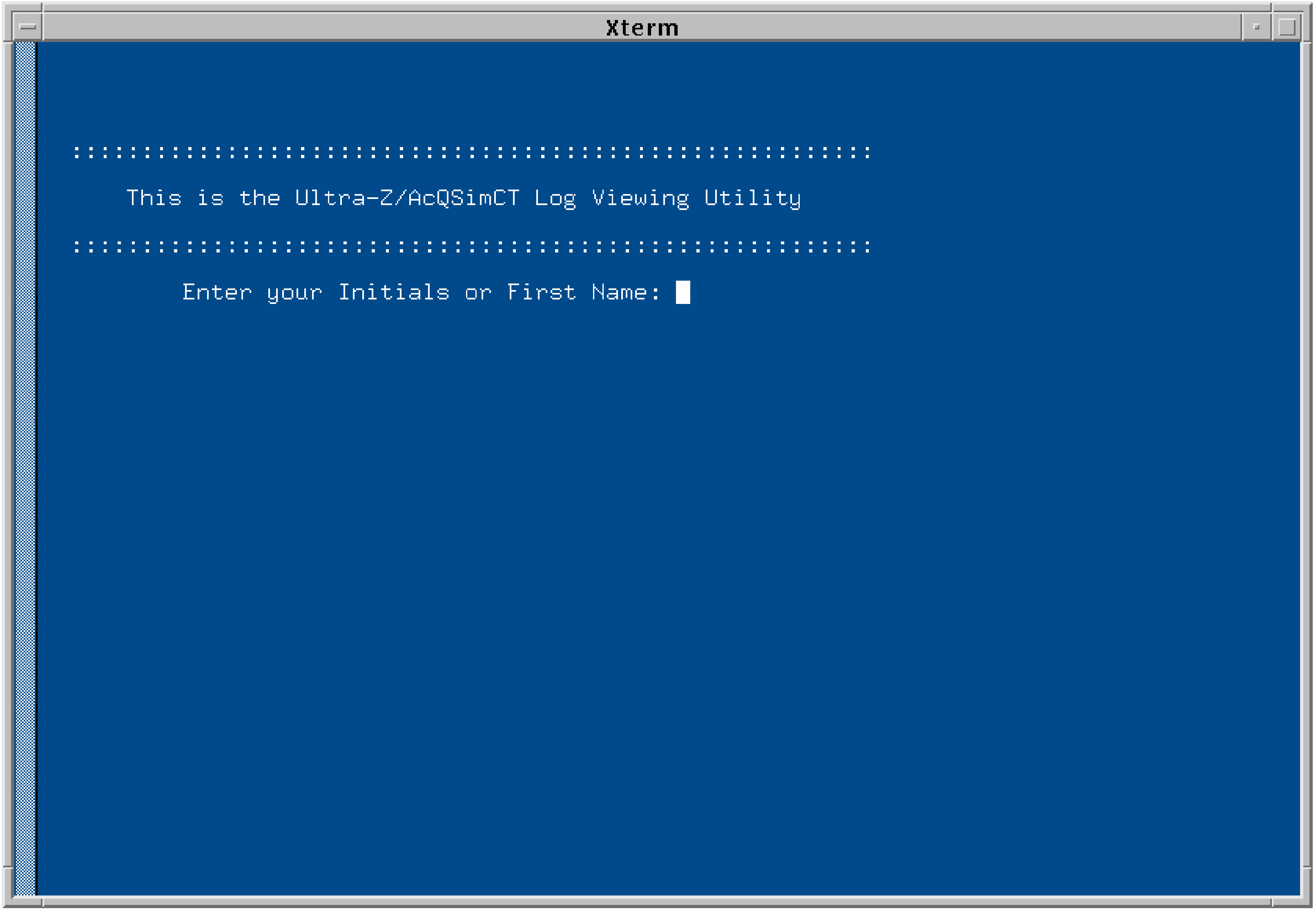
- Error log viewing (Similar to erp on the IQ/PQ).
- Service log for FSE (Used like the Service Log in the IQ/PQ showlogs).
- Tube Log for recording tube change data.
- Calibration/Performance Test log viewing.
- Bad Detector log viewing.

LAUNCHING Z LOGS

Z Logs can be launched from the Xterm window.

1. Open an Xterm window (double click the Xterm icon in the bottom right hand corner of the screen).
2. Type **zlogs** at the prompt.

3. The sign on screen appears in the figure below.



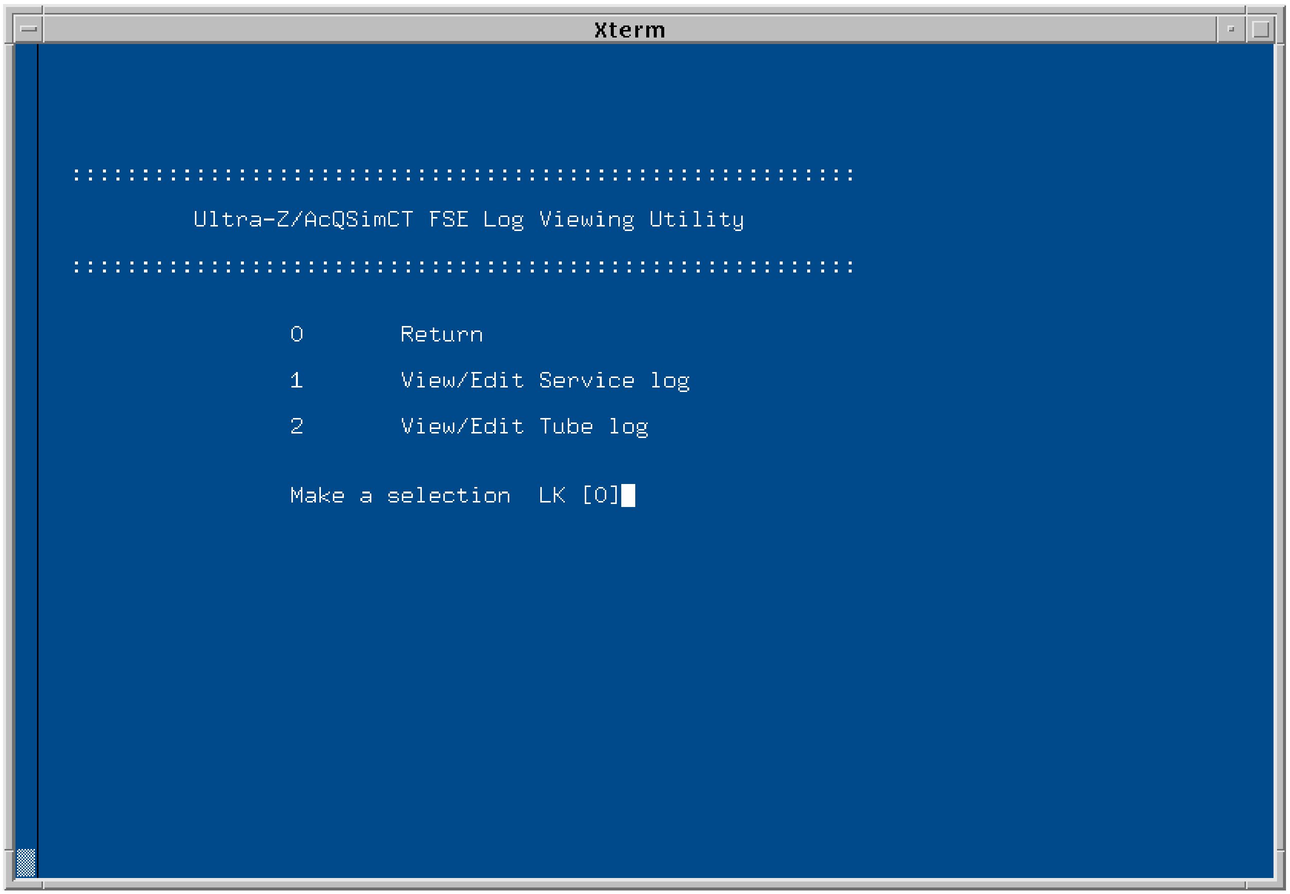
4. After the initial sign on, the main menu will appear as in the figure below.

USING Z LOGS

The usage of Z Logs is very basic and self–explanatory. Type the number of the log you wish view.   The following information is provided as a reference.

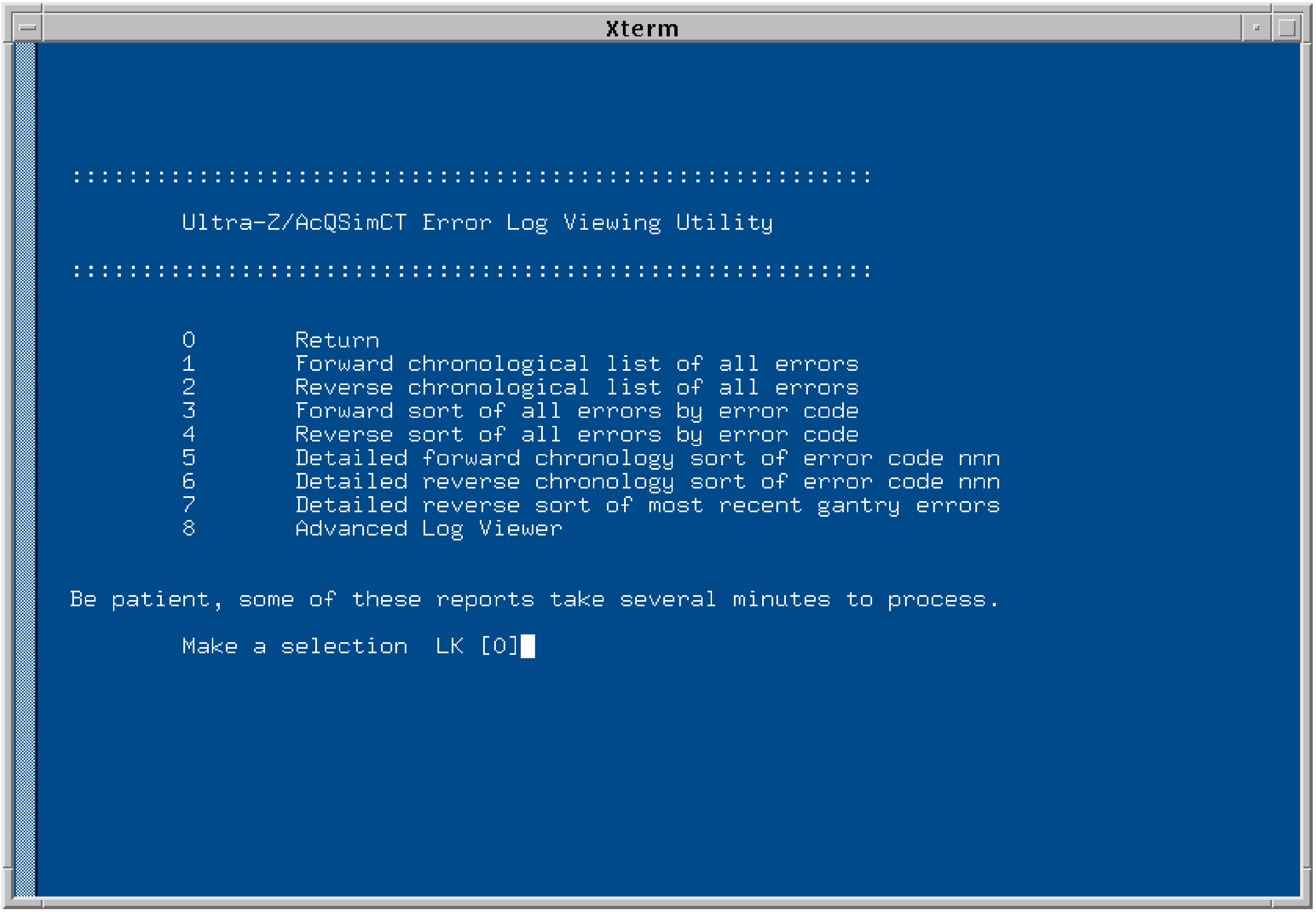
FSE LOGS

Selecting 1 from the main menu opens the FSE Logs menu. From here you can enter the Service Log or the Tube Log. These are text logs used for the purpose of keeping notes from service calls and tube changes.



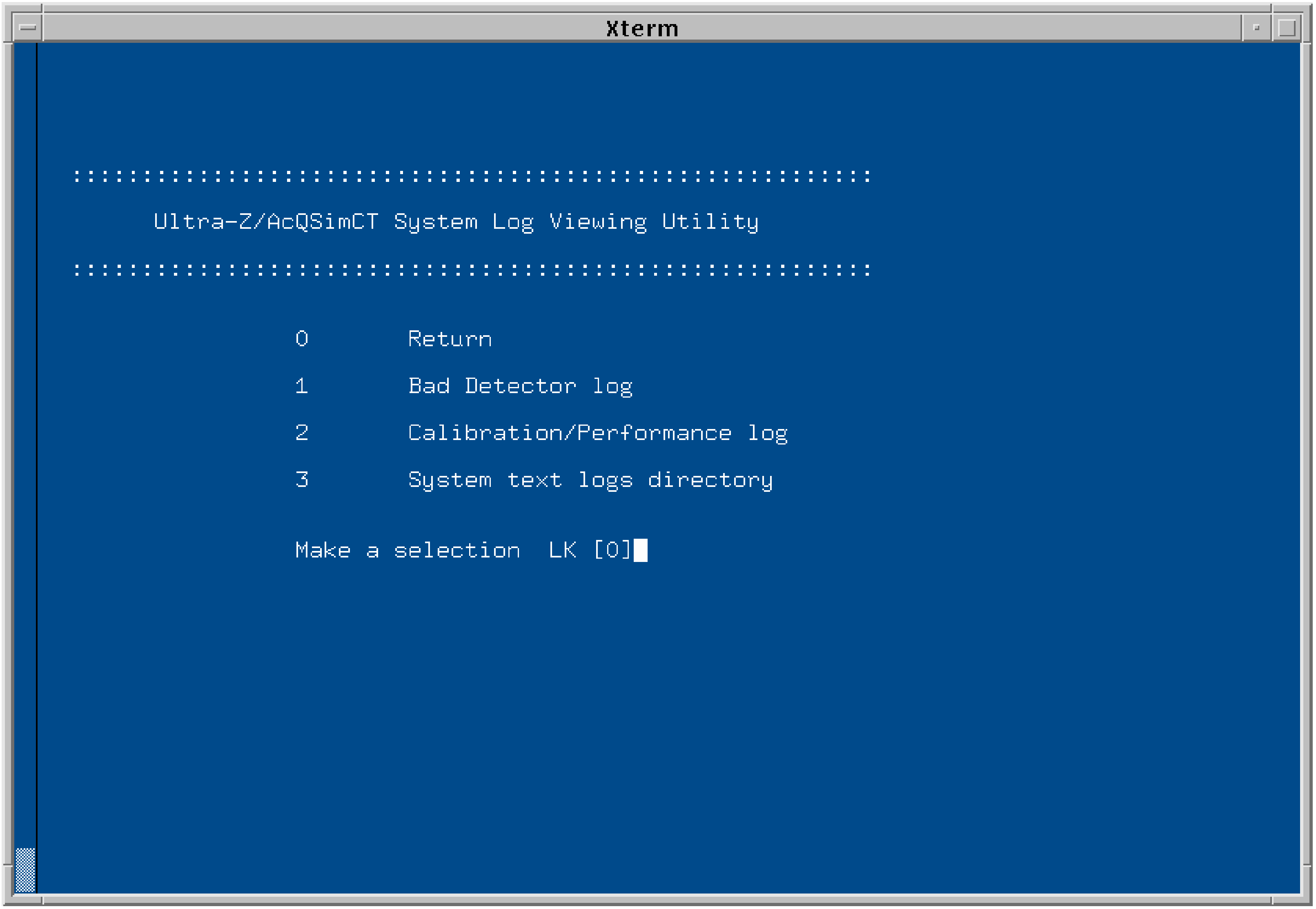
ERROR LOGS

Selecting 2 from the main menu opens the Error Log menu. System error logs can be viewed from this menu. Advanced Log viewer allows you to view individual error log files, select a error log file from a directory and change display option for the error log viewer.



SYSTEM TEXT LOGS

Selecting 3 from the main menu allows you to view the System Text Logs. This includes the Bas Detector Log, Calibration/Performance Log and other system logs that can be selected from the System Log Directory.

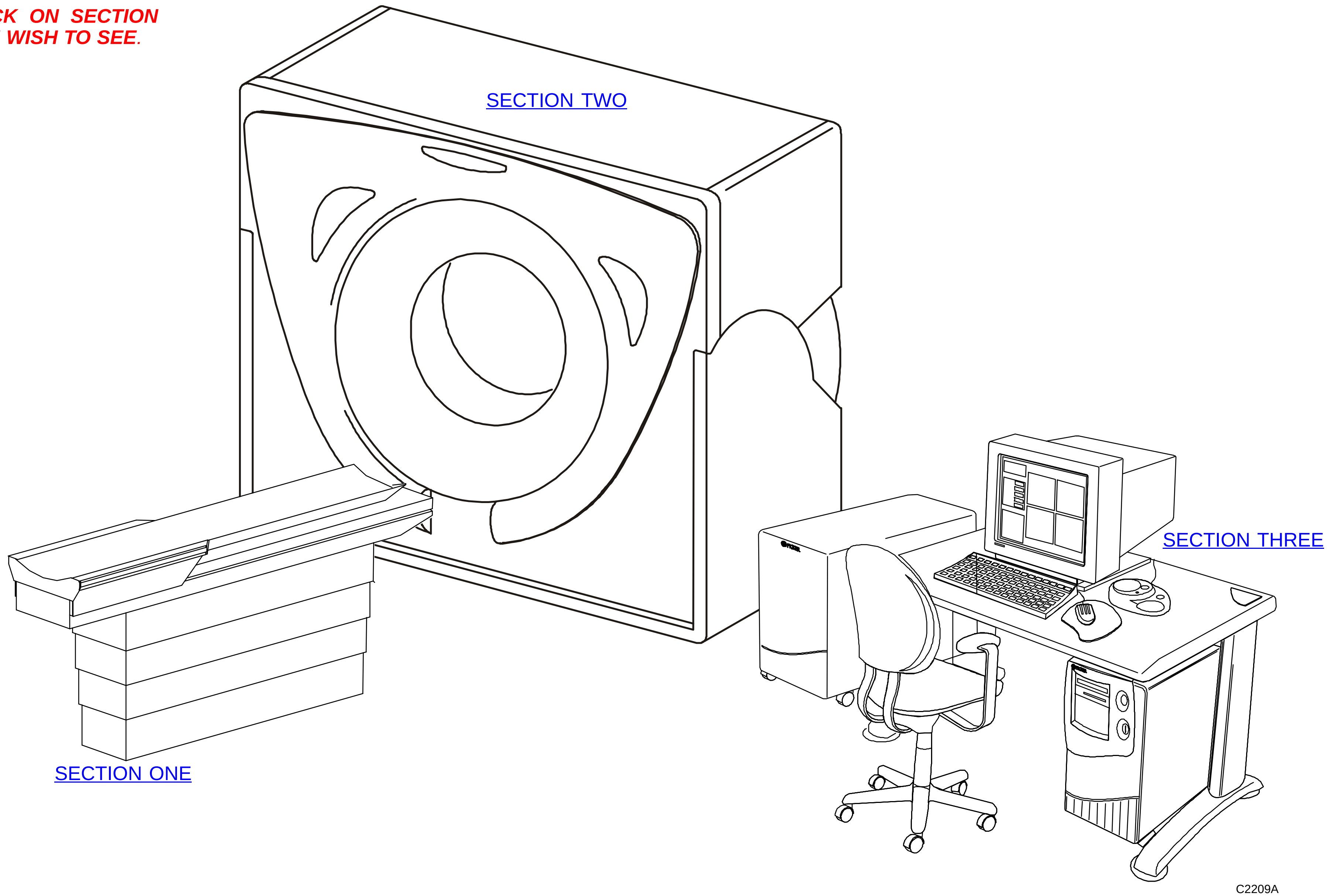


ADVANCED CAL

Advanced Calibration (Autocalibration) is run via the Service Application.

ACQSIM CT PARTS MANUAL

CLICK ON SECTION
YOU WISH TO SEE.



Patient Support Parts

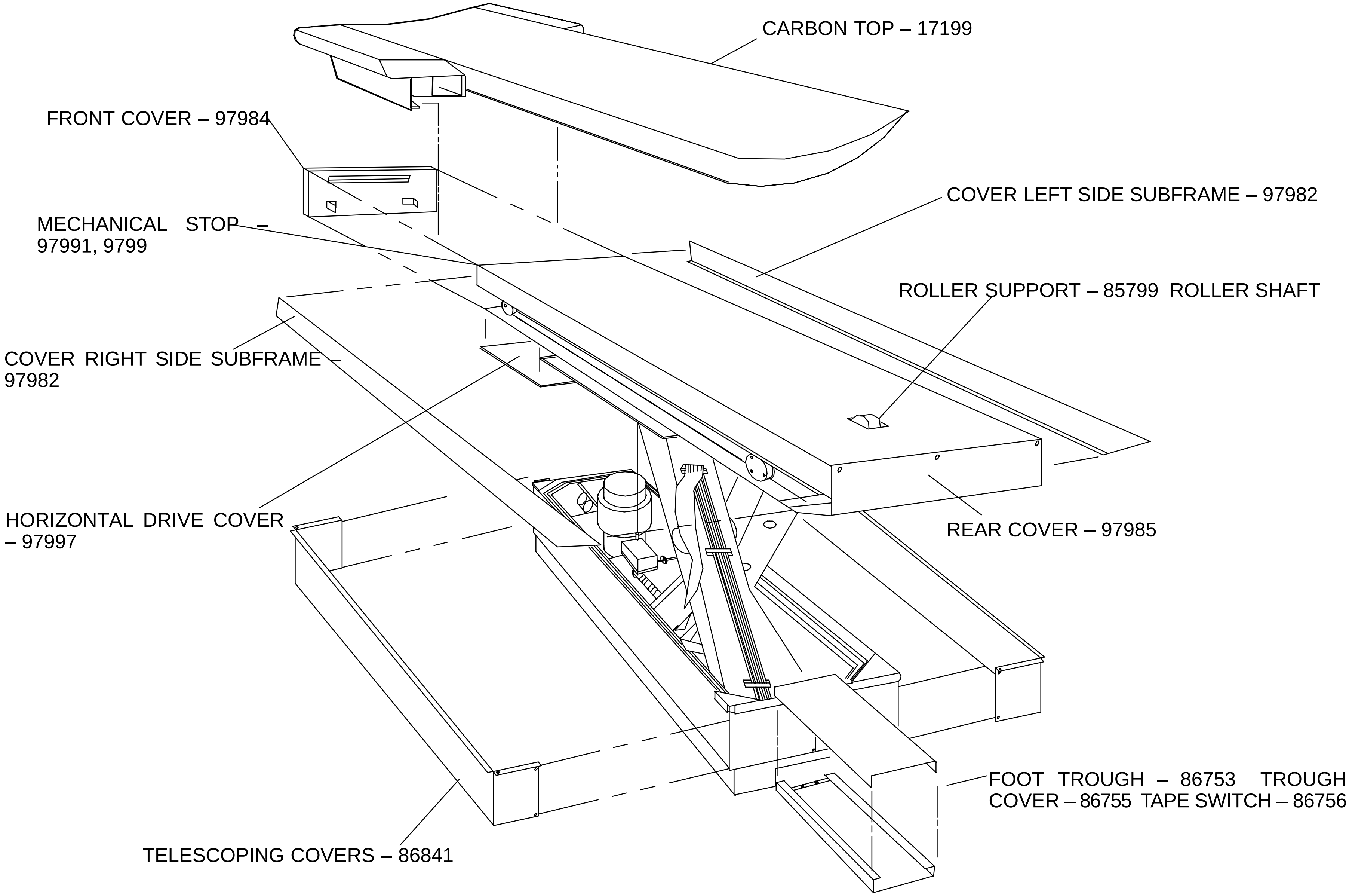


FIGURE 153 PATIENT SUPPORT ASSEMBLY

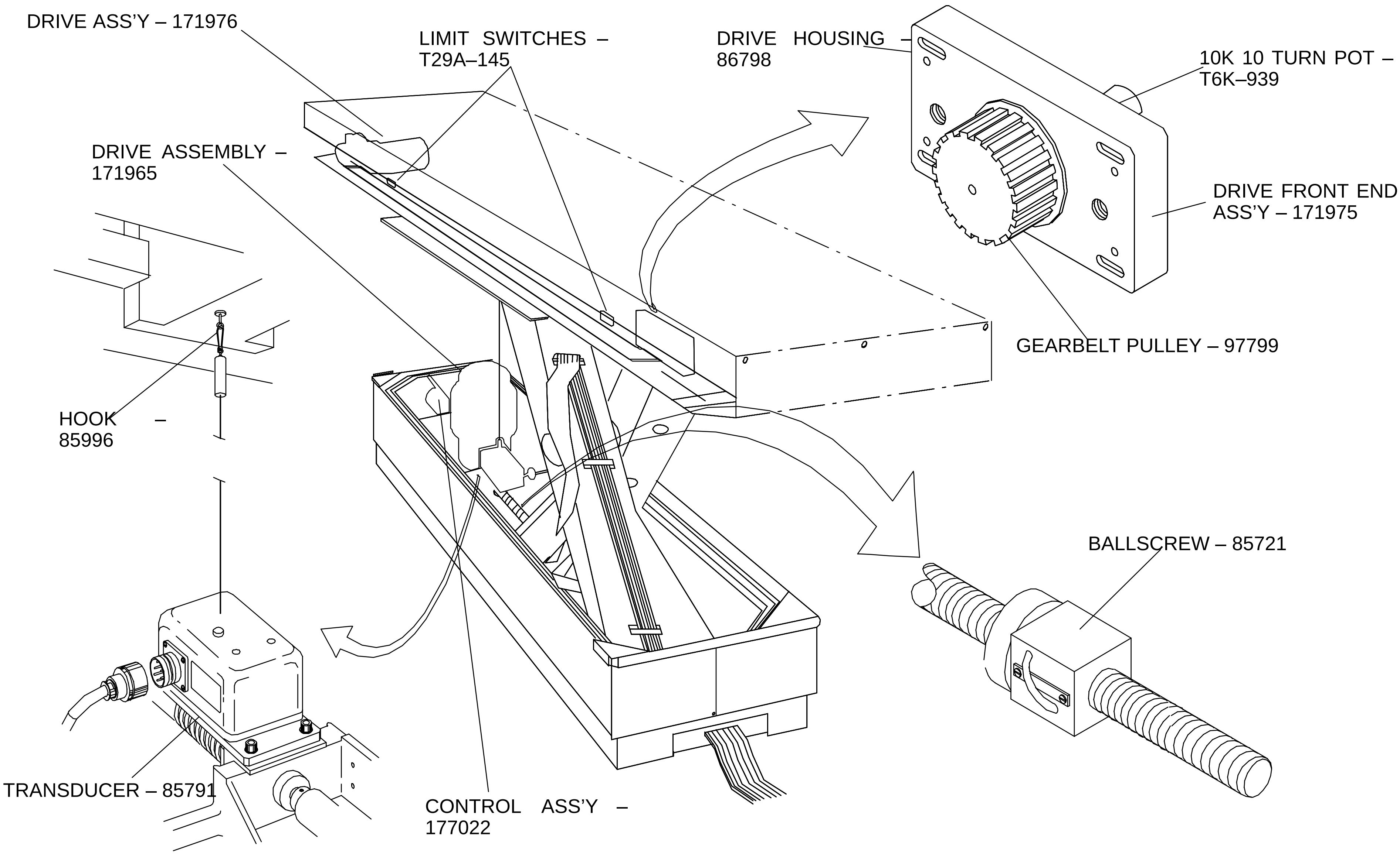


FIGURE 154 PATIENT SUPPORT ASSEMBLY – 171850A

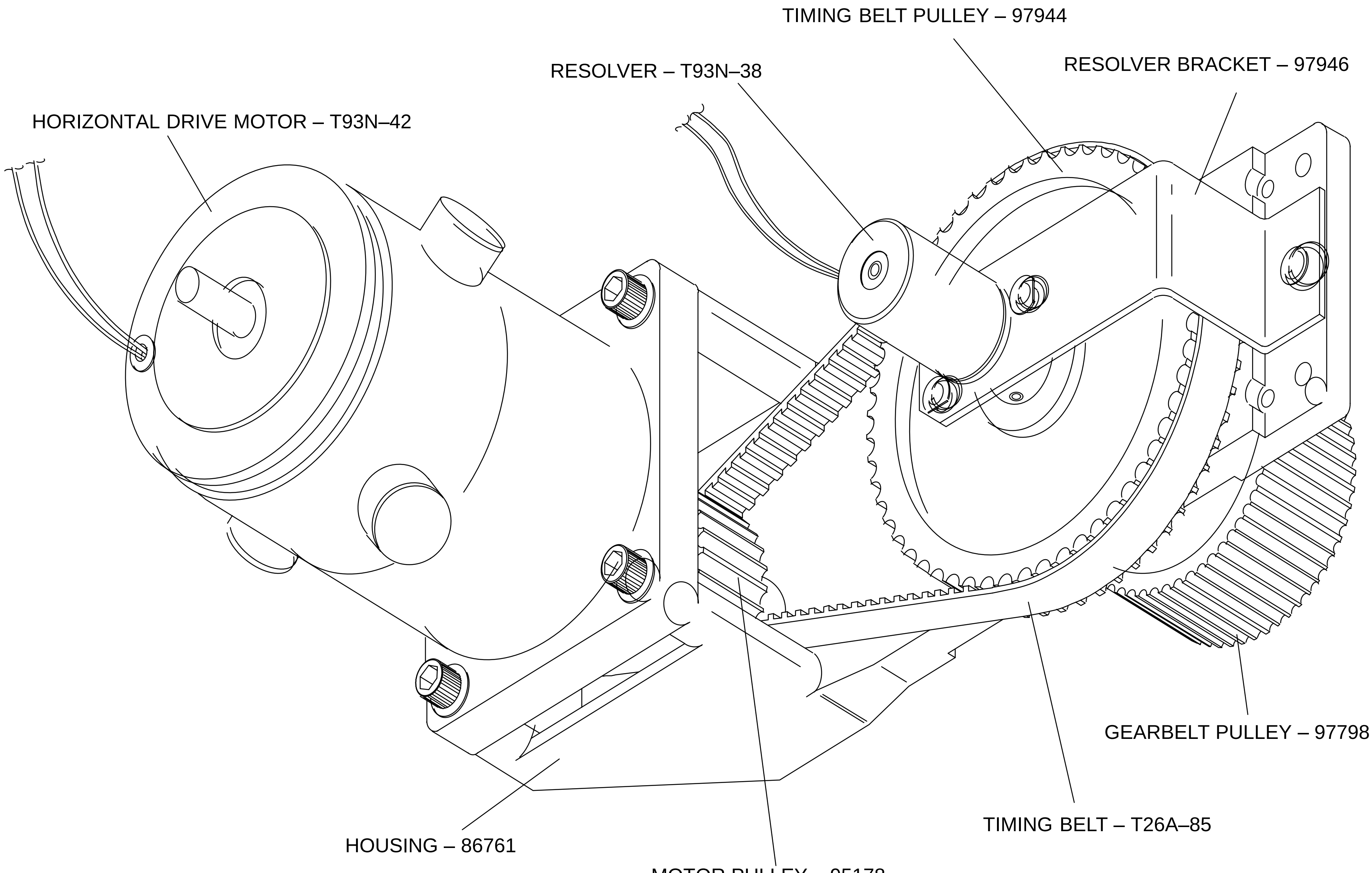


FIGURE 155 HORIZONTAL DRIVE ASSEMBLY – 171976

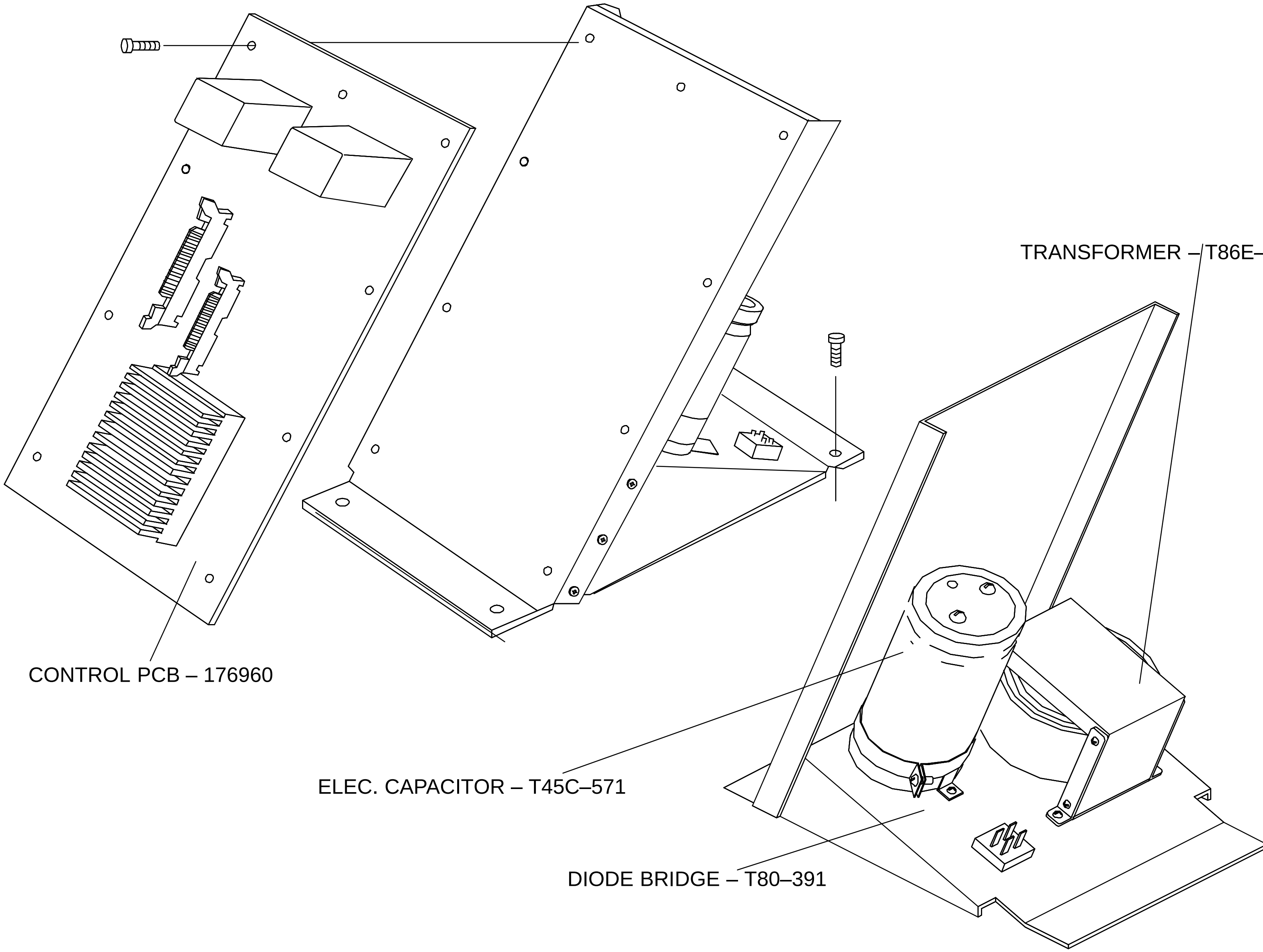


FIGURE 156 PATIENT SUPPORT CONTROL ASSEMBLY – 177022

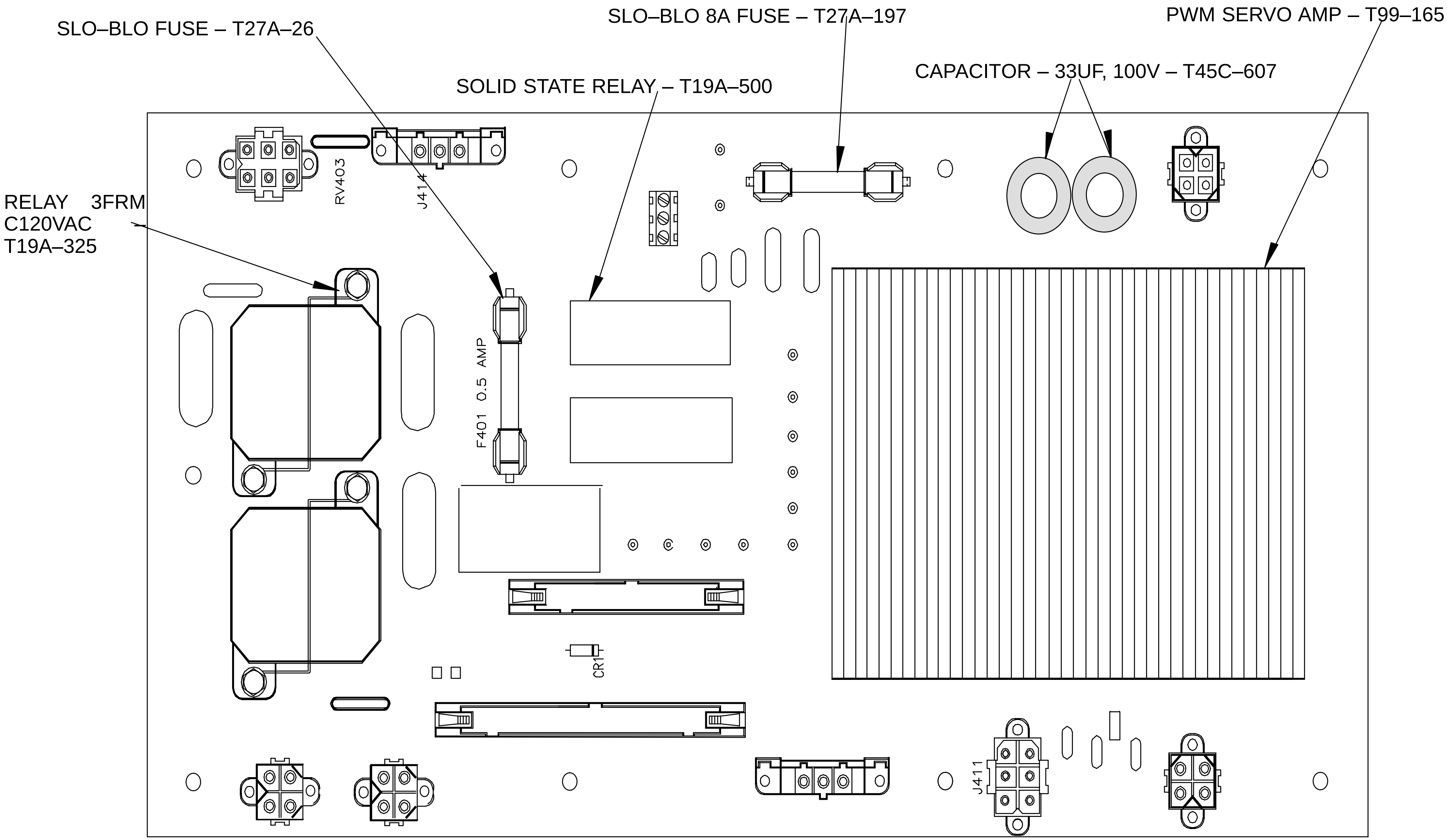


FIGURE 157 PATIENT SUPPORT CONTROL PCB ASSEMBLY – 176960

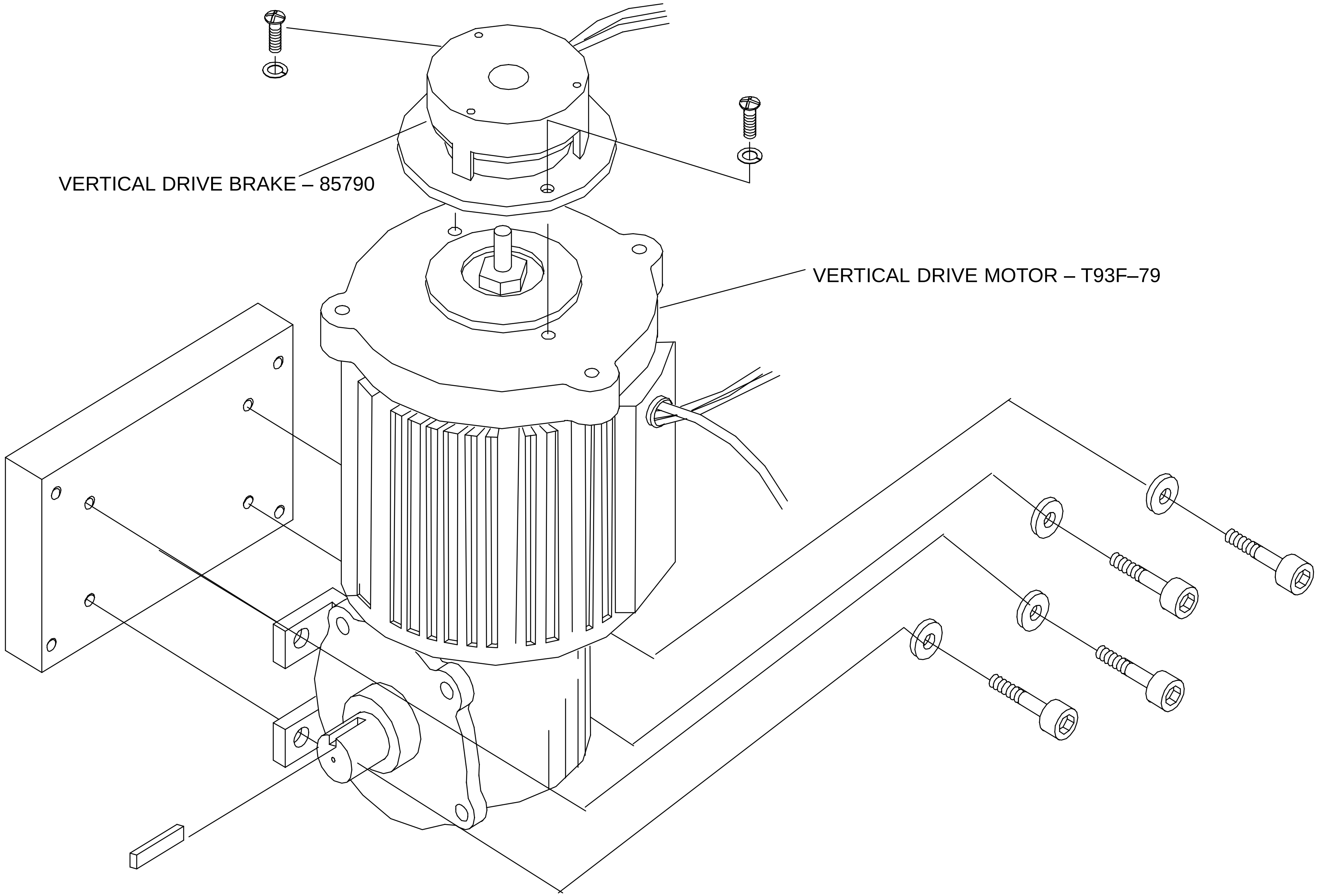


FIGURE 158 VERTICAL DRIVE ASSEMBLY – 171965

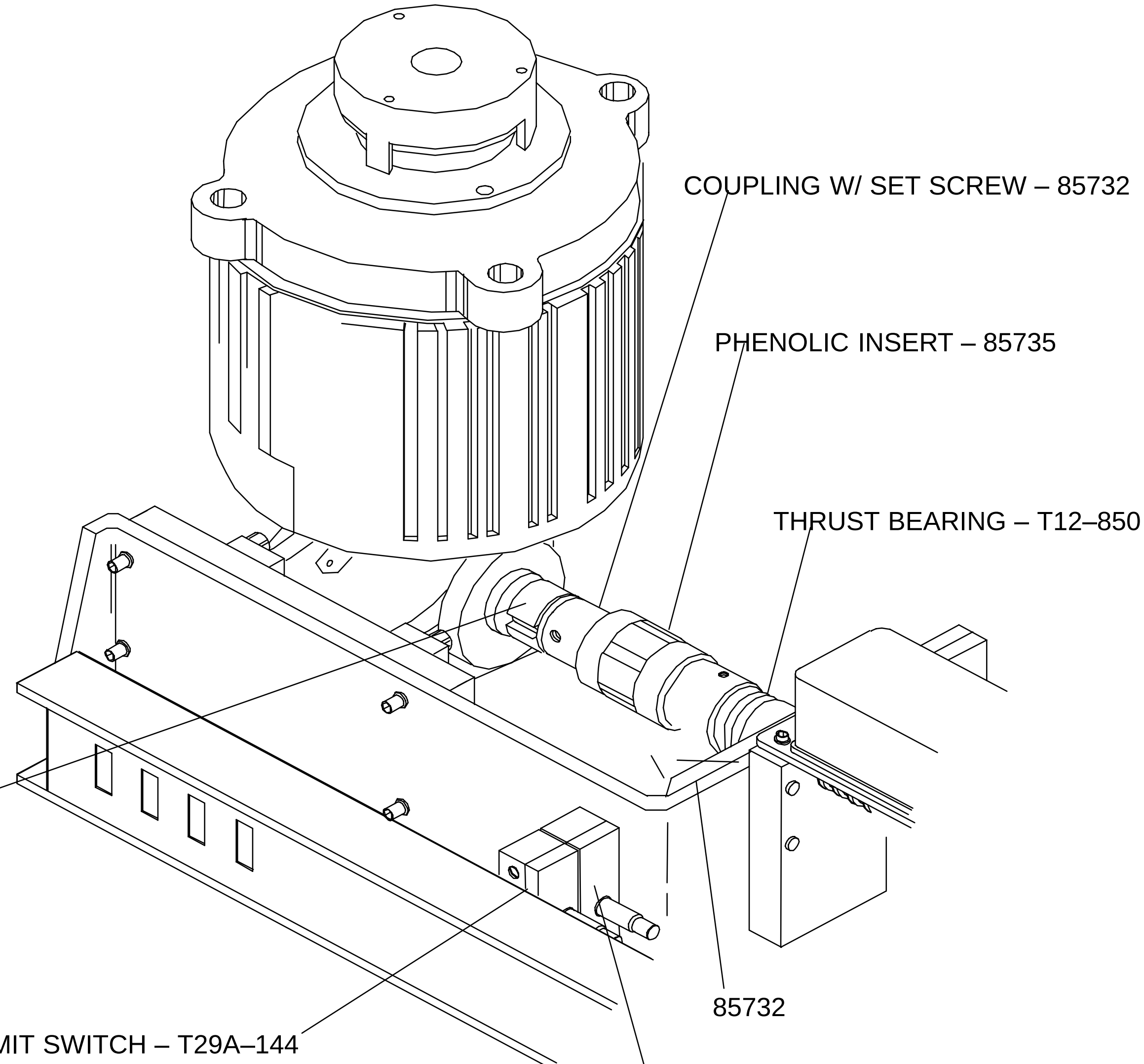


FIGURE 159 VERTICAL MOTOR COUPLING

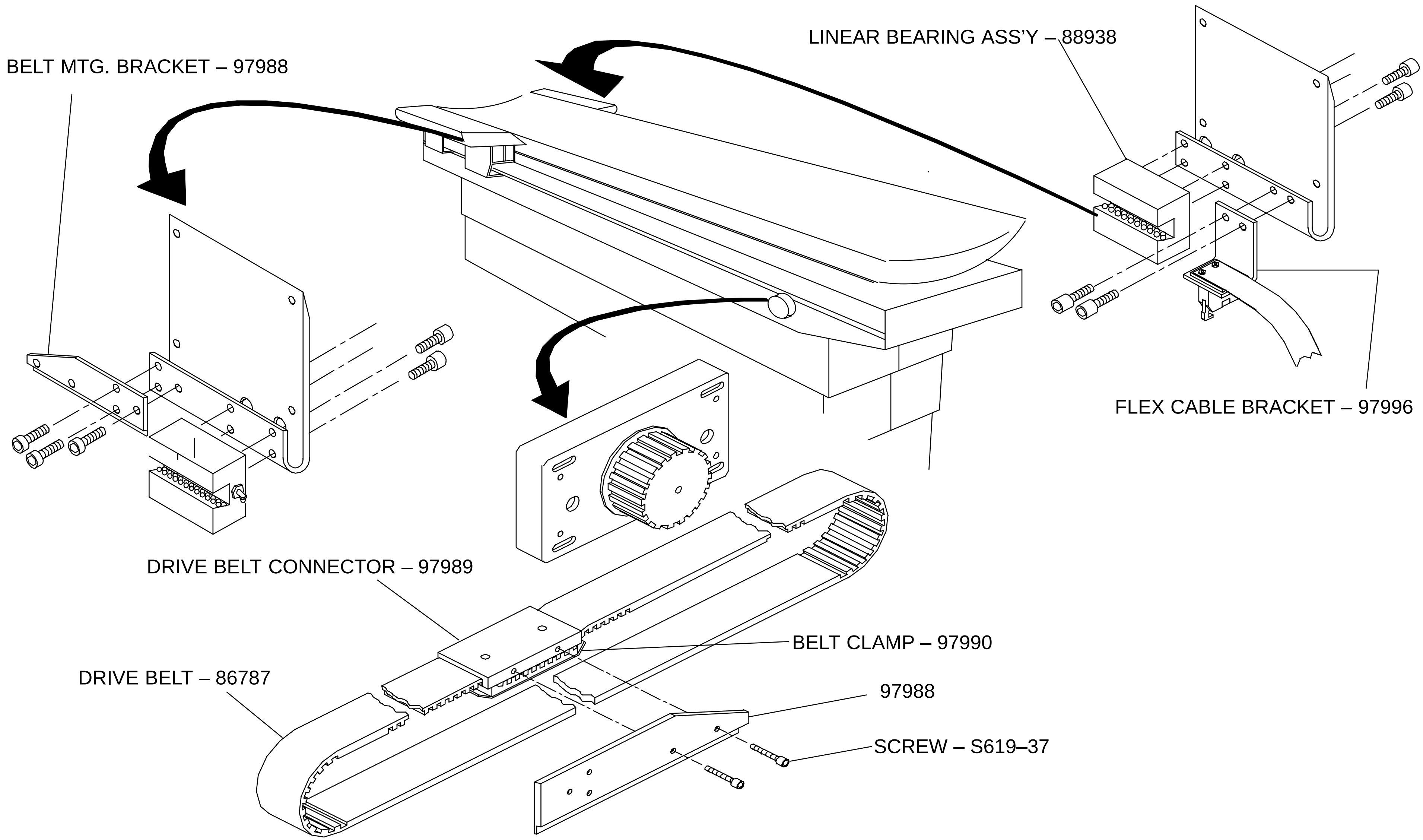


FIGURE 160 PATIENT SUPPORT ASSEMBLY – 171850A

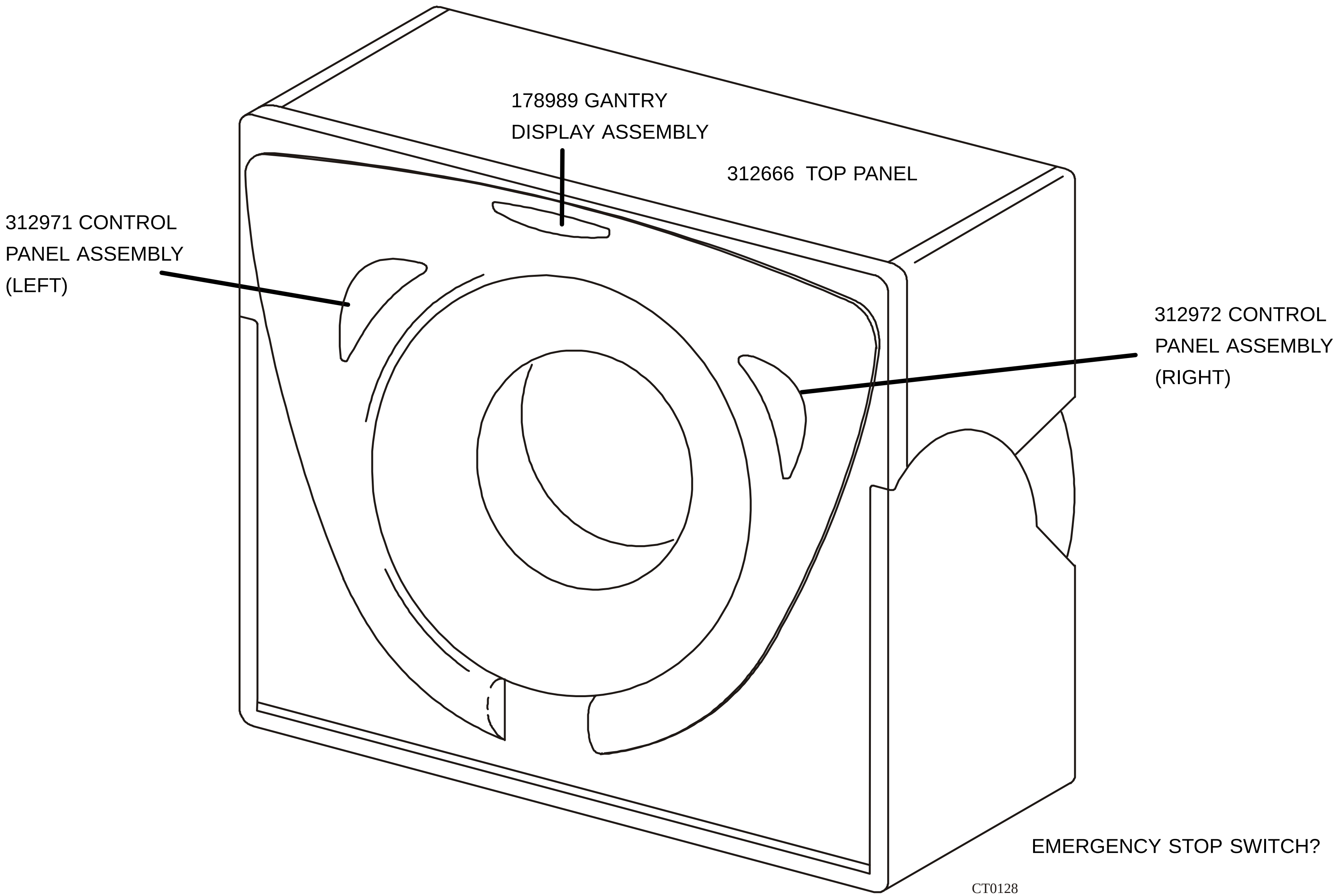


FIGURE 161 GANTRY TOUCH PANEL CONTROLS

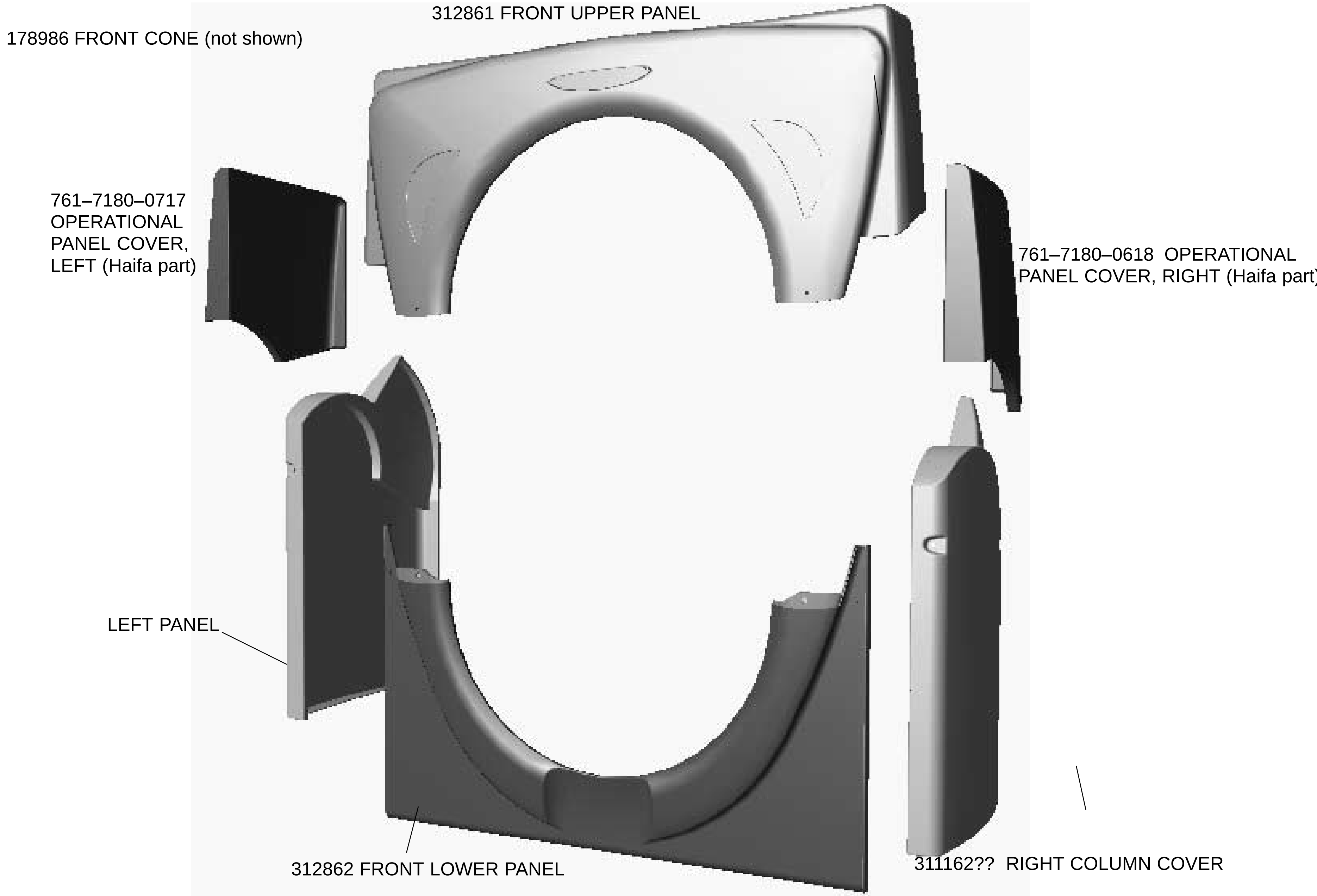


FIGURE 162 GANTRY COVERS

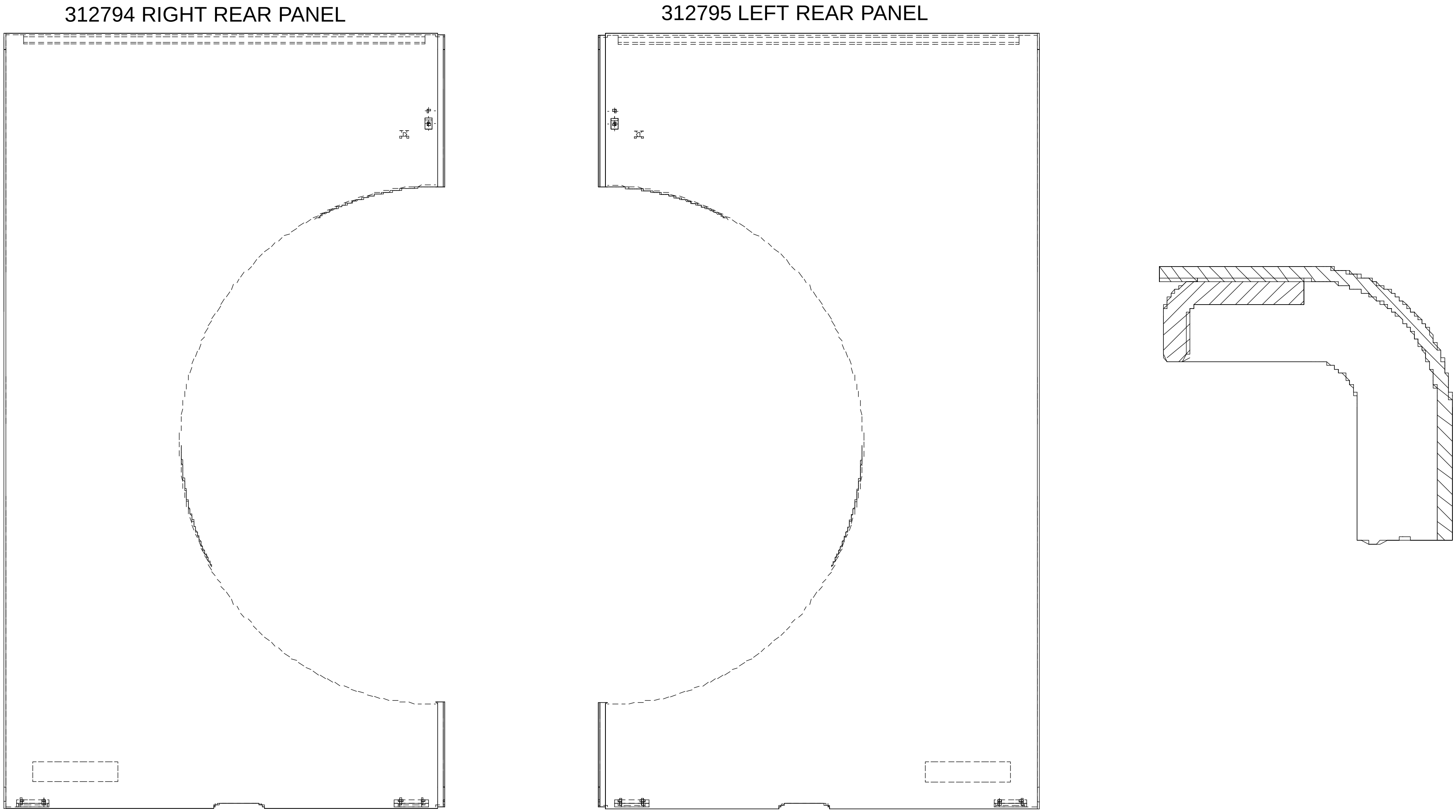


FIGURE 163 REAR PANELS

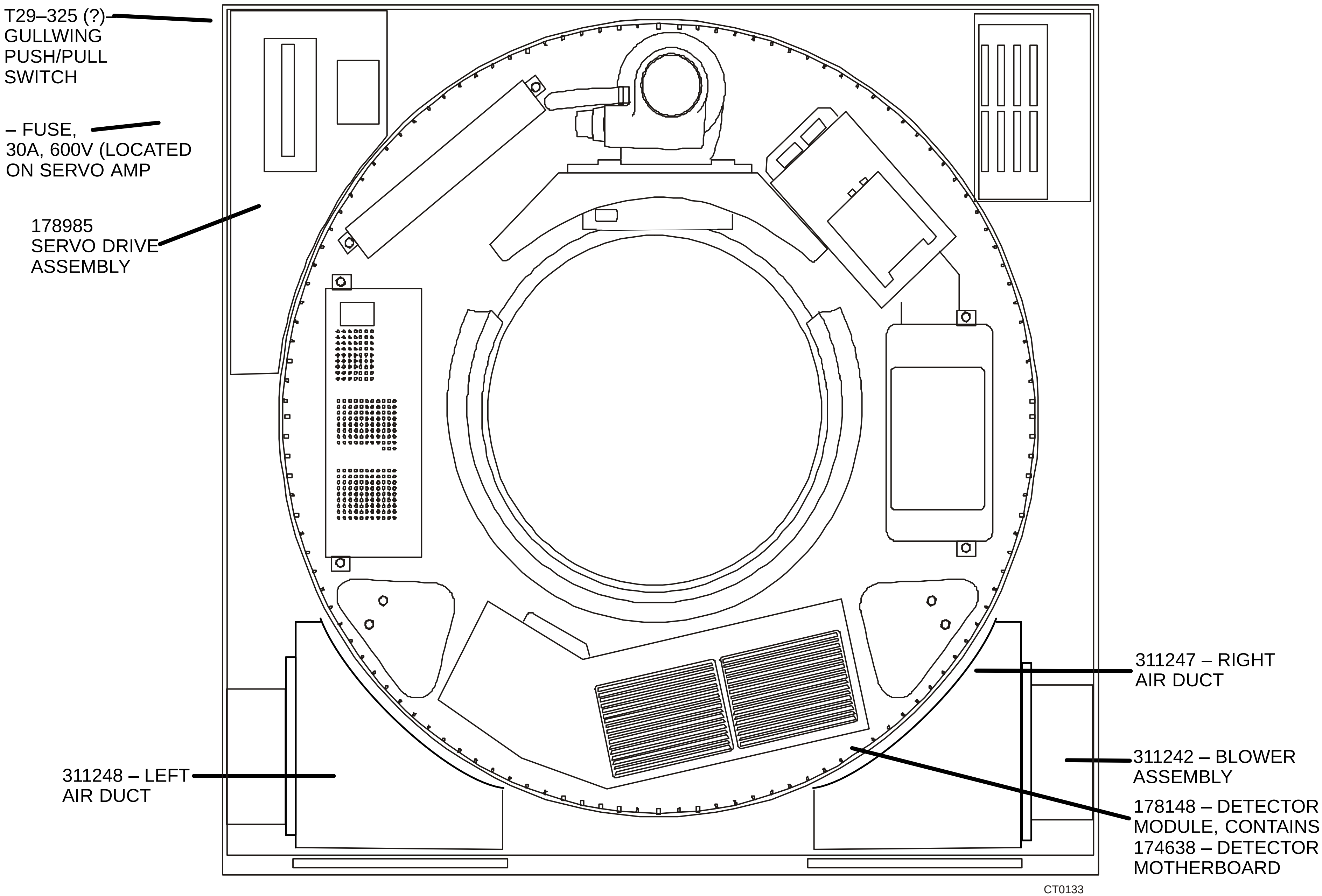


FIGURE 164 FRONT VIEW OF ACQSIM CT GANTRY

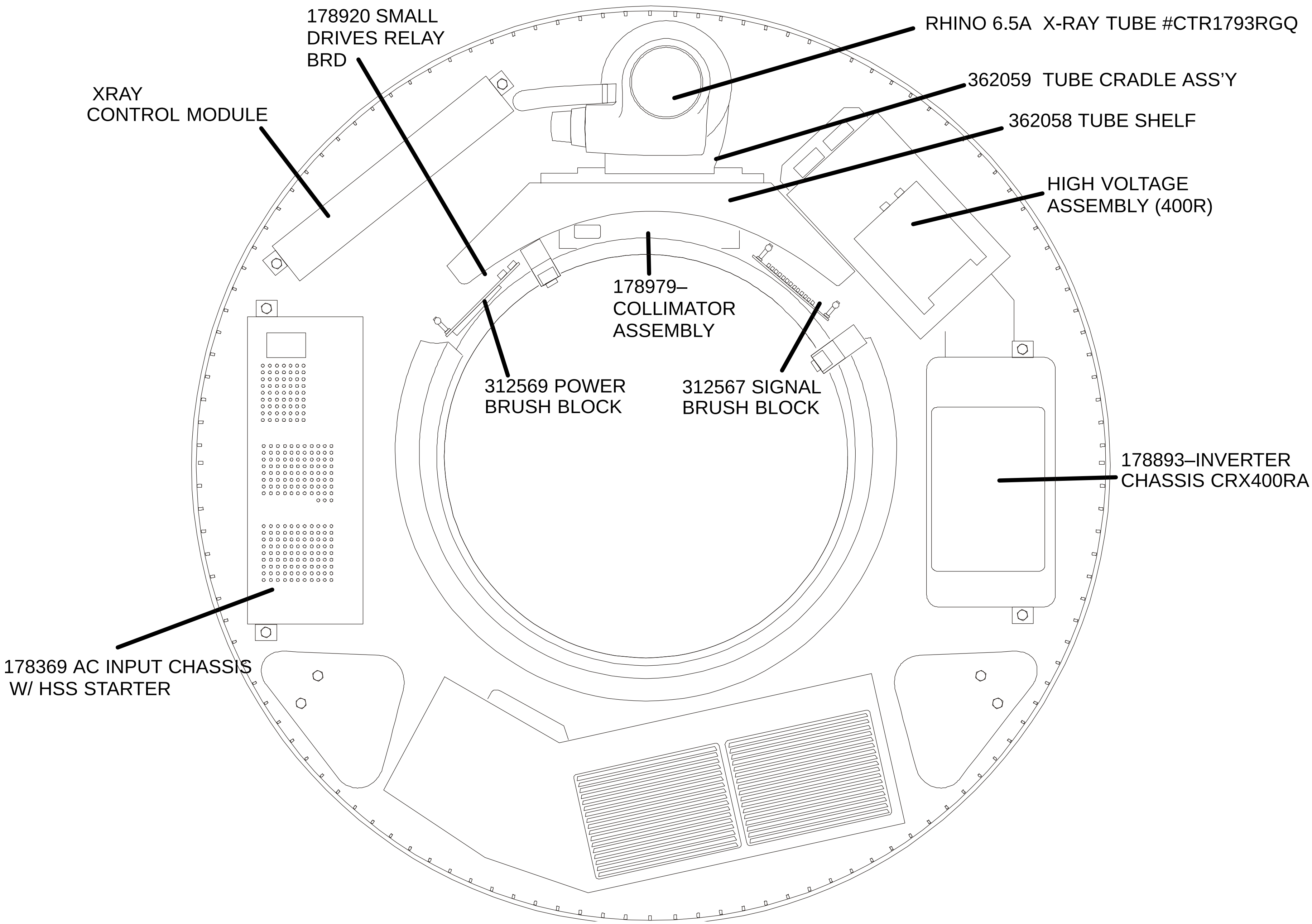


FIGURE 165 ACQSIM CT GANTRY – ROTATING SECTION

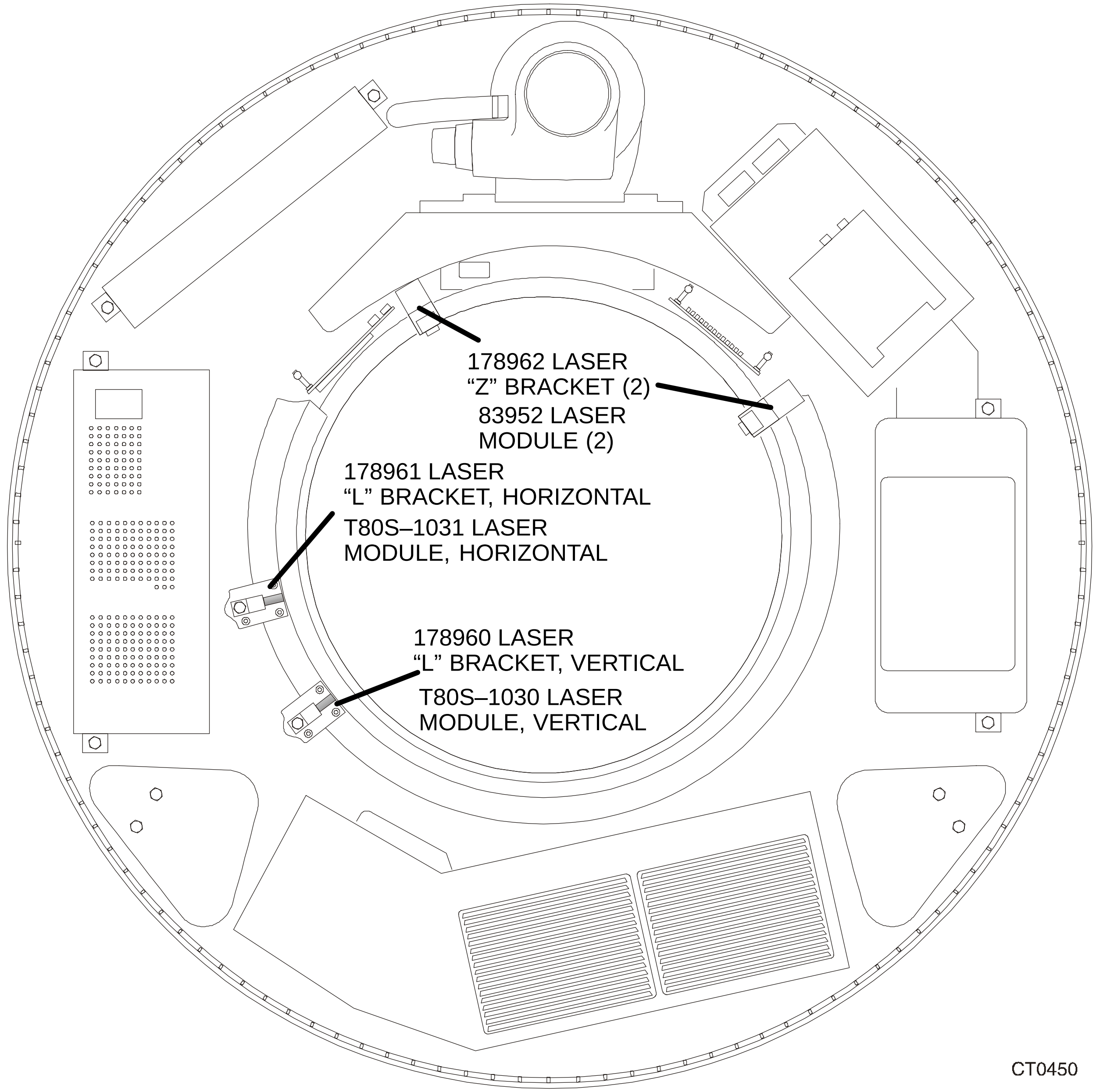


FIGURE 166 ACQSIM CT LASERS

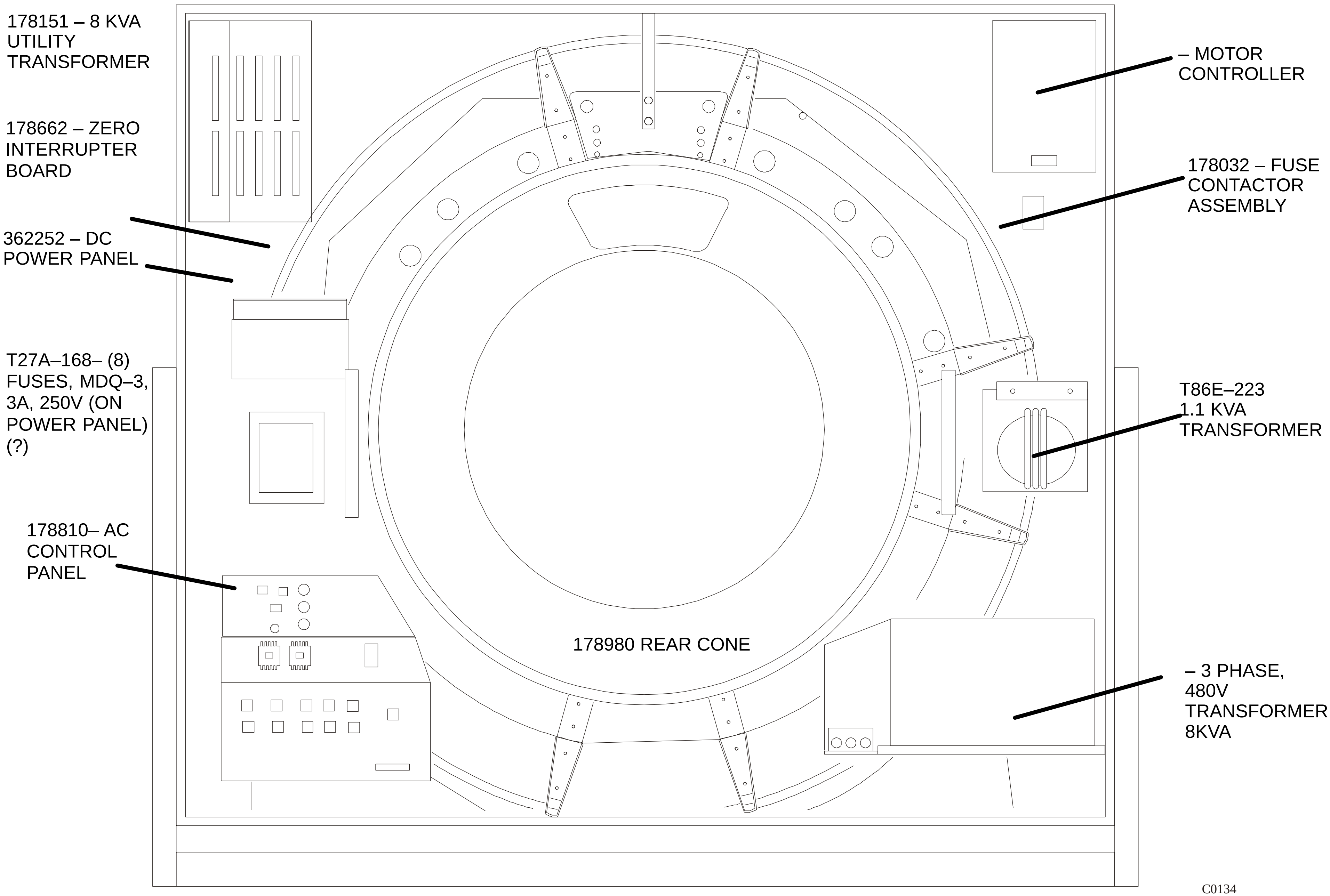
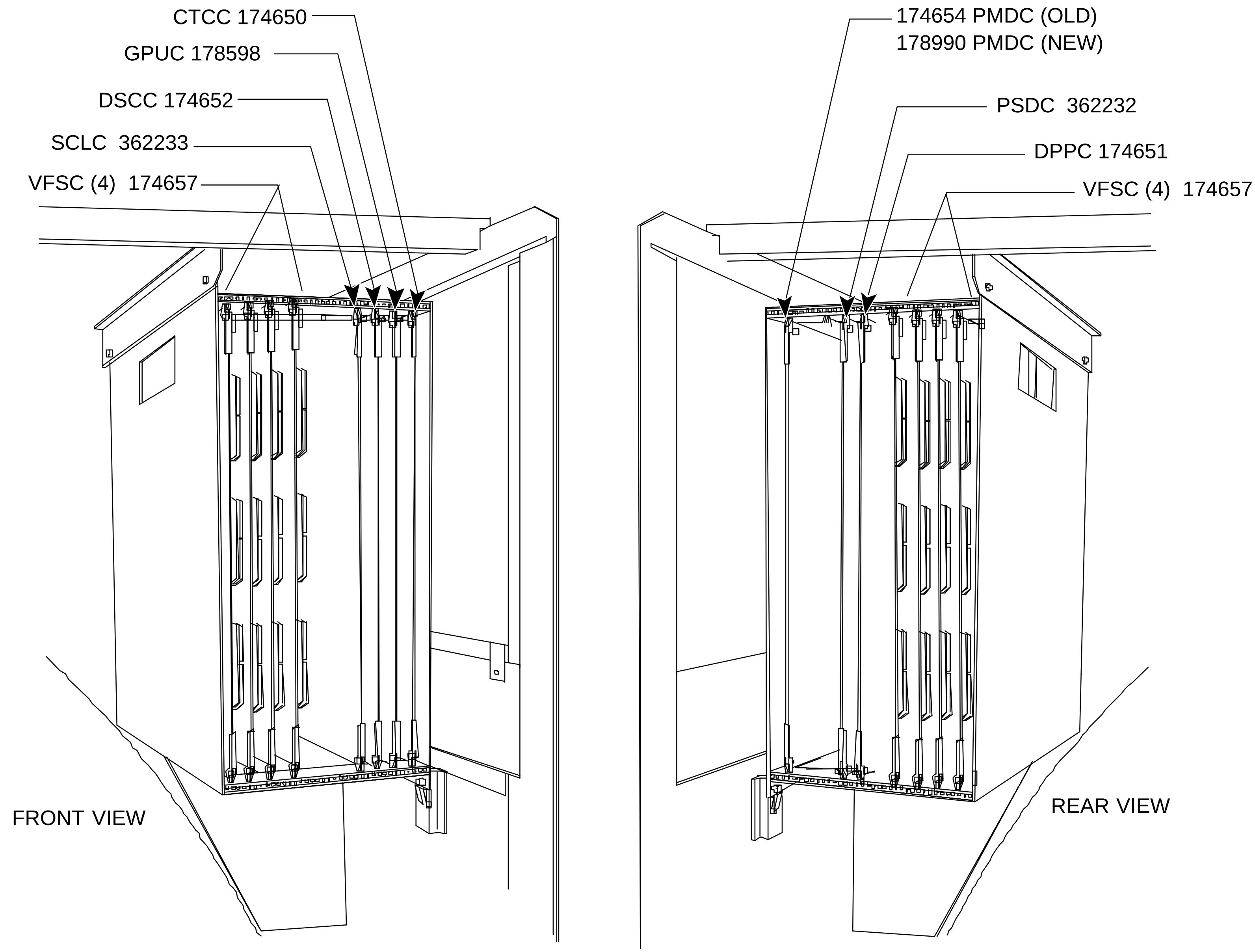


FIGURE 167 GANTRY – REAR



Note: Remove boards with PC board ejector, part number 312874. —>

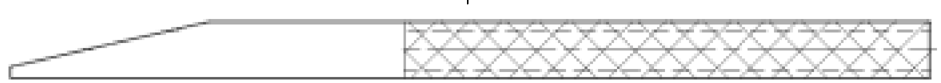


FIGURE 168 ACQSIM CT GANTRY CARDFILE ASSEMBLY – 362226

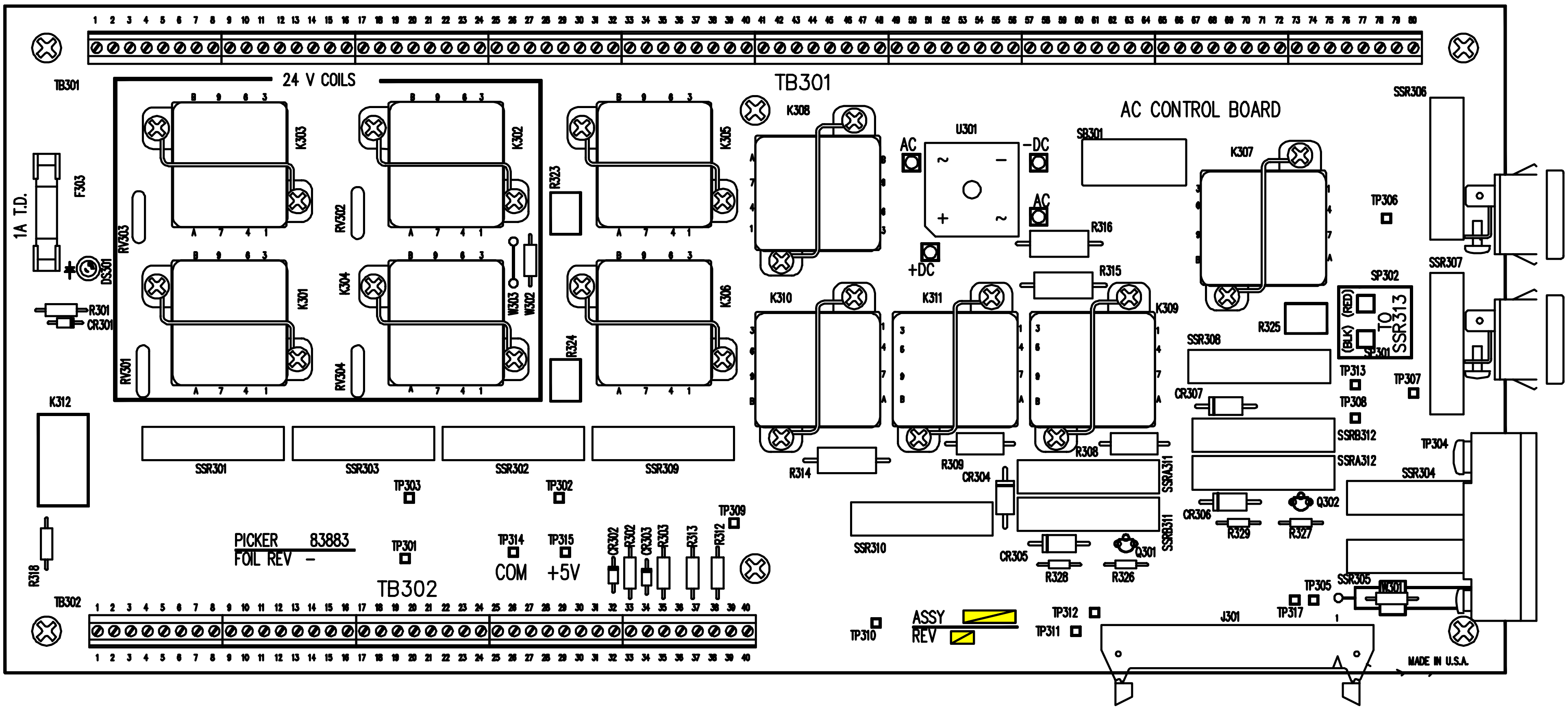


FIGURE 169 AC CONTROL PCB – 178253

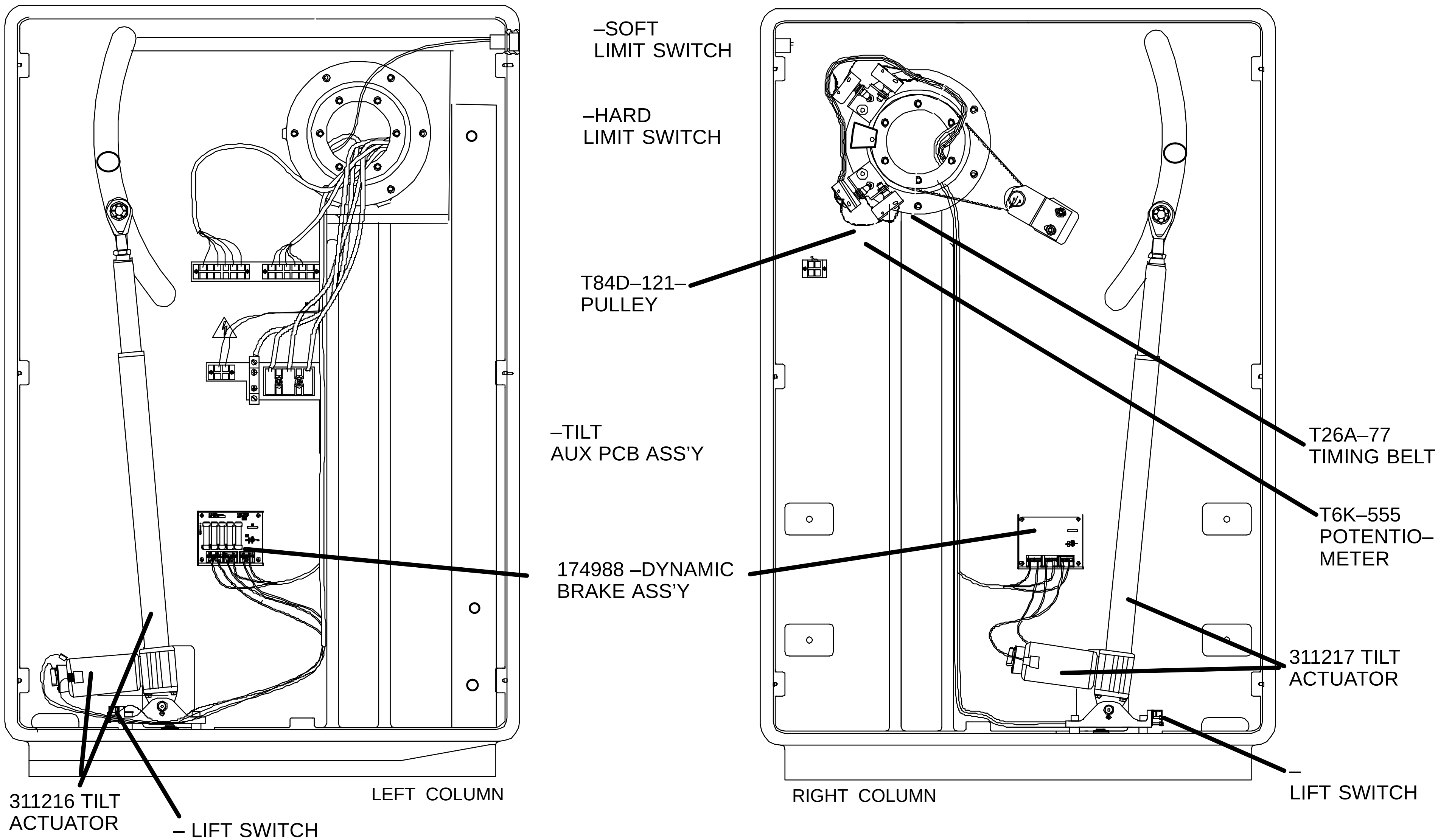


FIGURE 170 TILT ASSEMBLY

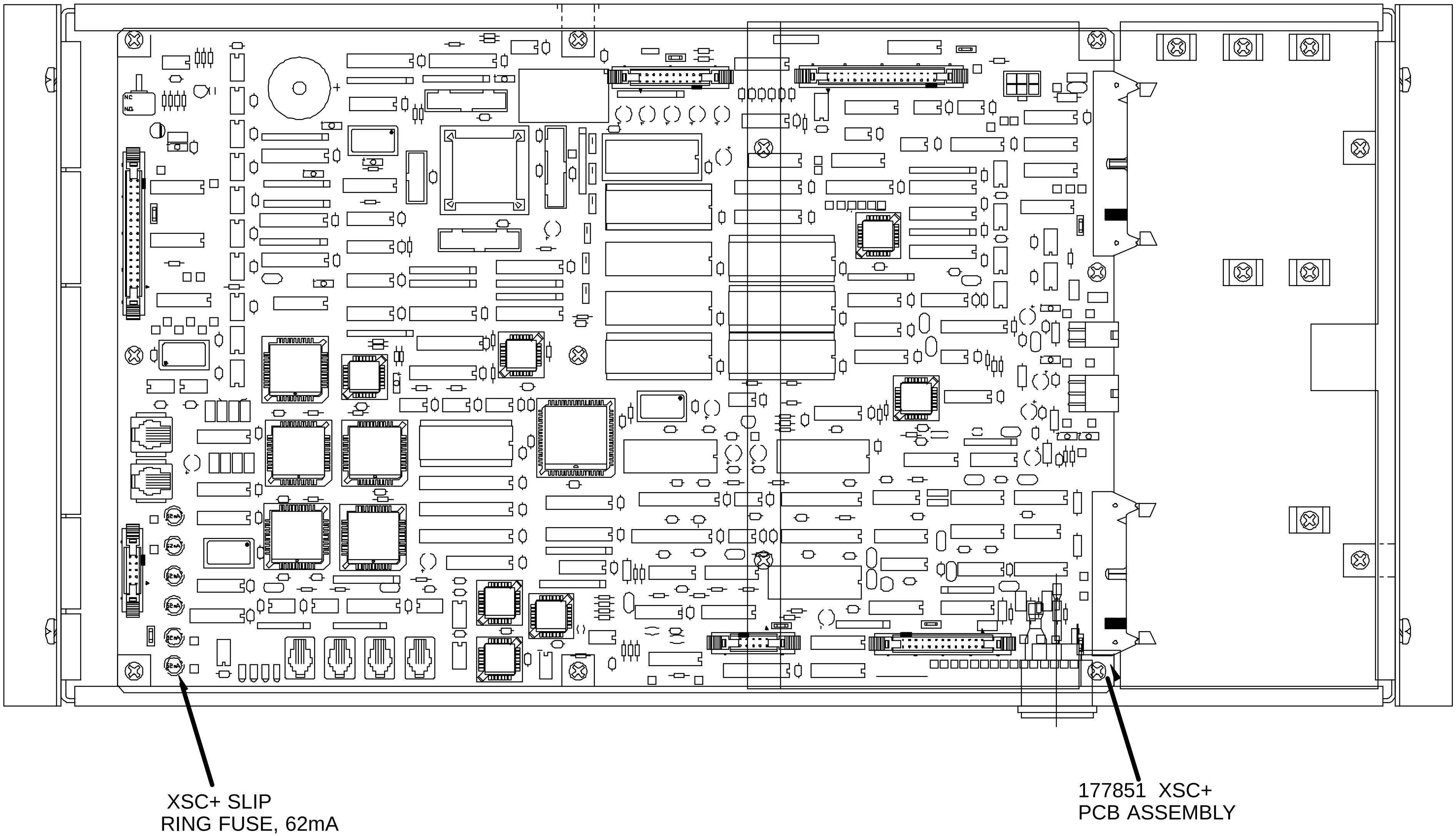


FIGURE 171 XSC PLUS CONTROL MODULE (PART A)

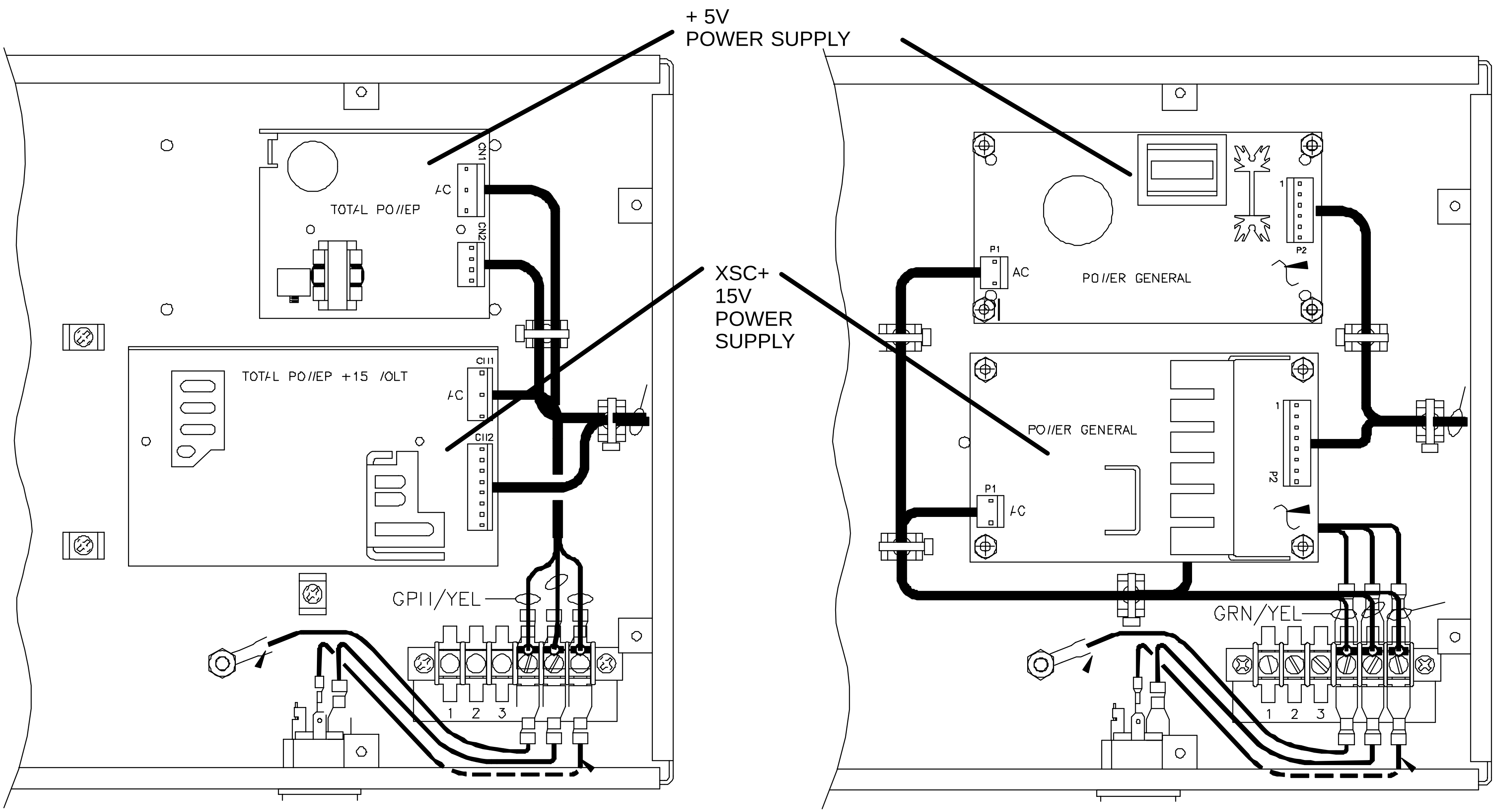


FIGURE 172 XSC PLUS CONTROL MODULE (PART B)

AcQSim CT Operator Console Parts

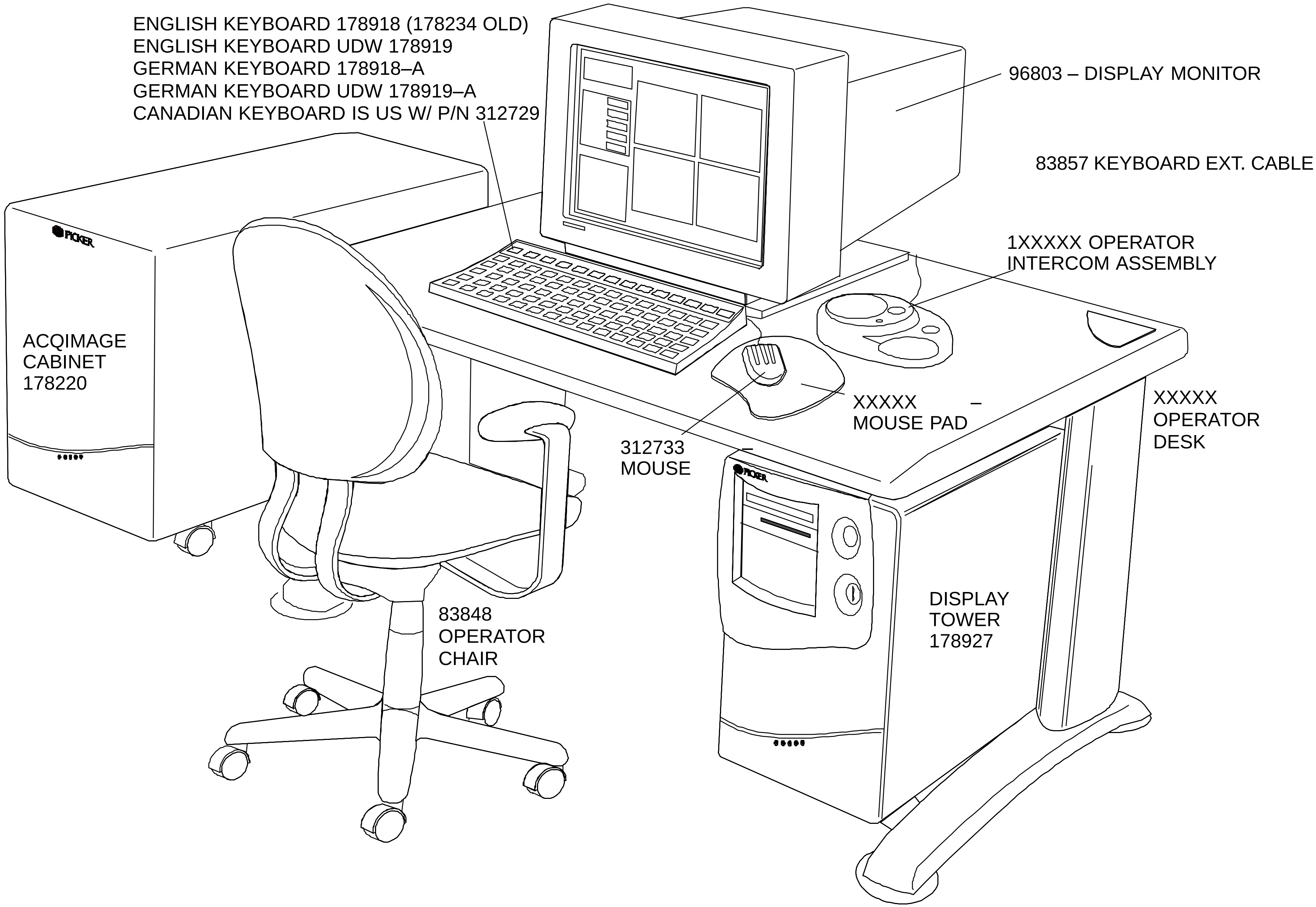


FIGURE 173 SECTION THREE – COMPLETE OPERATOR CONSOLE

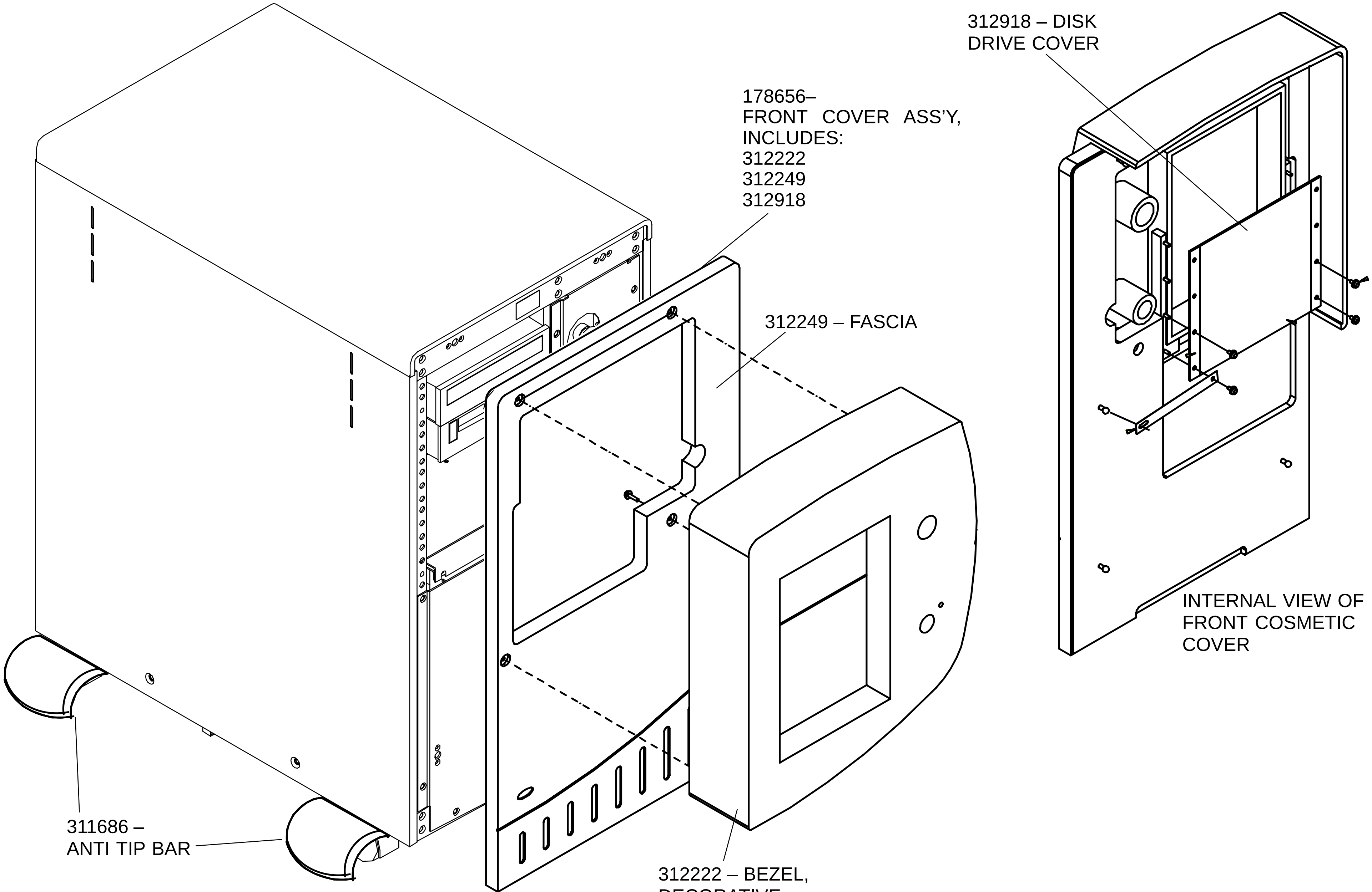


FIGURE 174 DISPLAY TOWER, (178927) – FRONT COSMETIC COVER ASSEMBLY

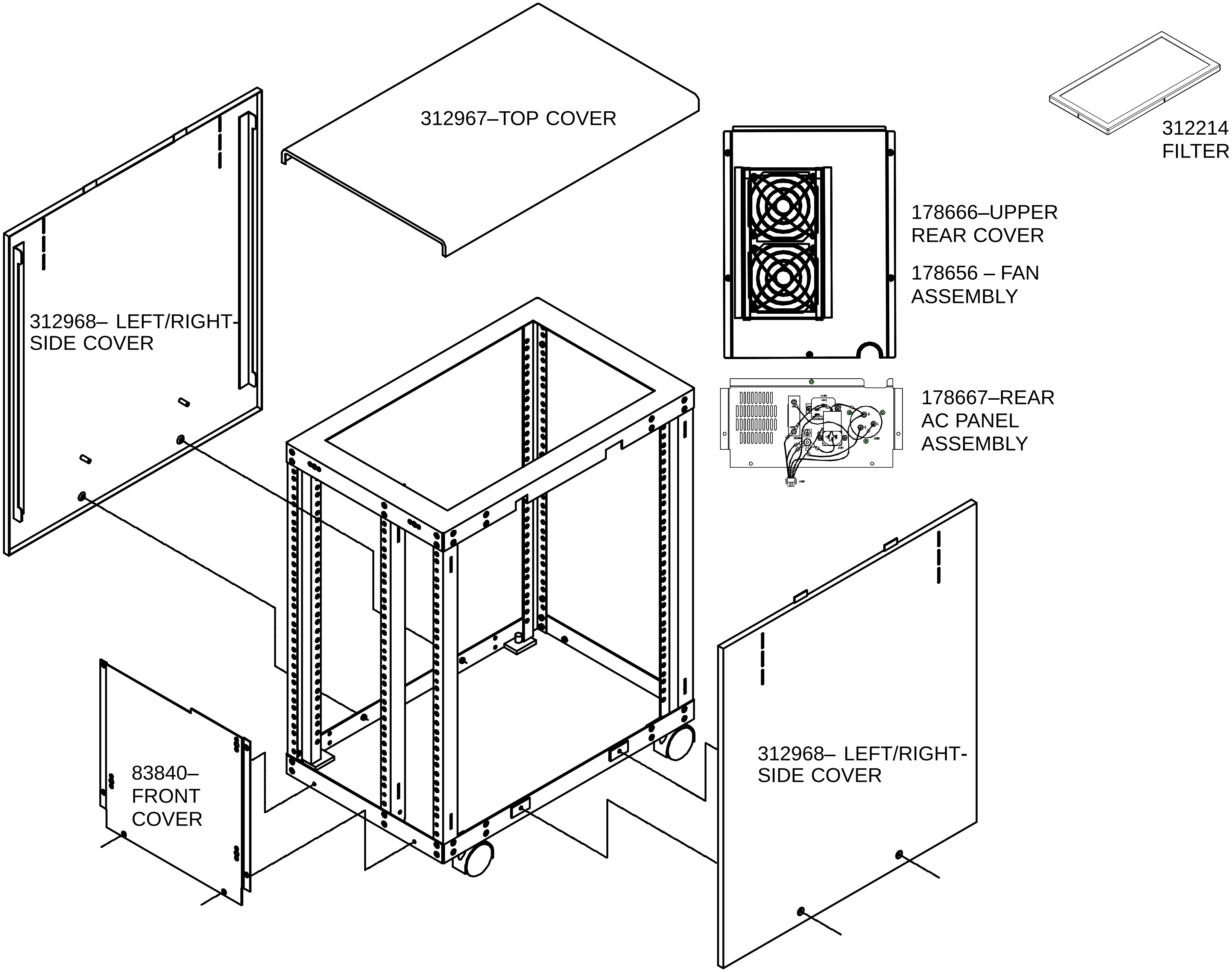


FIGURE 175 DISPLAY TOWER COVERS (no components shown, excluding cosmetic cover)

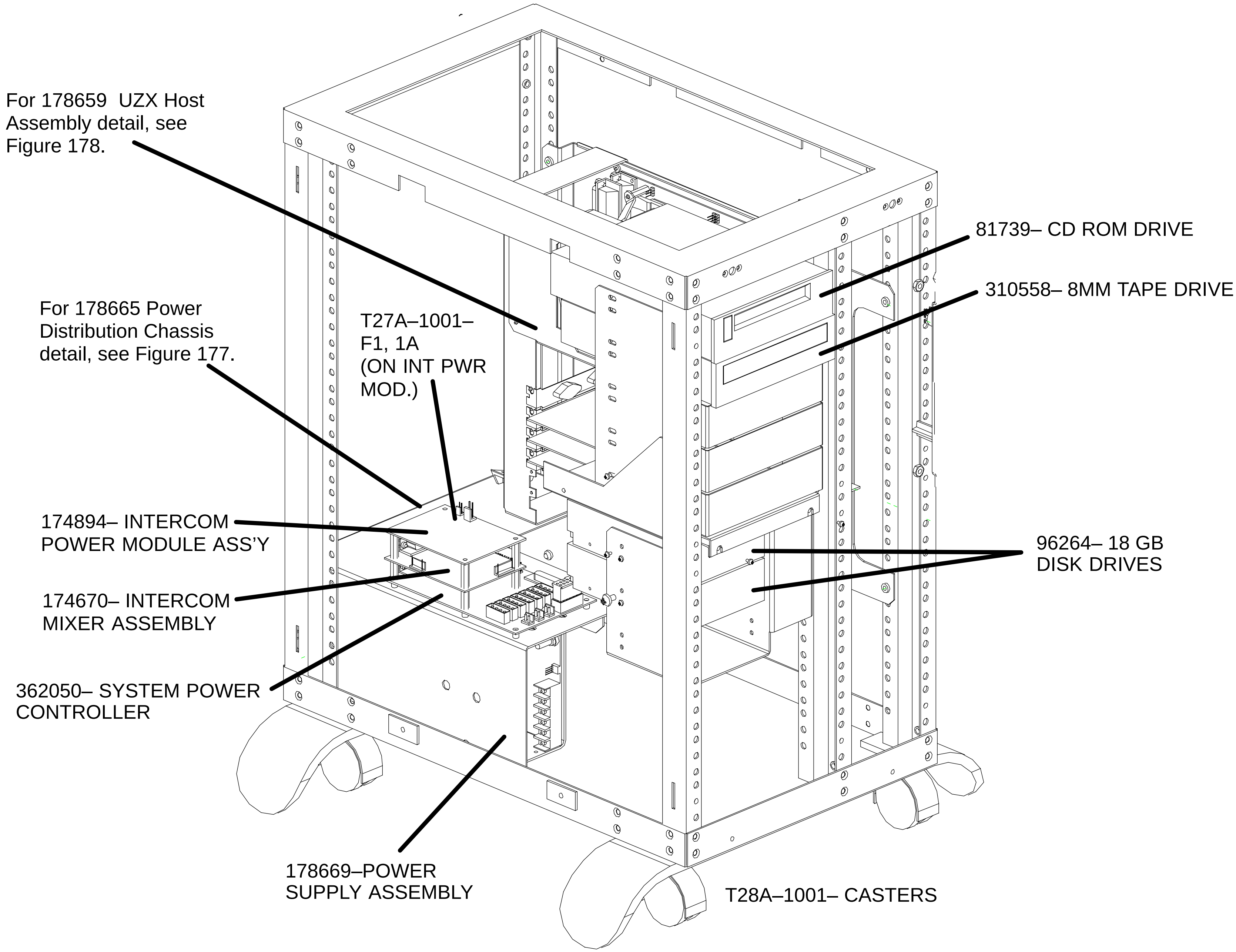


FIGURE 176 DISPLAY TOWER – SIDE, INNER FRONT AND REAR COVERS REMOVED

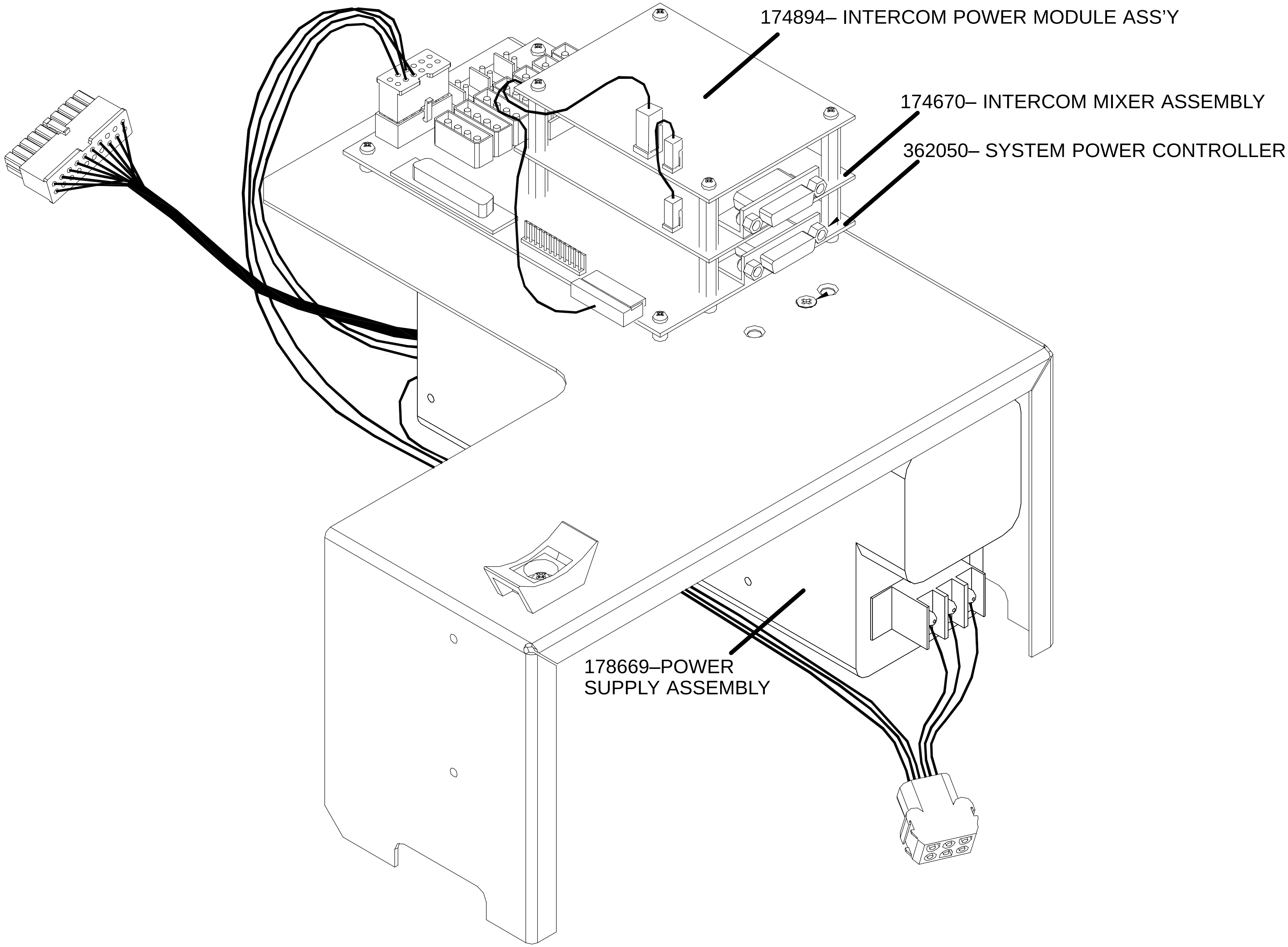


FIGURE 177 POWER DISTRIBUTION CHASSIS, 178665

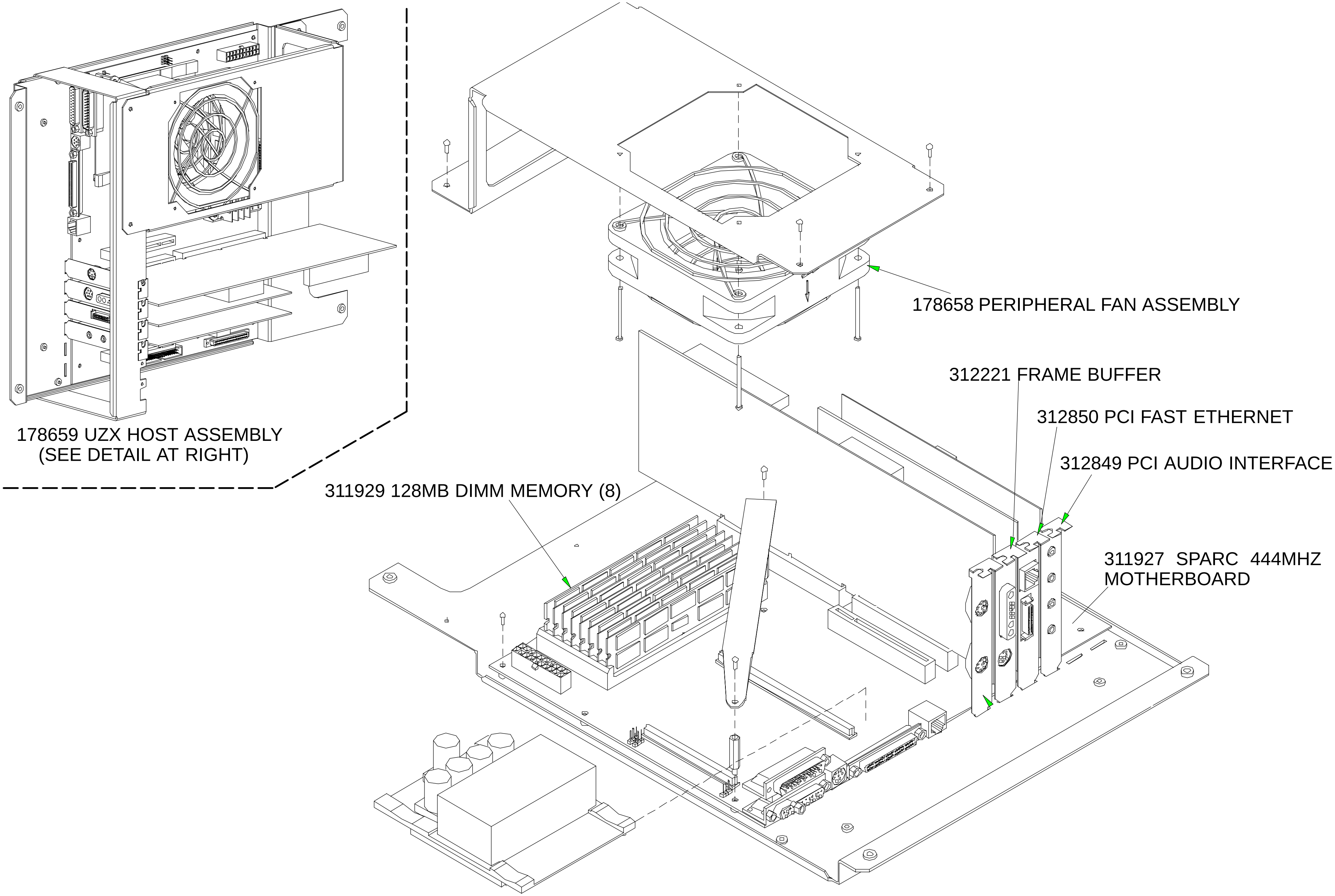


FIGURE 178 DISPLAY TOWER – 178659 ULTRA ZX HOST ASSEMBLY AND DETAIL

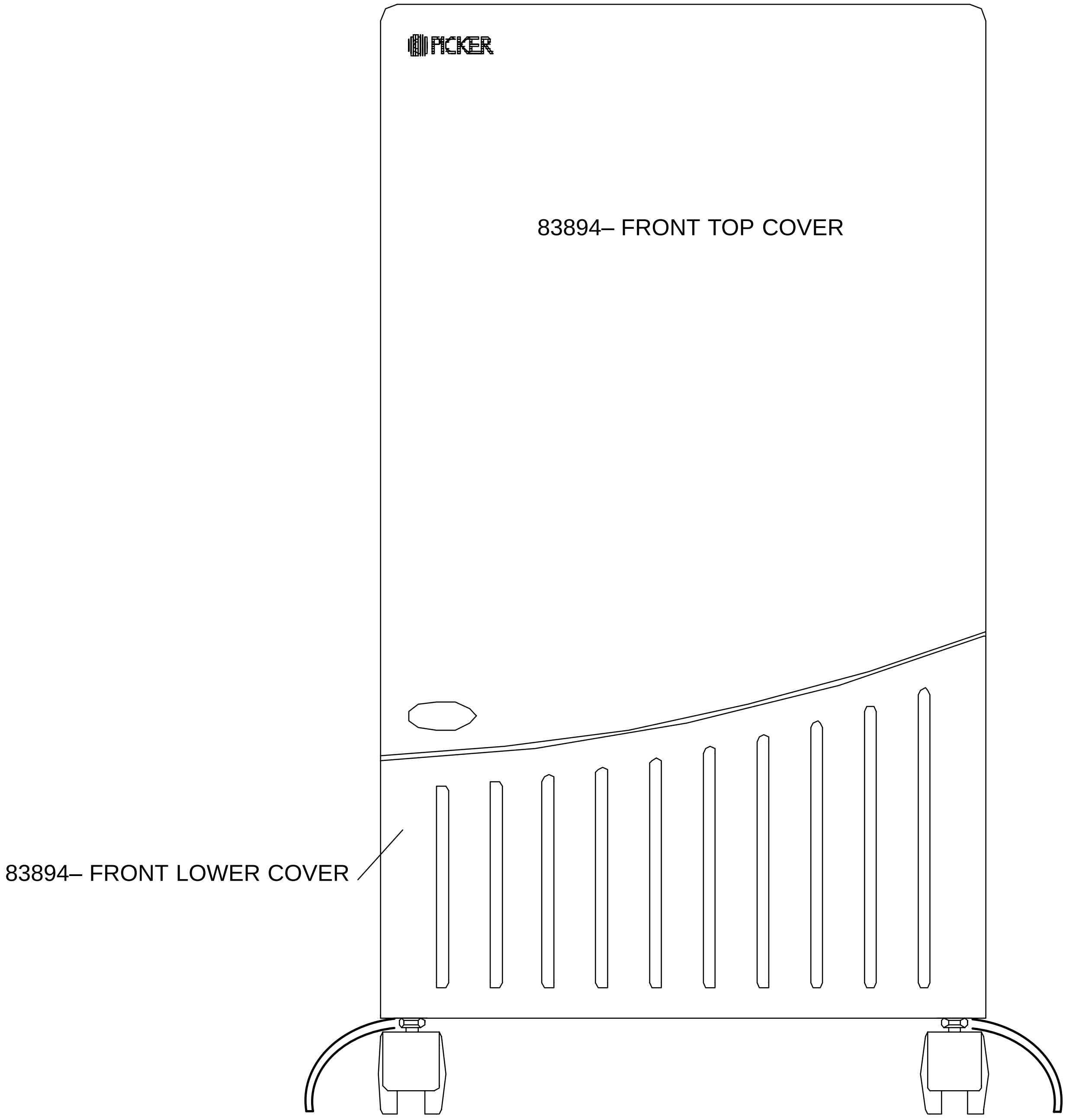


FIGURE 179 ACQIMAGE CABINET – FRONT

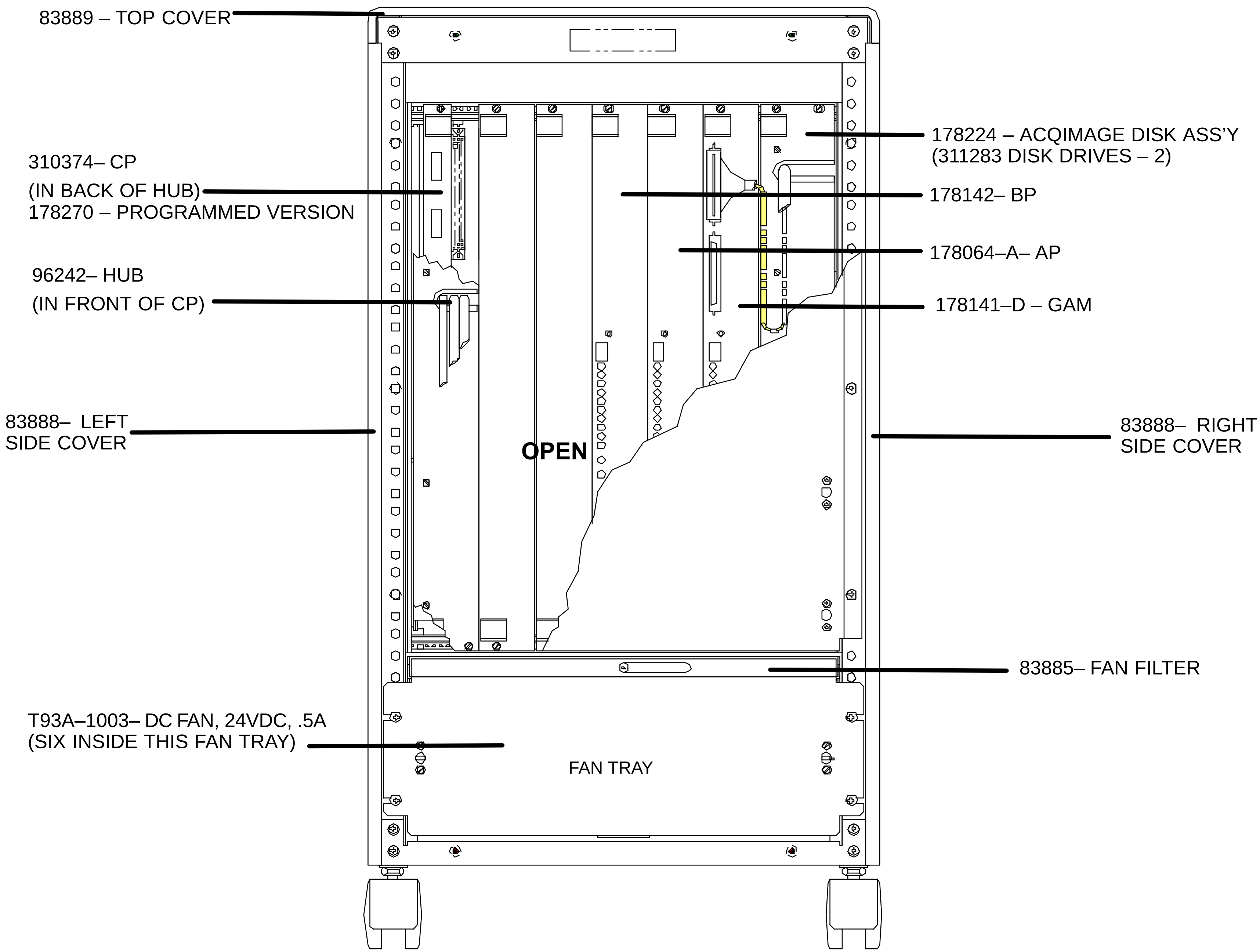


FIGURE 180 ACQIMAGE CABINET – ACQIMAGE BOARDS – (TYPICAL ASSEMBLY)

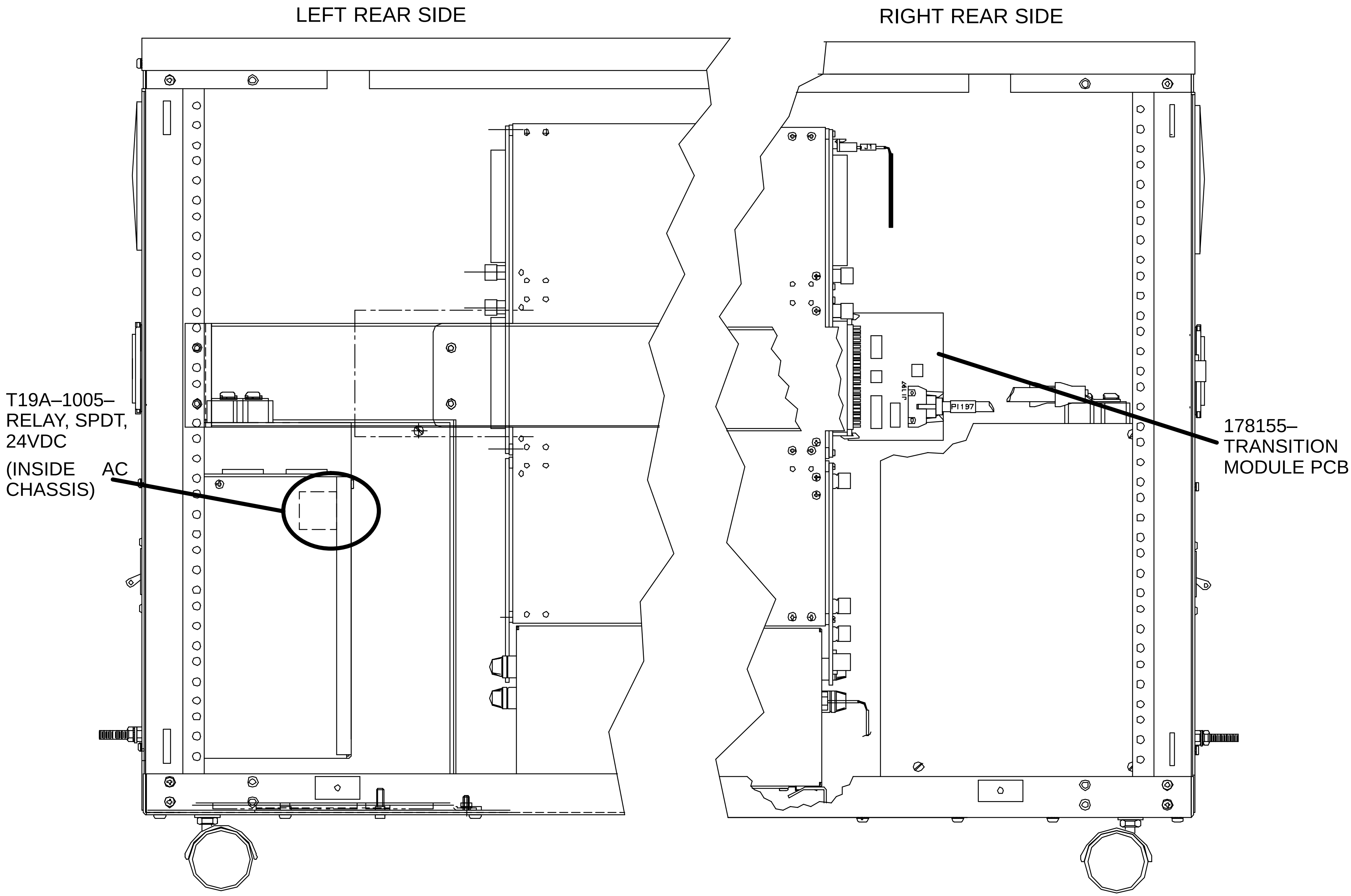


FIGURE 181 RECON TOWER – RIGHT AND LEFT REAR, INSIDE

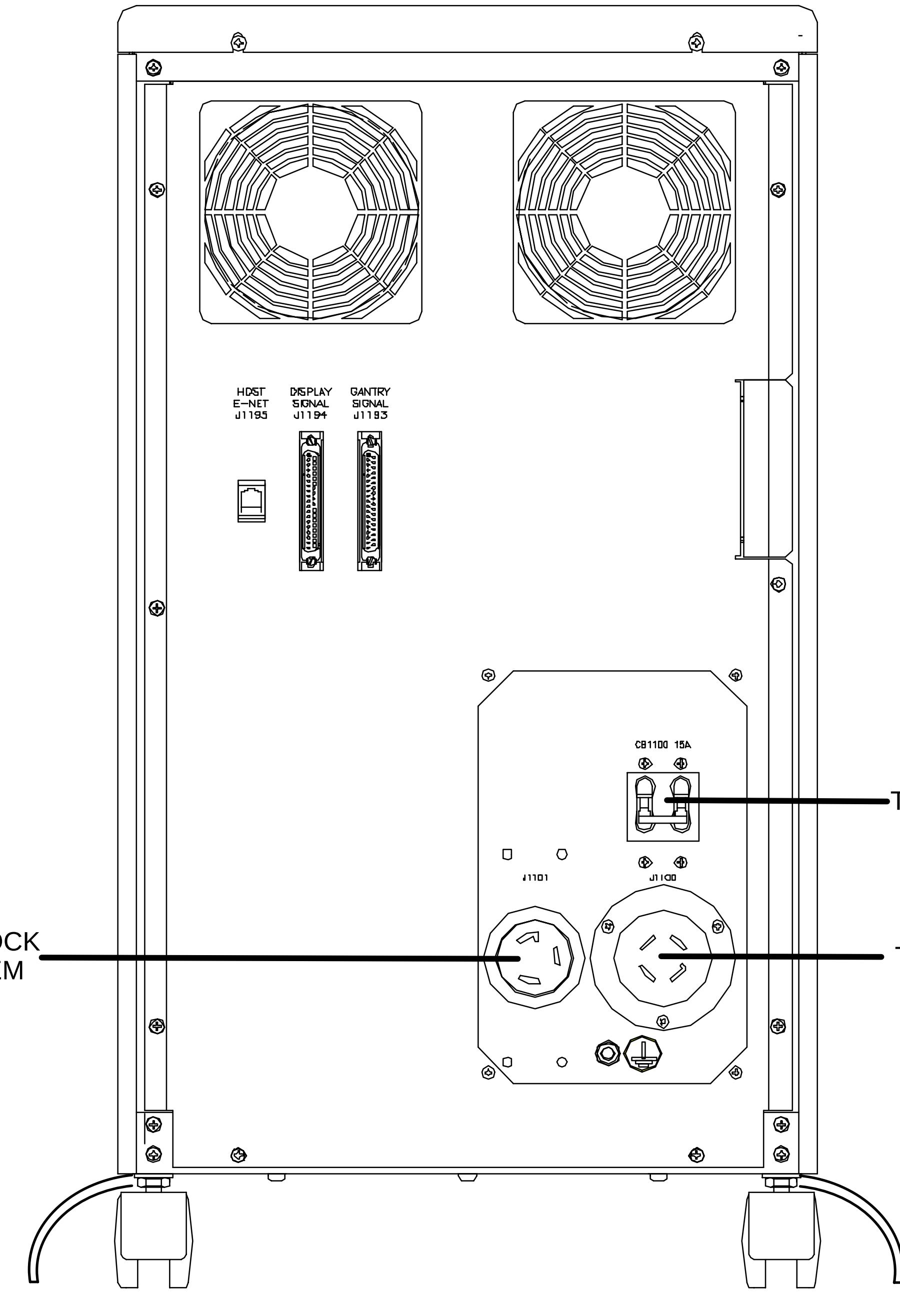


FIGURE 182 RECON TOWER – REAR OUTSIDE

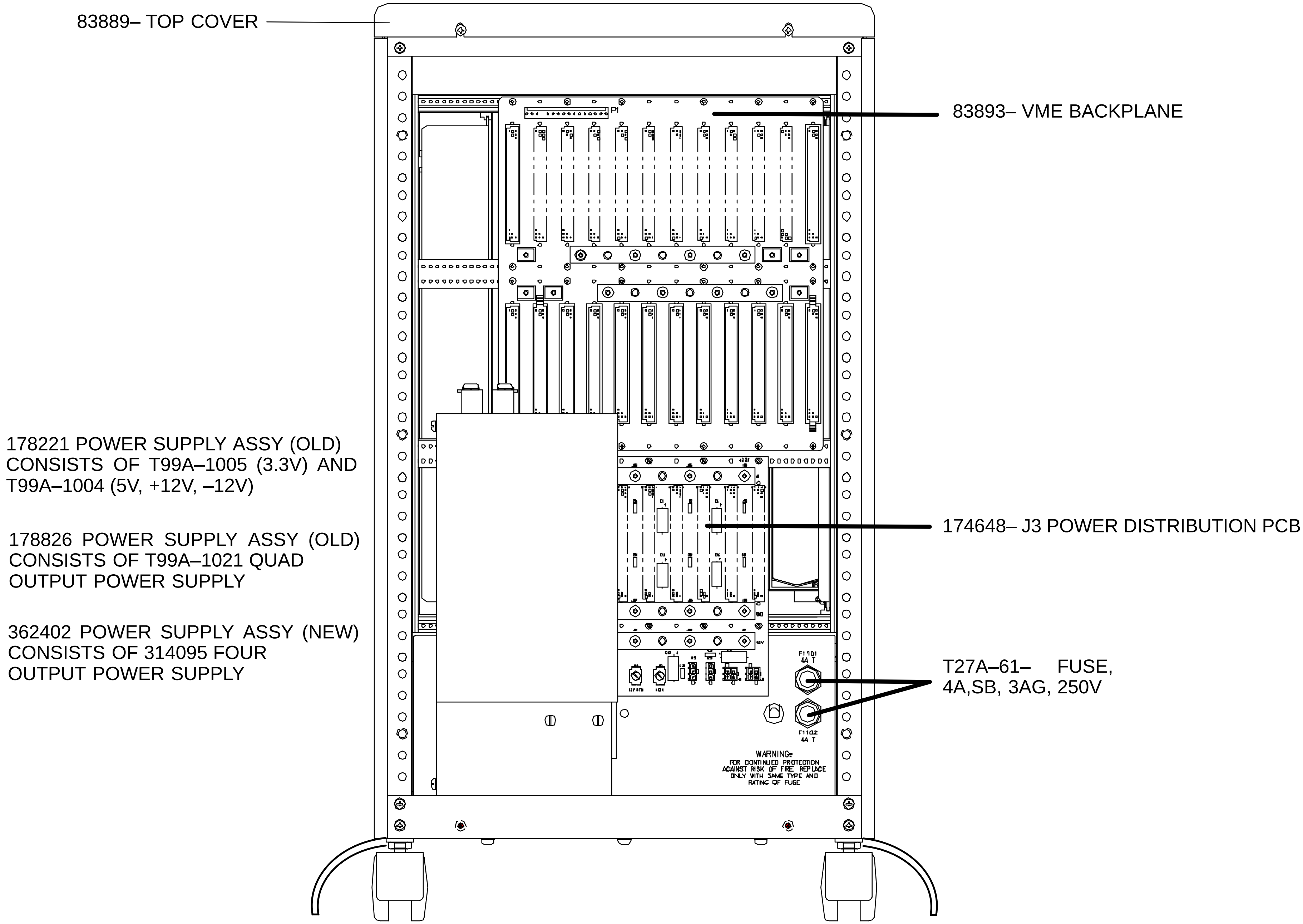


FIGURE 183 RECON TOWER – REAR INSIDE

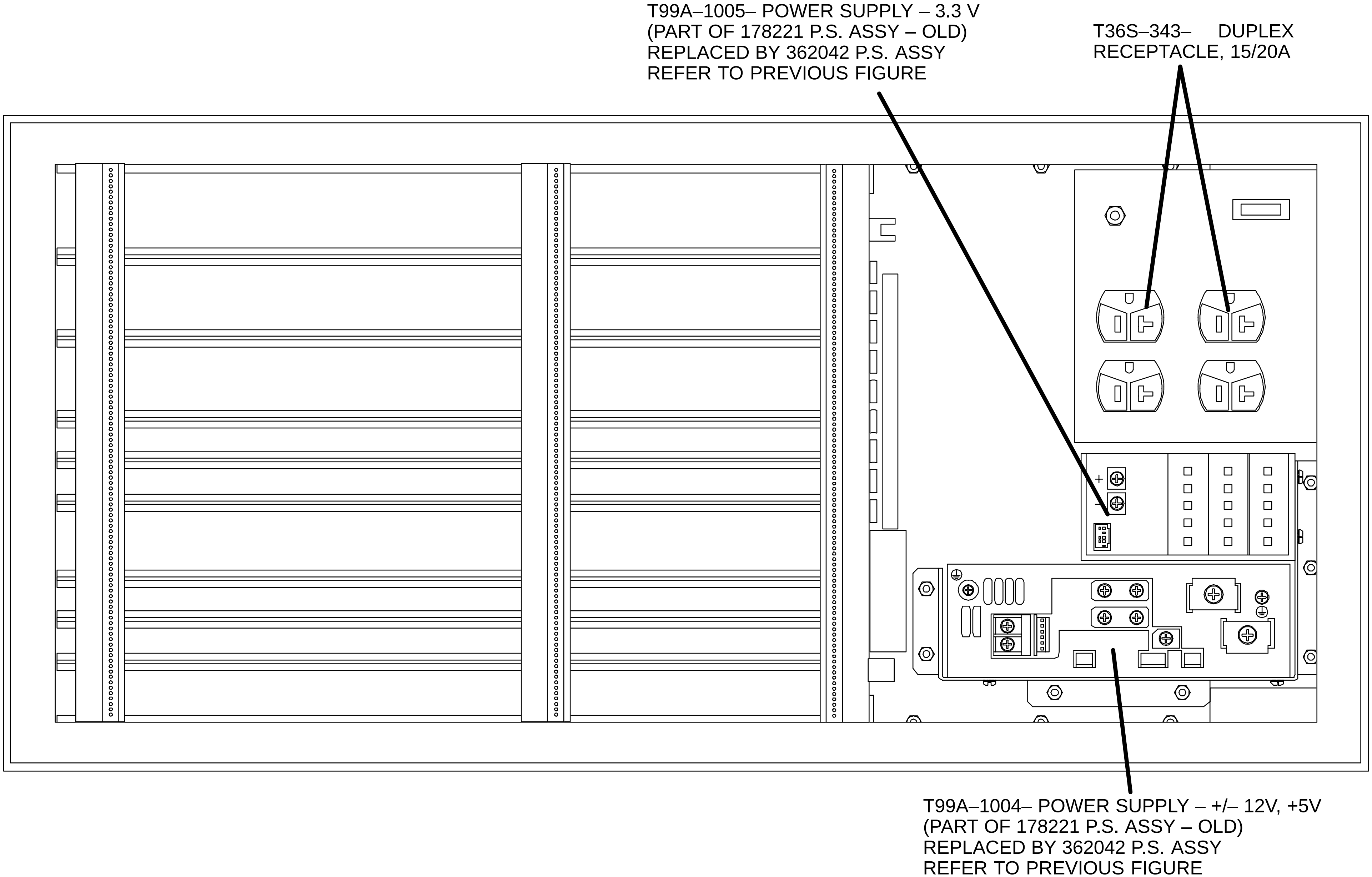


FIGURE 184 RECON TOWER, INSIDE, FROM TOP

CONSOLE ASSEMBLY CABLES

NOTE

Below is a list of external cables and AcQImage internal cables.
Cabling connection details can be found in the Installation manual.

CABLE NUMBER	CABLE LENGTH	CABLE DESCRIPTION	FROM LOCATION AND CONNECTOR	TO LOCATION AND CONNECTOR
L11038	50 FT.	Gantry Signal Cable	Gantry Backplane, J103	AcQImage CabinetJ1193 (Use P1193 end)
L11038–A	25 FT.			
L11036	50 FT.	Gantry Data (Ribbon)	Gantry Backplane, J101	AcQImage Cabinet Backplane, #11 on J2
L11036–A	25 FT.			
L11037	50 FT.	Gantry Power Cable	Gantry Left Column, TB12	AcQImage Cabinet, J1100
L11037–A	25 FT.			
L11039	50 FT.	Ethernet Cable –100 Base T4	AcQImage Cabinet, J1195	Display Cabinet, SUN Motherboard, J1252 (Run inside cabinet)
L11039–A	10 FT.			
L11040	50 FT.	Power Cable	AcQImage Cabinet, J1101	Display Cabinet, J1200
L11040–A	10 FT.			
L11041	50 FT.	Display Signal Cable	AcQImage Cabinet, J1194 (Use P1194 end)	Display Cabinet, J1211
L11041–A	10 FT.			
L7161	12 FT.	Couch Signal Cable	Couch Control Assembly PCB, P402	Gantry Backplane, J108
L7272	16 FT.	Couch Power Cable	Couch Control Assembly PCB, P417	Gantry Left Column, TB13
	FT.	RAMiX to I/O	CP – bottom connector	Inside of J1195 I/O port
L11043	FT.	Internal Ethernet 10MB	GAM, AP, BP front	Ethernet Hub
L11051	FT.	SCSI Data (GAM)	External GAM connector	Disk Drive
312257	—	SCSI Terminator	Physical end of SCSI bus	—

Detector Cable Assembly

Ultra Z	AcQSim CT
MB#8, 5, 2, 11	**MB#8, 4, 12**
J14 V/F_4 (J2)	(J2) V/F_8 (J2)
J13 V/F_4 (J1)	J13 V/F_8 (J1)
J12 V/F_3 (J2)	J12 V/F_7 (J2)
J11 V/F_3 (J1)	J11 V/F_7 (J1)
MB#7, 4, 1, 10	**MB#7, 3, 11**
J14 V/F_2 (J2)	J14 V/F_6 (J2)
J13 V/F_2 (J1)	J13 V/F_6 (J1)
J12 V/F_1 (J2)	J12 V/F_5 (J2)
J11 V/F_1 (J1)	J11 V/F_5 (J1)
MB#6, 3, 12, 9	**MB#6, 2, 10**
J14 V/F_6 (J2)	J14 V/F_4 (J2)
J13 V/F_6 (J1)	J13 V/F_4 (J1)
J12 V/F_5 (J2)	J12 V/F_3 (J2)
J11 V/F_5 (J1)	J11 V/F_3 (J1)
	MB#5, 1, 9
	J14 V/F_2 (J2)
	J13 V/F_2 (J1)
	J12 V/F_1 (J2)
	J11 V/F_1 (J1)

Spare Parts Kit – part number 174983

Part Number	Description	Qty	Part Number	Description	Qty
T27A–212	Fuse, 2.5A	5	T80MB–358	Air Flow Detector	1
T27A–232	Fuse, 35A	3	XT27A–12/01	Fuse, 10A 250V Ceramic SBlpkg	1
T27A–233	Fuse, 4A	5	XT27A–15/01	Fuse, 2A (5/pkg)	1
T27A–234	Fuse, 1A	5	XT27A–18/01	Fuse, 15A (5/pkg)	1
T27A–235	Fuse, 40A 600V DE TD	3	XT27A–192/01	Fuse, 5A (5/pkg)	1
T27A–237	Fuse, KLKD–30A	3	XT27A–26/01	Fuse, 0.5A (5/pkg)	1
T29–325	Switch Front InterIock	2	XT27A–31/01	Fuse, 1A 250V Glass SB (5/pkg)	1
T29A–27	Switch	1	XT27A–42/01	Fuse, 0.25A (5/pkg)	1
T29F–26	Switch Thermo	1	XT27A–57/01	Fuse, 2A 250V Glass SB (5/pkg)	1
T29F–29	Switch Thermo	1	XT27A–60/01	Fuse, 3A 250V Glass SB (5/pkg)	1
T29J–149	Switch Lockwash (5/pkg)	5	XT27A–96/01	Fuse, 10A (10/pkg)	1
T29J–156	Switch Limit 1 Pole	1	171956	Phantom Water Bottle	1
T36S–45	Connector	1	375270	Plano Box #1152	1
T48S–1	Wire 30GA Blue 50ft	50ft	398853	Daisy Chain (25pcs/pkg)	1
T66–200	Cable Clamp #3484–1000	10	77683	DC#4 Compound 5.3oz.	1
T66G–1	Tie–Wraps (Cable Ties 100/pkg)	1	81731	Velcro Tie–wraps	1
T66G–2	Tie–Wraps (Wire Ties 100/pkg)	1	87678	Shim Kit	1
T66G–35	Mounting Plate	25	S301–115	Adhesive, Loctite #609	1
T66G–36	Mounting Plate	25	S301–145	Adhesive, Loctite #242	1
T66G–38	Tie–Wraps (100/pkg)	1	S301–181	Adhesive, Loctite #416	1
T66G–44	Tie–Wrap Clamp	10	S301–196	Adhesive, Loctite #425	1
T66G–48	Tie–Wrap Holder	25	T19A–295	Relay pbkup 14a15 24v	1
T80–391	Bridge Rectifier	1	T19A–325	Relay kup I 4a3 5	2
T80–546	Bridge 2A 200V	1	T19A–352	Solid State Relay	1
T80–555	Bridge	1	T19A–386	Solid State Relay ct6	1

Part Number	Description	Qty	Part Number	Description	Qty
T19A–492	Solid State Relay 25amp	1	T27A–168	Fuse, 3A 500V DE TD	5
T19A–494	Aux Contact	1	T27A–174	Fuse, 3.15A 250V TD	5
T19A–498	Relay Contact 3c	1	T27A–175	Fuse, 1A 250V Quick Act	5
T19A–500	Solid State Relay	1	T27A–176	Fuse, 30A 600V	2
T19A–502	Relay DIP 4	1	T27A–197	Fuse, 8A 250V Glass	3
T19A–503	Relay DIP 4	1			
T19A–535	Relay 115VAC	1			
T19A–540	Relay Contact	1			
T19A–551	Solid State Relay	1			
T2–935	Screw, Self–Locking	5			
T27–66	Fuse, Tubular 0.062	5			
T27–89	Fuse, Micro 0.062A	5			
T27–90	Fuse, Micro 0.25A	5			
T27A–96	Fuse, I OA	5			
T27A–100	Fuse, I OA T.D.	5			
T27A–114	Fuse, Holder 30A	2			
T27A–117	Fuse, 5A 250V Glass SB	5			
T27A–122	Fuse, 5A 250V SB	5			
T27A–143	Fuse, 0.075A 250V Glass SB	5			
T27A–145	Fuse, 6.25A	5			
T27A–148	Fuse, 15A 500V TD	5			
T27A–149	Fuse, 5A 500V TD	1			
T27A–157	Fuse, 25A 500V DE TD	5			
T27A–159	Fuse, 6A 500V non delay	5			
T27A–161	Fuse, 7A 250V Clip Type	5			
T27A–166	Fuse, 20A 60OVDC non tim	4			

REPAIR / REPLACEMENT

AcQSIM CT – Patient Support Components Repair

PATIENT SUPPORT COVERS

Introduction: This procedure is used to remove and replace all Patient Support covers.

> **NOTE**
>
> Two service persons are recommended to perform these procedures.

Tools Required: 1. FSE Tool Kit

Estimated Time: 45 min.

TELESCOPING BASE COVER REMOVAL (FULL DOWN POSITION; PREFERRED METHOD):

> ⚠ **WARNING**
>
> **THE TELESCOPING BASE COVERS ARE HEAVY AND ARE HELD ONTO THE SUBFRAME BY FOUR SCREWS. REMOVE THE MOUNTING SCREWS ONLY WHEN THE PATIENT SUPPORT IS AT ITS LOWEST POSITION OR ONLY AFTER THE TELESCOPING BASE COVERS ARE PROPERLY AND SECURELY BLOCKED. FAILURE TO COMPLY CAN RESULT IN DAMAGE TO EQUIPMENT AND/OR SERIOUS INJURY TO PERSONNEL.**

> **NOTE**
>
> For most Removal/Replacement and Calibration procedures, only the top section of the Telescoping Base Covers need be removed.

1. Apply Gantry power. Refer to the PowerUp/PowerDown procedure.
2. Lower the Patient Support to its lowest position with the Gantry Control Panel Switches.

> ⚠ **WARNING**
>
> **USE EXTREME CAUTION WHEN OPERATING OR WORKING ON EQUIPMENT WITH COVERS REMOVED. DANGEROUS VOLTAGES ARE EXPOSED WHEN THE GANTRY COVERS ARE REMOVED. FAILURE TO COMPLY CAN CAUSE SERIOUS INJURY OR DEATH TO PERSONNEL.**

3. Remove the four screws and Washers that attach the Telescoping Base Cover Panels to the Subframe. Refer to Figure 185.

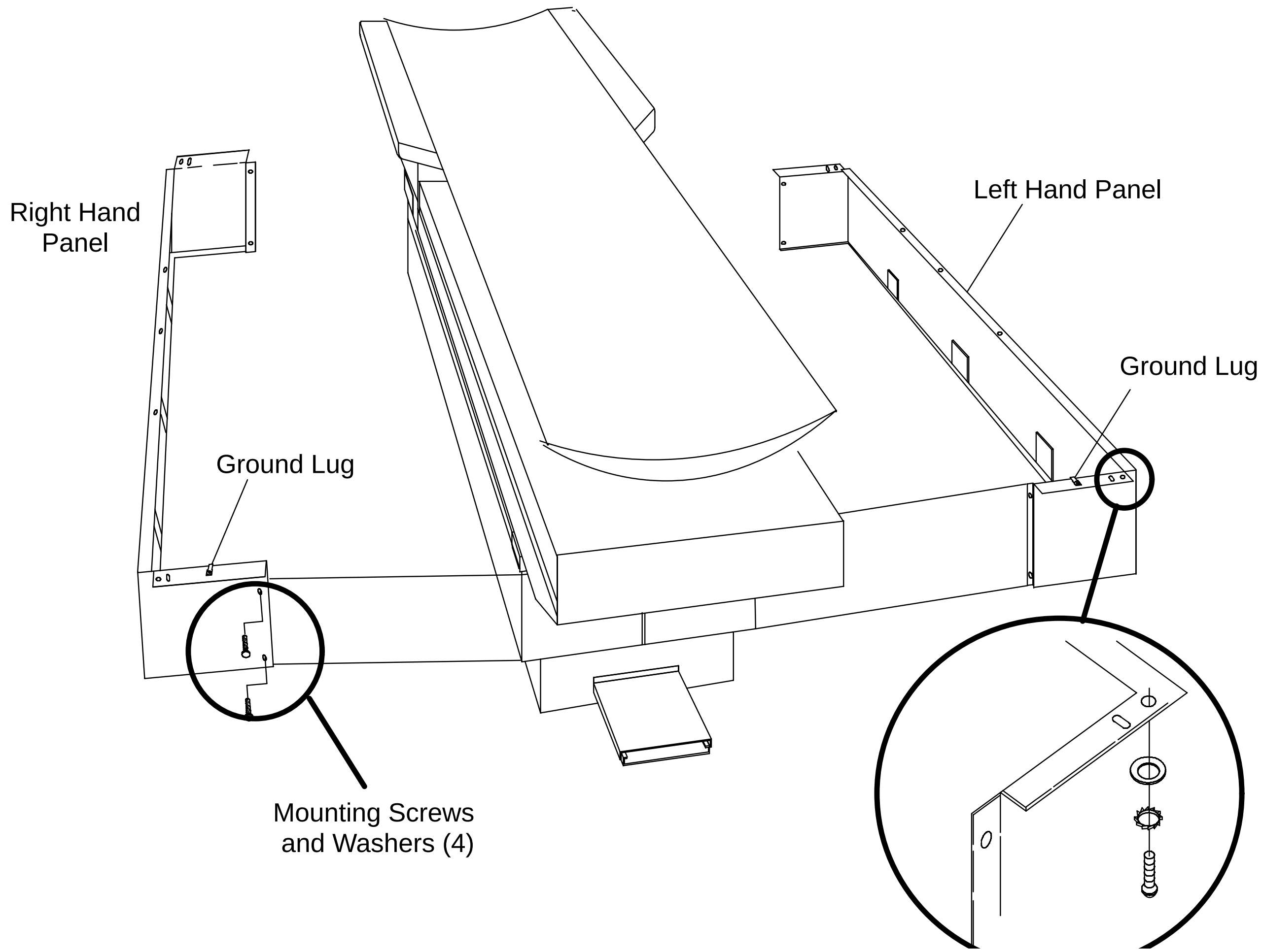


FIGURE 185 PATIENT SUPPORT TELESCOPING BASE COVERS

4. Using the Gantry Control Panel Switches, raise the Patient Support approximately 10 inches to allow access to internal cables.

5. Remove Gantry power. Refer to the PowerUp/PowerDown procedure.

6. Remove the two screws from both Outer Cover ends (front & rear, 4 total). Refer Figure 185.

7. Disconnect the two Ground Wires connected to the inside Panel surfaces. Refer to Figure 186. Tag the Panels left and right. Carefully separate the Panels and slide them out to the sides. Store the Covers in a secure area until reinstallation is required.

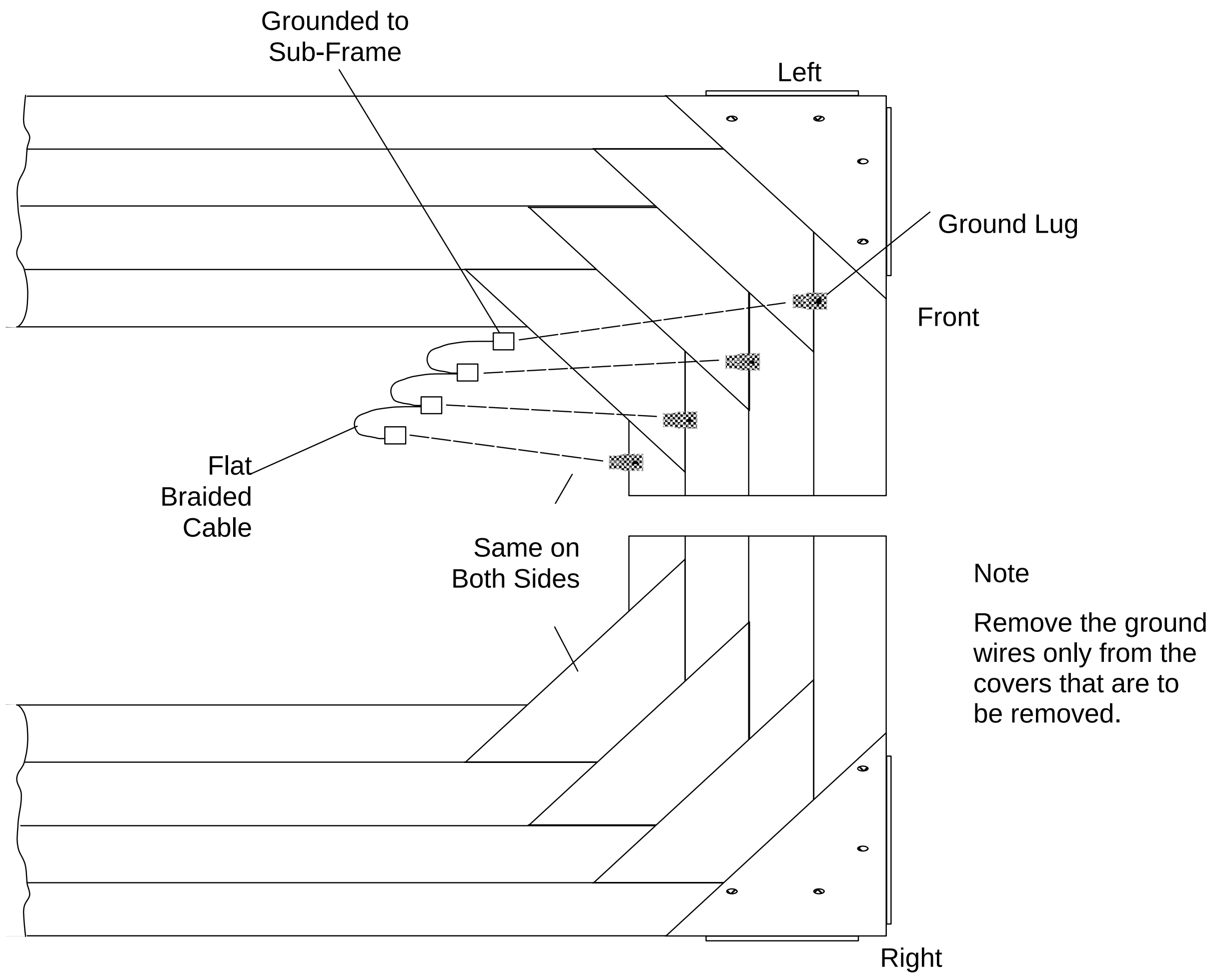


FIGURE 186 TELESCOPING BASE COVERS – GROUND CONNECTIONS

8. Repeat steps 6. and 7. to remove the remaining cover sections.

TELESCOPING BASE COVER REMOVAL (AT ANY RAISED POSITION OTHER THAN FULL DOWN):

9. Remove Gantry power. Refer to the PowerUp/PowerDown procedure.
10. Block the Telescoping Panels under the second Panel and the third Panel from the top. Place Blocks at the front and rear of each side Figure 187 .

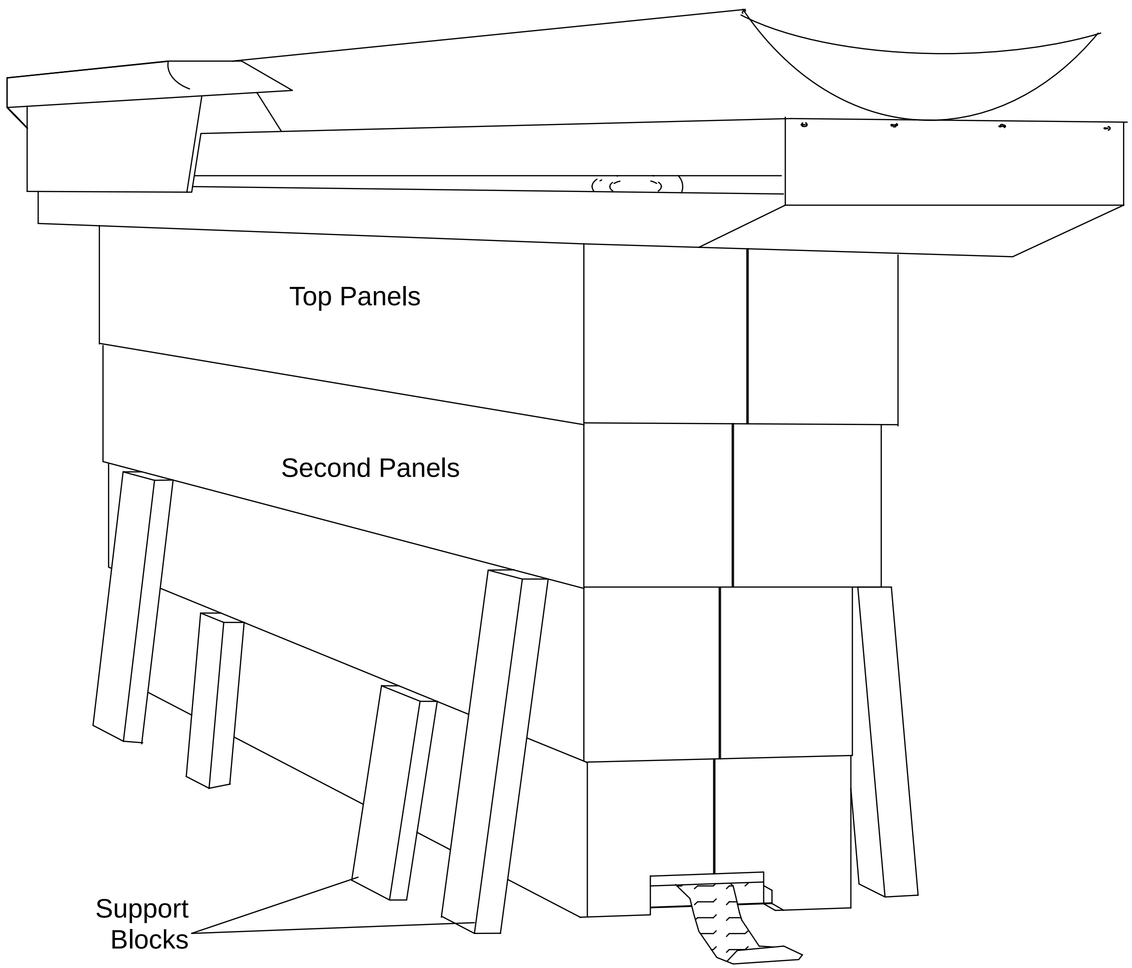


FIGURE 187 TELESCOPING PANELS BLOCKED BEFORE REMOVAL

11. Remove the four Screws and Washers that attach the Telescoping Base Cover Panels to the Subframe. Refer Figure 185.
12. Remove the two Screws from both ends of the Top Panels (front & rear, 4 total). Figure 185.

13. Carefully separate the Panels and remove Ground Wires connected to the inside Panel surfaces. Tag the Panels left and right and store them in a safe place.
14. Repeat steps 12. and 13. for the second Panel. Carefully slide the Panels out to the sides removing the Blocks for that layer in the process. DO NOT remove the blocks supporting the next lower layer.
15. Repeat steps 12. through 15. for the remaining Panels to be removed.

> ⚠ **CAUTION**
>
> GROUND STRAP CABLES MAY BE SECURED TO THE INSIDE OF THE PATIENT SUPPORT COVERS. DURING COVER ASSEMBLY, RE–SECURE ALL CABLES AND WIRES EXACTLY AS THEY WERE. FAILURE TO COMPLY MAY RESULT IN SEVERE DAMAGE TO THE EQUIPMENT.

TELESCOPING BASE COVER REPLACEMENT:

16. Install the Patient Support Telescoping Base Covers by reversing the steps of Telescoping Base Cover Removal (at the full down or raised position).

FRONT COVER REMOVAL:

17. Remove the two Phillips Head Screws from the underside of the front Cover. See Figure 188.
18. Remove the four Phillips Head Screws on the front side of the front Cover while holding the Cover.

FRONT COVER REPLACEMENT:

19. Install the Patient Support front Cover by reversing steps 17. and 18.

REAR COVER REMOVAL:

20. Remove the two Phillips Head Screws from underneath the Sub–frame that secure the rear Cover. See Figure 188.
21. Remove the rear Cover by pulling down and then out.

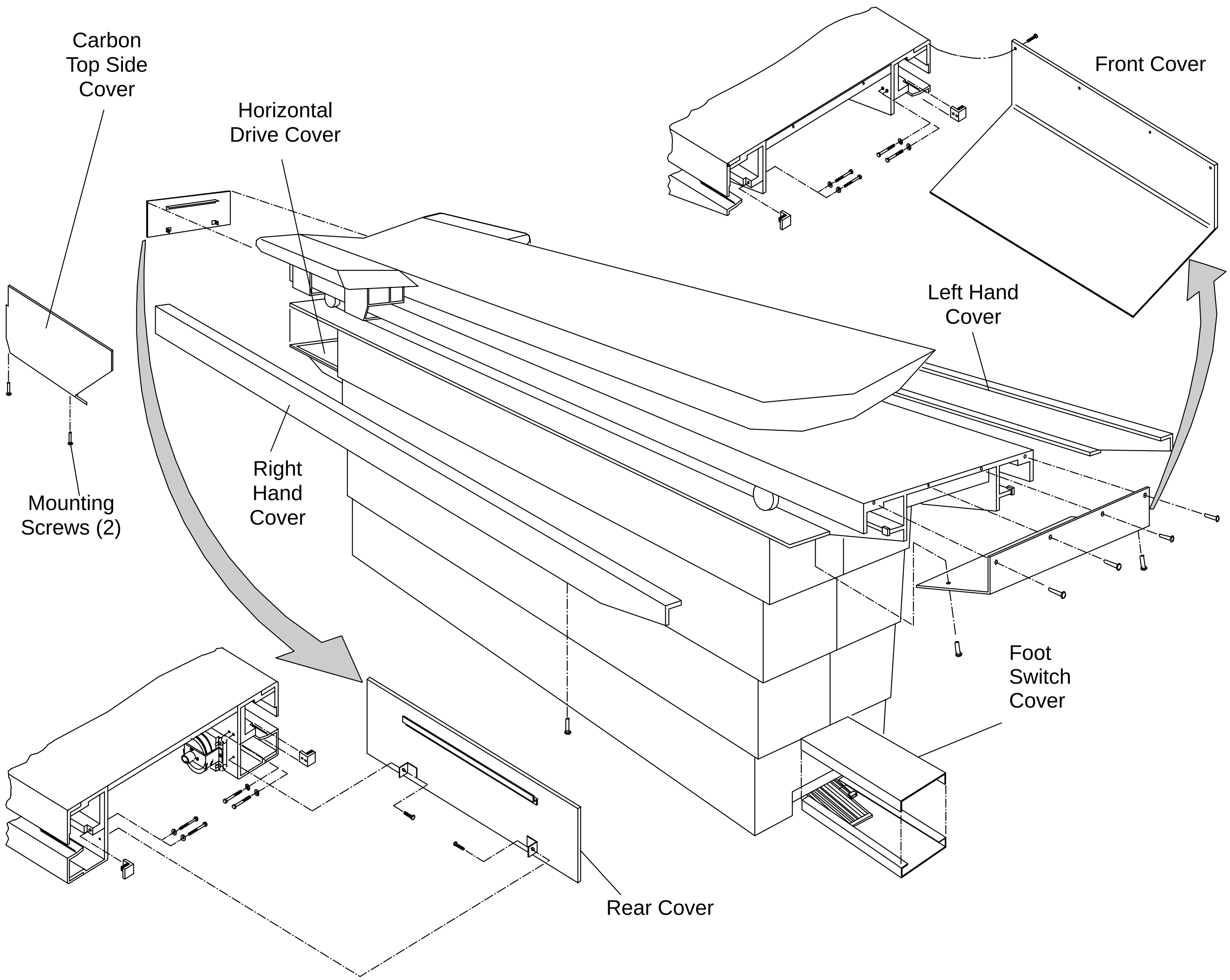


FIGURE 188 PATIENT SUPPORT COVERS

REAR COVER REPLACEMENT:

22. Install the Patient Support rear Cover by reversing steps 20. and 21.

HORIZONTAL DRIVE COVER REMOVAL:

23. Remove the four Pan Head Screws, Lock Washers, and Flat Washers from underneath the Table at the rear end, to gain access to the Horizontal Drive. Refer to Figure 188.

HORIZONTAL DRIVE REPLACEMENT:

24. Install the Horizontal Drive Cover by reversing step 23.

CARBON TOP SIDE COVERS (LEFT AND RIGHT) REMOVAL:

25. Position the Carbon Top by hand fully into the Gantry. This is required for some older Side Cover designs.
26. Remove the two Mounting Screws that attach the Carbon Top Side Cover to the J–Bracket. Refer to Figure 188.
27. Remove the Carbon Top Side Cover.

CARBON TOP SIDE COVERS REPLACEMENT:

28. Replace the Carbon Top Side Cover by reversing steps 25. through 27. Refer to the Horizontal Drive Belt repair procedure.

RAIL COVERS (LEFT AND RIGHT) REMOVAL:

29. Remove the rear–most front cover Screw that secures the front Cover and Side Rail to the Sub–frame. Refer to Figure 188.
30. Move the Carbon top by hand until it is against its front Mechanical Stop.
31. Remove the Patient Support rear Cover. Refer to Rear Cover Removal in this procedure.
32. Remove the Flat Head Screws from the underside of the Rail Cover. If both Rail Covers are removed at the same time, tag them left and right. See Figure 188.
33. Slide the Rail Cover toward the rear until it is clear of the overhanging Carbon top Side Cover.
34. Remove the Rail Cover.

CARBON TOP SIDE REPLACEMENT:

35. Install the Patient Support Rail Covers by reversing steps 29. through 34.

FOOT SWITCH COVER REMOVAL:

36. Apply Gantry power. Refer to the PowerUp/PowerDown procedure.
37. Tilt the Gantry to +5° (degrees) with the Gantry Control Panel Switches.
38. Remove Gantry power. Refer to the PowerUp/PowerDown procedure.
39. Remove the two screws and Washers that attach the Foot Switch Cover. Firmly grasp the Foot Switch with both hands and pull the Cover straight up. See Figure 189.

> ⚠ **WARNING**
>
> **USE EXTREME CAUTION WHEN OPERATING OR WORKING ON EQUIPMENT WITH COVERS REMOVED. DANGEROUS VOLTAGES ARE EXPOSED WHEN THE GANTRY COVERS ARE REMOVED. FAILURE TO COMPLY CAUSE SERIOUS INJURY OR DEATH TO PERSONNEL.**

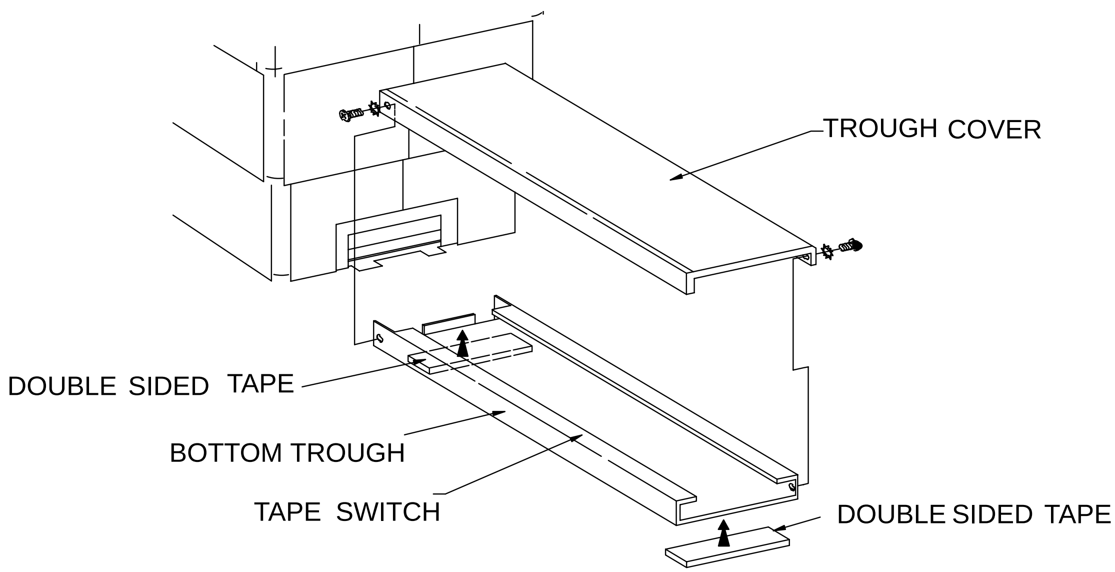


FIGURE 189 FOOT SWITCH COVER

FOOT SWITCH COVER REPLACEMENT:

40. Install the Foot Switch Cover by reversing step 39.
41. Apply Gantry power. Refer to the PowerUp/PowerDown procedure.
42. Activate the Footswitch and manually move the Table to verify Footswitch operation.
43. Using the Gantry Control Panel Switches, tilt the Gantry to zero degrees.
44. Remove Gantry power. Refer to the PowerUp/PowerDown procedure.

CARBON TOP

Introduction: This procedure is used to remove and replace the Patient Support Carbon Top.

Tools Required: 1. FSE tool kit

Estimated Time: 30 min.

CARBON TOP REMOVAL PROCEDURE:

1. Prepare a padded area with soft material to rest the Carbon Top after it is removed.
2. Apply Gantry power. Refer to the PowerUp/PowerDown procedure.

> ⚠ **WARNING**
>
> **USE EXTREME CAUTION WHEN OPERATING OR WORKING ON EQUIPMENT WITH COVERS REMOVED. DANGEROUS VOLTAGES ARE EXPOSED WHEN THE GANTRY COVERS ARE REMOVED. FAILURE TO COMPLY CAN CAUSE SERIOUS INJURY OR DEATH TO PERSONNEL.**

3. Raise the Patient Support to its maximum height with the Gantry Control Panel Switches.
4. Remove Gantry power. Refer to the PowerUp/PowerDown procedure.
5. Manually position the Carbon Top to the front Stop.
6. Tag and remove the left and right Carbon Top Side Covers. Refer to the Patient Support Covers Removal/Replacement procedure and Figure 190.

> ⚠ **WARNING**
>
> **SCISSOR–ACTION MECHANISM IS EXPOSED WHEN THE PATIENT SUPPORT TELESCOPING BASE COVERS ARE REMOVED. KEEP EQUIPMENT AND PERSONNEL AWAY FROM THE MECHANICAL ACTION OF THESE PARTS. FAILURE TO COMPLY CAN CAUSE SERIOUS DAMAGE TO EQUIPMENT AND/OR SERIOUS INJURY OR DEATH TO PERSONNEL.**

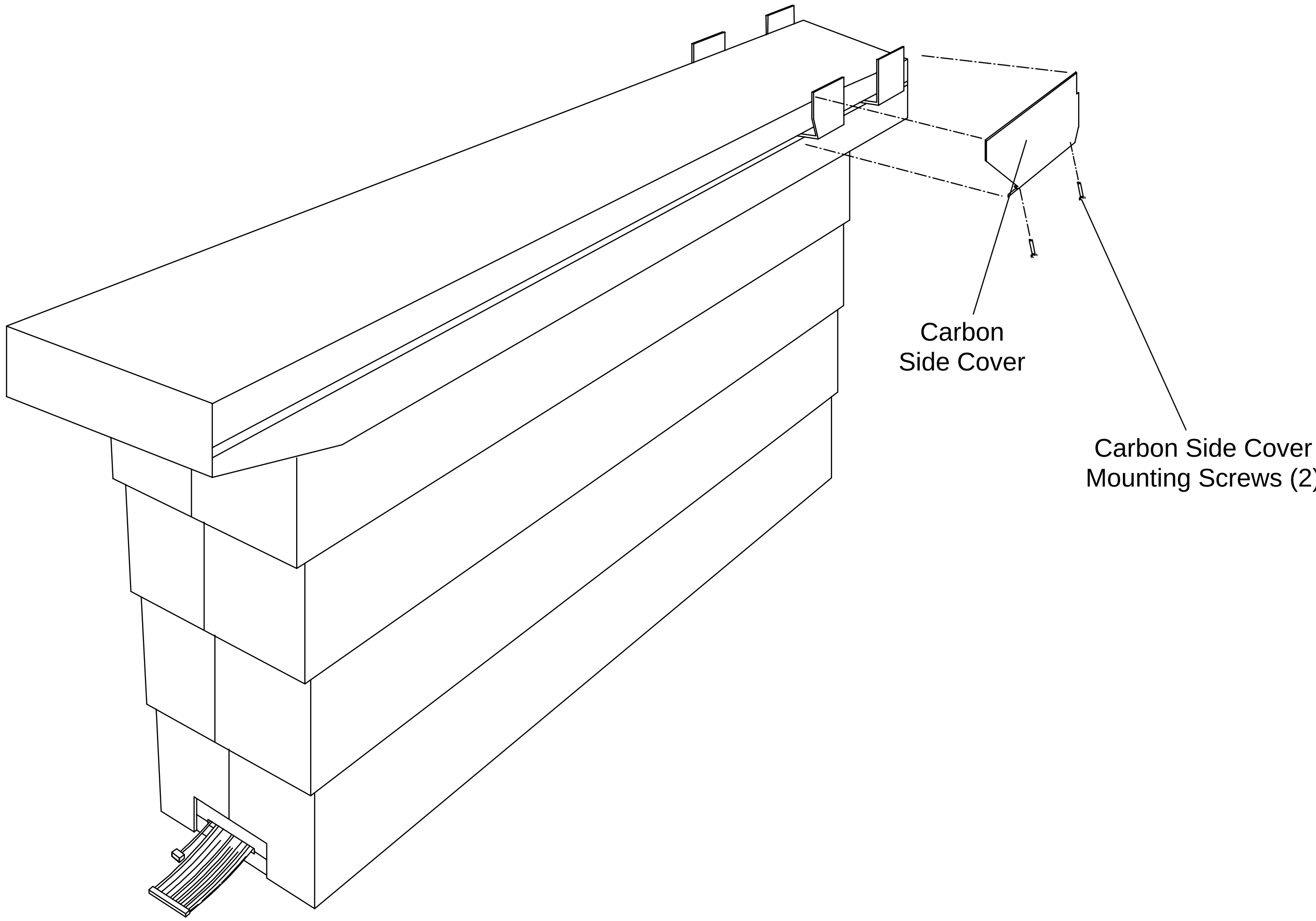


FIGURE 190 CARBON TOP SIDE COVERS

7. Disconnect the Tape Switch from the Ribbon Cable Assembly on the left side of the Carbon Top. See Figure 191.
8. Cut the three Tie Wraps that hold the Tape Switch Wires to the side of the J–Block.

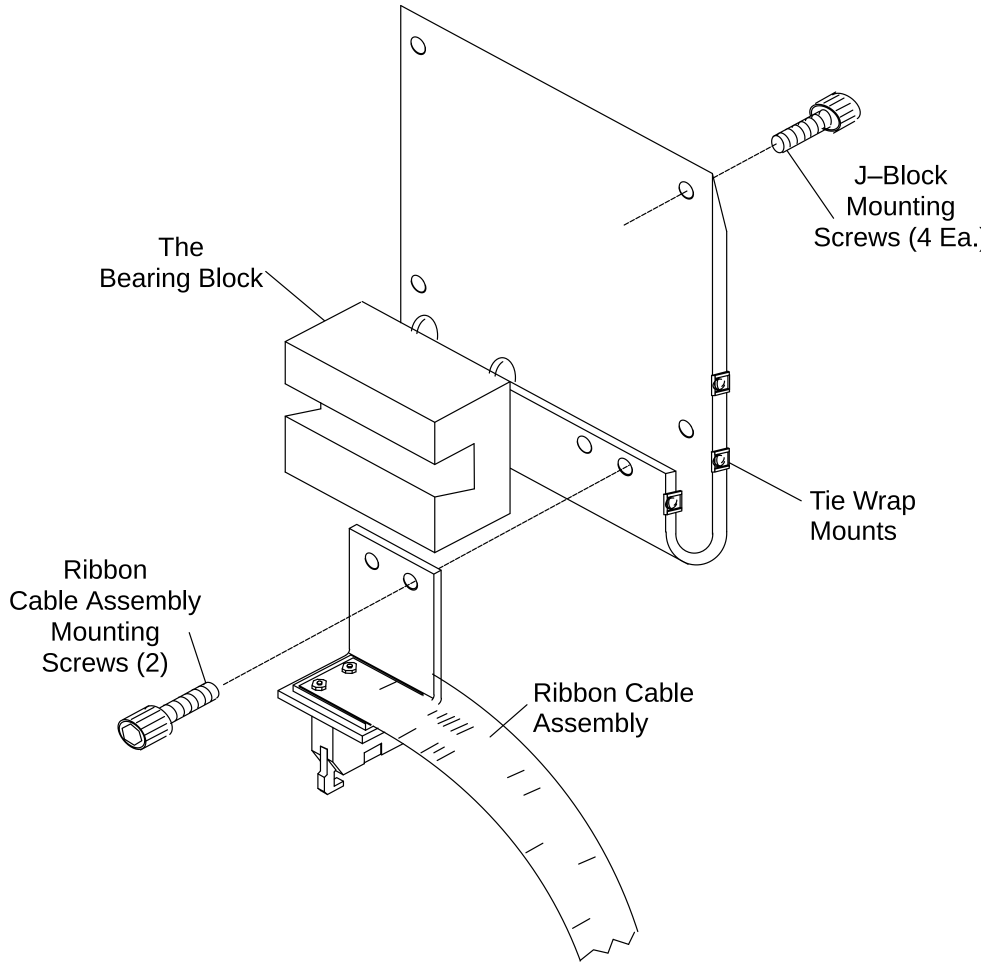


FIGURE 191 J Block

9. Remove the sixteen screws that mount the Carbon Top to the J–Blocks.

10. Use two persons to lift the Carbon Top off the J–Blocks. Place it on the soft surface prepared earlier.

CARBON TOP REPLACEMENT:

11. Install the Patient Support Carbon Top by reversing steps 2. through 10.

TAPE SWITCH – CARBON TOP

Introduction: This procedure is used to remove and replace the Patient Support Carbon Top Tape Switch(es).

Tools Required: 1. FSE tool kit

Material Required: 1. Double–Sided Tape 1/2 in. wide (P/N 76950)
Butt Splices (P/N T18A–34, 22–16 AWG)
Scraping Tool (e.g, razor blade)

Estimated TIme: 1.0 HR.

TAPE SWITCH REMOVAL PROCEDURE:

1. Apply Gantry power. Refer to the PowerUp/PowerDown procedure.
2. Manually position the Carbon Top against its rear Mechanical Stop.
3. Using the Gantry Control Panel Switches, raise the Patient Support to its highest position.
4. Remove Gantry power. Refer to the PowerUp/PowerDown procedure.
5. Remove the Carbon Top. Refer to the Carbon Top Removal/Replacement procedure.
6. Locate the Splices where the defective Tape Switch is joined to the Wire Harness. Cut the defective Switch Wires at the Splices. See Figure 192.
7. Remove the defective Tape Switch from the Carbon Top. The Tape Switch is held onto the Carbon Top by double–sided tape. It may be necessary to pull the defective Switch while scraping the adhesive with a razor blade.

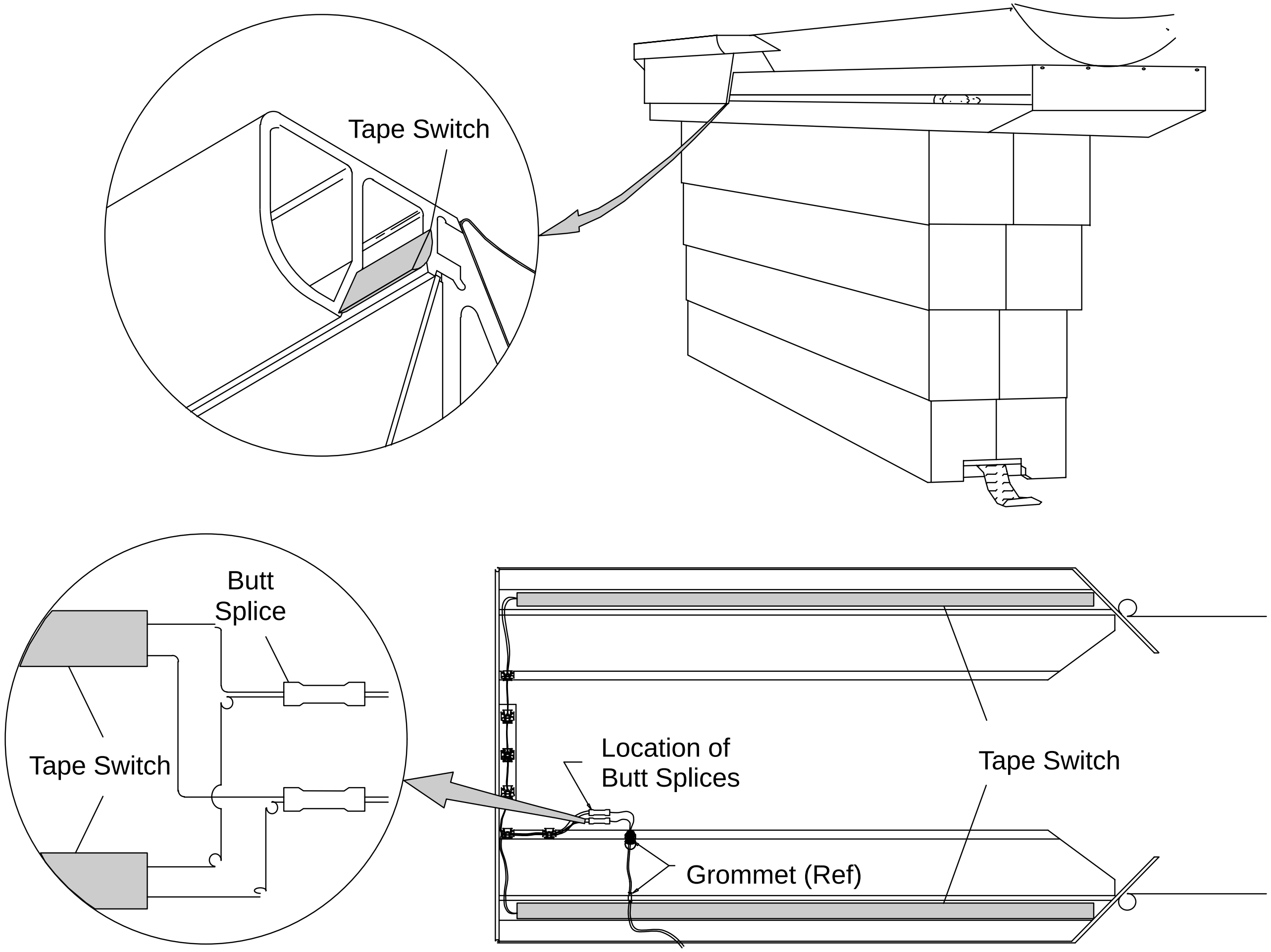


FIGURE 192 BOTTOM VIEW OF CARBON TOP

TAPE SWITCH REPLACEMENT:

8. Install the replacement Tape Switch using double–sided tape to fasten it to the Carbon Top.
9. Route the Wires from the replacement Tape Switch along the same path as the defective Tape Switch.
10. Strip 1/4” of insulation from the ends of the Tape Switch Wires and the Harness Wires.
11. Connect the Tape Switch Wires to the Harness Wires using Butt Splices.

12. Tie and secure the replacement Tape Switch Wires in the same manner as the original Tape Switch Wires.
13. Install the Carbon Top. Refer to the Carbon Top Removal/Replacement procedure.

SUPPORT ROLLER

Introduction: This procedure is used to remove and replace the Patient Support Carbon Top Support Roller.

Tools Required:

1. FSE tool kit
2. Snap Ring Pliers (90 degree bend)

Estimated Time: 20 min.

SUPPORT ROLLER REMOVAL PROCEDURE:

1. Apply Gantry power. Refer to the PowerUp/PowerDown procedure.
2. Raise the Patient Support to its highest position with the Gantry Control Panel Switches if not already in this position.
3. Remove Gantry power. Refer to the PowerUp/PowerDown procedure.
4. Move the Carbon Top to the front of the Patient Support so that it does not rest on the Support Roller.
5. Remove the four Mounting Bolts and Washers from the Support Roller Bracket that fasten the Bracket to the sub frame. See Figure 193.
6. Remove one of the Snap Rings holding the Support Roller Pin.
7. Support the Roller and push or pull the Pin out.

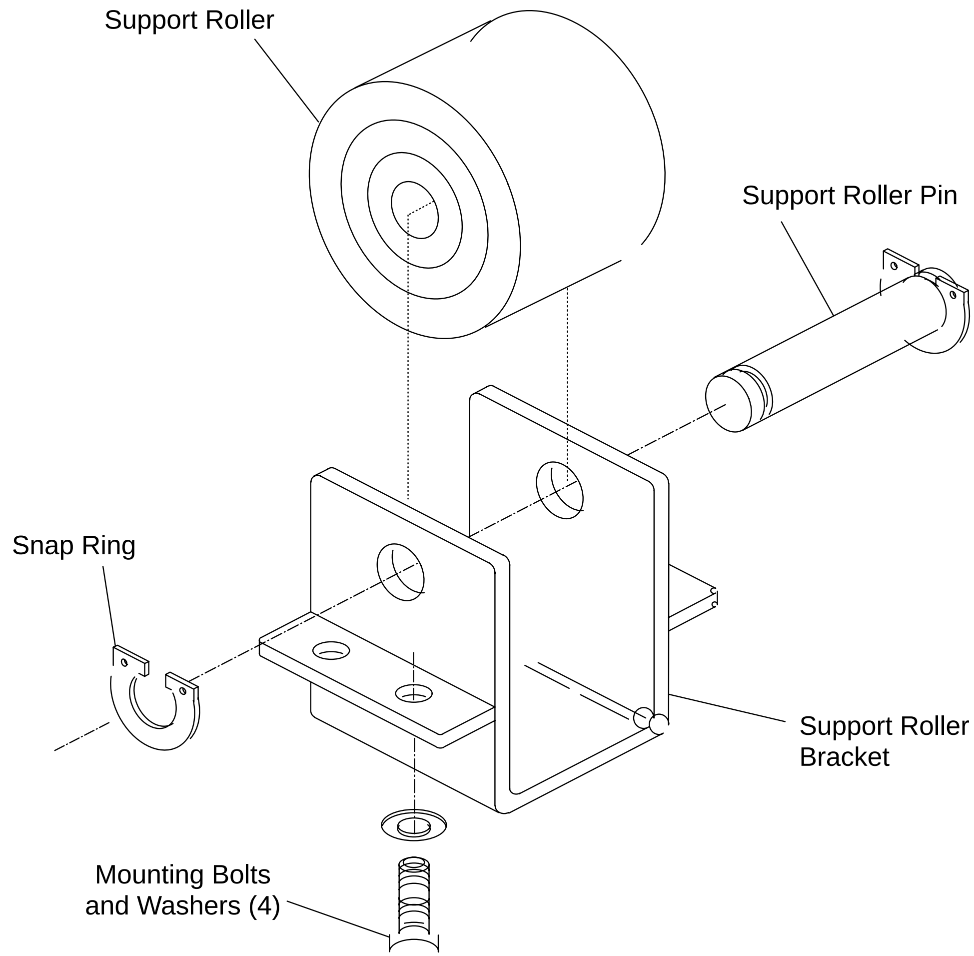


FIGURE 193 SUPPORT ROLLER ASSEMBLY

SUPPORT ROLLER REPLACEMENT:

8. Place the new Roller in position and slide the Pin through the Bracket and Roller.

9. Replace the Snap Ring onto the Pin. Ensure Snap Ring is completely seated in the groove of the Pin.

10. Replace the four Mounting Bolts and Washers for the Support Roller Bracket and fasten the Bracket to the Carbon Top.

11. Check Roller operation by moving the Carbon Top over the Roller to ensure that the Roller moves freely.

BEARING – CARBON TOP

Introduction: This procedure is used to remove and replace the Patient Support Carbon Top Bearings.

Tools Required: 1. FSE Tool kit

Estimated TIme: 1 HR.

BEARING – CARBON TOP REMOVAL PROCEDURE:

1. Remove power and the Carbon Top. Refer to the Carbon Top Removal/Replacement procedure.

> **NOTE**
>
> The Patient Support should be at its maximum height with power to the system removed due to following the Carbon Top Removal procedure.

2. Remove the Patient Support front, rear, Rail, and Horizontal Drive Covers. Refer to the Patient Support Covers Removal/Replacement procedure.
3. Remove the front or rear Mechanical Stops (as appropriate for the bearings being replaced). Refer to Figure 194.

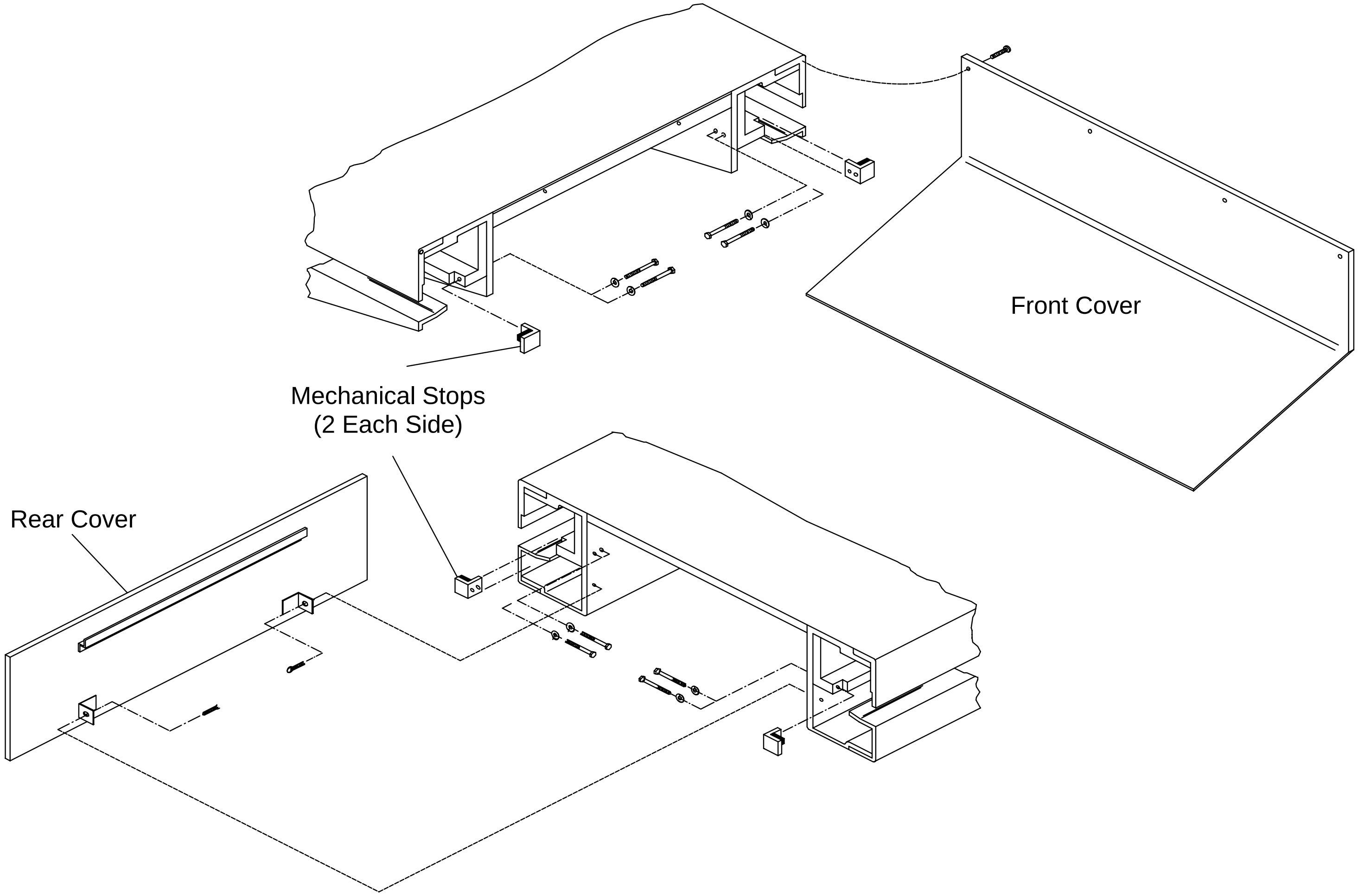


FIGURE 194 MECHANICAL STOPS

4. Disconnect the Horizontal Drive–Belt Connector to Belt–Hook Mounting Block from the J–block (if required). See Figure 195.
5. Remove the J–Block with the defective Bearing by sliding it off the end of the Linear Bearing Rail.

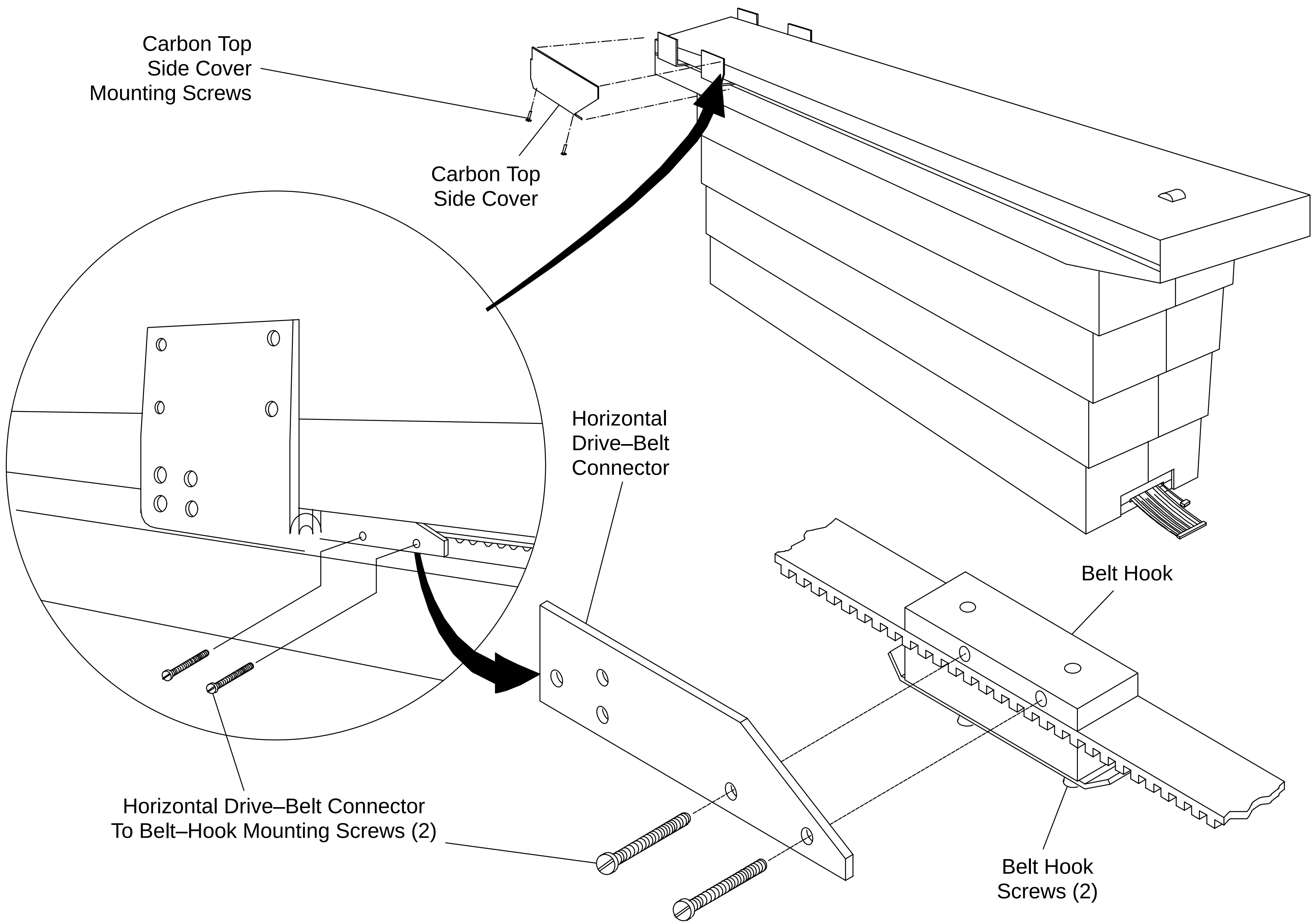


FIGURE 195 BELT BLOCK LOCATION ON CARBON TOP

6. Remove the defective Bearing by taking out the four Screws and pulling the THK Bearing Block from the J–Block. See Figure 196 and Figure 197.

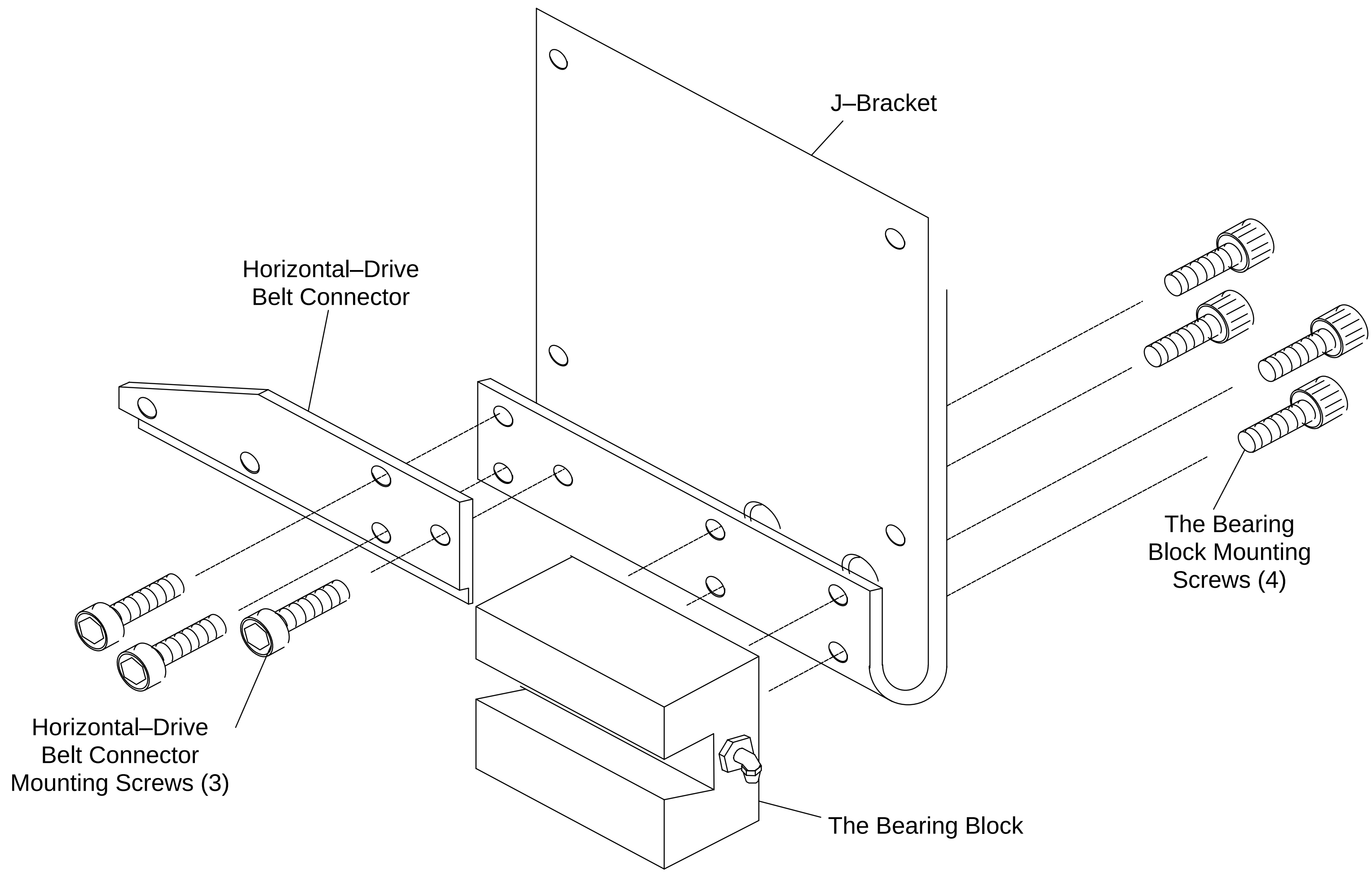


FIGURE 196 RIGHT REAR MOUNTING BLOCK ASSEMBLY

BEARING – CARBON REPLACEMENT:

7. Install the replacement Bearing by reversing the process in step 6.

8. Replace the J–Block by <u>CAREFULLY</u> sliding it back onto the Linear Bearing Rail. Care must be taken not to jar the ball bearings out of the channels.

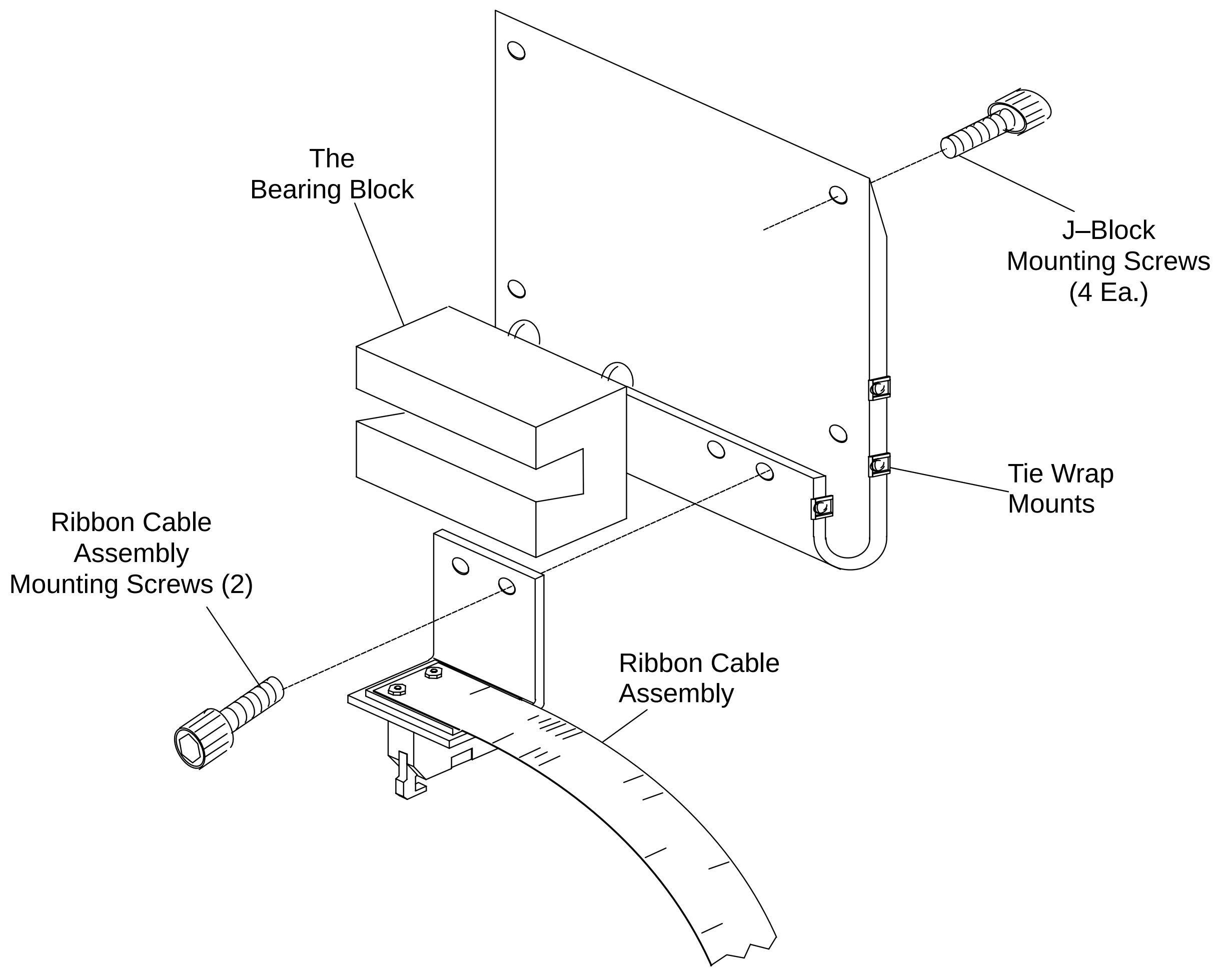


FIGURE 197 LEFT REAR MOUNTING BLOCK ASSEMBLY

9. Replace the Carbon Top. Refer to the Carbon Top Removal/Replacement procedure.
10. Move the Carbon Top by hand to verify smooth operation.
11. Install the Patient Support front and rear covers, Rail Covers and the Horizontal Drive Cover. Refer to the Patient Support Covers Removal/Replacement procedure .

TAPE SWITCH – FOOT

Introduction:	This procedure is used to remove and replace the Patient Support foot actuated Tape Switch.
Tools Required:	1. FSE tool kit
Material Required:	1. Double Sided Tape (P/N 76950) Butt Splices (P/N T18A–34, 22–16 AWG) Scraping Tool (e.g., razor blade) 3. Crimp tool
Estimated Time:	20 min.

TAPE SWITCH REMOVAL PROCEDURE:

1. Apply Gantry power. Refer to the PowerUp/PowerDown procedure.
2. Tilt the Gantry +5 degrees with the Gantry Control Panel Switches.
3. Remove Gantry power. Refer to the PowerUp/PowerDown procedure.
4. Remove the two screws and Washers that attach the Foot Switch Cover to its base. Grasp both sides of the Cover and pull it straight off. See Figure 198.
5. The Tape Switch is held on by double–sided tape. Remove the defective Tape Switch by pulling it from the Base. Remove any tape residue that may be on the Switch Base.
6. Cut the wires of the defective Switch(es) at the Switch Body.

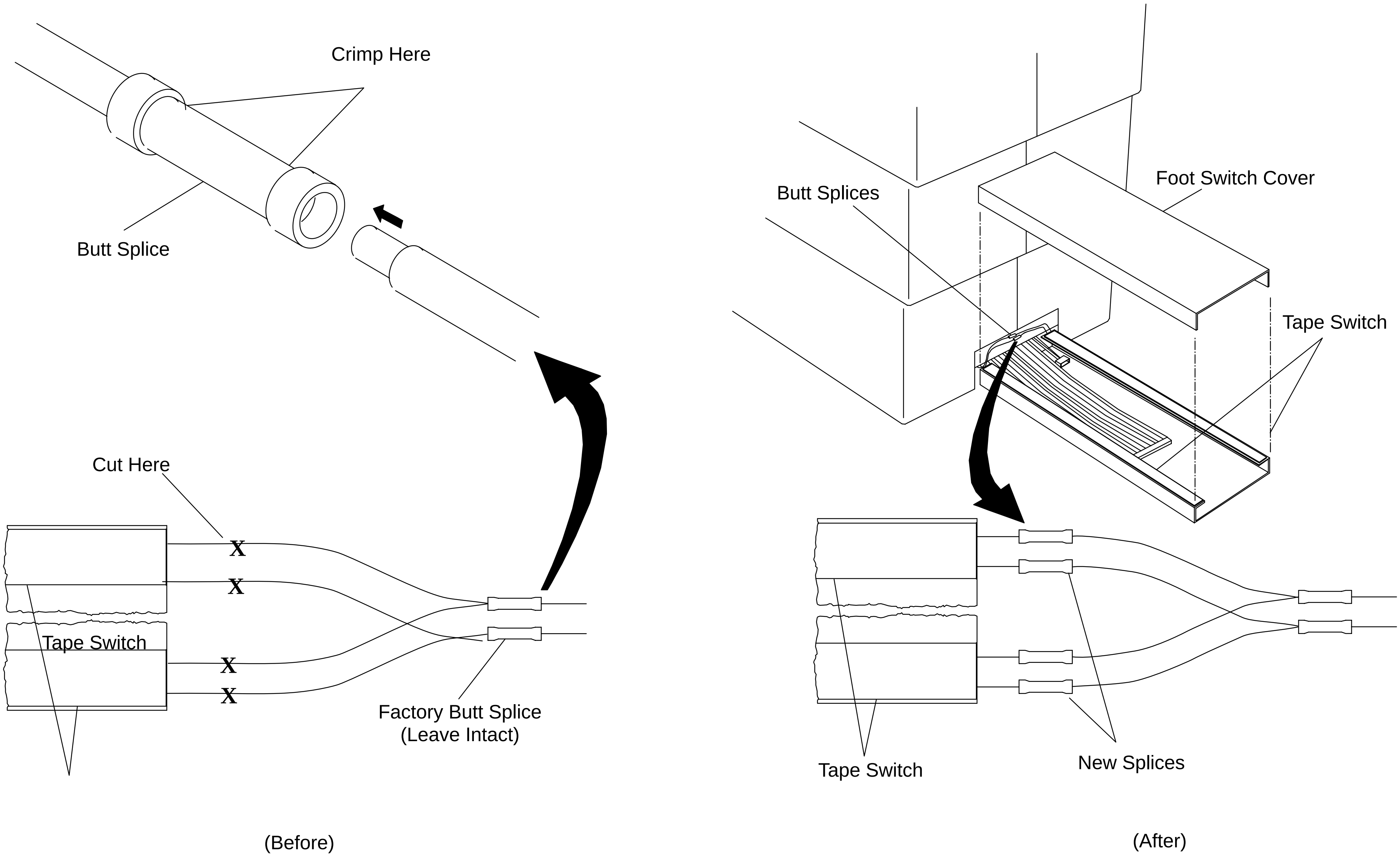


FIGURE 198 PATIENT SUPPORT FOOT SWITCH

TAPE SWITCH REPLACEMENT:

7. Strip 1/4” of insulation from the cut leads coming from the Patient Support.
8. Strip 1/4” of insulation from the wires on the new replacement Tape Switch(es).
9. Connect the cut leads from the Patient Support cable to the leads on the replacement Tape Switch with Butt Splices.
10. Use tie wraps to secure the wires away from the Cover.
11. Apply Gantry power. Refer to the PowerUp/PowerDown procedure.
12. Verify that both sides of the Foot Switch operate to allow free–floating movement of the Carbon Top.
13. Replace the Foot Switch Cover by reversing the process in step 4.

HORIZONTAL DRIVE MOTOR

Introduction: This procedure is used to remove and replace the Patient Support Horizontal Drive Motor.

Tools Required:
1. FSE tool kit
2. Flat thickness gauge

Material Required:
1. Tie–wraps
2. Butt Splices (P/N T–18A–34, 22–16 AWG)

Estimated Time: 30 min.

DRIVE MOTOR REMOVAL PROCEDURE:

1. Apply Gantry power. Refer to the PowerUp/PowerDown procedure.
2. Raise the Patient Support to its highest position with the Gantry Control Panel Switches.
3. Remove Gantry power. Refer to the PowerUp/PowerDown procedure.
4. Remove the rear, right Rail, and Horizontal Drive Covers. Refer to the Patient Support Covers Removal/Replacement procedure.
5. Note the Cable routing and cut all Tie Wraps that secure the Motor Cables.
6. Disconnect the Horizontal Drive Motor Cable.
7. Support the Horizontal Drive Motor and remove the four Socket Head Cap Screws and the Washers that hold the Assembly to its Mounting Bracket. See Figure 199.
8. Remove the Horizontal Drive Motor.

> **NOTE**
>
> Measure the distance between the Motor Mounting Plate and the edge of the Motor Pulley.

9. Remove the Pulley from the Motor Shaft by loosening the two Set Screws. Note the location and orientation of the Pulley on the Motor Shaft.

10. Attach the Pulley to the new Motor Shaft in the same location as the old Pulley.

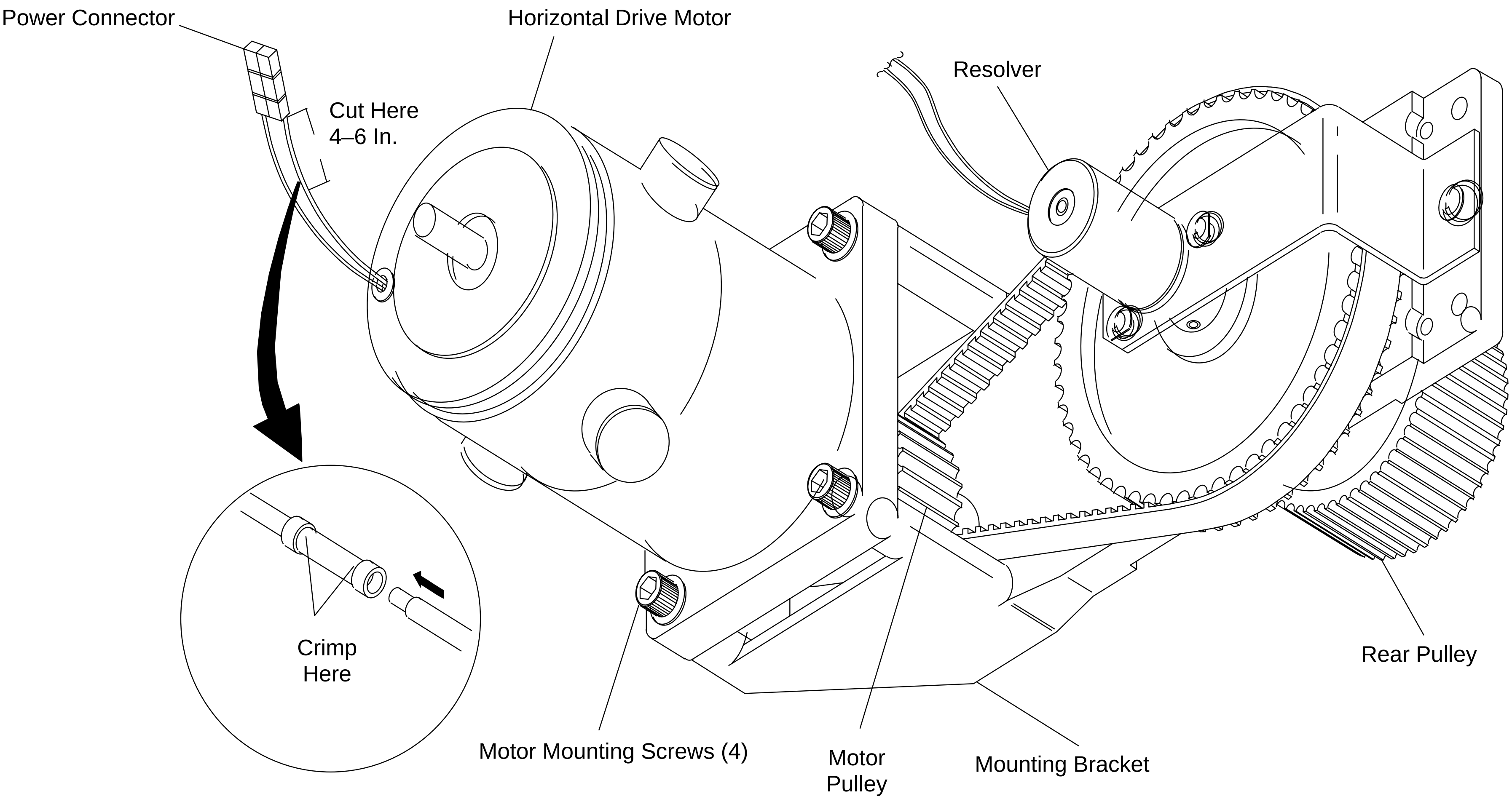


FIGURE 199 HORIZONTAL MOTOR ASSEMBLY

DRIVE MOTOR REPLACEMENT

> **NOTE**
>
> If the replacement Motor has a connector already attached steps 11. through 13. may be omitted.

11. Cut the Power Connector off the old Motor as shown Figure 199.
12. Strip 1/4” of insulation from the cut leads on both the replacement Motor and the Power Connector.
13. Connect the cut Leads from the new Motor and the Connector with Butt Splices.
14. Place the Timing Belt around the Pulley and mount the replacement Horizontal Drive Motor to the Mounting Bracket with the four (4) Socket Head Capscrews.
15. Attach the Horizontal Drive Motor Wires to the Horizontal Drive Motor Power Cable.
16. Apply Gantry power. Refer to the PowerUp/PowerDown procedure.
17. Check the operation of the Horizontal Motor using the Gantry Control Panel Switches.

> ⚠ **WARNING**
>
> **USE EXTREME CAUTION WHEN OPERATING OR WORKING ON EQUIPMENT WITH COVERS REMOVED. DANGEROUS VOLTAGES ARE EXPOSED WHEN THE GANTRY COVERS ARE REMOVED. FAILURE TO COMPLY CAN CAUSE SERIOUS INJURY OR DEATH TO PERSONNEL.**

18. Route the Horizontal Drive Motor Cable as it was in step 5. and replace all Tie Wraps that were removed.
19. Remove Gantry power. Refer to the PowerUp/PowerDown procedure in the Introduction section.
20. Replace the rear, right Rail, and Horizontal Drive Covers. Refer to the Patient Support Covers Removal/Replacement procedure.

HORIZONTAL DRIVE BELT

Introduction:	This procedure is used to remove and replace the Patient Support Horizontal Drive Belt.
Tools Required:	1. FSE tool kit
Estimated Time:	45 min.

DRIVE BELT REMOVAL PROCEDURE:

1. Apply Gantry power. Refer to the PowerUp/PowerDown procedure.
2. Raise the Patient Support to its highest position with the Gantry Control Panel Switches.
3. Remove Gantry power. Refer to the PowerUp/PowerDown procedure.
4. Remove the right Side Rail Cover and the right Carbon Top Side Cover. Refer to the Patient Support Covers Removal/Replacement procedure.
5. Manually move the Carbon Top to position the Belt Hook approximately midway between the Belt Pulleys. See Figure 200.

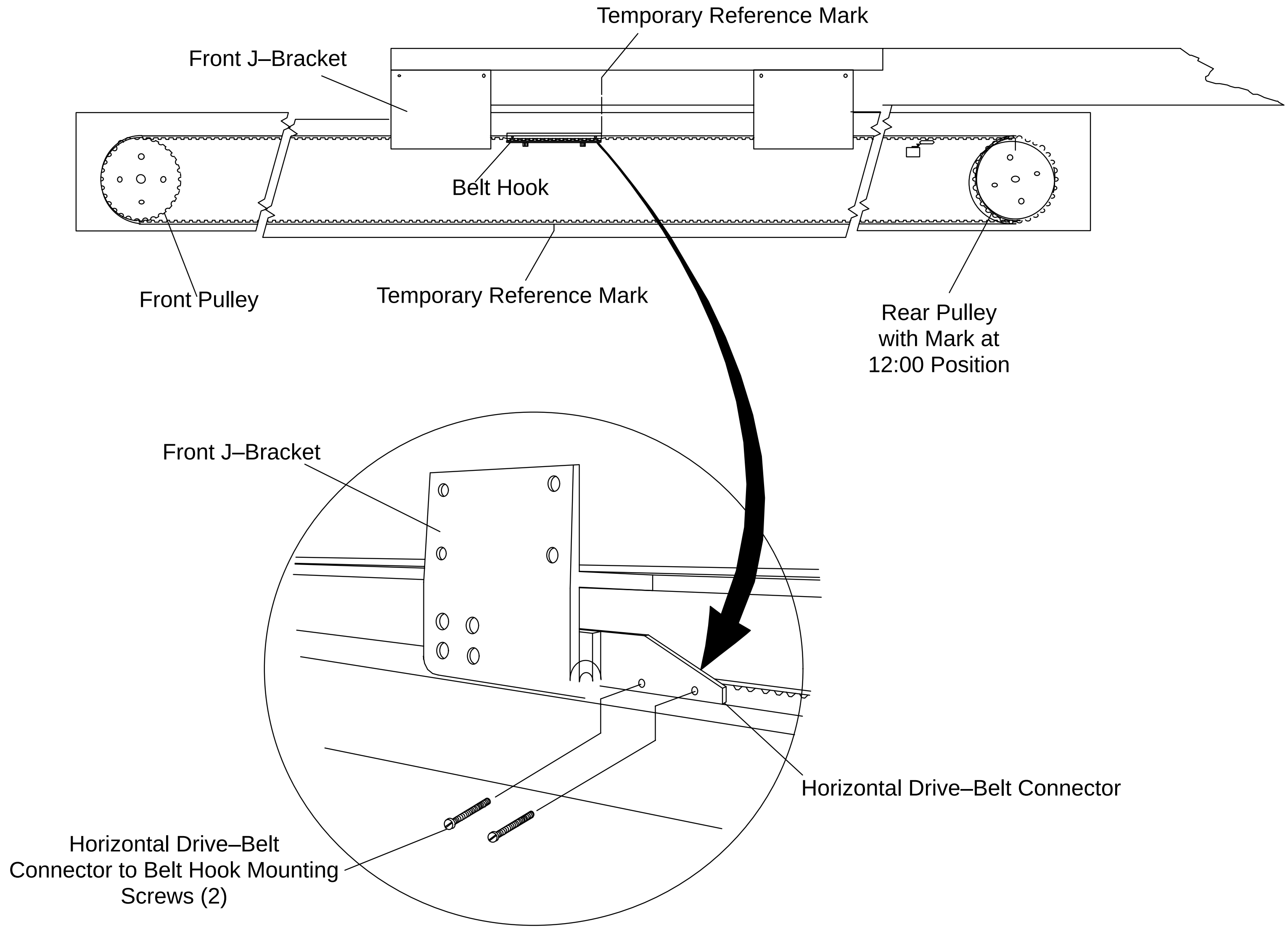


FIGURE 200 HORIZONTAL DRIVE–BELT ASSEMBLY

6. Place a temporary reference mark on the Sub–Frame in line with the front edge of the Belt Hook. Place a mark on the front Pulley in the 12:00 position. Refer to Figure 200.

7. Place a temporary reference mark on the edge of the lower Belt half at the center (not a critical dimension) and a corresponding mark on the side Channel.
8. Disconnect the Belt Hook from the Horizontal Drive–Belt Connector by removing the two Mounting screws and Washers. Refer to Figure 200.
9. Loosen the four front Pulley Mounting Plate Screws. Loosen Belt tension by screwing the Horizontal Drive Belt Adjusting Bolt into the Thrust Block until the Belt Hook Assembly rests on the lower Belt surface. See Figure 201.

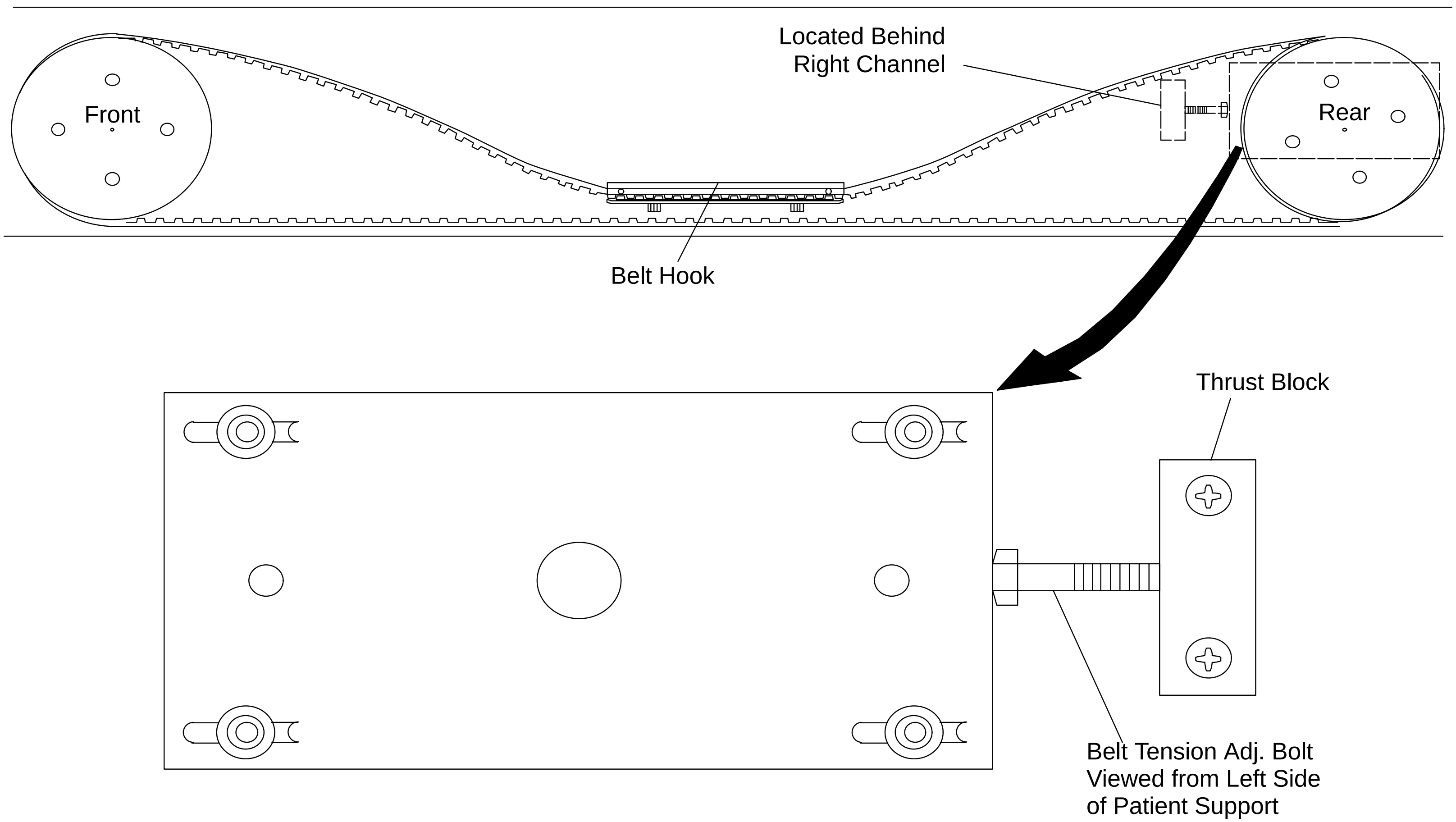


FIGURE 201 BELT TENSION ADJUST

10. Carefully remove the Belt and Belt Hook Assembly from both Pulleys.

11. Remove the two Screws that attach the top and bottom clamp of the Belt Hook Assembly. See Figure 202.

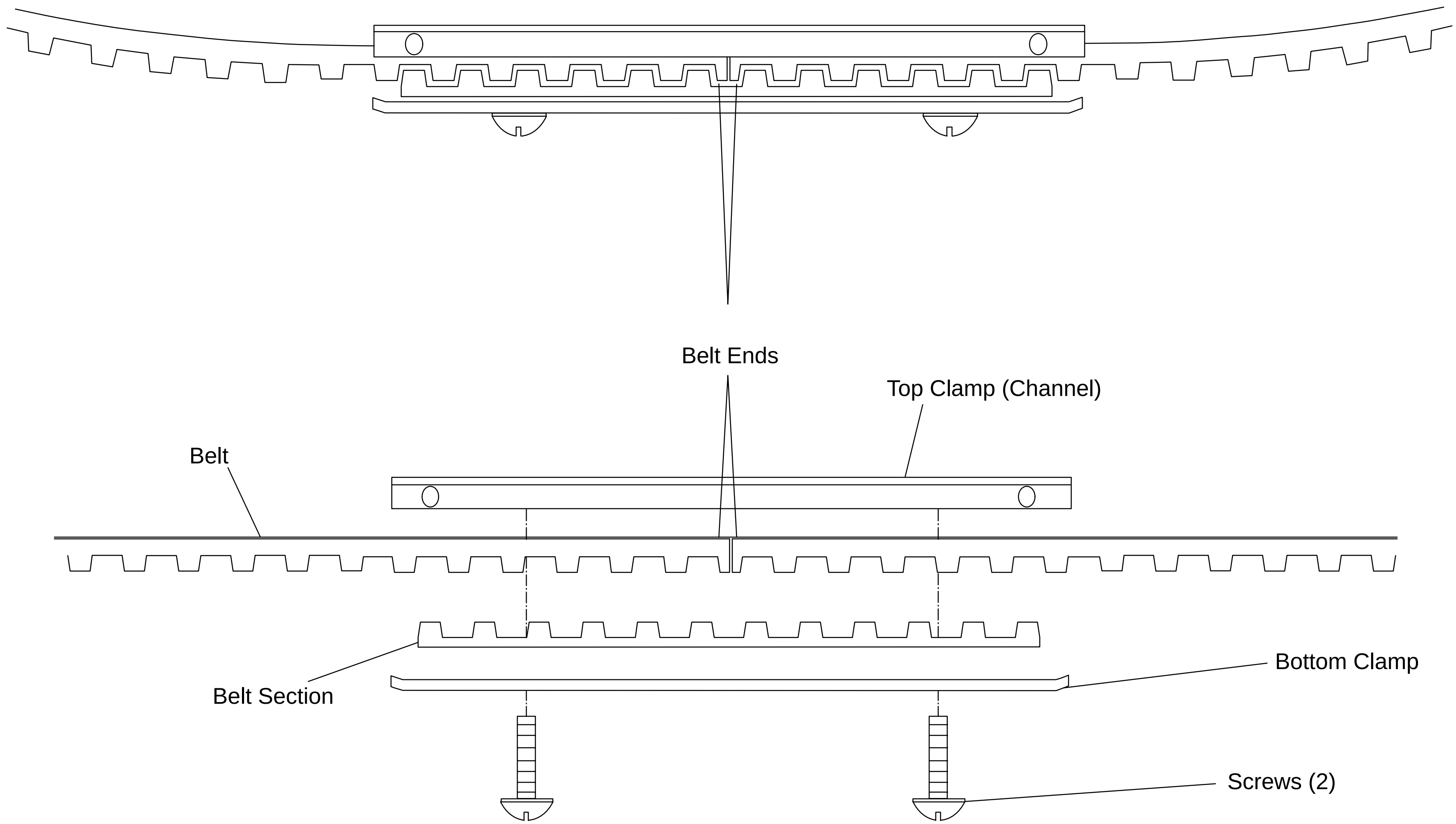


FIGURE 202 Belt Hook Assembly

DRIVE BELT REPLACEMENT:

12. Lay the replacement Belt and the defective Belts on a flat surface side–by–side with the ends aligned.

13. Place a temporary mark on the edge of the replacement Belt to align with the ”near center” mark on the defective Belt.

14. Assemble the replacement Belt and Belt Hook.

15. Position the replacement Belt Assembly in the right Channel so the ”near center” marks align.
16. Keeping the ”near center” marks aligned, carefully loop the Belt over the end Pulleys. Verify that the mark on the front Pulley is in the 12:00 position.
17. Tighten (or loosen) the Belt Tension Adjusting Bolt until the Belt Hook Assembly just rests on the lower Belt Surface. Place a temporary reference mark on the Bolt Head. Turn the Tension Adjusting Bolt two and one half (2 1/2) complete turns tightening the Belt. This is the correct Belt tension setting.
18. Manually move the Carbon Top until the Horizontal Drive–Belt Connector aligns with the holes in the Belt Hook Assembly. The front of the Belt Hook should be in line with the temporary reference mark made in step 6.
19. Secure the Belt Hook Assembly to the Horizontal Drive–Belt Connecting Bracket with the two Horizontal Drive–Belt Connector to Belt Hook Mounting Screws.
20. Calibrate Horizontal Position using the CC (Couch Calibration) in GPUSPY manual.
21. Replace the right Side Rail and right Carbon Top Side Covers. Refer to the Patient Support Covers Removal/Replacement procedure.

HORIZONTAL RESOLVER

Introduction: This procedure is used to remove and replace the Patient Support Horizontal Resolver. Return the information sheet and resolver from system in the included packaging. Resolver may terminate with either connector contacts or bare leads. Follow the appropriate procedure.

> **NOTE**
>
> Two service persons are recommended to perform this procedure.

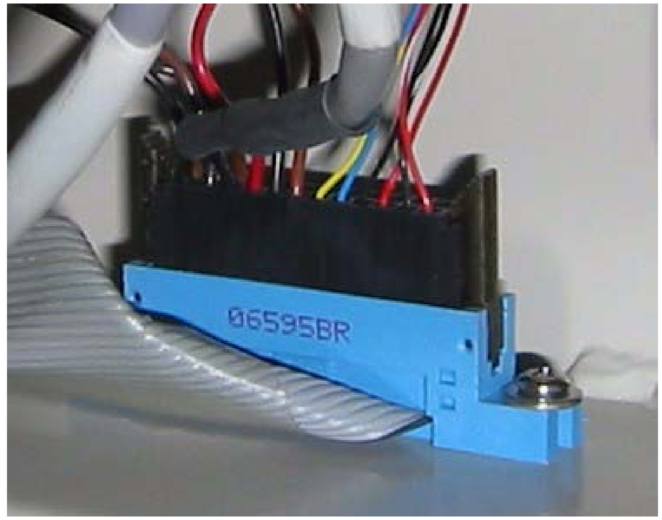
Tools Required: FSE Toolkit

Material Required: Resolver

Time Estimate: Three hours for resolver replacement and couch calibration

INITIAL CONDITIONS:

- Patient Support rear end, right Side Rail and Horizontal Drive Covers removed. Refer to the Patient Support Covers Removal/Replacement procedure.
- Lower telescoping covers removed. Refer to the Telescoping Base Cover Removal Procedures.
- Patient Support raised to its highest position with the Gantry Control Panel Switches.
- Gantry power removed. Refer to the PowerUp/PowerDown procedure.
- Identify and record the Resolver Wires by color and their position in Connector P408. Refer to Figure 203.
- Document locations of ty–wraps and cable routing.



RESOLVER WIRING

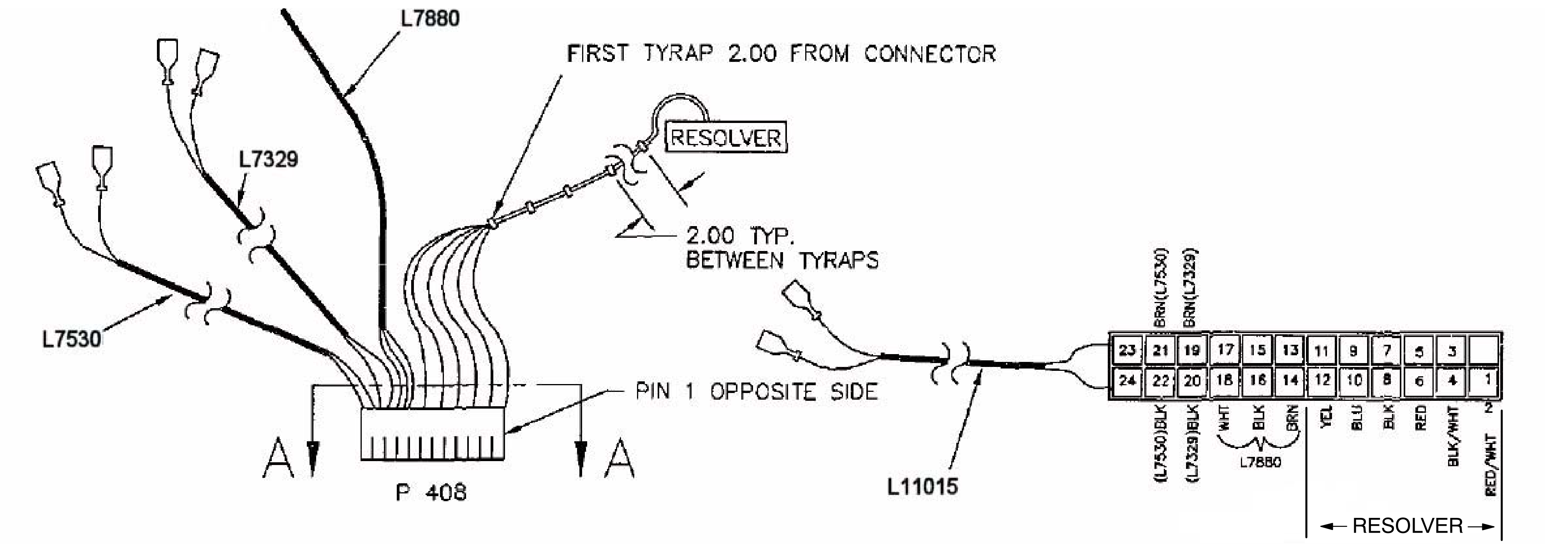
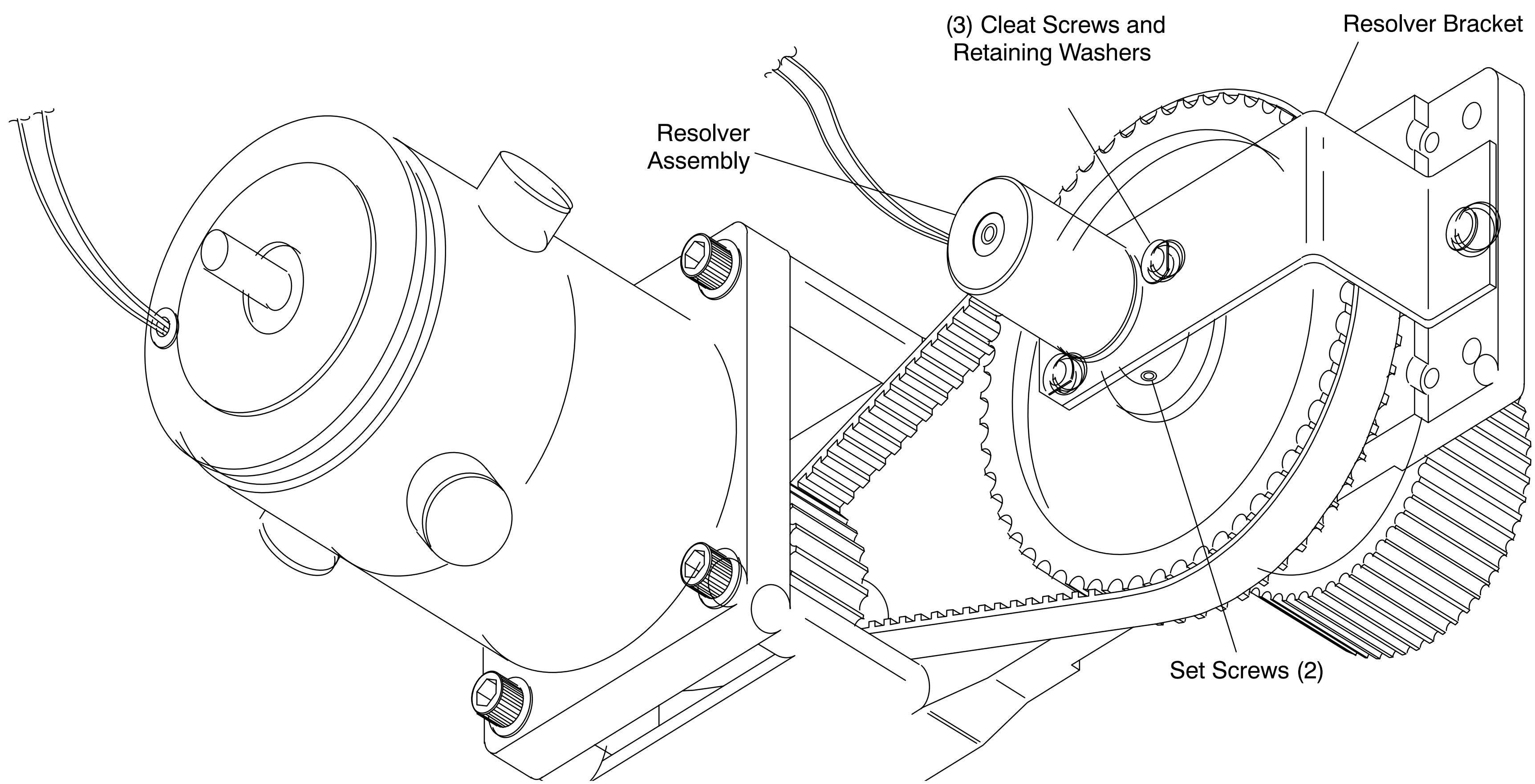


FIGURE 203 RESOLVER WIRING DIAGRAM

REMOVAL PROCEDURE FOR RESOLVER P/N T93N–38 (BARE LEADS)

> **NOTE**
>
> For resolver p/n 4535 671 04171 go to step 6 of this procedure.

1. Cut ty–wraps used to bind the Resolver Wires to other cables or structure.
2. Select a convienient location for splicing all six wires between connector P408 and the resolver. Cut the wires, leaving the P408 contacts intact.
3. Loosen, but do not remove, the three (3) Cleat Screws holding the Resolver in place. See figure below.



4. Turn the Cleat Retaining Washers so the flat sides face the Resolver, then tighten the Cleat Screws.

> ⚠ **CAUTION**
>
> EXERCISE EXTREME CARE WHILE REMOVING AND INSTALLING RESOLVER TO PREVENT ANY BENDING OR MISALIGNMENT ON RESOLVER SHAFT.

5. Loosen the two (2) Set Screws on the Resolver Pulley. Remove the Resolver from the Pulley and away from the Bracket. Continue at resolver installation step 1.
6. Applicable for resolver p/n 4535 671 04171. Cut ty–wraps used to bind the resolver wires to other cables or structure.
7. Remove the Resolver Wires from Connector P408. To release the Socket Contact Retention Blade, press against it with the point of a small object, while pulling on the Wire until the Contact is clear of the connector shell. Refer to Figure 204.
8. Loosen, but do not remove, the three (3) Cleat Screws holding the Resolver in place.
9. Turn the Cleat Retaining Washers so the flat sides face the Resolver, then tighten the Cleat Screws.
10. Loosen the two (2) Set Screws on the Resolver Pulley. Remove the Resolver from the Pulley and away from the Bracket.

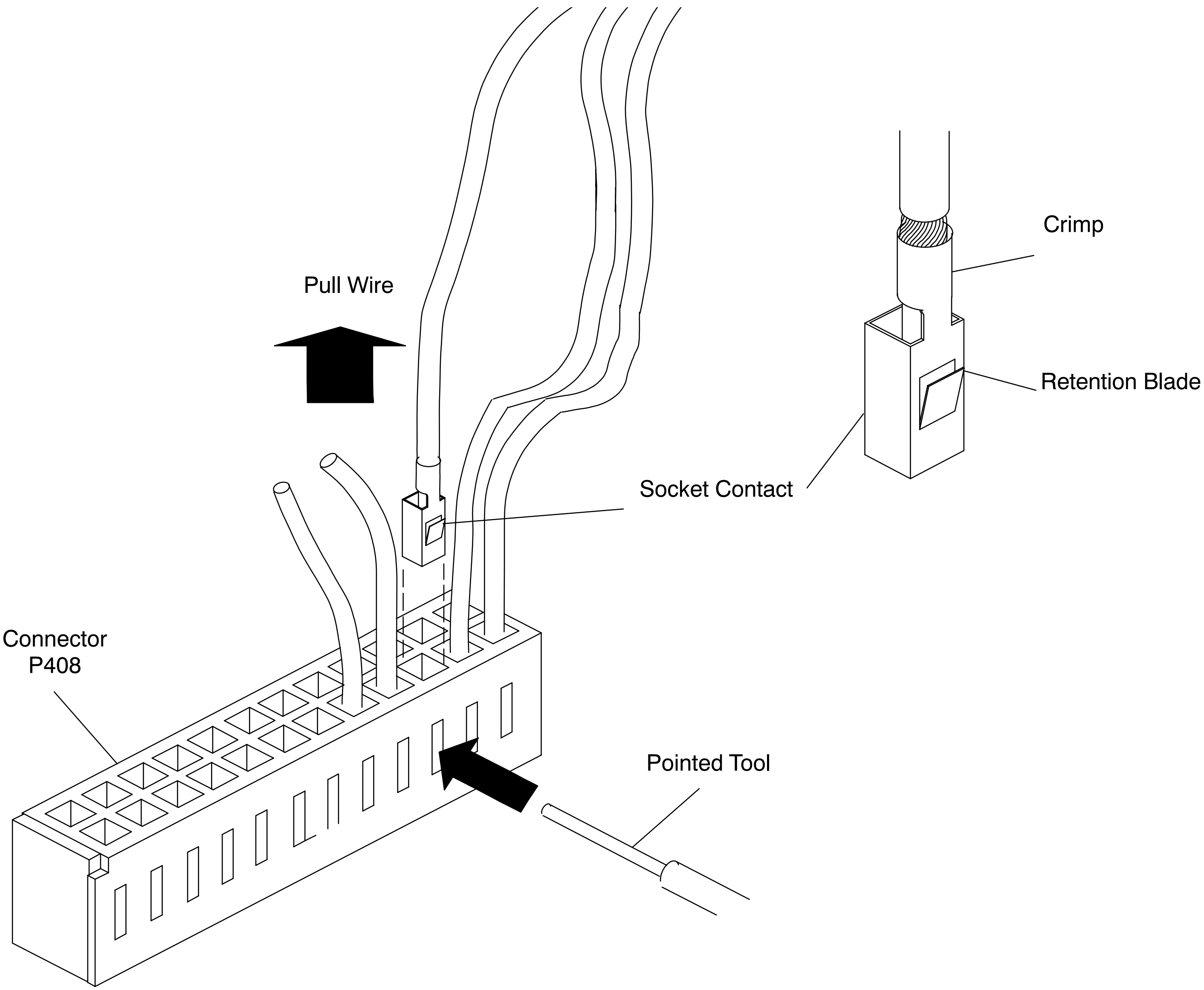


FIGURE 204 REMOVING RESOLVER PINS FROM THE CONNECTOR

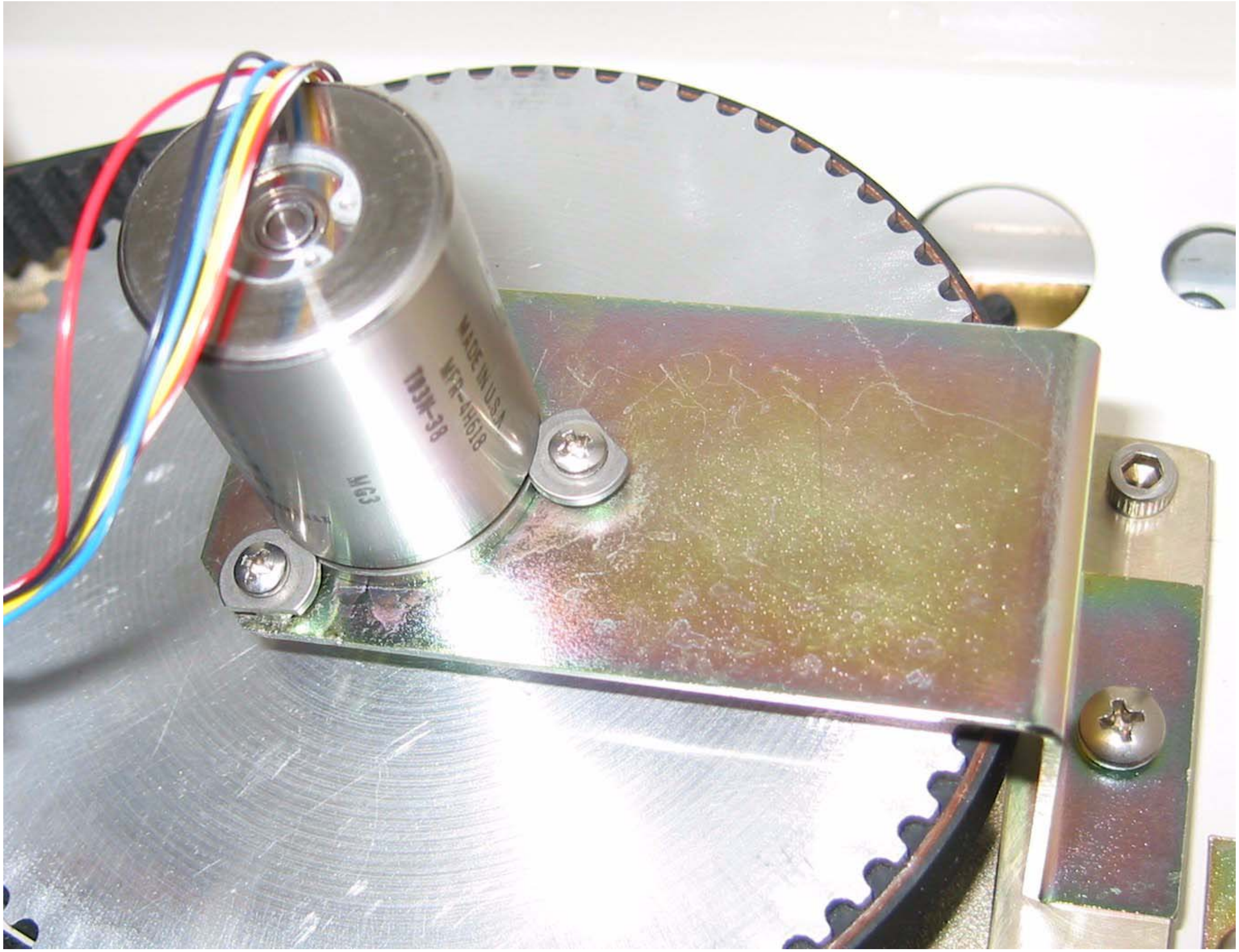
RESOLVER INSTALLATION:

> ⚠ **CAUTION**
>
> EXERCISE EXTREME CARE WHILE REMOVING AND INSTALLING RESOLVER TO PREVENT ANY BENDING OR MISALIGNMENT ON RESOLVER SHAFT.

1. Place the Resolver into the Resolver Bracket opening.
2. Carefully insert the shaft of the replacement Resolver into the Resolver Pulley. Verify that the resolver flange is completely contacting the mounting bracket. Equally tighten the Pulley Set Screws just enough to keep the Pulley on the shaft.

> ⚠ **CAUTION**
>
> DO NOT BUMP OR MOVE THE RESOLVER DURING THE FOLLOWING STEPS.

3. Loosen the three Cleat Screws and turn the Cleat Retaining Washers so the flat edge faces away from the resolver. Ensure that the Cleat Retaining Washers fit into the groove on the side of the Resolver. Refer to photograph on next page.
4. Turn the Resolver so the leads are at approximately the 10 o’clock position (as shown in photo).
5. Slightly tighten each cleat screw to assure resolver is uniformly seated to resolver bracket.



CLEAT RETAINING WASHERS INSTALLATION

6. Tighten the pulley set screws to 10 in–lbs.

> **NOTE**
>
> During the Horizontal Couch Calibration, the resolver may need to be rotated (reference figure 7). Tighten resolver cleat screws sufficiently to prevent resolver housing rotation until calibration is completed.

7. Route resolver wires as recorded in Initial Conditions paragraph (before removal). Secure wires away from moving parts with ty–wraps.
8. Refer to the Wire color and locations shown in Figure 203.

   For p/n 4535 671 10171 – install the resolver leads into connector P408

   For p/n T93N–38 – splice six wires to each respective wire clipped in removal step 3.
9. Calibrate Carbon Top Horizontal Position using CC (Couch Calibration).
10. When couch calibration is completed, verify Gantry power is removed. Refer to the PowerUp/PowerDown procedure.
11. Apply loctite to (3x) cleat screws at bracket interface and torque to 3–4 in–lbs.
12. Replace any remaining Patient Support Covers. Refer to the Patient Support Covers Removal/Replacement procedure.

HORIZONTAL POSITION POTENTIOMETER

Introduction: This procedure is used to remove and replace the Patient Support Horizontal Position Potentiometer.

Tools Required:
1. FSE tool kit
2. Crimper

Material Required
1. Insert Contacts (P/N T18–195, 24–20 AWG)

Estimated Time: 30 min.

POTENTIOMETER REMOVAL PROCEDURE:

1. Apply Gantry power. Refer to the PowerUp/PowerDown procedure.
2. Raise the Patient Support to its highest position with the Gantry Control Panel Switches.
3. Remove Gantry power. Refer to the PowerUp/PowerDown procedure.
4. Remove the Patient Support front end Covers. Refer to the Patient Support Covers Removal/ Replacement procedure.
5. Loosen and remove the Pan Head Screw to the Retainer Bracket and the two Set Screws to the Position Pot. Refer to Figure 205.
6. Remove the Position Pot from the Gearbelt Pulley and disconnect Connector J420.
7. Remove the Retainer Bracket from the Position Pot and cut the Tiewrap that anchors the Cable. See Figure 206.

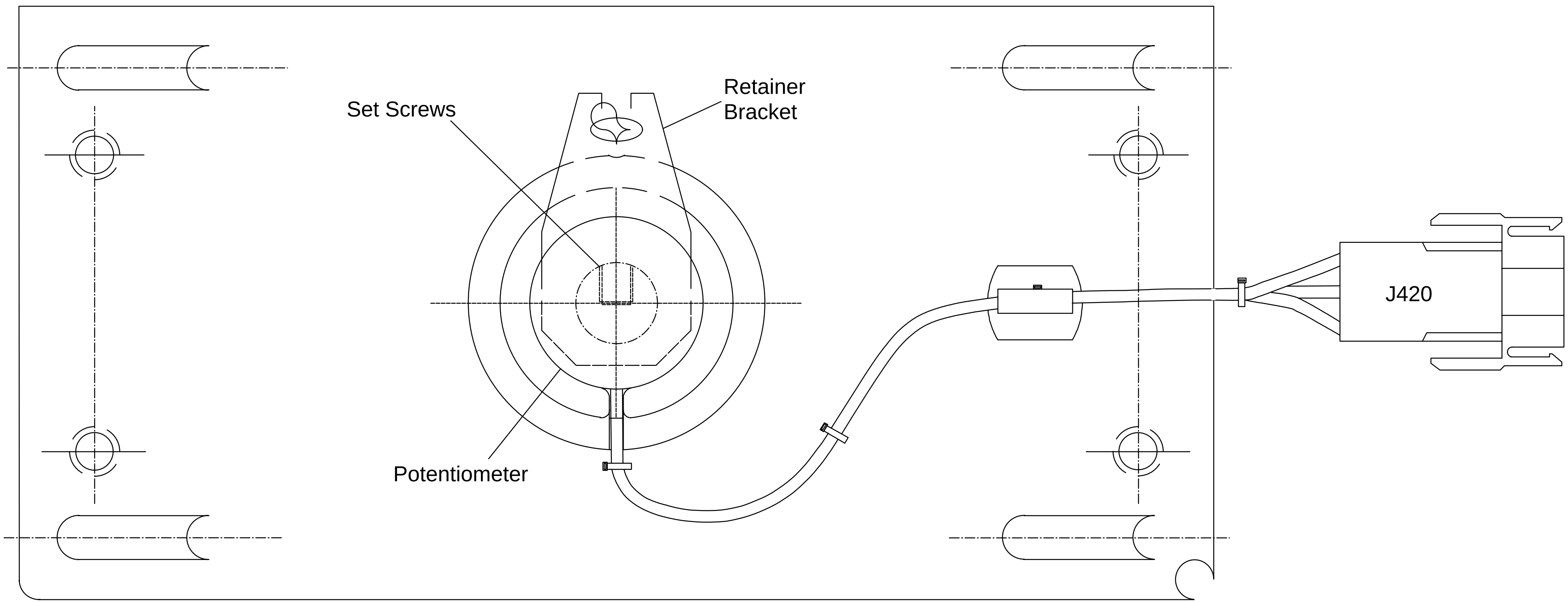


FIGURE 205 HORIZONTAL DRIVE GANTRY END ASSEMBLY

NOTE

Ensure that the Pan Head Screw that passes through the Retainer Bracket does not press against the Bracket. The purpose of this Screw is only to prevent the Retaining Bracket from rotating. Leave a small gap during replacement.

Note

The pan head screw is there to ensure that the bracket does not rotate. Leave a small gap between the head of the screw and the bracket.

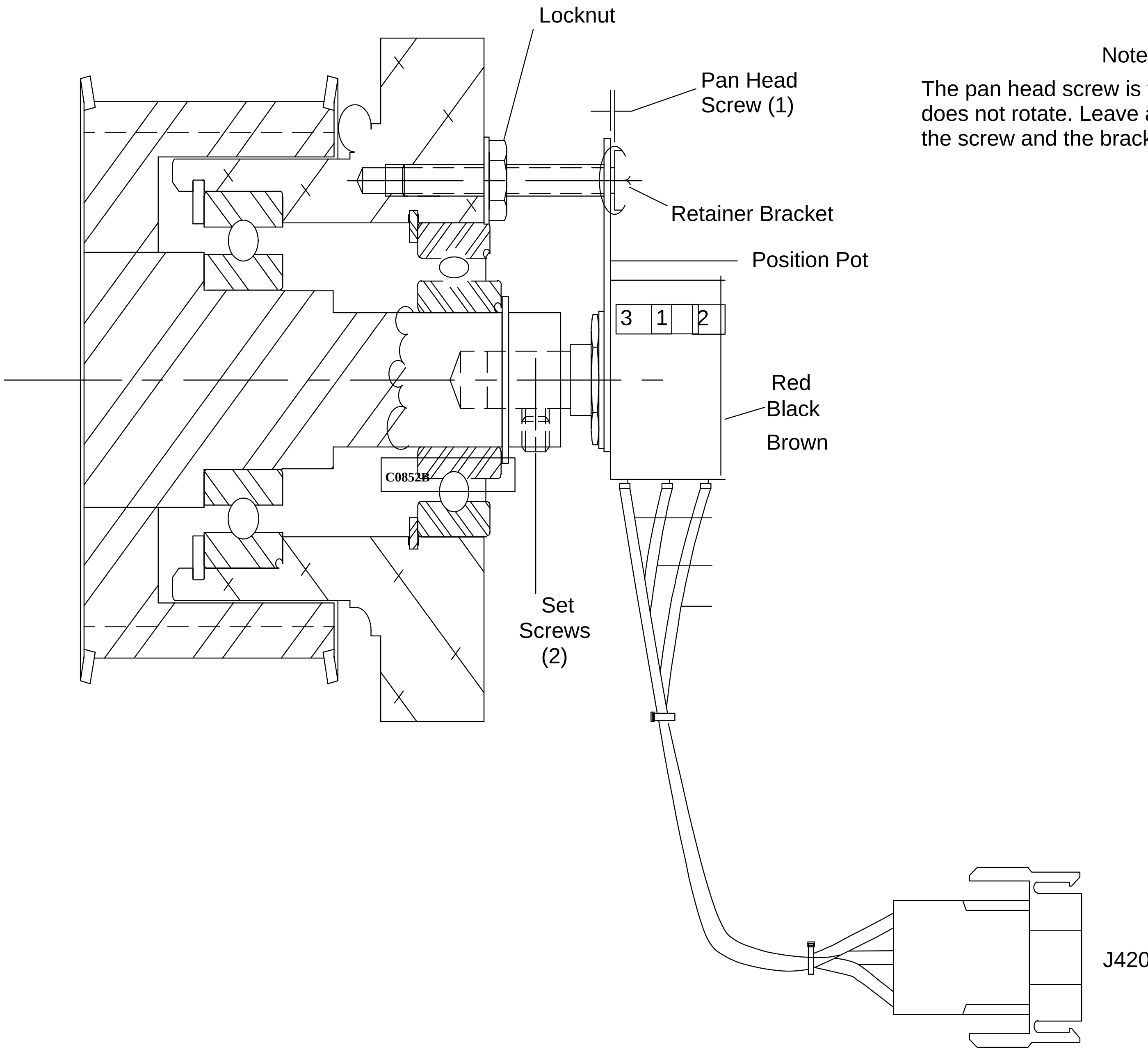


FIGURE 206 REMOVING POSITION POTENTIOMETER

POTENTIOMETER REPLACEMENT:

8. Trim the replacement Potentiometer leads to the proper length to reach Connector J420. Strip the ends and crimp an Insert Contact onto each lead.
9. Install the replacement Position Pot by reversing the process in step 5. through 7.
10. Calibrate the Carbon Top Horizontal Position using the CC (Couch Calibration) section of the GPUSPY manual.
11. Remove Gantry power. Refer to the PowerUp/PowerDown procedure .
12. Replace the Patient Support Covers. Refer to the Patient Support Removal/ Replacement procedure.

HORIZONTAL TIMING BELT

Introduction:	This procedure is used to remove and replace the Patient Support Horizontal Timing Belt.
Tools Required:	1. FSE tool kit
Estimated Time:	(30 min.)

TIMING BELT REMOVAL PROCEDURE:

1. Apply Gantry power. Refer to the PowerUp/PowerDown procedure.
2. Raise the Patient Support to its highest position with the Gantry Control Panel Switches.
3. Remove Gantry power. Refer to the PowerUp/PowerDown procedure.
4. Remove the Patient Support Gantry end and right Side Rail Covers. Refer to the Patient Support Covers Removal/Replacement procedure, or hange the resolver bracket assembly so there is no tension on the wires.
5. Remove the Resolver as outlined in the Resolver Removal/Replacement procedure.
6. Loosen the Resolver Bracket Mounting Screw and remove the Resolver Bracket Assembly from the Horizontal Drive Motor Housing Assembly. Refer to Figure 207.

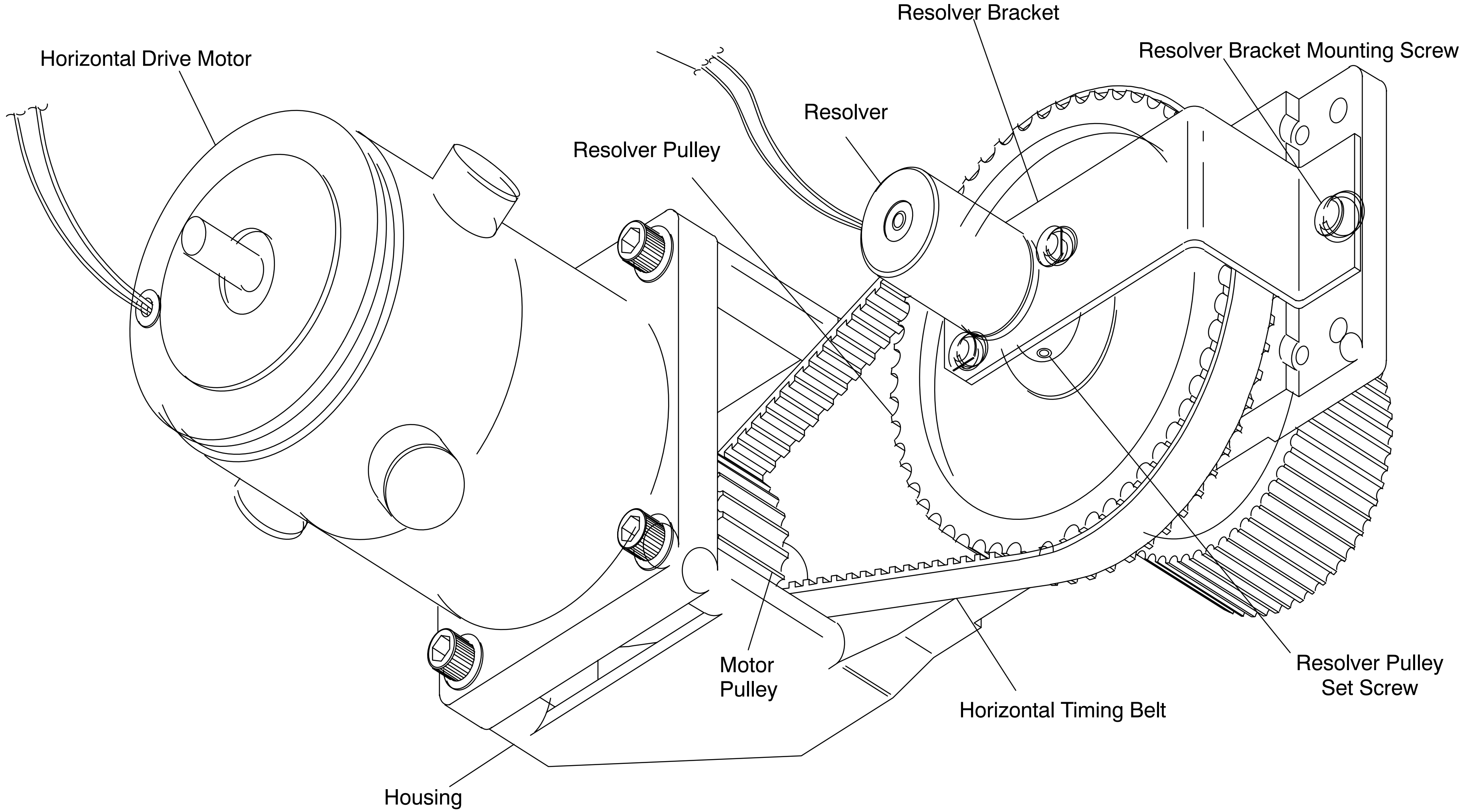


FIGURE 207 Horizontal Drive Assembly

7. Slide the Timing Belt off the Resolver Pulley and remove it from the Motor Pulley.

TIMING BELT REPLACEMENT

> **NOTE**
>
> Timing Belt tension is factory–set. Do not loosen the Horizontal Drive Motor Mounting Screws unless absolutely necessary.

8. Slide the replacement Timing Belt over the Motor Pulley and onto the Resolver Pulley. Refer to Figure 207.
9. Install the Resolver and Bracket Assembly by reversing steps 5. and 6.
10. Calibrate Horizontal Position using the CC (Couch Calibration) section of the GPUSPY manual.
11. Remove Gantry power. Refer to the PowerUp/PowerDown procedure.
12. Replace the Patient Support Covers. Refer to the Patient Support Covers Removal/Replacement procedure.

HORIZONTAL "HARD" LIMIT SWITCH

Introduction: This procedure is used to remove and replace the Patient Support Horizontal Travel "Hard" Limit Switches.

Tools Required: 1. FSE tool kit

Estimated Time: 15 min.

LIMIT SWITCH REMOVAL PROCEDURE:

1. Apply Gantry power. Refer to the PowerUp/PowerDown procedure.
2. Raise the Patient Support to its highest position with the Gantry Control Panel Switches.
3. Remove Gantry power. Refer to the PowerUp/PowerDown procedure.
4. Remove the Patient Support right Side Rail Cover. Refer to the Patient Support Covers Removal/ Replacement procedure.
5. Manually move the Carbon Top until the Belt Hook is centered between the front and rear Pulleys.
6. Tag (if needed) and remove the Wires connected to the Switch(es) to be removed. See Figure 208.

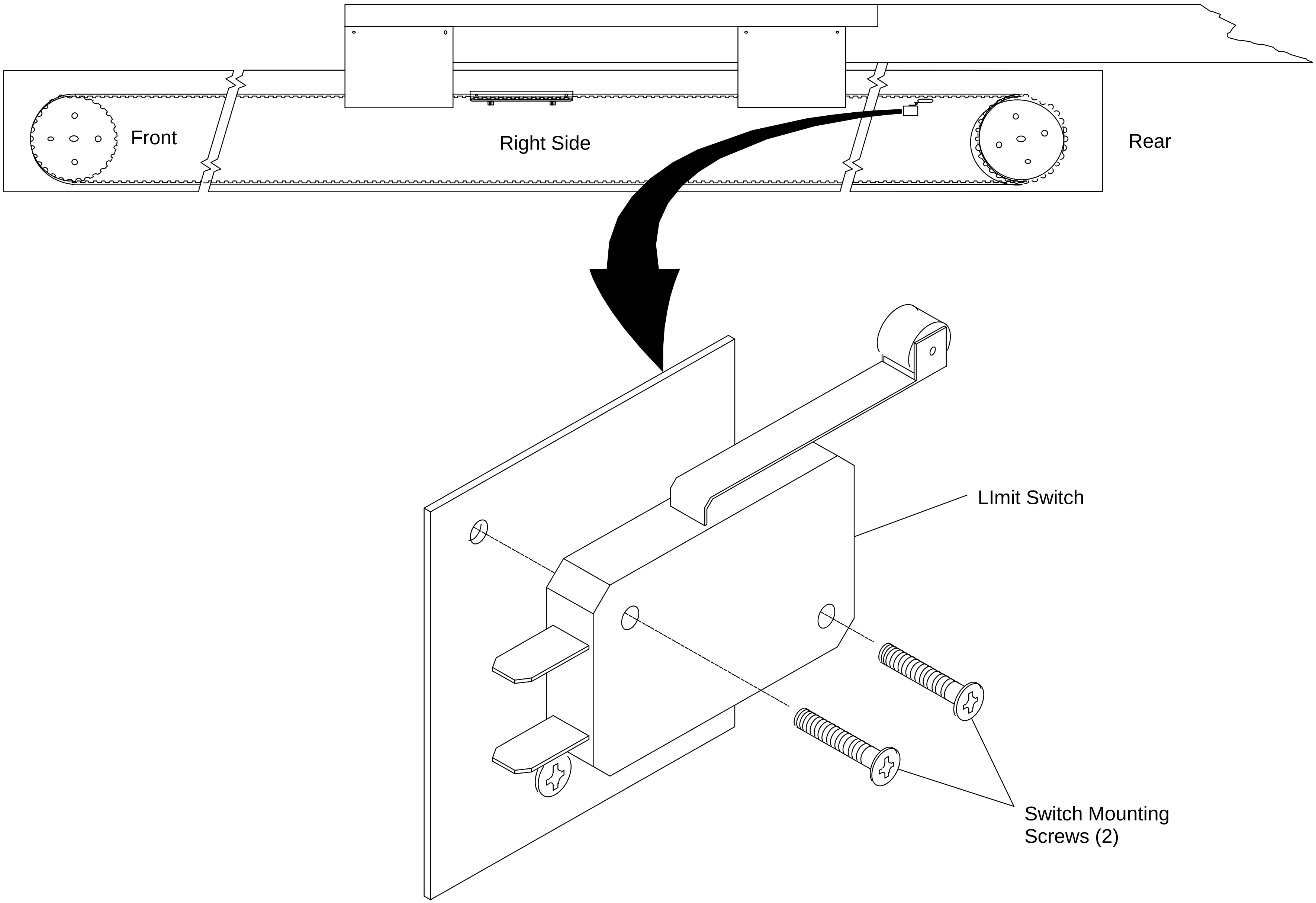


FIGURE 208 HORIZONTAL LIMIT SWITCH ASSEMBLY

7. Remove the two Switch Mounting Screws.
8. Remove the Switch(es).

LIMIT SWITCH REPLACEMENT

9. Install the Switch(es) using the two Mounting Screws.
10. Connect the Wires to the Limit Switch(es).
11. Apply Gantry power. Refer to the PowerUp/PowerDown procedure.
12. Calibrate the Horizontal Travel Limit Switch(es). Refer to the Horizontal Travel Limit Switches procedure in the Mechanical Calibration Manual.
13. Remove Gantry power. Refer to the PowerUp/PowerDown procedure in the Introduction section.
14. Replace the Patient Support right Side Rail Cover. Refer to the Patient Support Covers Removal/Replacement procedure.

VERTICAL DRIVE MOTOR

INTRODUCTION: This procedure is used to remove and replace the patient support vertical drive motor.

TOOLS REQUIRED: 1. FSE tool kit

MATERIAL REQUIRED: 1. Butt splices (P/N T18A–34, 22–16 AWG)

ESTIMATED TIME: 1 HR.

> ⚠ **WARNING**
>
> **DO NOT PERFORM THIS PROCEDURE IF THE VERTICAL SUPPORT BRACE IS NOT INSTALLED. REFER TO MANDATORY LETTER #376. FAILURE TO COMPLY MAY RESULT IN SERIOUS INJURY AND/OR DEATH TO SERVICE PERSONNEL.**

VERTICAL DRIVE MOTOR REMOVAL PROCEDURE:

1. Apply gantry power. Refer to the Powerup/Powerdown procedure.
2. Lower the patient support to its lowest position with the gantry control panel switches. This step is required for cover removal by the preferred method. If the patient support cannot be lowered, continue to step 3.

> ⚠ **WARNING**
>
> **USE EXTREME CAUTION WHEN OPERATING OR WORKING ON EQUIPMENT WITH COVERS REMOVED. DANGEROUS VOLTAGES ARE EXPOSED WHEN THE GANTRY COVERS ARE REMOVED. ACCIDENTAL CONTACT WITH THESE VOLTAGES MAY CAUSE SERIOUS INJURY AND/OR DEATH TO SERVICE PERSONNEL.**

3. Remove the patient support telescoping base covers. Refer to the Patient Support Covers Removal/ Replacement procedure.

⚠ WARNING

SCISSOR–ACTION MECHANISM IS EXPOSED WHEN THE PATIENT SUPPORT TELESCOPING BASE COVERS ARE REMOVED. KEEP EQUIPMENT AND PERSONNEL AWAY FROM THE MECHANICAL ACTION OF THESE PARTS. FAILURE TO COMPLY MAY RESULT IN EQUIPMENT DAMAGE AND/OR SERIOUS INJURY AND/OR DEATH TO SERVICE PERSONNEL.

4. Raise the patient support to its maximum height with the gantry control panel switches.
5. Locate the green safety vertical support brace mounted to face of scissor. Cut the tie–wrap that confines the brace cable and remove the brace from the holder. Install the brace as shown in Figure 209.

⚠ WARNING

DO NOT DRIVE THE SCISSORS CARRIAGE INTO THE VERTICAL SUPPORT BRACE. FAILURE TO COMPLY WILL CAUSE SERIOUS DAMAGE TO THE BALLSCREW.

6. Lower the patient support slowly. Stop the scissor carriage just before it touches the brace. Do not drive carriage into brace as this may cause misalignment of the vertical drive mechanism. See Figure 209.
7. Remove gantry power. Refer to the PowerUp/PowerDown procedure.
8. Disconnect P415 from the control pcb on the patient support control assembly.
9. Remove the vertical motor ground wire (green/yellow) from the patient support control assembly.

⚠ CAUTION

THE SET SCREWS IN THE MOTOR MOUNTING PLATE ARE FACTORY ADJUSTED. DO NOT ATTEMPT TO ADJUST THESE SET SCREWS. FAILURE TO COMPLY MAY CAUSE MISALIGNMENT AND SERIOUS DAMAGE TO THE MOTOR AND DRIVE MECHANISM.

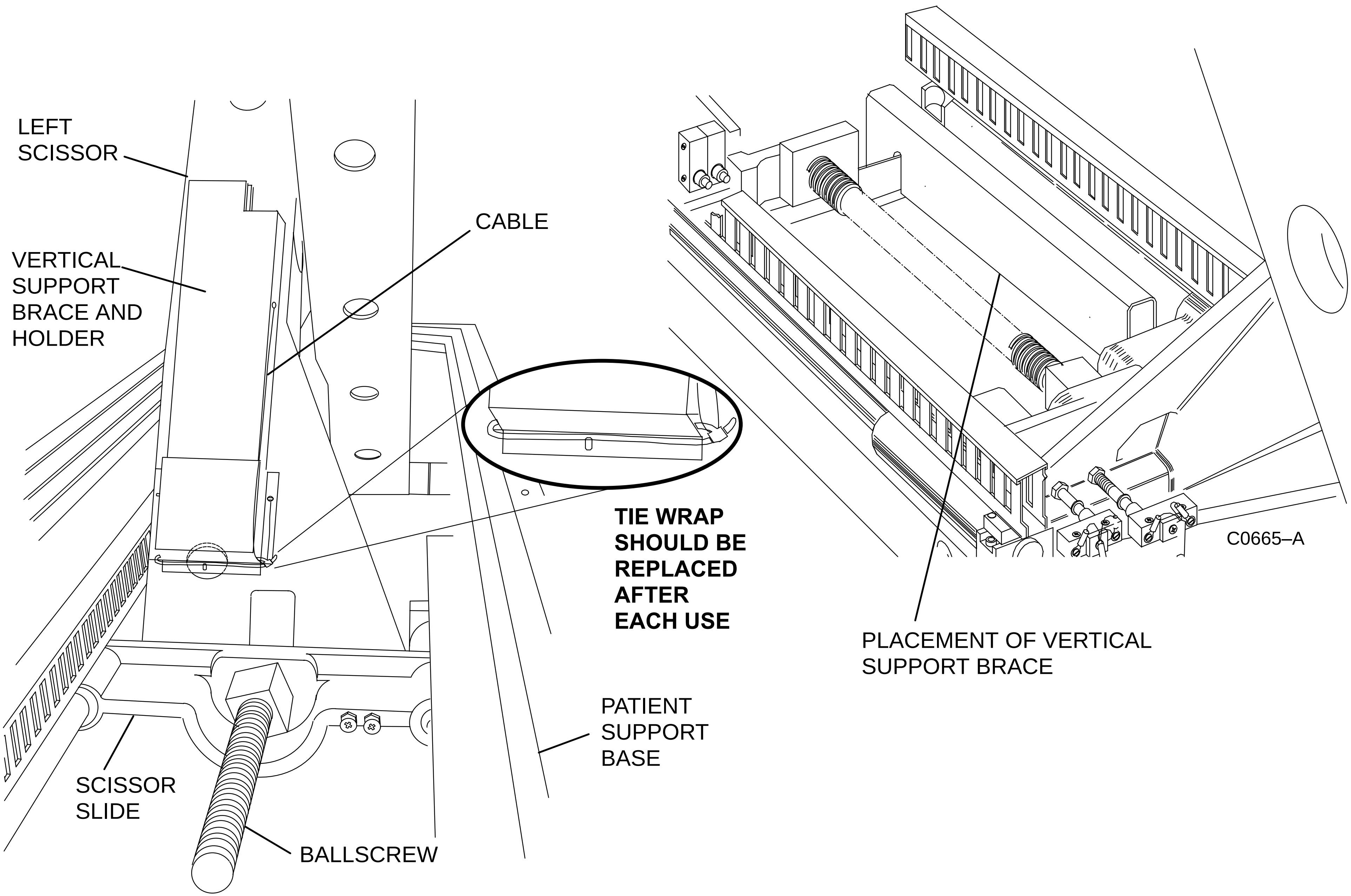


FIGURE 209 VERTICAL SUPPORT BRACE HOLDER & PLACEMENT

10. Remove the four mounting plate screws and washers. See Figure 210.

> ⚠ **WARNING**
>
> **WITH THE VERTICAL MOTOR REMOVED, THE PATIENT SUPPORT IS UNCONTROLLED AND FREE TO FALL QUICKLY. SECURE THE SCISSORS MECHANISM WITH THE VERTICAL SUPPORT BRACE TO PREVENT BALL SCREW MOVEMENT AND THE SUPPORT TOP FROM DROPPING. FAILURE TO COMPLY MAY RESULT IN EQUIPMENT DAMAGE, SERIOUS INJURY OR DEATH TO SERVICE PERSONNEL.**

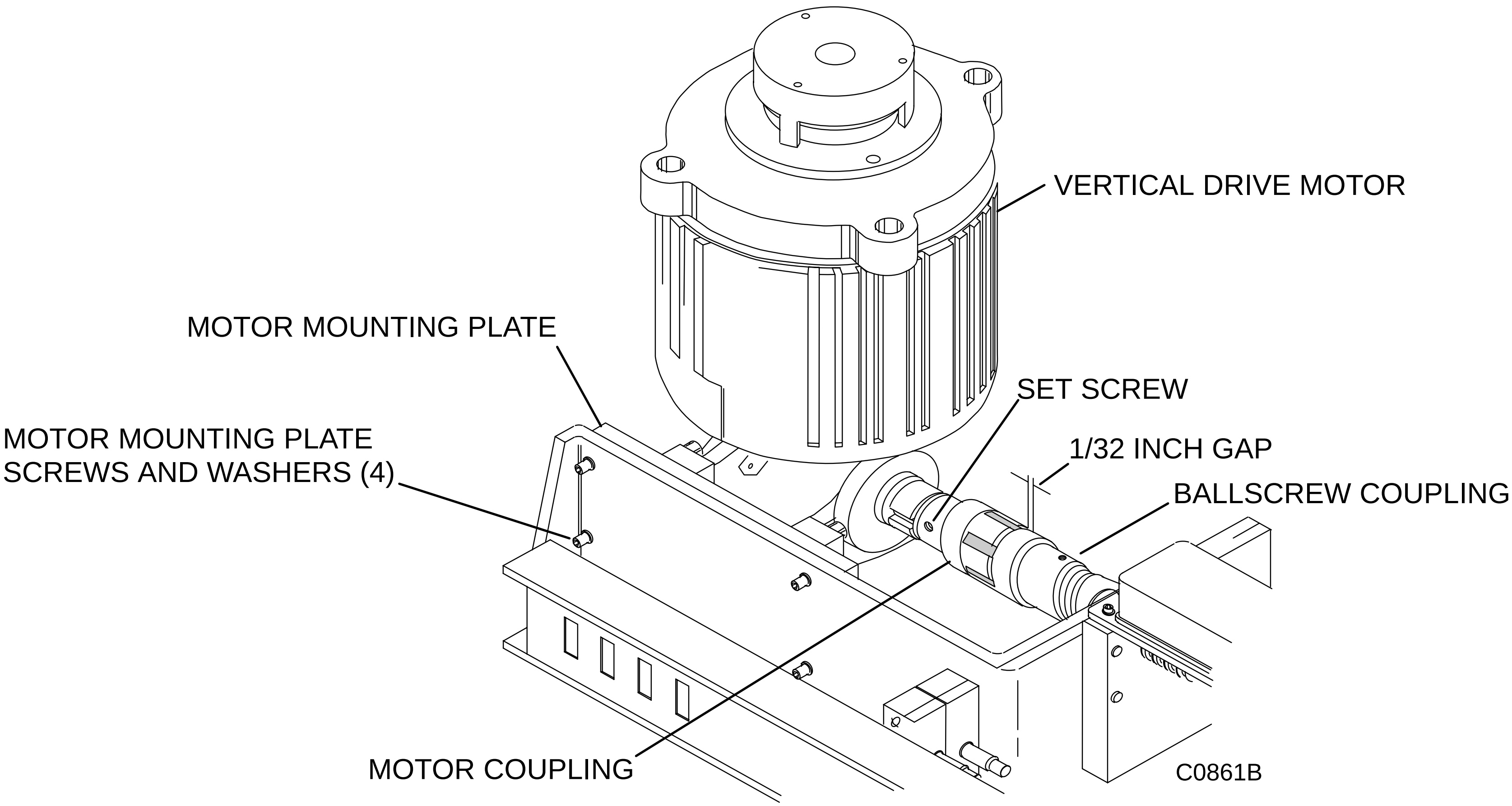


FIGURE 210 VERTICAL DRIVE MOTOR MOUNTING PLATE SCREWS

11. Grasp the vertical motor, rotate it slightly away from the frame, and slide it to the rear of the patient support until the couplings disengage (shaft keys, plastic insert, end pieces). Remove the motor.

12. Remove the four mounting bolts and washers from the motor side of the vertical motor mounting plate. See Figure 211.

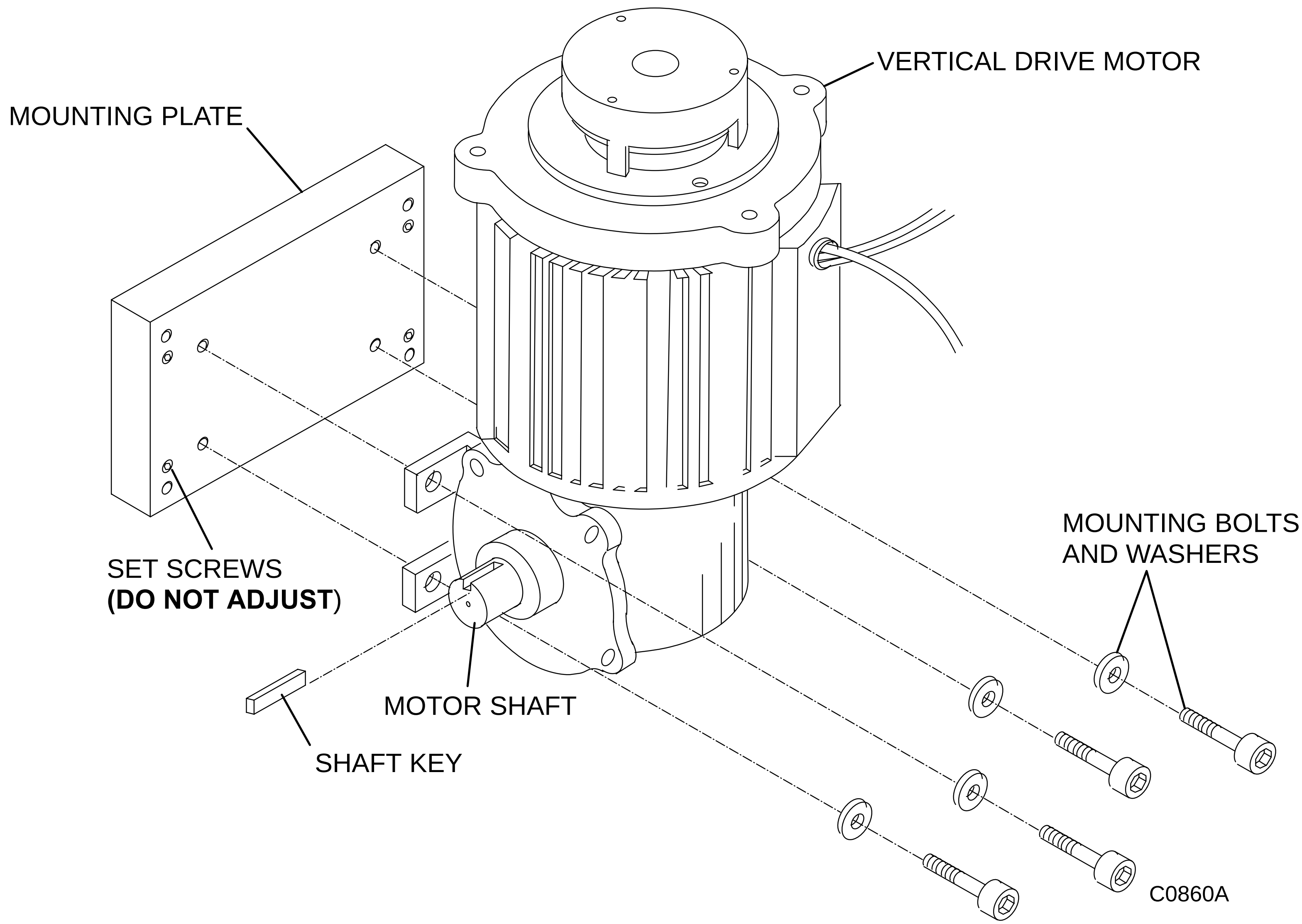


FIGURE 211   VERTICAL MOTOR MOUNTING PLATE

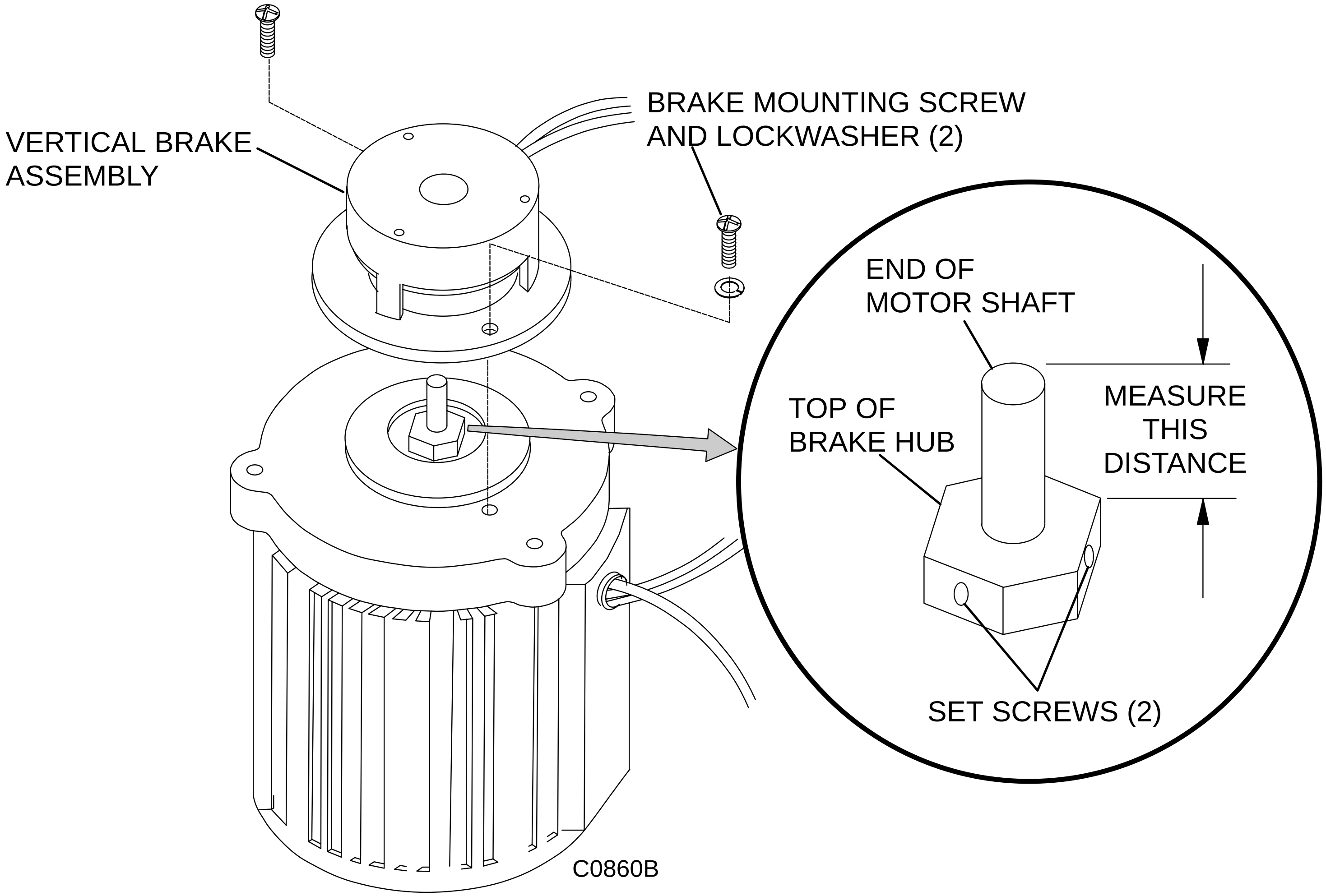


FIGURE 212 VERTICAL BRAKE ASSEMBLY

13. Remove the vertical brake assembly and the gasket (if present). DO NOT cut the brake wires as stated in the brake procedure. Refer to the Vertical Brake removal/replacement procedure and Figure 212.
14. Measure and record the distance from the top of the brake hub to the end of the motor shaft. Refer to Figure 212.
15. Remove the brake hub at the top of the motor.
16. Cut the motor wires 6” from connector P415. See Figure 214.
17. Loosen the set screw on the motor coupling and remove the coupling from the shaft. See Figure 213.

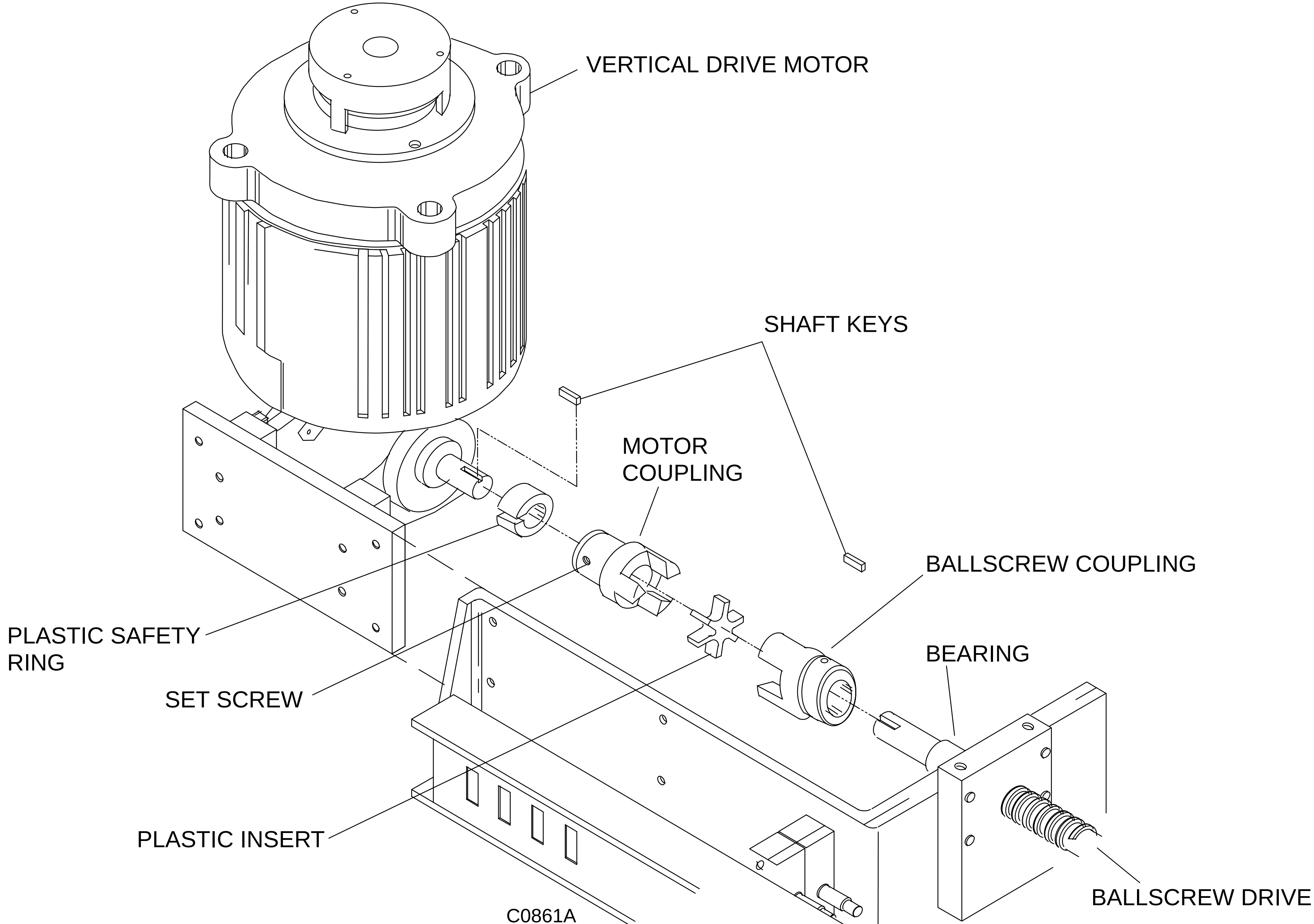


FIGURE 213 COUPLING EXPLODED VIEW

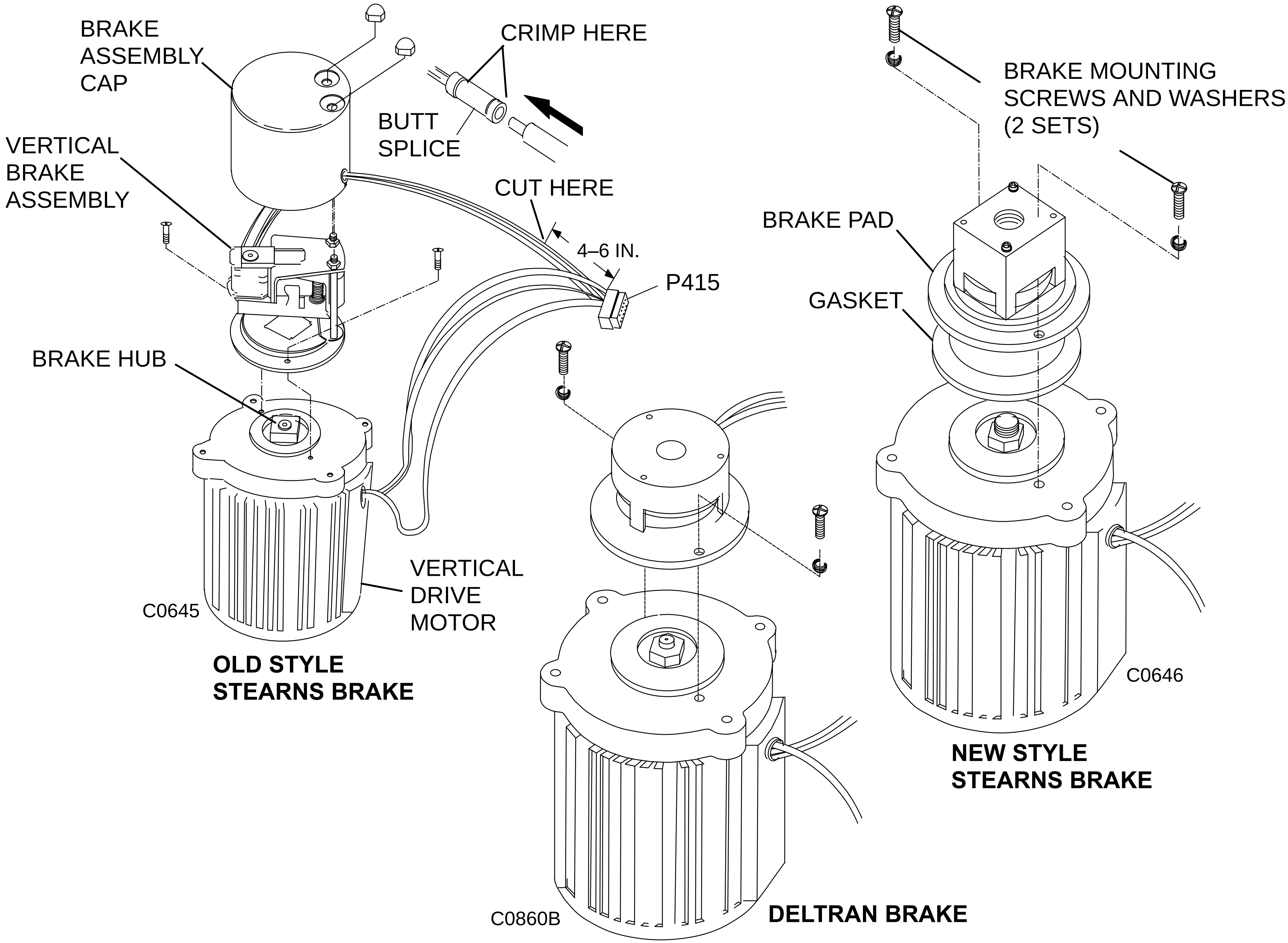


FIGURE 214 VERTICAL MOTOR AND BRAKE

VERTICAL DRIVE MOTOR REPLACEMENT:

18. Install the brake hub onto the top shaft of the replacement motor in the same location as recorded in step 14. Refer to Figure 212.
19. Install the brake assembly onto the replacement motor. Refer to the Vertical Brake Removal/Replacement procedure.
20. Connect the wires from the replacement motor to the wires that were cut from connector P415 using crimp–type butt splices.
21. Secure the motor to the mounting plate with the four mounting bolts and their respective washers. Refer to Figure 211.
22. Slide the plastic safety ring, shaft key and motor coupling onto the motor shaft. Do not tighten the motor coupling set screw at this time. Refer to Figure 213.
23. Place the plastic insert inside the jaw of the ballscrew shaft coupling.

> ⚠ **CAUTION**
>
> CRITICAL ALIGNMENT. DO NOT USE EXCESSIVE FORCE WHEN TIGHTENING THE MOUNTING PLATE SCREWS. FAILURE TO COMPLY MAY CAUSE MISALIGNMENT AND SERIOUS DAMAGE TO THE MOTOR AND DRIVE MECHANISM.

24. Maneuver the motor assembly into position and secure the motor mounting plate to the patient support frame with the four motor mounting plate screws and their washers. Refer to Figure 210.
25. On the motor shaft, slide the motor coupling along the shaft until there is a 1/32 inch gap between the motor coupling and the ballscrew coupling. Refer to Figure 210.
26. Tighten the set screws on the motor coupling half.
27. Connect cable P415 at the control pcb.

> ⚠ **WARNING**
>
> **USE EXTREME CAUTION WHEN OPERATING OR WORKING ON EQUIPMENT WITH COVERS REMOVED. DANGEROUS VOLTAGES ARE EXPOSED WHEN THE GANTRY COVERS ARE REMOVED. ACCIDENTAL CONTACT WITH THESE VOLTAGES MAY CAUSE SERIOUS INJURY AND/OR DEATH TO SERVICE PERSONNEL.**

28. Apply gantry power. Refer to the PowerUp/PowerDown procedure.
29. Raise the patient support to its maximum height with the gantry control panel switches.

> ⚠ **WARNING**
>
> **LOWERING THE PATIENT SUPPORT WITH THE VERTICAL SUPPORT BRACE IN PLACE WILL DAMAGE THE BALLSCREW DRIVE. FAILURE TO REMOVE THE VERTICAL SUPPORT BRACE WILL RESULT IN EQUIPMENT DAMAGE AND/OR SERIOUS INJURY AND/OR DEATH TO SERVICE PERSONNEL.**
>
> **SCISSOR–ACTION MECHANISM IS EXPOSED WHEN THE PATIENT SUPPORT TELESCOPING BASE COVERS ARE REMOVED. KEEP EQUIPMENT AND PERSONNEL AWAY FROM THE MECHANICAL ACTION OF THESE PARTS. FAILURE TO COMPLY MAY RESULT IN EQUIPMENT DAMAGE AND/OR SERIOUS INJURY AND/OR DEATH TO SERVICE PERSONNEL.**

30. Remove the vertical support brace and return it to its holder on the face of the scissor. Refer to Figure 209.
31. Secure the brace cable with an 8 inch long tie–wrap fed through the bottom of the support brace holder. Refer to Figure 209.
32. Raise and lower the patient support to verify operation of the motor with the gantry control panel switches. If an audible knocking is detected in the vertical movement, loosen the four motor mounting plate screws by one turn each.

> ⚠ **WARNING**
>
> **LOOSEN MOTOR MOUNTING PLATE SCREWS ONE TURN ONLY. FAILURE TO COMPLY MAY CAUSE THE MOTOR AND BALLSCREW COUPLINGS TO DISENGAGE, RESULTING IN SERIOUS INJURY OR DEATH DUE TO COLLAPSE OF THE PATIENT SUPPORT.**

33. Move the patient support up and down with the gantry control panel switches, allowing motor to align itself. Hold motor in place and tighten the four motor mounting plate screws.
34. Repeat this process as necessary until the knocking is resolved.
35. Lower patient support to its lowest position with the gantry control panel switches.
36. Remove gantry power. Refer to the PowerUp/PowerDown procedure.
37. Replace the patient support telescoping base covers. Refer to the Patient Support Covers Removal/ Replacement procedure.

VERTICAL BRAKE

INTRODUCTION: This procedure is used to remove and replace the brake on the patient support vertical drive motor.

TOOLS NEEDED: 1. FSE Tool Kit

ESTIMATED TIME: 30 min

> ⚠ **WARNING**
>
> **DO NOT PERFORM THIS PROCEDURE IF THE VERTICAL SUPPORT BRACE IS NOT INSTALLED. REFER TO MANDATORY LETTER #376. FAILURE TO COMPLY MAY RESULT IN SERIOUS INJURY AND/OR DEATH TO SERVICE PERSONNEL.**

VERTICAL BRAKE REMOVAL PROCEDURE:

1. Apply gantry power. Refer to the PowerUp/PowerDown.
2. Position the carbon top by hand against its front mechanical stop.
3. Lower the patient support to its lowest position with the gantry control panel switches. This step is required for cover removal by the preferred method. If the patient support cannot be lowered, continue to step 4.

> ⚠ **WARNING**
>
> **USE EXTREME CAUTION WHEN OPERATING OR WORKING ON EQUIPMENT WITH COVERS REMOVED. DANGEROUS VOLTAGES ARE EXPOSED WHEN THE GANTRY COVERS ARE REMOVED. ACCIDENTAL CONTACT WITH THESE VOLTAGES MAY CAUSE SERIOUS INJURY AND/OR DEATH TO SERVICE PERSONNEL.**

4. Remove the patient support telescoping base covers. Refer to the Patient Support Covers Removal/Replacement procedure.

⚠ **WARNING**

SCISSOR–ACTION MECHANISM IS EXPOSED WHEN THE PATIENT SUPPORT TELESCOPING BASE COVERS ARE REMOVED. KEEP EQUIPMENT AND PERSONNEL AWAY FROM THE MECHANICAL ACTION OF THESE PARTS. FAILURE TO COMPLY MAY RESULT IN EQUIPMENT DAMAGE AND/OR SERIOUS INJURY AND/OR DEATH TO SERVICE PERSONNEL.

5. Raise the patient support to its highest position with the gantry control panel switches.
6. Locate the green safety vertical support brace mounted to face of scissor. Cut the tie–wrap that confines the brace cable and remove the brace from the holder. Install the brace as shown in Figure 215.

⚠ **WARNING**

DO NOT DRIVE THE SCISSORS CARRIAGE INTO THE VERTICAL SUPPORT BRACE. FAILURE TO COMPLY WILL CAUSE SERIOUS DAMAGE TO THE BALLSCREW.

7. Lower the patient support slowly. Stop the scissor carriage just before it touches the brace. Do not drive carriage into brace as this may cause misalignment of the vertical drive mechanism. See Figure 215.
8. Remove gantry power. Refer to the PowerUp/PowerDown.
9. Remove the connector from J415 on the patient support control pcb.
10. Cut tie wraps as required to completely free the brake cable. Note the wire orientation; it will be important during replacement.
11. Cut the two wires in the cable that connect to the brake assembly. Cut the wires four to six inches from connector P415. See Figure 216.

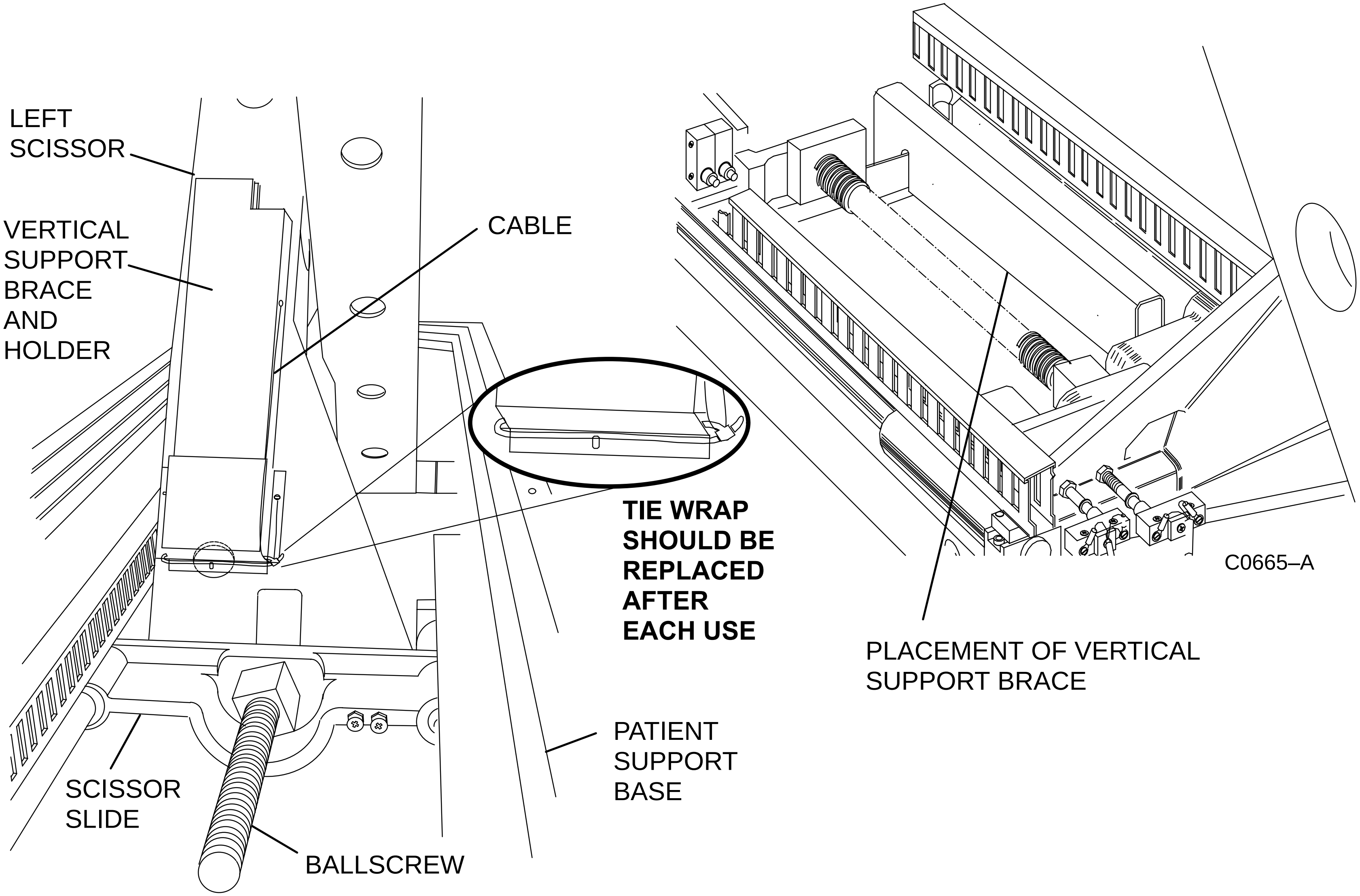


FIGURE 215 VERTICAL SUPPORT BRACE HOLDER & PLACEMENT

> **NOTE**
>
> Some new style Stearns brakes have an in–line plug. If the faulty and new brakes both have this feature, simply unplug the old brake and plug in the new one.

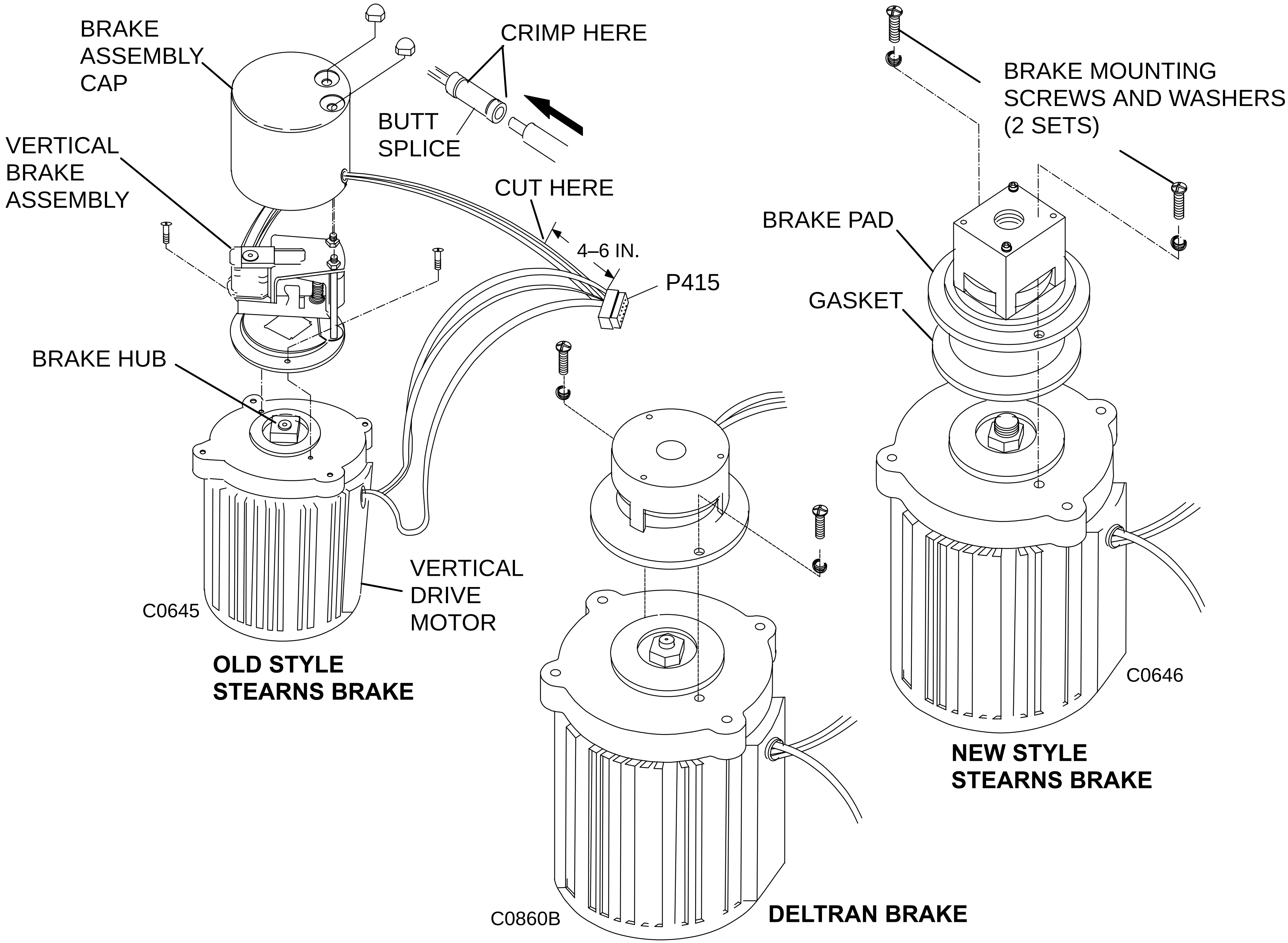


FIGURE 216 VERTICAL MOTOR AND BRAKE

Old Style Stearns Brake:

12. Remove the two cap nuts which secure the brake assembly cover. Remove the brake cover.
13. Remove the two screws that hold the brake assembly to the motor casing. Note the brake orientation and remove the brake from the motor.
14. Loosen the set screws on the brake hub and remove the hub from the motor shaft.

New Style Stearns Brake and Deltran Brake:

15. Remove the two screws and lockwashers that hold the brake assembly to the motor casing. Note the brake orientation and remove the brake from the motor.
16. If the replacement brake is different than the original brake, loosen the set screws and remove the hub from the motor shaft.

VERTICAL BRAKE REPLACEMENT:

17. Strip 1/4” of insulation from the end of the leads on the replacement brake.
18. Strip 1/4” of insulation from the end of the cut leads originating at connector P415.
19. Use a crimp–type butt splice to connect one wire from connector P415 to one wire from the brake assembly.
20. Splice the remaining wire from P415 to the remaining wire from the brake assembly using a butt splice.

New Style Stearns Brake:

21. Ensure the keystock is in place. Install the new brake hub onto the motor shaft by locating the bottom of the hub 40 to 50 thousandths above the shoulder of the motor, tighten set screws. See Figure 217.
22. Place the new brake assembly onto the motor in the same orientation as noted in step 13. or 15.
23. Secure the brake assembly to the motor with the two screws and lockwashers. Refer to Figure 216.

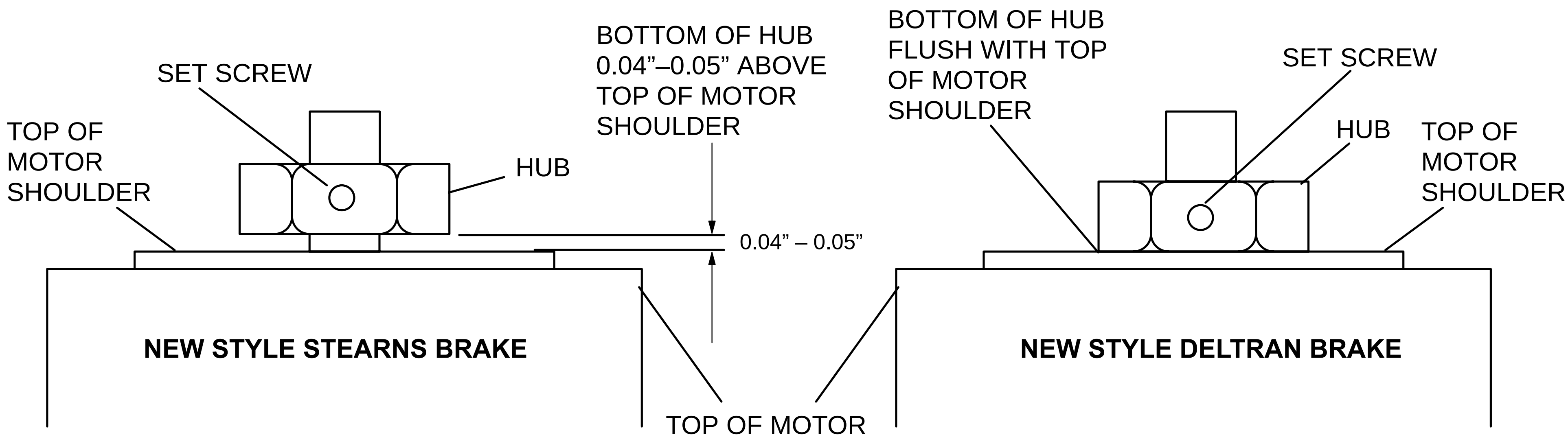


FIGURE 217 NEW STYLE STEARNS AND DELTRAN HUB LOCATION

New Style Deltran Brake:

24. Ensure the keystock is in place. Install the new brake hub onto the motor shaft by locating the bottom of the hub flush with the shoulder of the motor, tighten set screws. Refer to Figure 217.

25. Place the new brake assembly onto the motor in the same orientation as noted in step 13. or 15.

26. Secure the brake assembly to the motor with the two screws and lockwashers. Refer to Figure 216.

27. Connect P415 to J415 on patient support control pcb.

28. Temporarily secure the wires away from the mechanical moving parts until initial power checks are completed.

29. Apply gantry power. Refer to the PowerUp/PowerDown in the introduction section.

⚠ WARNING

USE EXTREME CAUTION WHEN OPERATING OR WORKING ON EQUIPMENT WITH COVERS REMOVED. DANGEROUS VOLTAGES ARE EXPOSED WHEN THE GANTRY COVERS ARE REMOVED. ACCIDENTAL CONTACT WITH THESE VOLTAGES MAY CAUSE SERIOUS INJURY AND/OR DEATH TO SERVICE PERSONNEL.

⚠ WARNING

SCISSOR–ACTION MECHANISM IS EXPOSED WHEN THE PATIENT SUPPORT TELESCOPING BASE COVERS ARE REMOVED. KEEP EQUIPMENT AND PERSONNEL AWAY FROM THE MECHANICAL ACTION OF THESE PARTS. FAILURE TO COMPLY MAY RESULT IN EQUIPMENT DAMAGE AND/OR SERIOUS INJURY AND/OR DEATH TO SERVICE PERSONNEL.

30. Raise the patient support to its maximum height with the gantry control panel switches.

⚠ WARNING

LOWERING THE PATIENT SUPPORT WITH THE VERTICAL SUPPORT BRACE IN PLACE WILL DAMAGE THE BALLSCREW DRIVE. FAILURE TO REMOVE THE VERTICAL SUPPORT BRACE WILL RESULT IN EQUIPMENT DAMAGE AND/OR SERIOUS INJURY AND/OR DEATH TO SERVICE PERSONNEL.

31. Remove the vertical support brace and return it to its holder on the face of the scissor. Refer to Figure 215.
32. Secure the brace cable with an 8 inch long tie–wrap fed through the bottom of the support brace holder. Refer to Figure 215.

⚠ WARNING

LOOSEN MOTOR MOUNTING PLATE SCREWS ONE TURN ONLY. FAILURE TO COMPLY MAY CAUSE THE MOTOR AND BALLSCREW COUPLINGS TO DISENGAGE, RESULTING IN SERIOUS INJURY OR DEATH DUE TO COLLAPSE OF THE PATIENT SUPPORT.

33. Use the gantry control panel switches to check the operation of the brake. First raise the patient support a few centimeters, then lower it a few centimeters. Verify that vertical motion stops almost immediately when the switches are released. Vertical down may coast about 3.5mm.

NOTE

If the brake makes an excessive amount of noise or the patient support does not stop immediately, check the wiring connections and the position of the brake hub. If the brake hub is not properly aligned, the brake will make a loud chattering noise and may fail to stop the patient support.

34. Tie wrap all wires and cables in same manner and position as they were originally.
35. Lower the patient support to its lowest vertical position using the gantry control panel switches.
36. Remove gantry power. Refer to the PowerUp/PowerDown.
37. Replace the telescoping bas covers. Refer to the Patient Support Covers Removal/Replacement procedure.

VERTICAL TRANSDUCER

Introduction: This procedure is used to remove and replace the Vertical Transducer in the Patient Support.

Tools Required: 1. FSE tool kit

Estimated Time: 20 min

VERTICAL TRANSDUCER REMOVAL

1. Position the Carbon Top by hand against its rear Mechanical Stop.
2. Apply Gantry power. Refer to the PowerUp/PowerDown procedure.
3. Lower the Patient Support to its lowest position with the Gantry Control Panel Switches.

> ⚠ **WARNING**
>
> **USE EXTREME CAUTION WHEN OPERATING OR WORKING ON EQUIPMENT WITH COVERS REMOVED. DANGEROUS VOLTAGES ARE EXPOSED WHEN THE GANTRY COVERS ARE REMOVED. FAILURE TO COMPLY CAN CAUSE SERIOUS INJURY OR DEATH TO PERSONNEL.**

4. Remove the top section only of the Patient Support Telescoping Base Covers. Refer to the Patient Support Covers Removal/Replacement procedure.

> ⚠ **WARNING**
>
> **SCISSOR–ACTION MECHANISM IS EXPOSED WHEN THE PATIENT SUPPORT TELESCOPING BASE COVERS ARE REMOVED. KEEP EQUIPMENT AND PERSONNEL AWAY FROM THE MECHANICAL ACTION OF THESE PARTS. FAILURE TO COMPLY CAN CAUSE SERIOUS DAMAGE TO EQUIPMENT AND/OR SERIOUS INJURY OR DEATH TO PERSONNEL.**

5. Raise the Patient Support to its highest position with the Gantry Control Panel Switches.
6. Remove Gantry power. Refer to the PowerUp/PowerDown procedure.

> **⚠ CAUTION**
>
> THE TRANSDUCER WIRE IS UNDER SPRING TENSION. WHEN DISCONNECTING OR CONNECTING THE WIRE, HOLD THE END SECURELY AND CAREFULLY LOWER OR RAISE IT SO THAT THE CLIP DOES NOT SNAP INTO THE TRANSDUCER HOUSING. FAILURE TO COMPLY MAY RESULT IN DAMAGE TO THE EQUIPMENT.

7. Remove the Transducer Hook from the Eyebolt on the bottom of the Subframe. See Figure 218.
8. Remove the Connector from J406 on the Transducer Assembly.

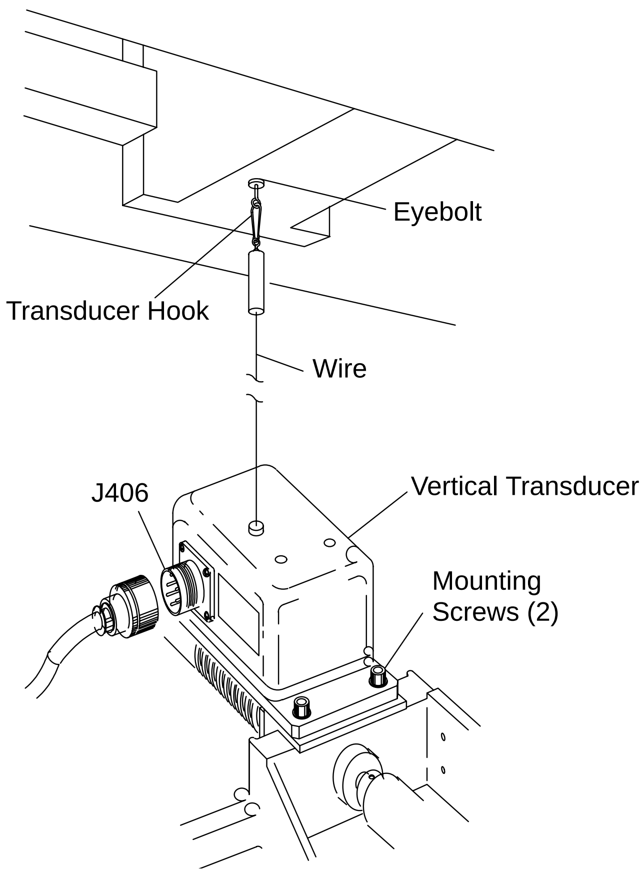


FIGURE 218 Patient Support Vertical Transducer

9. Remove the two Socket Head Cap Screws from the Transducer base. Remove the Transducer from the Patient Support.

VERTICAL TRANSDUCER REPLACEMENT

10. Position replacement Transducer on Base support and install the two Socket Head Mounting Screws.
11. Reconnect the Cable to J406 on the Transducer.
12. Attach the Transducer Hook to the Eyebolt on the bottom of the Subframe.
13. Apply Gantry power. Refer to the PowerUp/PowerDown procedure.

> ⚠ **WARNING**
>
> **USE EXTREME CAUTION WHEN OPERATING OR WORKING ON EQUIPMENT WITH COVERS REMOVED. DANGEROUS VOLTAGES ARE EXPOSED WHEN THE GANTRY COVERS ARE REMOVED. FAILURE TO COMPLY CAN CAUSE SERIOUS INJURY OR DEATH TO PERSONNEL.**

> ⚠ **WARNING**
>
> **SCISSOR–ACTION MECHANISM IS EXPOSED WHEN THE PATIENT SUPPORT TELESCOPING BASE COVERS ARE REMOVED. KEEP EQUIPMENT AND PERSONNEL AWAY FROM THE MECHANICAL ACTION OF THESE PARTS. FAILURE TO COMPLY CAN CAUSE SERIOUS DAMAGE TO EQUIPMENT AND/OR SERIOUS INJURY OR DEATH TO PERSONNEL.**

14. Check the operation of the Vertical Transducer by raising and lowering the Patient Support. Verify that the Vertical Position Display increases or decreases in value in accordance with the motion.
15. Calibrate the Vertical Transducer. Refer to the Patient Support Vertical Position procedure in the Mecanical Calibration manual
16. Position the Patient Support to its lowest vertical position with the Gantry Control Panel Switches.
17. Remove Gantry power. Refer to the PowerUp/PowerDown procedure in the Introduction section.
18. Replace the Patient Support Telescoping Base Covers. Refer to the Patient Support Covers Removal/Replacement procedure.

VERTICAL LIMIT SWITCH

Introduction:	This procedure is used to remove and replace the Patient Support Vertical Limit Switches.
Tools Required:	1. FSE tool kit
	2. Offset Philips Screwdriver
Estimated Time	20 min.

Vertical Limit Switch Removal Procedure:

1. Apply Gantry power. Refer to the PowerUp/PowerDown procedure.
2. Lower the Patient Support to its lowest position using the Gantry Control Panel Switches.

> ⚠ **WARNING**
>
> **USE EXTREME CAUTION WHEN OPERATING OR WORKING ON EQUIPMENT WITH COVERS REMOVED. DANGEROUS VOLTAGES ARE EXPOSED WHEN THE GANTRY COVERS ARE REMOVED. FAILURE TO COMPLY CAN CAUSE SERIOUS INJURY OR DEATH TO PERSONNEL.**

3. Disconnect the Patient Support Telescoping Base Covers from the Subframe. Refer to the Patient Support Covers Removal/Replacement procedure.

> ⚠ **WARNING**
>
> **SCISSOR–ACTION MECHANISM IS EXPOSED WHEN THE PATIENT SUPPORT TELESCOPING BASE COVERS ARE REMOVED. KEEP EQUIPMENT AND PERSONNEL AWAY FROM THE MECHANICAL ACTION OF THESE PARTS. FAILURE TO COMPLY CAN CAUSE SERIOUS DAMAGE TO EQUIPMENT AND/OR SERIOUS INJURY OR DEATH TO PERSONNEL.**

4. Raise the Patient Support to the center of its vertical travel with the Gantry Control Panel Switches.

5. Remove Gantry power. Refer to the PowerUp/PowerDown procedure.
6. Before removing a Switch, record the color and terminal connection for all wires attached to the Switch.
7. Remove the two Switch Mounting Screws. See Figure 219.
8. Remove the Limit Switch(es).

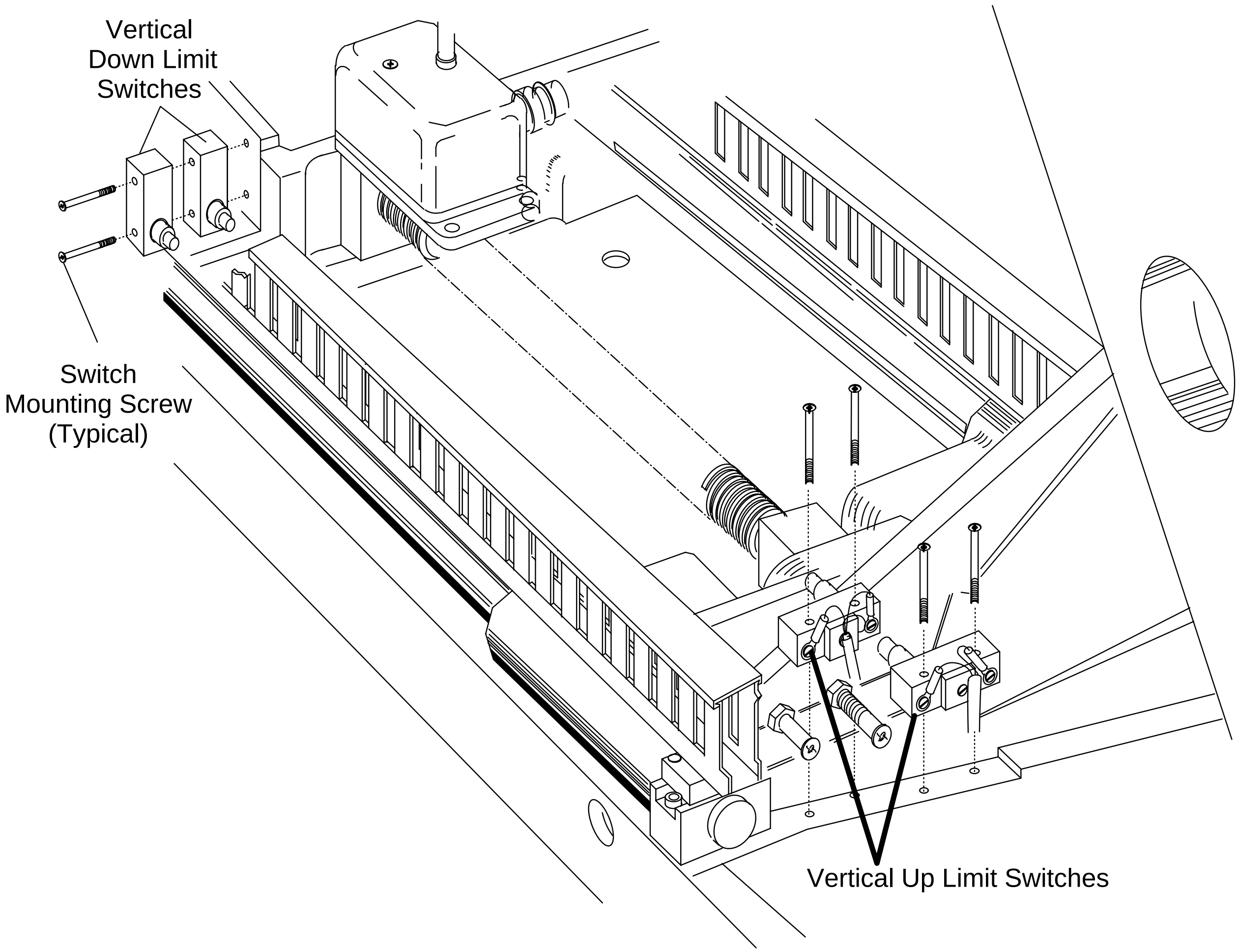


FIGURE 219 Vertical Limit Switches

Vertical Limit Switch Replacement:

9. Mount the replacement Switch and install the two Mounting Screws.

10. Attach the Switch wires according to the colors and terminals recorded in step 6.

> ⚠ **WARNING**
>
> **USE EXTREME CAUTION WHEN OPERATING OR WORKING ON EQUIPMENT WITH COVERS REMOVED. DANGEROUS VOLTAGES ARE EXPOSED WHEN THE GANTRY COVERS ARE REMOVED. FAILURE TO COMPLY CAN CAUSE SERIOUS INJURY OR DEATH TO PERSONNEL.**

11. Apply Gantry power. Refer to the PowerUp/PowerDown procedure.
12. Adjust the Limit Switch(es). Refer to Vertical Travel Limit Switches procedure in the Mechanical Calibration manual.
13. Lower the Patient Support to its lowest position with the Gantry Control Panel Switches.
14. Remove Gantry power. Refer to the PowerUp/PowerDown procedure.

> ⚠ **WARNING**
>
> **SCISSOR–ACTION MECHANISM IS EXPOSED WHEN THE PATIENT SUPPORT TELESCOPING BASE COVERS ARE REMOVED. KEEP EQUIPMENT AND PERSONNEL AWAY FROM THE MECHANICAL ACTION OF THESE PARTS. FAILURE TO COMPLY CAN CAUSE SERIOUS DAMAGE TO EQUIPMENT AND/OR SERIOUS INJURY OR DEATH TO PERSONNEL.**

15. Connect the Patient Support Telescoping Base Covers to the Subframe. Refer to the Patient Support Covers Removal/Replacement procedure.

PATIENT SUPPORT CONTROL PCB

Introduction: This procedure is used to remove and replace the Patient Support Control Circuit Board.

Tools Required:
1. FSE tool kit
2. Offset Philips Screwdriver

Estimated Time: 30 min.

PCB REMOVAL PROCEDURE:

1. Apply Gantry power. Refer to the PowerUp/PowerDown procedure.
2. Position the Patient Support to its lowest position with the Gantry Control Panel Switches. This step is to prepare for Cover removal. If the Patient Support cannot be lowered, continue to step 3.
3. Remove Gantry power. Refer to the PowerUp/PowerDown procedure.
4. Disconnect the Patient Support Telescoping Base Covers from the Subframe. Refer to the Patient Support Covers Removal/Replacement procedure.

> ⚠ **WARNING**
>
> **USE EXTREME CAUTION WHEN OPERATING OR WORKING ON EQUIPMENT WITH COVERS REMOVED. DANGEROUS VOLTAGES ARE EXPOSED WHEN THE GANTRY COVERS ARE REMOVED. FAILURE TO COMPLY CAN CAUSE SERIOUS INJURY OR DEATH TO PERSONNEL.**

5. Apply Gantry power. Refer to the PowerUp/PowerDown procedure.

> ⚠ **WARNING**
>
> **SCISSOR–ACTION MECHANISM IS EXPOSED WHEN THE PATIENT SUPPORT TELESCOPING BASE COVERS ARE REMOVED. KEEP EQUIPMENT AND PERSONNEL AWAY FROM THE MECHANICAL ACTION OF THESE PARTS. FAILURE TO COMPLY CAN CAUSE SERIOUS DAMAGE TO EQUIPMENT AND/OR SERIOUS INJURY OR DEATH TO PERSONNEL.**

6. Raise the Patient Support to its highest position with the Gantry Control Panel Switches.
7. Remove Gantry power. Refer to the PowerUp/PowerDown procedure.
8. Tag and remove all Cables connected to the Patient Support Control PCB.
9. Disconnect the Vertical Motor Ground Wire (GRN/YEL) from the Control Assembly Ground Stud next to the Transformer.
10. Remove the nine Mounting Screws that secure the PCB to the Control Assembly. See Figure 220.
11. Remove the Patient Support Control PCB.

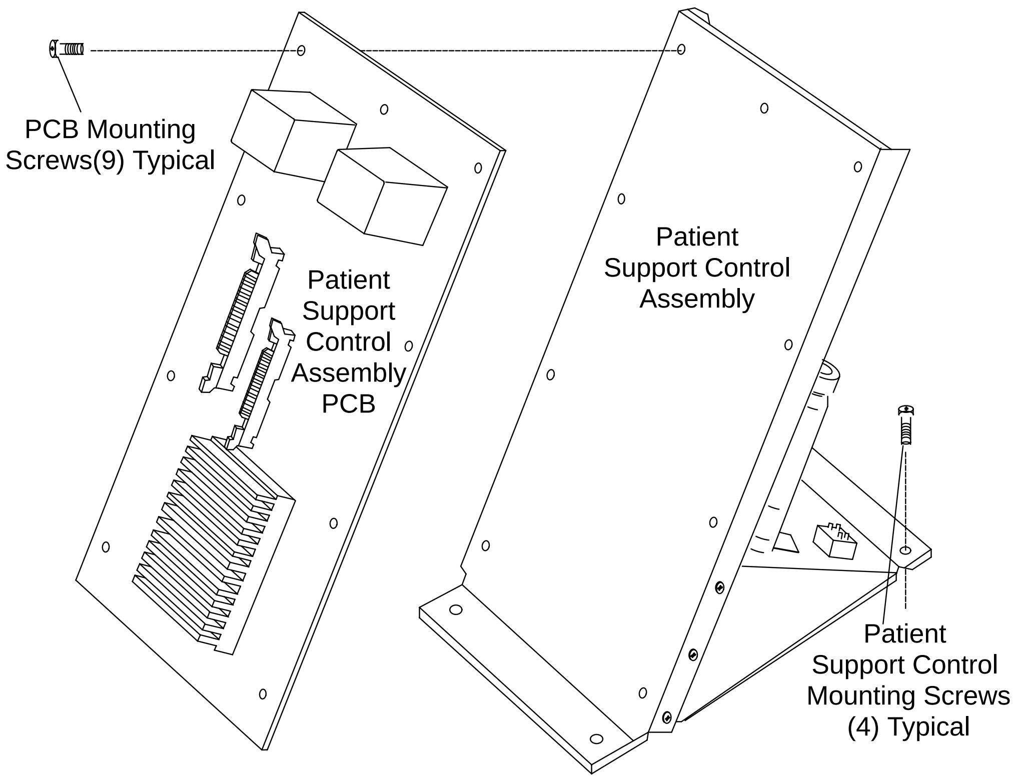


FIGURE 220 Patient Support Control PCB

PCB REPLACEMENT:

12. Fasten the Patient Support Control PCB to the Control Assembly using nine (9) Mounting Screws.
13. Connect the Cables to the Patient Support Control PCB.

> ⚠ **WARNING**
>
> **USE EXTREME CAUTION WHEN OPERATING OR WORKING ON EQUIPMENT WITH COVERS REMOVED. DANGEROUS VOLTAGES ARE EXPOSED WHEN THE GANTRY COVERS ARE REMOVED. FAILURE TO COMPLY CAN CAUSE SERIOUS INJURY OR DEATH TO PERSONNEL.**

> ⚠ **WARNING**
>
> **SCISSOR–ACTION MECHANISM IS EXPOSED WHEN THE PATIENT SUPPORT TELESCOPING BASE COVERS ARE REMOVED. KEEP EQUIPMENT AND PERSONNEL AWAY FROM THE MECHANICAL ACTION OF THESE PARTS. FAILURE TO COMPLY CAN CAUSE SERIOUS DAMAGE TO EQUIPMENT AND/OR SERIOUS INJURY OR DEATH TO PERSONNEL.**

14. Apply Gantry power. Refer to the PowerUp/PowerDown procedure.
15. Check the operation of the Patient Support by moving it vertically and horizontally with the Gantry Control Panel Switches.
16. Lower the Patient Support to its lowest position with the Gantry Control Panel Switches.
17. Remove Gantry power. Refer to the PowerUp/PowerDown procedure in the Introduction section.
18. Connect the Patient Support Telescoping Base Covers to the Subframe. Refer to the Patient Support Covers Removal/Replacement procedure.

BALLSCREW

INTRODUCTION: This procedure is used to remove and replace the patient support ballscrew mechanism.

TOOLS REQUIRED:

1. Standard FSE Toolkit
2. 1 1” Box or Crescent Wrench
3. #18 Drill bit, Drill
4. Gear Puller

Material Required:

1. 5/16”–18 x 1” Dog point set screw (P/N S620–124)
2. Mobile 28 grease (P/N 95749) or #7 Aeroshell (P/N 95617)

Estimated Time: 90 min.

> ⚠ **WARNING**
>
> **DO NOT PERFORM THIS PROCEDURE IF THE VERTICAL SUPPORT BRACE IS NOT INSTALLED. REFER TO MANDATORY LETTER #376. FAILURE TO COMPLY MAY RESULT IN SERIOUS INJURY AND/OR DEATH TO SERVICE PERSONNEL.**

1. Apply gantry power. Refer to the PowerUp\PowerDown procedure.

2. Lower the patient support to its lowest position with the gantry control panel switches. This step is required for cover removal by the preferred method. If the patient support cannot be lowered, continue to step 3.

3. Remove the patient support telescoping base covers. Refer to the Patient Support Covers Removal/Replacement procedure.

> ⚠ **WARNING**
>
> **USE EXTREME CAUTION WHEN OPERATING OR WORKING ON EQUIPMENT WITH COVERS REMOVED. DANGEROUS VOLTAGES ARE EXPOSED WHEN THE GANTRY COVERS ARE REMOVED. FAILURE TO COMPLY CAN CAUSE SERIOUS INJURY OR DEATH TO PERSONNEL.**

> ⚠ **WARNING**
>
> **SCISSOR–ACTION MECHANISM IS EXPOSED WHEN THE PATIENT SUPPORT TELESCOPING BASE COVERS ARE REMOVED. KEEP EQUIPMENT AND PERSONNEL AWAY FROM THE MECHANICAL ACTION OF THESE PARTS. FAILURE TO COMPLY CAN CAUSE SERIOUS DAMAGE TO EQUIPMENT AND/OR SERIOUS INJURY OR DEATH TO PERSONNEL.**

4. Raise the patient support to its maximum height with the gantry control panel switches.

5. Locate the green safety vertical support brace mounted to the face of scissor mechanism. Cut the tie–wrap that confines the brace cable and remove the brace from the holder. Install the brace as shown in Figure 221.

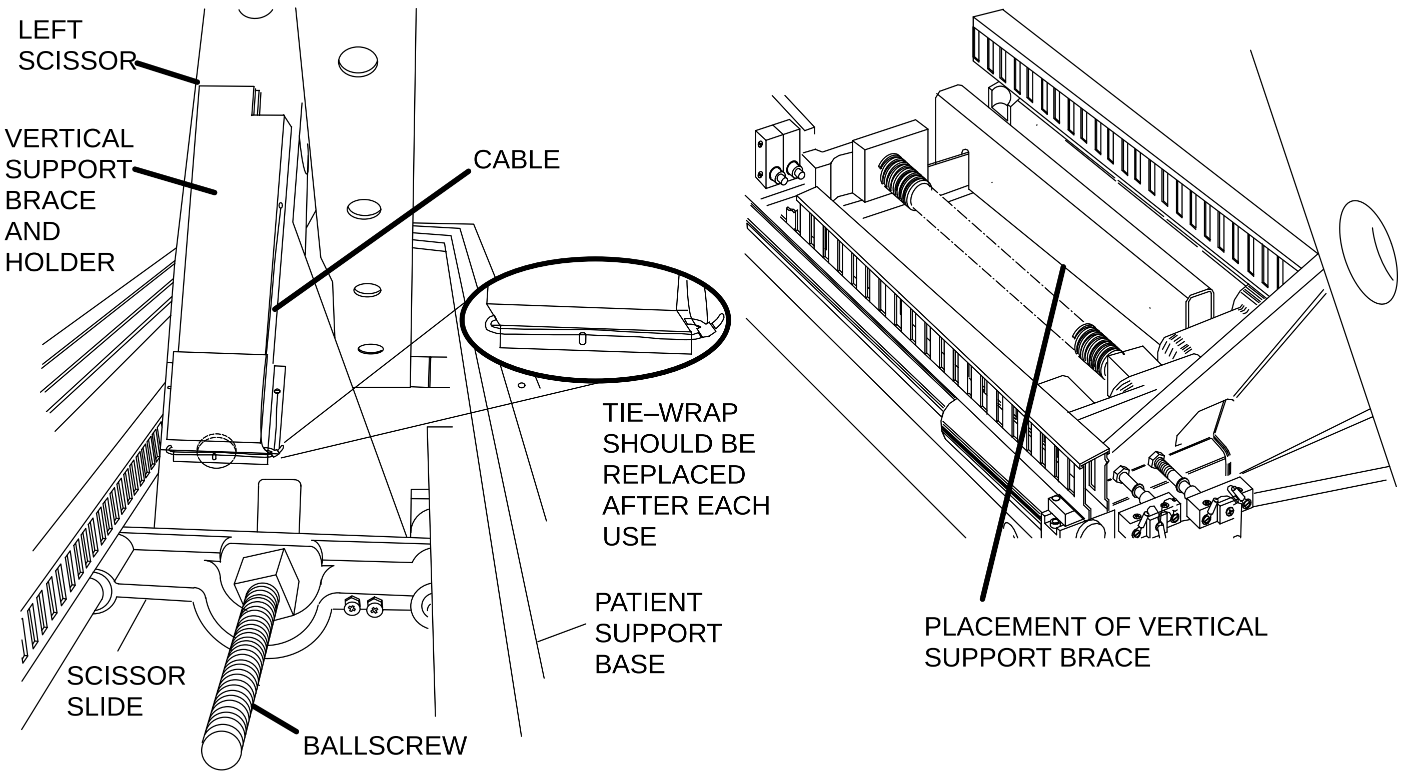


FIGURE 221 Metal Brace in Scissor Mechanism

⚠ WARNING

DO NOT DRIVE THE SCISSORS CARRIAGE INTO THE METAL BRACE. FAILURE TO COMPLY WILL CAUSE SERIOUS DAMAGE TO BRACE OR OTHER PATIENT SUPPORT COMPONENTS.

6. Lower the patient support slowly. Stop the scissor carriage just before it touches the brace. Do not drive carriage into brace as this may cause misalignment of the vertical drive mechanism.
7. Remove gantry power. Refer to the PowerUp/PowerDown procedure. **POWER OFF AT WALLBOX**.
8. Disconnect P415 from the control pcb on the patient support control assembly.
9. Remove the vertical motor ground wire (green/yellow) from the patient support control assembly.
10. Remove the four mounting plate screws and washers. See Figure 222.

⚠ WARNING

WITH THE VERTICAL MOTOR REMOVED, THE PATIENT SUPPORT IS UNCONTROLLED AND FREE TO FALL QUICKLY. THE PATIENT SUPPORT WILL COAST DOWN UNTIL THE SCISSORS SLIDE RESTS AGAINST THE VERTICAL SUPPORT BRACE. EXCERCISE EXTREME CAUTION WHILE THE BALLSCREW AND COUPLING ARE TURNING. FAILURE TO COMPLY MAY RESULT IN EQUIPMENT DAMAGE AND/OR SERIOUS INJURY OR DEATH TO SERVICE PERSONNEL.

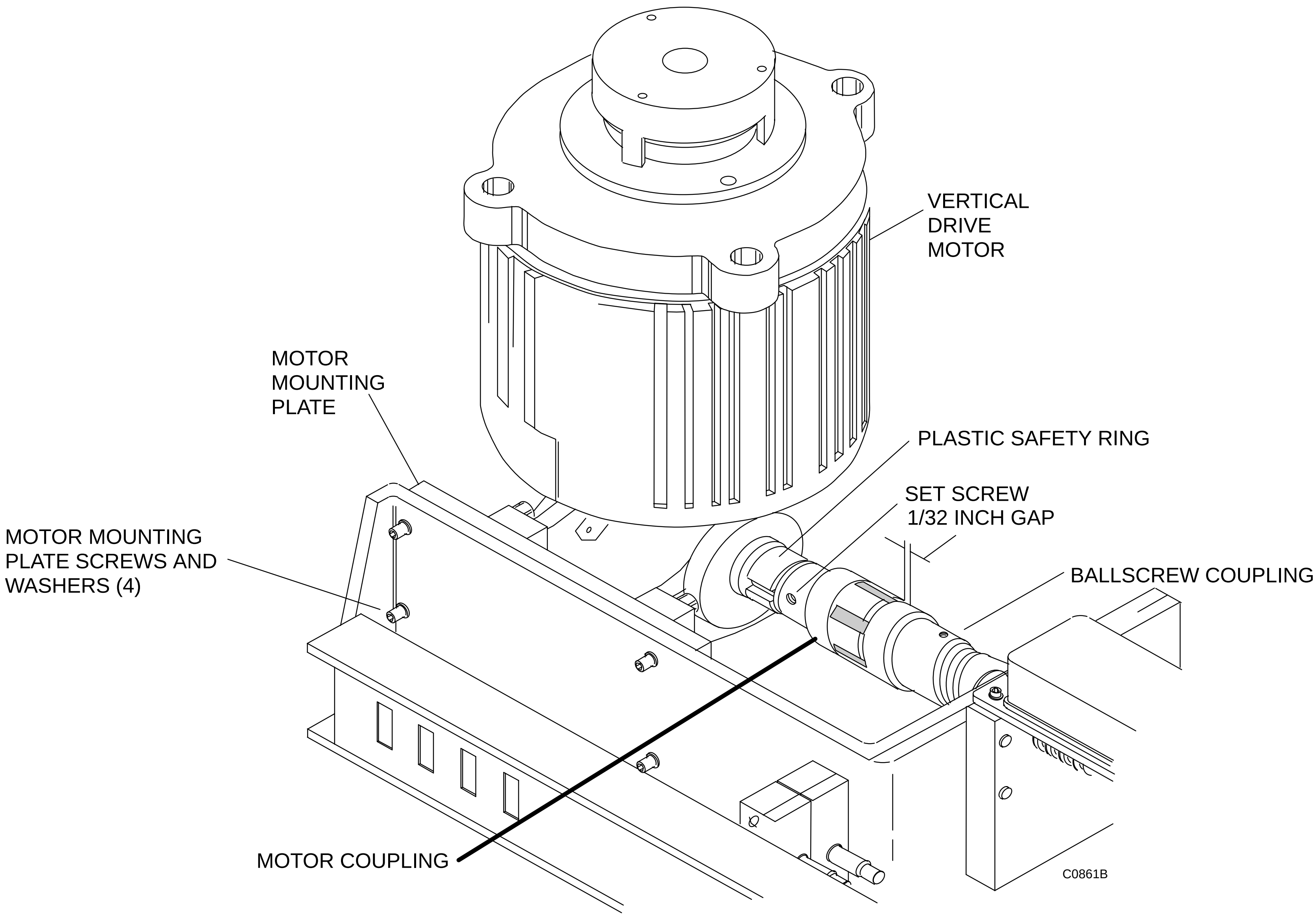


FIGURE 222 Vertical Drive Motor Mounting Plate Screws

11. Grasp the vertical motor, rotate it slightly away from the frame, and slide it to the rear of the patient support until the couplings disengage (shaft keys, plastic insert, end pieces). Remove the motor.

12. Remove coupling plastic insert if still in place.
13. Loosen set screw from coupling on end of ballscrew and remove coupling.

> **NOTE**
>
> A gear puller may be necessary to remove the coupling.

14. Remove key from end of ballscrew.
15. Remove first thrust bearing from end of ballscrew (3 PIECES). Refer to Figure 223.
16. Turn threaded wormgear portion of the ballscrew clockwise by hand until motor end of ballscrew clears bearing retainer.
17. Remove second thrust bearing from end of ballscrew (3 PIECES).
18. Remove square ballscrew retaining nut from scissors slide by unscrewing it with a wrench.
19. Remove defective ballscrew by maneuvering it free of the base.

BALLSCREW REPLACEMENT

20. Insert replacement ballscrew and thread square retaining nut into scissors slide.
21. Turn ballscrew by hand to position shown in Figure 224.

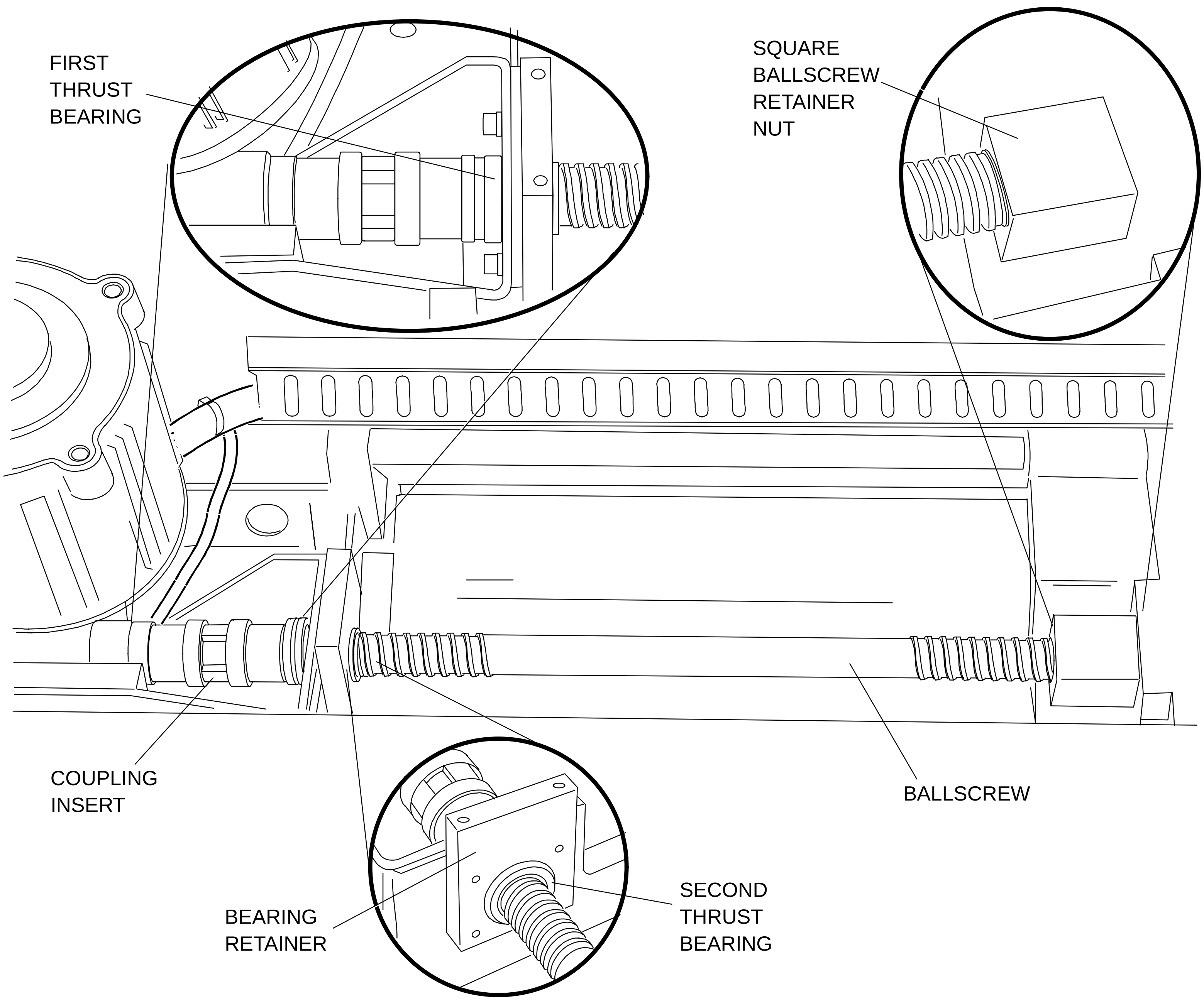


FIGURE 223 Ballscrew, Couplings, Thrust Bearings and Retainer Nut

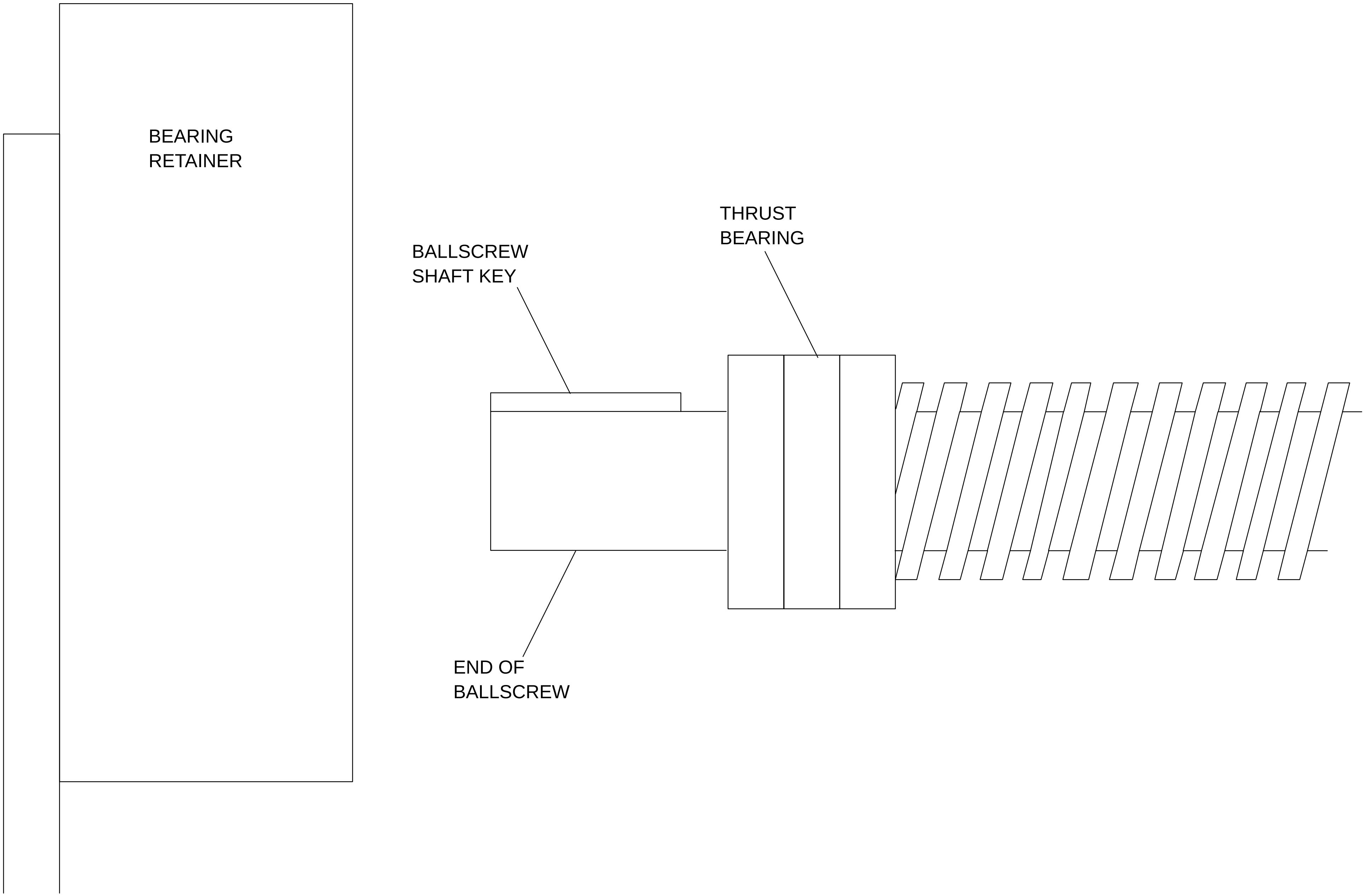


FIGURE 224 2nd Thrust Bearing Replacement Position

22. Replace 2nd thrust bearing over end of ballscrew.
23. Turn threaded wormgear portion of ballscrew counterclockwise by hand until thrust bearing is fully seated inside bearing retainer.

> **NOTE**
>
> From this point forward, refer to Figure 223 for proper location of items.

24. Replace remaining thrust bearing over end of ballscrew.
25. Replace ballscrew key.
26. Remove coupling set screw and discard.
27. Slide coupling over end of replacement ballscrew.
28. Rotate coupling/ballscrew by hand such that the threaded coupling set screw hole faces up (on top).
29. Push coupling and thrust bearing against vertical wall of bearing retainer so that no gaps are present and secure (clamp).
30. Drill a hole in end of ballscrew with #18 bit 3/16” to 1/4” in depth using the coupling’s set screw hole as the pilot hole.
31. Completely remove all debris from hole.
32. Fully insert the S620–124 dog point set screw into coupling. **<u>END OF SET SCREW SHOULD BE FLUSH WITH OUTER DIAMETER OF COUPLING</u>**.
33. Place the plastic coupling insert into the jaws of the ballscrew coupling.
34. Install the vertical drive motor by engaging the motor coupling into the ballscrew coupling. Leave a 1/32” gap between the jaws of the couplings. Refer to Figure 222.
35. Secure the motor mounting plate to the base with the four motor mounting plate screws and washers. Refer to Figure 222 .
36. Connect P415 from the motor/brake assembly to the patient support control pcb.
37. Reinstall the vertical motor ground wire (green/yellow) to the patient support control pcb.
38. Apply gantry power. Refer to the PowerUp/PowerDown procedure in the introduction section.
39. Raise the patient support to its maximum height with the gantry control panel switches.

> ⚠ **WARNING**
>
> **LOWERING THE PATIENT SUPPORT WITH THE VERTICAL SUPPORT BRACE IN PLACE WILL DAMAGE THE BALLSCREW DRIVE. FAILURE TO REMOVE THE VERTICAL SUPPORT BRACE WILL RESULT IN EQUIPMENT DAMAGE AND/OR SERIOUS INJURY AND/OR DEATH TO SERVICE PERSONNEL.**

40. Remove the vertical support brace and return it to its holder on the face of the scissor. Refer to Figure 221.
41. Secure the brace cable with an 8 inch long tie–wrap fed through the bottom of the support brace holder. Refer to Figure 221.
42. Apply Mobile 28 grease or #7 AeroShell grease to ballscrew. Raise and lower patient support, adding grease until ballscrew retainer nut is packed with grease.

> ⚠ **WARNING**
>
> **LOOSEN MOTOR MOUNTING PLATE SCREWS ONE TURN ONLY. FAILURE TO COMPLY MAY CAUSE THE MOTOR AND BALLSCREW COUPLINGS TO DISENGAGE, RESULTING IN SERIOUS INJURY OR DEATH DUE TO COLLAPSE OF THE PATIENT SUPPORT.**

43. Move the patient support up and down with the gantry control panel switches. If an audible knocking is detected in the vertical movement, loosen the four motor mounting plate screws by **one** turn each.
44. Move the patient support up and down with the gantry control panel switches, allowing motor to align itself. Hold motor in place and tighten the four motor mounting plate screws.
45. Repeat this process as necessary until the knocking is resolved.
46. Lower patient support to its lowest position with the gantry control panel switches and replace the patient support telescoping base covers. Refer to the Patient support covers removal/replacement procedure.

POWER UP / DOWN PROCEDURE

This section describes the system start–up/shutdown procedures. **Power–UP** is accomplished via the keyswitch on the Display Tower. **Power DOWN** of the entire system is accomplished via the General Utilities GUI. This is accessed via the “Service” option from the opening bootup GUI. This procedure covers the Service Application mode. *Scanning Application mode procedure is in the OP manual*.

SYSTEM POWER UP

1. Make sure the Gantry keyswitch is in the normal position. Turn the Display Tower keyswitch to the **ON** position. After the system boots, you will be prompted with the startup application display (or “4 Button Screen.”) You have **4 boot options**. Refer to Figure 225.:
   A.) Select Clinical Scanning (or do nothing) and the system will start and connect to the acquisition server. The acquisition server will then power up the AcQImage Cabinet, and if that is done successfully, the acquisition server will power up the gantry and X–ray system.
   B.) Select the Service Application option (within 15 seconds) by clicking on the Service button with the mouse pointer.
   C.) Reboot the system. **D**.) Start the Diagnostic GUI (within 15 seconds) by clicking once in the bottom left corner, then entering PICKER <CR>. (All caps). The AcQImage board LEDs will display a power up test sequence. Refer to the LED Interpretations.

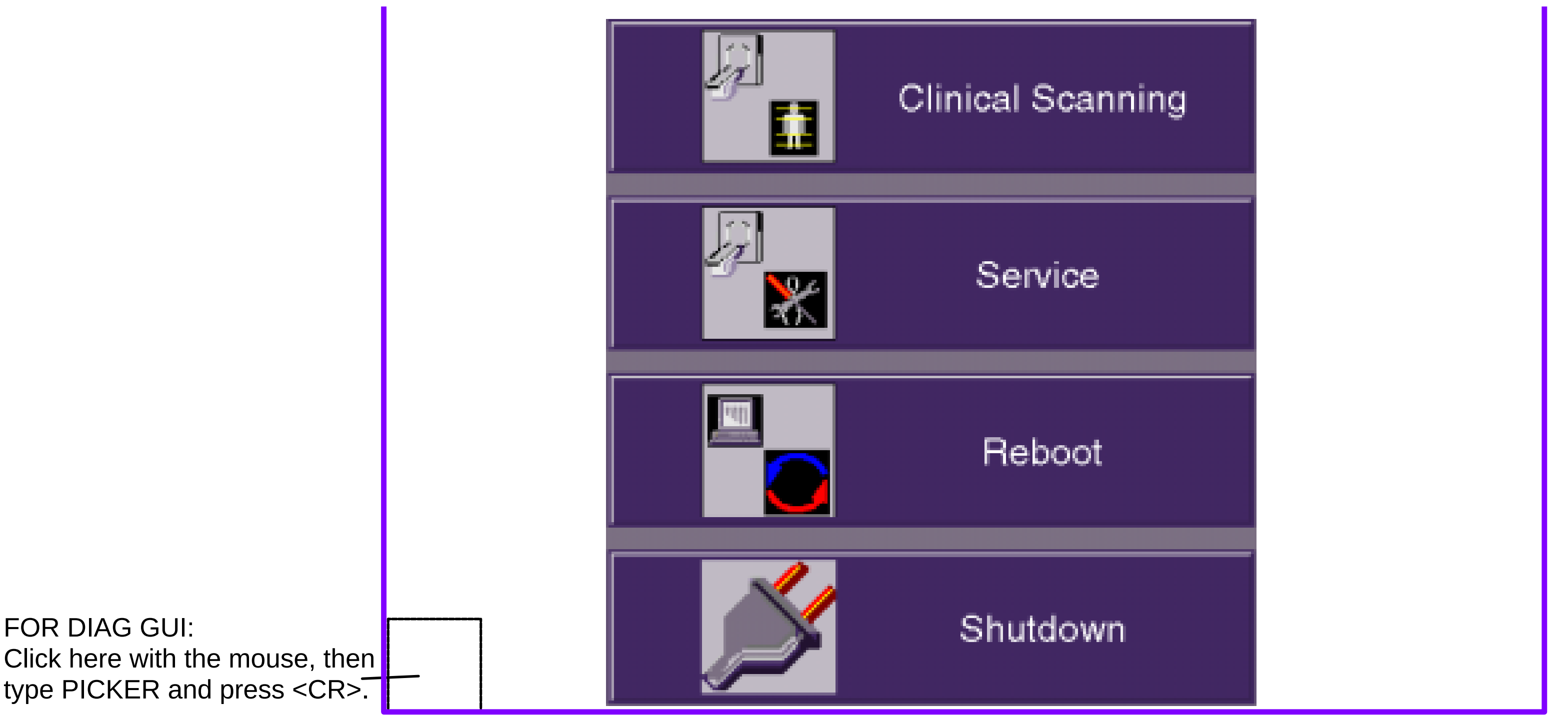


FIGURE 225 4 BUTTON SCREEN

PREPARE THE SYSTEM FOR USE (X–Ray Tube Warmup)

2. The X–Ray system needs to be properly warmed up prior to executing a scan procedure. X–Ray System Warmup must be completed at the start of every scan day or when the tube heat is below 6%. Tube heat is displayed in the Tube Warmup window.

3. To perform a tube warm–up, click on the Tube Warm Up button with the mouse pointer. Refer to Figure 225. This will initiate the warm–up sequence and **X–RAYS WILL BE EMITTED**. The tube warm–up status is displayed is displayed in the Tube Warmup Window. Refer to Figure 226.

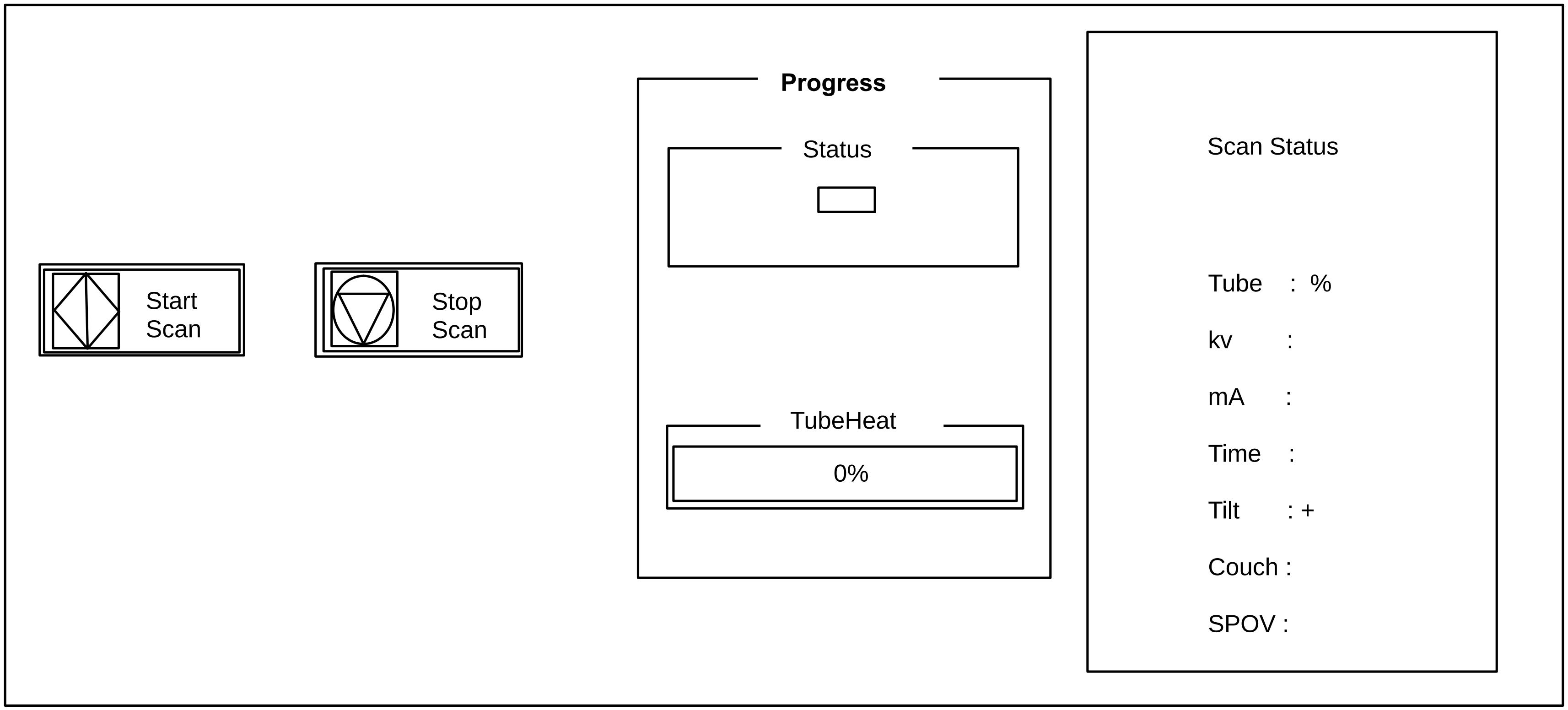


FIGURE 226 TUBE WARM–UP STATUS WINDOW

SYSTEM POWER DOWN

1. From the 4 Button Screen (Figure 225) click on SHUTDOWN with the mouse pointer.

> ⚠ **CAUTION**
>
> DO NOT POWER DOWN THE SYSTEM WITH THE TUBE HEAT AT OR ABOVE 35%. FAILURE TO COMPLY MAY RESULT IN SHORTENED TUBE LIFE AND/OR ERRATIC TUBE BEHAVIOR.

System shutdown ensures that all images on disk are properly closed, all image maintenance functions properly terminated, the gantry parked, the X–Ray system sufficiently cooled and all films printed.

There are three shutdown selections: Shutdown Gantry, Shutdown Console, Shutdown System and Switch to Console as seen in Figure 227.

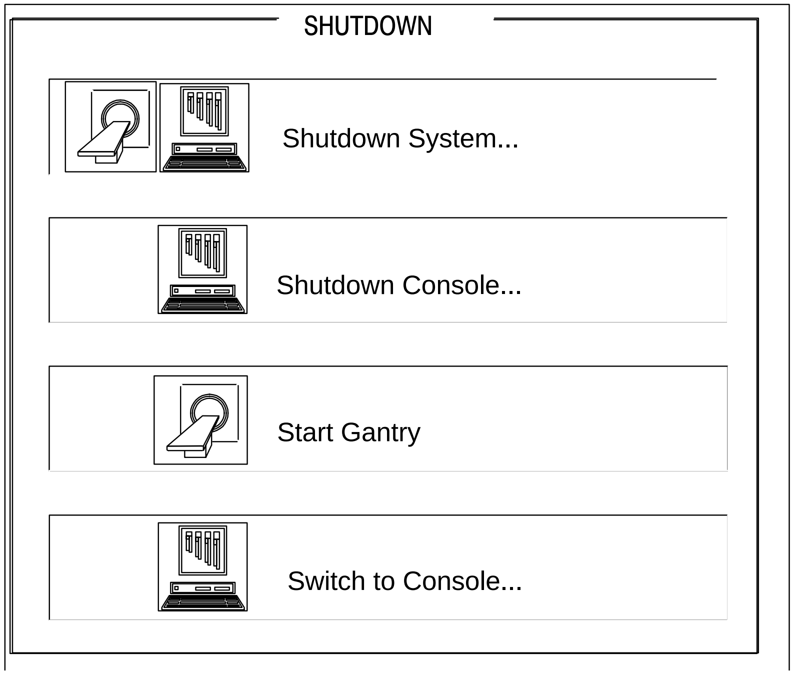


FIGURE 227 SHUTDOWN SCREEN

Shutdown Gantry:	This option should be used, if possible, in place of the e–stop switches. An orderly shutdown of the gantry will be performed. Display Tower and/or AcQImage Cabinet operation may continue after selected. To cancel shutdown, click the Abort button.
Shutdown System:	Performs an orderly shutdown of the entire scanner system. To cancel shutdown, click the Abort button.
Shutdown Console:	This option shuts down only the console units. An orderly shutdown of the image archiver, filming and file system will be done.
Start Gantry:	Toggles Start/Shutdown Gantry depending on status of Gantry power.
Switch to Console:	Changes the GUI to the Console Application (Operator) screen.

⚠ WARNING ⚠

THE GUI SHUTDOWN REMOVES POWER FROM THE SYSTEM, BUT NOT AT THE INCOMING POWER LINES. DANGEROUS VOLTAGES ARE EXPOSED WHEN THE GANTRY COVERS ARE REMOVED. BEFORE SERVICING EQUIPMENT, REMOVE POWER FROM THE WALL BOX. FAILURE TO COMPLY CAN CAUSE SERIOUS INJURY OR DEATH TO PERSONNEL.

2. When prompted via a dialog box, turn the Display Tower keyswitch to the **OFF** position.
3. On the AcQImage Cabinet and Display Tower, ensure the rear breaker switch is turned off.

AcQImage Boards – Power Up Test LED Interpretations

```
AP, PB, GAM Board Diagnostic Flash Version 1.1.2
```

```
0 Hardware Initialization
1 CPU Version Check
2 Diagnostic Flash CRC Check
3 Operating System Flash CRC Check
4 CPU Memory Test
5 Data Cache Test
6 Unexpected Exception Test
7 IBC Test
8 NVRAM Single Location Test
9 PCI Bus Select Test
A Ethernet Loopback Test
B UART Loopback Test
C apmem for AP, pbmem for PB, gamem for GAM

C CIO Test (CP Board Diagnostic Flash Version 1.1.2)
D Synchronous Serial Chip Loopback Test (CP Board Diagnostic Flash Version 1.1.2)
```

AcQSim CT – Gantry Components Repair

GANTRY COVERS

The following procedure shows how to open/close, install, remove/replace and adjust the various AcQSim CT Gantry covers.

Front Upper Cover

1. Loosen the captive screws on the front cover, lift up and secure. Refer to Figure 228.

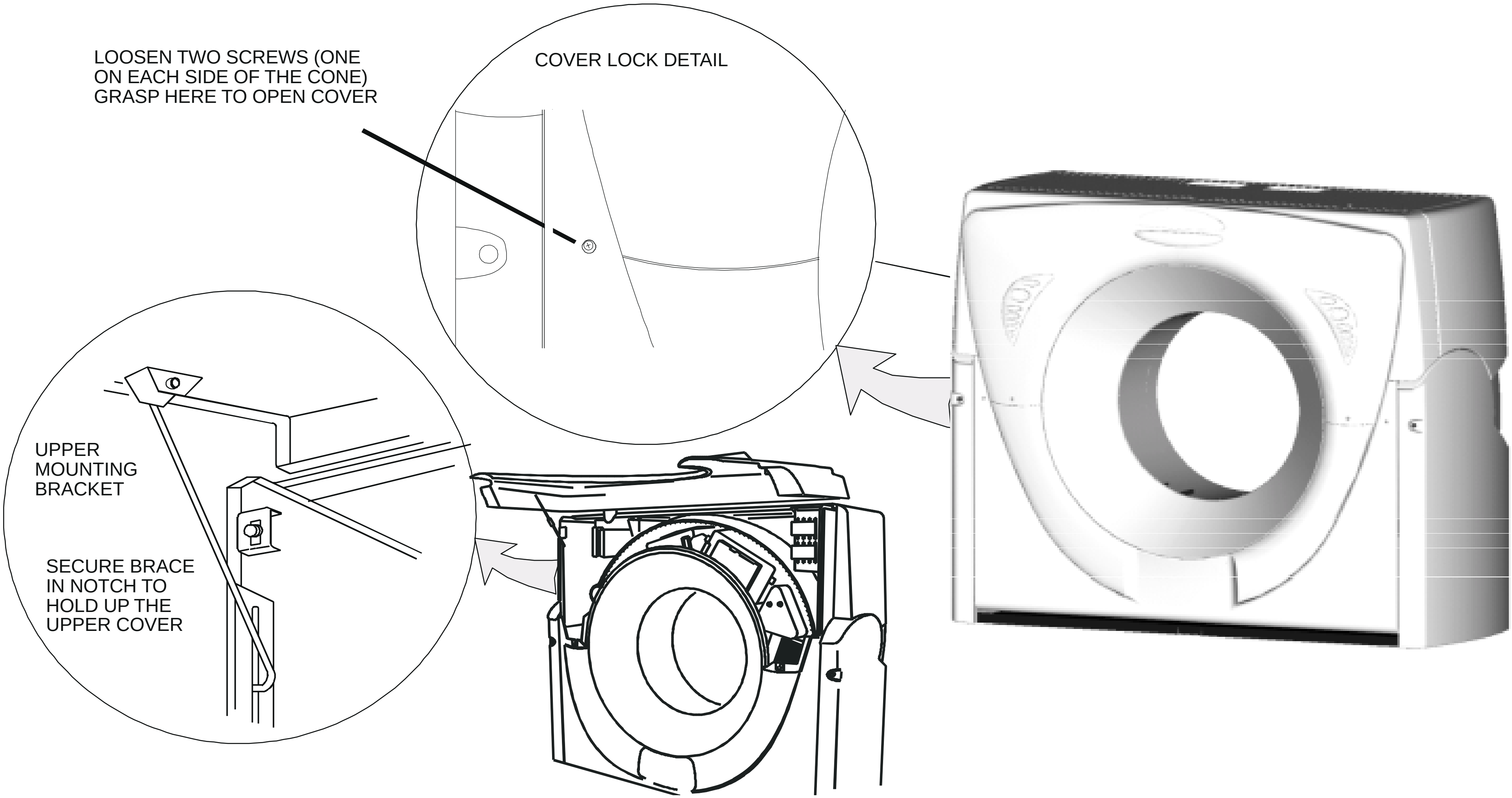


FIGURE 228 GANTRY UPPER PANEL OPENING

Lower Front Cover

1. Remove the screws that secure the lower Gantry cover. Lift the cover up and off the bottom latches and slide the cover away from Gantry. Store the cover in a secure place.. Refer to Figure 229.

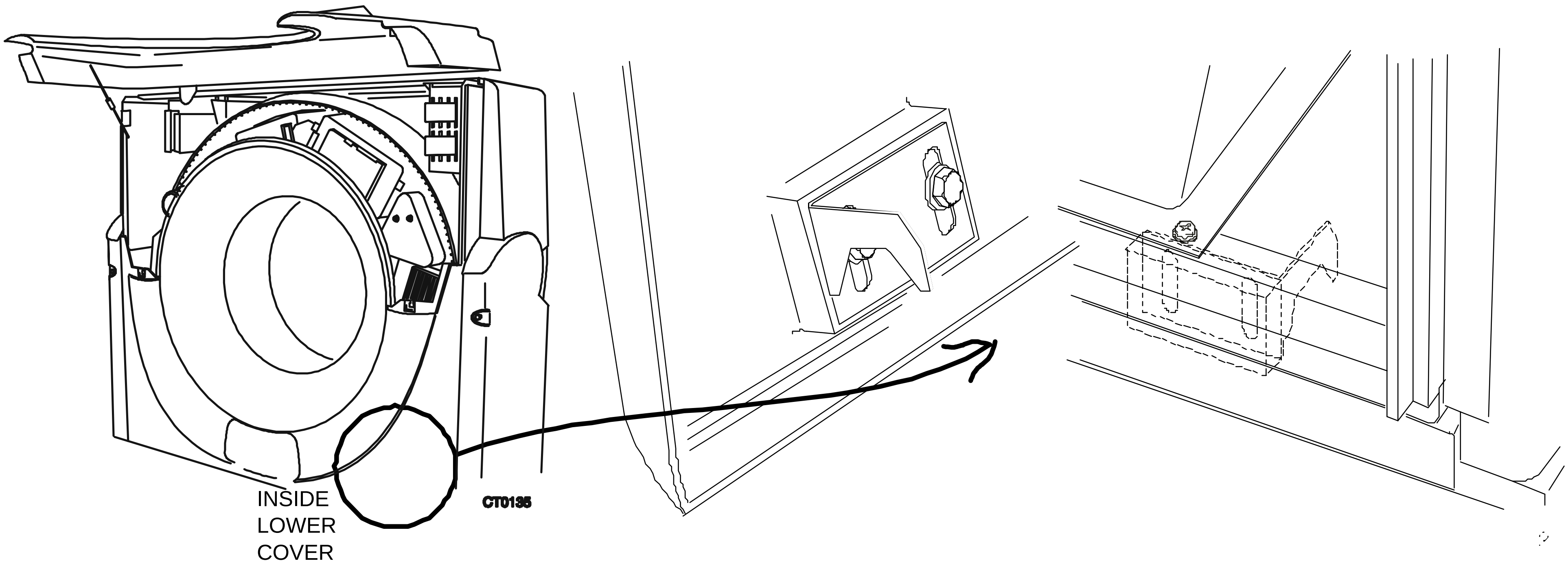


FIGURE 229 GANTRY FRONT LOWER COVER – DETAIL

Rear Covers

> ⚠ **CAUTION**
>
> Before removing the rear covers, unplug the two connectors from the cables that lead to the speaker and x–ray light. Failure to do this will result in damage to the cables and/or connectors.

1. Remove the screws that secure the rear Gantry covers. Lift up and away on each of the covers and set aside in a safe place. Refer to Figure 230.

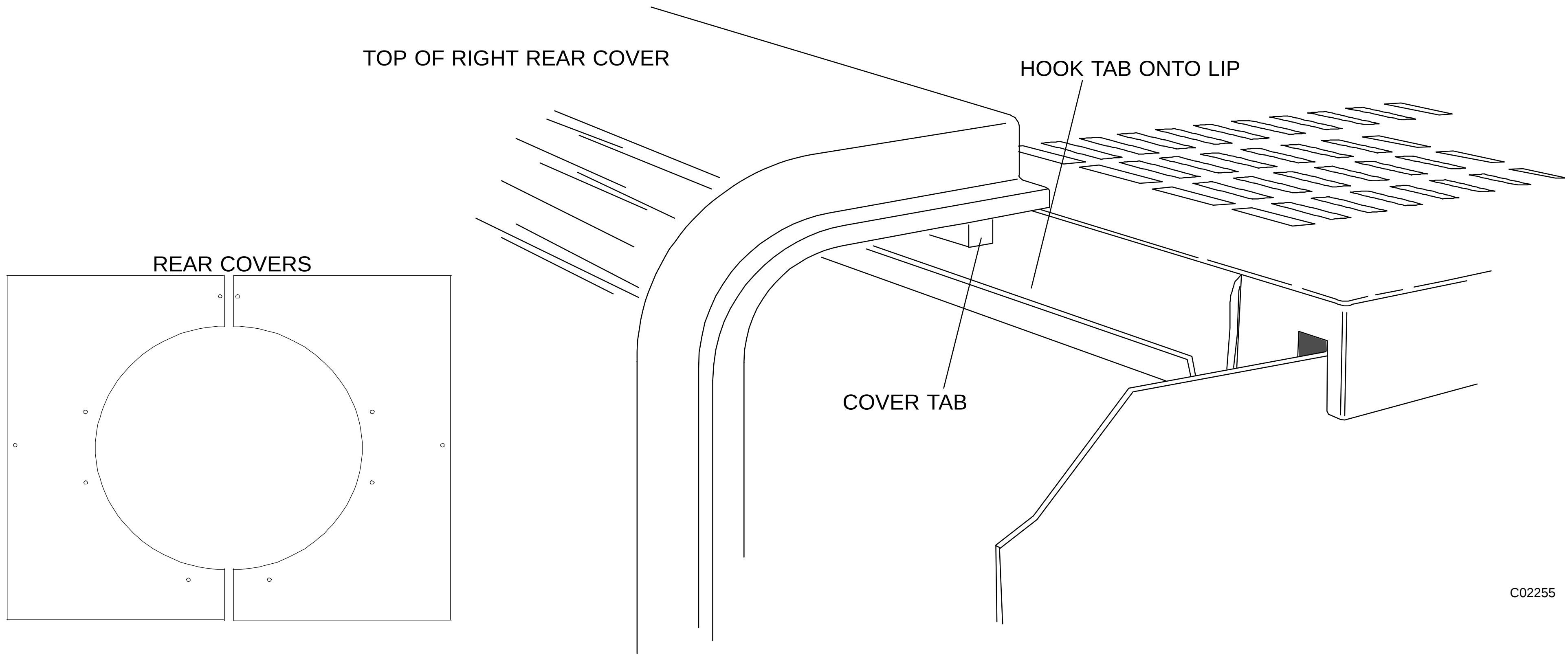


FIGURE 230 REAR COVER REMOVAL

Lower Side Cover Installation

1. Remove Lower Right Column Cover (p/n 312863) from shipping box. Remove the small bag attached to the inside surface of the cover, which contains the mounting screws and washers. Set it aside.
2. Open the Front Upper Cover and lock in place.
3. Uninstall the E–Stop Switch located on the right side column.
4. Install the E–Stop Switch in the Lower Right Side Column. Refer to Figure 231.

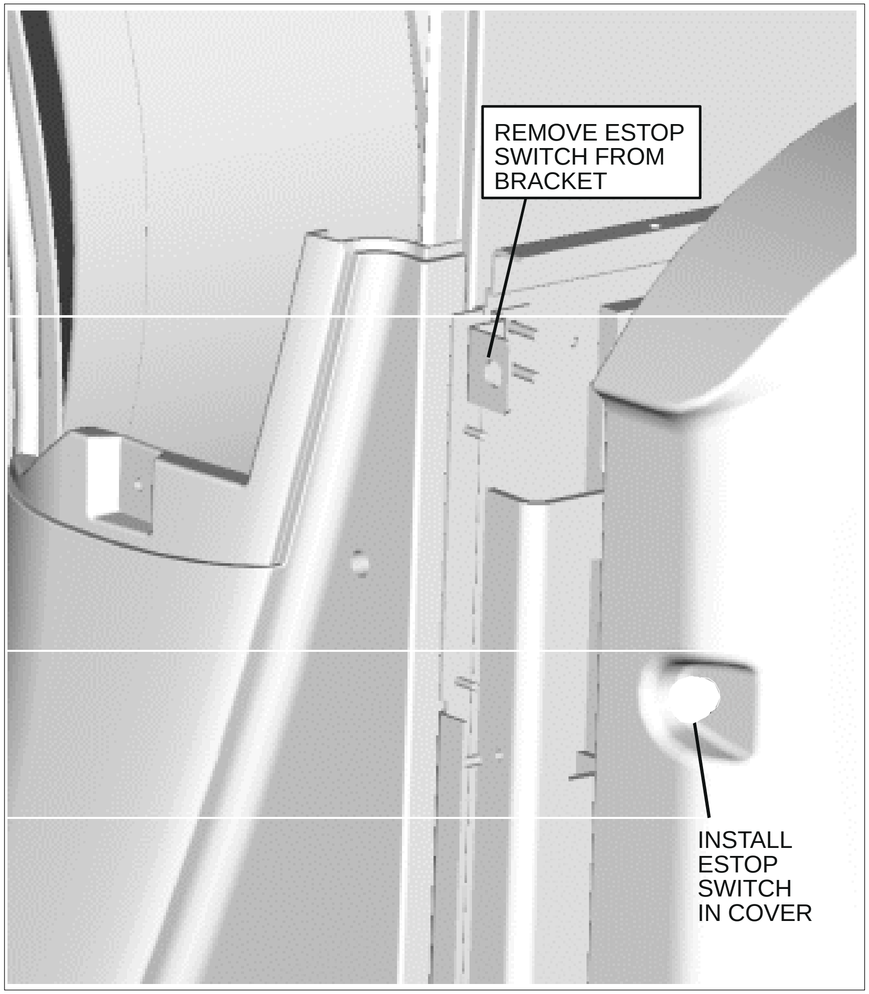


FIGURE 231   ESTOP SWITCH LOCATION

5. Install the Lower Right Column Cover onto the right side column. Align and insert Locating pins on Lower Right Column Cover with the V groove features on the Side Column. Refer to Figure 232.

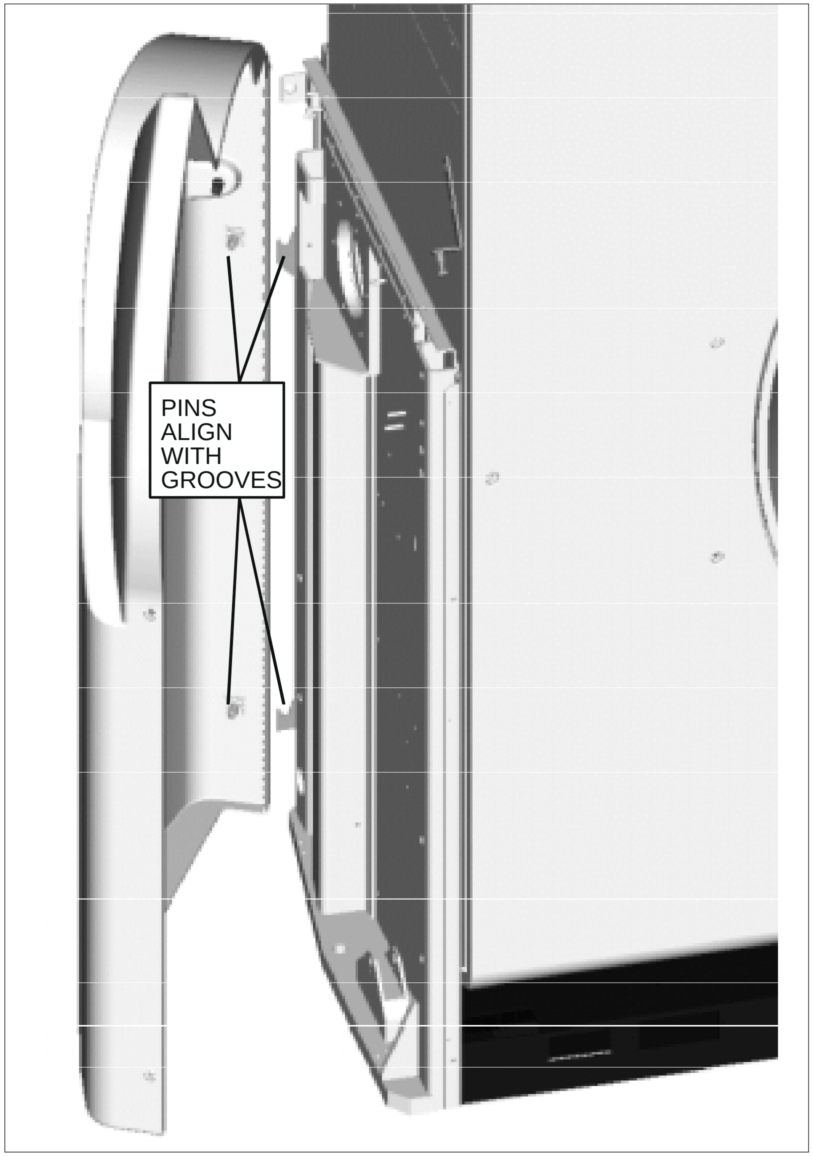


FIGURE 232 LOWER RIGHT COLUMN COVER

6. Fasten the Lower Right Column Cover to the Gantry Column using the two screws (1/4”–20 x 3/4”, Pan Head) and two nylon washers provided. Maintain a gap between the top of the Column Cover and Gantry Side Panel in the range of 0.050 to 0.250 inch. Refer to Figure 233.

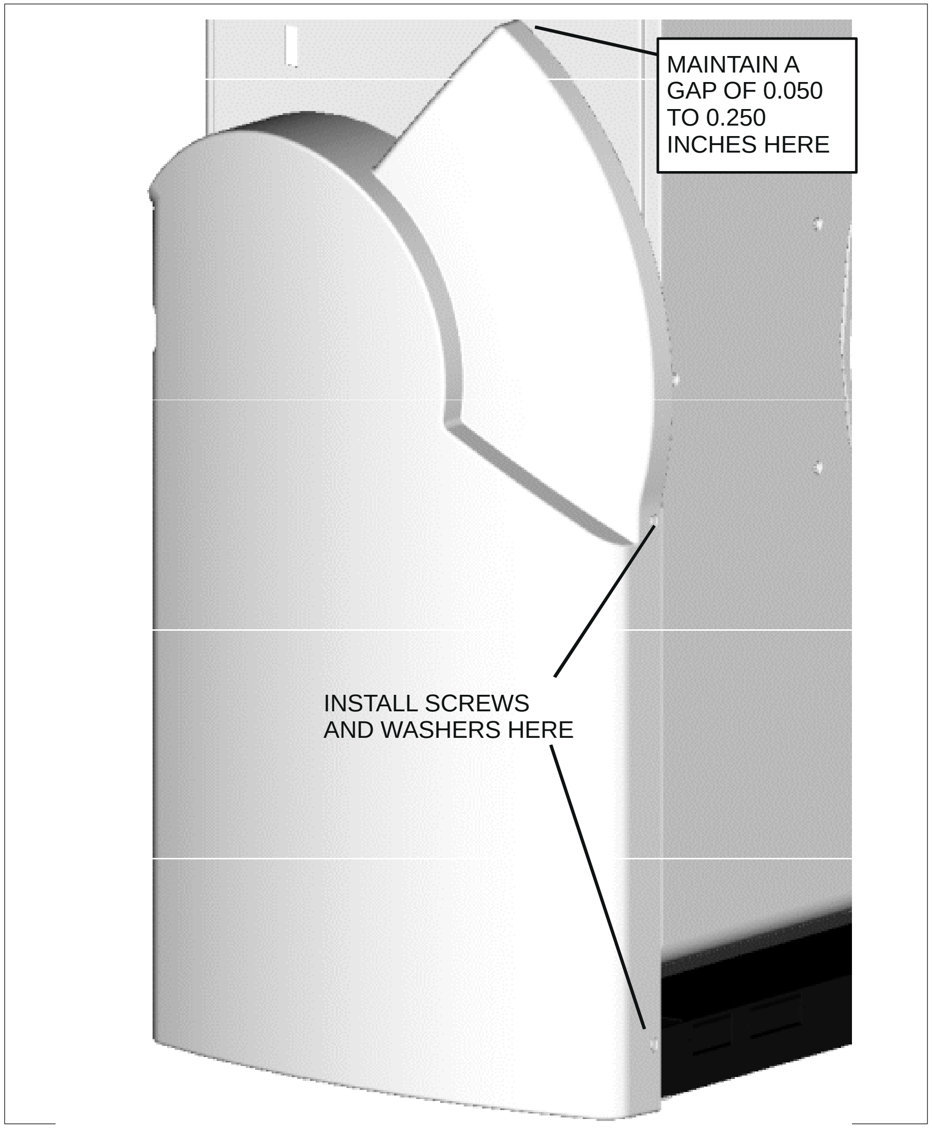


FIGURE 233 FASTENING LOWER SIDE COVER

7. Lower the Front Upper Cover. Make sure the Front Upper Cover is seated firmly against the Lower Front Cover.

8. Check the clearance between the Lower Right Column Cover and the Front Upper Cover as shown in Figure 234. It should measure between 0.100 to 0.250 inch. If the gap measures within the allowable range, assemble the Lower Left Side Cover (p/n 312865) in the same fashion as the right cover. If not, continue to step 9.

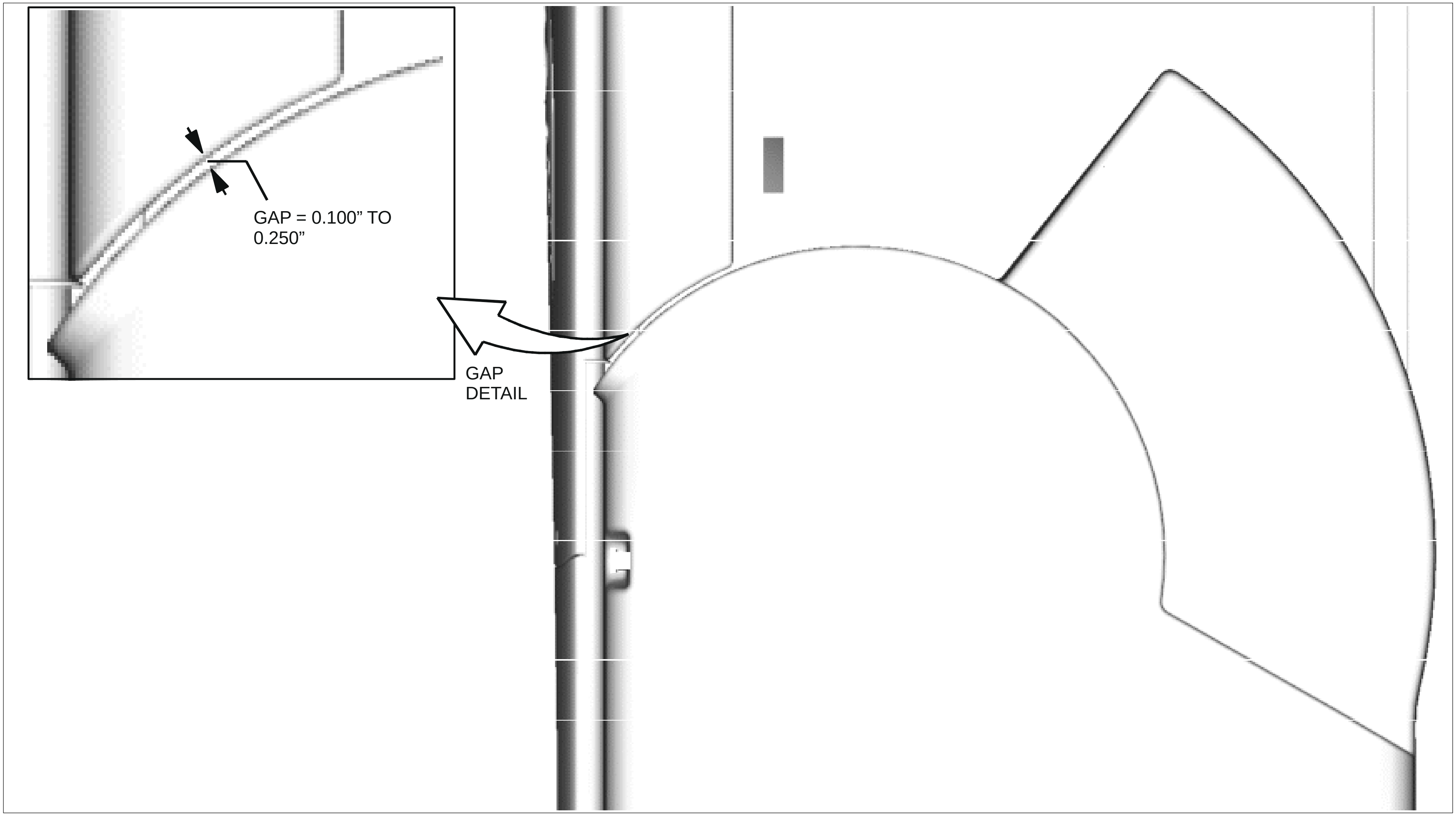


FIGURE 234 GAP BETWEEN SIDE COVER AND FRONT UPPER COVER

9. If the gap is not within the allowable range, raise the Front Upper Cover, remove the Lower Right Column Cover, and adjust the height of the V–groove Mounting Bracket. Refer to Figure 235. Continue to check the clearance between the Lower Right Column Cover and the Front Upper Cover and make sure it is within the allowable range of 0.100 to 0.250 inch.

10. Prior to final installation of Lower Right Side Cover, make sure wires are reconnected to E–Stop Switch.

11. Assemble the Lower Left Side Cover (p/n 312865) in the same fashion as the right cover.

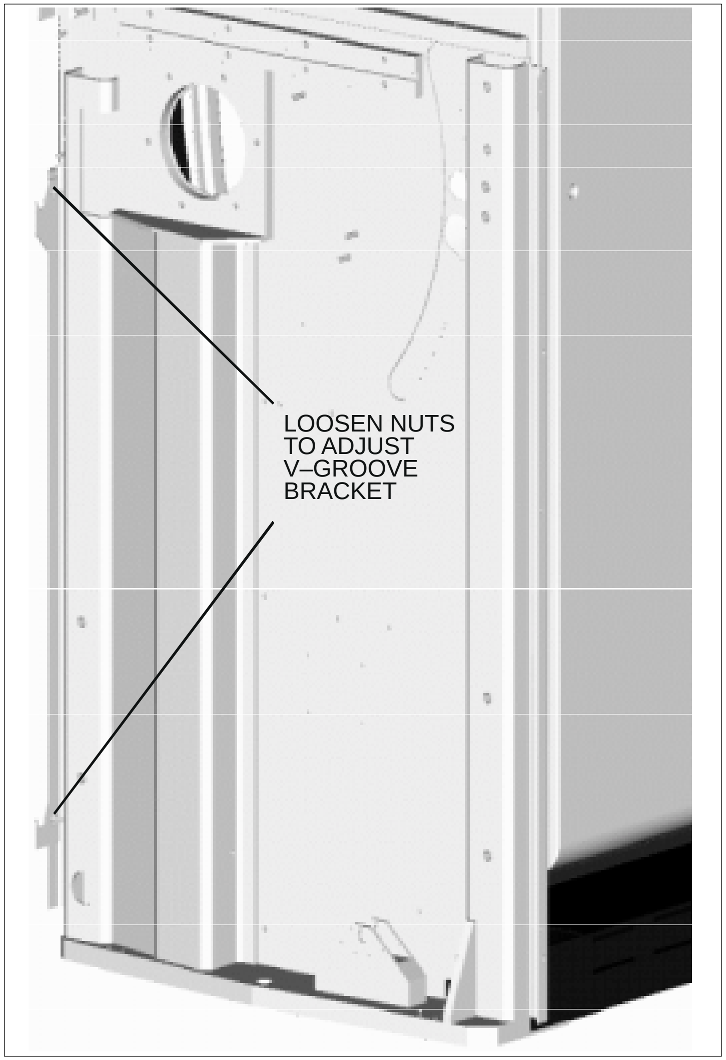


FIGURE 235 V–GROOVE BRACKET ADJUST

UPPER SIDE COVER INSTALLATION

1. Remove Upper Right Column Cover (p/n 362096) from shipping box.
2. Align and insert the pins on the Upper Right Column Cover into the holes on the Cover Mounting Bracket which is a attached to the Gantry Side Panel. Refer to Figure 236.

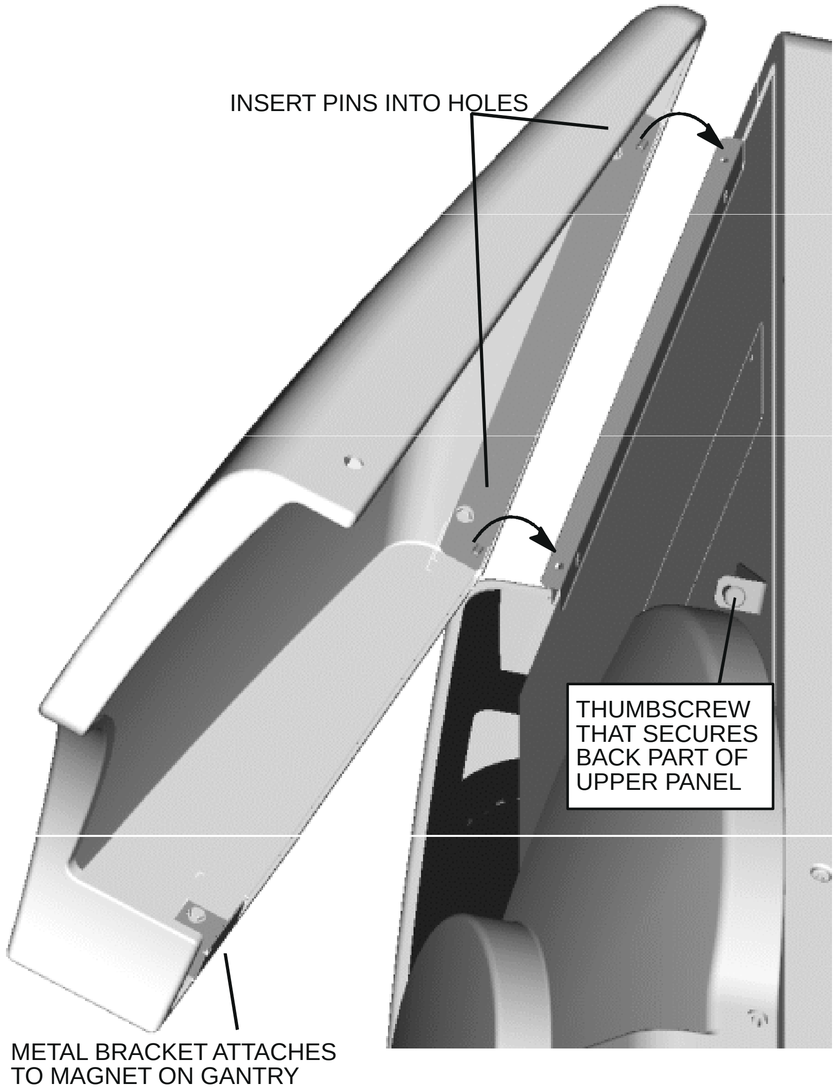


FIGURE 236   UPPER SIDE COVER

3. Swing the lower side of the Upper Right Column Cover down until it comes into contact with the Gantry Side Panel. A magnetic latch should attract and hold the cover in place.

4. Visually locate the thumbscrew through the hole on the back side of the Upper Right Column Cover. Insert a long, narrow slotted screwdriver into the hole and lightly tighten the thumbscrew. This is a blind connection, so ensure you get the right “feel” of the captive screw starting correctly. Refer to Figure 237.

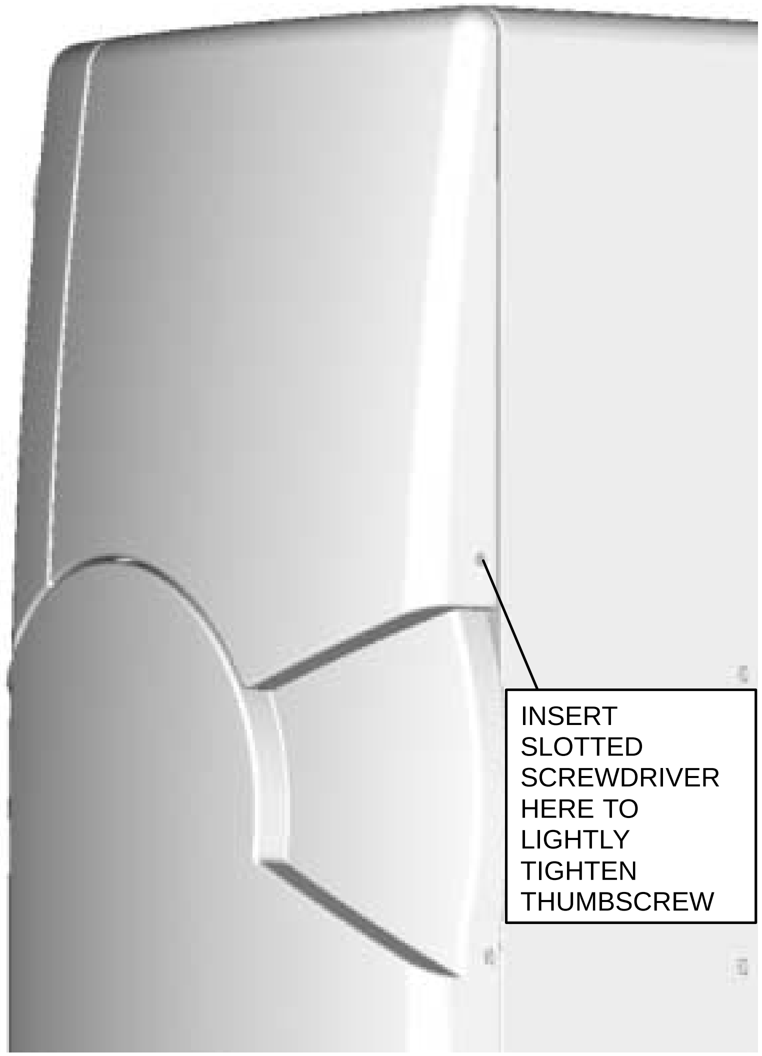


FIGURE 237 THUMBSCREW LOCATION

5. Check clearance ”A” between the Upper Right Column Cover and Lower Right Column Cover. Also check the uniformity of the gap between the Upper Right Column Cover and the Upper Front Cover. Refer to Figure 238.

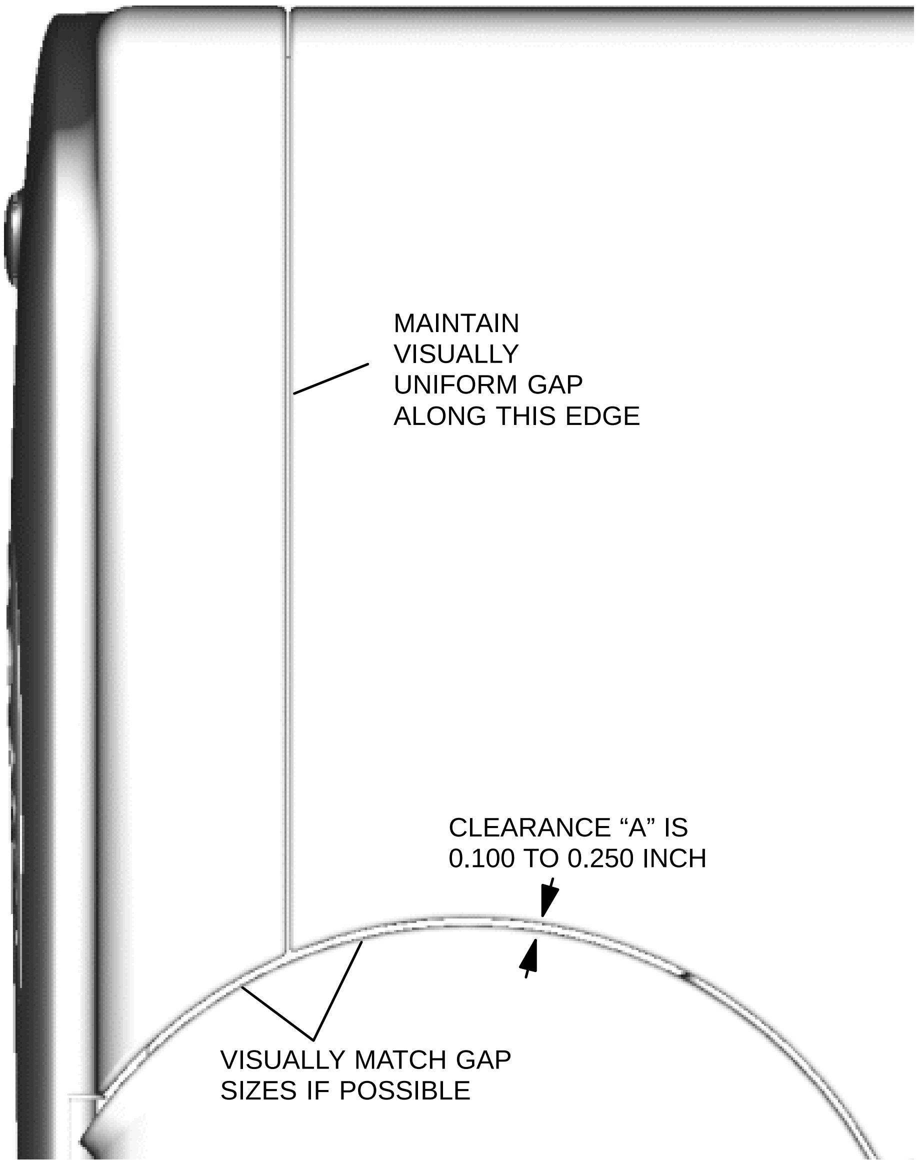


FIGURE 238 CLEARANCE “A” CHECK

6. Check clearance ”B”, between the Upper Right Column Cover and the Lower Right Column Cover. Refer to Figure 239.

7. If clearances ”A” and ”B” are within the allowable ranges and the gap uniformity looks good, than skip to the Hole Plug Installation procedure. If the clearances are not within tolerance, continue to the next page.

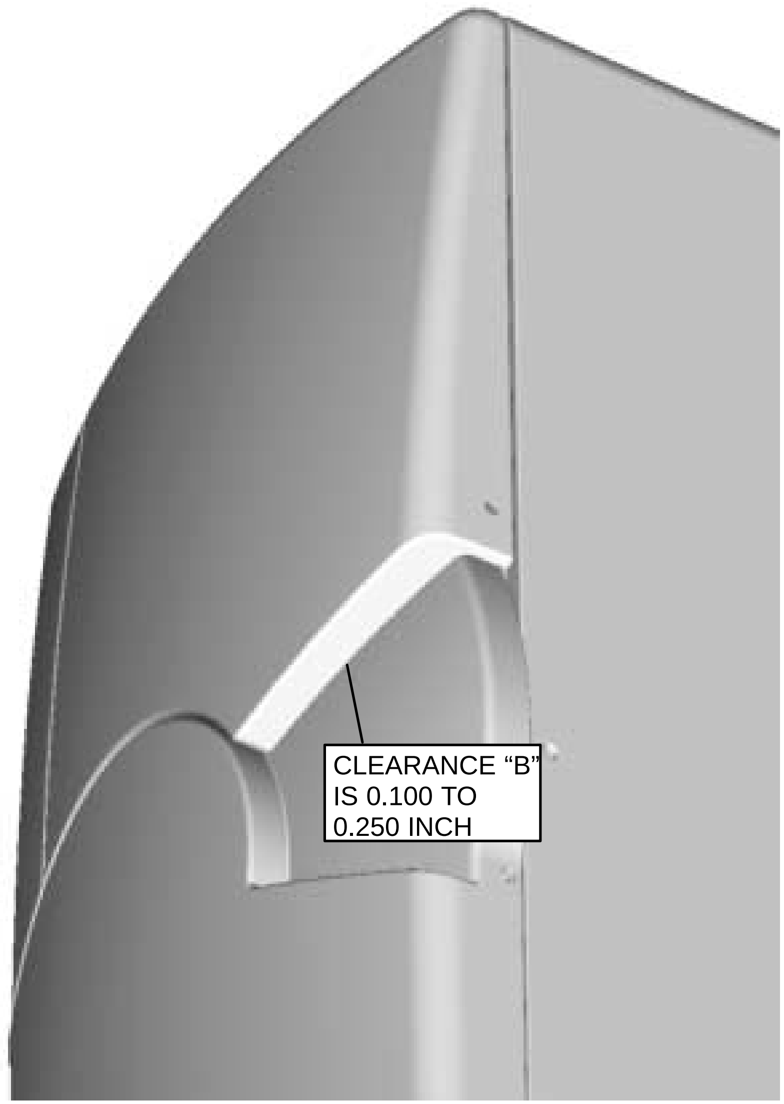


FIGURE 239 CLEARANCE “B” CHECK

8. If clearance ”A” is outside the allowable range or the gap between the Upper Right Column Cover and the Upper Front Cover does not visually appear uniform, remove the Upper Right Column Cover, and adjust the Cover Mounting Bracket by loosening the two bolts as shown below.

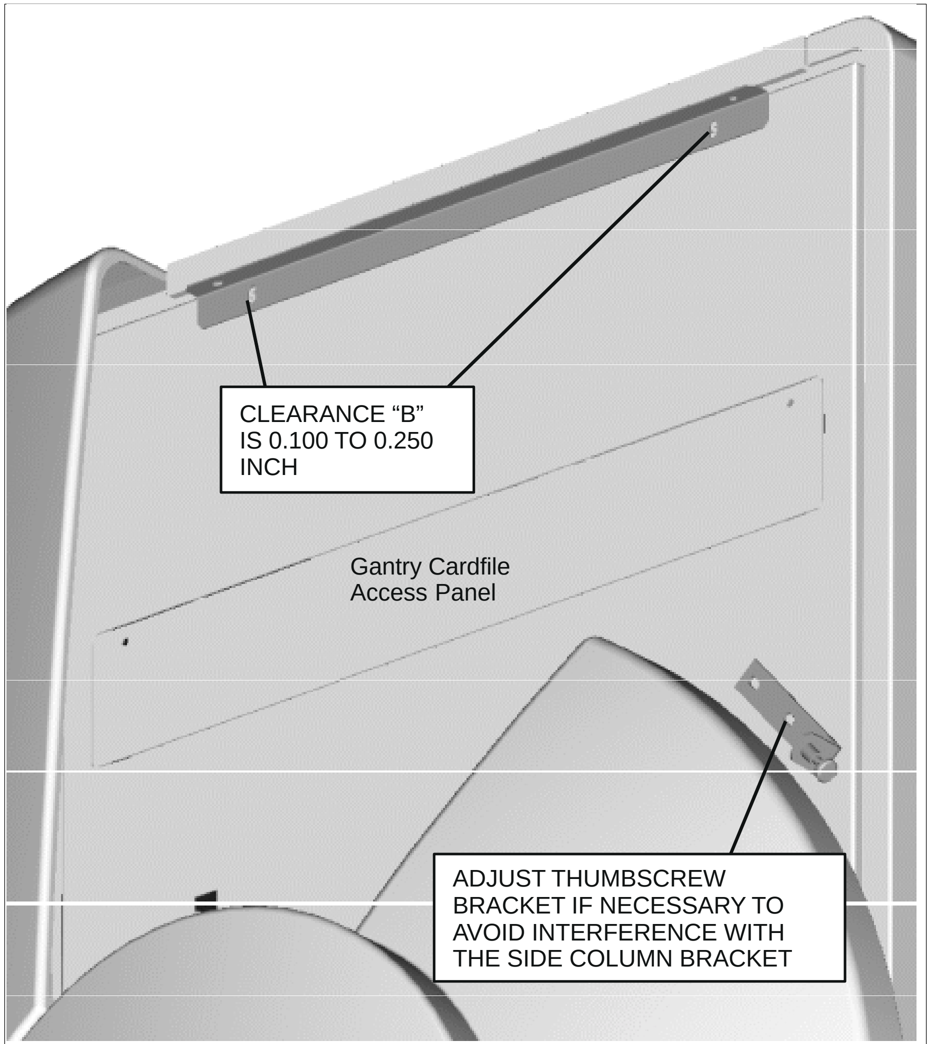


FIGURE 240 ADJUSTING CLEARANCE “A”

9. If clearance ”B” is outside the allowable range, than remove the Upper Right Column Cover, and adjust the Mounting Brackets located inside the Upper Right Column Cover by loosening the bolts as shown below.
10. Assembly of the Upper Left Column Cover (p/n 362097) follows this same procedure.

FIGURE 241 ADJUSTING CLEARANCE “B”

HOLE PLUG INSTALLATION

1. Install the Hole Plugs (p/n T39A–1013) at the four locations shown in Figure 242.

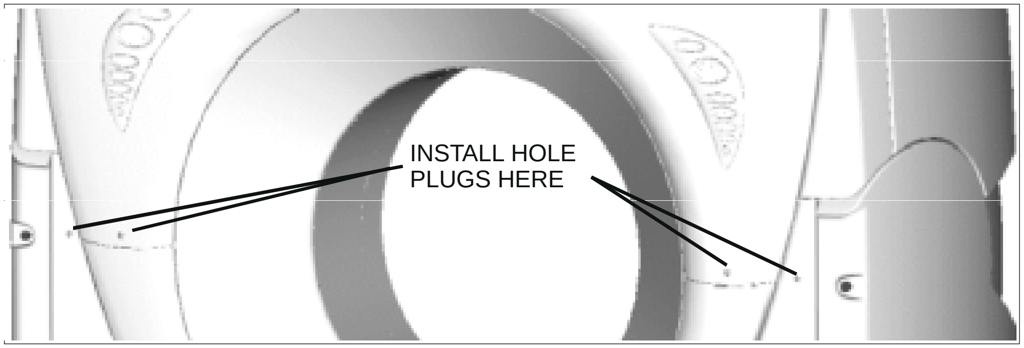


FIGURE 242 FRONT HOLE PLUG INSTALLATION

2. Install the Hole Plugs (p/n T39A–1013) at the two locations shown in Figure 243.

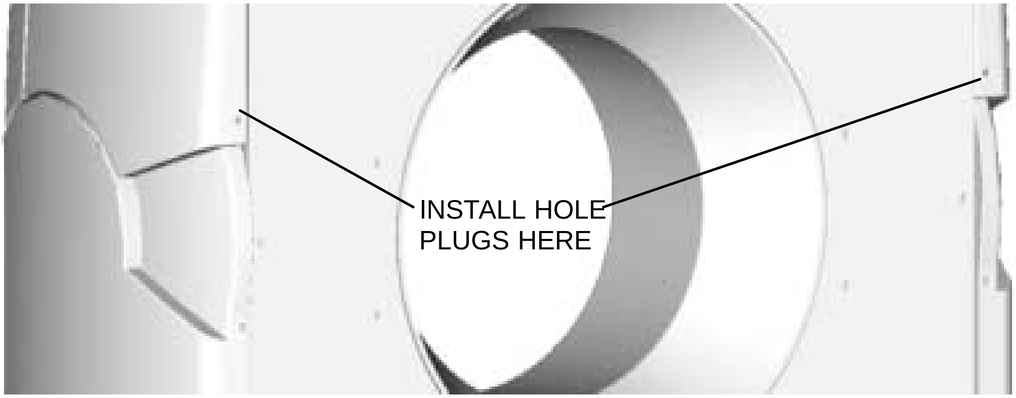


FIGURE 243 REAR HOLE PLUG INSTALLATION

GANTRY COVER INTERLOCK

Introduction: This procedure is used to remove and replace the Gantry Front Cover Interlock Switch.

Tools Required: 1. None

Estimated Time: 20 Min.

GANTRY COVER INTERLOCK REMOVAL PROCEDURE:

1. Remove Gantry power. Refer to the PowerUp/PowerDown procedure.
2. Open the Gantry upper Front Cover.
3. Tag and remove the two wires connected to the Interlock Switch. See Figure 244.

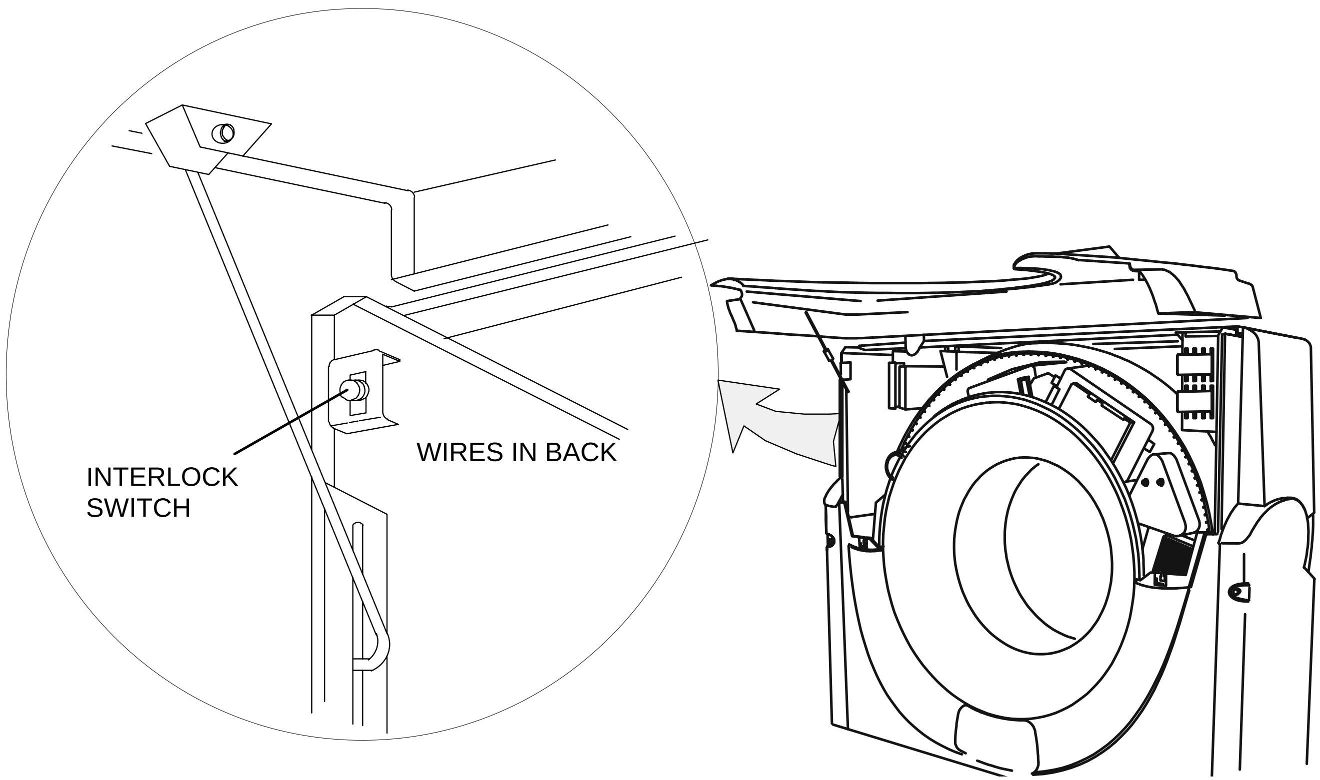


FIGURE 244 Interlock Switch Location

4. Depress the two latches and remove the switch toward the Gantry front.

GANTRY COVER INTERLOCK REPLACEMENT:

5. Install the Interlock Switch by reversing steps 3. and 4.
6. Apply Gantry power. Refer to the PowerUp/PowerDown procedure.

> ⚠ **WARNING**
>
> **USE EXTREME CAUTION WHEN OPERATING OR WORKING ON EQUIPMENT WITH COVERS REMOVED. DANGEROUS VOLTAGES ARE EXPOSED WHEN THE GANTRY COVERS ARE REMOVED. FAILURE TO COMPLY CAN CAUSE SERIOUS INJURY OR DEATH TO PERSONNEL.**

> ⚠ **WARNING**
>
> **ROTATING MACHINERY. STAND CLEAR OF ROTATING FRAME WHEN EQUIPMENT IS OPERATED. FAILURE TO COMPLY MAY RESULT IN SERIOUS INJURY OR DEATH TO PERSONNEL.**

7. Check the Interlock Switch operation by attempting to rotate the Scan Frame by pressing the laser button while the upper Front Cover is open. STAY CLEAR OF THE SCAN FRAME. The Scan Frame should not move. Pull the Switch and check operation. The Scan Frame should rotate.
8. Push the scan frame a bit CCW to get it off of “home” position. Now return the Switch to its center position, close and secure the upper Front Cover and again attempt to rotate the Scan Frame. The Scan Frame should rotate as commanded.

GANTRY CONTROL PANEL

Introduction: The following procedure is used to remove and replace Gantry Control Panels.

Tools Required: 1. #1 Tip Phillips Screwdriver

Estimated Time: 15 Min.

GANTRY CONTROL PANEL REMOVAL:

1. Remove Gantry power. Refer to the PowerUp/PowerDown procedure.
2. Open the Gantry Front Cover and secure it in the open position.
3. Remove the ground wire, ribbon cable and the nine screws that secure the control panel assembly to the inside of the upper cover. Refer to Figure 245.

> ⚠ **WARNING**
>
> **USE EXTREME CAUTION WHEN SERVICING EQUIPMENT WHILE UPPER GULL WING PANEL IS OPEN. THE PANEL HAS SHARP EDGES. FAILURE TO COMPLY COULD RESULT IN SERIOUS INJURY TO PERSONNEL.**

> ⚠ **WARNING**
>
> **USE EXTREME CAUTION WHEN OPERATING OR WORKING ON EQUIPMENT WITH COVERS REMOVED. DANGEROUS VOLTAGES ARE EXPOSED WHEN THE GANTRY COVERS ARE REMOVED. FAILURE TO COMPLY CAN CAUSE SERIOUS INJURY OR DEATH TO PERSONNEL.**

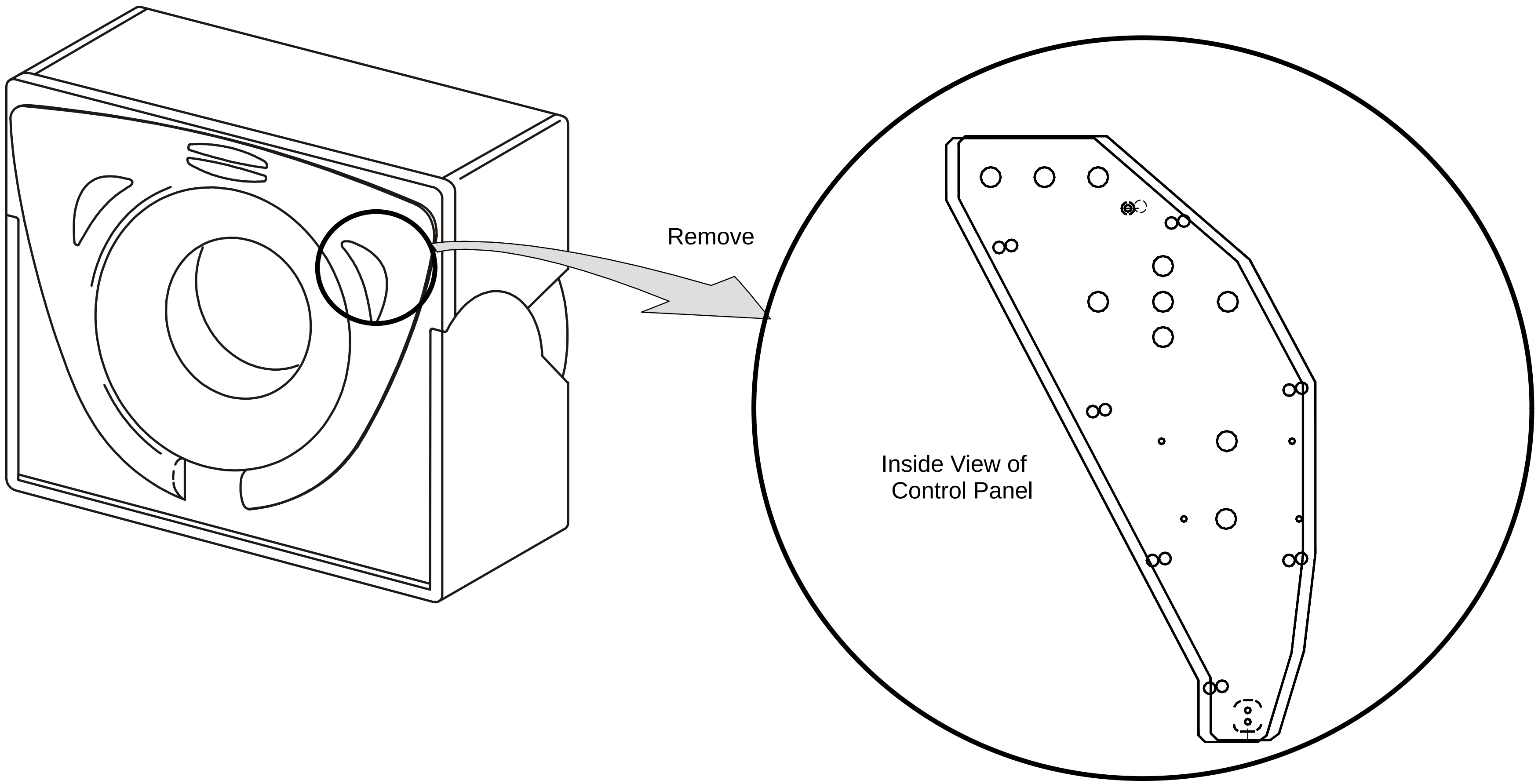


FIGURE 245 Gantry Control Panel

4. Remove the Control Panel from the inside of the Gantry Front Cover.

GANTRY CONTROL PANEL REPLACEMENT:

5. Place the new control panel where the old one was removed.
6. Attach the nine screws to secure the panel. Attach the ribbon cable and groud cable. Close the upper cover.
7. Apply power and verify the Control Panel's operation. Drive the Patient Support vertical or horizontal, activate the Laser, or tilt the Gantry as appropriate.

GANTRY DISPLAY ASSEMBLY

Introduction: The following procedure is used to remove and replace the gantry display assembly, part no. 178989.

Tools Required: #1 Tip Phillips Screwdriver

Estimated Time: 15 Minutes

GANTRY DISPLAY ASSEMBLY REMOVAL:

1. Remove Gantry power. Refer to the PowerUp/PowerDown procedure.
2. Open the Gantry Front Cover and secure it in the open position.

> ⚠ **WARNING**
>
> **USE EXTREME CAUTION WHEN SERVICING EQUIPMENT WHILE UPPER GULL WING PANEL IS OPEN. THE PANEL HAS SHARP EDGES. FAILURE TO COMPLY COULD RESULT IN SERIOUS INJURY TO PERSONNEL.**

> ⚠ **WARNING**
>
> **USE EXTREME CAUTION WHEN OPERATING OR WORKING ON EQUIPMENT WITH COVERS REMOVED. DANGEROUS VOLTAGES ARE EXPOSED WHEN THE GANTRY COVERS ARE REMOVED. FAILURE TO COMPLY CAN CAUSE SERIOUS INJURY OR DEATH TO PERSONNEL.**

3. Remove cable connectors from display board plugs P1, P2, P3, P4, P5, and P6. Remove cable connector from front auto–lite. Remove screw and flat washer securing two ground lugs to display board, and remove ground lugs. Refer to Figure 246.
4. Remove six screws, flat washers, and standoffs securing the front display shroud to the display board, and remove shroud from the front of the gantry cover. Refer to Figure 247.
5. Remove three remaining screws and flat washers securing the display board to the gantry “U” brackets, and remove display board.

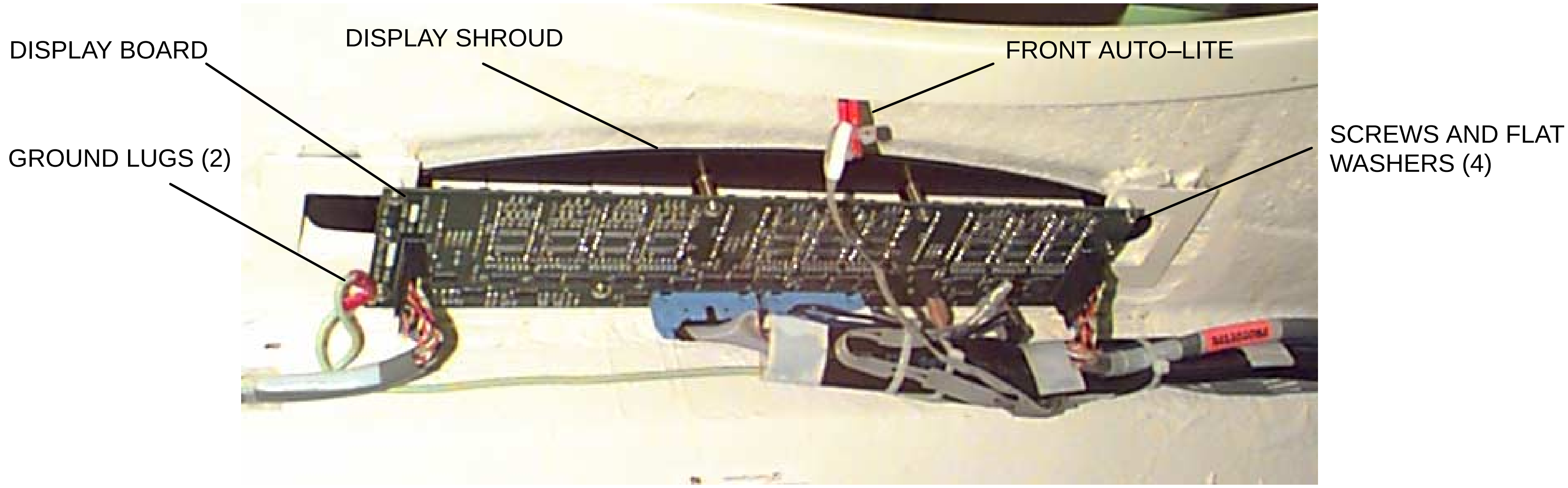


FIGURE 246 Gantry Display Assembly (as viewed from underneath gantry front cover).

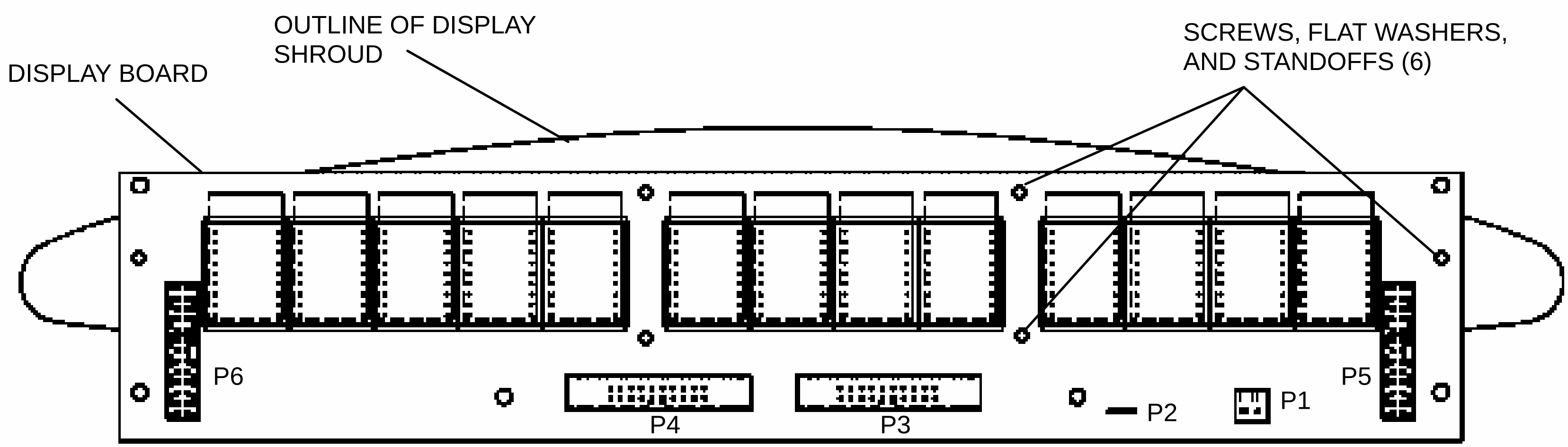


FIGURE 247 Display Shroud Positioning on Display Board (as viewed from underneath gantry front cover).

GANTRY DISPLAY ASSEMBLY REPLACEMENT:

1. Place the new display board where the old board was removed.
2. Install four screws and flat washers securing the display board to the gantry “U” brackets. Install two ground lugs to display board with screw and flat washer. Refer to Figure 246.
3. Position front display shroud over gantry front cover, and install six screws, flat washers, and standoffs securing the shroud to the display board. Refer to Figure 247.
4. Install cable connectors to display board connectors P1, P2, P3, P4, P5, and P6. Install cable connector to front auto–lite.
5. Close the gantry front cover.
6. Apply power and verify the display assembly operation. Drive the Patient Support vertical or horizontal, activate the Laser, or tilt the Gantry as appropriate.

X–RAY TUBE AND HEAT EXCHANGER

The process for replacing the X–ray tube and heat exchanger is as follows:

- Perform physical replacement of X–ray tube and heat exchanger (this paragraph)
- Perform Preheat Calibration procedure from MANEXP program
- Perform Tube Seasoning procedure from MANEXP program
- Perform Mechanical Tube Alignment procedure from MANEXP program
- Perform Lateral Centering procedure from Service Applications
- Perform Image Calibration procedure from Service Applications
- Perform Find_BadDets procedure from Service Applications
- Perform Check_Performance procedure from Service Applications
- Perform KVHVL procedure from Service Applications

INTRODUCTION:

The following procedure is used to remove and replace the X–Ray Tube and Heat Exchanger.

Tools Required:	1. 9/16”, 3/4”Socket, Ratchet
	2. 5/32” Hex Key
	3. Hoist Assembly
	4. Tube Mounting Tray Assembly
Material Required:	1. Transformer oil – 10cc
Tube Weight:	132 pounds (7” Tube with Cradle)
Heat Exchanger Wt:	45 pounds
Estimated Time	1 hour

REMOVAL PROCEDURE:

> ⚠ **CAUTION**
>
> DO NOT ATTEMPT TO REPLACE THE X–RAY TUBE UNTIL TUBE HEAT IS AT 35% OR LOWER ANODE HEAT.

1. Remove Gantry power. Refer to the PowerUp/PowerDown procedure.
2. Open the Gantry Front Covers and remove the Front Cone. Refer to the Gantry Covers procedure.

> ⚠ **WARNING**
>
> **USE EXTREME CAUTION WHEN SERVICING EQUIPMENT WHILE UPPER GULL WING PANEL IS OPEN. THE PANEL HAS SHARP EDGES. FAILURE TO COMPLY COULD RESULT IN SERIOUS INJURY OR TO PERSONNEL.**

3. Rotate the Scan Frame counterclockwise by hand until the X–Ray tube is at the 12:00 position. Lock the scan frame with the locking pin. Pull the top pin out from the bottom cover mounting bracket. Push the bracket down to allow room for heat exchanger removal. See Figure 248.

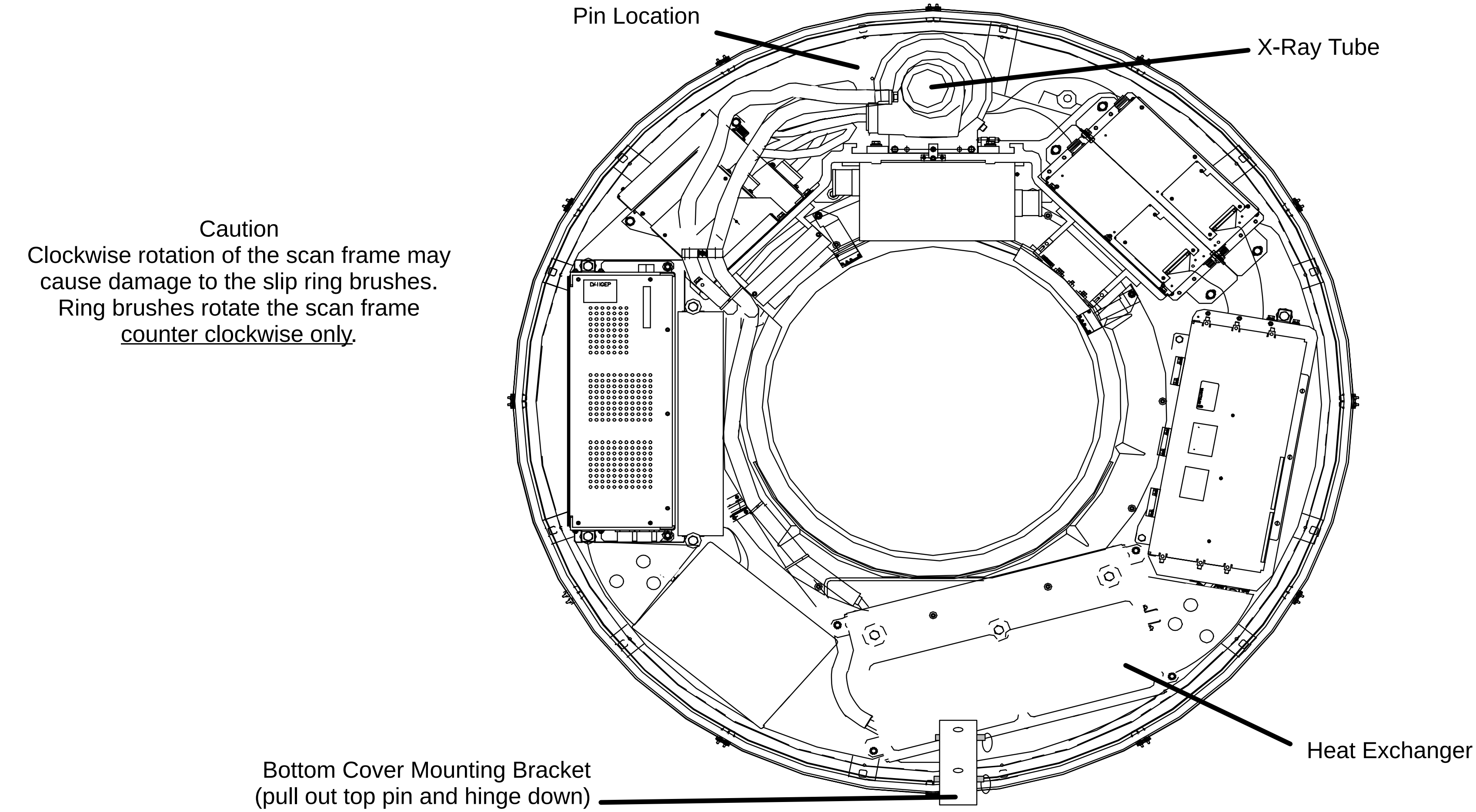


FIGURE 248 SCANNED FRAME PINNED WITH TUBE AT 12:00 POSITION

4. Secure Hoist Assembly to Patient Support. Make sure Clamp Base is seated firmly in position on Patient Support and that Knob Bolts are screwed in securely. See Figure 249.

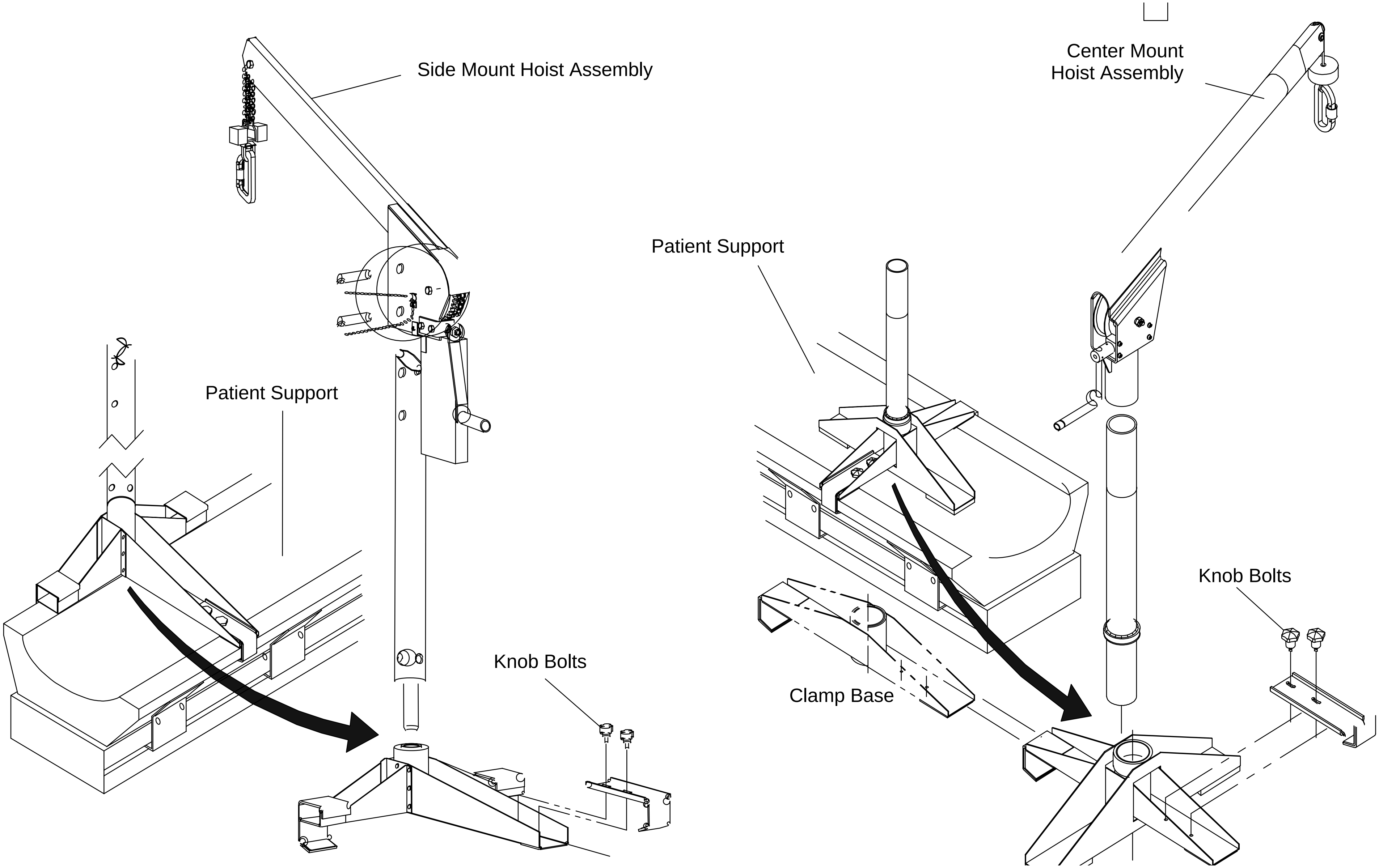


FIGURE 249 HOIST ASSEMBLY

5. Loosen and remove the four Cradle Locking Hex Head Bolts. See Figure 250.

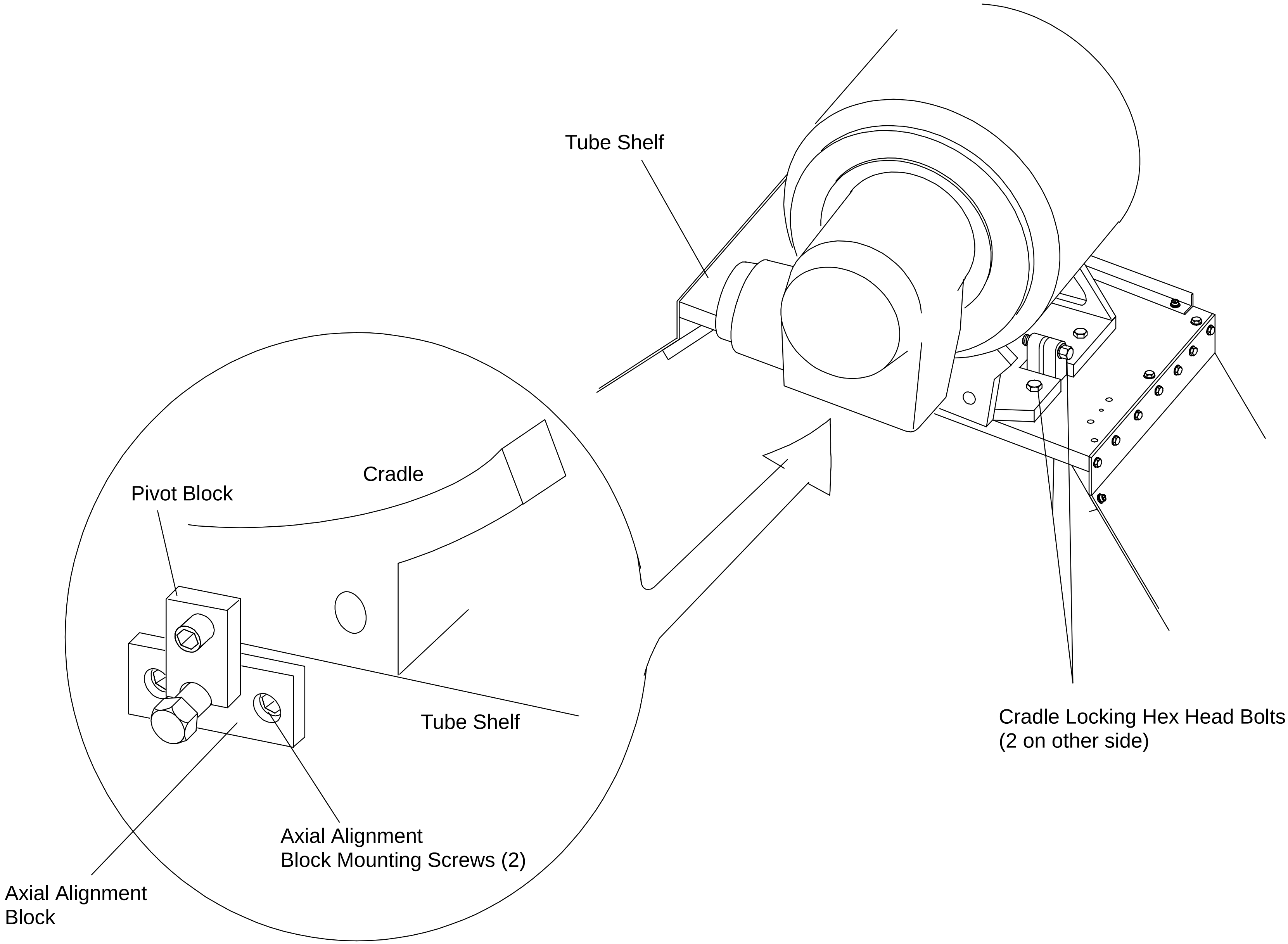


FIGURE 250 XRAY TUBE MOUNTING

6. Install the tube mounting tray in front of the xray tube and secure with the appropriate bolts. Refer to Figure 251.

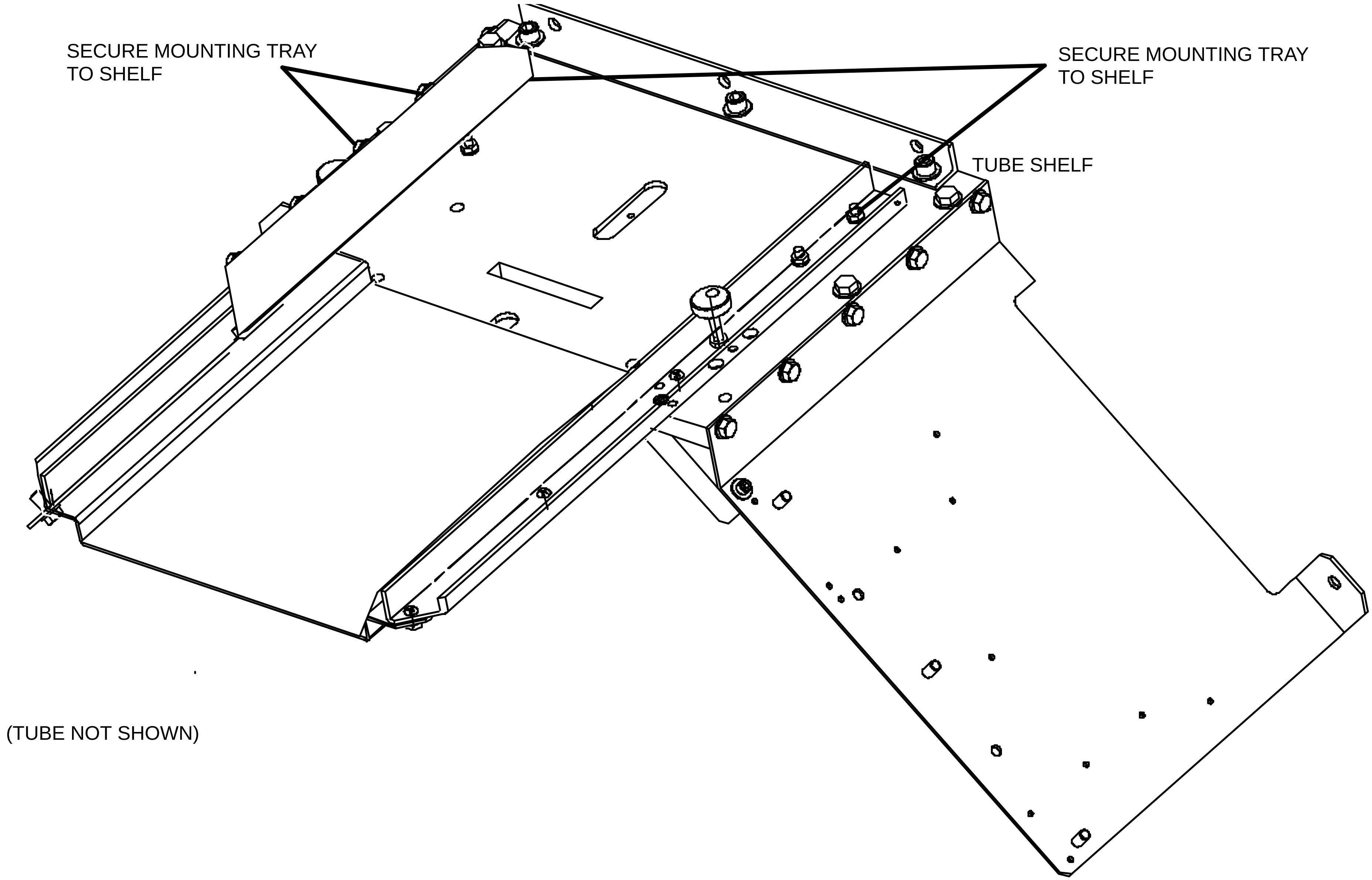


FIGURE 251 XRAY TUBE MOUNTING TRAY ON TUBE SHELF

7. Tag and disconnect the Quick Disconnects labeled J4 (Rotor Cable) and J6 (Thermal Switch Cable) that are located just behind and to the left of the X–Ray Tube.

8. Tag and remove High Voltage Federal Connectors from Anode and Cathode High Voltage Modules. Touch each High Voltage Federal Connector conductor to ground to bleed off any residual capacitive charge.

> ⚠ **WARNING**
>
> **MAKE NO ATTEMPT TO REMOVE THE HIGH VOLTAGE FEDERAL CONNECTORS FROM THE X–RAY TUBE. THEY ARE PERMANENTLY ATTACHED AND STAY WITH THE TUBE. REMOVING THESE CONNECTORS WILL CAUSE SERIOUS DAMAGE AND/OR DESTRUCTION OF THE TUBE.**

9. Loosen one of the two Axial Alignment Block Mounting Screws and remove the other. Let the Block swing down to allow removal of the X–Ray Tube Cradle. Refer to Figure 250.

> **NOTE**
>
> This method minimizes the need for lateral adjustment of the X–Ray Tube if the Axial Pivot Block is not loosened. However, care must be exercised not to damage the Patient Support. Place a protective cover on the Carbon Top before setting the Tube and Cradle on it.

10. Remove Hose Clamps and any Tie Wraps securing Oil Hoses to Scan Frame. Pull X–Ray Tube out along mounting tray (6”–10” max) in order to route Lifting Strap through Cradle Holes. See Figure 252.

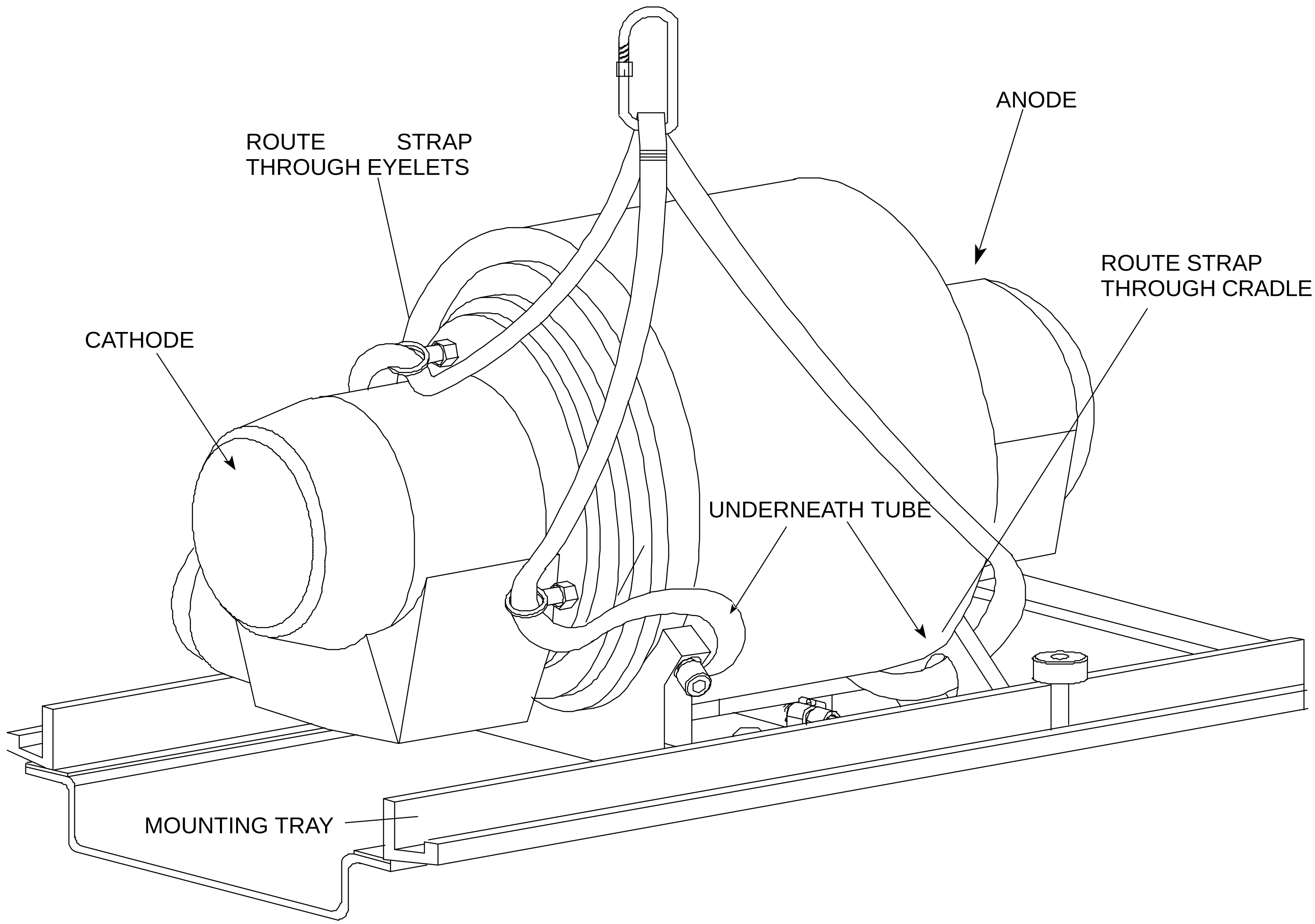


FIGURE 252 LIFTING STRAP ROUTING

11. Route Lifting Strap through one eyelet, under X–Ray Tube and up other side through other eyelet. Route second Lifting Strap through one Cradle hole at other end of X–Ray Tube, under X–Ray Tube and through hole on opposite side. Refer to Figure 252.

12. Place Lifting Strap ends onto the Hoist Clip and lock Clip. Remove slack and carefully pull Tube and Cradle Assembly to the end of Mounting Rails.
13. Carefully pull Tube Assembly from Mounting Rails. Brace Tube Assembly to keep it from rocking.
14. Carefully move Patient Support away from Gantry, swing X–Ray Tube to the right and lower Tube to floor. Lay Tube on its back with Shutter side up.
15. Remove Lifting Straps from Tube and Cradle Assembly and then remove Cradle from Tube.
16. Tag and disconnect the Quick Disconnect labeled J5 (Heat Exchanger Power) at the High Speed Starter.
17. Remove the two bolts on the heat exchanger supports. Slide the heat exchanger out and away from the gantry. See Figure 253.

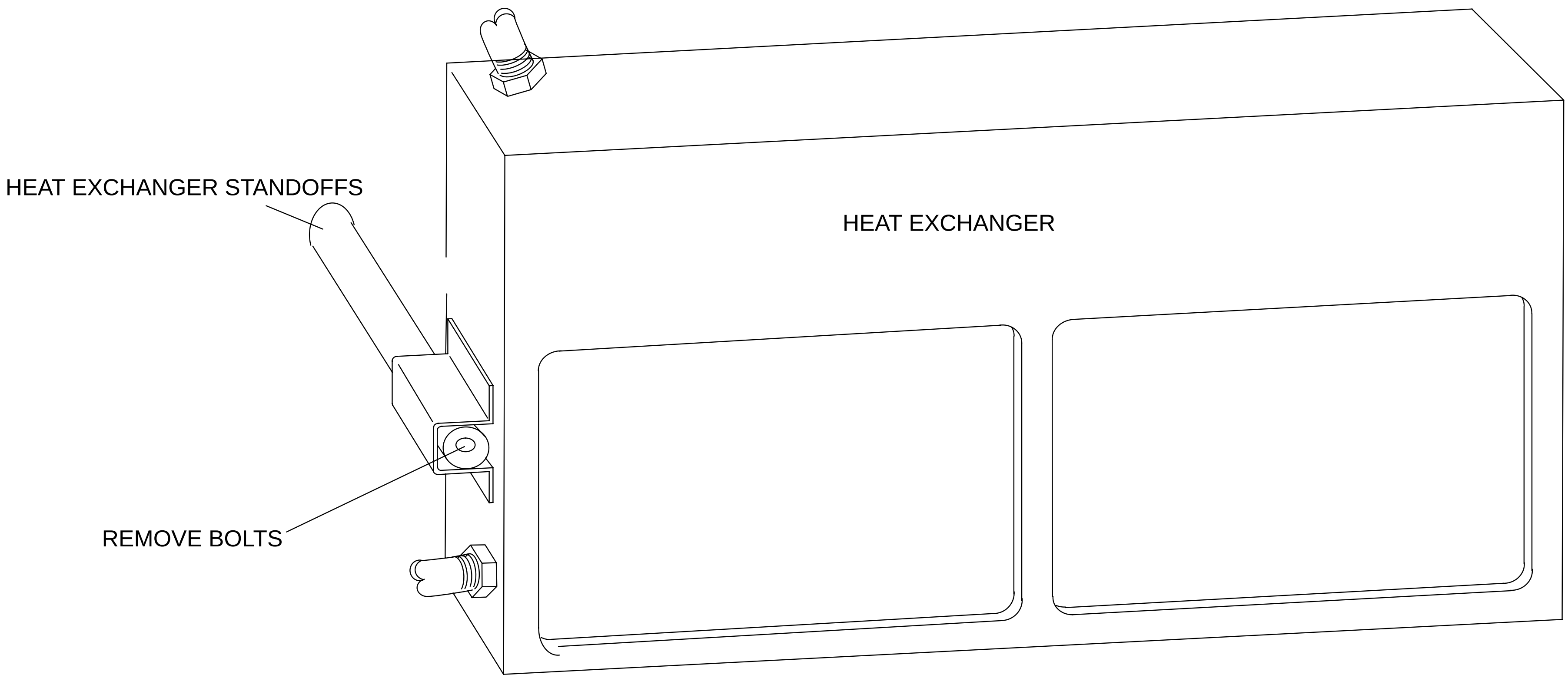


FIGURE 253 HEAT EXCHANGER MOUNTING HARDWARE

⚠ CAUTION

THE HEAT EXCHANGER WEIGHS 45 LBS. USE CAUTION WHEN HANDLING THIS ITEM. FAILURE TO COMPLY MAY RESULT IN EQUIPMENT DAMAGE AND/OR SERIOUS INJURY TO PERSONNEL.

18. Remove the Heat Exchanger Mounting bolts. Set them aside for later installation on the replacement unit.

XRAY TUBE REPLACEMENT:

19. Place box with replacement X–Ray Tube on left side of Patient Support. Open shipping container of replacement Tube and remove packaging material from container.
20. Lift Heat Exchanger from container and place it on floor near front of Gantry.

NOTE

Cover the Patient Support before placing the Tube on it or the Carbon Top may be damaged by the heavy Tube.

21. Place Lifting Straps through eyelets of Tube. Lift the X–Ray Tube from its vertical position within its container and position it with the Cradle Side up on the Patient Support.
22. With Tube Port facing upwards, remove protective metal covering. Check for the plastic covering over Tube Port opening and check for dirt/dust. Remove any debris found on the plastic. **DO NOT REMOVE PLASTIC**

23. Attach Cradle Assembly to Xray Tube. Follow illustration in Figure 254 for proper installation. Torque Mounting Hex Head and Socket Head Screws to 15 ft–lbs.lignment Pins.

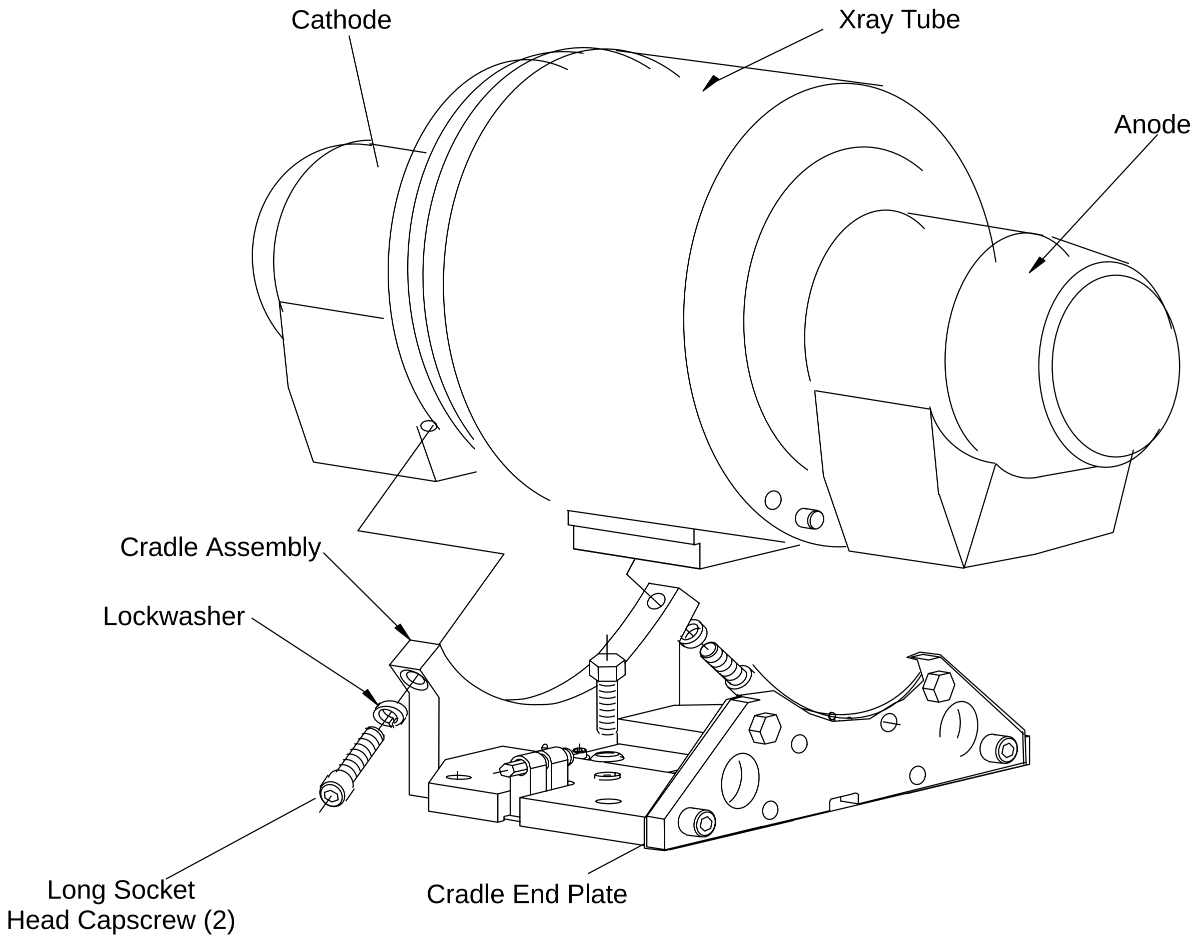


FIGURE 254 Cradle Assembly

24. Remove Packing Slips and Cable Ties from the Tube. Wipe off the Tube with cloth and check for container–debris.

25. (If possible) Rotate the Tube so that its Cradle side is facing down and its Oil Hoses facing left on the Patient Support.

NOTE

Ensure the Carbon Top has a protective cover to prevent damage by the Axial Alignment Pivot Block attached to the Cradle.

26. Remove the Lifting Straps from the eyelets. Reposition one strap through one eyelet, under X–Ray Tube and up other side through other eyelet. Route second strap through one cradle hole, under X–Ray Tube and through hole on opposite side. Refer to Figure 252.
27. Hoist the Tube to the tube tray. Push tube onto tray until back of cradle assembly is flush with front of Tube Shelf.

⚠ **WARNING**

USE EXTREME CAUTION AND VERIFY THAT THE X–RAY TUBE IS FULLY SUPPORTED BY THE MOUNTING RAILS BEFORE REMOVING THE LIFTING STRAPS. FAILURE TO COMPLY MAY RESULT IN EQUIPMENT DAMAGE AND/OR SERIOUS INJURY TO PERSONNEL.

28. Remove Lifting Straps from around Tube and move the hoist assembly out of the way.
29. Check Cables for clearance around the Tube. Push the Tube all the way back onto the Tube Shelf, ensuring the Cradle Assembly is seated on the Key Block. Refer to Figure 252.
30. Replace the Cradle Locking Bolts and fasten the Cradle down enough to relieve pressure from the tube tray.
31. Remove tube tray and secure Axial Alignment Block in its original position with the Socket Head Capscrew removed earlier.
    - Apply Loc–Tite to Alignment Block Screws and tighten to hand–tight.
    - Tighten Cradle hardware to 15 ft–lbs.
32. Position the Heat Exchanger on standoffs and push it back into position. Place the Cables behind the Heat Exchanger. Secure with the two bolts. Refer to Figure 253.
33. Secure Oil Hoses with Hose Clamps. DO NOT OVERTIGHTEN HEX NUTS.

NOTE

Align the Hose Clamps with the labels on the Hoses.

34. Reconnect Quick Disconnects J4 (Rotor Cable), J5 (Heat Exchanger Power Cable), and J6 (Thermal Switch Cable).
35. Inspect the Federal Connector for any sign of damage.
36. Using a clean, lint–free wipe and 90% alcohol, wipe the surface clean of dirt or dust.
37. Pour enough Transformer Oil into the Federal Connector receptacles to coat the entire surface of the cable connector after insertion into the module (10–15 cc). The amount will vary due to slight differences in the connectors.
38. Replace the Anode and Cathode High Voltage Federal Connectors.
39. Rotate the Scan Frame counterclockwise by hand while checking for loose Cables. Also check for clearance over the entire range of travel between the Rotating Frame Cables and the Stationary Frame.

> ⚠ **CAUTION**
>
> CLOCKWISE ROTATION OF THE SCAN FRAME MAY CAUSE DAMAGE TO THE SLIP RING BRUSHES. ROTATE THE SCAN FRAME COUNTERCLOCKWISE ONLY.

40. Align the X–Ray Tube. Refer to X–Ray Tube Mechanical Alignment in the Mechanical Calibration Manual.
41. Apply Gantry power. Refer to the PowerUp/PowerDown procedure.

> ⚠ **WARNING**
>
> **USE EXTREME CAUTION WHEN OPERATING OR WORKING ON EQUIPMENT WITH COVERS REMOVED. DANGEROUS VOLTAGES ARE EXPOSED WHEN THE GANTRY COVERS ARE REMOVED. FAILURE TO COMPLY CAN CAUSE SERIOUS INJURY OR DEATH TO PERSONNEL.**

42. Push the bottom cover mounting bracket back up and secure with the locking pin removed earlier. Replace the Gantry Front Cone. Refer to Front Cone Replacement in the Gantry Covers procedure.

> **NOTE**
>
> Step 42. is necessary to avoid artifacts in the test scans.

43. Pull the Front Cover Interlock Switch Plunger out to enable operation while the Front Covers are open.
44. To verify proper System operation, run Full–field Water Phantom Scans and allow the System to process and display an Image.
45. Return the Front Cover Interlock Switch Plunger to its center position.
46. Remove Gantry power. Refer to the PowerUp/PowerDown procedure.
47. Replace the Gantry lower Front Cover, close the upper Front Cover. Refer to Upper Front Gantry Covers Closing procedure in the Gantry Covers procedure.

X–RAY CONTROL CHASSIS

INTRODUCTION: This procedure is used to remove and replace the XSC+.

TOOLS REQUIRED:

1. Slotted Screwdriver
2. 3/4” Socket, Ratchet
3. Torque wrench (75 ft–lb)

ESTIMATED TIME: 45 Min.

CHASSIS WEIGHT: 17 pounds

XRAY CONTROL CHASSIS REMOVAL:

1. Remove Gantry power. Refer to the PowerUp/PowerDown procedure.
2. Open the Gantry Front Covers and remove the Front Cone. Refer to the Gantry Covers procedure.

> ⚠ **WARNING**
>
> **USE EXTREME CAUTION WHEN SERVICING EQUIPMENT WHILE THE UPPER GULL WING PANEL IS OPEN. THE PANEL HAS SHARP EDGES. FAILURE TO COMPLY COULD RESULT IN SERIOUS INJURY TO PERSONNEL.**

3. Rotate the Scan Frame counterclockwise by hand until the X–Ray Tube is at the 12:00 position. Lock the Scan Frame with the Locking Pin. See Figure 255.

> ⚠ **CAUTION**
>
> CLOCKWISE ROTATION OF THE SCAN FRAME MAY CAUSE DAMAGE TO THE SLIP RING BRUSHES. ROTATE THE SCAN FRAME COUNTERCLOCKWISE ONLY.

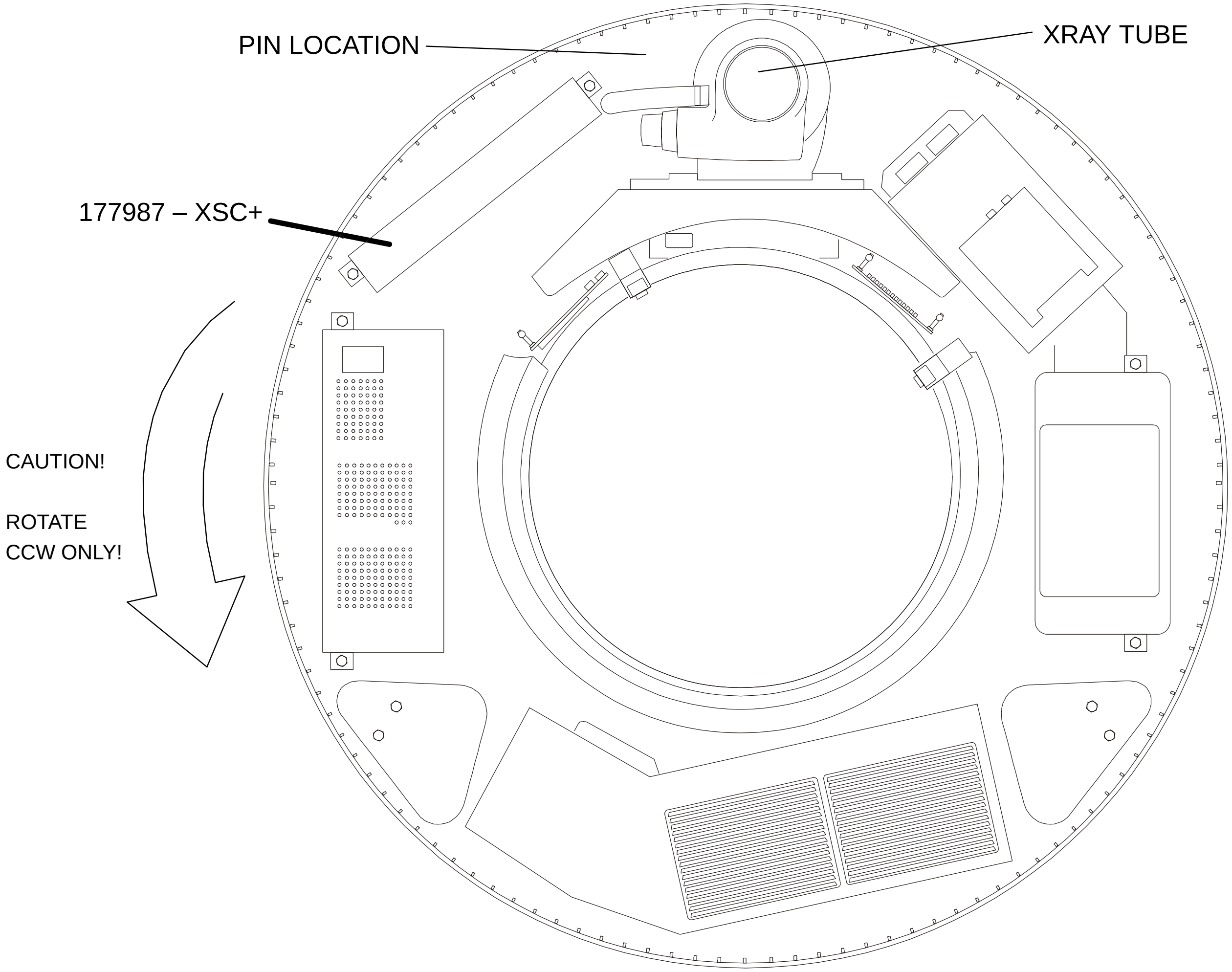


FIGURE 255 Scan Frame Pinned with Tube at 12:00 O’clock

CRX–400 SYSTEMS:

177987

4. Remove Hose Clamp from side of the XSC+ Chassis and move Hoses out of the way. See Figure 256.
5. Remove the power cable from the front of the XSC+.
6. Remove the two mounting bolts, their washers and lock washers that secure the XSC+ to the scan frame. Screw extender pins into the holes – tighten hand tight only. See Figure 256.

7. Unscrew the two Captive Slot Head Cover Screws and remove the Cover. See Figure 256.

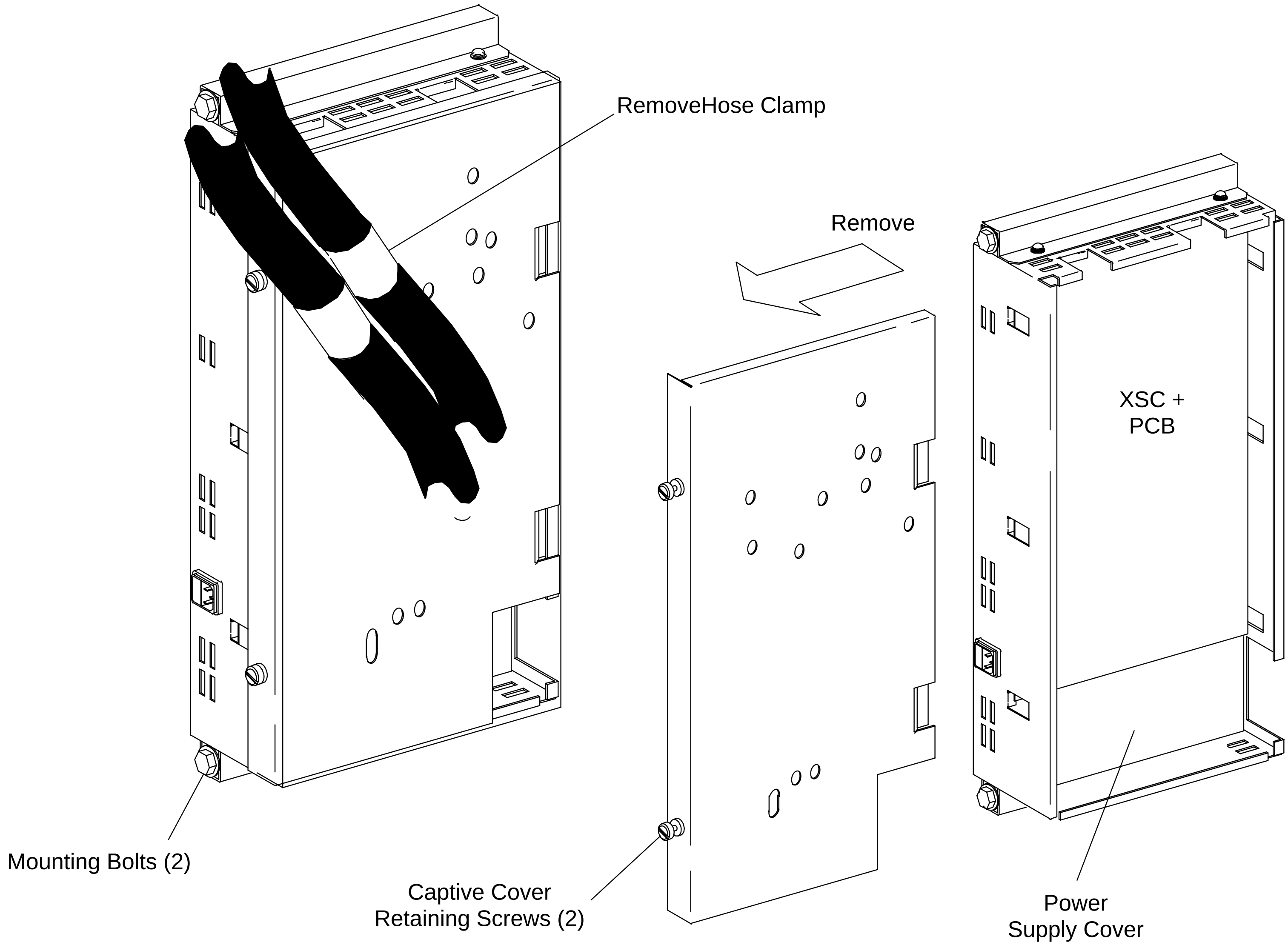


FIGURE 256 XSC+ Chassis Cover and Hose Clamp Removal

8. Tag and remove Cables connected to J2, J3, J5, J9, J10, J11, J14, J15, J21 and J22. Cut and remove any Tie Wraps necessary to remove Cables. See Figure 257.

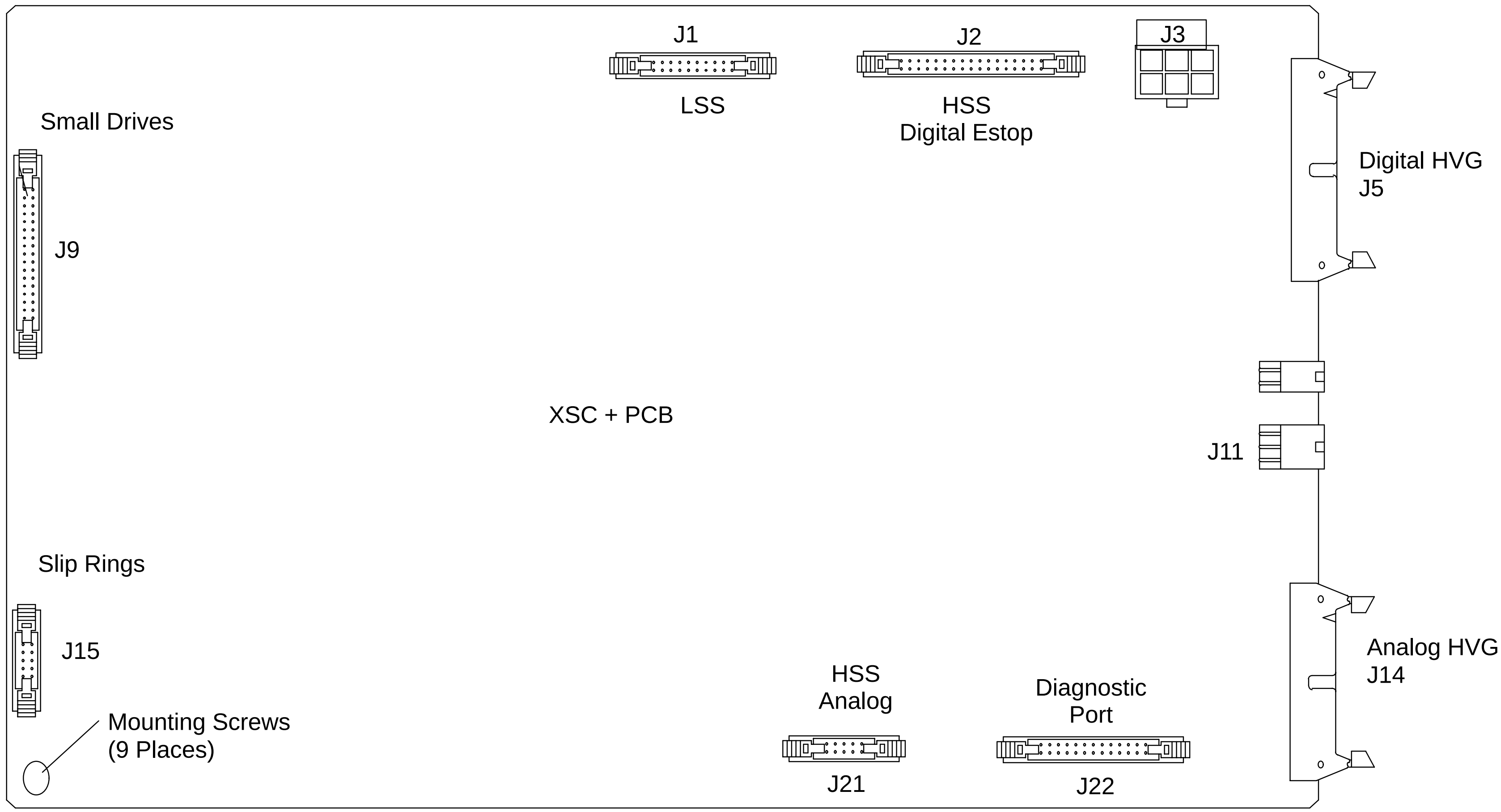


FIGURE 257 XSX+ PCB

9. Secure all Cables safely out of the way.
10. Replace the Cover and secure (only one screw required).
11. Slide the XSC+ Chassis off the Extender Pins removing it from the Gantry.

XSC+ CHASSIS REPLACEMENT:

12. Install the X–Ray Control Chassis by reversing the steps of the above section .

    Torque the Mounting Bolts to 15 ft–lb

13. Remove the Scan Frame Locking Pin. Rotate the Scan Frame counterclockwise by hand while checking for loose Cables. Also check for clearance over the entire range of travel between the Rotating Frame Cables and the Stationary Frame.

> ⚠ **CAUTION**
>
> CLOCKWISE ROTATION OF THE SCAN FRAME MAY CAUSE DAMAGE TO THE SLIP RING BRUSHES. ROTATE THE SCAN FRAME COUNTERCLOCKWISE ONLY.

14. Apply Gantry power. Refer to the PowerUp/PowerDown procedure.

> ⚠ **WARNING**
>
> **USE EXTREME CAUTION WHEN OPERATING OR WORKING ON EQUIPMENT WITH COVERS REMOVED. DANGEROUS VOLTAGES ARE EXPOSED WHEN THE GANTRY COVERS ARE REMOVED. FAILURE TO COMPLY CAN CAUSE SERIOUS INJURY OR DEATH TO PERSONNEL.**

15. Replace the Gantry Front Cone. Refer to the Gantry Covers procedure. Pull the Front Cover Interlock Switch Plunger out to enable operation while the Front Covers are open.
16. Push the Laser Button on the Gantry Control Panel to check the operation of the Laser.

> ⚠ **WARNING**
>
> **ROTATING MACHINERY. STAND CLEAR OF ROTATING FRAME WHEN EQUIPMENT IS OPERATED. FAILURE TO COMPLY MAY RESULT IN SERIOUS INJURY OR DEATH TO PERSONNEL.**

17. Using the diagnostic manexp, check the operation of the X–Ray Generator, Collimator, Compensator, and Shutter/Filter, by making an exposure from the Operator Console.
18. Return the Front Cover Interlock Switch Plunger to its center position. Remove Gantry power. Refer to the PowerUp/PowerDown procedure.
19. Replace the Gantry lower Front Cover, close the upper Front Cover. Refer to the Gantry Covers procedure.

COLLIMATOR ASSEMBLY

Introduction:	The following procedure is used to remove and replace the X–Ray Beam Collimator Assembly.
Tools Required:	1. 3/16” and 7/64” Hex Keys
	2. 5/16” Socket, Ratchet
	3. Torque wrench (75 ft–lb)
Material Required:	1. Loc–Tite (S301–145)
Chassis Weight:	29 lb.
Estimated Time:	25 MIN.

COLLIMATOR ASSEMBLY REMOVAL:

1. Apply Gantry Power. Refer to the PowerUp/PowerDown procedure.
2. Using the Gantry Control Panel switches, raise the Patient Support to its highest position.
3. Remove Gantry Power. Refer to the PowerUp/PowerDown procedure.
4. Open the Gantry Front Covers and remove the Front Cone. Refer to the Gantry Covers procedure.

> ⚠ **WARNING**
>
> **USE EXTREME CAUTION WHEN SERVICING EQUIPMENT WHILE THE UPPER GULL WING IS OPEN. THE PANEL HAS SHARP EDGES. FAILURE TO COMPLY COULD RESULT IN SERIOUS INJURY TO PERSONNEL.**

> ⚠ **WARNING**
>
> **USE EXTREME CAUTION WHEN OPERATING OR WORKING ON EQUIPMENT WITH COVERS REMOVED. DANGEROUS VOLTAGES ARE EXPOSED WHEN THE GANTRY COVERS ARE REMOVED. FAILURE TO COMPLY CAN CAUSE SERIOUS INJURY OR DEATH TO PERSONNEL.**

NOTE

The Collimator Assembly has been factory aligned. During Removal/Replacement procedures, be sure to not move the eccentric alignment bolt.

5. Place a reference mark on the Eccentric Adjustment Bolt and Gantry Frame. If the Eccentric Bolt should get moved, use the reference marks to realign it. Refer to Figure 258.
6. Loosen (but do not remove) the four Collimator Assembly Mounting Screws. See Figure 258.

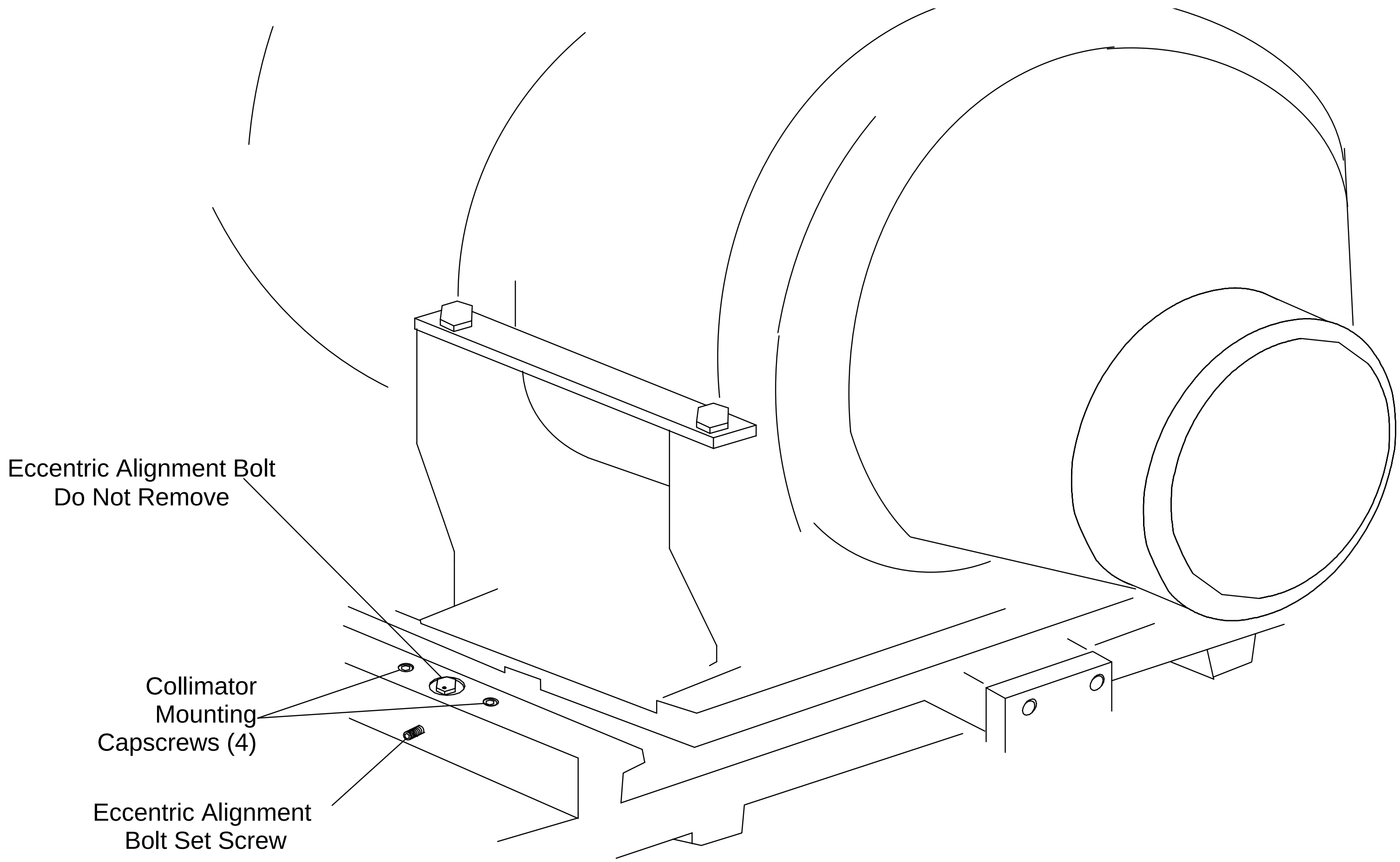


FIGURE 258 Collimator Mounting Screws

7. Rotate the Gantry Frame counterclockwise by hand until the Collimator Assembly is at the bottom center (6:00 position). The Scan Frame **cannot** be locked in this position. See Figure 259.

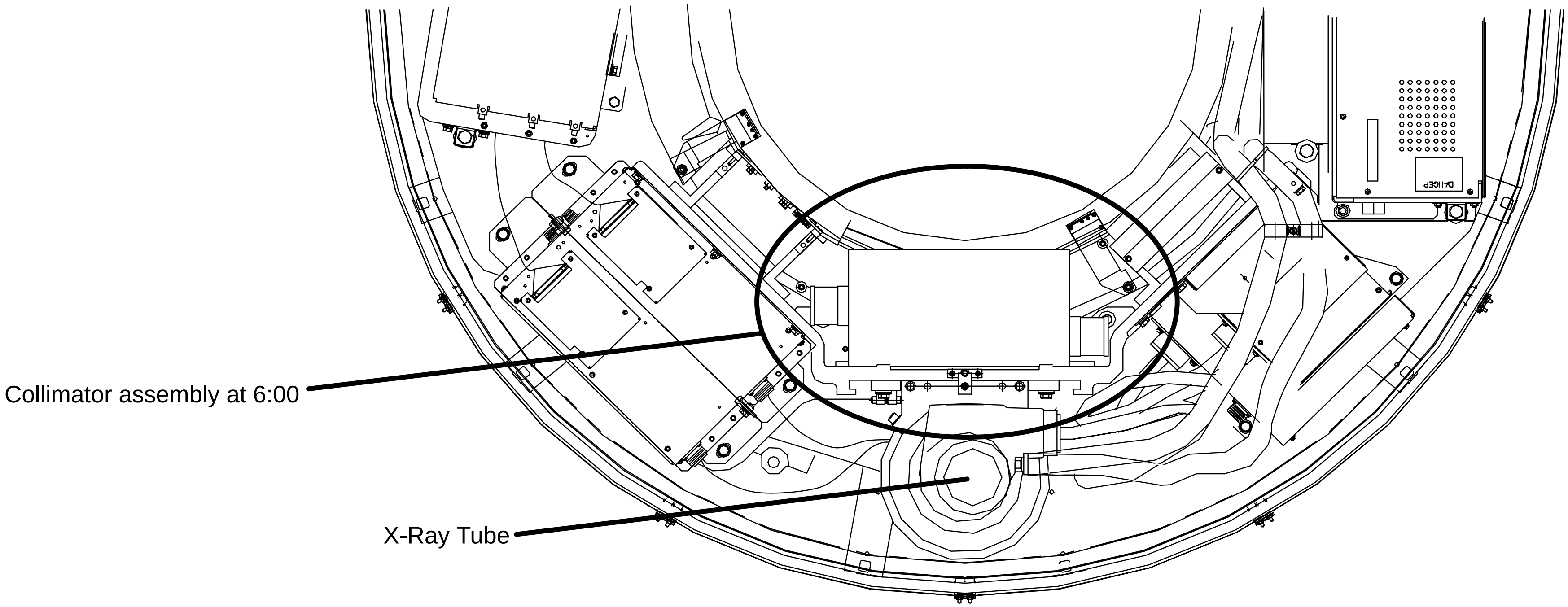


FIGURE 259 Collimator Assembly at 6:00

> ⚠ **CAUTION**
>
> CLOCKWISE ROTATION OF THE SCAN FRAME MAY CAUSE DAMAGE TO THE SLIP RING BRUSHES. ROTATE THE SCAN FRAME COUNTERCLOCKWISE ONLY.

8. Tag and disconnect all Collimator Assembly Cables from the Collimator Relay Control Board.

9. Carefully remove the four Collimator Mounting Capscrews. Refer to Figure 258.

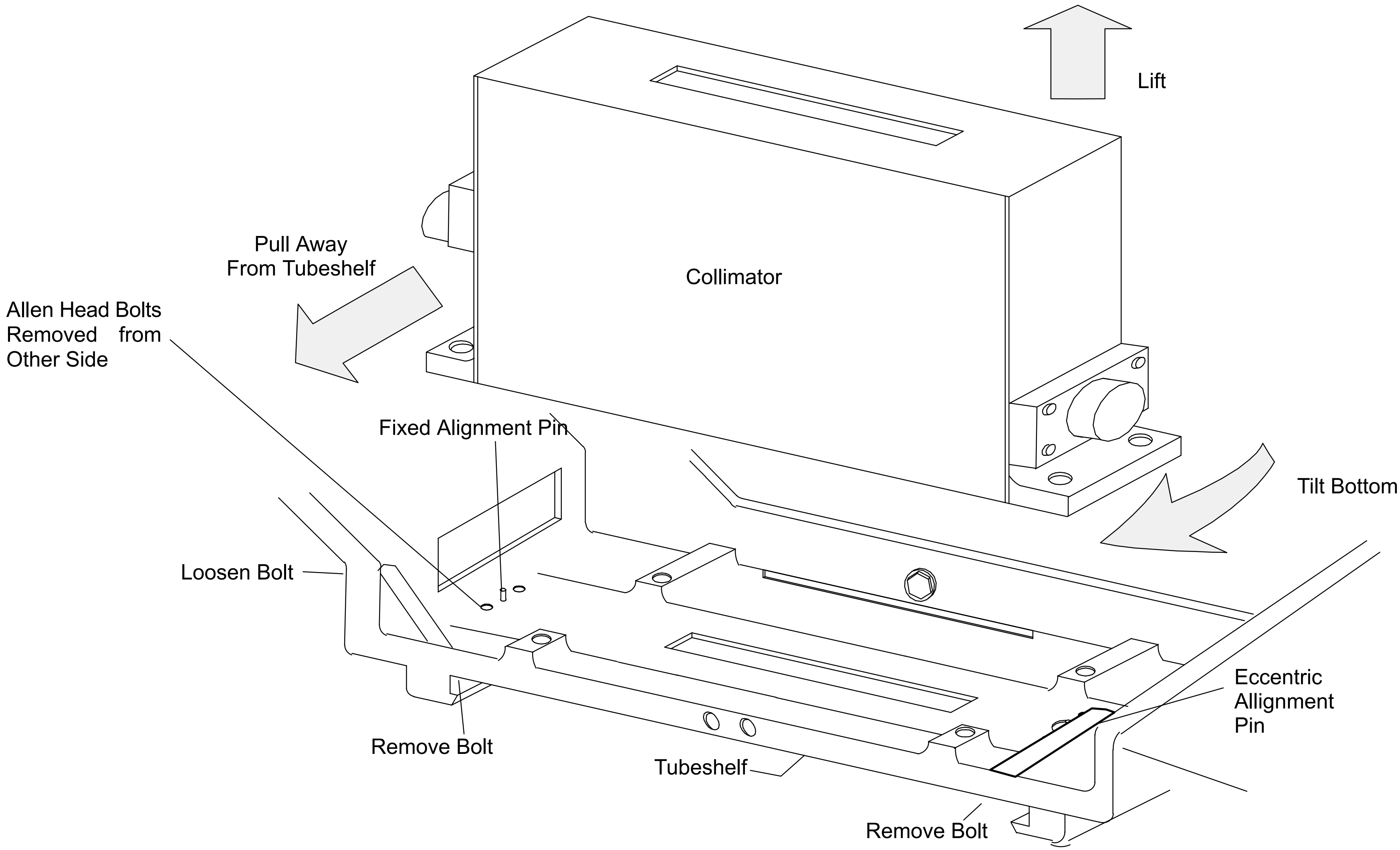


FIGURE 260 Collimator Assembly Removal

COLLIMATOR ASSEMBLY REPLACEMENT:

10. Inspect the mounting surfaces on the replacement Collimator Assembly and the Tube Shelf for dirt or foreign particles and clean if necessary.

NOTE

When installing the collimator assembly, ensure that the reference mark made in step 4. is lined up.

11. Install Collimator Assembly by reversing steps 9. and 8. Apply Loc–Tite to the Mounting Screw threads and torque to 13 ft–lb.
12. Rotate the Scan Frame counterclockwise by hand checking for loose Cables. Also check for clearance over the entire range of travel between the Rotating Frame Cables and the Stationary Frame.

⚠ **CAUTION**

CLOCKWISE ROTATION OF THE SCAN FRAME MAY CAUSE DAMAGE TO THE SLIP RING BRUSHES. ROTATE THE SCAN FRAME COUNTERCLOCKWISE ONLY.

13. Replace the Ganrty Front Cone. Refer to the Gantry Covers procedure.
14. Apply Gantry power. Refer to the PowerUp/PowerDown procedure.

⚠ **WARNING**

USE EXTREME CAUTION WHEN OPERATING OR WORKING ON EQUIPMENT WITH COVERS REMOVED. DANGEROUS VOLTAGES ARE EXPOSED WHEN THE GANTRY COVERS ARE REMOVED. FAILURE TO COMPLY CAN CAUSE SERIOUS INJURY OR DEATH TO PERSONNEL.

15. Pull the Front Cover Interlock Switch Plunger out to enable operation while the Front Covers are open.
16. Verify correct movement of the Compensator, Beam Filter and Collimator using gscope.
17. To verify proper system operation, run half–field and full–field water phantom scans and allow the system to process and display an image.
18. Remove Gantry power. Refer to the PowerUp/PowerDown procedure. Return the Front Cover Interlock Switch Plunger to its center position.
19. Replace the Gantry lower Front Cover, close the upper Front Cover. Refer to the Gantry Covers procedure.

SMALL DRIVES RELAY PCB

Introduction: The following procedure is used to remove and replace the Small Drives Relay Printed Circuit Board.

Tools Required:
1. 7/64” Ball Driver
2. 5/16” Socket, Ratchet

Material Required:
1. Loc–Tite (thread locker)

Estimated Time: 25 Min.

SMALL DRIVES RELAY PCB REMOVAL:

1. Remove Gantry power. Refer to the PowerUp/PowerDown procedure.
2. Open the Gantry Front Covers and remove the Front Cone. Refer to the Gantry Covers procedure.

> ⚠ **WARNING**
>
> **USE EXTREME CAUTION WHEN SERVICING EQUIPMENT WHILE THE UPPER GULL WING IS OPEN. THE PANEL HAS SHARP EDGES. FAILURE TO COMPLY COULD RESULT IN SERIOUS INJURY TO PERSONNEL.**

> ⚠ **WARNING**
>
> **USE EXTREME CAUTION WHEN OPERATING OR WORKING ON EQUIPMENT WITH COVERS REMOVED. DANGEROUS VOLTAGES ARE EXPOSED WHEN THE GANTRY COVERS ARE REMOVED. FAILURE TO COMPLY CAN CAUSE SERIOUS INJURY OR DEATH TO PERSONNEL.**

3. Rotate the Scan Frame counterclockwise by hand until the Small Drives Relay PCB is at the 6:00 position. The scan frame **cannot** be locked in this position. See Figure 261.

> ⚠ **CAUTION**
>
> **CLOCKWISE ROTATION OF THE SCAN FRAME MAY CAUSE DAMAGE TO THE SLIP RING BRUSHES. ROTATE THE SCAN FRAME COUNTERCLOCKWISE ONLY.**

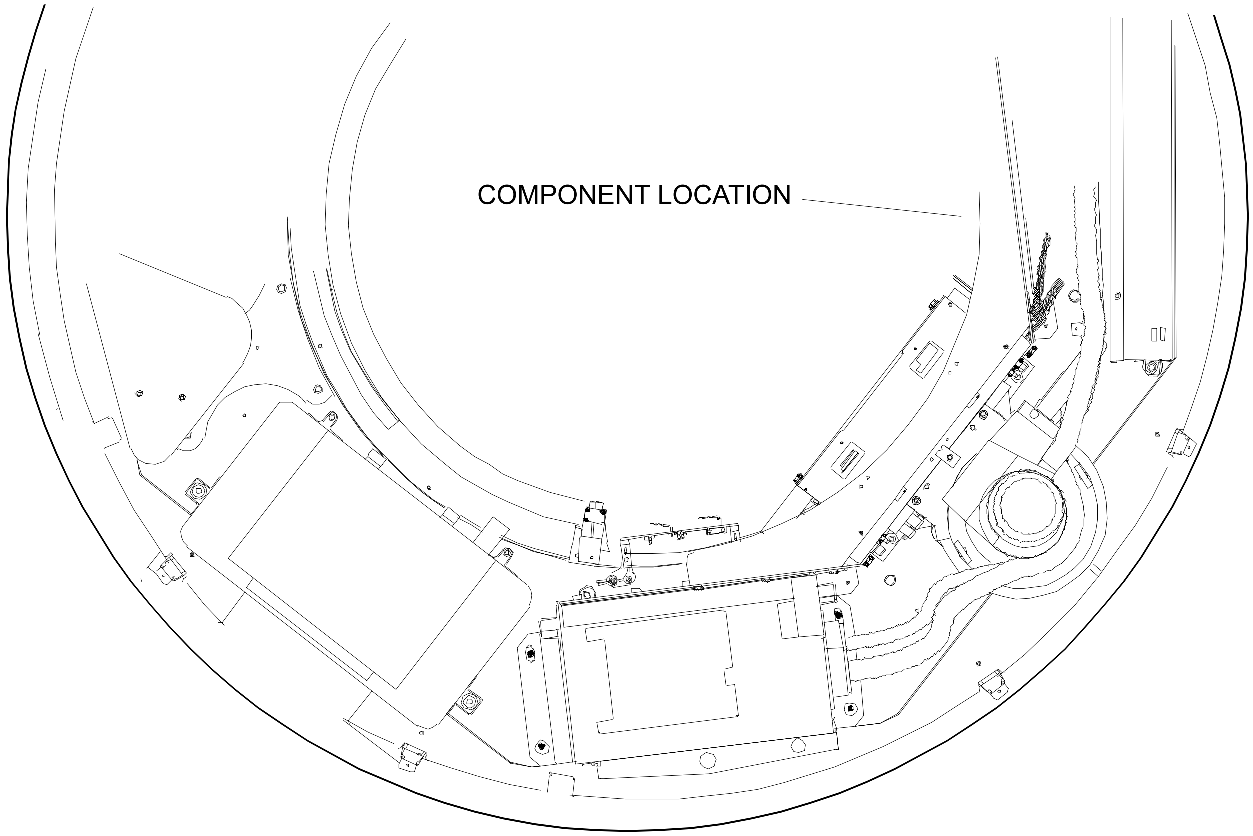


FIGURE 261 Small Drives Relay PCB at 6:00

4. Tag and disconnect all Wires/Cables exiting from Small Drives Relay PCB.

> ⚠ **CAUTION**
>
> **STATIC ELECTRICITY. CIRCUIT BOARDS MUST BE HANDLED WITH ANTI–STATIC TECHNIQUES. DISCHARGE YOUR BODY BEFORE HANDLING ANY CIRCUIT BOARD. KEEP ALL CIRCUIT BOARDS IN ANTI–STATIC BAGS OR ON CONDUCTIVE FOAM WHEN NOT INSTALLED IN THE EQUIPMENT. FAILURE TO COMPLY MAY CAUSE PERMANENT DAMAGE TO COMPONENTS ON CIRCUIT BOARDS.**

5. Support the Small Drives Relay PCB and remove six Mounting Socket Head Capscrews and the Threaded Rod and Sleeve. Remove the Small Drives Relay Board. See Figure 262.

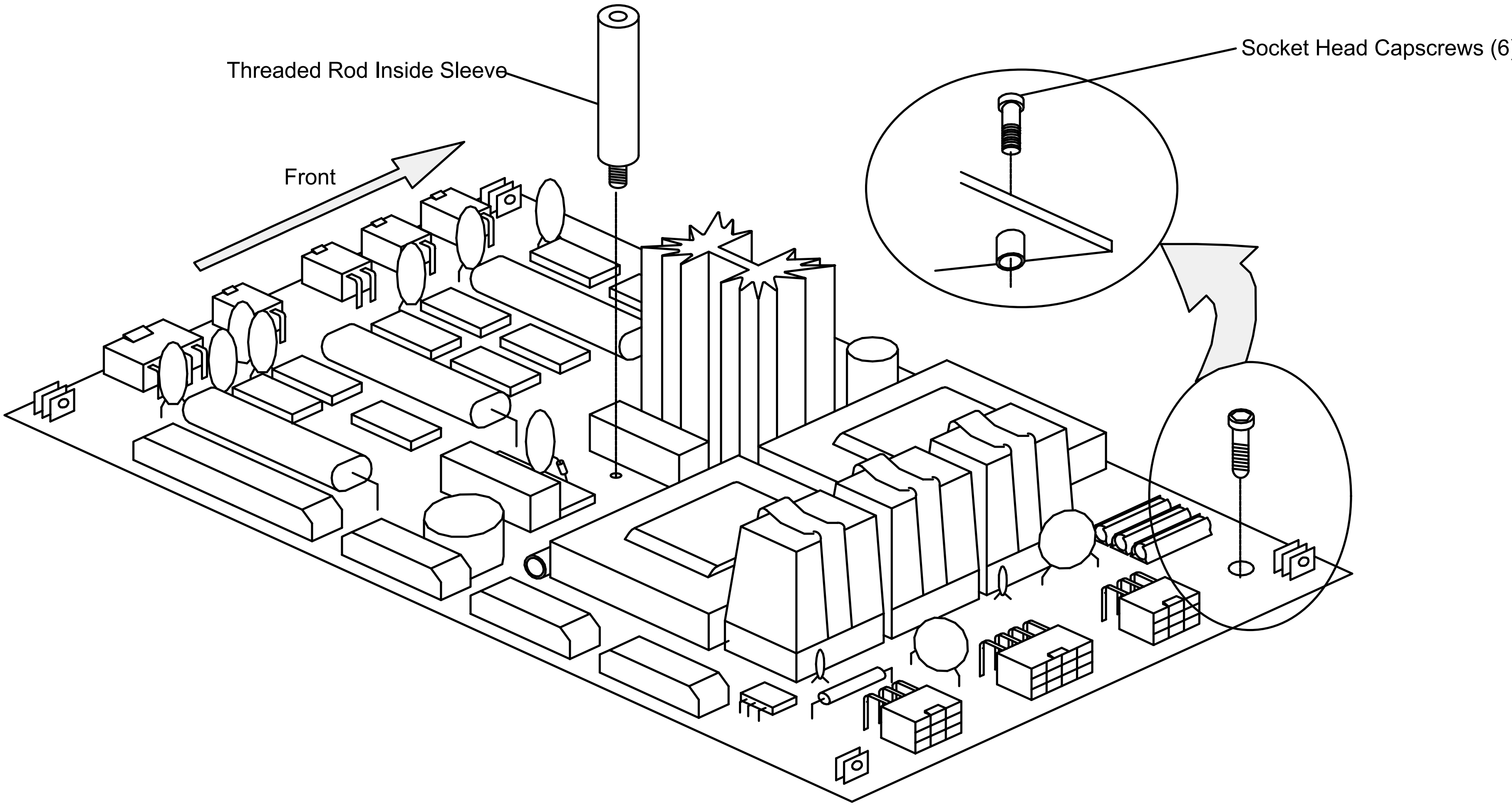


FIGURE 262 Small Drives Relay Printed Circuit Board

SMALL DRIVES RELAY PCB REPLACEMENT

6. Install the Small Drives Relay PCB by reversing steps 4. through 5.

   Apply Loc–Tite to the Protective Cover Standoff only on the side that screws into the Tubeshelf and to the six Mounting Socket Head Capscrews.

⚠ **CAUTION**

CLOCKWISE ROTATION OF THE SCAN FRAME MAY CAUSE DAMAGE TO THE SLIP RING BRUSHES. ROTATE THE SCAN FRAME COUNTERCLOCKWISE ONLY.

7. Rotate the scan frame counterclockwise by hand while checking for loose cables. Also check for clearance over the entire range of travel between the rotating frame cables and the stationary frame.
8. Replace the Gantry Front Cone. Refer to Gantry Covers procedure. Apply Gantry power. Refer to the PowerUp/PowerDown procedure.
9. Pull the Front Cover Interlock Switch Plunger out to enable operation while the Front Covers are open.

⚠ **WARNING**

ROTATING MACHINERY. STAND CLEAR OF ROTATING FRAME WHEN EQUIPMENT IS OPERATED. FAILURE TO COMPLY MAY RESULT IN SERIOUS INJURY OR DEATH TO PERSONNEL.

10. Check the operation of the Small Drives Relay PCB by making several Water Phantom Scans using different positions of the Filter, Compensator, and Collimator. Also, check the Laser operation by using the Gantry Control Panel Switches.
11. Return the Front Cover Interlock Switch Plunger to its center position. Remove Gantry power. Refer to PowerUp/PowerDown procedures.
12. Replace the Gantry lower Front covers, close the upper Front Cover. Refer to the Gantry Covers procedure.

HIGH VOLTAGE MODULE (UPDATE PER NEW MODULE, #178366)

Introduction: The following procedure is used to remove and replace the High Voltage Module.

Tools Required:
1. 1/2” and 9/16” Sockets, Ratchet
2. Torque Wrench (75 ft–lb)

Material Required:
1. Transformer oil
2. Dow Corning #4 Compound

Estimated TIme: 45 Min.

Chassis Weight: 42.0 pounds

HIGH VOLTAGE MODULE REMOVAL:

1. Remove Gantry power. Refer to the PowerUp/PowerDown procedure.
2. Open Gantry Front Covers and remove the Front Cone. Refer to the Gantry Covers procedure.

> ⚠ **WARNING**
>
> **USE EXTREME CAUTION WHEN SERVICING EQUIPMENT WHILE THE UPPER GULL WING PANEL IS OPEN. THE PANEL HAS SHARP EDGES. FAILURE TO COMPLY COULD RESULT IN SERIOUS INJURY TO PERSONNEL.**

> ⚠ **WARNING**
>
> **USE EXTREME CAUTION WHEN OPERATING OR WORKING ON EQUIPMENT WITH COVERS REMOVED. DANGEROUS VOLTAGES ARE EXPOSED WHEN THE GANTRY COVERS ARE REMOVED. FAILURE TO COMPLY CAN CAUSE SERIOUS INJURY OR DEATH TO PERSONNEL.**

3. Rotate the Scan Frame counterclockwise by hand until the X–Ray Tube is at the 12:00 position. Lock the Scan Frame with the Locking Pin. See Figure 263.

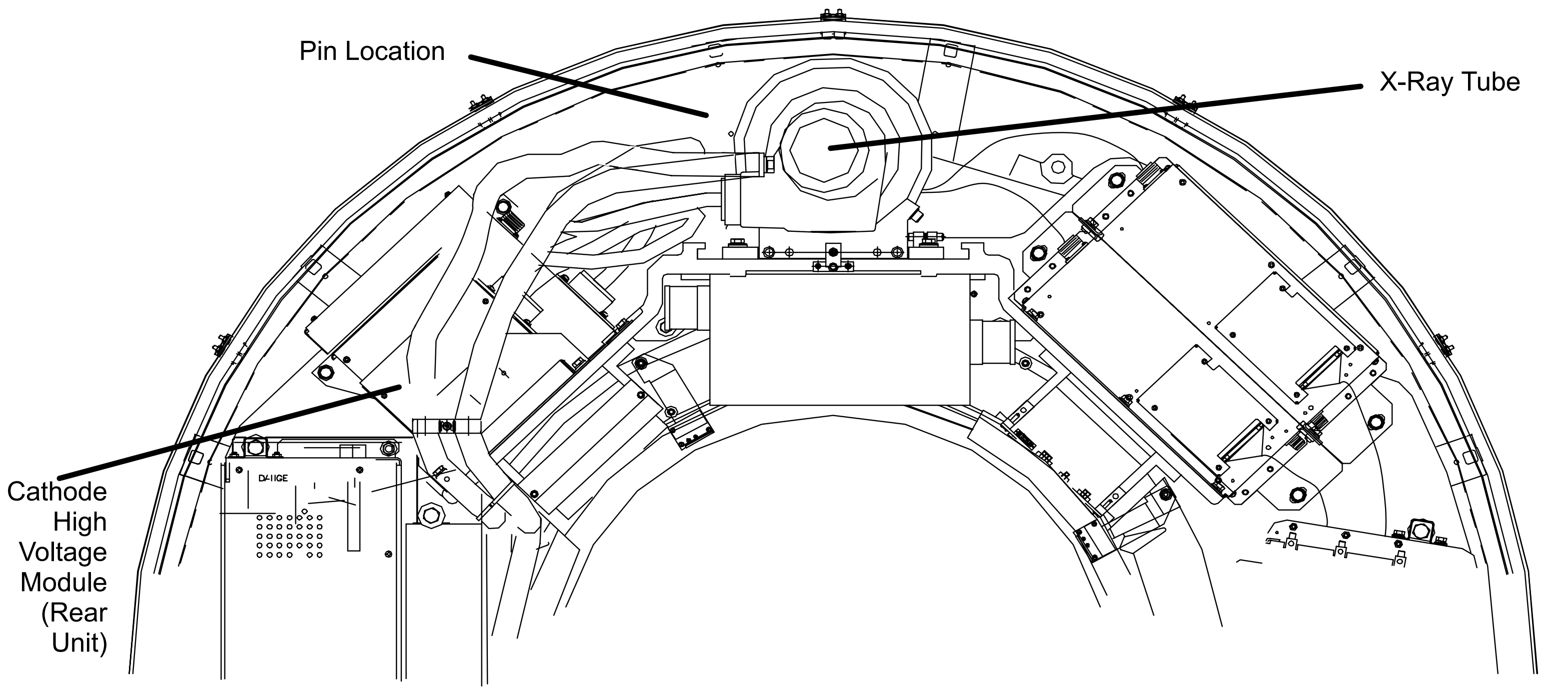


FIGURE 263   Scan Frame Pinned with X–Ray Tube at 12:00

4. Remove the Anode High Voltage Module. Refer to the Anode High Voltage Module Removal procedure.

5. Tag and disconnect all Cables on the Cathode High Voltage Module. See Figure 264.

> ⚠ **WARNING**
>
> **THE CATHODE HV MODULE WEIGHS 49 LBS. USE EXTREME CAUTION WHEN HANDLING THIS MODULE. FAILURE TO COMPLY CAN CAUSE SERIOUS DAMAGE TO THE EQUIPMENT AND/OR SERIOUS INJURY TO PERSONNEL.**

6. Remove the two Hex Head Bolts, Washers, and Lock washers that secure the Cathode High Voltage Module to the Gantry Scan Frame. Refer to Figure 264.

7. Remove the two Hex Head Bolts, Washers and Lockwashers that secure the Cathode High Voltage Module to the Tube Shelf and remove the module. Refer to Figure 264.

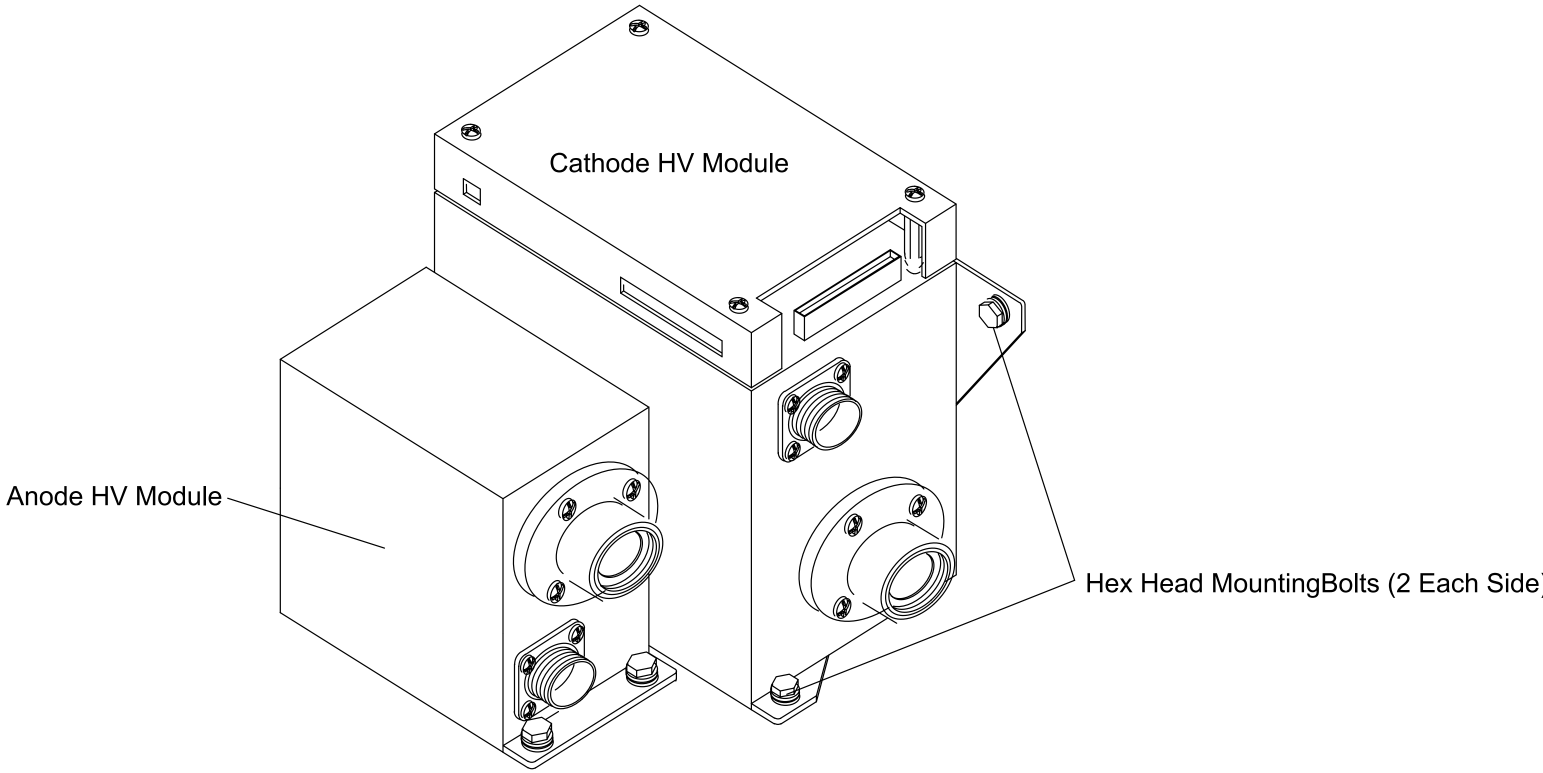


FIGURE 264 Anode and Cathode HV Modules

CATHODE HIGH VOLTAGE MODULE REPLACEMENT:

8. Install the replacement Cathode High Voltage Module by reversing steps 5. through 7.

9. Torque the Hex Head Mounting Bolts to 15 ft–lbs.

10. Replace the Spellman High Voltage and High Tension Federal Connectors. Refer to the Cable Cleaning and Installation procedure.

11. Replace the Anode High Voltage Module. Refer to the Anode High Voltage Module Replacement procedure.

12. Remove the Scan Frame Locking Pin.

13. Rotate the Scan Frame counterclockwise by hand while checking for loose cables. Also check for clearance over the entire range of travel between the Rotating Frame cables and the Stationary Frame.

⚠ CAUTION

CLOCKWISE ROTATION OF THE SCAN FRAME MAY CAUSE DAMAGE TO THE SLIP RING BRUSHES. ROTATE THE SCAN FRAME COUNTERCLOCKWISE ONLY.

14. Replace the Gantry Front Cone. Refer to the Gantry Covers procedure.
15. Apply Gantry power. Refer to the PowerUp/PowerDown procedure.

⚠ WARNING

USE EXTREME CAUTION WHEN OPERATING OR WORKING ON EQUIPMENT WITH COVERS REMOVED. DANGEROUS VOLTAGES ARE EXPOSED WHEN THE GANTRY COVERS ARE REMOVED. FAILURE TO COMPLY CAN CAUSE SERIOUS INJURY OR DEATH TO PERSONNEL.

16. Pull the Front Cover Interlock Switch Plunger out to enable operation while the Front Covers are open.
17. To verify system operation, run several full–field and half–field water phantom scans and allow the system to process and display an image.
18. Return the Front Cover Interlock Switch Plunger to its center position.
19. Remove Gantry power. Refer to the PowerUp/PowerDown procedure.
20. Replace the Gantry lower Front Cover, close the upper Front Cover. Refer to the Gantry Covers procedure.

INVERTER CHASSIS

Introduction:	The following procedure is used to remove and replace the Inverter Chassis. Allow five minutes for the Rail voltages to discharge after Gantry power has been removed.
Tools Required:	1. 7/16", 9/16" and 3/4" Sockets, Ratchet
	2. #1 tip Phillips Screwdriver
	3. 3/16" Hex Key
	4. Torque wrench (75 ft–lb)
	5. Hoist Assembly
Material Required:	None
Chassis Weight:	112 pounds
Estimated Time:	1.0 Hr.

INVERTER CHASSIS REMOVAL:

1. Remove Gantry power. Refer to the PowerUp/PowerDown procedure.
2. Open Gantry Front Covers and remove the Front Cone. Refer to the Gantry Covers procedure.

> ⚠ **WARNING** ⚠
>
> **USE EXTREME CAUTION WHEN OPERATING OR WORKING ON EQUIPMENT WITH COVERS REMOVED. DANGEROUS VOLTAGES ARE EXPOSED WHEN THE GANTRY COVERS ARE REMOVED. FAILURE TO COMPLY CAN CAUSE SERIOUS INJURY OR DEATH TO PERSONNEL.**

> ⚠ **WARNING**
>
> **USE EXTREME CAUTION WHEN SERVICING EQUIPMENT WHILE THE UPPER GULL WING IS OPEN. THE PANEL HAS SHARP EDGES. FAILURE TO COMPLY COULD RESULT IN SERIOUS INJURY TO PERSONNEL.**

3. Rotate the Scan Frame counterclockwise by hand until the X–Ray Tube is at the 12:00 position. Lock the Scan Frame with the Locking Pin. See Figure 265.

> ⚠ **CAUTION**
>
> CLOCKWISE ROTATION OF THE SCAN FRAME MAY CAUSE DAMAGE TO THE SLIP R BRUSHES. ROTATE THE SCAN FRAME COUNTERCLOCKWISE ONLY.

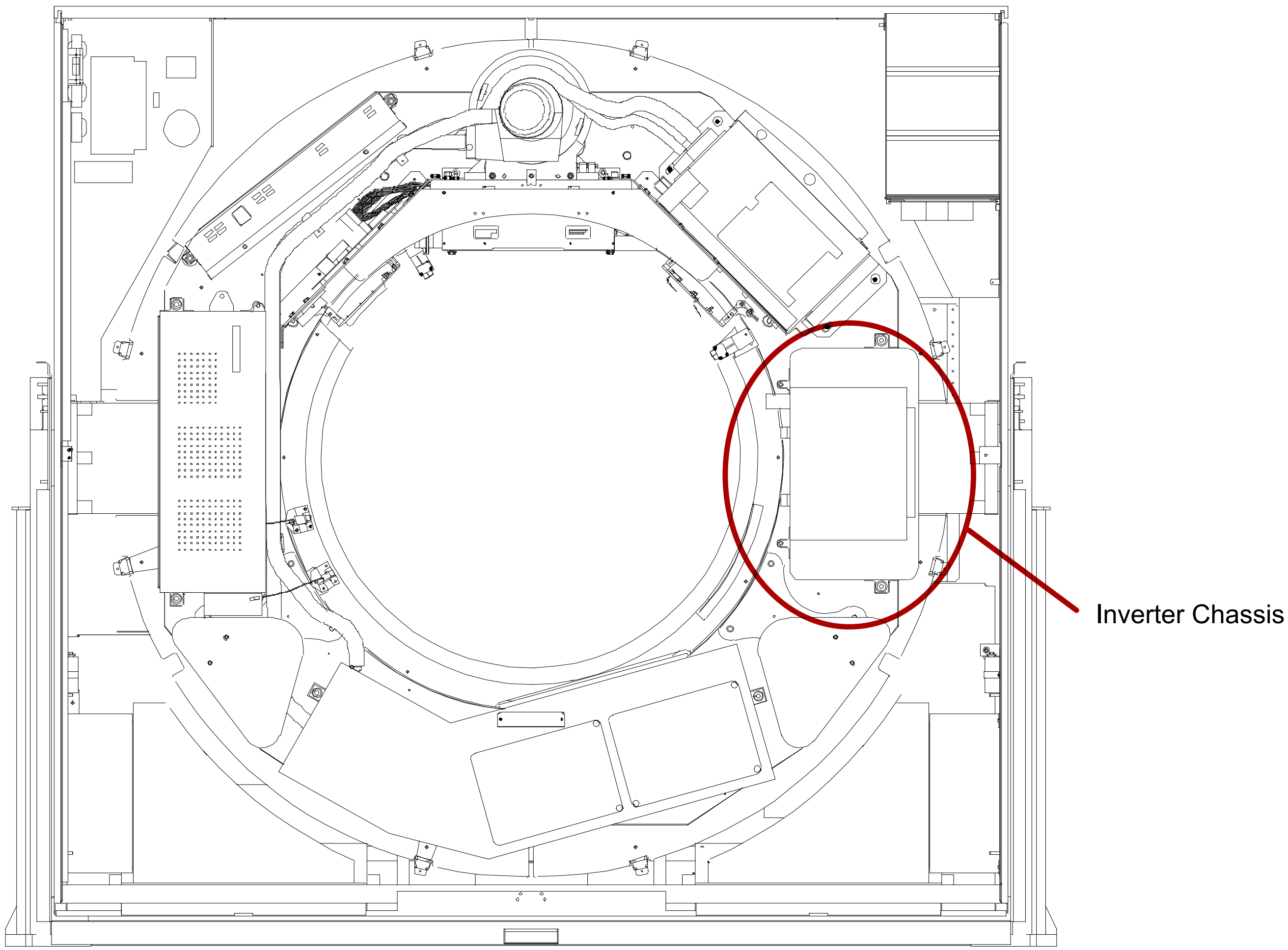


FIGURE 265 Scan Frame Pinned with Tube at 12:00

4. Remove the High Voltage Generator Control Chassis. Refer to the High Voltage Generator Control Chassis Removal procedure.

> **NOTE**
>
> The CRX–400 High Voltage Generator Control Chassis cannot be removed from the Inverter Chassis.

> **⚠ WARNING ⚠**
>
> **HAZARDOUS VOLTAGES ARE PRESENT ON THE INTERNAL VOLTAGE RAILS. ALLOW A MINIMUM OF FIVE MINUTES FOR THE RAIL VOLTAGES TO DISCHARGE AFTER GANTRY POWER IS REMOVED. FAILURE TO COMPLY MAY RESULT IN SEVERE EQUIPMENT DAMAGE AND/OR SERIOUS INJURY TO SERVICE PERSONNEL.**

5. Remove triangular shaped weights located in front of the inverter chassis. Refer to Figure 265.
6. Tag and remove all Cables at the Inverter Chassis. Use care when handling Ribbon Cables. Cut any tie wraps necessary to free the Cables.
7. Remove the Mounting Hardware securing the Inverter Chassis to the Scan Frame. See Figure 266.
8. Remove the two Hex Head Bolts and their Washers. Screw Extender Pins into the holes vacated by the two Hex Head Bolts. Tighten the Extender Pins only hand tight. Refer to Figure 266.

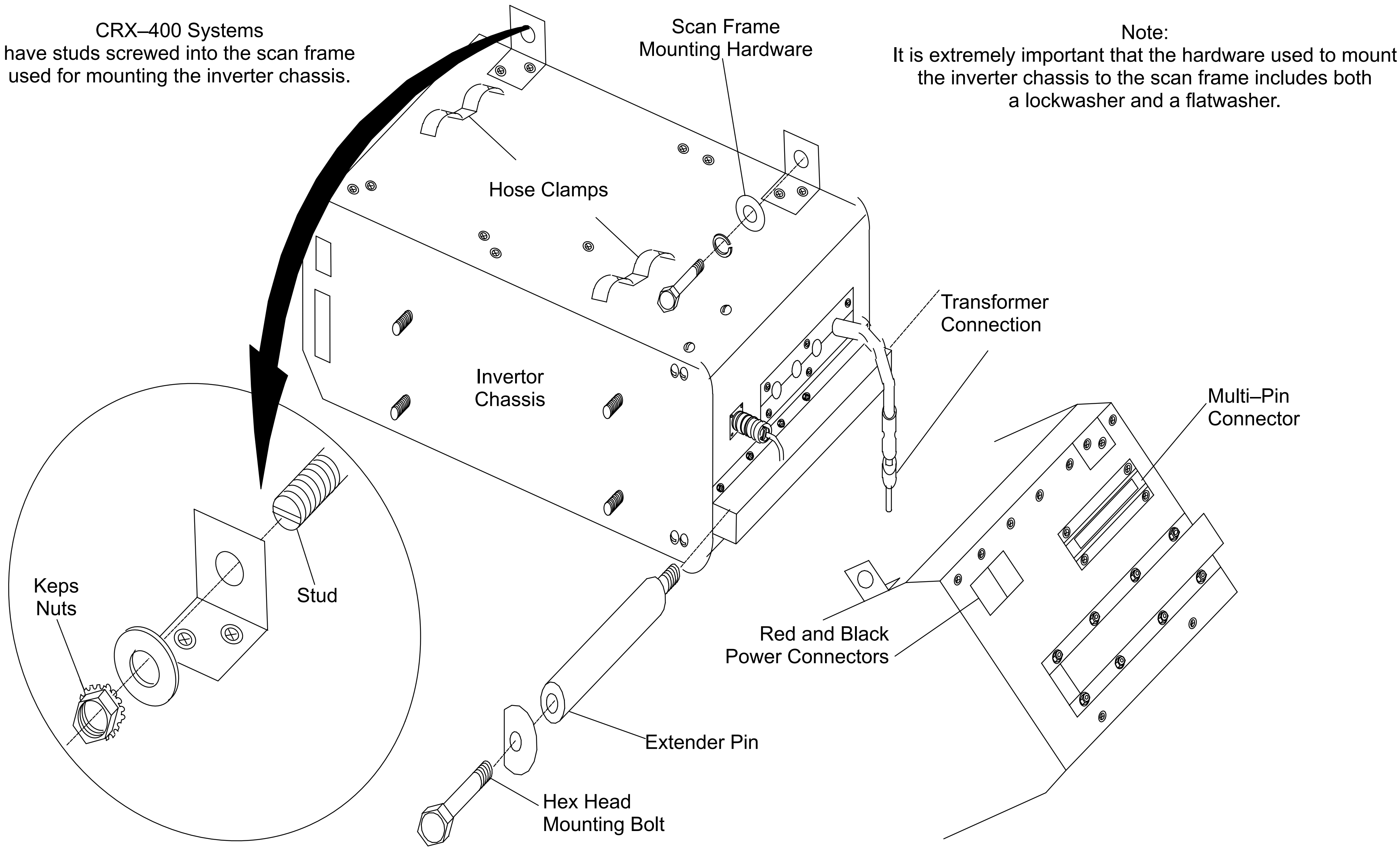


FIGURE 266 Inverter Chassis

⚠ WARNING

THE INVERTER CHASSIS WEIGHS 112 LBS AND IS TOP HEAVY. USE EXTREME CAUTION WHEN HANDLING THIS ITEM. FAILURE TO COMPLY CAN CAUSE SEVERE DAMAGE TO EQUIPMENT AND/OR SERIOUS INJURY TO PERSONNEL.

⚠ CAUTION

TWO PEOPLE MAY BE REQUIRED TO MOVE THE INVERTER CHASSIS FROM THE SCAN FRAME IF IT DOES NOT HAVE A LIFTING BRACKET.

9. Secure the Hoist Assembly to the Patient Support. Make sure that the clamp base is seated firmly in position on the Patient Support and that the Knob Bolts are screwed in securely. See Figure 267.
10. Connect the Hoist Clip to the Lifting Bracket on top of the Inverter Chassis and lock the Clip. Remove slack.
11. Slide the Inverter Chassis along the Extender Pins and remove it from the Gantry.

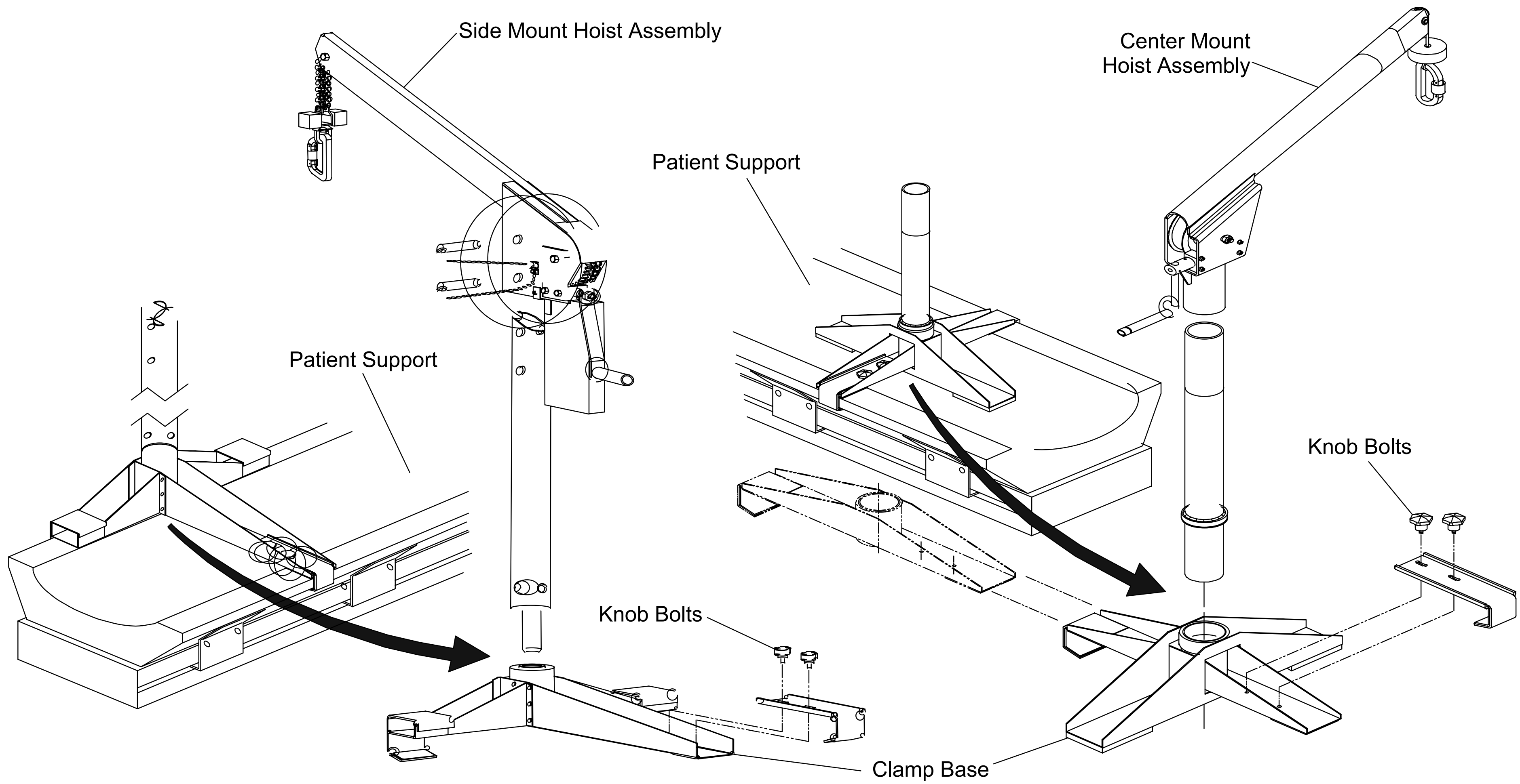


FIGURE 267 Hoist Assembly

INVERTER CHASSIS REPLACEMENT:

12. Clean the back surface of the replacement Inverter Chassis and the Scan Frame surface.
13. Connect the Hoist Clip to the Lifting Bracket on top of the Inverter Chassis and lock the Clip. Lift Chassis with Hoist.
14. Slide the replacement Inverter Chassis along the Extender Pins and Inverter Support Rods until it is firmly against the Scan Frame.

15. Remove the Extender Pins and replace the two (2) Hex Head Bolts, their Washers and the Scan Frame Mounting Hardware. Refer to Figure 266.

    Torque all Inverter Chassis Mounting Hardware to 15 ft–lbs.

16. Replace Cables and secure the Heat Exchanger Oil Lines with the Hose Clamps.
17. Install the triangular weights removed earlier. Remove the Locking Pin from the Scan Frame.
18. Rotate the Scan Frame counterclockwise by hand while checking for loose Cables. Also check for clearance over the entire range of travel between the Rotating Frame Cables and the Stationary Frame.
19. Replace the Gantry Front Cone. Refer to Front Cone Replacement in the Gantry Covers Removal/Replacement procedure.

⚠ CAUTION

CLOCKWISE ROTATION OF THE SCAN FRAME MAY CAUSE DAMAGE TO THE SLIP RING BRUSHES. ROTATE THE SCAN FRAME COUNTERCLOCKWISE ONLY.

20. Apply Gantry power. Refer to the PowerUp/PowerDown procedure.
21. Pull the Front Cover Interlock Switch Plunger out to enable operation while the Front Covers are open.

⚠ WARNING ⚠

USE EXTREME CAUTION WHEN OPERATING OR WORKING ON EQUIPMENT WITH COVERS REMOVED. DANGEROUS VOLTAGES ARE EXPOSED WHEN THE GANTRY COVERS ARE REMOVED. FAILURE TO COMPLY CAN CAUSE SERIOUS INJURY OR DEATH TO PERSONNEL

22. To verify proper system operation, run several full–field and half–field water phantom scans and allow the system to process and display an image.
23. Return the Front Cover Interlock Switch Plunger to its center position.
24. Remove Gantry power. Refer to the PowerUp/PowerDown procedure in the Introduction section.
25. Replace the Gantry lower Front Cover, close the upper Front Cover. Refer to the Gantry Covers procedure.

INVERTER MODULES

INTRODUCTION: The following procedure is used to remove and replace the inverter modules. Allow three minutes for the rail voltages to discharge after gantry power has been removed.

TOOLS REQUIRED:
1. #1,#2 tip phillips and 1"x4" slotted screwdrivers
2. 3/8" and 5/16" sockets, ratchet

MATERIALS REQUIRED:
1. Loc–Tite (thread locker)

ESTIMATED TIME: 50 Min.

INVERTER MODULE REMOVAL:

1. Remove gantry power **AT THE WALLBOX**. Refer to the PowerUp/PowerDown procedure.
2. Open gantry front covers and remove the front cone. Refer to the gantry covers procedure.

> ⚠ **WARNING**
>
> **USE EXTREME CAUTION WHEN SERVICING EQUIPMENT WHILE UPPER GULL WING PANELS ARE OPEN. THE PANELS HAVE SHARP EDGES. FAILURE TO COMPLY COULD RESULT IN SERIOUS INJURY TO SERVICE PERSONNEL.**

> ⚠ **WARNING** ⚠
>
> **USE EXTREME CAUTION WHEN OPERATING OR WORKING ON EQUIPMENT WITH COVERS REMOVED. DANGEROUS VOLTAGES ARE EXPOSED WHEN THE GANTRY COVERS ARE REMOVED. ACCIDENTAL CONTACT WITH THESE VOLTAGES MAY CAUSE SERIOUS INJURY OR DEATH TO SERVICE PERSONNEL.**

3. Rotate the scan frame counterclockwise by hand until the x–ray tube is at the 12:00 position. Lock the Scan frame with the locking pin. See Figure 268.

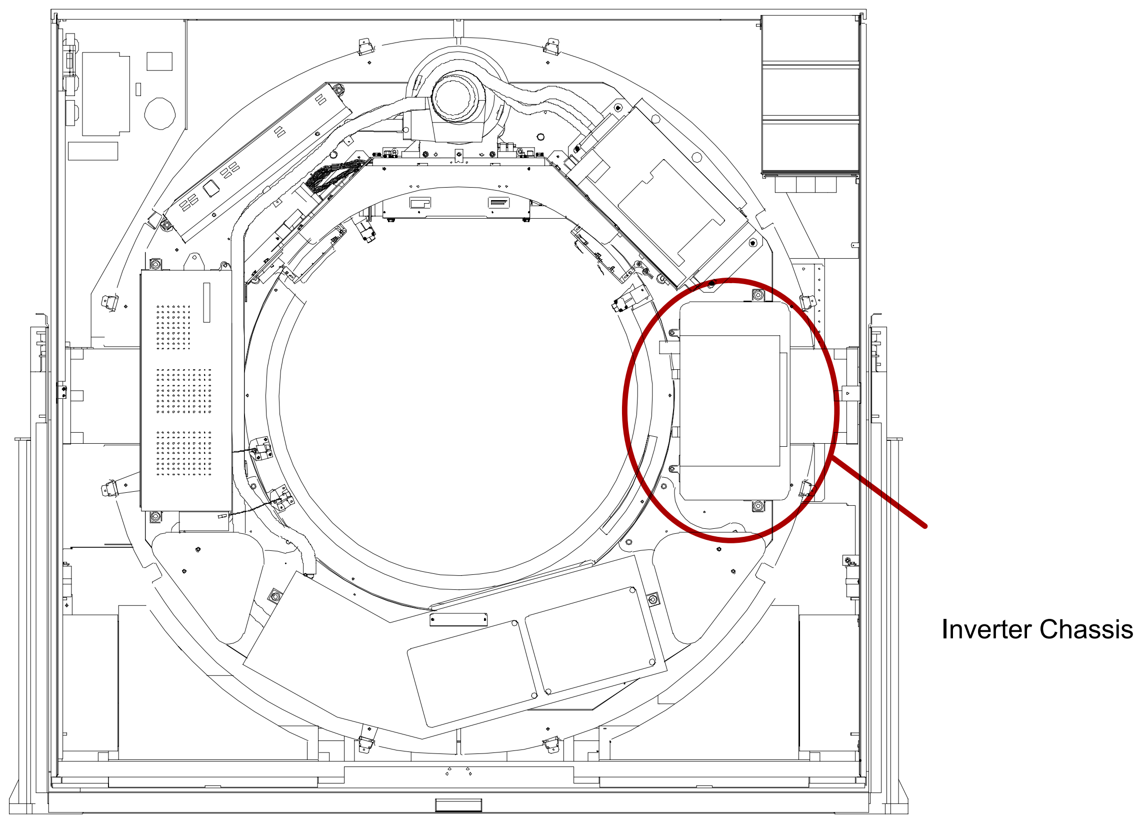


FIGURE 268 SCAN FRAME PINNED WITH TUBE AT 12:00

4. Tag and remove all wires/cables from the high voltage generator control chassis (hvgcc).

> **NOTE**
>
> The hvgcc on CRX–400 and 400R systems is an integral part of the inverter chassis and cannot be removed from the chassis.

5. Open the hvgcc cover. Tag and remove the ribbon cable connected from the hvgcc control board to the inverter modules. See Figure 269.

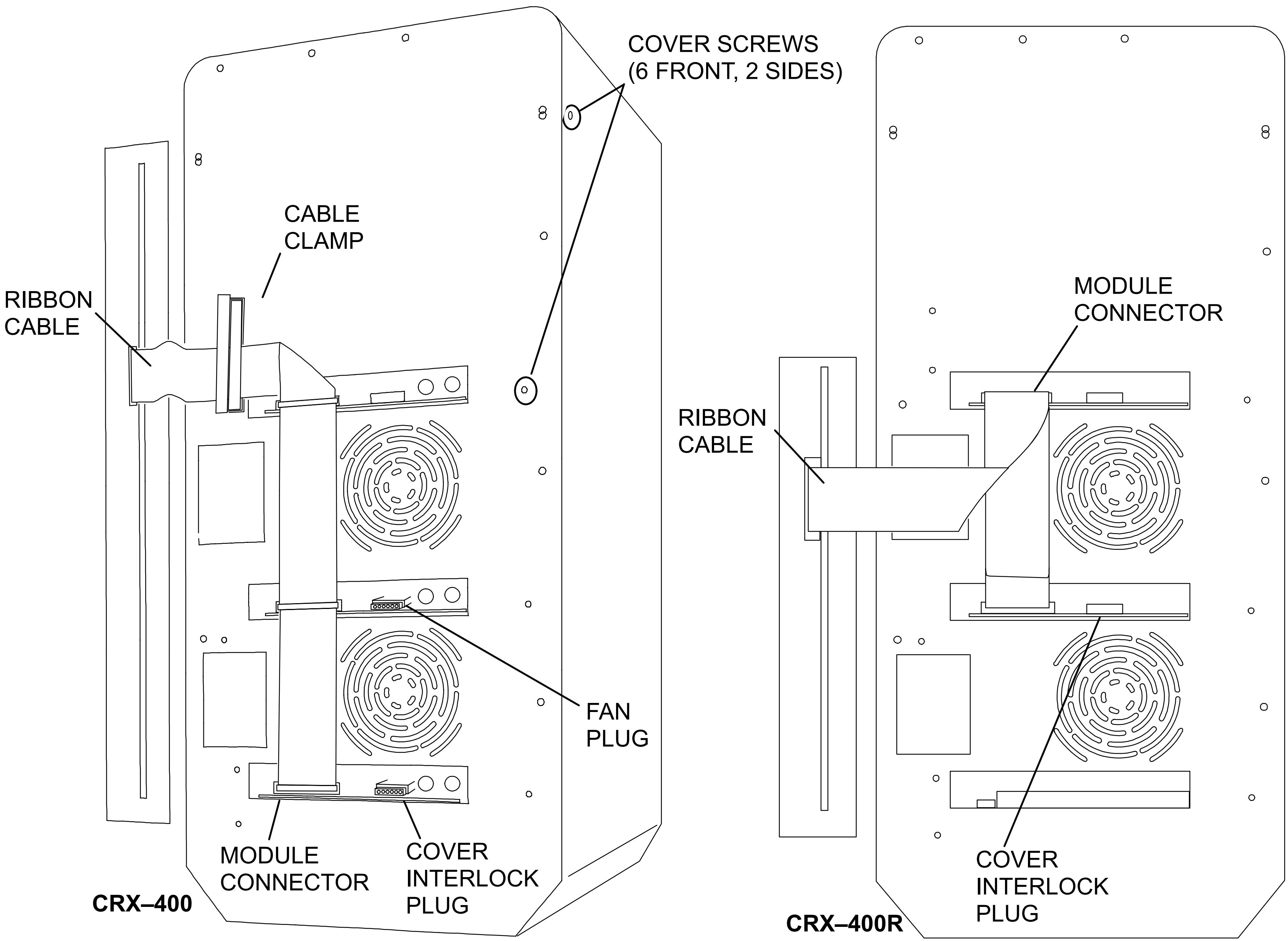


FIGURE 269 INVERTER CHASSIS COVER SCREWS

6. Remove the fan plugs from the front of the inverter modules.
7. Remove the six phillips head screws on the front and two phillips head screws on the side that secure the inverter cover to the inverter chassis. Refer to Figure 269.
8. Remove the inverter chassis cover. Refer to Figure 269.

> **NOTE**
>
> The inverter modules have heavy heat sinks on the back that are spring–loaded against the scan frame. **Relieve module spring tension by first loosening the inverter mounting bolts**. Protect the back surface of the heat sink because scratching will reduce the surface area in contact with the scan frame.

Bottom Module

9. Tag and remove the cover interlock connector from the bottom module and secure it out of the way.
10. Tag and unplug all red and black power rail connectors.

> **NOTE**
>
> The transformer connectors must be supported by hand when loosening/tightening the hex head nut during removal/replacement.

11. Tag and remove the four transformer connections from the module.
12. Remove the four slot head spring–loading screws to relieve spring compression on the module.

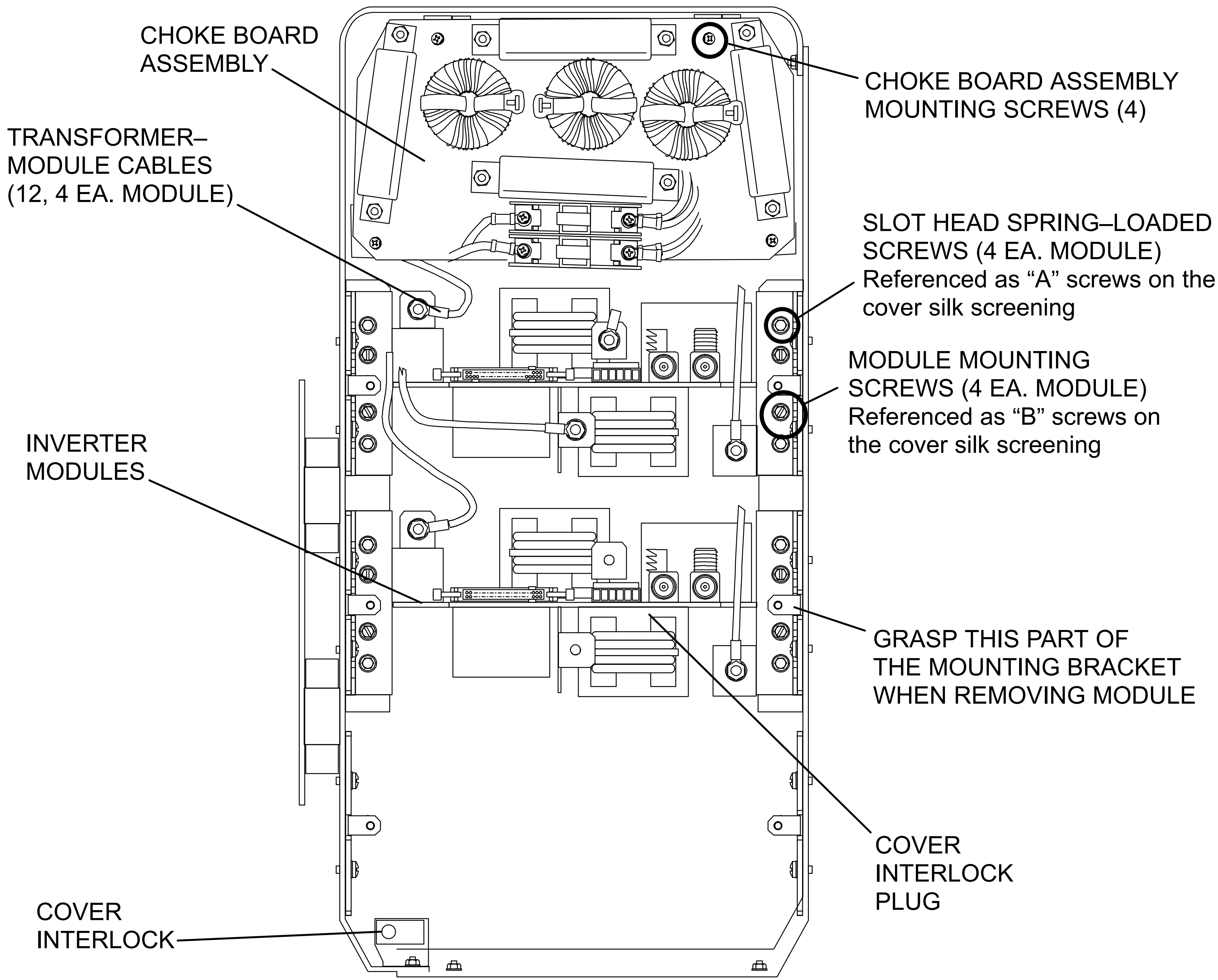


FIGURE 270 CRX–400R INVERTER COMPONENTS

13. Remove the four phillips head mounting screws. Grasp the mounting bracket and carefully pull the module out of the inverter chassis.

Middle Module:

14. **CRX–400R**: Tag and remove the cover interlock connector and secure it out of the way.
15. Tag and unplug the power rail connectors from the middle and top modules.

> **NOTE**
>
> The transformer connectors must be supported by hand when loosening/tightening the hex head nut during removal/replacement.

16. **CRX–400R**: Tag and remove the four transformer connections from the middle module.
17. Remove the four slot head spring–loading screws to relieve spring compression on the module.
18. Remove the four phillips head mounting screws. Grasp the mounting bracket and carefully pull the module out of the inverter chassis.

Top Module:

19. Tag and unplug the power rail connectors from the top module.

> **NOTE**
>
> The transformer connectors must be supported by hand when loosening/tightening the hex head nut during removal/replacement.

20. Tag and remove the four transformer connections from all modules.
21. Remove the four slot head spring–loading screws to relieve spring compression on the module.
22. Remove the four phillips head mounting screws. Grasp the mounting bracket and carefully pull the module out of the inverter chassis.

INVERTER MODULE REPLACEMENT:

CRX–400 and 400R Systems:

23. Configure the inverter module(s) as shown in Figure 271.
24. Install the replacement inverter module and secure the inverter chassis by reversing the steps of CRX–400 and 400R systems removal and the appropriate module sub–section above.

    Apply 30 in–lbs to the module mounting and spring–loading screws.

    Apply 10 in–lbs to the inverter cover mounting screws.

    Re–tighten the four Inverter Chassis mounting bolts loosened to relieve spring tension. Torque the bolts to 15 ft–lbs.
25. Remove the locking pin from the scan frame.
26. Rotate the scan frame counterclockwise by hand while checking for loose cables. Also check for clearance over the entire range of travel between the rotating frame cables and the stationary frame.

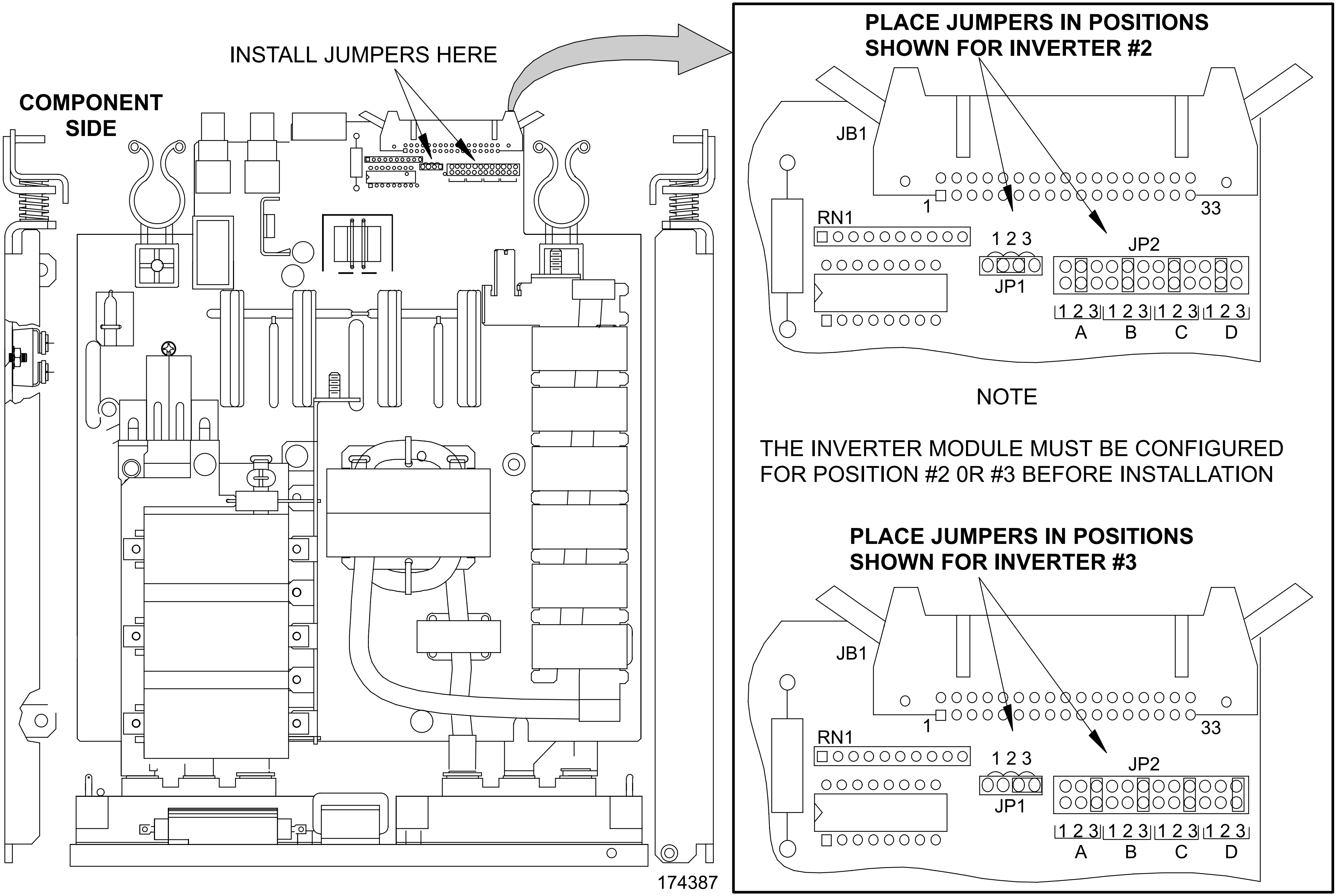


FIGURE 271 INVERTER MODULE #2 AND #3 JUMPER POSITIONS

CAUTION

CLOCKWISE ROTATION OF THE SCAN FRAME MAY CAUSE DAMAGE TO THE SLIP RING BRUSHES. ROTATE THE SCAN FRAME COUNTERCLOCKWISE ONLY.

27. Replace the gantry front cone. Refer to gantry covers procedure.
28. Apply gantry power. Refer to the PowerUp/PowerDown procedure.

WARNING

USE EXTREME CAUTION WHEN OPERATING OR WORKING ON EQUIPMENT WITH COVERS REMOVED. DANGEROUS VOLTAGES ARE EXPOSED WHEN THE GANTRY COVERS ARE REMOVED. ACCIDENTAL CONTACT WITH THESE VOLTAGES MAY CAUSE SERIOUS INJURY OR DEATH TO SERVICE PERSONNEL.

29. Pull the front cover interlock switch plunger out to enable operation while the front covers are open.
30. To verify proper system operation, run full–field and half–field water phantom scans and allow the system to process and display an image.
31. Return the front cover interlock switch plunger to its center position.
32. Remove gantry power. Refer to the PowerUp/PowerDown procedure.
33. Replace the gantry lower front cover, close the upper front cover. Refer to the gantry covers procedure.

HIGH VOLTAGE TRANSFORMER

INTRODUCTION: The following procedure is used to remove and replace the high voltage transformer. Allow five minutes for the rail voltages to discharge after gantry power is off.

TOOLS REQUIRED:
1. Slotted, #1 and #2 tip phillips screwdrivers
2. Small diagonal cutters

MATERIAL REQUIRED: Loc–Tite (Thread Locker)

CHASSIS WEIGHT: 25.7 pounds (CRX–400)
XXX pounds (CRX–400R)

ESTIMATED TIME: 1.0 Hr.

HIGH VOLTAGE TRANSFORMER REMOVAL:

1. Remove gantry power **AT THE WALLBOX**. Refer to the PowerUp/PowerDown procedure.
2. Open gantry front covers and remove the front cone. Refer to the gantry covers procedure.

> ⚠ **WARNING**
>
> **USE EXTREME CAUTION WHEN SERVICING EQUIPMENT WHILE UPPER GULL WING PANELS ARE OPEN. THE PANELS HAVE SHARP EDGES. FAILURE TO COMPLY COULD RESULT IN SERIOUS INJURY TO SERVICE PERSONNEL.**

> ⚠ **WARNING** ⚠
>
> **USE EXTREME CAUTION WHEN OPERATING OR WORKING ON EQUIPMENT WITH COVERS REMOVED. DANGEROUS VOLTAGES ARE EXPOSED WHEN THE GANTRY COVERS ARE REMOVED. ACCIDENTAL CONTACT WITH THESE VOLTAGES MAY CAUSE SERIOUS INJURY OR DEATH TO SERVICE PERSONNEL.**

3. Rotate the scan frame counterclockwise by hand until the x–ray tube is at the 12:00 position. Lock the scan frame with the locking pin. See Figure 272.

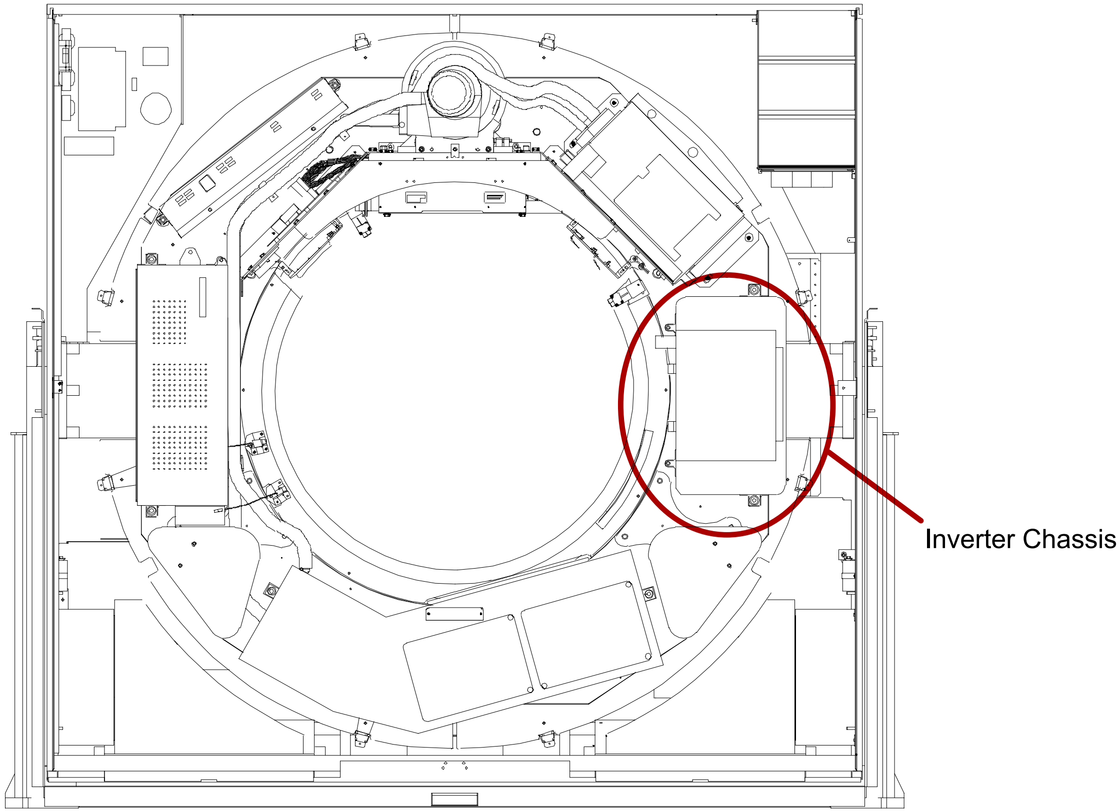


FIGURE 272 SCAN FRAME PINNED WITH TUBE AT 12:00

4. Remove the inverter chassis. Refer to the inverter chassis removal procedure.

> **NOTE**
>
> The inverter modules have heavy heat sinks on the back that are spring–loaded against the scan frame. They are difficult to handle and will sag down during removal. Protect the back surface of the heat sink because scratching will reduce the surface area in contact with the scan frame.

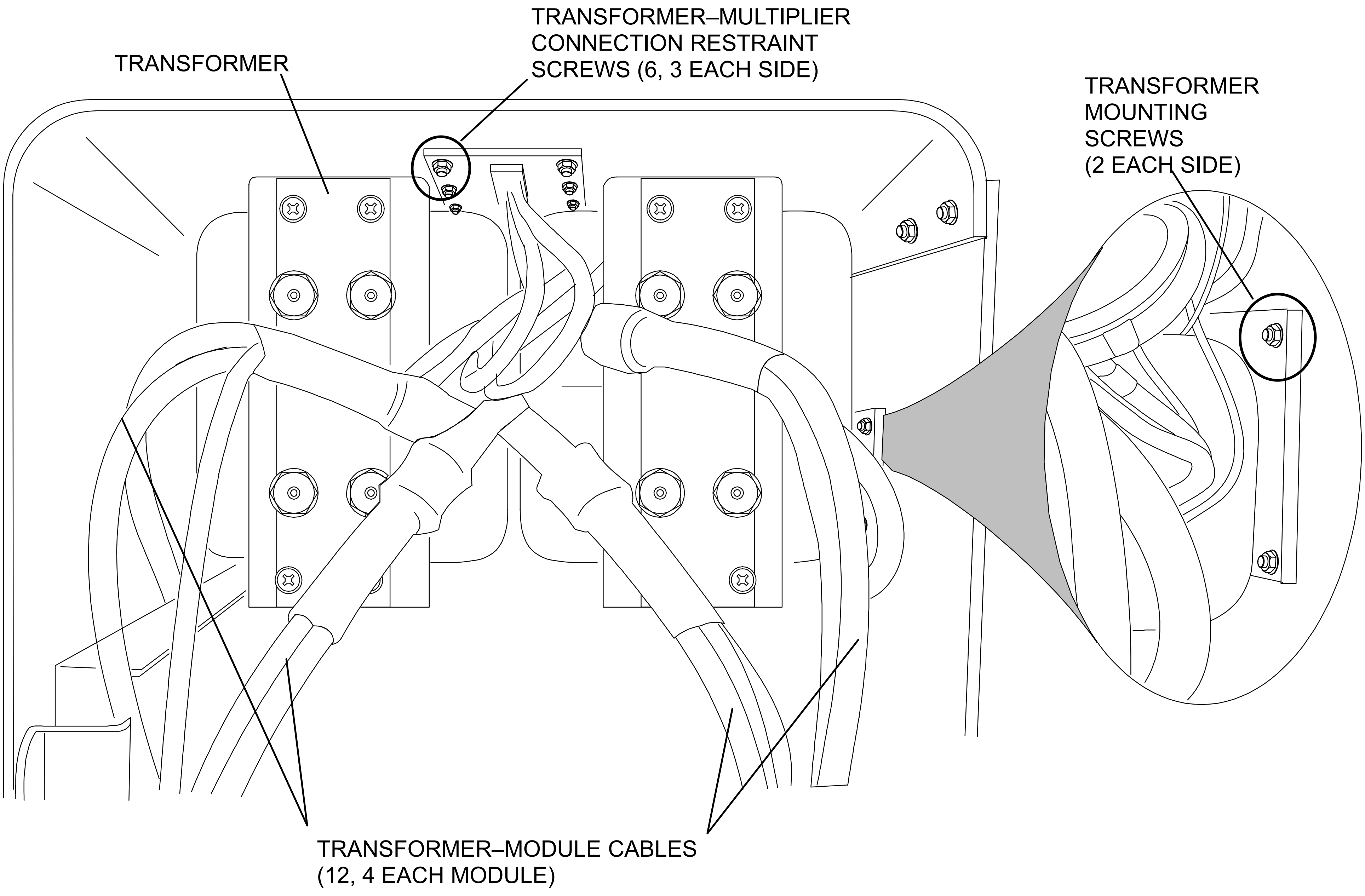


FIGURE 273 HIGH VOLTAGE TRANSFORMER

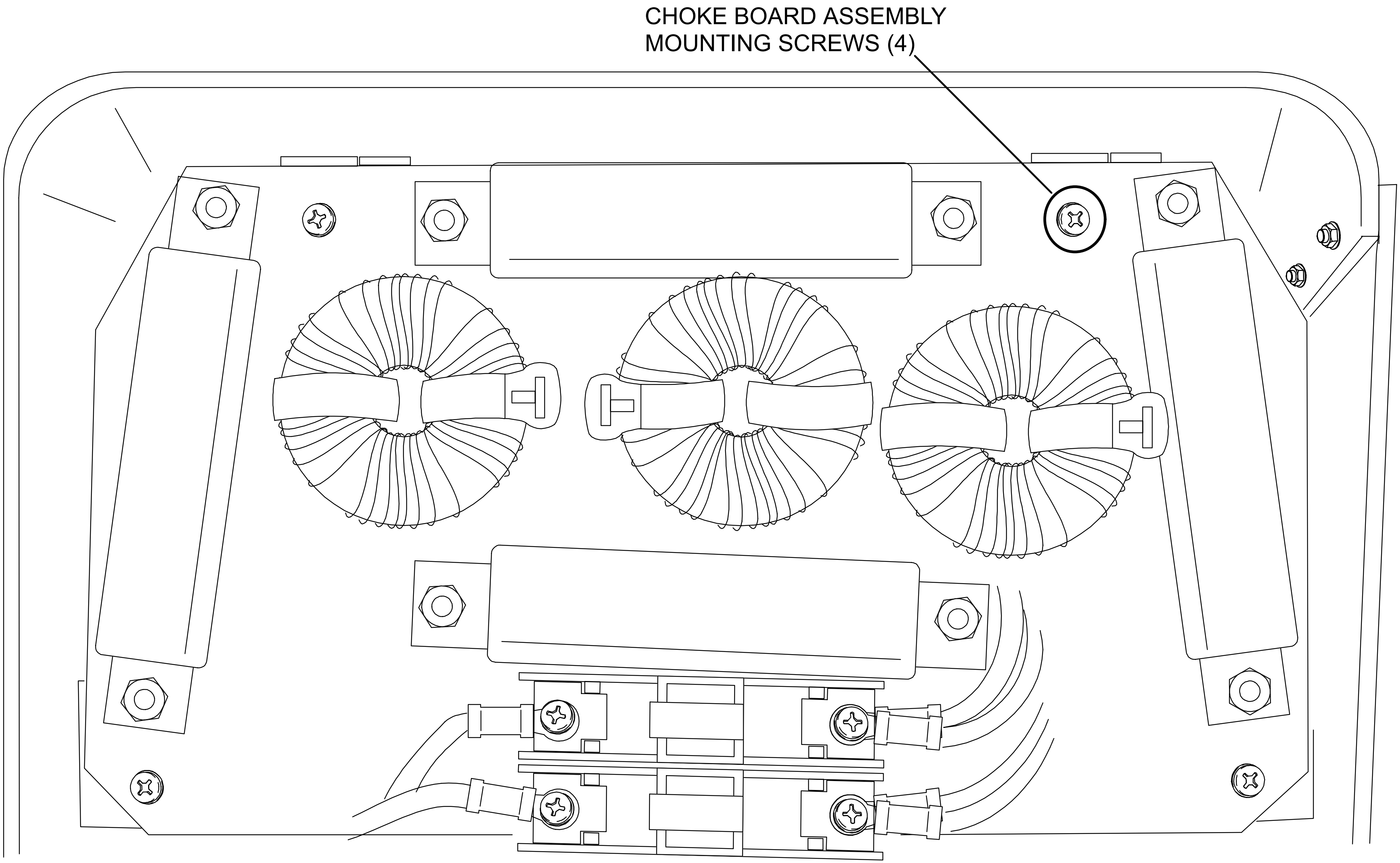


FIGURE 274 CHOKE BOARD ASSEMBLY

5. Place the inverter chassis on a bench (or the floor) with the cover side up and remove the cover. Refer to the appropriate steps of the inverter module removal procedure and to Figure 269.
6. Remove the transformer–module cables from all modules and free them from the twist–type cable ties. Note cable tags in order to ensure proper installation of the new transformer. Refer to Figure 273.

> **NOTE**
>
> The transformer connectors must be supported by hand when loosening/tightening the hex head nut during removal/replacement.

7. Remove the four choke board assembly mounting screws and lift up the choke board assembly. Refer to Figure 274. With the choke board assembly lifted out of the way, remove the anderson connector from the side of the inverter chassis.
8. Lay the choke board assembly up and over the inverter modules to allow access to the high voltage transformer beneath.
9. Remove the six transformer–multiplier cable restraint screws and release the cable restraints. Refer to Figure 273.
10. Remove the six transformer mounting screws and remove the transformer from the inverter chassis.

HIGH VOLTAGE TRANSFORMER REPLACEMENT:

11. Install the replacement high voltage transformer by reversing the steps of CRX–400 and 400R systems high voltage transformer removal procedure.

    Apply 20 in–lbs to transformer mounting screws.
12. Remove the locking pin from the scan frame.
13. Rotate the scan frame counterclockwise by hand while checking for loose cables. Also check for clearance over the entire range of travel between the rotating frame cables and the stationary frame.

> **⚠ CAUTION**
>
> CLOCKWISE ROTATION OF THE SCAN FRAME MAY CAUSE DAMAGE TO THE SLIP RING BRUSHES. ROTATE THE SCAN FRAME COUNTERCLOCKWISE ONLY.

14. Replace the gantry front cone. Refer to the gantry covers procedure.
15. Apply gantry power. Refer to PowerUp/PowerDown procedure.

16. Pull the front cover interlock switch plunger out to enable operation while the front covers are open.
17. To verify proper system operation, run full–field and half–field water phantom scans and allow the system to process and display an image.
18. Return the front cover interlock switch plunger to its center position.
19. Remove gantry power. Refer to the PowerUp/PowerDown procedure.
20. Replace the gantry lower front cover, close the upper front cover. Refer to the gantry covers procedure.

HIGH VOLTAGE GENERATOR CONTROL BOARDS

INTRODUCTION: The following procedure is used to remove and replace the high voltage generator system control board.

TOOLS REQUIRED: 1. Slotted and #2 tip phillips screwdrivers

MATERIAL REQUIRED: 1. Static–protective bags or conductive foam suitable to protect gantry circuit boards.

ESTIMATED TIME: 45 Min.

HIGH VOLTAGE GENERATOR CONTROL BOARD REMOVAL:

1. Remove gantry power **AT THE WALLBOX**. Refer to the PowerUp/PowerDown procedure.
2. Open the gantry front covers and remove the front cone. Refer to the gantry covers procedures.

> ⚠ **WARNING**
>
> **USE EXTREME CAUTION WHEN SERVICING EQUIPMENT WHILE UPPER GULL WING PANELS ARE OPEN. THE PANELS HAVE SHARP EDGES. FAILURE TO COMPLY COULD RESULT IN SERIOUS INJURY TO SERVICE PERSONNEL.**

> ⚠ **WARNING** ⚠
>
> **USE EXTREME CAUTION WHEN OPERATING OR WORKING ON EQUIPMENT WITH COVERS REMOVED. DANGEROUS VOLTAGES ARE EXPOSED WHEN THE GANTRY COVERS ARE REMOVED. ACCIDENTAL CONTACT WITH THESE VOLTAGES MAY CAUSE SERIOUS INJURY OR DEATH TO SERVICE PERSONNEL.**

3. Rotate the scan frame counterclockwise by hand until the x–ray tube is at the 12:00 position. Lock the scan frame with the locking pin. See Figure 275.

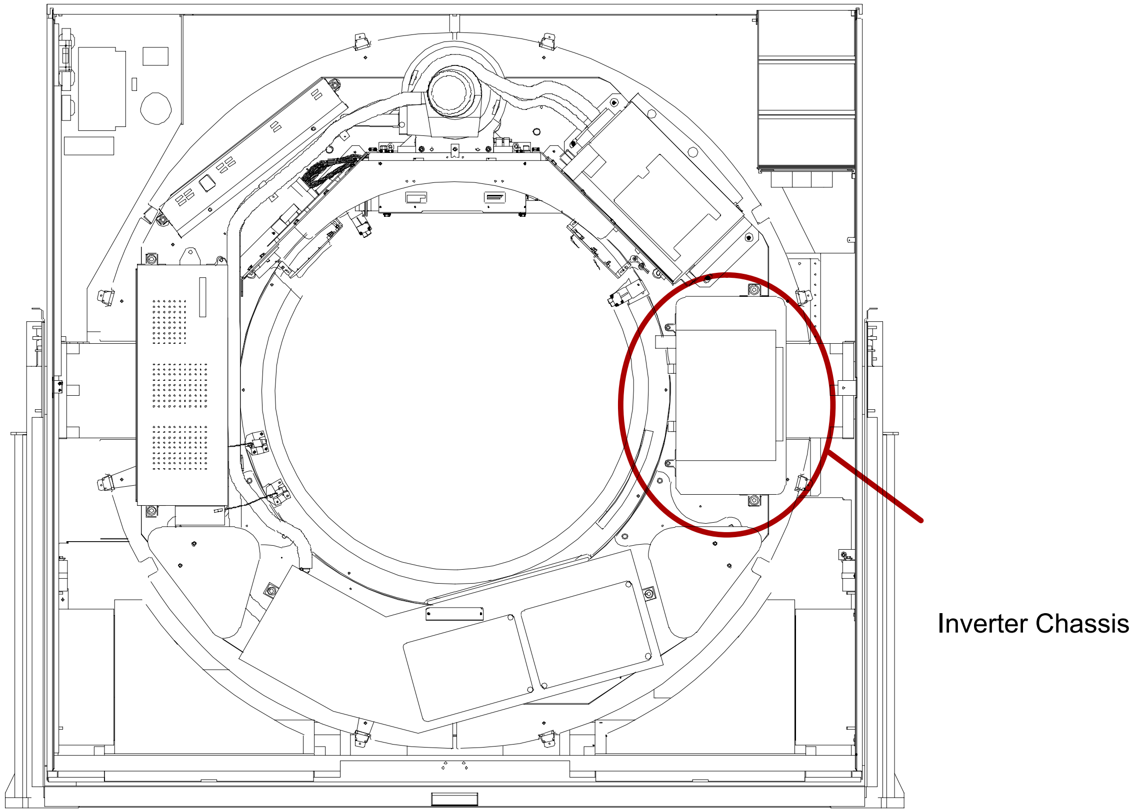


FIGURE 275 SCAN FRAME PINNED WITH TUBE AT 12:00

System Control Board

4. Loosen the two captive cover screws and open the cover. Refer to NO TAG.

5. Tag and remove all cables from the system control pcb at the control chassis.

⚠ **CAUTION**

STATIC ELECTRICITY MAY BE PRESENT. CIRCUIT BOARDS MUST BE HANDLED WITH STATIC–PROTECTIVE TECHNIQUES. DISCHARGE YOUR BODY BEFORE HANDLING ANY CIRCUIT BOARD. KEEP ALL CIRCUIT BOARDS IN STATIC–PROTECTIVE BAGS OR ON CONDUCTIVE FOAM WHEN NOT INSTALLED IN THE EQUIPMENT. FAILURE TO COMPLY MAY CAUSE PERMANENT DAMAGE TO COMPONENTS ON PRINTED CIRCUIT BOARDS. SEE XR4009 FOR DETAILS.

6. **CRX–400R**: Remove the two retaining screws which secure the system control board to the control chassis. Slide the board on its mountings to release it from the slide–fit slots and lift out the board. Store the circuit board in a static–protective bag. See Figure 276.

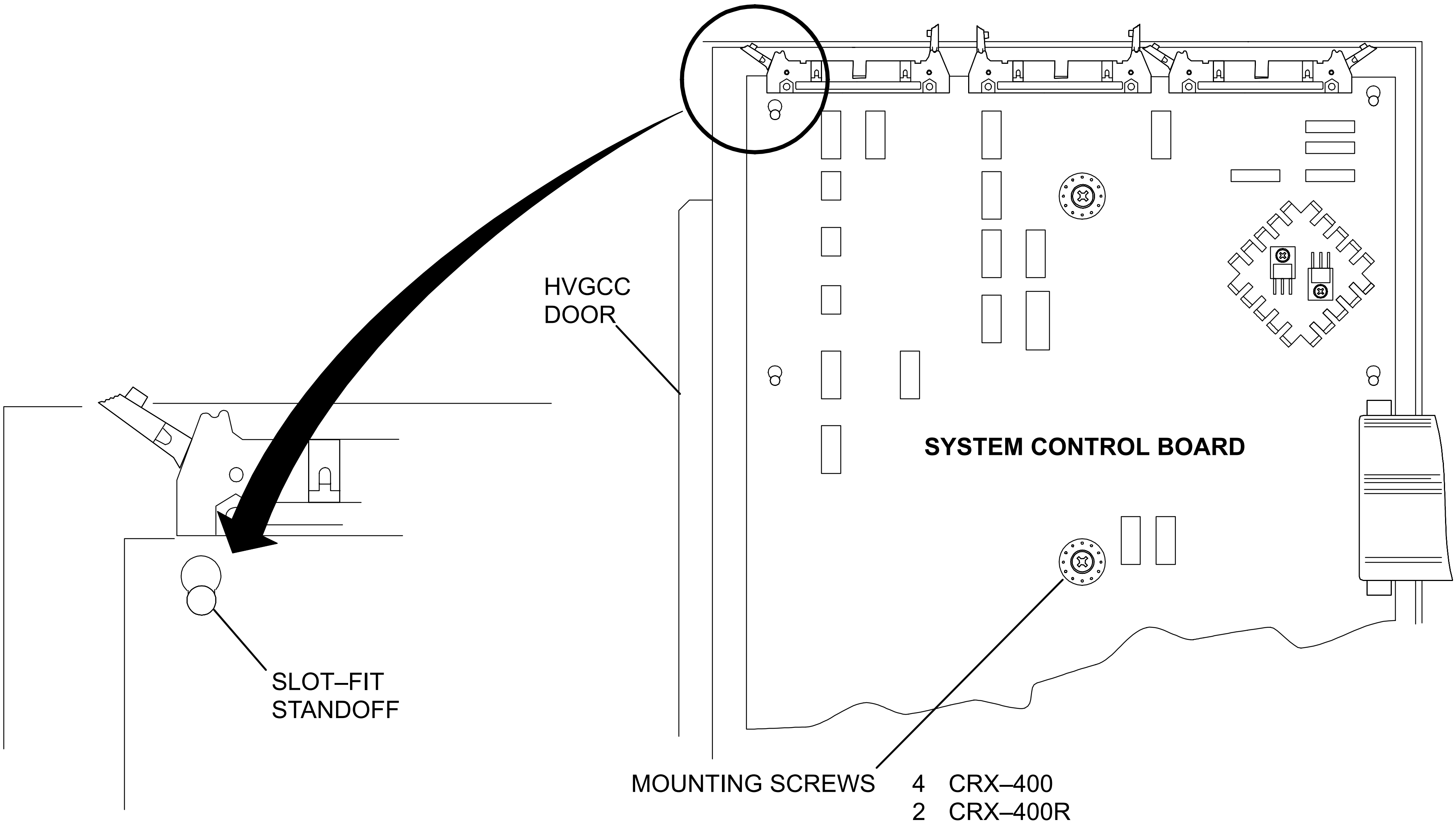


FIGURE 276 CRX–400R SYSTEM CONTROL BOARD

REPLACEMENT:

7. Install the replacement analog or system control board by reversing the steps of the appropriate removal section above.
8. Remove the scan frame locking pin.
9. Rotate the scan frame counterclockwise by hand while checking for loose cables. Also check for clearance over the entire range of travel between the rotating frame cables and the stationary frame.

> ⚠ **CAUTION**
>
> CLOCKWISE ROTATION OF THE SCAN FRAME MAY CAUSE DAMAGE TO THE SLIP RING BRUSHES. ROTATE THE SCAN FRAME COUNTERCLOCKWISE ONLY.

10. Replace the gantry front cone. Refer to the gantry covers procedure.
11. Apply gantry power. Refer to the PowerUp/PowerDown procedure.
12. Pull the front cover interlock switch plunger out to enable operation while the front covers are open.
13. To verify proper system operation, run full–field and half–field water phantom scans and allow the system to process and display an image.
14. Return the front cover interlock switch plunger to its center position.

> ⚠ **WARNING**
>
> **USE EXTREME CAUTION WHEN OPERATING OR WORKING ON EQUIPMENT WITH COVERS REMOVED. DANGEROUS VOLTAGES ARE EXPOSED WHEN THE GANTRY COVERS ARE REMOVED. ACCIDENTAL CONTACT WITH THESE VOLTAGES MAY CAUSE SERIOUS INJURY OR DEATH TO SERVICE PERSONNEL.**

15. Remove gantry power. Refer to the PowerUp/PowerDown procedure.
16. Replace the gantry lower front cover, close the upper front cover. Refer to the gantry covers procedure.

AC INPUT CHASSIS

Introduction: The following procedure is used to remove and replace the AC Input Chassis.

Tools Required:

1. 3/4” and 9/16” Sockets, Ratchet
2. Torque wrench (75 ft–lb)
3. Hoist Assembly

Estimated Time: 1 Hr.

AC INPUT CHASSIS REMOVAL:

1. Remove Gantry power. Refer to the PowerUp/PowerDown procedure.
2. Open the Gantry Front Covers and remove the Front Cone. Refer to the Gantry Covers procedure.

> ⚠ **WARNING**
>
> **USE EXTREME CAUTION WHEN SERVICING EQUIPMENT WHILE THE UPPER GULL WING IS OPEN. THE PANEL HAS SHARP EDGES. FAILURE TO COMPLY COULD RESULT IN SERIOUS INJURY TO PERSONNEL.**

3. Rotate the Scan Frame counterclockwise by hand until the X–Ray Tube is at the 12:00 position. Lock the Scan Frame with the Locking Pin. See Figure 277.

> ⚠ **WARNING** ⚠
>
> **USE EXTREME CAUTION WHEN OPERATING OR WORKING ON EQUIPMENT WITH COVERS REMOVED. DANGEROUS VOLTAGES ARE EXPOSED WHEN THE GANTRY COVERS ARE REMOVED. FAILURE TO COMPLY CAN CAUSE SERIOUS INJURY OR DEATH TO PERSONNEL.**

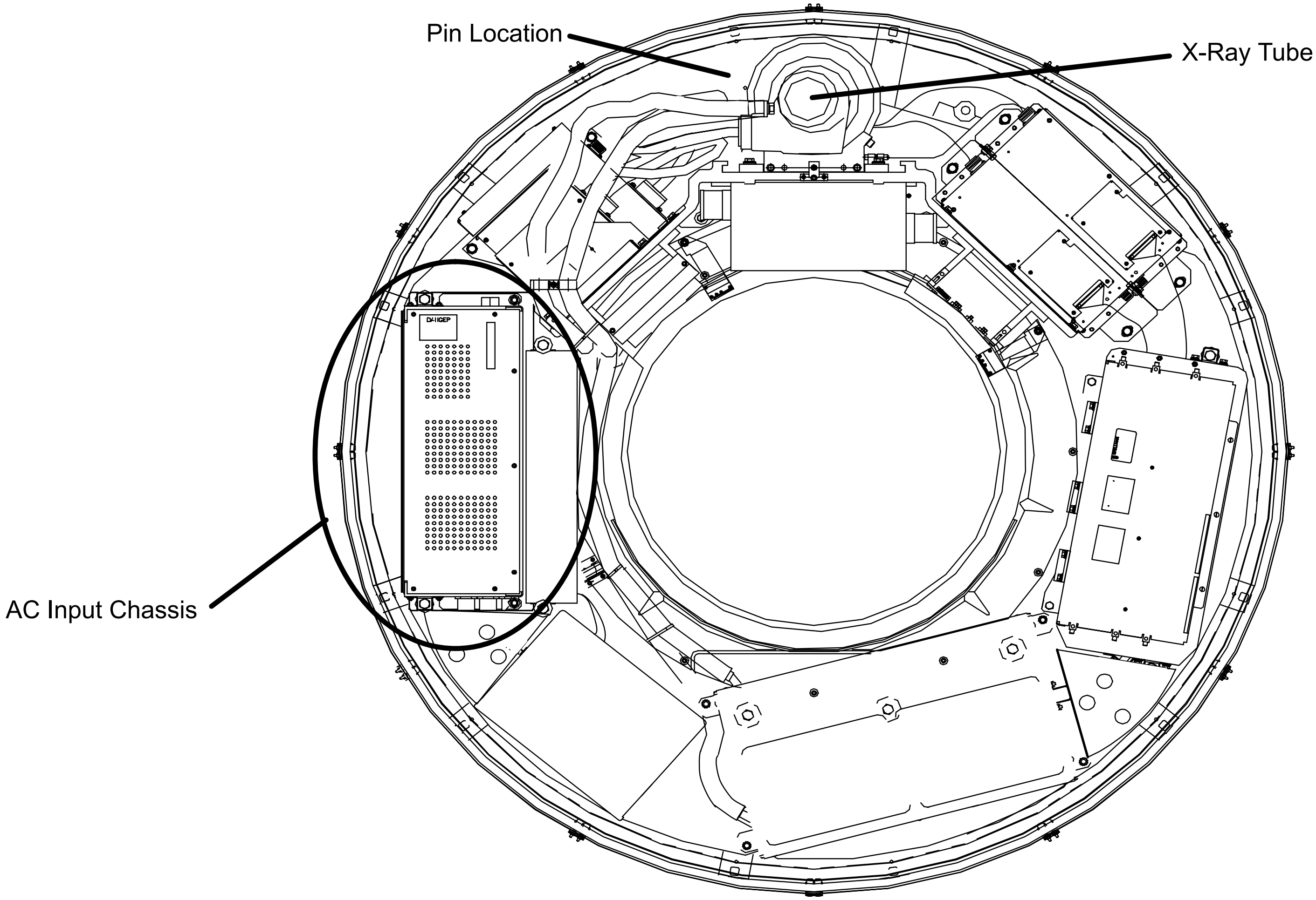


FIGURE 277 Scan Frame Pinned with Tube at 12:00

4. Tag/disconnect all Wires/Cables exiting the AC Input Chassis (top and bottom). Tie and secure them.

5. Remove the two Hex Head Mounting Bolts and screw Extender Pins into the vacated holes. See Figure 278.

6. Remove the two Hex Mounting Nuts and the Washers that secure the AC Input Chassis to the Gantry. See Figure 278.

⚠ WARNING

THE AC INPUT CHASSIS WEIGHS 106 LBS. USE CAUTION WHEN HANDLING THIS ITEM. FAILURE TO COMPLY COULD RESULT IN EQUIPMENT DAMAGE AND/OR SERIOUS INJURY OR DEATH TO PERSONNEL.

⚠ CAUTION

TWO PEOPLE MAY BE REQUIRED TO REMOVE THE AC INPUT CHASSIS FROM THE SCAN FRAME IF THE SYSTEM IS CRX–400 AND DOES NOT HAVE A LIFTING BRACKET.

7. Secure the Hoist Assembly to the Patient Support. Make sure that the Clamp Base is seated firmly in position on the Patient Support and that the Knob Bolts are screwed in securely. See Figure 279.

Note
It is extremely important that the hardware used to mount the AC input chassis to the scan frame includes both a lockwasher and a flat washer.

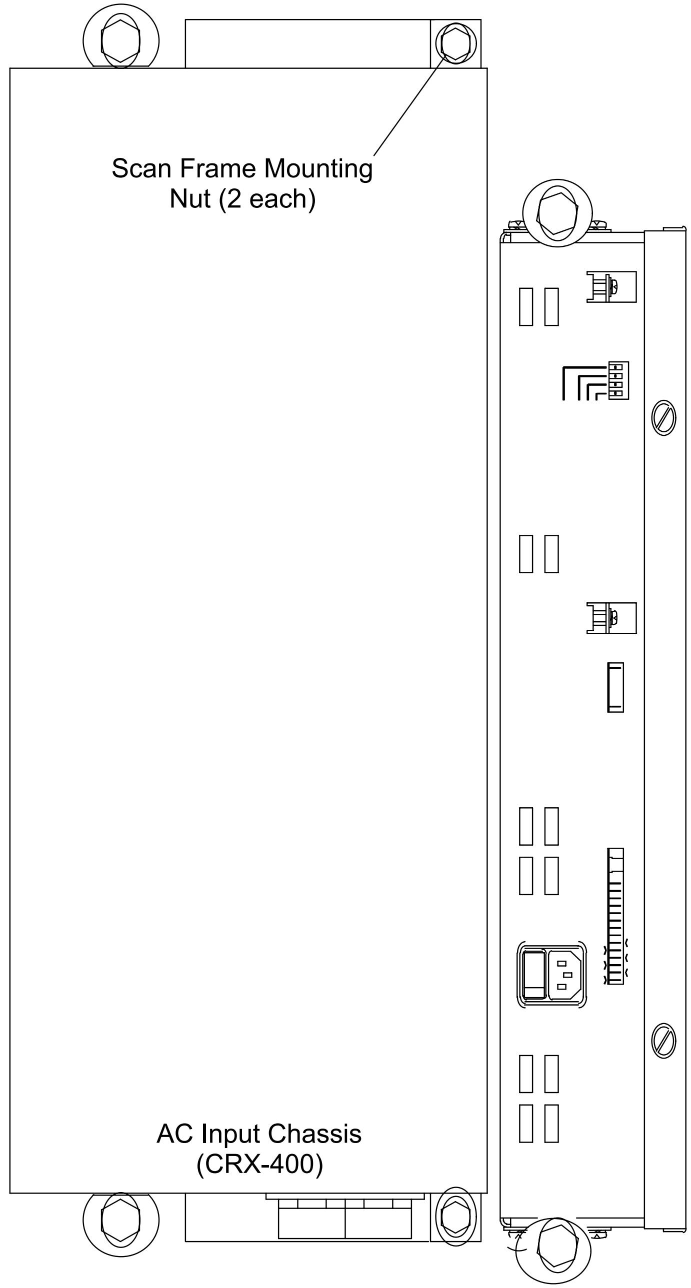


FIGURE 278 CRX400 AC INPUT CHASSIS

8. Connect the Hoist Clip to the Lifting Bracket on top of the AC Input Chassis and lock the Clip. Remove slack.

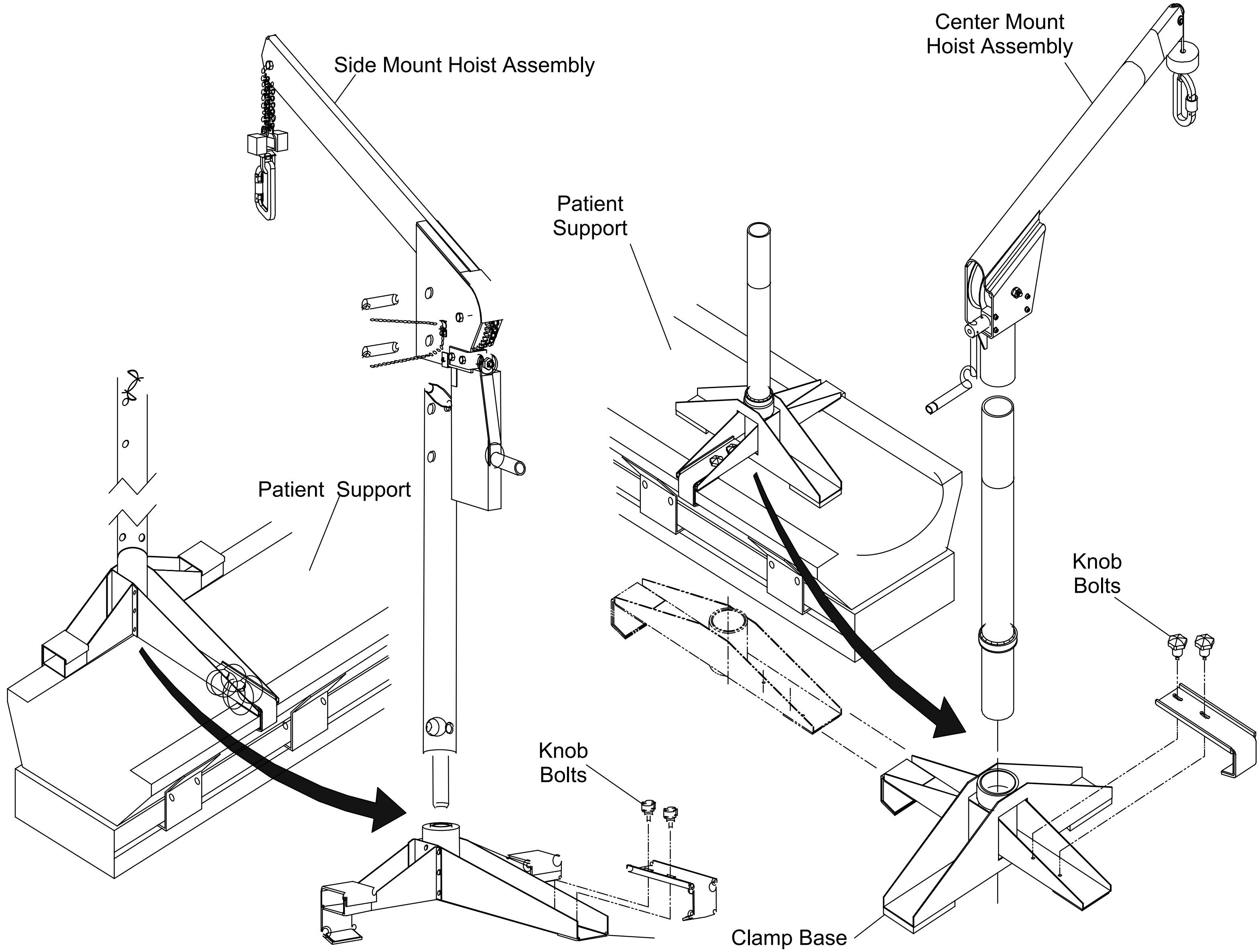


FIGURE 279 HOIST ASSEMBLY

9. Slide the AC Input Chassis along the Extender Pins and remove it from the Gantry.

AC INPUT CHASSIS REPLACEMENT:

10. Install the AC Input Chassis and the X–ray Control Chassis by reversing steps 4. through 9.

    Torque Mounting Bolts to 15 ft–lb.

11. Remove the Scan Frame Locking Pin.
12. Rotate the Scan Frame counterclockwise by hand while checking for loose Cables. Also check for clearance over the entire range of travel between the Rotating Frame Cables and the Stationary Frame.

> ⚠ **CAUTION**
>
> CLOCKWISE ROTATION OF THE SCAN FRAME MAY CAUSE DAMAGE TO THE SLIP RING BRUSHES. <u>ROTATE THE SCAN FRAME COUNTERCLOCKWISE ONLY</u>.

13. Replace the Gantry Front Cone. Refer to the Gantry Covers procedure.
14. Apply Gantry power. Refer to the PowerUp/PowerDown procedure.

> ⚠ **WARNING** ⚠
>
> **USE EXTREME CAUTION WHEN OPERATING OR WORKING ON EQUIPMENT WITH COVERS REMOVED. DANGEROUS VOLTAGES ARE EXPOSED WHEN THE GANTRY COVERS ARE REMOVED. FAILURE TO COMPLY CAN CAUSE SERIOUS INJURY OR DEATH TO PERSONNEL.**

15. Pull the Front Cover Interlock Switch Plunger out to enable operation while the Front Covers are open.
16. To verify proper System operation, run Full–field and Half–field Water Phantom Scans and allow the System to process and display an Image.
17. Return the Front Cover Interlock Switch Plunger to its center position.
18. Remove Gantry power. Refer to the PowerUp/PowerDown procedure.
19. Replace the Gantry lower Front Cover, close the upper Front Cover. Refer to the Gantry Covers procedure.

AC DIAGNOSTIC BOARD

Introduction:	The following procedure is used to remove and replace the AC Diagnostic Board located inside the AC Input Chassis.
Tools Required:	1. #2 tip Phillips screwdriver
	2. 1/4” hex key
Materials Required:	1. Loc–Tite (thread locker)
Eastimated Time:	35 min.

AC DIAGNOSTIC BOARD REMOVAL:

1. Remove Gantry power. Refer to the PowerUp/PowerDown procedure.
2. Open the Gantry Front Covers and remove the Front Cone. Refer to the Gantry Covers procedure.

> ⚠ **WARNING**
>
> **USE EXTREME CAUTION WHEN SERVICING EQUIPMENT WHILE THE UPPER GULL WING IS OPEN. THE PANEL HAS SHARP EDGES. FAILURE TO COMPLY COULD RESULT IN SERIOUS INJURY TO PERSONNEL.**

3. Rotate the Scan Frame counterclockwise by hand until the X–Ray Tube is at the 12:00 position. Lock the Scan Frame with the Locking Pin. See Figure 280.

> ⚠ **WARNING** ⚠
>
> **USE EXTREME CAUTION WHEN OPERATING OR WORKING ON EQUIPMENT WITH COVERS REMOVED. DANGEROUS VOLTAGES ARE EXPOSED WHEN THE GANTRY COVERS ARE REMOVED. FAILURE TO COMPLY CAN CAUSE SERIOUS INJURY OR DEATH TO PERSONNEL.**

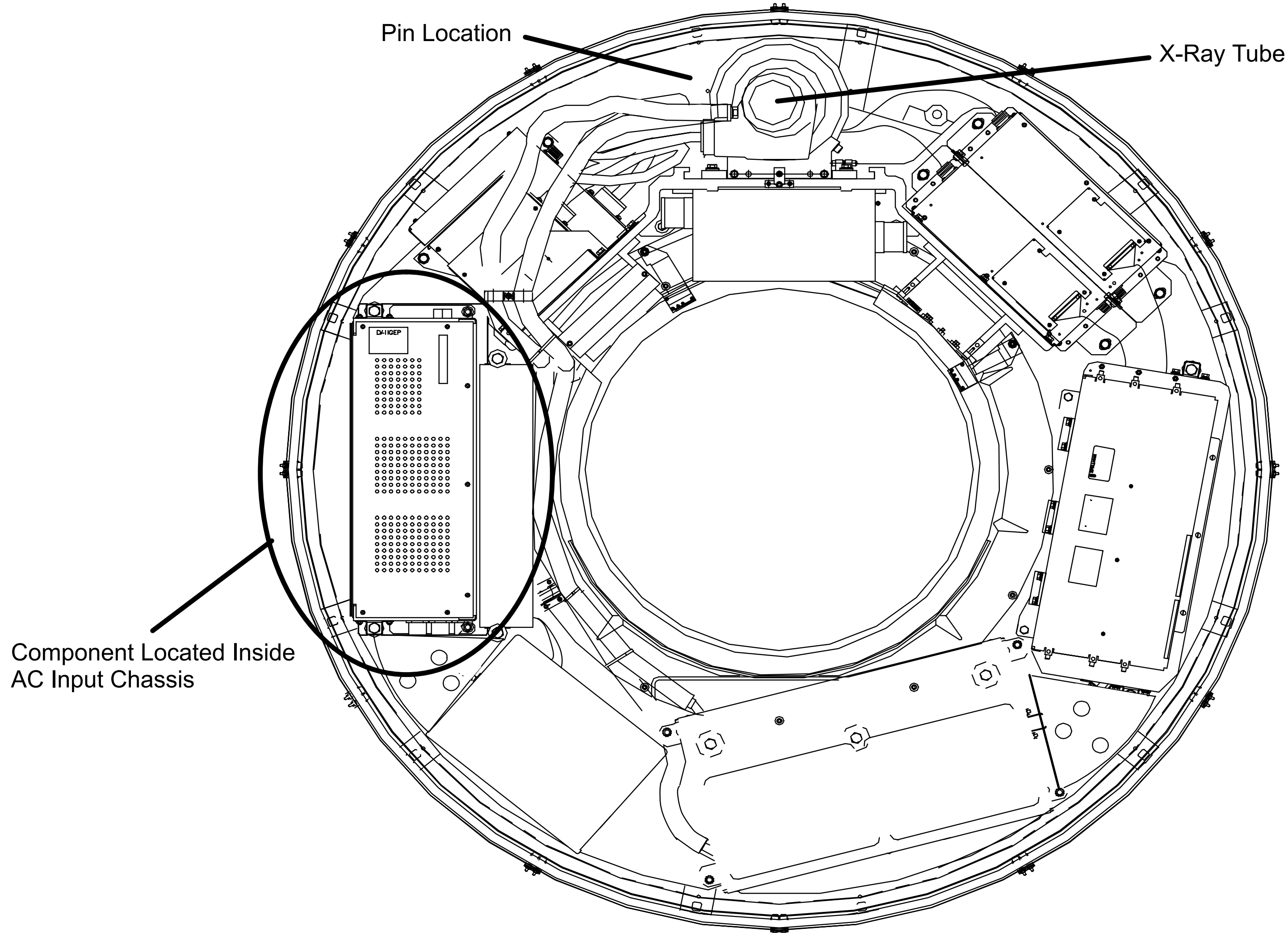


FIGURE 280 Scan Frame Pinned with Tube at 12:00

4. Remove the seven Cover Screws on the AC Input Chassis. See Figure 281.

> **NOTE**
>
> Some systems will not be equipped with Straps depending on the type of Main Bearing installed. For these systems the next two steps may be omitted.

5. Remove the Socket Head Capscrew securing the Left Strap to the Tube Shelf Bracket. See Figure 282.

6. Remove the Pivot AC Bracket. See Figure 282.

7. Remove the AC Input Chassis Cover.

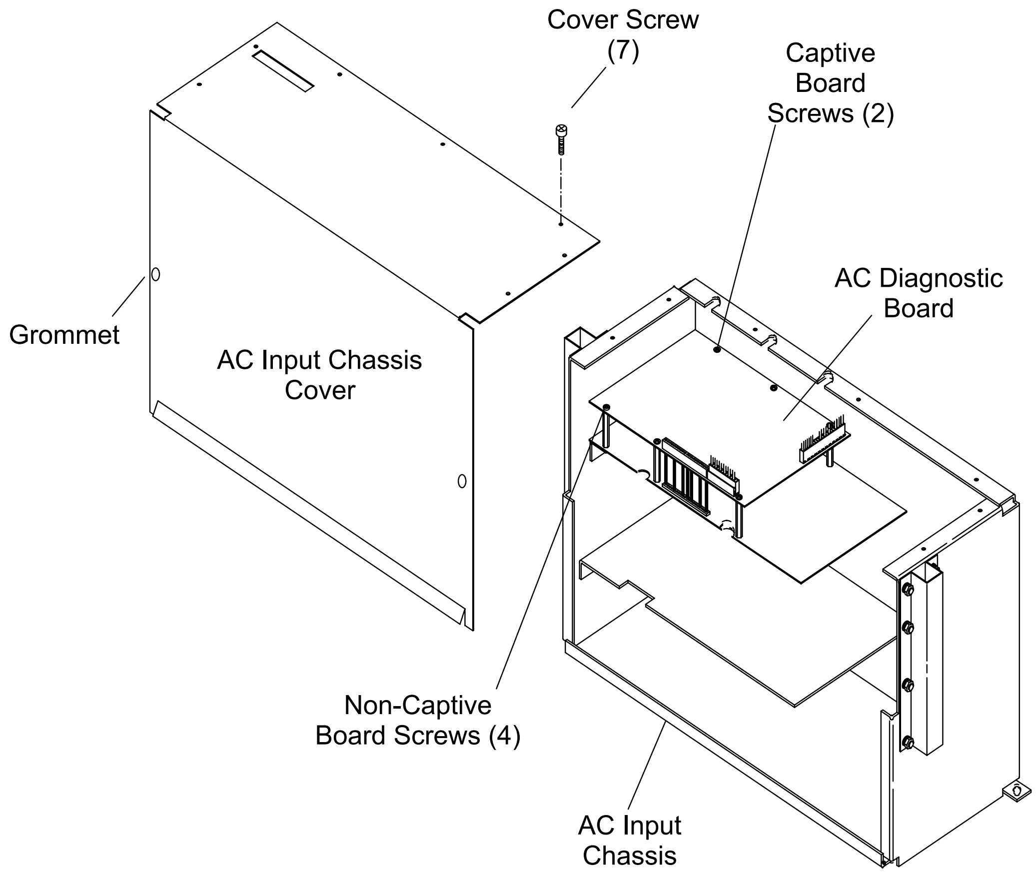


FIGURE 281 AC Input Chassis Cover and Diagnostic Board

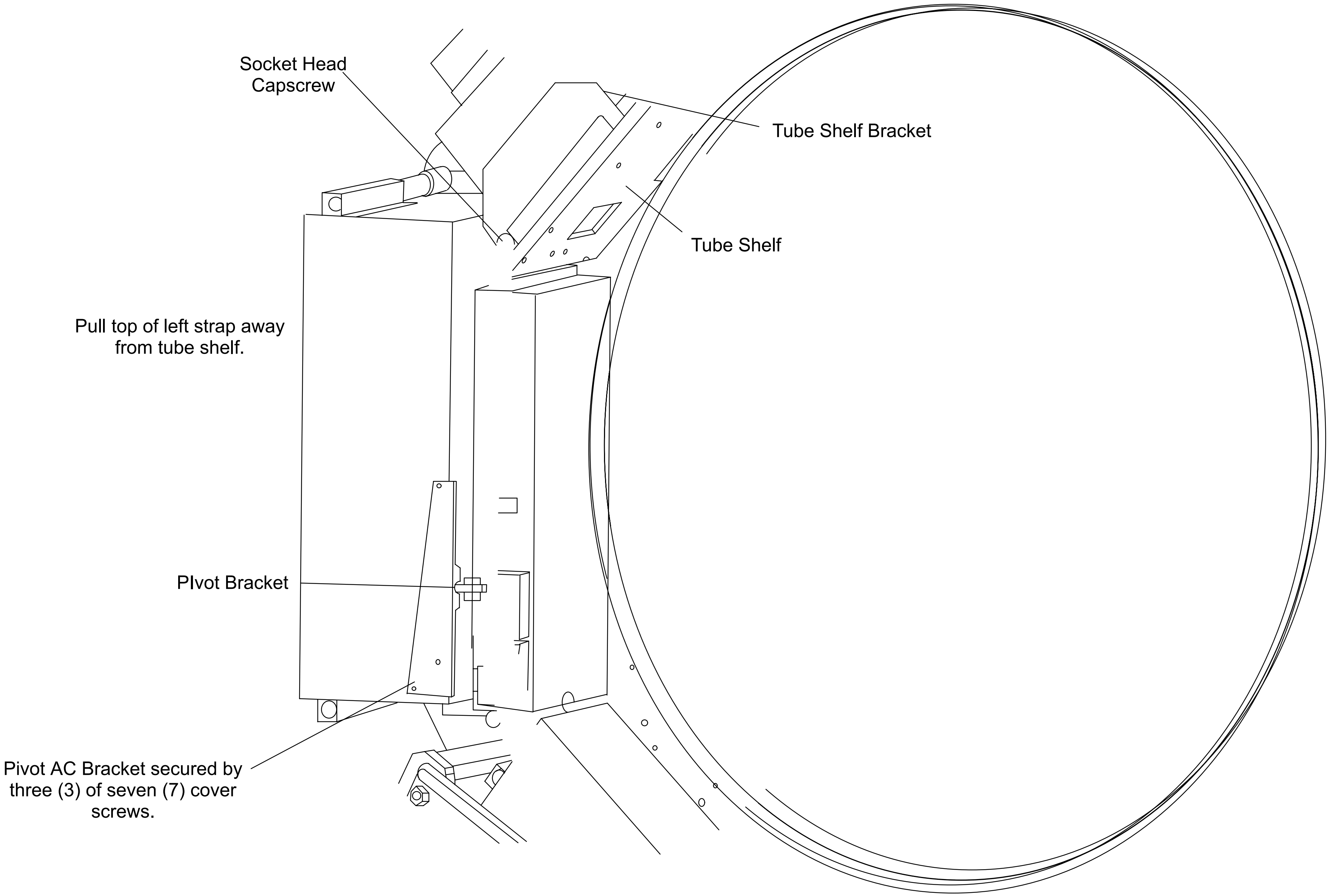


FIGURE 282 Left Strap and Pivot AC Bracket

8. Tag and disconnect all Wires/Cables connected to the AC Diagnostic Circuit Board.
9. Remove the six PCB Retaining Screws that secure the AC Diagnostic Board and remove the Board by carefully disengaging the rear–entry Card Connector Pins (JB5) and then working the PCB out of the Chassis. Note: it might be necessary to slightly bend the PCB to effect removal. Refer to Figure 281.

AC DIAGNOSTIC BOARD REPLACEMENT:

10. Install the AC Diagnostic Board by reversing step 8. and 9.
11. Replace and secure the AC Input Chassis Cover, AC Pivot Bracket and the Socket Head Capscrew for the Left Strap. Apply Loc–Tite to Cover Screws.
12. Remove the Scan Frame Locking Pin.
13. Rotate the Scan Frame counterclockwise by hand while checking for loose Cables. Also check for clearance over the entire range of travel between the Rotating Frame Cables and the Stationary Frame.

> ⚠ **CAUTION**
>
> CLOCKWISE ROTATION OF THE SCAN FRAME MAY CAUSE DAMAGE TO THE SLIP RING BRUSHES. ROTATE THE SCAN FRAME <u>COUNTERCLOCKWISE ONLY.</u>

14. Replace the Gantry Front Cone. Refer to the Gantry Covers procedure.
15. Apply Gantry power. Refer to the PowerUp/PowerDown procedure.

> ⚠ **WARNING** ⚠
>
> **USE EXTREME CAUTION WHEN OPERATING OR WORKING ON EQUIPMENT WITH COVERS REMOVED. DANGEROUS VOLTAGES ARE EXPOSED WHEN THE GANTRY COVERS ARE REMOVED. FAILURE TO COMPLY CAN CAUSE SERIOUS INJURY OR DEATH TO PERSONNEL.**

16. Pull the Front Cover Interlock Switch Plunger out to enable operation while the Front Covers are open.
17. To verify proper System operation, run Full–field and Half–field Water Phantom Scans and allow the System to process and display an Image.

18. Return the Front Cover Interlock Switch Plunger to its center position.
19. Remove Gantry power. Refer to the PowerUp/PowerDown procedure.
20. Replace the Gantry lower Front Cover, close the upper Front Cover. Refer to the Gantry Covers procedure.

3KVA SCAN DRIVE TRANSFORMER

Introduction:	This procedure is used to remove and replace the Scan Drive Motor Controller 3 KVA isolation transformer. Two people are recommended for this procedure due to the weight of the chassis.
Tools Required:	1. 1/4" x 4" slotted screwdriver 2. 9/16" socket, ratchet 3. Small offset Phillips screwdriver
Chassis Weight:	80 LBS.
Estimated Time:	1.5 Hrs.

TRANSFORMER REMOVAL:

1. Remove Gantry power. Refer to the PowerUp/PowerDown procedure.
2. Open the Gantry Front Covers and remove the Front Cone. Refer to the Gantry Covers procedure.

> ⚠ **WARNING**
>
> **USE EXTREME CAUTION WHEN SERVICING EQUIPMENT WHILE THE UPPER GULL WING IS OPEN. THE PANEL HAS SHARP EDGES. FAILURE TO COMPLY COULD RESULT IN SERIOUS INJURY TO PERSONNEL.**

> ⚠ **WARNING** ⚠
>
> **USE EXTREME CAUTION WHEN OPERATING OR WORKING ON EQUIPMENT WITH COVERS REMOVED. DANGEROUS VOLTAGES ARE EXPOSED WHEN THE GANTRY COVERS ARE REMOVED. FAILURE TO COMPLY CAN CAUSE SERIOUS INJURY OR DEATH TO PERSONNEL.**

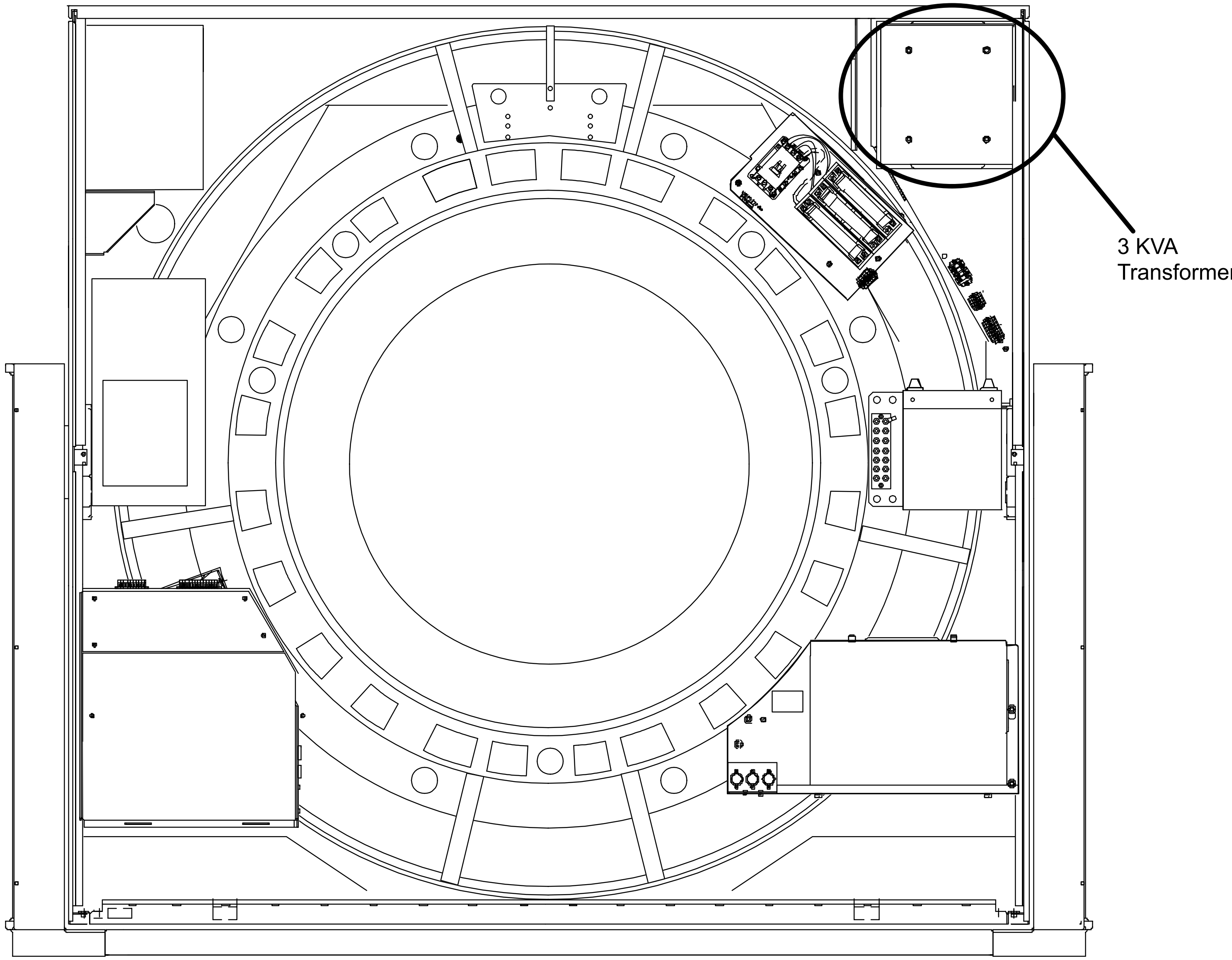


FIGURE 283 Scan Drive Power Transformer Location – Rear of Gantry

3. Remove the six Power Transformer Cover Mounting Screws and remove the Cover.
4. Tag and remove all connections from the load side of the Transformer Terminal Strip. See Figure 284.

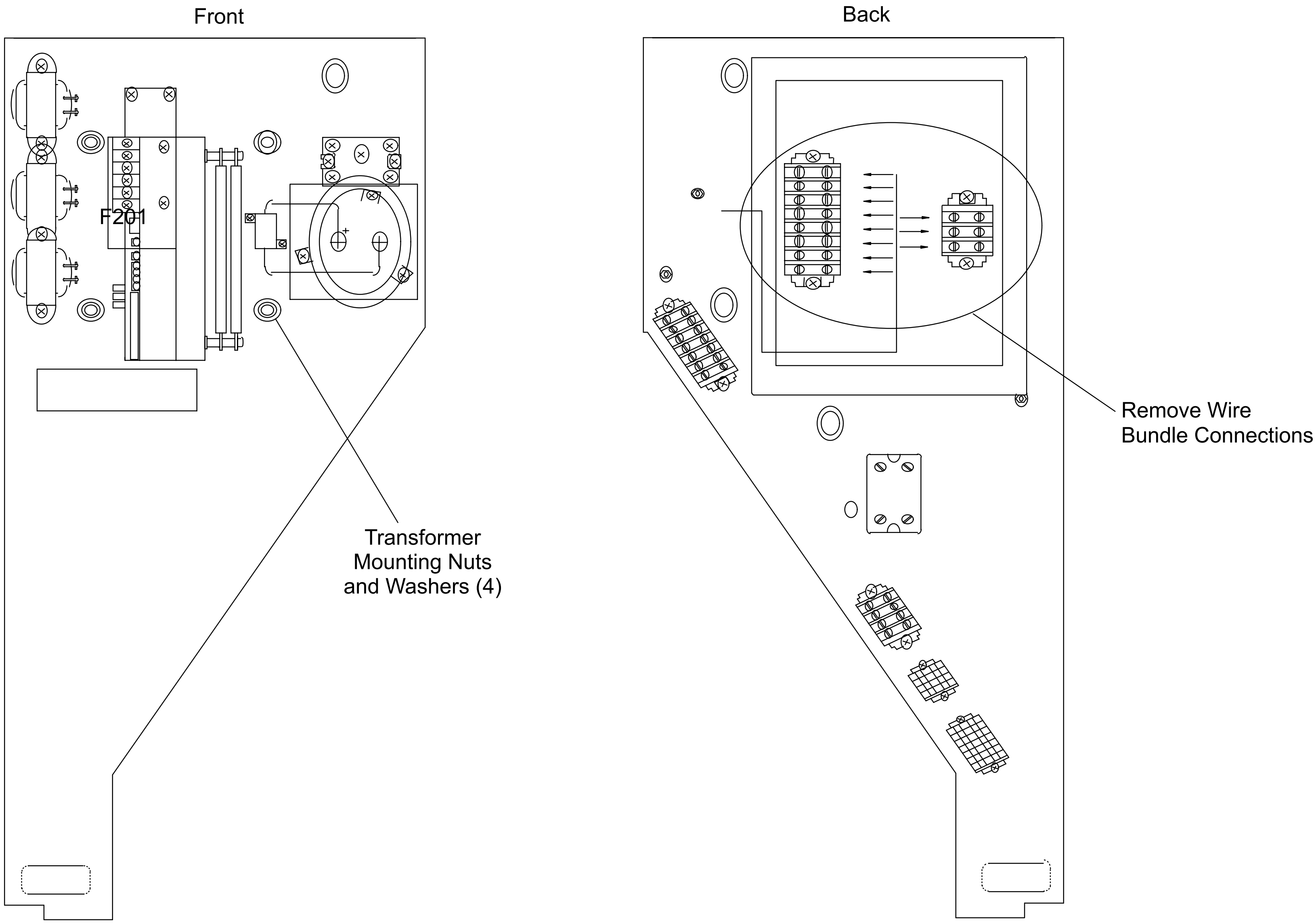


FIGURE 284 3 KVA Transformer Mounting and Connections

5. Loosen the top two Transformer Mounting Nuts and remove the Bottom two Nuts and Washers. Refer to Figure 284.

WARNING

USE TWO PERSONS TO HANDLE THE 3 KVA TRANSFORMER. THE 3KVA TRANSFER WEIGHS 80 LBS. FAILURE TO COMPLY COULD RESULT IN EQUIPMENT DAMAGE AND/OR SERIOUS INJURY TO PERSONNEL.

6. One person fully support the Transformer weight and the other person remove the upper two Mounting Nuts and Washers.
7. Remove the Transformer from the Gantry.

TRANSFORMER REPLACEMENT:

8. Install the replacement Transformer by reversing steps 4. through 7.
9. Replace the Gantry Front Cone. Refer to Front Cone Replacement in the Gantry Covers Removal/Replacement procedure.

WARNING

USE EXTREME CAUTION WHEN OPERATING OR WORKING ON EQUIPMENT WITH COVERS REMOVED. DANGEROUS VOLTAGES ARE EXPOSED WHEN THE GANTRY COVERS ARE REMOVED. FAILURE TO COMPLY CAN CAUSE SERIOUS INJURY OR DEATH TO PERSONNEL.

10. Apply Gantry power. Refer to the PowerUp/PowerDown procedure.
11. Verify 230 VAC 3 power on L1, L2 and L3 at the Terminal Block on the front of the Transformer.
12. Replace the Transformer Cover and the six (6) Mounting Screws
13. Pull the Front Cover Interlock Switch Plunger out to enable operation while the Front Covers are open.
14. To verify proper System operation, run Full–field and Half–field Water Phantom Scans and allow the System to process and display an Image.
15. Return the Front Cover Interlock Switch Plunger to its center position. Remove Gantry power. Refer to the PowerUp/PowerDown procedure.
16. Replace the Gantry lower Front Cover, close the upper Front and Rear Covers. Refer the Gantry Covers procedure.

SERVO AMPLIFIER

Introduction: This procedure is used to remove/replace the scan drive motor cntrl servo amp.

Tools Required: 1. #1 tip Phillips and 1/4” x 4” slottedscrewdriver

Estimated Time: 35 min.

SERVO AMPLIFIER REMOVAL:

1. Remove Gantry power. Refer to the PowerUp/PowerDown procedure.
2. Open the Gantry Front Covers and remove the Front Cone. Refer to the Gantry Covers procedures.

> ⚠ **WARNING**
>
> **USE EXTREME CAUTION WHEN SERVICING EQUIPMENT WHILE THE UPPER GULL WING IS OPEN. THE PANEL HAS SHARP EDGES. FAILURE TO COMPLY COULD RESULT IN SERIOUS INJURY TO PERSONNEL.**

3. Unscrew the two Phillips Mounting Screws and remove the Clear Lexan Shield from the front of the Servo Amplifier. See Figure 285.

> ⚠ **WARNING** ⚠
>
> **USE EXTREME CAUTION WHEN OPERATING OR WORKING ON EQUIPMENT WITH COVERS REMOVED. DANGEROUS VOLTAGES ARE EXPOSED WHEN THE GANTRY COVERS ARE REMOVED. FAILURE TO COMPLY CAN CAUSE SERIOUS INJURY OR DEATH TO PERSONNEL.**

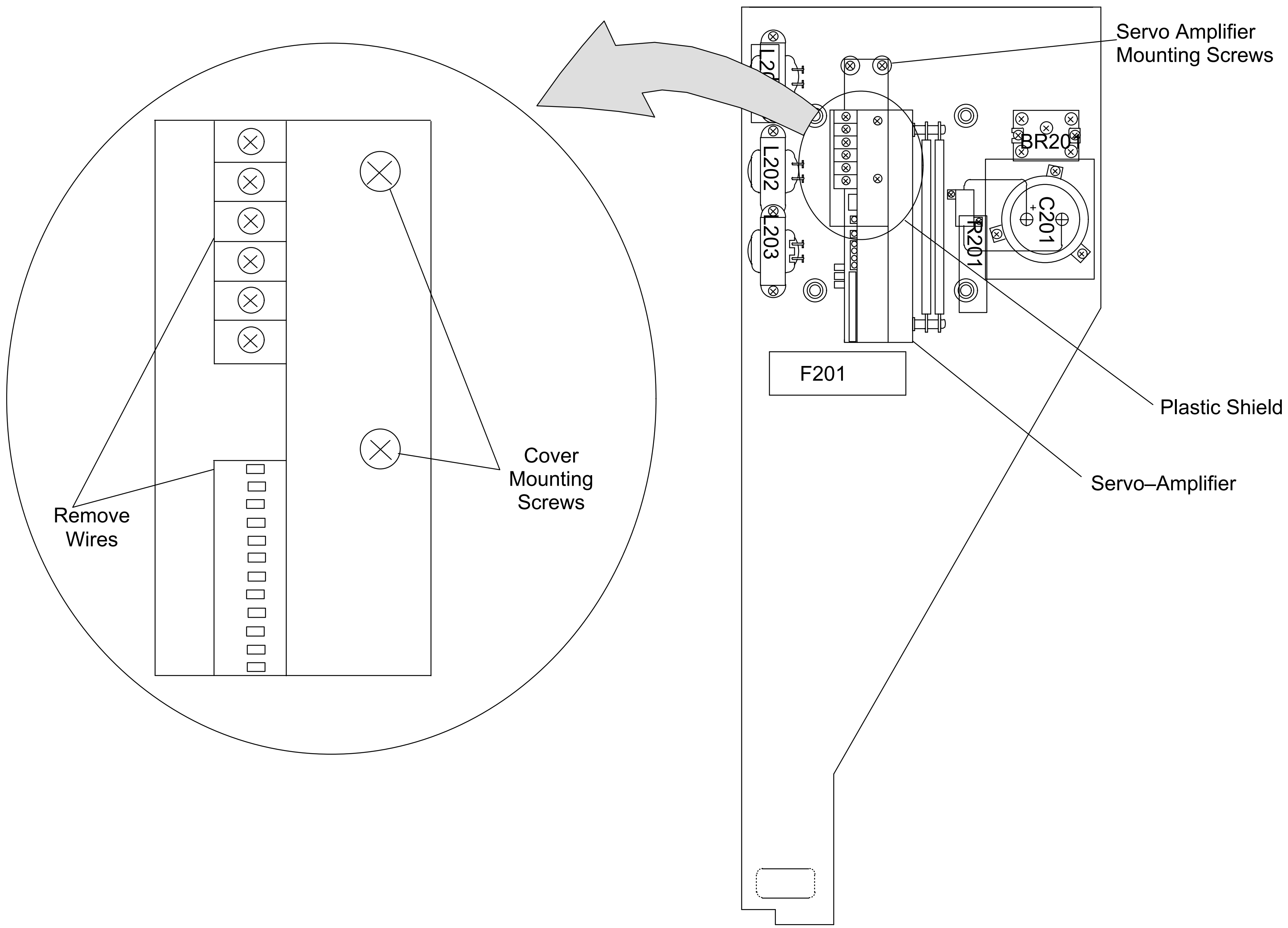


FIGURE 285 Servo Amplifier Assembly

4. Tag and remove all wires connected from the Gantry frame to the top terminal block. Keep all hardware for reuse during installation.

5. Loosen the three Servo Amplifier Mounting Screws (2 top and 1 bottom). Refer to Figure 285.
6. Support the weight of the Amplifier and remove both top Mounting Screws.
7. Tilt the top of the Servo Amplifier away from the Backplane and slide the bottom to the left to clear the bottom Mounting Screw.
8. Remove the Servo Amplifier from the Gantry.

SERVO AMPLIFIER REPLACEMENT:

9. Replace the Servo Amplifier by reversing steps 3. through 8.
10. Replace the Gantry Front Cone. Refer to the Gantry Covers procedure.
11. Apply Gantry power. Refer to the PowerUp/PowerDown procedure.
12. Pull the Front Cover Interlock Switch Plunger out to enable operation while the Front Covers are open.

> ⚠ **WARNING** ⚠
>
> **USE EXTREME CAUTION WHEN OPERATING OR WORKING ON EQUIPMENT WITH COVERS REMOVED. DANGEROUS VOLTAGES ARE EXPOSED WHEN THE GANTRY COVERS ARE REMOVED. FAILURE TO COMPLY CAN CAUSE SERIOUS INJURY OR DEATH TO PERSONNEL.**

13. To verify proper System operation, run Full–field and Half–field Water Phantom Scans and allow the System to process and display an Image.
14. Return the Front Cover Interlock Switch Plunger to its center position.
15. Remove Gantry power. Refer to the PowerUp/PowerDown procedure.
16. Replace the Gantry lower Front Cover, close the upper Front Cover. Refer to the Gantry Covers procedure.

SCAN DRIVE CONTROL: SOLID STATE RELAY & BRIDGE RECTIFIER

Introduction: This procedure is used to remove and replace the Solid State Relay (SSR) mounted on the back of the Scan Drive Control Chassis and the Bridge Rectifier mounted on the front of the Chassis.

Tools Required: 1. 5/16” x 6” slotted and #1 tip Phillips screwdrivers

Estimated Time: 1 Hr.

Solid State Relay Removal:

1. Remove Gantry power. Refer to the PowerUp/PowerDown procedure.
2. Open the Gantry Front Covers and remove the Front Cone. Refer to the Gantry Covers procedure.

> ⚠ **WARNING**
>
> **USE EXTREME CAUTION WHEN SERVICING EQUIPMENT WHILE THE UPPER GULL WING IS OPEN. THE PANEL HAS SHARP EDGES. FAILURE TO COMPLY COULD RESULT IN SERIOUS INJURY TO PERSONNEL.**

> ⚠ **WARNING** ⚠
>
> **USE EXTREME CAUTION WHEN OPERATING OR WORKING ON EQUIPMENT WITH COVERS REMOVED. DANGEROUS VOLTAGES ARE EXPOSED WHEN THE GANTRY COVERS ARE REMOVED. FAILURE TO COMPLY CAN CAUSE SERIOUS INJURY OR DEATH TO PERSONNEL.**

3. Tag and remove the four wires connected to the SSR terminals. See Figure 286.

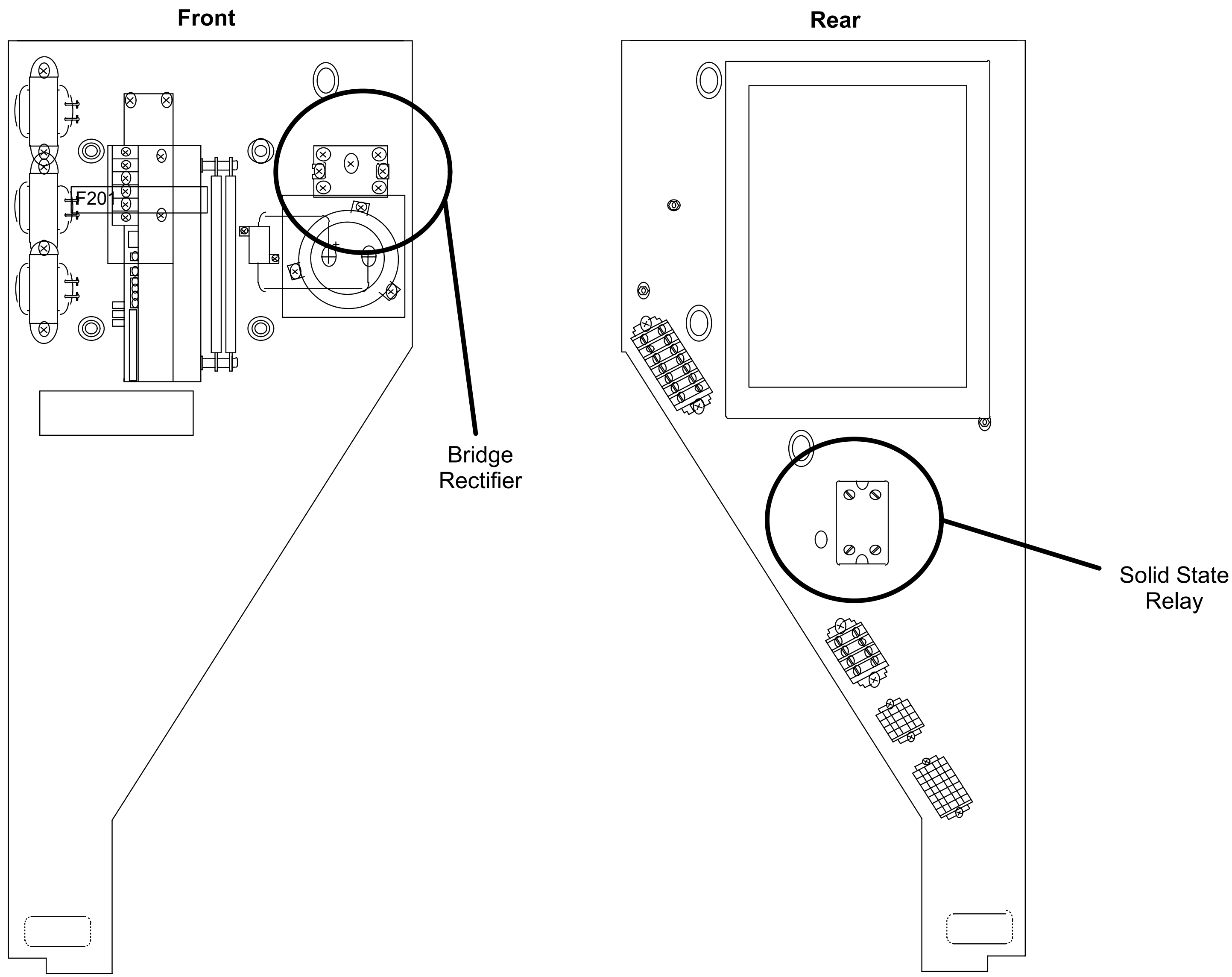


FIGURE 286 Bridge Rectifier and Solid State Relay Locations

4. Support the SSR and remove the two Mounting Screws. See Figure 287.

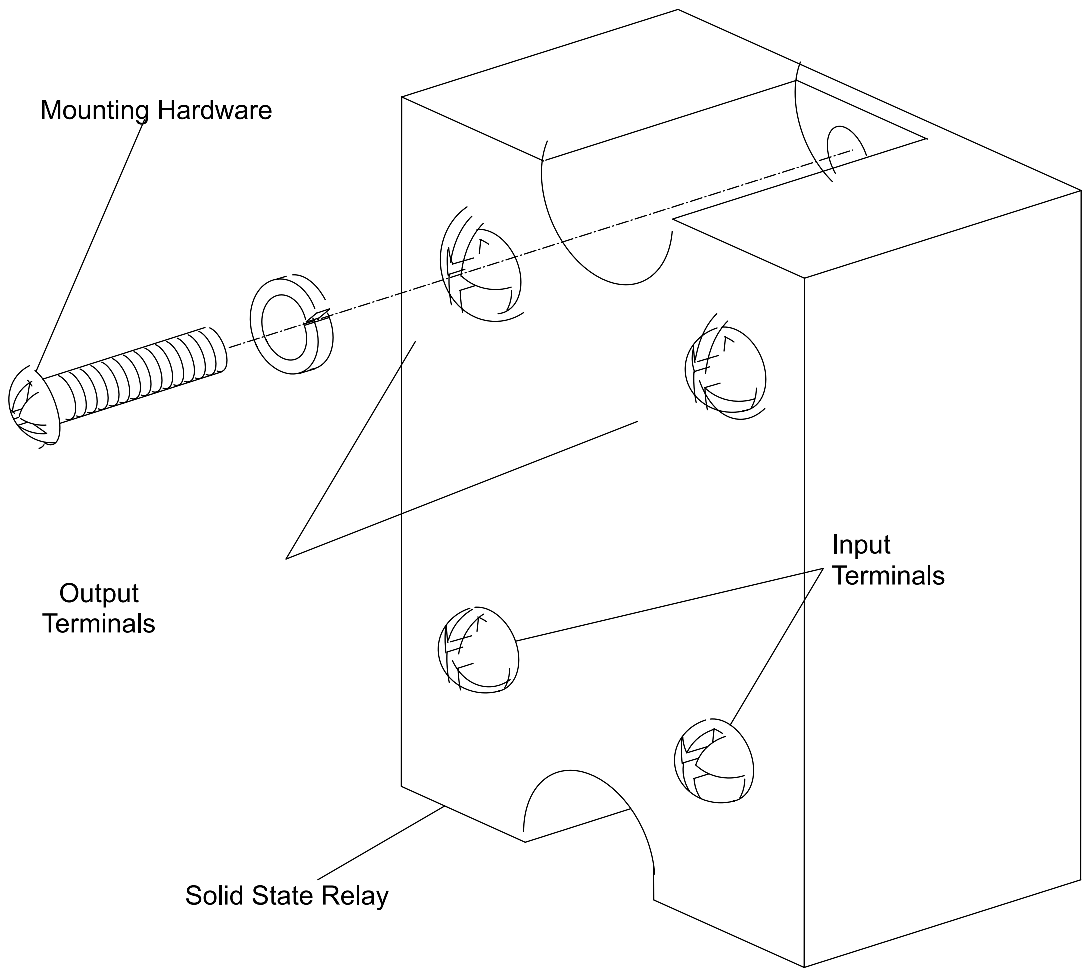


FIGURE 287 Solid State Relay and Mounting Hardware

5. Remove the SSR from the Gantry.

Bridge Rectifier Removal:

1. Remove Gantry power. Refer to the PowerUp/PowerDown procedure.
2. Open the Gantry Front Covers and remove the Front Cone. Refer to the Gantry Covers procedure.
3. Tag and remove the 5 wires connected to the Bridge Rectifier terminals (three AC input and two DC output). See Figure 288.

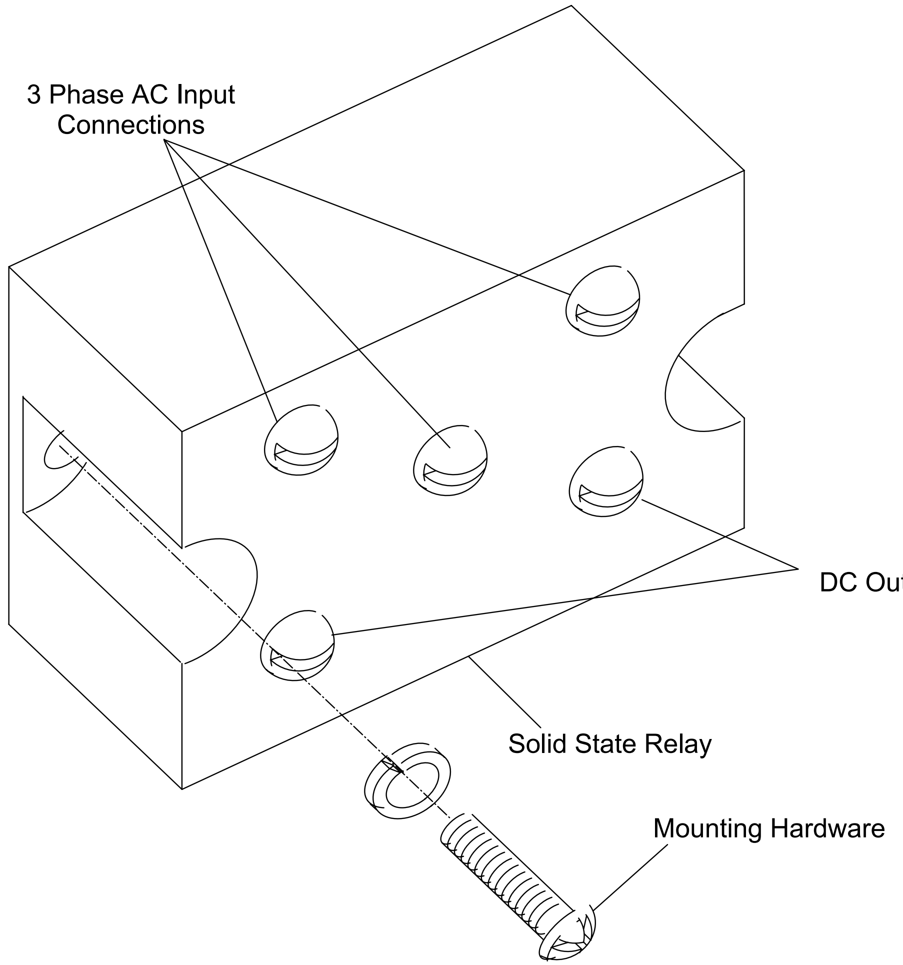


FIGURE 288 Bridge Rectifier

4. Support the Bridge Rectifier and remove the two Mounting Screws.
5. Remove the Bridge Rectifier from the Gantry.

Solid State Relay Replacement:

6. Install the replacement Relay by reversing steps 3. to 5. of Solid State Relay removal.

Bridge Rectifier Replacement:

7. Install the replacement Rectifier by reversing steps 3. to 5. of Bridge Rectifier removal.
8. Apply Gantry power. Refer to the PowerUp/PowerDown procedure.

> ⚠ **WARNING** ⚠
>
> **USE EXTREME CAUTION WHEN OPERATING OR WORKING ON EQUIPMENT WITH COVERS REMOVED. DANGEROUS VOLTAGES ARE EXPOSED WHEN THE GANTRY COVERS ARE REMOVED. FAILURE TO COMPLY CAN CAUSE SERIOUS INJURY OR DEATH TO PERSONNEL.**

9. Pull the Front Cover Interlock Switch Plunger out to enable operation while the Front Covers are open.
10. Turn on the Laser and verify that the Scan Frame moves.

> ⚠ **WARNING**
>
> **ROTATING MACHINERY. STAND CLEAR OF ROTATING FRAME WHEN EQUIPMENT IS OPERATED. FAILURE TO COMPLY MAY RESULT IN SERIOUS INJURY OR DEATH TO PERSONNEL.**

11. To verify proper System operation, run Full–field and Half–field Water Phantom Scans and allow the System to process and display an Image.
12. Return the Front Cover Interlock Switch Plunger to its center position.
13. Remove Gantry power. Refer to the PowerUp/PowerDown procedure.
14. Close the upper Front and Rear Covers. Refer to the Gantry Covers procedure.

SERVO–CONTROLLER FAN

Introduction: This procedure is used to remove and replace the Servo–Controller Fan.

Tools Required: 1. 5/16" open box end wrench

Estimated Time: 30 min.

SERVO–CONTROLLER FAN REMOVAL:

1. Remove Gantry power. Refer to the PowerUp/PowerDown procedure.
2. Open the Gantry Front Covers and remove the Front Cone. Refer to the Gantry Covers procedure.

> ⚠ **WARNING**
>
> **USE EXTREME CAUTION WHEN SERVICING EQUIPMENT WHILE THE UPPER GULL WING IS OPEN. THE PANEL HAS SHARP EDGES. FAILURE TO COMPLY COULD RESULT IN SERIOUS INJURY TO PERSONNEL.**

> ⚠ **WARNING** ⚠
>
> **USE EXTREME CAUTION WHEN OPERATING OR WORKING ON EQUIPMENT WITH COVERS REMOVED. DANGEROUS VOLTAGES ARE EXPOSED WHEN THE GANTRY COVERS ARE REMOVED. FAILURE TO COMPLY CAN CAUSE SERIOUS INJURY OR DEATH TO PERSONNEL.**

3. Tag and disconnect the black and white wires from the Fan Power Terminals.
4. Cut and remove any Tie–Wraps securing the Wire Bundle to the Fan.
5. Remove the four Hex Nuts and Washers that secure the Fan to the Servo–Controller Figure 289.

> **NOTE**
>
> Two of the four Hex Nuts are located behind the 3 KVA Scan Drive Transformer. The Nuts and Washers are small and can be easily dropped if not handled with care.

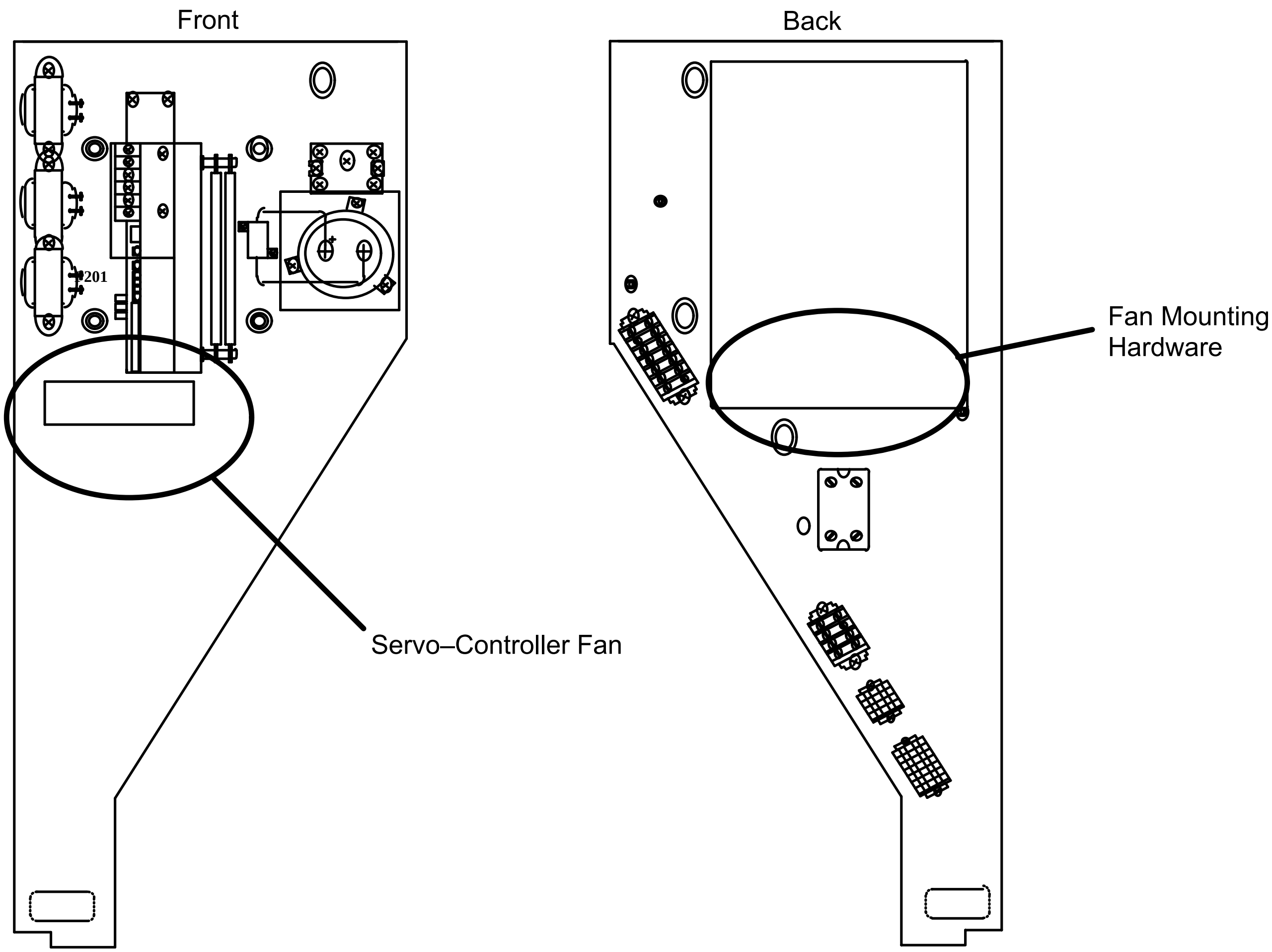


FIGURE 289 Servo–Controller Fan and Mounting Hardware

SERVO–CONTROLLER FAN REPLACEMENT

6. Install the replacement Fan by reversing steps 3. to 5.
7. Apply Gantry power. Refer to the PowerUp/PowerDown procedure.
8. Block the air flow from the Gantry Air Blowers and check that the Fan stops rotating.
9. Pull the Front Cover Interlock Switch Plunger out to enable operation while the Front Covers are open.

10. With the air flow blocked, check that the Fan is operating.
11. Return the Front Cover Interlock Switch Plunger to its center position.
12. Remove Gantry power. Refer to the PowerUp/PowerDown procedure.
13. Close the upper Front and Rear Covers. Refer to the Gantry Covers procedure.

GANTRY AIR BLOWER

Introduction: This procedure is used to remove and replace the Gantry Air Blowers. Since both Blowers are the same, this procedure will apply in either case.

Tools Required:

1. 7/16” socket, ratchet
2. #1 tip Phillips screwdriver
3. 3/16” hex key

Estimated Time: 1.0 hr.

BLOWER ASSEMBLY REMOVAL:

1. Remove Gantry power. Refer to the PowerUp/PowerDown procedure.
2. Open the Gantry Front Covers and remove the Front Cone. Refer to the Gantry Covers procedure.

> **⚠ WARNING**
>
> **USE EXTREME CAUTION WHEN SERVICING EQUIPMENT WHILE THE UPPER GULL WING IS OPEN. THE PANEL HAS SHARP EDGES. FAILURE TO COMPLY COULD RESULT IN SERIOUS INJURY TO PERSONNEL.**

> **⚠ WARNING**
>
> **USE EXTREME CAUTION WHEN OPERATING OR WORKING ON EQUIPMENT WITH COVERS REMOVED. DANGEROUS VOLTAGES ARE EXPOSED WHEN THE GANTRY COVERS ARE REMOVED. FAILURE TO COMPLY CAN CAUSE SERIOUS INJURY OR DEATH TO PERSONNEL.**

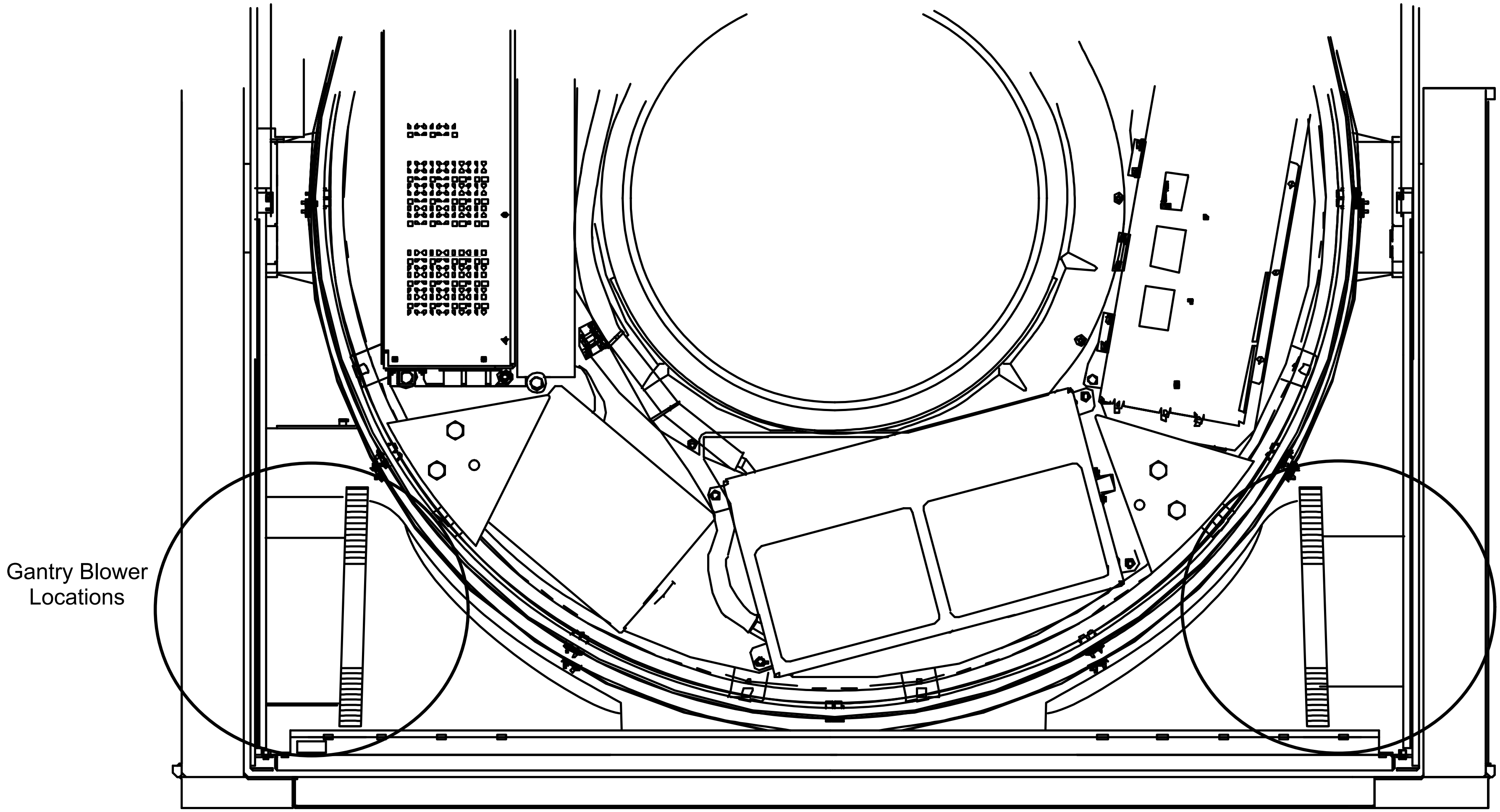


FIGURE 290 Gantry Air Lower Locations

3. Note routing of Blower Cable and cut Cable Ties securing it to the Gantry.
4. Tag and remove the three in–line connectors on the blower cable assembly connected outside the AC Control Chassis.
5. Remove the four Mounting Screws and remove the Air Inlet Duct. See Figure 291.

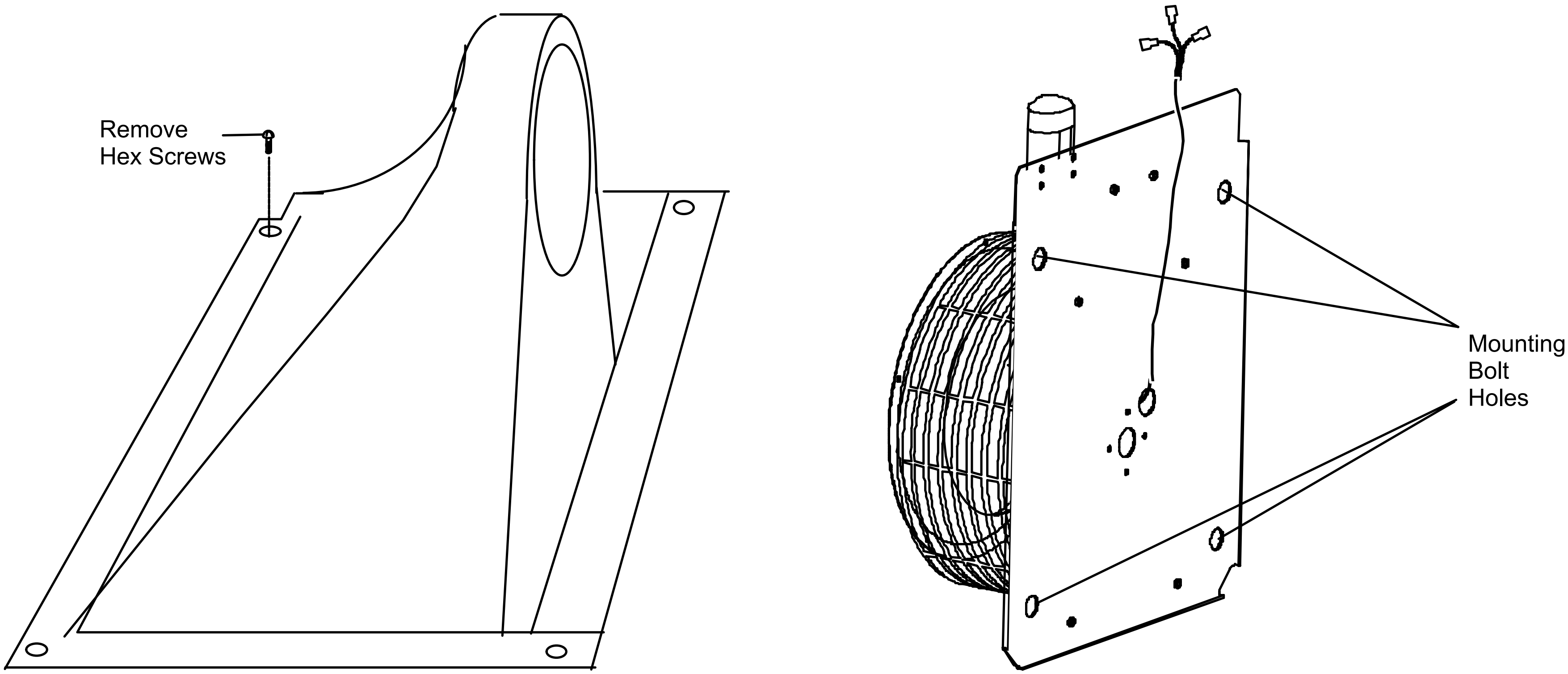


FIGURE 291 Gantry Air Duct (right) / Gantry Blower Assembly

6. Remove the four blower mounting Bolts, Nuts and Washers that secure the Blower Assembly to the Gantry column. See Figure 291.

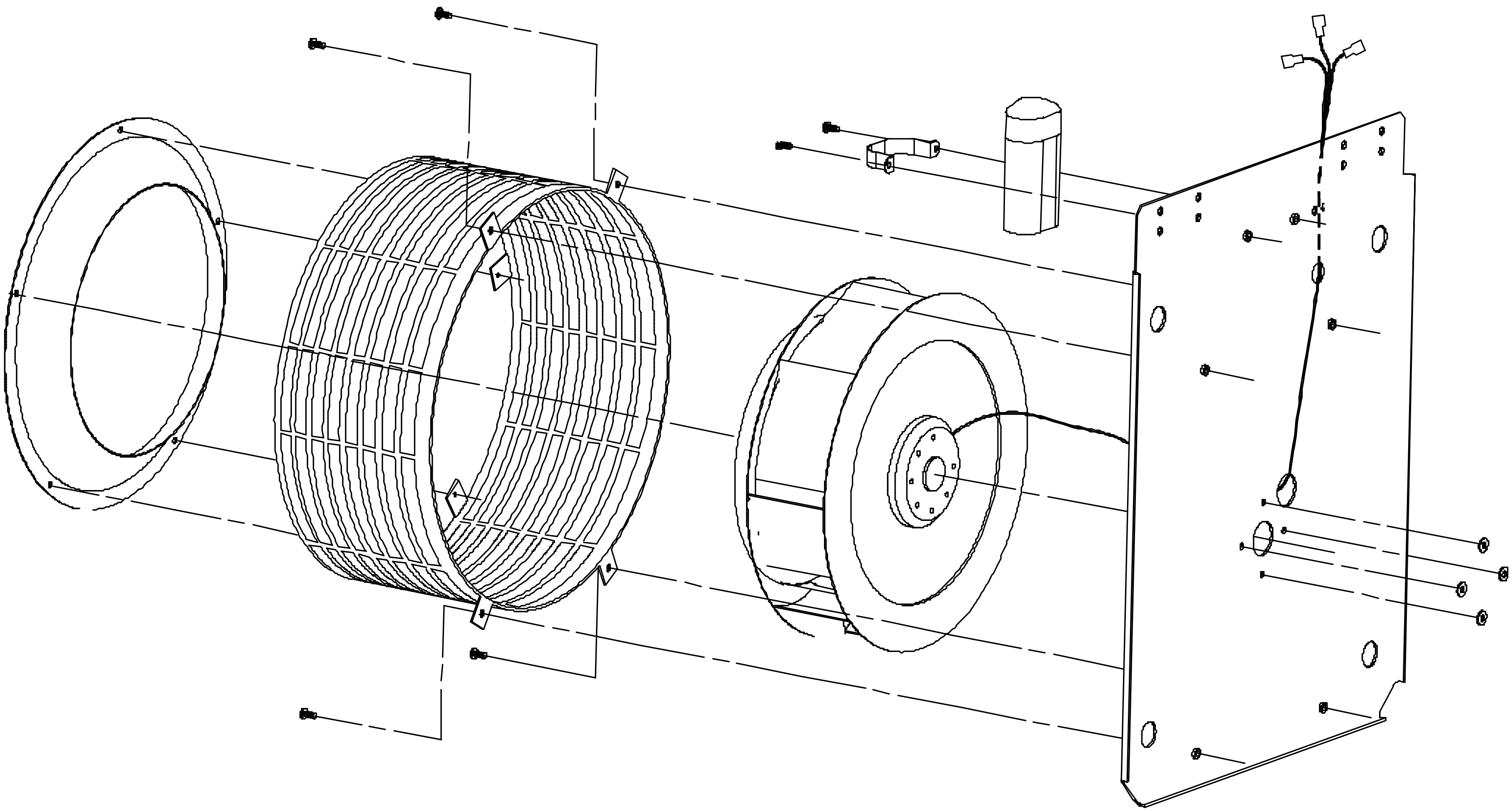


FIGURE 292 Gantry Air Blower Assembly Detail

7. Support the blower assembly, lift it off the mounting studs and remove it.

BLOWER REMOVAL:

8. Remove the four blower mounting nuts and washers that fasten the Blower to the Mounting Plate. Refer to Figure 292.

REPLACEMENT:

9. Install the replacement Blower by reversing step 8. above and install the Blower Assembly into the Gantry by reversing steps 3. through 7.

10. Replace the Gantry Front Cone. Refer to the Gantry Covers procedure.
11. Apply Gantry power. Refer to the PowerUp/PowerDown procedure.

> ⚠ **WARNING** ⚠
>
> **USE EXTREME CAUTION WHEN OPERATING OR WORKING ON EQUIPMENT WITH COVERS REMOVED. DANGEROUS VOLTAGES ARE EXPOSED WHEN THE GANTRY COVERS ARE REMOVED. FAILURE TO COMPLY CAN CAUSE SERIOUS INJURY OR DEATH TO PERSONNEL.**

12. Check Blower operation by turning the system ON at the the Console and feel the air flow from the Blower.
13. Pull the Front Cover Interlock Switch Plunger out to enable operation while the Front Covers are open.

> ⚠ **WARNING**
>
> **ROTATING MACHINERY. STAND CLEAR OF ROTATING FRAME WHEN EQUIPMENT IS OPERATED. FAILURE TO COMPLY MAY RESULT IN SERIOUS INJURY OR DEATH TO PERSONNEL.**

14. To verify proper System operation, run Full–field and Half–field Water Phantom Scans and allow the System to process and display an Image.
15. Return the Front Cover Interlock Switch Plunger to its center position.
16. Remove Gantry power. Refer to the PowerUp/PowerDown procedure.
17. Replace the Gantry lower Front and Rear Covers, close the upper Front and Rear Covers. Refer to the Gantry Covers procedure.

GANTRY AIR FILTER

Introduction: This procedure is used to remove and replace the Gantry air filters.

Tools Required: 1. #1 tip Phillips screwdriver

Materials Required: Two air filters

Estimated Time: 10 min.

AIR FILTER REMOVAL:

1. Remove Gantry power. Refer to the PowerUp/PowerDown procedure.
2. Open the Gantry Front Cover. Refer to the Gantry Covers procedure.
3. Remove the Air Filters by sliding them out. See Figure 293.

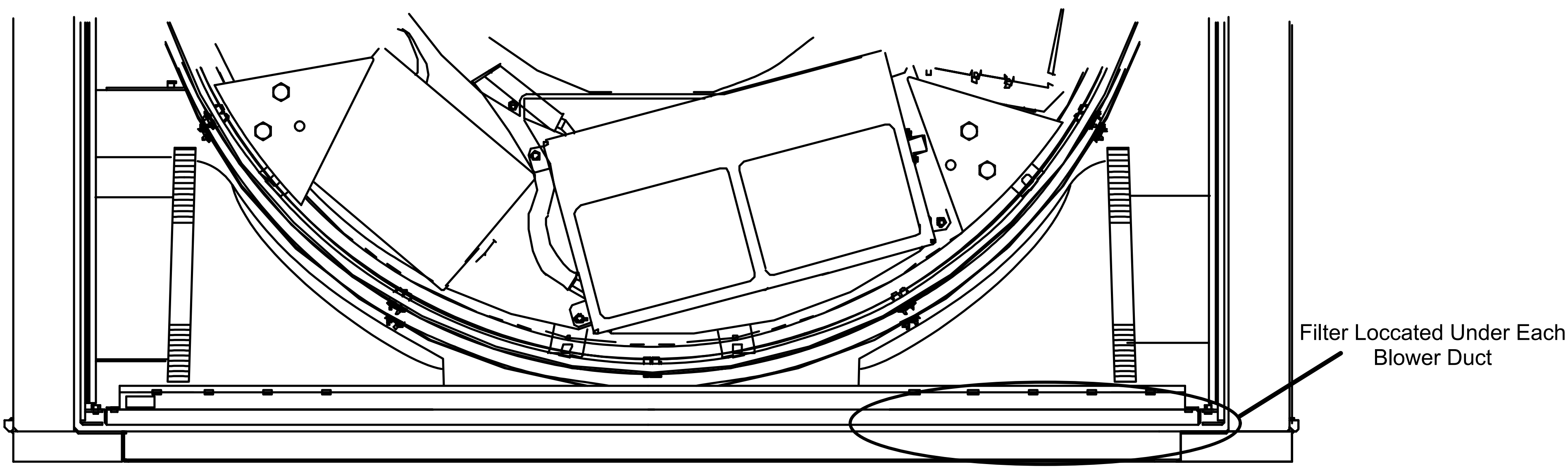


FIGURE 293 Gantry Air Filter Location

AIR FILTER REPLACEMENT:

4. Be sure that the air flow arrow points up (^) away from the floor before replacing the Air Filters. Position the Air Filter into the slot and slide the Filter into the Gantry until it is fully in position.
5. Apply Gantry power. Refer to the PowerUp/PowerDown procedure.

> ⚠ **WARNING** ⚠
>
> **USE EXTREME CAUTION WHEN OPERATING OR WORKING ON EQUIPMENT WITH COVERS REMOVED. DANGEROUS VOLTAGES ARE EXPOSED WHEN THE GANTRY COVERS ARE REMOVED. FAILURE TO COMPLY CAN CAUSE SERIOUS INJURY OR DEATH TO PERSONNEL.**

6. Check that air flows out of the Air Duct in the lower Front Cover (feel air flow with the hand).
7. Remove Gantry power. Refer to the PowerUp/PowerDown procedure.
8. Close and secure the Front Covers. Refer to the Gantry Covers procedure.

INDIVIDUAL CARDFILE CIRCUIT BOARD

Introduction: The purpose of this procedure is to remove and replace a typical Gantry Cardfile Circuit Board.

Tools Required: 1. Small Slotted Screwdriver

Estimated Time: 20 Min.

CIRCUIT BOARD REMOVAL:

1. Remove Gantry power. Refer to the PowerUp/PowerDown procedure.

> **WARNING**
>
> **USE EXTREME CAUTION WHEN OPERATING OR WORKING ON EQUIPMENT WITH COVERS REMOVED. DANGEROUS VOLTAGES ARE EXPOSED WHEN THE GANTRY COVERS ARE REMOVED. FAILURE TO COMPLY CAN CAUSE SERIOUS INJURY OR DEATH TO PERSONNEL.**

2. Remove the gantry cardfile covers by loosening the four captive screws on each cover. See Figure 294.

> **NOTE**
>
> The threaded ends of the gantry cardfile cover screw holes are plasitc. This helps to hold the screws in place, and are therefore labeled “captive.” However, the screws can come completely out if loosened too far.

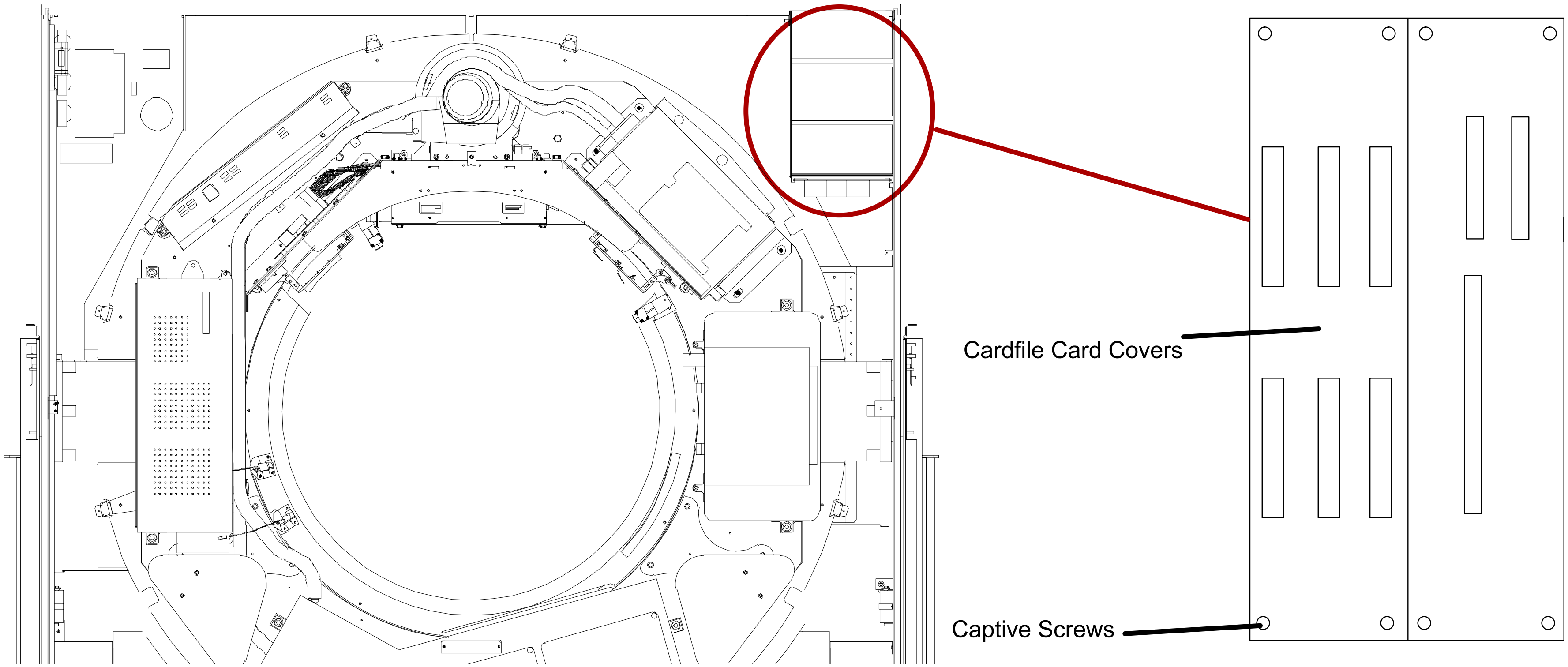
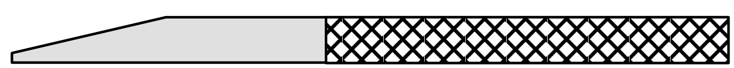


FIGURE 294 Cardfile Cover Removal

> ⚠ **CAUTION**
>
> STATIC ELECTRICITY. CIRCUIT BOARDS MUST BE HANDLED WITH ANTI–STATIC TECHNIQUES. DISCHARGE YOUR BODY BEFORE HANDLING ANY CIRCUIT BOARD. KEEP ALL CIRCUIT BOARDS IN ANTI–STATIC BAGS OR ON CONDUCTIVE FOAM WHEN NOT INSTALLED IN THE EQUIPMENT. FAILURE TO COMPLY MAY CAUSE PERMANENT DAMAGE TO COMPONENTS ON CIRCUIT BOARDS.

3. Provide an Anti–static Bag or piece of Conductive Foam large enough to protect the Circuit Card to be removed.
4. Discharge your body static electricity. Use a Wrist Strap and Ground Wire connected to the grounded Chassis. Board locations can be seen in Figure 295.
5. Use the pry tool to disengage the card from the cardfile: 

CIRCUIT BOARD REPLACEMENT:

6. Slide the Circuit Board out of the Cardfile and immediately place it in the Anti–static Bag or circuit side down on the Conductive Foam.
7. Install the new circuit board into the Gantry Cardfile by sliding it into the guides in the Cardfile. Push down on the ejectors to seat the board. Then, firmly push on the top edge of the Circuit Board until the Connector is firmly seated (the ejectors do not seat the board completely in).
8. Return the cardfile covers back to their original position.
9. Apply Gantry power. Refer to the PowerUp/PowerDown procedure.
10. To verify proper System operation, run Full–field and Half–field Water Phantom Scans and allow the System to process and display an Image.
11. Remove Gantry power. Refer to the PowerUp/PowerDown procedure.

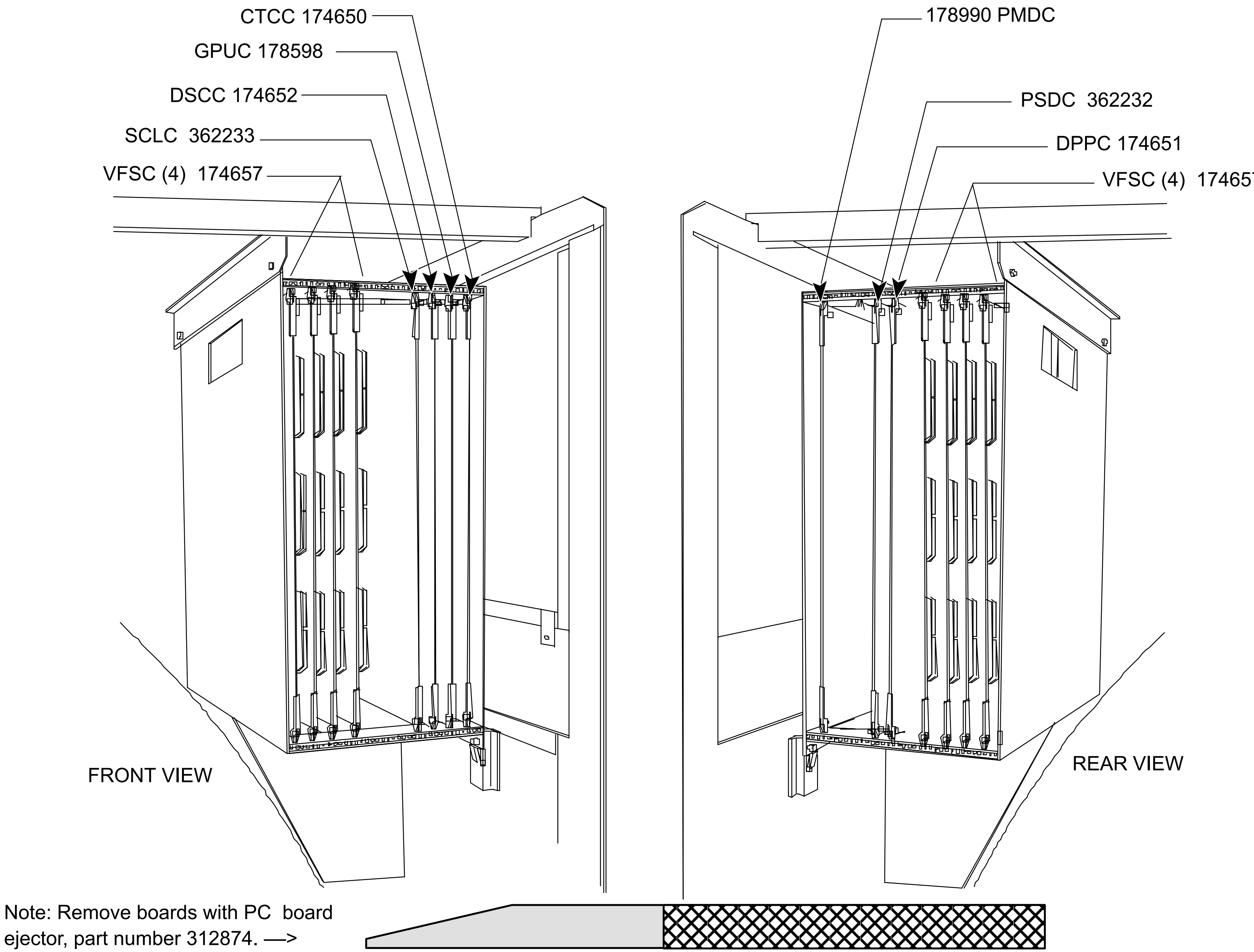


FIGURE 295 AcQSim CT Board Locations

GANTRY TILT ACTUATOR

Introduction: This procedure is used to remove and replace the Gantry Tilt Actuators. The procedure applies to both Actuators.

Tools Required:
1. #1 tip Phillips Screwdriver
2. 10” Adjustable Wrench
3. 6” Long Chain Nose Plier

Estimated Time: 50 min.

> **NOTE**
>
> Compressive force on either Actuator cannot be relieved. Expect the Gantry to sag when the Actuator is removed.

TILT ACTUATOR REMOVAL:

1. Apply Gantry power. Refer to the PowerUp/PowerDown procedure.
2. Remove both Gantry Column Covers. Refer to the Cover Removal procedure.

> ⚠ **WARNING** ⚠
>
> **USE EXTREME CAUTION WHEN OPERATING OR WORKING ON EQUIPMENT WITH COVERS REMOVED. DANGEROUS VOLTAGES ARE EXPOSED WHEN THE GANTRY COVERS ARE REMOVED. FAILURE TO COMPLY CAN CAUSE SERIOUS INJURY OR DEATH TO PERSONNEL.**

3. Disconnect the Speed Sensor from the operating Actuator. Refer to Figure 296.
4. Tilt the Gantry forward until the Lift Switch on the defective side stops Gantry motion. This will transfer all the force to the operating Actuator.

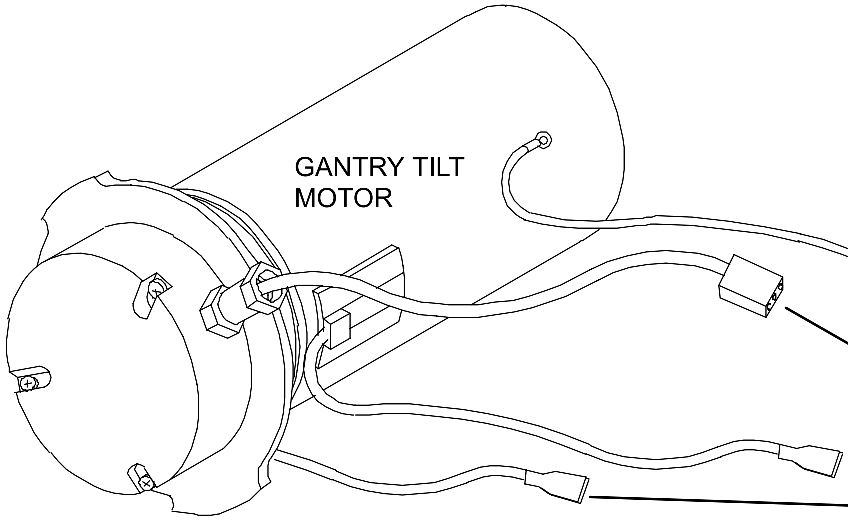


FIGURE 296 GANTRY TILT MOTOR

> ⚠ **CAUTION**
>
> MAKE SURE THAT THE ACTUATOR NOT CARRYING THE LOAD IS THE ONE YOU REMOVE.

> ⚠ **CAUTION**
>
> EXERCISE EXTREME CAUTION WHEN TILTING THE GANTRY WITH ONLY ONE ACTUATOR OPERATIONAL. THE GANTRY IS IN AN UNSTABLE STATE AND ITS MOVEMENT SHOULD BE MINIMIZED UNDER THESE CONDITIONS

> **NOTE**
>
> There will still be some pressure on the Actuator due to spring tension in the Mounting Shoe. If more tilt is required for Actuator removal, place a Tie–Wrap between the Gantry Coulmn Base and the defective Tilt Actuator's Limit Switch Activating Arm. This will allow for a small amount of additional lift of the defective unit.

5. Remove Gantry power at the wallbox. Refer to the Power Up/Down procedure.
6. Tag and remove all Cables/Wires connected to the defective Tilt Actuator.
7. Remove the Cotter Pin at the top of the Motor Drive Screw and remove the Castle Nut and Flat Washer. See Figure 297.

> **NOTE**
>
> Ultra Z lef and right Tilt Actuators CANNOT be interchanged.

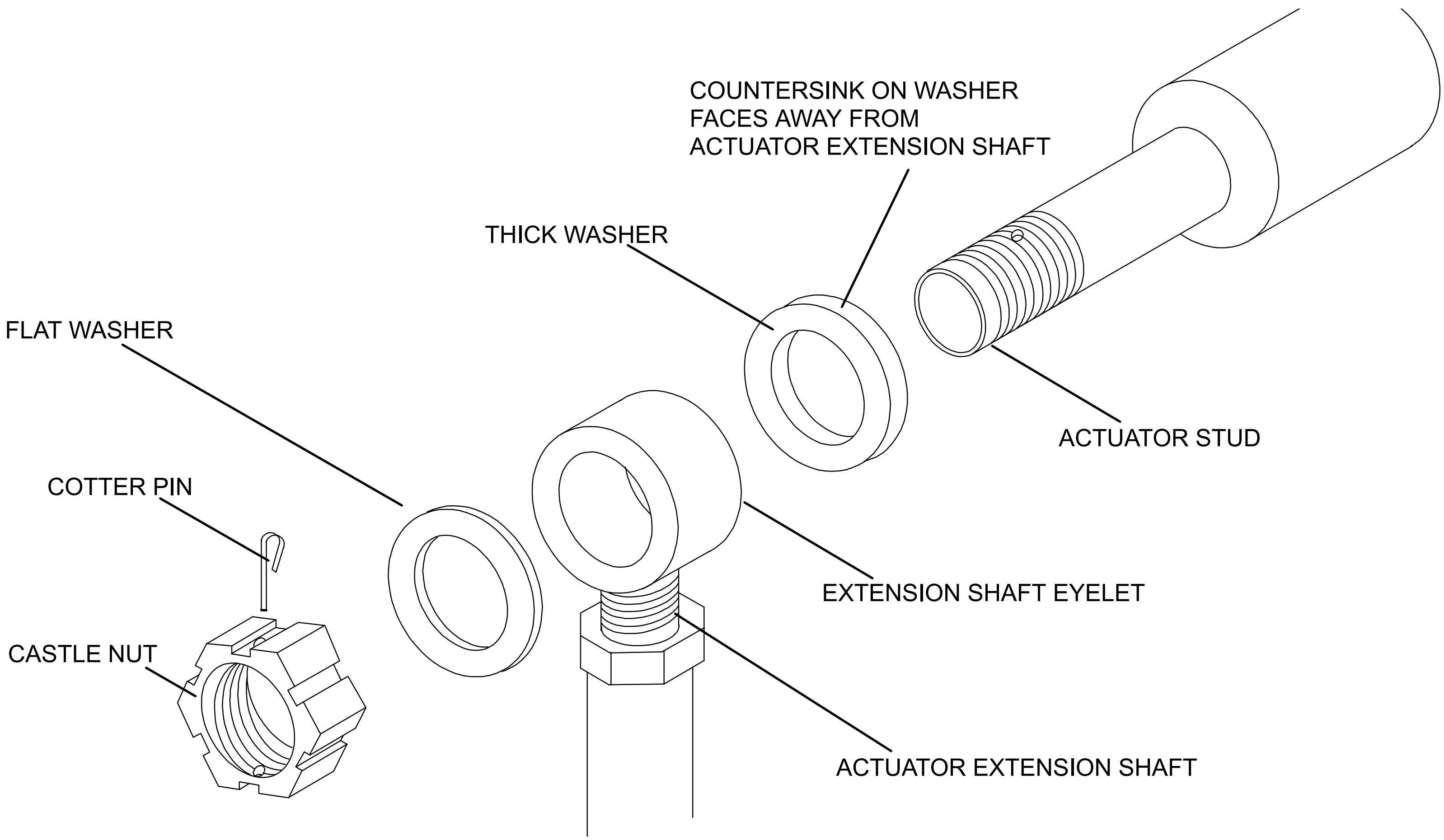


FIGURE 297 Extension Shaft Removal

8. Remove the Actuator Extension Shaft and the thick Washer from the Gantry Pin. Note the the bevel cut in the center of the thick Washer and how it is assembled onto the Pin.

9. Remove the actuator switch at the base of the tilt arm. This will expose the pin that secures the bottom of the tilt arm. Remove the pin. See Figure 298.

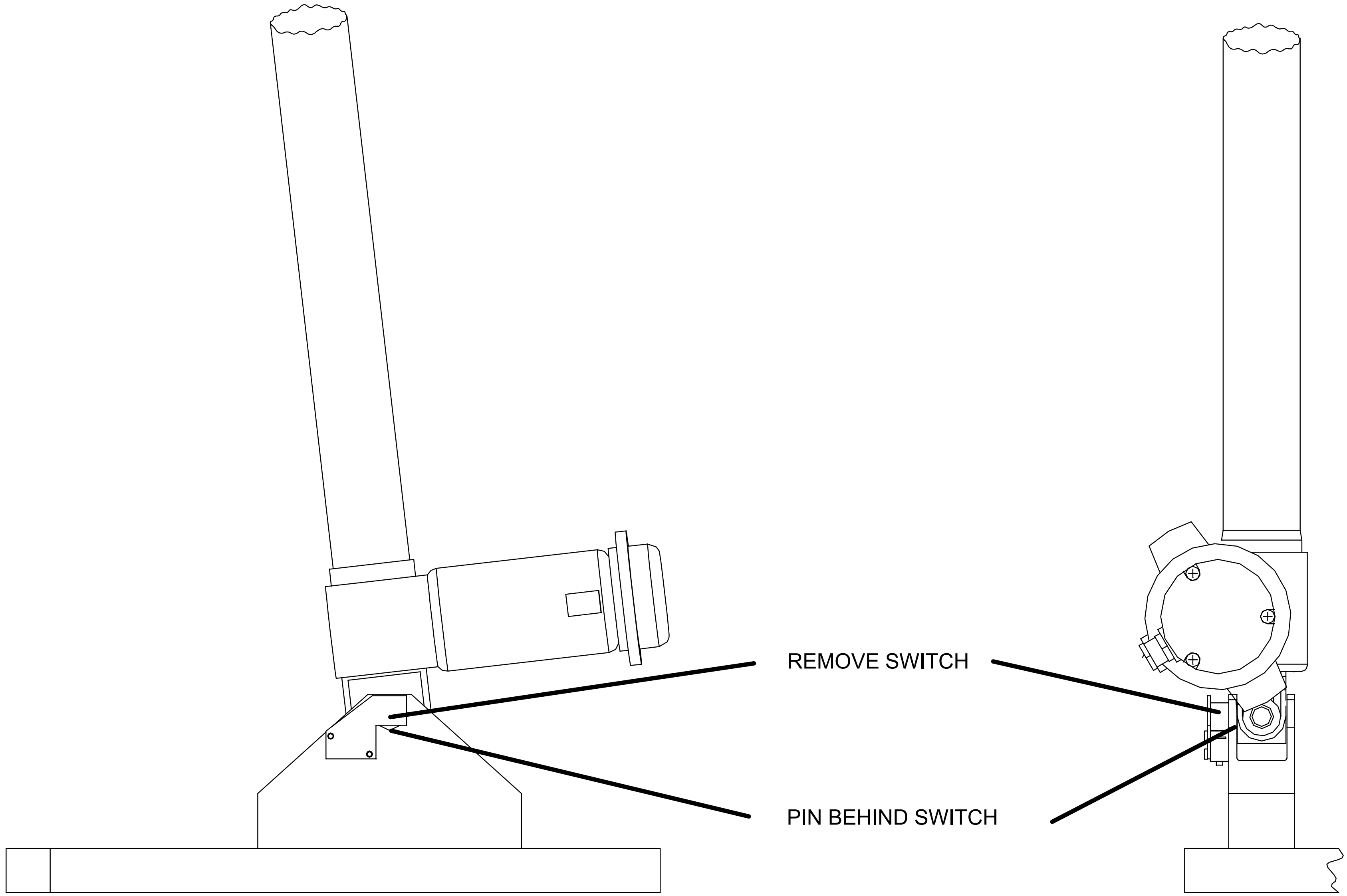


FIGURE 298 Tilt Arm– Pin Removal

10. Install a 10–32 x 1 inch long Screw into the threaded hole in the end of the Pivot Shaft. Pull on the Screw to remove the Pivot Pin.
11. Rotate the bottom of the Actuator Assembly out of the Mounting Shoe and remove the Actuator Assembly.

TILT ACTUATOR REPLACEMENT:

12. Extend the Actuator Extension Shaft by hand to the length required to match the mounting points on the Shoe Pivot and the Gantry Pin.
13. Position the Actuator into the Shoe and install the Pivot Pin. Install the Pivot Pin Retaining Screw and Washer. Refer to Figure 298. DO NOT connect the Actuator Extension Shaft to the Gantry at this time.

> **NOTE**
>
> Before replacing the Pivot Pin, apply a thin coat of Lubriplate grease to the Pin. This will minimize metallic chattering noise sometimes heard when the Gantry is being tilted.

14. Connect Actuator Power.
15. Apply Gantry power. Refer to the Power Up/Down procedure.

> **⚠ WARNING**
>
> **USE EXTREME CAUTION WHEN OPERATING OR WORKING ON EQUIPMENT WITH COVERS REMOVED. DANGEROUS VOLTAGES ARE EXPOSED WHEN THE GANTRY COVERS ARE REMOVED. FAILURE TO COMPLY CAN CAUSE SERIOUS INJURY OR DEATH TO PERSONNEL.**

> **NOTE**
>
> If the opposite Actuator is operational, remove its power and Speed Sensor connections prior to performing step 1.

16. Place a temporary mark on the Actuator Extension Shaft and using the Gantry Control Panel Switches, tilt the Gantry 2 or 3 degrees positive (+). Verify that the Extension Shaft on the replacement Actuator has extended. If the Actuator has retracted, reverse the Actuator Motor Leads.

17. Remove Gantry power. Refer to the PowerUp/PowerDown procedure in the Introduction section.
18. By hand, return the Extension Shaft to the correct length for installation as indicated by the reference mark on the Shaft.
19. Install the Extension Shaft onto the Gantry Pin. Refer to Figure 297.
20. Connect the Speed Sensor Cables (Hall effect) on both Actuators.
21. Apply Gantry power. Refer to the PowerUp/PowerDown procedure.

> ⚠ **WARNING** ⚠
>
> **USE EXTREME CAUTION WHEN OPERATING OR WORKING ON EQUIPMENT WITH COVERS REMOVED. DANGEROUS VOLTAGES ARE EXPOSED WHEN THE GANTRY COVERS ARE REMOVED. FAILURE TO COMPLY CAN CAUSE SERIOUS INJURY OR DEATH TO PERSONNEL.**

22. Tilt the Gantry in the positive (+) direction with the Gantry Control Panel Switches until the Actuator Mounting Shoe is fully seated.

> **NOTE**
>
> If a Tie–Wrap was used to allow more tilt than that allowed by the Lift Switch, it must be used again to allow seating of the Mounting Shoe.

23. Tilt the Gantry to plus and minus 30 degrees with the Gantry Control Panel Switches. Verify that the Actuator will operate over the entire range, that the Limit Switches stop the tilt motion, and that the Display indicates the correct angle.
24. Calibrate the Gantry tilt if required. Refer to the Gantry Tilt Calibration in the Mechanical Calibration Manual.
25. Remove Gantry power. Refer to the PowerUp/PowerDown procedure.
26. Replace the Gantry Column Covers. Refer to the Cover Removal procedure.

GANTRY TILT POTENTIOMETER BELT

Introduction:	This procedure is used to remove and replace the Tilt Potentiometer Drive Belt in the Gantry.
Estimated Time:	60 Min.
Tools Required:	Small Slotted Screwdriver

TILT POTENTIOMETER BELT REMOVAL:

1. Remove Gantry power. Refer to the PowerUp/PowerDown procedure.
2. Remove the Gantry Right Column Cover. Refer to the Gantry Covers procedure.
3. Tag and remove the four wires connected to the Gantry Column E–stop Switch.

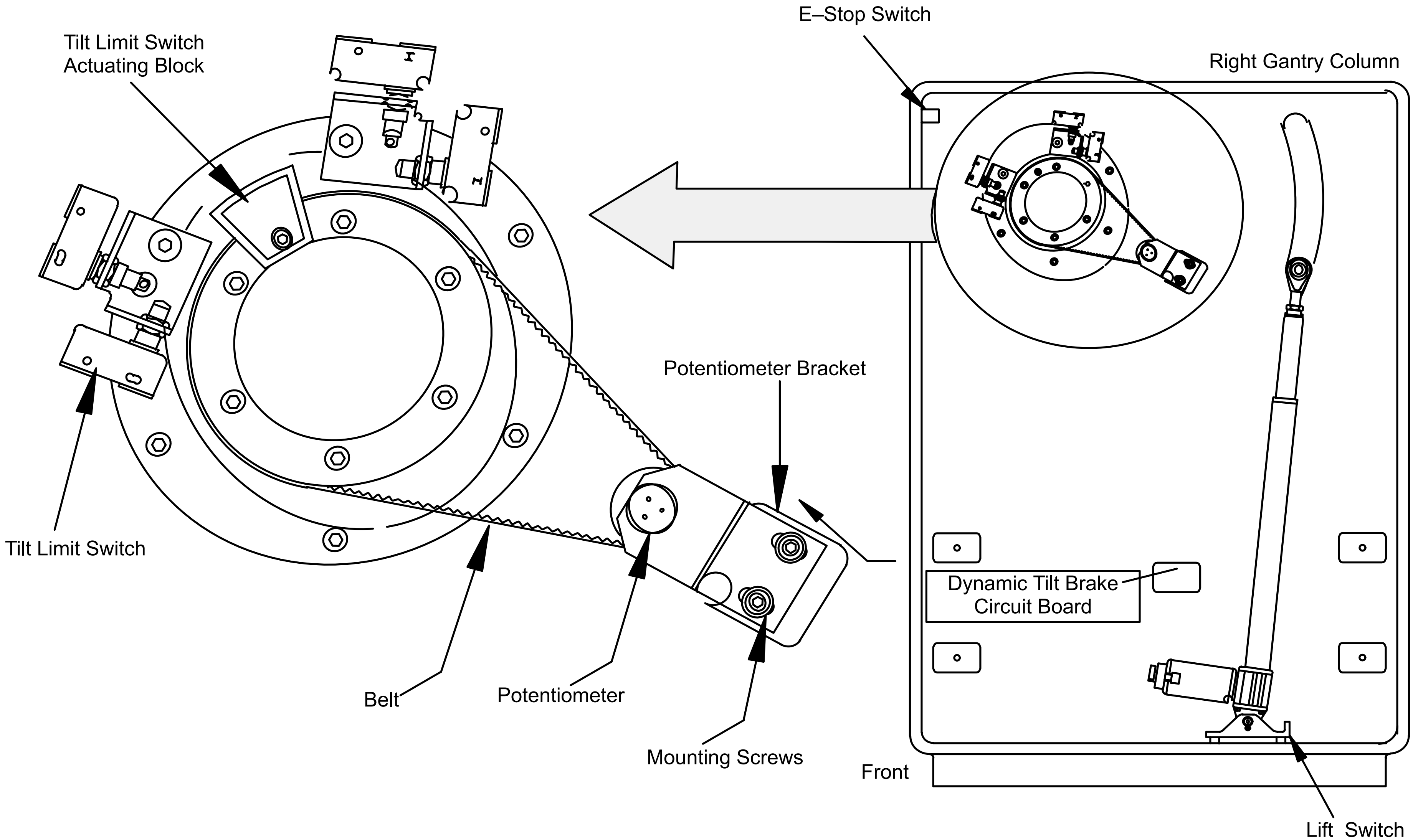


FIGURE 299 Tilt Potentiometer Belt

4. Tag and remove the eight Tilt Limit Switch Wires (2 Wires on each Switch).

5. Tag and remove the Tilt Actuator Power and Feedback Cables at the Motor.
6. Tag and remove the Tilt Actuator Lift Switch Cable (at the switch).
7. Tag and remove the Wires connected to the Dynamic Tilt Brake Circuit Board.
8. Tag and remove the Wires connected to the Door Switch Interlock Terminal Board.
9. Remove the Cables connected to the Gantry Cardfile. Unlace these Cables and pass them through the Column Spindle.
10. Remove the Column Ground Wire.
11. Loosen the two Potentiometer Mounting Plate Screws and slide the Tilt Potentiometer Assembly to remove tension on the Drive Belt. Slide the Belt off the Pulley.
12. Slide the Belt toward the Gantry until it clears the Tilt Limit Switch Actuating Block. Lift the Belt over the Block and remove.

TILT POTENTIOMETER BELT REPLACEMENT:

13. Install the Tilt Potentiometer Drive Belt and Cabling by reversing steps 3. through 12.
14. Calibrate the Tilt Potentiometer. Refer to the Gantry Tilt Calibration procedures in the Calibration Manual.
15. Replace the Gantry Right Column Cover. Refer to the Gantry Covers procedure.

GANTRY TILT POT & PULLEY

Introduction: This procedure is used to remove and replace the Gantry Tilt Potentiometer and the Potentiometer Pulley.

Tools Required: 1. 5/64” and 3/16” Hex Keys

Estimated Time: 30 Min.

POTENTIOMETER ASSEMBLY REMOVAL:

1. Remove Gantry power. Refer to the PowerUp/PowerDown procedure.
2. Remove the right Gantry Column Cover. Refer to the Gantry Covers procedure.
3. Tag and unsolder the three wires connected to the Tilt Pot. See Figure 300.

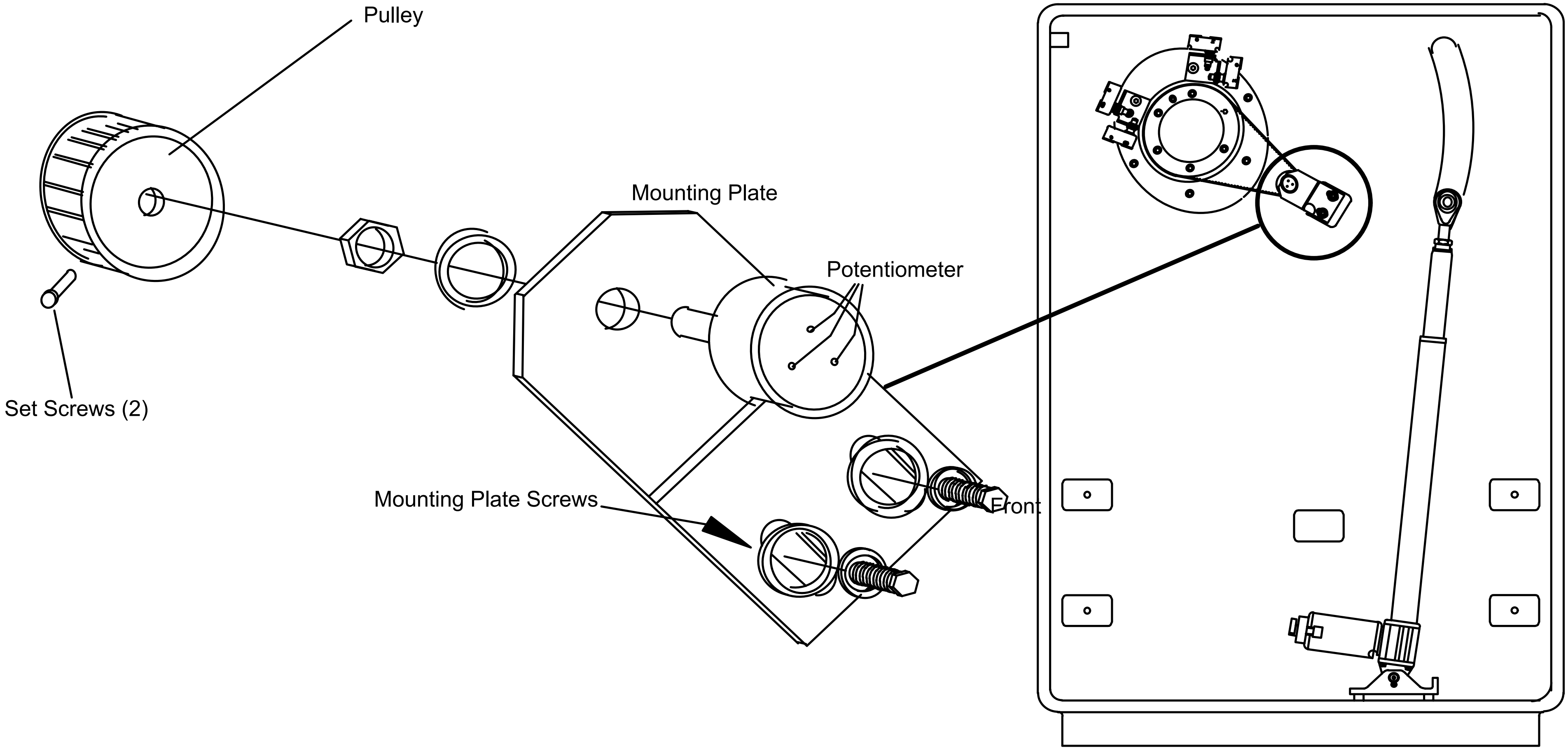


FIGURE 300 Potentiometer Assembly – Exploded

4. Loosen the two Pot Mounting Plate Screws.
5. Slide the Tilt Pot Assembly to remove tension on the Drive Belt and slide the Belt off the Pulley.
6. Remove the two Pot Mounting Plate Screws and Washers.
7. Remove the Tilt Pot Assembly.

POTENTIOMETER PULLEY REMOVAL:

8. Loosen the two Pulley Set Screws.

9. Slide the Pulley off the Pot Shaft.

POTENTIOMETER REMOVAL:

10. Remove the Potentiometer Pulley. Refer to steps 3. through 9.
11. Remove the Potentiometer Bushing Nut and Washer.
12. Remove the Potentiometer.

POTENTIOMETER REPLACEMENT:

13. Install the Potentiometer onto the Mounting Bracket by reversing steps 11. and 12.

POTENTIOMETER PULLEY REPLACEMENT:

14. Install the Pulley onto the Potentiometer Shaft by reversing steps 8. and 9.

POTENTIOMETER ASSEMBLY REPLACEMENT:

15. Install the Tilt Pot Assembly by reversing steps 4. through 7.
16. Apply Gantry power. Refer to the PowerUp/PowerDown procedure.

> ⚠ **WARNING** ⚠
>
> **USE EXTREME CAUTION WHEN OPERATING OR WORKING ON EQUIPMENT WITH COVERS REMOVED. DANGEROUS VOLTAGES ARE EXPOSED WHEN THE GANTRY COVERS ARE REMOVED. FAILURE TO COMPLY CAN CAUSE SERIOUS INJURY OR DEATH TO PERSONNEL.**

17. Calibrate the Tilt Pot. Refer to Gantry Tilt Calibration in the Calibration Manual.
18. Remove Gantry power. Refer to the PowerUp/PowerDown procedure.
19. Replace the right Gantry Column Cover. Refer to the Gantry Covers procedure.

GANTRY TILT LIMIT SWITCHES

Introduction: This procedure is used to remove and replace the Gantry Tilt Limit Switches. The procedure is written for a typical switch since all will follow the same sequence.

Tools Required:
1. 9/16” and 11/16” Open End Wrenches
2. 3/16”x3” Slotted Screwdriver

Estimated Time: 20 Min.

LIMIT SWITCH REMOVAL:

1. Remove Gantry power. Refer to the PowerUp/PowerDown procedure.
2. Remove the right Gantry Column Cover. Refer to the Gantry Covers procedure.
3. Tag and remove the two wires connected to the switch. See Figure 301.

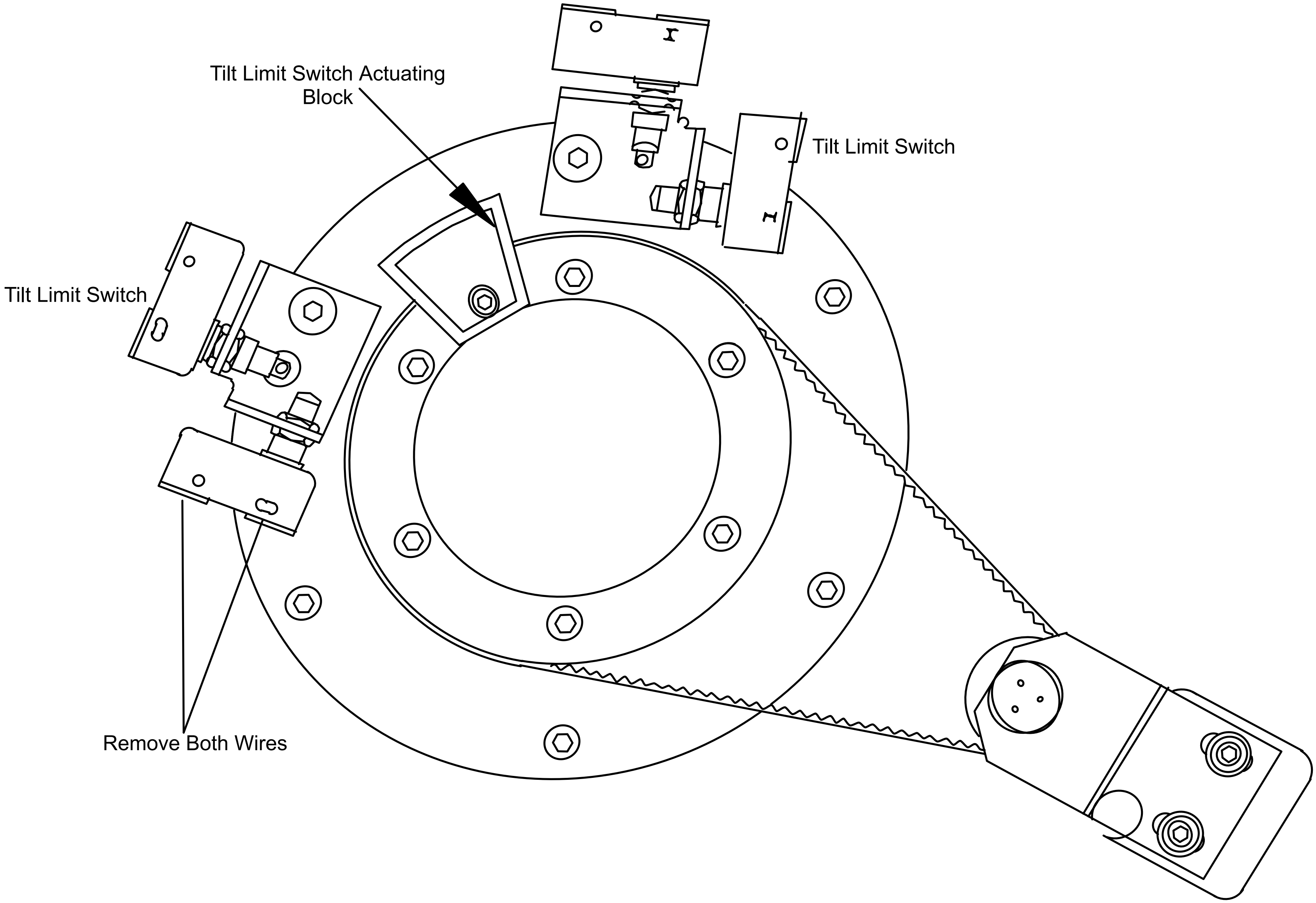


FIGURE 301 Tilt Limit Switches

4. Remove the Switch Mounting Nut.
5. Remove the Switch from the Bracket.

NOTE

If the replacement Switch is not supplied with Adjustment Nuts and Washers, remove them from the defective Switch and install them on the replacement Switch.

LIMIT SWITCH REPLACEMENT:

6. Install the replacement Tilt Limit Switch by reversing steps 3. through 5.
7. Apply Gantry power. Refer to the PowerUp/PowerDown procedure.

WARNING

USE EXTREME CAUTION WHEN OPERATING OR WORKING ON EQUIPMENT WITH COVERS REMOVED. DANGEROUS VOLTAGES ARE EXPOSED WHEN THE GANTRY COVERS ARE REMOVED. FAILURE TO COMPLY CAN CAUSE SERIOUS INJURY OR DEATH TO PERSONNEL.

8. Adjust the Tilt Limit Switch. Refer to the Gantry Tilt Calibration section in the Calibration Manual.
9. Remove Gantry power. Refer to the PowerUp/PowerDown procedure.
10. Replace the right Gantry Column Cover. Refer to the Gantry Covers procedure.

LASER ASSEMBLY

Introduction:	This procedure is used to remove and replace a Laser Assembly.
Tools Required:	1. 7/16” Open End Wrench
	2. 3/16”, 5/32” and 7/64” Hex Keys
	3. Ratchet and 5/16” Socket
	4. 3/16”x3” Slotted and #1 Tip Phillips Screwdrivers
	5. Torque wrench (0–250 in–lb)
Material Required:	None
Estimated Time:	30 Min.

LASER ASSEMBLY REMOVAL:

1. Remove Gantry power. Refer to the PowerUp/PowerDown procedure.
2. Open the Front Gantry Covers and remove the Front Cone. Refer to the Gantry Covers procedure.

> ⚠ **WARNING**
>
> **USE EXTREME CAUTION WHEN SERVICING EQUIPMENT WHILE THE UPPER GULL WING IS OPEN. THE PANEL HAS SHARP EDGES. FAILURE TO COMPLY COULD RESULT IN SERIOUS INJURY TO PERSONNEL.**

3. Move the Scan Frame by hand until the X–Ray Tube is at the 12:00 position. Lock the Scan Frame with the Locking Pin. See Figure 302.

> ⚠ **CAUTION**
>
> CLOCKWISE ROTATION OF THE SCAN FRAME MAY CAUSE DAMAGE TO THE SLIP RING BRUSHES. ROTATE THE SCAN FRAME COUNTERCLOCKWISE ONLY.

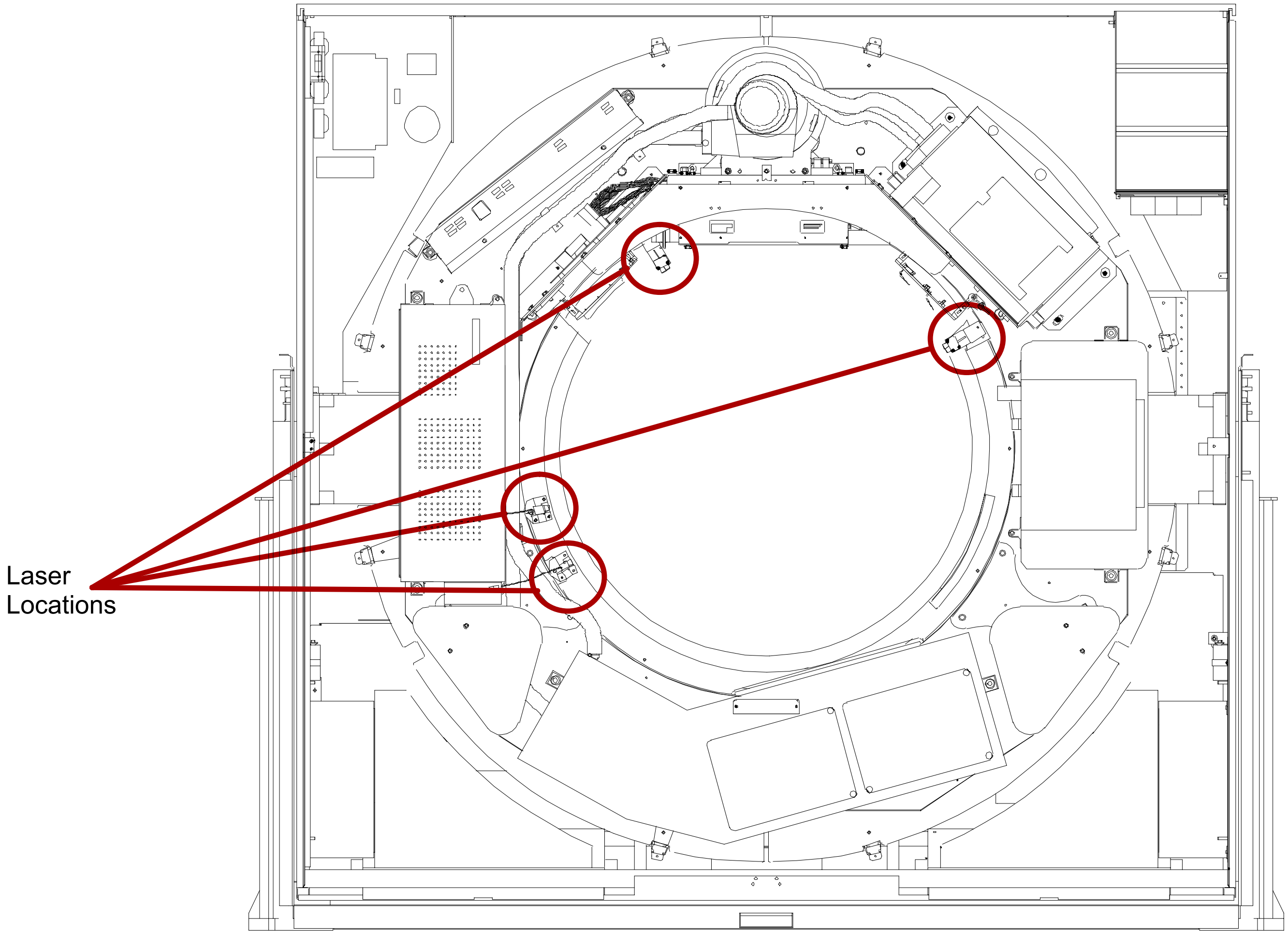


FIGURE 302 Scan Frame Pinned at 12:00 / Laser Locations

4. Disconnect the two Laser Power Connectors.

5. Carefully remove the hardware securing the Laser Assembly to the Mounting Shaft. Remove the Laser Assembly from the Gantry. See Figure 303.

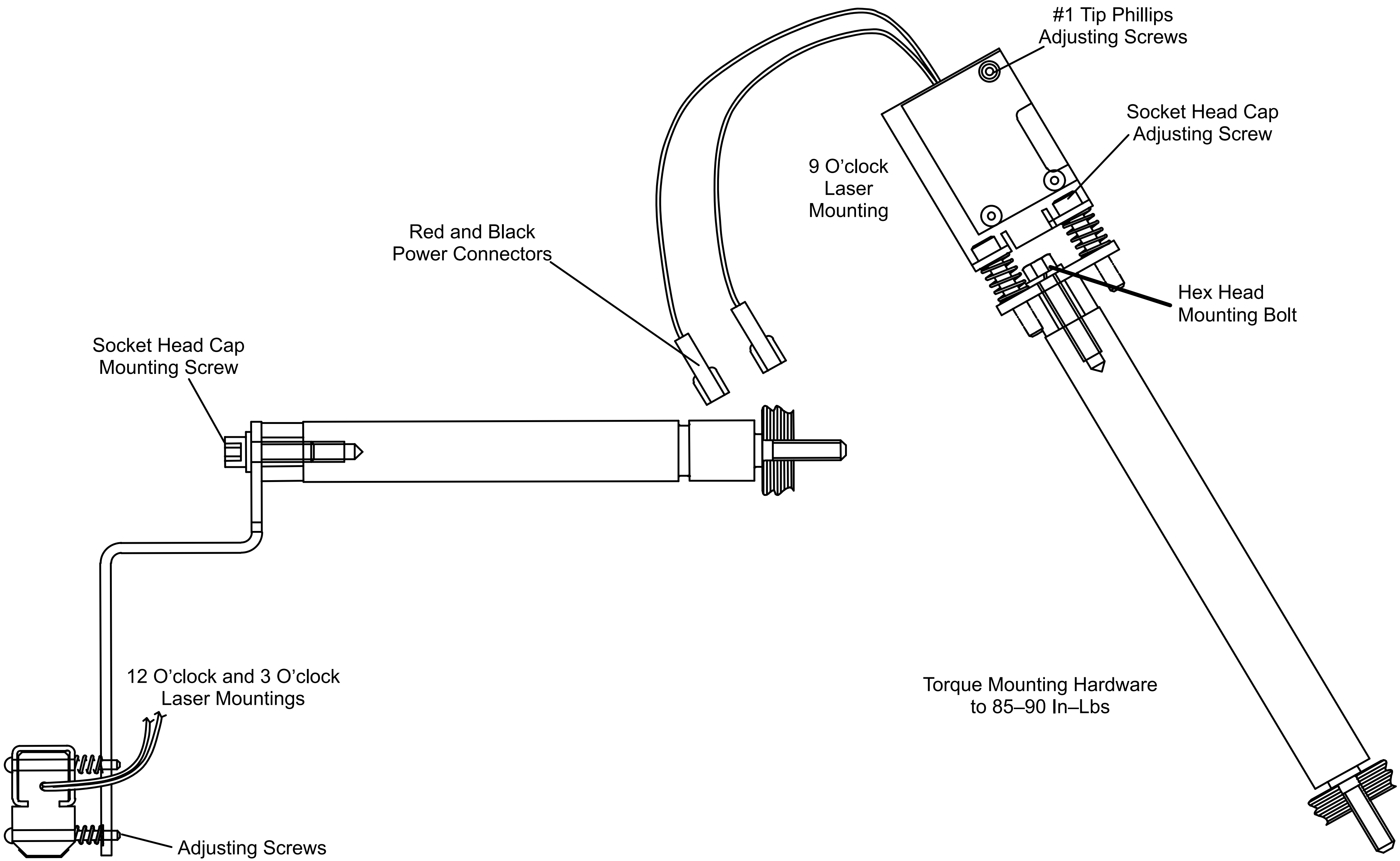


FIGURE 303 Laser Assembly Mounting – "Z" and "L" Models

LASER ASSEMBLY REPLACEMENT:

6. Reverse steps 4. and 5. making sure red wires are connected to red wires and black to black.
7. Remove the Locking Pin from the Scan Frame.
8. Rotate the Scan Frame counterclockwise by hand while checking for loose Cables. Also check for clearance over the entire range of travel between the Rotating Frame Cables and the Stationary Frame.

> ⚠ **CAUTION**
>
> CLOCKWISE ROTATION OF THE SCAN FRAME MAY CAUSE DAMAGE TO THE SLIP RING BRUSHES. <u>ROTATE THE SCAN FRAME COUNTERCLOCKWISE ONLY</u>.

9. Apply Gantry power. Refer to the PowerUp/PowerDown procedure.

> ⚠ **WARNING** ⚠
>
> **USE EXTREME CAUTION WHEN OPERATING OR WORKING ON EQUIPMENT WITH COVERS REMOVED. DANGEROUS VOLTAGES ARE EXPOSED WHEN THE GANTRY COVERS ARE REMOVED. FAILURE TO COMPLY CAN CAUSE SERIOUS INJURY OR DEATH TO PERSONNEL.**

10. Pull the Front Cover Interlock Switch Plunger out to enable operation while the Front Covers are open.
11. Place the Front Cover in its half–open position.
12. Check Laser operation by using the Gantry Control Panel Switches.

> ⚠ **WARNING**
>
> **ROTATING MACHINERY. STAND CLEAR OF ROTATING FRAME WHEN EQUIPMENT IS OPERATED. FAILURE TO COMPLY MAY RESULT IN SERIOUS INJURY OR DEATH TO PERSONNEL.**

13. Align the Laser Beam Assembly. Refer to Laser Alignment in the Calibration Manual.

14. Return the Front Cover Interlock Switch Plunger to its center position. Remove Gantry power. Refer to the PowerUp/PowerDown procedures.
15. Place the Front Cover in its full–open position.
16. Replace the Front Cone and the lower Gantry Covers. Close the upper Gantry Cover. Refer to the Gantry Covers procedure.

HIGH SPEED STARTER

Introduction: The following procedure is used to remove and replace the High Speed Starter, located in the AC Input Chassis.

Tools Required:
1. #1 tip Long Phillips Screwdriver
2. 3/4” Socket, Ratchet
3. Torque wrench (75 ft – lbs)

Material Required: Loc–Tite (Thread Locker)

Estimated Time: 1.0 Hr.

HIGH SPEED STARTER CHASSIS REMOVAL:

1. Remove Gantry power. Refer to the PowerUp/PowerDown procedure.
2. Open Gantry Front Covers and remove the Front Cone. Refer to the Gantry Covers procedure.

> ⚠ **WARNING**
>
> **USE EXTREME CAUTION WHEN SERVICING EQUIPMENT WHILE THE UPPER GULL WING IS OPEN. THE PANEL HAS SHARP EDGES. FAILURE TO COMPLY COULD RESULT IN SERIOUS INJURY TO PERSONNEL.**

> ⚠ **WARNING** ⚠
>
> **USE EXTREME CAUTION WHEN OPERATING OR WORKING ON EQUIPMENT WITH COVERS REMOVED. DANGEROUS VOLTAGES ARE EXPOSED WHEN THE GANTRY COVERS ARE REMOVED. FAILURE TO COMPLY CAN CAUSE SERIOUS INJURY OR DEATH TO PERSONNEL.**

> ⚠ **CAUTION**
>
> CLOCKWISE ROTATION OF THE SCAN FRAME MAY CAUSE DAMAGE TO THE SLIP RING BRUSHES. ROTATE THE SCAN FRAME COUNTERCLOCKWISE ONLY.

3. Rotate the Scan Frame counterclockwise by hand until the X–Ray Tube is at the 12:00 position. Lock the Scan Frame with the Locking Pin. See Figure 304.

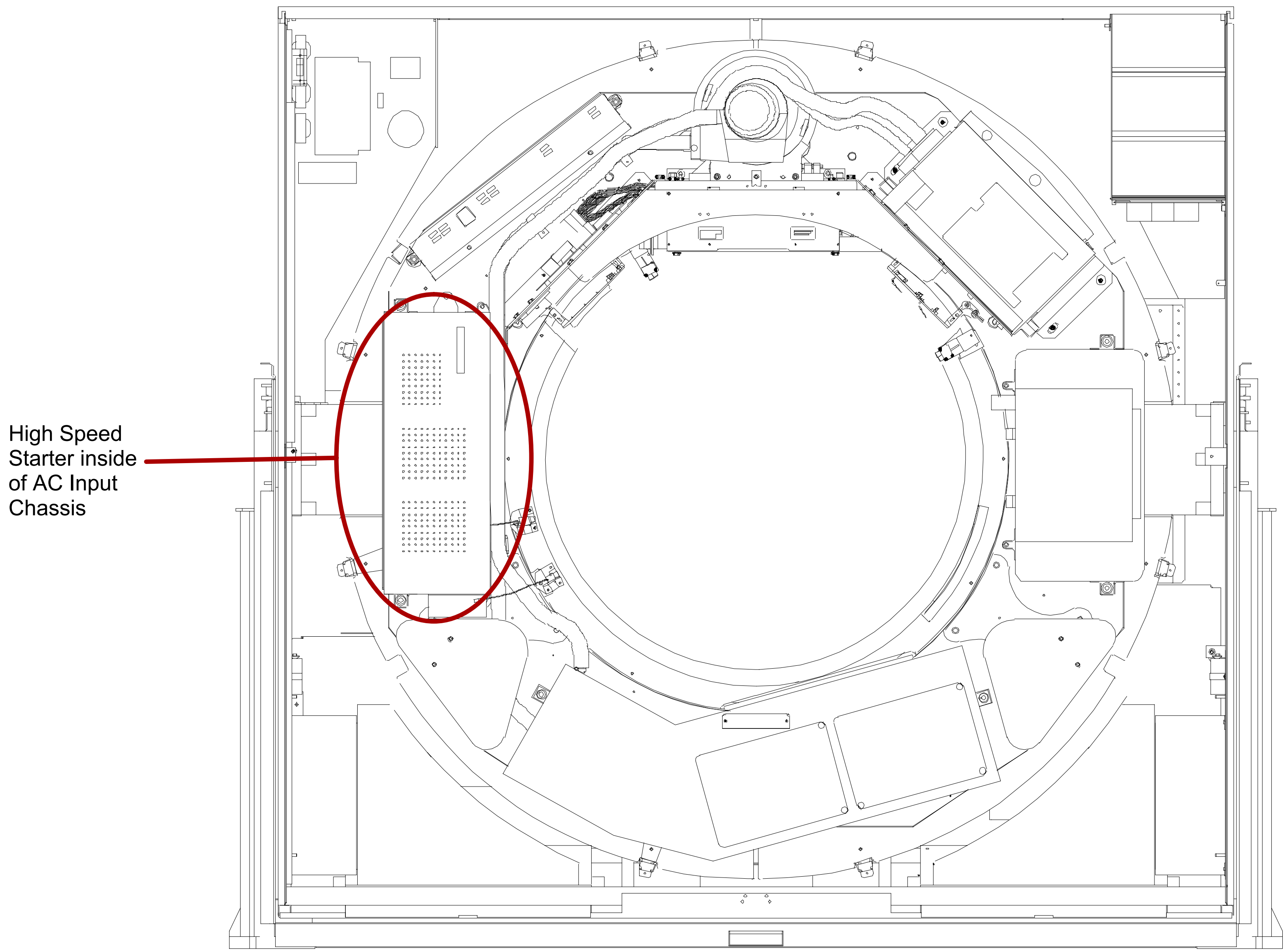


FIGURE 304   Scan Frame Pinned with Tube at 12:00 / HSS Location

4. Remove the Phillips head cover screws and remove the AC Input Chassis Cover. The High Speed Starter will be located near the top, on one side of the cabinet. Refer to Figure 305 for two views of the High Speed Starter.

5. Remove the red and black power wires and the cables connected to the printed circuit board.

6. With a long phillips head screwdriver, loosen the six captive screwsHigh Speed Starter. See Figure 305.

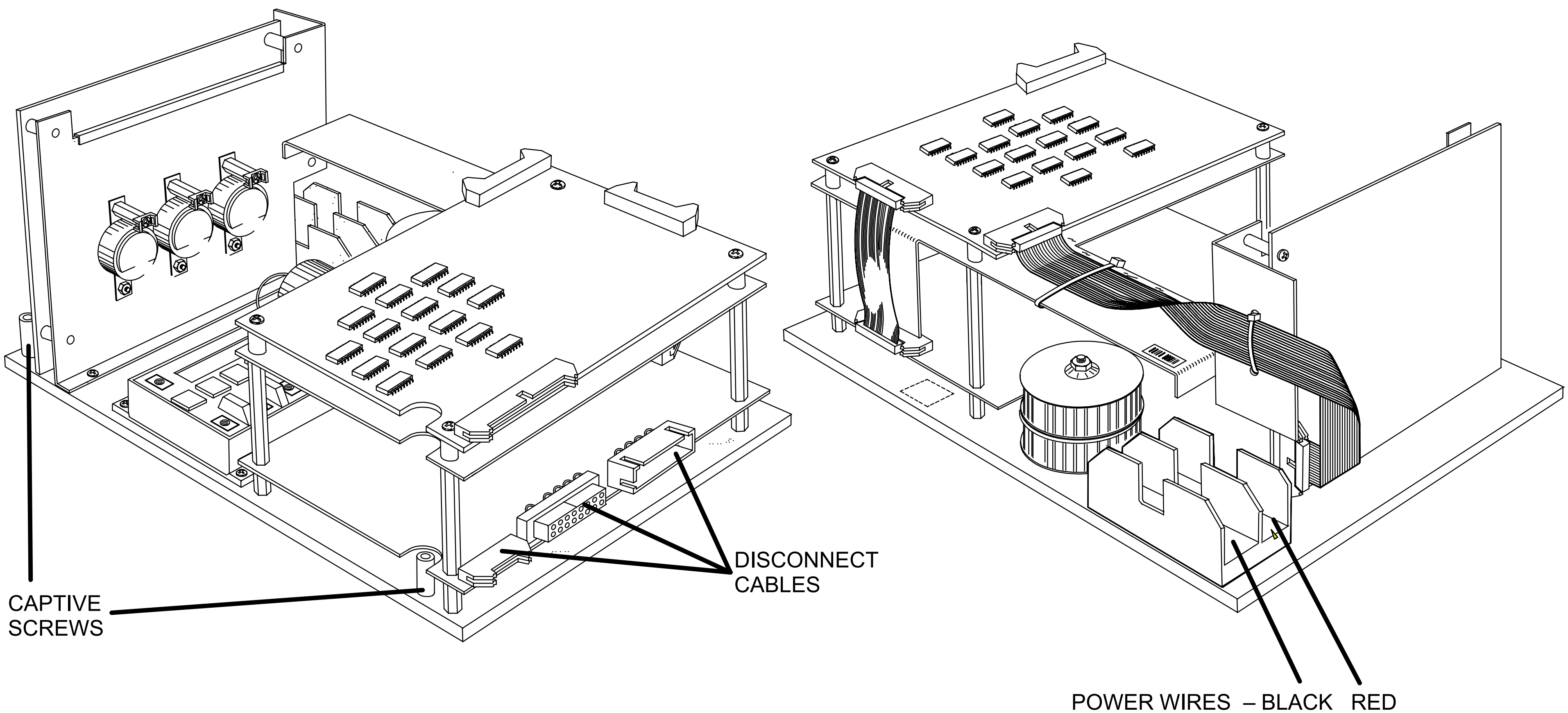


FIGURE 305 High Speed Starter – Two Different Views

7. Slide the HSS assembly so the cable connectors clear the AC Input Chassis holes, then lift the assembly up and out.

HIGH SPEED STARTER CHASSIS REPLACEMENT:

8. Reverse steps 4. through 7.
9. Remove the Scan Frame Locking Pin.
10. Rotate the Scan Frame counterclockwise by hand while checking for loose Cables. Also check for clearance over the entire range of travel between the Rotating Frame Cables and the Stationary Frame.

> ⚠ **CAUTION**
>
> CLOCKWISE ROTATION OF THE SCAN FRAME MAY CAUSE DAMAGE TO THE SLIP RING BRUSHES. <u>ROTATE THE SCAN FRAME COUNTERCLOCKWISE ONLY</u>.

11. Replace the Gantry Front Cone. Refer to the Gantry Covers procedure.
12. Apply Gantry power. Refer to the PowerUp/PowerDown procedure.

> ⚠ **WARNING** ⚠
>
> **USE EXTREME CAUTION WHEN OPERATING OR WORKING ON EQUIPMENT WITH COVERS REMOVED. DANGEROUS VOLTAGES ARE EXPOSED WHEN THE GANTRY COVERS ARE REMOVED. FAILURE TO COMPLY CAN CAUSE SERIOUS INJURY OR DEATH TO PERSONNEL.**

13. Pull the Front Cover Interlock Switch Plunger out to enable operation while the Front Covers are open.
14. To verify proper System operation, run Full–field and Half–field Water Phantom Scans and allow the System to process and display an Image.
15. Return the Front Cover Interlock Switch Plunger to its center position.
16. Remove Gantry power. Refer to the PowerUp/PowerDown procedure.
17. Replace the Gantry lower Front Cover, close the upper Front Cover. Refer to the Gantry Covers procedure.

BRUSH BLOCK

Introduction: This procedure is used to remove and replace Slip Ring Brush Blocks (power and signal).

Tools Required:
1. 7/64”, 3/16” Hex Keys
2. 9/16 and 5/16” Socket, Ratchet
3. Torque Wrench (75 ft–lb)

Estimated Time: 1.0 Hr.

POWER BRUSH BLOCK REMOVAL:

1. Remove Gantry power. Refer to the PowerUp/PowerDown procedure.
2. Open Gantry Front Covers and remove the Front Cone. Refer to the Gantry Covers procedure.

> ⚠ **WARNING**
>
> **USE EXTREME CAUTION WHEN SERVICING EQUIPMENT WHILE THE UPPER GULL WING IS OPEN. THE PANEL HAS SHARP EDGES. FAILURE TO COMPLY COULD RESULT IN SERIOUS INJURY TO PERSONNEL.**

> ⚠ **WARNING** ⚠
>
> **USE EXTREME CAUTION WHEN OPERATING OR WORKING ON EQUIPMENT WITH COVERS REMOVED. DANGEROUS VOLTAGES ARE EXPOSED WHEN THE GANTRY COVERS ARE REMOVED. FAILURE TO COMPLY CAN CAUSE SERIOUS INJURY OR DEATH TO PERSONNEL.**

3. Rotate the Scan Frame counterclockwise by hand until the X–Ray Tube is at the 12:00 position. Lock the Scan Frame with the Locking Pin. See Figure 306.

> ⚠ **CAUTION**
>
> CLOCKWISE ROTATION OF THE SCAN FRAME MAY CAUSE DAMAGE TO THE SLIP RING BRUSHES. <u>ROTATE THE SCAN FRAME COUNTERCLOCKWISE ONLY.</u>

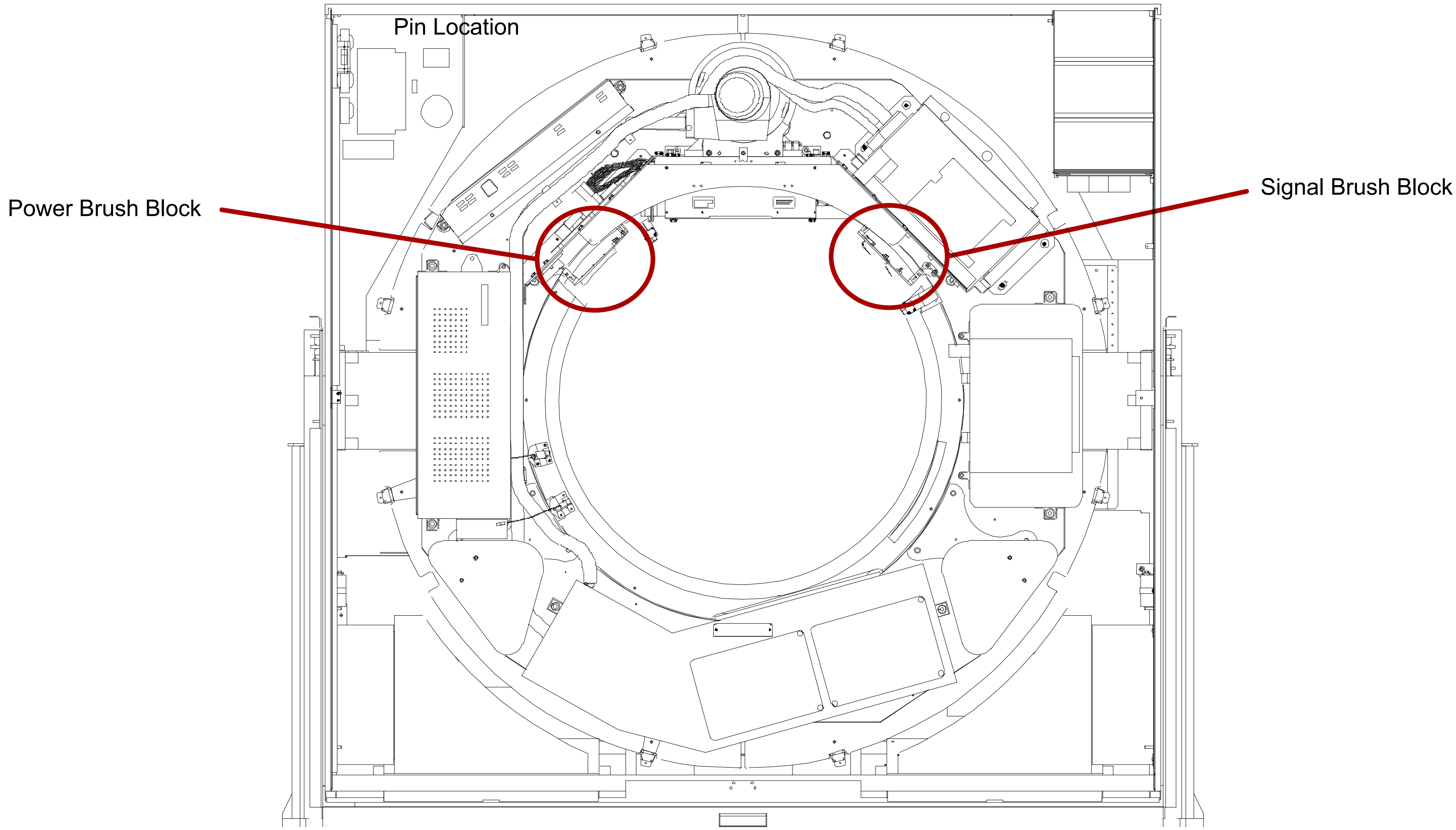


FIGURE 306 Scan Frame at 12:00 / Brush Block Locations

4. Tag and remove all Wires/Cables exiting the Power Brush Block.
5. Support the Power Brush Block while loosening the two Socket Head Mounting screws holding the Brush Block to the Mounting Bracket and remove the Block. See Figure 307.

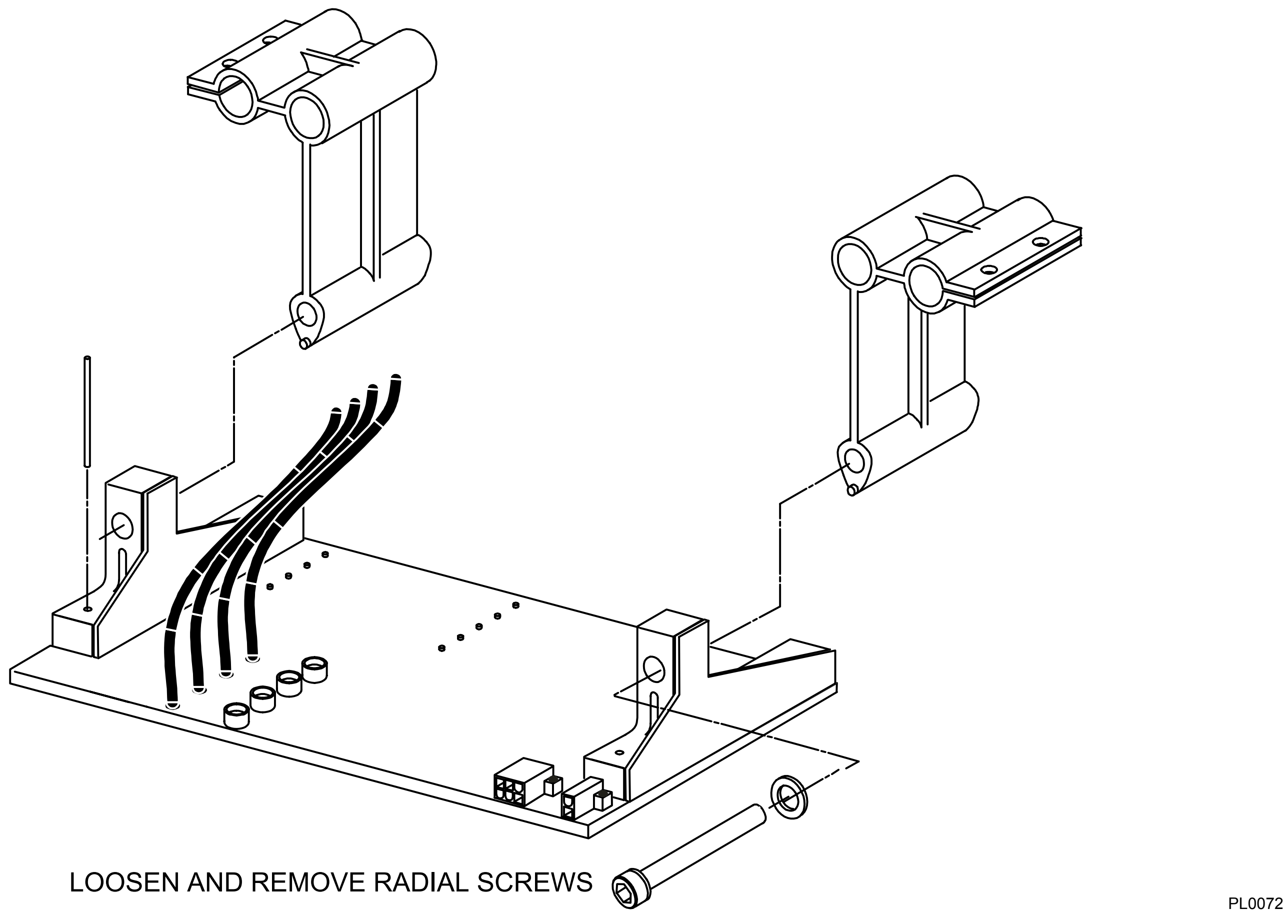


FIGURE 307 Slip Ring Brush Block Mounting

POWER BRUSH BLOCK REPLACEMENT:

6. Remove the Scan Frame Locking Pin.
7. Clean the Slip Ring with a soft cloth and 90% Isopropyl alcohol. Access to the Slip Ring is gained through a removable Cover on the Slip Ring Dust Cover.
8. Install the new Power Brush Block by reversing steps 4. and 5. of Power Brush Block Removal.
9. Align the Brush Block in the Slip Ring grooves. Refer to the Brush Block Alignment procedure in the Calibration Manual.
10. If only the power brush block is being removed, skip to step 20.

===

SIGNAL BRUSH BLOCK REMOVAL:

11. Perform steps 1., 2., and 3. of the Power Brush Block Removal Procedure.
12. Remove the Collimator Control Relay PCB. Refer to the Small Drives Relay removal/replacement procedure.
13. Tag and disconnect all Wires/Cables connected to the Signal Brush block.
14. Remove the two Socket Head Mounting Screws that attach the Brush Block to the Mounting Brackets. Refer to Figure 308.
15. Carefully lift the Brush Block up and remove it from the Gantry.

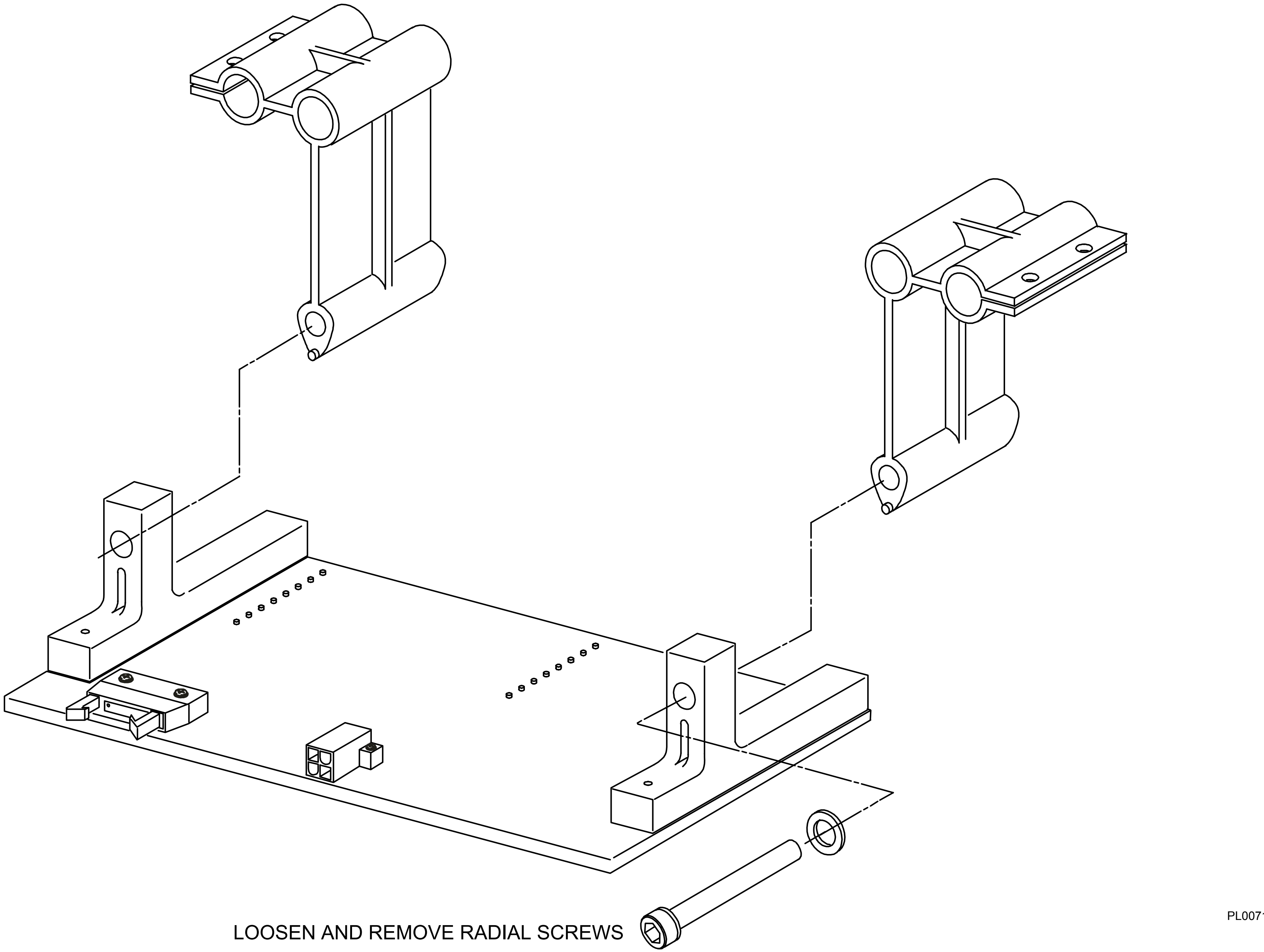


FIGURE 308 SIGNAL BRUSH BLOCK

SIGNAL BRUSH BLOCK REPLACEMENT:

16. Remove the Scan Frame Locking Pin. Clean the Slip Ring with a soft cloth and 90% Isopropyl alcohol. Access to the Slip Ring is gained through a removable Cover on the Slip Ring Dust Cover.
17. Install the new Signal Brush Block by reversing steps 13. through 15. Use Lock Washers on all Mounting Hardware.
18. Align the Brush Block in the Slip Ring grooves. Refer to the Brush Block Alignment procedure.

> **NOTE**
>
> Normal Alignment when replacing the Brush Block should not require moving the Brush Block Mounting Posts. However, if Axial alignment with the drill blanks cannot be achieved, the posts must be loosened. If the Mounting Posts must be moved, the 3 O'clock Laser Rod must be removed and Laser calibration subsequently performed.

> **NOTE**
>
> Perform step 19. only if the Signal Brush Block was removed.

19. Install the Collimator Control Relay PCB. Refer to the Small Drives Relay PCB removal/replacement procedure.
20. Replace the Gantry Front Cone. Refer to the Gantry Covers procedure.
21. Apply Gantry power. Refer to the PowerUp/PowerDown procedure.

> ⚠ **WARNING** ⚠
>
> **USE EXTREME CAUTION WHEN OPERATING OR WORKING ON EQUIPMENT WITH COVERS REMOVED. DANGEROUS VOLTAGES ARE EXPOSED WHEN THE GANTRY COVERS ARE REMOVED. FAILURE TO COMPLY CAN CAUSE SERIOUS INJURY OR DEATH TO PERSONNEL.**

22. Pull the Front Cover Interlock Switch Plunger out to enable operation while the Front Covers are open.
23. To verify proper System operation, run Full–field and Half–field Water Phantom Scans and allow the System to process and display an Image.

24. Return the Front Cover Interlock Switch Plunger to its center position.
25. Remove Gantry power. Refer to the PowerUp/PowerDown procedure.
26. Replace the Gantry lower Front Cover, close the upper Front Cover. Refer to the Gantry Covers procedure.

SLIP RING DUST COVER

Introduction: This procedure is used to remove and replace the Gantry slip ring dust cover.

Tools Required:

1. 7/64”, 3/16” Hex Keys
2. 9/16 and 5/16” Socket, Ratchet
3. Torque Wrench (75 ft–lb)

Estimated Time: 1.0 Hr.

DUST COVER REMOVAL:

1. Remove Gantry power. Refer to the PowerUp/PowerDown procedure.
2. Open Gantry Front Covers and remove the Front Cone. Refer to the Gantry Covers procedure.

> **⚠ WARNING**
>
> **USE EXTREME CAUTION WHEN SERVICING EQUIPMENT WHILE THE UPPER GULL WING IS OPEN. THE PANEL HAS SHARP EDGES. FAILURE TO COMPLY COULD RESULT IN SERIOUS INJURY TO PERSONNEL.**

> **⚠ WARNING** ⚠
>
> **USE EXTREME CAUTION WHEN OPERATING OR WORKING ON EQUIPMENT WITH COVERS REMOVED. DANGEROUS VOLTAGES ARE EXPOSED WHEN THE GANTRY COVERS ARE REMOVED. FAILURE TO COMPLY CAN CAUSE SERIOUS INJURY OR DEATH TO PERSONNEL.**

3. Rotate the Scan Frame counterclockwise by hand until the X–Ray Tube is at the 12:00 position. Lock the Scan Frame with the Locking Pin. See Figure 309.

> **⚠ CAUTION**
>
> CLOCKWISE ROTATION OF THE SCAN FRAME MAY CAUSE DAMAGE TO THE SLIP RING BRUSHES. ROTATE THE SCAN FRAME COUNTERCLOCKWISE ONLY.

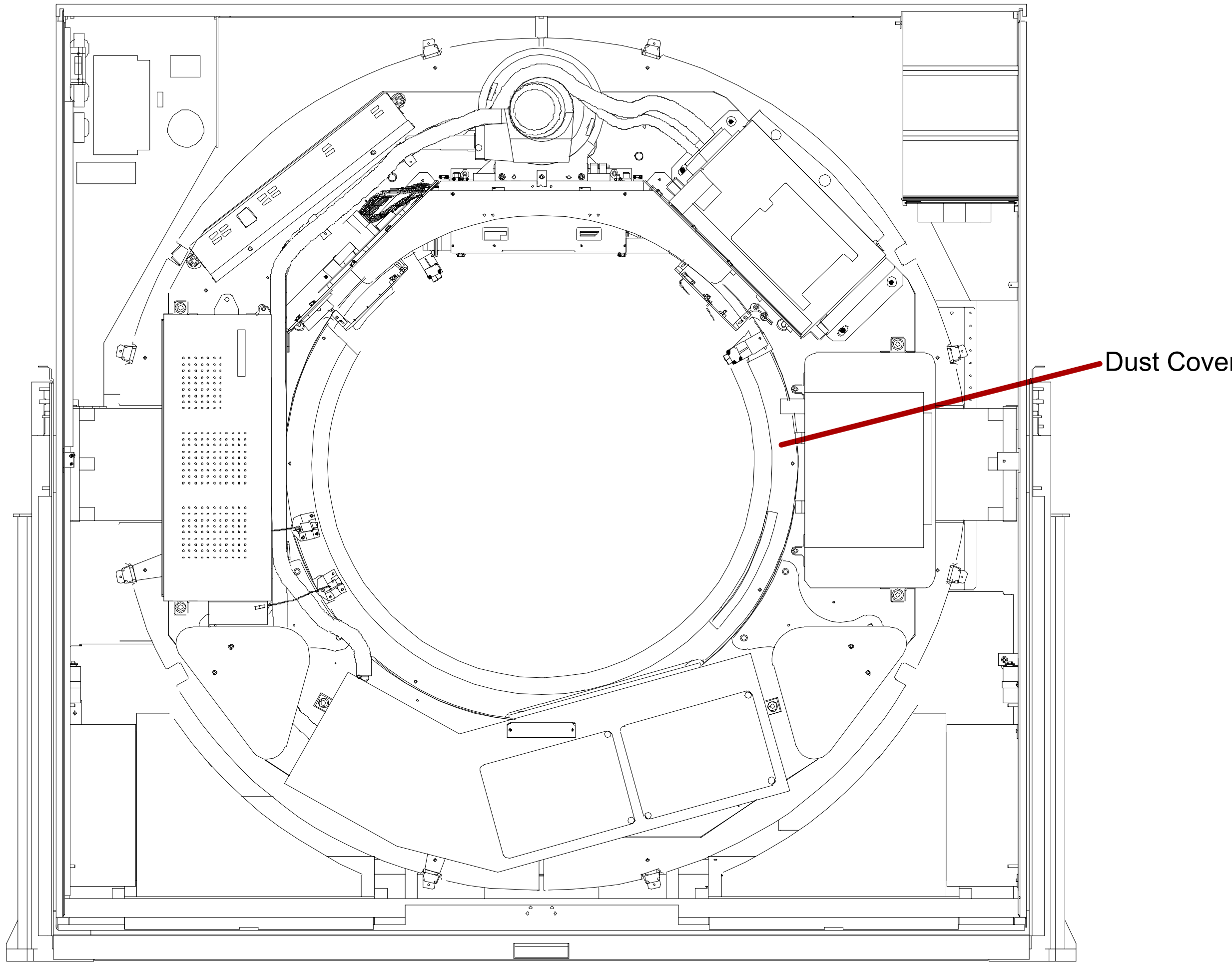


FIGURE 309 Gantry Pinned with Tube at 12:00 / Dust Cover Location

4. Remove the Socket Head Cap Screws holding the Slip Ring Dust Cover to the Scan Frame. See Figure 310.
5. Loosen and swing the lasers out of the way.
6. The Heat Exchanger cables need to be pushed out of the way. Carefully remove the slip ring dust cover (this is a difficult step.)

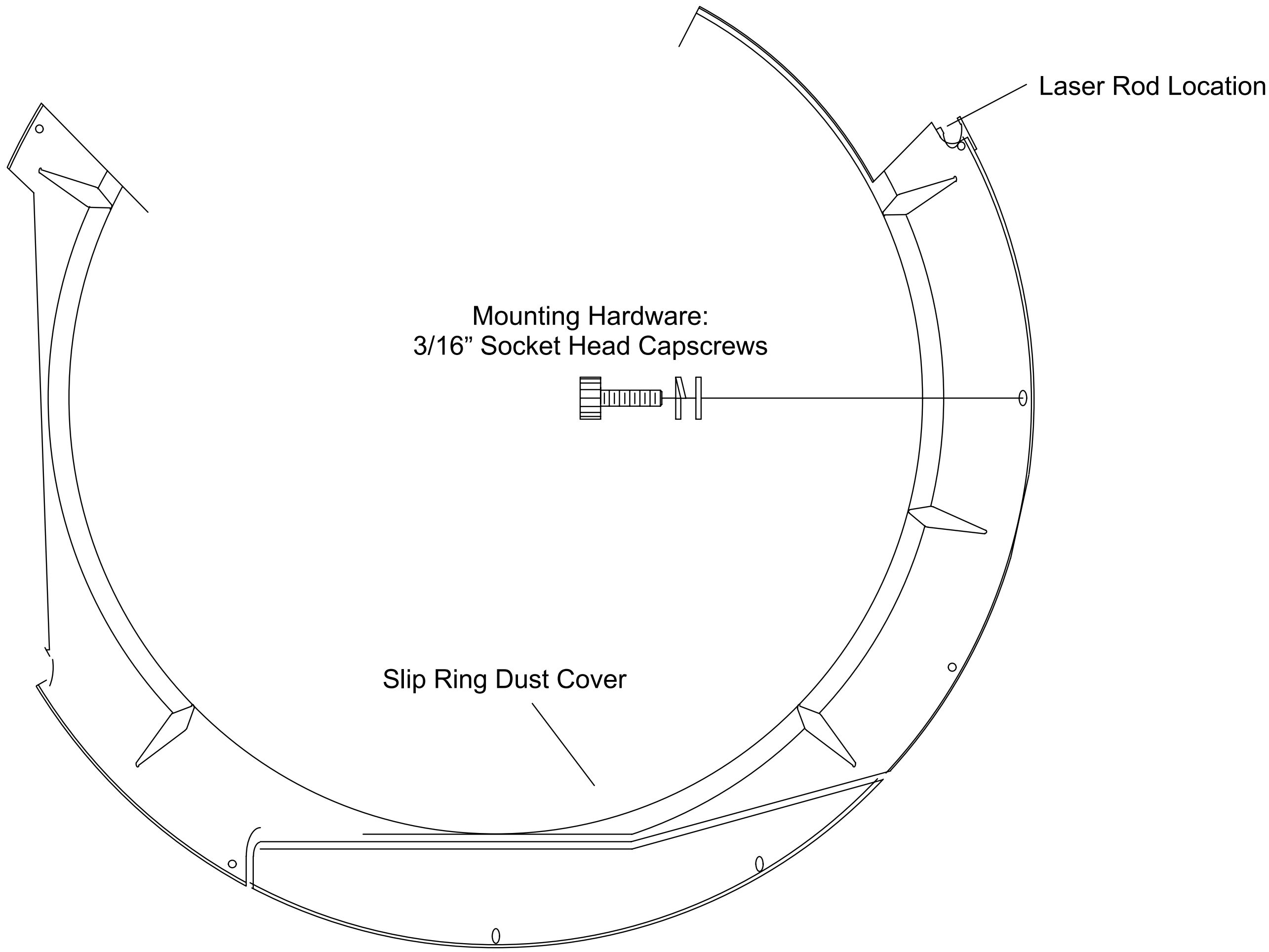


FIGURE 310 Sllip Ring Dust Cover

DUST COVER REPLACEMENT:

7. Install the Slip Ring Dust Cover by reversing steps NO TAG to 6.
8. Remove the Scan Frame Locking Pin.
9. Rotate the Scan Frame counterclockwise by hand while checking for loose Cables. Also check for clearance over the entire range of travel between the Rotating Frame Cables and the Stationary Frame.

> ⚠ **CAUTION**
>
> CLOCKWISE ROTATION OF THE SCAN FRAME MAY CAUSE DAMAGE TO THE SLIP RING BRUSHES. <u>ROTATE THE SCAN FRAME COUNTERCLOCKWISE ONLY</u>.

10. Replace the Gantry Front Cone and lower Front Cover, close the upper Front Cover. Refer to the Gantry Covers procedure.

POWER MONITOR / CARDFILE FILTER BOARD / DC POWER SUPPLY

Introduction: The following procedure is used to remove and replace the Gantry DC Power Panel Assembly, Power Monitor board and/or the Cardfile Filter Assembly Board.

Tools Required:
1. Hex Keys
2. Phillips and slotted screwdrivers

Estimated Time: 25 min.

POWER MONITOR / CARDFILE FILTER BOARD / POWER SUPPLY REMOVAL PROCEDURE:

1. Remove Gantry power. Refer to the PowerUp/PowerDown procedure.
2. Remove the left rear gantry cover. Refer to the Gantry Covers procedure.

> ⚠ **WARNING** ⚠
>
> **USE EXTREME CAUTION WHEN OPERATING OR WORKING ON EQUIPMENT WITH COVERS REMOVED. DANGEROUS VOLTAGES ARE EXPOSED WHEN THE GANTRY COVERS ARE REMOVED. FAILURE TO COMPLY CAN CAUSE SERIOUS INJURY OR DEATH TO PERSONNEL.**

3. The power supply assembly is located at the middle left side of the gantry. A detailed view is shown in Figure 311.

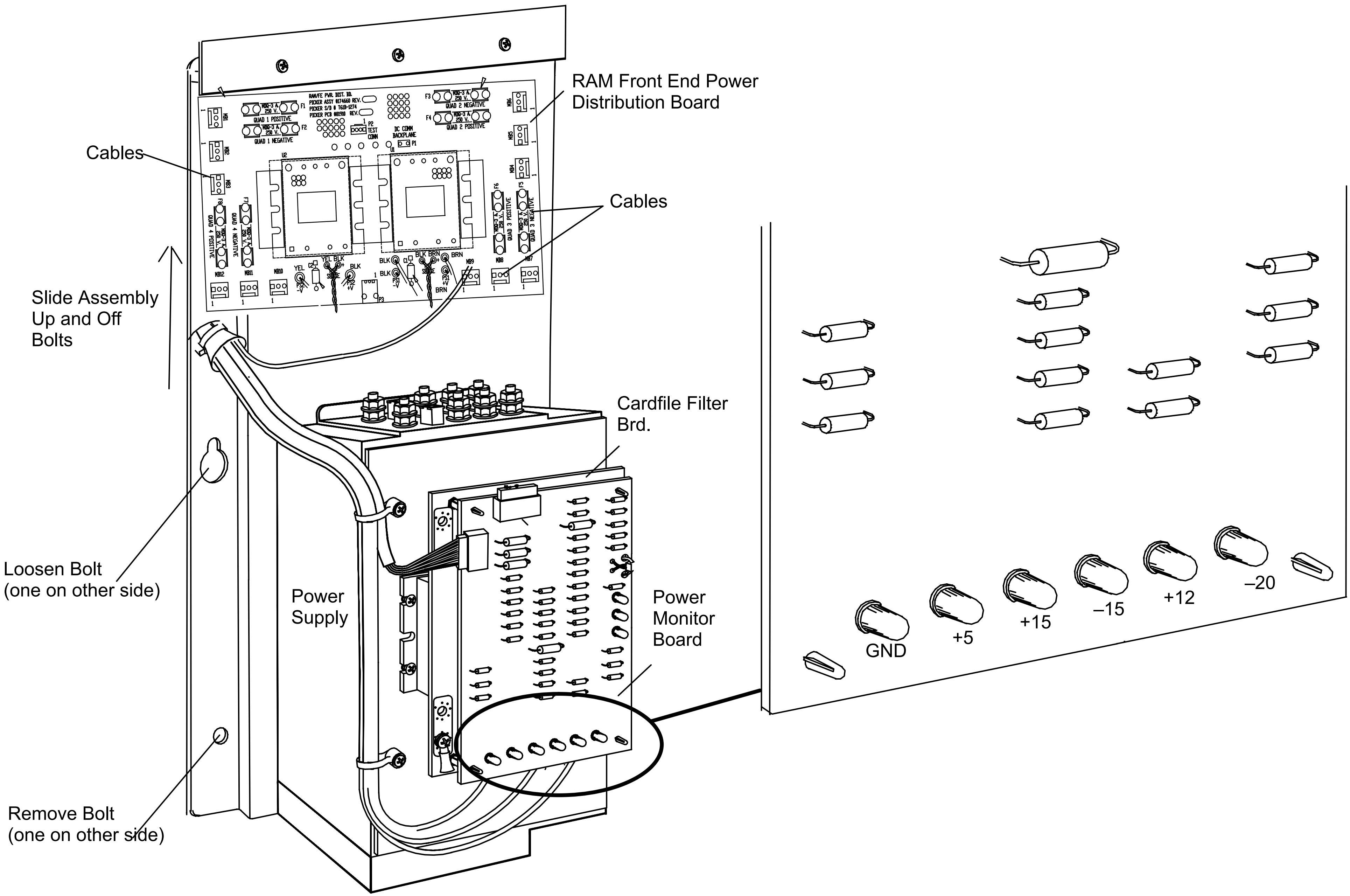


FIGURE 311 DC Power Supply / Power Monitor Board / CardFile Filter PCB / Voltage Test Points

> **NOTE**
>
> The level of Service on the Power Supply Assembly is to remove the entire assembly and replace. The Fault Sense Board can be replaced by itself, if properly determined to be faulty.

4. **POWER MONITOR BOARD**: Tag and remove all wires connected to the Fault Sense board. Remove tie wraps as necessary.
5. The board is held on with plastic tabs. Carefully pull up on the PCB just behind the tab to release the board. Repeat at each tab until the board is able to be lifted off.
6. **POWER SUPPLY**: Tag and remove all wires connected to the Power Supply. Remove tie wraps as necessary. The Power Monitor, Cardfile Filter and RAM Front End Power Distribution boards are attached to the power supply and will be removed along with the entire assembly.
7. Remove the two lower hex head bolts. Loosen the two upper bolts. Refer to Figure 311.
8. Grasp the power supply and raise up and off of the loosened hex bolts. Remove entire assembly.

POWER SUPPLY ASSEMBLY REPLACEMENT

9. **POWER SUPPLY**: Place the entire power supply assembly over the upper two head head bolts loosened earlier. Insert the two hex head bolts removed earlier. Tighten all four bolts.
10. Attach all of the wires removed earlier.
11. **POWER MONITOR BOARD**: (If removed by itself) Align the holes in the PCB with the plastic tabs. Carefully press the PCB onto each of the tabs. Re–attach the wires removed earlier.
12. Apply Gantry power. Refer to the PowerUp/PowerDown procedure.
13. Verify the +5, +15, –15, +12 and –20 VDC voltages with respect to ground. Refer to Figure 311.

> **NOTE**
>
> There is no calibration adjustment for the DC Power Supply. If the voltages are not in range, the entire Power Supply must be replaced.

14. Remove Gantry power. Refer to the PowerUp/PowerDown procedure.
15. Replace the left Gantry cover. Refer to the Gantry Covers procedure.

FUSE/CONTACTOR CHASSIS

Introduction: The following procedure is used to remove and replace the Fuse/Contactor Chassis.

Tools Required:
1. Phillips and Slotted Screwdrivers
2. Hex Key
3. Nut Driver

Estimated Time: 40 min.

FUSE/CONTACTOR REMOVAL:

1. Remove Gantry power. Refer to the PowerUp/PowerDown procedure.
2. Remove the left and right rear Gantry covers. Refer to the Gantry Covers procedure.

> ⚠ **WARNING** ⚠
>
> **USE EXTREME CAUTION WHEN OPERATING OR WORKING ON EQUIPMENT WITH COVERS REMOVED. DANGEROUS VOLTAGES ARE EXPOSED WHEN THE GANTRY COVERS ARE REMOVED. FAILURE TO COMPLY CAN CAUSE SERIOUS INJURY OR DEATH TO PERSONNEL.**

3. Remove the two Philips Head Screws and remove the clear Fuse/Contactor Cover. See Figure 313.

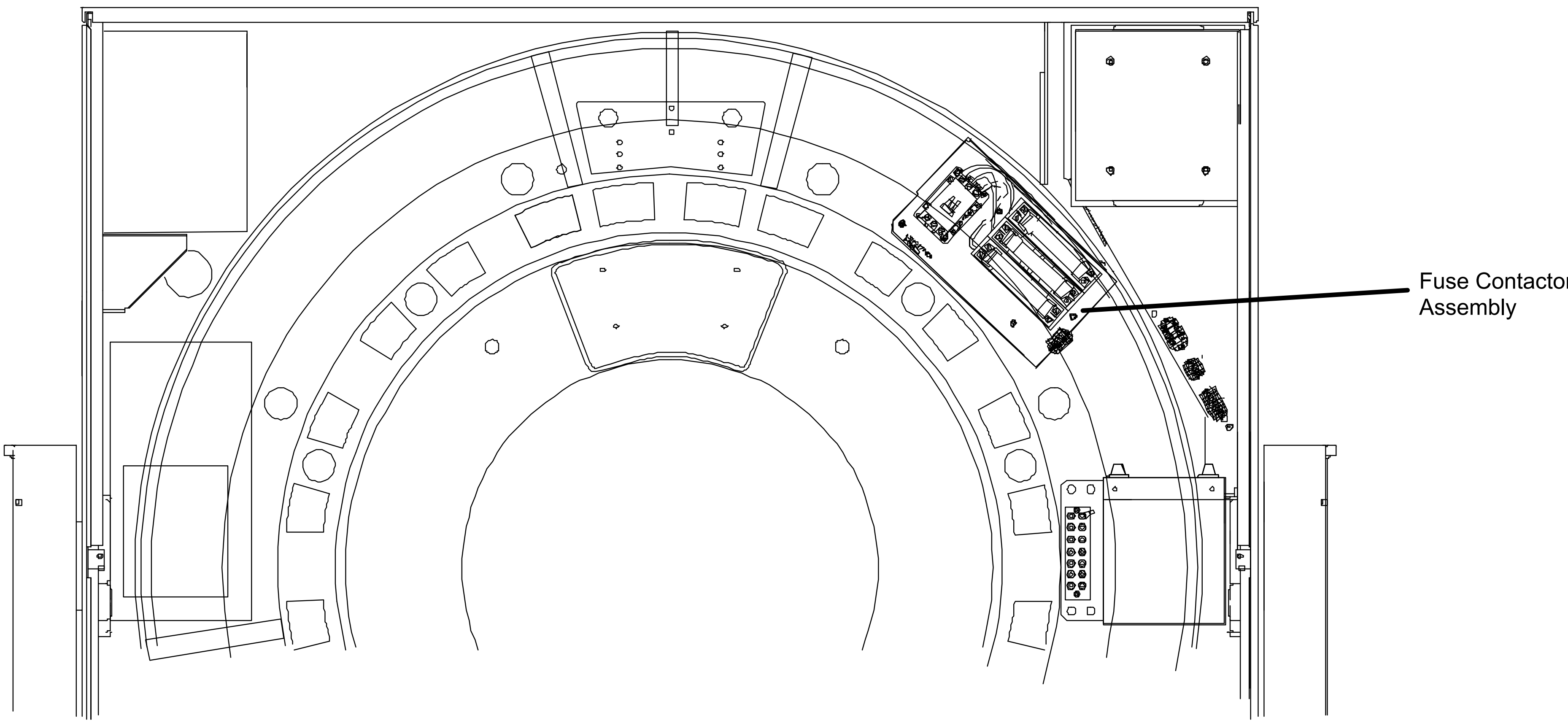


FIGURE 312 Fuse/Contactor Assembly – Upper Right Rear of Gantry

4. Tag and disconnect all Wires/Cables attached to the Fuse/Contactor chassis. Refer to Figure 313.
5. Remove the two hex mounting Bolts and remove the Chassis assembly. Refer to Figure 313.

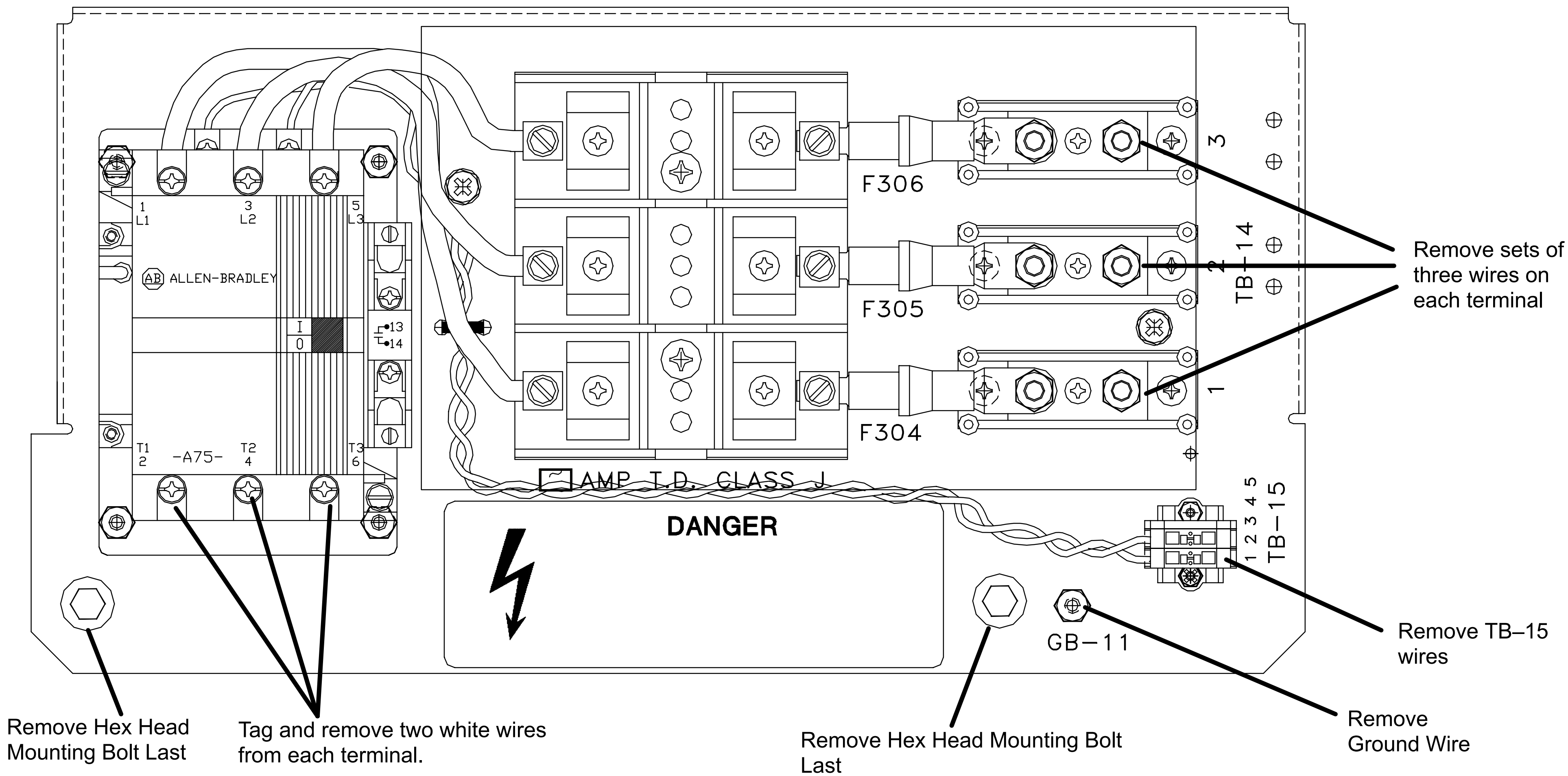


FIGURE 313 Fuse/Contactor Removal

FUSE/CONTACTOR REPLACEMENT:

6. Install the Fuse/Contactor Chassis by reversing steps 3. through 5.

7. Apply Gantry power. Refer to the Power Up/Down procedures.

> ⚠ **WARNING** ⚠
>
> **USE EXTREME CAUTION WHEN OPERATING OR WORKING ON EQUIPMENT WITH COVERS REMOVED. DANGEROUS VOLTAGES ARE EXPOSED WHEN THE GANTRY COVERS ARE REMOVED. FAILURE TO COMPLY CAN CAUSE SERIOUS INJURY OR DEATH TO PERSONNEL.**

8. To verify proper System operation, run Full–field and Half–field Water Phantom Scans and allow the System to process and display an Image.
9. Replace the rear Gantry covers. Refer to the Gantry Covers procedure.

AC CONTROL CHASSIS PRINTED CIRCUIT BOARD

Introduction: This procedure is used to remove and replace the Gantry AC Control Chassis Circuit Board Assembly.

Tools Required: 1. #2 Phillips Screwdrivers

Estimated Time: 1.0 hr.

PRINTED CIRCUIT BOARD REMOVAL:

1. Remove Gantry power. Refer to the PowerUp/PowerDown procedure. Remove power at wall box.
2. Remove the rear left gantry cover. Refer to the Gantry Covers procedure.

> ⚠ **WARNING** ⚠
>
> **USE EXTREME CAUTION WHEN OPERATING OR WORKING ON EQUIPMENT WITH COVERS REMOVED. DANGEROUS VOLTAGES ARE EXPOSED WHEN THE GANTRY COVERS ARE REMOVED. FAILURE TO COMPLY CAN CAUSE SERIOUS INJURY OR DEATH TO PERSONNEL.**

3. Remove the ribbon cable from J301. Refer to Figure 314.
4. Tag and unplug the two black wires that are attached to the small board attached to the AC Control Chassis PCB.
5. Remove the black and red wires from the SSR313 terminals. Cut tie wraps as necessary.
6. Remove the two Philips head screws from the small board attached to the AC Control Chassis PCB and remove.
7. Remove the six Philips Screws that hold the AC Control Chassis PCB in place. See Figure 314.

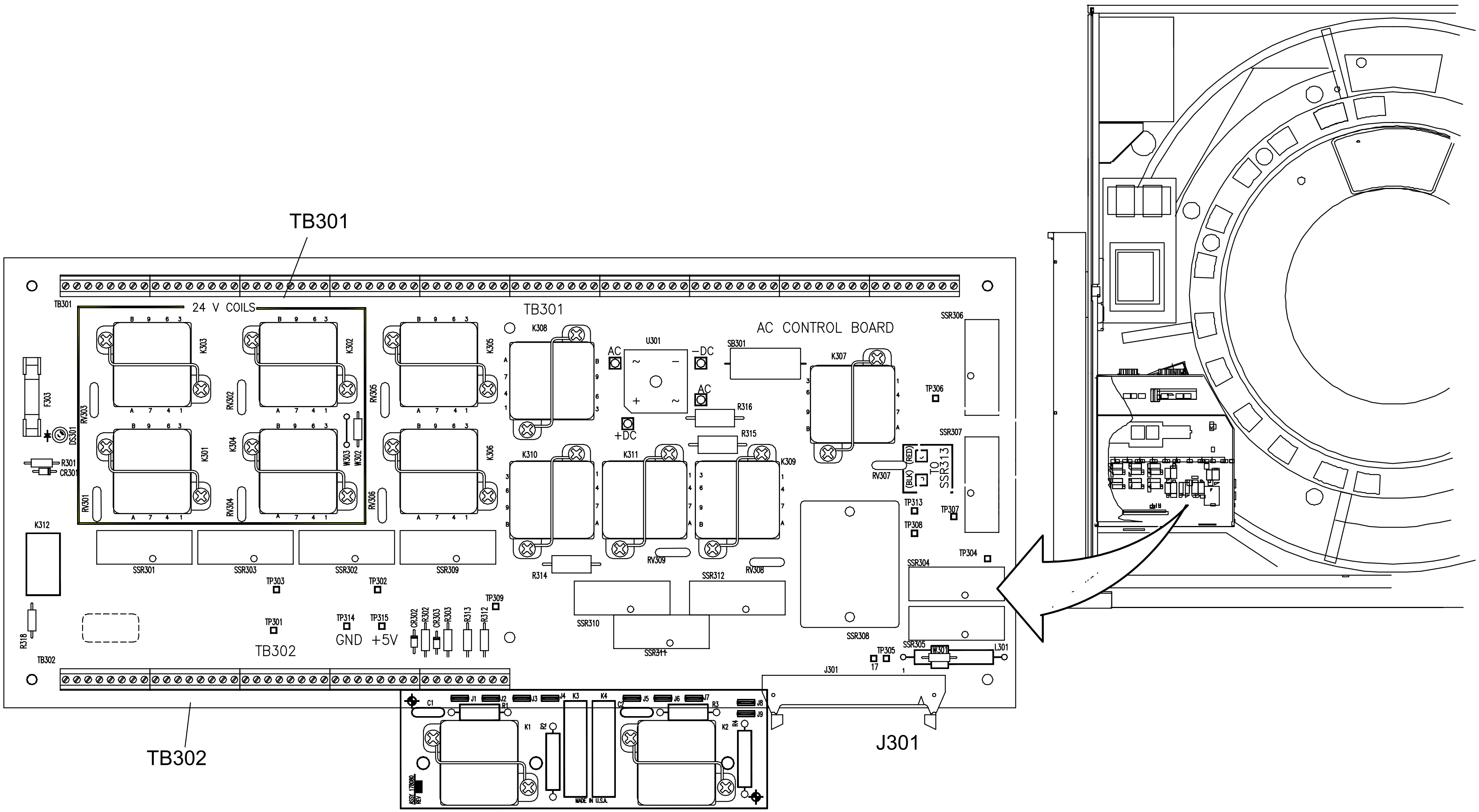


FIGURE 314 AC Control Chassis PCB Location

8. Carefully pull the Board away from the Chassis. Cut harness ties as required.

PRINTED CIRCUIT BOARD REPLACEMENT:

9. Hold replacement AC Control Board in front of the old one, with the new TB302 just above old TB302.
10. Transfer each wire, one at a time, from the old TB302 to the new TB302.
11. Move the replacement AC Control Board so that the new TB301 is just below the old TB301.
12. Transfer each wire, one at a time, from the old TB301 to the new TB301.
13. Carefully pull the bottom of the replacement AC Control Board out far enough to slide the old Board out from the bottom.
14. Secure the replacement AC Control Board to the Chassis with six Philips Screws removed earlier.
15. Replace Harness Ties as required.
16. Plug the Ribbon Cable into J301.
17. Replace the left rear gantry cover. Refer to the Gantry Covers procedure.
18. Apply Gantry power. Refer to the PowerUp/PowerDown procedure.
19. To verify proper System operation, run Full–field and Half–field Water Phantom Scans and allow the System to process and display an Image.

DETECTOR MODULE AND DETECTOR MOTHERBOARD

Introduction: This procedure is used to remove and replace Detector Module Assemblies and Detector Module Motherboards.

Tools Required:
1. Phillips Screwdriver, small slotted screwdriver
2. Anti–Static wrist band and bags

DETECTOR MODULE REMOVAL:

1. Remove Gantry power. Refer to the PowerUp/PowerDown procedure.
2. Open the upper front Gantry cover. Remove the lower front gantry cover if the detector being removed is located in the bottom half of the gantry. Refer to the Gantry covers procedure.

> ⚠ **WARNING**
>
> **USE EXTREME CAUTION WHEN SERVICING EQUIPMENT WHILE THE UPPER GULL WING PANEL IS OPEN. THE PANEL HAS SHARP EDGES. FAILURE TO COMPLY COULD RESULT IN SERIOUS INJURY TO PERSONNEL.**

> ⚠ **WARNING** ⚠
>
> **USE EXTREME CAUTION WHEN OPERATING OR WORKING ON EQUIPMENT WITH COVERS REMOVED. DANGEROUS VOLTAGES ARE EXPOSED WHEN THE GANTRY COVERS ARE REMOVED. FAILURE TO COMPLY CAN CAUSE SERIOUS INJURY OR DEATH TO PERSONNEL.**

3. Locate the Detector Module to be removed.

> ⚠ **CAUTION**
>
> ANTI–STATIC WRIST BAND AND ANTI–STATIC BAGS MUST BE USED IN THE FOL-LOWING PROCEDURES

4. Remove the slotted head screw from the front of the detector module being removed. You may have to rotate the gantry (ccw) a bit to gain better access to the head screws. Firmly grasp the front edge of the Detector Module and pull it straight out the front of the Gantry. See Figure 315.

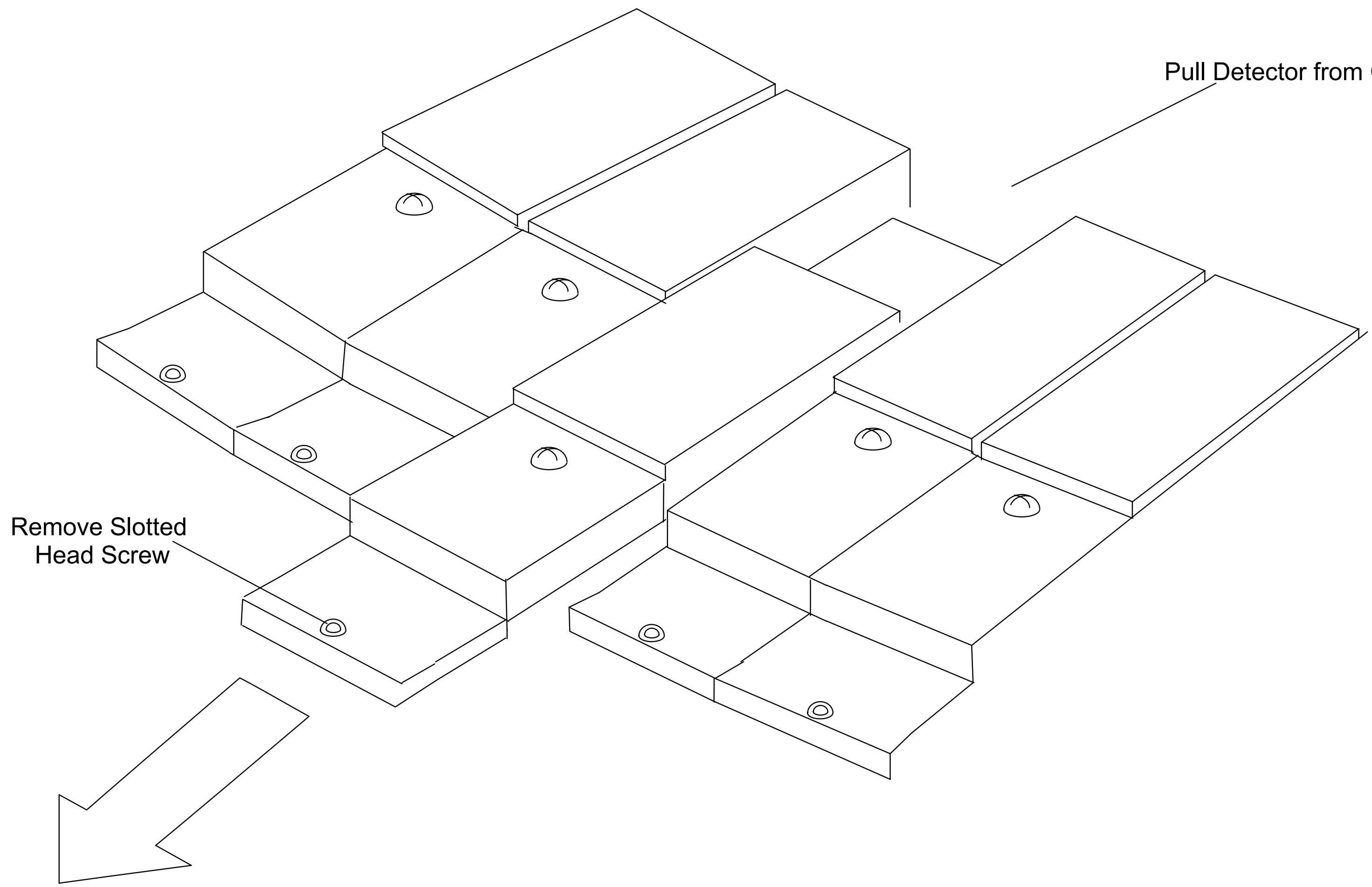


FIGURE 315 Detector Module Removal

DETECTOR MODULE REPLACEMENT:

5. Replace Detector Module by reversing step 4.
6. Replace the lower Gantry cover if removed earlier. Close the upper cover. Refer to the Gantry Covers procedure.
7. Apply Gantry power. Refer to the PowerUp/PowerDown procedure.
8. After allowing five minutes for Detector stabilization, run AutoCalibration. Refer to the AutoCalibration Manual.

DETECTOR MODULE MOTHERBOARD REMOVAL:

9. Follow steps 1. through 4. of Detector Module removal to remove all Detector Modules from defective Motherboard.
10. Remove four detector cables and power and mux connectors
11. Place Detector Modules into Anti–Static Bags and store in a safe place until replacement.
12. Remove the ten Philips Screws from the Motherboard. Carefully bend mother board up slightly on each end to help desengage the four connectors located underneath. These connectors plug into the gantry ring. Refer to Figure 316.

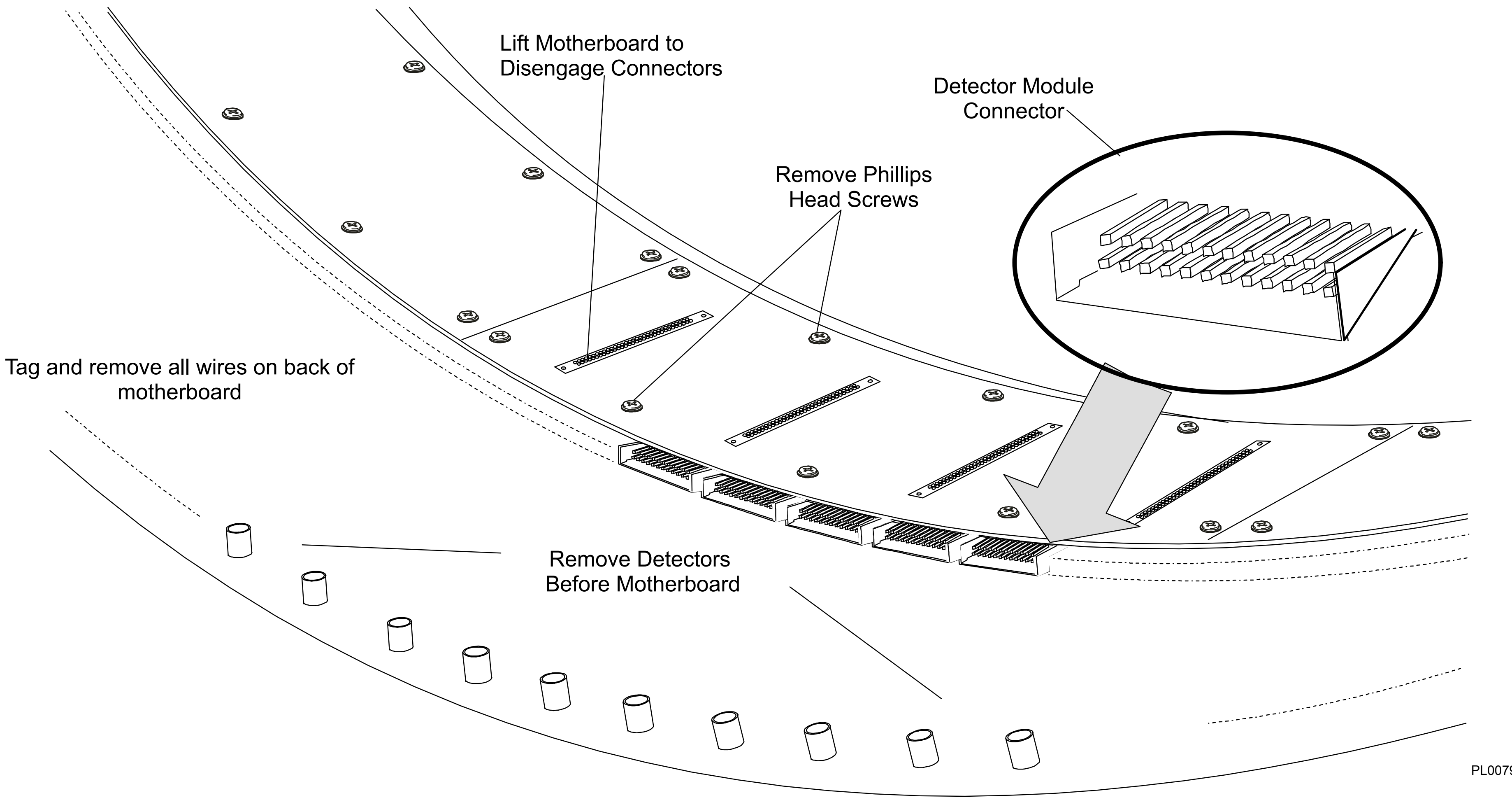


FIGURE 316 Motherboard Removal

DETECTOR MODULE MOTHERBOARD REPLACEMENT:

13. Install Motherboard by reversing steps 10. through 12. of Motherboard removal.

> **NOTE**
>
> When replacing Motherboards, the gaps on each side must be exactly the same. Otherwise the detector modules will not fit properly.

14. Replace and secure the Detector Modules by inserting the new detectors into their respective location. The detector should go in opposite of that shown in Figure 315.
15. Replace the bottom cover if removed earlier. Close the upper cover. Refer to the Gantry Covers procedure.
16. Apply Gantry power. Refer to the PowerUp/PowerDown procedure.
17. After allowing five minutes for Detector stabilization, verify/perform AutoCalibration. Refer to the AutoCalibration Manual for detailed information.

GANTRY COLUMN E–STOP SWITCH

Introduction: This procedure is used to remove and replace the Gantry Column Emergency Stop (E–stop) Switches.

Tools Required: 1. Philips Screwdriver

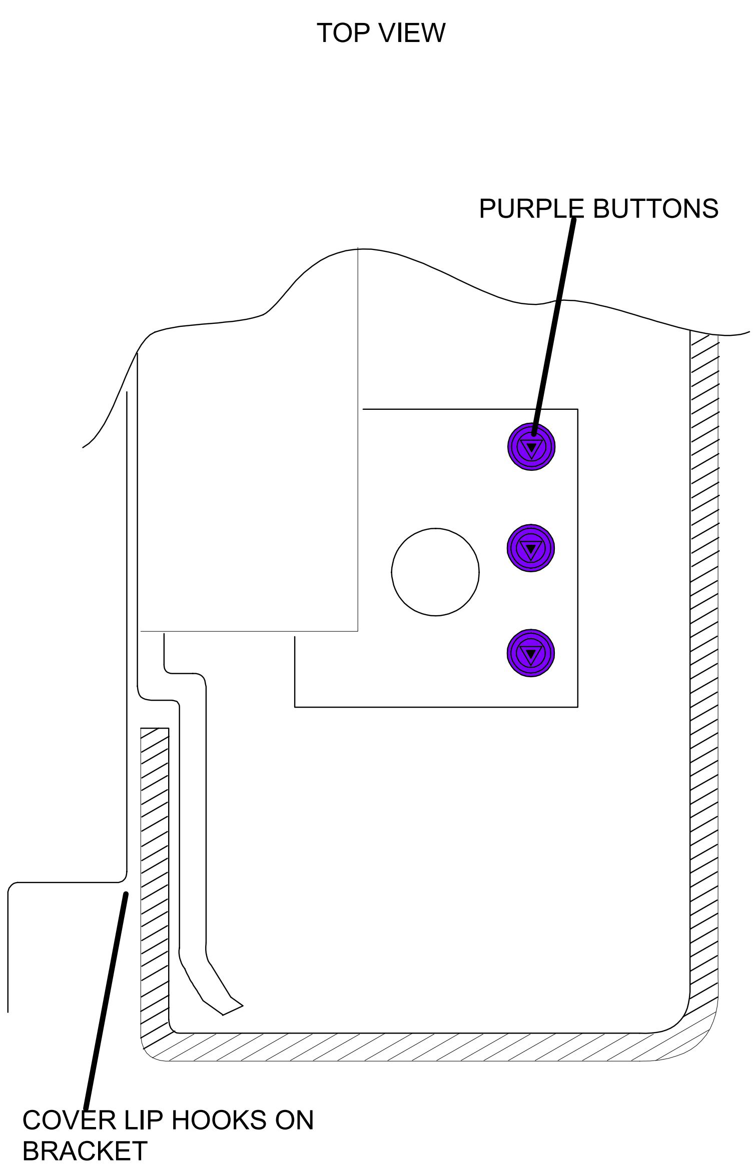
Estimated Time: 10 Min.

E–STOP SWITCH REMOVAL:

1. Remove Gantry power. Refer to the PowerUp/PowerDown procedure.
2. Above each gantry column e–stop switch is a small purple button (with a point–down triangle). Grab the edges with your fingernails and remove. This will expose a phillips head screw –remove the screw. Refer to NO TAG.
3. Pull the curved purple cover that the keyswitch is mounted out a bit. Grasp the square portion of the e–stop switch and pull out so the two pieces disconnect. **Note the orientation of the two pieces as they are not keyed**. You can now set the curved purple cover aside.
4. Unscrew the Switch Collar located on the back side of the purple cover. Remove the switch.

E–STOP SWITCH REPLACEMENT:

5. Install the new column E–Stop switch and secure with the switch collar.
6. Plug the front portion of the switch into the rear portion, exactly as is was removed.
7. Attach the purple cover back to the column and secure with the philips head screw.
8. Apply Gantry power. Refer to the PowerUp/PowerDown procedure.
9. Check the operation of the E–Stop Switch by pressing it and observing that the Scanner shuts down.



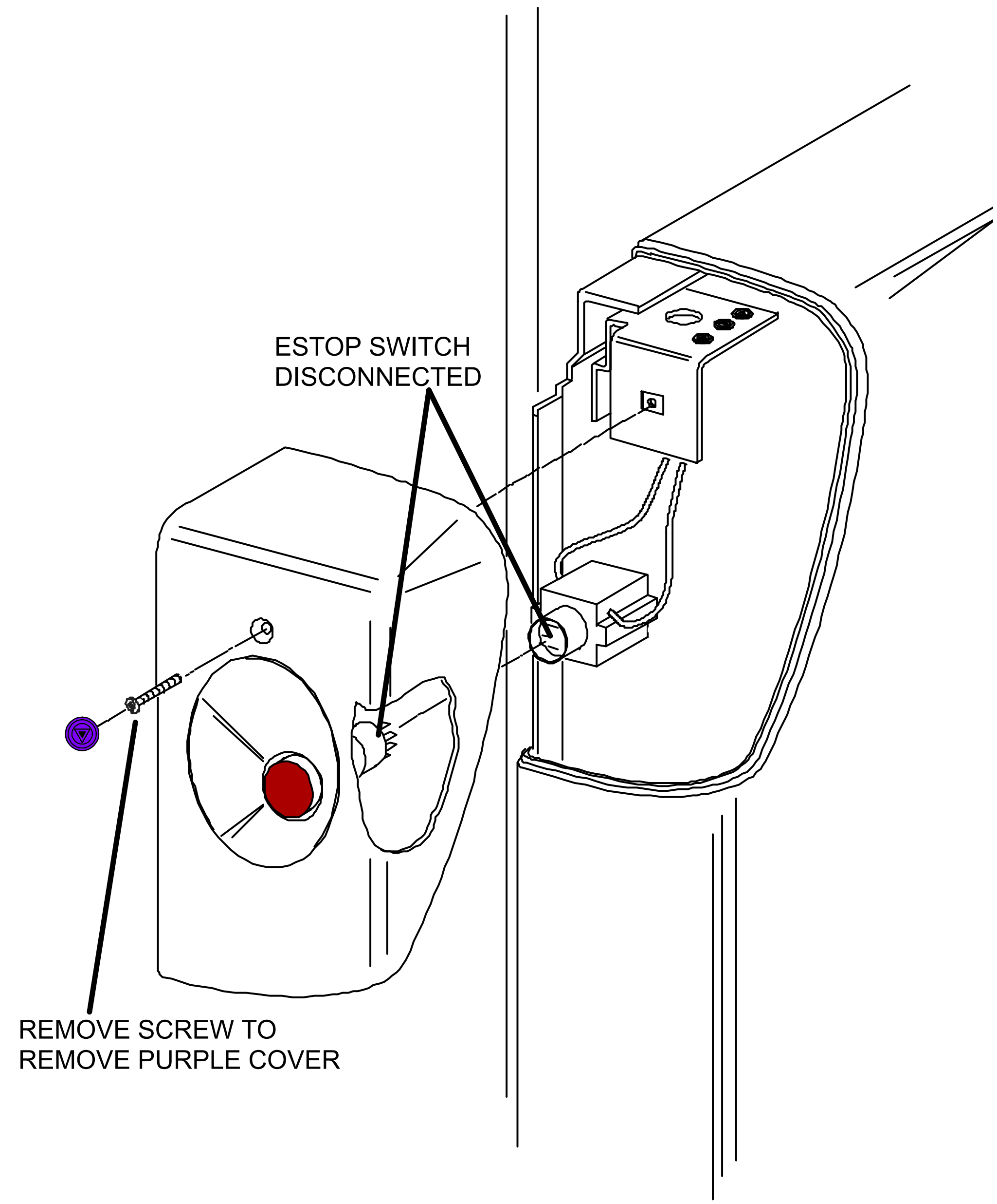


FIGURE 317 Gantry Column E–Stop Switch

1.1 KVA TRANSFORMER

Introduction:	This procedure is used to remove and replace the 1.1 KVA Auxiliary Transformer in the Gantry.
Tools Required:	1. Phillips and Slotted Screwdrivers
	2. 1/4” Hex Key
Chassis Weight:	25 pounds
Estimated TIme:	25 min.

TRANSFORMER REMOVAL:

1. Remove Gantry power. Refer to the PowerUp/PowerDown procedure.
2. Remove the rear gantry covers. Refer to the Gantry Covers procedure.

> ⚠ **WARNING** ⚠
>
> **USE EXTREME CAUTION WHEN OPERATING OR WORKING ON EQUIPMENT WITH COVERS REMOVED. DANGEROUS VOLTAGES ARE EXPOSED WHEN THE GANTRY COVERS ARE REMOVED. FAILURE TO COMPLY CAN CAUSE SERIOUS INJURY OR DEATH TO PERSONNEL.**

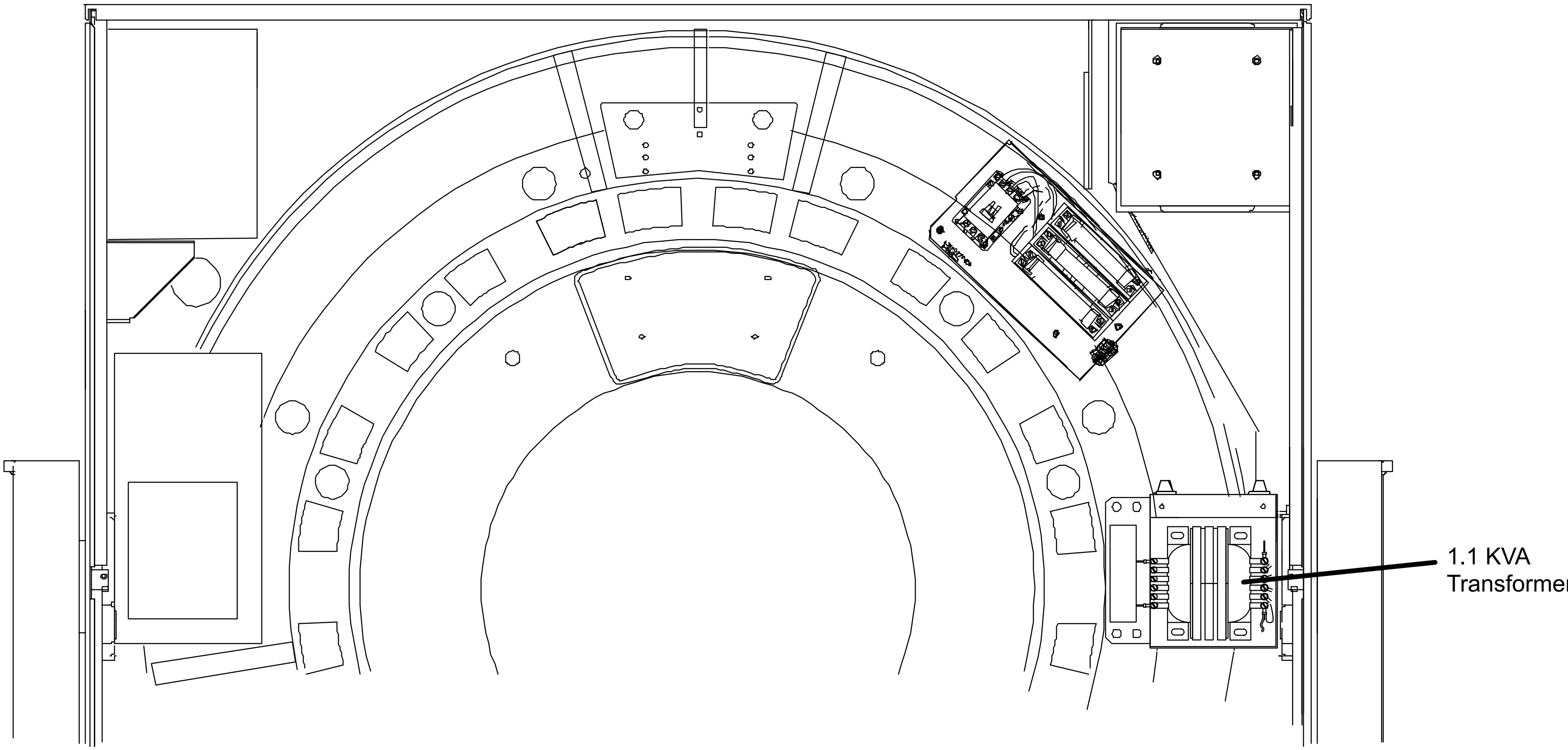


FIGURE 318 1.1 KVA Transformer Location

3. Remove the two transformer cover mounting screws. Remove the transformer cover. See Figure 319.

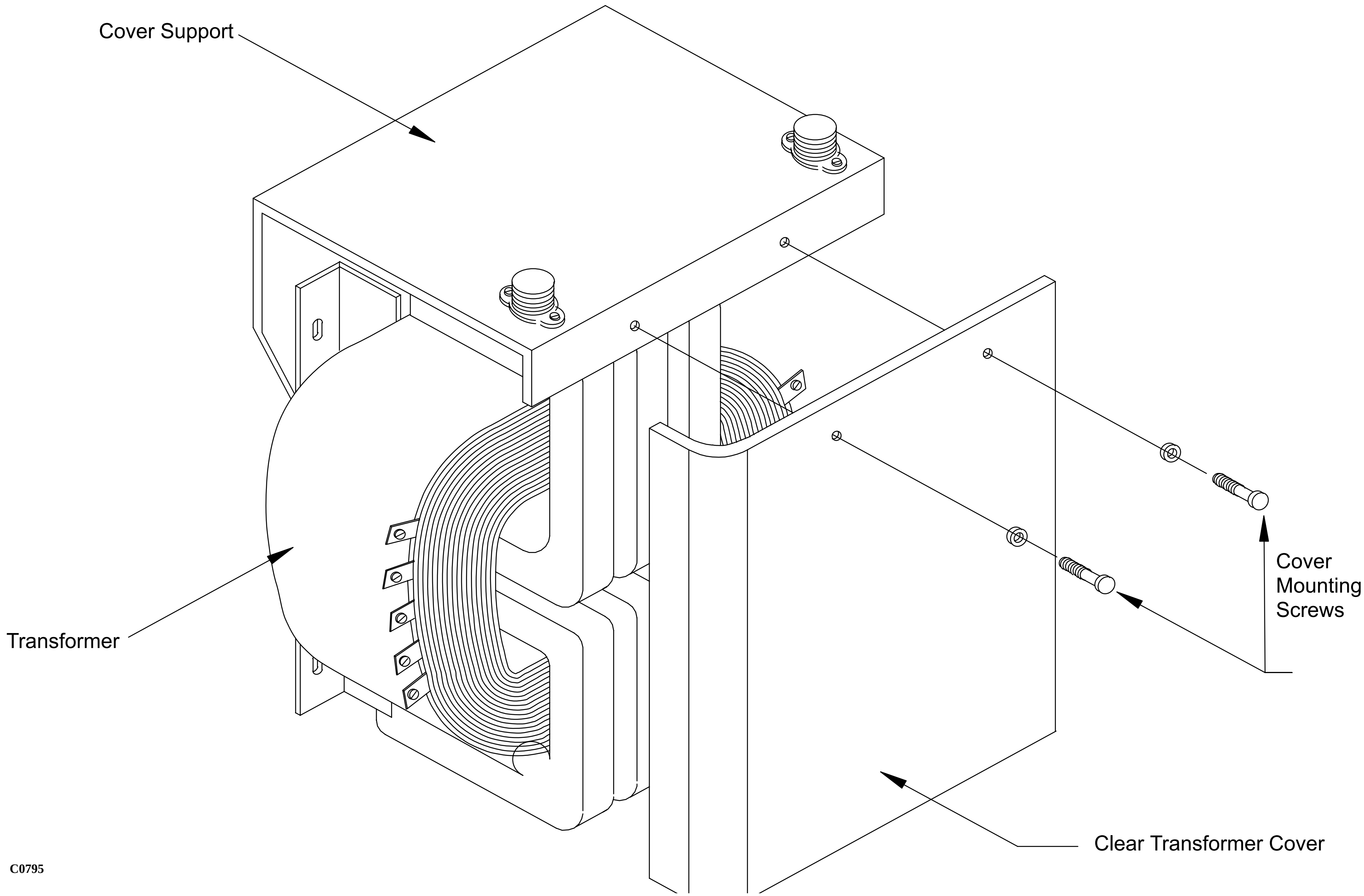


FIGURE 319 1.1 KVA Auxiliary Transformer Clear Cover

4. Tag and remove the eight transformer connections. See Figure 320.

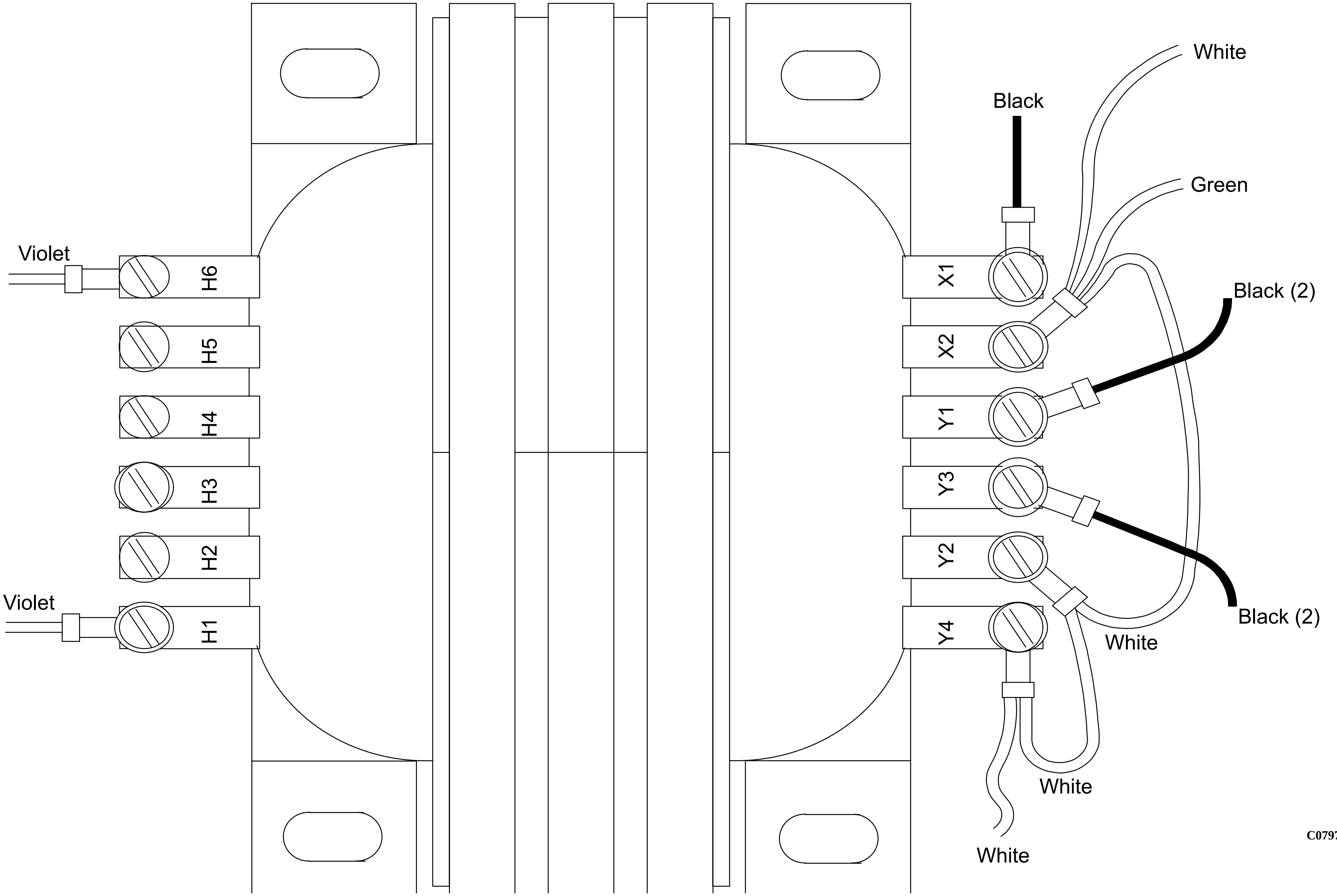


FIGURE 320 1.1 KVA Transformer Connections

⚠ WARNING

YOU MAY NEED TO USE TWO PERSONS TO HANDLE THE 1.1 KVA TRANSFORMER. THE 1.1 KVA TRANSFORMER WEIGHS 25 LBS. FAILURE TO COMPLY CAN RESULT IN EQUIPMENT DAMAGE AND/OR SERIOUS INJURY TO PERSONNEL.

NOTE

THE TRANSFORMER COVER SUPPORT AND SPACING WASHERS WILL FALL IF THEY ARE NOT PROPERLY SUPPORTED WHEN THE TRANSFORMER MOUNTING SCREWS ARE REMOVED.

5. Support the Transformer weight and remove the four Transformer Mounting bolts with the hex head key. Carefully remove the Transformer, Cover Support, and Spacer Washers. See Figure 321.

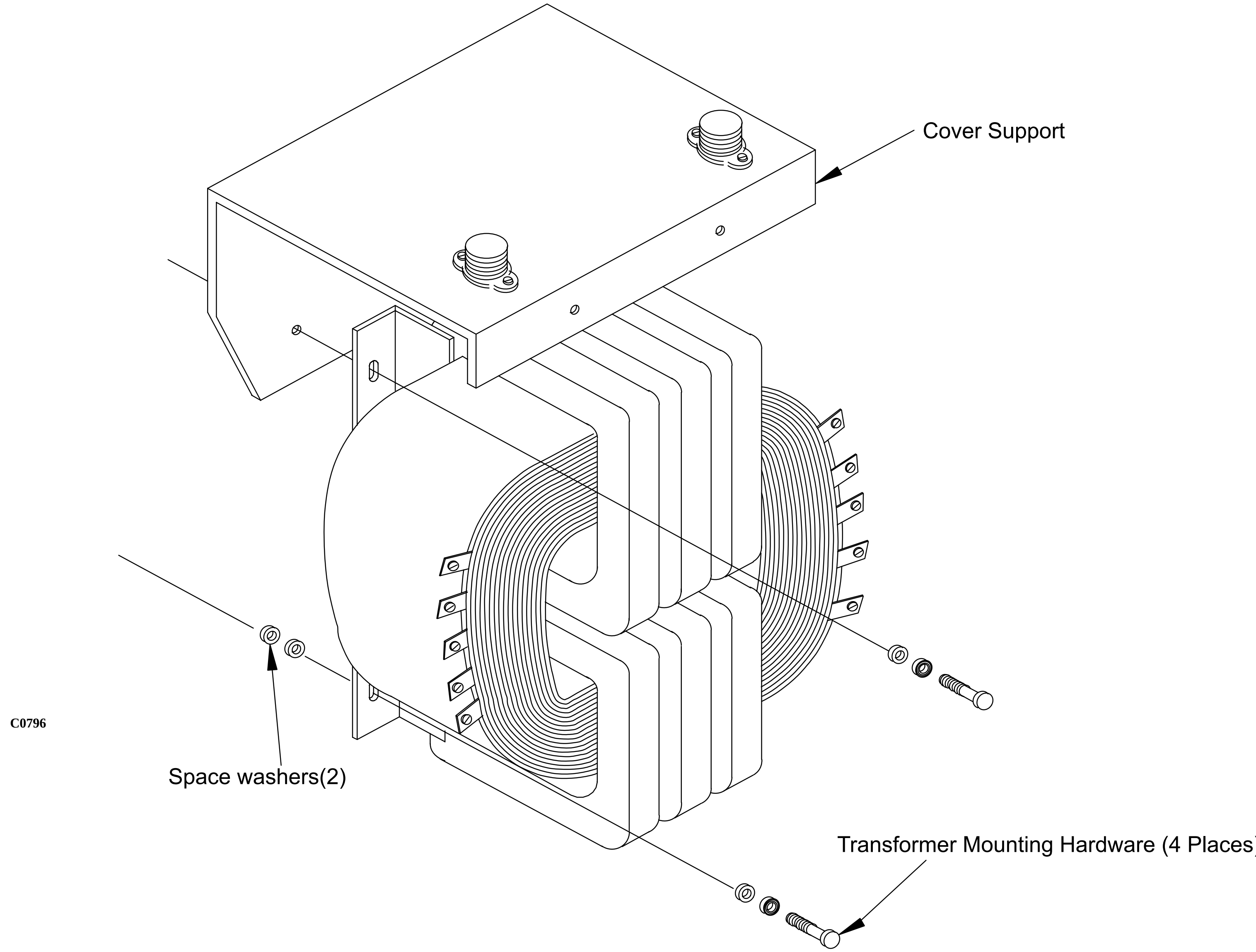


FIGURE 321 1.1 KVA Transformer Mounting Screws

TRANSFORMER REPLACEMENT:

6. Install the 1.1 KVA Transformer by reversing steps 5.and 4.

7. Apply Gantry power. Refer to the PowerUp/PowerDown procedure.

> **⚠ WARNING**
>
> **USE EXTREME CAUTION WHEN OPERATING OR WORKING ON EQUIPMENT WITH COVERS REMOVED. DANGEROUS VOLTAGES ARE EXPOSED WHEN THE GANTRY COVERS ARE REMOVED. FAILURE TO COMPLY CAN CAUSE SERIOUS INJURY OR DEATH TO PERSONNEL.**

8. Use a DVM to verify that there is 480 VAC power at terminals H1 and H6 on the 1.1 KVA Transformer.
9. Replace the Transformer Cover by reversing step 3.
10. To verify proper System operation, run Full–field and Half–field Water Phantom Scans and allow the System to process and display an Image.
11. Remove Gantry power. Refer to the PowerUp/PowerDown procedure.
12. Replace the rear covers. Refer to the Gantry Covers procedure.

8 KVA TRANSFORMER

Introduction: This procedure is used to remove and replace the 8 KVA Utility Transformer in the Gantry. Two people are recommended for this procedure due to the Transformer weight.

Tools Required:
1. Philips and Slotted Screwdrivers
2. 1/4” hex key

Chassis Weight: 130 pounds

Estimated TIme: 35 Min.

8 KVA TRANSFORMER REMOVAL:

1. Remove Gantry power. Refer to the PowerUp/PowerDown procedure. Remove power at the wallbox.
2. Remove the gantry rear covers. Refer to the Gantry Covers procedure.
3. Loosen the two Phillips Cover Screws and remove the Transformer Cover. See Figure 322.

> ⚠ **WARNING** ⚠
>
> **USE EXTREME CAUTION WHEN OPERATING OR WORKING ON EQUIPMENT WITH COVERS REMOVED. DANGEROUS VOLTAGES ARE EXPOSED WHEN THE GANTRY COVERS ARE REMOVED. FAILURE TO COMPLY CAN CAUSE SERIOUS INJURY OR DEATH TO PERSONNEL.**

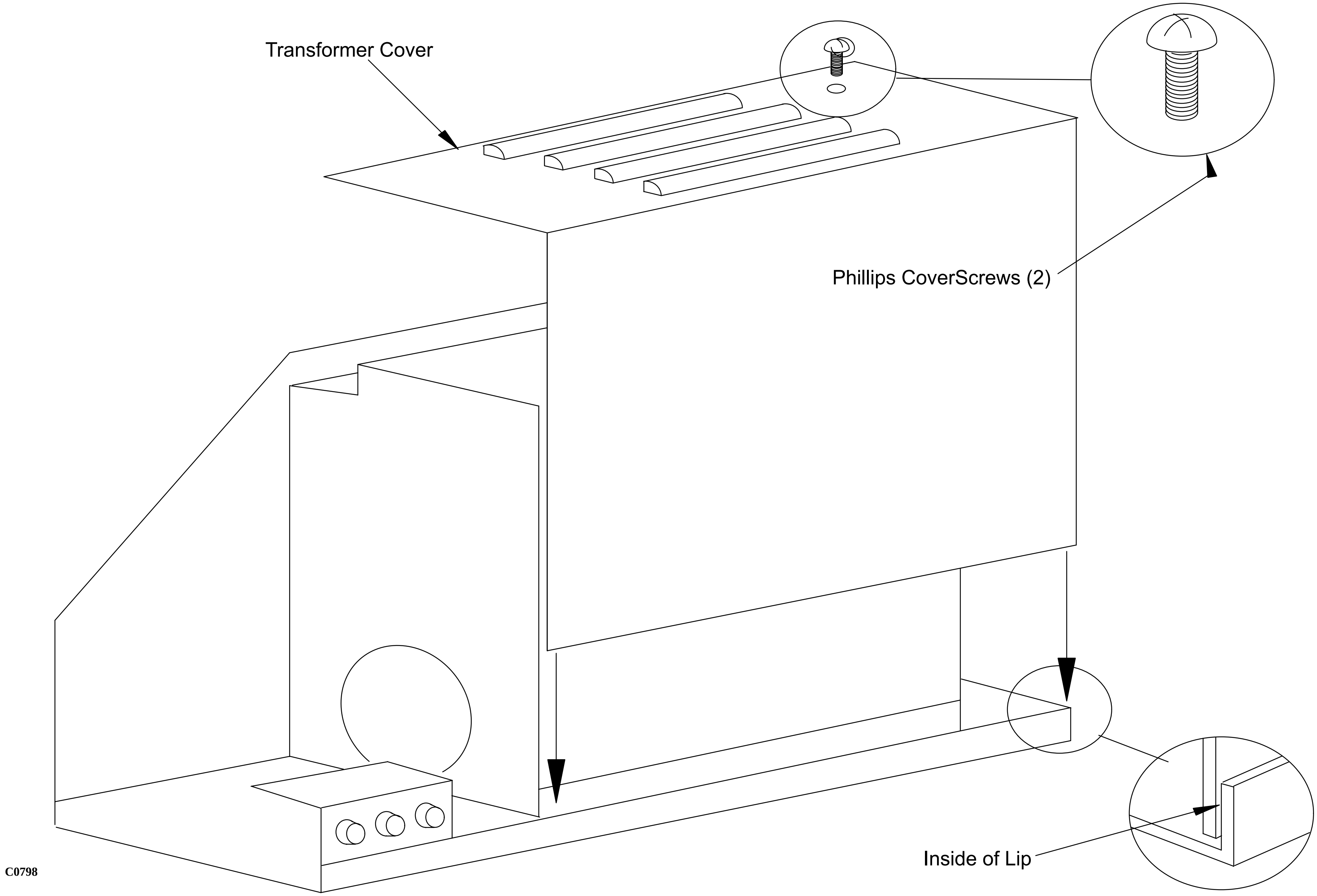


FIGURE 322 8 KVA Transformer Cover

4. Remove the two screws from the terminal block cover. See Figure 323.

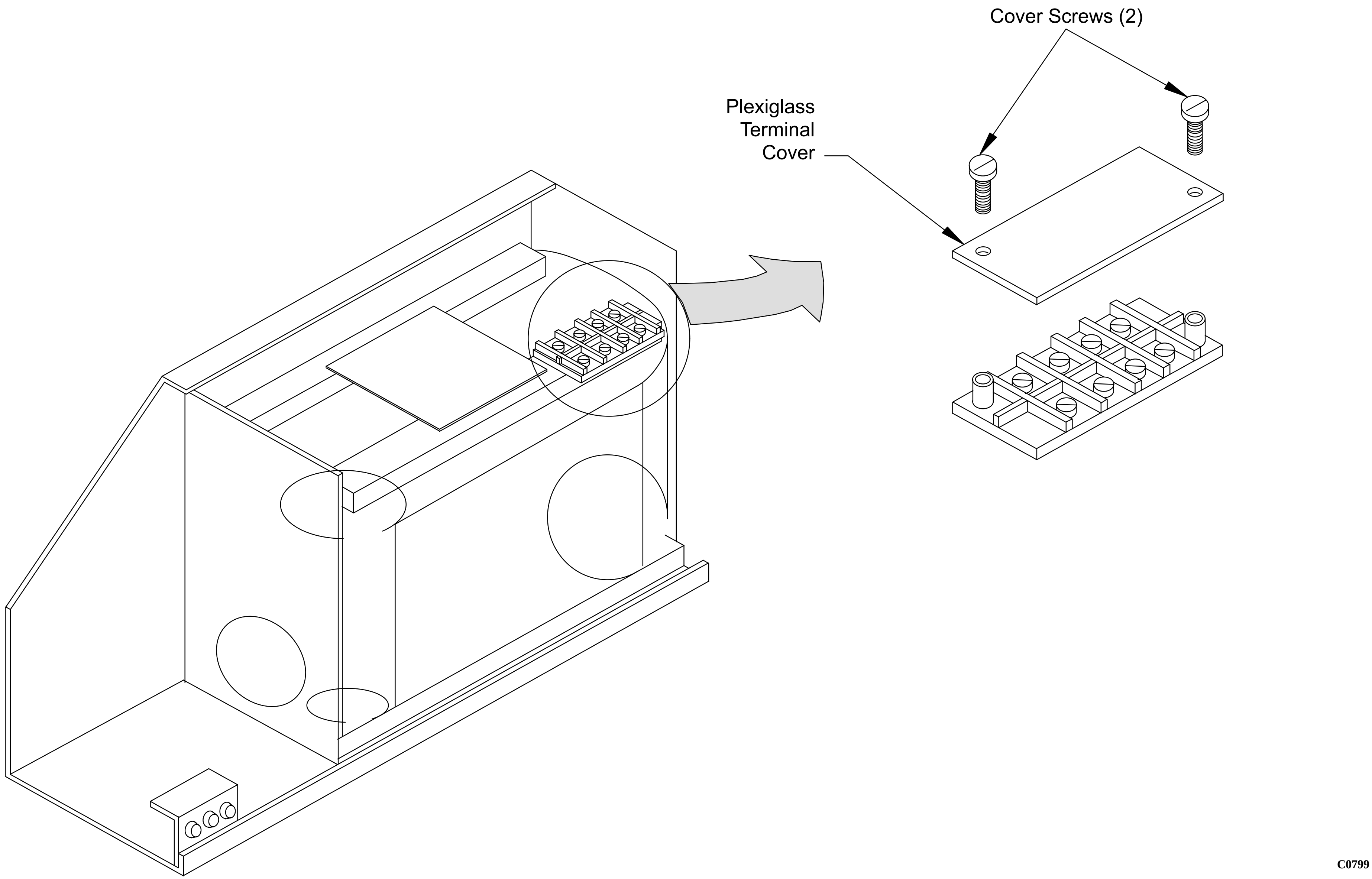


FIGURE 323 Terminal Block Cover

5. Tag and remove the five Transformer Wires connected to Terminals X0, X1, X2, and X3 on the Transformer Terminal Block.

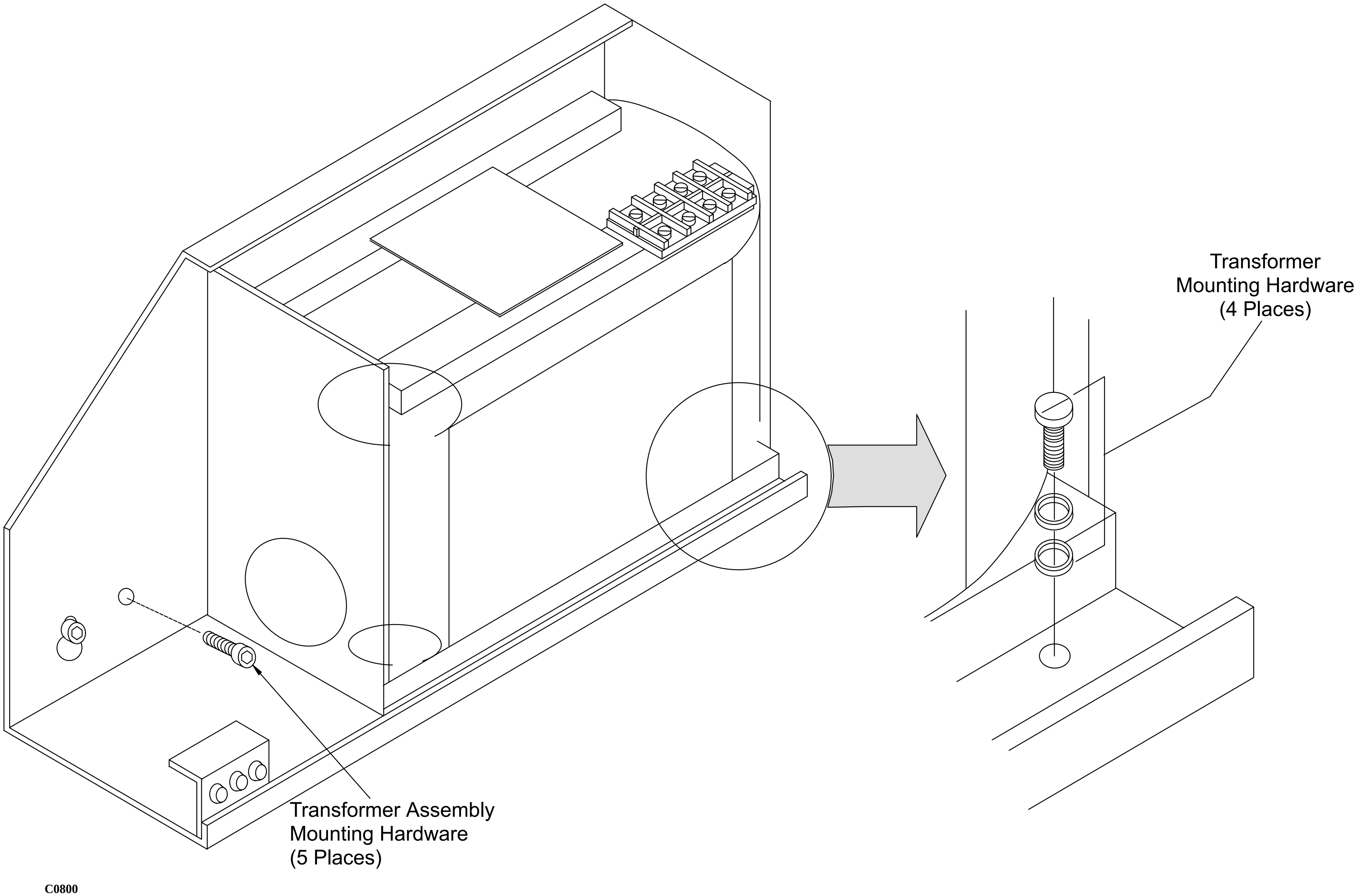


FIGURE 324 8 KVA Transformer Mounting Bolts

6. Tag (if needed) and remove the following Wires from the Fuse Side Terminals:

WIRE	FUSE
H1	F307
H2	F308
H3	F309

⚠ WARNING

USE TWO PERSONS TO HANDLE THE 8 KVA TRANSFORMER. THE 8 KVA TRANSFORMER WEIGHS 130 POUNDS. FAILURE TO COMPLY CAN RESULT IN EQUIPMENT DAMAGE AND/OR SERIOUS INJURY OR DEATH TO PERSONNEL.

7. Remove the Transformer Mounting Hardware and remove the Transformer from the Gantry.

8 KVA TRANSFORMER REPLACEMENT:

8. Install the replacement 8 KVA Transformer by reversing steps 3. through 7.
9. Turn the wallbox power back on. Apply Gantry power. Refer to the PowerUp/PowerDown procedure.
10. Use a DVM to verify that 208 VAC (nominal) 3 phase power is present at terminals X1, X2, and X3 on the Transformer Terminal Block.
11. Replace the Terminal Block Plastic Cover. Refer to Figure 322.
12. Replace the Transformer Housing Cover. Refer to Figure 322.
13. To verify proper System operation, run Full–field and Half–field Water Phantom Scans and allow the System to process and display an Image.
14. Remove Gantry power. Refer to the PowerUp/PowerDown procedure.
15. Replace the gantry rear covers. Refer to the Gantry Covers procedure.

POWER UP / DOWN PROCEDURE

This section describes the system start–up/shutdown procedures. **Power–UP** is accomplished via the keyswitch on the Display Tower. **Power DOWN** of the entire system is accomplished via the General Utilities GUI. This is accessed via the “Service” option from the opening bootup GUI. This procedure covers the Service Application mode. *Scanning Application mode procedure is in the OP manual*.

SYSTEM POWER UP

1. Make sure the Gantry keyswitch is in the normal position. Turn the Display Tower keyswitch to the **ON** position. After the system boots, you will be prompted with the startup application display (or “4 Button Screen.”) You have **4 boot options**. Refer to Figure 325.:
   A.) Select Clinical Scanning (or do nothing) and the system will start and connect to the acquisition server. The acquisition server will then power up the AcQImage Cabinet, and if that is done successfully, the acquisition server will power up the gantry and X–ray system.
   B.) Select the Service Application option (within 15 seconds) by clicking on the Service button with the mouse pointer.
   C.) Reboot the system. **D**.) Start the Diagnostic GUI (within 15 seconds) by clicking once in the bottom left corner, then entering PICKER <CR>. (All caps). The AcQImage board LEDs will display a power up test sequence. Refer to the LED Interpretations.

FOR DIAG GUI:
Click here with the mouse, then type PICKER and press <CR>.

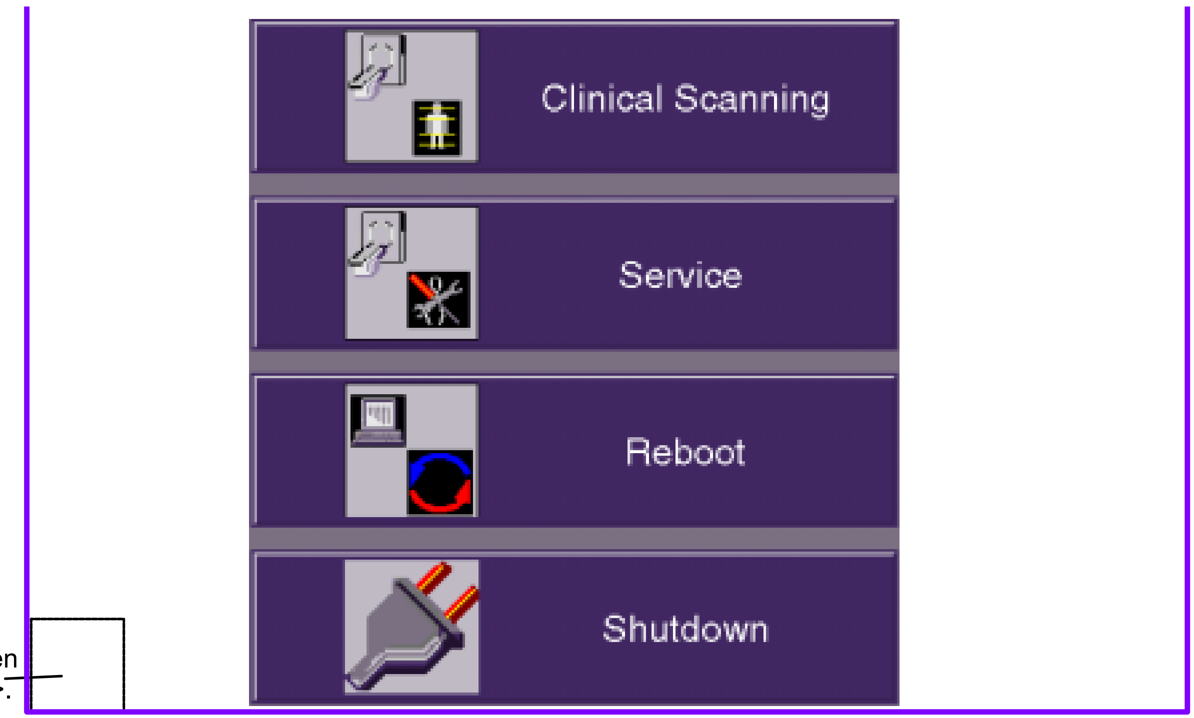


FIGURE 325 4 BUTTON SCREEN

PREPARE THE SYSTEM FOR USE (X–Ray Tube Warmup)

2. The X–Ray system needs to be properly warmed up prior to executing a scan procedure. X–Ray System Warmup must be completed at the start of every scan day or when the tube heat is below 6%. Tube heat is displayed in the Tube Warmup window.

3. To perform a tube warm–up, click on the Tube Warm Up button with the mouse pointer. Refer to Figure 325. This will initiate the warm–up sequence and **X–RAYS WILL BE EMITTED**. The tube warm–up status is displayed is displayed in the Tube Warmup Window. Refer to Figure 326.

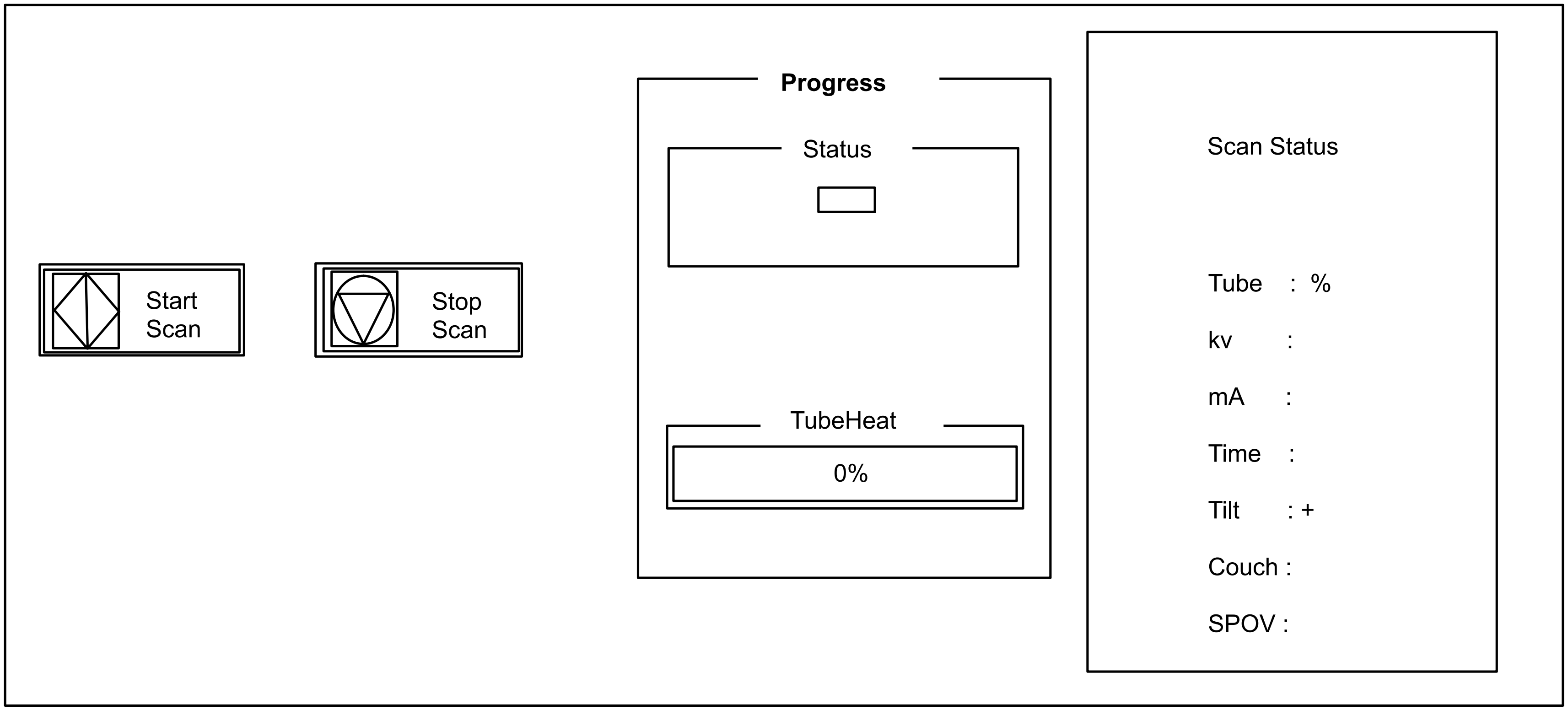


FIGURE 326 TUBE WARM–UP STATUS WINDOW

SYSTEM POWER DOWN

1. From the 4 Button Screen (Figure 325) click on SHUTDOWN with the mouse pointer.

> ⚠ **CAUTION**
>
> DO NOT POWER DOWN THE SYSTEM WITH THE TUBE HEAT AT OR ABOVE 35%. FAILURE TO COMPLY MAY RESULT IN SHORTENED TUBE LIFE AND/OR ERRATIC TUBE BEHAVIOR.

System shutdown ensures that all images on disk are properly closed, all image maintenance functions properly terminated, the gantry parked, the X–Ray system sufficiently cooled and all films printed.

There are three shutdown selections: Shutdown Gantry, Shutdown Console, Shutdown System and Switch to Console as seen in Figure 327.

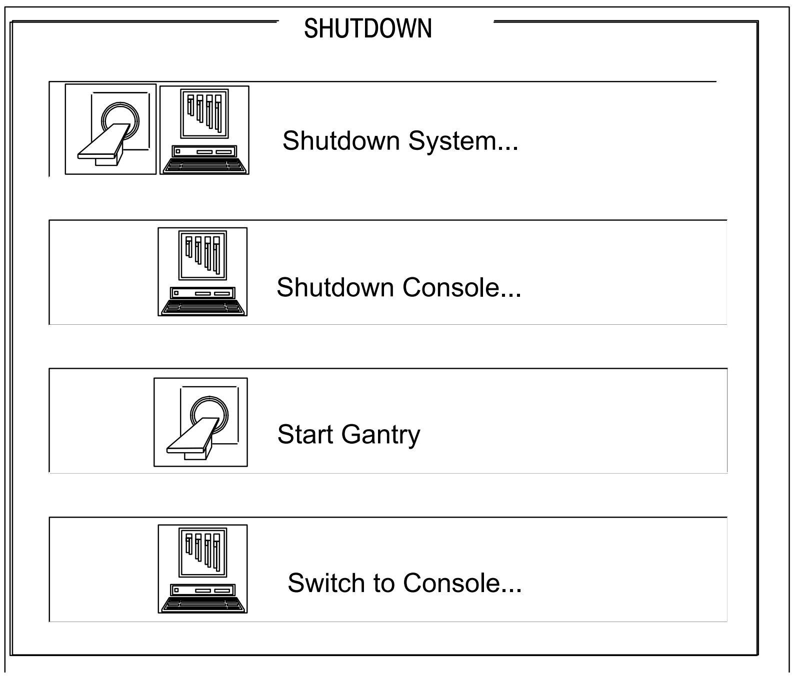


FIGURE 327 SHUTDOWN SCREEN

Shutdown Gantry: This option should be used, if possible, in place of the e–stop switches. An orderly shutdown of the gantry will be performed. Display Tower and/or AcQImage Cabinet operation may continue after selected. To cancel shutdown, click the Abort button.

Shutdown System: Performs an orderly shutdown of the entire scanner system. To cancel shutdown, click the Abort button.

Shutdown Console: This option shuts down only the console units. An orderly shutdown of the image archiver, filming and file system will be done.

Start Gantry: Toggles Start/Shutdown Gantry depending on status of Gantry power.

Switch to Console: Changes the GUI to the Console Application (Operator) screen.

⚠ **WARNING** ⚠

THE GUI SHUTDOWN REMOVES POWER FROM THE SYSTEM, BUT NOT AT THE INCOMING POWER LINES. DANGEROUS VOLTAGES ARE EXPOSED WHEN THE GANTRY COVERS ARE REMOVED. BEFORE SERVICING EQUIPMENT, REMOVE POWER FROM THE WALL BOX. FAILURE TO COMPLY CAN CAUSE SERIOUS INJURY OR DEATH TO PERSONNEL.

2. When prompted via a dialog box, turn the Display Tower keyswitch to the **OFF** position.
3. On the AcQImage Cabinet and Display Tower, <u>ensure the rear breaker switch is turned off</u>.

AcQImage Boards – Power Up Test LED Interpretations

```
AP, PB, GAM Board Diagnostic Flash Version 1.1.2
```

```
0 Hardware Initialization
1 CPU Version Check
2 Diagnostic Flash CRC Check
3 Operating System Flash CRC Check
4 CPU Memory Test
5 Data Cache Test
6 Unexpected Exception Test
7 IBC Test
8 NVRAM Single Location Test
9 PCI Bus Select Test
A Ethernet Loopback Test
B UART Loopback Test
C apmem for AP, pbmem for PB, gamem for GAM

C CIO Test (CP Board Diagnostic Flash Version 1.1.2)
D Synchronous Serial Chip Loopback Test (CP Board Diagnostic Flash Version 1.1.2)
```

CONSOLE COMPONENTS REPAIR/REPLACEMENT

ACQIMAGE CABINET COVERS

REMOVAL / REPLACEMENT

Introduction:	The following procedure is used to open, remove, re–install or replace AcQImage Cabinet covers.
Tools Required:	Phillips screwdriver
Estimated Time:	5 min.

TOP COVER REMOVAL:

1. Remove AcQImage Cabinet power. Refer to the PowerUp/PowerDown procedure.

> **WARNING**
>
> **USE EXTREME CAUTION WHEN OPERATING OR WORKING ON EQUIPMENT WITH COVERS REMOVED. DANGEROUS VOLTAGES ARE EXPOSED WHEN THE COVERS ARE REMOVED. FAILURE TO COMPLY CAN CAUSE SERIOUS INJURY OR DEATH TO PERSONNEL.**

> **NOTE**
>
> If the system needs to be moved, release the lock on the wheels before moving. Ensure wheels are locked after moving.

2. Remove the two Phillips screws from the upper rear of the top cover.
3. Lift the cover up a bit and slide the cover toward the rear, then lift upward.

SIDE COVER REMOVAL:

1. Remove the two lower Phillips screws from the bottom of the side cover.
2. Lift up slightly from the bottom and pull the bottom of the cover toward you. The top will follow after the small tabs have cleared. Refer to Figure 329.

REAR COVER REMOVAL:

1. Remove the eight Phillips screws – three on each side and two on the bottom. Note that the AC distribution box is attached to the lower half of the rear cover on the internal side and has cables to remove.

2. While holding the large brass ground connector bolt, slide the cover down a little and pull outward from the bottom. Disconnect the cable that leads to the Transition Module. Refer to Figure 330. Set the cover aside as seen in Figure 329.

FRONT COVER REMOVAL: (EXTERNAL COVERS)

1. The two front decorative cover sections are held on with ball stud and spring clips. Grasp the sides of the bottom portion and pull outward and downward. Refer to Figure 328.
2. Grasp the middle bottom of the upper portion and pull outward. Refer to Figure 328.

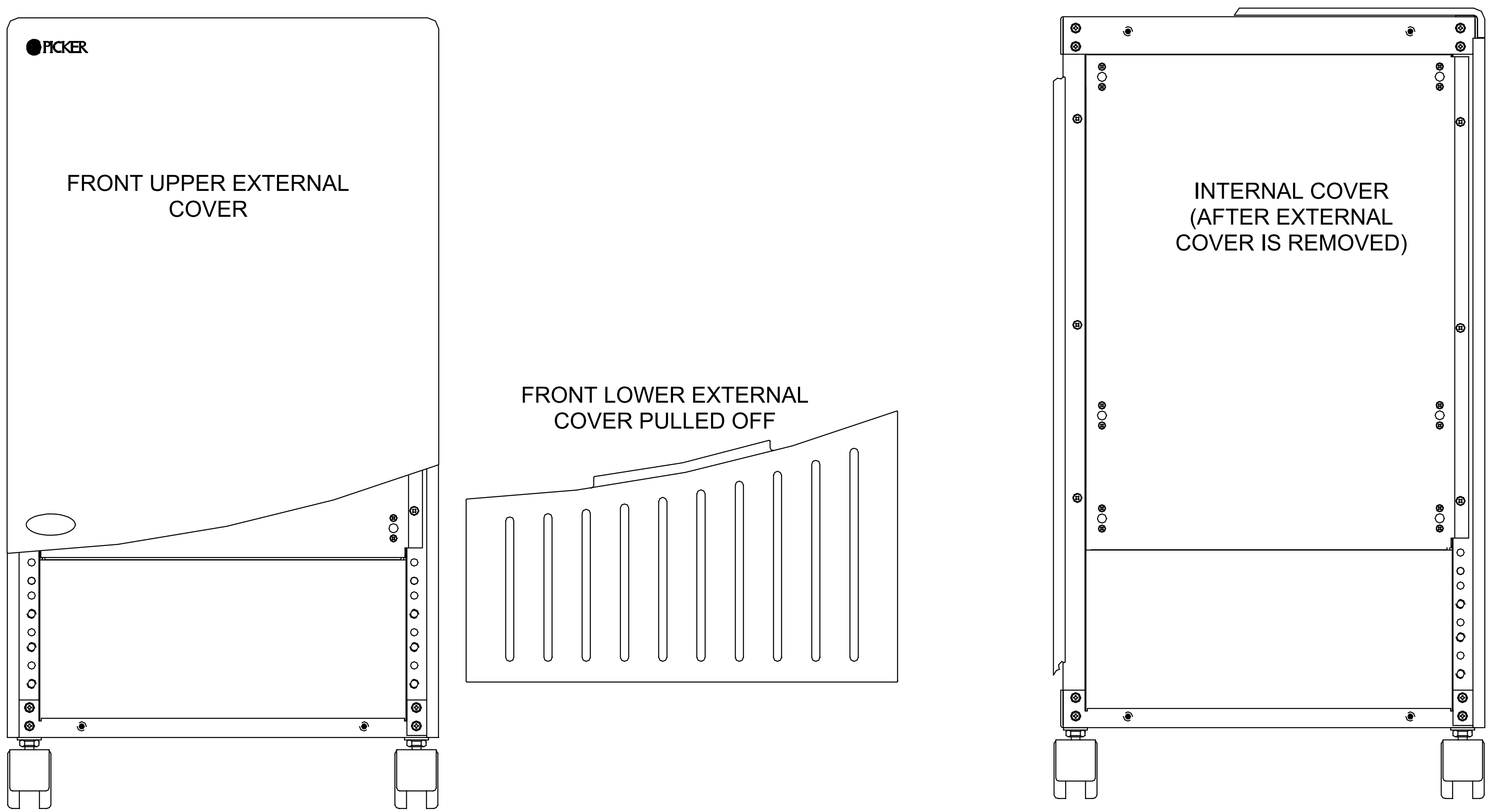


FIGURE 328 AcQImage Cabinet – Front Covers, External and Internal

FRONT COVER REMOVAL: (INTERNAL COVER)

1. The front covers removed here are mainly cosmetic. To access the recon boards (for removal, diagnostics, etc) you must remove the **internal** front internal steel panel by removing the six screws. Refer to Figure 328.

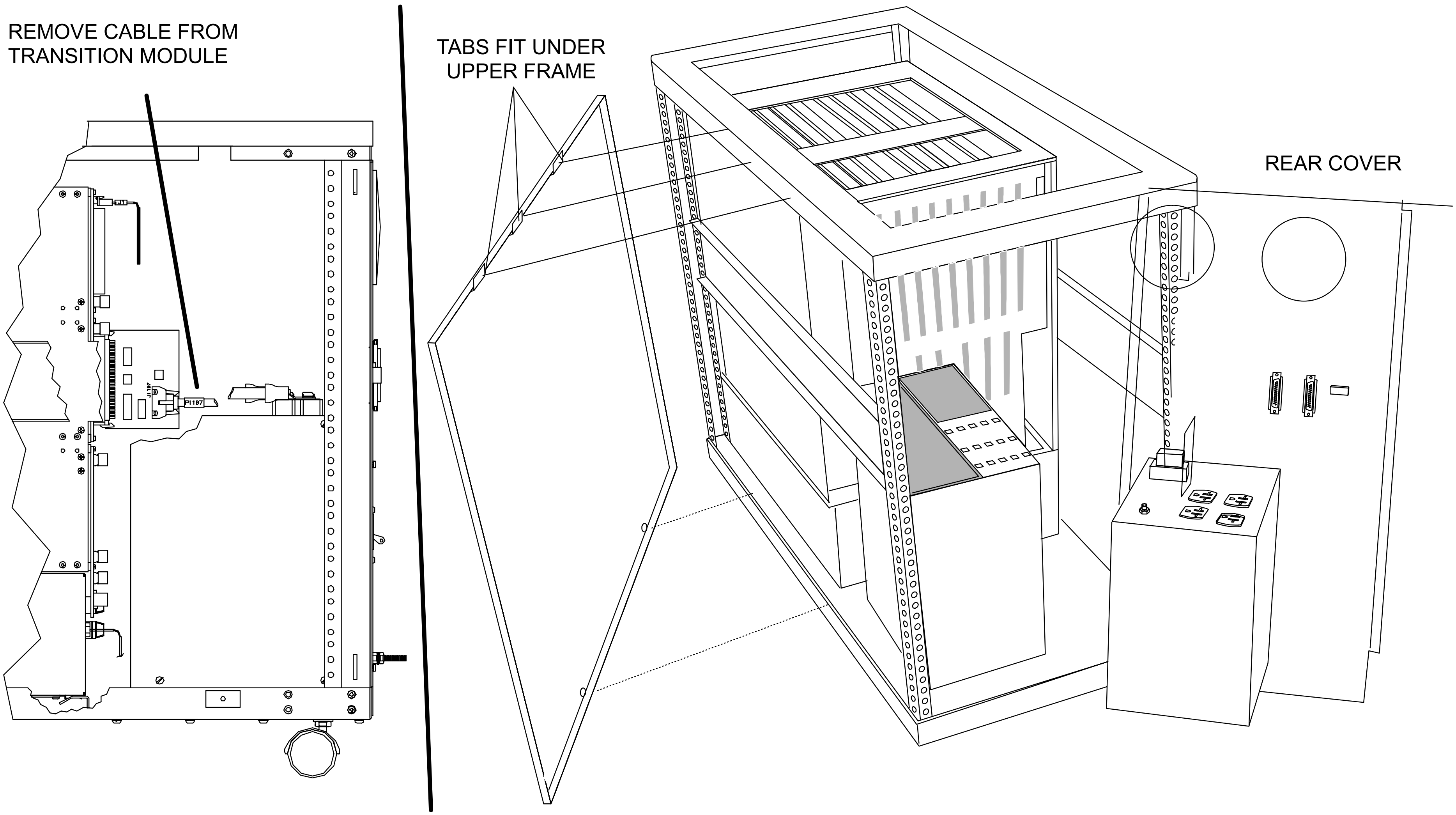


FIGURE 329 AcQImage Cabinet – Top, Side, Rear Covers Removed / Transition Module

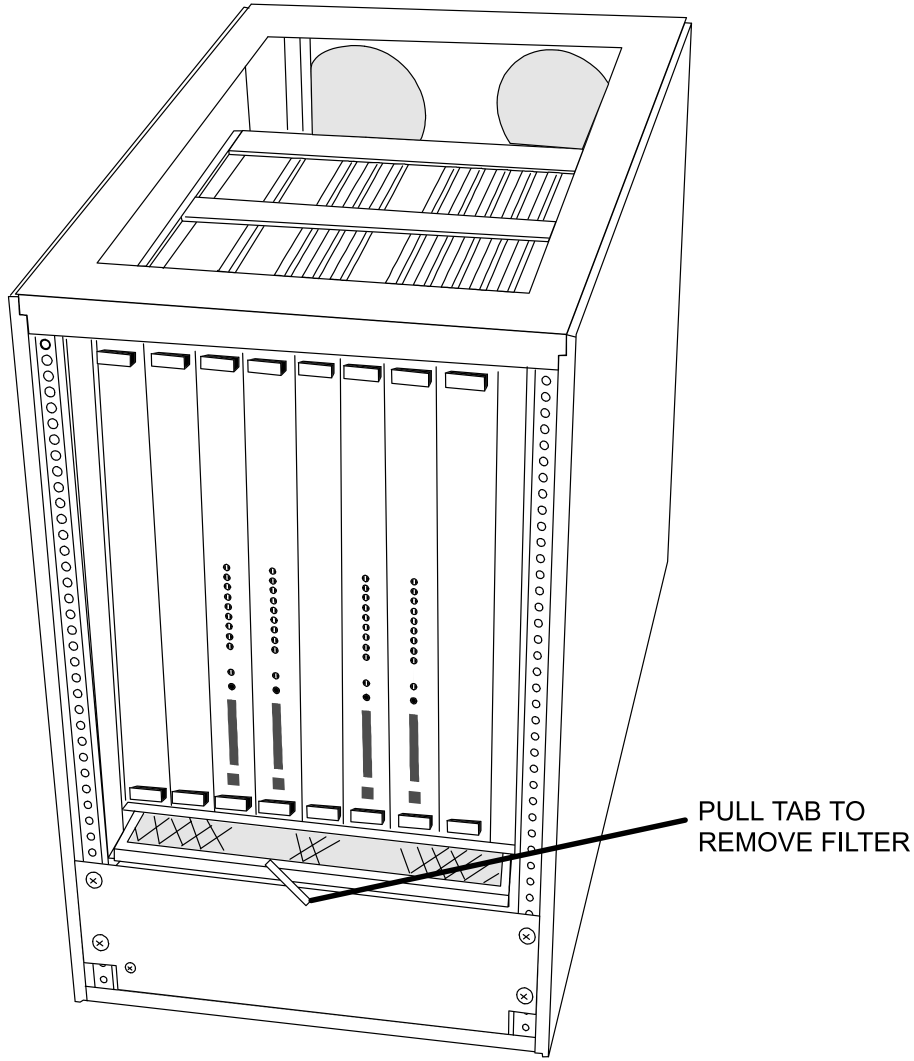


FIGURE 330 AcQImage Cabinet – Inside Covers, Fan Filter Removal

AIR FILTER REMOVAL/REPLACEMENT/CLEANING

1. Grasp the tab on the air filter and slide out to remove. Either replace or wash the filter. Ensure filter is dry before sliding back into the AcQImage cabinet. Refer to Figure 330. (Internal cover does not have to be removed to service the air filter.)

TOP COVER REPLACEMENT:

1. Position the rear of the top cover over the two screw holes.
2. Slide the cover forward until it seats. Install and tighten the two screws removed earlier.

SIDE COVER REPLACEMENT:

1. Insert the three upper tabs into the slots and swing the bottom portion in toward the tower.
2. Install the two screws removed earlier.

REAR COVER REPLACEMENT:

1. Ensure any cables removed earlier are put back.
2. While holding the brass ground lug, slide the top portion up under the frame and align the screw holes.
3. Install the eight screws removed earlier.

FRONT COVER REPLACEMENT (INTERNAL):

1. Align the steel inner panel with the six screw holes and install the six screws removed earlier.

FRONT COVER REPLACEMENT (EXTERNAL):

1. Line up the ball stud and spring clips of the top decorative cover and push in until you feel them “pop” in. Repeat this with the bottom portion.

ACQIMAGE CABINET FAN ASSEMBLY

REMOVAL / REPLACEMENT

Introduction:	The following procedure is used to remove and replace the fan assembly in the event of a failure.
Tools Required:	Phillips Screwdriver
Estimated Time:	10 Min.

FAN ASSEMBLY REMOVAL:

1. Remove the AcQImage Cabinet power. Refer to the PowerUp/PowerDown procedure.

> ⚠ **WARNING** ⚠
>
> **USE EXTREME CAUTION WHEN OPERATING OR WORKING ON EQUIPMENT WITH COVERS REMOVED. DANGEROUS VOLTAGES ARE EXPOSED WHEN THE COVERS ARE REMOVED. FAILURE TO COMPLY CAN CAUSE SERIOUS INJURY OR DEATH TO PERSONNEL.**

> **NOTE**
>
> If the system needs to be moved, release the lock on the wheels before moving. Ensure wheels are locked after moving.

2. Remove the lower front decorative cover and left side cover first. Refer to the AcQImage Cabinet Covers Removal procedure.
3. From the left side, follow the red and black wires from the fan assembly strain relief and unplug from the bottom of the backplane. Remove the ground wire from the internal ground stud.
4. From the front, remove the 4 screws that hold the fan assembly in place. Slide the entire fan assembly out. Refer to Figure 331.

FAN ASSEMBLY REPLACEMENT:

1. From the front, feed the red, black and ground wires into the assembly. Slide the assembly part way in.
2. From the side, pull the wires so they do not get caught under the fan assembly. Pull the fan assembly back and rest it up on the small ledge.

3. Attach the connector and ground wires removed earlier. From the front, replace the four screws. Secure the front decorative cover and side cover. Refer to the AcQImage Cabinet Covers Removal procedure.

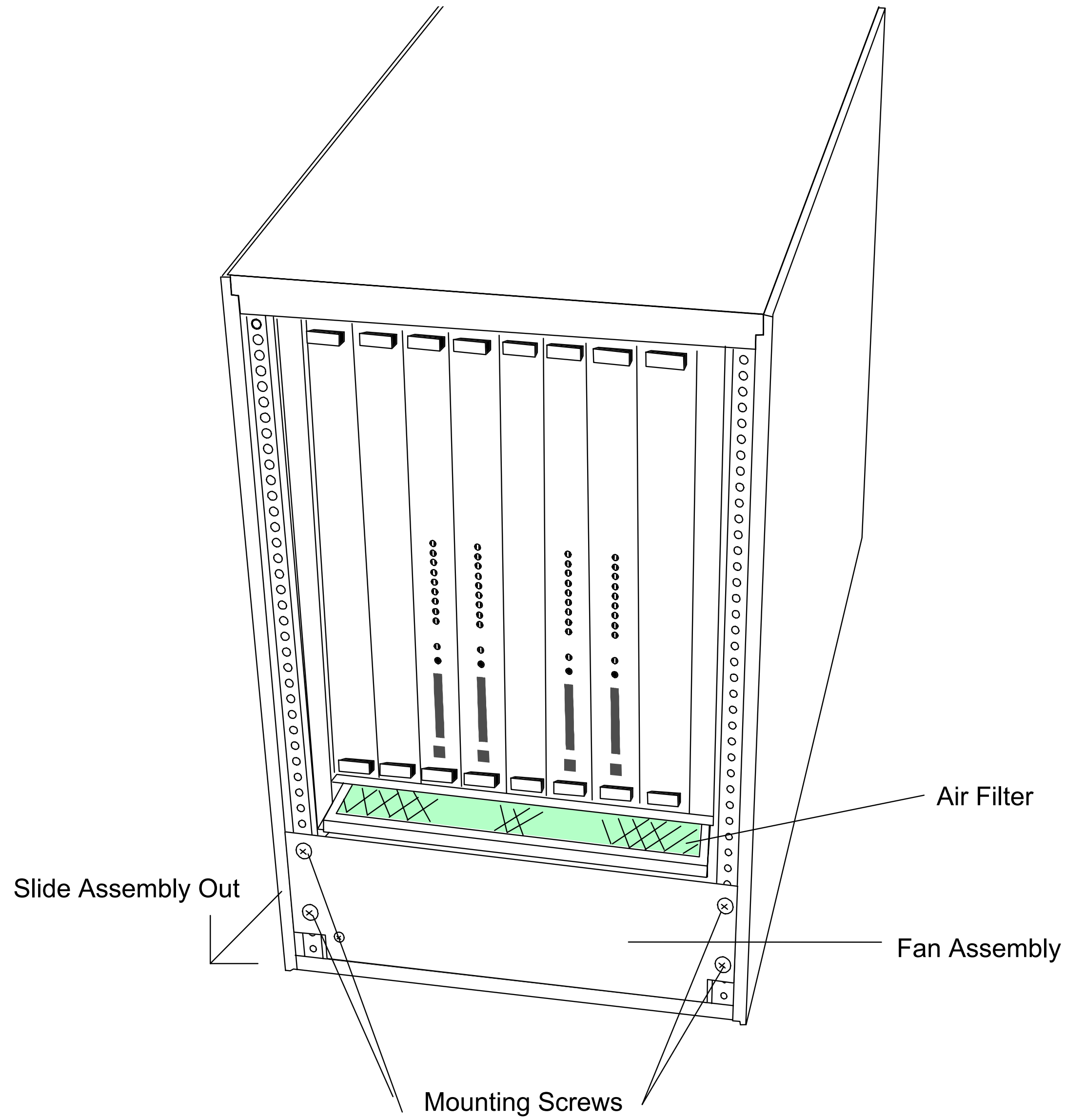


FIGURE 331 Fan Assembly Location

ACQIMAGE CABINET HARD DISK ASSEMBLY

REMOVAL / REPLACEMENT

Introduction:	The following procedure is used to remove and replace a hard disk assembly in the event of a failure.
Tools Required:	Phillips Screwdriver
Estimated Time:	15 Min.

FAN ASSEMBLY REMOVAL:

1. Remove the AcQImage Cabinet power. Refer to the Power Up/Power Down procedure.

> ⚠ **WARNING** ⚠
>
> **USE EXTREME CAUTION WHEN OPERATING OR WORKING ON EQUIPMENT WITH COVERS REMOVED. DANGEROUS VOLTAGES ARE EXPOSED WHEN THE COVERS ARE REMOVED. FAILURE TO COMPLY CAN CAUSE SERIOUS INJURY OR DEATH TO PERSONNEL.**

> **NOTE**
>
> If the system needs to be moved, release the lock on the wheels before moving. Ensure wheels are locked after moving.

2. Remove the front decorative covers. Refer to the AcQImage Cabinet Covers Removal procedure.
3. Remove the front steel panel by removing the six screws. The hard disk assembly is located on the far right board.
4. Disconnect the SCSI cable attached to the GAM board. Refer to Figure 332.
5. Loosen the four captive screws that secure the hard disk assembly in place. Grasp the handles and the slide the assembly part way out. Refer to Figure 332.
6. Disconnect the power and SCSI connectors from the drive being replaced.
7. While holding the drive, loosen the four screws that secure the drive to the slide–out frame.

HARD DISK ASSEMBLY REPLACEMENT:

1. The drive may need to be configured. Refer to the Peripherals Installation and Configuration manual.
2. Bolt the new drive in where the old one was.
3. Attach the power and SCSI cables to the new drive.
4. Slide the assembly back into the card cage and secure the captive screws.

NOTE

It may be necessary to “dress” the power cable back into place. To do this, remove the right side cover and pull excess cable out of card cage as disk assembly is pushed back into place.

5. Plug the SCSI cable back into the GAM board.
6. Install the internal steel cover and decorative covers removed earlier.

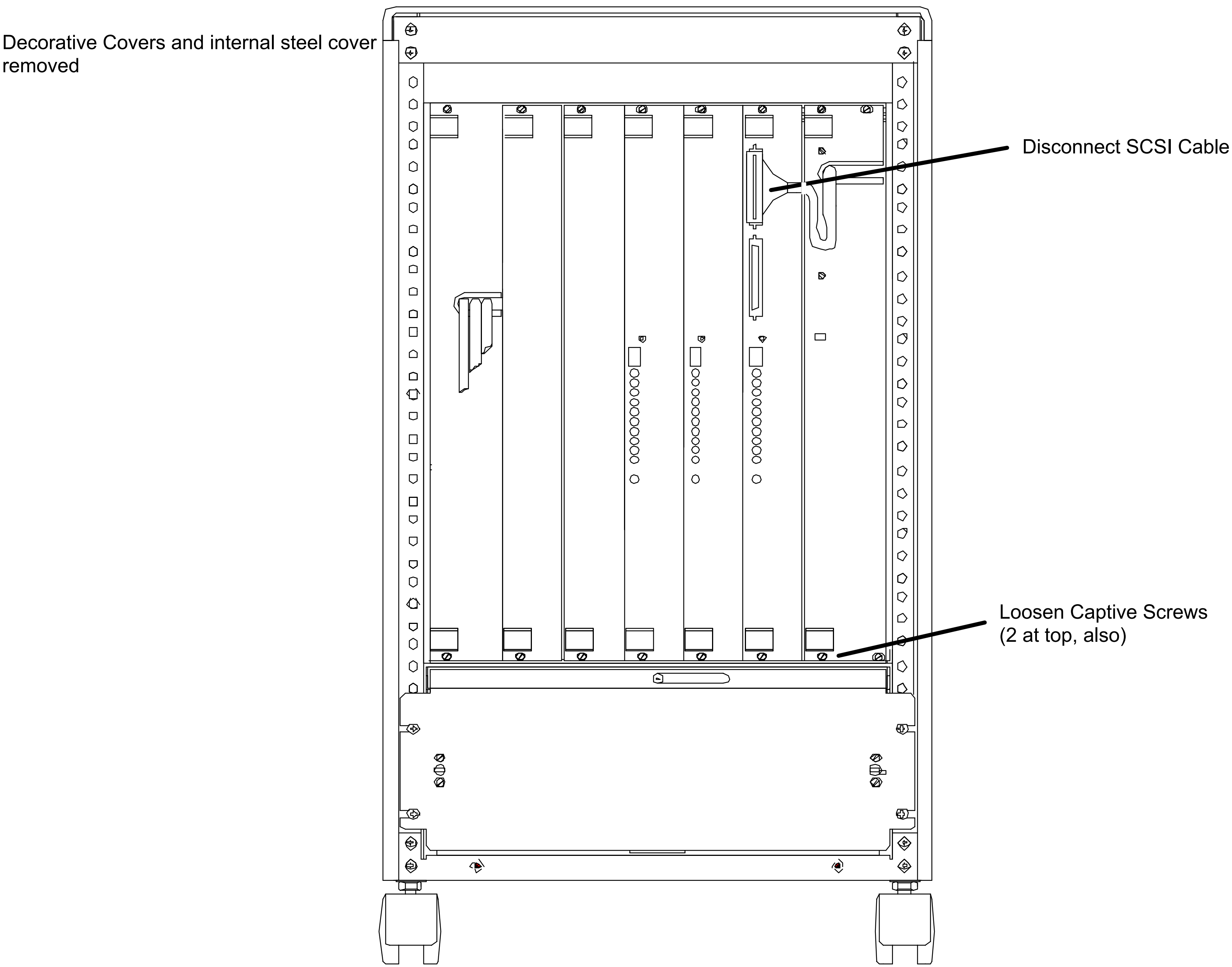


FIGURE 332 AcQImage Cabinet hard Disk Assembly Location

ACQIMAGE CABINET HUB ASSEMBLY

REMOVAL / REPLACEMENT

Introduction: The following procedure is used to remove and replace the AcQImage Cabinet hub assembly in the event of a failure.

Tools Required: Phillips Screwdriver, slotted screwdriver, 3/8” Nutdriver

Estimated Time: 15 Min.

HUB REMOVAL:

1. Remove the AcQImage Cabinet power. Refer to the PowerUp/PowerDown procedure.

> ⚠ **WARNING** ⚠
>
> **USE EXTREME CAUTION WHEN OPERATING OR WORKING ON EQUIPMENT WITH COVERS REMOVED. DANGEROUS VOLTAGES ARE EXPOSED WHEN THE COVERS ARE REMOVED. FAILURE TO COMPLY CAN CAUSE SERIOUS INJURY OR DEATH TO PERSONNEL.**

> **NOTE**
>
> If the system needs to be moved, release the lock on the wheels before moving. Ensure wheels are locked after moving.

2. Remove the front decorative cover and internal steel cover. Remove the top cover. Refer to the [AcQImage Cabinet Covers Removal](#) procedure. The Hub assembly is located on the far left side of the card cage. Refer to Figure 333.
3. Loosen the captive screws that secure the hub assembly in place. Grasp the handles and slide the assembly out part way.
4. Remove the blank panel to the right of the Hub assembly for easier access. Disconnect and label all of the ethernet cables from the Hub assembly.
5. Disconnect the power cable and ground cable fromt the Hub assembly.
6. Cut the tie–wrap that secures the power cable to the bottom of the Hub assembly.
7. Slide the entire Hub assembly out of the card cage.
8. On a secure surface, remove the four bracket nuts that hold the Hub to the slide–out panel.

HUB ASSEMBLY REPLACEMENT:

1. Place the new Hub assembly where the old one was removed. Secure with the bracket and four nuts.
2. Slide the Hub assembly back into the card rack part way. Attach a new tie–wrap to secure the power cord to the bottom of the Hub assembly.
3. Reconnect the power and ground cables to the Hub assembly.
4. Make sure all external switch settings match those on the original hub.
5. Reconnect the ethernet cables to the Hub. Install the blank panel removed earlier.
6. Slide the Hub assembly in and secure the captive screws.
7. Attach the steel cover with the six screws. Attach the top and bottom decorative covers.

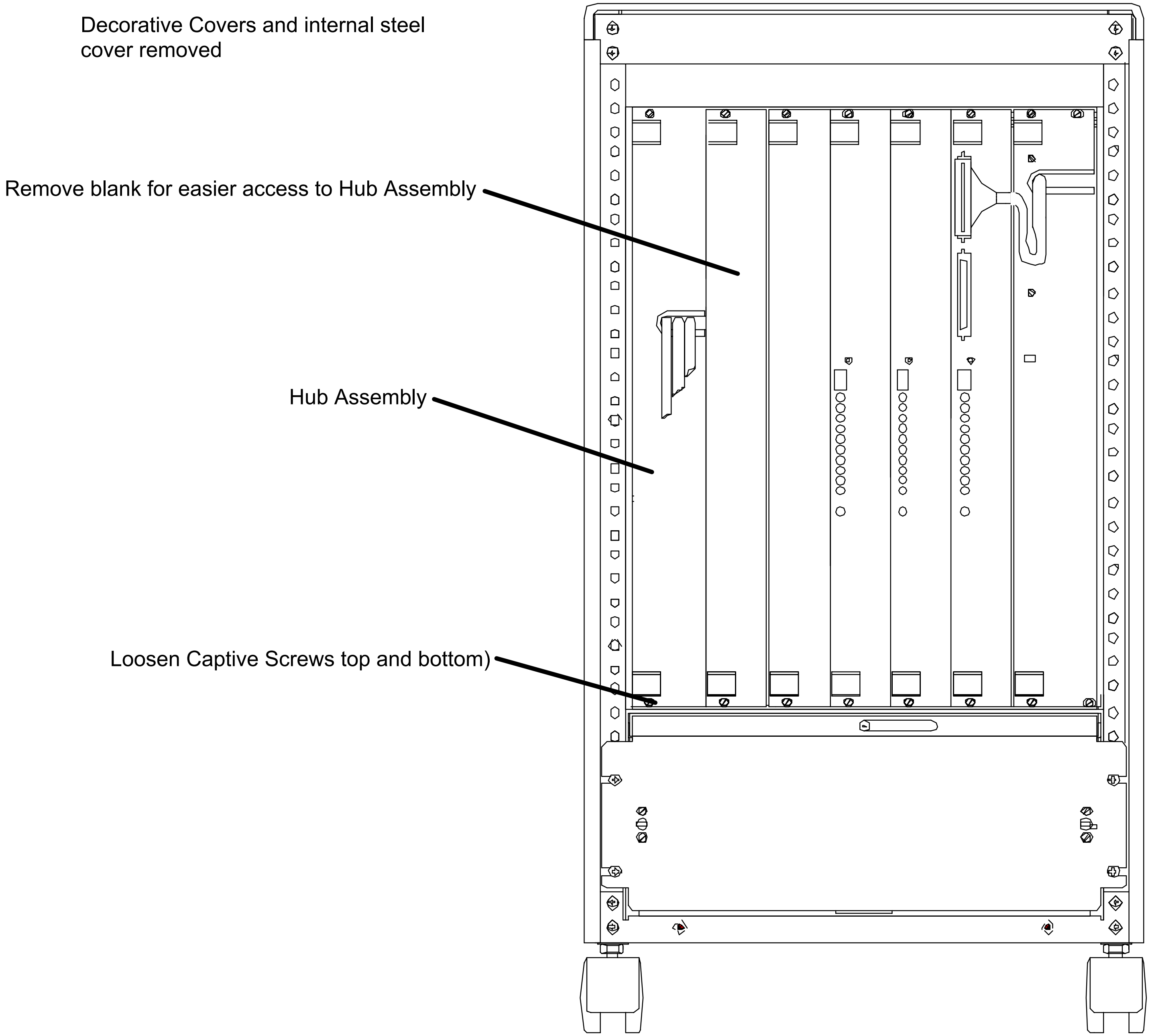


FIGURE 333 Hub Assembly Location

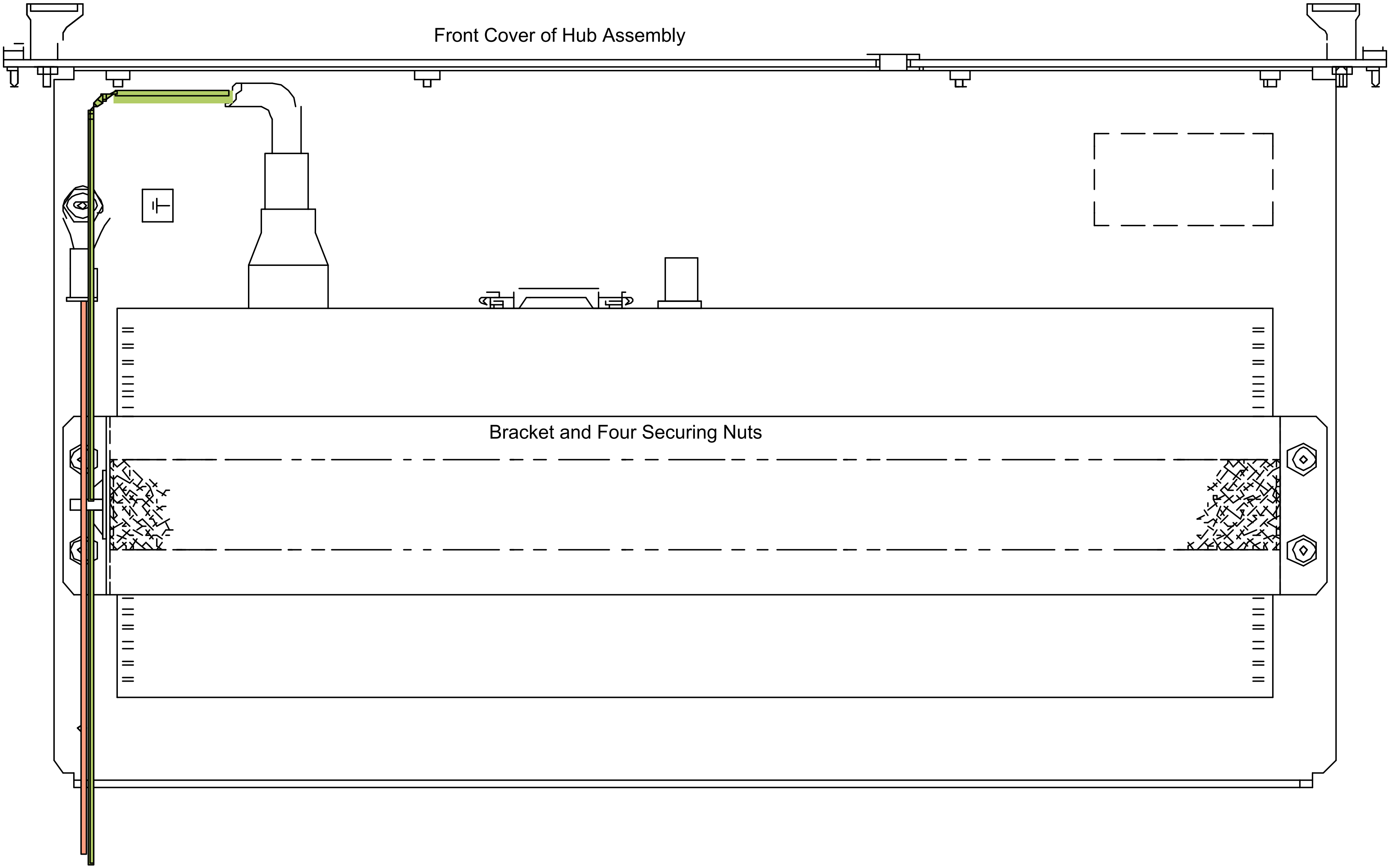


FIGURE 334 Detail of Hub Assembly

ACQIMAGE CABINET BOARDS

REMOVAL / REPLACEMENT

Introduction:	The following procedure is used to remove and replace any of the AcQImage boards in the event of a failure.
Tools Required:	Phillips Screwdriver
Estimated Time:	20 Min.

BOARD REMOVAL:

1. Remove the AcQImage Cabinet power. Refer to the PowerUp/PowerDown procedure.

> ⚠ **WARNING** ⚠
>
> **USE EXTREME CAUTION WHEN OPERATING OR WORKING ON EQUIPMENT WITH COVERS REMOVED. DANGEROUS VOLTAGES ARE EXPOSED WHEN THE COVERS ARE REMOVED. FAILURE TO COMPLY CAN CAUSE SERIOUS INJURY OR DEATH TO PERSONNEL.**

> **NOTE**
>
> If the system needs to be moved, release the lock on the wheels before moving. Ensure wheels are locked after moving.

2. The GAM, AP and BP boards may be in different slot locations, with the most common configuration being from left right: BP, AP and GAM. When the boards are properly booted, the LED status indicator on the front will read “4” for BP, “3” for AP, and “2” for GAM. Refer to Figure 335.

3. Loosen the top and bottom captive screws from the board you wish to remove. In order to remove the CP, the Hub Assembly must be removed first. Refer to the Hub Assembly Removal procedure. Refer to Figure 335. Remove and label any cables attached to the CP board.

4. Push up on the top tab and down on the bottom tab to release the board. Carefully slide the board out until it is free of the enclosure.

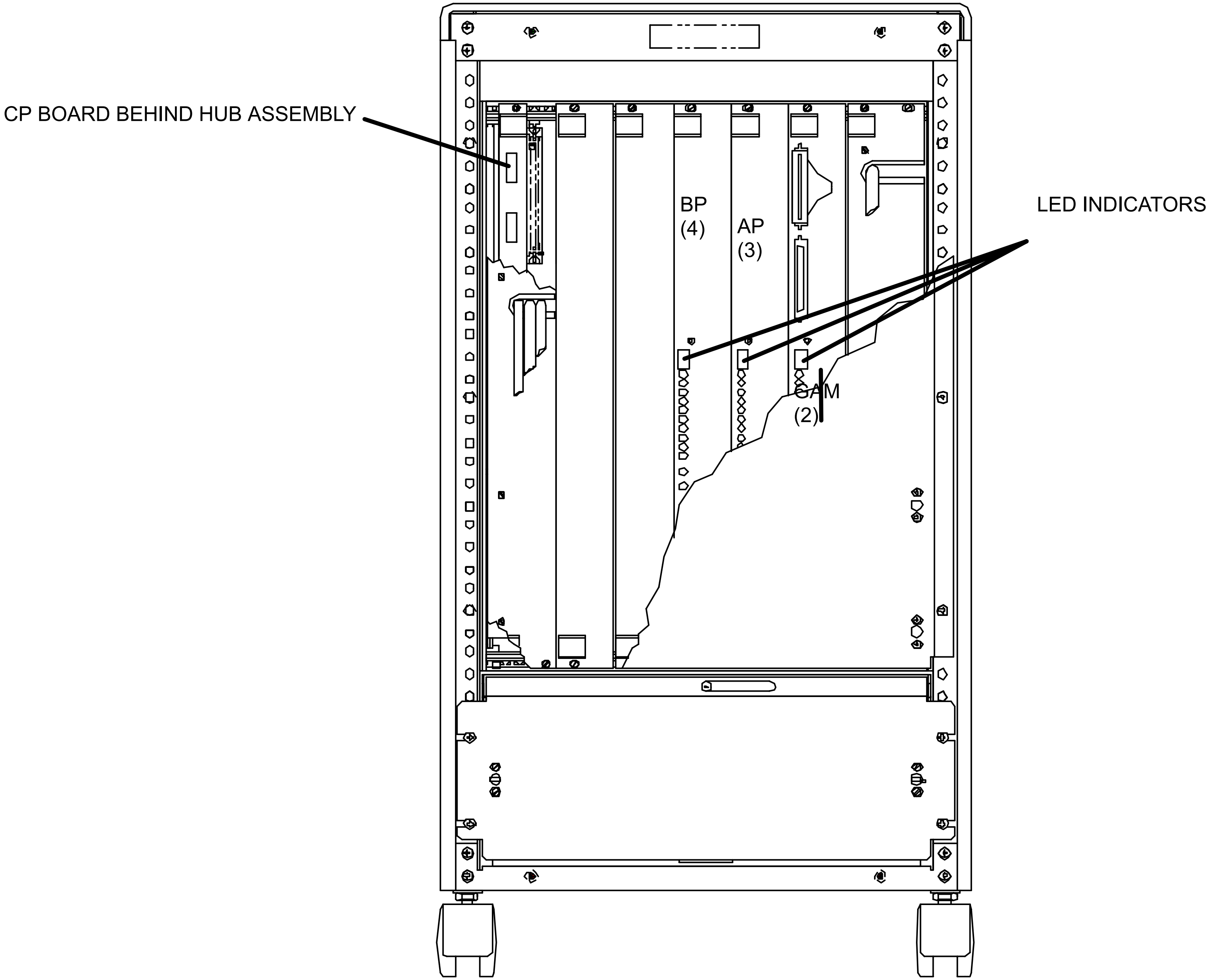


FIGURE 335 ACQIMAGE CABINET BOARD LOCATIONS

BOARD REPLACEMENT:

> ⚠ **CAUTION**
>
> THE PLASTIC BOARD GUIDES ARE HELD INTO PLACE BY SMALL PLASTIC TABS AND CAN EASILY BE REMOVED. ENSURE THAT EACH OF THE GUIDES IS SEPARATED BY TWO HOLES. MISALIGNMENT OF THE GUIDES CAN LEAD TO BOARD/ VME BACKPLANE DAMAGE. REFER TO FIGURE 336.

1. Slide the new board in where the old one was removed and secure the board with the captive screws.
2. Re–attach any cables removed earlier from the old CP.
3. Refer to the Hub Assembly Replacement procedure to put the Hub Assembly back.

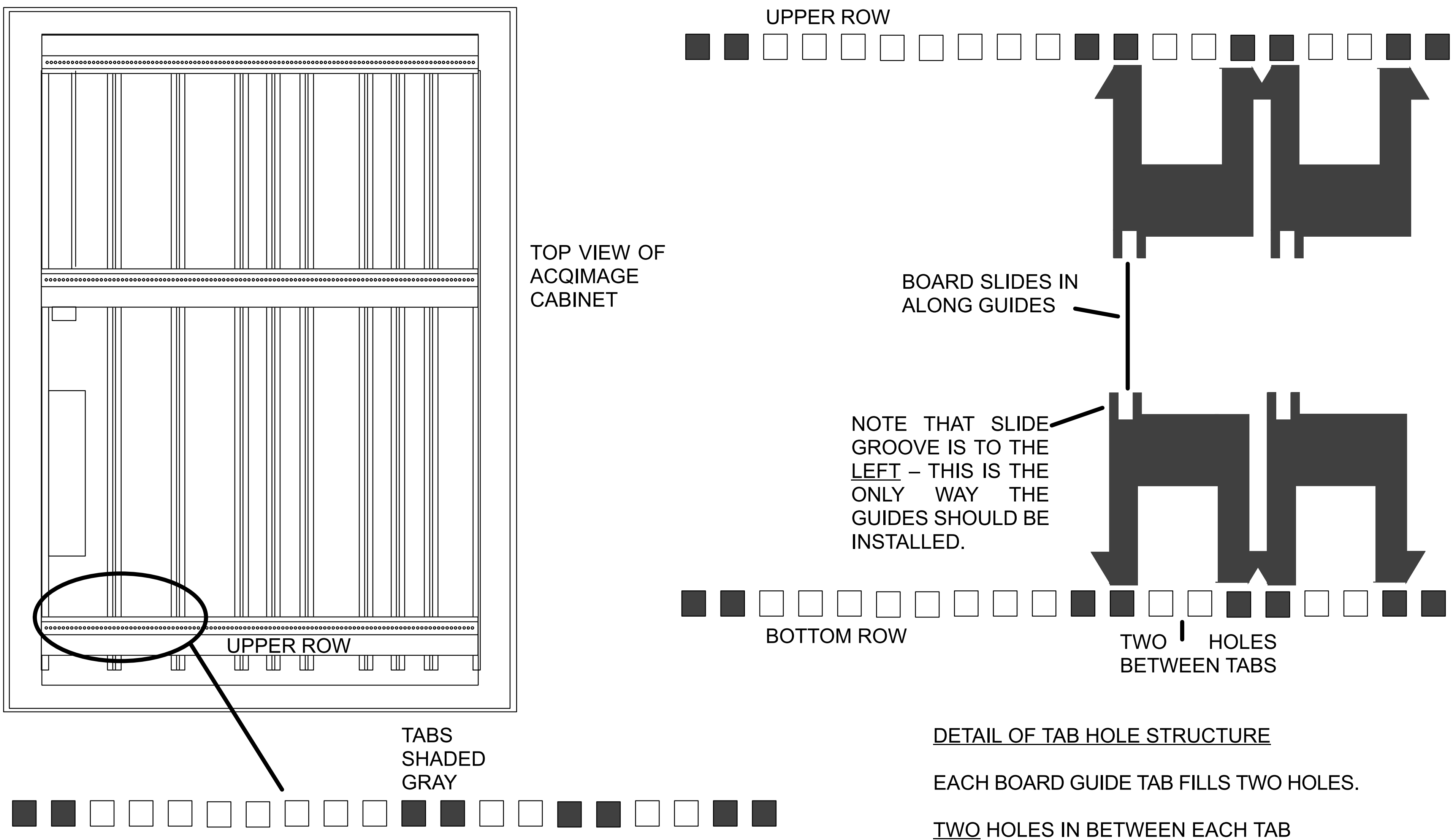


FIGURE 336 PROPER BOARD GUIDE ALIGNMENT

ACQIMAGE CABINET AC DISTRIBUTION CHASSIS COMPONENTS

REMOVAL / REPLACEMENT

Introduction: The following procedure is used to remove and replace AC distribution Chassis components in the event of a failure.

Tools Required: Phillips Screwdriver

Estimated Time: 25 Min.

AC DISTRIBUTION REMOVAL:

1. Remove the AcQImage Cabinet power. Refer to the PowerUp/PowerDown procedure.

> ⚠ **WARNING** ⚠
>
> **USE EXTREME CAUTION WHEN OPERATING OR WORKING ON EQUIPMENT WITH COVERS REMOVED. DANGEROUS VOLTAGES ARE EXPOSED WHEN THE COVERS ARE REMOVED. FAILURE TO COMPLY CAN CAUSE SERIOUS INJURY OR DEATH TO PERSONNEL.**

> **NOTE**
>
> If the system needs to be moved, release the lock on the wheels before moving. Ensure wheels are locked after moving.

2. Remove the rear cover. Refer to the Rear Cover Removal procedure.
3. Disconnect the gray cable that leads to the transition module located on Slot 1 (right side) of J2. Remove the three line cords, two logic cables and the green/yellow cable from the top of the AC Distribution Chassis.
4. Swing the rear panel out as shown in Figure 337.
5. Remove the six screws from the cover of the AC distribution box. Turn the rear panel as required to get all of the screws out.
6. Once the AC distribution box is open, any or all of the components can be removed and replaced by simply removing their respective nuts and bolts. Tag and remove any cables associated with the component to be replaced.

AC DISTRIBUTION BOX REPLACEMENT:

1. Once the component(s) has/have been removed and replaced, install the cover and secure with the six screws.
2. Ensure that any cables removed earlier are plugged back in.
3. Install the rear cover. Refer to the Rear Cover Replacement procedure.
4. Ensure that the ground cable attached to the AC distribution box is tight.

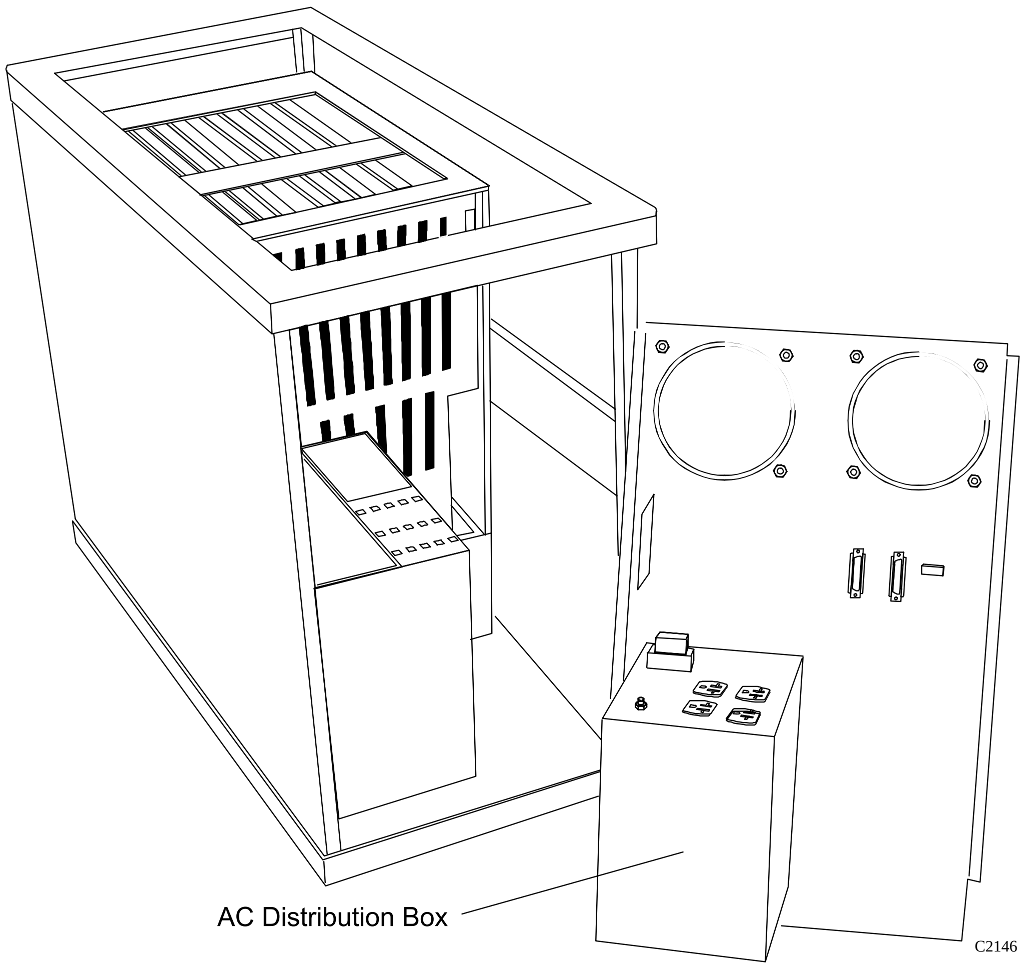


FIGURE 337 AC Distribution Box Location

ACQIMAGE CABINET POWER SUPPLY ASSEMBLY

REMOVAL / REPLACEMENT

Introduction: The following procedure is used to remove and replace the new power supply assembly (362402) or remove the older power supply assemblies (178826 or 178221) in the event of a failure. Power supply assemblies 178826 and 178221 are obsolete, and should be replaced with power supply assembly 362402. If replacing old power supply assembly 178221, resistor assembly 178834 must also be installed.

Tools Required: Phillips screwdriver, slotted screwdriver

Estimated Time: 35 Min.

POWER SUPPLY REMOVAL (APPLICABLE TO ALL VERSIONS):

1. Remove system power. Refer to the PowerUp/PowerDown procedure.

> ⚠ **WARNING** ⚠
>
> **USE EXTREME CAUTION WHEN OPERATING OR WORKING ON EQUIPMENT WITH COVERS REMOVED. DANGEROUS VOLTAGES ARE EXPOSED WHEN THE COVERS ARE REMOVED. FAILURE TO COMPLY CAN CAUSE SERIOUS INJURY OR DEATH TO PERSONNEL.**

> **NOTE**
>
> If the system needs to be moved, release the lock on the wheels before moving. Ensure wheels are locked after moving.

2. Remove the rear and right side covers. Refer to the AcQImage Cabinet Cover Removal procedure.. (Right side when facing the front of the AcQImage Cabinet.)
3. Disconnect the Signal I/O panel cable (gray cable no. L11048) plug P1197 from transition module 178155 connector J1197 located on the right side of the backplane. Swing the rear panel out as shown in Figure 338.

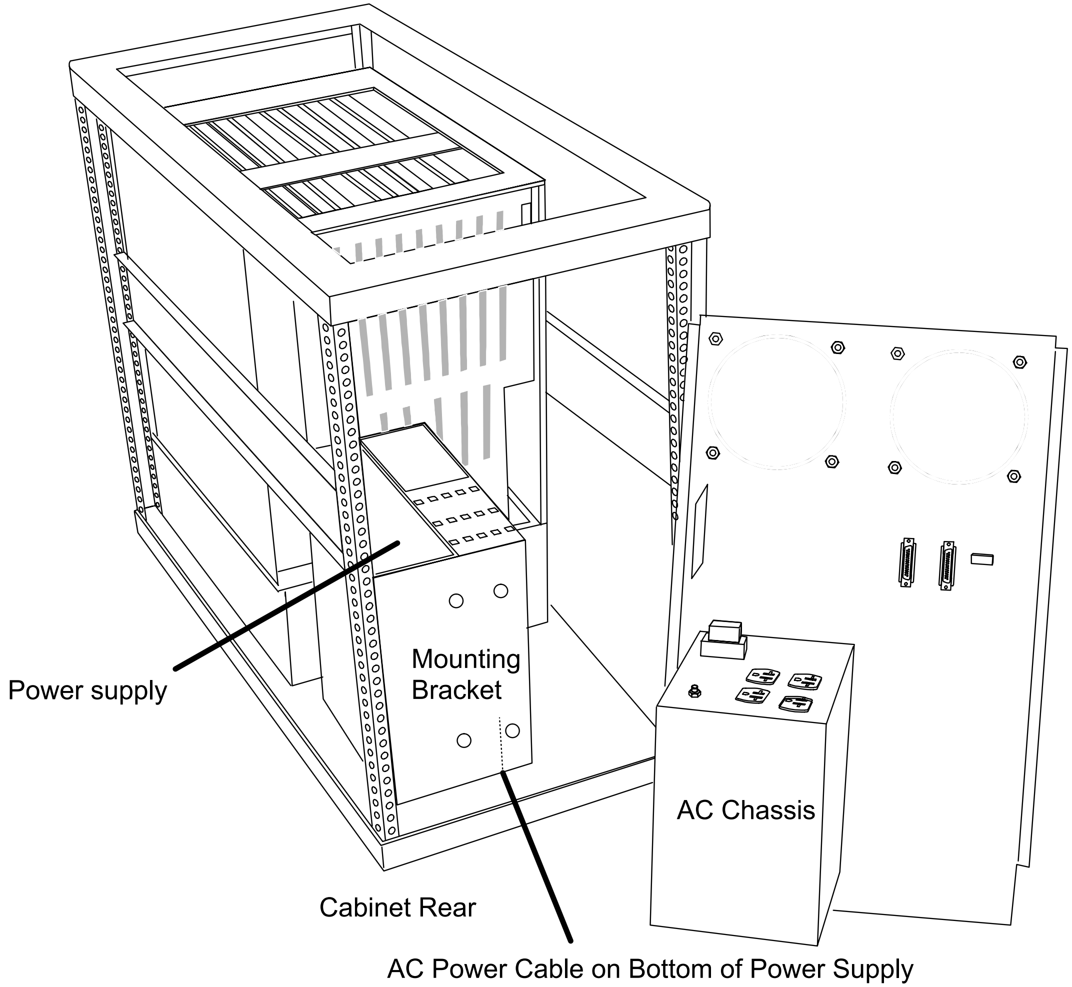


FIGURE 338 Power Supply Location on AcQImage Cabinet

OLDER POWER SUPPLY ASSEMBLY REMOVAL (178221):

In the event of an older power supply assembly failure (178221) a new assembly (362402) should replace the entire older assembly. NOTE: *Due to the gauge of the wire and size of the terminals of the older supply, the existing power wires cannot be used with the new power supply.*

Therefore, the wires and cabling from the power supply to their destinations must be removed. The new power supply assembly will have the proper wires attached at the supply side. You will have to route the supply wires to their proper destination. Refer to Figure 339 for backplane detail.

1. If you have not done so, perform the "Power Supply Removal" procedure from the beginning of this section. The power supply is located on the far left side when facing the rear of the tower.
2. Disconnect all of the cables from the backplane connections of the power supply assembly. Unplug the power cord that leads to the AC Chassis plug J1107.
3. Disconnect connector J1 of cable L11046 from the top of the backplane. Disconnect the black and white wires on the other end from their backplane destinations (BB3–3 and BB4–2). Carefully clip the tie wraps that secure L11046 and remove it.
4. Remove P1194 from J1194 on the AC Distribution Box.
5. Loosen and remove the seven nuts that secure the power supply assembly mounting bracket to the bottom of the cabinet. Lift the entire assembly out of the cabinet.

OLDER POWER SUPPLY ASSEMBLY REMOVAL (178826):

In the event of an older power supply assembly failure (178826) a new assembly (362402) should replace the entire older assembly. NOTE: *Due to the gauge of the wire and size of the terminals of the older supply, the existing power wires cannot be used with the new power supply.*

Therefore, the wires and cabling from the power supply to their destinations must be removed. The new power supply assembly will have the proper wires attached at the supply side. You will have to route the supply wires to their proper destination. Refer to Figure 339 for backplane detail.

1. If you have not done so, perform the "Power Supply Removal" procedure from the beginning of this section. The power supply is located on the far left side when facing the rear of the tower.
2. Disconnect all of the cables from the backplane connections of the power supply assembly. Unplug the power cord that leads to the AC Chassis plug J1107.

178834 Resistor Cable – From J3 BB to GND BB

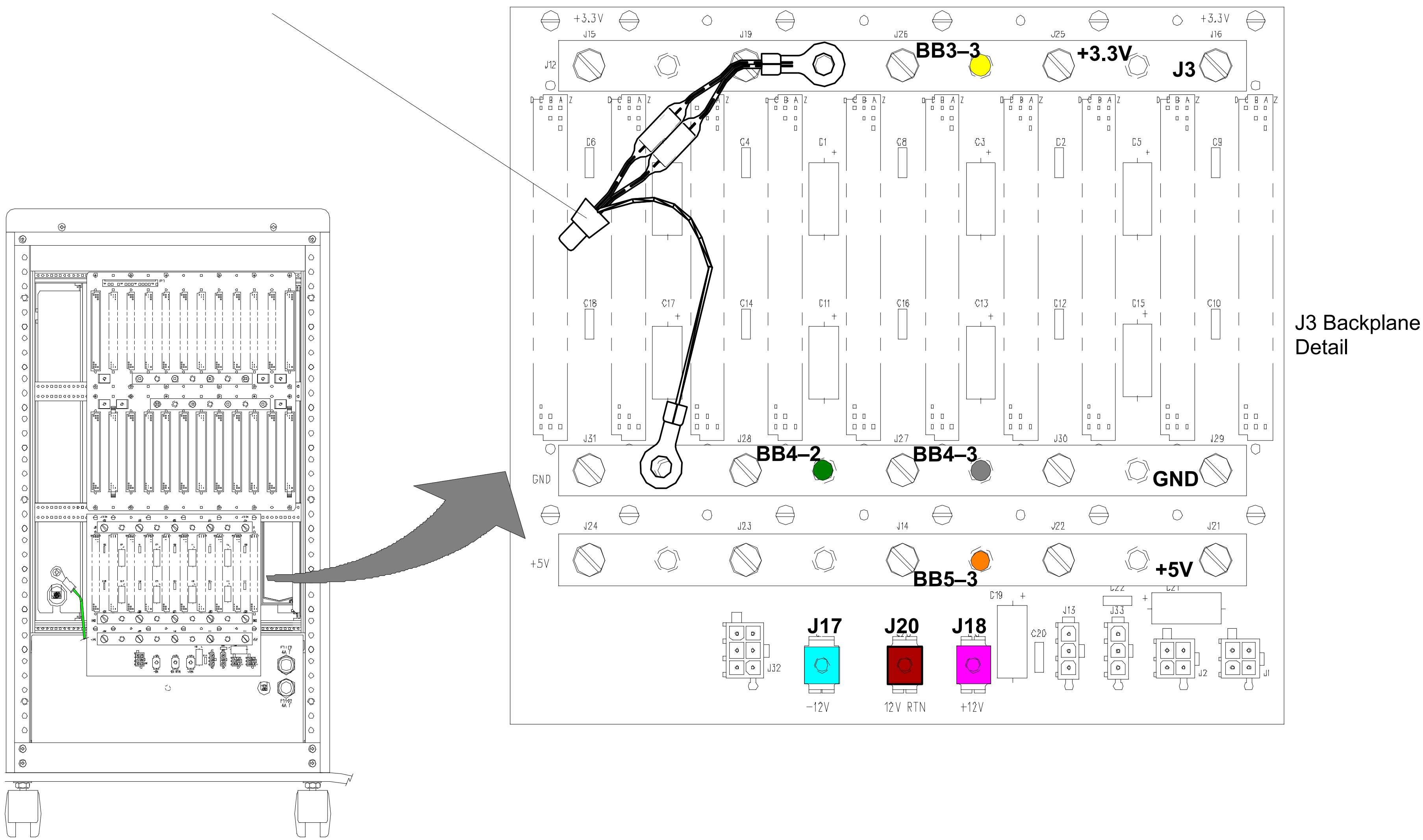


FIGURE 339 RESISTOR ASSEMBLY 178834 INSTALLATION
(ONLY PERFORMED IF REPLACING POWER SUPPLY ASSEMBLY 178221).

3. Disconnect connector J1 of cable L11581 from connector P1 on the top of the backplane. Carefully clip the tie wraps that secure L11581 and remove it.
4. Loosen and remove the seven nuts that secure the power supply assembly mounting bracket to the bottom of the cabinet. Lift the entire assembly out of the cabinet.

NEW POWER SUPPLY ASSEMBLY REMOVAL (362402):

1. If you have not done so, perform the "Power Supply Removal" procedure from the beginning of this section. The power supply is located on the far left side when facing the rear of the tower.
2. Disconnect all of the cables from the backplane connections of the power supply assembly. Refer to Figure 339 for backplane detail. Unplug the power cord that leads to the AC Chassis plug J1107.
3. Remove connector J1 of cable L22680 from connector P1 on the top of the backplane. Carefully clip the tie wraps that secure L22680 and remove it.
4. Loosen and remove the seven nuts that secure the power supply assembly mounting bracket to the bottom of the cabinet. Lift the entire assembly out of the cabinet.

NEW POWER SUPPLY ASSEMBLY INSTALLATION (362402):

1. Install power supply cover plate 314117 over the standoffs at the bottom of the cabinet. Lower the power supply assembly mounting bracket 362402 onto the cover plate with exhaust fan positioned in cover plate cutout and secure with the seven nuts removed earlier.
2. If replacing old power supply assembly 178221, install the resistor assembly 178834 on backplane as shown in Figure 339. Resistor assembly 178834 is already installed if power supply assemblies 178826 or 362402 are being replaced.
3. Install cable L22679 on the backplane as follows (refer to Figure 339 for backplane detail):
   3.3 V RTN (three black leads) to BB4–2, GND Bus Bar
   +3.3 V (three white leads) to BB3–3, +3.3 V Bus Bar
   +12 V (two blue leads) to J18, +12 V
   12 V RTN (two yellow leads) to J20, 12 V RTN
   –12 V (two gray leads) to J17, –12 V
4. Install cable L22678 on the backplane as follows:
   BKPLN – (black lead) to BB4–3, GND Bus Bar
   BKPLN + (red lead) to BB5–3, + 5 V Bus Bar

5. Install the AC power cord L22677 that is attached to new power supply into J1107 on the AC Chassis.
6. Install connector J1 of Control cable L22680 into the 12–pin connector P1 on the top of the J1/J2 backplane. Secure with tie wraps. Refer to Figure 340.
7. Install Signal I/O panel cable L11048 plug P1197 to transition module connector J1197 located on the right side of the backplane.
8. Re–install the side and rear covers on the cabinet. Refer to the AcQImage Cabinet Cover Removal procedure.
9. Verify that all connections are correct. Prior to returning the AcQImage Cabinet to service, carefully unplug the GAM, DSP–AP, Projector, and CP cards from the backplane. Remove the DC power connectors to both of the GAM image drives (you will need to unscrew the drive tray from the AcQImage Cabinet to do this).
10. Power up the new supply and verify all voltages at the appropriate terminals and at the test points on the front panel of the GAM drive tray assembly. Note that before an accurate voltage measurement can be taken on the new power supply, it may be necessary to introduce a nominal load to the +5V section (eg. 5. , 5W resistor) in order to bring the entire supply into regulation.
11. Assuming all the supply stages are present and within tolerance (+5V, +3.3V, +12V, and –12V), remove power to the AcQImage Cabinet, reinstall the PCB's into the backplane, and reconnect the GAM drive power connectors.

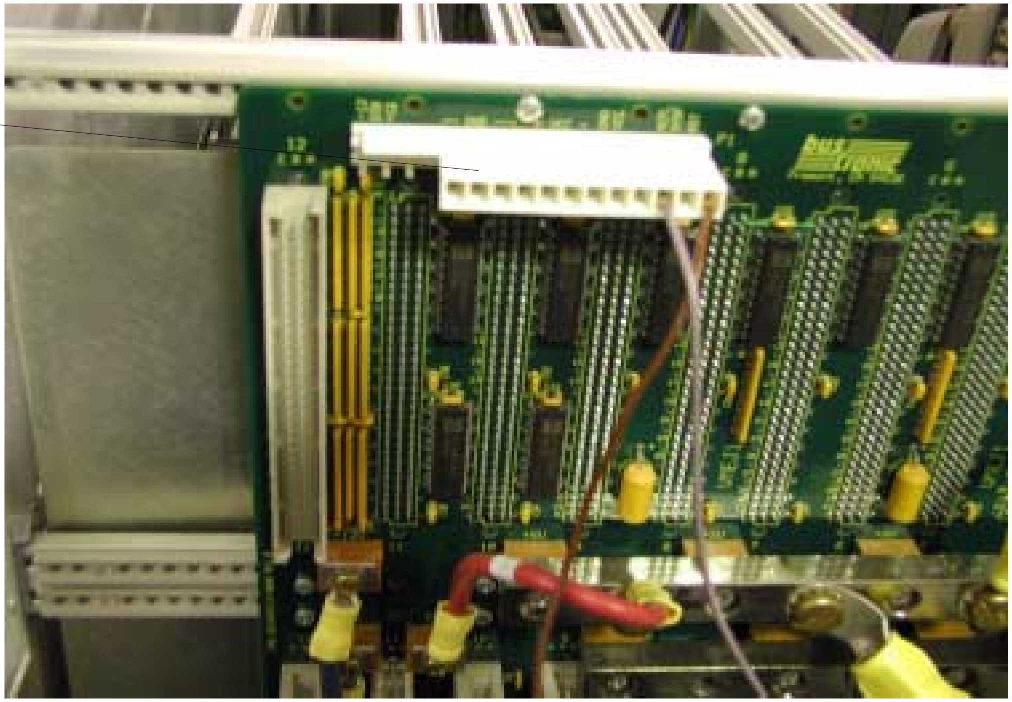


FIGURE 340 CONTROL CABLE L22680 CONNECTION TO J1/J2 BACKPLANE

UZX DISPLAY TOWER COVERS

REMOVAL / REPLACEMENT

Introduction:	The following procedure is used to open, reinstall or replace the Display Tower covers.
Tools Required:	Phillips screwdriver
Estimated Time:	10 min.

1. Remove Display Tower power. Refer to the PowerUp/PowerDown procedure. **Refer to Figure 342 for all of the cover removal / replacement procedures.**

WARNING

USE EXTREME CAUTION WHEN OPERATING OR WORKING ON EQUIPMENT WITH COVERS REMOVED. DANGEROUS VOLTAGES ARE EXPOSED WHEN THE COVERS ARE REMOVED. FAILURE TO COMPLY CAN CAUSE SERIOUS INJURY OR DEATH TO PERSONNEL.

NOTE

If the system needs to be moved, release the lock on the wheels before moving. Ensure wheels are locked after moving.

FRONT COVER REMOVAL:

1. Refer to Figure 341. The decorative front cover is held on with ball stud and spring clips. Remove the key(s) first. Grasp the hand hold in the center bottom and pull outward then pull on the sides to remove the entire front decorative cover.
2. The decorative cover removed here is only cosmetic. To access the disk drives and the lower front portion, you must remove the lower front steel panel by removing the six screws. Refer to Figure 342.

TOP COVER REMOVAL:

1. (The front cover does not have to be removed) Remove the two Phillips screws from the upper rear of the top cover.
2. Slide the cover toward the rear a little and lift the cover upward.

SIDE COVER REMOVAL:

1. (Front decorative cover must be removed first for the left panel.) Remove the two lower Phillips screws from the bottom of the side cover.
2. Lift up slightly from the bottom and pull the bottom of the cover toward you. The top will follow after the small tabs have cleared the frame.

REAR COVER REMOVAL:

1. **Top Rear Cover:** Remove the five Phillips screws and move the cover slightly out, being careful not to pull the fan wires. If you want to move the top panel farther away, push the tab and remove the two power cables from the SPC.

2. **Bottom Rear Cover:** Support the rear lower cover by the brass ground screw and remove the four Phillips screws and place the cover on the floor. The bottom rear cover holds the AC chassis, which can be disconnected with the cable connector.

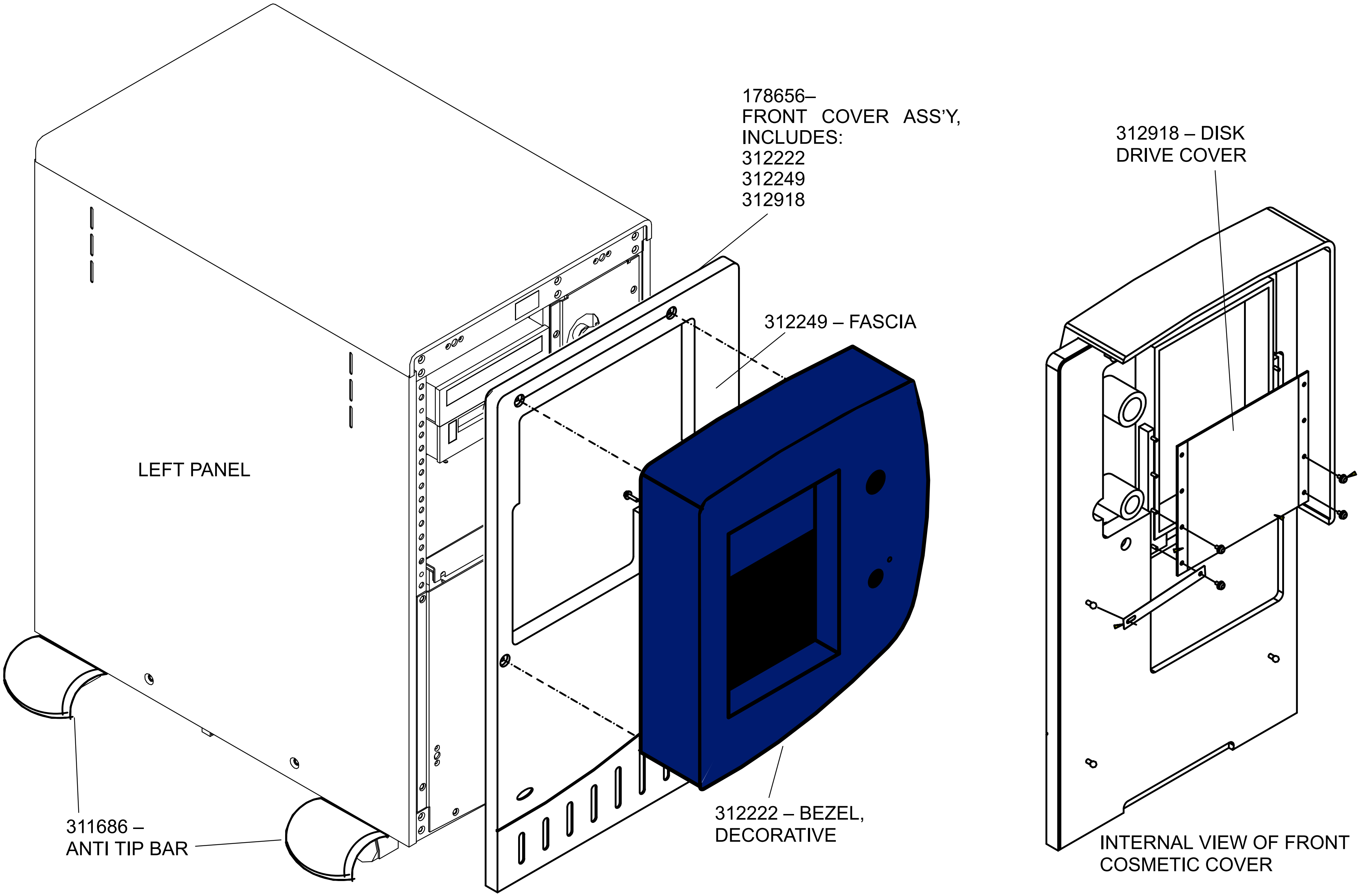


FIGURE 341 COSMETIC COVER REMOVAL

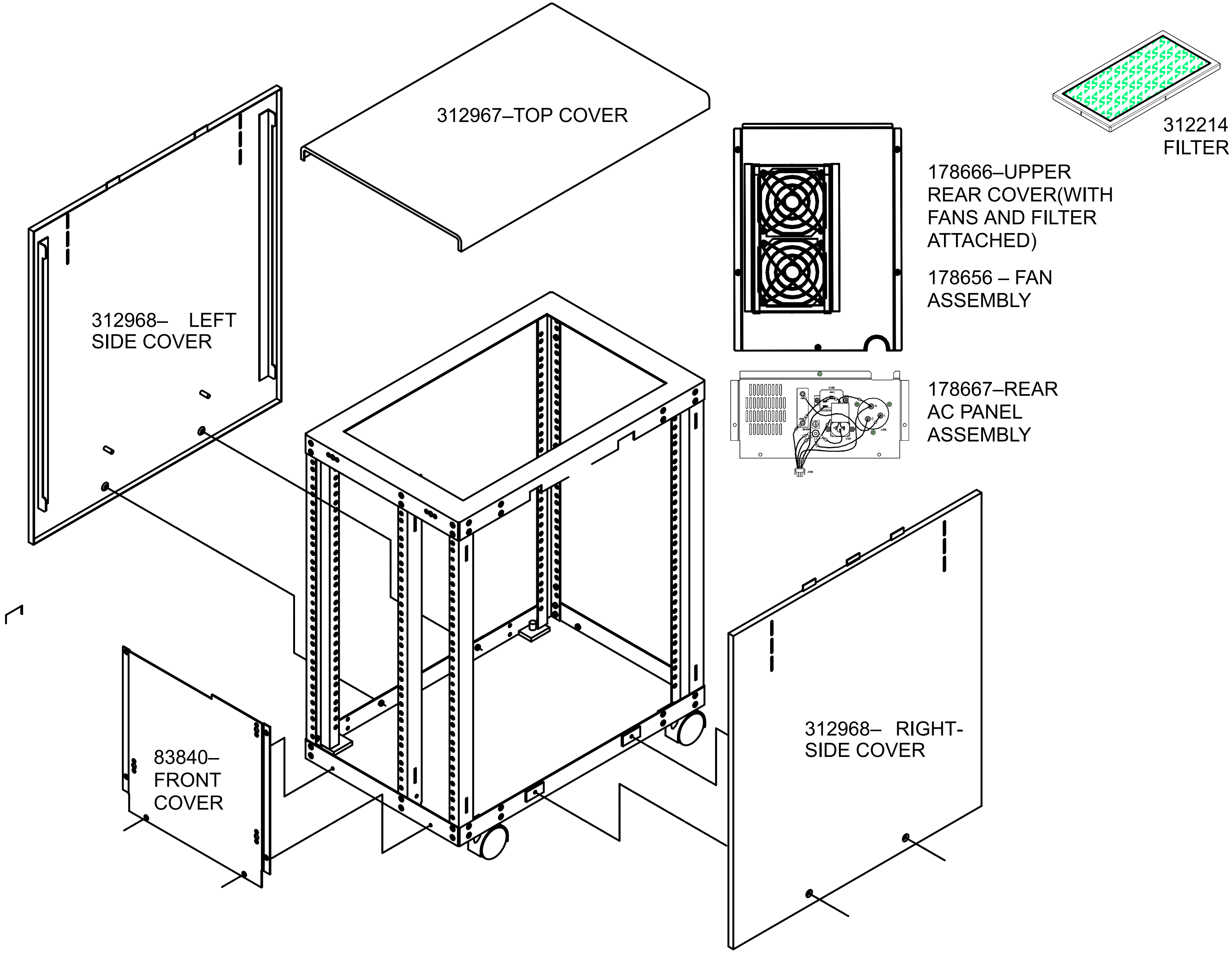


FIGURE 342 DISPLAY TOWER – ALL COVERS REMOVED (Internal Components not Shown)

TOP COVER REPLACEMENT:

1. Position the rear of the top cover over the two screw holes.
2. Slide the cover forward until it seats. Install and tighten the two screws removed earlier.

SIDE COVER REPLACEMENT:

1. Insert the upper tabs into the slots and swing the bottom portion in toward the tower. Align the vertical tabs with the slot on the chassis and push inward.
2. Install the two screws removed earlier.

REAR COVER REPLACEMENT:

1. **Bottom Rear Cover:** The lower rear cover must be put back first. Plug the AC Chassis cable connector back in if removed earlier.
2. Align the cover with the screw holes and install the four screws removed earlier.
3. **Top Rear Cover**:   If the fan power cables were disconnected, plug them back into the SPC.
4. While holding the brass ground lug, slide the top portion up under the frame and align the screw holes.
5. Install the five screws removed earlier.

FRONT COVER REPLACEMENT:

1. Inner steel panel: Align the steel inner panel with the six screw holes and install the six screws removed earlier.
2. Make sure key is removed before installing. Line up the ball stud and spring clips of the top decorative front cover and push in until you feel them “pop” in.

UZX DISPLAY TOWER POWER DISTRIBUTION ASSEMBLY AND ASSOCIATED COMPONENTS

REMOVAL / REPLACEMENT

INTRODUCTION: The following procedure is used to remove and replace the UZX Display Tower power supply and associated boards: Intercom Mixer and Power modules, System Power Controller and AC Chassis in the event of a failure.

TOOLS REQUIRED: Phillips Screwdriver

ESTIMATED TIME: 35 Min.

POWER SUPPLY REMOVAL:

1. Remove the Display Tower power. Refer to the PowerUp/PowerDown procedure.

> ⚠ **WARNING** ⚠
>
> **USE EXTREME CAUTION WHEN OPERATING OR WORKING ON EQUIPMENT WITH COVERS REMOVED. DANGEROUS VOLTAGES ARE EXPOSED WHEN THE COVERS ARE REMOVED. FAILURE TO COMPLY CAN CAUSE SERIOUS INJURY OR DEATH TO PERSONNEL.**

> **NOTE**
>
> If the system needs to be moved, release the lock on the wheels before moving. Ensure wheels are locked after moving.

2. Remove the front cosmetic cover, left side cover and the bottom rear cover. (If the AC Chassis needs to be replaced, it can be done with this procedure.) Lay the AC Chassis on the floor as in Figure 343 or disconnect the cable connector as shown in Figure 344 and move the bottom rear panel out of the way.
3. Remove the top rear cover as shown in Figure 343.
4. To remove the power supply, the power distribution chassis must first be removed. Refer to Figure 345. Several cables must be **tagged and removed** from the Power Distribution assembly:

TAG and REMOVE THE FOLLOWING CABLES TO REMOVE THE PDC:

All peripheral power cables L11057 (J220 through J226 – some may not be used)

Thick gray cable L11029 or ribbon cable L12052 from SPC J1214
(From parallel port on Motherboard)

Push the single tab and remove fan cables P1230, P1231, and P1232
from upper left of the SPC

Push tab and remove power supply cable P1210 from the upper right of the SPC.
Remove multi–wire P1213 (Estop/K'switch/LED Assy)

Gray cable P1270 that comes from the Intercom board

Multi–wire power cable P1901 that leads to the Sparc motherboard

Loosen the strain relief that secures the video and keyboard cables

The power supply assembly is shipped with all harnesses connected to it so do not disconnect any cables from it.

FIGURE 343 DISPLAY TOWER WITH REAR COVERS REMOVED

5. With the fan power cables disconnected, remove the five screws from the top rear cover (with fans) and move it out of the way.
6. Once all the cables are disconnected, remove the two sets of screws that secure the Power Distribution assembly to the frame. Slide the Power Distribution assembly in a bit and then out the left side of the chassis. (The power supply will be easier to remove with the entire power assembly removed.) Refer to Figure 345.
7. Remove <u>one</u> of the two screws that secure the power supply to the power distribution chassis. This screw is located toward the rear of the Power Distribution chassis.

In order to remove the other screw that secures the power supply, the three boards must first be removed:

Intercom Power Module – Top Board

8. Loosen and remove the four screws that secure the board. If this board needs to be replaced, remove cables P1275 and P1276.

Intercom Mixer Module – Middle Board

9. With the Intercom Power Module removed or moved out of the way, loosen and remove the four standoffs. If this board needs to be replaced, remove cables P1271 and P1272.

System Power Controller – Lowest Board

10. WIth the top two Intercom boards removed or moved out of the way, remove the four standoffs and two screws that secure the SPC to the chassis. This will finally get you to the second power supply screw. Support the power supply at the bottom and remove the second screw. You should be able to pull the power supply out completely.

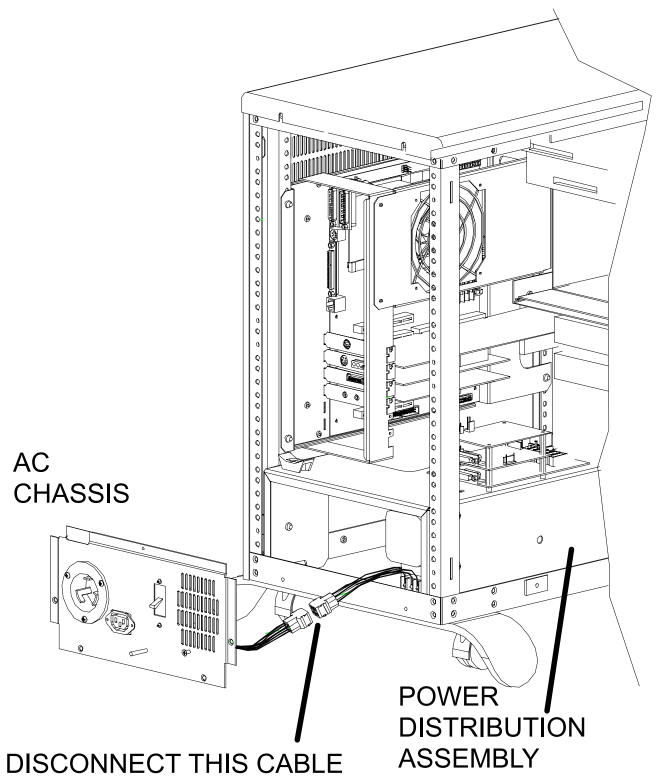


FIGURE 344 AC CHASSIS ASSEMBLY

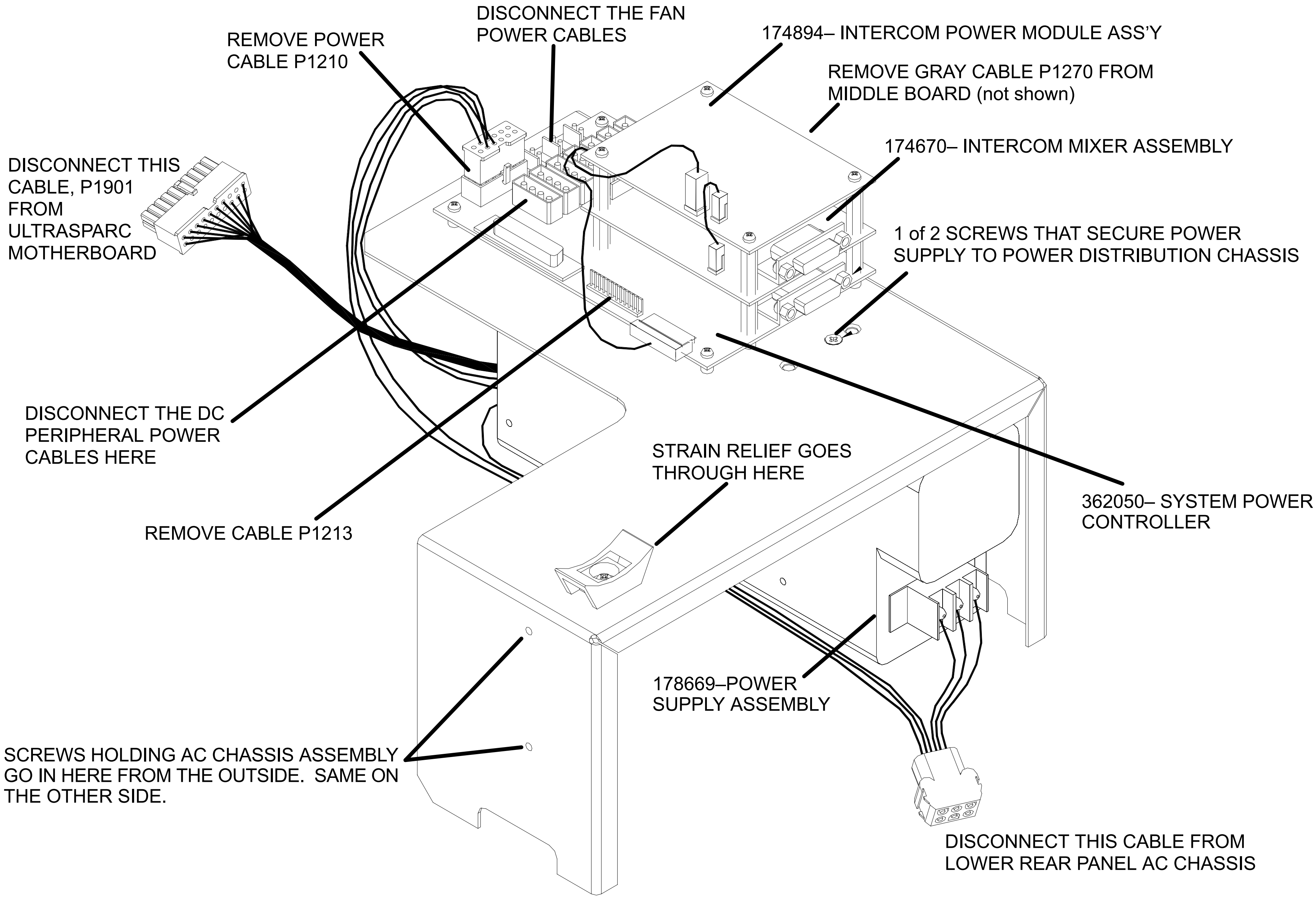


FIGURE 345 POWER DISTRIBUTION ASSEMBLY REMOVED

POWER SUPPLY REPLACEMENT:

1. Note: The new power supply comes with the cables attached to it.
2. Secure the new power supply to the Power Distribution Chassis with the two screws removed earlier.
3. Replace or re–install the SPC, Intercom Mixer and Intercom Power Module boards. Attach any cables that may have been removed from these boards if they were replaced.
4. Make sure all of the fan and peripheral power cables are pushed toward the front of the cabinet. Slide the Power Distribution Chassis (PDC) in from the left side and back a bit until it touches the back of the tower chassis. Secure the PDC with the four screws removed earlier.
5. Attach all of the cables removed earlier, with the exception of the two fan cables from the top rear cover.
6. Plug the lower rear cover AC Chassis cable back in. Secure the chassis to the tower with the four small screws removed earlier.
7. Before securing the upper rear cover to the tower chassis, remember to plug in the two fan cables to the SPC. Secure the cover with the five captive screws.
8. Re–install the left cover and cosmetic cover, roll back into place and secure the wheels.

UZX DISPLAY TOWER SPARC MOTHERBOARD OR MOTHERBOARD COMPONENTS

REMOVAL / REPLACEMENT

Introduction: The following procedure is used to remove and replace the UZX Display Tower UltraSparc motherboard in the event of a failure. This procedure also covers removal of the motherboard components: Video board, audio board, ethernet board and CPU fan assembly.

Tools Required: Phillips Screwdriver

Estimated Time: 30 Min.

1. Remove the Display Tower power. Refer to the PowerUp/PowerDown procedure.

> **WARNING**
>
> **USE EXTREME CAUTION WHEN OPERATING OR WORKING ON EQUIPMENT WITH COVERS REMOVED. DANGEROUS VOLTAGES ARE EXPOSED WHEN THE COVERS ARE REMOVED. FAILURE TO COMPLY CAN CAUSE SERIOUS INJURY OR DEATH TO PERSONNEL.**

> **NOTE**
>
> If the system needs to be moved, release the lock on the wheels before moving. Ensure wheels are locked after moving.

SPARC MOTHERBOARD REMOVAL:

1. Remove the front cosmetic cover, then the upper rear and left side covers. Refer to the Display Tower Cover Removal procedure for details.

> **CAUTION**
>
> STATIC ELECTRICITY. CIRCUIT BOARDS MUST BE HANDLED WITH ANTI–STATIC TECHNIQUES. DISCHARGE YOUR BODY BEFORE HANDLING ANY CIRCUIT BOARD. KEEP ALL CIRCUIT BOARDS IN ANTI–STATIC BAGS OR ON CONDUCTIVE FOAM WHEN NOT INSTALLED IN THE EQUIPMENT. FAILURE TO COMPLY MAY CAUSE PERMANENT DAMAGE TO COMPONENTS ON CIRCUIT BOARDS.

2. Refer to Figure 346. Disconnect (and label if necessary) the cables from the rear of the UltraSparc motherboard. Disconnect the other cables from the front of the motherboard (SCSI, power cable to SPC, fan power cable, etc).
3. The UltraSparc is mounted on a bracket. You must remove this bracket with the motherboard attached first in order to remove the motherboard.

Observe static precautions and refer to Figure 347 for the following steps.

4. Loosen the four screws that secure the entire bracket assembly to the tower chassis and slide the entire assembly out through the rear of the unit. Place the unit on a flat surface.
5. **FAN ASSEMBLY**: Disconnect the fan power cable from the SPC. Remove the four screws that secure the fan assembly to the motherboard and remove the fan assembly.
6. **MOTHERBOARD COMPONENTS REMOVAL**: Remove the screws that secure each of the individual boards (ethernet, etc) from the motherboard. Take note of the location of each board and then remove each board. See the end of this procedure for more details on removing these individual components.
7. Remove the screws that secure the support bracket to the mounting bracket and remove.
8. Remove all of the screws that secure the motherboard to the mounting bracket and lift it off.

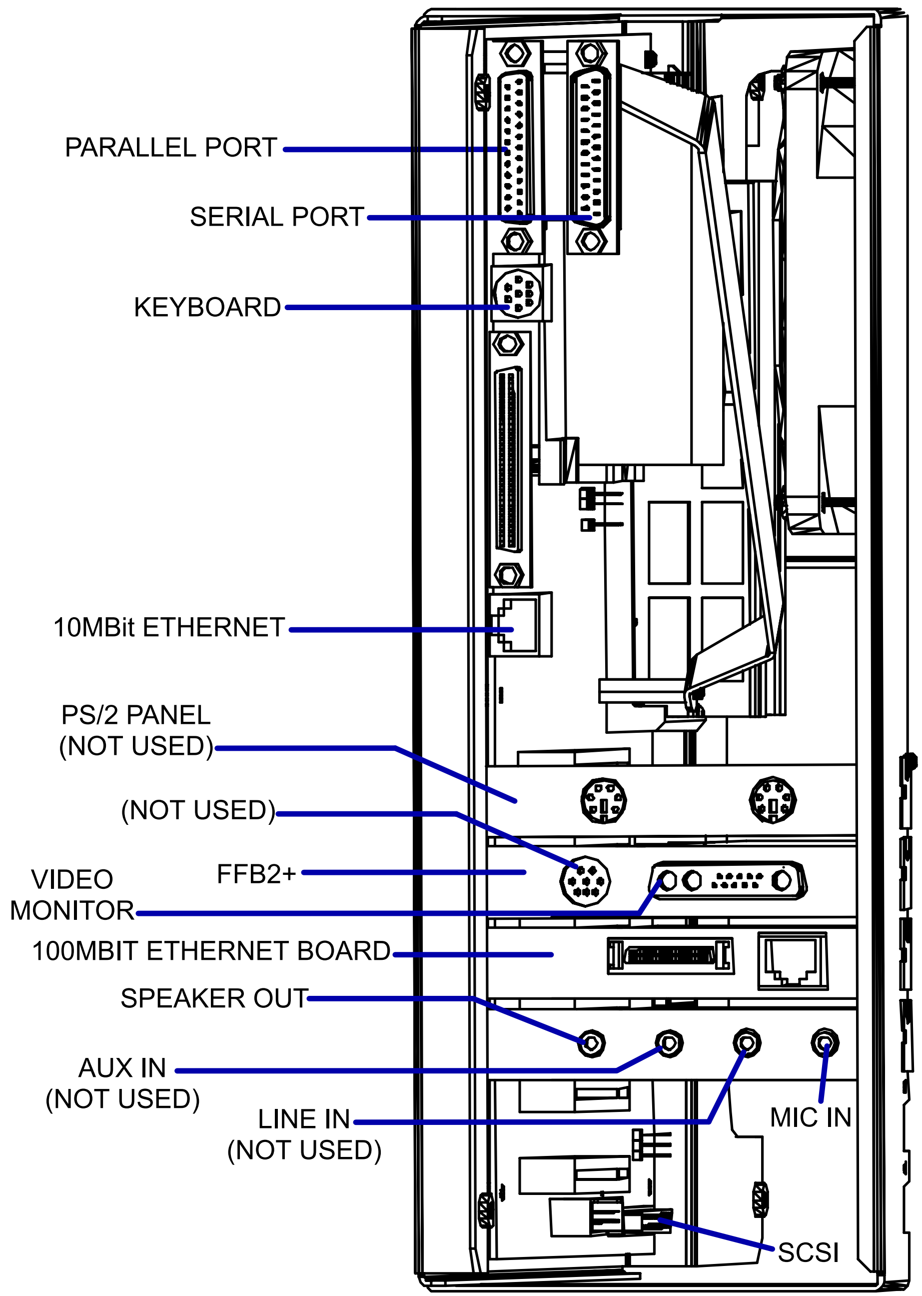


FIGURE 346 ULTRA SPARC MOTHERBOARD

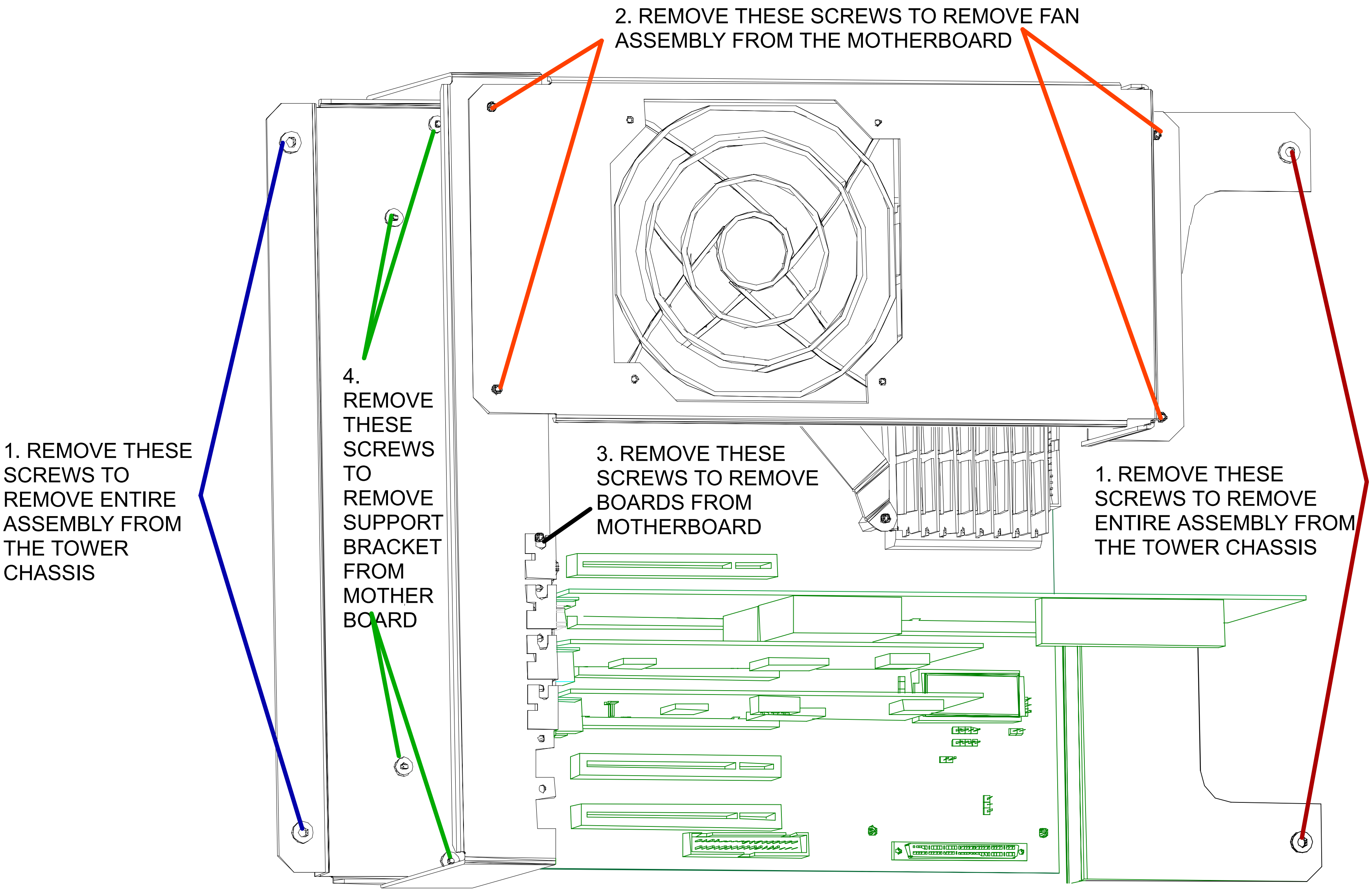


FIGURE 347 MOTHERBOARD REMOVAL

ULTRA SPARC MOTHERBOARD REPLACEMENT:

1. Place the new motherboard onto the mounting bracket and secure with the screws removed earlier.
2. Install the support bracket with the screws removed earlier.
3. Insert all of the individual boards back into their proper locations and secure.
4. Install the fan assembly back into its position with the screws removed earlier.
5. Slide the entire assembly into the tower chassis and secure with the four screws removed earlier.
6. Attach all cables that were removed. Refer to Figure 346 for locations.

ULTRA SPARC MOTHERBOARD COMPONENTS

To remove/replace an UltraSparc motherboard, the individual boards on the motherboard need to be removed. If one of these specific boards needs to be replaced, see steps 5. and 6. and refer to Figure 348.

> **NOTE**
>
> The Display Tower has three fans. One is on the Ultra Sparc assembly and the removal is covered in this procedure. The other two fans are connected to the upper rear cover. Simply remove the fan power cable and the screws that secure it to the back cover to replace them.

NVRAM FAILS OR IS REPLACED

If seeing: “IDPROM contents are invalid”, or “Timeout waiting for ARP/RARP packet”,

Then the NVRAM is either bad or not configured to boot from disk. To configure it to boot from disk, press **STOP–A** and get to an OK prompt.

Type **printenv** and look at the fields for boot–disk, and diag–device. They should be set to disk0, if they say disk net, then the system is trying to boot from the network and not from the local hard disk. To set this correctly type **setenv boot–device disk0** then **setenv diag–device disk0**. The system should now boot from disk.

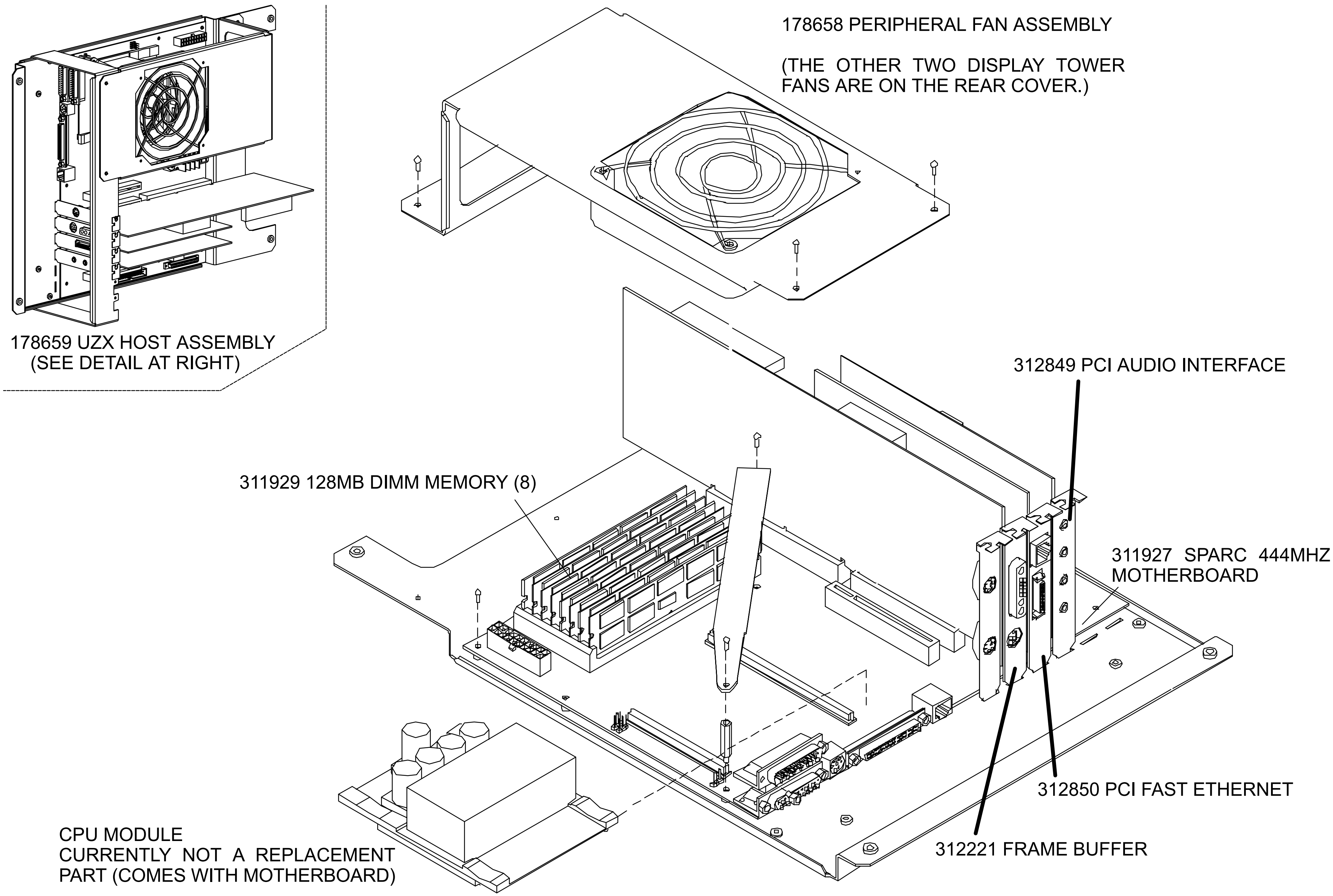


FIGURE 348 MOTHERBOARD COMPONENTS

POWER UP / DOWN PROCEDURE

This section describes the system start–up/shutdown procedures. **Power–UP** is accomplished via the keyswitch on the Display Tower. **Power DOWN** of the entire system is accomplished via the General Utilities GUI. This is accessed via the “Service” option from the opening bootup GUI. This procedure covers the Service Application mode. *Scanning Application mode procedure is in the OP manual*.

SYSTEM POWER UP

1. Make sure the Gantry keyswitch is in the normal position. Turn the Display Tower keyswitch to the **ON** position. After the system boots, you will be prompted with the startup application display (or “4 Button Screen.”) You have **4 boot options**. Refer to Figure 349.:
   A.) Select Clinical Scanning (or do nothing) and the system will start and connect to the acquisition server. The acquisition server will then power up the AcQImage Cabinet, and if that is done successfully, the acquisition server will power up the gantry and X–ray system.
   B.) Select the Service Application option (within 15 seconds) by clicking on the Service button with the mouse pointer.
   C.) Reboot the system. **D**.) Start the Diagnostic GUI (within 15 seconds) by clicking once in the bottom left corner, then entering PICKER <CR>. (All caps). The AcQImage board LEDs will display a power up test sequence. Refer to the LED Interpretations.

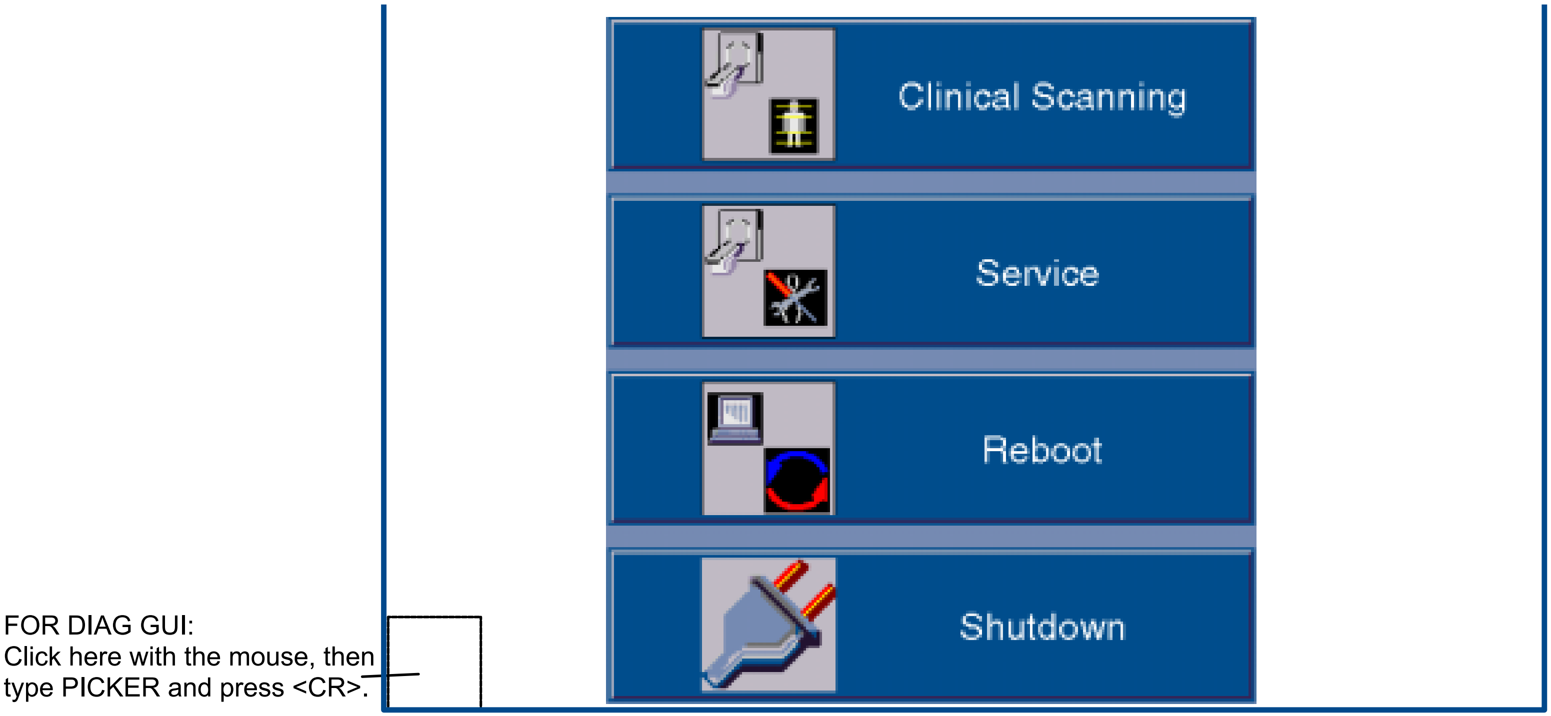


FIGURE 349 4 BUTTON SCREEN

PREPARE THE SYSTEM FOR USE (X–Ray Tube Warmup)

2. The X–Ray system needs to be properly warmed up prior to executing a scan procedure. X–Ray System Warmup must be completed at the start of every scan day or when the tube heat is below 6%. Tube heat is displayed in the Tube Warmup window.

3. To perform a tube warm–up, click on the Tube Warm Up button with the mouse pointer. Refer to Figure 349. This will initiate the warm–up sequence and **X–RAYS WILL BE EMITTED**. The tube warm–up status is displayed is displayed in the Tube Warmup Window. Refer to Figure 350.

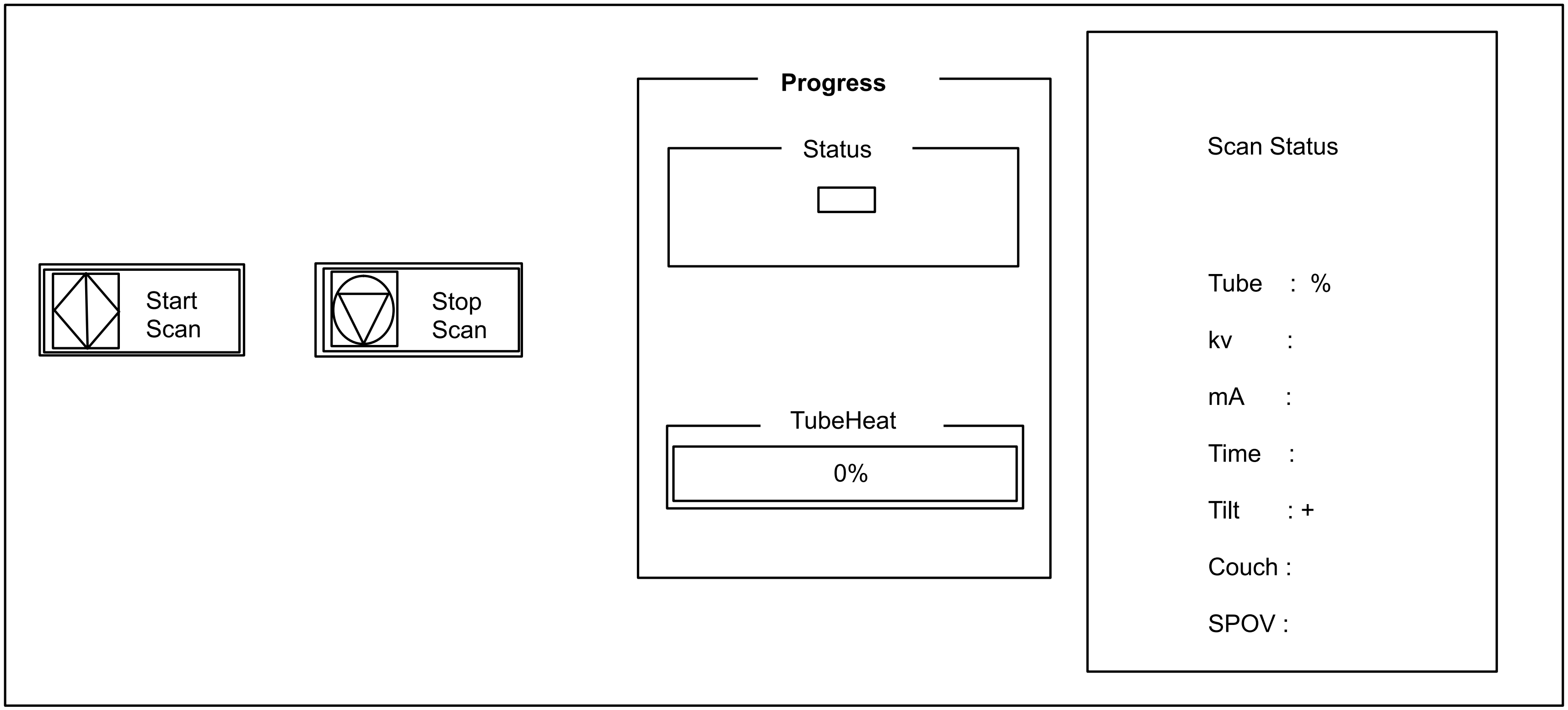


FIGURE 350 TUBE WARM–UP STATUS WINDOW

SYSTEM POWER DOWN

1. From the 4 Button Screen (Figure 349) click on SHUTDOWN with the mouse pointer.

> ⚠ **CAUTION**
>
> DO NOT POWER DOWN THE SYSTEM WITH THE TUBE HEAT AT OR ABOVE 35%. FAILURE TO COMPLY MAY RESULT IN SHORTENED TUBE LIFE AND/OR ERRATIC TUBE BEHAVIOR.

System shutdown ensures that all images on disk are properly closed, all image maintenance functions properly terminated, the gantry parked, the X–Ray system sufficiently cooled and all films printed.

There are three shutdown selections: Shutdown Gantry, Shutdown Console, Shutdown System and Switch to Console as seen in Figure 351.

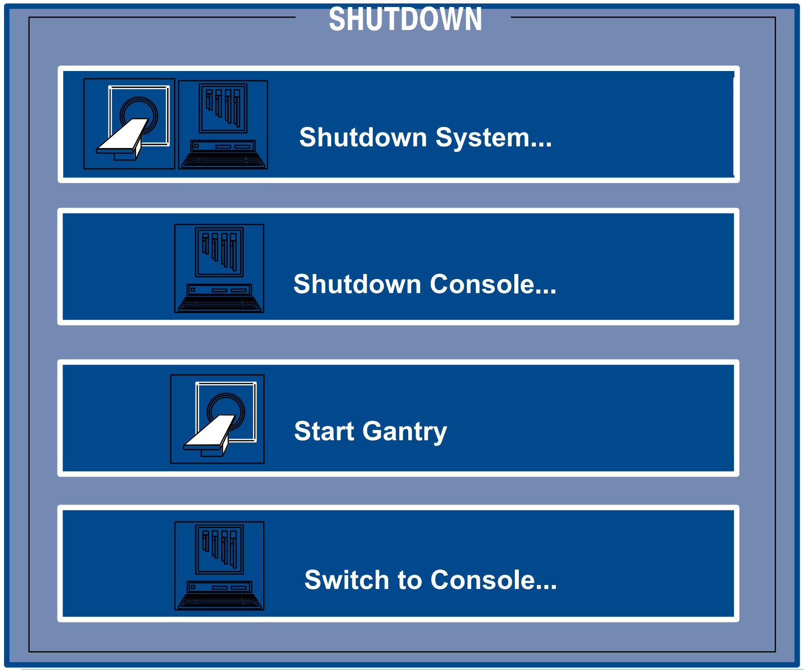


FIGURE 351 SHUTDOWN SCREEN

Shutdown Gantry: This option should be used, if possible, in place of the e–stop switches. An orderly shutdown of the gantry will be performed. Display Tower and/or AcQImage Cabinet operation may continue after selected. To cancel shutdown, click the Abort button.

Shutdown System: Performs an orderly shutdown of the entire scanner system. To cancel shutdown, click the Abort button.

Shutdown Console: This option shuts down only the console units. An orderly shutdown of the image archiver, filming and file system will be done.

Start Gantry: Toggles Start/Shutdown Gantry depending on status of Gantry power.

Switch to Console: Changes the GUI to the Console Application (Operator) screen.

WARNING

THE GUI SHUTDOWN REMOVES POWER FROM THE SYSTEM, BUT NOT AT THE INCOMING POWER LINES. DANGEROUS VOLTAGES ARE EXPOSED WHEN THE GANTRY COVERS ARE REMOVED. BEFORE SERVICING EQUIPMENT, REMOVE POWER FROM THE WALL BOX. FAILURE TO COMPLY CAN CAUSE SERIOUS INJURY OR DEATH TO PERSONNEL.

2. When prompted via a dialog box, turn the Display Tower keyswitch to the **OFF** position.
3. On the AcQImage Cabinet and Display Tower, ensure the rear breaker switch is turned off.

AcQImage Boards – Power Up Test LED Interpretations

```
AP, PB, GAM Board Diagnostic Flash Version 1.1.2
```

```
0 Hardware Initialization
1 CPU Version Check
2 Diagnostic Flash CRC Check
3 Operating System Flash CRC Check
4 CPU Memory Test
5 Data Cache Test
6 Unexpected Exception Test
7 IBC Test
8 NVRAM Single Location Test
9 PCI Bus Select Test
A Ethernet Loopback Test
B UART Loopback Test
C apmem for AP, pbmem for PB, gamem for GAM

C CIO Test (CP Board Diagnostic Flash Version 1.1.2)
D Synchronous Serial Chip Loopback Test (CP Board Diagnostic Flash Version 1.1.2)
```

PERIPHERALS / OPTIONS INDEX

SERVICE APPLICATIONS

Introduction

The Service Application gives Service the ability to easily set up the AcQSim CT system via a Graphical User Interface (GUI). The different setup menus include General Utilities, DITA, Calibrate/Perf. and Configure.

Service Applications Procedure

1. Boot the system. Select the SERVICE button from the opening screen.   Refer to Figure 352.   SEE NOTE below:

> **NOTE**
>
> The Diagnostic GUI, which initiates AcQImage Diagnostics, Gantry and other utilities, may be started at this time (before starting the Service App) by clicking in the bottom left corner (slightly smaller than 1” square) and typing **PICKER** <CR>. After exiting the Diagnostic GUI, the Service Application Selection screen will return (Figure 352).

Click once in the bottom left corner. Type PICKER (caps) and press <CR> to activate the Diagnostic GUI.

FIGURE 352   SERVICE APPLICATIONS SELECTION SCREEN

2. The opening screen defaults to “General Utilities”: At the bottom of the screen are the four major categories of Service Applications:

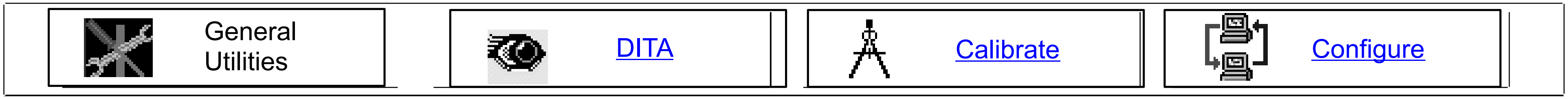


FIGURE 353 SERVICE APPLICATIONS – MAIN CATEGORIES

The following procedures detail each individual Service Application, based on Figure 353, going from left to right (General Utilities | DITA | Calibrate | Configure). Click on the links above for more information.

GENERAL UTILITIES

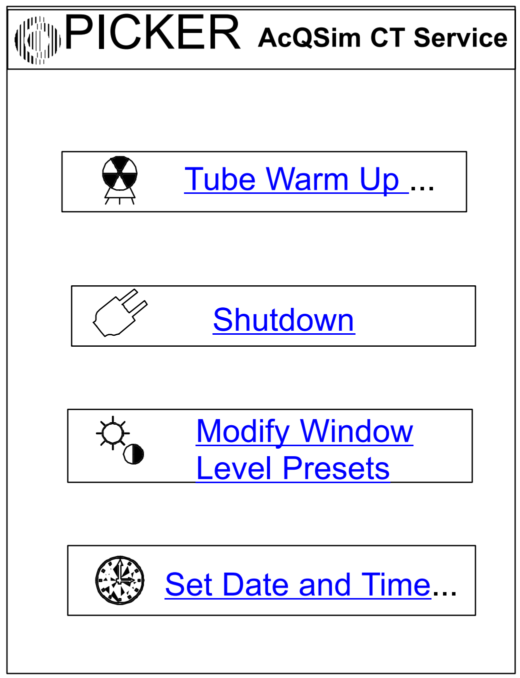


FIGURE 354 GENERAL UTILITIES WINDOW

The **General Utilities menu** includes the following options. Click on the respective buttons in Figure 354 for more information.

Tube Warm Up: Ensures proper tube heat necessary before taking a scan. Press the Start Scan button to begin warmup. Status is seen in the Progress and Scan Status windows.

Shutdown: Used to start/shutdown Gantry, Console –(Display Tower and AcQImage Cabinet) or shut down the entire system. Switch to Console launches the Console App.

NOTE: If the Gantry power is <u>on</u>, the third button will read “Shutdown Gantry.” If the Gantry power is <u>off</u>, the button will read “Start Gantry.”

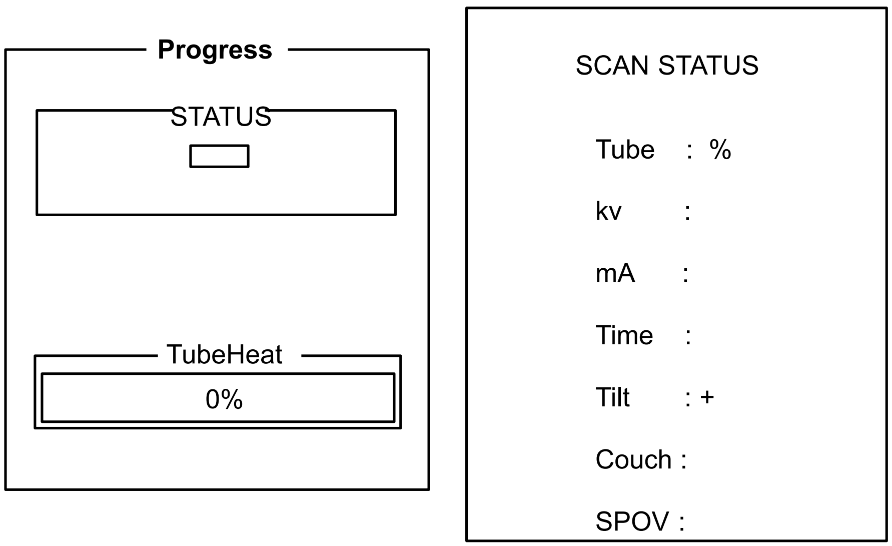


FIGURE 355 TUBE WARM UP

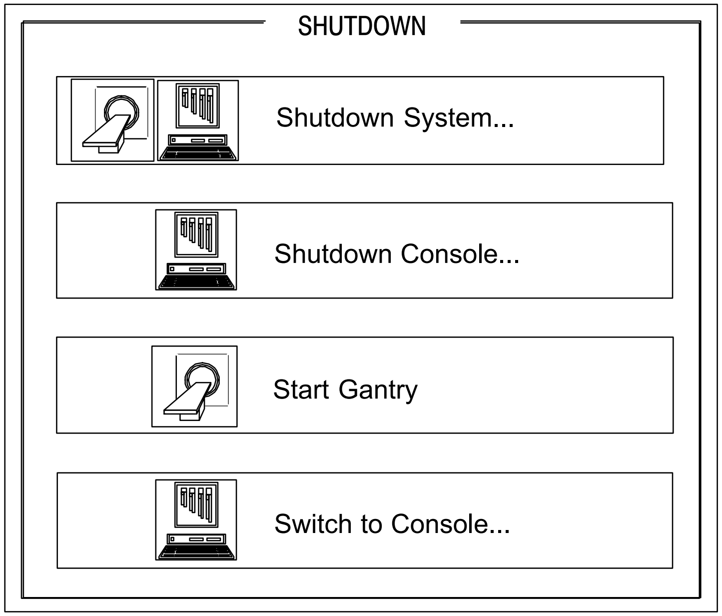


FIGURE 356 SHUTDOWN SCREEN

Modify Presets: Opens the Window/Level Presets Editor. You may add customized presets or delete existing ones. To add, click on the ADD button to highlight the next preset row. Click in the cell under the Preset column and type in a preset name. Repeat this in the window and level cells and click on SAVE. To delete, click on the preset value, then click on the DELETE button. If you make a mistake, click on the CANCEL button.

WINDOW LEVEL PRESETS EDITOR

Preset	Window Width	Window Level
Brain	180	45
Mediast.	400	20
Spine	350	70
Abdomen	350	50
Lung	1600	-300
Bone	2000	350
Pilot	800	500

FIGURE 357 MODIFY PRESETS

Set Time and Date: Allows you to set the time and date of the system, time zone, and 12/24 hours clock. Use either up/down arrows, toggle buttons or pulldown menus. When the time settings are correct, click on the SET TIME button. When the date settings are correct, click on the SET DATE button.

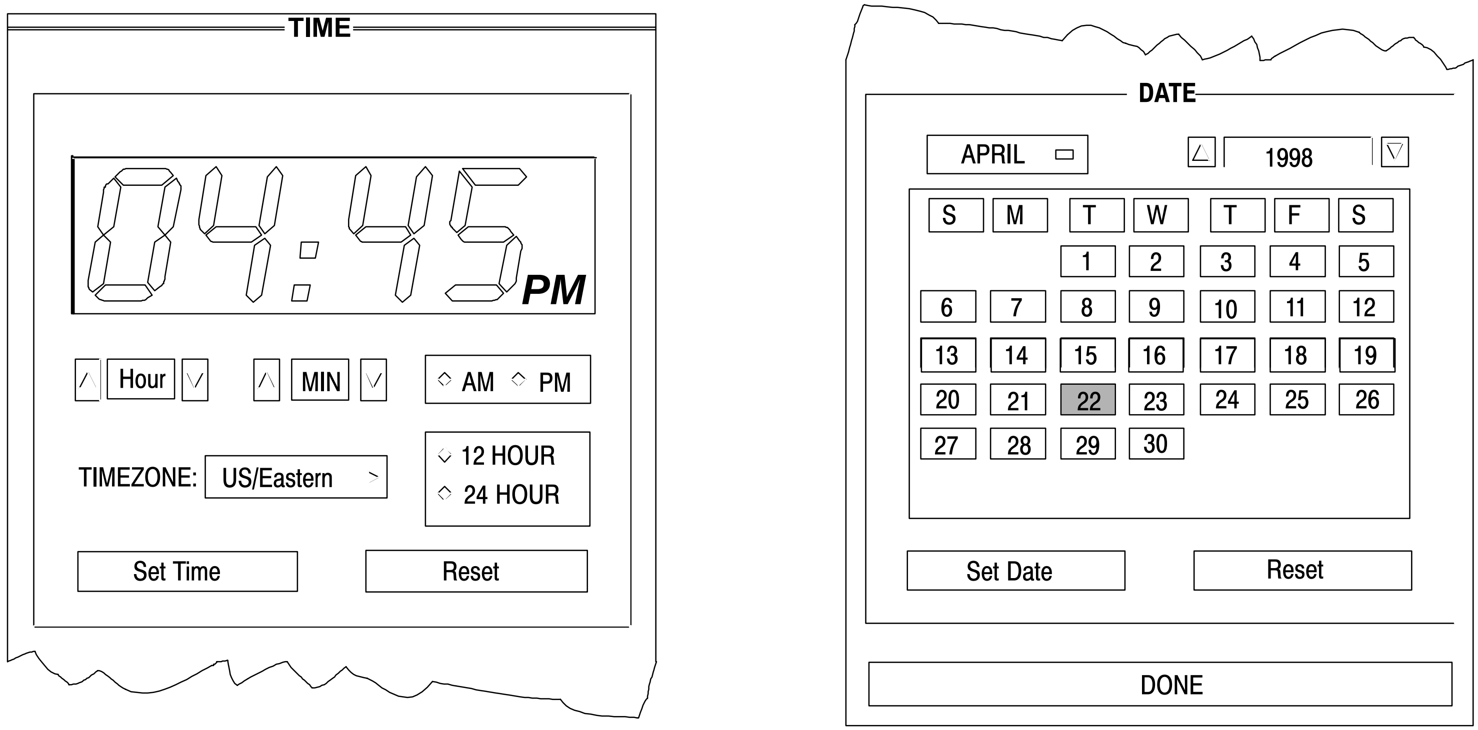


FIGURE 358 SET TIME and DATE

DITA

The AcQSim CT system supports the storage and display of data at many steps of the reconstruction process. A utility called DITA is used to display this information and can be used to isolate faults ranging from individual detector channels to failure of entire processing section such as the AP board. DITA opens with the study selection screen:

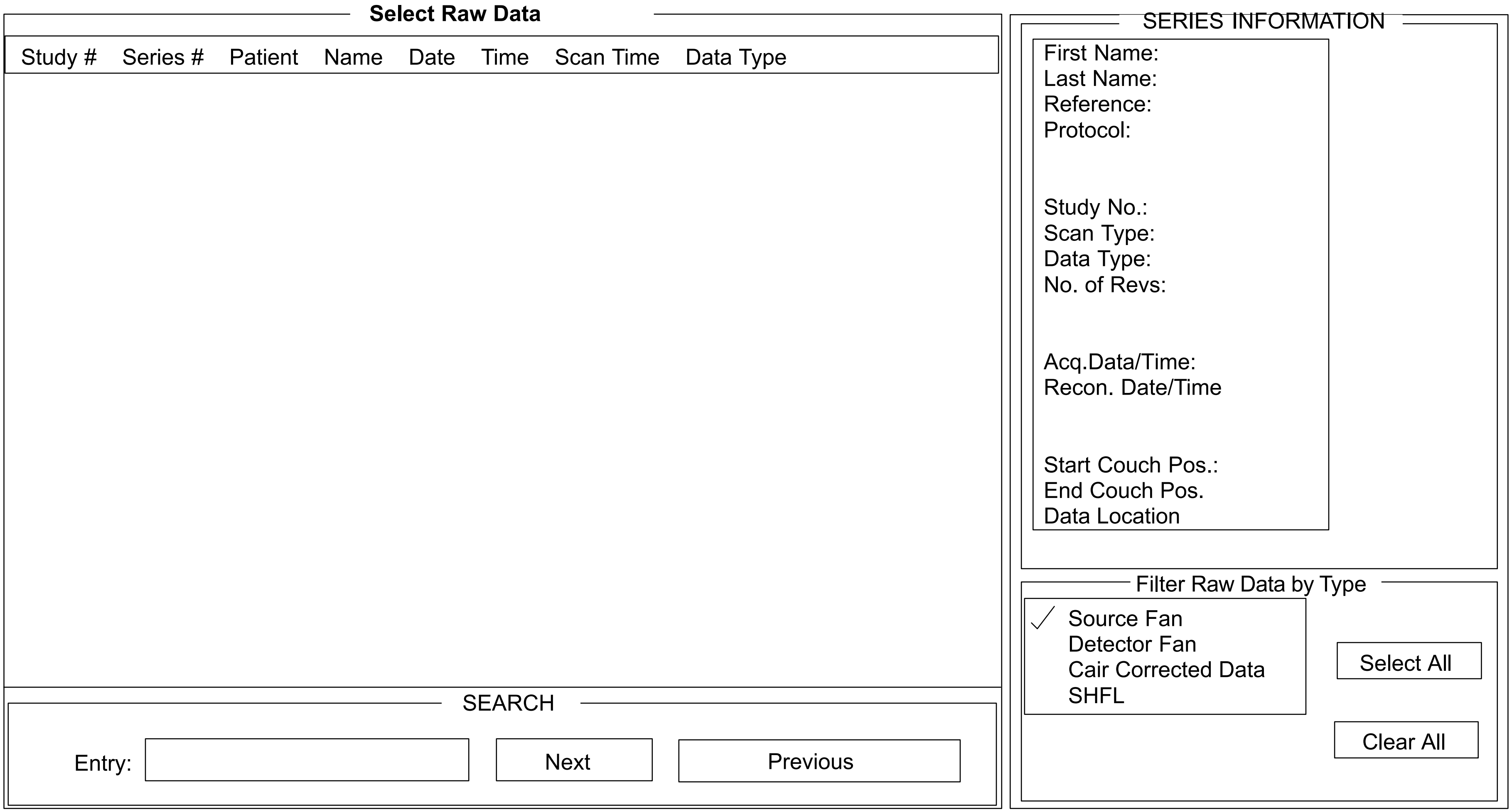


FIGURE 359 OPEN STUDY LISTING

To the right of this list is the Data Type selection menu. This menu allows you to list the different types of data, Source Fan, Raw Data, CAIR Correction Data or SHFL. Click on each type to toggle the check mark. Corresponding data will appear in the menu selection list. Highlight the study of choice, then click on OK.

Measure Selection Window

After the data loads, it will be displayed in the right window. The upper left window has three options and defaults to FILM, which would be used if filming the image was desired.

Click on the MEASURE tab. The Measure Selection box will appear:

Display Mode Selector: Choose the way you want the data to be displayed: Integer, Hex or Volts.

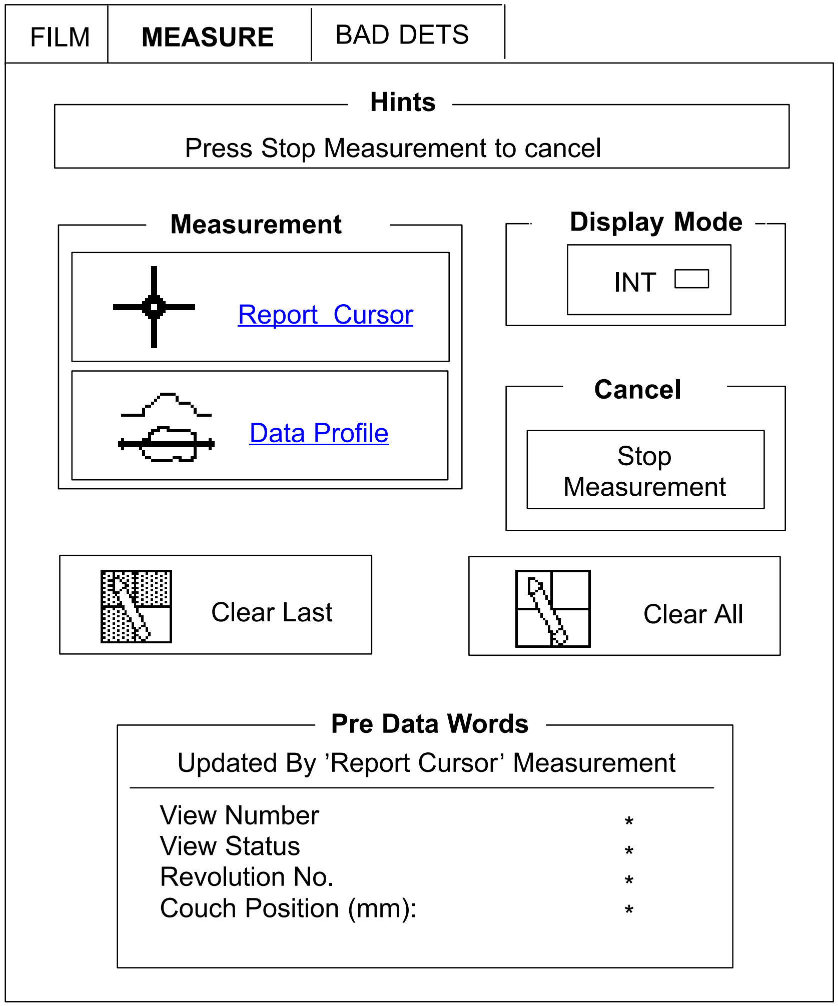
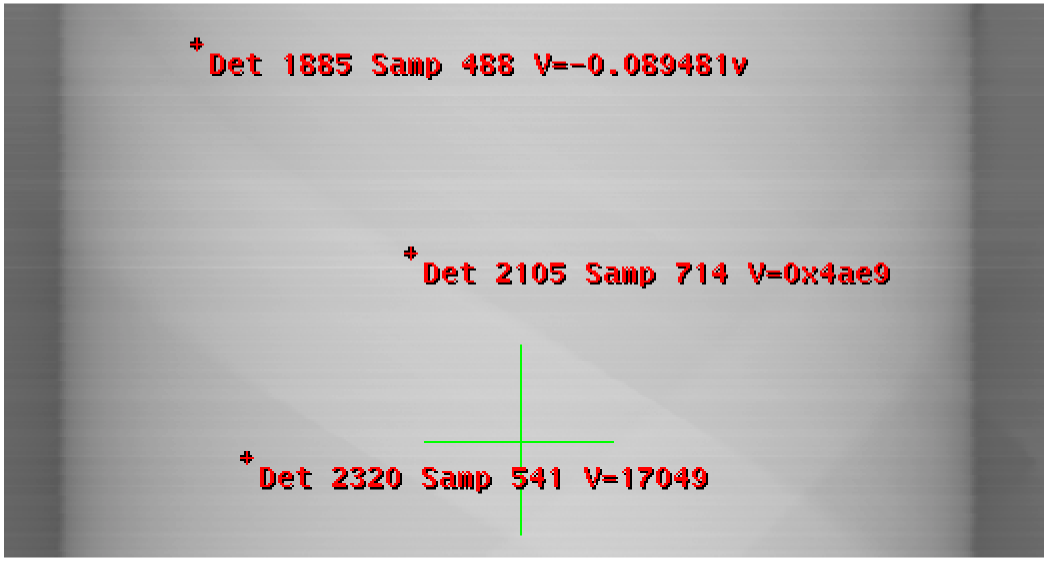


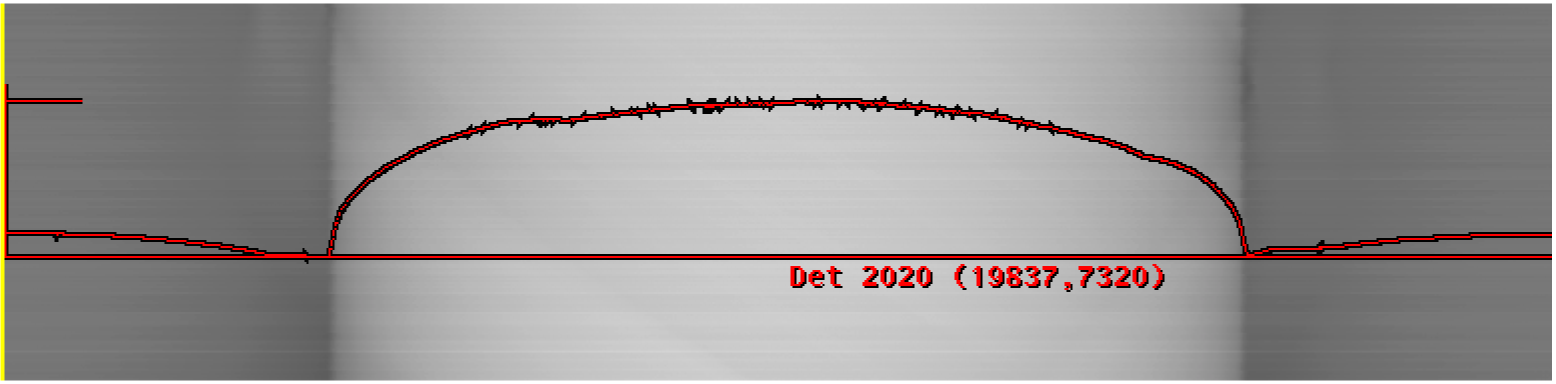
FIGURE 360 MEASURE WINDOW

The MEASURE SELECTION BOX contains the following measurement selections:

REPORT CURSOR: When the Report Cursor from Figure 360 is selected, you can move the cursor to the right window where the data is displayed. The cursor will change to look like a pencil point. This cursor is used to select and individual detector or area of the data. When you select a point on the data, a short report will appear. In this example, Detectors 1885, 2105 and 2320 were selected:

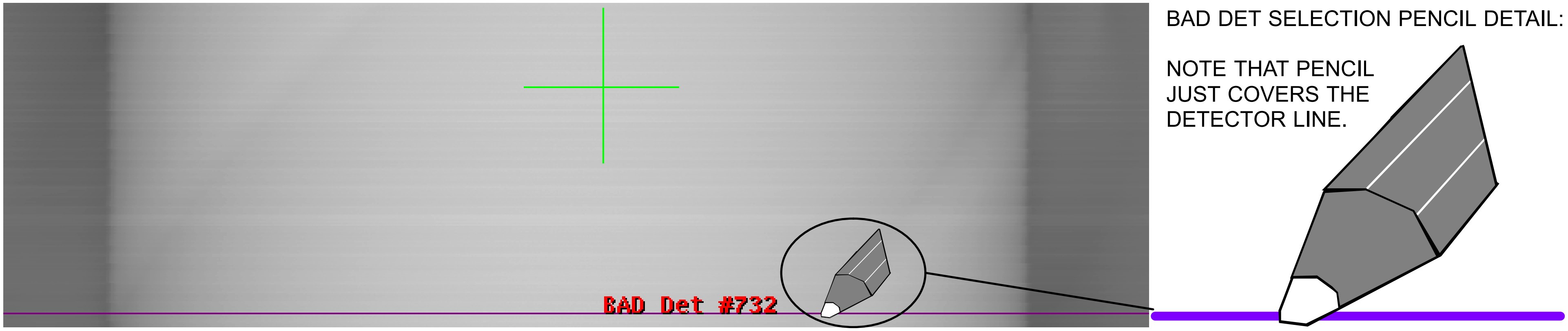


DATA PROFILE: When the Data Profile button in Figure 360 is selected, you can select an area of data that you want a profile on using the pencil point cursor. In this example, detector 2020 was selected:



Bad Dets Selection Window

To review the bad detector list, or add/remove a bad detector from the list, click on the BAD DETS tab. The Bad Dets selection window will be displayed. If you select the “Highlight Bad Dets” button, any listed bad detector will appear on the image data:



The Bad Detector List will update immediately after a detector is added or removed from the list. If changes are made to the list and you do not save it, you will be prompted when exiting DITA (see Figure 363).

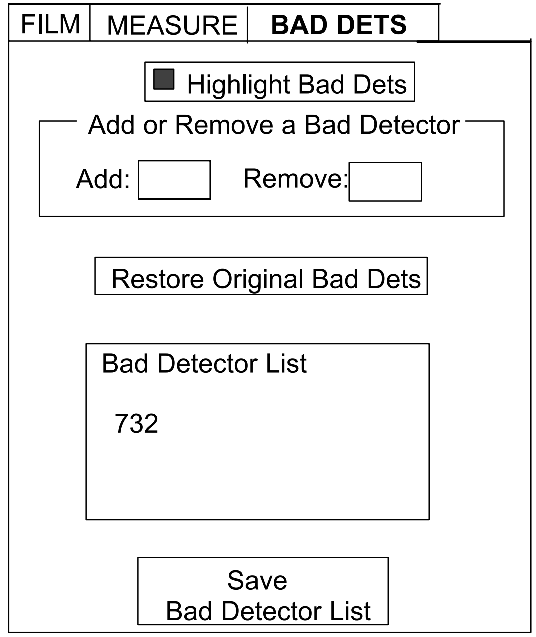


FIGURE 361 BAD DETS WINDOW

The bottom left window contains AcQuisition Information as seen in Figure 362.

Enter the Detector Number to move the displayed data within the window.

Pan and Zoom: Grab the purple box within the display window. Move the box around to pan, increase the size of the box by pulling one of the four corner "handles" outward to zoom.

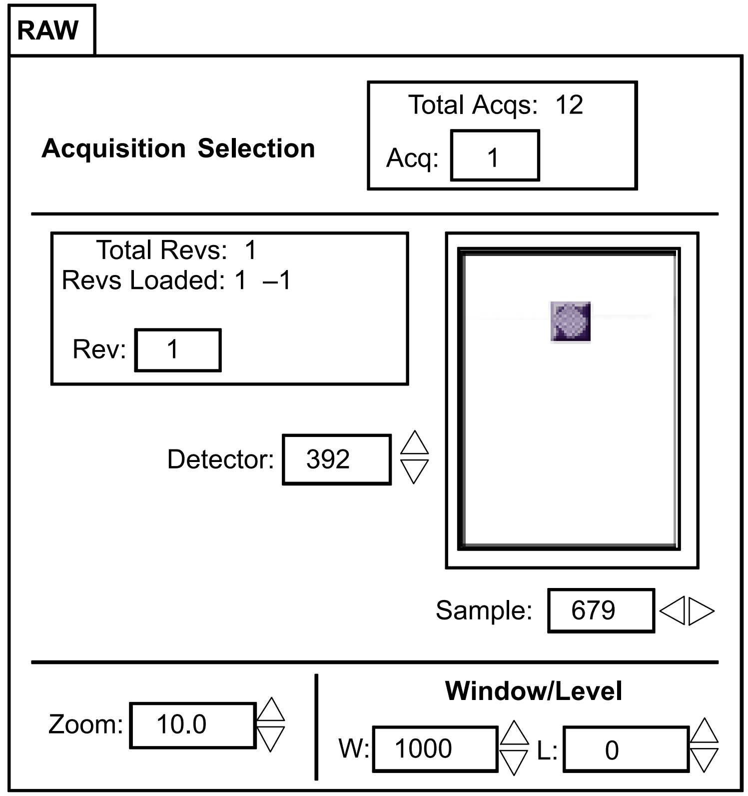


FIGURE 362   RAW DATA INFORMATION WINDOW

If any changes to the bad detector list were made and not saved, the following message will appear before the program terminates:

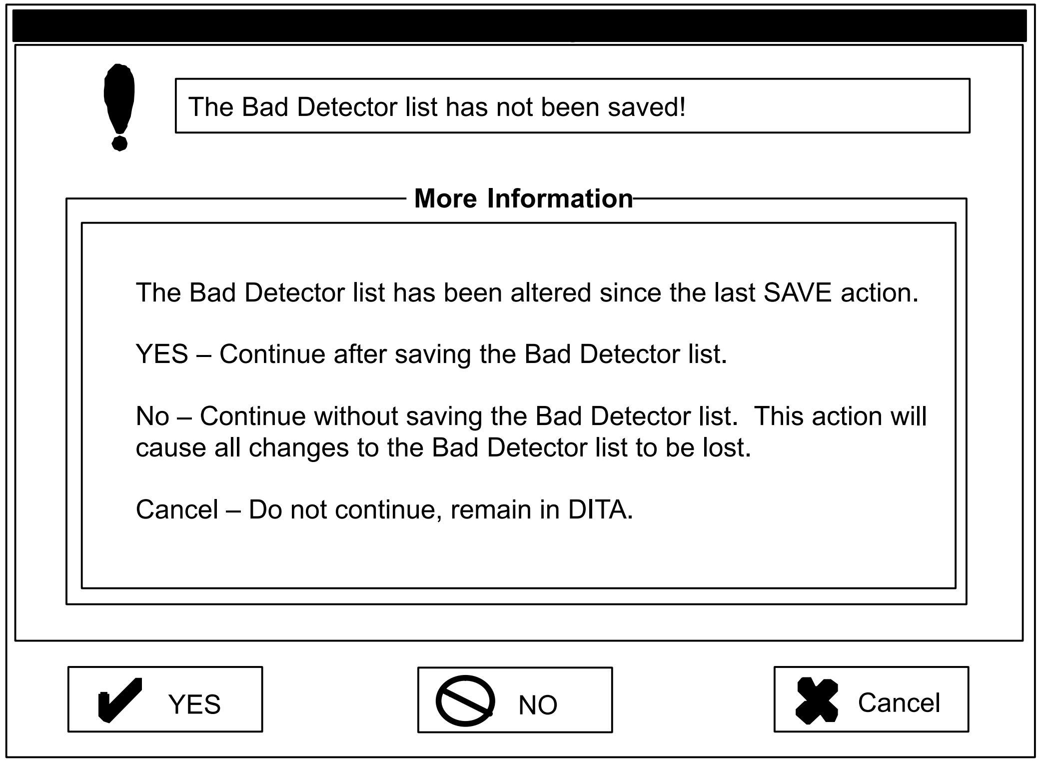


FIGURE 363 EXIT WINDOW

AUTO–CALIBRATION

Introduction

The **calibration** worksteps provide a way to calibrate the AcQSim CT scanner.   An automatic sequence will control couch movement, scanning with desired parameters, and running a desired analysis program. It also implements a **performance** assessment sub–system for determining the relative performance of a CT system.

The **User Interface** is no longer via the ELTP like the Q series scanners. All calibration workstep interfaces are GUI based, (Graphical User Interface) with selections made via a mouse “click.”   Each GUI has a cohesive purpose and is described below.

Calibration makes **Automatic Bad Detector Entries** in the vector database. Calibration will write a list that fails to meet spec (see below). Calibration will not remove any detectors from the bad list. Manual entry and removal of bad detectors is still available via DITA. Detector status is also available via DITA.

FIELD DETECTOR MODULE REJECTION CRITERIA

a. There can be no more than 16 bad detectors per system / else replace the module with the most bad detectors.

b. There can be no more than 3 bad channels per detector module / else replace the module.

c. There can be no more than 2 adjacent bad detectors, either in a module or in adjacent modules / else replace the module.

d. No defective channel can adversely effect other channels in a quadrature manner / else replace the defective component causing the adverse effect.

e. Any module with a defective ARDAS channel must be replaced.

Please promptly return the modules for repair with a description of the defects, date and scanner I.D. Knowledge gained from any reported defects will be applied to reduce the number of defects in future production.

The next several pages describe the GUI workstep interfaces. The first graphic shows the entire Calibration GUI to visualize where there individual workstep GUI areas are located. The are detailed in subsequent graphics.

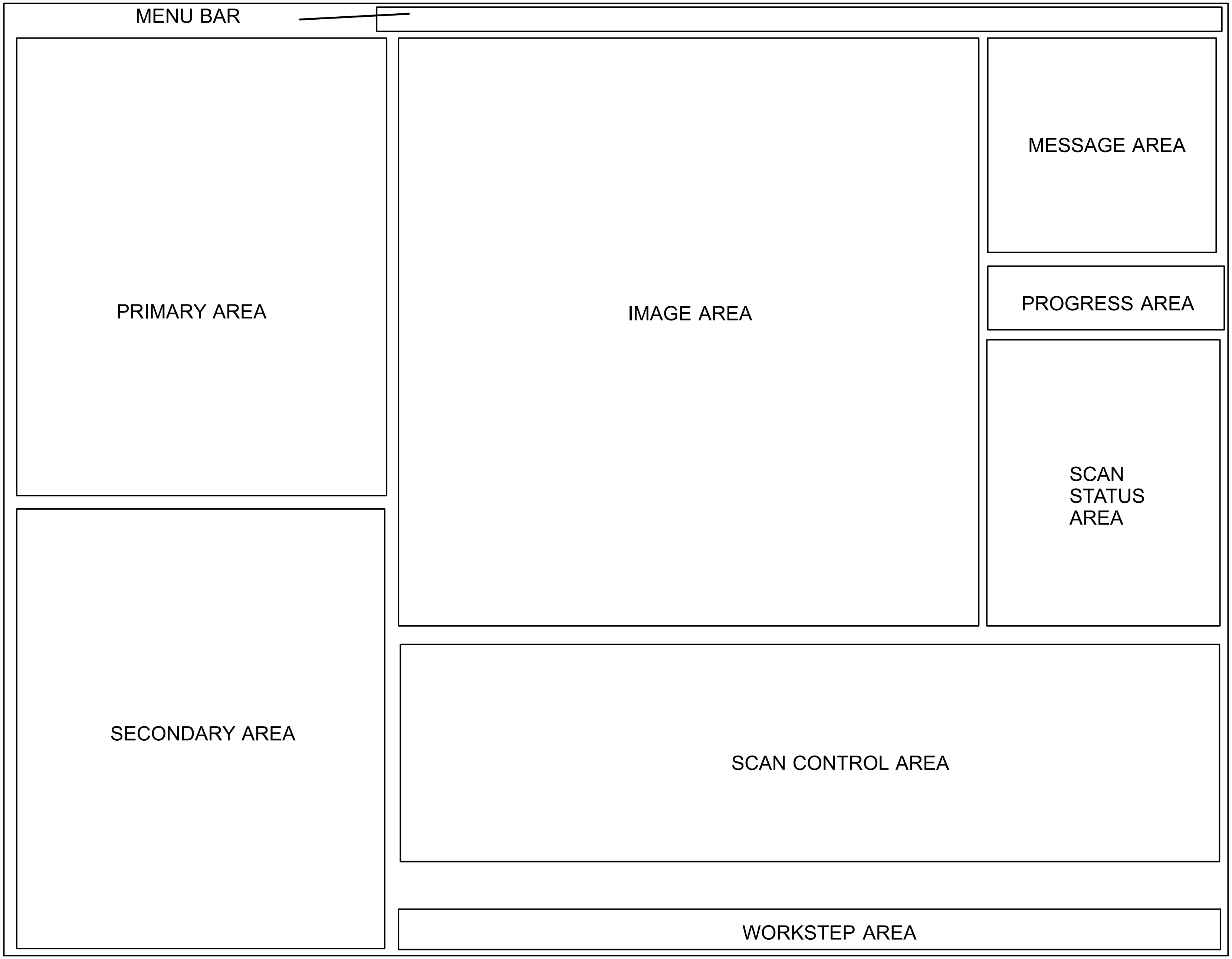


FIGURE 364 CALIBRATION WORKSTEP GUI LAYOUT – FULL SCREEN (NO DETAIL)

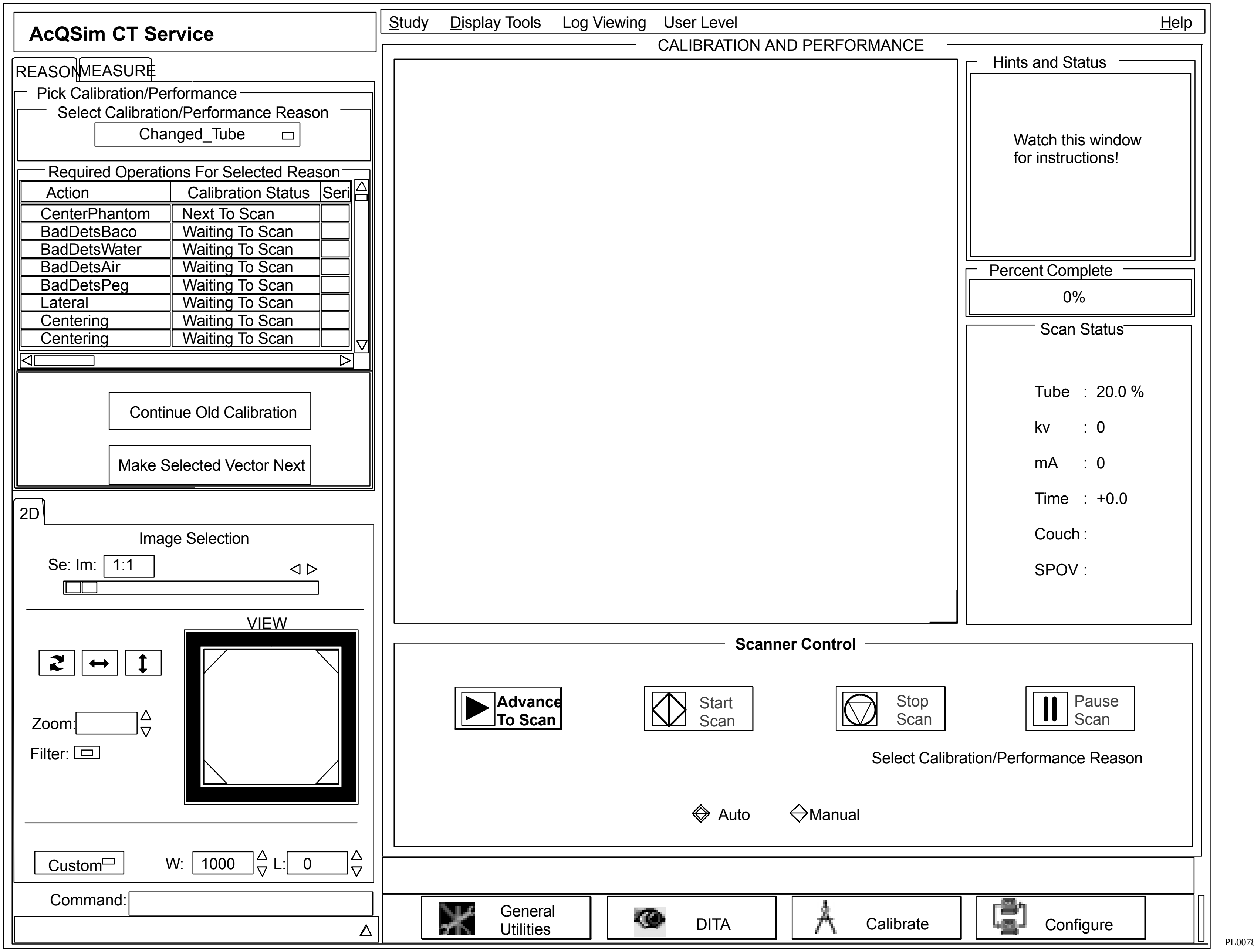


FIGURE 365 CALIBRATION WORKSTEP – FULL SCREEN (WITH DETAIL)

PRIMARY GUI AREA

The **PRIMARY AREA** contains all ”primary” functions. The calibration workstep divides this area into three major functions: REASON, MEASURE and FILM. Refer to Figure 366.

Reason GUI (Tab 1):The reason gui allows the user to select the reason for calibrating or doing performance of the system. The gui has two main parts: the Reason Pull–Down Menu and the Vector Grid:

> **Reason Pull–Down Menu**: This menu lists all possible reasons for calibrating. When starting a calibration or performance the user simply selects one of these reasons. The calibration workstep will determine all calibration or performance functions required by this reason. It will fill in the Vector Grid accordingly and then enable the advance to scan button of the X–ray Control GUI.

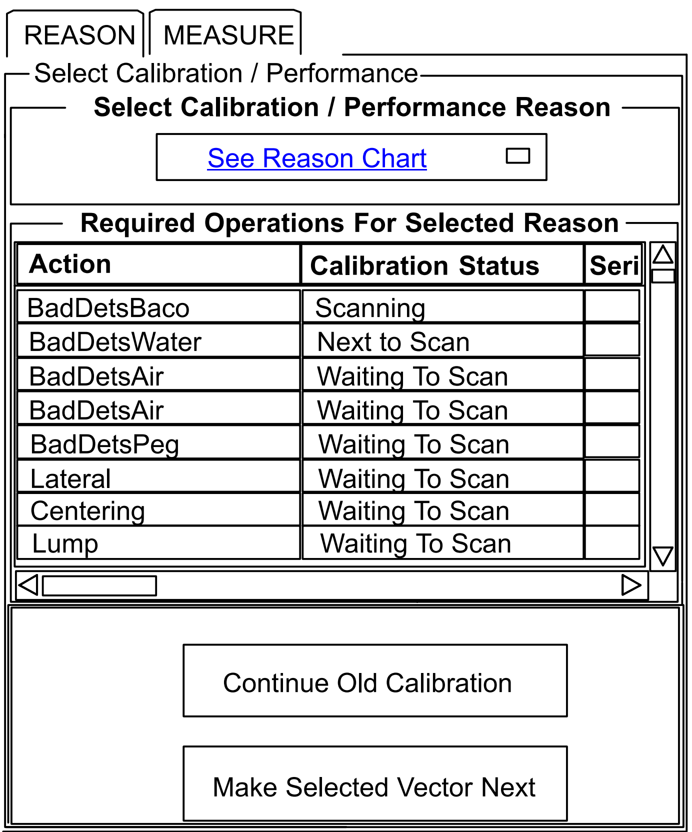


FIGURE 366 PRIMARY GUI AREA

Vector Grid: The vector grid shows one row for each vector that will be created or SPEC (Scanner Performance, Evaluation, and Correction) action to be taken. Each row lists the action, an event number, status and the Vector Grid parameters for creating that vector: KV, MA, etc. The user may (not recommended) start the calibration at any of these rows (vectors or actions) by simply double–clicking anywhere in that row. The status of each vector, as shown in column three of the vector grid, is very important when problems arise. You can watch the normal state of progression: `Waiting to Scan–>Next to Scan–>Scanning–>Calculating–>Complete`

There are other states that may appear FOR A SHORT TIME. Any other vector state which remains for longer than a minute indicates a problem. In manual mode the states are different.

Measure GUI (Tab 2): The measure gui operates the same as in the scanning workstep and thus will only work when an image is on the screen and preview is off.

Film GUI (Tab 3): Works the same as the scanning workstep.

Reason Pull Down Menu

REASON	DESCRIPTION OF ACTION PERFORMED
Changed_Tube	
Changed_Detector_Module	
Changed_VFSCCard	
Changed_HighVoltgeModule	
Changed_Collimator_Module	
Changed_Patient_Aperature	
Changed/Moved_Motherboard	
Changed_XSC	
Verify_Gantry	
Check_Performance	
Center_Phantom	
Custom_MTF	
Custom_Scan	
Clear_BadDetsList	
Find_BadDets	
Perform_Lateral_Centering	
Find_Centering	
Find_DAC	
Find_Air	
Find_MA	

Find_PilotAir	
Find_Flatts	
Find_Level	
Find_Level2	
TouchUp_Air	
Perform_MTF	
Perform_KVHVL	
Perform_LowCon	
Perform_CNA	
Perform_Uniformity	
Perform_SliceThickness	
Evaluate_ARDAS	
ManBadDets_Baco	
ManBadDets_Water	
ManBadDets_Air	
ManBadDets_Peg	
ManPerform_Lateral_Centering	
ManFind_Centering	
ManFind_DAC	
ManFind_Air	
ManFind_MA	
ManFind_PilotAir	
ManFind_FlattsFF	
ManFind_LevelFF	
ManFind_Level2FF	
ManPerform_MTF	
ManPerform_KVHVL1	
ManPerform_KVHVL2	
ManPerform_LowCon	
ManPerform_CNABaco	
ManPerform_CNAWater	
ManPerform_CNAResolver	
ManPerform_UniformityFF	
ManPerform_SliceThickness	

SECONDARY GUI AREA

The **Secondary Area** contains functions considered secondary:

> **2D View GUI**: Works the same as scanning and will only be visible if there is an image on the screen. Some functions only work when preview is off such as positioning to a new series.

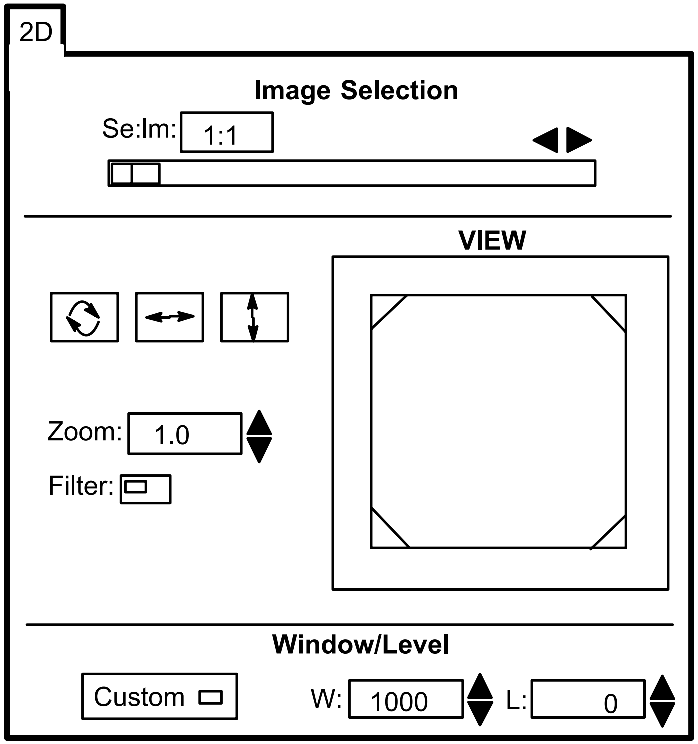


FIGURE 367 SECONDARY GUI

WORKSTEP FLOW AREA

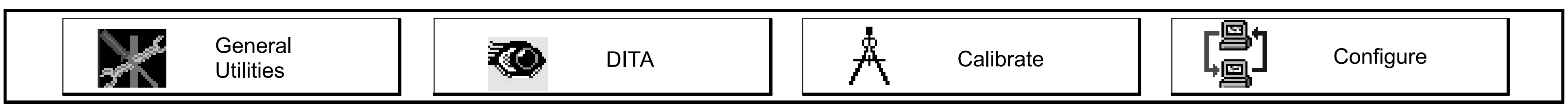


FIGURE 368 WORKSTEP FLOW AREA GUI

The **Workstep Flow area** contains the workstep flow buttons. While in the calibration workstep, if calibration is active in any way you **will not** be allowed to leave the calibration workstep until all activity has ended.

The **PREVIEW GUI** (Not shown yet) turns previewing of images ON or OFF.

SCANNER CONTROL AREA

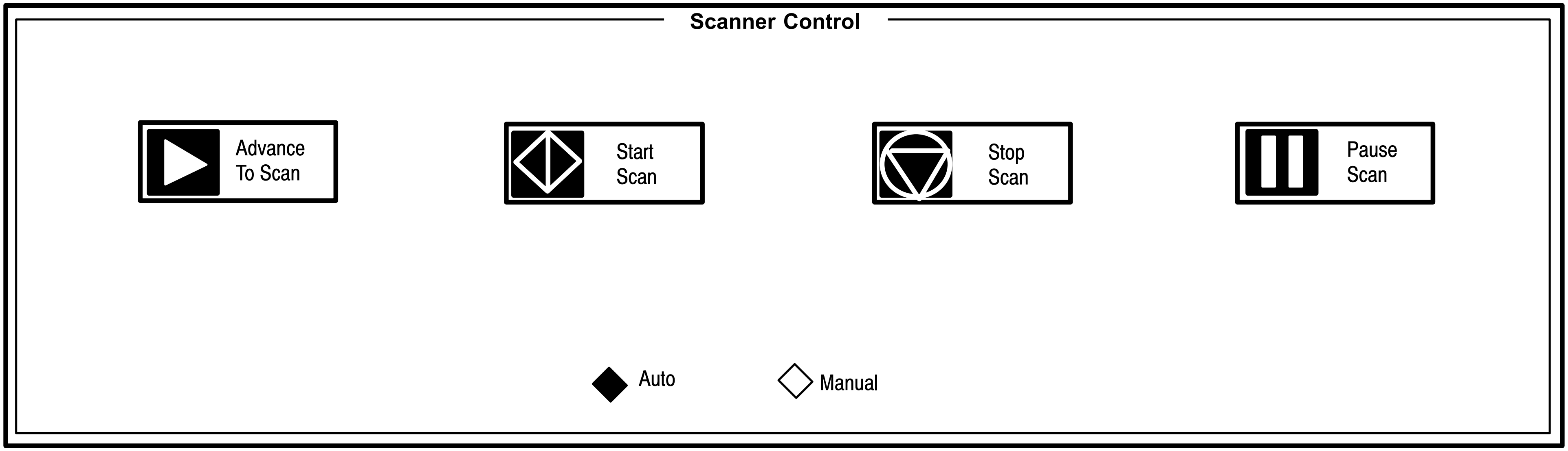


FIGURE 369 SCANNER CONTROL AREA GUI

The **Scanner Control area** has all scanner control buttons which are described relative to calibration in the next five sections. They include Advance to Scan, Start Scan, Stop Scan and Pause Scan:

> **NOTE**
>
> The buttons gray out when their meaning is no longer valid for the calibration workstep's state. Thus, the pause button is grayed out (inactive) if no reason has been selected.

Advance To Scan: Is enabled once a reason has been asserted and before calibration scanner activity starts. When pressed will cause the scanner to spin up the rotor and some other preliminary activities which are to be determined.

Start Scan: Actually starts the calibration or performance activity by scanning to collect data. This button is enabled only when we are ready to scan.

Stop Scanning: Will cause the calibration workstep to stop scanning and stop all math processes, but continues until all queued data has been consumed. When the math server finishes its calculations, all scanning stops and all plans are stopped the advance to button is enabled and so is the selection of a new calibration reason.

Pause Scanning: Will cause calibration to stop scanning after the current scan has finished.

Auto/Manual Radio Buttons: If auto is selected calibration proceeds in it's most automatic mode. That is, minimal operator intervention is needed. If in the manual mode then the start button must be pressed after each calibration vector action is taken (when a row is finished in the grid widget).

SCAN STATUS AREA

The **Scan Status** area shows the status of the Tube heat, tube kV and mA, time elapsed, degree of gantry tilt and the focal spot.

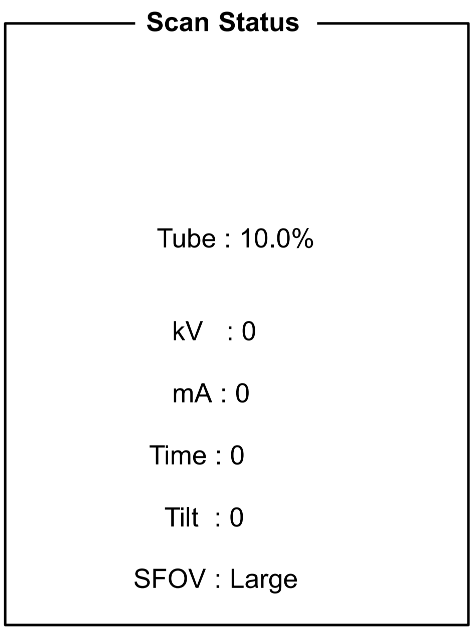


FIGURE 370 SCAN STATUS AREA GUI

CALIBRATION / PERFORMANCE PROGRESS

The **Calibration / Performance Progress Area** indicates the time calibration has been active, an estimate of how much longer it will take, and the percent complete.

NOTE: Errors and pauses will affect the accuracy of this widget.

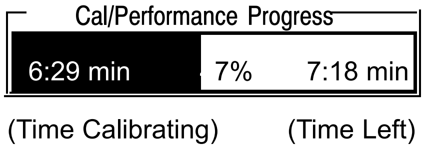


FIGURE 371 PROGRESS GUI

CALIBRATION MESSAGES

The **Calibration Messages** area displays operational messages to the user. These messages may help you determine what to do next or may simply give status.

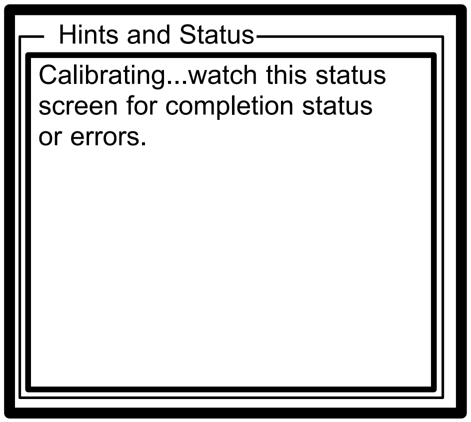


FIGURE 372 CALIBRATION MESSAGES GUI

DB GUI

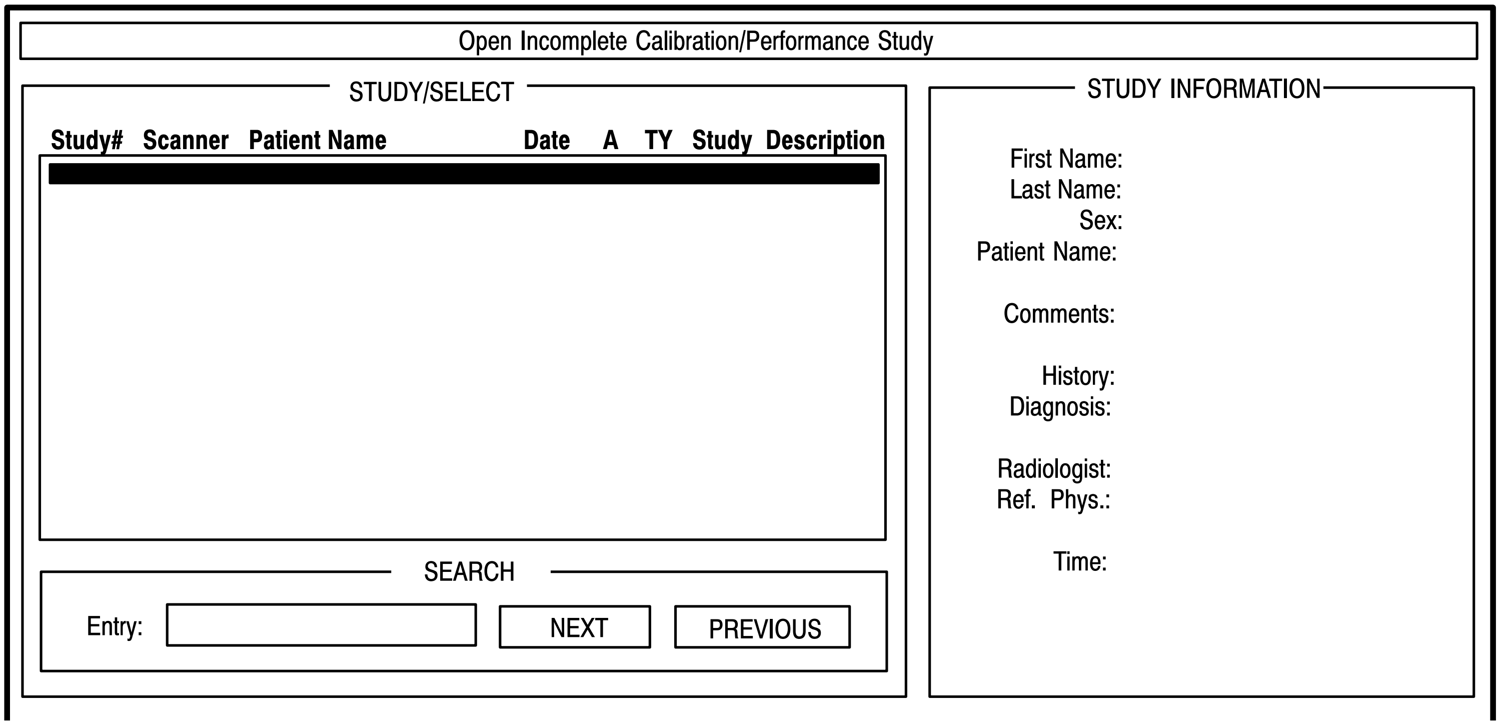


FIGURE 373 DB GUI

The **DB GUI** Appears when you select the “Continue old cal or perf” button in the primary area.

You are responsible for removing the studies created by calibration. Every study created by calibration will have the last name: CALIBRATION.

CALIBRATION PROCEDURE

Calibrating the AcQSim CT system has been greatly simplified. The Workstep GUIs show most of the information needed to perform a system calibration. A basic procedure is below:

1. Select a reason from the Primary Area pull–down menu, or select “Continue old Cal or Perf.”

    If an old study DB gui appears on screen or if an old calibration is in the DB and you select “continue”, from the DB area, then the reason selected will to be the old one.

    If you want manual mode, select it in the Scanner Control area.

2. Click on the “Advance to Scan” button in the Scanner Control area.
3. Click on the “Start Scan” button in the Scanner Control area.

   You may have to hit start each time if in manual mode or in lateral centering.

   You may click on “Preview Off“ to use the full features of measure gui.
4. Wait for completion of the calibration sequence, while noting the changing states in the Primary Area vector grid.

AUTOPERFORMANCE

AutoPerfromance is performed in the same manner as Autocalibration, selecting the Performance “Reason” from the REASON tab in the Primary GUI window. See the Reason Pull Down Menu Display for selections.

CONFIGURE

The different CONFIGURE menus for the SCANNER include Camera, Dicom, Network, Options, AcQImage, Boards, and Site configurations. Refer to Figure 374. UltraZ Diagnostic Workstation Configure follows.

Site	Boards	Network/Camera	AcQImage	Options	Generic

FIGURE 374 CONFIGURATION SELECTIONS

The following procedures detail each individual configuration, based on Figure 374, going from left to right.

You may click on the buttons in Figure 374 to jump to their respective screens.

Additional information on networking terms can be found in: Understanding IP Addresses, Subnetmasks, and Routing

Site: **Allows complete SITE configuration.**

FIGURE 375 SITE CONFIGURATION

Boards: Allows complete system BOARDS configuration.

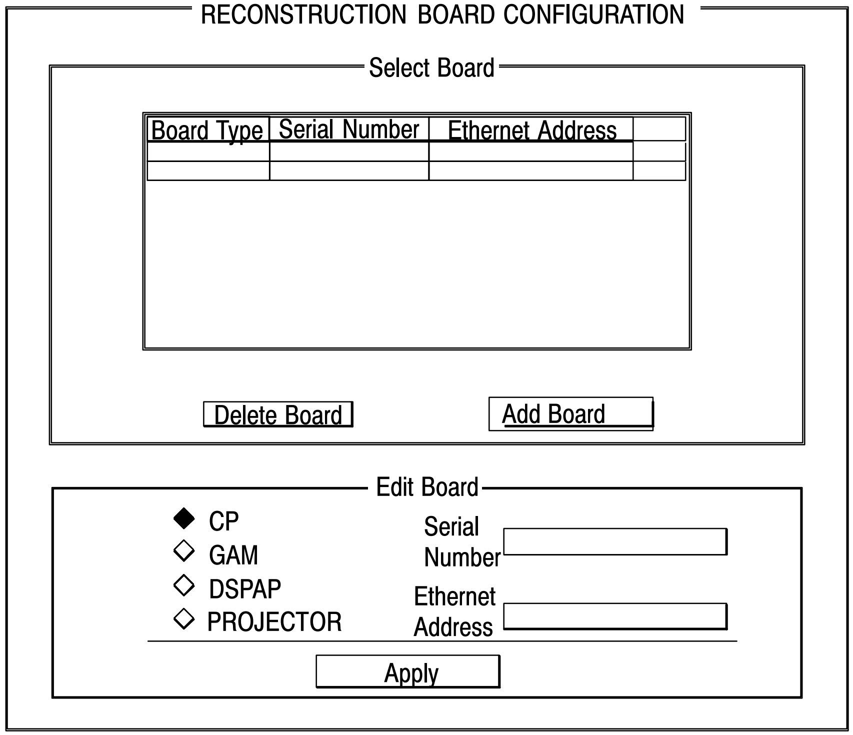


FIGURE 376 BOARDS CONFIGURATION

BOARDS – ETHERNET AND IP ADDRESS CONFIGURATION

There are 3 ethernet networks employed within the product. Refer to Figure 377.

Private internal ethernet network: Used for communication between boards in the ACQIMAGE CABINET. Configurations may be necessary.

Private internal 100 Mbit ethernet network: Used for communication between the ACQIMAGE CABINET and the DISPLAY CONSOLE. No configuration required.

Standard ethernet NETWORK PORT on the DISPLAY CONSOLE: Used for network connections to options and external hospital networks. Configuration is necessary.

NOTE

Refer to the Networking Glossary for assistance with definition of terms.

Systems from the factory will have ethernet addresses already assigned to the AcQImage Cabinet boards. If a board is added or changed in the field the new ethernet address assigned to the board(s) will need to be entered. Each board is physically labeled with a serial number and ethernet address. Values are entered in the Configure | Boards option of the Service Application. Refer to Figure 376.

Once the ethernet address(es) has(have) been updated, IP addressing for the board will automatically be generated and loaded into the configuration file as well. There is no need to establish IP addresses for the boards as they will be assigned according to a predetermined scheme. Default IP addresses will be assigned to the boards according to the order they are entered in the configuration:

192.168.2.10 – First board
192.168.2.20 – Second board
192.168.2.30 – Third board
192.168.2.40 – Fourth board, etc.

IP ADDRESSING – 100 MB ACQIMAGE / DISPLAYCONSOLE LINK

IP addresses for the 100 MB link are preset to default values by the applications software. The following addresses are assigned and need not to be changed:

192.168.1.1 DISPLAY CONSOLE 100 MB port,
192.168.1.2 ACQIMAGE CABINET 100MB port

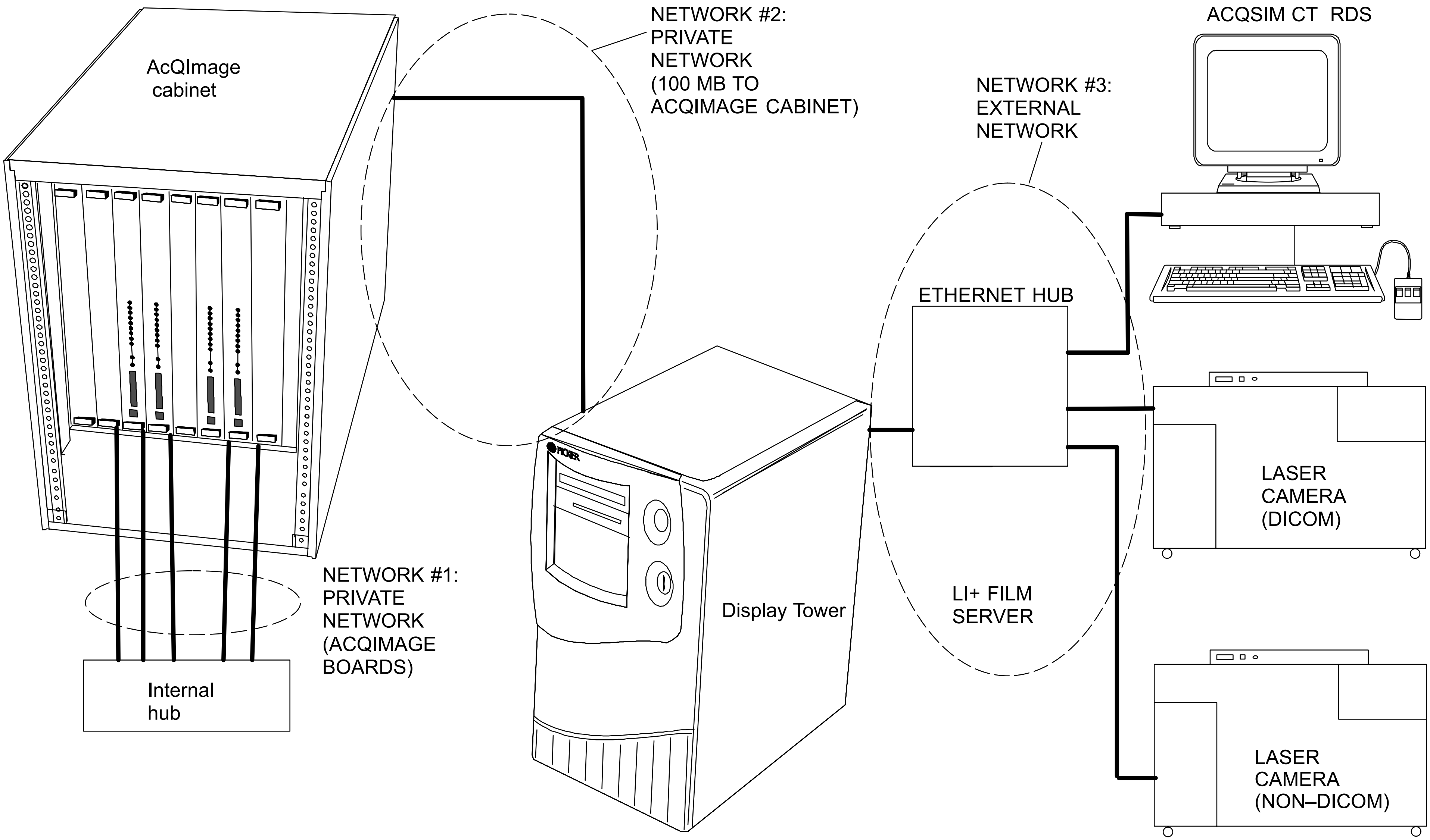


FIGURE 377 EXAMPLE OF PRODUCT'S THREE ETHERNET NETWORKS

NETWORK PORT CONFIGURATION

If there are any Picker option (network) devices or hospital network connections in the installation then configuration of the ethernet NETWORK PORT will be necessary. The NETWORK PORT is not configured as shipped from the factory. Each installation is **unique and must be done on site**. It will be necessary to enter IP address information as well as other parameters for the local DISPLAY CONSOLE and other devices connected to it via the network. This section refers to configuration details for the DISPLAY CONSOLE (and will also work for the UltraZ Diagnostic Workstation) . As a rule all other devices connected on the network will need to be configured locally on their own as well. Refer to individual installation manuals and instructions for these specific devices (e.g. UltraZ Diagnostic Workstation, Voxel–Q, Li +, Dicom Camera, etc.).

Values are entered in the Configure | Network/Camera option of the Service Application.

The Network/Camera Configure Option consists of *Local Parameters* and *Other Devices* tabs:

Network/Camera (Local Parameters): Allows complete <u>local</u> Network configuration:

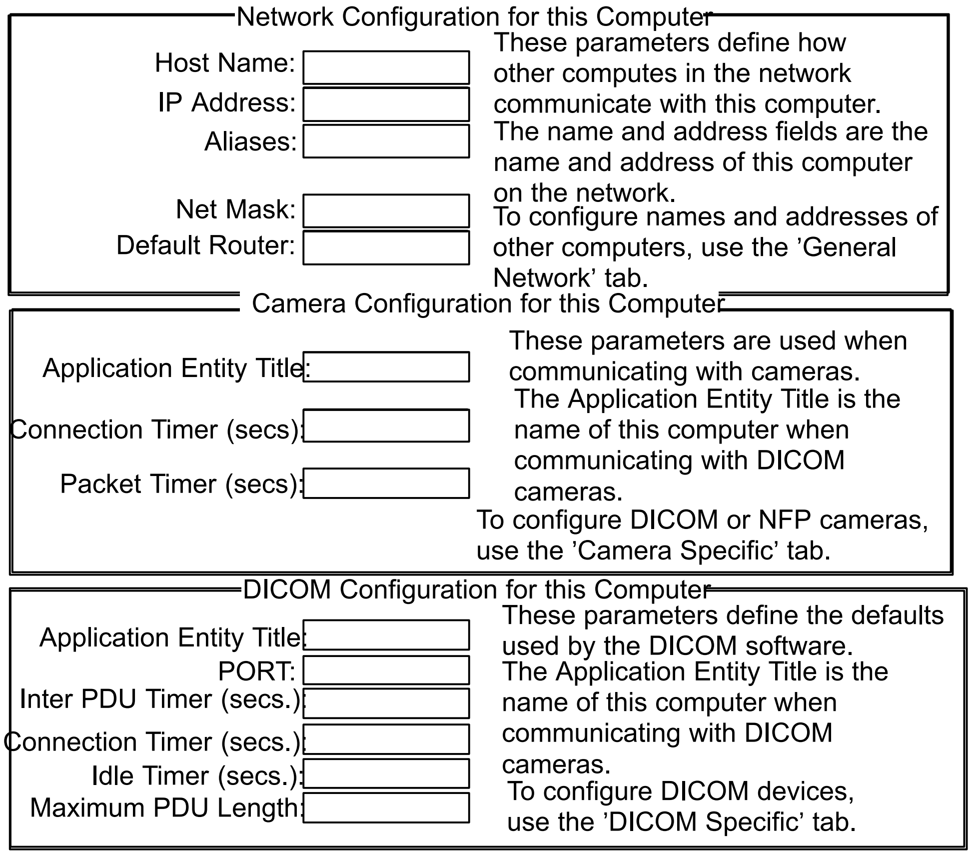


FIGURE 378 NETWORK/CAMERA (LOCAL PARAMETERS)

LOCAL PARAMETERS

LOCAL PARAMETERS consists of three subsets, *Network Configuration for this Computer*, *Camera Configuration for this Computer* and *DICOM Configuration for this Computer*. Refer to Figure 378.

Network Configuration for this Computer

Host Name: Used by other devices on the network to communicate with the DISPLAY CONSOLE. Examples of names might be st.mary1, oncology2, picker2077, etc. Names must be limited to 16 characters or less.

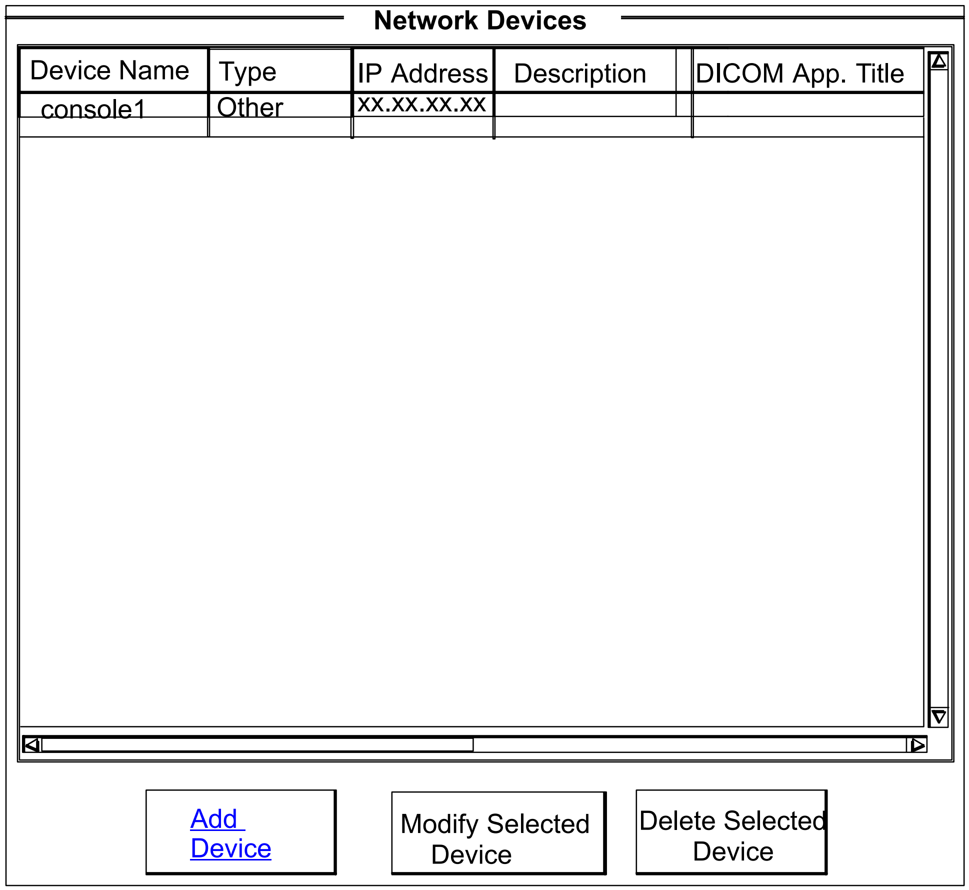
IP Address:	Associated with the Host Name. This will be the address through which the DISPLAY CONSOLE will communicate with network devices. If the scanner system and it's networked options are contained within a closed system and not connected to any outside hospital network, then you may choose a suitable IP address. You must choose an address that conforms with standard networking conventions (refer to NETWORK ADDRESSING CONVENTIONS for assistance). If the network is not connected as a closed local network, then you should consult the hospital network administrator or a field service networking specialist for assistance on the assignment of the IP address.
Aliases:	Any alternate names assigned to this host name are entered here. In general no aliases are required.
Net Mask:	A default value of 255.255.255.0 may be entered in case you have a closed network. **You must consult the network administrator or field service networking specialist for information on setting this value if you are connected to a hospital network.
Default Route:	Consult the network administrator or field service networking specialist first for information on setting this value if you are not part of a closed system. For a closed system, the default route should be a valid address for the subnet that is not assigned to any device. A default value equal to the first three fields of the IP address (subnet ID) followed by a 1 in the fourth field would be a good default for a simple closed network. This assumes there is no 1 device on the net. For example, a good default route value for the 192.168.10.xx subnet would be 192.168.10.1 (refer to NETWORK ADDRESSING CONVENTIONS for assistance).

Camera Configuration for this Computer

This box supplies configuration information for DICOM camera print support supplied by this computer. In the box accept the default timer values. These should not be changed unless instructed to do otherwise by a field specialist. The default title that appears for the Application Entity Title is OK as long as there are no other DICOM devices on the net with the same title. This title represents a logical association with the DICOM process on this computer that supports the camera function. The name chosen for this title is arbitrary in some sense but should be indicative of it's functionality for informational purposes. If you rename the title be advised that each Application Entity Title for a device on the net should be set to a unique, 16 character maximum name.

DICOM Configuration for this Computer

This box supplies configuration information for DICOM storage support supplied by this computer. In the box entitled *DICOM Configuration for this Computer*, accept the default timer, port, and PDU Length values. These should not be changed unless instructed to do otherwise by a field specialist. The default title that appears for the Application Entity Title is OK as long as there are no other DICOM devices on the net with the same title. This title represents a logical association with the DICOM process on this computer that supports the storage function. The name chosen for this title is arbitrary in some sense but should be indicative of it's functionality for informational purposes. If you rename the title be advised that each Application Entity Title for a device on the net should be set to a unique, 16 character maximum name.



PL0077

FIGURE 379 NETWORK/CAMERA (OTHER DEVICES)

OTHER DEVICES

In order that the DISPLAY CONSOLE properly recognize and communicate with other devices on the network it is necessary to provide configuration information on those devices. Selecting the *Other Devices* tab displays the screen summary context seen in Figure 379.

OTHER DEVICES consists of three subsets, *Device Type*, *Network Device Configuration* and either *DICOM Device Configuration.* or *Camera Device Configuration*. This screen (Figure 380) appears when you choose to Add, Modify or Delete a device from the summary context screen (Figure 379).

Device Type

Select the type of device to which you will be making changes (RDS/Scanner, Voxel Q, DICOM Network, DICOM Camera, LI+ Camera or other). The box just below the Network Device Configuration box will change according to the device selected. For example, if you selected RDS, then fill in as follows:

Network Device Configuration

Device Name:	This must match the name given to the RDS in it's own configuration procedure for the device to be recognized.
IP Address:	Must match IP address assigned to the RDS in it's own configuration procedure.
Device Description:	This is the name which you wish the user of the Display Console to see and use when referencing this remote network device. Examples might be rds_plus, rds_office, etc.
Aliases:	Alias names assigned to the RDS. In general, no aliases are required.

If a DICOM device is selected, you will need to supply additional information in the *DICOM Device Configuration* box. If this is a camera or Li+ you will also need to supply additional information.

> **NOTE**
>
> Figure 380 shows the DICOM Device Configuration when RDS/Scanner, Voxel Q or DICOM Net device types are selected.
> Figure 381 shows the Camera Device Configuration when DICOM Cameras or LI+ device types are are selected. When "other" is selected, a configuration box does not appear.

DICOM Device Configuration

In this box accept the default values if given. These should not be changed unless you are proficient in setting the values or are instructed to do otherwise by a field specialist. The one exception to this is the *Port* field. It must be configured to match what the Port value that the remote box has configured for itself. It will almost always be Port 104, but it can be changed. Refer to Figure 380.

Also it will be necessary to supply an *Application Entity Title*. This title represents a logical association with the process on that remote box that supports that particular DICOM function. The Application Entity Title needs to match the title that was configured on the remote box always (in this example the remote box being the RDS/Scanner).

When completed, select *Apply Changes* and the entry will be loaded.

Select *Done* and a summary context screen of Network Devices will be displayed again after a valid transaction has been completed. At that point you repeat this process for configuration information on additional devices (Voxel Q, DICOM Camera , etc.) until completed.

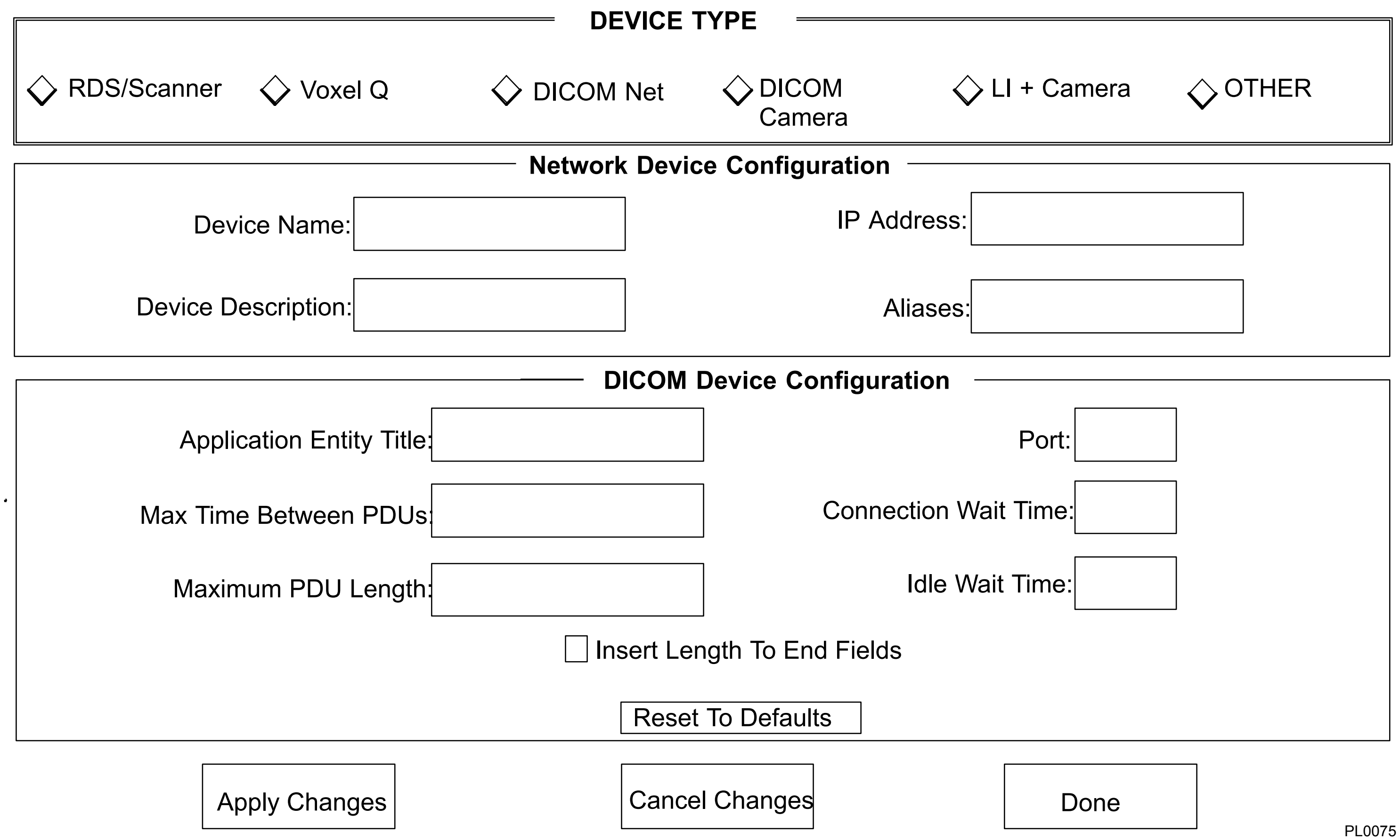


FIGURE 380 NETWORK/CAMERA, (DICOM DEVICE TYPE)

Camera Device Configuration

Follow a similar approach as the DICOM Device Configuration example to supply the information required. In addition, you must select a camera type by clicking, scrolling and selecting from the list. Refer to Figure 381. For the Li+ camera accept all default information unless you are proficient in setting the values or are instructed to do otherwise by a field specialist. In either case, remember to enter a valid Application Entity Title as previously discussed.

> **NOTE**
>
> Application Entity Title is NOT required for the LI Plus Film Server.

Other

For other devices that do not fall into the previous categories, complete a Network Device Configuration page from the Other device types for each device. Refer to Figure 382.

> **NOTE**
>
> If there are multiple devices of the same type on the network that perform similar functions, then each of those devices must be assigned a unique Application Entity Title. For example, if there were two unique UltraZ Diagnostic Workstation workstations on the network each would need a unique identifier. As an example if the first one had the title DICOM_STORAGE then the second one would need to be alternately named DICOM_STORAGE2 or STORAGE2 or something unique. In this case, you need to add each device separately and complete a separate configuration screen for each device.

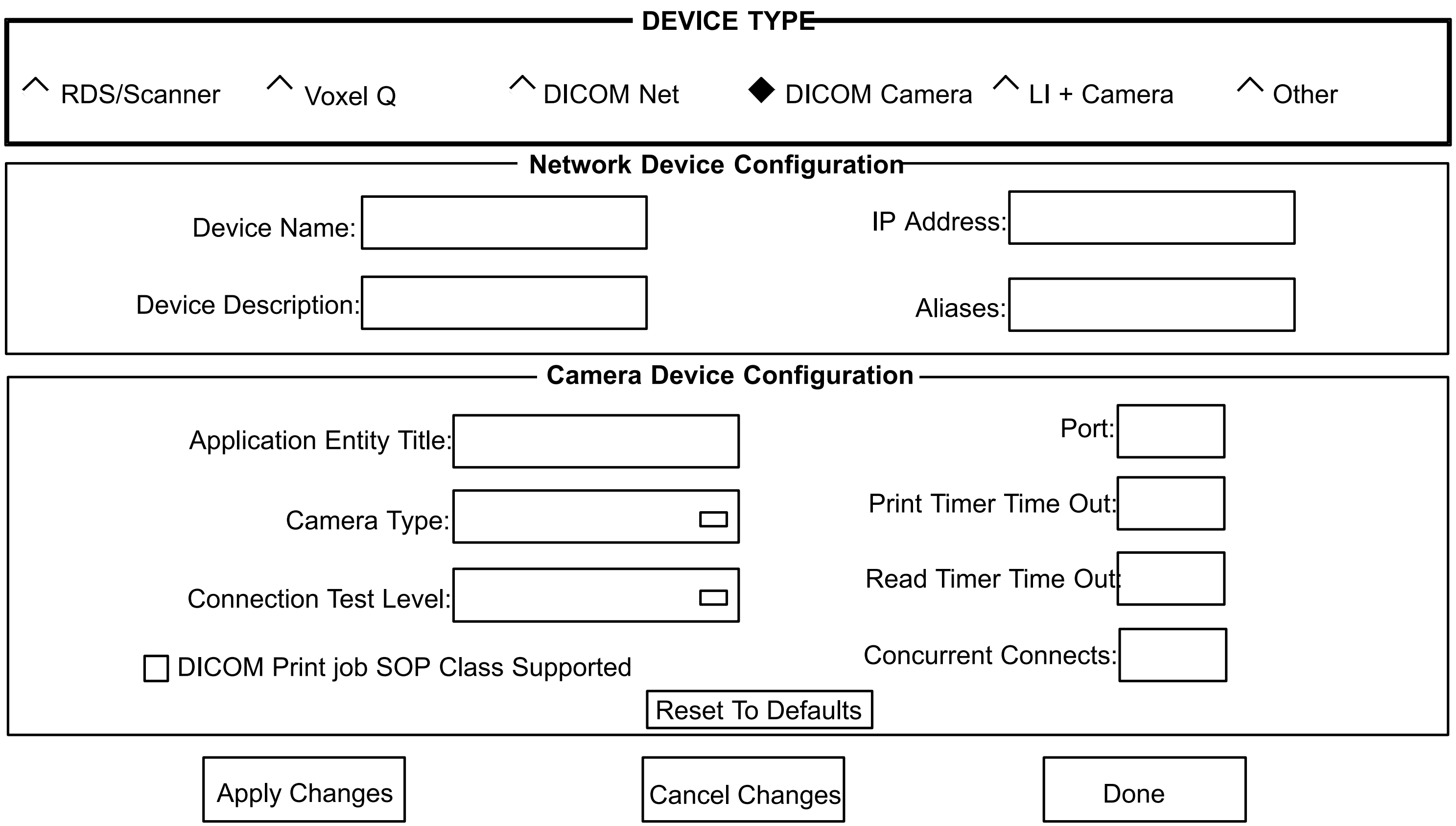


FIGURE 381 NETWORK/CAMERA, ADD DEVICE (CAMERA DEVICE TYPE)

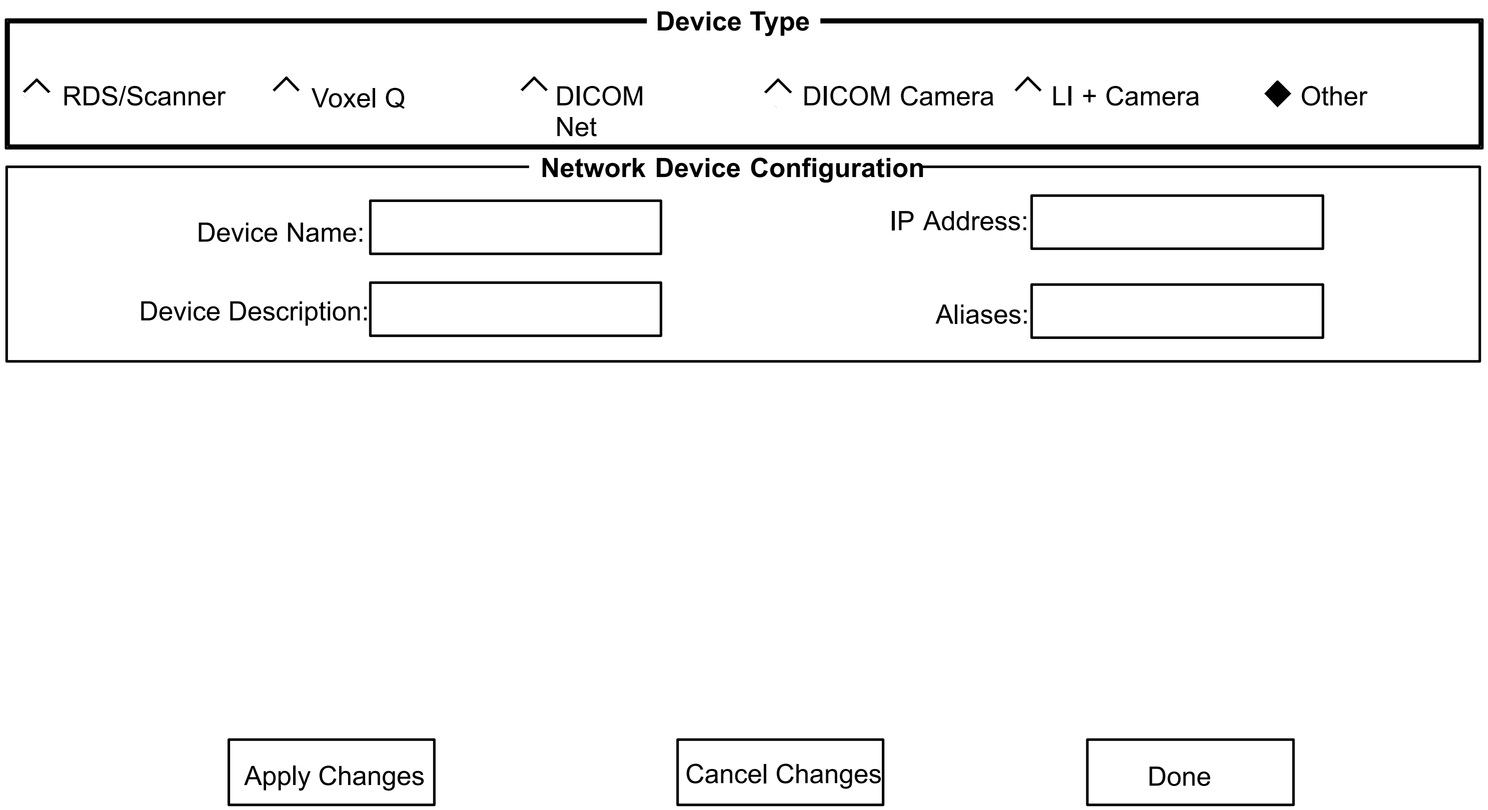


FIGURE 382 NETWORK/CAMERA, ADD DEVICE (OTHER DEVICE TYPE)

NETWORK ADDRESSING CONVENTIONS

As shown in Figure 383, the AcQSim CT product requires IP network addresses to be configured for the Display console as well as the networked components of an AcQSim CT system.

In the example shown the display tower needs connected to a RDS, a DICOM laser camera, and a Li Plus server connected to a non DICOM camera.

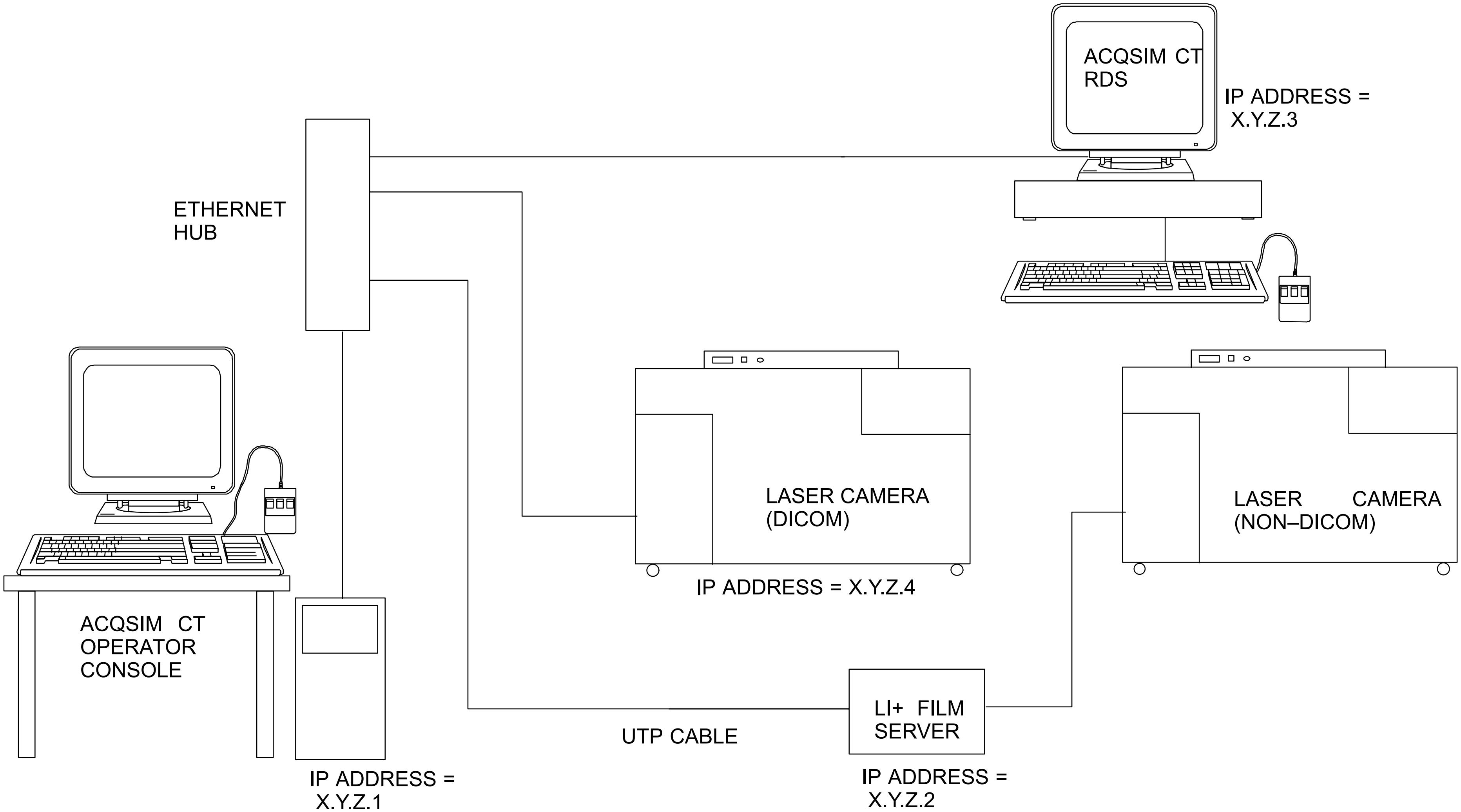


FIGURE 383 AcQSim CT “CLOSED” NETWORK CONFIGURATION

In the example the IP addresses are labeled X.Y.Z.# to indicate that the first three digits of the IP address for each component must be the same when on a closed network. X, Y, and Z can be almost any numbers (See next section).

Figure 383 shows a closed network of only AcQSim CT components and related items. Figure 384 shows a AcQSim CT network connected to another network.

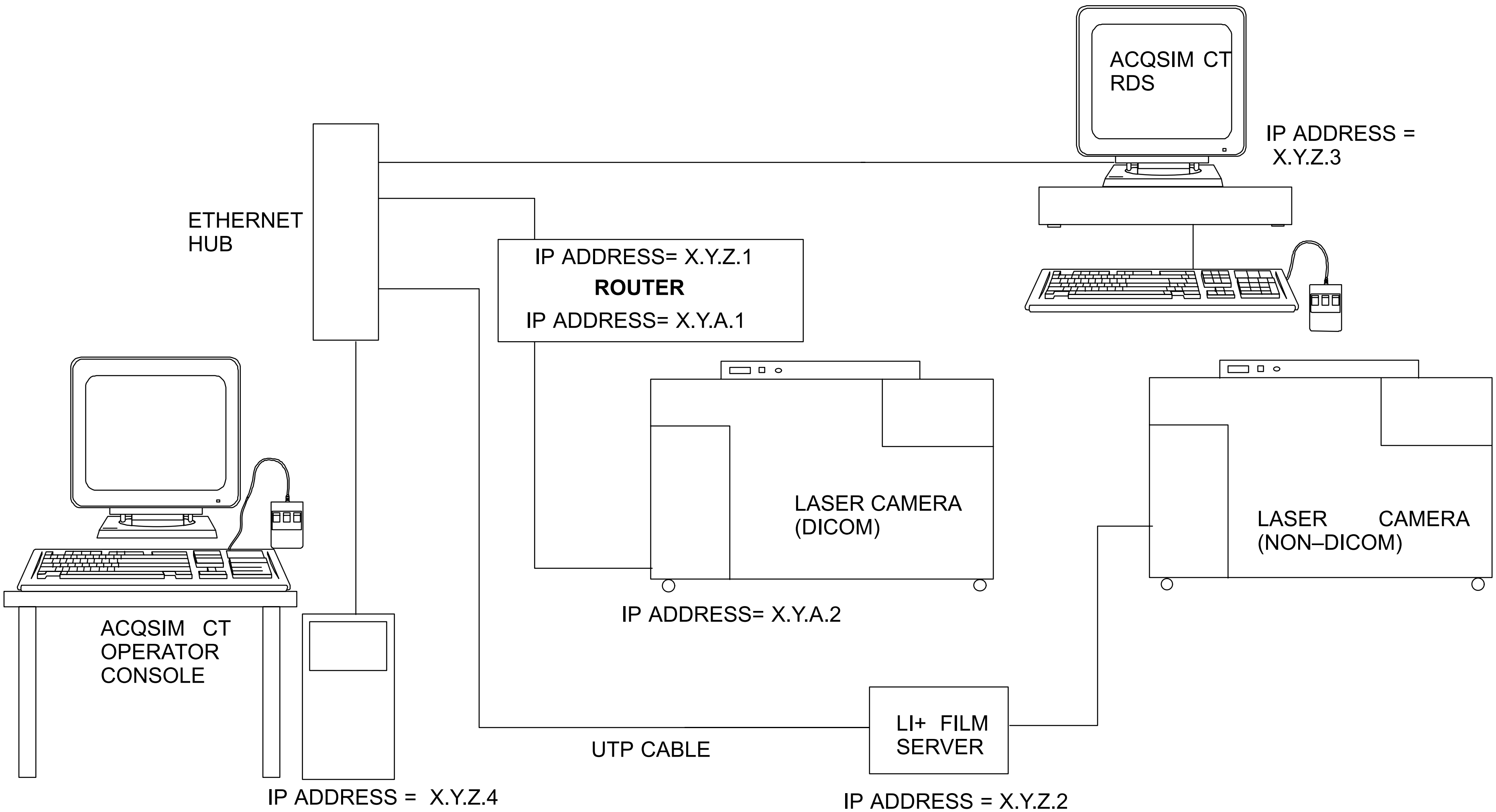


FIGURE 384 AcQSim CT CONNECTED TO ANOTHER NETWORK

In this example (Figure 384) the DICOM laser camera is connected to the X.Y.A network while the AcQSim CT system (and components) is connected to the X.Y.Z network. This example shows the use of a router to connect the two network segments together.

PICKING IP ADDRESSES

If a AcQSim CT system is connected to an external network as shown in the second example, a network administrator or other hospital personnel must provide the IP addresses.

If a AcQSim CT system is connected as closed network, as shown in the first example, and the hospital does not know or care what the IP addresses are, then any IP addresses can be used. The only stipulation is that the address of each node must have the same network ID.

Understanding IP Addresses, Subnetmasks, and Routing

Every node on a TCP/IP network is identified by a unique IP address. This address is used to identify a host on a network; it also specifies routing information on the network. The IP address identifies a computer 32–bit address that is unique across a TCP/IP network. An address is usually represented in dotted decimal notation, which depicts each octet (eight bits, or one byte) of an IP address as its decimal value and separates each octet with a period. For example: 138.238.128.28

> **NOTE**
>
> It is important to note that since an IP address identifies a node on a particular network, each node on the network must be assigned a unique IP address valid for its particular network. Therefore if a machine was assigned an IP address in the 138.238.135 subnet and is subsequently moved into the 138.238.128 subnet, a different IP address must be assigned.

Network ID and Host ID

The IP address contains two pieces of information: the network ID and the host ID for the individual computer.

The network ID identifies a group of computers and other devices that are all located on the same logical network.

The host ID identifies an individual computer within a particular network ID.

IP Address Classes

The Internet community has defined address classes to accommodate networks of varying sizes. Each network class can be discerned from the first octet of its IP address.

The table below shows the relationship between the first octet of a given address and its network ID and host ID fields. It also identifies the total number of network ID's and hosts ID's for each address class in the Internet addressing scheme. In the table a.b.c.d is used to designate the bytes of the IP address.

Class	Class Values	Network ID	Host ID	Available Networks	Available Hosts/Nets
A	1–126	a	b.c.d	126	16,777,214
B	128–191	a.b.	c.d	16, 384	65, 534
C	192–223	a.b.c	d	2,097,151	254

Note that the network addresses 0 and 127 are reserved. Addresses 224 and above are reserved for special protocols and cannot be used as network addresses.

Subnetting

CLASS A SUBNETTING

A class A address could be diagramed as:

```
  Network           Host
 +------+  +----------------------+
 |      |  |                      |
[0xxxxxxx][xxxxxxxxxxxxxxxxxxxxxxxx]
```

which shows the eight network bits followed by the 24 host bits.

In practice, people don't really attach 16 million hosts to a network so administrators of a Class A site often divide the host address portion into a (sub)network and host portion. (Subnetting is now supported by most operating systems.) Each Class A network number can support up to 65,534 subnets (network numbers 0.0 and 255.255 are reserved) with each having 254 hosts (host numbers 0 and 255 are reserved). This is done by using the 16 high –order bits of the host portion for the subnet number and the lower eight bits for the host as diagramed below.

```
 Network        Subnet         Host
 +------+  +--------------+  +------+
 |      |  |              |  |      |

[0xxxxxxx][xxxxxxxxxxxxxxxx][xxxxxxxx]
```

CLASS B SUBNETTING

The first two bits of a Class B address are 1 and 0, the next fourteen bits identify the network and the last sixteen the host, as diagramed:

```
      Network             Host
 +--------------+  +--------------+
 |              |  |              |
[10xxxxxxxxxxxxxx][xxxxxxxxxxxxxxxx]
```

As with the Class A address, we can divide the host portion of a Class B address into subnet and host parts. For instance, let's split our Class B network number on the byte boundary, that is, the eight MSB's of the host portion identifies the subnet and the remaining bits the host, as diagramed:

```
     Network        Subnet     Host
 +--------------+  +------+  +------+
 |              |  |      |  |      |
[10xxxxxxxxxxxxxx][xxxxxxxx][xxxxxxxx]
```

This arrangement allows 254 subnets (0,255 reserved) each with 254 hosts (0,255 reserved).

CLASS C SUBNETTING

The first three bits of a Class C address are 1, 1, and 0, the next 21 bits identify the network and the last eight the host, as diagramed:

```
        Network             Host
 +----------------------+  +------+
 |                      |  |      |
[110xxxxxxxxxxxxxxxxxxxxx][xxxxxxxx]
```

Class C networks can be subnetted by using 2 bits of the host address as the subnetwork with 6 bits left for host addresses as diagramed:

```
           Network           Subnet   Host
 +----------------------+    ++     +----+
 |                      |    ||     |    |
[110xxxxxxxxxxxxxxxxxxxxx]  [xx]   [xxxxxx]
```

This arrangement allows 2 subnets (0,4 reserved) each with 62 hosts (0,63 reserved)

SUBNET MASKS

Subnet masks are 32–bit values that allow the recipient of packets to distinguish the network ID portion of the IP address from the host ID. Like an IP address, the value of a subnet mask is frequently represented in dotted decimal notation.

A subnet mask (or number) is used to determine the number of bits used for the subnet and host portions of the address. The mask is a 32–bit value that uses one–bits for the network and subnet portions and zero–bits for the host portion. For example, below is a Class B address of 191.70.55.130 and apply some different subnet masks. A bitwise logical AND operation is performed between the IP address and the subnet mask as shown:

Here we use a mask that retains the default 16 network and host bits for a Class B address:

```
   191         70          55          130

1011 1111   1000 0110   0011 0111   1000 0010  IP address
1111 1111   1111 1111   0000 0000   0000 0000  Subnet mask

1011 1111   1000 0110   0000 0000   0000 0000  Result

   191         70           0           0
```

Employed here is a mask that divides the host portion into a subnet and host that are each eight bits wide:

```
  191          70           55           130

1011 1111    1000 0110    0011 0111    1000 0010  IP address
1111 1111    1111 1111    1111 1111    0000 0000  Subnet mask

1011 1111    1000 0110    0011 0111    0000 0000  Result

  191          70           55             0
```

This division allows 254 (256–2 reserved) subnets, each with 254 hosts. This division on a byte boundary makes it easy to determine the subnet and host from the dotted–decimal IP address.

The subnet–host boundary, however, can be at any bit position in the host portion of the IP address. Here, we use a mask that allows more subnets (512–2 reserved), but with the trade–off of fewer hosts (128–2) per subnet:

```
   191          70           55           130

1011 1111    1000 0110    0011 0111    1000 0010  IP address
1111 1111    1111 1111    1111 1111    1000 0000  Subnet mask

1011 1111    1000 0110    0011 0111    1000 0000  Result

   191          70           55           128
```

The table below shows default subnet masks for standard IP classes:

Address Class	Default Bits for Subnet Mask	Subnet Mask
Class A	11111111 00000000 00000000 00000000	255.0.0.0
Class B	11111111 11111111 00000000 00000000	255.255.0.0
Class C	11111111 11111111 11111111 00000000	255.255.255.0

NOTE

Not all devices support masking to any bit. Some require that the subnet mask be on byte boundaries only as shown above.

The subnet–host number tradeoff

Here’s a table that let’s you see at a glance the trade off between the number of subnets and hosts with different subnet masks for both Class B and Class C addresses. We’ve already subtracted two from the results in the last two columns to take the reserved network and host numbers into account:

Class B Subnetting:

# Mask Bits	Subnet Mask	# Subnets	# Hosts
2	255.255.192.0	2	16382
3	255.255.224.0	6	8190
4	255.255.240.0	14	4094
5	255.255.248.0	30	2046
6	255.255.252.0	62	1022
7	255.255.254.0	126	510
8	255.255.255.0	254	254
9	255.255.255.128	510	126
10	255.255.255.192	1022	62
11	255.255.255.224	2046	30
12	255.255.255.240	4094	14
13	255.255.255.248	8190	6
14	255.255.255.252	16382	2

Class C Subnetting:

# Mask Bits	Subnet Mask	# Subnets	# Hosts
2	255.255.255.192	2	62
3	255.255.255.224	6	30
4	255.255.255.240	14	14
5	255.255.255.248	30	6
6	255.255.255.252	62	2

Routing

The most common implementation of subnetting is placing routers between the subnets. Routers are physical pieces of equipment that connect two or more subnetworks.

A router will have two or more ethernet connections (1 connection per subnetwork) and provides the ability to route network packets to the appropriate subnetwork.

In example 2, the router has connections to the subnetworks X.Y.Z and X.Y.A The router has an IP address of X.Y.Z.1 for connecting to the X.Y.Z subnetwork as well as and IP address of X.Y.A.1 for connection for the X.Y.A network.

If the AcQSim CT display console sends images to the RDS all of the data is contained in the X.Y.Z network. The router is not used. If the display console sends images to the DICOM camera then the router is used. The router uses the IP address and subnetmask to determine where to forward the network packets.

What’s It All Mean?

When installing a AcQSim CT system in a closed network environment and the site does not have a specific IP address required, use a class C address. Do not use .0 or .255 in any of your addressing schemes as they are reserved for special functions. A closed network in this case is one in which the UltraZ and peripherals like camera, UDW, etc are only connected to each other (i.e., no connections exist to the hospital or outside network.)

You should use the following scheme for a closed system:

```
IP Address                      Device

192.168.10.10                   Display Console (Scanner)
192.168.10.20                   RDS +
192.168.10.30                   DICOM Camera
192.168.10.40                   Li + Server

        and so on
```

> **NOTE**
>
> The 192.168.XX.XX IP address range is a universally accepted standard for closed networks.

Please note that if the site at a later date wants the equipment connected to a another network, the IP addresses must be changed.

The hospital must provide the IP addresses to use for the AcQSim CT components when connecting the scanner to external networks.

> ⚠ **WARNING**
>
> Never use the IP addresses 144.54.xx.xx as network addresses as they are reserved for in–house Picker/Philips networks. Use of these addresses could cause serious network conflicts

CONFIGURE – ACQSIM CT DIAGNOSTIC WORKSTATION

The different CONFIGURE menus for the AcQSim CT Diagnostic Workstation include Site, Network/Camera, Options and Generic configurations. Refer to Figure 385.

Site	Network/Camera	Options	Generic

FIGURE 385 CONFIGURATION SELECTIONS

The following procedures detail each individual configuration, based on Figure 385, going from left to right. ** Only **SITE** and **Network/Camera** and **Options** are pertinent for UltraZ Diagnostic Workstation.

You may click on the buttons in Figure 385 to jump to their respective screens.

Site: Allows complete SITE configuration.

FIGURE 386 SITE CONFIGURATION

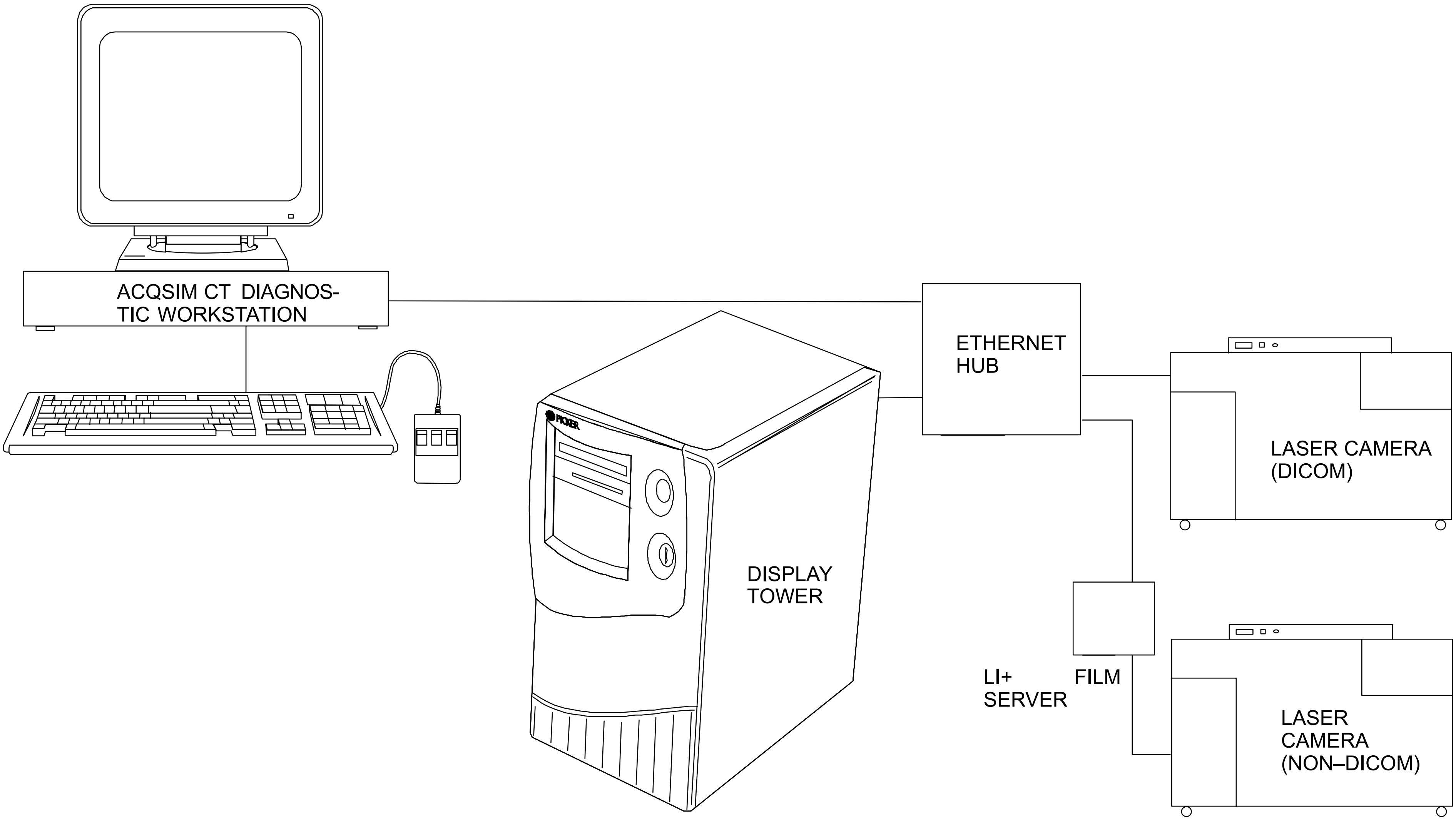


FIGURE 387 EXAMPLE OF ETHERNET NETWORK WITH UltraZ Diagnostic Workstation

NETWORK PORT CONFIGURATION

The UltraZ Diagnostic Workstation is a networked device and configuration must be performed for it to properly communicate with other network devices. Refer to Figure 387 for an example. The NETWORK PORT is not configured as shipped from the factory. Each installation is **unique and must be done on site**. It will be necessary to enter IP address information as well as other parameters for the UltraZ Diagnostic Workstation and other devices connected to it via the network. This section refers to configuration details for the UltraZ Diagnostic Workstation. As a rule, all other devices connected on the network will need to be configured locally on their own as well. Refer to individual installation manuals and instructions for these specific devices (e.g. scanner, other UltraZ Diagnostic Workstation, Voxel–Q, Li +, Dicom Camera, etc.).

Values are entered in the Configure | Network/Camera option of the Service Application.

The Network/Camera Configure Option consists of *Local Parameters* and *Other Devices* tabs:

> **NOTE**
>
> Refer to the Networking Glossary for assistance with definition of terms.

Network/Camera (Local Parameters): Allows complete local Network configuration:

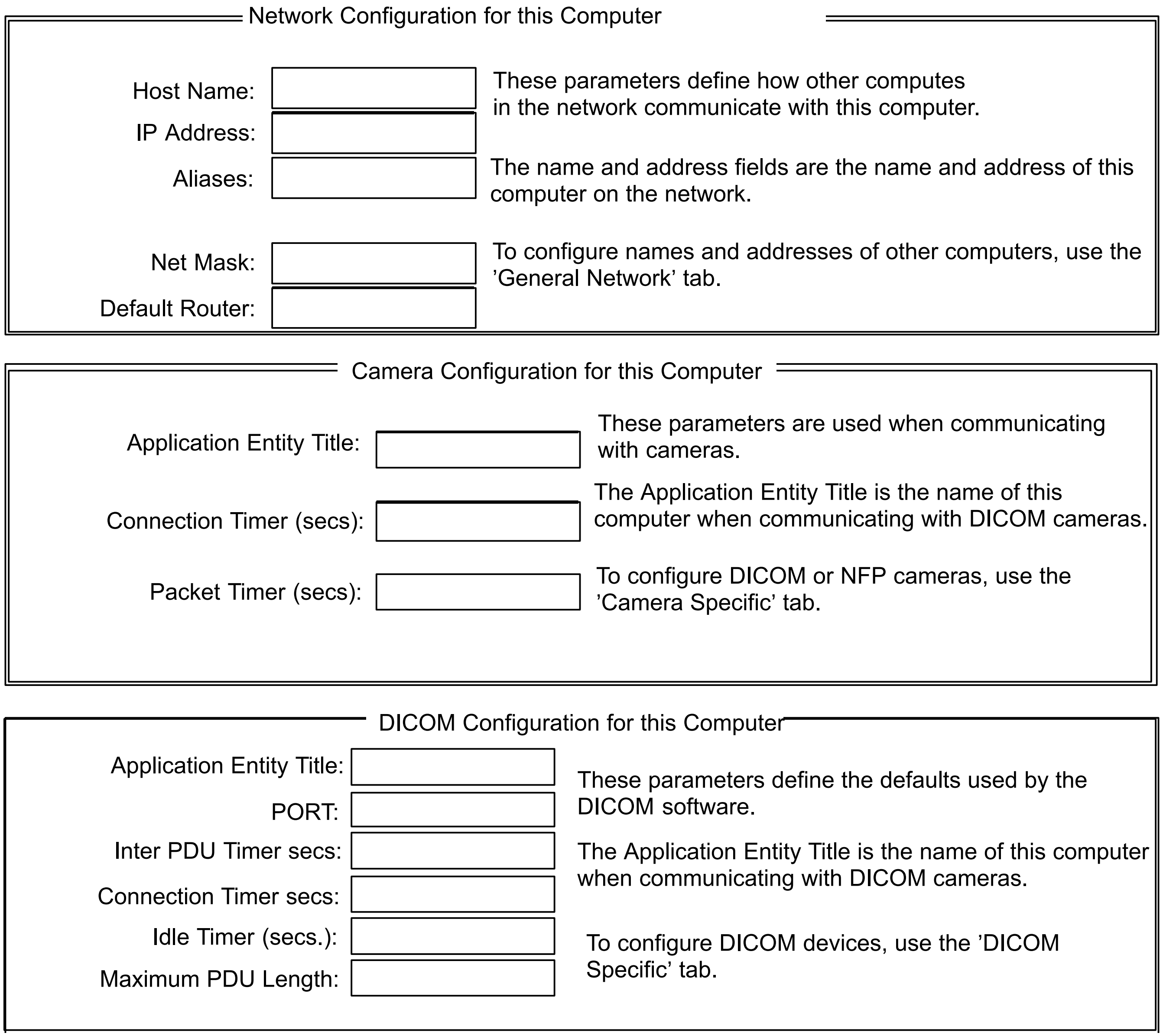


FIGURE 388 NETWORK/CAMERA (LOCAL PARAMETERS)

LOCAL PARAMETERS

LOCAL PARAMETERS consists of three subsets, *Network Configuration for this Computer*, *Camera Configuration for this Computer* and *DICOM Configuration for this Computer*. Refer to Figure 388.

Network Configuration for this Computer

Host Name: Used by other devices on the network to communicate with the UltraZ Diagnostic Workstation. Examples of names might be st.mary1, oncology2, picker2077, etc. Names must be limited to 16 characters or less.

IP Address: Associated with the Host Name. This will be the address through which the UltraZ Diagnostic Workstation will communicate with network devices. If the UltraZ Diagnostic Workstation and networked devices are contained within a closed system and not connected to any outside hospital network, then you may choose a suitable IP address. You must choose an address that conforms with standard networking conventions (refer to NETWORK ADDRESSING CONVENTIONS for assistance). If the network is not connected as a closed local network, then you should consult the hospital network administrator or a field service networking specialist for assistance on the assignment of the IP address.

Aliases: Any alternate names assigned to this host name are entered here. In general no aliases are required.

Net Mask: A default value of 255.255.255.0 may be entered in case you have a closed network. **You must consult the network administrator or field service networking specialist for information on setting this value if you are connected to a hospital network.

Default Route: Consult the network administrator or field service networking specialist first for information on setting this value if you are not part of a closed system. For a closed system, the default route should be a valid address for the subnet that is not assigned to any device. A default value equal to the first three fields of the IP address (subnet ID) followed by a 1 in the fourth field would be a good default for a simple closed network. This assumes there is no 1 device on the net. For example, a good default route value for the 192.168.10.xx subnet would be 192.168.10.1 (refer to NETWORK ADDRESSING CONVENTIONS for assistance).

Camera Configuration for this Computer

This box supplies configuration information for DICOM camera print support supplied by this computer. In the box accept the default timer values. These should not be changed unless instructed to do otherwise by a field specialist. The default title that appears for the Application Entity Title is OK as long as there are no other DICOM devices on the net with the same title. This title represents a logical association with the DICOM process on this computer that supports the camera function. The name chosen for this title is arbitrary in some sense but should be indicative of it's functionality for informational purposes. If you rename the title be advised that each Application Entity Title for a device on the net should be set to a unique, 16 character maximum name.

DICOM Configuration for this Computer

This box supplies configuration information for DICOM storage support supplied by this computer. In the box entitled *DICOM Configuration for this Computer*, accept the default timer, port, and PDU Length values. These should not be changed unless instructed to do otherwise by a field specialist. The default title that appears for the Application Entity Title is OK as long as there are no other DICOM devices on the net with the same title. This title represents a logical association with the DICOM process on this computer that supports the storage function. The name chosen for this title is arbitrary in some sense but should be indicative of it's functionality for informational purposes. If you rename the title be advised that each Application Entity Title for a device on the net should be set to a unique, 16 character maximum name.

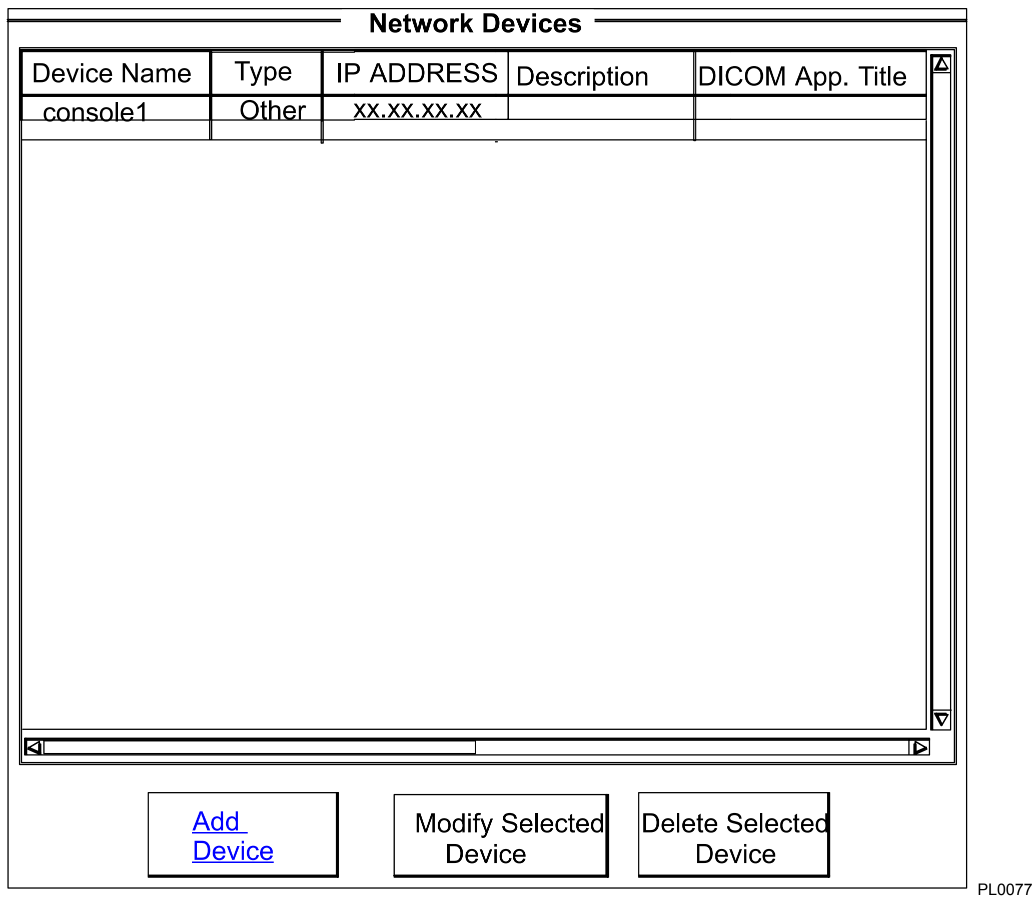


FIGURE 389 NETWORK/CAMERA (OTHER DEVICES)

OTHER DEVICES

In order that the UltraZ Diagnostic Workstation properly recognize and communicate with other devices on the network it is necessary to provide configuration information on those devices. Selecting the *Other Devices* tab displays the screen summary context seen in Figure 379*.*

OTHER DEVICES consists of three subsets, *Device Type*, *Network Device Configuration* and either *DICOM Device Configuration.* or *Camera Device Configuration*. This screen (Figure 390) appears when you choose to Add, Modify or Delete a device from the summary context screen (Figure 389).

Device Type

Select the type on which you will be making changes (RDS/Scanner, Voxel Q, DICOM Network, DICOM Camera, LI+ Camera, other). The box just below the Network Device Configuration box will change according to the device selected, e.g., if you selected RDS/Scanner (for the scanner), then fill in as follows:

Network Device Configuration

Device Name:	This must match the name given to the scanner in it's own configuration procedure for the device to be recognized.
IP Address:	Must match IP address assigned to the scanner in it's configuration procedure.
Device Description:	This is the name which you wish the user of the UltraZ Diagnostic Workstation to see and use when referencing this remote network device. Examples might be scanner, scanner_oncology, etc.
Aliases:	Alias names assigned to the scanner. In general, no aliases are required.

If a DICOM device is selected, you will need to supply additional information in the *DICOM Device Configuration* box. If this is a camera or Li+ you will also need to supply additional information.

> **NOTE**
>
> Figure 390 shows the DICOM Device Configuration when RDS/Scanner, Voxel Q or DICOM Net device types are selected.
> Figure 391 shows the Camera Device Configuration when DICOM Cameras or LI+ device types are are selected. When "other" is selected, a configuration box does not appear.

DICOM Device Configuration

In this box accept the default values if given. These should not be changed unless you are proficient in setting the values or are instructed to do otherwise by a field specialist. The one exception to this is the *Port* field. It must be configured to match what the Port value that the remote box has configured for itself. It will almost always be Port 104, but it can be changed. Refer to Figure 390.

It will be necessary to supply an *Application Entity Title*. This title represents a logical association with the process on that remote box that supports that particular DICOM function. The Application Entity Title needs to match the title that was configured on the remote box (in this example, the remote box being the scanner).

When completed, select *Apply Changes* and the entry will be loaded.

Select *Done* and a summary context screen of Network Devices will be displayed again after a valid transaction has been completed. At that point you repeat this process for configuration information on additional devices (Voxel Q, DICOM Camera , etc.) until completed.

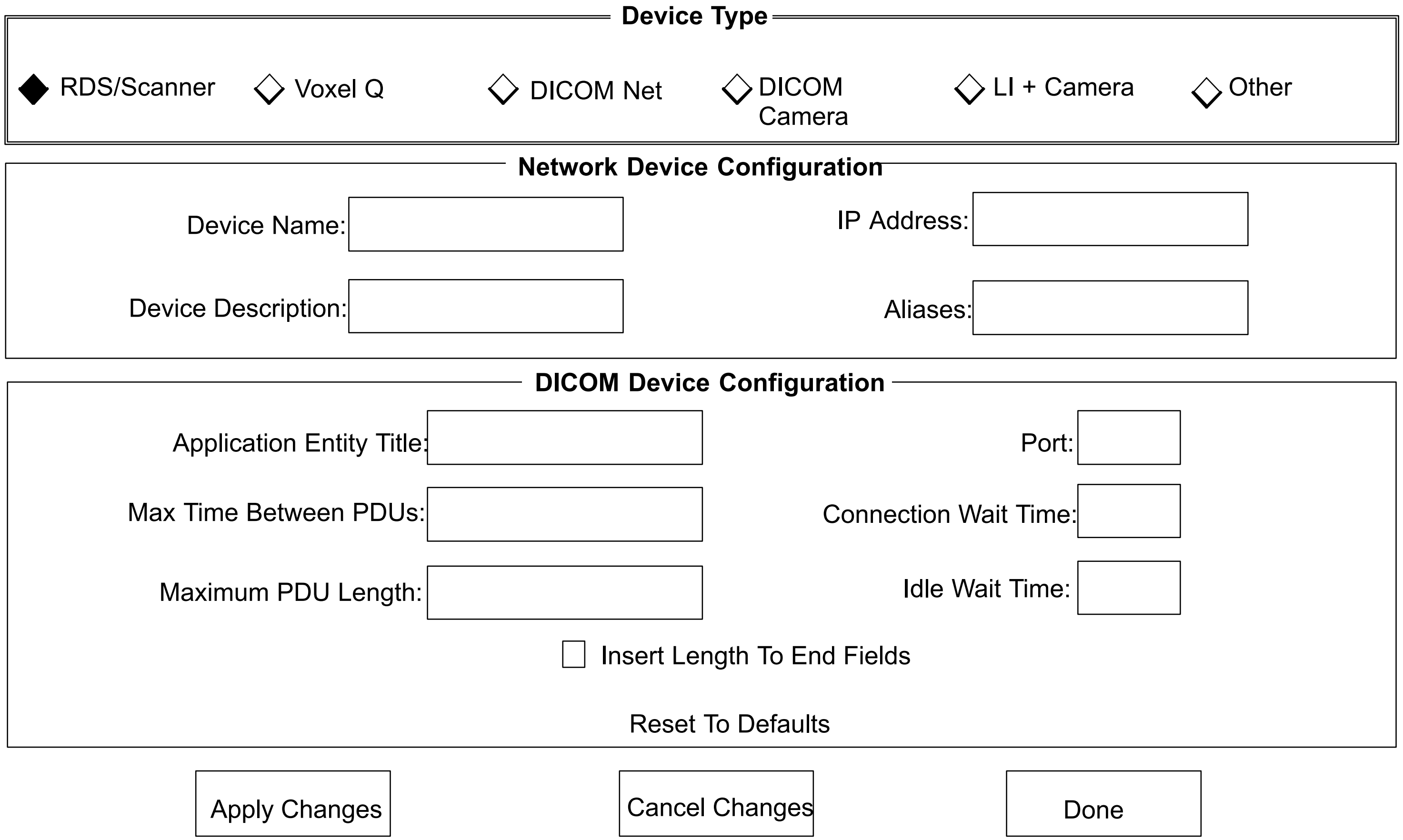


FIGURE 390 NETWORK/CAMERA, ADD DEVICE (DICOM DEVICE TYPE)

Camera Device Configuration

Follow a similar approach as the DICOM Device Configuration example to supply the information required. In addition, you must select a camera type by clicking, scrolling and selecting from the list. Refer to Figure 391. For the Li+ camera accept all default information unless you are proficient in setting the values or are instructed to do otherwise by a field specialist. In either case, remember to enter a valid Application Entity Title as previously discussed.

Other

For other devices that do not fall into the previous categories, complete a Network Device Configuration page from the Other device types for each device. Refer to Figure 392.

NOTE

If there are multiple devices on the network, then each of those devices must be assigned a unique Application Entity Title. For example, if there were two unique UltraZ Diagnostic Workstation workstations on the network each would need a unique identifier. As an example if the first one had the title DICOM_STORAGE then the second one would need to be alternately named DICOM_STORAGE2 or STORAGE2 or something unique. In this case, you need to add each device separately and complete a separate configuration screen for each device.

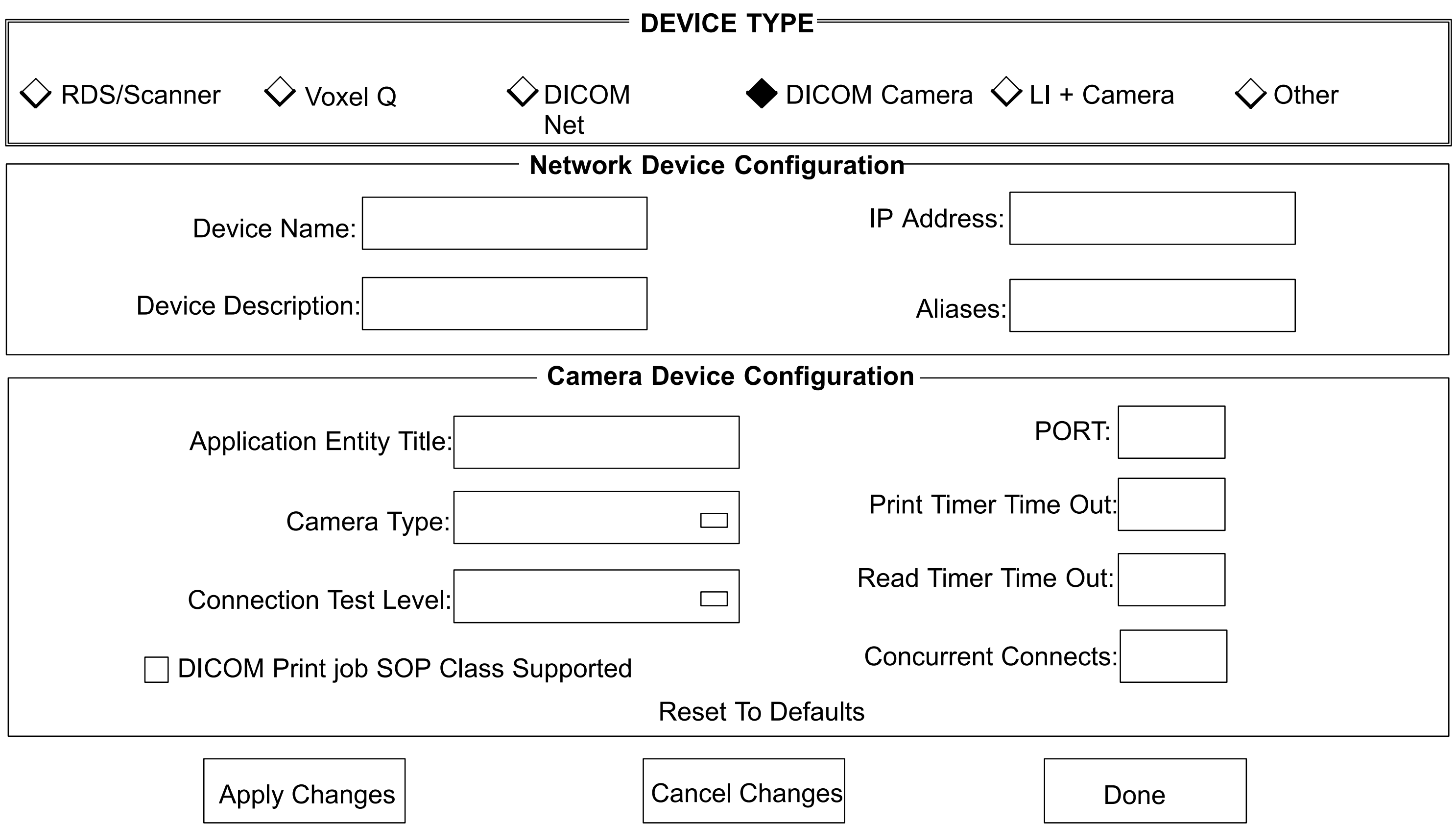


FIGURE 391 NETWORK/CAMERA, ADD DEVICE (CAMERA DEVICE TYPE)

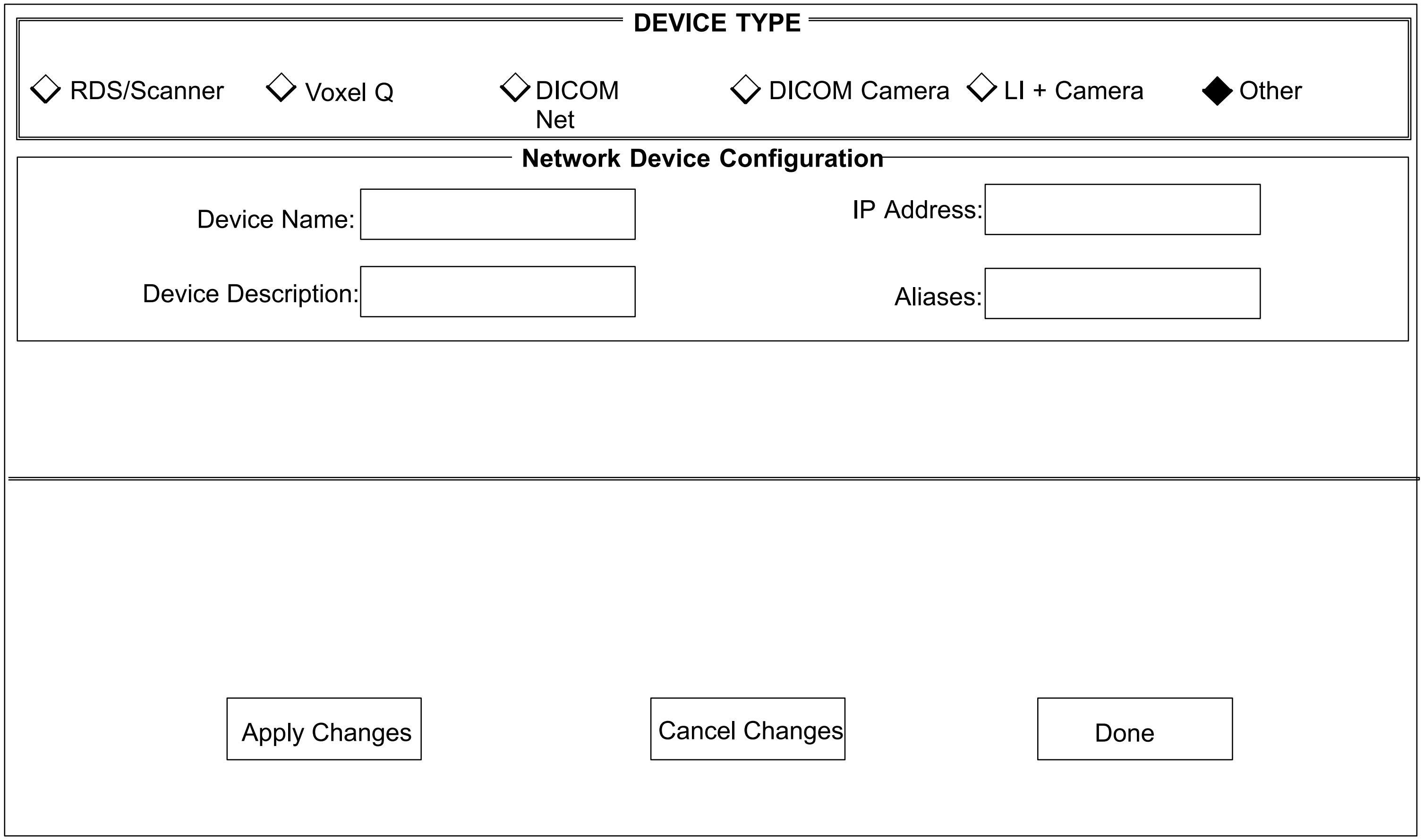


FIGURE 392 NETWORK/CAMERA, ADD DEVICE (OTHER DEVICE TYPE)

NETWORKING GLOSSARY

10Base–2 — a transmission medium specified by IEEE 802.3 that carries information at rates up to 10Mbps in baseband form using low–cost coaxial cable over distances up to 185 meters (607 ft). Also called thin Ethernet, thinnet, thin coax, or cheapernet.

10Base–5 — a transmission medium specified by IEEE 802.3 that carries information at rates up to 10Mbps in baseband form using 50 ohm coaxial cable over distances up to 500 meters (1,640 ft). Also called thick Ethernet or thicknet or thick coax, the cable is commonly referred to as yellow cable. Thick Ethernet cable is typically used as a trunk or backbone path of the network.

10Base–FL — IEEE 802.3 Fiber Optic Ethernet. A fiber optic standard that allows up to 2,000 meters (6,560 ft.) of multimode duplex fiber optic cable in a point–to–point link.

10Base–T — a transmission medium specified by IEEE 802.3 that carries information at rates up to 10Mbps in baseband form using twisted pair conductors. Also called unshielded twisted pair (UTP) wire. Using low cost Level 3 or better UTP wiring, 100 meters (328 ft.) of point–to–point link segments are possible. Uses RJ45 connectors and sometimes 50–pin AMP connectors to a patch panel.

100Base–TX — A particular alternative within the 100Base–TX CSMA/CD proposals before the IEEE 802.3 for a 100 Mb/s Ethernet that specifies two pair of UTP5.

100Base–T — A generic name for 100 Mb/s twisted pair CSMA/CD proposals before the IEEE 802.3. Specific proposals include 100Base–Tx and 00Base–T4.

A

Address — a number uniquely identifying each node in a network.

ARP — (Address Resolution Protocol) ¿ that dynamically maps Internet addresses to physical (hardware) addresses on a LAN.

ATM — (Asynchronous Transfer Mode) a new type of cell switching technology which uses fixed–length packets to transmit data from source to destination. ATM uses fixed–length 53–byte cell–switching to transmit data, voice and video over both LANs and WANs. Also referred to as BISDN and Cell Relay.

AUI
(Attachment Unit Interface) the branch cable interface located between a MAU (transceiver) and a DTE (typically a workstation). Includes a 15–pin D–sub connector and sometimes a 15–conductor twisted pair cable. Maximum length is 50 meters (164 ft.).

B

Backbone
any network considered to provide interconnection among subnetworks.

Bridge
a LAN interconnection device used to link two or more local or remote LANs. Bridges are used extensively in LAN systems to extend their physical dimensions or modify their performance. C

Collision
an unwanted condition in which two packets are being transmitted over a medium at the same time, resulting in destruction of the data.

CSMA
(Carrier Sense Multiple Access) ¿
A station wishing to transmit first senses the medium and transmits only if the medium is idle.

CSMA/CD
(Carrier Sense Multiple Access with Collision Detection) – a refinement of CSMA in which a station ceases transmission if it detects a collision.

E

Ethernet
the LAN technology that uses CSMA/CD physical access method and 10 Mbps digital transmission. The forerunner of the IEEE 802.3 CSMA/CD standard.

F

FDDI
(Fiber Distributed Data Interface) ¿
and the topology is a dual–attached, counter–rotating Token Ring. The FDDI protocol has also been adapted to run over traditional copper wires.

FOIRL
(Fiber Optic Inter Repeater Link) – A fiber optic signaling method based on the IEEE 802.3 standard governing fiber optics. Allows up to 1,000 meters (3,280 ft.) of multimode duplex fiber optic cable in a point–to–point link.

H

Hub/Repeater
the central signal distributor, used in a wiring topology consisting of several point–to–point segments originating from a central point. The term hub is often used interchangeably with the term repeater. Multiport 10BASE–T, 10BASE2, and fiber optic (10BASE–FL, FOIRL) repeaters are considered hubs.

I

IEEE 802.3	a physical layer standard that uses the CSMA/ CD access method on a bus topology LAN.
IEEE 802.5 (Token Ring)	the IEEE committee and its specification that defined a LAN protocol suite. Originated by IBM, now an IEEE standard for a token–passing, ring network that can be configured in a star topology. Token Ring cards are available in 4 Mb/s and 16 Mb/s versions. Subsequent upgrades for fiber are specified in ANSI X3T9.
IP (Internet Protocol)	a connectionless protocol which provides best–effort delivery of datagrams across an internet (the network layer protocol of the TCP/IP protocol suite).
IPX	(Internetwork Packet Exchange) a network layer protocol developed by Novell, Inc. and used in NetWare implementations.
L	LAN (Local Area Network) – a network system that provides a relatively small area with high–speed data transmission at a low error rate. May include PCs, printers, minicomputers, and mainframes linked by a transmission medium such as a coaxial cable or twisted pair wiring.
Learning bridge	a bridge which automatically learns the topology of the LAN addresses of each node as it receives packets. Requires little or no setup at time of installation.

M

MAU	(medium attachment unit) – a device used to attach a processing node to a network at the physical level. An example is the transceiver used to attach devices to an Ethernet cable.
Medium	a physical conduit for data transmission, e.g., coaxial cable or radio waves.
Multiport repeater	a repeater that collects signals from one transmission channel and, after performing the standard repeater functions, retransmits the signals to more than one new transmission channel.
Multiport transceiver	a transceiver that allows a number of devices to be attached to one LAN transceiver attachment on the backbone network.

N

Node	a point where one or more functional units interconnect transmission lines.

P

Packet	the basic unit of data transfer in LANs. A chain of one or more buffers that compose a network message.
Packet Internet Groper	(Ping) – A program used to test reachability of destinations by sending them an ICMP echo request and waiting for a reply.
Protocol	the rules or conventions used to govern the exchange of information between networked nodes.
R	
Repeater	a hardware device that regenerates LAN signals to extend the length, topology or interconnectivity of the network, or converts signals between media at the same time.
Rmon	Remote network monitoring probe – a device that was designed to help perform network management on a network segment.
RJ45	this connector is a 10BASE–T standard for connecting UTP cabling. They are inexpensive and easy to install onto UTP cable.
Router	a hardware/software product which receives network layer datagrams and forwards them to their destination based on the network layer address in the packet.
S	
SNMP	(Simple Network Management Protocol) – a set of rules for performing network management functions. Approved for use with TCP/IP in UNIX environments. Created within the Internet community using the RFC process.
Subnetwork	a network that has been connected to a larger and more powerful network system by a bridge or router.
T	
TCP	(Transmission Control Protocol) the transport protocol offering a connection–oriented transport service in the Internet suite of protocols.
TCP/IP	the internetworking protocols developed by the U.S. government's Advanced Research Project Agency (ARPA). Widely adopted and supported by computer and software manufacturers as a standard computer networking protocol.
Thicknet	see 10BASE5

Thinnet – see 10BASE2

Token Ring (IEEE 802.5) the IEEE committee and its specification that defined a LAN protocol suite. Originated by IBM, now an IEEE standard for a token–passing, ring network that can be configured in a star topology. Token Ring cards are available in 4 Mb/s and 16 Mb/s versions. Subsequent upgrades for fiber are specified in ANSI X3T9.

Topology the physical arrangement of devices in a network, regardless of their logical relationships. Types include star, ring, and bus.

Transceiver the attachment hardware connecting the controller interface to the transmission cable in IEEE 802.3 networks.

Twisted pair wiring two insulated wires twisted together and used for transmission (the twisting creates a low level of noise elimination).

U

UTP Unshielded twisted pair – see 10BASE–T

W

WAN (wide area network) – a network that includes nodes distributed over a larger geographic area than can be served by a LAN.

This ends the Network/Camera section of the CONFIGURE option from the main Service Apps menu.

NOTE

The next section displays the AcQImage section of the CONFIGURE option from the main Service Apps menu. The AcQImage selections have defaults set at the factory. The following screens have been provided to show the numerous settings/options.

If you wish to change a setting and are unsure of the terminology, contact Service Engineering.

After selecting the AcQImage button from the CONFIGURE menu. Submenus consist of Scan, General, Recon, Detector, Hardware.

AcQImage (Scan): Shows the AcQImage Scan configuration. (The opening screen defaults to SCAN:)

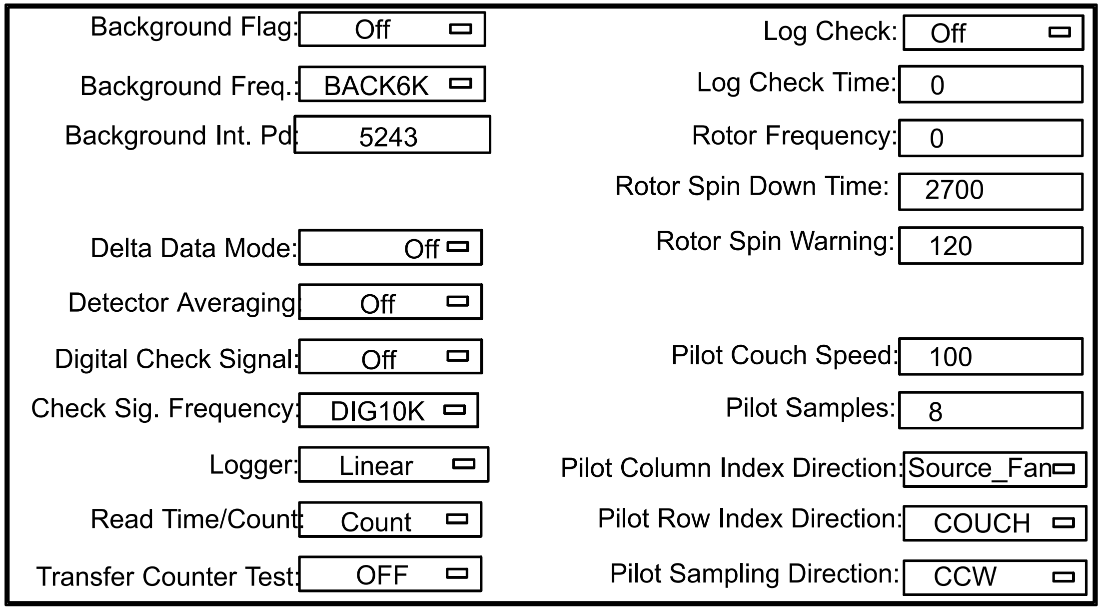


FIGURE 393 ACQIMAGE CONFIGURATION – **SCAN**

AcQImage (General): Shows complete AcQImage General configuration:

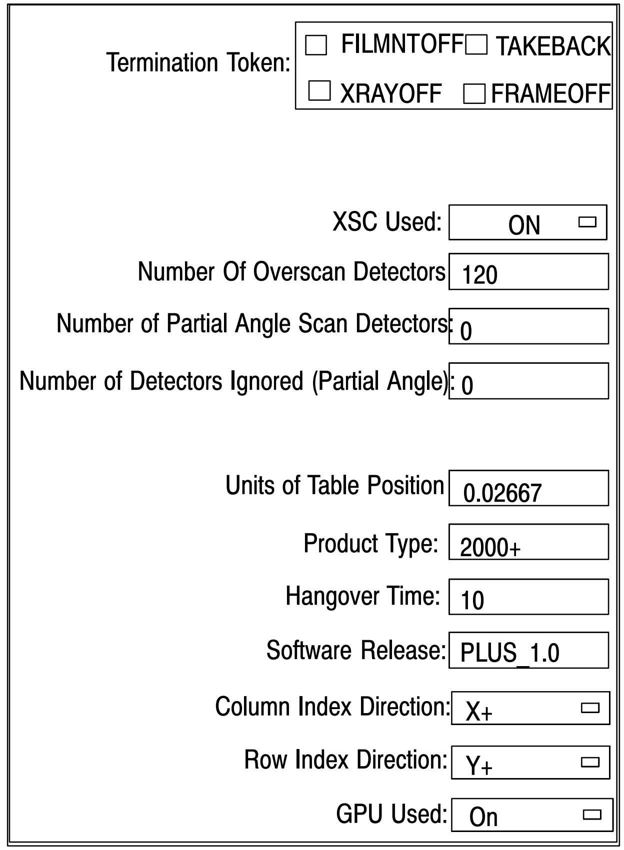


FIGURE 394 ACQIMAGE CONFIGURATION – **GENERAL**

AcQImage (Recon): Shows complete AcQImage Recon configuration:

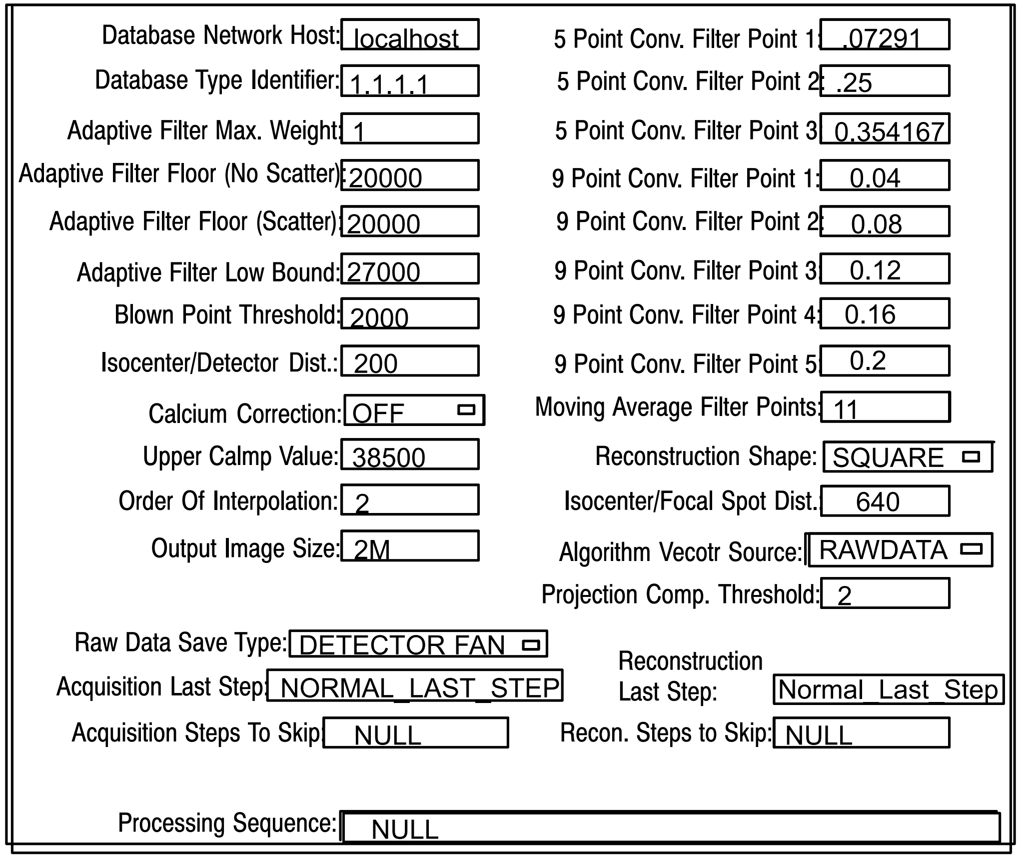


FIGURE 395 ACQIMAGE CONFIGURATION – **RECON**

AcQImage (Detector): Shows complete AcQImage Detector configuration:

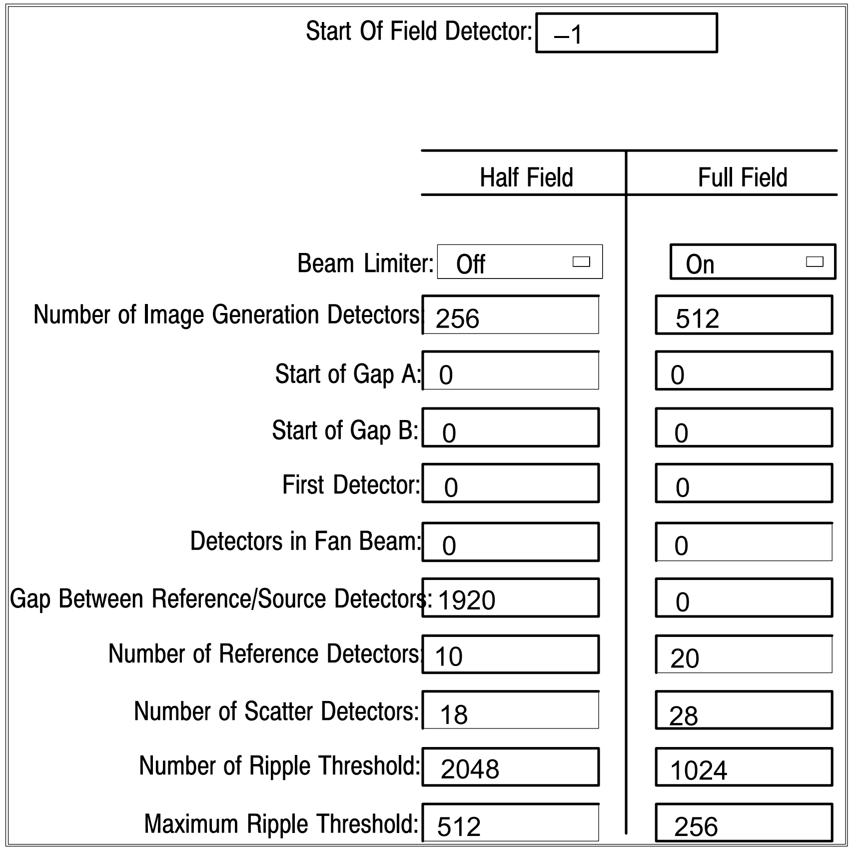


FIGURE 396 ACQIMAGE CONFIGURATION – **DETECTOR**

AcQImage (Hardware): Shows complete AcQImage Detector configuration:

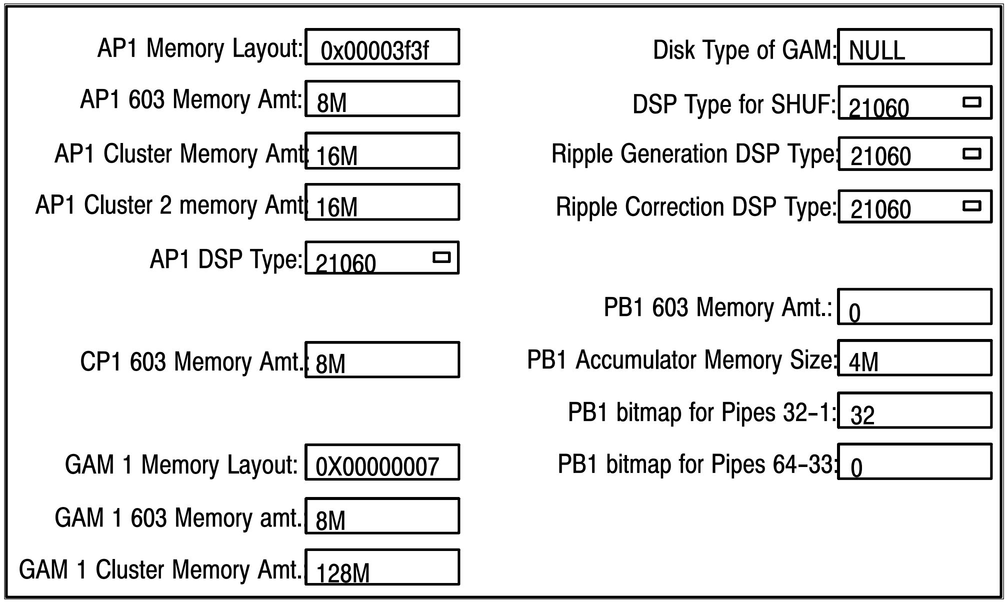


FIGURE 397 ACQIMAGE CONFIGURATION – **HARDWARE**

Options: Allows complete OPTIONS configuration:

Enabling Key	Option	Expiration
Enabling key strings will be in these locations		
	Picker Service	
	Extra Disks	
	Host ID Change Validation	
	Xray Tube Type: RHINO5_0	
	Auto Send to RDS Support	
	Auto Send to Voxel Q Support	
	MO Drive Attachment	
	Second Tape Drive Attachment	
	Dumb Injectors Support	
	C–Arm Present	
	SVIP Support	
	CTA Support	
	MSSI	
	Pinpoint	

FIGURE 398 OPTIONS CONFIGURATION

Generic: Allows _______ configuration:

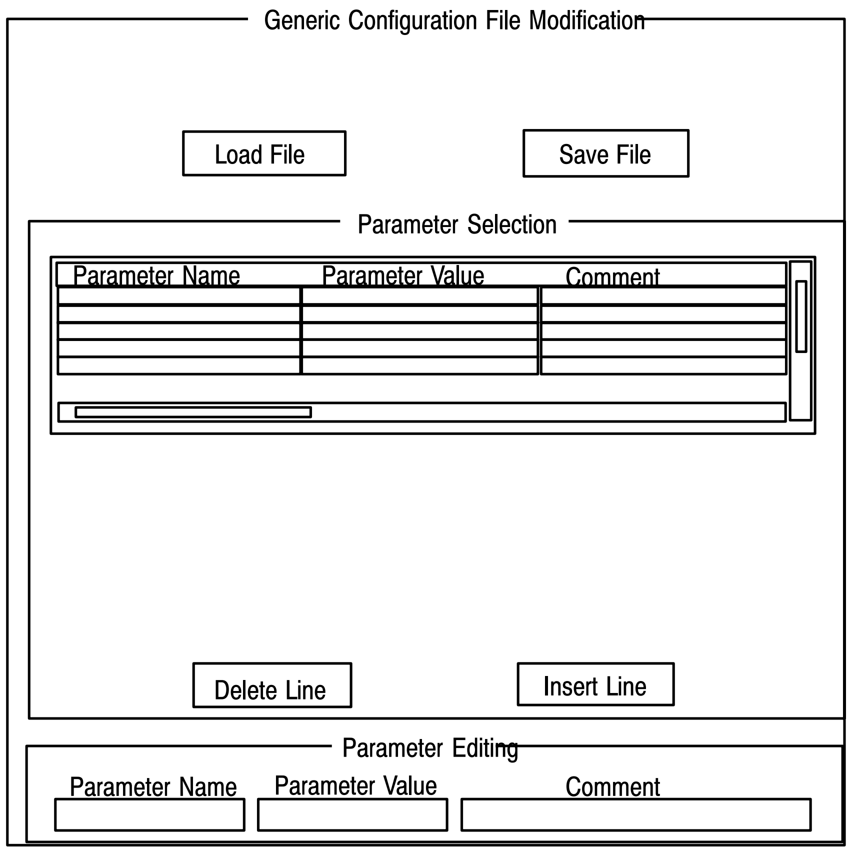


FIGURE 399 GENERAL CONFIGURATION

Update: After Configuration is complete, click on the "Update Configuration Files" button:

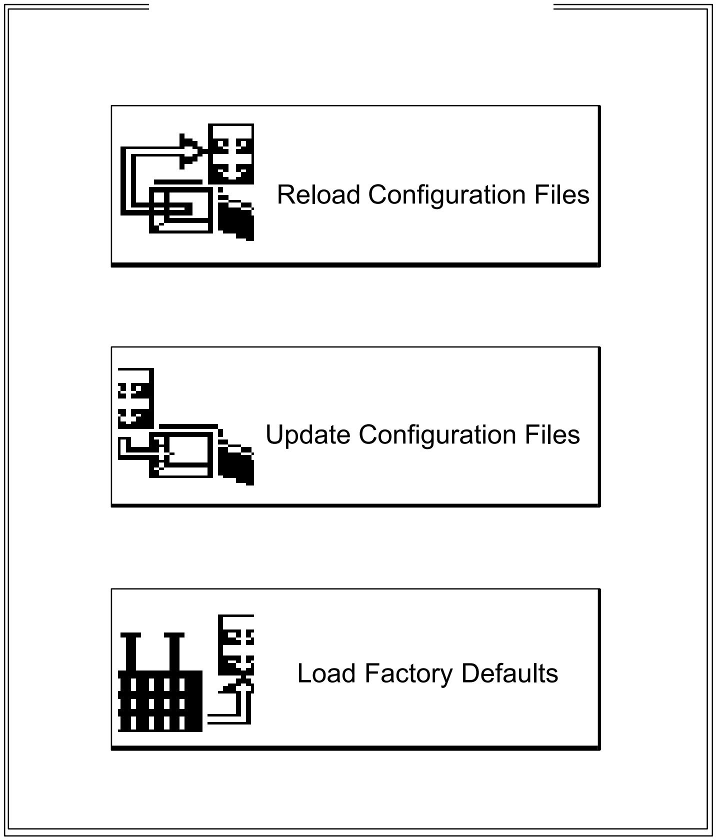


FIGURE 400 UPDATE CONFIGURATION FILES

ERROR LOGGER

Introduction

Errors created during abnormal system operation are recorded in the Error Logger. The contents of the Error Logger can be displayed from the main Service Application pull down menu, called LOG VIEWING.

Study	Display Tools	**Log Viewing**	User Level
		Service Log Forward Chrono	
		Service Log Reverse Chrono	
		Calibration Log	
		Bad Detector Log	
		All Logs	
		AcQImage Log	
		PICR Log	

FIGURE 401 LOG VIEWING PULLDOWN MENU

You may select any of the options show in Figure 401. To see samples of the first three error logs, click on the respective heading. Each of the log listing windows allows you to search for specific words or numbers and includes page up, page down and done buttons.

> **NOTE**
>
> The Study and Display Tools pull down menus are covered in the Operator's manual.

```
------------------------------------------------------------------------------
          SYSTEM:                            :     0
          FILE:       .  ()
          DATE:          ,
          INFO:       Scan/Recon failure, Summary error:MODSUMMARY_INTERNAL_ERR, Location
          SYSNAME:

------------------------------------------------------------------------------
          SYSTEM:                            :     0
          FILE:       .  ()
          DATE:          ,
          INFO:       Scan/Recon failure, Summary error:MODSUMMARY_INTERNAL_ERR, Location
          SYSNAME:

------------------------------------------------------------------------------
          SYSTEM:                            :     0
          FILE:       .  ()
          DATE:          ,
          INFO:       Scan/Recon failure, Summary error:MODSUMMARY_INTERNAL_ERR, Location
          SYSNAME:

------------------------------------------------------------------------------
          SYSTEM:                            :     0
          FILE:       .  ()
          DATE:          ,
          INFO:       Scan/Recon failure, Summary error:MODSUMMARY_DEPENDENT_ERR, Locatio
          SYSNAME:

------------------------------------------------------------------------------
          SYSTEM:                            :     0
          FILE:       .  ()
```

FIGURE 402 SERVICE LOG FORWARD CHRONO – SAMPLE OUTPUT

```
---------------------------------------------------------------------------------
11:12:16  SYSTEM:    SCAN           ERROR:   517   APEID:   0xFFFFFFFF
          FILE:      Errr.cpp (707)
          DATE:      Tue  Nov 17,  1998
          INFO:      Bad GSERV state, state is -65536, expected Manual exposure active
          SYSNAME:   bay13-1
          Data Type: 0x00000001

   SYSTEM: GSRV     TRACER: ffff8000   PROG:         PC: 0

   GSERV STATUS: Bad GSERV state, state is -65536, expected Manual exposure active
   GSRV: 8000   SCNS:     0   FUNS:     0   CONT:     1   ROTR:    78   FLMT:     0
   XRAY:    0   FLTR:     0   COMP:     0   COLL:    14   LASR:     0   SHUT:     0
   SPOT:    1   BLIM:     0     KV:     6     MA:     1   HEAT:     e   FSPD:   190
   FPOS: 4afa   CSPD:     0   CHOR:  2040   CVER:   302   TILT:    80   BACK:     0

   GPU STATUS: NO ERROR
   G4-0:   18   G4-1:    18   G4-2:    80   G4-3:    10   G4-4:   200   G4-5:   400
   G4-6:  800   G4-7:  ff80   G4-8:  8030   G4-9:     0   G4-A:     0   G4-B:  ff00
   G4-C:    0   G4-D:     0   G4-E:  ff03   G4-F:   ff0   G5-0:  1388   G8-0:  5421
   G8-1: 997a   G8-2:  8603   G8-4:  2040   G8-5:   302   G8-7:  1080   G8-8:  a000
   G8-9:    0   G8-A:  c7a8   G8-B:  4050   G8-C:     0   G9-0:    ff   MTMP:    19
   CTMP:   1d   GPUS:     0   FMOD:     2   FSPD:   190   CMOD:     1   CSPD:     0

   SCAN TABLE:
   CTYP:          APE: ffffffff             GPUF:  ffff   XSCF:  ffff   BACO:  ffff
   DPPC: ffff   CNTW:  ffff   TIMW:  ffff     KV:  ffff     MA:  ffff   FSPD:  ffff
   XTIM: ffff   HTIM:  ffff   RSPD:  ffff   SPOT:  ffff   FLTR:  ffff   COMP:  ffff
   COLL: ffff   BLIM:  ffff   SAMP:  ffff   ANGL:  ffff   DELY:  ffff   CSPD:  ffff
   ACIN: ffffffff             POIN:  ffff   REVB:  ffff   INTG:  ffff   FDET:  ffff
    SOF: ffff    EOF:  ffff    FAN:  ffff   SCAT:  ffff   #REV:  ffff   GAPA:  ffff
```

FIGURE 403 SERVICE LOG REVERSE CHRONO – SAMPLE OUTPUT

```
------------------------------------------------------------------------------
09:23:12.319  SYSTEM:                              :     1
         FILE:       BadDetBaco.m (1)
         DATE:       Fri  Nov 13,  1998
         INFO:       SYSTEM PERFORMANCE: --------------------
         SYSNAME:    Fri

------------------------------------------------------------------------------
09:23:12.588  SYSTEM:                              :     1
         FILE:       BadDetBaco.m (1)
         DATE:       Fri  Nov 13,  1998
         INFO:       SYSTEM PERFORMANCE: 13-Nov-1998 09:23:12
         SYSNAME:    Fri

------------------------------------------------------------------------------
09:23:12.600  SYSTEM:                              :     1
         FILE:       BadDetBaco.m (1)
         DATE:       Fri  Nov 13,  1998
         INFO:       SYSTEM PERFORMANCE: BACO avg offset: -3.03e-02 v (12111 hz)
         SYSNAME:    Fri

------------------------------------------------------------------------------
09:23:12.611  SYSTEM:                              :     1
         FILE:       BadDetBaco.m (1)
         DATE:       Fri  Nov 13,  1998
         INFO:       SYSTEM PERFORMANCE: BACO max offset: -3.28e-02 v (13120 hz)
         SYSNAME:    Fri

------------------------------------------------------------------------------
09:23:12.623  SYSTEM:                              :     1
         FILE:       BadDetBaco.m (1)
```

FIGURE 404 CALIBRATION LOG

PERIPHERALS – INSTALLATION AND CONFIGURATION

DISPLAY TOWER – OVERVIEW

This section covers removal, replacement and configuration of the various peripherals contained in the AcQSim CT. The Display Tower contains the SCSI devices for the display system, including optical disk, 8mm tape, CD–ROM and hard disks. The SCSI (Small Computer System Interface) bus located within the AcQSim CT Display Tower originates at the Sparc Host Board (p/n 311927). Any peripheral attached to the SCSI bus must conform to the SCSI–2 standards for single ended devices. Up to seven devices can be connected.

There are two connectors located on the host board for connection to the SCSI bus, one on the rear panel (HD–50 connector, unused). All peripherals within the Display Tower are interconnected using a 50 conductor non–shielded flat ribbon cable. Connectors are spaced along the cable to allow a daisy–chain of the devices. Hard drives are wide SCSI–2 and are connected to a 68 pin connector. The SCSI cable has an adaptor to convert 68 pin to 50 pin. The internal SCSI cable (P/N L11031) originates at the host and is terminated at the primary 8mm tape drive using an in–line terminator (P/N 310576). Figure 405 shows the Display Tower SCSI drives configuration. The SCSI ID Table identifies the SCSI ID number assignments for the Display Tower peripherals:

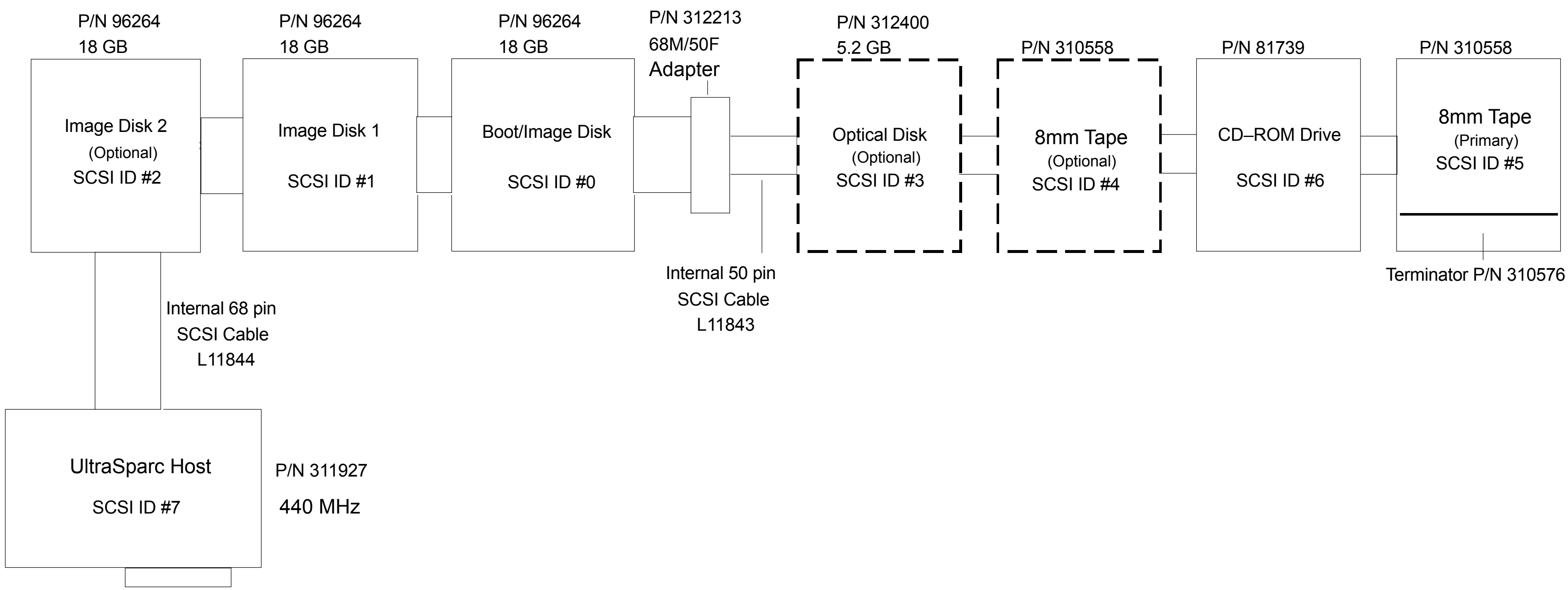


FIGURE 405 DISPLAY TOWER SCSI BLOCK DIAGRAM

The **SCSI ID Table** below identifies the SCSI ID number assignments for the Display Tower peripherals:

SCSI ID	AcQSim CT Display Tower Peripherals
0	Hard Drive – Boot Disk
1	Hard Drive – Image Disk 1
2	Hard Drive – Image Disk 2 (optional)
3	Optical Drive (optional)
4	Tape Archive, 8mm (optional)
5	Tape Archive, 8mm (primary)
6	CD–ROM Drive, 4x Speed or higher
7	Host

DISPLAY TOWER DRIVES – CONFIGURATION AND INSTALLATION

Display Tower Removal / Replacement

Introduction: The following procedure is used to remove and replace any of the drives (tape, CD–ROM, hard disk, etc) in the event of a failure, upgrade or to check configuration. This procedure also instructs how to align the Keyswitch/Emergency Stop assembly and the Peripheral Bay Assembly.

Tools Required: Phillips Screwdriver

Estimated Time: 20 Min.

DISPLAY TOWER DRIVE REMOVAL:

1. Remove the Display Tower power. Refer to the PowerUp/PowerDown procedure.

> ⚠ **WARNING** ⚠
>
> **USE EXTREME CAUTION WHEN OPERATING OR WORKING ON EQUIPMENT WITH COVERS REMOVED. DANGEROUS VOLTAGES ARE EXPOSED WHEN THE COVERS ARE REMOVED. FAILURE TO COMPLY CAN CAUSE SERIOUS INJURY OR DEATH TO PERSONNEL.**

> **NOTE**
>
> If the system needs to be moved, release the lock on the wheels before moving. Ensure wheels are locked after moving.

2. For user peripheral work (CD ROM, etc), remove the front cosmetic cover. To access the hard disk drives, also remove the front cover. Refer to the Display Tower Covers Removal procedure.
3. Remove the two screws from the bottom of the assembly where the drive to be removed is located. Refer to Figure 406.
4. Slide the assembly out enough to gain access to the screws on the side of the assembly.
5. Disconnect the data and power cables from the drive or drives being removed.
6. Remove the two screws on each side of the drive or drives being removed and pull the drive from the assembly.

All peripherals are mounted in a drive mounting bracket assembly, consisting of: 1) a drive mounting bracket (P/N 81977), a bottom plate (P/N 81980) and a chassis support (P/N 81981). The chassis support is mounted to the vertical rails of the Display Tower frame. Two screws are used to retain the drive mounting bracket to the bottom plate in the Display Tower frame. Removal of the two screws allows the drive mounting bracket to be slid out from the chassis support to service the devices.

The hard drive mounting brackets can hold up to three half height 3 1/2” form factor devices. Half height filler panels (P/N 83994) are used to fill the space below the last device in the user–accessible peripheral bay. These filler panels are required as they restrict access by non–service personnel in the event that the front facade assembly (P/N 178656) is removed from the Display Tower.

> **IMPORTANT NOTE**
>
> After a CD ROM, 8mm Tape or Optical Disk Drive has been replaced or added, the disk must be configured into the system. To do this, reboot the system and while the memory is being initialized, press **STOP A** You will get an “ok” prompt. At the ok prompt, type **boot –rv** and press <cr>.

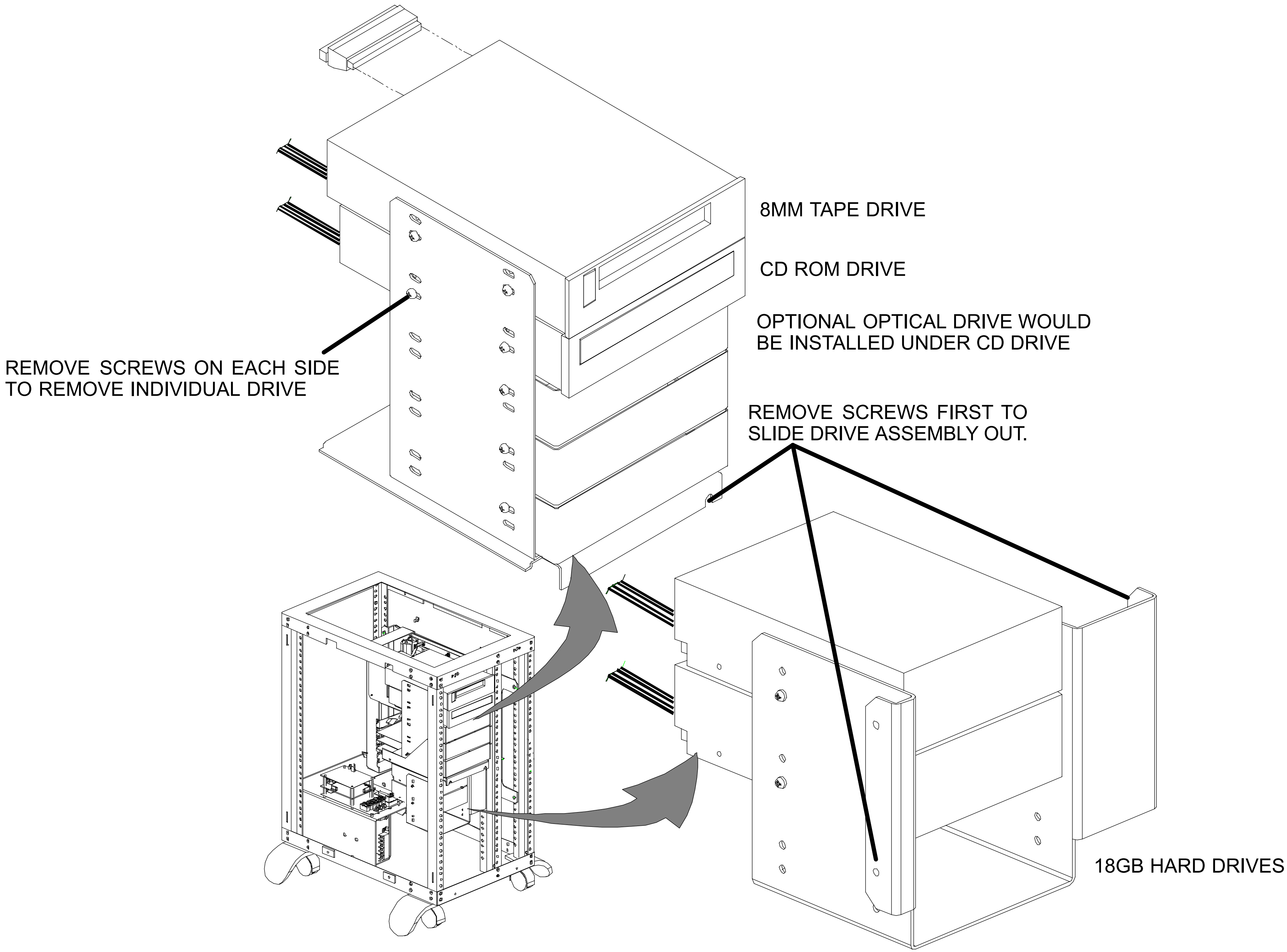


FIGURE 406 DISPLAY TOWER DRIVE LOCATIONS

DISPLAY TOWER DRIVE REPLACEMENT:

1. Ensure the drive is configured correctly. Refer to the Configuration Details for specific drives.
2. Insert the new drive in the assembly where the old one was removed. Line up the holes in the frame and install the four screws removed earlier. Reconnect the data and power cables.
3. Slide the entire assembly back in, align the holes in the bottom of the respective frame and install the two screws removed earlier. If the user peripheral assembly was removed, you must re–align the bracket to the center of the access holes on the cosmetic cover.
4. If only a hard drive was replaced, then re–install the covers, roll back into place and secure the wheels. If an additional 8mm tape, CD ROM or Optical disk drive was installed, the filler panel on the cosmetic cover must be adjusted before replacing the cover.

> **NOTE**
>
> The copper ESD strip (used to conduct static away from the peripherals) is attached to the front of the cosmetic cover and the unpainted area of the filler panel. Ensure that the strip is still attached after moving the filler panel.

5. Remove the four screws that secure the filler panel. For each 1/2 height drive added, move the panel down one set of screw holes. Secure the filler panel with the four screws. Before pushing the cosmetic cover onto the Display Tower, visually align the cover to ensure proper placement of the filler panel – it should now be just below the lowest drive. Refer to Figure 407.
6. After the alignment is correct, you may push the the cosmetic cover back on to the Display Tower.

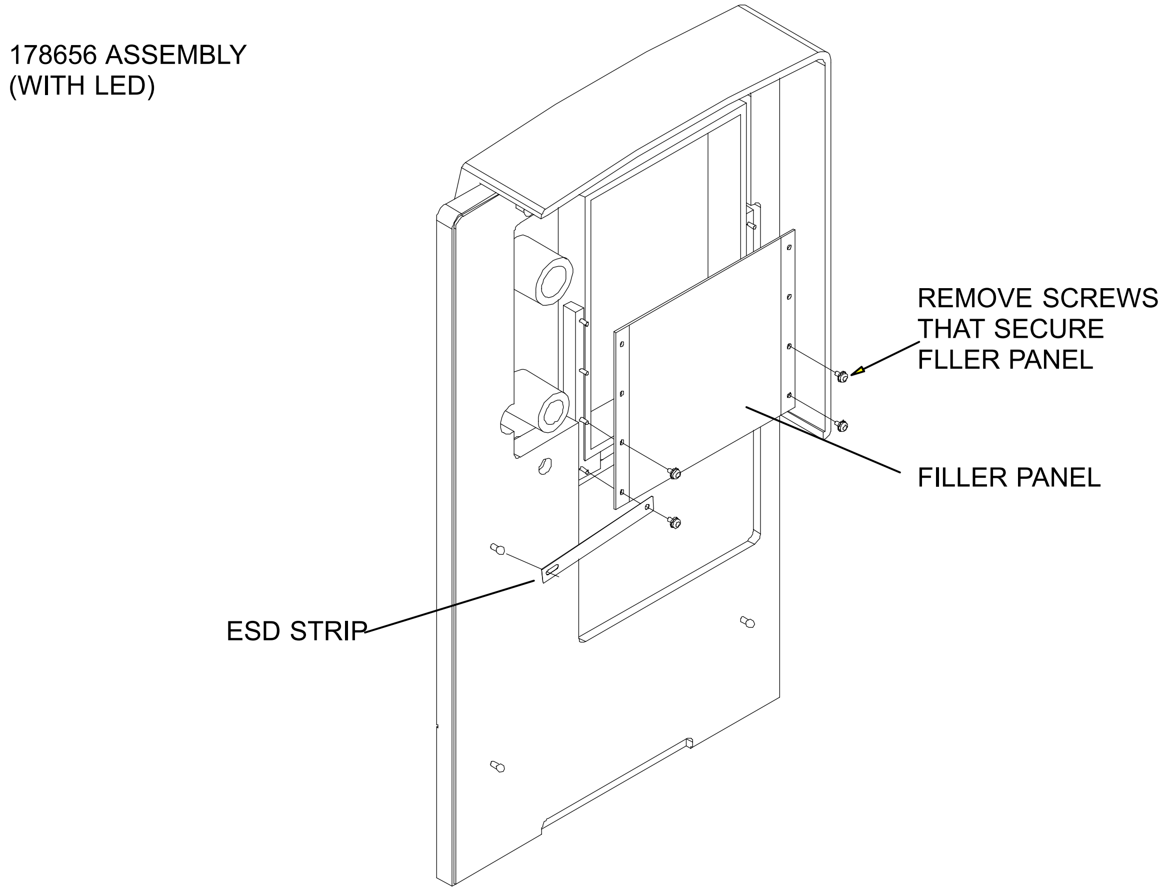


FIGURE 407 ADJUSTING FILLER PANEL

DISPLAY TOWER FACADE ALIGNMENT PROCEDURE

The alignment procedure provides clearance to operate the emergency stop switch without binding. It also provides clearance around the user bay peripheral assembly to allow easy removal and installation of the facade assembly.

Keyswitch / Emergency Stop Assembly Alignment

1. Remove the right side cover. Refer to the Display Tower Covers Removal procedure.
2. Loosen the four nuts on the back of the Keyswitch/Emergency stop assembly. Do not remove the nuts.
3. Install the facade assembly onto the Display Tower.

4. Adjust the position of the Keyswitch/Emergency Stop assembly until the Emergency Stop switch can be pressed and released without binding.
5. Tighten the four nuts on the rear of the Keyswitch/Emergency Stop assembly.
6. Press and twist to release if installed the Emergency Stop several times to ensure no binding exists.
7. Install the right side cover. Refer to the Display Tower Covers Removal procedure.

Alignment of User Peripheral Bay Assembly

1. Remove the facade assembly from the Display Tower.
2. Loosen both of the screws that lock in the peripheral bay assembly. Refer to Figure 406.
3. The peripheral bay should be moved horizontally and verically until it appears to be centered. Hold in the position as you tighten the screws.
4. Install the facade assembly making sure that the peripheral bay assembly does not hit the facade when it is completely installed. The peripheral bay should have an equal amount of clearance on either side.
5. If the facade is still not clearing the bay, then repeat steps 2. through 4. until the facade can be installed without interferring with the user peripheral bay assembly.

CONFIGURATION DETAILS FOR SPECIFIC DRIVES – DISPLAY TOWER

The following pages diagram the jumper and dip switch settings to properly set configuration parameters and SCSI ID numbers for drives. Refer to the SCSI ID Table for correct assignment of ID numbers.

The following links will take you to the drive of your choice:

Figure 408 – SEAGATE ST318275LW Hard Drive

Figure 409 – QUANTUM QM318100KN–LW 18 GB Hard Drive

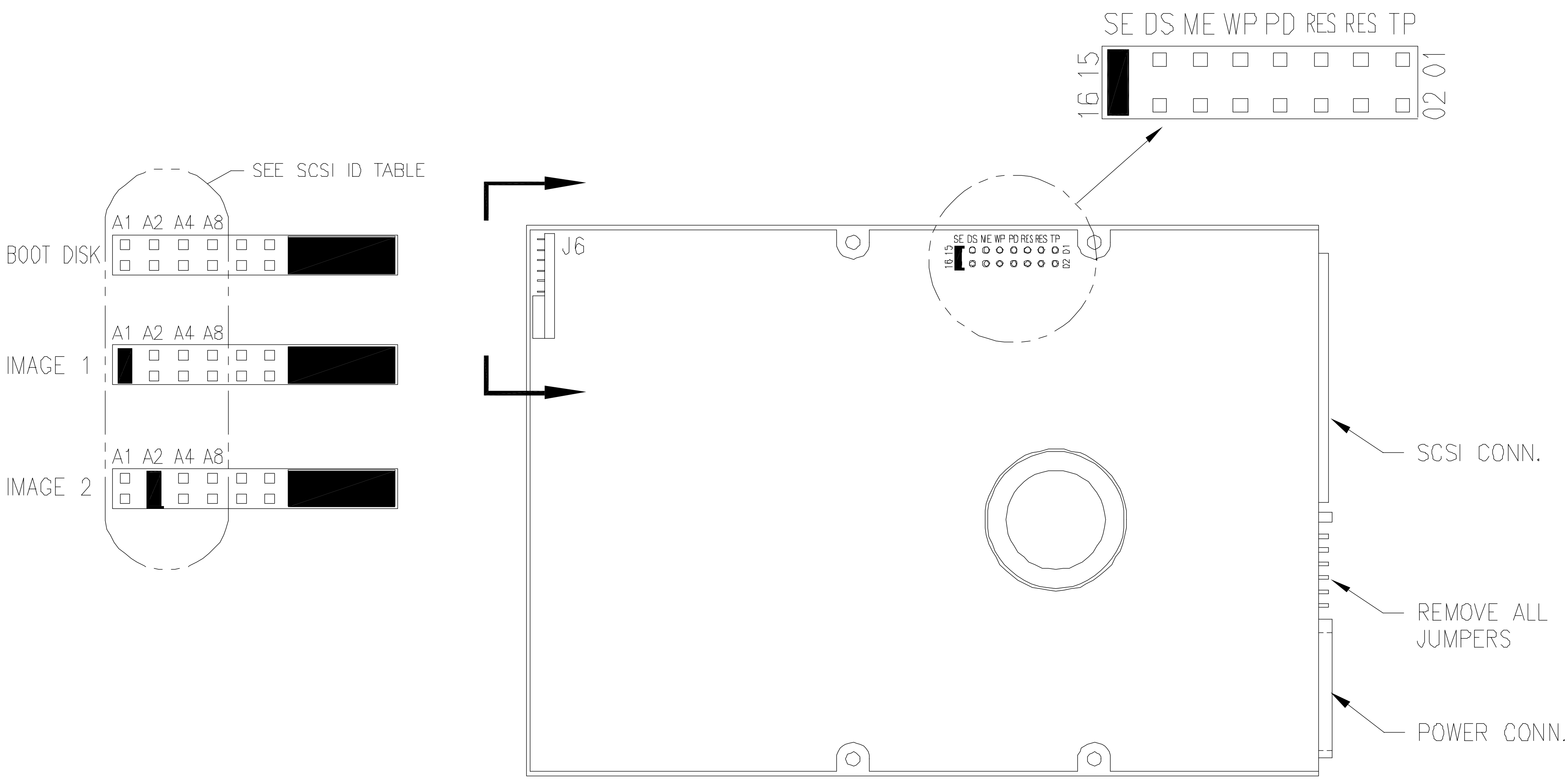
Figure 411 – IBM 25L1900   18 GB Hard Drive

Figure 412 – IBM 07N3110   18GB Hard Drive

Figure 412 – PLEXTOR PX–40TSI   40x CD–ROM Drive

Figure 413 – 8mm TAPE DRIVE EXABYTE ELIANT 820 Tape Drive

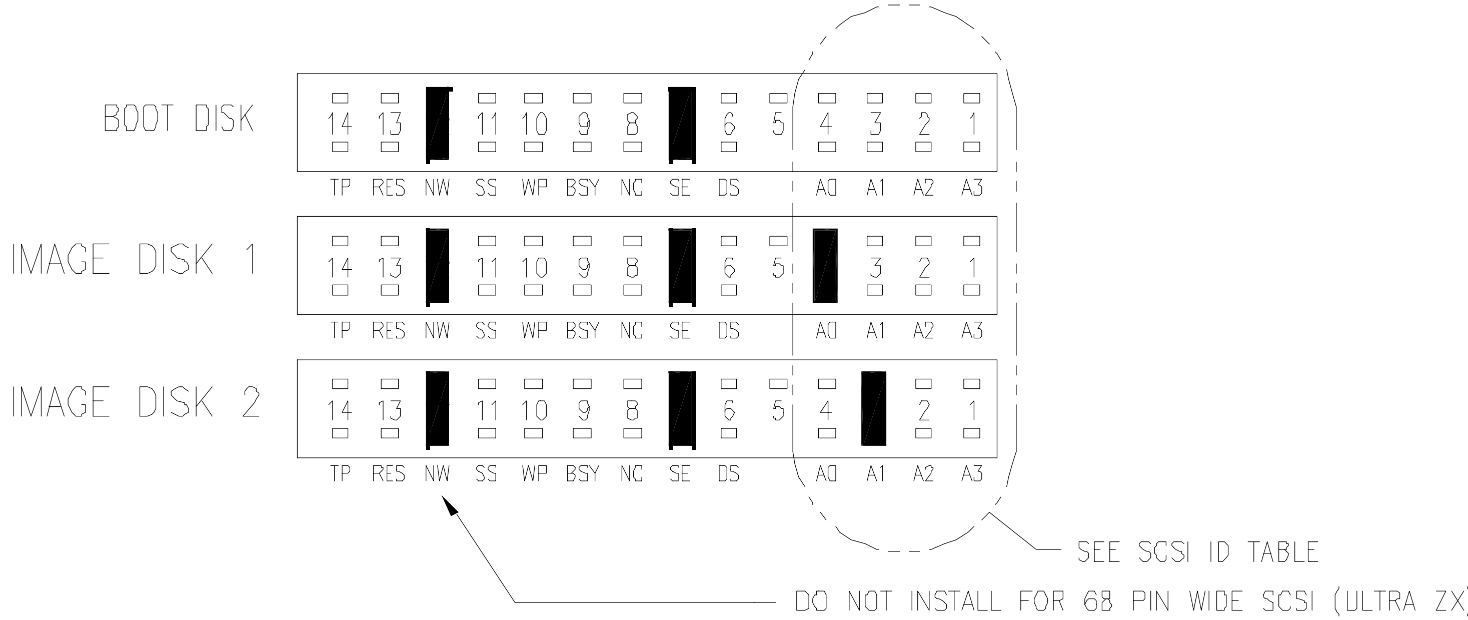
Figure 414 – HEWLETT PACKARD C1113J Optical Drive



SCSI ID TABLE

SCSI ID	POSITION				DRIVE CONFIG.
	A1	A2	A4	A8	
0	OPEN	OPEN	OPEN	OPEN	BOOT DISK
1	SHORT	OPEN	OPEN	OPEN	IMAGE 1
2	OPEN	SHORT	OPEN	OPEN	IMAGE 2

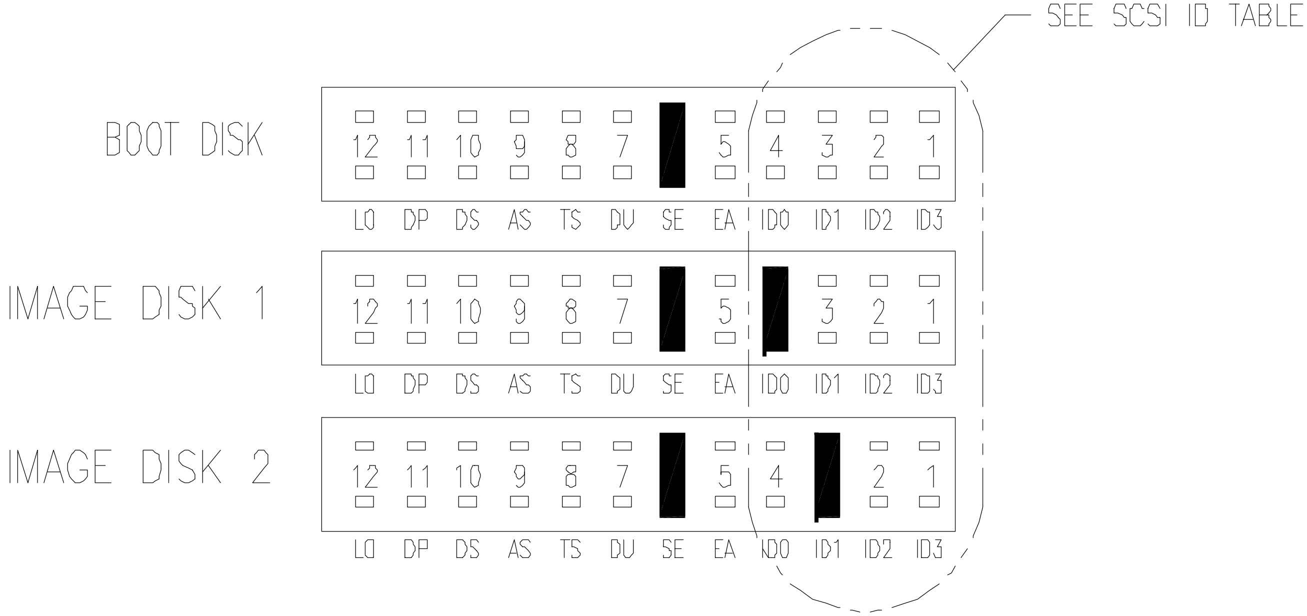
FIGURE 408 SEAGATE ST318275LW 18 GB HARD DRIVE



SCSI ID TABLE

SCSI ID	POSITION A0	POSITION A1	POSITION A2	POSITION A3	DRIVE CONFIG.
0	OPEN	OPEN	OPEN	OPEN	BOOT DISK
1	SHORT	OPEN	OPEN	OPEN	IMAGE DISK 1
2	OPEN	SHORT	OPEN	OPEN	IMAGE DISK 2

FIGURE 409 QUANTUM QM318100KN–LW 18GB HARD DRIVE



SCSI ID TABLE

SCSI ID	POSITION ID0	POSITION ID1	POSITION ID2	POSITION ID3	DRIVE CONFIG.
0	OPEN	OPEN	OPEN	OPEN	BOOT DISK
1	SHORT	OPEN	OPEN	OPEN	IMAGE DISK 1
2	OPEN	SHORT	OPEN	OPEN	IMAGE DISK 2

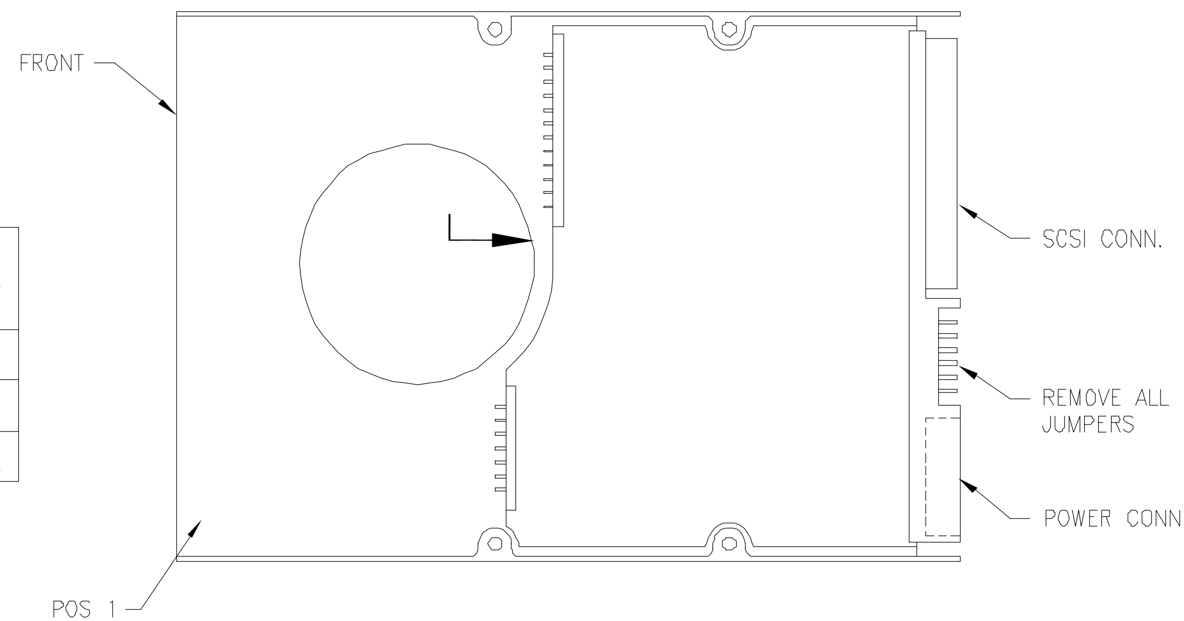


FIGURE 410 IBM 25L1900 18GB HARD DRIVE

SCSI ID TABLE

SCSI ID	POSITION ID0	POSITION ID1	POSITION ID2	POSITION ID3	DRIVE CONFIG.
0	OPEN	OPEN	OPEN	OPEN	BOOT DISK
1	SHORT	OPEN	OPEN	OPEN	IMAGE DISK 1
2	OPEN	SHORT	OPEN	OPEN	IMAGE DISK 2

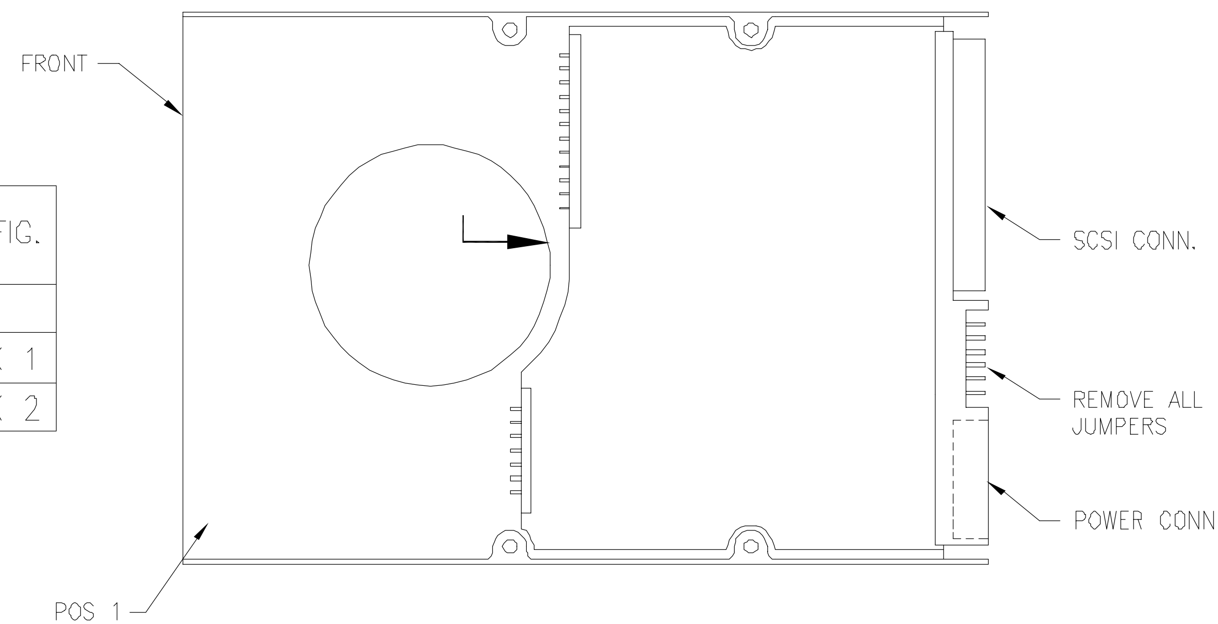
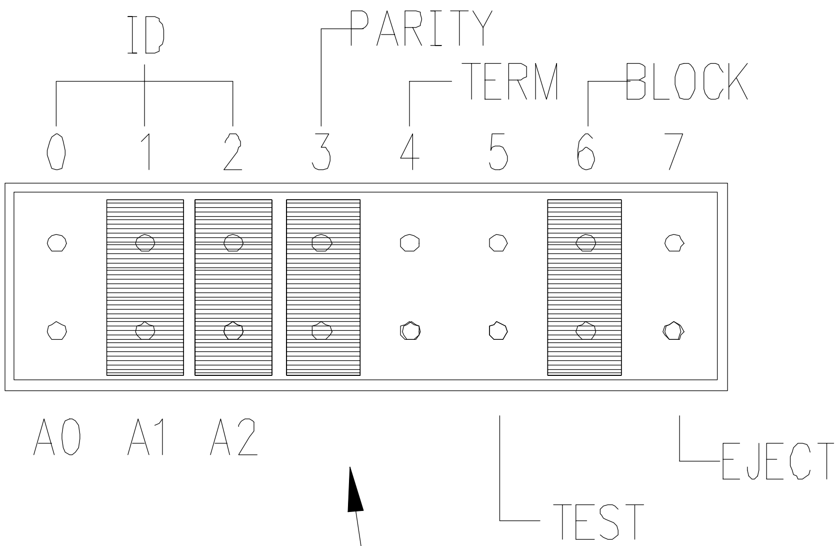


FIGURE 411 IBM 07N3110 18GB HARD DRIVE



SCSI ID TABLE

SCSI ID	POSITION			DRIVE CONFIG.
	A0	A1	A2	
6	OPEN	SHORT	SHORT	CD–ROM

FIGURE 412 PLEXTOR PX–40TSI 40x CD–ROM DRIVE

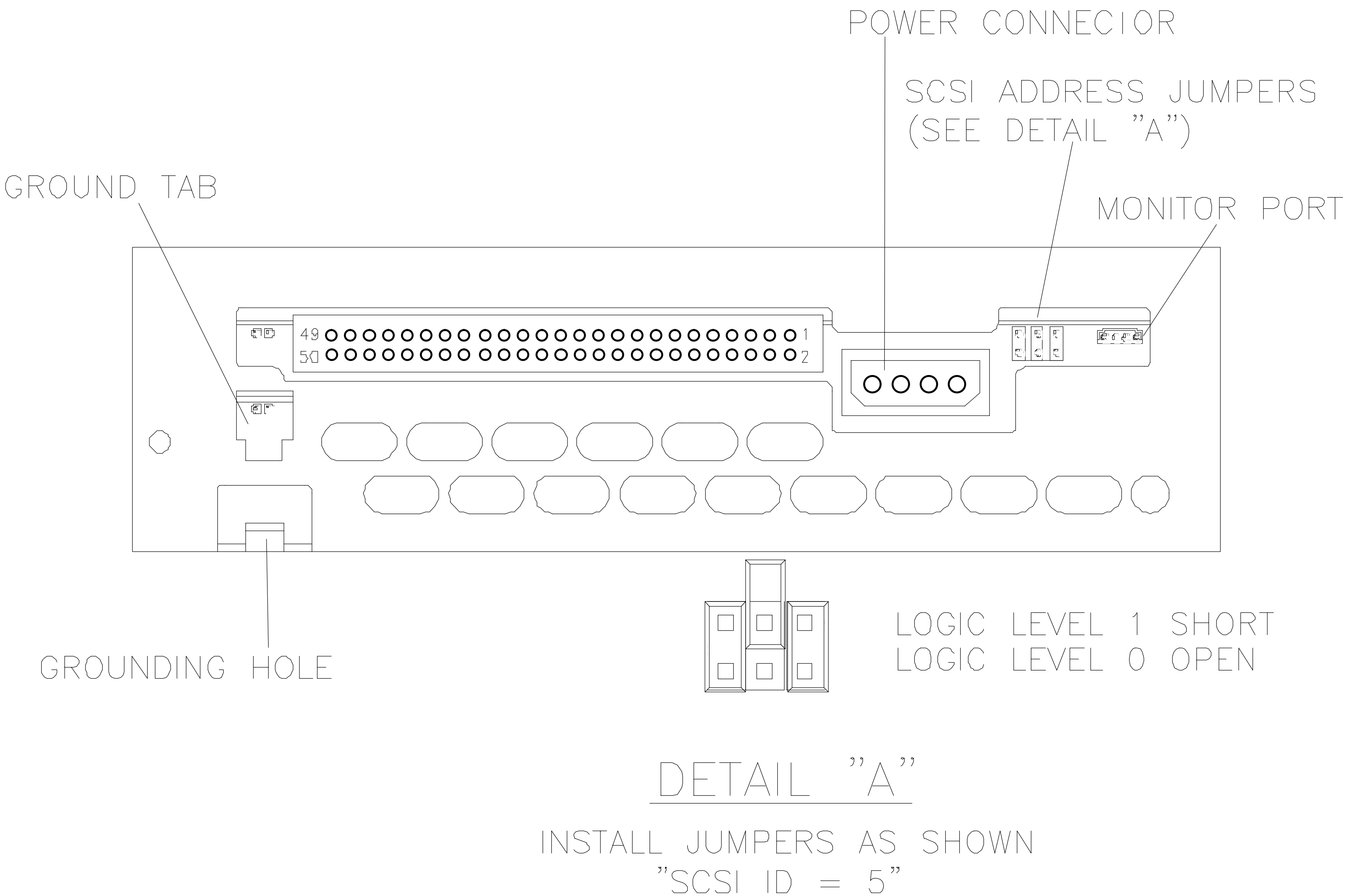
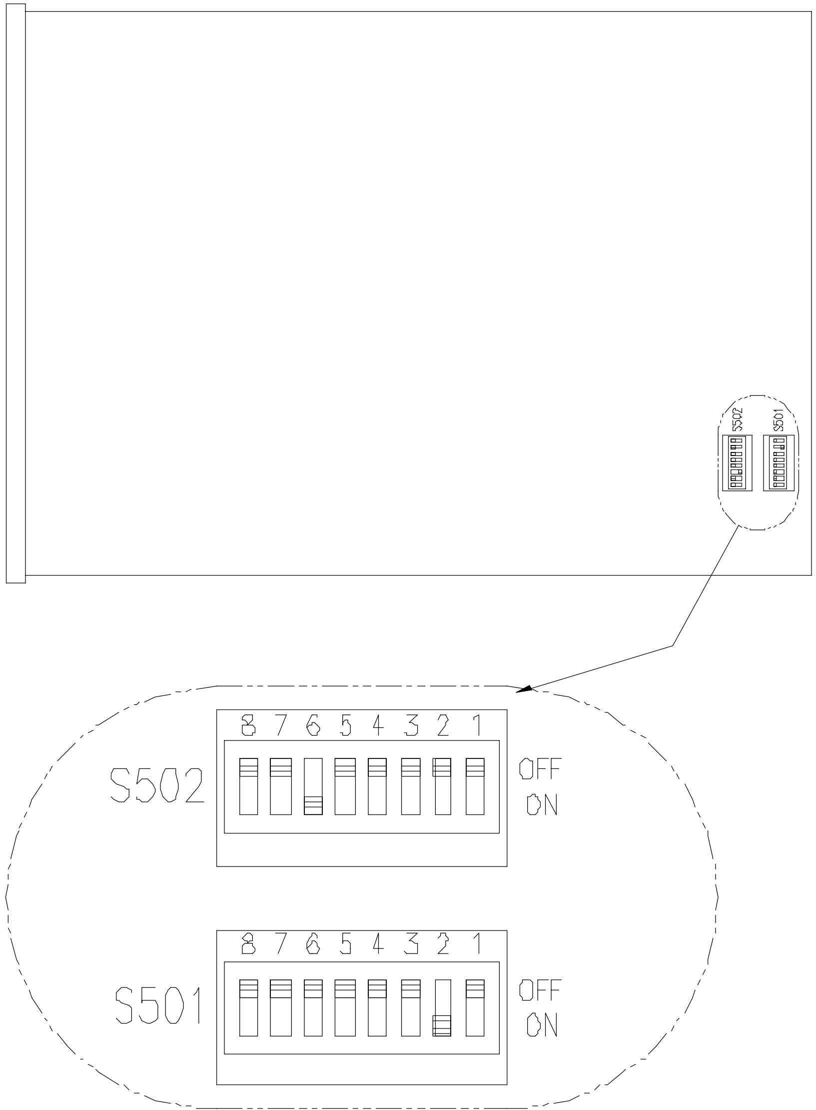
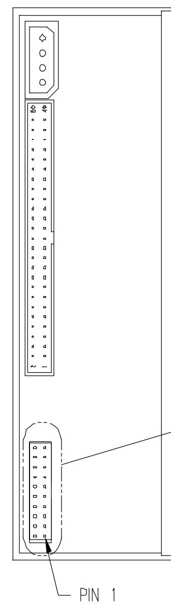


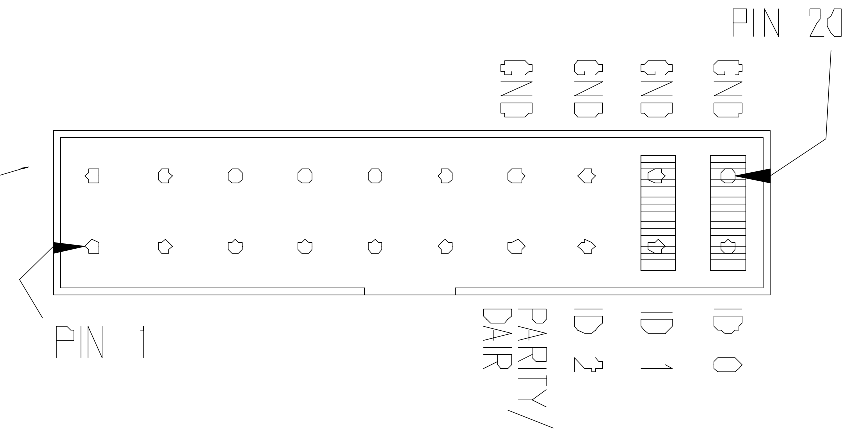
FIGURE 413 EXABYTE RELIANT 820 8mm TAPE DRIVE





SCSI ID TABLE

SCSI ID	POSITION			DRIVE CONFIG.
	A2	A1	A0	
3	OPEN	SHORT	SHORT	REMOVABLE MEDIA



DIP SW S501	**DIP SW S502**
1 Jukebox Eject Distance Override	Reserved for Production
2 Dair Enabled	Reserved for Production
3 Reserved	Reserved for Production
4 Reserved	Reserved for Production
5 Reserved for Debug	Reserved
6 Reserved for Debug	Force Verify, Write Operation
7 Reserved for Debug	Fast SCSI Enable
8 Reserved for Debug	Reserved

FIGURE 414 HEWLETT PACKARD C1113J OPTICAL DRIVE

ACQIMAGE CABINET DRIVES – OVERVIEW

The AcQImage cabinet contains two SCSI disk drives for storage of scan data. The two disk drives are contained on a single board that is located on the right side (front view) of the AcQImage Cabinet card rack. Refer to Figure 415 and Figure 416.

ACQIMAGE CABINET HARD DISK ASSEMBLY –CONFIGURATION AND INSTALLATION

AcQImage Cabinet Removal / Replacement

Introduction:	The following procedure is used to remove and replace an AcQImage Cabinet hard disk drive in the event of a failure.
Tools Required:	Phillips Screwdriver
	Small slotted screwdriver
Estimated Time:	15 Min.

AcQImage Cabinet Hard Disk Removal:

1. Remove the AcQImage Cabinet power. Refer to the PowerUp/PowerDown procedure.

> ⚠ **WARNING** ⚠
>
> **USE EXTREME CAUTION WHEN OPERATING OR WORKING ON EQUIPMENT WITH COVERS REMOVED. DANGEROUS VOLTAGES ARE EXPOSED WHEN THE COVERS ARE REMOVED. FAILURE TO COMPLY CAN CAUSE SERIOUS INJURY OR DEATH TO PERSONNEL.**

> **NOTE**
>
> If the system needs to be moved, release the lock on the wheels before moving. Ensure wheels are locked after moving.

2. Remove the front cosmetic covers and rear panel first. Refer to the AcQImage Cabinet Covers Removal procedure.
3. Remove the internal front steel panel by removing the six screws. The hard disk assembly is located on the far right side of the card cage. Refer to Figure 415.

4. Disconnect the SCSI cable attached to the GAM board.
5. Loosen the 4 captive screws that secure the hard disk drive assembly in place. Grasp the handles and slide the assembly out part way. Refer to Figure 415.
6. Disconnect the power and SCSI connectors from the drive. Refer to Figure 416.
7. While holding the drive, loosen the four screws that secure the drive to the slide out frame.

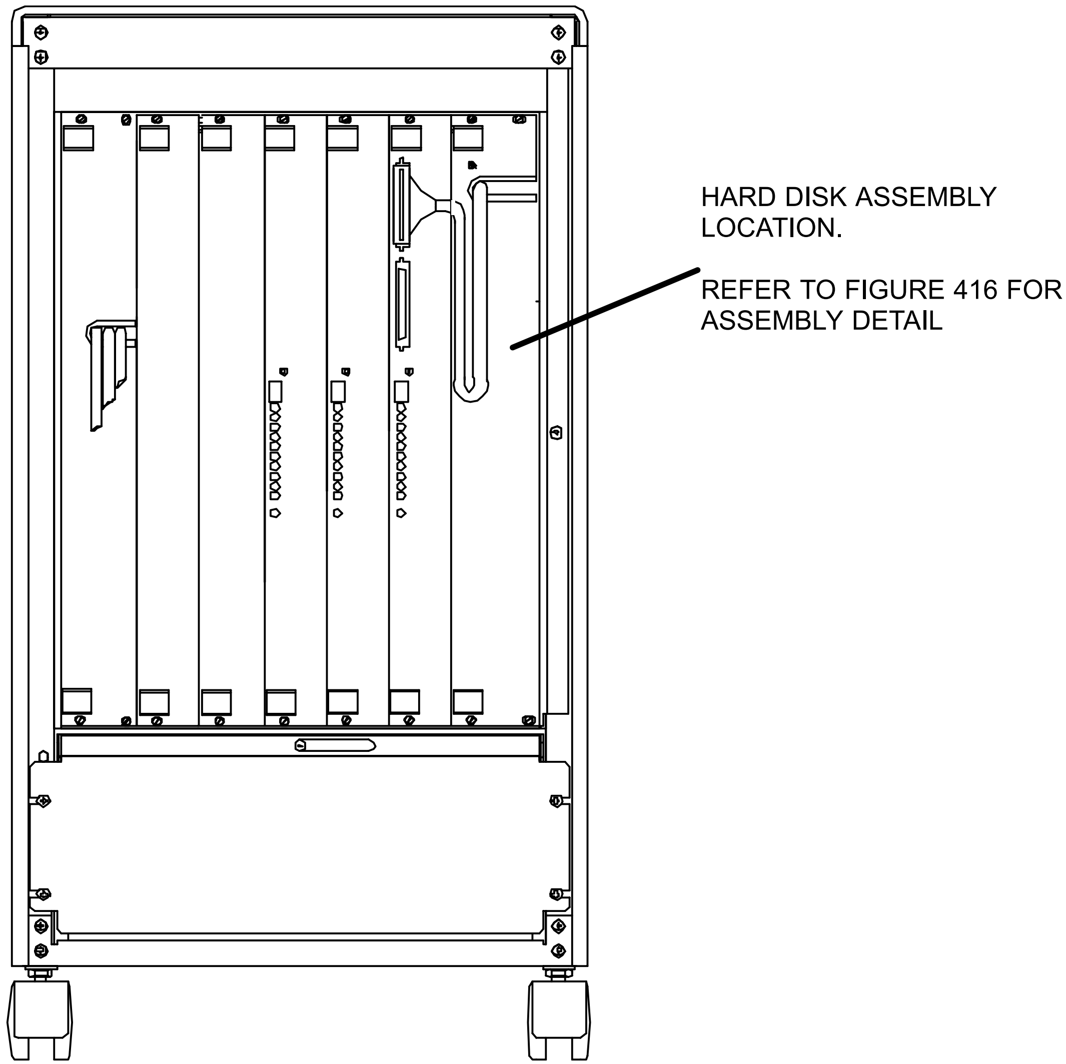


FIGURE 415 ACQIMAGE CABINET HARD DISK DRIVE ASSEMBLY LOCATION

DISK 2 HAS TERMINATOR,

DISK 1 DOES NOT.

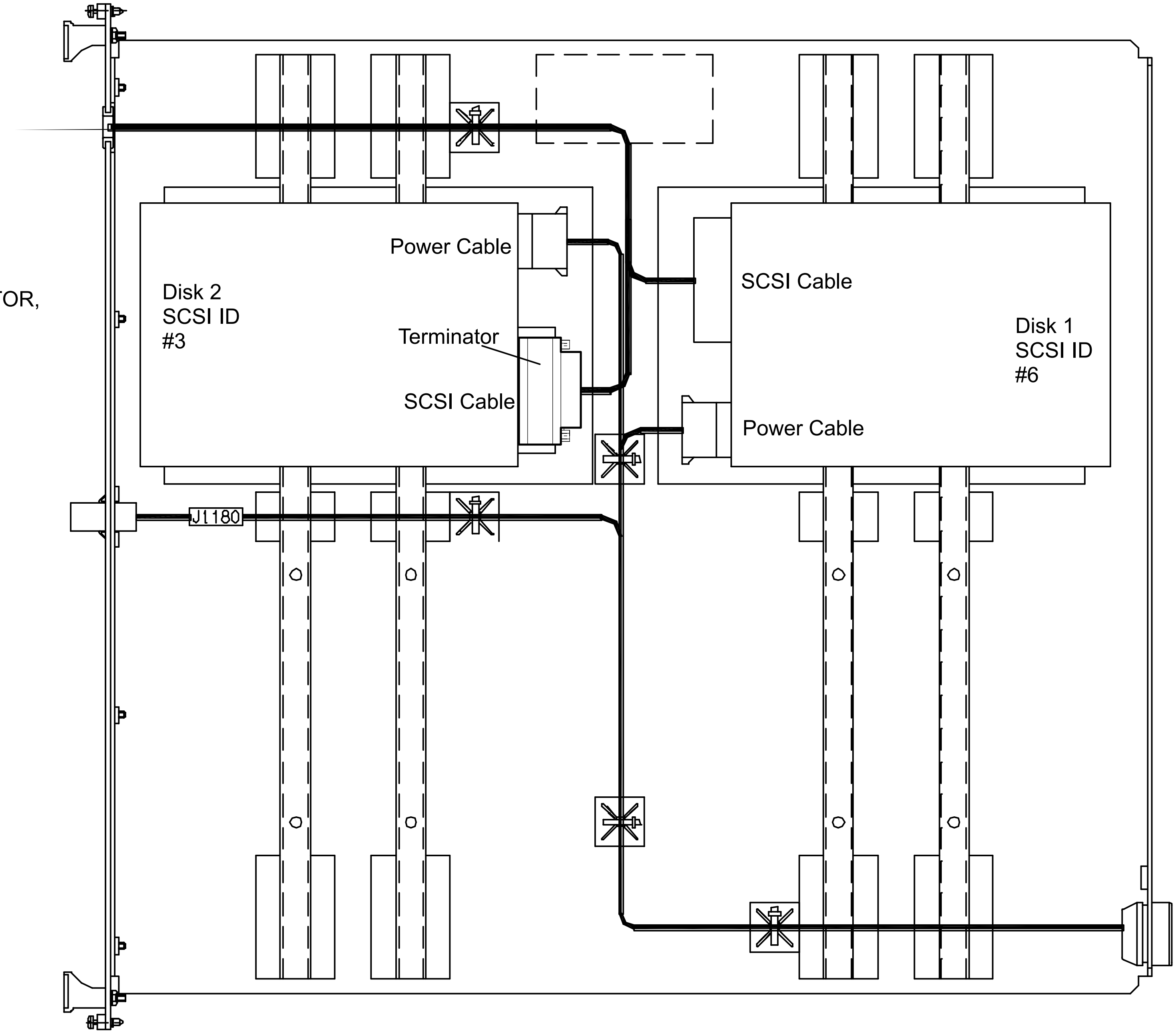


FIGURE 416 LOCATION OF ACQIMAGE DRIVES ON REMOVABLE BOARD

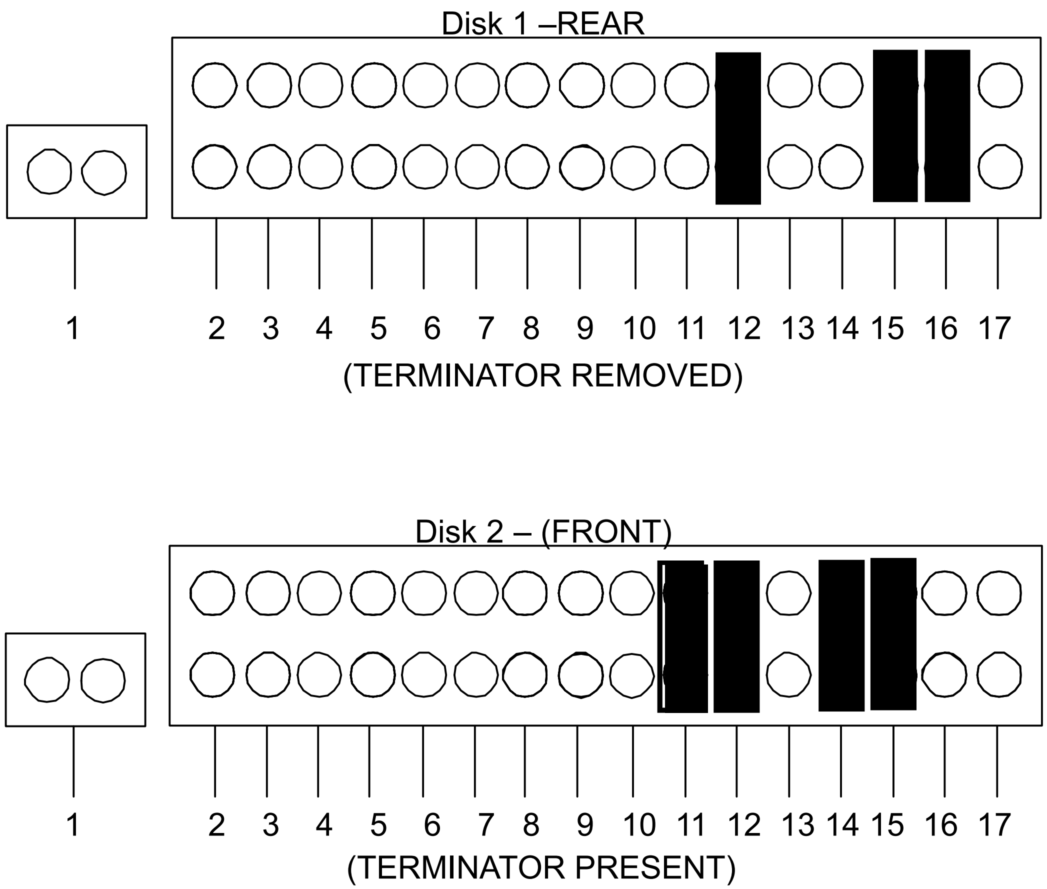


FIGURE 417 ACQIMAGE DISK DRIVE CONFIGURATIONS

AcQImage Cabinet Hard Disk Replacement:

1. Ensure the drive is configured properly, according to Figure 417.
2. Bolt the new drive in where the old one was.
3. Attach the power and SCSI connectors to the new drive.
4. Slide the assembly back into the card cage and secure the captive screws.
5. Plug the SCSI cable back into the GAM board.

ACQSIM CT 1.0 SOFTWARE INSTALLATION INSTRUCTIONS

The following kit instructions should be used when installing 1.0 software onto an AcQSim CT system.

> **NOTE**
>
> This procedure in intended for the first series of software installations on the AcQSim CT and is not intended to be part of a software upgrade kit.

1. Make a Site tape immediately before installation. If in the Console or Service Application, select Shutdown Console from the General Utilities button. When the main menu appears, click in the lower left corner of the Main Startup menu and type the word “**site**” (lower case) and press <CR>.
2. After the Save Site / Restore Site menu appears, push the tape in, wait for the green light to stop blinking and select “1” to Save Site. The system will eject the tape when finished.
3. Exit the Save Site Utility. When the Shutdown button becomes available, press it and wait for the system to shut down.
4. Open the CD–ROM drive and insert CD #1. Close the drive.
5. When the ok prompt appears, type **boot cdrom** and press <CR>. After a few minutes, the password prompt will appear. Type **picker** and press <CR>.
6. The system then checks the disk configuration and prompts you with:

```
Enter the type of system installation:

1. RDS                  (Note: “RDS” refers to an Ultra Z Diagnostic Workstation, UDW)
2. Scanner Console

q. quit
```

7. Select option “2,” *Scanner Console* from the menu. You will see the following menu.

```
Enter a Console Installation option:

1. Install the OS and Application Software
2. Configure a Replaced Disk(s)
3. Upgrade Application Software Only

q. quit
```

8. Select option “1,” *Install the OS and Application Software* from the menu. Answer “**y**” to the next question. The following will appear:

Main Menu Option #1 – *Reformat and Configure entire system:*

```
Enter IP Address [default xxx.xxx.xx.xx]:
Enter Netmask [default 255.255.255.0]:
Enter Router [default xxx.xx.xx.xx]:
Enter Nodename [default consolexxxx]:
```

9. Press <CR> to accept the default values. Make sure the Netmask number is 255.255.255.0. After the information is entered, the system will display your choices as you entered them:

```
IP Address:
Netmask:
Router:
Nodename:
```

10. After the "Ready to run the installation. Are you satisfied? (y/n) [n]?" prompt appears:
    If you answer NO to "satisfied," the installation will cancel and take you back to the first sub–menu where you can quit the installation program, or continue and adjust the network information.

    If installation information is correct, choose "y" to "satisfied."

11. The software will now begin installing from the first CD. This will take about 20 minutes. When the "serial/parallel" change menu appears, choose "q" and press <CR>. The following menu will appear:

```
1. Install SCANNER Console Application Software
2. Eject the CD.
```

Select option "2," *Eject the CD*. Remove the first CD and put the second CD in, then close the drive door.

12. Select option "1," to continue with the installation. You will be prompted with the "satisfied" question again. Select "y" and the system will continue with the software installation. After the installation is complete, a menu will appear:

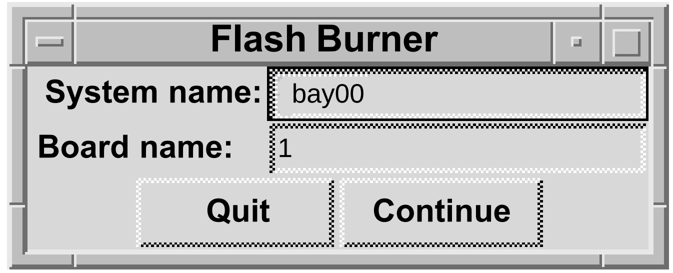
```
1. View the installation error log  [/cd_errlog]
2. Eject the CD  (it's OK to eject it now)
3. Continue booting the machine

Choice:
```

> **NOTE**
>
> After the installation has completed, additional messages may be displayed on the screen, pushing the above menu out of view. Simply press <CR> to see it again.

13. View the error log by choosing option **1**. There should not be any errors in the error log at this time. After viewing, the menu will be displayed again. Select option **2** to eject the CD. Remove the CD and close the drive door. Then select option **3** to continue booting to the application software.

14. Perform a Restore Site.

15. After the 1.0 software installation is done and the system has booted, start the Diagnostic GUI. (To start Diag GUI: Go to General Utilities, select Shutdown Console to get to the Startup men. Click on the lower left hand corner, type PICKER and press <CR>.)



16. Run the Flash Burner from the Diag GUI. The flash burner is a utility for transferring executable code from a directory on the SPARC disk to the Flash of the CP, AP, BP or GAM.

17. Enter the board number (1 for CP).

18. An Xterm will be displayed. Select option 1.) Transfer OS. When completed, the board will reboot.

19. Restart the the Flash burner from the GUI again on the same board. Then select option 2.) Transfer Diags.

20. When completed, the board will reboot. Now one board is done. Repeat this process on the remaining three boards. Board numbers are: 2 for GAM, 3 for DSP/AP, 4 for BP. You can view the date of the Flash OS via a Telnet session. At the pSH+> prompt, type **romver** and press <CR>.

21. This completes the 1.0 scanner software portion of the upgrade. After the initialization is complete, run a pilot, axial and spiral scan to further ensure system functionality. This completes the 1.0 installation. Reboot the system.

ACQSIM CT 2.0 SOFTWARE UPGRADE INSTRUCTIONS

The following kit should be used when upgrading a 1.0 software based AcQSim CT system with 2.0 software.

Kit # 4535 670 06581		
Part Number	**Description**	**Qty.**
4535 670 06281	AcQSim CT 2.0 Operating System Software CD (2 CD’s in set)	1
CT–9369	AcQSim CT 2.0 Software Upgrade Technical Note	1
T55Y–1013	AcQSim CT DICOM Conformance Statement	1
4535 670 06921	AcQSim CT 2.0 Software Upgrade Kit Instructions	1
179971	AcQSim CT 2.0 User Documentation Package	1

1. Be sure all images are archived before starting this upgrade. Make a site tape immediately before installation. If in the Console or Service Application, select Shutdown Console from the General Utilities button. When the Main Startup menu appears, click in the lower left hand corner and type the word **site** (lower case) and press <CR>.
2. After the Save Site / Restore Site menu appears, push the tape in the tape drive, wait for the green light to stop blinking and select “1” to Save Site. The system will eject the tape when finished.
3. Exit the Save Site Utility. When the Shutdown button becomes available, press it and wait for the system to shut down.
4. Open the CD–ROM drive and insert CD #1. Close the drive.
5. When the ok prompt appears, type **boot cdrom** (lower case) and press <CR>. After a few minutes, the password prompt will appear. Type **picker** (lower case) and press <CR>.

6. The system then checks the disk configuration and prompts you with:

```
Enter the type of system installation:

1. RDS                  (Note: “RDS” refers to an Ultra Z Diagnostic Workstation, UDW)
2. Scanner Console

q. quit
```

7. Select option “2,” *Scanner Console* from the menu and press <CR>. You will see the following menu.

```
Enter a Console Installation option:

1. Install the OS and Application Software
2. Configure a Replaced Disk(s)
3. Upgrade Application Software Only

q. quit
```

8. Select option “1,” *Install the OS and Application Software* from the menu and press <CR>. Answer “**y**” to the next question. The following will appear:

Main Menu Option #1 – *Reformat and Configure entire system:*

```
Enter IP Address [default xxx.xxx.xx.xx]:
Enter Netmask [default 255.255.255.xxx]:
Enter Router [default xxx.xx.xx.xx]:
Enter Nodename [default consolexxxx]:
```

9. Press <CR> to accept the default values. After the information is entered, the system will display your choices as you entered them:

```
IP Address:
Netmask:
Router:
Nodename:
```

10. After the “Ready to run the installation. Are you satisfied? (y/n) [n]?” prompt appears:
    If you answer NO to “Are you satisfied?”, the installation will cancel and take you back to the first sub–menu where you can quit the installation program, or continue and adjust the network information.

    If installation information is correct, choose “y” to “Are you satisfied?” and press <CR>.

11. The software will now begin installing from the first CD. Some time will pass during installation (approximately 25 minutes). **DO NOT PRESS ANY KEYS ON THE KEYBOARD.** The following menu will then appear:

```
1. Install SCANNER Console Application Software
2. Eject the CD.
```

Select option “2,” *Eject the CD*. Remove the first CD and put the second CD in, then close the drive door.

12. Select option “1,” to continue with the installation. You will be prompted with the “Are you satisfied?” question again. Select “y”, press <CR>, and the system will continue with the software installation (approximately 20 to 30 minutes). After the installation is complete, a menu will appear:

```
1. View the beginning of the installation error log
2. Eject the CD  (it’s OK to eject it now)
3. Continue booting the machine

Choice:
```

> **NOTE**
>
> After the installation has completed, additional messages may be displayed on the screen, pushing the above menu out of view. Simply press <CR> to see it again.

13. View the error log by choosing option **1**. There should not be any errors in the error log at this time. After viewing, the menu will be displayed again. Select option **2** to eject the CD. Remove the CD and close the drive door. Then select option **3** to continue booting to the application software and press <CR>.

14. Perform a Restore Site. When the Main Startup menu appears, click in the lower left corner and type the word **site** (lower case) and press <CR>.

15. After the Save Site / Restore Site menu appears, push the tape in the tape drive, wait for the green light to stop blinking and select “2” to Restore Site. The system will eject the tape when finished.

16. Exit the Save Site utility.

17. After the 1.2 software installation is complete and the system has booted, start the Diagnostics GUI. At the Main Startup menu, click in the lower left hand corner, type **PICKER** (upper case) and press <CR>.
18. At the Diagnostics Software Test Menu, click on the “Utilities” button and select “Flash Burner”. The flash burner is a utility for transferring executable code from a directory on the SPARC disk to the Flash of the CP, DSP/AP, BP and GAM boards.
19. At the Flash Burner dialog box, enter the number of the board selected in the “Board name” category (1 for CP, 2 for GAM, 3 for DSP/AP, and 4 for BP) and click “Continue”.

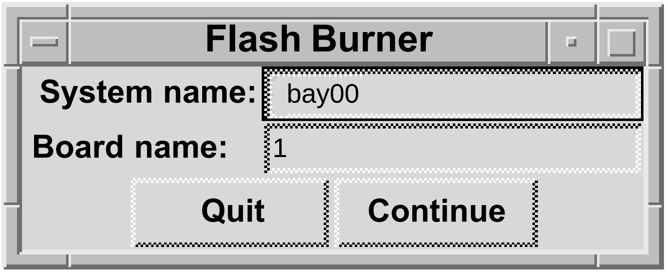


FIGURE 1 FLASH BURNER DIALOG BOX

20. An Xterm window will be displayed. Click the mouse in the Xterm window to activate it. Select option 1.) Transfer OS.
21. At the next menu, select 0.) Start transfer. When completed, the board will reboot.
22. Restart the Flash burner from the GUI again on the same board. Then select option 2.) Transfer Diags.
23. When completed, the board will reboot. The message “Rebooting the board, I/O session is dead” will be displayed. Press <CR> and repeat steps 18 thru 23 on the remaining three boards. Board numbers are: 2 for GAM, 3 for DSP/AP, 4 for BP. You can view the date of the Flash OS via a Telnet session. At the pSH+> prompt, type **romver** and press <CR>. When flash burner is complete on all boards, select “Exit” to exit the Diagnostics GUI.

Site Protocol Update Program Instructions

Proceed with the following instructions to perform the site protocol update program.

1. Install the AcQSim CT software and restore the site information from tape.
2. At the four–button GUI, click the mouse in the lower left–hand corner. Type **pptools** and press <CR>.
3. Click in the window that comes up. Type **update_protocols** and press <CR>.

4. Type exit and press <CR>.
5. Test a sample of the site defined protocols by scanning.
6. When testing is complete, create a site tape with the new data.
7. This completes the 2.0 scanner software portion of the upgrade. After the initialization is complete, run a pilot, axial and spiral scan to further ensure system functionality. This completes the 2.0 software upgrade. Reboot the system before turning it over to the customer.

Philips Medical Systems

AcQSim CT 3.1 Software Upgrade Kit Instructions

Part Number: 4535 670 52251

Revision: A

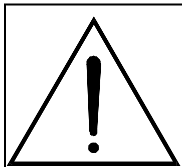
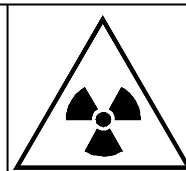
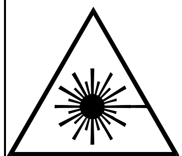
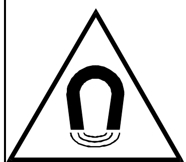
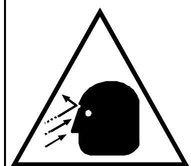
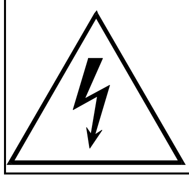
PHILIPS

Service Publication, Philips Medical Systems (Cleveland) Inc. Highland Hts., OH 44143

PHILIPS

Philips Medical Systems (Cleveland), Inc. Highland Hts., OH 44143

SYMBOL DESCRIPTIONS

	Attention symbol.		Radiation warning symbol.
	Laser warning symbol.		Biohazard warning symbol.
	Magnetism warning symbol.		Projectile warning symbol.
	Electrical warning symbol.		

REVISION HISTORY

REVISION	DATE	COMMENTS
–1	11/14/03	First draft
–2	12/15/03	Final review version
A	03/01/04	Release at revision A

NOTICE

THE INFORMATION CONTAINED IN THIS MANUAL CONFORMS WITH THE CONFIGURATION OF THE EQUIPMENT AS OF THE DATE OF MANUFACTURE. REVISIONS TO THE EQUIPMENT SUBSEQUENT TO THE DATE OF MANUFACTURE WILL BE ADDRESSED IN SERVICE UPDATES DISTRIBUTED TO PHILIPS MEDICAL SYSTEMS TECHNICAL SERVICE ORGANIZATION.

TO THE USER OF THIS MANUAL

THE USER OF THIS MANUAL IS DIRECTED TO READ AND CAREFULLY REVIEW THE INSTRUCTIONS, WARNINGS AND CAUTIONS CONTAINED HEREIN PRIOR TO BEGINNING INSTALLATION OR SERVICE ACTIVITIES. WHILE YOU MAY HAVE PREVIOUSLY INSTALLED OR SERVICED EQUIPMENT SIMILAR TO THAT DESCRIBED IN THIS MANUAL, CHANGES IN DESIGN, MANUFACTURE OR PROCEDURE MAY HAVE OCCURRED WHICH SIGNIFICANTLY AFFECT THE PRESENT INSTALLATION OR SERVICE.

THE INSTALLATION AND SERVICE OF EQUIPMENT DESCRIBED HEREIN IS TO BE PREFORMED BY AUTHORIZED, QUALIFIED PHILIPS MEDICAL SYSTEM PERSONNEL. ASSEMBLERS AND OTHER PERSONNEL NOT EMPLOYED BY NOR DIRECTLY AFFILIATED WITH PHILIPS MEDICAL SYSTEMS TECHNICAL SERVICES ARE DIRECTED TO CONTACT THE LOCAL PHILIPS MEDICAL SYSTEMS OFFICE BEFORE ATTEMPTING INSTALLATION OR SERVICE PROCEDURES.

INSTALLATION AND ENVIRONMENT

EXCEPT FOR INSTALLATIONS REQUIRING CERTIFICATION BY THE MANUFACTURER PER FEDERAL STANDARDS, SEE THAT A RADIATION PROTECTION SURVEY IS MADE BY A QUALIFIED EXPERT IN ACCORDANCE WITH NCRP 102, SECTION 7, AS REVISED OR REPLACED IN THE FUTURE. PERFORM A SURVEY AFTER EVERY CHANGE IN EQUIPMENT, WORKLOAD, OR OPERATING CONDITIONS WHICH MIGHT SIGNIFICANTLY INCREASE THE PROBABILITY OF PERSONS RECEIVING MORE THAN THE MAXIMUM PERMISSIBLE DOSE EQUIVALENT.

Diagnostic Imaging Systems – MECHANICAL-ELECTRICAL WARNING

ALL OF THE MOVEABLE ASSEMBLIES AND PARTS OF THIS EQUIPMENT SHOULD BE OPERATED WITH CARE AND ROUTINELY INSPECTED IN ACCORDANCE WITH THE MANUFACTURER'S RECOMMENDATIONS CONTAINED IN THE EQUIPMENT MANUALS.

ONLY PROPERLY TRAINED AND QUALIFIED PERSONNEL SHOULD BE PERMITTED ACCESS TO ANY INTERNAL PARTS. LIVE ELECTRICAL TERMINALS ARE DEADLY; BE SURE LINE DISCONNECTS ARE OPENED AND OTHER APPROPRIATE PRECAUTIONS ARE TAKEN BEFORE OPENING ACCESS DOORS, REMOVING ENCLOSURE PANELS, OR ATTACHING ACCESSORIES.

DO NOT UNDER ANY CIRCUMSTANCES, REMOVE THE FLEXIBLE HIGH TENSION CABLES FROM THE X-RAY TUBE HOUSING OR HIGH TENSION GENERATOR AND/OR THE ACCESS COVERS FROM THE GENERATOR UNTIL THE MAIN AND AUXILIARY POWER SUPPLIES HAVE BEEN DISCONNECTED. FAILURE TO COMPLY WITH THE ABOVE MAY RESULT IN SERIOUS OR FATAL BODILY INJURIES TO THE OPERATOR OR THOSE IN THE AREA.

ELECTRICAL-GROUNDING INSTRUCTIONS

THE EQUIPMENT MUST BE GROUNDED TO AN EARTH GROUND BY A SEPARATE CONDUCTOR. THE NEUTRAL SIDE OF THE LINE IS NOT TO BE CONSIDERED THE EARTH GROUND. ON EQUIPMENT PROVIDED WITH A LINE CORD, THE EQUIPMENT MUST BE CONNECTED TO PROPERLY GROUNDED, THREE-PIN RECEPTACLE. DO NOT USE A THREE-TO-TWO PIN ADAPTER.

Diagnostic Imaging Systems – RADIATION WARNING

X-RAY AND GAMMA-RAYS ARE DANGEROUS TO BOTH OPERATOR AND OTHERS IN THE VICINITY UNLESS ESTABLISHED SAFE EXPOSURE PROCEDURES ARE STRICTLY OBSERVED.

THE USEFUL AND SCATTERED BEAMS CAN PRODUCE SERIOUS OR FATAL BODILY INJURIES TO ANY PERSONS IN THE SURROUNDING AREA IF USED BY AN UNSKILLED OPERATOR. ADEQUATE PRECAUTIONS MUST ALWAYS BE TAKEN TO AVOID EXPOSURE TO THE USEFUL BEAM, AS WELL AS TO LEAKAGE RADIATION FROM WITHIN THE SOURCE HOUSING OR TO SCATTERED RADIATION RESULTING FROM THE PASSAGE OF RADIATION THROUGH MATTER.

THOSE AUTHORIZED TO OPERATE, PARTICIPATE IN OR SUPERVISE THE OPERATION OF THE EQUIPMENT MUST BE THOROUGHLY FAMILIAR AND COMPLY COMPLETELY WITH THE CURRENT ESTABLISHED SAFE EXPOSURE FACTORS AND PROCEDURES DESCRIBED IN PUBLICATIONS, SUCH AS: SUBCHAPTER J OF TITLE 21 OF THE CODE OF FEDERAL REGULATIONS, "DIAGNOSTIC X-RAY SYSTEMS AND THEIR MAJOR COMPONENTS", AND THE NATIONAL COUNCIL ON RADIATION PROTECTION (NCRP) NO. 102, "MEDICAL X-RAY AND GAMMA-RAY PROTECTION FOR ENERGIES UP TO 10 MEV-EQUIPMENT DESIGN AND USE", AS REVISED OR REPLACED IN THE FUTURE.

THOSE RESPONSIBLE FOR PLANNING OF X-RAY AND GAMMA-RAY EQUIPMENT INSTALLATIONS MUST BE THOROUGHLY FAMILIAR AND COMPLY COMPLETELY WITH NCRP NO. 49, "STRUCTURAL SHIELDING DESIGN AND EVALUATION FOR MEDICAL OF X-RAYS AND GAMMA-RAYS OF ENERGIES UP TO 10 MEV", AS REVISED AND REPLACED IN THE FUTURE. FAILURE TO OBSERVE THESE WARNINGS MAY CAUSE SERIOUS OR FATAL BODILY INJURIES TO THE OPERATOR OR THOSE IN THE AREA.

TABLE OF CONTENTS

ACQSIM CT 3.1 SOFTWARE UPGRADE INSTRUCTIONS

The following kit should be used when upgrading an AcQSim CT system with 3.1 software.

Kit # 4535 670 52811		
Part Number	**Description**	**Qty.**
4535 670 52711	AcQSim CT 3.1 System Software CD (2 CD’s in set)	1
4535 670 52841	AcQSim CT 3.1 Software Release Notes	1
4535 670 51481	AcQSim CT 3.1 DICOM Conformance Statement	1
4535 670 52251	AcQSim CT 3.1 Software Upgrade Kit Instructions	1
4535 670 17151	AcQSim CT 3.1 User Documentation Package	1

> **NOTE**
>
> If the customer has purchased the gating option or localization option, it is recommended that the activation key be obtained prior to upgrading the software, if needed.

> **NOTE**
>
> During the upgrade, the hard drive will be reformatted and all patient images will be lost.

1. Be sure all images are archived before starting this upgrade. Make a site tape immediately before installation. If in the Console or Service Application, select Shutdown Console from the General Utilities button. When the Main Startup menu appears, click in the lower left hand corner and type the word **site** (lower case) and press <enter>.
2. After the Save Site / Restore Site menu appears, push the tape in the tape drive, wait for the green light to stop blinking and select “1” to Save Site. The system will eject the tape when finished.
3. Exit the Save Site Utility. When the Shutdown button becomes available, press it and wait for the system to shut down.
4. Open the CD–ROM drive and insert CD #1. Close the drive.
5. When the ok prompt appears, type **boot cdrom** (lower case) and press <enter>. After a few minutes, the password prompt will appear. Type **picker** (lower case) and press <enter>.

6. The system then checks the disk configuration and prompts you with:

```
Enter the type of system installation:

1. RDS                  (Note: “RDS” refers to an Ultra Z Diagnostic Workstation, UDW)
2. Scanner Console

q. quit
```

7. Select option “2,” *Scanner Console* from the menu and press <enter>. You will see the following menu.

```
Enter a Console Installation option:

1. Install the OS and Application Software
2. Configure a Replaced Disk(s)
3. Upgrade Application Software Only

q. quit
```

8. Select option “1,” *Install the OS and Application Software* from the menu and press <enter>. Answer “**y**” to the next question. The following will appear:

```
Please confirm the following network information:

Enter IP Address [default xxx.xxx.xx.xx]:
Enter Netmask [default 255.255.255.xxx]:
Enter Router [default xxx.xx.xx.xx]:
Enter Nodename [default consolexxxx]:
```

9. Press <enter> to accept the default values. After the information is entered, the system will display your choices as you entered them:

```
IP Address:
Netmask:
Router:
Nodename:
```

10. After the “Ready to run the installation. Are you satisfied? (y/n) [n]?” prompt appears:
    If you answer NO to “Are you satisfied?”, the installation will cancel and take you back to the first sub–menu where you can quit the installation program, or continue and adjust the network information.

    If installation information is correct, choose “y” to “Are you satisfied?” and press <enter>.

11. The software will now begin installing from the first CD. Some time will pass during installation (approximately 25 minutes). **DO NOT PRESS ANY KEYS ON THE KEYBOARD.** The following menu will then appear:

```
1. Install SCANNER Console Application Software
2. Eject the CD.
```

Select option “2,” *Eject the CD,* and press <enter>. Remove the first CD and put the second CD in, then close the drive door.

12. Select option “1,” and press <enter> to continue with the installation. You will be prompted with the “Are you satisfied?” question again. Select “y”, press <enter>, and the system will continue with the software installation (approximately 20 to 30 minutes). After the installation is complete, the system will reboot and the following menu will appear:

```
1. View the beginning of the installation error log
2. Eject the CD  (it's OK to eject it now)
3. Continue booting the machine

Choice:
```

> **NOTE**
>
> After the installation has completed, additional messages may be displayed on the screen, pushing the above menu out of view. Simply press <enter> to see it again.

13. View the error log by choosing option **1**. Errors may exist related to pdf help files. Other than these, there should be no errors in the error log at this time. After viewing, the menu will be displayed again. Select option **2** and press <enter> to eject the CD. Remove the CD and close the drive door. Then select option **3** to continue booting to the application software and press <enter>.

> **NOTE**
>
> A serial number mismatch error may occur. Close the error window.

14. Perform a Restore Site. When the Main Startup menu appears, click in the lower left corner and type the word **site** (lower case) and press <enter>.

15. After the Save Site / Restore Site menu appears, push the tape in the tape drive, wait for the green light to stop blinking and select “2” to Restore Site. The system will eject the tape when finished.
16. Exit the Save Site utility.

> **NOTE**
> A valid diagnostics key is required to continue. If you can not enter Diagnostics as instructed in step 17., obtain a Diagnostics key per appropriate procedures.

17. After the 3.1 software installation is complete and the system has booted, start the Diagnostics GUI. At the Main Startup menu, click in the lower left hand corner, type **PICKER** (upper case) and press <enter>.
18. At the Diagnostics Software Test Menu, click on the “Utilities” button and select “Flash Burner”. The flash burner is a utility for transferring executable code from a directory on the SPARC disk to the Flash of the CP, DSP/AP, BP and GAM boards.
19. At the Flash Burner dialog box, enter the number of the board selected in the “Board name” category (1 for CP, 2 for GAM, 3 for DSP/AP, and 4 for BP) and click “Continue”.

> **NOTE**
> The Acqimage Box must be enabled in the Hardware Status window before clicking on “Continue”.

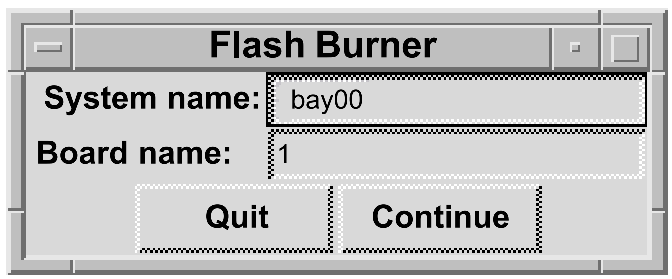


FIGURE 1 FLASH BURNER DIALOG BOX

20. An Xterm window will be displayed. Click the mouse in the Xterm window to activate it. Select option 1.) Transfer OS.
21. At the next menu, select 0. Start transfer. When completed, the board will reboot. The message “Rebooting the board, I/O session is dead” will be displayed. Press <enter> to continue. There may be a delay before the Flash Burner window closes.

22. Restart the Flash burner from the GUI again on the same board. Then select option 2.) Transfer Diags.

23. When completed, the board will reboot. The message “Rebooting the board, I/O session is dead” will be displayed. Press <enter> and repeat steps 18 thru 23 on the remaining three boards. Board numbers are: 2 for GAM, 3 for DSP/AP, 4 for BP. You can view the date of the Flash OS via a Telnet session. At the pSH+> prompt, type **romver** and press <enter>. When flash burner is complete on all boards, select “Exit” to exit the Diagnostics GUI.

Site Protocol Update Program Instructions

If upgrading from software version 1.0, proceed with the following instructions to perform the site protocol update program.

1. Restore the site information from tape (If the site information tape has already been restored, it is not necessary to restore it again).
2. At the four–button GUI, click the mouse in the lower left–hand corner. Type **pptools** and press <enter>.
3. Click in the window that comes up. Type **update_protocols** and press <enter>.
4. Type **exit** and press <enter>.
5. Test a sample of the site defined protocols by scanning.
6. When testing is complete, create a site tape with the new data.
7. This completes the 3.1 scanner software portion of the upgrade. After the initialization is complete, run a pilot, axial and spiral scan to further ensure system functionality. This completes the 3.1 software upgrade. Reboot the system before turning it over to the customer.

Manufacturer Name Update Instructions

To ensure that the manufacturer name reported on the scanner is ”Philips”, proceed with the following instructions.

1. Restore the site information tape (If the site information tape has already been restored, it is not necessary to restore it again).
2. At the four–button GUI, click the mouse in the lower left–hand corner. Type **pptools** and press <enter>.
3. Click in the window that comes up. Type **update_mfg_name** and press <enter>.
4. Type **exit** and press <enter>.
5. Verify that the manufacturer name is now ”Philips” by entering the Service application from the four–button GUI and checking the name reported in the Configure Workstep.
6. When the verification is complete, create a site information tape.

Security Vulnerability Update

Security vulnerabilities have been identified in the AcQSim CT product. The Solaris Operating System commands **(sadmind, in.talkd, rpc.walid, and sendmail)** contain security weaknesses that can be exploited. The instructions below edit a configuration file to disable Solaris files that have been identified as security vulnerabilities. For sendmail, the permissions will be modified to disable outside access of this command.

1. Get to the four button screen from which the options are ”Clinical Scanning”, ”Service”, ”Reboot”, and ”Shutdown”. If currently in ”Clinical Scanning” or ”Service”, exit the console by selecting ”Shutdown Console” from the General Utilities Shutdown option.
2. Move the mouse pointer into the lower left–hand corner of the four button screen, left click once, and type **pptools** <enter>. Note that you will not see anything appearing on the screen as you type. This will launch an xterm window.
3. Click inside the x–term window. From the command prompt, type the following commands followed by <enter>. The commands to type appear in bold below. You will be prompted for a password upon pressing <enter> following the “**su**” command below. Use ”**picker**”. (Unix is case sensitive so make sure that Caps Lock is off).

   su <enter>
   picker <enter>
   cd /etc <enter>
   cat inetd.conf > inetd.bak <enter>

   NOTE: The following command contains (” ”) and (|) that are required to be typed.

   The pipe symbols (|) within the command are the two vertical lines above one another and commonly looks like a full vertical line. This symbol is found on the same keyboard key as the backslash (\ ) key. It is not the letter ” L or l ” or the number ” 1 ”. Make sure there is a space between the second quotation mark and the term inetd.bak.

   egrep –v ”walld|talkd|sadmind” inetd.bak > inetd.conf <enter>

   NOTE: The following command contains three consecutive zeros ”000” not the letters ”OOO”.

   chmod 000 /usr/lib/sendmail <enter>

4. The update is complete. Type ”exit” at the # prompt to exit the shell. Type ”exit” again in the xterm window and select ”Reboot” from the four button screen to restart the scanner

Gating Option Activation Instructions

If customer has purchased Gating option, activate as follows.

1. From the four button main menu screen, select **Service**.
2. Select **configure > options**. At the Options Configuration screen, highlight the Gating option.
3. Gain access to Servecom. Follow the menu items, Service, CT, Key Codes, AcQSim CT/Ultra Z Diagnostic Keys. Input the AcQSim CT serial number to obtain the Gating Key.
4. Enter enabling key information exactly as given to you.
5. Select **Update Configuration Files** to save information. Note the Gating button at the bottom of the screen will become active. See Figure 2.

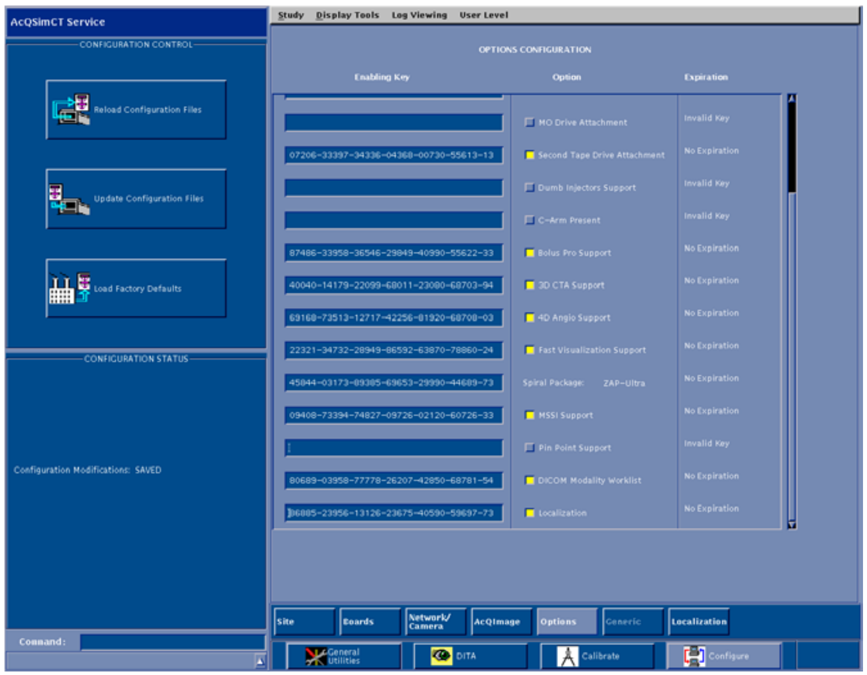


FIGURE 2 GATING AND LOCALIZATION

Localization Option Activation Instructions

If customer has purchased localization option, activate as follows.

1. From the four button main menu screen, select **Service**.
2. Select **configure > options**. At the Options Configuration screen, highlight the Localization option.
3. Gain access to Servecom. Follow the menu items, Service, CT, Key Codes, AcQSim CT/Ultra Z Diagnostic Keys. Input the AcQSim CT serial number to obtain the Localization Key.
4. Enter enabling key information exactly as given to you.
5. Select **Update Configuration Files** to save information. Note the Localization button at the bottom of the screen will become active. See Figure 2.

Connectivity to other Systems

The customers RTP system may require configuration of a “dummy” treatment machine to import the isocenter (RT Plan) from AcQSim CT 3.1. The treatment machine can be changed within the RTP system after import. Refer to the AcQSim CT 3.1 DICOM Conformance Statement for details on what tags and values are sent.

Localization Option Reference Information

To access the Laser Calibration Data, select the Localization button at the bottom of the service applications mode. At the Laser Calibration Data screen, the recommended values are listed in the table. Measure the laser distance for each point when the laser installation is done to determine the actual value. It is recommended that these values be verified by on–site hospital personnel. The following reference information is provided for the Laser Calibration Data screen (see figure 3):

The Treatment Machine name is the name of the linear accelerator (linac) data is sent to. The CT Sim offset is site specific. The CT Sim offset is the shift between the scan plane and the Laser Marking Plane. The CT Sim offset range of valid values is from 0 to 800.

Select Limit isocenter to slice boundary option to **Yes** if treatment planning system requires beam isocenters to be on a slice plane. When this option is set to yes, beam isocenters that are computed or moved in Localization will jump to the nearest slice plane. If you leave this option set to **No**, beam isocenters can be computed or moved to locations between slice planes.

The maximum contour points selection represents the maximum number of points per contour stored on the system and sent out through DICOM transfer. Some treatment planning systems require fewer than the default 512 contour points. This field can be used to limit the contour points to the amount required by these systems. The maximum contour points range of valid values is 16 to 512.

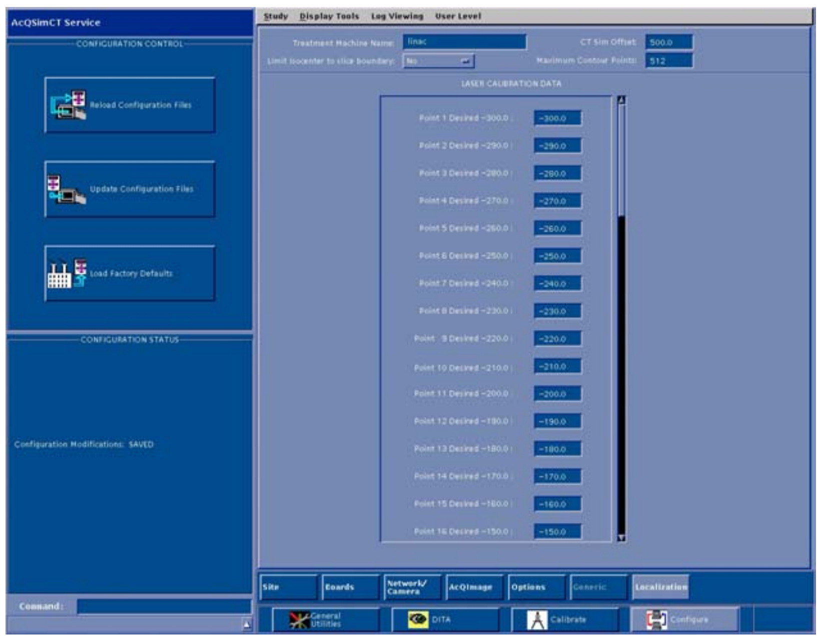


FIGURE 3 LASER CALIBRATION DATA SCREEN

Network/Camera Service Tab Reference Information

In the service application mode, some modifications have been made to the menu screen for the **Network/Camera** tab (see figure 4):

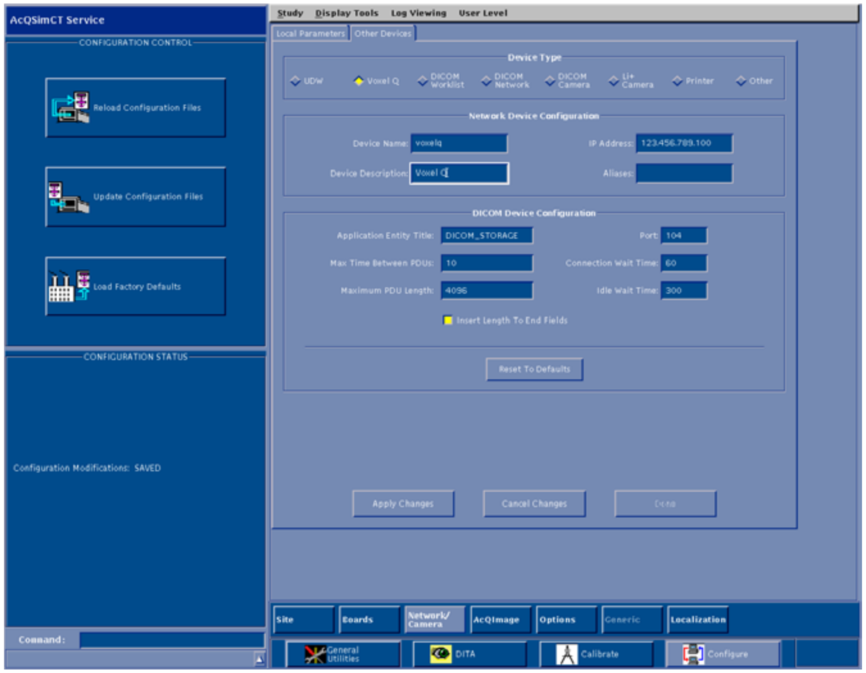


FIGURE 4 NETWORK/CAMERA TAB SCREEN

Under the **Device** type, the available devices have been updated to include:

UDW (Ultra Z Diagnostic Workstation)
Voxel Q
DICOM Worklist
DICOM Network
DICOM Camera
Li+ Camera
Printer
Other

Printer Setup for Worksheet Printing

In order to print a worksheet, the software must know about one or more network printers. If there is more than one printer, one of them must be designated as the default printer.

1. Printer configuration is accomplished by selecting **Service** from the top level AcQSim CT Startup menu.
2. Once in the service application mode, select **Configure**, and then the **Network/Camera** button.
3. Select the **Other Devices** tab to bring up a list of network devices.
4. Select the **Printer** radio button.
5. Select either **Add Device** to add a printer or **Modify Selected Device**. See figure 5.

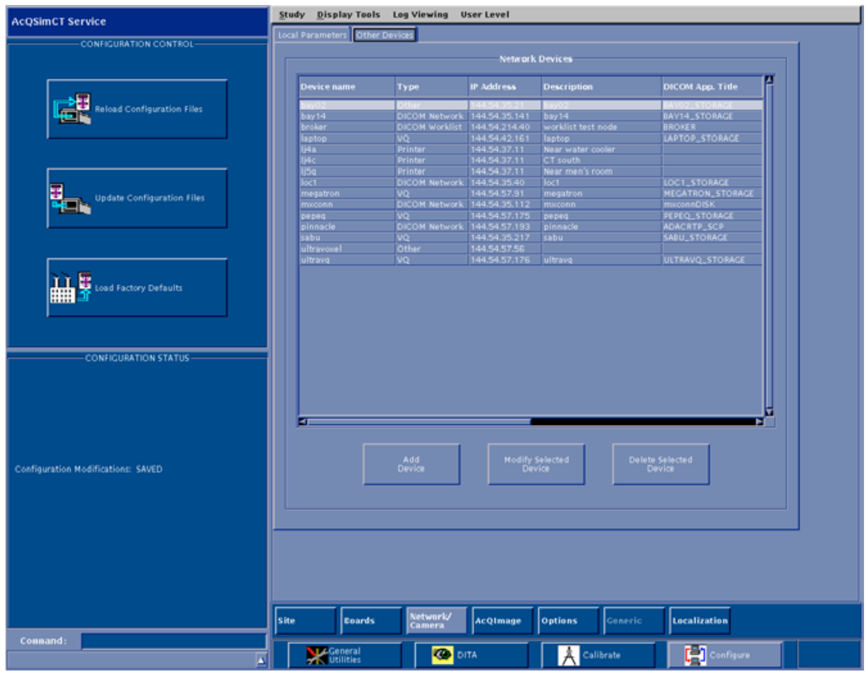


FIGURE 5   NETWORK DEVICES SCREEN

If **Add Device** is selected, the screen in figure 6 is displayed.

At this screen, enter the device name, its description, and the IP address of the printer's server.

There are two radio buttons displayed for selection, one to indicate if it is a PostScript printer and another to indicate whether the printer is the default printer. The default printer is the printer that will be used when the **Worksheet** button is selected on the Isocenter menu screen.

> **NOTE**
>
> There can be only one default printer.

Once printer properties have been set as desired, select **Done** and then select **Update Configuration Files** (located on the left hand side of the service application mode main screen) to save the new or modified printer configuration information. It is recommended to update the site tape after printer has been configured.

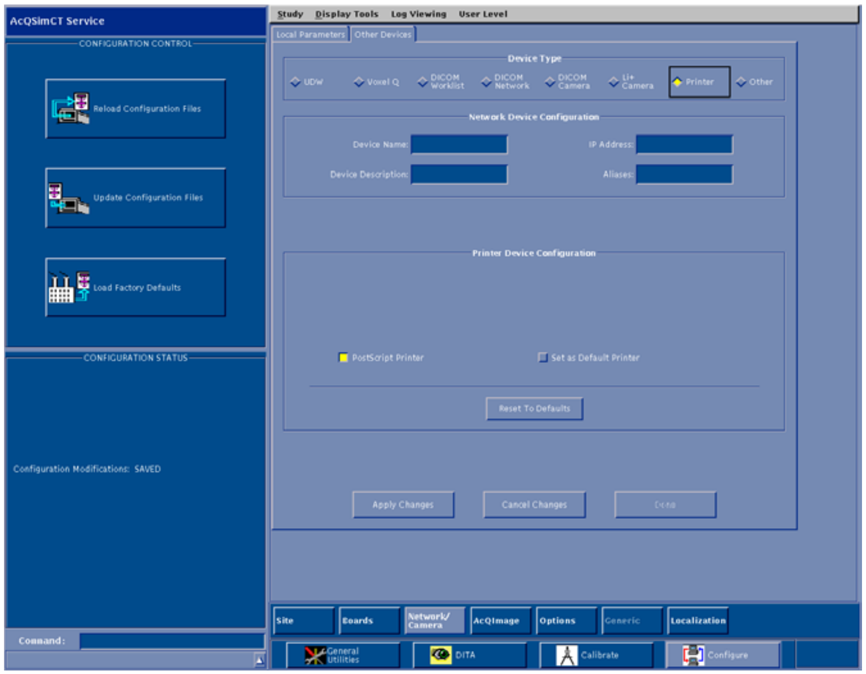


FIGURE 6   PRINTER DEVICE CONFIGURATION SCREEN

Voxel Q 4.9.1 Software Update Instructions

If the customer is using the Voxel Q workstation, it must be running software version 4.9.1 or higher in order to import LOC images from the AcQSim CT running version 3.1 software. If the customer has purchased the LOC package, Voxel Q 4.9.1 software is required.

1. Determine if the Voxel Q is running version 4.9.1 software by launching the vx application.
2. If the Voxel Q is running version 4.9.1 software, you do not need to update the Voxel Q software.
3. If 4.9.1 software is not installed on the Voxel Q, install the Voxel Q V4.9.1 Software Upgrade Kit 4535 670 49101 for AcQPlan units, or 4535 670 49091 for all other units. Follow the instructions supplied with the kit. This software should be installed on all Voxel Q systems the customer is using.

Philips Medical Systems

AcQSim CT 3.1.1 Software Upgrade Kit Instructions

Part Number: 4535 670 73151

Revision: A

PHILIPS

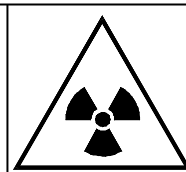
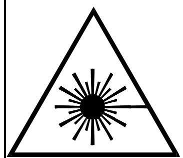
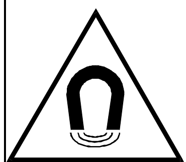
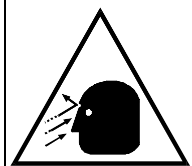
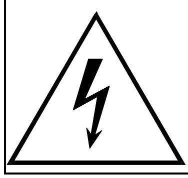
Service Publication, Philips Medical Systems (Cleveland) Inc. Highland Hts., OH 44143

PHILIPS

Philips Medical Systems (Cleveland), Inc. Highland Hts., OH 44143

This document prepared by the Service Publications Department.
Note any additions/corrections via an electronic Field Feedback Form.

SYMBOL DESCRIPTIONS

	Attention symbol.		Radiation warning symbol.
	Laser warning symbol.		Biohazard warning symbol.
	Magnetism warning symbol.		Projectile warning symbol.
	Electrical warning symbol.		

REVISION HISTORY

REVISION	DATE	COMMENTS
–1	07/13/04	Final review copy
A	07/15/04	Release at revision A

NOTICE

THE INFORMATION CONTAINED IN THIS MANUAL CONFORMS WITH THE CONFIGURATION OF THE EQUIPMENT AS OF THE DATE OF MANUFACTURE. REVISIONS TO THE EQUIPMENT SUBSEQUENT TO THE DATE OF MANUFACTURE WILL BE ADDRESSED IN SERVICE UPDATES DISTRIBUTED TO PHILIPS MEDICAL SYSTEMS TECHNICAL SERVICE ORGANIZATION.

TO THE USER OF THIS MANUAL

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THOSE AUTHORIZED TO OPERATE, PARTICIPATE IN OR SUPERVISE THE OPERATION OF THE EQUIPMENT MUST BE THOROUGHLY FAMILIAR AND COMPLY COMPLETELY WITH THE CURRENT ESTABLISHED SAFE EXPOSURE FACTORS AND PROCEDURES DESCRIBED IN PUBLICATIONS, SUCH AS: SUBCHAPTER J OF TITLE 21 OF THE CODE OF FEDERAL REGULATIONS, "DIAGNOSTIC X-RAY SYSTEMS AND THEIR MAJOR COMPONENTS", AND THE NATIONAL COUNCIL ON RADIATION PROTECTION (NCRP) NO. 102, "MEDICAL X-RAY AND GAMMA-RAY PROTECTION FOR ENERGIES UP TO 10 MEV-EQUIPMENT DESIGN AND USE", AS REVISED OR REPLACED IN THE FUTURE.

THOSE RESPONSIBLE FOR PLANNING OF X-RAY AND GAMMA-RAY EQUIPMENT INSTALLATIONS MUST BE THOROUGHLY FAMILIAR AND COMPLY COMPLETELY WITH NCRP NO. 49, "STRUCTURAL SHIELDING DESIGN AND EVALUATION FOR MEDICAL OF X-RAYS AND GAMMA-RAYS OF ENERGIES UP TO 10 MEV", AS REVISED AND REPLACED IN THE FUTURE. FAILURE TO OBSERVE THESE WARNINGS MAY CAUSE SERIOUS OR FATAL BODILY INJURIES TO THE OPERATOR OR THOSE IN THE AREA.

TABLE OF CONTENTS

ACQSIM CT 3.1.1 SOFTWARE UPGRADE INSTRUCTIONS

The following kit should be used when upgrading an AcQSim CT system with 3.1.1 software.

Kit # 4535 670 85281		
Part Number	**Description**	**Qty.**
4535 670 90591	AcQSim CT 3.1.1 System Software CD (2 CD's in set)	1
4535 670 73161	AcQSim CT 3.1.1 Software Release Notes	1
4535 670 89701	AcQSim CT 3.1.1 DICOM Conformance Statement	1
4535 670 73151	AcQSim CT 3.1.1 Software Upgrade Kit Instructions	1
4535 670 55541	AcQSim CT 3.1 User Documentation Package	1

> **NOTE**
>
> If the customer has purchased the gating option or localization option, it is recommended that the activation key be obtained prior to upgrading the software, if needed.

> **NOTE**
>
> During the upgrade, the hard drive will be reformatted and all patient images will be lost.

1. Be sure all images are archived before starting this upgrade. Make a site tape immediately before installation. If in the Console or Service Application, select Shutdown Console from the General Utilities button. When the Main Startup menu appears, click in the lower left hand corner and type the word **site** (lower case) and press <enter>.

> **NOTE**
>
> Make sure the site save tape is not write–protected.

2. After the Save Site / Restore Site menu appears, push the tape in the tape drive, wait for the green light to stop blinking and select "1" to Save Site. The system will eject the tape when finished.
3. Exit the Save Site Utility. When the Shutdown button becomes available, press it and wait for the system to shut down.

4. Open the CD–ROM drive and insert CD #1. Close the drive.

5. When the ok prompt appears, type **boot cdrom** (lower case) and press <enter>. After a few minutes, the password prompt will appear. Type **picker** (lower case) and press <enter>.

6. The system then checks the disk configuration and prompts you with:

```
Enter the type of system installation:

1. RDS                  (Note: “RDS” refers to an Ultra Z Diagnostic Workstation, UDW)
2. Scanner Console

q. quit
```

7. Select option “2,” *Scanner Console* from the menu and press <enter>. You will see the following menu.

```
Enter a Console Installation option:

1. Install the OS and Application Software
2. Configure a Replaced Disk(s)
3. Upgrade Application Software Only

q. quit
```

NOTE

If you have not archived patient data as instructed in step 1, step 8 will delete all data.

8. Select option “1,” *Install the OS and Application Software* from the menu and press <enter>. Answer “**y**” to the next question. The following will appear:

```
Please confirm the following network information:

Enter IP Address [default xxx.xxx.xx.xx]:
Enter Netmask [default 255.255.255.xxx]:
Enter Router [default xxx.xx.xx.xx]:
Enter Nodename [default consolexxxx]:
```

9. Press <enter> to accept the default values. After the information is entered, the system will display your choices as you entered them:

```
IP Address:
Netmask:
Router:
Nodename:
```

10. After the "Ready to run the installation. Are you satisfied? (y/n) [n]?" prompt appears:
    If you answer NO to "Are you satisfied?", the installation will cancel and take you back to the first sub–menu where you can quit the installation program, or continue and adjust the network information.

    If installation information is correct, choose "y" to "Are you satisfied?" and press <enter>.

11. The software will now begin installing from the first CD. Some time will pass during installation (approximately 25 minutes). **DO NOT PRESS ANY KEYS ON THE KEYBOARD.** The following menu will then appear:

```
1. Install SCANNER Console Application Software
2. Eject the CD.
```

Select option "2," *Eject the CD,* and press <enter>. Remove the first CD and put the second CD in, then close the drive door.

12. Select option "1," and press <enter> to continue with the installation. You will be prompted with the "Are you satisfied?" question again. Select "y", press <enter>, and the system will continue with the software installation (approximately 20 to 30 minutes). After the installation is complete, the system will reboot and the following menu will appear:

```
1. View the beginning of the installation error log
2. Eject the CD  (it's OK to eject it now)
3. Continue booting the machine

Choice:
```

> **NOTE**
>
> After the installation has completed, additional messages may be displayed on the screen, pushing the above menu out of view. Simply press <enter> to see it again.

> **NOTE**
>
> In the error log, errors may exist related to pdf help files. Errors may also exist related to matlab/etc/license.log. These errors do not affect operation and should be ignored. Other than these listed errors, there should be no other errors in the error log at this time.

13. View the error log by choosing option **1**. After viewing, the menu will be displayed again. Select option **2** and press <enter> to eject the CD. Remove the CD and close the drive door. Then select option **3** to continue booting to the application software and press <enter>.

> **NOTE**
>
> A serial number mismatch error may occur. Close the error window.

14. Perform a Restore Site. When the Main Startup menu appears, click in the lower left corner and type the word **site** (lower case) and press <enter>.
15. After the Save Site / Restore Site menu appears, push the tape in the tape drive, wait for the green light to stop blinking and select “2” to Restore Site. The system will eject the tape when finished.
16. Exit the Save Site utility.

> **NOTE**
>
> A valid diagnostics key is required to continue. If you can not enter Diagnostics as instructed in step 17., obtain a Diagnostics key per appropriate procedures.

17. After the 3.1.1 software installation is complete and the system has booted, start the Diagnostics GUI. At the Main Startup menu, click in the lower left hand corner, type **PICKER** (upper case) and press <enter>.
18. At the Diagnostics Software Test Menu, click on the “Utilities” button and select “Flash Burner”. The flash burner is a utility for transferring executable code from a directory on the SPARC disk to the Flash of the CP, DSP/AP, BP and GAM boards.

19. At the Flash Burner dialog box (refer to Figure 1), enter the number of the board selected in the “Board name” category (1 for CP, 2 for GAM, 3 for DSP/AP, and 4 for BP) and click “Continue”.

> **NOTE**
>
> The Acqimage Box must be enabled in the Hardware Status window before clicking on “Continue”.

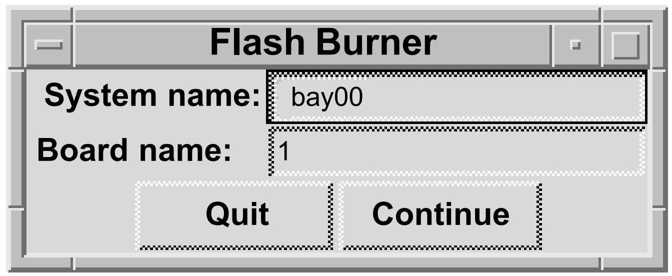


FIGURE 1 FLASH BURNER DIALOG BOX

20. An Xterm window will be displayed. Click the mouse in the Xterm window to activate it. Select option 1.) Transfer OS.

21. At the next menu, select 0. Start transfer. When completed, the board will reboot. The message “Rebooting the board, I/O session is dead” will be displayed. Press <enter> to continue. There may be a delay before the Flash Burner window closes.

22. Restart the Flash burner from the GUI again on the same board. Then select option 2.) Transfer Diags.

23. When completed, the board will reboot. The message “Rebooting the board, I/O session is dead” will be displayed. Press <enter> and repeat steps 18 thru 23 on the remaining three boards. Board numbers are: 2 for GAM, 3 for DSP/AP, 4 for BP. You can view the date of the Flash OS via a Telnet session. At the pSH+> prompt, type **romver** and press <enter>. When flash burner is complete on all boards, select “Exit” to exit the Diagnostics GUI.

Site Protocol Update Program Instructions

If upgrading from software version 1.0, proceed with the following instructions to perform the site protocol update program.

1. Restore the site information from tape (If the site information tape has already been restored, it is not necessary to restore it again).
2. At the four–button GUI, click the mouse in the lower left–hand corner. Type **pptools** and press <enter>.
3. Click in the window that comes up. Type **update_protocols** and press <enter>.
4. Type **exit** and press <enter>.
5. Test a sample of the site defined protocols by scanning.
6. When testing is complete, create a site tape with the new data.
7. This completes the 3.1.1 scanner software portion of the upgrade. After the initialization is complete, run a pilot, axial and spiral scan to further ensure system functionality. This completes the 3.1.1 software upgrade. Reboot the system before turning it over to the customer.

Manufacturer Name Update Instructions

To ensure that the manufacturer name reported on the scanner is ”Philips”, proceed with the following instructions.

1. Restore the site information tape (If the site information tape has already been restored, it is not necessary to restore it again).
2. At the four–button GUI, click the mouse in the lower left–hand corner. Type **pptools** and press <enter>.
3. Click in the window that comes up. Type **update_mfg_name** and press <enter>.
4. Type **exit** and press <enter>.
5. Verify that the manufacturer name is now ”Philips” by entering the Service application from the four–button GUI and checking the name reported in the Configure Workstep.
6. When the verification is complete, create a site information tape.

Security Vulnerability Update

Security vulnerabilities have been identified in the AcQSim CT product. The Solaris Operating System commands **(sadmind, in.talkd, rpc.walid, and sendmail)** contain security weaknesses that can be exploited. The instructions below edit a configuration file to disable Solaris files that have been identified as security vulnerabilities. For sendmail, the permissions will be modified to disable outside access of this command.

1. Get to the four button screen from which the options are ”Clinical Scanning”, ”Service”, ”Reboot”, and ”Shutdown”. If currently in ”Clinical Scanning” or ”Service”, exit the console by selecting ”Shutdown Console” from the General Utilities Shutdown option.
2. Move the mouse pointer into the lower left–hand corner of the four button screen, left click once, and type **pptools** <enter>. Note that you will not see anything appearing on the screen as you type. This will launch an xterm window.
3. Click inside the x–term window. From the command prompt, type the following commands followed by <enter>. The commands to type appear in bold below. You will be prompted for a password upon pressing <enter> following the “**su**” command below. Use ”**picker**”. (Unix is case sensitive so make sure that Caps Lock is off).

   su <enter>
   picker <enter>
   cd /etc <enter>
   cat inetd.conf > inetd.bak <enter>

   NOTE: The following command contains (” ”) and (|) that are required to be typed.

   The pipe symbols (|) within the command are the two vertical lines above one another and commonly looks like a full vertical line. This symbol is found on the same keyboard key as the backslash (\ ) key. It is not the letter ” L or l ” or the number ” 1 ”. Make sure there is a space between the second quotation mark and the term inetd.bak.

   egrep –v ”walld|talkd|sadmind” inetd.bak > inetd.conf <enter>

   NOTE: The following command contains three consecutive zeros ”000” not the letters ”OOO”.

   chmod 000 /usr/lib/sendmail <enter>
4. The update is complete. Type ”exit” at the # prompt to exit the shell. Type ”exit” again in the xterm window and select ”Reboot” from the four button screen to restart the scanner

Gating Option Activation Instructions

If customer has purchased Gating option, activate as follows.

1. From the four button main menu screen, select **Service**.
2. Select **configure > options**. At the Options Configuration screen, highlight the Gating option.
3. Gain access to Servecom. Follow the menu items, Service, CT, Key Codes, AcQSim CT/Ultra Z Diagnostic Keys. Input the AcQSim CT serial number to obtain the Gating Key.
4. Enter enabling key information exactly as given to you.
5. Select **Update Configuration Files** to save information. Note the Gating button at the bottom of the screen will become active. See Figure 2.

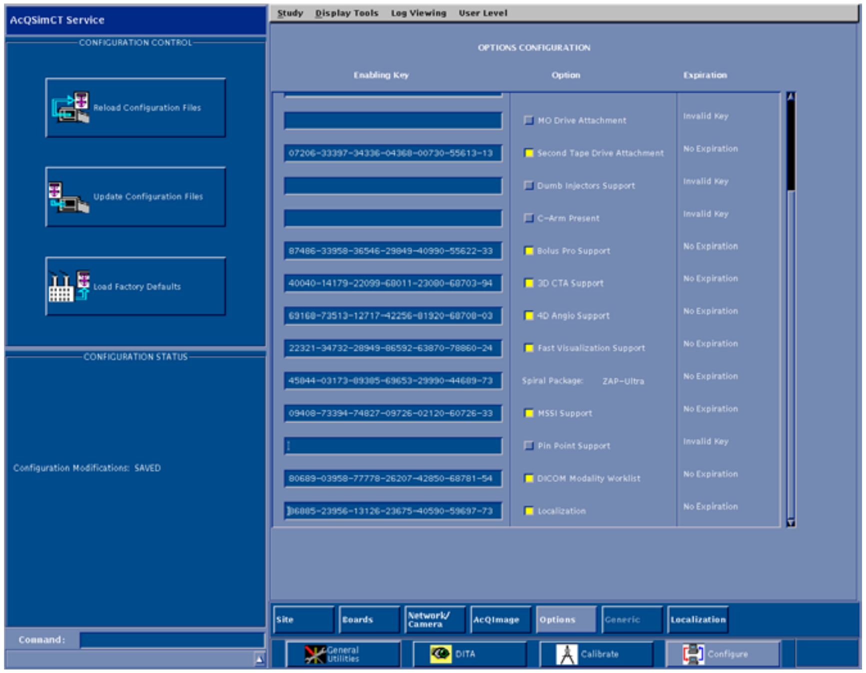


FIGURE 2   GATING AND LOCALIZATION

Localization Option Activation Instructions

If customer has purchased localization option, activate as follows.

1. From the four button main menu screen, select **Service**.
2. Select **configure > options**. At the Options Configuration screen, highlight the Localization option.
3. Gain access to Servecom. Follow the menu items, Service, CT, Key Codes, AcQSim CT/Ultra Z Diagnostic Keys. Input the AcQSim CT serial number to obtain the Localization Key.
4. Enter enabling key information exactly as given to you.
5. Select **Update Configuration Files** to save information. Note the Localization button at the bottom of the screen will become active. See Figure 2.

Connectivity to other Systems

The customers RTP system may require configuration of a "dummy" treatment machine to import the isocenter (RT Plan) from AcQSim CT 3.1.1. The treatment machine can be changed within the RTP system after import. Refer to the AcQSim CT 3.1.1 DICOM Conformance Statement for details on what tags and values are sent.

Localization Option Reference Information

To access the Laser Calibration Data, select the Localization button at the bottom of the service applications mode. At the Laser Calibration Data screen, the recommended values are listed in the table. Measure the laser distance for each point when the laser installation is done to determine the actual value. It is recommended that these values be verified by on–site hospital personnel. The following reference information is provided for the Laser Calibration Data screen (see figure 3):

The Treatment Machine name is the name of the linear accelerator (linac) data is sent to. The CT Sim offset is site specific. The CT Sim offset is the shift between the scan plane and the Laser Marking Plane. The CT Sim offset range of valid values is from 0 to 800.

Select Limit isocenter to slice boundary option to **Yes** if treatment planning system requires beam isocenters to be on a slice plane. When this option is set to yes, beam isocenters that are computed or moved in Localization will jump to the nearest slice plane. If you leave this option set to **No**, beam isocenters can be computed or moved to locations between slice planes.

The maximum contour points selection represents the maximum number of points per contour stored on the system and sent out through DICOM transfer. Some treatment planning systems require fewer than the default 512 contour points. This field can be used to limit the contour points to the amount required by these systems. The maximum contour points range of valid values is 16 to 512.

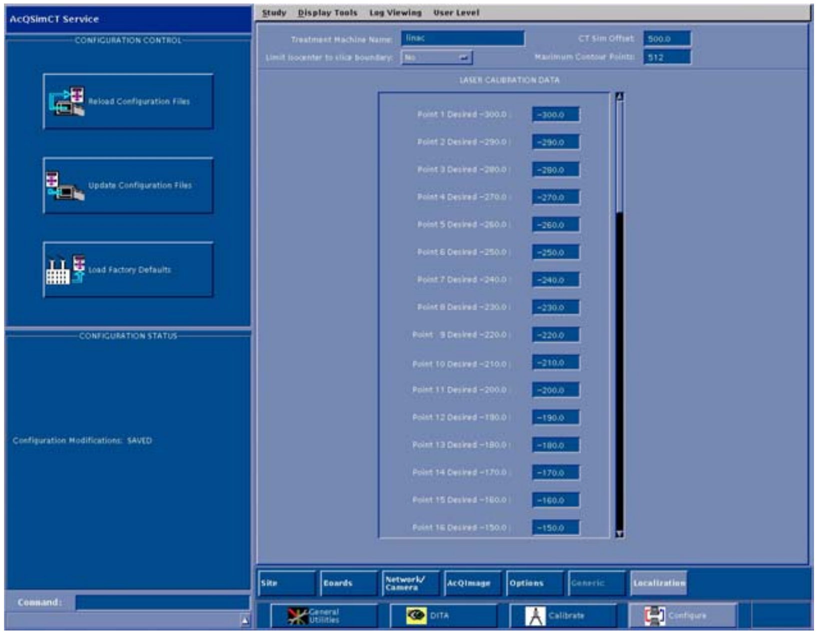


FIGURE 3 LASER CALIBRATION DATA SCREEN

Network/Camera Service Tab Reference Information

In the service application mode, some modifications have been made to the menu screen for the **Network/Camera** tab (see figure 4):

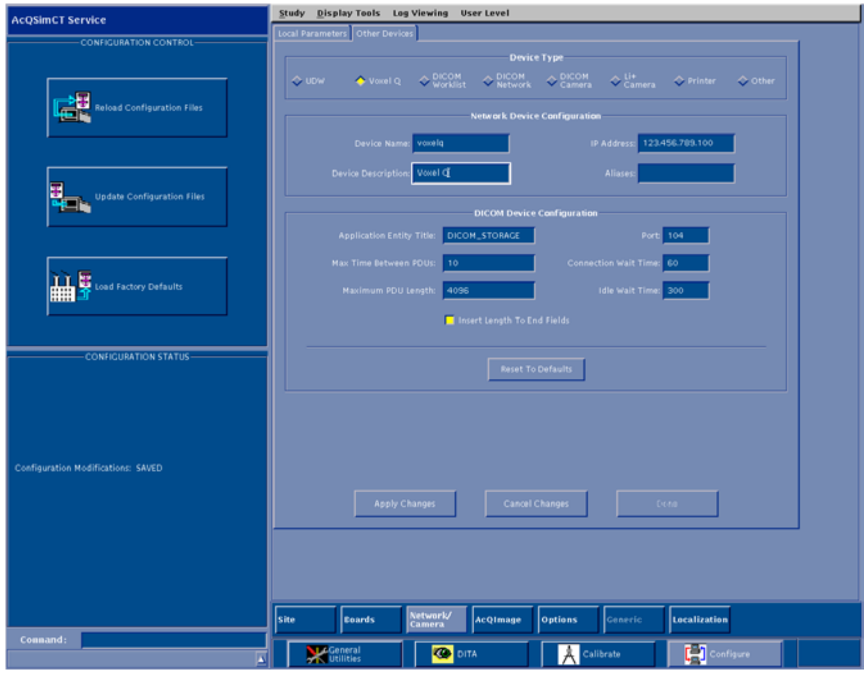


FIGURE 4 NETWORK/CAMERA TAB SCREEN

Under the **Device** type, the available devices have been updated to include:

UDW (Ultra Z Diagnostic Workstation)
Voxel Q
DICOM Worklist
DICOM Network
DICOM Camera
Li+ Camera
Printer
Other

Printer Setup for Worksheet Printing

In order to print a worksheet, the software must know about one or more network printers. If there is more than one printer, one of them must be designated as the default printer.

1. Printer configuration is accomplished by selecting **Service** from the top level AcQSim CT Startup menu.
2. Once in the service application mode, select **Configure**, and then the **Network/Camera** button.
3. Select the **Other Devices** tab to bring up a list of network devices.
4. Select the **Printer** radio button.
5. Select either **Add Device** to add a printer or **Modify Selected Device**. See figure 5.

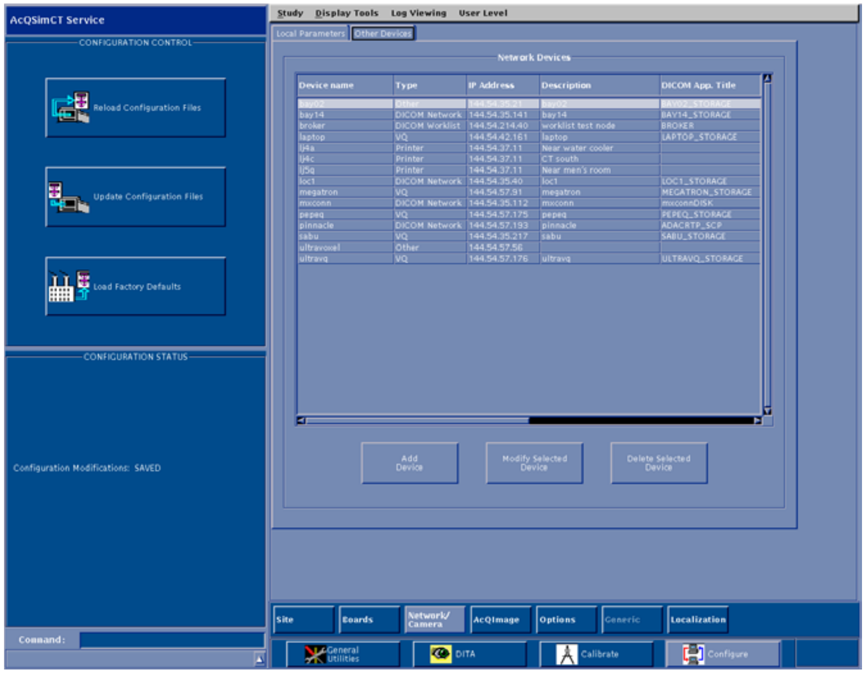


FIGURE 5 NETWORK DEVICES SCREEN

If **Add Device** is selected, the screen in figure 6 is displayed.

At this screen, enter the device name, its description, and the IP address of the printer's server.

There are two radio buttons displayed for selection, one to indicate if it is a PostScript printer and another to indicate whether the printer is the default printer. The default printer is the printer that will be used when the **Worksheet** button is selected on the Isocenter menu screen.

> **NOTE**
>
> There can be only one default printer.

Once printer properties have been set as desired, select **Done** and then select **Update Configuration Files** (located on the left hand side of the service application mode main screen) to save the new or modified printer configuration information. It is recommended to update the site tape after printer has been configured.

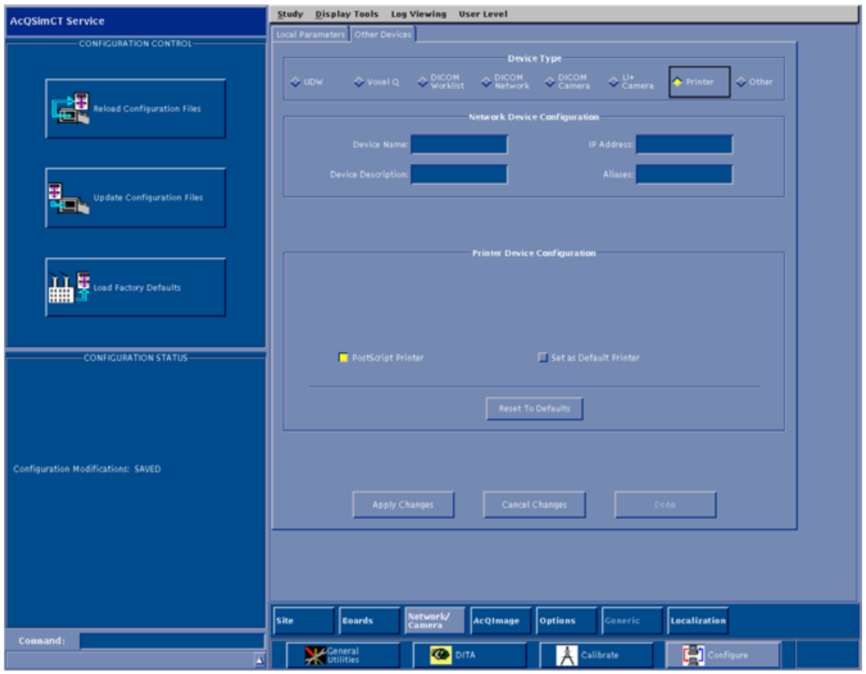


FIGURE 6 PRINTER DEVICE CONFIGURATION SCREEN

Voxel Q 4.9.1 Software Update Instructions

If the customer is using the Voxel Q workstation, it must be running software version 4.9.1 or higher in order to import LOC images from the AcQSim CT running version 3.1.1 software. If the customer has purchased the LOC package, Voxel Q 4.9.1 software is required.

1. Determine if the Voxel Q is running version 4.9.1 software by launching the vx application.
2. If the Voxel Q is running version 4.9.1 software, you do not need to update the Voxel Q software.
3. If 4.9.1 software is not installed on the Voxel Q, install the Voxel Q V4.9.1 Software Upgrade Kit 4535 670 49101 for AcQPlan units, or 4535 670 49091 for all other units. Follow the instructions supplied with the kit. This software should be installed on all Voxel Q systems the customer is using.

Philips Medical Systems

AcQSim CT 4.0 Software Upgrade Kit Instructions

Part Number: 4535 670 49781

Revision: A

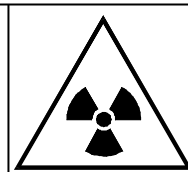
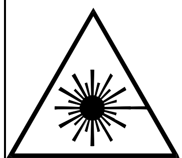
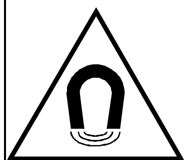
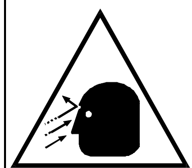
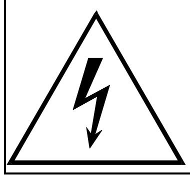
PHILIPS

Service Publication, Philips Medical Systems (Cleveland) Inc. Highland Hts., OH 44143

PHILIPS

Philips Medical Systems (Cleveland), Inc. Highland Hts., OH 44143

SYMBOL DESCRIPTIONS

	Attention symbol.		Radiation warning symbol.
	Laser warning symbol.		Biohazard warning symbol.
	Magnetism warning symbol.		Projectile warning symbol.
	Electrical warning symbol.		

REVISION HISTORY

REVISION	DATE	COMMENTS
–1	07/22/04	Final review copy
A	07/26/04	Release at revision A

NOTICE

THE INFORMATION CONTAINED IN THIS MANUAL CONFORMS WITH THE CONFIGURATION OF THE EQUIPMENT AS OF THE DATE OF MANUFACTURE. REVISIONS TO THE EQUIPMENT SUBSEQUENT TO THE DATE OF MANUFACTURE WILL BE ADDRESSED IN SERVICE UPDATES DISTRIBUTED TO PHILIPS MEDICAL SYSTEMS TECHNICAL SERVICE ORGANIZATION.

TO THE USER OF THIS MANUAL

THE USER OF THIS MANUAL IS DIRECTED TO READ AND CAREFULLY REVIEW THE INSTRUCTIONS, WARNINGS AND CAUTIONS CONTAINED HEREIN PRIOR TO BEGINNING INSTALLATION OR SERVICE ACTIVITIES. WHILE YOU MAY HAVE PREVIOUSLY INSTALLED OR SERVICED EQUIPMENT SIMILAR TO THAT DESCRIBED IN THIS MANUAL, CHANGES IN DESIGN, MANUFACTURE OR PROCEDURE MAY HAVE OCCURRED WHICH SIGNIFICANTLY AFFECT THE PRESENT INSTALLATION OR SERVICE.

THE INSTALLATION AND SERVICE OF EQUIPMENT DESCRIBED HEREIN IS TO BE PREFORMED BY AUTHORIZED, QUALIFIED PHILIPS MEDICAL SYSTEM PERSONNEL. ASSEMBLERS AND OTHER PERSONNEL NOT EMPLOYED BY NOR DIRECTLY AFFILIATED WITH PHILIPS MEDICAL SYSTEMS TECHNICAL SERVICES ARE DIRECTED TO CONTACT THE LOCAL PHILIPS MEDICAL SYSTEMS OFFICE BEFORE ATTEMPTING INSTALLATION OR SERVICE PROCEDURES.

INSTALLATION AND ENVIRONMENT

EXCEPT FOR INSTALLATIONS REQUIRING CERTIFICATION BY THE MANUFACTURER PER FEDERAL STANDARDS, SEE THAT A RADIATION PROTECTION SURVEY IS MADE BY A QUALIFIED EXPERT IN ACCORDANCE WITH NCRP 102, SECTION 7, AS REVISED OR REPLACED IN THE FUTURE. PERFORM A SURVEY AFTER EVERY CHANGE IN EQUIPMENT, WORKLOAD, OR OPERATING CONDITIONS WHICH MIGHT SIGNIFICANTLY INCREASE THE PROBABILITY OF PERSONS RECEIVING MORE THAN THE MAXIMUM PERMISSIBLE DOSE EQUIVALENT.

Diagnostic Imaging Systems – MECHANICAL-ELECTRICAL WARNING

ALL OF THE MOVEABLE ASSEMBLIES AND PARTS OF THIS EQUIPMENT SHOULD BE OPERATED WITH CARE AND ROUTINELY INSPECTED IN ACCORDANCE WITH THE MANUFACTURER'S RECOMMENDATIONS CONTAINED IN THE EQUIPMENT MANUALS.

ONLY PROPERLY TRAINED AND QUALIFIED PERSONNEL SHOULD BE PERMITTED ACCESS TO ANY INTERNAL PARTS. LIVE ELECTRICAL TERMINALS ARE DEADLY; BE SURE LINE DISCONNECTS ARE OPENED AND OTHER APPROPRIATE PRECAUTIONS ARE TAKEN BEFORE OPENING ACCESS DOORS, REMOVING ENCLOSURE PANELS, OR ATTACHING ACCESSORIES.

DO NOT UNDER ANY CIRCUMSTANCES, REMOVE THE FLEXIBLE HIGH TENSION CABLES FROM THE X-RAY TUBE HOUSING OR HIGH TENSION GENERATOR AND/OR THE ACCESS COVERS FROM THE GENERATOR UNTIL THE MAIN AND AUXILIARY POWER SUPPLIES HAVE BEEN DISCONNECTED. FAILURE TO COMPLY WITH THE ABOVE MAY RESULT IN SERIOUS OR FATAL BODILY INJURIES TO THE OPERATOR OR THOSE IN THE AREA.

ELECTRICAL-GROUNDING INSTRUCTIONS

THE EQUIPMENT MUST BE GROUNDED TO AN EARTH GROUND BY A SEPARATE CONDUCTOR. THE NEUTRAL SIDE OF THE LINE IS NOT TO BE CONSIDERED THE EARTH GROUND. ON EQUIPMENT PROVIDED WITH A LINE CORD, THE EQUIPMENT MUST BE CONNECTED TO PROPERLY GROUNDED, THREE-PIN RECEPTACLE. DO NOT USE A THREE-TO-TWO PIN ADAPTER.

Diagnostic Imaging Systems – RADIATION WARNING

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4535 670 49781	AcQSim CT 4.0 Software Upgrade Kit Instructions	1
4535 670 96251	AcQSim CT 4.0 User Documentation Package	1

NOTE

During the upgrade, the hard drive will be reformatted and all patient images will be lost.

1. Be sure all images are archived before starting this upgrade. Make a site tape immediately before installation. If in the Console or Service Application, select Shutdown Console from the General Utilities button. When the Main Startup menu appears, click in the lower left hand corner and type the word **site** (lower case) and press <enter>.

NOTE

Make sure the site save tape is not write–protected.

2. After the Save Site / Restore Site menu appears, push the tape in the tape drive, wait for the green light to stop blinking and select “1” to Save Site. The system will eject the tape when finished.
3. Exit the Save Site Utility. When the Shutdown button becomes available, press it and wait for the system to shut down.
4. Open the CD–ROM drive and insert CD #1. Close the drive.
5. When the ok prompt appears, type **boot cdrom** (lower case) and press <enter>. After a few minutes, the password prompt will appear. Type **picker** (lower case) and press <enter>.

6. The system then checks the disk configuration and prompts you with:

```
Enter the type of system installation:

1. RDS                  (Note: “RDS” refers to an Ultra Z Diagnostic Workstation, UDW)
2. Scanner Console

q. quit
```

7. Select option “2,” *Scanner Console* from the menu and press <enter>. You will see the following menu.

```
Enter a Console Installation option:

1. Install the OS and Application Software
2. Configure a Replaced Disk(s)
3. Upgrade Application Software Only

q. quit
```

> **NOTE**
>
> If you have not archived patient data as instructed in step 1, step 8 will delete all data.

8. Select option “1,” *Install the OS and Application Software* from the menu and press <enter>. Answer “**y**” to the next question. The following will appear:

```
Please confirm the following network information:

Enter IP Address [default xxx.xxx.xx.xx]:
Enter Netmask [default 255.255.255.xxx]:
Enter Router [default xxx.xx.xx.xx]:
Enter Nodename [default consolexxxx]:
```

9. Press <enter> to accept the default values. After the information is entered, the system will display your choices as you entered them:

```
IP Address:
Netmask:
Router:
Nodename:
```

10. After the “Ready to run the installation. Are you satisfied? (y/n) [n]?” prompt appears:
    If you answer NO to “Are you satisfied?”, the installation will cancel and take you back to the first sub–menu where you can quit the installation program, or continue and adjust the network information.

    If installation information is correct, choose “y” to “Are you satisfied?” and press <enter>.

11. The software will now begin installing from the first CD. Some time will pass during installation (approximately 25 minutes). **<u>DO NOT PRESS ANY KEYS ON THE KEYBOARD</u>.** The following menu will then appear:

```
1. Install SCANNER Console Application Software
2. Eject the CD.
```

Select option “2,” *Eject the CD,* and press <enter>. Remove the first CD and put the second CD in, then close the drive door.

12. Select option “1,” and press <enter> to continue with the installation. You will be prompted with the “Are you satisfied?” question again. Select “y”, press <enter>, and the system will continue with the software installation (approximately 20 to 30 minutes). After the installation is complete, the system will reboot and the following menu will appear:

```
1. View the beginning of the installation error log
2. Eject the CD  (it’s OK to eject it now)
3. Continue booting the machine

Choice:
```

> **NOTE**
>
> After the installation has completed, additional messages may be displayed on the screen, pushing the above menu out of view. Simply press <enter> to see it again.

NOTE

In the error log, errors may exist related to pdf help files. Errors may also exist related to matlab/etc/license.log. These errors do not affect operation and should be ignored. Other than these listed errors, there should be no other errors in the error log at this time.

13. View the error log by choosing option **1**. After viewing, the menu will be displayed again. Select option **2** and press <enter> to eject the CD. Remove the CD and close the drive door. Then select option **3** to continue booting to the application software and press <enter>.

NOTE

A serial number mismatch error may occur. Close the error window.

14. Perform a Restore Site. When the Main Startup menu appears, click in the lower left corner and type the word **site** (lower case) and press <enter>.
15. After the Save Site / Restore Site menu appears, push the tape in the tape drive, wait for the green light to stop blinking and select "2" to Restore Site. The system will eject the tape when finished.
16. Exit the Save Site utility.

NOTE

A valid diagnostics key is required to continue. If you can not enter Diagnostics as instructed in step 17., obtain a Diagnostics key per appropriate procedures.

17. After the 4.0 software installation is complete and the system has booted, start the Diagnostics GUI. At the Main Startup menu, click in the lower left hand corner, type **PICKER** (upper case) and press <enter>.
18. At the Diagnostics Software Test Menu, click on the "Utilities" button and select "Flash Burner". The flash burner is a utility for transferring executable code from a directory on the SPARC disk to the Flash of the CP, DSP/AP, BP and GAM boards.

19. At the Flash Burner dialog box (refer to Figure 1), enter the number of the board selected in the “Board name” category (1 for CP, 2 for GAM, 3 for DSP/AP, and 4 for BP) and click “Continue”.

> **NOTE**
>
> The Acqimage Box must be enabled in the Hardware Status window before clicking on “Continue”.

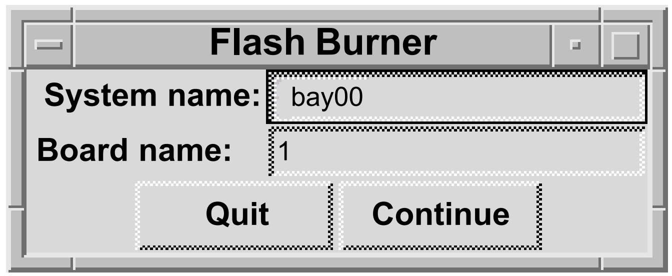


FIGURE 1 FLASH BURNER DIALOG BOX

20. An Xterm window will be displayed. Click the mouse in the Xterm window to activate it. Select option 1.) Transfer OS.

21. At the next menu, select 0. Start transfer. When completed, the board will reboot. The message “Rebooting the board, I/O session is dead” will be displayed. Press <enter> to continue. There may be a delay before the Flash Burner window closes.

22. Restart the Flash burner from the GUI again on the same board. Then select option 2.) Transfer Diags.

23. When completed, the board will reboot. The message “Rebooting the board, I/O session is dead” will be displayed. Press <enter> and repeat steps 18 thru 23 on the remaining three boards. Board numbers are: 2 for GAM, 3 for DSP/AP, 4 for BP. You can view the date of the Flash OS via a Telnet session. At the pSH+> prompt, type **romver** and press <enter>. When flash burner is complete on all boards, select “Exit” to exit the Diagnostics GUI.

Site Protocol Update Program Instructions

If upgrading from software version 1.0, proceed with the following instructions to perform the site protocol update program.

1. Restore the site information from tape (If the site information tape has already been restored, it is not necessary to restore it again).
2. At the four–button GUI, click the mouse in the lower left–hand corner. Type **pptools** and press <enter>.
3. Click in the window that comes up. Type **update_protocols** and press <enter>.
4. Type **exit** and press <enter>.
5. Test a sample of the site defined protocols by scanning.
6. When testing is complete, create a site tape with the new data.
7. This completes the 4.0 scanner software portion of the upgrade. After the initialization is complete, run a pilot, axial and spiral scan to further ensure system functionality. This completes the 4.0 software upgrade. Reboot the system before turning it over to the customer.

Manufacturer Name Update Instructions

To ensure that the manufacturer name reported on the scanner is ”Philips”, proceed with the following instructions.

1. Restore the site information tape (If the site information tape has already been restored, it is not necessary to restore it again).
2. At the four–button GUI, click the mouse in the lower left–hand corner. Type **pptools** and press <enter>.
3. Click in the window that comes up. Type **update_mfg_name** and press <enter>.
4. Type **exit** and press <enter>.
5. Verify that the manufacturer name is now ”Philips” by entering the Service application from the four–button GUI and checking the name reported in the Configure Workstep.
6. When the verification is complete, create a site information tape.

Security Vulnerability Update

Security vulnerabilities have been identified in the AcQSim CT product. The Solaris Operating System commands **(sadmind, in.talkd, rpc.walid, and sendmail)** contain security weaknesses that can be exploited. The instructions below edit a configuration file to disable Solaris files that have been identified as security vulnerabilities. For sendmail, the permissions will be modified to disable outside access of this command.

1. Get to the four button screen from which the options are ”Clinical Scanning”, ”Service”, ”Reboot”, and ”Shutdown”. If currently in ”Clinical Scanning” or ”Service”, exit the console by selecting ”Shutdown Console” from the General Utilities Shutdown option.
2. Move the mouse pointer into the lower left–hand corner of the four button screen, left click once, and type **pptools** <enter>. Note that you will not see anything appearing on the screen as you type. This will launch an xterm window.
3. Click inside the x–term window. From the command prompt, type the following commands followed by <enter>. The commands to type appear in bold below. You will be prompted for a password upon pressing <enter> following the “**su**” command below. Use ”**picker**”. (Unix is case sensitive so make sure that Caps Lock is off).

   su <enter>
   picker <enter>
   cd /etc <enter>
   cat inetd.conf > inetd.bak <enter>

   NOTE: The following command contains (” ”) and (|) that are required to be typed.

   The pipe symbols (|) within the command are the two vertical lines above one another and commonly looks like a full vertical line. This symbol is found on the same keyboard key as the backslash (\ ) key. It is not the letter ” L or l ” or the number ” 1 ”. Make sure there is a space between the second quotation mark and the term inetd.bak.

   egrep –v ”walld|talkd|sadmind” inetd.bak > inetd.conf <enter>

   NOTE: The following command contains three consecutive zeros ”000” not the letters ”OOO”.

   chmod 000 /usr/lib/sendmail <enter>
4. The update is complete. Type ”exit” at the # prompt to exit the shell. Type ”exit” again in the xterm window and select ”Reboot” from the four button screen to restart the scanner

Localization Option Activation Instructions

If customer has purchased localization option, activate as follows.

1. From the four button main menu screen, select **Service**.
2. Select **configure > options**. At the Options Configuration screen, highlight the Localization option.
3. Gain access to Servecom. Follow the menu items, Service, CT, Key Codes, AcQSim CT/Ultra Z Diagnostic Keys. Input the AcQSim CT serial number to obtain the Localization Key.
4. Enter enabling key information exactly as given to you.
5. Select **Update Configuration Files** to save information. Note the Localization button at the bottom of the screen will become active. See Figure 2.

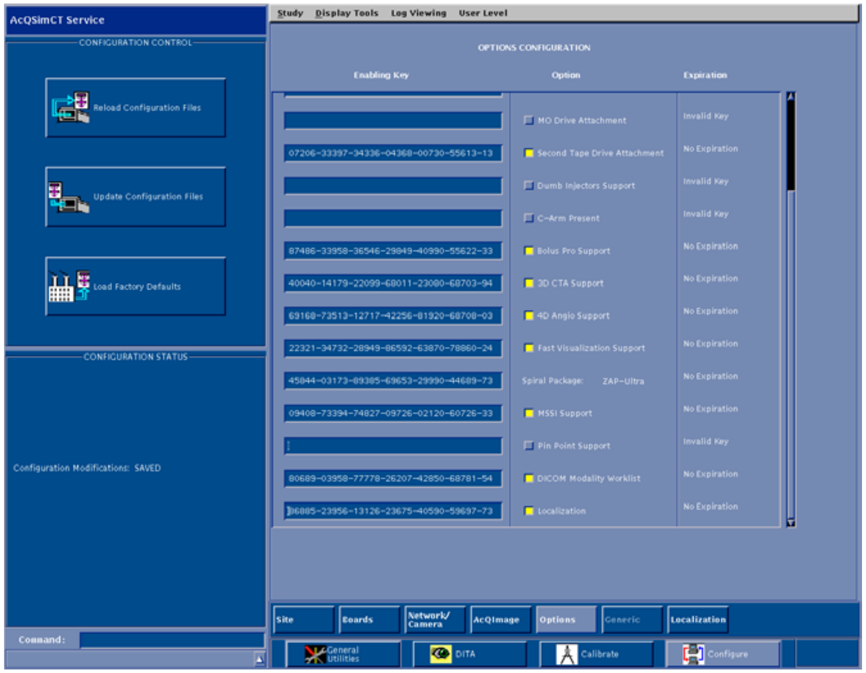


FIGURE 2 LOCALIZATION

Connectivity to other Systems

The customers RTP system may require configuration of a "dummy" treatment machine to import the isocenter (RT Plan) from AcQSim CT 4.0. The treatment machine can be changed within the RTP system after import. Refer to the AcQSim CT 4.0 DICOM Conformance Statement for details on what tags and values are sent.

Localization Option Reference Information

To access the Laser Calibration Data, select the Localization button at the bottom of the service applications mode. At the Laser Calibration Data screen, the recommended values are listed in the table. Measure the laser distance for each point when the laser installation is done to determine the actual value. It is recommended that these values be verified by on–site hospital personnel. The following reference information is provided for the Laser Calibration Data screen (see figure 3):

The Treatment Machine name is the name of the linear accelerator (linac) data is sent to. The CT Sim offset is site specific. The CT Sim offset is the shift between the scan plane and the Laser Marking Plane. The CT Sim offset range of valid values is from 0 to 800.

Select Limit isocenter to slice boundary option to **Yes** if treatment planning system requires beam isocenters to be on a slice plane. When this option is set to yes, beam isocenters that are computed or moved in Localization will jump to the nearest slice plane. If you leave this option set to **No**, beam isocenters can be computed or moved to locations between slice planes.

The maximum contour points selection represents the maximum number of points per contour stored on the system and sent out through DICOM transfer. Some treatment planning systems require fewer than the default 512 contour points. This field can be used to limit the contour points to the amount required by these systems. The maximum contour points range of valid values is 16 to 512.

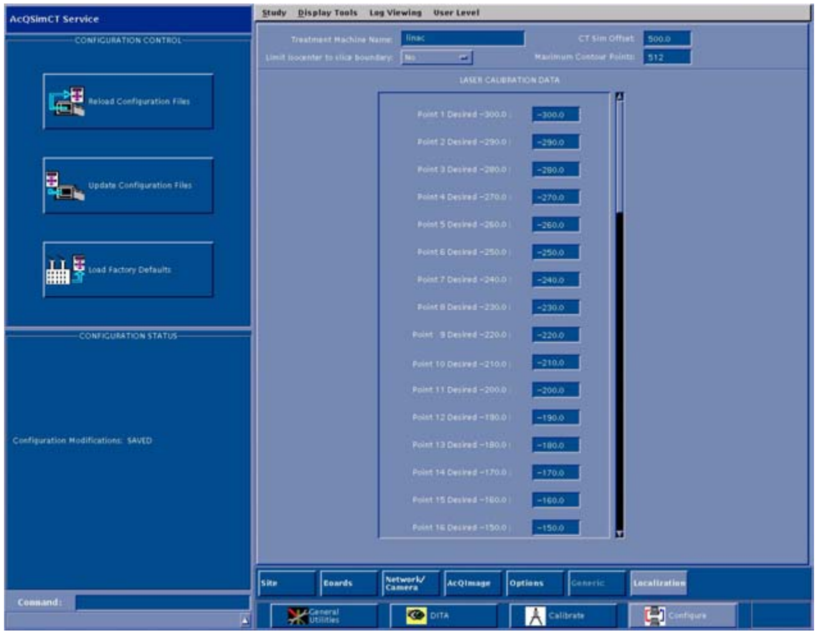


FIGURE 3 LASER CALIBRATION DATA SCREEN

Network/Camera Service Tab Reference Information

In the service application mode, some modifications have been made to the menu screen for the **Network/Camera** tab (see figure 4):

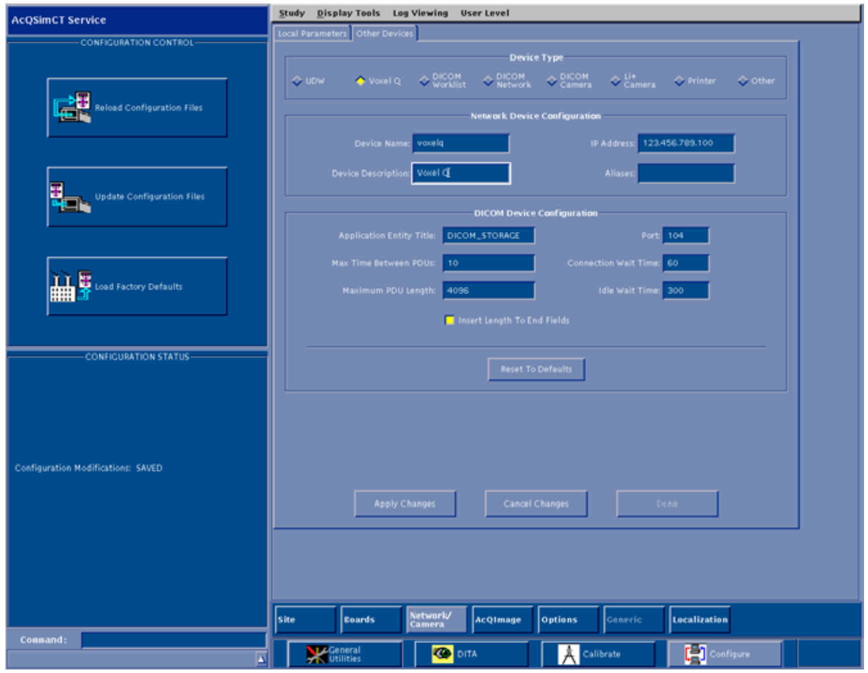


FIGURE 4 NETWORK/CAMERA TAB SCREEN

Under the **Device** type, the available devices have been updated to include:

UDW (Ultra Z Diagnostic Workstation)
Voxel Q
DICOM Worklist
DICOM Network
DICOM Camera
Li+ Camera
Printer
Other

Printer Setup for Worksheet Printing

In order to print a worksheet, the software must know about one or more network printers. If there is more than one printer, one of them must be designated as the default printer.

1. Printer configuration is accomplished by selecting **Service** from the top level AcQSim CT Startup menu.
2. Once in the service application mode, select **Configure**, and then the **Network/Camera** button.
3. Select the **Other Devices** tab to bring up a list of network devices.
4. Select the **Printer** radio button.
5. Select either **Add Device** to add a printer or **Modify Selected Device**. See figure 5.

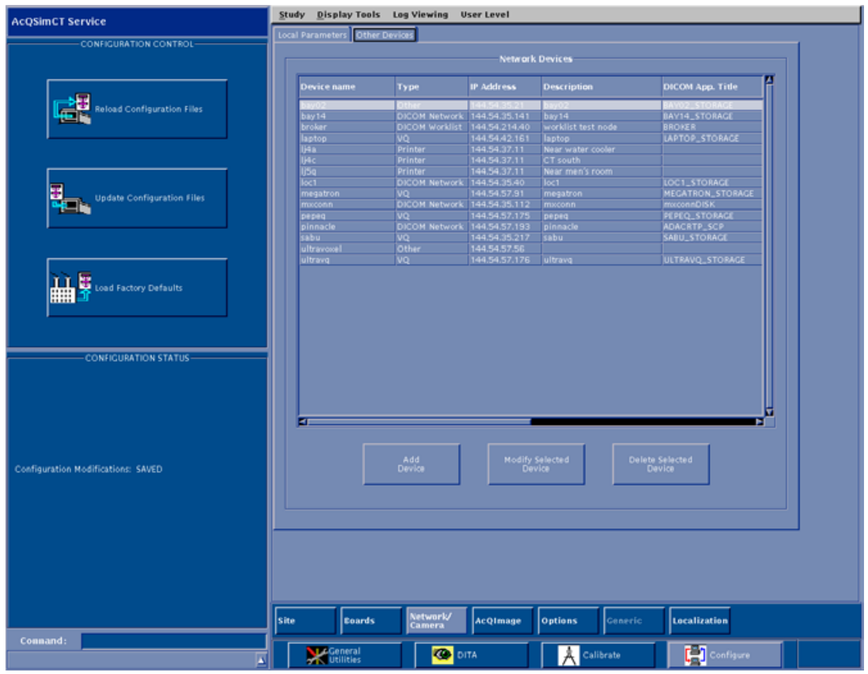


FIGURE 5   NETWORK DEVICES SCREEN

If **Add Device** is selected, the screen in figure 6 is displayed.

At this screen, enter the device name, its description, and the IP address of the printer's server.

There are two radio buttons displayed for selection, one to indicate if it is a PostScript printer and another to indicate whether the printer is the default printer. The default printer is the printer that will be used when the **Worksheet** button is selected on the Isocenter menu screen.

> **NOTE**
>
> There can be only one default printer.

Once printer properties have been set as desired, select **Done** and then select **Update Configuration Files** (located on the left hand side of the service application mode main screen) to save the new or modified printer configuration information. It is recommended to update the site tape after printer has been configured.

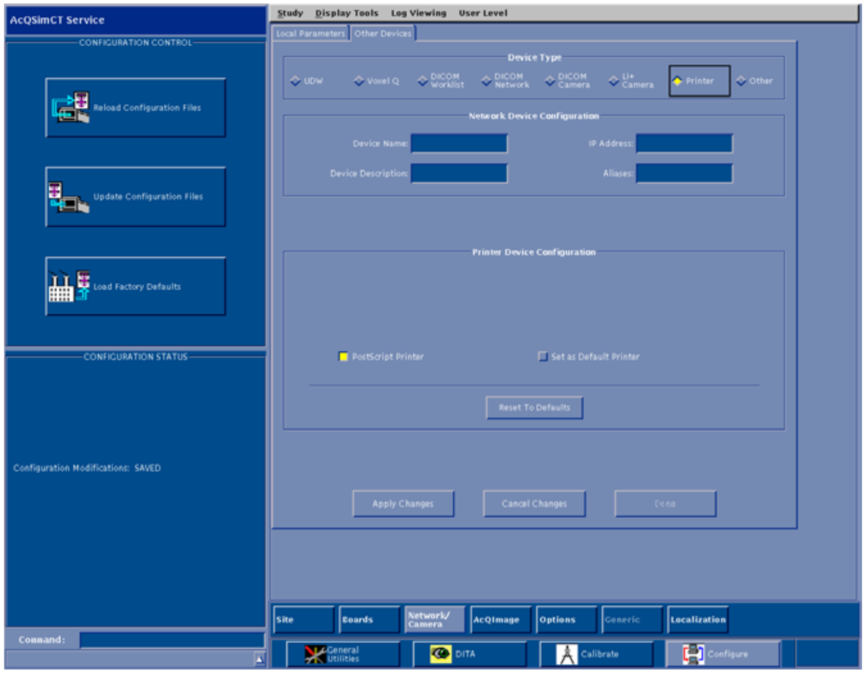


FIGURE 6   PRINTER DEVICE CONFIGURATION SCREEN

Voxel Q 4.9.1 Software Update Instructions

If the customer is using the Voxel Q workstation, it must be running software version 4.9.1 or higher in order to import LOC images from the AcQSim CT running version 4.0 software. If the customer has purchased the LOC package, Voxel Q 4.9.1 software is required.

1. Determine if the Voxel Q is running version 4.9.1 software by launching the vx application.
2. If the Voxel Q is running version 4.9.1 software, you do not need to update the Voxel Q software.
3. If 4.9.1 software is not installed on the Voxel Q, install the Voxel Q V4.9.1 Software Upgrade Kit 4535 670 49101 for AcQPlan units, or 4535 670 49091 for all other units. Follow the instructions supplied with the kit. This software should be installed on all Voxel Q systems the customer is using.

Philips Medical Systems

AcQSim CT

Patient Support Horizontal Resolver Replacement

Kit Instructions

Part Number: 4535 671 04181 Revision: A

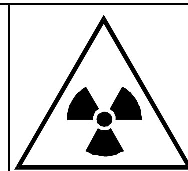
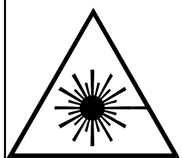
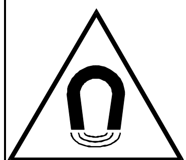
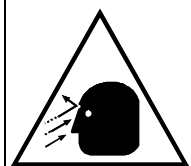
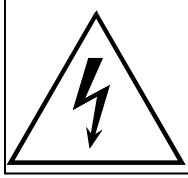
PHILIPS

Service Publication, Philips Medical Systems (Cleveland) Inc. Highland Hts., OH 44143

PHILIPS

Philips Medical Systems (Cleveland), Inc. Highland Hts., OH 44143

SYMBOL DESCRIPTIONS

	Attention symbol.		Radiation warning symbol.
	Laser warning symbol.		Biohazard warning symbol.
	Magnetism warning symbol.		Projectile warning symbol.
	Electrical warning symbol.		

REVISION HISTORY

REVISION	DATE	COMMENTS
–1	08/12/04	Final review copy
A	08/19/04	Release at revision A

NOTICE

THE INFORMATION CONTAINED IN THIS MANUAL CONFORMS WITH THE CONFIGURATION OF THE EQUIPMENT AS OF THE DATE OF MANUFACTURE. REVISIONS TO THE EQUIPMENT SUBSEQUENT TO THE DATE OF MANUFACTURE WILL BE ADDRESSED IN SERVICE UPDATES DISTRIBUTED TO PHILIPS MEDICAL SYSTEMS TECHNICAL SERVICE ORGANIZATION.

TO THE USER OF THIS MANUAL

THE USER OF THIS MANUAL IS DIRECTED TO READ AND CAREFULLY REVIEW THE INSTRUCTIONS, WARNINGS AND CAUTIONS CONTAINED HEREIN PRIOR TO BEGINNING INSTALLATION OR SERVICE ACTIVITIES. WHILE YOU MAY HAVE PREVIOUSLY INSTALLED OR SERVICED EQUIPMENT SIMILAR TO THAT DESCRIBED IN THIS MANUAL, CHANGES IN DESIGN, MANUFACTURE OR PROCEDURE MAY HAVE OCCURRED WHICH SIGNIFICANTLY AFFECT THE PRESENT INSTALLATION OR SERVICE.

THE INSTALLATION AND SERVICE OF EQUIPMENT DESCRIBED HEREIN IS TO BE PREFORMED BY AUTHORIZED, QUALIFIED PHILIPS MEDICAL SYSTEM PERSONNEL. ASSEMBLERS AND OTHER PERSONNEL NOT EMPLOYED BY NOR DIRECTLY AFFILIATED WITH PHILIPS MEDICAL SYSTEMS TECHNICAL SERVICES ARE DIRECTED TO CONTACT THE LOCAL PHILIPS MEDICAL SYSTEMS OFFICE BEFORE ATTEMPTING INSTALLATION OR SERVICE PROCEDURES.

INSTALLATION AND ENVIRONMENT

EXCEPT FOR INSTALLATIONS REQUIRING CERTIFICATION BY THE MANUFACTURER PER FEDERAL STANDARDS, SEE THAT A RADIATION PROTECTION SURVEY IS MADE BY A QUALIFIED EXPERT IN ACCORDANCE WITH NCRP 102, SECTION 7, AS REVISED OR REPLACED IN THE FUTURE. PERFORM A SURVEY AFTER EVERY CHANGE IN EQUIPMENT, WORKLOAD, OR OPERATING CONDITIONS WHICH MIGHT SIGNIFICANTLY INCREASE THE PROBABILITY OF PERSONS RECEIVING MORE THAN THE MAXIMUM PERMISSIBLE DOSE EQUIVALENT.

Diagnostic Imaging Systems – MECHANICAL-ELECTRICAL WARNING

ALL OF THE MOVEABLE ASSEMBLIES AND PARTS OF THIS EQUIPMENT SHOULD BE OPERATED WITH CARE AND ROUTINELY INSPECTED IN ACCORDANCE WITH THE MANUFACTURER'S RECOMMENDATIONS CONTAINED IN THE EQUIPMENT MANUALS.

ONLY PROPERLY TRAINED AND QUALIFIED PERSONNEL SHOULD BE PERMITTED ACCESS TO ANY INTERNAL PARTS. LIVE ELECTRICAL TERMINALS ARE DEADLY; BE SURE LINE DISCONNECTS ARE OPENED AND OTHER APPROPRIATE PRECAUTIONS ARE TAKEN BEFORE OPENING ACCESS DOORS, REMOVING ENCLOSURE PANELS, OR ATTACHING ACCESSORIES.

DO NOT UNDER ANY CIRCUMSTANCES, REMOVE THE FLEXIBLE HIGH TENSION CABLES FROM THE X-RAY TUBE HOUSING OR HIGH TENSION GENERATOR AND/OR THE ACCESS COVERS FROM THE GENERATOR UNTIL THE MAIN AND AUXILIARY POWER SUPPLIES HAVE BEEN DISCONNECTED. FAILURE TO COMPLY WITH THE ABOVE MAY RESULT IN SERIOUS OR FATAL BODILY INJURIES TO THE OPERATOR OR THOSE IN THE AREA.

ELECTRICAL-GROUNDING INSTRUCTIONS

THE EQUIPMENT MUST BE GROUNDED TO AN EARTH GROUND BY A SEPARATE CONDUCTOR. THE NEUTRAL SIDE OF THE LINE IS NOT TO BE CONSIDERED THE EARTH GROUND. ON EQUIPMENT PROVIDED WITH A LINE CORD, THE EQUIPMENT MUST BE CONNECTED TO PROPERLY GROUNDED, THREE-PIN RECEPTACLE. DO NOT USE A THREE-TO-TWO PIN ADAPTER.

Diagnostic Imaging Systems – RADIATION WARNING

X-RAY AND GAMMA-RAYS ARE DANGEROUS TO BOTH OPERATOR AND OTHERS IN THE VICINITY UNLESS ESTABLISHED SAFE EXPOSURE PROCEDURES ARE STRICTLY OBSERVED.

THE USEFUL AND SCATTERED BEAMS CAN PRODUCE SERIOUS OR FATAL BODILY INJURIES TO ANY PERSONS IN THE SURROUNDING AREA IF USED BY AN UNSKILLED OPERATOR. ADEQUATE PRECAUTIONS MUST ALWAYS BE TAKEN TO AVOID EXPOSURE TO THE USEFUL BEAM, AS WELL AS TO LEAKAGE RADIATION FROM WITHIN THE SOURCE HOUSING OR TO SCATTERED RADIATION RESULTING FROM THE PASSAGE OF RADIATION THROUGH MATTER.

THOSE AUTHORIZED TO OPERATE, PARTICIPATE IN OR SUPERVISE THE OPERATION OF THE EQUIPMENT MUST BE THOROUGHLY FAMILIAR AND COMPLY COMPLETELY WITH THE CURRENT ESTABLISHED SAFE EXPOSURE FACTORS AND PROCEDURES DESCRIBED IN PUBLICATIONS, SUCH AS: SUBCHAPTER J OF TITLE 21 OF THE CODE OF FEDERAL REGULATIONS, "DIAGNOSTIC X-RAY SYSTEMS AND THEIR MAJOR COMPONENTS", AND THE NATIONAL COUNCIL ON RADIATION PROTECTION (NCRP) NO. 102, "MEDICAL X-RAY AND GAMMA-RAY PROTECTION FOR ENERGIES UP TO 10 MEV-EQUIPMENT DESIGN AND USE", AS REVISED OR REPLACED IN THE FUTURE.

THOSE RESPONSIBLE FOR PLANNING OF X-RAY AND GAMMA-RAY EQUIPMENT INSTALLATIONS MUST BE THOROUGHLY FAMILIAR AND COMPLY COMPLETELY WITH NCRP NO. 49, "STRUCTURAL SHIELDING DESIGN AND EVALUATION FOR MEDICAL OF X-RAYS AND GAMMA-RAYS OF ENERGIES UP TO 10 MEV", AS REVISED AND REPLACED IN THE FUTURE. FAILURE TO OBSERVE THESE WARNINGS MAY CAUSE SERIOUS OR FATAL BODILY INJURIES TO THE OPERATOR OR THOSE IN THE AREA.

TABLE OF CONTENTS

Return this data sheet along with the removed resolver, (re)using the kit shipping materials.

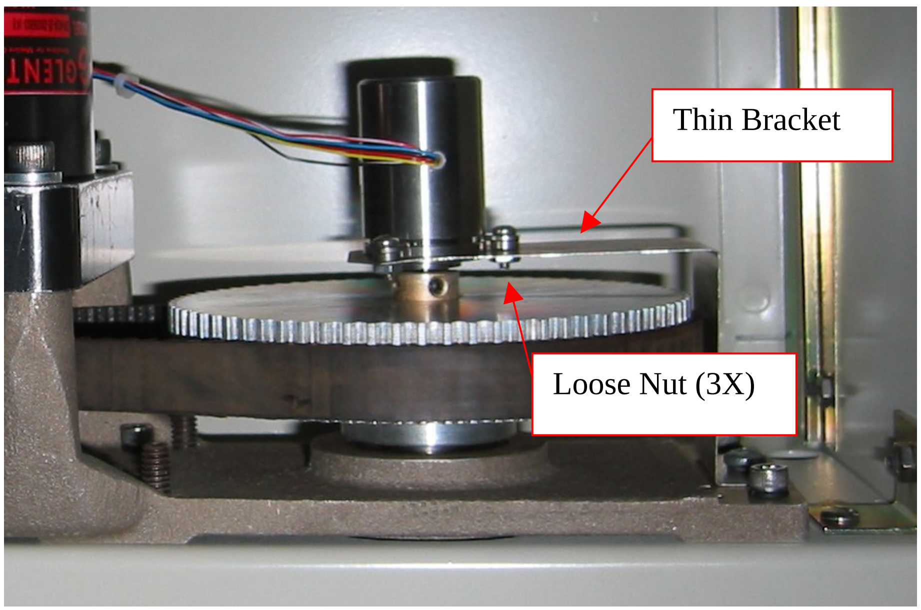
SYSTEM DATA SHEET

Date:____________ **System I.D.:**________________

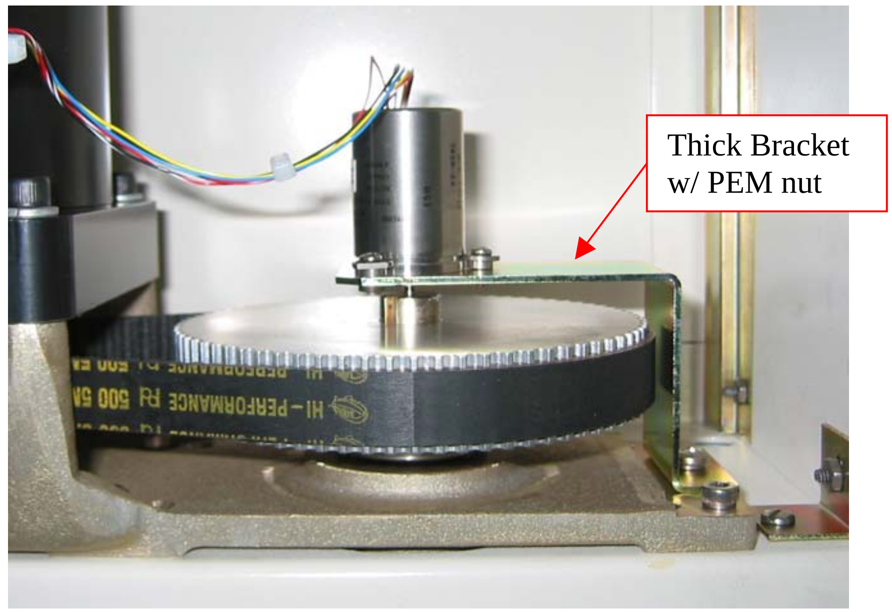
FSE: ____________________________________
(PLEASE PRINT YOUR NAME)

Voice: (____)__________ e-mail:__________________@philips.com
Cell: (____)__________
(OPTIONAL)

1. Identify the bracket which most closely resembles system configuration **(check only one)**:



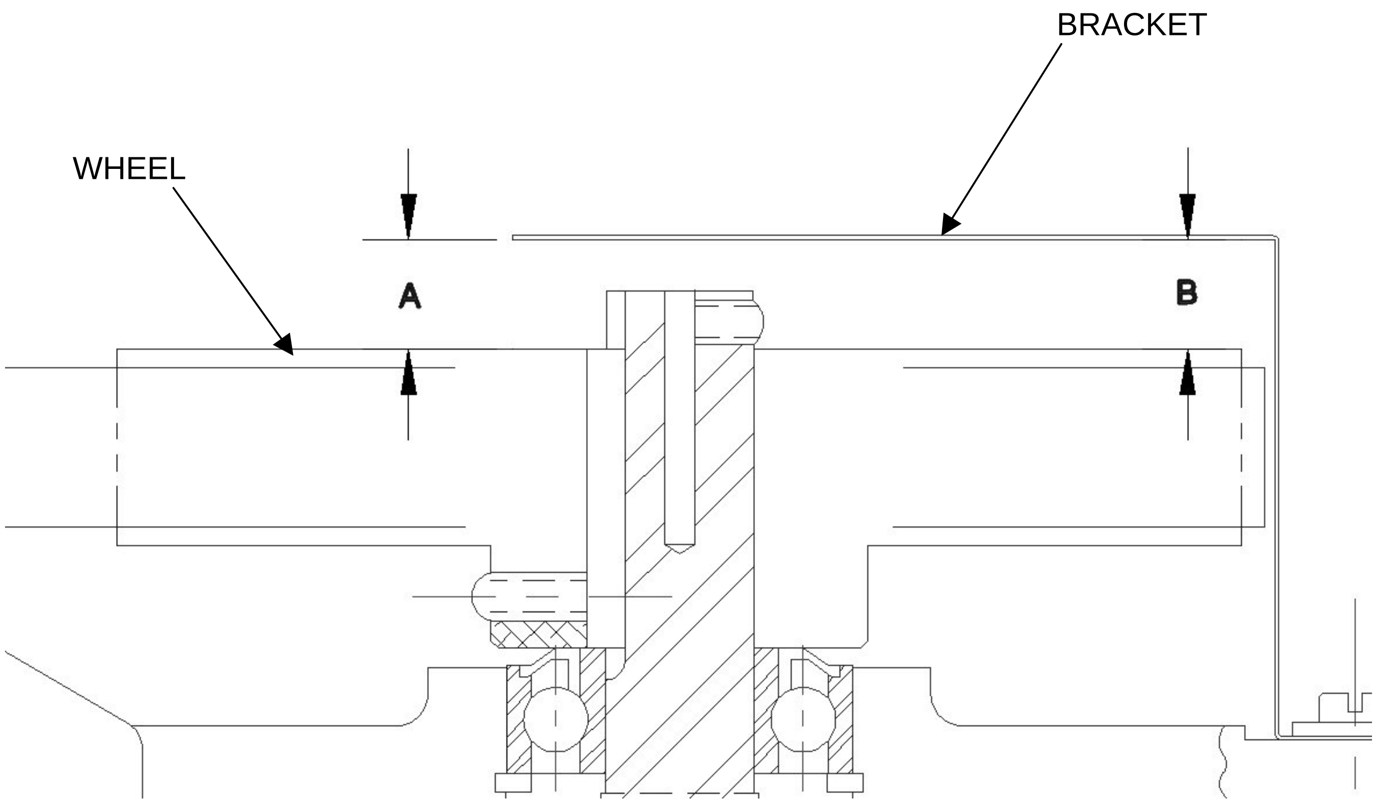
Type 1 - 97946 (Thin)



Type 2 - 83882 (Thick)

Return this data sheet along with the removed resolver, (re)using the kit shipping materials.

2. Measure distance between Resolver Bracket and Drive Wheel at locations A and B:



	BEFORE	AFTER	COMMENTS
DIMENSION A			
DIMENSION B			

Return this data sheet along with the removed resolver, (re)using the kit shipping materials.

3. List any service history related to the horizontal drive module for this system.
   Attach documentation, job numbers, defect / resolution, findings, customer comments, etc. (or list below).
   Please include dates wherever possible.

HORIZONTAL RESOLVER

Introduction: This procedure is used to remove and replace the Patient Support Horizontal Resolver. Return the information sheet and resolver from system in the included packaging.

> **NOTE**
>
> Two service persons are recommended to perform these procedures.

Tools Required: FSE Toolkit, #2 Phillips screwdriver, flat screwdriver, wire cutter, steel scale 1/64” resolution, 5/64” hex key

Material Required: Resolver, ty–wraps, data sheet (pages i through iii of this document)

Time Estimate: Three hours for resolver replacement and couch calibration

RESOLVER REMOVAL PROCEDURE:

INITIAL CONDITIONS:

- Patient Support rear end, right Side Rail and Horizontal Drive Covers removed. Refer to the Patient Support Covers Removal/Replacement procedure.
- Lower telescoping covers removed. Refer to the Telescoping Base Cover Removal Procedures.
- Patient Support raised to its highest position with the Gantry Control Panel Switches.
- Gantry power removed. Refer to the PowerUp/PowerDown procedure.

1. Identify and record bracket type on information sheet.
2. Identify and record the Resolver Wires by color and their position in Connector P408. Refer to Figure 1 and Figure 2.
3. Document locations of ty–wraps and cable routing.
4. Measure distance between wheel and resolver bracket in two places and record on data sheet as dimension A and dimension B.
5. Cut ty–wraps used to bind the Resolver Wires to other cables or structure.

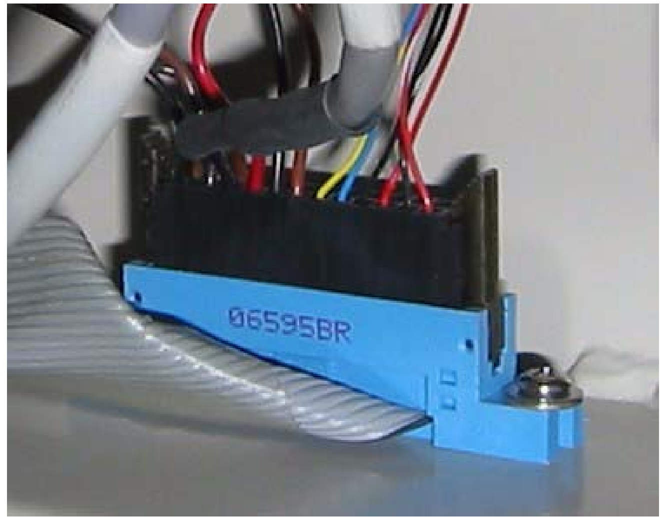


FIGURE 1 RESOLVER WIRING

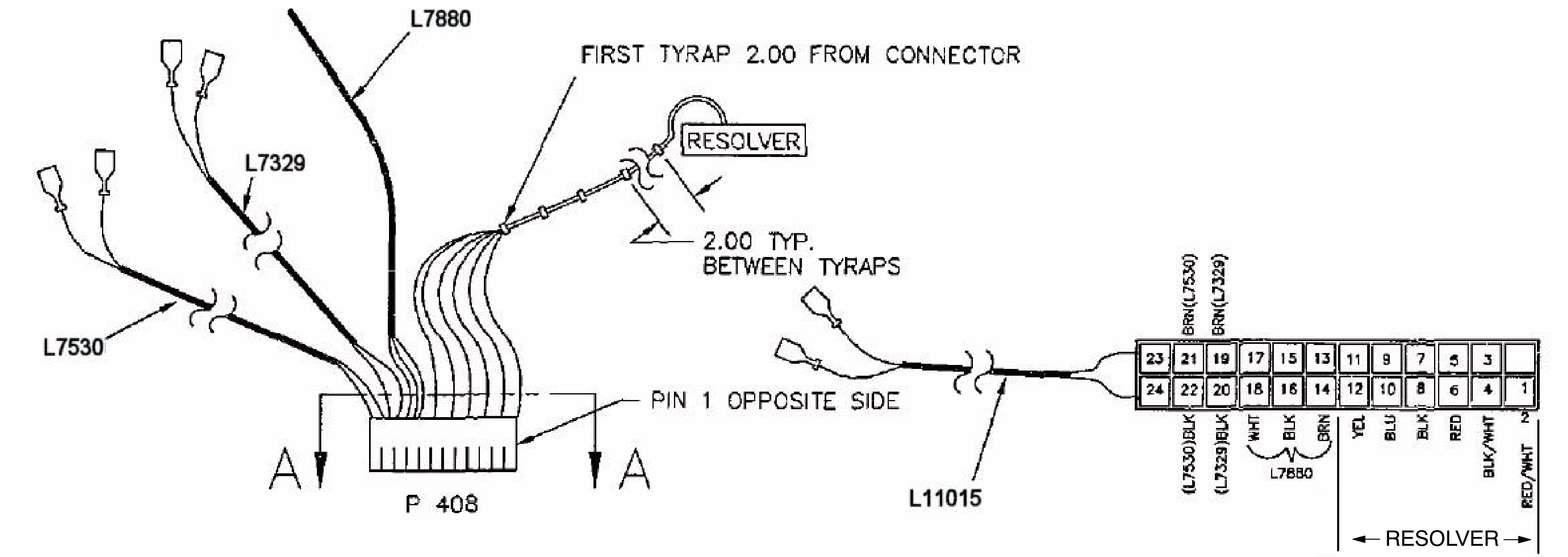


FIGURE 2 RESOLVER WIRING DIAGRAM

6. Remove the Resolver Wires from Connector P408. To release the Socket Contact Retention Blade, press against it with the point of a small object, while pulling on the Wire until the Contact is clear of the connector shell. Refer to Figure 3.

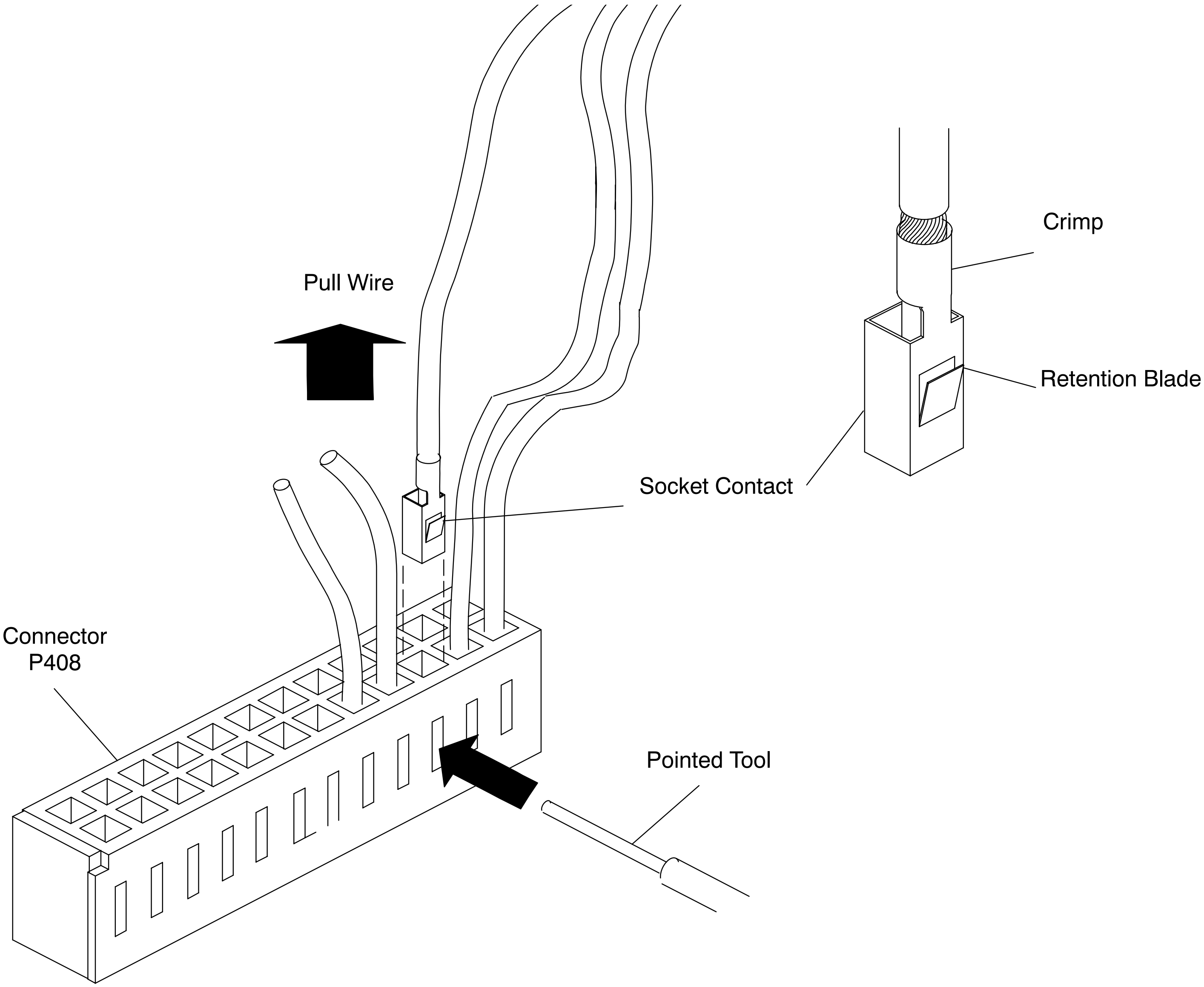


FIGURE 3 REMOVING RESOLVER PINS FROM THE CONNECTOR

7. Loosen the three (3) Cleat Screws holding the Resolver in place. See Figure 4.

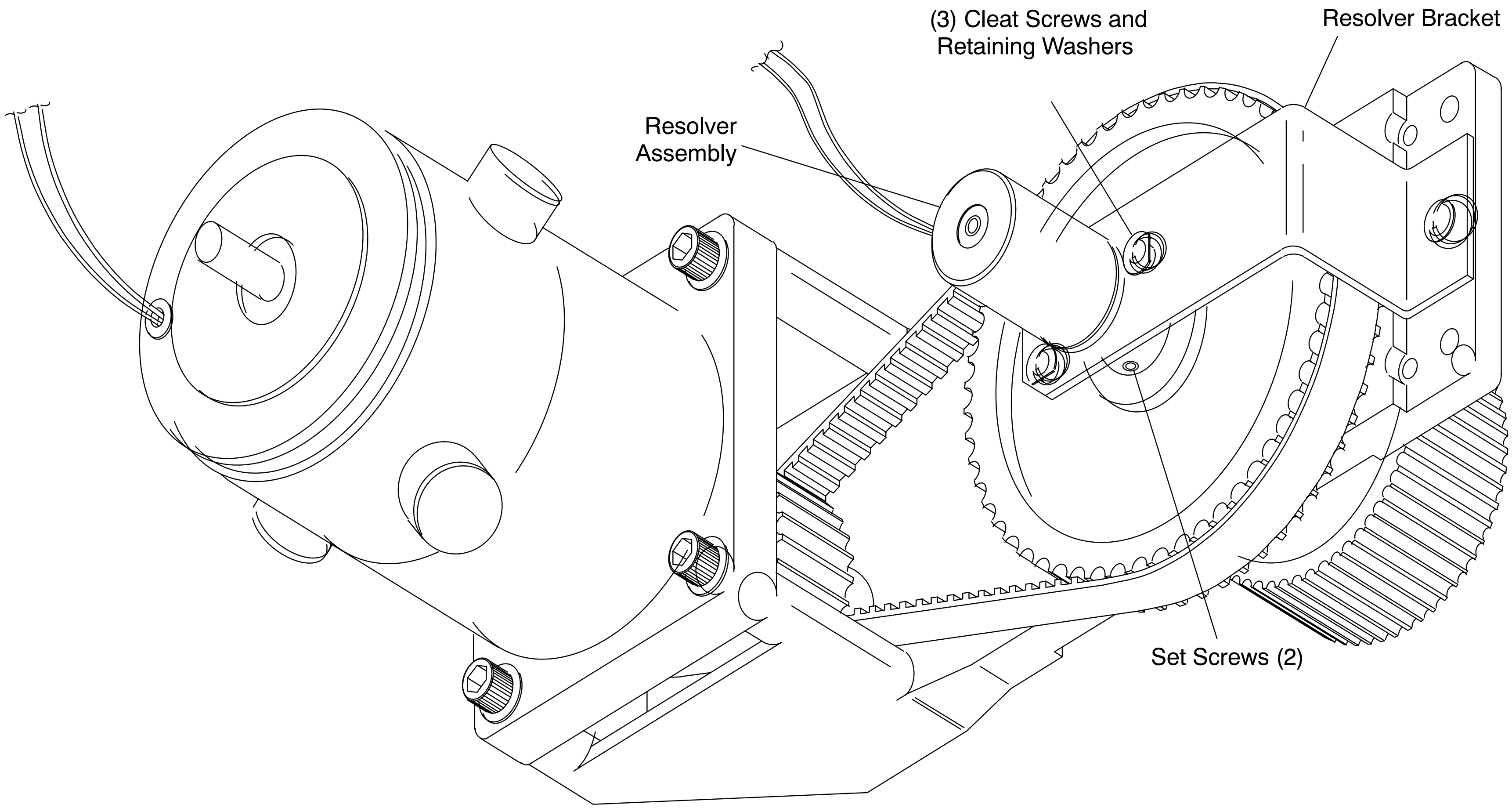


FIGURE 4 HORIZONTAL DRIVE ASSEMBLY

8. Turn the Cleat Retaining Washers so the flat sides face the Resolver, then tighten the Cleat Screws.

> ⚠ **CAUTION**
>
> EXERCISE EXTREME CARE WHILE REMOVING AND INSTALLING RESOLVER TO PREVENT ANY BENDING OR MISALIGNMENT ON RESOLVER SHAFT.

9. Loosen the two (2) Set Screws on the Resolver Pulley with a 5/64” Hex Key. Remove the Resolver from the Pulley and away from the Bracket.
10. Measure distance between wheel and resolver bracket in two places and record on data sheet as dimension A and dimension B.

RESOLVER INSTALLATION:

1. Place the Resolver into the Resolver Bracket opening.
2. Carefully insert the shaft of the replacement Resolver into the Resolver Pulley. Verify that the resolver flange is completely contacting the mounting bracket. Equally tighten the Pulley Set Screws just enough to keep the Pulley on the shaft.

> ⚠ **CAUTION**
>
> DO NOT BUMP OR MOVE THE RESOLVER DURING THE FOLLOWING STEPS.

3. Loosen the three Cleat Screws and turn the Cleat Retaining Washers so the flat edge faces away from the resolver. Ensure that the Cleat Retaining Washers fit into the groove on the side of the Resolver. Refer to Figure 5.
4. Turn the Resolver so the leads are at approximately the 10 o’clock position (as shown in Figure 4).
5. Slightly tighten each cleat screw to assure resolver is uniformly seated to resolver bracket.

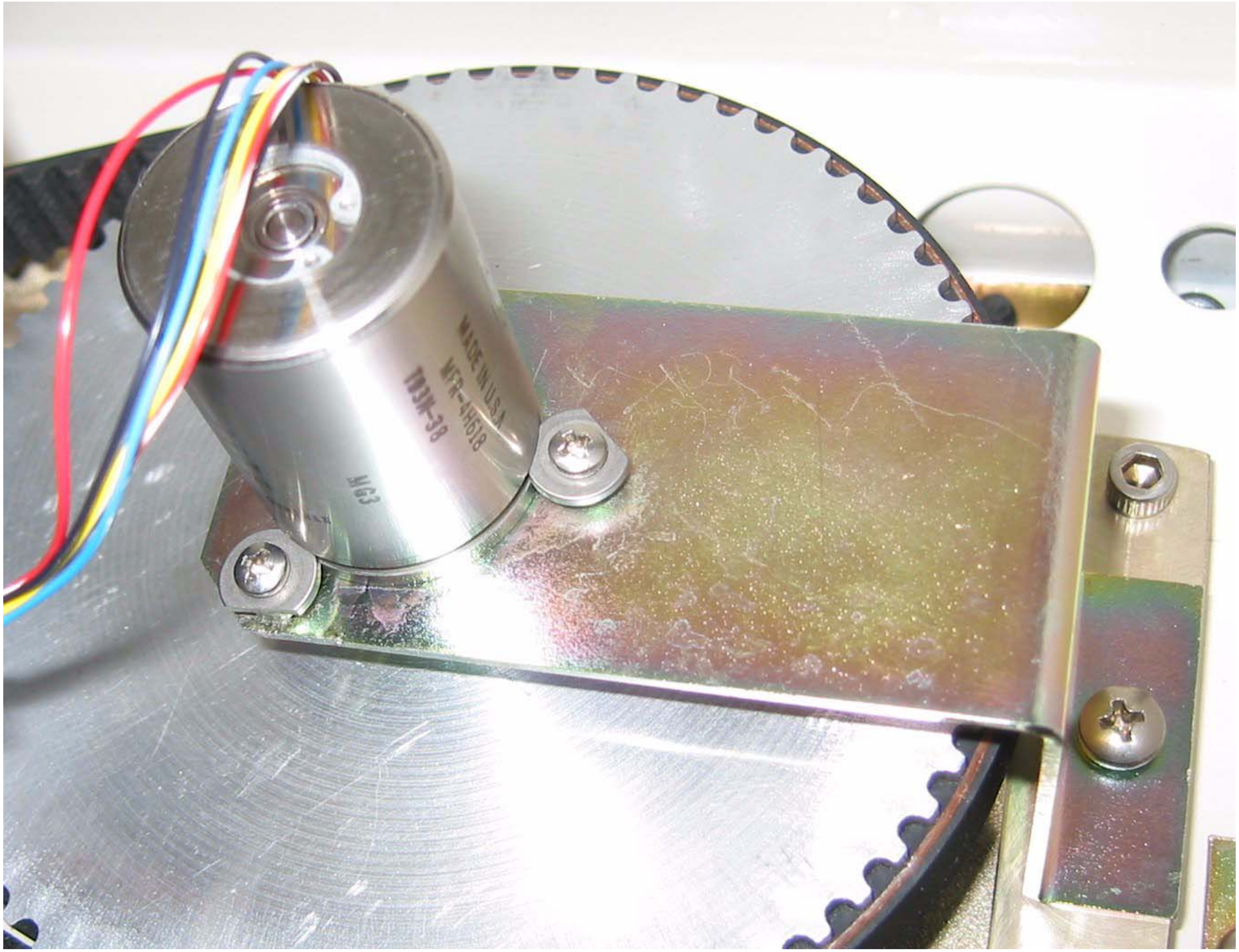


FIGURE 5 CLEAT RETAINING WASHERS INSTALLATION

6. Tighten the pulley set screws to 10 in–lbs.

> **NOTE**
>
> During the Horizontal Couch Calibration, the resolver may need to be rotated (reference figure 7). Tighten resolver cleat screws sufficiently to prevent resolver housing rotation until calibration is completed.

7. Route resolver wires per step 3 (removal) above. Secure wires away from moving parts with ty–wraps.
8. Install the Resolver leads into Connector P408. Refer to the Wire color and locations shown in Figure 1 and Figure 2.
9. Calibrate Carbon Top Horizontal Position using CC (Couch Calibration) in the next section.
10. When couch calibration is completed, verify Gantry power is removed. Refer to the PowerUp/PowerDown procedure.
11. Apply loctite to (3x) 8–32 cleat screws at bracket interface and torque to 10 in–lbs.
12. Replace any remaining Patient Support Covers. Refer to the Patient Support Covers Removal/Replacement procedure.

CC (HORIZONTAL COUCH CALIBRATION)

INITIAL CONDITIONS:

- Lower the telescoping covers on the couch and ensure the panel at the underneath rear of the couch is removed. Refer to the telescoping base cover removal procedure.

1. Before this procedure can be run, you must first power down the system. Refer to the PowerUp/Down procedure.
2. Remove the bottom rear panel from the couch. Disconnect the red and black connector that is attached to the couch servo motor leads.
3. Remove the Gantry cardfile access cover.
4. Apply power to the system. Refer to the PowerUp/Down procedure. At the 3 Button Screen, start the Diagnostic GUI.
5. Select **Gputty+** from the GANTRY pulldown menu. Press the letter "**r**" to execute (Shift Control E to exit at any time).

> **NOTE**
>
> <sb cr> refers to pressing the Spacebar and Return key. Beeps indicate adjustment within range.

> **NOTE**
>
> A step that does not tell you to "<sb cr>" will automatically time out. Adjustments should be made quickly since you cannot go back a step.

6. Type **cc** at the prompt to display:

```
SPACE BAR AND RETURN STEPS THROUGH TESTS, YOU WILL SEE A <sb, cr> PROMPT
A —> SIGN PROMPTS YOU TO SETUP OR ADJUSTMENT
HINT:  0x BEFORE A NUMBER IS A HEXIDECIMAL NUMBER !!!!!
DISCONNECT OR DISABLE COUCH HORZ. MOTOR
!!!!!!!!!!!!!!!!!
DO YOU WISH TO CONTINUE ? 1=YES, 0= EXIT>
```

7. Press **1** to continue. Push the couch toward the scan frame and mark two positions from the back seam of the patient support with a pencil: 180mm and 1180mm. Refer to Figure 6.

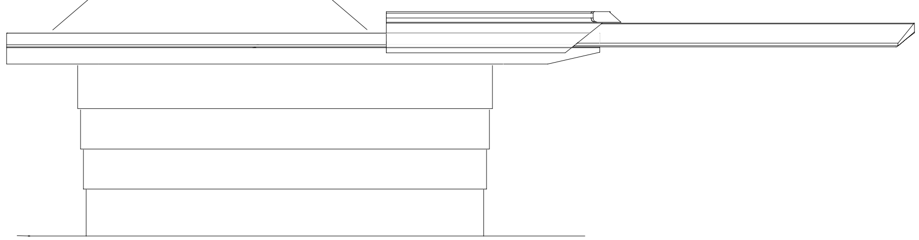


FIGURE 6 180mm and 1180mm COUCH MARKINGS

8. The pots are 20 turn pots. Turn them at least 20 times ccw or until you hear or feel them click against their stops.
9. Refer to Figure 8 for location of adjustment pots on the CTCC. **Follow the on–line instructions**:

```
TURN BOTH HORIZ ZERO (R12) AND
SCALE POTS (R11) FULLY CCW, THEN <sb cr>

SET THE MULT SCALE ADJ. SWITCHES (SW1, 2 AND 3) TO 0x800, THEN <sb cr>
```

10.

```
*** PUSH VERTICAL ENABLE ON GCP WHEN ADJUSTMENT IS COMPLETE
MOVE TOP SO DS5 READS BETWEEN 0xE80 AND 0x1180
AND ADJUST RESOLVER FEEDBACK (R7) FOR 2.0 V RMS
BEEPER WILL SOUND AT CORRECT ADJUSTMENT
            DS5         Vrms
            1038        1.97
```

11. Refer to Figure 7:

```
PUT END OF COUCH TOP 180MM (7 1/16 IN) FROM SEAM
THEN <sb cr>
PUSH VERTICAL ENABLE ON GCP TO LEAVE LOOP

SET HORZ POT TO -1.0 --> -1.5 VOLTS  THEN TIGHTEN
            -1.475


PUSH VERTICAL ENABLE ON GCP TO LEAVE LOOP
TURN RESOLVER TO READ ZERO +/- 0x10 AND RETURN
            DS5         10


PUSH VERTICAL ENABLE ON GCP WHEN ADJUSTMENT IS COMPLETE
ADJUST COARSE ZERO POT. DS4 SHOULD BE 0x10
            DS4         FF
```

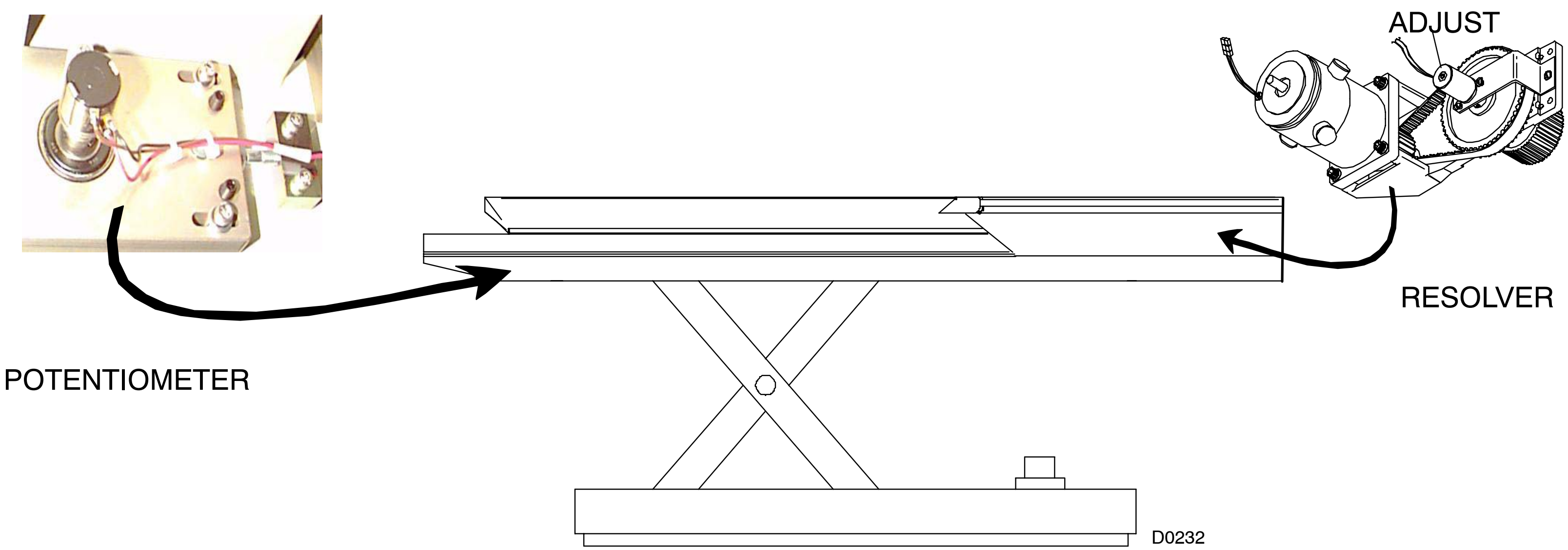


FIGURE 7 COUCH RESOLVER AND POTENTIOMETER LOCATION

This step will automatically time out (no <sb cr>).

```
MOVE COUCH IN UNTIL READOUT IS ABOUT –9.49
BEEP  –8.67  BEEPER WILL SOUND WHEN IN RANGE
```

This step will automatically time out (no <sb cr>).

```
NOW MOVE UNTIL READOUT IS ABOUT 0x0  +/– 0x20
            DS5         0
```

This step will automatically time out (no <sb cr>).

```
ADJUST HORIZ SCALE POT (R11) TO READ 0x90
            DS4         90
```

This step will automatically time out (no <sb cr>).

```
PUT END OF COUCH TOP 180MM (7 1/16 IN) FROM SEAM
THEN <sb cr>

ADJUST HORZ ZERO POT (R12)  DS4 SHOULD BE 0x10
            DS4         10
```

This step will automatically time out (no <sb cr>).

```
MOVE COUCH IN UNTIL READOUT IS ABOUT –9.47
DS5         –9.47

BEEP  –9.47  BEEPER WILL SOUND WHEN IN RANGE
```

This step will automatically time out (no <sb cr>).

```
NOW MOVE UNTIL READOUT IS ABOUT 0x0  +/– 0x20
            DS5         0
```

This step will automatically time out (no <sb cr>).

```
ADJUST HORIZ SCALE POT (R11) TO READ 0x90.
            DS4         90
```

This step will automatically time out (no <sb cr>).

```
MOVE TO EXACTLY 1180MM FROM SEAM (47 7/16 IN) THEN <sb cr>

SET MULT SCALE ADJ SWITCHES (SW1, 2 AND 3) TO xxxx THEN <sb cr>
```

Refer to Figure 8 for switch locations.

```
HORIZONTAL CAL COMPLETE
REMEMBER TO RECONNECT THE MOTOR
```

Refer to Figure 8.

- The "old" MSB hex switch has been hardwired to a value of "C." SW1 is the second MSB, SW2 and SW3 follow.

12. After the calibration procedure is complete, exit from the CC program. At the AcQSim CT> prompt, press Shift Control E to exit Gputty+.
13. Exit the Diagnostic GUI and remove power from the system. Refer to the PowerUp/Down procedure.
14. Attach the red and black connector that is attached to the couch servo motor leads.
15. Replace the telescoping couch base covers and rear panel. Refer to the telescoping base cover removal procedure.
16. Resume at step 10. of resolver installation.

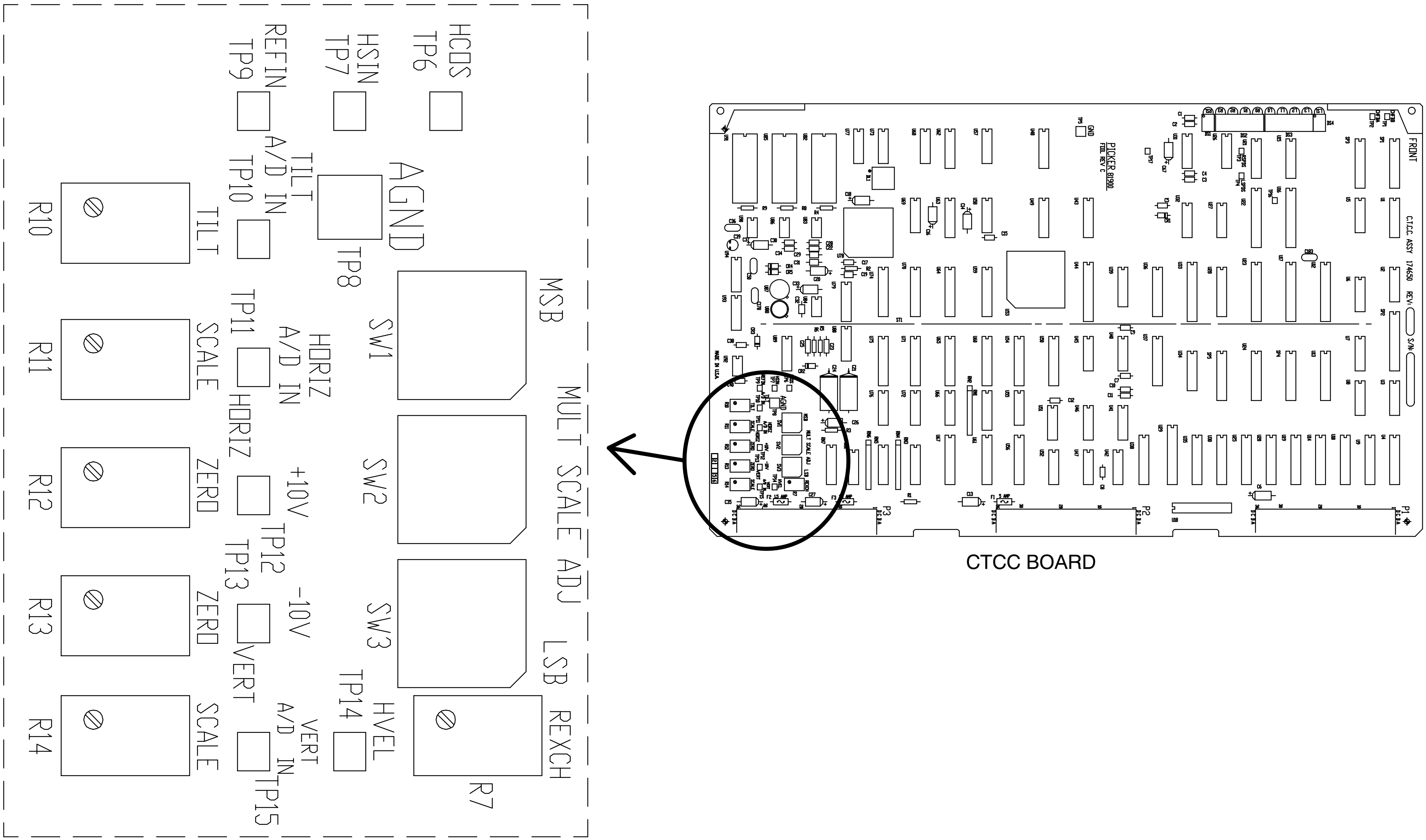


FIGURE 8 CTCC BOARD COMPONENT LOCATIONS DETAIL

PATIENT SUPPORT COVERS

Introduction: This procedure is used to remove and replace all Patient Support covers.

> **NOTE**
>
> Two service persons are recommended to perform these procedures.

Tools Required: 1. FSE Tool Kit

TELESCOPING BASE COVER REMOVAL (FULL DOWN POSITION; PREFERRED METHOD):

> ⚠ **WARNING**
>
> **THE TELESCOPING BASE COVERS ARE HEAVY AND ARE HELD ONTO THE SUBFRAME BY FOUR SCREWS. REMOVE THE MOUNTING SCREWS ONLY WHEN THE PATIENT SUPPORT IS AT ITS LOWEST POSITION OR ONLY AFTER THE TELESCOPING BASE COVERS ARE PROPERLY AND SECURELY BLOCKED. FAILURE TO COMPLY CAN RESULT IN DAMAGE TO EQUIPMENT AND/OR SERIOUS INJURY TO PERSONNEL.**

> **NOTE**
>
> For most Removal/Replacement and Calibration procedures, only the top section of the Telescoping Base Covers need be removed.

1. Apply Gantry power. Refer to the PowerUp/PowerDown procedure.
2. Lower the Patient Support to its lowest position with the Gantry Control Panel Switches.

> ⚠ **WARNING**
>
> **USE EXTREME CAUTION WHEN OPERATING OR WORKING ON EQUIPMENT WITH COVERS REMOVED. DANGEROUS VOLTAGES ARE EXPOSED WHEN THE GANTRY COVERS ARE REMOVED. FAILURE TO COMPLY CAN CAUSE SERIOUS INJURY OR DEATH TO PERSONNEL.**

3. Remove the four screws and Washers that attach the Telescoping Base Cover Panels to the Subframe. Refer to Figure 9.

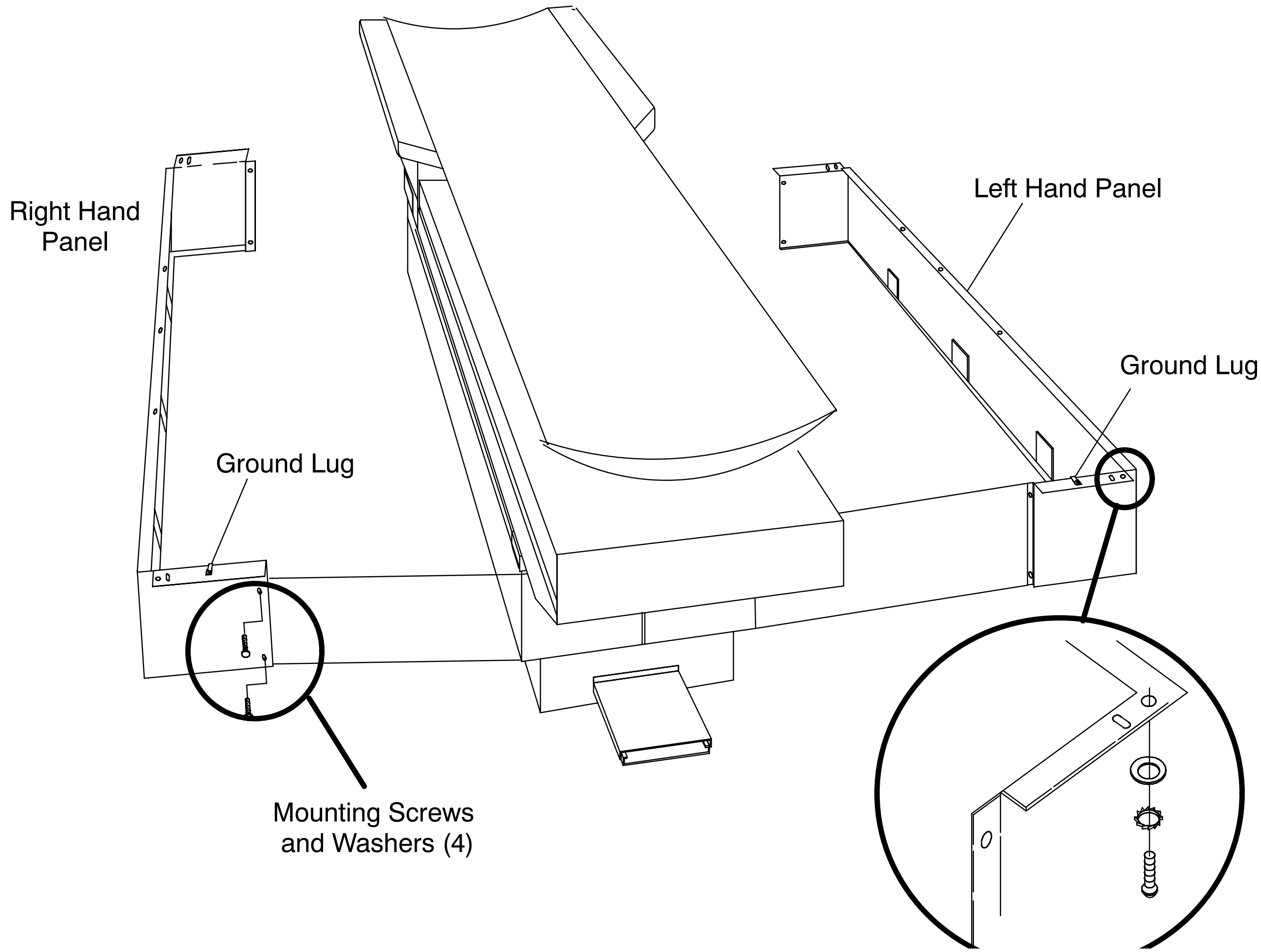


FIGURE 9 PATIENT SUPPORT TELESCOPING BASE COVERS

4. Using the Gantry Control Panel Switches, raise the Patient Support approximately 10 inches to allow access to internal cables.
5. Remove Gantry power. Refer to the PowerUp/PowerDown procedure.
6. Remove the two screws from both Outer Cover ends (front & rear, 4 total). Refer Figure 9.

7. Disconnect the two Ground Wires connected to the inside Panel surfaces. Refer to Figure 10. Tag the Panels left and right. Carefully separate the Panels and slide them out to the sides. Store the Covers in a secure area until reinstallation is required.

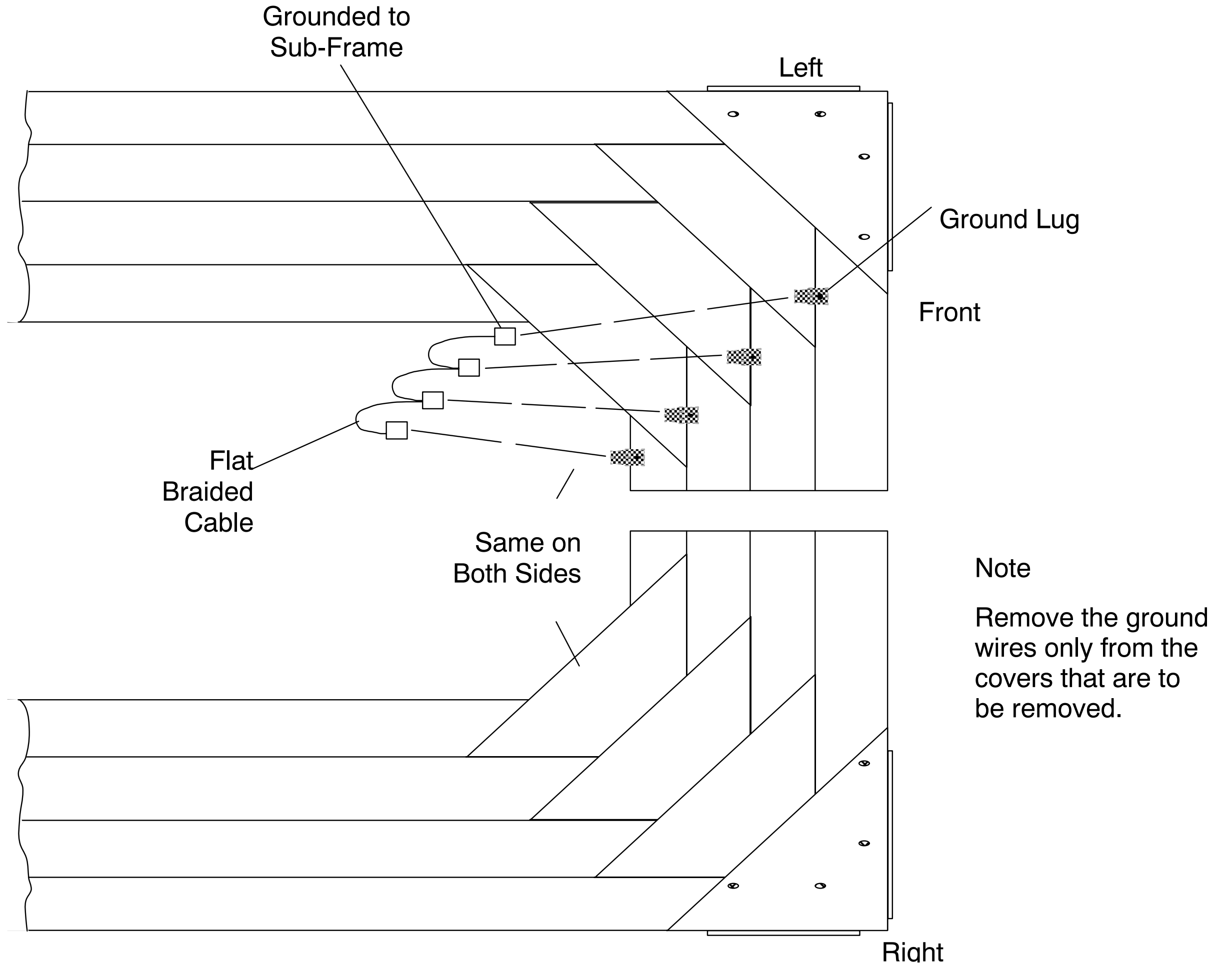


FIGURE 10 TELESCOPING BASE COVERS – GROUND CONNECTIONS

8. Repeat steps 6. and 7. to remove the remaining cover sections.

TELESCOPING BASE COVER REPLACEMENT:

9. Install the Patient Support Telescoping Base Covers by reversing the steps of Telescoping Base Cover Removal.

FRONT COVER REMOVAL:

10. Remove the two Phillips Head Screws from the underside of the front Cover. See Figure 11.
11. Remove the four Phillips Head Screws on the front side of the front Cover while holding the Cover.

FRONT COVER REPLACEMENT:

12. Install the Patient Support front Cover by reversing steps 10. and 11.

REAR COVER REMOVAL:

13. Remove the two Phillips Head Screws from underneath the Sub–frame that secure the rear Cover. See Figure 11.
14. Remove the rear Cover by pulling down and then out.

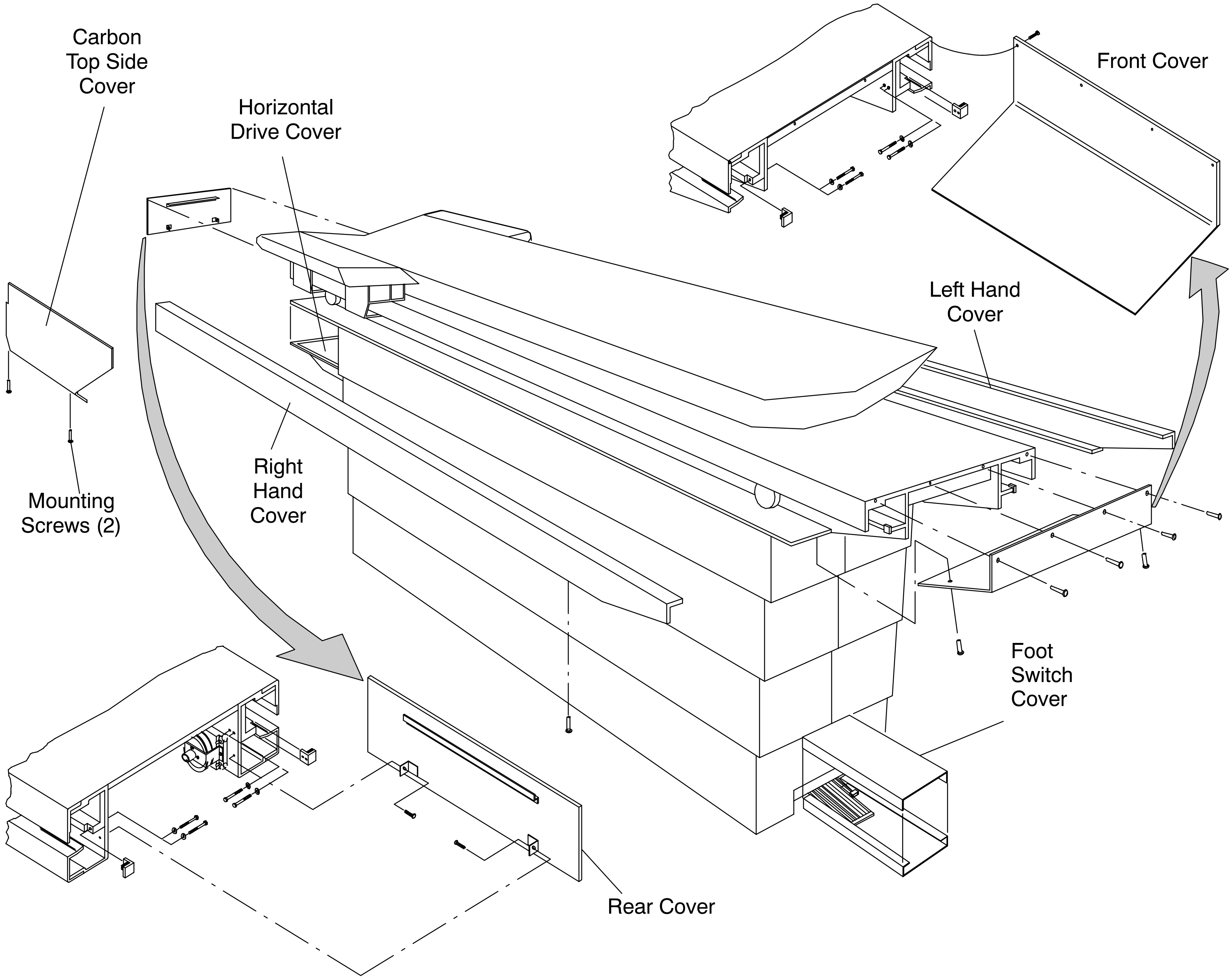


FIGURE 11 PATIENT SUPPORT COVERS

REAR COVER REPLACEMENT:

15. Install the Patient Support rear Cover by reversing steps 13. and 14.

HORIZONTAL DRIVE COVER REMOVAL:

16. Remove the four Pan Head Screws, Lock Washers, and Flat Washers from underneath the Table at the rear end, to gain access to the Horizontal Drive. Refer to Figure 11.

HORIZONTAL DRIVE REPLACEMENT:

17. Install the Horizontal Drive Cover by reversing step 16.

CARBON TOP SIDE COVERS (LEFT AND RIGHT) REMOVAL:

18. Position the Carbon Top by hand fully into the Gantry. This is required for some older Side Cover designs.
19. Remove the two Mounting Screws that attach the Carbon Top Side Cover to the J–Bracket. Refer to Figure 11.
20. Remove the Carbon Top Side Cover.

CARBON TOP SIDE COVERS REPLACEMENT:

21. Replace the Carbon Top Side Cover by reversing steps 18. through 20.

RAIL COVERS (LEFT AND RIGHT) REMOVAL:

22. Remove the rear–most front cover Screw that secures the front Cover and Side Rail to the Sub–frame. Refer to Figure 11.
23. Move the Carbon top by hand until it is against its front Mechanical Stop.
24. Remove the Patient Support rear Cover. Refer to Rear Cover Removal in this procedure.
25. Remove the Flat Head Screws from the underside of the Rail Cover. If both Rail Covers are removed at the same time, tag them left and right. See Figure 11.
26. Slide the Rail Cover toward the rear until it is clear of the overhanging Carbon top Side Cover.
27. Remove the Rail Cover.

CARBON TOP SIDE REPLACEMENT:

28. Install the Patient Support Rail Covers by reversing steps 22. through 27.

FOOT SWITCH COVER REMOVAL:

29. Apply Gantry power. Refer to the PowerUp/PowerDown procedure.
30. Tilt the Gantry to +5° (degrees) with the Gantry Control Panel Switches.
31. Remove Gantry power. Refer to the PowerUp/PowerDown procedure.
32. Remove the two screws and Washers that attach the Foot Switch Cover. Firmly grasp the Foot Switch with both hands and pull the Cover straight up. See Figure 12.

> **⚠ WARNING**
>
> **USE EXTREME CAUTION WHEN OPERATING OR WORKING ON EQUIPMENT WITH COVERS REMOVED. DANGEROUS VOLTAGES ARE EXPOSED WHEN THE GANTRY COVERS ARE REMOVED. FAILURE TO COMPLY CAUSE SERIOUS INJURY OR DEATH TO PERSONNEL.**

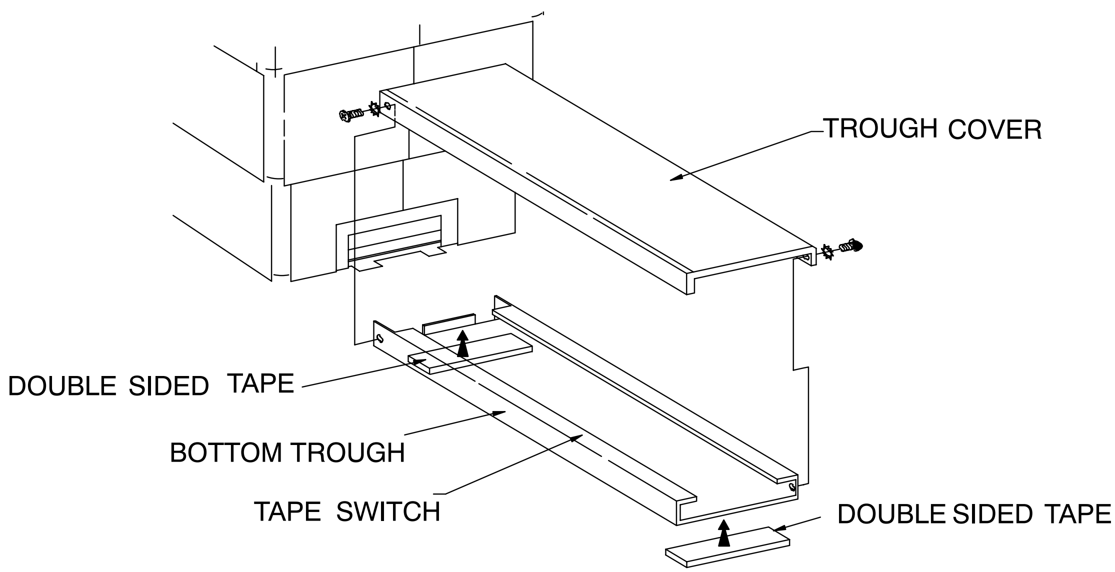


FIGURE 12 FOOT SWITCH COVER

FOOT SWITCH COVER REPLACEMENT:

33. Install the Foot Switch Cover by reversing step 32.
34. Apply Gantry power. Refer to the PowerUp/PowerDown procedure.
35. Activate the Footswitch and manually move the Table to verify Footswitch operation.
36. Using the Gantry Control Panel Switches, tilt the Gantry to zero degrees.
37. Remove Gantry power. Refer to the PowerUp/PowerDown procedure.

POWER UP / DOWN PROCEDURE

This section describes the system start–up/shutdown procedures. **Power–UP** is accomplished via the keyswitch on the Display Tower. **Power DOWN** of the entire system is accomplished via the General Utilities GUI. This is accessed via the “Service” option from the opening bootup GUI. This procedure covers the Service Application mode. *Scanning Application mode procedure is in the OP manual*.

SYSTEM POWER UP

1. Make sure the Gantry keyswitch is in the normal position. Turn the Display Tower keyswitch to the **ON** position. After the system boots, you will be prompted with the startup application display (or “4 Button Screen.”) You have **4 boot options**. Refer to Figure 13.:
   A.) Select Clinical Scanning (or do nothing) and the system will start and connect to the acquisition server. The acquisition server will then power up the AcQImage Cabinet, and if that is done successfully, the acquisition server will power up the gantry and X–ray system.
   B.) Select the Service Application option (within 15 seconds) by clicking on the Service button with the mouse pointer.
   C.) Reboot the system. **D**.) Start the Diagnostic GUI (within 15 seconds) by clicking once in the bottom left corner, then entering PICKER <CR>. (All caps). The AcQImage board LEDs will display a power up test sequence. Refer to the LED Interpretations.

FOR DIAG GUI:
Click here with the mouse, then type PICKER and press <CR>.

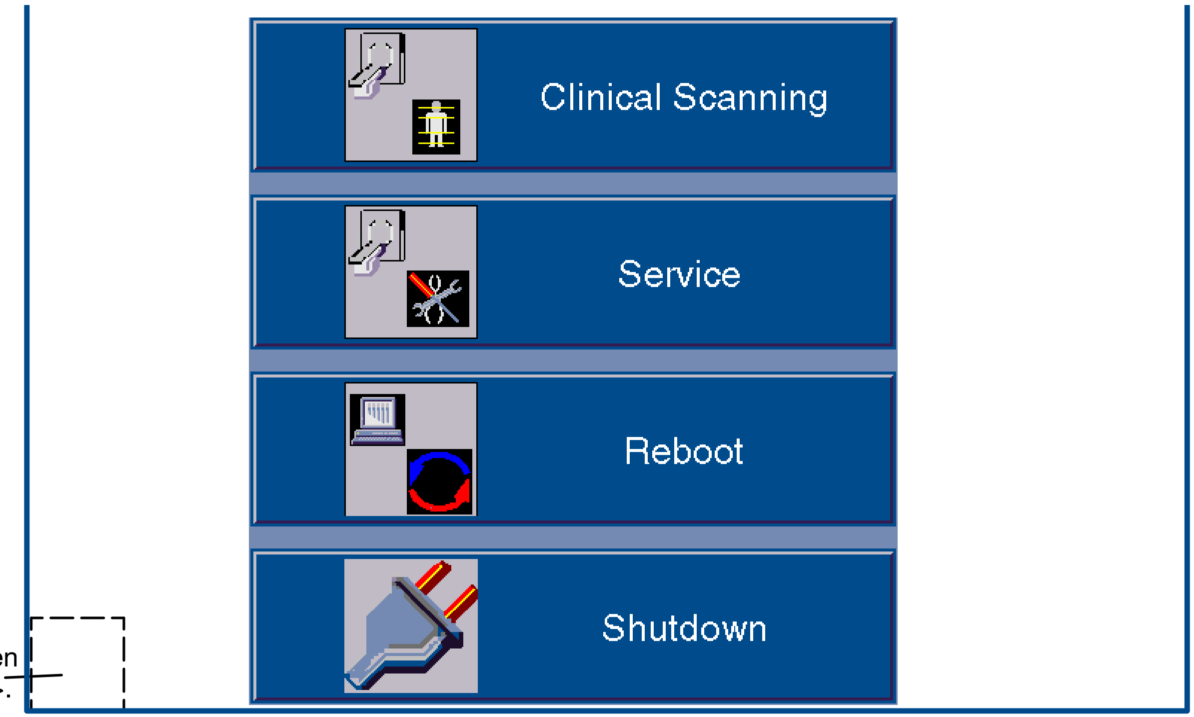


FIGURE 13 4 BUTTON SCREEN

PREPARE THE SYSTEM FOR USE (X–Ray Tube Warmup)

2. The X–Ray system needs to be properly warmed up prior to executing a scan procedure. X–Ray System Warmup must be completed at the start of every scan day or when the tube heat is below 6%. Tube heat is displayed in the Tube Warmup window.

3. To perform a tube warm–up, click on the Tube Warm Up button with the mouse pointer. Refer to Figure 13. This will initiate the warm–up sequence and **X–RAYS WILL BE EMITTED**. The tube warm–up status is displayed is displayed in the Tube Warmup Window. Refer to Figure 14.

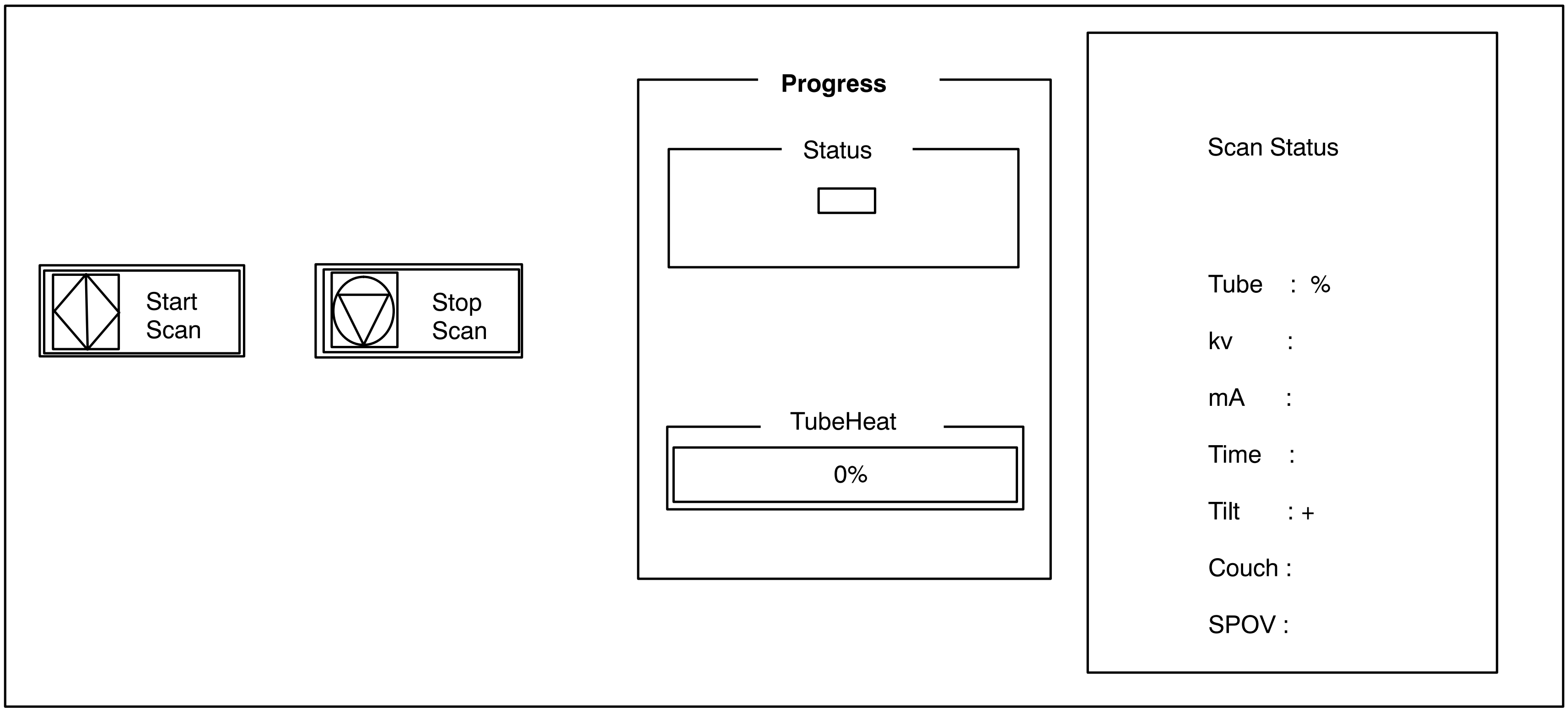


FIGURE 14 TUBE WARM–UP STATUS WINDOW

SYSTEM POWER DOWN

1. From the 4 Button Screen (Figure 13) click on SHUTDOWN with the mouse pointer.

> ⚠ **CAUTION**
>
> DO NOT POWER DOWN THE SYSTEM WITH THE TUBE HEAT AT OR ABOVE 35%. FAILURE TO COMPLY MAY RESULT IN SHORTENED TUBE LIFE AND/OR ERRATIC TUBE BEHAVIOR.

System shutdown ensures that all images on disk are properly closed, all image maintenance functions properly terminated, the gantry parked, the X–Ray system sufficiently cooled and all films printed.

There are three shutdown selections: Shutdown Gantry, Shutdown Console, Shutdown System and Switch to Console as seen in Figure 15.

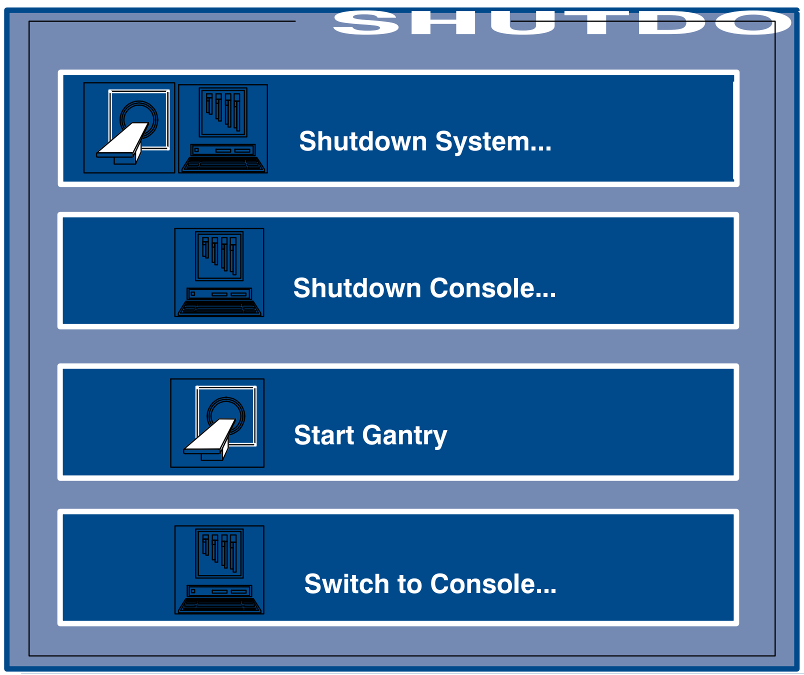


FIGURE 15 SHUTDOWN SCREEN

Shutdown Gantry: This option should be used, if possible, in place of the e–stop switches. An orderly shutdown of the gantry will be performed. Display Tower and/or AcQImage Cabinet operation may continue after selected. To cancel shutdown, click the Abort button.

Shutdown System: Performs an orderly shutdown of the entire scanner system. To cancel shutdown, click the Abort button.

Shutdown Console: This option shuts down only the console units. An orderly shutdown of the image archiver, filming and file system will be done.

Start Gantry: Toggles Start/Shutdown Gantry depending on status of Gantry power.

Switch to Console: Changes the GUI to the Console Application (Operator) screen.

⚠ WARNING ⚠

THE GUI SHUTDOWN REMOVES POWER FROM THE SYSTEM, BUT NOT AT THE INCOMING POWER LINES. DANGEROUS VOLTAGES ARE EXPOSED WHEN THE GANTRY COVERS ARE REMOVED. BEFORE SERVICING EQUIPMENT, REMOVE POWER FROM THE WALL BOX. FAILURE TO COMPLY CAN CAUSE SERIOUS INJURY OR DEATH TO PERSONNEL.

2. When prompted via a dialog box, turn the Display Tower keyswitch to the **OFF** position.
3. On the AcQImage Cabinet and Display Tower, ensure the rear breaker switch is turned off.

AcQImage Boards – Power Up Test LED Interpretations

```
AP, PB, GAM Board Diagnostic Flash Version 1.1.2

    0 Hardware Initialization
    1 CPU Version Check
    2 Diagnostic Flash CRC Check
    3 Operating System Flash CRC Check
    4 CPU Memory Test
    5 Data Cache Test
    6 Unexpected Exception Test
    7 IBC Test
    8 NVRAM Single Location Test
    9 PCI Bus Select Test
    A Ethernet Loopback Test
    B UART Loopback Test
    C apmem for AP, pbmem for PB, gamem for GAM

    C CIO Test (CP Board Diagnostic Flash Version 1.1.2)
    D Synchronous Serial Chip Loopback Test (CP Board Diagnostic Flash Version 1.1.2)
```

ACQSIM CT DICOM CONFORMANCE STATEMENT – REV. 2.0 SOFTWARE

Medical imaging devices claiming conformance to the DICOM standard must indicate in sufficient detail the service classes and information objects, as defined by the standard, to which they conform. This document details the conformance of Philips Medical Systems' AcQSim CT scanner family to the DICOM standard. The AcQSim CT scanner family requires software revision 1.0 as a minimum. This document does not attempt to detail any other Philips CT products or other medical imaging devices manufactured by Philips Medical Systems.

1 Implementation Model

This implementation provides for simple transfer of images to and from the AcQSim CT scanner family. The transfer may be started locally by an operator on the scanner, or remotely by a DICOM compliant node. This implementation also provides for filming of images to DICOM compliant printers. The AcQSim CT operator initiates a transfer of images to a remote DICOM node by selecting "Study" from the clinical scanning application's menu bar. The operator then selects "DICOM" and then "Send". A dialog box displaying a directory of the data base and the list of possible destination DICOM nodes will appear. The operator then may select any combination of individual studies, series, and images to send. The operator must also select the DICOM node to send the images to by clicking on it with the mouse. If the operator mouse clicks on the "Cancel" button, the dialog will be dismissed and no action will be taken. If the operator mouse clicks on the "Send" button, the dialog will be dismissed and the requested transfers will be started. The operator will be informed of the progress of the transfer by way of the status area in the lower left corner of the screen. An error message will be displayed if the transfers fail, and a message will be displayed as each individually selected study/series/image is transferred. Images are sent using the DICOM Storage Service Class.

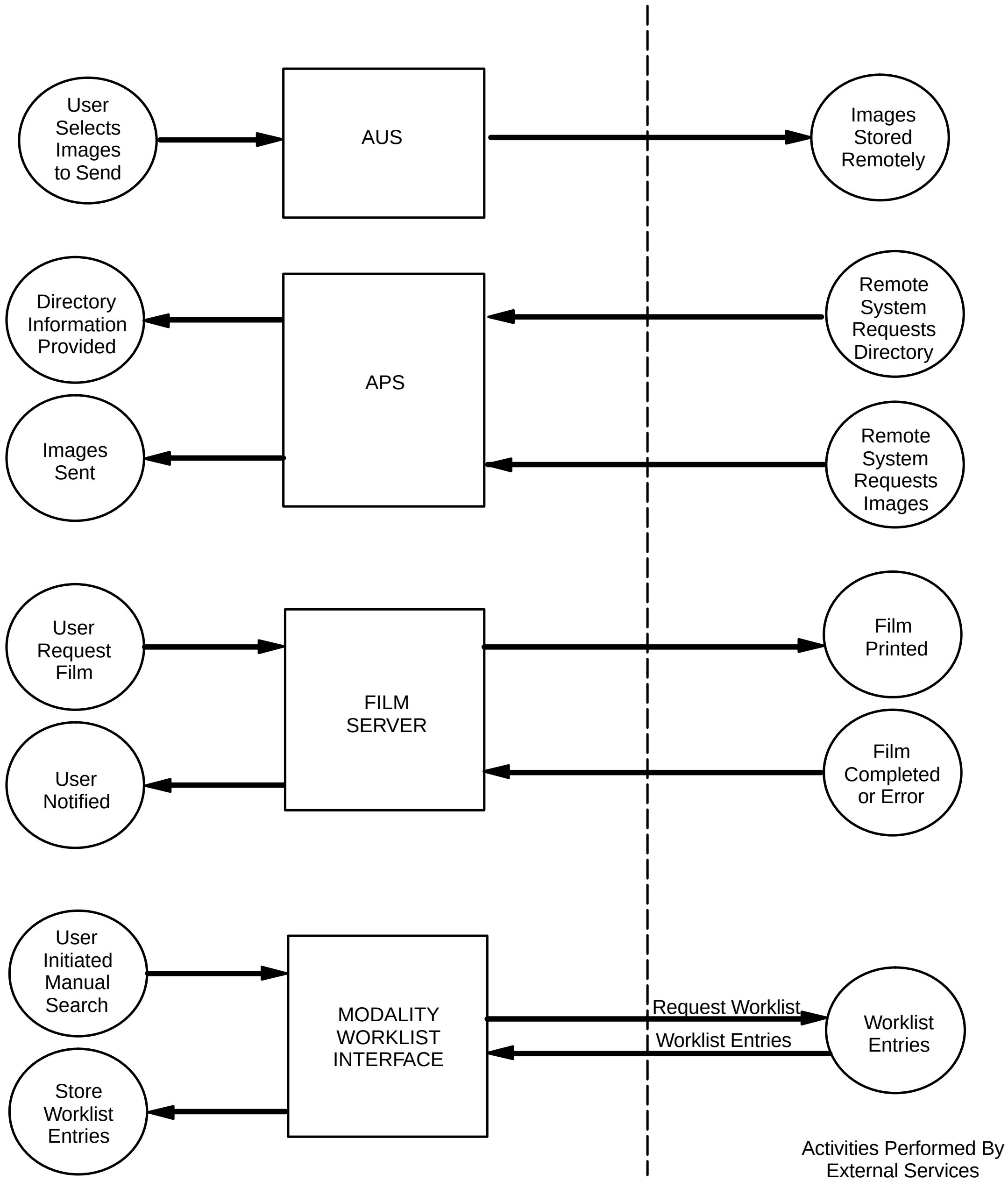
Other DICOM compliant nodes may retrieve images from the AcQSim CT using the DICOM Query/Retrieve Service class. No local operator action is required for a remote node to retrieve images.

To print to a DICOM compliant printer, the operator selects the printer from the list supplied in the "FILMING OPTIONS" dialog box. He then selects the images to print using the "Expose" key or clicking on a film cell in the film area on the left side of the screen. When the operator has selected all of the images he wishes to print, he may click on "Print" to print the images. Printing will also occur when a sheet of film is full.

Worklist requests are initiated when the operator specifically requests a worklist. The software package which runs to accomplish the worklist request and manage the subsequent return of the worklist records will be called the WORKLIST software.

To initiate a manual worklist request, the user must gain access to the worklist operations screen. From the "Study . DICOM" menu bar, the user selects the Worklist menu item to enter the "DICOM Modality Worklist" screen. From this screen, the user has the option of narrowing the search by restricting the acceptable returned values of any of the listed fields. Once the search parameters are established, the user enters the "SEARCH" softkey. The WORKLIST software sends the worklist request to the configured node. It then accepts and processes any received worklist entries.

2 Application Data Flow Diagram



One way image data is transferred from the AcQSim CT is when an operator selects a set of studies, series, or images to transfer. The software which runs to accomplish the transfer is known as the ACR–NEMA User Services (AUS).

Images may be transferred from the AcQSim CT by remote DICOM nodes using the standard DICOM Query/Retrieve Service classes. The software handles DICOM requests from remote nodes is known as the ACR–NEMA Provider Services (APS). The APS software is started at system start–up time, and will accept requests from other compliant nodes any time the system is running, subject to the availability of disk, memory, and network resources.

AcQSim CT operators may film to DICOM compliant printers. The software which controls this is known as the Film Server. The film server forms an association with a printer when an operator has signalled that a sheet of film should be printed. Once the association is made, the film server sends the required information to the printer.

2.1 Functional definitions of AE’s

The AUS software begins to run when a user selects DICOM send or DICOM receive from the menu bar. When the AUS software is invoked, an association will be established with the AE identified by the user. Images will be transferred one at a time until there are no more images to transfer.The APS software begins to run at system startup time. After initialization, the APS software waits for a connection at the port address configured for its Application Entity Title. When another node connects, the presentation and application contexts are checked to see if a valid context has been proposed. Once a valid context is proposed and the association is accepted, APS waits for C–ECHO, C–FIND and C–MOVE requests from the remote node. The association is maintained until the remote node initiates the release process or the association has been idle for a specified (configurable) amount of time.

The Film Server software also begins to run at system startup time. The Film Server does not attempt to make an association with a DICOM print device until it has all of the information needed to print a sheet of film. It then makes the association, sends the necessary information to the printer, waits for a completion status (if Print Job SOP Class is supported), and breaks the association. The Film Server may also attempt to make an association when a user selects a printer as a means of checking the printer’s availability.

When the WORKLIST software is invoked, an association will be established with the AE identified by the system configuration.

3 AE Specification

3.1 AUS software

The AUS software provides Standard Conformance to the following DICOM SOP Classes as an SCU:

SOP Class Name	SOP Class UID
Verification	1.2.840.10008.1.1
CT Image Storage	1.2.840.10008.5.1.4.1.1.2
SC Image Storage	1.2.840.10008.5.1.4.1.1.7

The AUS software never acts in the role of an SCP.

3.1.1 Association Establishment Policies

3.1.1.1 General

The AUS software attempts to establish an association when the operator has chosen a set of images to transfer. The association is maintained by AUS until the operator has pressed the cancel button, or the images specified by the operator have been transferred.

When transferring images to a remote node, the images are sent one at a time over the open association until either all of the images have transferred successfully, the remote node's Service Class Provider software reports an error, or the remote node breaks down the association. The AUS software waits for a response message after each image is sent.

The AUS software does not place any restrictions on the maximum PDU size. If the Service Class Provider for the association does not specify a maximum PDU size, the AUS software sends PDU's of not more than 4096 bytes.

3.1.1.2 Number of Associations

The AUS software creates one association to a remote DICOM node for each user request. It is possible to have more than one association to a particular remote node if the operator issues multiple requests to the DICOM software in a short period of time. There is no specific maximum to the number of concurrent associations allowed by the AUS software, but it is constrained by disk and memory resources of the PQ 2000+ scanner. Also, more than four concurrent connections may result in a performance degradation.

3.1.1.3 Asynchronous Nature

The AUS software waits for a response after each DICOM command is sent before sending the next command. Therefore, there is no asynchronous activity in this implementation.

The Asynchronous Operations Window negotiation is not supported.

3.1.1.4 Implementation Identifying Information

The AUS will provide a single Implementation Class UID which is 2.16.840.1.113662.2.4.

3.1.2 Association Initiation Policy

3.1.2.1 Associated Real World Activity

The AUS software attempts to initiate an association whenever the AcQSim CT operator selects a set of studies for transfer.

3.1.2.2 Proposed presentation contexts.

The AUS software will propose all of the following contexts whenever an association establishment is tried. However, objects will only be sent over contexts that have been agreed upon.

The implementation described here offers only the default transfer syntax (DICOM Implicit VR Little Endian)

Presentation Context Table					
Abstract Syntax		Transfer Syntax		Role	Ext. Neg.
Name	UID	Name	UID list		
CT Image Storage	1.2.840.10008.5.1.4.1.1.2	DICOM Implicit VR Big Endian	1.2.840.10008.1.2.2	SCU	None
		DICOM Implicit VR Little Endian	1.2.840.10008.1.2.1	SCU	None
		DICOM Implicit VR Little Endian	1.2.840.10008.1.2	SCU	None
SC Image Storage	1.2.840.10008.5.1.4.1.1.7	DICOM Implicit VR Big Endian	1.2.840.10008.1.2.2	SCU	None
		DICOM Implicit VR Little Endian	1.2.840.10008.1.2.1	SCU	None
		DICOM Implicit VR Little Endian	1.2.840.10008.1.2	SCU	None

3.1.2.3 SOP Specific Conformance Statement

3.1.2.3.1 Real World Activity 1 – Images Selected to Send

The operator is informed when a selected study, series, or image has been transferred. If an attempt to store an image on a remote DICOM node fails, the operator will be notified and the AUS software will attempt to send the next selected item in the list.

Warning: C–STORE Response statuses are treated the same as successful statuses.

The AcQSim CT scanner saves the following data elements for images created on the scanner. This is the set of optional elements which will be sent with an image generated on an AcQSim CT scanner. Note: Some of these fields may be absent if the data for them has not been supplied by the operator.

IOD	Attribute Name	Attribute Tag	Possible Values
Patient	Patient’s Name	(0010, 0010)	Patient’s Name or NULL
	Patient ID	(0010, 0020)	Patient ID
	Patient’s Birth Date	(0010, 0030)	NULL
	Patient’s Sex	(0010, 0040)	”M”, ”F”, ”O” or NULL
	Patient Birth Time	(0010, 1032)	NULL
	Ethnic Group	(0010, 2160)	NULL
	Patient Comments	(0010, 4000)	Comments
General Study	Study Instance UID	(0020, 000D)	UID
	Study Date	(0008, 0020)	Date
	Study Time	(0008, 0030)	Time
	Study ID	(0020, 0010)	Numeric ID for this Study
	Accession Number	(0008, 0050)	NULL
	Referring Physician’s Name	(0008, 0090)	Physician’s Name or NULL
	Name of Physician Reading Study	(0008, 1060)	Physician’s name or NULL
Patient Study	Admitting Diagnoses Description	(0008, 1080)	NULL
	Patient’s Age	(0010, 1010)	Patient’s age or NULL
	Patient’s Size	(0010, 1020)	NULL
	Patient’s Weight	(0010, 1030)	Patient’s weight or NULL
	Patient’s Occupation	(0010, 2180)	NULL
	Additional Patient’s History	(0010, 21B0)	NULL

General Series	**Modality**	**(0008, 0060)**	**"CT"**
	Series Instance UID	**(0020, 000E)**	**UID**
	Series Number	**(0020, 0011)**	**Number**
	Laterality	**(0020, 0060)**	**NULL**
	Series Date	**(0008, 0021)**	**Date**
	Series Time	**(0008, 0031)**	**Time**
	Protocol Name	**(0018, 1030)**	**Protocol Name**
	Series Description	**(0008, 103E)**	**NULL**
	Operator's Name	**(0008, 1070)**	**Operator's Name or NULL**
	Body Part Examined	**(0018, 0015)**	**Body Part**
General Equipment	**Manufacturer**	**(0008, 0070)**	**"Picker"**
	Institution Name	**(0008, 0080)**	**Name as configured by user**
	Institution Address	**(0008, 0081)**	**Address as configured by user**
	Station Name	**(0008, 1010)**	**Station name as configured by user**
	Institutional Department Name	**(0008, 1040)**	**Department name as configured by user**
	Manufacturer's Model Name	**(0008, 1090)**	**"Picker UltraZ"**
	Spatial Resolution	**(0018, 1050)**	**Number**
	Date of Last Calibration	**(0018, 1200)**	**Date**
	Time of Last Calibration	**(0018, 1201)**	**Time**
	Pixel Padding Value	**(0028, 0120)**	**0**

General Image	**Image Number**	**(0020, 0013)**	**Number**
	Patient Orientation	**(0020, 0020)**	**Combination of ”H”, ”F”, ”A”, ”P”, ”L”, and ”R”**
	Image Date	**(0008, 0023)**	**Date**
	Image Time	**(0008, 0033)**	**Time**
	Acquisition Number	**(0020, 0012)**	**Number**
	Acquisition Date	**(0008, 0022)**	**Date**
	Acquisition Time	**(0008, 0032)**	**Time**
	Derivation Description	**(0008, 2111)**	**NULL**
Image Plane	**Pixel Spacing**	**(0028, 0030)**	**Number pair**
	Image Orientation (Patient)	**(0020, 0037)**	**Direction cosines for orientation**
	Image Position (Patient)	**(0020, 0032)**	**x,y,z coordinates of first pixel**
	Slice Thickness	**(0018, 0050)**	**Number**
	Slice Location	**(0020, 1041)**	**Number**

Image Pixel	**Samples per Pixel**	**(0028, 0002)**	**1**
	Photometric Interpretation	**(0028, 0004)**	**”MONOCHROME2”**
	Rows	**(0028, 0010)**	**Pixel rows (512 or 1024 for CT images)**
	Columns	**(0028, 0011)**	**Pixel columns (512 or 1024 for CT images)**
	Bits allocated	**(0028, 0100)**	**16 (for CT Images)**
	Bits Stored	**(0028, 0101)**	**16 (for CT images)**
	High Bit	**(0028, 0102)**	**15 (for CT images)**
	Pixel Representation	**(0028, 0103)**	**1 (Two’s complement)**
	Pixel data	**(7FE0, 0010)**	**Pixel data**
	Pixel Aspect Ratio	**(0028, 0034)**	**Number pair**
	Smallest Image Pixel Value	**(0028, 0106)**	**Minimum actual pixel value in this image**
	Largest Image Pixel Value	**(0028, 0107)**	**Maximum actual pixel value in this image**
Contrast/ bolus	**Contrast/Bolus Agent**	**(0018, 0010)**	**Agent as provided by the operator**
	Contrast/Bolus Route	**(0018, 1040)**	**”IV”, ”GI”, or ”IV & GI”**
	Contrast/Bolus Volume	**(0018, 1041)**	**0**
	Contrast/Bolus Start Time	**(0018, 1042)**	**NULL**
	Contrast/Bolus Stop Time	**(0018, 1043)**	**NULL**
	Contrast/Bolus Total Dose	**(0018, 1044)**	**0**

CT Image	**Image Type**	**(0008, 0008)**	
	Samples per Pixel	**(0028, 0002)**	**1**
	Photometric Interpretation	**(0028, 0004)**	**"MONOCHROME2"**
	Bits allocated	**(0028, 0100)**	**16 (for CT images)**
	Bits Stored	**(0028, 0101)**	**16 (for CT images)**
	High Bit	**(0028, 0102)**	**15 (for CT images)**
	Rescale Intercept	**(0028, 1052)**	**0**
	Rescale Slope	**(0028, 1053)**	**1**
	KVP	**(0018, 0060)**	**Peak Voltage as specified by the protocol**
	Acquisition Number	**(0020, 0012)**	**Number**
	Scan Options	**(0018, 0022)**	**NULL**
	Reconstruction Diameter	**(0018, 1100)**	**Diameter in millimeters**
	Distance Source to Detector	**(0018, 1110)**	**Distance in millimeters**
	Distance Source to Patient	**(0018, 1111)**	**Distance in millimeters**
	Gantry/Detector Tilt	**(0018, 1120)**	**Tilt in degrees**
	Table Height	**(0018, 1130)**	**Height in millimeters**
	Rotation Direction	**(0018, 1140)**	**"CW"**
	Exposure Time	**(0018, 1150)**	**Exposure time in milliseconds**
	X–ray Tube Current	**(0018, 1151)**	**Current in mA**
	Exposure	**(0018, 1152)**	**Number**
	Generator Power	**(0018, 1170)**	**Power in kW**
VOI LUT	**Window Center**	**(0028, 1050)**	**Number**
	Window Width	**(0028, 1051)**	**Number**

SOP Common	**SOP Class UID**	**(0008, 0016)**	**1.2.840.10008.5.1.4.1.1.2 for CT** **1.2.840.10008.5.1.4.1.1.7 for SC**
	SOP Instance UID	**(0008, 0018)**	**UID**
	Specific Character Set	**(0008. 0005)**	**Character set used**
	Instance Creation Date	**(0008, 0012)**	**Date**
	Instance Creation Time	**(0008, 0013)**	**Time**

Additionally, CT images that were created by an AcQSim CT scanner will contain the following private elements.

Identifier	Attribute Tag	Type
Private Group Tag	(0011, 0010)	LO
Patient Locale	(0011, 1010)	CS
Patient Flags	(0011, 1011)	UL
Patient Trauma Flag	(0011, 1012)	UL
Patient UID	(0011, 1013)	UI
Study Locale	(0011, 1020)	CS
Study Folder	(0011, 1021)	CS
Series Locale	(0011, 1030)	CS
Series Protocol Modifications	(0011, 1031)	CS
Series Flags	(0011, 1032)	UL
Series Coronal Position	(0011, 1033)	CS
Series Frame of Reference Absolute Position	(0011, 1034)	FD
Series Frame of Reference Absolute Couch Height	(0011, 1035)	FD
Series Protocol ID	(0011, 1036)	UL
Image Compression	(0011, 1040)	CS
Image Locale	(0011, 1042)	CS

Image Flags	**(0011, 1043)**	**UL**
Pilot Tube Position (degrees)	**(0011, 1050)**	**FD**
Reconstruction filter	**(0011, 1051)**	**CS**
Spiral Interpolation filter	**(0011, 1052)**	**CS**
Reconstruction Horizontal Center	**(0011, 1053)**	**FD**
Reconstruction Vertical Center	**(0011, 1054)**	**FD**
Injector Flag	**(0011, 1055)**	**CS**
Spiral Revolutions	**(0011, 1056)**	**FD**
Pitch Value for Spiral	**(0011, 1057)**	**FD**
Pilot Edge Enhancement	**(0011, 1058)**	**CS**
Type of Pilot Scan Used	**(0011, 1059)**	**CS**
Sampling	**(0011, 105A)**	**FD**
Scan Type	**(0011, 105B)**	**CS**
CT Image Locale	**(0011, 105C)**	**CS**
CT Image Flags	**(0011, 105D)**	**UL**
Peristaltic Correction Flag	**(0011, 105E)**	**UL**
Off Focal Correction Flag	**(0011, 105F)**	**UL**
IBC Flag	**(0011, 1060)**	**UL**
MSSI Flag	**(0011, 1061)**	**UL**
Scan Angle in Degrees	**(0011, 1062)**	**FD**
MSSI Width in mm	**(0011, 1063)**	**FD**
The Slice Spacing from the Current Image Group	**(0011, 1064)**	**FD**
The ID of the Group that this Image Belongs To	**(0011, 1065)**	**UL**
The Scan Time	**(0011, 1066)**	**UL**

IBC Threshold	**(0011, 1067)**	**UL**
The Scan Field of View	**(0011, 1068)**	**CS**

3.1.3 Association Acceptance policy

The AUS software never accepts associations.

3.2 APS Specification

The APS software provides Standard Conformance to the following DICOM SOP Classes as an SCP:

SOP Class Name	SOP Class UID
Verification	1.2.840.10008.1.1
Patient Root Query/Retrieve Information Model – FIND	1.2.840.10008.5.1.4.1.2.1.1
Patient Root Query/Retrieve Information Model – MOVE	1.2.840.10008.5.1.4.1.2.1.2
Study Root Query/Retrieve Information Model – FIND	1.2.840.10008.5.1.4.1.2.2.1
Study Root Query/Retrieve Information Model – MOVE	1.2.840.10008.5.1.4.1.2.2.2
Patient/Study only Query/Retrieve Information Model – FIND	1.2.840.10008.5.1.4.1.2.3.1
Patient/Study only Query/Retrieve Information Model – MOVE	1.2.840.10008.5.1.4.1.2.3.2

The APS software never acts in the role of an SCU.

3.2.1 Association Establishment Policies

3.2.1.1 General

The APS software is started at system initialization time. After initializing, the APS waits for association requests. When a successful association is made, a new process is spawned to handle requests on the association. This process will receive one request at a time, process it, and send a response before reading the next request from the network. The process will close the association if it has been idle for more than the configured time, or if the Service Class User breaks down the connection.

The APS software does not place any restrictions on the maximum PDU size. If the Service Class User for the association does not specify a maximum PDU size, the APS software sends PDU's of not more than 4096 bytes.

3.2.1.2 Number of Associations

The APS software places no limits on the number of concurrent associations. However, more than 4 associations active concurrently will impact system performance. Other system resource limitations may also impact the maximum number of concurrent associations.

There are no additional restrictions on multiple simultaneous associations with a single AE.

3.2.1.3 Asynchronous Nature

Although there may be concurrent associations, images are processed in a serial fashion. The APS software processes each image in turn and sends a response before processing the next image. Therefore, there is no asynchronous activity in this implementation.

The Asynchronous Operation Window negotiation is not supported.

3.2.1.4 Implementation Identifying Information

APS will provide a single Implementation Class UID which is 2.16.840.1.113662.2.4

3.2.2 Association Initiation Policy

The APS software only initiates associations to perform C–STORE sub–operations for a C–MOVE request. It utilizes the AUS software described previously to make this association. Details of the association initiation policy may be seen in the AUS description above.

3.2.3 Association Acceptance Policy

There are two Real World Activities which may cause association establishment. These are: "Remote System Requests Directory", and "Remote System Requests Images". The APS software acts the same way in making the association regardless of the remote system's intent.   The association acceptance policy for both of these real world activities is grouped into "Remote connection to APS" below.

3.2.3.1 Remote connection to APS

3.2.3.1.1 Associated Real World Activity

The Associated Real World Activity is an attempt by a remote system to connect to the AcQSim CT for retrieval.

3.2.3.1.2 Acceptable presentation contexts.

The table below indicates which presentation contexts will be accepted by the APS software.

Presentation Context Table					
Abstract Syntax		Transfer Syntax		Role	Ext. Neg.
Name	UID	Name	UID List		
Patient Root Query/Retrieve Information Model – FIND	1.2.840.10008.5.1.4.1.2.1.1	DICOM Explicit VR Big Endian	1.2.840.1008.1.2.2	SCP	None
		DICOM Explicit VR Little Endian	1.2.840.10008.1.2.1		
		DICOM Implicit VR Little Endian	1.2.840.10008.1.2		
Patient Root Query/Retrieve Information Model – MOVE	1.2.840.10008.5.1.4.1.2.1.2	DICOM Explicit VR Big Endian	1.2.840.1008.1.2.2	SCP	None
		DICOM Explicit VR Little Endian	1.2.840.10008.1.2.1		
		DICOM Implicit VR Little Endian	1.2.840.10008.1.2		
Study Root Query/Retrieve Information Model – FIND	1.2.840.10008.5.1.4.1.2.2.1	DICOM Explicit VR Big Endian	1.2.840.1008.1.2.2	SCP	None
		DICOM Explicit VR Little Endian	1.2.840.10008.1.2.1		
		DICOM Implicit VR Little Endian	1.2.840.10008.1.2		
Study Root Query/Retrieve Information Model – MOVE	1.2.840.10008.5.1.4.1.2.2.2	DICOM Explicit VR Big Endian	1.2.840.1008.1.2.2	SCP	None
		DICOM Explicit VR Little Endian	1.2.840.10008.1.2.1		
		DICOM Implicit VR Little Endian	1.2.840.10008.1.2		
Patient/Study Only Query/ Retrieve Information Model – FIND	1.2.840.10008.5.1.4.1.2.3.1	DICOM Explicit VR Big Endian	1.2.840.1008.1.2.2	SCP	None
		DICOM Explicit VR Little Endian	1.2.840.10008.1.2.1		
		DICOM Implicit VR Little Endian	1.2.840.10008.1.2		
Patient/Study Only Query/ Retrieve Information Model – MOVE	1.2.840.10008.5.1.4.1.2.3.2	DICOM Explicit VR Big Endian	1.2.840.1008.1.2.2	SCP	None
		DICOM Explicit VR Little Endian	1.2.840.10008.1.2.1		
		DICOM Implicit VR Little Endian	1.2.840.10008.1.2		

3.2.3.1.2.1 SOP Specific Conformance to Verification SOP Class

APS provides standard conformance to the DICOM Verification Service Class.

3.2.3.1.2.2 SOP Specific Conformance to the Query/Retrieve Service Class

3.2.3.1.2.2.1 C–FIND Conformance

The APS software conforms to C–FIND behavior in the standard way for Patient Root Model, Study Root Model, and Patient/Study Only Model.

Relational Search is not supported.

Extended Negotiation is not supported.

These Optional Keys are supported when a C–FIND is requested on the appropriate hierarchy level:

Optional Keys Supported by APS Software		
Description	**Tag**	**Level**
Patient Sex	**(0010, 0040)**	**Patient**
Patient's Age	**(0010, 1010)**	**Patient**
Patient's Size	**(0010, 1020)**	**Patient**
Patient's Weight	**(0010, 1030)**	**Patient**
Ethnic Group	**(0010, 2160)**	**Patient**
Patient Comments	**(0010, 4000)**	**Patient**
Occupation	**(0010, 2180)**	**Patient**
Referring Physician's Name	**(0008, 0090)**	**Study**
Name of Physicians Reading Study	**(0008, 1060)**	**Study**
Admitting Diagnoses Description	**(0008,1080)**	**Study**
Patient's Age	**(0010, 1010)**	**Study**
Patient's Size	**(0010, 1020)**	**Study**

Patient's Weight	**(0010, 1030)**	**Study**
Occupation	**(0010, 2180)**	**Study**
Series Laterality	**(0020, 0060)**	**Series**
Series Date	**(0008, 0021)**	**Series**
Series Time	**(0008, 0031)**	**Series**
Performing Physician's Name	**(0008, 1050)**	**Series**
Protocol Name	**(0018, 1030)**	**Series**
Series Description	**(0008, 103E)**	**Series**
Operator's Name	**(0008, 1070)**	**Series**
Body Part Examined	**(0018, 0015)**	**Series**
Patient Position	**(0018, 5100)**	**Series**
Patient Orientation	**(0020, 0020)**	**Image**
Image Date	**(0008, 0023)**	**Image**
Image Time	**(0008, 0033)**	**Image**
Image Type	**(0008, 0008)**	**Image**
Acquisition Number	**(0020, 0012)**	**Image**
Acquisition Date	**(0008, 0022)**	**Image**
Acquisition Time	**(0008, 0032)**	**Image**
Derivation description	**(0008,2111)**	**Image**

3.2.3.1.2.2.2 C–MOVE Conformance

The APS software conforms to C–MOVE behavior in the standard way for Patient Root Model, Study Root Model, and Patient/Study Only Model.

These are the Storage Service Class SOP Classes under which the APS software supports C–STORE sub–operations generated by the C–MOVE:

SOP Class Name	SOP Class UID
CT Image Storage	1.2.840.10008.5.1.4.1.1.2
SC Image Storage	1.2.840.10008.5.1.4.1.1.7

3.2.3.1.3 Presentation Context Acceptance Criteria

APS will always accept any of the Presentation Contexts specified in section 2.2.3.1.2. As many as eleven contexts will be accepted per association.

3.2.3.1.4 Transfer Syntax Acceptance Criteria

APS accepts only the DICOM default transfer syntax (DICOM Implicit VR Little Endian).

3.3 Film Server Specification

The Film Server software provides Standard Conformance to the following DICOM SOP Classes as an SCU:

SOP Class Name	SOP Class UID
Basic Grayscale Print Management Meta SOP Class	1.2.840.10008.5.1.1.9
Basic Film Session SOP Class	1.2.840.10008.5.1.1.1
Basic Film Box SOP Class	1.2.840.10008.5.1.1.2
Basic Grayscale Image Box SOP Class	1.2.840.10008.5.1.1.4
Printer SOP Class	1.2.840.10008.5.1.1.16
Basic Color Print Management Meta SOP Class	1.2.840.10008.5.1.1.18
Basic Film Session SOP Class	1.2.840.10008.5.1.1.1
Basic Film Box SOP Class	1.2.840.10008.5.1.1.2
Basic Color Image Box SOP Class	1.2.840.10008.5.1.1.4.1
Printer SOP Class	1.2.840.10008.5.1.1.16
Print Job SOP Class	1.2.840.10008.5.1.1.14

The Film Server software never acts in the role of an SCP.

3.3.1 Association Establishment Policies

3.3.1.1 General

The Film Server software attempts to establish an association when it determines that it has the necessary data to print a sheet of film. The film server obtains data when the operator indicates that an image is to be filmed, or when images are being created and the "Auto Film" option is enabled. One or more images may be placed on a film, depending on the format selected by the user.

The Film Server software does not place any restrictions on the maximum PDU size. If the Service Class Provider for the association does not specify a maximum PDU size, the Film Server software sends PDU's of s not more than 4096 bytes.

3.3.1.2 Number of Associations

The number of associations the film server will maintain with a single printer is configurable. The default is one. The maximum number of associations the film server will support is the sum of the number of associations allowed for each configured printer.

3.3.1.3 Asynchronous Nature

For each sheet of film to be printed, the Film Server software creates an association, sets film attributes, prints the film, and releases the association. There is no asynchronous activity.

The Asynchronous Operations Window negotiation is not supported.

3.3.1.4 Implementation Identifying Information

The Film Server software will provide a single Implementation Class UID which is 2.16.840.1.113662.2.4

3.3.2 Association Initiation Policy

There are two Real World Activities which may cause association establishment. First is when an operator selects a printer, and second is when the operator has indicated to the Film Server software that a sheet of film should be printed. The implementation described here offers only the default transfer syntax (DICOM Implicit VR Little Endian).

3.3.2.1 Printer Selected

3.3.2.1.1 Associated Real World Activity

When the operator selects a printer for the first time, the film server consults its configuration to determine what level of connection testing should be done. If the configuration indicates a DICOM association should be attempted, the Film server software will request an association.

3.3.2.1.2 Proposed presentation contexts.

The Film Server software will propose Job SOP Class and either Grayscale or Color Meta SOP class based on the type of printer. Only the default transfer syntax (DICOM Implicit VR Little Endian) will be offered.

Presentation Context Table					
Abstract Syntax		Transfer Syntax		Role	Ext. Neg.
Name	UID	Name List	UID list		
Basic Grayscale Print Management Meta SOP Class	1.2.840.10008.5.1.1.9	DICOM Implicit VR Little Endian	1.2.840.10008.1.2	SCU	None
Basic Color Print Management Meta SOP Class	1.2.840.10008.5.1.1.18	DICOM Implicit VR LittleEndian	1.2.840.10008.1.2	SCU	**None**
Print Job SOP Class	**1.2.840.10008.5.1.1.14**	DICOM Implicit VR Little Endian	1.2.840.10008.1.2	SCU	**None**

3.3.2.1.3 SOP Specific conformance

As a result of this Real World Activity, the Film Server software will only create, then release the association. It does not exercise any of functionality of the SOP classes and so it provides standard conformance to the service classes it associates with as a Service Class User.

3.3.2.2 Film Sheet to be Printed.

3.3.2.2.1 Associated Real World Activity

The Film Server is informed that a sheet of film is to be printed. This may occur because the operator has depressed the ”Print” button, or enough images have been selected by the operator for a full sheet of film to be printed.

3.3.2.2.2 Proposed presentation contexts.

The Film Server software will propose Job SOP Class and either Grayscale or Color Meta SOP class based on the type of printer. Only the default transfer syntax (DICOM Implicit VR Little Endian) will be offered.

Presentation Context Table					
Abstract Syntax		Transfer Syntax		Role	Ext. Neg.
Name	UID	Name List	UID list		
Basic Grayscale Print Management Meta SOP Class	1.2.840.10008.5.1.1.9	DICOM Implicit VR Little Endian	1.2.840.10008.1.2	SCU	None
Basic Color Print Management Meta SOP Class	1.2.840.10008.5.1.1.18	DICOM Implicit VR Little Endian	1.2.840.10008.1.2	**SCU**	**None**
Print Job SOP Class	**1.2.840.10008.5.1.1.14**	DICOM Implicit VR Little Endian	1.2.840.10008.1.2	**SCU**	**None**

3.3.2.3 SOP Specific Conformance Statement

The following DIMSE Service Elements and optional attributes are used by the Film Server software:

Basic Film Session SOP Class			
DIMSE Services Used	**Optional Attribute Description**	**Tag**	**Possible Values**
N–CREATE			
	Number of Copies	(2000, 0010)	See Note Below
N–SET			
	Number of Copies	(2000, 0010)	See Note Below
N–DELETE			
N–ACTION			

Basic Film Box SOP Class			
DIMSE Services Used	**Optional Attribute Description**	**Tag**	**Possible Values**
N–CREATE			
	Magnification Type	(2010, 0060)	See Note Below
	Border Density	(2010, 0100)	See Note Below
	Trim	(2010, 0140)	See Note Below
N–SET			
	Magnification Type	(2010, 0060)	See Note Below
	Border Density	(2010, 0100)	See Note Below
	Trim	(2010, 0140)	See Note Below
N–DELETE			
N–ACTION			

Basic Grayscale Image Box SOP Class			
DIMSE Services Used	**Optional Attribute Description**	**Tag**	**Possible Values**
N–SET			
	Magnification Type	(2010, 0060)	See Note Below

Basic Color Image Box SOP Class			
DIMSE Services Used	**Optional Attribute Description**	**Tag**	**Possible Values**
N–SET			
	Magnification Type	(2010, 0060)	See Note Below

Print Job SOP Class			
DIMSE Services Used	**Optional Attribute Description**	**Tag**	**Possible Values**
N–EVENT–REPORT			
N–GET			

Basic Film Session SOP Class

Basic Film Box SOP Class

Basic Grayscale Image Box SOP Class

Basic Color Image Box SOP Class

Print Job SOP ClassNOTE: The film server software uses a file to determine what the valid values are for mandatory and optional attributes. A separate file exists for each printer type and is generated by Philips. The data in the file is based on the conformance statement provided by the manufacturer of the printer.

3.3.3 Association Acceptance policy

The Film Server software never accepts associations.

3.4 WORKLIST Specification

The WORKLIST software provides Standard Conformance to the following DICOM SOP Class as an SCU:

SOP Class Name	SOP Class UID
Modality Worklist Information Model – FIND	1.2.840.10008.5.1.4.31

The WORKLIST software never acts in the role of an SCP.

3.4.1 Association Establishment Policy

3.4.1.1 General

The WORKLIST software attempts to establish an association when the operator has chosen to search the network for worklist procedures through the access screens. Once the association is established, the WORKLIST software waits for returned worklist entries until the SCP sends an “end of worklist.”

3.4.1.2 Number of Associations

One WORKLIST application requests one association at a time.

3.4.1.3 Asynchronous Nature

No asynchronous capability.

3.4.1.4 Implementation Identifying Information

The WORKLIST software provides a single Implementation Class UID of 2.16.840.1.113662.2.1.1

3.4.2 Association Initiation Policy

The WORKLIST software attempts to initiate an associaton whenever the operator selects “SEARCH” from the PROfile Connect Operations menu.

The WORKLIST software never accepts an association request.

3.4.2.1 Search for Worklist Scheduled Procedure

3.4.2.1.1 Associated Real World Activity

The operator initiates the worklist request by selecting “SEARCH” from the WORKLIST entry under the Study menu bar.

3.4.2.1.2 Proposed Presentation Contexts

Each time an association is initiated, the WORKLIST software proposes a single Presentation Context to be used on that association.

Presentation Context Table					
Abstract Syntax		Transfer Syntax		Role	Ext. Neg.
Name	UID	Name List	UID list		
Modality Worklist Information Model – FIND	1.2.840.10008.5.1.4.31	DICOM Implicit VR Little Endian	1.2.840.10008.1.2	SCU	None

3.4.2.2 SOP Specific Conformance Statement

To request a worklist, the WORKLIST software sends a C–FIND request. This request contains the desired search criteria for determining the matching worklist entries. It also contains a specification of which fields are requested to be returned with the matching entires. The WORKLIST software then looks for one or more C–Find responses containing the matching worklist records.

For a Manual Search, the WORKLIST software requires that the operator remain in the WORKLIST screen until the search is resolved, successful or unsuccessful. If any returned procedures exist, the WORKLIST software displays the returned procedures to the operator. It is the operator's choice to select worklist entries for subsequent use on the scanner. If no procedures are returned, or a communications failure occurs, an appropriate status is displayed to the operator.

At any time during a manual search operation, the operator may abort the worklist request. The WORKLIST software cancels it's association, and the user screen closes.

Extended Negotiation is not supported.

3.4.2.2.1 Modality Worklist Information Model

The following table defines the WORKLIST software's usage of the attributes for the Modality Worklist Model. All type 1 and type 2 elements are sent as well as the listed type 3 elements.

WORKLIST FIELDS

Attribute Name	Tag	Type	Implementation Details	Displayed to user as
Scheduled Procedure Step Module				
Scheduled Procedure Step Sequence	(0040, 0100)	1	None	Not Displayed
> Scheduled Station AE Title	(0040, 0001)	1	Default to AE Title of Device*	Not Displayed
> Scheduled Procedure Step Start Date	(0040, 0002)	1		Scheduled Date
> Scheduled Procedure Step Start Time	(0040, 0003)	1		Scheduled Time
> Scheduled Procedure Step Location	(0040, 0011)	2		Scheduled Location

> Modality	(0008, 0060)	1	Default to “CT”**	Not Displayed
> Scheduled Performing Physician name	(0040, 0006)	2		Radiologist
> Scheduled Procedure Step Description	(0040, 0007)	1C		Examination Description
> Scheduled Procedure Step Action Item Code Sequence	(0040, 0008)	1C		Scheduled Action Item Code
>> Code Value	(0008, 0100)	1C		
>> Coding Scheme Designator	(0008, 0102)	1C		
>> Code Meaning	(0008, 0104)	3		
> Scheduled Procedure Step ID	(0040, 0009)	1		Scheduled Prodedure ID
Requested Procedure Module				
Requested Procedure ID	(0040, 1001)	1		Requested Procedure ID
Requested Procedure Description	(0032, 1060)	1C		Requested Procedure Description
Requested Procedure Code Sequence	(0032, 1064)	1C		Requested Procedure Code
> Code Value	(0008, 0100)	1C		
> Coding Scheme Designator	(0008, 0102)	1C		
> Code Meaning	(0008, 0104)	3		
Study Instance UID	(0020, 000D)	1		Not Displayed
Requested Procedure Comments	(0040, 1400)	3		Requested Procedure Comments
Names of Intended Recipients of Results	(0040, 1010)	3		Intended Recipients of Results

Imaging Service Request Module				
Imaging Service Request Comments	(0040, 2400)	3		Imaging Service Request Comments
Accession Number	(0008, 0050)	2		Accession Number
Requesting Physician	(0032, 1032)	2		Requesting Physician
Requesting Service	(0032, 1033)	3		Requesting Service
Referring Physician's Name	(0008, 0090)	2		Referring Physician
Visit Information Module				
Current Patient Location	(0038, 0300)	2		Current Patient Location
Patient Identification Module				
Patient's Name	(0010, 0010)	1		Last Name, First Name, Middle Initial
Patient ID	(0010, 0020)	1		Patient ID
Other Patient ID's	(0010, 1000)	3		Other Patient ID's
Patient Demographic Module				
Patient's Birth Date	(0010, 0030)	2		Birth Date
Patient's Sex	(0010, 0040)	2		Sex
Patient's Age	(0010, 1010)	3		Age
Patient's Weight	(0010, 1030)	2		Weight
Ethnic Group	(0010, 2160)	3		Ethnic Group
Patient Comments	(0010, 4000)	3		Patient Comments
Patient Medical Module				
Pregnancy Status	(0010, 21C0)	2		Pregnancy Status
Medical Alerts	(0010, 2000)	2		Medical Alerts
Additional Patient History	(0010, 21B0)	3		Additional Patient History

SOP Common Module				
Special Character Set	(0008, 0005)	1C	Default to Latin 1 character set***	Not Displayed

* This field shall not be stored in the database. This value shall be obtained from the network configuration.

** This field shall not be stored in the database. This value shall be hard–coded.

*** This field shall not be stored in the database. This value shall be obtained as a canned message.

4 Communications Profiles

4.1 Supported Communications Stacks (Parts 8,9)

The AUS, APS, and Film Server software provide DICOM 3.0 TCP/IP network communications support as defined in part 8 of the DICOM 3.0 standard.

4.2 TCP/IP Stack

The TCP/IP protocol stack is supported as implemented in Sun Microsystems Solaris 2.5

4.2.1 Physical Media Support

The following physical media connections are available:

10 Base T, 10 Base 2, 10 Base 5, 10 Base FOIRL, An AUI connector is provided for maximum flexibility.

5 Extensions/Specializations/Privatization

No extensions, privatization, or specializations are used in this implementation.

6 Configuration

6.1 AE Title/Presentation Address mapping

The mapping between AE Title and presentation context is done in two files, film_server.cfg and dicom.cfg. These files should only be modified using the service application on the AcQSim CT.

6.2 Configurable Parameters

The following local parameters are configurable using the AcQSim CT Service Application. Please consult your Philips Service representative for detailed information on using the Service Application. Default values for each field are shown in parentheses.

– Application Title used by the Film Server software (PICKER_CAM_SCU)

– Connection wait time when connecting to a printer. (60 seconds)

– Time to wait between PDUs when connected to a printer (60 seconds)

– Application title used by AUS and APS software. (DICOM_STORAGE)

– TCP/IP Port used by APS software. (104)

– Defaults to use when APS software is associated with a host that has not been configured.

– Default connection wait time. (60 seconds)

– Default maximum time to wait between PDUs. (10 seconds)

– Default maximum time to keep an idle connection up. (600 seconds)

– Default maximum PDU length. (4096 bytes)

– For each configured printer:

– Application Title (PRINT_SCP)

– Port (104)

– Camera Type

– Connection Test Level (Ping)

– DICOM Print Job SOP Class Support (No)

– Print Time Out (300 seconds)

– Read Time Out (60 seconds)

– Number of Concurrent Associations. (1)

– For each configured storage/query/retrieve device:

– Application Title (DICOM_STORAGE)

– Port (104)

– Maximum time to wait between PDUs (10 seconds)

– Maximum time to wait for a connection (60 seconds)

– Maximum time to keep up an idle association (300 seconds)

– Maximum PDU length (4096)

– Insert length to end elements (No)

The Worklist Device Configuraton allows the user to expand the date range a number of units before and/or after the current day. The date range is used to prevent scheduled procedure information not scheduled within the date range from being returned.

Please note: port numbers 6002, 6938, 6950, and 7020 are used for Philips CT proprietary protocols and should not be used for any of the DICOM ports.

In most cases, the system must be rebooted for configuration changes to take effect.

All UID's generated by the DICOM software on the AcQSim CT product are based on Philips's UID root. This root is 2.16.840.1.113662.

6.3 Support of Extended Character Sets

The AcQSim CT is capable of handling the Latin 1 character set (ISO–IR 100Latin alphabet No. 1, supplementary set). The AcQSim CT will accept, send, and display images which use this character set. The AcQSim CT will also accept and send images using other character sets, but data in them may not be displayed correctly.

SCHEMATICS

AcQSim CT schematics are available in 11x17” paper format. Order part number **T55E–2918**.